THE FINAL EXAM TEXTBOOK · ONE VOLUME, ALL NINE JSS SUBJECTS

The Final Exam Textbook

General Psychology, Psychological Assessment, Social Psychology, Personality, Physiological Psychology, Developmental Psychology, Clinical Psychology, Psychiatric Disorders and Research Methodology, in one book

Written for Sihan, for the JSS AHER MA Clinical Psychology entrance, 27 October 2026. Built from every paper and mock on the site.

COLOUR EDITION · A4, DOUBLE SIDED · 22 OCTOBER 2026

How this book works

This is the whole JSS syllabus in one book: one chapter for each of the nine subjects JSS names, in the order JSS lists them, cut so that all nine fit in under 1000 A4 pages. It is built from every past paper, question bank and practice mock on the site. Every question was sorted into the section that should teach it, and every option in it, wrong ones included, is explained somewhere in the book, because next year's paper can make a wrong option the right answer. Chapter 1 is the full-length General Psychology book (821 pages) cut down to its exam core; the full version stays in the Library for looking things up.

How to read it. Each chapter opens with a real case, lists its sections with a learning goal for each, and then teaches them in order, the way Pinel does: the question, the idea, the study behind it and who did it, then what it means. Read the High yield sections first if time is short (the tags are explained below), and do every Scan Your Brain before moving on. Each chapter ends with the case revisited, a summary of every section, a Who's who, the key terms, real past-paper questions with the answers explained, and a one-page High-yield sheet for your last read before the exam.

Key terms are in bold colour where they are defined, and the index at the back points to those pages. People are in bold where they are introduced, with their full name and years.

The boxes

EXAM TRAP

The confusable pair, the bait option, the famous misattribution. Read every one.

KEY STUDY

A classic experiment told as a story: the question, the method, the result, the criticism.

CLINICAL LINK

Where the idea matters to a clinical psychologist, and to an interview panel.

REMEMBER

The one idea in the section you must not leave without.

INDIAN CONTEXT

Indian research, Indian psychology and Indian applications.

CASE

A person whose story makes the concept concrete.

SCAN YOUR BRAIN

Fill-in-the-blank checks after every two or three sections, answers underneath. Cover the answers first.

The colour code: how often each topic is asked

Every numbered section carries a tag, and every subsection that has been asked carries dots. Both are counted from the past papers and practice mocks on the site: each question was matched to the section that teaches it, a real past-paper question scores 2 points and a mock or question-bank item 1 point.

HIGH YIELD · 23 QS ●●●	MEDIUM YIELD · 8 QS ●●	LOWER YIELD · 1 Q ●
The most-asked quarter of the book's sections (at least 62 points). Read these first and know every trap.	Asked regularly (at least 33 points). Read them fully.	Asked rarely or not yet. Still in the syllabus, so still taught in full.

"23 Qs" is the number of questions on the site that this section answers. The same tags sit next to each section in the Contents, and the page after this one shows where the marks are across the book.

Printing

Print A4, double sided, flip on the long edge. Every chapter opens on a right-hand page, the inner margin is wider for binding, and page numbers sit on the outer corner. The colour and the black and white editions have identical pages, page numbers and index, so either can be printed.

Where the marks are

3180 questions from the past papers, question banks and practice mocks on the site are taught in this book. This is how they spread across the chapters.

Chapter	Questions	From real papers	Share of the book	High-yield sections
1 General Psychology	815	402	25.6%	9 of 11
2 Psychological Assessment	336	136	10.6%	2 of 11
3 Social Psychology	136	93	4.3%	0 of 11
4 Personality	202	103	6.4%	1 of 10
5 Physiological Psychology	364	151	11.4%	2 of 11
6 Developmental Psychology	230	101	7.2%	1 of 11
7 Clinical Psychology	366	191	11.5%	4 of 10
8 Psychiatric Disorders	411	205	12.9%	4 of 11
9 Research Methodology	320	175	10.1%	2 of 11

The 12 most-asked sections

143 QS 1.6 Learning: Conditioning, Cognition and Observation 49

112 QS 1.2 Schools and Systems, the Cognitive Revolution and Indian Psychology 18

111 QS 1.7 Memory: Encoding, Storage, Retrieval and Forgetting 60

88 QS 7.4 Behaviour Therapy 659

79 QS 8.4 Schizophrenia Spectrum and Other Psychotic Disorders 766

74 QS 1.11 Emotion and Stress 94

74 QS 1.10 Motivation 86

69 QS 7.5 Cognitive and Cognitive-Behavioural Therapies, and the Third Wave 669

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CHAPTER 1

General Psychology

IN THIS CHAPTER

1.1 ●● Psychology as a Science: Roots, Founders and the First Schools: Explain how psychology grew out of philosophy and physiology into a laboratory science, and name the founders, methods and central ideas of Wundt's voluntarism, Titchener's structuralism and the functionalism of James and the American pioneers.

1.2 ●●● Schools and Systems, the Cognitive Revolution and Indian Psychology: Trace psychology's rival schools (behaviourism, Gestalt, psychoanalysis, humanistic and existential psychology) through the cognitive revolution to today's perspectives, and explain the core ideas of Indian psychology: its pioneers, Samkhya, the three gunas, Patanjali's yoga and the Vedantic self.

1.3 ●● Sensation and Psychophysics: Explain how physical energy becomes sensation, how thresholds, the psychophysical laws and signal detection theory measure it, and how vision, hearing, the chemical senses, the skin senses and the body senses work.

1.4 ●●● Perception: Organising and Interpreting the World: Explain how the brain organises and interprets sensory input, from the bottom-up and top-down theories, Gestalt laws, depth cues and constancies to illusions, motion, pattern recognition, perceptual set and perception without awareness.

1.5 ●●● Attention and Consciousness: Explain how attention selects, divides and sustains processing (the filter and attenuation models, automaticity, the Stroop effect and vigilance), and describe consciousness and its states: biological rhythms, sleep stages and disorders, dreams, hypnosis, meditation and psychoactive drugs.

1.6 ●●● Learning: Conditioning, Cognition and Observation: Define learning and explain habituation, classical and operant conditioning, the conditioning therapies, schedules of reinforcement, and cognitive and observational learning, naming who showed what and when.

1.7 ●●● Memory: Encoding, Storage, Retrieval and Forgetting: Explain how memories are encoded, stored, organised, retrieved, distorted and forgotten, using the multi-store and working memory models, the systems of long-term memory, the theories of forgetting and the classic amnesic cases.

1.8 ●●● Thinking, Problem Solving, Reasoning, Decision Making and Language: Explain how people form concepts, solve problems, reason, judge, decide and create, and how language is structured, acquired and related to thought, naming the classic studies and their traps.

1.9 ●●● Intelligence: Theories, Measurement, Heredity and Environment: Define intelligence, calculate ratio and deviation IQs, describe the Wechsler, Stanford-Binet, culture-fair and Indian tests, compare the factor, multiple, triarchic and emotional intelligence theories, interpret heritability correctly and classify intellectual disability and giftedness.

1.10 ●●● Motivation: Explain why people and animals act, from instinct, drive, arousal and incentive theories through Maslow's hierarchy, achievement motivation and self-determination theory, and apply these ideas to conflict, hunger, sex and work.

1.11 ●●● Emotion and Stress: Explain what an emotion is, compare the classic theories and brain systems of emotion, and describe how stress arises, how the body responds to it, and how coping, support and personality shape its effects on health.

Weightage. 815 of the 3180 questions mapped in this book (25.6%) test this chapter, 402 of them from real past papers. 9 of its 11 sections are High yield.

Leipzig, in the winter of 1879. In a small room of an old university building known as the Konvikt, a doctoral student named Max Friedrich sits in front of a Hipp chronoscope, a clockwork timer that measures intervals in thousandths of a second. A stimulus is presented; Friedrich presses a telegraph key. Then the rule changes: now he must press not the instant he senses the stimulus, but only once he has clearly grasped what it is. The second reaction takes longer, and the extra time, Friedrich and his supervisor reasoned, is the time the mind needs to *apperceive*, to bring an impression into the clear focus of attention. Trial after trial, the numbers go into a notebook.

His supervisor was a 47-year-old professor of philosophy, Wilhelm Wundt. When Wundt arrived in Leipzig four years earlier, the university had given him this room to store the apparatus he used in lecture demonstrations. By the winter of 1879 it had become something new: a place where students carried out original experiments on the mind and published them as research. Friedrich's study of the duration of apperception became the first dissertation to come out of that room, and historians who later combed the Leipzig archives dated the start of the work to the last weeks of 1879. That is why textbooks, and nearly every entrance paper in India, treat Leipzig in 1879 as the birth of psychology as a science.

Look closely and that one small act holds this whole chapter. The stimulus had to be sensed and then perceived as something; attention had to seize it; practice changed the times; the instruction had to be remembered; a decision released the press; and the readiness and strain of waiting were motivation and feeling in miniature. Remember Max Friedrich pressing his key. Each section that follows takes one step of that act and asks how it works.

1.1 Psychology as a Science: Roots, Founders and the First Schools

MEDIUM YIELD · 46 QS

"Psychology has a long past, but only a short history," wrote Hermann Ebbinghaus in 1908. The long past is philosophy and physiology; the short history, set out in Figure 1.1, begins in Wilhelm Wundt's Leipzig laboratory in 1879.

What psychology is ●●

The word joins the Greek *psyche* (soul) and *logos* (study); **Aristotle** (384 to 322 BCE) wrote the first treatise on the soul, *De Anima*. Later schools redefined the subject: mind for the philosophers, conscious experience for Wundt ("the Science of Mental Life" for William James in 1890), then behaviour for **William McDougall** (1905, 1908) and John B. Watson (1913). Woodworth's definition was the individual's activities in relation to the environment; the one Indian textbooks quote, "the scientific study of behaviour", is **Norman L. Munn**'s. Today psychology is *the scientific study of behaviour and mental processes*: behaviour is overt (speaking, blushing), **mental processes** internal and covert (perceiving, remembering).

Its goals are to describe, explain, predict and control (or influence); basic research seeks understanding for its own sake, applied research solves practical problems. Among its fields, **clinical psychology** assesses and treats psychological disorders, counselling psychology

helps people in the normal range with adjustment problems, and abnormal psychology studies the causes of disorders. Animal behaviour across species is the field of **comparative psychology**; **evolutionary psychology** explains behaviour as inherited adaptations shaped by natural selection, and differential psychology studies individual differences. Health, developmental, social, industrial-organisational, educational, forensic, experimental and neuropsychology study illness, the lifespan, people among others, work, schooling, the law, basic processes and the brain.

> EXAM TRAP
>
> **Mental processes** are internal and covert, not outward behaviour, overt actions or animal behaviour. The bank credits **Munn** with "the scientific study of behaviour" ("J. B. William" is no known psychologist). Animal behaviour is **comparative** psychology, not social, abnormal or differential. Clinical psychology is the **assessment and treatment of mental illness**, not industry or organisations. "Behaviour in relation to genetic inheritance" is keyed **evolutionary psychology**.

Figure 1.1 The short history of psychology at a glance. Notice how the roots (left) cluster in German physiology and British biology, and how quickly the institutions (right) followed once Wundt's laboratory opened in 1879.

Philosophical roots ●●

Philosophy asked where knowledge comes from: some is inborn, says **nativism**; all comes through the senses, says **empiricism**; reason is the chief route, says **rationalism**. **Plato** (about 427 to 347 BCE) saw learning as the recollection of inborn knowledge. Aristotle, an empiricist, gave the **laws of association**, similarity, contrast and contiguity: "Sachin" calls up "Tendulkar" by contiguity.

Rene Descartes (1596 to 1650) saw the body as a machine run by the reflex, yet gave humans a separate thinking mind that interacts with the body through the pineal gland: **dualism**. He also held **innate ideas** (God, self, infinity). His reflex led to behaviourism, and his nativism, through Kant, to Gestalt psychology.

John Locke (1632 to 1704) answered in 1690 that the mind at birth is "white paper", a **tabula rasa**, filled by sensation and reflection. Simple ideas combine into complex ones; primary qualities (size, shape, motion) lie in objects, secondary qualities (colour, taste, warmth) in the perceiver. Building all mental life from such links became the doctrine of **associationism**, founded by **David Hartley** (1749). **James Mill** (1773 to 1836) held a complex idea to be the sum of its parts, mental mechanics; his son **John Stuart Mill** (1806 to 1873) held that parts fuse into something new, as hydrogen and oxygen make water: **mental chemistry**. Before Locke, **Thomas Hobbes** (1588 to 1679) was a materialist; after him, **George Berkeley** (1685 to 1753) held that to be is to be perceived, **David Hume** (1711 to 1776) gave the laws of resemblance, contiguity and cause and effect, and **Alexander Bain** founded the journal *Mind* (1876).

German rationalists made the mind active. **Gottfried Wilhelm Leibniz** (1646 to 1716), who built reality from active units called monads, held that unnoticed *petites perceptions* (each wave in the sea's roar) add up to clear awareness, which he called apperception. **Johann Friedrich Herbart** (1776 to 1841) added a threshold of consciousness and the **apperceptive mass**, the existing ideas that must absorb a new idea for it to be understood. **Immanuel Kant** (1724 to 1804) gave the mind innate categories and denied that psychology could be a science.

On mind and body, interactionism (Descartes) lets each act on the other; **psychophysical parallelism** (Leibniz, Wundt) has them run in step without causation; epiphenomenalism (**Thomas Henry Huxley**, 1825 to 1895) makes mind a by-product of the brain; and monism admits one substance, or, in the double-aspect theory of **Baruch Spinoza** (1632 to 1677), one reality with two faces. Ontology asks what exists; epistemology, how we know.

> EXAM TRAP
>
> Tabula rasa is **Locke**; Plato, Descartes and Kant were nativists. Empiricists draw knowledge from **sensory experience**; logical reasoning and contemplation are rationalist routes, and theories are products of knowledge. A practice key says Descartes' dualism influenced **behaviourism and Gestalt psychology**, not empiricism and rationalism. Post-positivism treats knowledge as fallible; scientific realism holds unobservables real. The apperceptive mass (Herbart's term, credited to Wundt by IHBAS 2025) is **pure perception and pre-existing ideas**, not associated elements (associationism), "sensation and pure perception" or Helmholtz's unconscious inference.

Physiological and biological roots

The **Bell-Magendie law** (**Charles Bell**, 1811; **Francois Magendie**, 1822) split the spinal nerve roots into sensory (dorsal) and motor (ventral). By the doctrine of specific nerve energies of **Johannes Muller** (1801 to 1858; *Handbook of Human Physiology*, 1833 to 1840), a sensation's quality depends on the nerve excited: press on the eye and you see light. **Hermann von Helmholtz** (1821 to 1894) timed the nerve impulse at about 27 metres per second (1850), called perception unconscious inference, and built the 1802 idea of **Thomas Young** into the trichromatic theory of colour vision. Localisation ran from the phrenology of **Franz Joseph Gall** and **Johann Gaspar Spurzheim** (character read from skull bumps, about 1800) through the

ablation studies of **Pierre Flourens** (1820s) to **Paul Broca**'s speech area (1861), **Carl Wernicke**'s comprehension area (1874) and the motor cortex of **Gustav Fritsch** and **Eduard Hitzig** (1870).

Charles Darwin (1809 to 1882; *On the Origin of Species*, 1859; *The Expression of the Emotions*, 1872) gave psychology continuity between species, the question of function and an interest in variation. **Francis Galton** (1822 to 1911) measured that variation, founding the study of individual differences: *Hereditary Genius* (1869), twin studies, an anthropometric laboratory (1884), correlation, the questionnaire, word association and eugenics. **George John Romanes** founded comparative psychology on anecdotes (1882), and **Conwy Lloyd Morgan**'s canon (1894) demanded parsimony: never a higher faculty where a lower one will do.

> EXAM TRAP
>
> Helmholtz is keyed for the **trichromatic theory**; the JND is Weber's, the law of effect Thorndike's, the unconscious mind Freud's.

Measuring the mind ●●

Kant was answered at Leipzig. **Ernst Heinrich Weber** (1795 to 1878) found that the just noticeable difference is a constant fraction of the standard stimulus. **Gustav Theodor Fechner** (1801 to 1887), after an insight on 22 October 1850, made sensation grow with the logarithm of intensity ($S = k \log I$) and founded **psychophysics** in *Elements of Psychophysics* (1860). (Seeing a contour needs a minimum intensity and exposure time, not a minimum reaction time; Section 1.3.) In 1868 **Franciscus Cornelis Donders** (1818 to 1889) subtracted the time of a simple reaction from that of one with an added stage, such as identifying the stimulus, founding **mental chronometry**. **Hermann Ebbinghaus** (1850 to 1909), working alone, measured memory with nonsense syllables (1885; Section 1.7).

> EXAM TRAP
>
> The founder of psychophysics is **Fechner**, not Weber, Wundt or Ebbinghaus. Ebbinghaus, Galton and Freud each worked independently as pioneers and each devised new methods (nonsense syllables; the questionnaire and word association; free association), but Freud founded no laboratory. The Panwar bank keys (d), "(a) and (c)"; RML 2020 keyed (a), "pioneers", alone. Both statements are true, so prefer (d).

Wundt and the Leipzig laboratory ●●●

Wilhelm Maximilian Wundt (1832 to 1920), a physician who had assisted Helmholtz, announced a new science in *Principles of Physiological Psychology* (1873 to 1874), "physiological" meaning borrowed experimental method. At Leipzig his students began original research in 1879, the conventional birth of psychology, and the journal *Philosophische Studien* followed in 1881. Weber, Fechner and Helmholtz were precursors; Wundt founded the discipline by giving it a laboratory, a journal and students.

He studied **immediate experience**, sensations and simple feelings that the mind actively organises through attention and will; hence **voluntarism**. Contents in the broad field of consciousness are perceived, those in its focus apperceived: the ceiling fan is perceived, the question paper apperceived. By **creative synthesis** a whole gains new properties, as a melody ex-

ceeds its notes. His **tridimensional theory of feeling** (1896) has three axes: pleasure-displeasure, tension-relaxation and excitement-depression. His method was **introspection** under experimental control (*innere Wahrnehmung*), short judgements of signalled stimuli, not armchair self-observation (*Selbstbeobachtung*). Language, myth and custom he studied in the ten-volume **Volkerpsychologie** (folk psychology, 1900 to 1920). His students included Hall, Cattell, Titchener, Kulpe, Witmer, Munsterberg, **Emil Kraepelin** (1856 to 1926; psychiatric classification), **Charles Spearman** (1863 to 1945; factor analysis) and **Vladimir Bekhterev** (1857 to 1927; reflexology).

EXAM TRAP

The first laboratory is **Wundt's**, at **Leipzig**, not Vienna, Paris or Cambridge; Galton's (1884) was anthropometric, Weber founded none, Jung used word association clinically, and James's 1875 Harvard room was for demonstration. *The Principles of Psychology* (1890) is **James's**. Introspection belongs to **Wundt** (Watson attacked it; Wertheimer founded Gestalt) and to **structuralism** (functionalism added tests, behaviourism rejected it, psychoanalysis used free association). Spearman is to factor analysis as Wundt is to **introspection** (humanism: Maslow, Rogers). Folk psychology (RML 2024) is **Wundt's**; Pavlov found conditioning, Roentgen X-rays.

Structuralism and its rivals ●●

Edward Bradford Titchener (1867 to 1927) took a stricter version to Cornell and named it **structuralism** (1898): consciousness analysed into elements, aiming at a "periodic table" of the mind. Its elements were sensations, images and affections, varying in quality, intensity, duration, clearness and extensity; of Wundt's three feeling dimensions he kept only pleasantness-unpleasantness. Introspection was "a highly specialised form of self-observation"; its cardinal sin, the **stimulus error**, was naming the object ("an apple") instead of describing the experience (a glossy red patch). By his **context theory of meaning**, perception is a complex of sensory and imaginal processes. His society, the Experimentalists (1904), barred women.

At Wurzburg **Oswald Kulpe** (1862 to 1915) found **imageless thought**, thought without images, and showed that the task (*Aufgabe*) steers thought through what Narziss Ach called a determining tendency, creating a set (*Einstellung*): "Mysuru" brings "palace" if you want a building, "Karnataka" if a state. The act psychology of **Franz Brentano** (1874) studied mental acts and their intentionality, their pointing at objects; **Carl Stumpf** (1848 to 1936) taught the Gestalt founders; **Georg Elias Muller** (1850 to 1934) extended memory research at Gottingen. Structuralism died with Titchener in 1927: introspection could not settle disputes, and the school excluded animals, children and applied work. Its topics returned in cognitive psychology.

EXAM TRAP

Introspection is **self-observation**, not self-perception (a later social-psychology theory), self-thinking or attention. "Sensory and imaginal" perception is **Titchener**; Bruner and Schafer were "New Look" perception researchers, Kimble a learning theorist. Structuralism aimed at a **periodic table of elements**; "understanding consciousness" is too general and "behaviour, not mind" is behaviourism. Its heir is **cognitive psychology**, not neuroscience, behaviourism or functionalism.

Functionalism and the American pioneers ●●●

Americans asked what the mind is *for*. Their answer, **functionalism**, studies the mind as it functions in adapting the organism to its environment; it drew on Darwin, on pragmatism (ideas judged by their consequences) and on a demand for useful science.

William James (1842 to 1910), father of American psychology and inspiration of functionalism, wrote *The Principles of Psychology* (1890). It described the **stream of consciousness** ("It is nothing jointed; it flows"), personal, changing, continuous and selective; habit, "the enormous fly-wheel of society"; the self, split into the material, social and spiritual "Me" and the knowing "I" or pure ego; and, with **Carl Lange** (1885), the James-Lange theory of emotion (Section 1.11).

At Chicago, **John Dewey** (1859 to 1952) attacked the **reflex arc** in 1896: stimulus and response are not separate links, since a child's burn changes how the next flame is seen. **James Rowland Angell** (1869 to 1949) defined functional psychology (1906) and taught Watson; **Harvey A. Carr** (1873 to 1954) took the adaptive act as its unit. At Columbia, **James McKeen Cattell** (1860 to 1944) measured reaction time and individual differences and coined **mental test** (1890), **Robert Sessions Woodworth** (1869 to 1962) wrote *Dynamic Psychology* (1918) and added the organism to stimulus and response (S-O-R), and **Edward Lee Thorndike** (1874 to 1949) studied animal learning (Section 1.6).

Granville Stanley Hall (1844 to 1924) earned the first American doctorate in psychology (1878), opened the first American research laboratory (Johns Hopkins, 1883), founded the *American Journal of Psychology* (1887) and was first president of the American Psychological Association (APA, 1892); he called adolescence "storm and stress", explained by recapitulation (each child relives human evolution). **Lightner Witmer** (1867 to 1956) opened the first **psychological clinic** (Pennsylvania, 1896) and first printed "clinical psychology" (1907). **Hugo Munsterberg** (1863 to 1916) founded forensic and industrial psychology. **Mary Whiton Calkins** (1863 to 1930), refused a Harvard doctorate as a woman, became the first woman APA president (1905); **Margaret Floy Washburn** (1871 to 1939) was the first woman PhD in psychology (1894) and wrote *The Animal Mind*; **Leta Stetter Hollingworth** (1886 to 1939) disproved claims of female inferiority.

> **KEY STUDY: THE FIRST CASE AT THE FIRST CLINIC (1896)**
>
> A teacher brought Witmer a fourteen-year-old "chronic bad speller". Testing his vision, memory and reading rather than labelling him, Witmer found poor eyesight behind part of the problem; corrected vision and individual teaching helped. Assess first; seek the simplest cause.

Functionalism dissolved into the mainstream: Angell's student Watson dropped consciousness to found behaviourism (Section 1.2), and the functional question passed to evolutionary psychology.

EXAM TRAP

Functionalism ("how people work, play and adapt") is **William James**, not Titchener, Watson, Hall or "William Wundt"; Darwin inspired it but led no school, and Dewey or Angell is right only for the formal Chicago school. It studies the **mind** adapting the **organism**, grew from **Darwin's evolution** (not Freud, Pavlov or Gestalt laws) and gave rise to the **evolutionary perspective**. Dewey's arc is the **reflex arc**; "efferent" is only the outgoing half; "association" and "dynamic" arcs are invented. "Clinical psychology" in print: **Witmer**, not Freud, Erikson or Cattell. APA: **1892**, not 1896. First woman APA president: **Calkins**, not Washburn, Hollingworth or the psychoanalyst Karen Horney.

Zeitgeist, great persons and "fathers"

Why credit one man and one year? The **great-person theory** (**Thomas Carlyle**, 1841) credits exceptional individuals; the **Zeitgeist** ("spirit of the times") view, championed by **Edwin Garrigues Boring** (1886 to 1968), credits ripe times, pointing to multiple discovery, as when Darwin and **Alfred Russel Wallace** reached natural selection independently (1858). "Father" may mean founder, namer, chief influence or first in a country, so read the qualifier (Table 1.1). Psychology's arrival in India, through N. N. Sengupta at Calcutta (1915, 1916), is told in Section 1.2.

Table 1.1 Who is the "father" of what.

Father of	Usual answer	Common bait
Scientific, experimental psychology	Wundt	Aristotle; Fechner
Psychophysics	Fechner	Weber
Associationism	Hartley	Aristotle
Structuralism	Titchener	Wundt (in Indian texts)
American psychology; functionalism	James	Hall; Titchener (named it)
Behaviourism	Watson	Pavlov; Skinner
Clinical psychology	Witmer	Freud; Kraepelin
Individual differences	Galton	Cattell
Intelligence testing	**Alfred Binet** (1905)	Galton; Cattell
Modern psychology in India	N. N. Sengupta	G. Bose; M. V. Gopalaswamy

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Thinking and remembering are internal, covert _____ processes.
2. Locke called the mind at birth a tabula _____.
3. Herbart's store of existing ideas is the _____ mass.
4. Wundt's system, stressing will and attention, is _____.
5. Naming "an apple" instead of describing the experience is the _____ error.
6. Dewey's 1896 paper attacked the _____ concept.
7. The first psychological clinic (1896) was opened by Lightner _____.

Answers: (1) mental, (2) rasa, (3) apperceptive, (4) voluntarism, (5) stimulus, (6) reflex arc, (7) Witmer.

1.2 Schools and Systems, the Cognitive Revolution and Indian Psychology

HIGH YIELD · 112 QS

Section 1.1 ended with structuralism and functionalism. This section meets the rival schools that followed (about 1900 to 1960), the mind's return in the cognitive revolution, and the Indian traditions, which asked the same questions two thousand years earlier.

Schools, paradigms and the four forces ●●

A school of thought is a group of psychologists sharing assumptions about subject matter, method and explanation. **Thomas Samuel Kuhn** (1922 to 1996), in *The Structure of Scientific Revolutions* (1962), argued that a mature science works within one paradigm, a shared framework of problems and methods, until anomalies force a paradigm shift; a field of competing schools, as psychology was, is a **preparadigmatic science**. The humanists later numbered the schools as "forces": psychoanalysis first, behaviourism second, humanistic psychology third, transpersonal psychology fourth.

The schools divided on enduring issues, which the historian **Robert Irving Watson** (1909 to 1980) called **prescriptions** (1967): mind-body, nature-nurture, free will-determinism, mechanism-vitalism, molar-molecular, nomothetic-idiographic (general laws or the unique individual), objectivism-subjectivism and empiricism-rationalism. Rene Descartes fed two schools: his reflexes led to behaviourism, his innate ideas to Gestalt nativism.

EXAM TRAP

The **fourth force** is **transpersonal** psychology; "experiential" is not a force. **Two Watsons**: John B. founded behaviourism (1913); Robert I. proposed the prescriptions (1967).

Behaviourism: Watson, the neobehaviourists and Skinner ●●●

Behaviourism threw consciousness out of psychology altogether. Animal psychology (Thorndike's puzzle-box cats, 1898) and Russian objective psychology had prepared the ground: Pavlov's conditioned reflexes and the reflexology of **Vladimir Mikhailovich Bekhterev** (1857 to 1927), who conditioned movements.

KEY STUDY: WATSON'S MANIFESTO (1913)

John Broadus Watson (1878 to 1958) launched **behaviourism** with "Psychology as the Behaviorist Views It": psychology is "a purely objective experimental branch of natural science", its subject matter behaviour, not consciousness, and "introspection forms no essential part of its methods". Its goal is prediction and control: given the stimulus, to predict the response, and given the response, the stimulus. The same laws hold for humans and animals.

Watson treated thinking as subvocal speech, tiny movements of the larynx; this **peripheralism** puts mental life in the muscles rather than the brain. Such implicit (covert) responses cannot be seen but can be measured with instruments, so covert behaviour is "measurable but not observable", and the "mental processes" in today's definition of psychology are internal, covert processes. Infants, he held, have only three unlearned emotions, fear, rage and love; the rest are conditioned, as he and **Rosalie Rayner** tried to show with Little Albert (1920). His boast "give me a dozen healthy infants" (1925) is environmentalism at its purest.

The neobehaviourists (about 1930 to 1960) drew on logical positivism (only statements that observation can verify are meaningful) and on **operationism**, proposed by the physicist **Percy Williams Bridgman** (1882 to 1961) in 1927: a concept means the operations used to measure it, hence the operational definition. **Edward Chace Tolman** (1886 to 1959) studied molar, purposive acts and inferred intervening variables between stimulus and response, above all the **cognitive map**, a mental representation of the environment. His evidence was **latent learning**: rats that explored a maze unrewarded for ten days matched rewarded rats the day after food appeared (Tolman and Honzik, 1930). **Clark Leonard Hull** (1884 to 1952) built a hypothetico-deductive system of drive reduction in which reaction potential is habit strength multiplied by drive; **John Dollard** and **Neal Elgar Miller** (1950) recast Freud in his terms. **Edwin Ray Guthrie** (1886 to 1959) needed only contiguity and one-trial learning. **Burrhus Frederic Skinner** (1904 to 1990; *Walden Two*, 1948; *Verbal Behavior*, 1957; *Beyond Freedom and Dignity*, 1971), who studied operant behaviour, was a radical ("thoroughgoing") behaviourist: thoughts are real behaviour but not causes, whereas methodological behaviourism simply sets them aside. **Zing-Yang Kuo** (1898 to 1970) rejected instinct outright.

From the late 1950s Chomsky's attack (1959, below), instinctive drift (1961) and Kohler's insight weakened behaviourism; it left operational definitions and behaviour therapy, such as **Joseph Wolpe**'s systematic desensitisation by reciprocal inhibition (1958).

EXAM TRAP

Watson founded behaviourism (not rationalism) and defined psychology as the science of **behaviour** (not soul, mind or consciousness). One Panwar item keys **N. L. Munn**, whose textbooks popularised "the scientific study of behaviour": prefer Watson if offered, else Munn (not Woodworth, of the S-O-R formula, or Wundt). **Cognitive map and latent learning** are Tolman's, not Kohler's (insight), Harlow's (learning sets), Skinner's, Gould's (he only tested the idea in honeybees, 1986) or Bandura's. **Dollard and Miller** = early social learning; **Bandura and Walters** (1963) = later.

Gestalt psychology and Kurt Lewin ●●●

A second rebellion, in Germany, kept consciousness but denied that experience is built from elements.

KEY STUDY: WERTHEIMER'S PHI PHENOMENON (1912)

At Frankfurt, **Max Wertheimer** (1880 to 1943) flashed two lines in quick succession. About 30 milliseconds apart they looked simultaneous, about 200 apart successive, but at about 60 a single line seemed to move: the **phi phenomenon**. The motion lay in neither stimulus, so no sum of sensations could produce it. The paper founded Gestalt psychology.

Gestalt means an organised whole, and the school's slogan is that the whole is different from (popularly, greater than) the sum of its parts: a square has closedness and an inside that none of its four lines has. Field theory holds that perception, thought and behaviour are set by the whole field of forces present at the moment. Against the elements of structuralism and the reflexes of behaviourism, Gestalt studied wholes and relations and leaned nativist; its laws of organisation (Pragnanz, figure and ground) are in Section 1.4.

Wolfgang Kohler (1887 to 1967), stranded on Tenerife from 1913 to 1920, saw the chimpanzee Sultan join two sticks to reach a banana: **insight**, a sudden reorganisation of the perceptual field (German 1917; *The Mentality of Apes*, 1925). His hens learned the relation "darker" rather than a shade (**transposition**), and his isomorphism matched the structure of experience to brain fields. **Kurt Koffka** (1886 to 1941; *Growth of the Mind*, 1921; *Principles of Gestalt Psychology*, 1935) separated the geographical environment from the behavioural one, the world as perceived. Wertheimer's *Productive Thinking* (1945) set understanding against rote. Wertheimer, Kohler and Lewin fled the Nazis to America, where Koffka had settled in 1927.

Kurt Lewin (1890 to 1947) carried field theory into motivation and social life. In B = f(P, E), behaviour is a function of the person and the environment, together the **life space**: the totality of facts determining behaviour at a moment, the environment being psychological, as it exists for that person. Topological psychology maps its regions and boundaries, hodological space its paths; valences attract or repel, and vectors push. His students found that interrupted tasks are better recalled (**Bluma Zeigarnik**, 1927) and spontaneously resumed (**Maria Ovsiankina**, 1928). He coined group dynamics, compared autocratic, democratic and laissez-faire leaders (Social chapter), and founded **action research**, a cycle of planning, acting, observing and reflecting with the people affected; he is the father of modern social psychology.

Gestalt therapy is different: an experiential, here-and-now therapy of **Frederick Salomon Perls** (Fritz Perls, 1893 to 1970), Laura Perls and Paul Goodman, famous for the empty chair, in which the client addresses an imagined person or a disowned part of the self.

EXAM TRAP

Founders: Wertheimer, Kohler, Koffka (not Watson, James, Pavlov or Lewin); the school most important for perception. **Empty chair** = Gestalt therapy, not psychoanalysis, cognitive-behavioural or dialectical behaviour therapy, or person-centred therapy. Loose keys stretch "Gestalt" to cover the therapy: "here-and-now" is keyed to Gestalt psychology, and "Perls was a proponent of the Gestalt perspective" is accepted because "Gestalt focuses on wholeness in the present moment". A management stem adding that the parts relate to each other and to the whole means **systems theory**. **Life space** = person plus psychological environment, not groups. Lewin founded **action research**, not psychoanalysis, psychophysics or functionalism.

Psychoanalysis: the first force ●●

Sigmund Freud (1856 to 1939) began as a neurologist treating hysteria (paralysis or blindness without nerve damage). In Paris in 1885 **Jean-Martin Charcot** (1825 to 1893) showed him hysterical symptoms appearing and vanishing under hypnosis; **Pierre Janet** (1859 to 1947) explained hysteria as dissociation, and **Hippolyte Bernheim** (1840 to 1919) of Nancy saw hypnosis as suggestion. **Josef Breuer** (1842 to 1925) found that his patient Anna O. lost symptoms by talking about their origins, the "talking cure" or cathartic method, and their *Studies on Hysteria* (1895) began psychoanalysis. Freud turned to free association, called dreams "the royal road" to the unconscious (*The Interpretation of Dreams*, 1900) and lectured at Clark University in 1909.

His topographic model divides the mind into conscious, preconscious and unconscious; his **structural model** (1923) into the id (impulsive, irrational, seeking immediate gratification), the ego (reality) and the superego (conscience), the ego guarding against anxiety with defence mechanisms such as repression. (Freud's neurotic anxiety is the fear that id impulses will break through.) Dissenters and neo-Freudians (Personality chapter) stressed relationships and culture: Adler (individual psychology), Jung (analytical psychology, the collective unconscious), Karen Horney (**basic anxiety**, a pervasive loneliness and helplessness), Fromm (humanistic psychoanalysis), Sullivan (interpersonal psychiatry), Erikson (psychosocial stages), Anna Freud (ego defences) and Melanie Klein (object relations). From psychoanalysis, the first force, descends the psychodynamic perspective, which explains behaviour, and many disorders, by unconscious urges and the hidden struggle over their expression.

> EXAM TRAP
>
> Freud began as a **neurologist**. **Father of psychoanalysis**: Freud, not Bernheim, Adler or Jung. Psychoanalytic theory is Freud's, **psychosocial** theory Erikson's. Gratification, irrational, impulsive = **id**. Horney's **basic anxiety** is not Freud's neurotic anxiety, existential anxiety or "innate anxiety" (no such technical term).

Humanistic, existential and positive psychology ●●●

Against the determinism of psychoanalysis (instinct) and behaviourism (reward), **Abraham Harold Maslow** (1908 to 1970) launched a third force and studied self-actualisation, the fulfilment of one's potential (Section 1.10); **Carl Ransom Rogers** (1902 to 1987) described an innate actualising tendency that thrives on unconditional positive regard. **Charlotte Buhler** (1893 to 1974) added a lifespan view and **Rollo May** (1909 to 1994) existentialism. The Association for Humanistic Psychology formed in 1961 or 1962 (sources differ), and the Old Saybrook conference (1964) marked the movement's formal birth. Humanistic psychology emphasises free will, the whole person, an innate tendency to grow and reaching maximum potential, understood from inside the person's experience: **phenomenology**, after **Edmund Husserl** (1859 to 1938), describes experience by bracketing assumptions about its causes. Disorder means blocked growth; healing comes from acceptance, respect and belief in the person's potential, not from reinforcement or uncovering unconscious conflict. Critics call humanistic and phenomenological approaches vague and blind to the wider social and cultural contexts of experience.

Existential psychology starts from the givens no one escapes, which **Irvin David Yalom** (1980) named the four ultimate concerns: death, freedom, isolation and meaninglessness.

Soren Kierkegaard (1813 to 1855) called anxiety the dizziness of freedom; **Martin Heidegger** (1889 to 1976) named human existence **Dasein**, being-in-the-world (1927); **Jean-Paul Sartre** (1905 to 1980) held that existence precedes essence, so each person must keep recreating himself and meaning is never fixed. **Ludwig Binswanger** (1881 to 1966) and **Medard Boss** (1903 to 1990) built Daseinsanalysis on Heidegger. **Viktor Emil Frankl** (1905 to 1997) founded **logotherapy**: the will to meaning is the primary human motive, and its frustration brings an existential vacuum or a **noogenic neurosis**. May (*The Meaning of Anxiety*, 1950) separated normal, usable anxiety, existential anxiety included, from neurotic anxiety.

Maslow later foresaw a fourth force (1968), **transpersonal psychology**, the study of peak experiences, meditation and self-transcendence. Positive psychology, the science of positive experiences, traits and institutions, was launched by **Martin Elias Peter Seligman** (born 1942) as President of the American Psychological Association in 1998, with **Mihaly Csikszentmihalyi** (1934 to 2021), known for flow. Its themes include the PERMA model of well-being (Seligman, 2011; positive emotion, engagement, relationships, meaning and accomplishment), C. R. Snyder's hope (the will and the ways to reach a goal), which its interventions promote, and **savouring**, deliberately enhancing the enjoyment of positive experiences.

> EXAM TRAP
>
> **Third force** = humanism (Maslow, Rogers), not cognitivism or the biopsychosocial model. "Uncovering unconscious fears" is psychoanalytic, not existential. **Dasein** and **noogenic anxiety** belong to the existential school. **Will to meaning** = logotherapy, not Ellis's rational emotive behaviour therapy, acceptance and commitment therapy or the broader existential-humanistic therapy. Positive psychology was founded by **Seligman**, not Gardner (multiple intelligences), Pinker (language and evolutionary psychology) or Bandura; its interventions promote **hope**; savouring is not avoidance, relaxation or trauma work.

The cognitive revolution and today's perspectives ●●●

Other critics charged behaviourism with leaving out the mind. Tolman's maps, Kohler's insight and **Karl Spencer Lashley**'s (1890 to 1958) argument of 1948 that rapid sequences such as speech need central plans were anomalies, and the computer made inner states respectable. The symbolic birth year of the **cognitive revolution** is 1956: **Allen Newell** and **Herbert Alexander Simon**'s Logic Theorist modelled problem solving, **Noam Chomsky** (born 1928) presented formal models of language, **George Armitage Miller** (1920 to 2012) gave "the magical number seven, plus or minus two", and the Dartmouth project launched artificial intelligence. **Donald Eric Broadbent** (1926 to 1993) produced the first major information-processing theory (1958); Chomsky's 1959 review of Skinner's *Verbal Behavior* argued that children produce sentences never reinforced; and **Ulric Neisser** (1928 to 2012), the father of cognitive psychology, named the field in *Cognitive Psychology* (1967).

Cognitive psychology studies internal mental processes such as attention, memory and problem solving through the **information-processing approach**, which treats the mind as a computer passing information through stages of limited capacity. Its forerunners were **Frederic Charles Bartlett** (schemas, 1932) and **Jean Piaget** (1896 to 1980), who studied children's thinking from the 1920s. Cognitive neuroscience asks how the brain does it, and **connectionism**, or parallel distributed processing (David Rumelhart and James McClelland,

1986), stores knowledge as weights across many units working at once. **Albert Bandura**'s (1925 to 2021) self-efficacy (1977), belief in one's capacity to act, and the cognitive therapies of Albert Ellis and Aaron Beck, which target maladaptive thoughts and cognitive distortions, fed cognitive-behavioural therapy (CBT; Clinical chapter).

By the 1970s schools had given way to coexisting perspectives (Table 1.2, applied to Ravi, a Mysuru student anxious before every examination). Some lists add a developmental perspective, the study of change as people mature. Cross-cultural psychology compares processes across cultures, cultural psychology studies how culture and mind make each other up, indigenous psychology builds theory from within one culture, and **eclecticism** selects concepts and techniques from several schools to fit the problem.

Table 1.2 The contemporary perspectives, applied to one case of examination anxiety.

Perspective	Explains behaviour through	Ravi's anxiety comes from
Biological (neurobiological)	Brain, nervous system, genes	An over-reactive threat system
Evolutionary	Natural selection	An ancient alarm misfiring
Psychodynamic	Unconscious conflict, early childhood	Fear of parental disapproval
Behavioural	Stimulus-response conditioning	Failure paired with humiliation
Cognitive	Internal mental processes	Catastrophic thoughts
Humanistic	Free will and growth	Worth conditional on marks
Sociocultural	Culture and social context	Competitive family pressure
Biopsychosocial (George Engel, 1977)	All three levels together	All of these, interacting

EXAM TRAP

First major information-processing theory: Broadbent, 1958 (not Treisman, about 1960; Chomsky, 1957; or Watson, 1920). **Pioneer of cognitive psychology** against Kohler, Chomsky and Kohlberg (moral development): **Piaget**. "Behaviour is governed by the brain and nervous system" = **neurobiological**, not psychophysical (Fechner's stimulus-sensation measurement, Section 1.3) or neo-behaviouristic. Cultural differences are studied by **cross-cultural psychologists** (a Panwar key says social psychologists, the vaguer answer).

Indian psychology: the idea and its pioneers ●●●

Psychology in India is the discipline as practised here, long mostly Western psychology on Indian soil. **Indian psychology** is an approach that integrates the classical Indian traditions with psychological concepts. It is consciousness-centred; holistic, treating body, mind, spirit, family and community as one system; transformative; open to disciplined first-person methods such as meditation; and value-laden. In mental health it combines traditional practices such as yoga with modern therapy.

Modern psychology entered India at Calcutta. **Narendra Nath Sengupta** (1889 to 1944) set up the first laboratory there in 1915 and the first Department of Psychology in 1916, and founded the Indian Psychological Association (1925), whose *Indian Journal of Psychology* began in 1926; he is the father of modern psychology in India. **Girindrasekhar Bose** (1887 to 1953) founded the Indian Psychoanalytical Society (1922) and corresponded with Freud; the father of psychoanalysis in India, he proposed the **theory of opposite wishes**, in which every wish is

paired with its opposite, one expressed and one repressed. **M. V. Gopalaswamy** began psychology at Mysore in 1924; Madras (1943) and Pune (1950) followed.

Durganand Sinha (1922 to 1998) urged **indigenisation**, making psychology's concepts and methods fit Indian culture, either from without, by adapting Western tools (as in **Uma Chowdhury**'s Indian Thematic Apperception Test, about 1960), or from within, from the culture's own traditions. **Jai B. P. Sinha** proposed the **nurturant-task leader** (1980), whose warm care depends on hard work; **Udai Pareek** (1925 to 2010) studied role stress; and **Sudhir Kakar** (born 1938) applied psychoanalysis to Indian childhood and society. The Indian Psychology movement dates from the Pondicherry Manifesto (2002). **Matthijs Cornelissen** founded the Indian Psychology Institute; **Koneru Ramakrishna Rao**, **Anand C. Paranjpe** and **Ajit K. Dalal** edited the *Handbook of Indian Psychology* (2008); and Cornelissen, **Girishwar Misra** and **Suneet Varma** edited *Foundations of Indian Psychology* (2011).

> EXAM TRAP
>
> Indian psychology is not Western theory applied exclusively, folklore, or meditation without psychological context. Keys naming Bose (IHBAS) or Uma Chaudhuri (Panwar) for the **first laboratory** are wrong: it was Sengupta's; without Sengupta, choose Bose (Mangal and A. K. Singh are textbook authors; neither they nor Rathi founded a laboratory). **Sequence**: Calcutta (1916), Mysore (1924), journal (1926), Madras, Pune; a key putting the journal before Mysore is wrong. Bose's theory is **opposite wishes** (not repression, dharma-sankata or "guilt theory"). "Amitosh Mukherjee" garbles Asutosh Mookerjee, a Calcutta vice-chancellor. Kakar is keyed for Horney's basic anxiety only as the sole psychoanalyst offered ("Kamala Chaudhuri" there probably garbles Uma Chowdhury). "G. R. S. Rao" as father of Indian psychology (RML 2025) probably means K. Ramakrishna Rao.

Samkhya, the three gunas and the inner instrument ●●●

Indian philosophies (darshanas) are **astika** if they accept the Vedas (Nyaya, logic; Vaisheshika; Samkhya; Yoga; Mimamsa, ritual; Vedanta) and nastika if not (Buddhism, Jainism, the materialist Charvaka); Sufism, Islamic mysticism, lies outside the scheme.

Samkhya, attributed to the sage **Kapila** and set out in Ishvarakrishna's *Samkhya Karika*, is a dualism of **purusha**, pure consciousness, the passive, unchanging witness, and **prakriti**, primordial matter, unconscious, active and ever-changing, from which all forms of life emerge. In purusha's presence prakriti evolves, down to the five great elements (pancha mahabhutas), as Figure 1.2 shows. We suffer by mistaking prakriti's activity for the self ("I am angry"); discriminating the seer from the seen brings liberation, kaivalya.

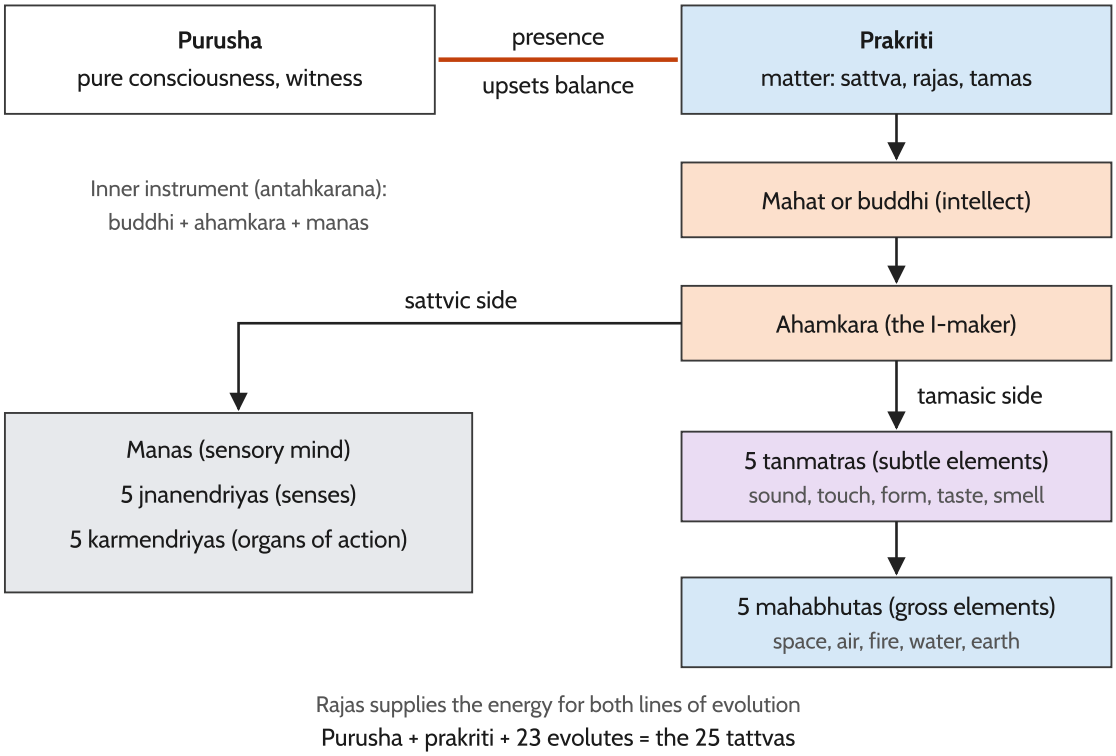

Figure 1.2 Samkhya's map of mind and matter. In the presence of purusha, prakriti evolves into intellect, the I-maker, the sensory mind, the ten organs and the elements. Notice that everything below the top row, including thinking, feeling and the sense of "I", is a product of prakriti; only purusha is conscious.

Prakriti is woven of three strands, the **gunas**, always mixed in varying proportions: sattva (clarity, calm, harmony, balance), rajas (activity, passion, desire, restlessness) and tamas (inertia, lethargy, darkness, dullness, ignorance). The *Bhagavad Gita* (chapters 14, 17 and 18) made them a psychology of personality. Before the same examination, Ananya keeps her plan and sleeps on time (sattva), Rohan flits between six guidebooks and lies awake racing (rajas), and Karthik sleeps till noon, telling himself there is no point (tamas). Nobody is pure; diet, routine and practice shift the balance.

The mind's working part is the **antahkarana**, the inner instrument: manas, the sensory mind that attends and doubts; ahamkara, the I-maker or ego; buddhi, the intellect that decides; and, in later Vedanta, chitta, the store of memory impressions.

EXAM TRAP

Purusha is **not** dynamic or ever-changing; prakriti is not pure consciousness, unchanging or the intelligent self. The gunas belong to **prakriti** (not purusha, atman, Brahman, samadhi or samskara) and come from **Samkhya** with the *Gita* (not Nyaya, Charvaka or Mimamsa). Bavisha, who has stopped eating and cannot act, shows **tamas alone** (yama is a yoga limb, not a state). Not gunas: vata (a dosha), ahimsa, prakriti, ahamkara, or "karma, dharma, moksha". The ego is **ahamkara**; the *Gita*'s three elements of mind are manas, buddhi and ahamkara, not "prakriti, purusha, kshetra" (kshetra, the field, is the body that the self knows) or "dharma, artha, kama".

Yoga, the self and other Indian paths ●●●

Samkhya says what the mind is made of; yoga says how to change it. **Patanjali**'s *Yoga Sutras* adopt Samkhya's map, and sutra 1.2 defines yoga as **chitta-vritti-nirodha**, the stilling of the mind's fluctuations; the goal is liberation (kaivalya). The five fluctuations (vrittis) are valid cognition, error, verbal imagination, sleep and memory. They are stilled by practice (abhyasa) and non-attachment (vairagya), and driven by five afflictions (kleshas): ignorance (avidya), egoism, attachment, aversion and clinging to life. The method is the eight limbs, in fixed order (Table 1.3).

Table 1.3 The eight limbs of Patanjali's yoga, in order.

Limb	Meaning
1. Yama	Ethical restraints, such as non-violence (ahimsa)
2. Niyama	Personal observances, such as purity
3. Asana	Steady, comfortable posture
4. Pranayama	Breath regulation
5. Pratyahara	Withdrawal of the senses inward
6. Dharana	Concentration on one point
7. Dhyana	Meditation: an unbroken flow of attention
8. Samadhi	Absorption in the object

A student meditating on a candle flame turns her senses away from the hostel's noise and smells (pratyahara), fixes on the flame and brings attention back whenever it drifts (dharana), finds it flowing unbroken (dhyana), and for a moment loses even "I am meditating" (samadhi).

The *Gita*'s principle of disinterestedness is **nishkama karma**, acting without attachment to the fruits. It drives karma yoga, beside the paths of knowledge (jnana) and devotion (bhakti), and answers Arjuna's dharma-sankata, a dilemma in which every action breaks some duty. The Upanishads call the true self **atman**, eternal consciousness distinct from the body and from passing states of mind; **Adi Shankaracharya**'s Advaita Vedanta (eighth century CE) makes it one with Brahman, ultimate reality, and calls the world of many things maya, appearance. The *Mandukya Upanishad*'s four states are waking (jagrat), dreaming (svapna), deep sleep (sushupti) and **turiya**, "the fourth", the consciousness that witnesses the other three. The *Taittiriya Upanishad*'s five sheaths (**panchakosha**), outer to inner, are annamaya (the food body), pranamaya (breath), manomaya (mind and senses, the sheath of perception), vijnanamaya (intellect) and anandamaya (bliss); the middle three form the subtle body (linga sharira).

Buddhism (**Siddhartha Gautama**, the Buddha, traditionally 563 to 483 BCE) teaches four Noble Truths: suffering, its cause in craving, its cessation and the Noble Eightfold Path. Its anatta denies a permanent self, its Abhidhamma catalogues states of mind, and vipassana (insight meditation) trains sati, mindfulness. Sufi psychology refines the nafs, the lower self, from the commanding ammara to the tranquil mutma'inna and the ego's passing away (fana). **Sri Aurobindo** (1872 to 1950) put the **psychic being**, the evolving soul, at the centre of integral yoga, behind an outer being of body, vital (life-energy, desire and emotion) and

mind. Ayurveda's three doshas (vata, pitta, kapha) govern the body. Karma's impressions (samskaras) work like habits, and dharma, the order of duties, heads the four aims of life (purusharthas): dharma, artha (wealth), kama (pleasure) and moksha (liberation).

CLINICAL LINK

Buddhist mindfulness underlies third-wave CBT, such as **Marsha Linehan**'s dialectical behaviour therapy, which blends CBT with Zen mindfulness and dialectical philosophy (Clinical chapter). Psychoanalysis is no wave of CBT.

EXAM TRAP

Goal of yoga: control of the mind and kaivalya, not fitness, cognitive enhancement or emotion regulation only. **Last four limbs**: pratyahara, dharana, dhyana, samadhi. One point = **dharana**; senses turned inward = **pratyahara** (not pranayama or niyama); karma and jnana are not limbs. Nishkama karma is the *principle*, karma yoga the path. **Atman** is not the body, a passing state or Jung's collective unconscious; turiya is not deep sleep. **Annamaya** is outermost, **manomaya** perceptual; the atman is never a sheath, so a key adding it to the subtle body is wrong.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Kuhn called a field of competing schools a _____ science.
2. Tolman's unrewarded rats showed _____ learning.
3. Lewin's life space is the person plus the _____ environment.
4. Frankl called neurosis from frustrated meaning _____ neurosis.
5. India's first psychology laboratory (1915) was set up by _____.
6. The guna of inertia and ignorance is _____.
7. The yoga limb that withdraws the senses inward is _____.

Answers: (1) preparadigmatic, (2) latent, (3) psychological, (4) noogenic, (5) Narendra Nath Sengupta, (6) tamas, (7) pratyahara.

1.3 Sensation and Psychophysics MEDIUM YIELD · 40 QS

Everything you know about the world arrives as physical energy. This section follows that energy into experience, shows how sensation is measured, and then takes each sense in turn.

From energy to experience ●●

In **sensation** the sense organs detect physical energy and convert it into neural signals; in **perception** the brain organises and interprets those signals. Optical illusions therefore distort perception, not sensation: in the Muller-Lyer figure the receptors register the lines correctly (Section 1.4). The structuralist **Edward Bradford Titchener** (1867 to 1927) defined a perception as a complex of sensory and imaginal elements.

A receptor is a cell specialised for one form of energy, and its conversion of that energy into neural signals is **transduction**. Because all impulses look alike, **Johannes Peter Muller** (1801 to 1858) proposed the **doctrine of specific nerve energies** (today, labelled-line coding): the quality of a sensation depends on which nerve is stimulated, not on what stimulates it, so pressing a closed eyelid makes you see light. Intensity is coded differently. By the **all-or-none law** an action potential fires at full size once threshold is reached; a stronger stimulus never makes a bigger impulse, and a sub-threshold one makes none, not a miniature or graded one. Intensity is therefore carried by rate coding (faster firing) and by recruiting more neurons. Every sense except smell relays to the cortex through the thalamus (vision via its lateral geniculate nucleus, hearing via the medial). Finally, **sensory adaptation** is the fading of response to an unchanging stimulus, which is why you soon stop feeling your wristwatch.

> EXAM TRAP
>
> Illusions distort *perception*; "translation" and "adaptation" are bait. "Sensory and imaginal" is *Titchener*, not Bruner (Section 1.4). Specific nerve energies is *Muller*, not Helmholtz. The relay for every sense but smell is the *thalamus*, not the hypothalamus (drives), medulla (breathing), corpus callosum (joins the hemispheres), basal ganglia (movement) or hippocampus (memory).

Thresholds and the psychophysical laws ●●

Gustav Theodor Fechner (1801 to 1887) founded **psychophysics**, the measurement of the relation between stimuli and sensations, with his *Elements of Psychophysics* (1860). The **absolute threshold** (or limen) is the minimum intensity a person can detect. It is not a switch: as a faint tone grows stronger, "yes" answers climb gradually along an S-shaped psychometric function, since neurons fire spontaneously. The threshold is therefore defined as the intensity detected on *50 percent* of trials; weaker stimuli are subliminal. **Eugene Galanter** (1962) gave the classic examples: a candle flame 48 kilometres away on a dark night, a watch ticking at 6 metres. The difference threshold, or **just noticeable difference** (JND), is the smallest change detected 50 percent of the time.

Ernst Heinrich Weber (1795 to 1878), comparing lifted weights in 1834, found the JND to be a constant fraction of the standard: **Weber's law**, $\Delta I/I = k$. If k for weights is 0.02, 500 grams needs 10 grams more to feel heavier, but 5 kilograms needs 100. The fraction is finest for pitch and among the coarsest for taste and smell. Fechner then assumed that every JND is an equal unit of sensation; counting JNDs gives **Fechner's law**, $S = k \log I$, so each doubling of light adds about one equal step of brightness. **Stanley Smith Stevens** (1906 to 1973) instead had people give each stimulus a number proportional to its strength (**magnitude estimation**) and found **Stevens' power law** (1957), $S = kI^n$: exponents below 1 compress (brightness, about 0.33), exponents above 1 expand (electric shock, about 3.5). Cross-modality matching (squeezing a hand grip as hard as a sound is loud) confirmed them (Table 1.4).

Table 1.4 The three psychophysical laws compared.

Law	Formula	Main limitation
Weber (1834)	$\Delta I/I = k$	Fails near threshold
Fechner (1860)	$S = k \log I$	Assumes equal JNDs; no expansion

Law	Formula	Main limitation
Stevens (1957)	$S = kI^{n}$	Swayed by range and number habits

EXAM TRAP

Minimum detectable stimulus: *absolute threshold*; smallest detectable change: *difference threshold*. "Sensory adaptation point", "perceptual boundary" and "signal detection of response" are invented; response bias is a signal detection term, and the phi phenomenon is apparent movement (Section 1.4). Weber's JND is a *constant fraction*, not a fixed amount. The equal-JND assumption is *Fechner's*, not Weber's. The father of psychophysics is Fechner, not Ernst Weber (whose JND he built on), Wilhelm Wundt (experimental psychology), Hermann Ebbinghaus (memory, inspired by Fechner) or Edward Tolman (learning).

Measuring thresholds and signal detection ●●

Fechner's **method of limits** changes the stimulus in steps, in alternating ascending and descending series. Its errors of *habituation* (keeping an answer too long) and *anticipation* (switching too early) are cancelled by counterbalancing. The **method of constant stimuli** presents fixed values many times in random order: most accurate, but slow. In the method of adjustment (average error) the observer sets the stimulus: fastest, but least precise. Modern staircases weaken the stimulus after each "yes" and strengthen it after each "no".

Yet a threshold mixes what an observer can sense with how readily the observer says "yes", as wartime radar operators showed; **signal detection theory** (SDT), set out by **David M. Green** and **John A. Swets** in 1966, separates the two. Noise alone and signal plus noise produce overlapping distributions of internal evidence, and the observer says "yes" above a cut-off, the criterion. "Yes" to a signal is a *hit*, "no" a *miss*; "yes" to noise alone is a *false alarm*, "no" a *correct rejection*. Sensitivity, or **d prime** (d′), is the distance between the distributions, *z*(hit rate) minus *z*(false alarm rate): zero when hits equal false alarms, larger as hits outnumber them. The criterion (β or *c*), or **response bias**, follows expectations and payoffs. A *liberal* observer (a smoke alarm) gives many hits and false alarms; a *conservative* one (a court) few false alarms but more misses. Plotting hit rate against false alarm rate as the criterion moves gives the **receiver operating characteristic** (ROC) curve. It is bowed, not straight as a fixed threshold predicts, which is why SDT replaced the threshold. Lie detection and claims of subliminal perception are judged the same way.

EXAM TRAP

Telling a signal from background noise is *sensitivity* (*d′*); willingness to say "yes" is the *criterion* (β); the ROC curve plots both; the JND and Weber's constant are classical psychophysics. SDT added the *response criterion*. "Receiver-operant characteristics in communication studies" (IHBAS 2025) means *audience factors influencing persuasion* (Social Psychology chapter), not the ROC curve.

Vision: light, the eye, rods and cones ●●●

The eye responds to wavelengths of about 380 to 760 *nanometres*; shorter waves are ultraviolet, longer ones infrared. Since speed equals frequency times wavelength, frequency and wavelength are *inversely* related, for light and sound alike. Wavelength is seen as hue, inten-

sity as brightness and purity as saturation. Light passes through the cornea, the pupil (sized by the iris) and the lens, which changes shape to focus (**accommodation**), to the retina. In the retina's macula lies the **fovea**, packed only with cones and giving the sharpest vision. The optic disc, where the optic nerve leaves, has no receptors and forms the **blind spot**.

About 120 million rods, with one pigment, rhodopsin, serve **scotopic vision**, vision in dim light: a rod responds to a single quantum of light, but many rods converge on each ganglion cell, so rod vision is blurred (and, with one pigment, colourless). Rods dominate the periphery and are absent from the fovea. About 6 to 7 million cones, with three pigments, serve photopic vision: colour and fine detail in moderate to bright light. Walk into a dark cinema and **dark adaptation** slowly raises your sensitivity to dim light, first in the cones and then, after the rod-cone break, in the rods as bleached rhodopsin regenerates (Figure 1.3). Light adaptation takes seconds to a minute; both are visual adaptation, adjustment to different light intensities. At dusk, as peak sensitivity shifts from cones (about 555 nanometres) to rods (500), reds darken before blues: the **Purkinje shift**, described by **Jan Evangelista Purkyne** (Purkinje) in 1825.

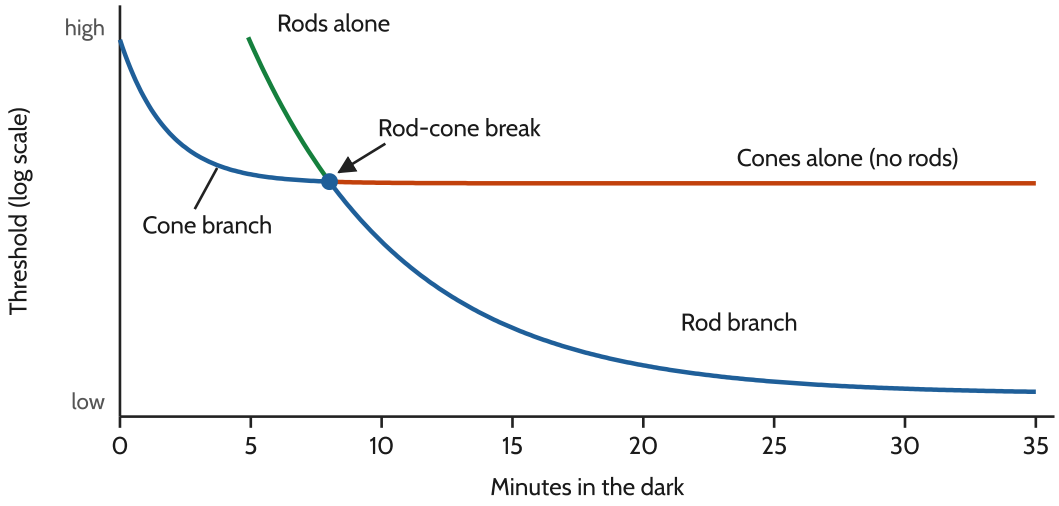

Figure 1.3 The dark adaptation curve. The threshold for detecting light falls rapidly as the cones adapt and levels off after a few minutes (cone branch). At the rod-cone break, about 7 to 10 minutes into darkness, the more sensitive rods take over and the threshold continues to fall for 20 minutes or more (rod branch). A person with no functioning rods would follow the dashed line; one with no functioning cones would show only a rod curve.

Visual acuity, the ability to see fine detail, peaks at the fovea. Rapid jumps between fixations, **saccades**, and tiny microsaccades keep the image moving on the retina; a stabilised image fades within seconds, so these movements keep perception constant. Illusory light and dark stripes at a light-dark edge, **Mach bands** (**Ernst Mach**, 1865), come from **lateral inhibition**: retinal cells suppress their neighbours, so a cell on the light side of the edge, having dark neighbours, is less inhibited and signals "brighter", exaggerating contrast.

EXAM TRAP

Scotopic vision is *rods*; the retina and iris are structures, not receptors. Cones peak in the *fovea centralis*, not the optic disc, the periphery, "the macula but not the fovea" (the fovea lies inside the macula) or the ora serrata (its front edge). "Cones give only monochromatic information" is false: rods do. Dark adaptation raises sensitivity to *low* illumination, not bright or coloured light; adjusting to light levels is *visual adaptation*, not accommodation or Piaget's assimilation. *Saccades* ensure constant perception; fixations are pauses. Mach bands come from *lateral inhibition*, not the opponent process (colour afterimages), the rod-cone dichotomy (day versus night vision) or temporal summation (summing weak signals over time).

Colour vision ●

Lights mix additively (red, green and blue light make white, as on a screen); paints mix subtractively (each absorbs wavelengths, so blue and yellow leave green). **Thomas Young** (1802) and **Hermann von Helmholtz** (1821 to 1894, in 1852) proposed the **trichromatic theory**: three receptor types whose ratio of activity codes colour, confirmed in 1964 as short-, medium- and long-wavelength cones. **Ewald Hering** (1834 to 1918) proposed the **opponent-process theory** (1878): opposed pairs of red versus green, blue versus yellow and black versus white. It explains why nobody sees a reddish green and why staring at red leaves a green negative afterimage. The dual-process theory unites them: trichromatic in the cones, opponent in later cells. In colour vision deficiency, anomalous trichromacy (one cone type shifted) is commonest; in dichromacy one cone type is missing (protanopia, long, and deuteranopia, medium, confuse red with green; tritanopia, short, blue with yellow); in monochromacy at most one works. Red-green defects are X-linked, so far commoner in men; the plates of **Shinobu Ishihara** (1917) screen for them.

EXAM TRAP

Helmholtz gave the trichromatic theory (with Young), the place theory of hearing and unconscious inference, not the JND (Weber and Fechner), the unconscious (Sigmund Freud) or the law of effect (Edward Thorndike).

Hearing ●●

A sound's frequency, in hertz (Hz), is heard as pitch (humans hear about 20 to 20,000 Hz); its amplitude as loudness, in decibels (dB), a logarithmic scale on which each 10 dB is a tenfold rise in intensity; its complexity as timbre. Sound passes from the pinna to the tympanic membrane, across the ossicles (malleus, incus, stapes) to the oval window and into the cochlea, where waves along the basilar membrane bend the hair cells of the organ of Corti. The bending opens channels mechanically, so potassium flows in and the cell depolarises.

Helmholtz's **place theory** (1863), confirmed by the travelling wave of **Georg von Bekesy** (Nobel Prize 1961), holds that each frequency peaks at its own place: high near the narrow base, low near the floppy apex. It best explains high pitches. In the **frequency theory** of **William Rutherford** (1886) the nerve fires in step with the sound, which suits low pitches, but no neuron fires much faster than 1,000 times a second. **Ernest Glen Wever** and **Charles William Bray** (1930) answered with the **volley principle**: fibres fire in staggered turns, like soldiers in rotation, following up to about 4,000 to 5,000 Hz. Today, timing codes low pitches and place

high ones. Sounds are located by interaural differences in time (low frequencies) and intensity (high). Outer or middle ear damage (wax, otosclerosis) causes conductive hearing loss; damage to hair cells or the auditory nerve (noise, ageing) causes permanent sensorineural loss.

> **EXAM TRAP**
>
> Frequencies flexing different parts of the basilar membrane is *place coding*; frequency coding is firing in time with the wave, rate coding signals loudness, and "pitch coding" is not a term. Place theory does not explain low pitches, localisation (comparing the ears) or loudness (firing rate). Hair cells transduce by *mechanical bending*; closing sodium channels happens in photoreceptors, and GABA and chemical signalling are wrong. The frequency theory is William, not Ernest, Rutherford's.

Taste and smell ●

Taste receptors sit in taste buds inside the **papillae** of the tongue (fungiform, foliate and circumvallate; filiform papillae have none and sense texture). The five basic tastes are sweet, salty, sour, bitter and **umami**, the savoury taste of glutamate named by **Kikunae Ikeda** (1908); the "tongue map" of taste zones is a myth. Smell receptors lie in the olfactory epithelium high in the nose. Pheromones, chemical signals between members of a species, are clear in many animals but doubtful in humans. Flavour is taste plus smell, which is why food tastes flat with a cold. **Hans Henning** (1916) arranged six primary odours on a smell prism and four tastes on a taste tetrahedron, but the hundreds of smell receptor types found since 1991 rule out a small set of primary odours.

> **EXAM TRAP**
>
> Papillae serve *taste*. The olfactory nerve (I) is the only sensory nerve whose pathway bypasses the thalamus; the optic (II), trigeminal (V) and vestibulocochlear (VIII) relay there.

Skin, pain and the body senses ●●

The skin senses are touch, pressure, temperature and pain. Pain is detected by free nerve endings, nociceptors, and ascends in the anterolateral system; fine touch, vibration and limb position take the dorsal columns. In the **gate control theory** of **Ronald Melzack** and **Patrick David Wall** (1965), a gate in the spinal cord's substantia gelatinosa is opened by small pain fibres, closed by large touch fibres (so rubbing a bump helps) and set from above by attention and emotion; endorphins, the body's own opiates, help to close it.

> **KEY STUDY: RAMACHANDRAN FINDS A HAND ON THE FACE (1990S)**
>
> Most amputees still feel the missing limb, a **phantom limb** (the term of **Silas Weir Mitchell**, 1871). **Vilayanur Subramanian Ramachandran** (born 1951) touched an arm amputee's face with a cotton bud, and the man felt it on specific fingers of the phantom hand. The face area of the somatosensory cortex borders the hand area, so phantoms reflect *cortical reorganisation*: neighbours invade the lost limb's territory.

Kinesthesis is the sense of limb position and movement, from proprioceptors in muscles, tendons and joints. The vestibular sense (balance) comes from the inner ear. The three **semicir-**

cular canals detect rotational acceleration, so after spinning the still-moving fluid makes you feel you are turning; the otolith organs (utricle and saccule), whose calcium carbonate crystals are pulled by gravity, detect tilt and linear acceleration.

EXAM TRAP

The anterolateral system carries *nociceptive* signals, not proprioception, vibration or touch. Phantoms are *cortical remapping*, not nerve regrowth, denial or cerebellar damage. Dizziness after spinning is the *semicircular canals*, which detect rotation in three planes, not static position or linear motion (otolith organs and their crystals), limb position (proprioceptors) or pitch (cochlea).

Senses in concert and senses deprived ●●

Sensory interaction is one sense shaping another: in the McGurk effect (Harry McGurk and John MacDonald, 1976), "ba" heard with lips saying "ga" is perceived as "da". Combining senses aids learning, so teaching for learning difficulties uses multisensory input with regular repetition. In **synaesthesia** a minority automatically and consistently experience a second sensation, such as coloured letters. Remove input and the brain makes its own: in the McGill laboratory of **Donald Olding Hebb** (Bexton, Heron and Scott, 1954), students in goggles and gloves quit within days, thought poorly and hallucinated, the classic sensory deprivation study. Likewise, **Charles Bonnet syndrome** (**Charles Bonnet**, 1760) is complex visual hallucinations in people with sight loss who are cognitively intact and keep insight.

EXAM TRAP

Charles Bonnet syndrome is not auditory hallucinations in dementia or deafness, tactile ones in alcohol withdrawal, multimodal ones in delirium or olfactory auras before seizures.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The absolute threshold is detected on _____ percent of trials.
2. SDT indexes sensitivity by _____.
3. Scotopic vision depends on the _____.
4. Mach bands are explained by lateral _____.
5. High frequencies peak near the _____ of the basilar membrane.
6. Rotational acceleration of the head is detected by the _____ canals.

Answers: (1) 50, (2) d′ (d prime), (3) rods, (4) inhibition, (5) base, (6) semicircular.

1.4 Perception: Organising and Interpreting the World HIGH YIELD · 38 QS

Section 1.3 followed light as far as the retina. This section asks what the brain does with it next, and why seeing, as its illusions show, is far less direct than it feels.

Inference or direct pickup: the theories ●●●

Perception is the selection, organisation and interpretation of sensory signals into meaningful objects and events: sensation gives you a pattern of light, perception your mother's face. **Bottom-up processing** (data-driven) builds the percept from features of the stimulus; **top-down processing** (conceptually driven) lets knowledge, expectation and context shape how those features are read. The theories differ in the weight they give each.

For **Hermann von Helmholtz** (1821 to 1894), perception was **unconscious inference**: rapid, automatic conclusions from sensory evidence plus past experience. **Richard Langton Gregory** (1923 to 2010) built this into the **constructivist approach**, the indirect or top-down theory (*Eye and Brain*, 1966), in which a percept is a best-guess hypothesis; that is why a hollow mask looks like a normal, convex face. The **Bayesian approach** puts the idea in probabilities: the percept combines prior beliefs with sensory evidence, and the weaker the evidence, the more the priors decide.

The classic bottom-up theory is the **direct perception** (ecological) approach of **James Jerome Gibson** (1904 to 1979). The light reaching a moving observer, he argued, specifies the world without inference: optic flow streams outwards from the point you are heading for, and objects offer **affordances**, possibilities for action perceived directly (a chair affords sitting). **Ulric Neisser** (1928 to 2012) joined the two directions in his perceptual cycle (1976): a schema directs exploration, and the information sampled modifies the schema.

> EXAM TRAP
>
> Unconscious inference and probabilistic reasoning define the **constructivist** approach. The computational approach (Marr, below) asks what each stage of vision computes, and the information-processing approach describes a chain of stages; neither centres on inference. Bayesian perception rests on **prior beliefs and sensory evidence**, not genes, culture or emotion. The bottom-up theorist, and the coiner of affordances, is **James J. Gibson**: Helmholtz and Gregory are top-down, Wertheimer is a Gestalt psychologist, and Gibson's wife Eleanor built the visual cliff. The apperceptive mass (Herbart's term, often credited to Wundt; Section 1.1) is not unconscious inference.

Gestalt organisation: figure, ground and grouping ●●●

No school has contributed more to perception than Gestalt psychology (Section 1.2), launched in 1912 by **Max Wertheimer** (1880 to 1943) with **Kurt Koffka** (1886 to 1941) and **Wolfgang Köhler** (1887 to 1967). Its claim is that the whole is different from the sum of its parts.

First the field splits into a **figure**, which stands out, and a ground behind it. As **Edgar Rubin** (1886 to 1951) described in 1915, the figure is thing-like (high in object character), clearly outlined, usually smaller and in front; the ground is shapeless, diffuse and seems farther away. Rubin's vase is a reversible figure: it flips into two faces although the stimulus never changes, so the perceiver does the organising.

Elements are then grouped (Figure 1.4). By the **law of proximity**, elements close together in space or time belong together. By **similarity**, alike elements group; by the **law of closure**, gaps are filled, so a broken outline is seen as a circle; by **good continuation** (continuity), elements on a smooth path group; by **common fate**, elements moving together group, like a

flock of birds. Symmetry, common region (a shared enclosure) and connectedness (joining lines) group elements too. The master law is **Prägnanz** (good figure, simplicity): every pattern is seen in the simplest, most complete organisation the conditions allow, so two overlapping rings look like two circles.

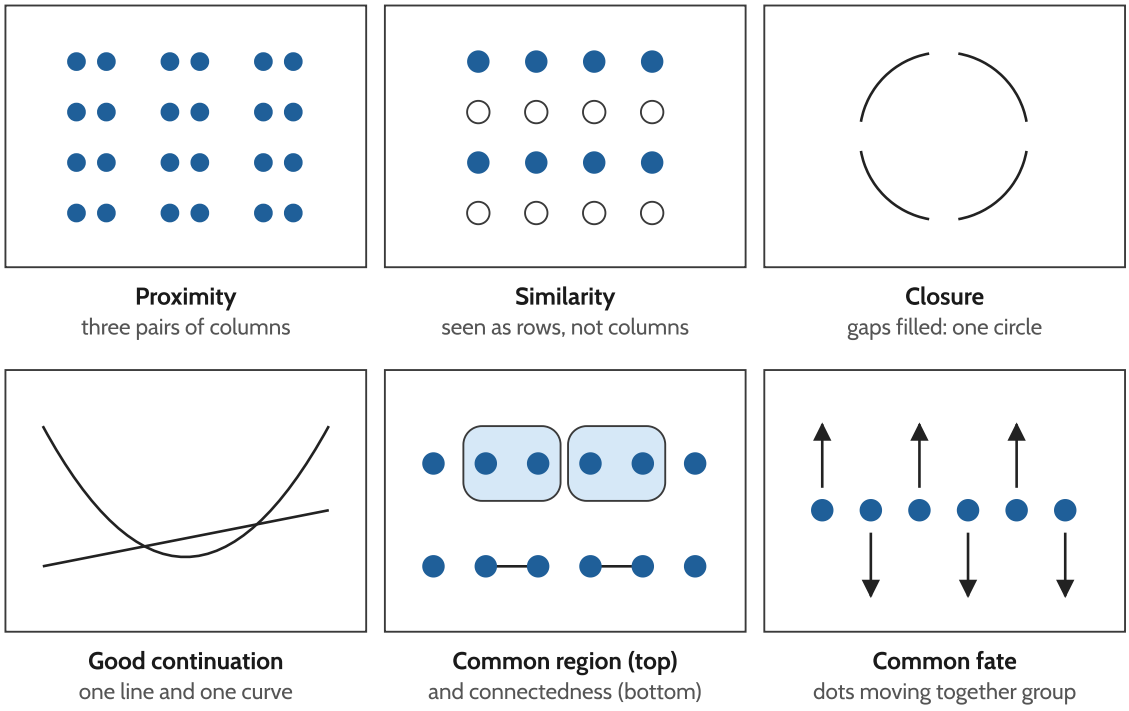

Figure 1.4 Six grouping laws you can watch in your own perception. In each panel the elements are physically identical or evenly spaced in some respect, yet you see definite groups; the grouping is supplied by your visual system.

EXAM TRAP

Close in **space and time** is **proximity**. The tendency to **complete** a figure is **closure**; continuation and continuity both name the smooth-path law. Organised "in the best possible whole form" is **Prägnanz**. The school that contributed most to perception is **Gestalt**, not structuralism, behaviourism or psychoanalysis. Interposition is a depth cue, interpolation fills in between known points, extrapolation extends beyond them; one paper keyed **structuring** for working out a rule from incomplete evidence and picking the items that fit it.

Depth cues and the visual cliff ●●●

How does a flat retinal image give you three dimensions? The brain combines depth cues, and only two are **binocular cues**, needing both eyes. In **retinal disparity**, the eyes, about 6 centimetres apart, receive slightly different views, and the nearer the object, the greater the difference; fusing the two gives the solid depth of stereopsis. In **convergence**, the eyes turn inwards to fixate a near object, and the brain registers how far they have turned.

All other cues are **monocular cues**, available to one eye (Table 1.5). Most are static pictorial cues, the ones a painter uses to put depth on a flat canvas. In the **texture gradient**, pebbles look smaller and more densely packed with distance; engineers exaggerate it by painting stripes ever closer together before a toll plaza or on an exit ramp, so that drivers feel their

speed and slow down. **Motion parallax** needs movement: from a train, nearby poles flash past while distant hills crawl.

Table 1.5 Depth cues classified.

Cue	Type	Depth signal
Retinal disparity	Binocular	Images differ more when near
Convergence	Binocular, oculomotor	Eyes turn in more when near
Accommodation	Monocular, oculomotor	Lens thickens for near objects
Motion parallax	Monocular, movement	Near objects sweep past faster
Interposition (overlap)	Monocular, pictorial	A partly hidden object is farther
Relative and familiar size	Monocular, pictorial	Smaller image, farther object
Linear perspective	Monocular, pictorial	Parallel lines converge
Texture gradient	Monocular, pictorial	Texture finer and denser
Aerial perspective	Monocular, pictorial	Distant objects hazy, bluish
Height in the field	Monocular, pictorial	Nearer the horizon, farther
Shading	Monocular, pictorial	Shadows reveal shape

KEY STUDY: THE VISUAL CLIFF (GIBSON AND WALK, 1960)

Question. Can infants perceive depth before learning it from falls? **Method. Eleanor Jack Gibson** (1910 to 2002) and **Richard Walk** (1920 to 1999) built a **visual cliff**: glass over a checkerboard lying just beneath one half (the shallow side) and far below the other (the deep side). Mothers called 36 infants aged 6 to 14 months across it. **Results.** Of the 27 who left the centre board, only three crept onto the deep side; newborn animals avoided it too. **Conclusion.** Depth perception is present by crawling age. **Criticism.** Crawlers have already had months of visual experience, so innateness is not proved.

EXAM TRAP

The binocular cues are only **retinal disparity** and **convergence**. Interposition, accommodation, motion parallax and aerial perspective are all monocular. Assimilation (Piaget's term) is no depth cue. The cliff tested **depth perception**, not object permanence, colour or imprinting, and showed it present **by crawling age**, not fully formed at birth or only after age two. One paper keyed **180 degrees** for each eye's visual field; the better figure is about 160 degrees, with about 200 for both eyes together.

Perceptual constancy ●●

Perceptual constancy is the perception of objects as stable despite changing sensory input. Hold one hand at arm's length and the other halfway to your face: the far hand's retinal image is half the size, yet both look alike. This **size constancy** follows **size-distance invariance**: perceived size is retinal size scaled by perceived distance, a rule that **Emmert's law** (1881) shows with afterimages, which look larger on a farther wall. Shape constancy keeps an opening door rectangular; brightness constancy keeps coal black in sunlight, because lightness depends on the ratio of an object's light to its surroundings'; colour constancy keeps a

banana yellow under any lamp; location constancy keeps the world still as the eyes jump in saccades. Tiny saccades also refresh the retinal image, which would otherwise fade, and so ensure constant perception.

> EXAM TRAP
>
> The outstretched hand is **size constancy**, not shape constancy, motion parallax or depth perception. Lateral inhibition (Section 1.3) **enhances** edges; it does not dull them. A fixation is the pause between saccades, and sensory adaptation the fading they prevent.

Illusions and what they reveal ●●

An **illusion** is a misinterpretation of a stimulus that really exists. The error lies in perception, not sensation: the retinal image is accurate. Indian textbooks contrast **personal illusions**, which vary with observer and occasion (a rope seen as a snake at dusk), with permanent or universal illusions, which everyone sees every time (railway tracks converging).

In the **Müller-Lyer illusion**, published in 1889 by **Franz Carl Müller-Lyer** (1857 to 1916), a shaft ending in outward fins looks longer than an equal shaft ending in arrowheads (Figure 1.5). Gregory (1963) blamed **inappropriate constancy scaling**: arrowheads suggest the near outside corner of a building, fins the far inside corner of a room, and size constancy enlarges the "farther" shaft on a flat page. **Marshall Segall**, **Donald Thomas Campbell** (1916 to 1996) and **Melville Jean Herskovits** (1895 to 1963) showed that experience matters (1963, 1966): by their **carpentered-world hypothesis**, people raised among rectangular buildings are more susceptible than people raised among round huts.

The Ponzo illusion, the Ames room (a distorted room that looks rectangular through a peephole, shrinking a person in its far corner) and the moon illusion (terrain makes the horizon moon seem farther) are misapplied size scaling too: whatever is registered as farther is seen as bigger. The Ebbinghaus illusion (Titchener circles) is size contrast, and the Zöllner, Poggendorff and Hering figures distort lines. The Necker cube is an ambiguous figure that flips in depth; the Penrose triangle is an impossible figure.

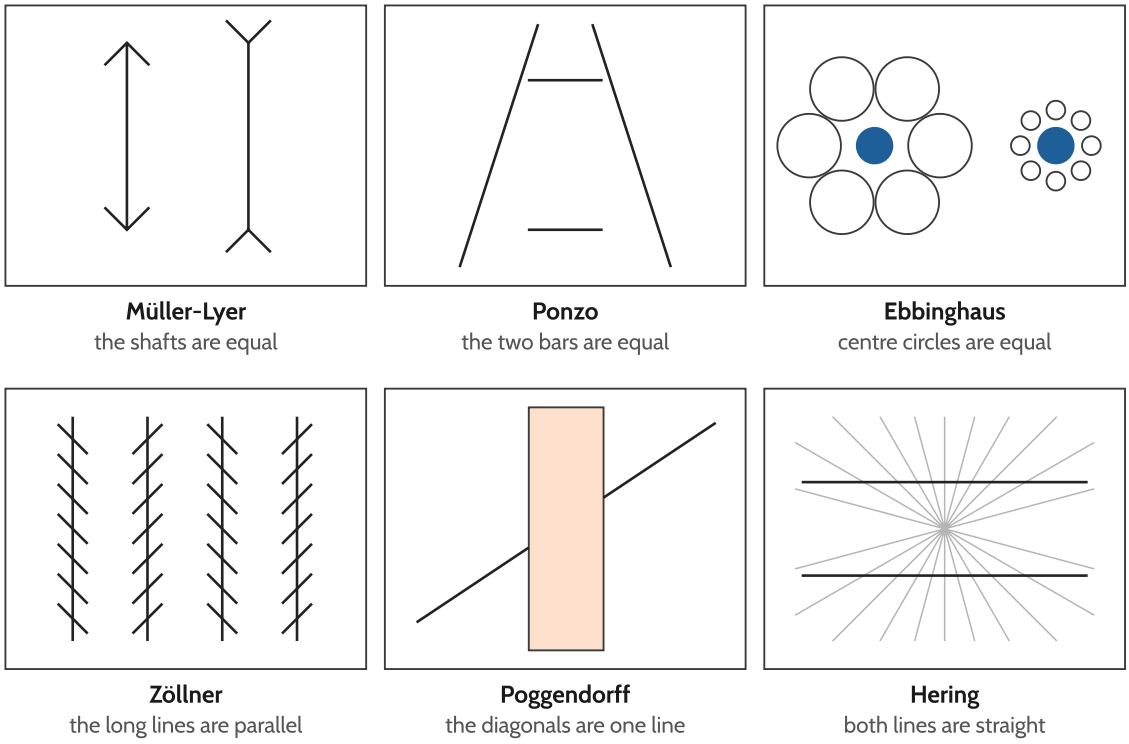

Figure 1.5 Six classic geometric illusions. In every panel the caption underneath is physically true, yet you see something else: the finned shaft looks longer, the upper bar looks longer, the circle among small circles looks larger, the long lines look tilted, the diagonal looks broken and the straight lines look bowed. Check them with a ruler.

> EXAM TRAP
>
> Rails converging for everyone is a **permanent** illusion (visual too, but the item tests the classification), not a personal one or the Müller-Lyer. Illusions distort **perception**, not sensation, adaptation or "translation" (transduction). The Müller-Lyer is strongest with **buildings with corners** and best explained by **misapplied size constancy**, not disparity (it works with one eye), lateral inhibition or figure-ground reversal.

Illusion, hallucination and delusion ●●

An illusion misinterprets a real external stimulus. A **hallucination** is a perception without an adequate external stimulus, such as a voice in an empty room. A delusion is not a perception at all but a false belief held with unshakeable conviction. Psychopathology adds the sensory distortion, a real object perceived but altered in size, intensity or quality (micropsia), and imagery, a picture in the mind's eye under voluntary control. Déjà vu is the illusion of having seen something before; déjà entendu, of having heard it before.

EXAM TRAP

Illusions are a **misinterpretation of stimuli arising from an external object**, or **distortions of existing stimuli**. "Without an adequate external stimulus" and "not based on external reality" are **hallucinations**; "a real object perceived in a distorted way" is a sensory distortion; "vivid mental images" is imagery; "innate mechanisms" fails because learning shapes illusions. Clear memory of the past is hypermnesia; involuntary repeated speech is palilalia (Psychiatric Disorders chapter).

Motion perception

Real motion is signalled by direction-selective neurons, concentrated in area MT (V5). Apparent motion is seen where nothing moves: in Wertheimer's phi phenomenon (1912) two lines flashed in quick succession look like one line moving (strictly, smooth object motion is beta movement), and films give stroboscopic motion. In induced motion a still object seems to move because its frame moves, like the moon behind racing clouds. In the **autokinetic effect** a still light in darkness seems to wander for lack of a frame of reference; **Muzafer Sherif** (1906 to 1988) used it in 1935 to study how group norms form. The motion aftereffect (waterfall illusion) is a still scene drifting opposite to a motion just watched.

Pattern and object recognition •

Recognising an A in any handwriting defeats **template matching**, comparison with exact stored copies, so feature analysis uses parts instead. In the **Pandemonium** model of **Oliver Selfridge** (1959), image demons record the pattern, feature demons shout when they find their line or curve, cognitive demons (one per letter) shout in proportion to their matching features, and a decision demon picks the loudest. **David Hubel** and **Torsten Wiesel** found such feature detectors, cells tuned to lines of one orientation, in cat visual cortex from 1959. A **prototype**, in the work of **Eleanor Rosch** (born 1938), is an idealised average member of a category. The **recognition-by-components** theory of **Irving Biederman** (1987) builds objects from about 36 three-dimensional geons, and **David Marr** (1945 to 1980) traced vision from a primal sketch through a viewer-centred 2.5-D sketch to an object-centred 3-D model. Prosopagnosia (face blindness) is a selective loss of face recognition. In the word superiority effect (**Gerald Reicher**, 1969), a letter is identified better in a word than alone: knowledge helps from the top.

Set, culture, subliminal perception and ESP ••

A **perceptual set** is a readiness to perceive in a particular way. **Jerome Seymour Bruner** (1915 to 2016) and **Leigh Minturn** (1955) found one shape read as B among letters and as 13 among numbers, a context effect. Bruner's New Look stressed needs: with **Cecile Goodman** in 1947 he found that poor ten-year-olds overestimated coin sizes more than rich ones, and **Elliott McGinnies** (1949) found that taboo words needed longer exposures (perceptual defence). Experience reshapes perception. Perceptual learning sharpens discrimination; **George Malcolm Stratton** (1865 to 1957) showed perceptual adaptation by adjusting to lenses that turned his world upside down (1896, 1897); and S.B., given sight at 52 and studied by Gregory and **Jean Wallace** (1963), could not at first judge distance. Culture matters too, as the carpentered world showed. **Herman Witkin** (1916 to 1979) found a perceptual style, **field de-**

pendence-independence (rod-and-frame and embedded figures tests): field-dependent people are swayed by the surrounding field, field-independent people separate items from it.

Subliminal perception is the processing of stimuli too faint or brief to be noticed. **James Vicary**'s 1957 "Eat Popcorn" cinema study was a hoax, admitted in 1962, but in masked semantic priming (**Anthony Marcel**, 1983) an unseen word speeds responses to a related one, and in blindsight (**Lawrence Weiskrantz**, 1974) patients with damaged primary visual cortex deny seeing yet locate targets above chance. **Extrasensory perception** claims perception with no sensory input: telepathy (mind to mind), clairvoyance (hidden objects) and precognition (the future), studied with psychokinesis (mind over matter) by parapsychology. **Joseph Banks Rhine** (1895 to 1980) used Zener cards at Duke (chance is 5 hits in 25), and later researchers the Ganzfeld procedure, guessing a target picture in a uniform sensory field. The results do not replicate, so most scientists remain sceptical.

EXAM TRAP

Semantic priming studies **subliminal perception**, not field dependence-independence (Witkin), locus of control (a personality variable) or signal detection (Section 1.3). Failing to notice obvious changes is change blindness, an attention topic (Section 1.5), not a failure of constancy. Deprivation and coin size is **Bruner and Goodman**, not Segall, Herskovits, or **K. D. Broota** and **H. C. Ganguli** (Indian work on cultural differences in perceptual selectivity, 1970s).

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Processing driven by knowledge, expectation and context is called _____ processing.
2. Gibson's term for the possibilities for action an object offers is _____.
3. The Gestalt tendency to fill the gaps in a figure is the law of _____.
4. The two binocular depth cues are retinal disparity and _____.
5. Gregory explained the Müller-Lyer illusion as inappropriate constancy _____.
6. A perception without an adequate external stimulus is a _____.
7. Biederman's recognition-by-components theory builds objects from about 36 _____.
8. Bruner and Goodman found that poor children overestimated the size of _____ more than rich children did.

Answers: (1) top-down, (2) affordances, (3) closure, (4) convergence, (5) scaling, (6) hallucination, (7) geons, (8) coins.

1.5 Attention and Consciousness HIGH YIELD · 61 QS

Attention selects the small part of the world that reaches awareness, and consciousness, the awareness itself, changes state many times a day and night: in sleep and dreams, under hypnosis, in meditation and on drugs.

What attention is: span and determinants ●

William James (1842 to 1910) defined attention in 1890 as the mind "taking possession … in clear and vivid form, of one out of what seem several simultaneously possible objects or trains of thought": selection plus concentration, needed because the brain's capacity is limited. Classical texts give it five characteristics: *selectivity*, *shifting* to a new object, a limited *span*, *fluctuation* (attention to an unchanging object waxes and wanes, as a faint ticking watch seems to fade and return) and *division* between tasks.

The **span of attention** is the number of objects apprehended in one brief glance. **William Stanley Jevons** (1871), glancing at handfuls of beans, was never wrong with three or four; **tachistoscope** studies, using displays flashed for a fraction of a second, put the span at about six to eight unrelated items. **External determinants** of attention are properties of the stimulus (intensity, size, contrast, change, movement, novelty, repetition); **internal determinants** are properties of the person (motives, interests, attitudes and **mental set**, a temporary readiness to perceive something).

> EXAM TRAP
>
> Number plates carry about four digits plus a few letters because of the **span of attention**, not selective, divided or sustained attention. Miller's memory span of seven (Section 1.7) is a different process.

Selective attention: filters, attenuators and spotlights ●●●

The British engineer **Colin Cherry** (1953) named the cocktail party problem: how do we follow one conversation among many? Hearing your own name from a conversation you were ignoring is the **cocktail party phenomenon**.

> KEY STUDY: CHERRY'S SHADOWING EXPERIMENTS (1953)
>
> In **dichotic listening** a different message goes to each ear, and the listener repeats one aloud as it arrives (**shadowing**). People recalled almost nothing of the unattended message's meaning, missing even a switch from English to German, yet noticed physical changes such as a male voice turning female. **Neville Moray** (1959) then found that a word list repeated 35 times in that ear left no memory, yet a third of listeners noticed their own name there.

Donald Broadbent's **filter model** (1958), the first major information-processing theory, passes one message on a physical feature (ear, pitch) and blocks the rest before meaning is analysed: **early selection**, with the filter as a **bottleneck**. **Anne Treisman**'s **attenuation model** (1964) turns unattended messages down, not off. Each word in a mental dictionary has a threshold, and important words such as your name have permanently low thresholds, so they fire even from the weakened channel. **J. Anthony Deutsch** and **Diana Deutsch** (1963) instead proposed **late selection**, at the response, after every message is analysed for meaning (Table 1.6).

Table 1.6 Theories of selective auditory attention.

Theory	Theorist and year	Where selection happens
Filter (switch)	Broadbent, 1958	Early; the rest is blocked

Theory	Theorist and year	Where selection happens
Attenuation	Treisman, 1964	Early but leaky
Late selection	Deutsch and Deutsch, 1963	At the response, after meaning
Pertinence	**Donald Norman**, 1968	Late; signal strength plus relevance
Multimode	**William Johnston** and **Steven Heinz**, 1978	Flexible; later costs more effort
Perceptual load	**Nilli Lavie**, 1994 to 1995	Early if the task is demanding, late if easy

Vision has the same limits. **Michael Posner** likened covert attention, moved without moving the eyes, to a **spotlight** (others prefer a **zoom lens**, wider but less detailed). In his cueing task (1980) a central arrow directs voluntary **endogenous** attention, a peripheral flash captures reflexive **exogenous** attention, and after about 300 milliseconds the flashed spot is answered more slowly (**inhibition of return**). In Treisman and **Garry Gelade**'s **feature integration theory** (1980), features register in parallel at a preattentive stage, so a red letter pops out among green ones, but binding them into objects needs serial focused attention, so conjunction search slows with every added item; unattended features can combine wrongly (illusory conjunctions). Missing a person in a gorilla suit while counting basketball passes (**Daniel Simons** and **Christopher Chabris**, 1999) is **inattentional blindness**, missing an unexpected object in plain view; missing a large change made across a blink or flicker is **change blindness**; missing a second target 200 to 500 milliseconds after a first is the **attentional blink**.

> EXAM TRAP
>
> "Filter-attenuation theory" is **Treisman** (1964), not Broadbent, Johnston or Heinz, and her attenuator model, not Broadbent's switch model, best explains the cocktail party effect. A bottleneck is a narrow processing stage, not the cocktail party phenomenon. Failing to notice obvious changes is **change blindness**, not inattentional blindness or perceptual constancy (Section 1.4).

Divided attention, automaticity, Stroop and vigilance ●●●

Divided attention asks how far we can do two things at once. **Daniel Kahneman**'s **capacity model** (*Attention and Effort*, 1973) treats attention as one limited pool of mental effort; **Christopher Wickens**'s **multiple resource theory** (1984) adds separate pools by modality, code and stage, so dissimilar tasks interfere less: driving combines better with the radio than with texting, though even a hands-free phone conversation impairs it.

Practice helps because some processes stop needing capacity. An **automatic process** runs without intention, awareness or interference with other activity (Posner and Charles Snyder, 1975). **Richard Shiffrin** and **Walter Schneider** (1977) contrasted fast, parallel, effortless and inflexible automatic processing, learned through consistent practice, with slow, serial, effortful and flexible **controlled processing**. In **Patricia Devine**'s model (1989), stereotypes are activated automatically in everyone, but low-prejudiced people use controlled processing to refute them (Chapter 3).

KEY STUDY: STROOP'S COLOUR-WORD EXPERIMENTS (1935)

John Ridley Stroop (1897 to 1973) asked people to read colour words printed in a conflicting ink (RED printed in green) or to *name the ink colour* of such words. Reading was barely slowed, but naming the inks took about 74 per cent longer than naming plain colour patches, and errors rose.

The **Stroop effect** is this slowing of colour naming when word and ink conflict. Reading is automatic, so the word cannot be shut out, and its name competes with the controlled naming of the ink at the response stage. The effect thus shows a limit of **selective attention**: interference in reaction time from conflicting information, one attribute of the stimulus (the word) dominating the other (the ink).

Watching radar for hours for a rare blip demands **sustained attention**, or **vigilance**. **Norman Mackworth** (1948) had observers watch a clock pointer for occasional double jumps; detection fell across the two-hour watch, the **vigilance decrement**. In an oddball task, where a rare target appears among frequent non-targets, the brain produces a positive event-related potential about 300 milliseconds later, the **P300**, which indexes the allocation of attention to the target and is reduced in schizophrenia and Alzheimer's disease. Attention fails in ADHD, in hemispatial neglect (ignoring the left side of space after right parietal damage) and in delirium (Chapter 8).

EXAM TRAP

The Stroop effect is an **attention** phenomenon (selective attention; interference in reaction time from conflicting information), not long-term memory retrieval, working memory, procedural memory, perception or language development. RML 2024 asks what it is "an account of" and keys **stimulus dominance**: the printed word dominates its ink colour. "Semantic dominance" tempts because the word is meaningful; "instructional" and "conceptual" dominance are not terms used for it. Air traffic controllers and radar readers exemplify **sustained attention**, not selective, divided or "perceptual" attention.

Consciousness: James's stream and Freud's levels ●●●

Consciousness is our awareness of ourselves and our environment at a given moment. Its level, from coma to alert waking, depends on the brainstem reticular activating system and is scored on the **Glasgow Coma Scale** (eye, verbal and motor responses; 3 to 15). The structuralists analysed its contents into elements by introspection (Section 1.1); James objected that consciousness "does not appear to itself chopped up in bits … it flows": it is a **stream of consciousness** (1890), personal, ever changing, continuous and selective. Its most researched feature is **daydreaming**, which **Jerome L. Singer** showed to be normal and nearly universal.

Sigmund Freud (1856 to 1939) held that most mental life never enters the stream. His **topographic model** (1900) has three levels: the conscious, the small tip of present awareness; the **preconscious**, accessible memories recallable at will (your mobile number); and the much larger **unconscious**, repressed wishes known only indirectly, through dreams, symptoms and **Freudian slips**, errors of speech that betray unconscious impulses. By **psychic determinism**, no mental event is accidental. In his structural model (1923) the id works on the pleasure principle and the ego on the reality principle (Chapter 4). The **psychodynamic perspective** holds that behaviour is driven by unconscious inner forces largely beyond our con-

trol and shaped in early childhood. Research has not supported the Oedipus complex, but it does support the concept of an unconscious mind.

Modern psychology adds the **cognitive unconscious**, processing outside awareness that is automatic, not repressed, such as the unconscious stereotyping measured by the Implicit Association Test (Anthony Greenwald and colleagues, 1998). Processes never open to awareness, such as blood-pressure control, are **nonconscious**. **David Chalmers** (1995) called the question of why brain processing is accompanied by subjective experience at all the **hard problem of consciousness**. An **altered state of consciousness**, a term made standard by **Charles Tart** (1969), is a temporary, qualitative change in the overall pattern of mental functioning, arising naturally (sleep), psychologically (hypnosis, meditation) or chemically (drugs).

EXAM TRAP

"Stream of consciousness" is **William James** (1890), not Carl Jung, Wilhelm Wundt or B. F. Skinner; its most researched feature is **daydreaming**, not gender differences, brain-wave patterns or eye movements. Memories recallable at will are **preconscious**, not unconscious. The three levels form the **topographic** model (1900); id, ego and superego the **structural** model (1923).

INDIAN CONTEXT

The *Mandukya Upanishad* names four states: jagrat (waking), svapna (dreaming), sushupti (dreamless sleep) and **turiya**, "the fourth", pure consciousness witnessing all three, identified with the **Atman**, the true self or soul, eternal consciousness (Section 1.2).

Biological rhythms and the suprachiasmatic nucleus ●●

A **circadian rhythm** is a biological cycle of about 24 hours, such as sleep and waking or body temperature. Without time cues, in caves, human rhythms persist but drift slightly beyond 24 hours, so they come from an internal clock, reset daily by **zeitgebers** ("time givers"), above all light. The clock is the **suprachiasmatic nucleus** (SCN) of the hypothalamus, just above the optic chiasm: animals without it sleep the usual total amount but at random times. Light reaches it directly along the **retinohypothalamic tract**, and it drives the pineal gland to secrete **melatonin** in darkness. **Jet lag** and shift work put the clock out of step with the day. **Ultradian rhythms** are shorter than a day (the 90-minute sleep cycle), **infradian rhythms** longer (the menstrual cycle).

EXAM TRAP

The SCN is the **master circadian pacemaker**; its lesions disrupt circadian rhythms, not total sleep. It is not the hunger and satiety centre (lateral and ventromedial hypothalamus), the thermoregulation centre (preoptic area), the paraventricular stress nucleus, the controller of anterior pituitary hormones, or a seat of memory (hippocampus, mammillary bodies), language or pain perception.

The stages and cycles of sleep ●●●

Sleep opened to study after **Hans Berger** published the first recordings of the human electroencephalogram (EEG) in 1929: neurons firing independently give small, fast, **desynchronised** waves, neurons firing together large, slow, **synchronised** ones. In 1953 **Eugene Aserinsky** and **Nathaniel Kleitman** found recurring bursts of rapid eye movement with a waking-like EEG, from which woken sleepers usually reported vivid dreams; **William Charles Dement** and Kleitman (1957) then described the cycle of stages. Sleep thus divides into **REM sleep** and NREM sleep, whose old stages 1 to 4 became N1, N2 and N3 in 2007 (Table 1.7).

Table 1.7 EEG waves and the stages of sleep.

State	EEG	What marks it
Alert waking	Beta, 13 to 30 Hz	Active thinking
Relaxed, eyes closed	Alpha, 8 to 12 Hz	Rest before sleep
N1	Theta, 4 to 7 Hz	Drowsy transition
N2 (about half the night)	Sleep spindles, K-complexes	Light but definite sleep
N3 (old stages 3 and 4)	Delta, 0.5 to 4 Hz, high amplitude	Deepest, slow-wave sleep; growth hormone peak
REM (20 to 25 per cent)	Fast, low voltage, desynchronised; sawtooth waves	Rapid eye movements, atonia, dreams

REM is also called **paradoxical sleep**, because the brain looks awake while the body is limp: the eyes dart, the core muscles are almost paralysed (**REM atonia**), breathing turns irregular and genital arousal accompanies nearly every REM period. One run of NREM stages followed by REM is a **sleep cycle** of about 90 minutes, so seven to eight hours hold **four to six cycles**; N3 dominates the early cycles, and REM periods lengthen toward morning (Figure 1.6). A newborn sleeps about 16 hours, half of it REM, and older adults about six, with little N3 (**Howard Roffwarg** and colleagues, 1966). Slow-wave sleep is switched on by GABA neurons of the **ventrolateral preoptic area** (VLPO), which silence the arousal systems; REM is generated in the pons.

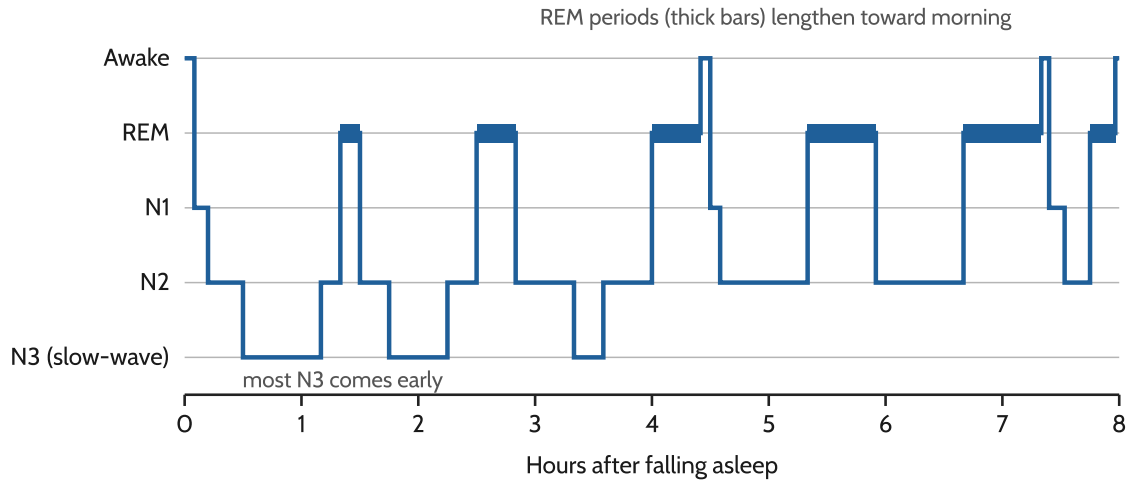

Figure 1.6 A typical night of young adult sleep (a hypnogram). Cycles of about 90 minutes repeat four to six times; N3 is concentrated early in the night, and REM periods become longer toward morning.

EXAM TRAP

REM sleep is **not** marked by EEG synchrony, delta waves or a growth hormone surge (all N3). Cycles per night: **four to six** (RML 2024); RML 2022, offering no 4 to 6, keys **4 to 7**, the option containing it. Slow-wave sleep begins when the **VLPO** becomes active; the SCN is the clock, and the sublaterodorsal nucleus (REM-on) and ventrolateral periaqueductal grey (REM-off) govern REM.

Why we sleep, and sleep disorders ●●

Recuperation theories say sleep restores what waking runs down; **adaptation theories** say evolution timed sleep to keep animals safe and save energy; **memory consolidation** theories note that sleep after learning reduces forgetting (Section 1.7). Sleep loss brings sleepiness, low mood and lapses of sustained attention, including **microsleeps** of a few seconds. Volunteers woken at every REM onset tried to enter REM ever more often, then showed **REM rebound**, extra REM (Dement, 1960). In **cataplexy**, part of narcolepsy, strong emotion triggers a sudden loss of muscle tone in full wakefulness: REM atonia while awake. Sleep disorders fall into insomnia, **hypersomnia** (excessive sleepiness, as in narcolepsy), breathing-related disorders and the **parasomnias**, abnormal behaviours during sleep: NREM parasomnias (sleepwalking and sleep terrors from N3, and bedwetting) and REM parasomnias (nightmares, REM sleep behaviour disorder). Table 1.8 lists the main ones (Chapter 8).

Table 1.8 The main sleep disorders in brief.

Disorder	Key features	Stage or cause
Insomnia	Trouble falling or staying asleep	Stress, anxiety, poor habits
Narcolepsy	Daytime sleep attacks, cataplexy, sleep paralysis	REM intrudes into waking
Sleep apnoea	Breathing stops repeatedly; snoring	Airway collapse
Sleepwalking (somnambulism)	Sitting, walking or complex acts while asleep	N3, early in the night
Sleep terrors	Screaming fear, inconsolable, little recall	N3, early in the night

Disorder	Key features	Stage or cause
Nightmares	Frightening dream, vivid recall	REM, late in the night
REM sleep behaviour disorder	Acting out dreams	Loss of REM atonia
Restless legs syndrome	Leg discomfort, urge to move	Delays sleep onset

EXAM TRAP

Somnambulism is sitting, walking or complex behaviour while asleep; screaming fear is sleep terrors, acting out nightmares is REM sleep behaviour disorder, and leg discomfort is restless legs syndrome. Sleepwalking arises from **N3**, not REM: a sleeper in REM is paralysed. Treatments for insomnia, such as stimulus control and Viktor Frankl's paradoxical intention (trying to stay awake), are in Chapter 7.

Dreams: Freud, Jung and the brain ●●●

Freud's *The Interpretation of Dreams* (1900) called dreams "the royal road to a knowledge of the unconscious". Each dream is a **wish fulfilment**, the disguised fulfilment of a repressed wish. The **manifest content** is the dream as remembered, the **latent content** its hidden meaning, and the **dream work** disguises one as the other by condensation (several ideas in one image), displacement (feeling shifted to a trivial object), symbolisation (thoughts made into images) and secondary revision (tidying it into a story).

Carl Gustav Jung (1875 to 1961) saw dreams as undisguised symbolic messages that compensate for a one-sided conscious attitude. He held that the unconscious has two levels: a personal (individual) unconscious of forgotten and repressed experience and, beneath it, the **collective unconscious**, an inherited layer shared by all humanity, which he explained by evolution, through the similar brain structure of all human races. It holds **archetypes**, universal images recurring in myths: the persona (social mask), the shadow (disowned dark side), the **anima** (the feminine side of a man) and the **animus** (the masculine side of a woman). Jung also devised the word association test, in which hesitations reveal unconscious complexes (Chapter 4).

In the **activation-synthesis hypothesis** of **J. Allan Hobson** and **Robert McCarley** (1977), the pons fires randomly during REM and the cortex synthesises a story from the activation, which is why dreams are visual, bizarre and soon forgotten. Information-processing views link dreams to the overnight consolidation of recent experience, and **Antti Revonsuo**'s **threat-simulation theory** (2000) says dreams evolved to rehearse responses to danger. In a **lucid dream** the dreamer knows he is dreaming, confirmed by pre-arranged eye-movement signals from REM sleep (**Stephen LaBerge**). **Hypnagogic imagery** comes while falling asleep, **hypnopompic imagery** while waking.

EXAM TRAP

The Interpretation of Dreams: **1900** (printed November 1899). The collective unconscious is from **Carl Jung**, not Sigmund Freud, Alfred Adler or Karen Horney; his two levels are **collective and individual**, not anima and animus or extraversion and introversion.

Hypnosis, meditation and psychoactive drugs ●●●

Hypnosis is a social interaction in which one person's suggestions of changes in perception, memory or behaviour are experienced by the other as real and involuntary. Its crux is **suggestibility**, and it is not sleep. **Franz Anton Mesmer**'s **animal magnetism** (1770s) was put down to imagination by a royal commission in 1784. The Scottish surgeon **James Braid** coined "hypnotism" (*Neurypnology*, 1843). **Jean-Martin Charcot** took hypnotisability for a sign of hysteria; the **Nancy school** of **Ambroise-Auguste Liébeault** and **Hippolyte Bernheim** showed it was normal suggestion. Freud learned from both, published the mechanism of hysteria with **Josef Breuer** (1893; *Studies on Hysteria*, 1895), then replaced hypnosis with **free association**, the patient on the couch reporting whatever comes to mind.

Susceptibility is measured by the **Stanford Hypnotic Susceptibility Scales** (**André Weitzenhoffer** and **Ernest Ropiequet Hilgard**, 1959). Hypnosis can relieve pain and produce posthypnotic suggestions (acted on at a cue after hypnosis ends) and posthypnotic amnesia, but it cannot give superhuman strength or recover memories reliably, since it raises false recall and confidence together. Hilgard's **neodissociation theory** holds that hypnosis splits consciousness, a **hidden observer** registering pain that the hypnotised person denies; the **social-cognitive theory** of **Theodore Xenophon Barber** and **Nicholas Spanos** says hypnotised people enact an expected role.

Meditation trains attention. **Concentrative meditation** holds it on one object, such as the repeated mantra of **transcendental meditation**, introduced by **Maharishi Mahesh Yogi** in the 1950s; **mindfulness meditation** (opening-up) observes whatever arises without judging it, from Buddhist vipassana. **Herbert Benson** showed that meditation's slowed heart rate and breathing, the **relaxation response** (1975), follow from any similar technique. Surveys showing that regular meditators are less stressed are correlational, however: without random assignment, calmer people may simply meditate more, and only an experiment can show cause (Chapter 9). In McGill University's **sensory deprivation** studies (1954), students lying in a cubicle with goggles and gloves mostly quit within two or three days, and many hallucinated. A **psychoactive drug** alters perception, mood or consciousness by acting on the nervous system (Table 1.9). With repeated use, **tolerance** means a larger dose is needed for the same effect, **dependence** means the drug is needed to function, and **withdrawal** is the unpleasant reaction, usually opposite to the drug's effects, on stopping (Chapter 8).

Table 1.9 The main classes of psychoactive drugs.

Class	Examples	Main effects
Depressants	Alcohol, barbiturates, benzodiazepines	Enhance GABA; sedation, sleep
Stimulants	Caffeine, nicotine, amphetamines, cocaine	Arousal, euphoria, then a crash
Opiates	Morphine, heroin, codeine	Pain relief; mimic endorphins
Hallucinogens	LSD, psilocybin, mescaline	Distorted perception
Cannabis	Ganja, charas, bhang (THC)	Relaxation, slowed time sense

EXAM TRAP

The crux of hypnosis is **suggestibility**, not the analytical mind or the logical mind. Freud and Breuer, not Freud and Charcot, published the mechanism of hysteria; Freud, not Bernheim, fathered psychoanalysis. Valium, Xanax, Rohypnol and Halcion are **benzodiazepines**: "sedatives" and "hypnotics" name uses, not the class, and they are not barbiturates.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The number of objects apprehended in one brief glance is the _____ of attention.
2. In the Stroop task, automatic _____ slows naming of the ink colour.
3. In Freud's model, memories recallable at will are _____.
4. The master circadian clock is the _____ nucleus.
5. Delta waves mark stage _____.
6. Jung's inherited layer of the psyche is the _____ unconscious.

Answers: (1) span, (2) reading, (3) preconscious, (4) suprachiasmatic, (5) N3, (6) collective.

1.6 Learning: Conditioning, Cognition and Observation HIGH YIELD · 143 QS

Almost everything that makes you a particular person was learned, from your language to the way your stomach tightens in an exam hall. This section runs from the simplest learning, through conditioning, to the maps, insights and imitation that conditioning alone could not explain.

What learning is ●●

The standard definition descends from **Gregory Kimble** (1961): **learning** is a relatively permanent change in behaviour, or in the potential for behaviour, that results from experience. "Potential" matters because learning is inferred from *performance*, what an organism actually does, and the two can come apart. "Experience" rules out *maturation*, change on a biological timetable (a baby walks when her body is ready), as well as fatigue, drugs, motivation and instinct. **Nikolaas Tinbergen** (1907 to 1988) showed that a male stickleback attacks any crude model with a red belly: a *fixed action pattern*, an unlearned, stereotyped sequence triggered by a sign stimulus and run to completion.

The simplest learning involves one stimulus and is called non-associative. In **habituation** the response to a repeated, harmless stimulus decreases: trains outside a new hostel room wake you the first night but not by the end of the week. It is stimulus-specific and shows dishabituation (a novel strong stimulus restores the response), proof that it is learning, not fatigue. Its near opposite, **sensitisation**, is an increased response to many stimuli after a strong or noxious one. **Eric Kandel** (born 1929) traced both to synapses of the sea slug *Aplysia* (Physiological Psychology chapter). In infants, habituation reveals discrimination before speech (**Robert Fantz**, 1964): a baby bored with one picture who looks longer at a new one can tell them apart.

Between instinct and ordinary learning lies **imprinting**, the rapid, unrewarded and largely irreversible attachment to the first suitable moving object, possible only within a *critical period*: in 1935 **Konrad Lorenz** (1903 to 1989) reported incubator-hatched greylag goslings following him as their mother. Everything else here is associative learning: learning that two events go together.

> EXAM TRAP
>
> Fixed action patterns are **innate**, not products of classical or operant conditioning. The near opposite of habituation is **sensitisation**; "habitation" and "inhabitation" are sound-alikes.

Pavlov and classical conditioning ●●●

Ivan Petrovich Pavlov (1849 to 1936), a physiologist who won the 1904 Nobel Prize for work on digestion, collected his dogs' saliva through a tube in the cheek. In the late 1890s he noticed that they salivated before food arrived, at the sight of the dish or the attendant's footsteps. From about 1897 he studied these "psychic secretions": a metronome was sounded, meat powder followed, and after several pairings the metronome alone produced salivation. This is **classical conditioning** (Pavlovian or respondent conditioning): learning in which a neutral stimulus, repeatedly paired with a stimulus that already produces a response, comes to produce a similar response itself.

Five terms describe it. The **unconditioned stimulus** (UCS) produces a response without learning (food in the mouth), and that unlearned response is the unconditioned response (UCR: salivation to food). A neutral stimulus (NS), the metronome on day one, does not produce it. After pairing, the metronome is a **conditioned stimulus** (CS), and its learned response is the conditioned response (CR: salivation to the metronome). So the metronome is an NS before conditioning and a CS after it, and salivation is a UCR to food but a CR to the metronome. The fundamental element is association, the pairing of two stimuli close in time. Table 1.10 adds three cases; each response is involuntary.

Table 1.10 Three classical conditioning situations broken into their parts.

Situation	UCS	UCR	CS	CR
Little Albert	Loud clang	Startle and fear	White rat	Fear of the rat
Student who lost a match	Defeat and humiliation	Distress	The sports ground	Anxiety at the ground
Cancer patient	Chemotherapy drug	Nausea	Hospital smell	Nausea on arrival

Timing matters. In delay conditioning the CS begins first and is still on when the UCS arrives; this is the most effective arrangement, because the CS works as a warning. Trace conditioning, the other forward arrangement, leaves a gap between CS and UCS and is weaker; simultaneous conditioning gives no warning and is weak; backward conditioning (UCS first) is the weakest of all.

EXAM TRAP

The bell is a **neutral stimulus before** pairing and a **conditioned stimulus after**; never a UCS, never an "operant stimulus". It becomes a CS by **repeated pairing with a UCS**, not by reward, punishment or random occurrence. "Delay" does not mean slow: delay conditioning is the most effective; trace has the gap.

The CR grows during acquisition, quickly at first and then more slowly. If the CS is then presented repeatedly without the UCS, the CR fades: **extinction**. (Pavlov called the UCS the "reinforcement", so withdrawing reinforcement means omitting the UCS.) Extinction suppresses rather than erases learning (Pavlov saw it as active inhibition): after a rest the CR returns with no new pairings (**spontaneous recovery**); a novel distracting stimulus restores it briefly (disinhibition); so does the UCS alone (reinstatement) or a return to the original context (renewal). Hence fears can relapse after therapy.

Responding to stimuli similar to the CS, the more so the more similar they are (a generalisation gradient), is **stimulus generalisation**; responding only to the CS, learned when a similar stimulus is never followed by the UCS, is **stimulus discrimination**. A dog whose circle-versus-ellipse discrimination **Shenger-Krestovnikova** (1921) made impossible became agitated and lost even easy discriminations: *experimental neurosis*, the first laboratory model of anxiety. In *higher-order conditioning* an established CS acts as the UCS for a new stimulus (a black square paired only with the metronome comes to produce salivation), which is how the word "exam" alone can make a heart race. In *sensory preconditioning* (**W. J. Brogden**, 1939) two neutral stimuli are paired first; when one is later conditioned, the other also produces the CR.

EXAM TRAP

Omitting the UCS produces **extinction**. Spontaneous recovery needs only time, disinhibition a novel stimulus, reinstatement the UCS alone and reconditioning new pairings; sensitisation is no comeback. Avoiding a black cat but playing with a tabby is **discrimination**, not extinction, successive approximation (shaping) or negative transfer.

Conditioned fear and its therapies ●●●

John Broadus Watson (1878 to 1958), the founder of behaviourism, set out with **Rosalie Rayner** to show that emotions are conditioned. In 1920 they paired a white rat seven times with the clang of a steel bar behind the head of an eleven-month-old infant, "Albert B.". Afterwards the rat alone made him cry and crawl away, and the fear generalised to a rabbit, a dog and a fur coat but not to his wooden blocks: a conditioned emotional response. With one infant and no control, the study would be unethical today.

Can fear be unconditioned? In 1924 **Mary Cover Jones** (1897 to 1987) treated Peter, a boy who feared rabbits: while he ate a favourite food, a caged rabbit was brought a little closer each day until he let it nibble his fingers. This is **counterconditioning**: replacing a conditioned response with an incompatible one, so that the client learns the stimulus is no longer threatening. **Joseph Wolpe** (1915 to 1997) borrowed a physiologist's term for the principle, **reciprocal inhibition**: a response incompatible with anxiety, made to occur in the presence of the feared stimulus, weakens the bond between them. His **systematic desensitisation**

(*Psychotherapy by Reciprocal Inhibition*, 1958) has three steps: relaxation training, usually the progressive muscle relaxation of **Edmund Jacobson** (1888 to 1983); an anxiety hierarchy, its items rated on Wolpe's 0 to 100 Subjective Units of Disturbance Scale (SUDS); and desensitisation proper, in which the relaxed client imagines each item in turn, moving up only when it no longer arouses anxiety. A relaxed student who pictures himself on the stage, then introducing himself, then giving his feared talk, is being desensitised.

In **flooding**, another conditioning therapy, the client stays in the most feared situation at full intensity, with no steps or relaxation, until anxiety subsides: extinction with escape blocked. Implosive therapy (**Thomas Stampfl**) is flooding in imagination. Aversion therapy pairs a cue such as the taste of alcohol with a nausea-producing drug, and in covert sensitisation (**Joseph Cautela**, 1967) the behaviour and an unpleasant outcome are both imagined. Exposure and response prevention (**Victor Meyer**, 1966) exposes a person with obsessive-compulsive disorder (OCD) to the feared cue and blocks the compulsion, so that anxiety falls by habituation and extinction (Clinical Psychology chapter).

> EXAM TRAP
>
> Systematic desensitisation is **Wolpe's** and rests on **reciprocal inhibition**, not operant extinction, shaping, habituation, the law of effect, catharsis or vicarious reinforcement. Relaxation is **Jacobson's**; hierarchy ratings use **SUDS**, not the Fear Survey Schedule, State-Trait Anxiety Inventory or Hamilton Anxiety Rating Scale. Graded steps with relaxation is desensitisation, full intensity is flooding, and imagined behaviour plus imagined unpleasantness is covert sensitisation. Exposure and response prevention rests on **habituation and extinction**; "graded exposure" is a procedure, not a principle. **Symptom substitution** is not a technique but the psychoanalytic prediction that a removed symptom will be replaced.

Prediction and preparedness ●●

Conditioning was long thought to need only contiguity, the CS and UCS close together in time. In 1968 **Robert Rescorla** (1940 to 2020) showed that rats did not fear a tone if shocks were as likely without it as with it: conditioning depends on contingency, how well the CS predicts the UCS. In blocking (**Leon Kamin**, 1969), rats already afraid of a noise learned almost nothing about a redundant light added to it. In overshadowing the most salient part of a compound CS takes most of the learning; in latent inhibition (**Robert Lubow** and **A. U. Moore**, 1959) a stimulus often met alone beforehand conditions slowly.

The Rescorla-Wagner model (Rescorla and **Allan Wagner**, 1972), $\Delta V = \alpha\beta(\lambda - \Sigma V)$, says that an animal learns only when surprised. A CS gains associative strength (ΔV) in proportion to the prediction error, the gap between the most the UCS can support (λ) and what all stimuli present already predict (ΣV), scaled by salience and learning rate ($\alpha\beta$). The error shrinks as the UCS becomes expected, so each trial adds less; in blocking it is zero from the start. In the 1990s **Wolfram Schultz** found dopamine neurons of the ventral tegmental area signalling exactly this error. The CS is thus a signal that predicts the UCS, not, as Pavlov's stimulus substitution theory held, a substitute for it.

KEY STUDY: GARCIA AND KOELLING'S "BRIGHT, NOISY, TASTY WATER" (1966)

Question. Are all stimuli equally easy to associate with all consequences? **Method. John Garcia** (1917 to 2012) and **Robert Koelling** let rats drink saccharin-flavoured water whose every lick also produced a flash and a click; then some were made ill, others shocked. **Result.** Ill rats avoided the taste but not the light and click; shocked rats avoided the light and click but not the taste. **Conclusion.** Taste "belongs" with illness, and sights and sounds with pain; the finding so contradicted theory that Garcia struggled to publish it.

Such **conditioned taste aversion** forms in a single trial, even when illness follows the taste by hours, and refutes equipotentiality, the assumption that any stimulus can be associated with any consequence equally easily. **Martin Seligman** (born 1942) called the principle **preparedness** (1970): each species is biologically prepared to learn some associations easily, which is why phobias of snakes and heights are common and phobias of electric sockets rare. Chemotherapy patients are given a novel "scapegoat" food to protect their usual diet, and advertisers pair a brand with a beloved film star so that the liking transfers.

EXAM TRAP

Taste aversion is **classical conditioning** (taste = CS, illness = UCS), not operant, second-order conditioning or a "spurious association" (a statistical term). It illustrates **biological preparedness**, defying equipotentiality and contiguity. Prediction error is signalled by **dopamine neurons in the ventral tegmental area**, not serotonin, GABA or acetylcholine. Latent *inhibition* is not Tolman's latent *learning*.

Thorndike, Skinner and consequences ●●●

Most behaviour, however, is emitted rather than elicited, and its consequences decide whether it recurs. **Edward Lee Thorndike** (1874 to 1949) shut hungry cats in puzzle boxes (*Animal Intelligence*, 1898). Each cat scrambled until it hit the release by accident, and over trials its escape time fell gradually, not suddenly: trial-and-error learning. His **law of effect** (1911) states that responses followed by satisfaction are strengthened and those followed by discomfort weakened. His connectionism, which treats learning as forming stimulus-response bonds, added the law of exercise (use strengthens a bond), later abandoned, and the law of readiness (acting when ready is satisfying).

The radical behaviourist **Burrhus Frederic Skinner** (1904 to 1990), in *The Behavior of Organisms* (1938), separated respondent behaviour, elicited by a stimulus, from operant behaviour, emitted by the organism and operating on the environment. In **operant conditioning** the probability of a response is changed by its consequences (Thorndike's trial-by-trial version is often called instrumental conditioning). In the Skinner box a rat pressed a lever freely while a cumulative recorder plotted its rate. A discriminative stimulus, such as a chai stall's "Open" board, signals that a response will be reinforced, bringing it under stimulus control. Applied behaviour analysis and functional analysis, which ask what a problem behaviour gets the person or lets him escape, rest on these principles; merely recording a behaviour often changes it (reactivity).

Two questions classify every consequence (Table 1.11): does the behaviour increase (reinforcement) or decrease (punishment), and is a stimulus added (positive) or removed (negative)? Positive and negative mean plus and minus, not good and bad. A friend's warm

agreement with an opinion you voice is positive reinforcement, so the attitude strengthens. Pressing a lever to switch off a shock already on is **negative reinforcement**. A fine for jumping a red light is negative punishment. A parent who buys a sweet to end a tantrum positively reinforces the tantrum and is negatively reinforced for giving in.

Table 1.11 The four kinds of operant consequence.

Effect on behaviour	Stimulus added (positive)	Stimulus removed (negative)
Increases (reinforcement)	Positive reinforcement: something wanted is given (praise)	Negative reinforcement: something unpleasant is taken away (a headache ends)
Decreases (punishment)	Positive punishment: something unpleasant is given (a scolding)	Negative punishment: something wanted is taken away (time-out, response cost)

Food is a primary reinforcer, effective without learning. Money and praise are **secondary reinforcers**, neutral stimuli that became reinforcing through association with primary ones; one linked with many primary reinforcers, as money is, is a generalised reinforcer. (Harry Harlow's infant monkeys preferred a cloth mother to a wire one that fed them: attachment rests on contact comfort, not secondary reinforcement.) In the 1960s **Teodoro Ayllon** and **Nathan Azrin** built the **token economy** at Anna State Hospital, Illinois: long-stay psychiatric patients earned tokens for self-care and ward work and exchanged them for privileges (*The Token Economy*, 1968).

Escape learning terminates an aversive stimulus already present; avoidance learning prevents its arrival. **Orval Hobart Mowrer** (1907 to 1982) explained avoidance with his **two-factor theory** (1947): a warning signal comes to elicit fear by classical conditioning, and avoiding it is negatively reinforced by fear reduction, an operant process that also keeps the fear from extinguishing. Compulsive washing in OCD is such an escape or safety behaviour. Time-out (brief removal from a setting where reinforcement is available) and **response cost** (loss of a reinforcer already held) are negative punishment. As **William Kaye Estes** (1919 to 2011) showed in 1944, punishment suppresses a response without erasing it; it also breeds fear of the punisher and models aggression.

New behaviour is built by **shaping**, reinforcing successive approximations to a target: facing the lever, approaching it, touching it, finally pressing it. Shaping works on a response's speed and force and needs no words. Chaining links responses into a sequence; a prompt adds a cue and fading withdraws it. Operant extinction, the fading of a response no longer reinforced, often begins with an extinction burst of harder, more varied responding (jabbing at a lift button), after which responding falls back to its operant level, the rate before reinforcement.

> EXAM TRAP
>
> The law of effect is **Thorndike's** (not Skinner's, Pavlov's or Watson's), not modelling, insight or unconscious drive. Operant conditioning works through **reinforcement and punishment**, not modelling or cognitive dissonance. **Negative reinforcement is not punishment**, and it too increases behaviour, though one real item keyed only positive reinforcement. Removing a desirable stimulus to reduce behaviour is **response cost**, not negative reinforcement, positive punishment or extinction; time-out is **negative punishment**. OCD compulsions are **escape (avoidance) behaviours**, not conditioned stimuli, unconditioned responses or positively reinforced acts.

EXAM TRAP

The token economy is **operant** (secondary reinforcement), the therapy **not** derived from classical conditioning, and **Ayllon and Azrin's** (Amity 2023 keyed Skinner, who supplied only the principles). Stickers that buy a homework exemption are **tokens**, not primary or generalised reinforcers or "prepotent" (dominant) responses. Ignoring tantrums is **extinction**, not "operant escape" (another wrong Amity 2023 key). Shaping is not chaining, prompting, fading or the law of effect; RML 2024 keyed its principles as speed, power and non-verbal.

Schedules and other operant effects ●●●

A **schedule of reinforcement** is the rule deciding which responses are reinforced. Continuous reinforcement, after every correct response, teaches fastest but extinguishes fastest; partial reinforcement makes behaviour persist far longer once reward stops (the partial reinforcement extinction effect). Reward may depend on the number of responses (ratio) or on time elapsed (interval), fixed or variable, giving the four schedules that **Charles Ferster** and Skinner mapped in 1957 (Figure 1.7).

- Fixed ratio (FR), as in piece-rate pay: high, steady responding with a brief post-reinforcement pause.
- Variable ratio (VR), as in slot machines: the highest, steadiest rate and the greatest resistance to extinction.
- Fixed interval (FI), as in checking a dish due out of the oven: a pause, then faster and faster responding, the scallop.
- Variable interval (VI), as in checking for a reply to a message: moderate, very steady responding.

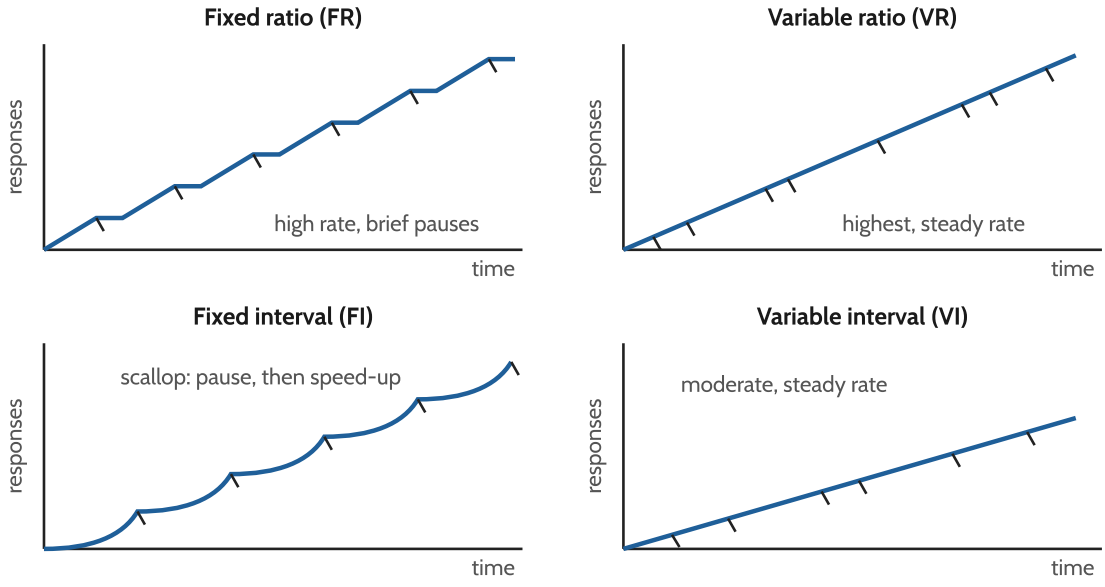

Figure 1.7 Stylised cumulative records for the four basic schedules. The steeper the line, the faster the responding; short diagonal ticks mark reinforcers. Notice the flat steps after each reinforcer on FR, the unbroken steep line on VR, the scallops on FI and the steady, shallower line on VI.

EXAM TRAP

"Number" means ratio, "time" interval, "unpredictable" variable. Reinforcement after every response is **continuous**, the least resistant to extinction. Sheena's chocolate "for every 45 minutes" of study is keyed **fixed interval**.

Differential reinforcement extinguishes a problem behaviour while reinforcing other behaviour, that is, its absence (DRO), an incompatible (DRI) or alternative (DRA) behaviour, or a low rate of it (DRL). The **Premack principle** of **David Premack** (1959) holds that a more probable activity can reinforce a less probable one: homework first, then video games. Skinner's pigeons, fed every 15 seconds whatever they did (1948), developed odd rituals: superstitious behaviour, strengthened by accidental reinforcement.

KEY STUDY: LEARNED HELPLESSNESS (SELIGMAN AND MAIER, 1967)

Martin Seligman and **Steven Maier** gave dogs shocks they could switch off, identical shocks they could not control (the yoked group), or none. Later, in a shuttle box, most yoked dogs passively took shocks they could have escaped by jumping a barrier. Lack of control, not shock, produced **learned helplessness**: passive acceptance of aversive events, even avoidable ones, after repeated unavoidable ones. It became a model of depression, later recast in attributional terms (Psychiatric Disorders chapter).

Biology limits operant learning too: **Keller Breland** and **Marian Breland** (1961) found raccoons trained to bank coins rubbing them together as if washing food (instinctive drift). **Richard Herrnstein**'s matching law (1961) states that responses are divided among options in proportion to the reinforcement each yields. Rats that **Leo Crespi** (1942) shifted to a medium reward ran slower if shifted down, and faster if shifted up, than rats always given it: contrast effects, too abrupt to be learning, showing that reward size changes motivation. In **biofeedback** a bodily process is measured and displayed so that changes in the desired direction are reinforced, following **Neal Miller**'s (1909 to 2002) claim that autonomic responses can be operantly conditioned. Electromyographic (EMG) biofeedback suits tension headache, and **Arnold Kegel**'s perineometer (1948) was an early form.

EXAM TRAP

Premack: **high-probability behaviour reinforces low-probability behaviour**, never the reverse. Learned helplessness is **Seligman and Maier's**, not Beck's, Sullivan's or Kahneman's. The Crespi effect is a **change in motivation when reward size changes**, not retroactive inhibition, extinction or insight. Autonomic control by operant conditioning is **biofeedback**, not hypnosis, sublimation or desensitisation.

Table 1.12 compares the two conditionings. They are not "entirely different" but interact constantly: Albert's fear was classical, but had he crawled away and felt relief, his avoidance would have been operant. The hallmark of behaviour therapy is the use of both to eliminate unwanted behaviour, not a search for underlying causes, insight, cognitive restructuring, dream analysis or free association.

Table 1.12 Classical versus operant conditioning.

Feature	Classical conditioning	Operant conditioning
Key figures	Pavlov, Watson	Thorndike, Skinner
What is associated	Two stimuli (CS and UCS)	A response and its consequence
Behaviour	Elicited, involuntary, emotional	Emitted, voluntary, goal-directed
Extinction	CS presented without the UCS	Response no longer reinforced

EXAM TRAP

The main difference: **classical pairs two stimuli; operant relies on reinforcement and punishment**, not "classical relies on reinforcement" (backwards) or "only operant involves cognition".

Latent learning, insight and learning sets ●●●

Against the view that learning is only stimulus-response connections stamped in by reward, **Edward Chace Tolman** (1886 to 1959), a purposive behaviourist, argued that animals learn knowledge.

KEY STUDY: TOLMAN AND HONZIK'S LATENT LEARNING EXPERIMENT (1930)

Tolman and **Charles Honzik** ran three groups of hungry rats through a maze daily for seventeen days: one always found food in the goal box, one never did, and one found food only from day 11. The third group made as many errors as the unrewarded rats for ten days, then at once matched the always-rewarded rats. It had learned the maze all along; reward merely gave it a reason to show it, so learning and performance differ.

This is **latent learning**: learning without obvious reinforcement that stays hidden until there is an incentive to show it. Tolman (1948) said the rats had formed a **cognitive map**, a mental representation of the layout; rats learn places and expectancies of what leads to what, not only turns. **Wolfgang Köhler** (1887 to 1967), a founder of Gestalt psychology (Section 1.2), studied chimpanzees on Tenerife (1913 to 1920). Sultan, unable to reach a banana with either of two short sticks, later fitted one into the other and at once raked it in; others stacked boxes. Such **insight learning**, the sudden grasp of a solution by perceiving the relations among a problem's elements, is abrupt, smooth once found, well retained and readily transferred; its flash of understanding is the aha experience. **Harry Harlow** (1905 to 1981) found that monkeys given hundreds of discrimination problems came to solve each new one in a single trial: a learning set, or learning to learn (1949), suggesting that apparent insight rests on earlier, gradual learning. Concept formation (Section 1.8) runs from observing instances, through discriminating them from non-instances and generalising to all that share the essential features, to abstracting the defining properties.

EXAM TRAP

Latent learning and the cognitive map are **Tolman's**, not Bandura's, Skinner's, Köhler's or Harlow's (Gould studied honeybee navigation). Sultan means **insight**, and the realisation is the **"aha" experience**, not the "light bulb"; insight graded in a mental status examination is psychiatric. Concept formation runs **observation, discrimination, generalisation, abstraction**; assimilation is Piaget's term.

Observational learning ●●●

Most complex skills, from rolling a chapati to a cover drive, are learned largely by watching. This is **observational learning** (social learning, modelling): learning by watching others, called models, and the consequences of their behaviour. **John Dollard** (1900 to 1980) and Neal Miller (1941) held that imitation must itself be reinforced, the early social learning theory; **Albert Bandura** (1925 to 2021) showed that it need not be.

KEY STUDY: BANDURA, ROSS AND ROSS'S BOBO DOLL EXPERIMENT (1961)

Bandura, **Dorothea Ross** and **Sheila Ross** let 72 nursery-school children watch an adult punch, kick and hammer an inflatable Bobo doll. Left later with the doll, these children copied the model's specific acts and words far more than children who saw a quiet model or none, with no reinforcement at all. In 1965 children who saw the model punished imitated less, until offered a reward for showing what the model had done, when all groups did equally well. **Criticism.** Bobo is made to be hit, and only short-term effects were measured.

The model's consequences, then, affect performance, not learning: seeing a model rewarded increases imitation (vicarious reinforcement), and seeing one punished reduces it. Bandura's four processes are attention (noticing the model), retention (storing the act in images and words), motor reproduction (turning the memory into action) and motivation (a reason to perform). The first three govern learning and the fourth performance, so reinforcement is not necessary, and the model must actually demonstrate the target behaviour. Modelling is also a therapy (Clinical Psychology chapter).

With **Richard Walters** (1963) Bandura developed later social learning theory, renamed social cognitive theory in 1986. Its **reciprocal determinism** (1978) holds that the person (beliefs, expectations), behaviour and environment influence one another in both directions, against Skinner's view that environment alone shapes behaviour. Self-regulation means that most behaviour is guided by self-set standards and self-reward rather than external rewards. The theory's central motivational construct is **self-efficacy** (1977), a person's belief in his capability to carry out the actions a specific task requires. (**Julian Rotter**'s separate social learning theory of 1954 belongs to the Personality chapter: behaviour potential, the likelihood of a behaviour in a given situation in relation to a reinforcement, depends on expectancy and reinforcement value. Social learning frameworks in general explain behaviour by expectancies, situational demands and incentives rather than by reinforcement history alone.)

EXAM TRAP

The four processes are **attention, retention, motor reproduction and motivation**, not sensory-perceptual or memory stages; the one NOT necessary is **reinforcement**, a view that is **Bandura's**, not Skinner's, Tolman's or Miller and Dollard's. Reciprocal causation belongs to **social cognitive theory**, not two-factor, drive-reduction or constructivist theory, and is neither reciprocal inhibition nor reciprocal altruism. Self-efficacy (Bandura, not Beck, Ellis or Skinner) is not self-esteem, external locus of control or Adlerian striving; it, not goal difficulty, rewards or drives, is the motivational core of social cognitive theory. Most behaviour stems from **self-regulation**, not "reciprocal dualism" or "behavioural determinism". Match: Dollard and Miller, early social learning; Bandura and Walters, later; Skinner, radical behaviourism; Maslow, self-actualisation.

Other theorists and learning in practice ●●

The grand theories divided on two questions: is what is learned a stimulus-response connection, as the stimulus-response (S-R) behaviouristic approach holds, or knowledge (Tolman, Köhler); and does learning need only contiguity, or reinforcement too? **Edwin Ray Guthrie** (1886 to 1959) chose contiguity: a stimulus combination that accompanied a movement will tend to be followed by it again, at full strength after one trial. A habit is broken by attaching a new response to its cue: present the cue too weakly to trigger the habit and strengthen it gradually, as with a light blanket before a saddle (threshold method); repeat the response to exhaustion (fatigue method); or present the cue while a competing response occurs (incompatible response method). **Clark Leonard Hull** (1884 to 1952) chose reinforcement, defined as drive reduction: reaction potential equals habit strength multiplied by drive (${}_{S}E_{R} = {}_{S}H_{R} \times D$). Table 1.13 adds Estes and **Robert Mills Gagné** (1916 to 2002). Mowrer and his wife devised the bell-and-pad for bed-wetting (1938), and constructivists (Piaget, Vygotsky, Bruner) hold that learners actively build knowledge.

Table 1.13 Other learning theorists at a glance.

Theorist	Family	Central idea	Key work
Guthrie	S-R contiguity	One-trial learning; three ways to break habits	*The Psychology of Learning* (1935)
Hull	S-R reinforcement	Drive reduction; habit strength × drive	*Principles of Behavior* (1943)
Estes	Mathematical	Stimulus sampling: each trial conditions a fraction (θ) of stimulus elements	Stimulus sampling theory (1950)
Gagné	Instructional	Eight types in a hierarchy: signal, S-R, chaining, verbal association, multiple discrimination, concept, principle, problem solving	*The Conditions of Learning* (1965)

A learning curve plots performance against practice: usually negatively accelerated (fast early gains, then diminishing returns), sometimes positively accelerated (a slow start on a hard task) or S-shaped. **William Lowe Bryan** and **Noble Harter** (1897, 1899) found telegraphers stalling on a plateau, a temporary flat stretch while letter habits became automatic, before word habits lifted them again; it is not the final physiological limit. Overlearning, practice beyond the first errorless recall, aids retention (**William C. F. Krueger**, 1929). Distributed (spaced) practice usually beats massed practice; whole learning suits short, meaningful material and part learning long, difficult material. Knowledge of results (feedback) is essential,

and a specific goal makes practice purposeful (**Edwin Locke** and **Gary Latham**). Learners with learning difficulties need multisensory input and regular repetition.

Transfer of training, the effect of one task on learning another, is positive (cycling helps with a motorcycle), negative (an old habit interferes) or zero. **William James** (1890) and Thorndike refuted formal discipline, the belief that Latin trains the mind; Thorndike's identical elements theory ties transfer to elements the tasks share. **Sidney Leavitt Pressey** built the first teaching machine (1920s); Skinner's programmed instruction (1954) used small steps, active responses, immediate feedback and self-pacing.

> **EXAM TRAP**
>
> Learning through S-R associations is the **stimulus-response behaviouristic approach**, not the cognitive, psychoanalytic or existential approach. "Learning always leads to efficacy" given **a specific goal**, not ability, aptitude or knowledge. A plateau is not the physiological limit or extinction.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The near opposite of habituation is _____.
2. Pavlov's bell was a neutral stimulus before pairing and a _____ stimulus after it.
3. Wolpe's systematic desensitisation rests on _____.
4. Removing an unpleasant stimulus to increase a behaviour is _____ reinforcement.
5. The schedule with the highest rate and greatest resistance to extinction is _____.
6. Bandura's four processes are attention, retention, _____ and motivation.

Answers: (1) sensitisation, (2) conditioned, (3) reciprocal inhibition, (4) negative, (5) variable ratio and (6) motor reproduction.

1.7 Memory: Encoding, Storage, Retrieval and Forgetting HIGH YIELD · 111 QS

Memory, the capacity to retain information and use it later, is not one thing. After encephalitis destroyed both his hippocampi in 1985, the British musician **Clive Wearing** could follow a conversation only while it lasted and had lost almost every episode of his past, yet he could still sight-read and conduct. To see how, begin with how memory was first measured.

Ebbinghaus and the measurement of memory ●●

Cognitive psychology treats the mind as an information processor (Section 1.2) and divides remembering into *encoding* (turning information into a code the system can hold), *storage* and *retrieval*. Encoding needs attention: fewer than half of the adults tested by **Raymond Nickerson** and **Marilyn Adams** (1979) could pick out the correct drawing of a coin they had handled thousands of times. Details never attended to are never encoded (*encoding failure*) and so cannot be retrieved.

The first memory experiments were made by **Hermann Ebbinghaus** (1850 to 1909), his own sole participant. To escape the associations of real words he invented the *nonsense syllable* (DAX, BUP), learned lists to one perfect recitation and relearned them later. Relearning was faster even when nothing could be recalled. The **savings method** expresses this as original learning time minus relearning time, divided by original learning time, multiplied by 100: a list learned in 20 readings and relearned in 12 shows (20 minus 12) divided by 20, or 40 per cent, savings. His *forgetting curve* (*Über das Gedächtnis*, 1885) falls steeply, losing more than half within an hour and about two thirds within a day, then levels off without reaching zero. He also first noted the spacing effect and the advantage of a list's beginning and end.

Retention is measured by *recall*, which is free (any order), serial (original order) or cued (from a hint); *paired-associate learning*, introduced by **Mary Whiton Calkins** in 1894, is cued recall of pairs, such as a Kannada word and its English meaning. *Recognition*, picking old items from new ones as in a multiple-choice question, is easier than recall. Relearning is the most sensitive measure. The *reconstruction method* hands back the presented items shuffled for the person to put in order.

EXAM TRAP

Items presented, scrambled and put back in order (RML 2024) is the reconstruction method, a measuring technique, not Bartlett's reconstructive memory. Ebbinghaus, **Francis Galton** and Sigmund Freud were independent pioneers, not laboratory founders, and each devised a new method (nonsense syllables; questionnaires and word association; free association). RML 2020 keys "independent pioneers" alone, as Ebbinghaus's method was experimental, not observational; a bank version keys "(a) and (c)", adding "novel observational methods".

The multi-store model ●●●

In 1968 **Richard C. Atkinson** and **Richard M. Shiffrin** proposed the **multi-store model**, also called the modal or stage model: attended information passes from sensory memory to short-term memory, and rehearsed information passes on to long-term memory, from which it is retrieved when needed. Attention and rehearsal are *control processes* that the person chooses.

The first store, **sensory memory** (the sensory register), briefly holds a large amount of raw, pre-categorical information, one store per sense. **Ulric Neisser** named the visual store *iconic memory* and the auditory store *echoic memory* in 1967; a touch store, haptic memory, is little studied.

KEY STUDY: SPERLING'S PARTIAL REPORT (1960)

Shown twelve letters for 50 milliseconds, people report only four or five but insist they saw more. **George Sperling** sounded a tone just after the flash to signal which of the three rows to report. People gave about three of the four cued letters, so about nine of the twelve had been briefly available. Delaying the tone by about one second removed the advantage: the icon fades within roughly 250 to 500 milliseconds.

Echoic memory lasts two to four seconds, as the "three-eared man" study of **Christopher Darwin**, **Michael Turvey** and **Robert Crowder** (1972) showed.

Short-term memory (STM) holds a little information for seconds unless it is rehearsed. **George Armitage Miller** put its span at "the magical number seven, plus or minus two" (1956) meaningful units, or chunks. Grouping items into larger meaningful units is **chunking**: I N D I A I I T I S R O is twelve letters but three chunks (INDIA, IIT, ISRO). Without chunking and rehearsal, capacity is nearer four chunks (**Nelson Cowan**, 2001). In the Brown-Peterson technique (**John Brown**, 1958; **Lloyd Peterson** and **Margaret Peterson**, 1959), counting backwards blocked rehearsal, and recall of a consonant trigram fell from about 80 per cent at 3 seconds to about 10 per cent at 18, hence a duration of 18 to 30 seconds; **Geoffrey Keppel** and **Benton Underwood** (1962) traced the loss to interference, not decay. STM codes mainly by sound: **Reuben Conrad** (1964) found people writing V for B, and **Alan Baddeley** showed in 1966 that sound-alike words confuse immediate recall and meaning-alike words delayed recall. *Long-term memory* (LTM) codes mainly by meaning (Table 1.14).

Table 1.14 The three stores of the multi-store model compared.

Feature	Sensory memory	Short-term memory	Long-term memory
Capacity	Large	7 ± 2 items (Miller); about 4 chunks (Cowan)	No known limit
Duration	Icon 0.25 to 0.5 s; echo 2 to 4 s	About 18 to 30 s unrehearsed	Minutes to a lifetime
Main code	Raw sensory	Acoustic	Semantic
Main cause of loss	Decay and masking	Displacement and interference	Interference and retrieval failure

> **EXAM TRAP**
>
> "Largest capacity but shortest duration" (JEMAS 2025) is sensory memory; "largest capacity" alone in the Atkinson and Shiffrin model is long-term memory. "Icon and echo" are always sensory memory, and the store that "lasts only a few seconds" is sensory memory, not eidetic imagery.

Free recall gives the U-shaped **serial position effect**, or primacy-recency effect (**Bennet Murdock**, 1962; Figure 1.8). The *primacy effect* reflects extra rehearsal of early items into LTM; the *recency effect* reflects the last items still sitting in STM. **Murray Glanzer** and **Anita Cunitz** (1966) found that 30 seconds of counting before recall abolished recency but spared primacy.

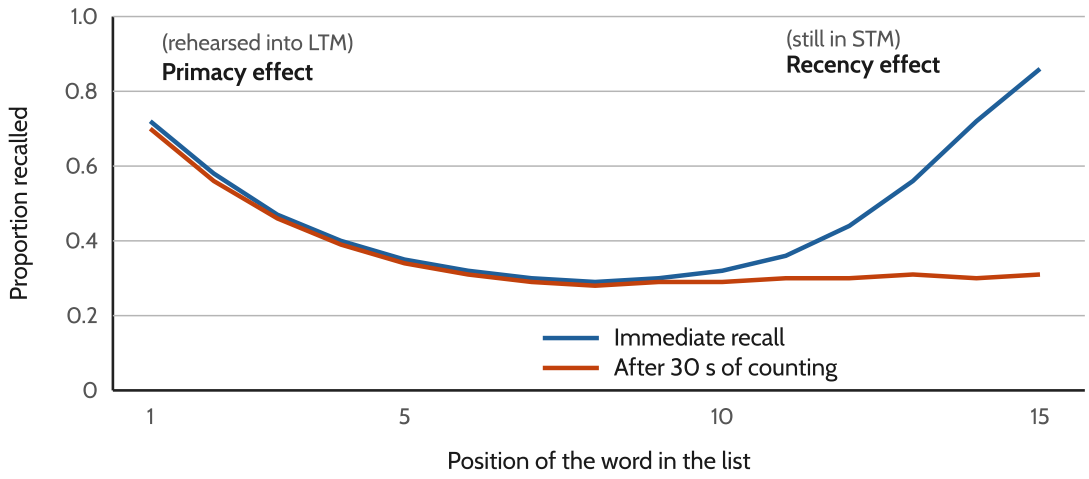

Figure 1.8 The serial position curve in free recall (schematic, with the pattern reported by Glanzer and Cunitz, 1966). Immediate recall (solid line) shows both primacy and recency. A 30-second filled delay (dashed line) removes the recency effect but leaves primacy intact.

Amnesic patients such as H.M., met below, have a normal span and recency effect but no new long-term learning, while patient K.F. (**Tim Shallice** and **Elizabeth Warrington**, 1970) had a two-item span and normal long-term learning: a double dissociation that supports separate stores. Yet neither store proved unitary, and mere repetition does little for LTM.

> EXAM TRAP
>
> The stage model has three stores; working memory is NOT one of them (RML 2025), and information must pass through short-term (sometimes "short-term (working)") memory to reach LTM. RML 2024 keys "McCray and Hunter" (**J. W. McCrary** and **W. S. Hunter**, 1953) for first observing the primacy-recency effect, although Ebbinghaus noted it in 1885. Better recall of the last items is recency, not the Flynn effect (rising IQ scores) or halo effect (a rating bias). As order effects in a psychophysics experiment, primacy and recency are controlled by counterbalancing (Section 1.3).

Working memory ●●●

Multiply 24 by 7 in your head and you must hold numbers, calculate and store a partial result at once. Alan Baddeley and **Graham Hitch** (1974) therefore recast STM as **working memory**, a limited-capacity system for holding and manipulating information during reasoning and comprehension. Using the dual-task method, they found that people holding six digits still reasoned accurately, so STM must have several parts (Figure 1.9).

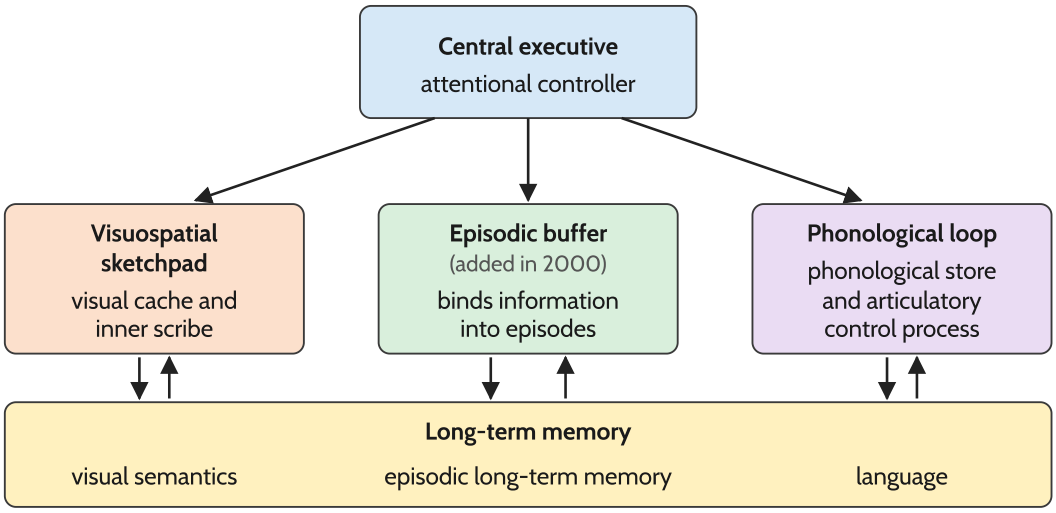

Figure 1.9 Baddeley's working memory model (Baddeley and Hitch, 1974; episodic buffer added by Baddeley, 2000). The central executive directs attention but stores nothing; the three subsidiary systems hold information and exchange it with long-term memory.

The **phonological loop** stores and rehearses auditory-verbal information: a phonological store (inner ear) holds speech sounds for about two seconds, and an articulatory control process (inner voice) refreshes them. Similar-sounding items are confused, spans are larger for short words (the word-length effect), and *articulatory suppression* (saying "the, the, the") abolishes the word-length effect. The **visuospatial sketchpad** holds visual and spatial information, such as a mentally rotated object; **Robert Logie** (1995) split it into a visual cache and an inner scribe.

The **central executive**, an attentional controller that stores nothing, focuses, divides and switches attention and inhibits unwanted responses. Random number generation loads it, and dorsolateral prefrontal damage causes a dysexecutive syndrome. In the Stroop task (**John Ridley Stroop**, 1935), naming the ink colour of RED printed in blue is slow because automatic reading competes with controlled colour naming, an interference in selective attention (Section 1.5). The **episodic buffer**, added by Baddeley in 2000, holds about four chunks and binds information from the loop, the sketchpad and LTM into single, coherent episodes. STM is storage alone (forward span); working memory is storage plus processing, tested by complex span and by backward digit span.

> EXAM TRAP
>
> Random number generation loads the central executive (RML 2022); repeating a word loads the loop, a spatial configuration the sketchpad, irrelevant speech the phonological store. A "visual executive loop" (Amity 2023) does not exist, and the Stroop effect is selective attention, not memory.

Levels of processing ●●

Fergus Craik and **Robert Lockhart** (1972) argued that memory is a by-product of how deeply information is processed. Their **levels of processing** run from shallow structural processing (is the word in capitals?) through phonemic processing (does it rhyme with WEIGHT?) to

deep semantic processing (does it fit the sentence?), and deeper processing leaves more durable traces, as Craik and Endel Tulving (1975) confirmed. Maintenance rehearsal, mere repetition, adds little to LTM; elaborative rehearsal, linking an item to meaning, builds it.

Words judged for "Does this describe you?" are recalled about twice as well as words judged for meaning: the **self-reference effect** (**T. B. Rogers**, **Nicholas Kuiper** and **W. S. Kirker**, 1977). Words you generate also beat words you merely read (the generation effect). Critics called depth circular, since it is defined by the memory it explains, and **Donald Morris**, **John Bransford** and **Jeffery Franks** (1977) found that rhyme processing wins on a rhyming test: memory depends on the match between study and test (transfer-appropriate processing).

> EXAM TRAP
>
> The self-reference effect is Rogers, Kuiper and Kirker's (keyed "Rogers and Kuiper", RML 2024), not Craik and Lockhart's, **Allan Paivio**'s (dual coding theory, 1971: verbal plus imagery codes) or **Michael Posner**'s (attention). It means better recall of self-related information (IHBAS 2020), not self-concept, self-schema or self-awareness (Personality chapter).

The systems of long-term memory ●●●

Clive Wearing kept skills and facts but lost his past, so long-term memory is not one store. In 1972 **Endel Tulving** (1927 to 2023) separated **episodic memory**, personal events with their what, where and when (your first day of college), from **semantic memory**, the network of facts and concepts that make up our knowledge of the world, also called generic memory. Episodic recall carries autonoetic consciousness, a sense of travelling back in time; semantic memory carries only noetic knowing. Knowing how (riding a bicycle) is **procedural memory**. Tulving's patient K.C. kept his general knowledge but lost every episode of his life.

Explicit memory involves conscious recollection, tested by recall and recognition; **implicit memory** is an influence of past experience on behaviour, feelings or thought without conscious recollection (**Peter Graf** and **Daniel Schacter**, 1985), roughly **Larry Squire**'s declarative and nondeclarative memory. Declarative memory needs the hippocampus; nondeclarative memory covers procedural skills (striatum), priming, classical conditioning (amygdala, cerebellum) and habituation. In *priming*, earlier exposure eases processing of the same item (repetition priming) or a related one: BUTTER is recognised faster after BREAD (semantic priming; **David Meyer** and **Roger Schvaneveldt**, 1971), even with a masked, unseen prime, so semantic priming is a tool for studying subliminal perception.

> EXAM TRAP
>
> Implicit memory does NOT involve conscious recollection (RML 2020); it "affects performance without conscious awareness" (RML 2025), and limited capacity describes STM. Procedural memory is one kind of implicit memory.

Autobiographical memory holds the specific episodes of one's own life; older adults recall a surplus from ages 10 to 30, the *reminiscence bump* (**David Rubin** and colleagues, 1986). *Prospective memory*, remembering to act in future (a tablet at 8 p.m.), contrasts with retrospective memory. **Roger Brown** and **James Kulik** (1977) named the **flashbulb memory**, a vivid memory of the circumstances in which one learned of a shocking event, and credited it to a "Now Print!" mechanism.

KEY STUDY: NEISSER AND HARSCH, THE CHALLENGER DISASTER (1992)

The morning after the Challenger shuttle exploded in January 1986, Ulric Neisser and **Nicole Harsch** had students write down how they heard the news. Two and a half years later, a quarter of 44 were wrong on every major detail, yet confident. After 11 September 2001, **Jennifer Talarico** and David Rubin (2003) found that flashbulb and everyday memories lost consistency equally, but flashbulb memories stayed more vivid and more confidently believed.

Flashbulb memories are thus special in vividness and confidence, not accuracy, and arise from emotional arousal at encoding plus rehearsal. *Eidetic imagery*, a visual image that lingers half a minute or more, occurs in a few children and rarely in adults. *Metamemory* (**John Flavell**, 1971), knowledge and control of one's own memory, is part of metacognition and includes the feeling of knowing, a sense that one could recognise an answer one cannot recall. In the **tip-of-the-tongue state**, which Roger Brown and **David McNeill** (1966) produced with definitions of rare words (sextant), people knew the first letter and syllable count: the word was stored but not retrievable.

EXAM TRAP

A vivid recollection of an important event is a flashbulb memory (RML 2025), not an engram, hindsight bias (the "knew it all along" distortion of memory) or equipotentiality. Its confidence stays very high as accuracy falls (NIMHANS 2026 recall), and arousal plus rehearsal explains it, not Now Print!, lack of interference or eidetic imagery. Tip-of-the-tongue is a retrieval failure (RML 2024), not storage, repression or aphasia (Wernicke's or Broca's, where language itself is damaged). Planning, monitoring and controlling memory is metacognition (IHBAS 2020), not attention, inhibition or a knowledge base.

Networks, schemas and reconstruction ●●●

In the hierarchical network of **Allan Collins** and **Ross Quillian** (1969), concepts are nodes (ANIMAL, BIRD, CANARY) and, by cognitive economy, each property is stored once at the highest node it fits, so "A canary has skin" takes longer to verify than "A canary can sing". In the spreading activation model of Collins and **Elizabeth Loftus** in 1975, activation spreads to related concepts and primes them. In **connectionism**, or parallel distributed processing (**David Rumelhart**, **James McClelland** and colleagues, 1986), a concept is a pattern of simultaneous activation across many interconnected units, and knowledge lies in the connection weights. A proposition is the smallest unit of knowledge that can be true or false.

EXAM TRAP

"A network of interconnected nodes" (IHBAS 2024) keys the connectionist model, although Collins and Quillian's network also has nodes; "associative" and "linkage" describe any network rather than name a model. Parallel distributed processing means associative networks and simultaneous activation (IHBAS 2025), not serial processing.

A **schema** is organised knowledge, built from experience, that guides perception and memory; the schema approach studies how such knowledge is used to understand behaviour. A script, a schema for a routine (the restaurant script of **Roger Schank** and **Robert Abelson**,

1977: enter, order, eat, pay, leave), belongs to generic memory. The schema idea came from **Frederic Charles Bartlett** (1886 to 1969). In *Remembering* (1932), English students retelling the Native American tale "The War of the Ghosts" shortened it and emphasised a few details (levelling and sharpening, in **Gordon Allport** and **Leo Postman**'s later terms), explained away puzzling events (rationalisation) and turned canoes into boats. Remembering is an "effort after meaning": *reconstructive memory* rebuilds the past from fragments and schemas.

Asked how fast filmed cars were going when they "smashed" rather than "hit" each other, students gave higher speeds and a week later more often recalled broken glass that was never shown (32 against 14 per cent; Elizabeth Loftus and **John Palmer**, 1974). This is the **misinformation effect**: memory distorted by leading questions or misleading later information. Whole false events can be implanted (being lost in a shopping mall as a child), and people who study bed, rest, tired and dream recall the absent "sleep" (the Deese-Roediger-McDermott paradigm). In **source monitoring** (**Marcia Johnson** and colleagues, 1993) we attribute a memory to its origin from its contextual details; reality monitoring, perceived versus imagined, is one kind. Source errors cause source amnesia (a fact remembered without its source), cryptomnesia (another's idea recalled as one's own) and false eyewitness identifications. *Déjà vu* is the illusion of recognition, jamais vu its reverse, and pseudologia fantastica pathological lying. In the recovered memory controversy Loftus warned that suggestive therapy can implant false memories of abuse, though real abuse can also be forgotten and recalled.

> EXAM TRAP
>
> Failing to recall the relevant elements of an autobiographical experience is a source-monitoring deficit (RML 2022); "personal-" and "social-monitoring" are invented. Defence lawyers fear source errors (Amity 2023), not forgetting or poor eyesight; confidence does not mean accuracy. Schema theory is Bartlett's, not Wundt's, Titchener's or Watson's, and his rationalisation is not the defence mechanism. Scripts come from generic memory (RML 2025).

Retrieval and forgetting ●●●

Much forgetting is a failure to find what is still stored: given category names as cues, people recall far more words (Endel Tulving and **Zena Pearlstone**, 1966). By the **encoding specificity principle** (Tulving and **Donald Thomson**, 1973), a cue works to the extent that it was encoded with the memory; recognition beats recall because the test supplies the item. In **context-dependent memory** the physical setting must match: divers recalled words best where they had learned them, underwater or on the beach (**Duncan Godden** and Alan Baddeley, 1975). In **state-dependent memory** the internal state must match: material learned while drunk is recalled better when drunk again (**Donald Goodwin** and colleagues, 1969). **Gordon Bower** (1981) added mood-dependent memory and mood-congruent memory, the easier recall of material matching one's current mood.

EXAM TRAP

Rahul, who expects to do better in his usual classroom (IHBAS 2024), shows encoding specificity, keyed because context-dependent memory is not offered; "the principle of congruity" is a distractor. Offered both (JEMAS 2025), choose context-dependent for surroundings and state-dependent for internal states. Alcohol-induced amnesia is state-dependent memory (RML 2022).

Forgetting is useful: it clears out unneeded detail and reduces interference, and the more is retained, the less is forgotten. First, **decay theory**, credited to **Edward Lee Thorndike**'s (1874 to 1949) law of disuse, says unused traces fade with time. But **John Jenkins** and **Karl Dallenbach** (1924) found that after eight hours of sleep students recalled over half of their nonsense syllables, and after eight waking hours about a tenth: events, not time, caused the loss. Table 1.15 lists other named effects on retention.

Interference theory blames other memories. In **proactive interference** earlier learning disrupts newer learning; in **retroactive interference** newer learning disrupts older memories (PRO: old disrupts new; RETRO: new disrupts old). Dialling your old phone number instead of the new one is proactive, as is earlier psychology intruding on a later sociology test; losing the old number to the new one is retroactive. After lists A-B and then A-C, recalling A-B measures retroactive and recalling A-C proactive interference. **Arthur Melton** and **J. M. Irwin** (1940) blamed response competition and unlearning.

EXAM TRAP

RML 2024 keys an English speaker applying English grammar to Spanish as "retroactive". The key is wrong: old disrupts new, so it is proactive. "Interference from new information" (IHBAS 2024) is retroactive. Decay theory is keyed to Thorndike (RML 2024), not Ebbinghaus, Tulving or Melton and Irwin.

In retrieval failure, or cue-dependent forgetting (Tulving, 1974), the trace survives but the cues do not match. Retrieval can even cause forgetting: practising some items of a category (FRUIT-orange) makes the unpractised ones (FRUIT-banana) harder to recall than items of unpractised categories (**Michael Anderson**, **Robert Bjork** and **Elizabeth Bjork**, 1994), through inhibition of competitors. **Sigmund Freud** (1856 to 1939) proposed motivated forgetting through **repression**, the unconscious exclusion of threatening memories from awareness; suppression is conscious, and evidence for repression is weak. Last, consolidation (**Georg Elias Müller** and **Alfons Pilzecker**, 1900) can fail: a head injury or electroconvulsive therapy erases recent memories, and by Ribot's law (**Théodule Ribot**, 1881) the new perishes before the old.

EXAM TRAP

Retrieval-induced forgetting (RML 2020) is keyed "practised information remembered better than unpractised", but its defining half is that related unpractised items suffer. Repression is unconscious, suppression conscious (RML 2025); motivated forgetting covers both, and inhibition weakens competitors. "Less retention leads to less forgetting" is the incorrect statement (RML 2025).

Table 1.15 Classic named effects on retention.

Effect	Who and when	What happens
Zeigarnik effect	**Bluma Zeigarnik**, 1927	Interrupted tasks are recalled better
Ovsiankina effect	**Maria Ovsiankina**, 1928	Interrupted tasks tend to be resumed
Reminiscence	**Philip Ballard**, 1913	Recall rises after a delay without study
Von Restorff effect	**Hedwig von Restorff**, 1933	A distinctive list item is remembered better
Spacing effect	Ebbinghaus, 1885	Distributed practice beats massed practice
Testing effect	**Arthur Gates**, 1917	Retrieval practice beats restudy

Amnesia and the biology of memory ●●●

Amnesia is a pathological loss of memory. In **anterograde amnesia** new memories cannot be formed after an injury; in **retrograde amnesia** memories from before it are lost, usually sparing the remote past. Of the other "a-" disorders, aphasia (with alexia and acalculia) affects language, apraxia skilled movement and agnosia recognition; dementia is a progressive decline in memory, judgement and abstraction.

In 1953 the neurosurgeon **William Beecher Scoville** removed both medial temporal lobes, with most of the hippocampus and amygdala, from **Henry Gustav Molaison** (1926 to 2008), known as H.M., to relieve his epilepsy. Scoville and **Brenda Milner** reported in 1957 a profound anterograde amnesia: he reread magazines with fresh interest and never learned his new address. Yet his intelligence, digit span and childhood memories were intact, and over three days he improved at tracing a star in a mirror while denying that he had ever done it.

> EXAM TRAP
>
> H.M.'s main deficit was anterograde amnesia for declarative memory (Real PYQ 2024); procedural, short-term and old memories were spared. Both medial temporal lobes were removed, not frontal, parietal or occipital cortex. The hippocampus is the structure damaged in anterograde amnesia and critical for consolidation (JEMAS 2025, RML 2024, RML 2023), not the hypothalamus, pituitary, cortex, amygdala (emotional memory), prefrontal cortex (working memory) or cerebellum (motor learning).

Korsakoff's syndrome (Korsakoff's psychosis; **Sergei Korsakoff**, late 1880s) results mainly from thiamine (vitamin B1) deficiency in chronic alcoholism, often after Wernicke's encephalopathy (**Carl Wernicke**, 1881: confusion, ataxia, nystagmus); together they form the Wernicke-Korsakoff syndrome, which damages the mammillary bodies and mediodorsal (dorsomedial) thalamus. It brings dense anterograde amnesia, extensive retrograde amnesia and **confabulation**, sincere false stories that fill memory gaps, unlike malingering (deliberate faking), delirium (acute, fluctuating consciousness) or confusion. Alzheimer's disease, by contrast, first erodes recent episodic memory, and Huntington's and Parkinson's diseases impair procedural learning.

> **EXAM TRAP**
>
> Memory deficits, confabulation and nystagmus in a chronic drinker (Real PYQ 2024) is Wernicke-Korsakoff syndrome, not Alzheimer's, vascular or frontotemporal dementia. The alcohol-related illness with profound retrograde amnesia (RML 2024) is Korsakoff's psychosis; the vitamin is B1, not B12, D or folate.

Infantile amnesia, the lack of memories from the first three or four years, reflects an immature hippocampus and the lack of language. Transient global amnesia is a sudden anterograde amnesia lasting hours. Dissociative amnesia, a loss of personal memories with no medical cause (generalised when it covers the whole life), belongs to the Psychiatric Disorders chapter.

Richard Semon (1904) called the physical memory trace the **engram**. **Karl Spencer Lashley** (1890 to 1958) cut away rats' cortex after maze learning and found that memory loss depended on how much was removed (mass action), not where, with areas substituting for one another (equipotentiality): memories are distributed across the cortex ("In search of the engram", 1950). **Donald Olding Hebb** (1904 to 1985) proposed in 1949 that neurons active together strengthen their links, and **Timothy Bliss** and **Terje Lømo** (1973) found such **long-term potentiation** (LTP), a lasting strengthening of hippocampal synapses after high-frequency stimulation and the leading cellular candidate for memory. The hippocampus consolidates new memories into the neocortex, partly during sleep, so remote memories survive its damage.

Improving memory

A mnemonic adds organisation, meaning or imagery. The method of loci, credited to **Simonides of Ceos**, places vivid images along a familiar route; the peg-word system links items to rhyming number pegs (one is a bun); the keyword method (Richard Atkinson, 1975) links a foreign word to a sound-alike image (Spanish "pato", duck: a duck in a pot). An acronym is a word of first letters (VIBGYOR); an acrostic is a sentence of first letters ("Every Good Boy Deserves Fruit"). SQ3R (**Francis P. Robinson**, 1946) is Survey, Question, Read, Recite, Review; PQRST is Preview, Question, Read, Self-recitation, Test. Testing and spacing beat rereading, overlearning gives diminishing returns, and sleep consolidates.

> **EXAM TRAP**
>
> A first-letter sentence is an acrostic, not an acronym, peg-word or keyword. The "S" of PQRST is self-recitation (RML 2024), not storing, sequencing or semantics.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Learning in 30 trials and relearning in 18 gives savings of _____ per cent.
2. A 30-second filled delay before recall abolishes the _____ effect.
3. Baddeley added the _____ to working memory in 2000.
4. Flashbulb memories keep their vividness and _____ as accuracy falls.
5. Distortion of a memory by a leading question asked afterwards is the _____ effect.
6. Dialling your old phone number instead of the new one is _____ interference.
7. Korsakoff's syndrome follows deficiency of _____ (vitamin B1).

Answers: (1) 40, (2) recency, (3) episodic buffer, (4) confidence, (5) misinformation, (6) proactive, (7) thiamine.

1.8 Thinking, Problem Solving, Reasoning, Decision Making and Language

HIGH YIELD · 53 QS

Thinking is the mental manipulation of internal representations to form new ones, as in solving, reasoning, deciding or creating. Seeing the chai stall is not thinking; working out whether you can afford a second cup is. Banished by Watson, thought returned with the cognitive revolution (Section 1.2): cognitive psychology studies the structuralists' topics (consciousness, imagery, thought) by objective measures.

Elements, Imagery and Types of Thought •

Thought uses five elements: *images*, *concepts*, *symbols*, *propositions* (the smallest units of knowledge that can be true or false) and *language*. **Roger Newland Shepard** and Jacqueline Metzler (1971) found that the time to judge two rotated block figures identical rose linearly with the angle between them, as if the image were turned at a steady speed: **mental rotation**. **Stephen Michael Kosslyn**'s image-scanning studies (1978) suggested that images depict space, against **Zenon Walter Pylyshyn**'s view (1973) that knowledge is stored as propositions. **Allan Paivio**'s **dual-coding theory** (1971) gives pictures and concrete words both a verbal and an imagery code, so they are remembered better than abstract words. Table 1.16 lists types of thinking.

Table 1.16 Types of thinking.

Type	Associated with	What it is
Directed versus undirected	Würzburg school; Freud	Goal-guided and reality-checked versus drifting by association (daydreaming)
Autistic (dereistic)	**Eugen Bleuler**, 1911	Wish-driven fantasy that ignores reality
Convergent versus divergent	J. P. Guilford	One correct answer versus many varied answers
Critical versus creative	Robert Ennis; Guilford	Evaluating claims versus generating novel, useful ideas
Reflective	**John Dewey**, 1910	Careful weighing of a belief and its grounds

Type	Associated with	What it is
Productive versus reproductive	**Max Wertheimer**, 1945	Restructuring a problem versus reapplying a method

> EXAM TRAP
>
> Dual coding favours **visual** information, not verbal, auditory or semantic. Autistic thinking is fantasy, not autism. Loosening of associations, circumstantiality and word salad are disorders of the *form* of thought (Psychiatric Disorders chapter).

Concepts: Rules and Prototypes ●●

A **concept** is a mental category of things sharing properties, formed through observation, discrimination, generalisation and abstraction. The classical view defines it by necessary and sufficient features, which suits *artificial concepts* set by a rule: *conjunctive* (all attributes: a voter is a citizen, adult *and* enrolled), *disjunctive* (any one: "out" in cricket) or *relational* ("taller than"). **Jerome Seymour Bruner**, Jacqueline Goodnow and George Austin (1956) found that people learn them by testing hypotheses, by *scanning* or *focusing*; *conservative focusing*, changing one attribute at a time, works best, and *focus gambling* changes several.

Natural concepts are fuzzier. **Ludwig Wittgenstein** (1953) saw that games share only a **family resemblance**, and **Eleanor Rosch** showed in the 1970s that typicality is graded: "a robin is a bird" is verified faster than "a penguin is a bird". We classify by similarity to a **prototype**, the abstract, averaged best example, and learn the *basic level* (chair, not furniture) first; *exemplar theory* stores many instances instead. How concepts are networked in memory is told in Section 1.7.

> EXAM TRAP
>
> Artificial concepts by hypothesis testing = **Bruner**, not Rosch (natural concepts), **Clark Leonard Hull** (gradual associative abstraction, 1920) or Watson. A prototype is **abstract and averaged**, not a template (exact copy), a feature (a part) or a geon (a simple solid).

Problem Solving: Strategies and Obstacles ●●●

A problem is a gap between an initial state and a goal state, well defined (a sudoku) or ill defined ("choose a career"). **Allen Newell** and **Herbert Alexander Simon** (*Human Problem Solving*, 1972; the General Problem Solver) treated solving as search through a **problem space** of states linked by operators (legal moves). An **algorithm** guarantees the solution but may be slow (trying every padlock code); a **heuristic** is a quick rule of thumb with no guarantee. **Means-end analysis** sets subgoals that reduce the biggest difference from the goal (to reach Bengaluru, first reach the station); *working backward* starts from the goal; **hill climbing** always takes the step that gets closest, so it stalls when a backward step is needed; *analogy* reuses an earlier solution, though people rarely notice one unprompted (**Mary Gick** and **Keith Holyoak**'s radiation problem, 1980); *trial and error* is the most primitive. Stage models range from Dewey (1910) to Sternberg's problem-solving cycle.

The Gestaltists stressed **insight**, a sudden restructuring of the problem: **Wolfgang Köhler**'s chimpanzee Sultan suddenly joined two sticks to reach a banana (1917), and **Norman Maier**'s nine-dot problem (1930) defeats people until they let the lines leave the square. Janet

Metcalfe and David Wiebe (1987) found "warmth" ratings flat until just before an insight solution, but rising steadily on algebra. *Incubation*, setting a problem aside, often helps. Experience can also block a solution.

KEY STUDY: DUNCKER'S CANDLE PROBLEM (1945)

Karl Duncker asked people to fix a candle to a wall with matches and a box of drawing pins. Few thought of pinning up the emptied box as a shelf, but many more did when the box came empty. Knowing an object's usual use (top-down processing) blinded them to its other uses: **functional fixedness**.

Functional fixedness has a strategic cousin, **mental set** (Einstellung): in **Abraham Luchins**'s water jars (1942), people drilled on B minus A minus 2C kept using it when A minus C was far simpler. Confirmation bias and irrelevant information also obstruct.

EXAM TRAP

A heuristic is "a general rule that is usually right": efficient, **not guaranteed**, not more accurate than an algorithm. Functional fixedness concerns **objects** (a matchbox seen only as a container, not a platform) and reflects overactive top-down processing; mental set concerns **strategies**. As fixedness is a special case of set, one paper keyed **mental set** "the most basic and common obstacle"; *fixation* is the umbrella word, "mental barrier" no term, trial and error a strategy. In a mental status examination, insight is awareness of illness, graded 1 to 6; **intellectual insight** is grade 5.

Reasoning: Deduction and Induction ●●●

Deductive reasoning moves from general premises to a conclusion that must be true if they are ("All Mysuru autos must use the meter; this is one; so it must"). **Inductive reasoning** moves from specific observations to a probable generalisation that the next case may overturn ("every auto I took overcharged, so all do"). An argument is *valid* if its conclusion follows from its form, *sound* if its premises are also true. **Charles Sanders Peirce** (1839 to 1914) called inference to the best explanation, as in diagnosis, **abductive reasoning**. In research, deduction tests theory against data (quantitative work), induction builds theory from data (qualitative work), and the hypothetico-deductive method of **Karl Raimund Popper** (1902 to 1994) is the full cycle of hypothesis, deduced prediction and test.

EXAM TRAP

"Starts with a theory and tests it with observations" = **deductive**. "Gathers data, looks for patterns, then generates a theory", "specific observations to broader generalisations", "theory development" and "qualitative research" = **inductive**. Keys prefer "deductive reasoning" to "hypothetico-deductive method" and rest deductive validity on **the soundness of the initial theory**. **Heuristics** was keyed as NOT reflecting induction; probabilistic conclusions, decision making and concept formation do.

In syllogisms (two premises, one conclusion), **belief bias** makes people accept invalid arguments with believable conclusions, and the *atmosphere effect* (Robert Woodworth and Saul Sells, 1935) lets a "some" or "no" premise suggest a matching conclusion. For "if p, then q",

modus ponens (p, so q) and *modus tollens* (not q, so not p) are valid; *affirming the consequent* (q, so p) and *denying the antecedent* (not p, so not q) are fallacies. In **Peter Cathcart Wason**'s selection task (1966), under one person in ten turns the right cards (Figure 1.10), yet a permission rule ("if drinking beer, must be over 19") makes the same logic easy.

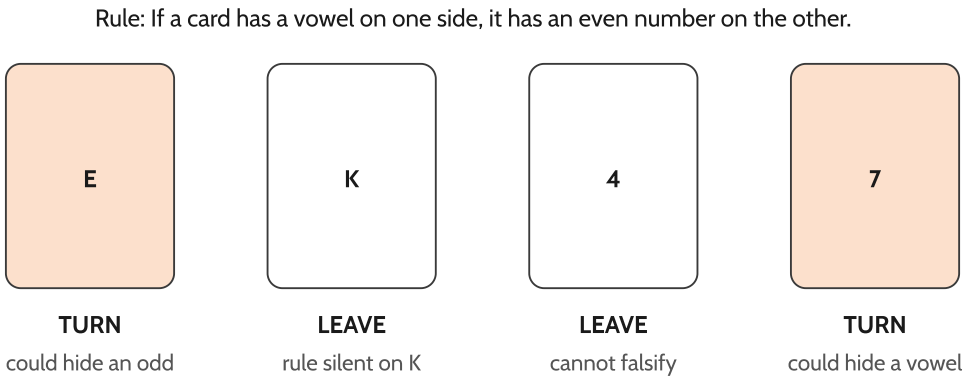

Figure 1.10 The abstract Wason selection task. The two cards with heavy outlines (E and 7) are the only ones that could reveal a counterexample. The popular choice of the 4 card tests for confirmation, which can never prove a general rule.

Philip Nicholas Johnson-Laird (1983) argued that we reason with *mental models* of possible situations, not formal rules. In Wason's 2-4-6 task (1960), people guessed "rising by two" and tested only fitting sequences (the rule was any increasing numbers): **confirmation bias**, seeking and favouring evidence for what one already believes.

> EXAM TRAP
>
> Confirmation bias was keyed as "closely related to" **representativeness**, a debatable key (recognition is least related). Acting on an expectation until it comes true is a **self-fulfilling prophecy**, not confirmation bias, the base-rate fallacy or "motivational reaction" (no term). Piaget's formal operations bring hypothetico-deductive (a capacity, not a stage), counterfactual and metacognitive reasoning; **dialectical** reasoning, integrating contradictions, is postformal (Developmental Psychology chapter).

Judgement and Decision Making ●●

Normative models say how a rational agent should decide, maximising expected utility; *descriptive* models say how people actually decide. Simon's **bounded rationality** holds that information, mental capacity and time limit choice, so people practise **satisficing**, taking the first good-enough option (Nobel Prize, 1978). **Gerd Gigerenzer** (born 1947) adds that fast and frugal heuristics, such as the *recognition heuristic* (choosing the option you recognise), are often ecologically rational.

Amos Tversky and **Daniel Kahneman** (Kahneman's Nobel Prize, 2002) catalogued the heuristics of judgement and their biases (Table 1.17). Representativeness invites *base-rate neglect*: a description that "sounds like" an engineer is judged an engineer whether engineers are 30 or 70 in 100. Most people judged Linda, a socially concerned philosophy graduate, likelier to be "a feminist bank teller" than "a bank teller", although a conjunction cannot be likelier than its parts: the **conjunction fallacy**.

Table 1.17 The classic heuristics and their typical biases.

Heuristic	Proposed	We judge by	Typical error
Availability	1973	Ease of recall	More words seem to begin with K than have K third
Representativeness	1972	Similarity to the typical case	Base-rate neglect; conjunction fallacy; gambler's fallacy (after many reds, black seems "due")
Anchoring and adjustment	1974	Adjusting from a starting value	Estimates stay near an arbitrary anchor

In the Asian disease problem (1981), most people preferred saving 200 of 600 lives for sure to a gamble, but chose the gamble when the sure option read "400 people will die": risk aversion for gains, risk seeking for losses. **Prospect theory** (1979) explains this *framing effect*, the same choice decided differently when worded as gains or as losses: outcomes are gains or losses from a reference point, and losses loom about twice as large as equal gains (**loss aversion**), which also drives the endowment effect (overvaluing what we own), status quo bias (preferring no change) and the sunk-cost fallacy (persisting because of past investment). **Hindsight bias**, the "I knew it all along" effect (**Baruch Fischhoff**, 1975), and overconfidence (confidence exceeding accuracy) distort self-knowledge. Dual-process theory contrasts fast, intuitive *System* 1, home of the heuristics, with slow, deliberate *System* 2.

> EXAM TRAP
>
> Bounded rationality = **Herbert Simon**, not Kahneman, Tversky or Howard Gardner. Representativeness errs when people **ignore base rates and focus solely on similarity**; the other options describe availability, anchoring and confirmation bias. Overrating how well we would have predicted events = **hindsight**, not the just-world hypothesis or "decisiveness".

Creativity ●●

Creativity is producing work that is both novel and appropriate. **Joy Paul Guilford**, whose 1950 American Psychological Association address launched its study, equated it with divergent production, scored by *fluency* (number of ideas), *flexibility* (number of categories), *originality* (statistical rarity) and *elaboration* (detail). **Graham Wallas** (*The Art of Thought*, 1926) described four stages: *preparation* (defining and working on the problem), *incubation* (setting it aside while work continues outside awareness), *illumination* (the sudden insight) and *verification* (testing and refining it).

> EXAM TRAP
>
> Order: **preparation, incubation, illumination, verification** ("Please Incubate Ideas Very carefully"). Sensitivity, fluency, flexibility and originality are Guilford's abilities; analytical, creative and practical are Sternberg's intelligences; "problem-finding, brainstorming, selection, implementation" is not Wallas.

Creativity is measured by the *Torrance Tests of Creative Thinking* (**Ellis Paul Torrance**, 1966) and **Sarnoff Mednick**'s *Remote Associates Test* (1962): which word links cottage, Swiss and cake? (Cheese.) **Edward de Bono**'s *lateral thinking* (1967) challenges assumptions; **Alex Faickney Osborn**'s brainstorming (*Applied Imagination*, 1953) defers judgement and seeks

quantity. By the threshold hypothesis, creativity and IQ correlate only up to an IQ of about 120. In **Robert Jeffrey Sternberg** and Todd Lubart's investment theory (1991), creative people "buy low and sell high". The creative personality is open to experience, independent and tolerant of ambiguity.

The Structure of Language ●●

Language combines arbitrary symbols by rules into unlimited messages. **Charles Francis Hockett** (1960) listed its design features, notably semanticity (meaning), arbitrariness (nothing in "dog" is doglike), displacement (talking about what is absent), generativity (endless new sentences) and duality of patterning (meaningless sounds build meaningful units). The **phoneme** is the smallest unit of sound that distinguishes words (/p/ and /b/ in pat and bat) and has no meaning; the **morpheme** is the smallest unit of meaning, free (cat) or bound (-s), so "cats" has two. Syntax governs how words combine, *semantics* is meaning, **pragmatics** is use in social context ("Can you pass the salt?" is a request) and **prosody** is rhythm and intonation. **Noam Chomsky**'s transformational grammar (*Syntactic Structures*, 1957; *Aspects*, 1965) separates *surface structure*, the words as spoken, from **deep structure**, who did what to whom: "Visiting relatives can be boring" has one surface and two deep structures. *Competence* is tacit knowledge of the rules, *performance* their imperfect use.

> EXAM TRAP
>
> Smallest unit of meaning = **morpheme**; basic unit of sound = **phoneme**; a syllable is a unit of pronunciation, a consonant one kind of phoneme. Emotional prosody depends on **right-hemisphere regions homotopic to the language areas**, not Broca's area, motor cortex or occipital lobe. **Recursive thought** ("I think that you think...") is perspective taking, not reversibility (Piaget's mental undoing of an action), referential style (below) or reconstruction (rebuilding a memory).

Acquiring Language ●●●

Burrhus Frederic Skinner's *Verbal Behavior* (1957) explained language by reinforcement of verbal operants such as the *mand* (a request) and the *tact* (naming). Chomsky's 1959 review replied that children produce sentences they never heard, make errors no adult models ("I goed") and are rarely corrected for grammar. He proposed an innate **language acquisition device** (LAD) holding a *universal grammar*, argued from the **poverty of the stimulus**: children's input is too sparse to explain the grammar they reach. Bruner's *language acquisition support system* (LASS, 1983) stresses adult scaffolding and child-directed speech (motherese): higher and wider pitch, slower rate, more pauses, shorter utterances. **Dan Isaac Slobin**'s *language-making capacity* (1985) is a set of cognitive and perceptual operating principles specialised for language. **Eric Heinz Lenneberg** (1967) proposed a **critical period** from about two to puberty, now usually called a sensitive period; Genie, isolated until 13, learned words but never grammar.

Infants coo at about 2 months and babble at about 6; at about 12 months a single word, a **holophrase**, does a sentence's work; at 18 to 24 months comes what **Roger Brown** (1925 to 1997) called **telegraphic speech**, two words without function words ("more milk"); and at 2 to 3 years **overregularisation** ("two foots"). Children overextend words ("dog" for every animal), underextend them ("dog" only for their own) and learn new words from one exposure

(fast mapping); **Katherine Nelson** (1973) found some children's first words mostly object names, a *referential* style. Language is most often compromised in autism spectrum disorder.

> EXAM TRAP
>
> Chomsky's innate capacity is the **LAD** (an inborn processor activated by verbal input, holding universal grammar, the rules common to all languages), not the phonological loop (working memory), the zone of proximal development (Vygotsky) or a schema; LASS = **Bruner**, LMC = **Slobin**. Irreversible deficits after early deprivation support **sensitive periods**, not semantics or syntax (levels of language) or nativism (the broader theory that language is innate).

Language and Thought ●●●

Benjamin Lee Whorf (1897 to 1941), a student of **Edward Sapir** (1884 to 1939), argued that language shapes thought. The strong version, linguistic determinism, has failed: **Brent Berlin** and **Paul Kay** (1969) found colour terms entering languages in a near-universal order, and Rosch found that the Dani, with two colour terms, remember focal colours as English speakers do. The weak version, **linguistic relativity** (language influences thought), survives: Hindi kinship terms (*chacha*, *mama*, *phupha*) make speakers track which side an uncle is on. **Steven Pinker** (1994) holds that we think in "mentalese".

For **Jean Piaget** (1896 to 1980), thought comes first and language expresses it; preschoolers' egocentric speech simply fades. For **Lev Semyonovich Vygotsky** (*Thinking and Speech*, 1934), thought and speech have separate roots that merge at about two, after which thought becomes verbal; social speech becomes self-guiding private speech and then **inner speech**. He traced concepts from heaps through complexes to true concepts in adolescence, tied to society's tasks, social institutions and self-reflection.

> EXAM TRAP
>
> **Vygotsky: thought depends on language; Piaget: language depends on thought.** "One studied early, the other later language" is wrong: both studied the same years. Private speech **guides behaviour and is internalised**; it is not regression, imitation, disorder or fading egocentrism. Gender plays no part in Vygotsky's account. "Language and thought involve..." expects **simultaneous** processes; "independent" fits only their early roots. Linguistic relativity = **Whorf**, not Chomsky, Piaget or Pinker.

Bilingualism is the use of two languages, and a *polyglot* speaks many (a linguist studies language scientifically); bilingual children reach milestones on time. Apes such as Washoe, Nim Chimpsky and Kanzi learn meaningful symbols but little grammar. Speech perception is *categorical* (/ba/ switches abruptly to /pa/ as voice onset time rises) and audiovisual: in the McGurk effect, a heard "ba" dubbed onto lips saying "ga" is heard as "da". Reading builds most directly on spoken language, above all awareness of its sounds, not on theory of mind, imagination or mental representation. **Aphasia** is an acquired disorder of language from brain damage, not of intelligence, memory or abstract thinking (Broca's aphasia impairs production, Wernicke's comprehension: Physiological Psychology chapter).

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Seeing a box only as a container shows _____.
2. Reasoning from specific observations to a probable generalisation is _____.
3. Wallas's second stage is _____.
4. The smallest unit of meaning is the _____.
5. "More milk" at 18 to 24 months is _____ speech.
6. For Vygotsky, private speech becomes _____ speech.

Answers: (1) functional fixedness, (2) induction, (3) incubation, (4) morpheme, (5) telegraphic, (6) inner.

1.9 Intelligence: Theories, Measurement, Heredity and Environment

HIGH YIELD · 63 QS

Everyone thinks they recognise intelligence, yet psychologists still argue over what it is, whether it is one thing or many, and how much of it a test can capture.

Defining Intelligence ●●

Alfred Binet (1857 to 1911) put judgement at the centre: "to judge well, to comprehend well, to reason well." **Edwin Garrigues Boring** (1886 to 1968) retorted in 1923 that intelligence is "what the tests test", an **operational definition** (one framed by its measuring procedure). **David Wechsler** (1896 to 1981) gave the most quoted definition (1944): "the aggregate or global capacity of the individual to act purposefully, to think rationally and to deal effectively with his environment", aggregate because it comprises many abilities, global because they work as one. **Robert Jeffrey Sternberg** (born 1949), who re-polled the experts with Douglas Detterman in 1986, stresses adapting to, selecting and shaping real-world environments.

Achievement tests look back at what a specific course taught (a board examination); **aptitude** tests look ahead to the capacity to learn a specific skill with training (the Differential Aptitude Tests), so they anchor guidance; intelligence tests predict learning in general. Among lay (implicit) theories, **Carol Susan Dweck** (born 1946) contrasted an entity theory, or fixed mindset, with an incremental theory, or **growth mindset** (*Mindset*, 2006), the belief that effort develops intelligence. Fixed-mindset children avoid challenges; children praised for effort, not cleverness, persist after failure.

EXAM TRAP

The WAIS is an *intelligence* test, not an aptitude, achievement or attitude (evaluative stance) test. RML 2022 keyed the aptitude test as "the heart of the guidance programme", over the case study (a life history), the interview and the cumulative record (a running school file); textbooks give the phrase to counselling, so choose it if offered.

From Galton to Binet: The First Tests and the IQ ●●

Francis Galton (1822 to 1911), author of *Hereditary Genius* (1869) and father of differential psychology, measured reaction time and sensory acuity in his London anthropometric laborato-

ry (1884). **James McKeen Cattell** (1860 to 1944) coined the term **mental test** (1890) for a similar battery, which failed to predict college grades (Clark Wissler, 1901). Asked by the French ministry to pick out children needing special schooling, Binet and the physician **Théodore Simon** (1873 to 1961) measured judgement and reasoning instead, keeping items that older children passed more often than younger ones. Their 1905 scale had 30 items of rising difficulty; the 1908 revision grouped them by age, a child's **mental age** being the age level of the items passed; the 1911 revision reached adults. (**Jean Piaget** standardised tests in Simon's laboratory around 1920 but did not build the first test.)

Henry Herbert Goddard (1866 to 1957) translated the scale into English in 1908. In 1912 **William Stern** (1871 to 1938) divided mental age by chronological age, coining the **intelligence quotient (IQ)**; **Lewis Madison Terman** (1877 to 1956) multiplied it by 100 in his 1916 **Stanford-Binet**. So the **ratio IQ** = (MA ÷ CA) × 100: a 10-year-old with a mental age of 8 scores 80. In 1917 **Robert Mearns Yerkes** (1876 to 1956) led the first group tests, for army recruits: the verbal Army Alpha and, for the illiterate, the pictorial Army Beta.

> EXAM TRAP
>
> Stern coined IQ; Terman popularised it. Inverting the ratio (10 ÷ 8 × 100 = 125) is the usual slip: a mental age below chronological age always gives an IQ below 100. James McKeen Cattell coined "mental test"; Raymond Cattell, unrelated, gave us fluid and crystallised intelligence and the 16PF personality questionnaire.

Deviation IQ and What a Score Means ●●●

The ratio IQ fails for adults: mental age stops growing in the late teens, so an ordinary 40-year-old with a mental age of 16 would score (16 ÷ 40) × 100 = 40. The Wechsler-Bellevue scale (1939) dropped mental age and compared each person with norms from a large representative sample of the same age. In this **deviation IQ** (the "standard IQ"; loosely, the actual score relative to the expected mean score for one's age) every age group has a mean of 100 and a standard deviation (SD), the typical distance of scores from the mean, of 15: IQ = 100 + 15z (z: distance from the mean in SDs).

Because IQs are roughly normally distributed, any IQ converts to a percentile, the percentage of people scoring below it (Figure 1.11): 100 is the 50th percentile and 85, one SD below, the 16th. Stanford-Binet deviation IQs used an SD of 16 from 1960 until the fifth edition (2003), so there an IQ of 136 gives z = 36 ÷ 16 = 2.25, about the 98.7th percentile. WAIS-IV bands: 130 or above very superior, 90 to 109 average, 80 to 89 low average, 70 to 79 borderline ("very low" in newer editions), 69 or below extremely low. DSM-IV-TR's borderline intellectual functioning (71 to 84) straddles two bands, so 70 to 84 is "low average or borderline, depending on the edition".

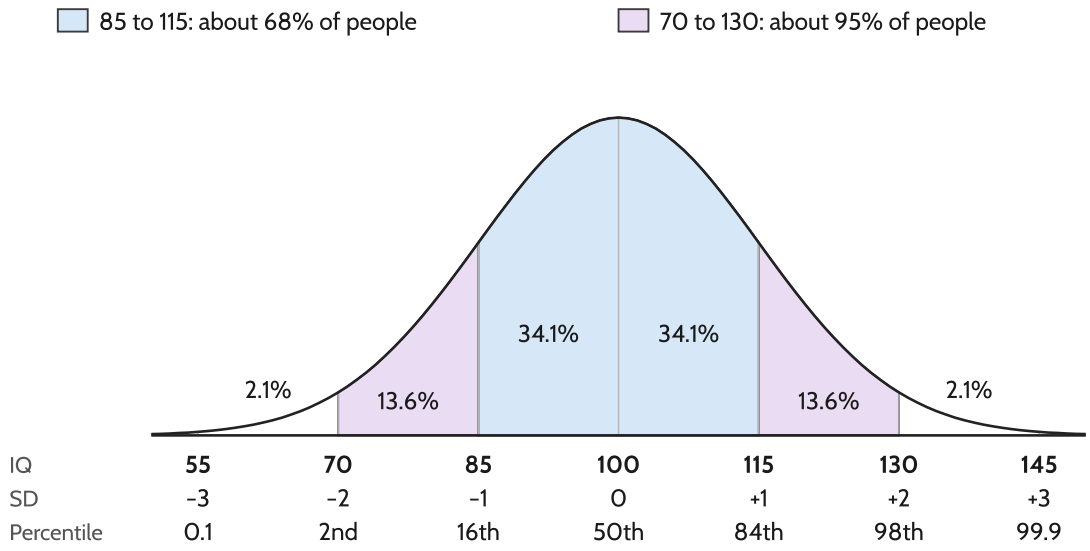

Figure 1.11 The distribution of deviation IQ (mean 100, SD 15). Read across the bottom rows to convert an IQ into standard deviations and percentiles: 85 is the 16th percentile, 115 the 84th, 130 about the 98th, and 70 about the 2nd.

> **EXAM TRAP**
>
> The original Stanford-Binet used the ratio IQ. The deviation IQ is not "only for adults", not built from unstandardised raw scores, and no less reliable than the ratio IQ, whose meaning shifts with age.

The Wechsler Scales, the SB5 and Other Tests ●●●

Wechsler, at New York's Bellevue Hospital, set out to fix the Stanford-Binet's faults: little appeal for adults, too much word manipulation, too much weight on speed and reliance on mental age. His tests add non-verbal performance subtests and give a profile as well as a total. The family is the WAIS (1955; WAIS-IV 2008, ages 16 to 90), the WISC (1949; WISC-V 2014, 6 years to 16 years 11 months) and the WPPSI (from 2 years 6 months). In the WAIS-IV, four index scores (Table 1.18) replaced the Verbal and Performance IQs and combine into the Full Scale IQ (FSIQ), with an optional General Ability Index; the WISC-V and WAIS-5 (2024) have five.

Table 1.18 The WAIS-IV (2008): 10 core subtests in four indices.

Index	Core subtests	Supplemental
Verbal Comprehension (VCI)	Similarities, Vocabulary, Information (3)	Comprehension
Perceptual Reasoning (PRI)	Block Design, Matrix Reasoning, Visual Puzzles (3)	Picture Completion, Figure Weights
Working Memory (WMI)	Digit Span, Arithmetic	Letter-Number Sequencing
Processing Speed (PSI)	Symbol Search, Coding	Cancellation

EXAM TRAP

The Working Memory Index reflects attention, concentration and mental control. "VIQ, PIQ, FSIQ and memory quotient" mixes the old WAIS with the old Wechsler Memory Scale (1945), a memory test whose single memory quotient became five WMS-IV (2009) indices: Auditory Memory, Visual Memory, Visual Working Memory, Immediate Memory and Delayed Memory. Letter-Number Sequencing, keyed as core in the WAIS-IV but not the WISC-V, is strictly supplemental in one and secondary in the other; it is still the best answer, as the other options are core in both.

The Stanford-Binet Fifth Edition (SB5; Gale Roid, 2003; ages 2 to 85 and over; mean 100, SD 15) measures five factors drawn from Carroll's model (below) and Alan Baddeley's working memory: Fluid Reasoning, Knowledge, Quantitative Reasoning, Visual-Spatial Processing and Working Memory, each verbal and non-verbal. The Kaufman Brief Intelligence Test (KBIT-2) has a Verbal (crystallised) and a Nonverbal (fluid) scale. For infants, the Bayley Scales (cognitive, language and motor development up to 42 months) yield a developmental quotient (DQ); the McCarthy Scales cover about 2½ to 8½ years.

The WAIS-IV FSIQ is highly reliable (about .98). Content validity asks whether items sample the domain, **concurrent validity** whether scores agree with a criterion measured at the same time (an established test given alongside), predictive validity whether they forecast a later one, and construct validity whether the test behaves as theory predicts. IQ correlates about .5 with school grades and job performance (Frank Schmidt and John Hunter, 1998). The WAIS documents intellectual disability or decline, as for a disability allowance; children's special educational needs call for the WISC.

Culture-Fair and Indian Tests ●●●

Because vocabulary and information depend on schooling, psychologists sought a culture-free test; that proved impossible, so a **culture-fair test** instead minimises language, reading and specific knowledge. The classic is **Raven's Progressive Matrices** (Standard form 1938) by Spearman's student **John Carlyle Raven** (1902 to 1970): each of its 60 progressively harder items is a pattern with a piece missing, solved by inferring the rule. It is the classic test of fluid intelligence and the best single marker of g, widely used with Indian norms. Raymond Cattell's figure-only Culture Fair Intelligence Test works the same way.

In India, **V. V. Kamat**'s Binet-Kamat Test (1930s) adapted the Stanford-Binet and is heavily verbal; **Arthur J. Malin**'s Intelligence Scale for Indian Children (MISIC, 1969) adapts the WISC and is half verbal; **C. M. Bhatia**'s Battery of Performance Tests of Intelligence (1955) has five performance subtests (Kohs block design, Alexander's pass-along, pattern drawing, immediate memory, picture construction), needs no reading and has rural and urban norms; normed on 11- to 16-year-olds, it is also used with illiterate adults. So for an illiterate rural 10-year-old, where the WISC-IV and SB5 rest on schooling and foreign norms, Bhatia's Battery best minimises cultural and educational bias.

EXAM TRAP

Raven's measures *fluid* intelligence: prefer it to crystallised, verbal, social or even "general intelligence". It is culture-fair, not culture-free. The Battery is C. M. Bhatia's, not that of Girishwar Misra, Durganand Sinha or J. B. P. Sinha (cultural, indigenous and organisational psychologists).

Factor Theories: From g to CHC ●●●

Factor analysis asks how few underlying factors explain a table of correlations among tests. Its inventor, **Charles Edward Spearman** (1863 to 1945), found in 1904 that schoolchildren's marks and sensory scores all correlated positively, the **positive manifold**, and explained it by a **general factor (g)** common to every test plus a specific factor (s) peculiar to each: the **two-factor theory**. **Louis Leon Thurstone** (1887 to 1955) found seven **primary mental abilities** (1938): verbal comprehension, word fluency, number, space, associative memory, perceptual speed and reasoning. Because these correlate, g returned at the top of hierarchies such as **Philip Ewart Vernon**'s (1950) and **John Bissell Carroll**'s three strata (1993); Kevin McGrew merged Carroll's model with Cattell and Horn's into the CHC theory behind most modern tests (Table 1.19).

Joy Paul Guilford (1897 to 1987) went the other way. His **structure of intellect** (1967) crosses five operations (cognition, memory, divergent production, convergent production, evaluation), four contents (figural, symbolic, semantic, behavioural) and six products, giving 5 × 4 × 6 = 120 abilities, later 150 and then 180.

Raymond Bernard Cattell (1905 to 1998), later with **John Leonard Horn** (1928 to 2006), split g in two (1963). **Fluid intelligence (Gf)** is the ability to reason and solve novel problems independently of acquired knowledge; measured by matrices, it peaks in the early twenties and then declines. **Crystallised intelligence (Gc)** is the repertoire of information, cognitive skills and strategies acquired through education and experience; measured by vocabulary and information, it holds or rises into the sixties. In Cattell's investment theory, Gf invested in learning crystallises as Gc; Horn later extended the list to about nine broad abilities.

Table 1.19 Other factor theories and the hierarchies.

Theorist	Year	Theory	Key structure
Edward Thorndike	1920s	Multifactor ("anarchic")	Abstract, mechanical, social; CAVD test; no g
Godfrey Thomson	1916	Sampling theory	Tests sample overlapping sets of many small bonds
P. E. Vernon	1950	Hierarchical	g; v:ed and k:m; minor; specific
Arthur Jensen	1969	Level I and II	Associative (rote) versus conceptual
J. B. Carroll	1993	Three strata	g; 8 broad; about 70 narrow

EXAM TRAP

g is general and s specific; neither means group factors (Thurstone, Vernon) or a genetic predisposition. Guilford's cube has three dimensions but 120 abilities; RML 2021 keyed "120 dimensions", so give 120 there. "Ability to deal with new problems" is Gf; block design, matrix reasoning and coding do not measure Gc, and processing speed and working memory are correlates, not definitions, of it.

Gardner, Sternberg, PASS and Emotional Intelligence ●●

Howard Gardner (born 1943) built his theory from brain damage and prodigies, not factor analysis. *Frames of Mind* (1983) proposed seven relatively independent **multiple intelligences**: linguistic, logical-mathematical, spatial, musical, bodily-kinaesthetic, interpersonal and intrapersonal; naturalist was added in 1999 and existential only considered. Critics note that most correlate with one another and with g.

Sternberg's **triarchic theory** (*Beyond IQ*, 1985) has three parts. The componential or analytical part is the mental machinery: metacomponents that plan and monitor, performance components that execute, and knowledge-acquisition components that learn. The experiential or creative part copes with novelty and automatises the familiar. The contextual or practical part adapts to the environment, shapes it or selects a better one: a clerk who learns her office's unwritten rules adapts, one who changes a pointless procedure shapes, one who leaves for a better job selects. In the late 1990s this grew into **successful intelligence**, the ability to achieve success in life by one's own standards and within one's culture through a balance of analytical, creative and practical abilities; it rests partly on tacit knowledge, the unspoken know-how a setting demands.

> EXAM TRAP
>
> Componential = analytical, experiential (not "experimental", and not Dweck's "incremental") = creative, contextual = practical. "Verbal, performance, full scale" is Wechsler; "fluid, crystallised" is Cattell; "linguistic, logical-mathematical, spatial" is Gardner.

The **PASS model** of the Odisha-born **Jagannath Prasad Das** (born 1931), **Jack A. Naglieri** and **John R. Kirby** (1994), built on **Alexander Romanovich Luria**'s (1902 to 1977) functional units of the brain, has four processes: Planning, Attention, Simultaneous (parts into a whole) and Successive (serial order), measured by the Cognitive Assessment System. **Stephen J. Ceci**'s bioecological view (1990) ties each cognitive potential to its context.

Edward Lee Thorndike's (1874 to 1949) social intelligence (1920), the ability to understand and manage people, foreshadowed **emotional intelligence**, defined by **Peter Salovey** and **John D. Mayer** (1990) as the ability to monitor and use one's own and others' emotions; their four-branch ability model (perceiving, using, understanding and managing emotions) is measured by the MSCEIT. **Daniel Goleman** (born 1946) popularised a mixed model (1995): knowing one's emotions, managing them, motivating oneself, recognising others' emotions and handling relationships (relabelled in 1998 as self-awareness, self-regulation, motivation, empathy and social skills). **Reuven Bar-On**'s EQ-i (emotional quotient, 1997) and trait EI are self-report versions; **Danah Zohar** proposed spiritual intelligence (2000).

> EXAM TRAP
>
> Salovey and Mayer coined emotional intelligence; Goleman popularised it. His 1998 labels rename the same five parts, so match the question's wording, and if both lists appear choose the original 1995 one ("knowing one's emotions" to "handling relationships"). "IQ, EQ, SQ, CQ" is no one's model of EI.

Heredity, Environment and the Brain •

Higher IQ goes with faster reaction and inspection times, larger working memory, slightly larger brains and, in Richard Haier's 1988 study, lower brain glucose use (neural efficiency); intelligence is polygenic. Family studies compare relatives of differing closeness, adoption studies compare children with their biological and adoptive parents, and **twin studies** compare monozygotic (MZ) twins, who share virtually all genes, with dizygotic (DZ) twins, who share half, most powerfully when reared apart. In **Thomas J. Bouchard Jr.** and Matt McGue's 1981 review, MZ twins correlated .86 on IQ when reared together and .72 when reared apart (about .70 in Bouchard's Minnesota Study of Twins Reared Apart, from 1979), DZ twins .60.

Heritability is the proportion of *variation* in a trait, within one population at one time, associated with genetic variation; it says nothing about one individual. It rises with age, and **Eric Turkheimer** (2003) found it near .10 in the poorest families, a gene-environment interaction. **Richard Lewontin**'s seeds (1970) show why it cannot explain group gaps: varied seed differs genetically within each tray, yet the gap between a rich-soil and a poor-soil tray is wholly environmental. In **Irving Gottesman**'s **reaction range** (1963), genes set limits and environment decides where within them a person lands.

Iodine, nutrition and schooling raise IQ; early intervention (Head Start, 1965; the Milwaukee Project; the Abecedarian Project, 1972, from infancy) brings IQ gains that often fade but lasting school and life benefits. **James Robert Flynn** (1934 to 2020) showed in 1984 that average scores rose about 3 points a decade, the **Flynn effect**, too fast for genes. Neither **Arthur Robert Jensen** (1923 to 2012), who argued in 1969 that group IQ gaps are partly genetic, nor *The Bell Curve* (**Richard J. Herrnstein** and **Charles Murray**, 1994) has direct support, and **Claude Steele** and **Joshua Aronson** (1995) showed that **stereotype threat**, fear of confirming a group stereotype, depresses scores.

> EXAM TRAP
>
> Individual differences come from many genetic and environmental factors together, not from genes, environment or income alone. Linkage studies, which track DNA markers through families, are neither twin nor adoption designs.

Intellectual Disability and Giftedness •••

Intellectual disability (DSM-IV-TR's mental retardation) requires deficits in intellectual functions, an IQ about two SDs below the mean (about 70; 85 is merely low average); deficits in **adaptive functioning**, the conceptual, social and practical skills of independent living; and onset in the developmental period. DSM-IV-TR graded severity by IQ (Table 1.20); DSM-5 grades it by adaptive functioning. **Global developmental delay** is a provisional DSM-5 label for children under 5 who cannot yet be fully assessed, so an assessed 7-year-old with an IQ of 65 and severe adaptive deficits has intellectual disability itself. Causes include Down syndrome, fragile X and phenylketonuria.

Table 1.20 Levels of intellectual disability (DSM-IV-TR).

Level	IQ range	Historical label
Mild (about 85% of cases)	50 to 55 to about 70	Moron (Goddard, 1910)

Level	IQ range	Historical label
Moderate	35 to 40 to 50 to 55	Imbecile
Severe	20 to 25 to 35 to 40	Imbecile or idiot
Profound	Below 20 or 25	Idiot

Adaptive behaviour is measured by **Edgar Arnold Doll**'s (1889 to 1968) **Vineland Social Maturity Scale** (1935), which yields a social age and a social quotient = (SA ÷ CA) × 100, and by the Vineland Adaptive Behavior Scales (communication, daily living skills, socialisation, motor). India's National Trust Act (1999) covers autism, cerebral palsy, mental retardation and multiple disabilities, but not specific learning disability.

At the other tail, an IQ of 130 or above is the usual cut-off for giftedness, though **Joseph Renzulli** (born 1936) required three rings (1978): above-average ability, task commitment and creativity.

KEY STUDY: TERMAN'S GIFTED CHILDREN (FROM 1921)

Question. Are bright children frail and odd? **Method.** Terman followed 1,528 California children with IQs of about 135 or more (mean about 151), the "Termites", for life. **Result.** They were healthier, better adjusted and more successful than average; social and behavioural problems clustered among those with childhood IQs of about 180 or more. **Criticism.** The mostly white, middle-class sample missed two future Nobel physicists.

Savant syndrome pairs a developmental disability, often autism, with an island of exceptional skill such as calendar calculation; a prodigy excels without disability. Creativity is treated with thinking (Section 1.8).

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The term intelligence quotient was coined in 1912 by _____.
2. On a Wechsler scale, an IQ of 85 lies at the _____ percentile.
3. For an illiterate rural Indian child, the best choice is _____ Battery of Performance Tests.
4. Cattell called crystallised intelligence a repertoire of information, cognitive skills and _____.
5. Adapting to, shaping and selecting environments make up Sternberg's _____ subtheory.
6. Heritability is the proportion of _____ in a trait, within a population, associated with genetic variation.

Answers: (1) William Stern, (2) 16th, (3) Bhatia's, (4) strategies, (5) contextual, (6) variation.

1.10 Motivation HIGH YIELD · 74 QS

Why does a student study until 2 a.m., or a monkey dismantle a puzzle that hides no food? Each of psychology's answers below grew from the failures of the last.

What motivation is ●●●

In psychology, **motivation** (Latin *movere*, to move) is the set of processes that start, direct, energise and sustain behaviour.

Take thirst. A **need** is a state of deficiency (the body lacks water, felt or not). A **drive** is the aroused, tense state the need produces, which energises behaviour. A **motive** is any internal state directing behaviour towards a goal, usually a learned one such as achievement. An **incentive** is an external object that attracts or repels; the **goal** is the end state pursued. The **motivational cycle** runs need, drive, instrumental behaviour, goal, relief, until the need returns.

The **primary motives** (biological, unlearned, mostly homeostatic) include hunger, thirst, sleep, pain avoidance, sex and the maternal drive; the **secondary motives** (psychosocial, learned), such as achievement, affiliation, power and gregariousness, grow from experience. Henry Murray's **viscerogenic** (bodily) and **psychogenic** (mental) needs draw the same line.

Doing something for its own sake is **intrinsic motivation**; doing it for a separable outcome such as marks or praise is **extrinsic motivation**. A toddler who struggles to open a box she knows is empty is intrinsically motivated: no bodily imbalance (homeostasis), payoff (extrinsic) or learned social motive (secondary) is involved. Malingering (feigning for an external reward) and factitious disorder (feigning without one) differ on this line.

> EXAM TRAP
>
> **Gregariousness** is a **social motive**, though McDougall called it an instinct ("psychological" and "personal" motives are not standard classes). Among the perspectives (Section 1.2), inner forces beyond individual control are **psychodynamic**; free will and an innate tendency to grow to one's maximum potential are **humanistic** (Maslow, Rogers).

Instinct theories ●●

An **instinct** is a complex, unlearned, stereotyped pattern of behaviour shared by a whole species. **William James** (1842 to 1910) held in 1890 that humans have more instincts than animals, not fewer. **William McDougall** (1871 to 1938), in *An Introduction to Social Psychology* (1908), made instincts, each paired with an emotion (flight with fear, pugnacity with anger), the prime movers of all behaviour; he later listed eighteen. His **hormic psychology** (Greek *horme*, urge) holds that behaviour is purposive.

Sigmund Freud (1856 to 1939) divided the instincts (*Trieb*) in 1920 into the **life instincts** (Eros), whose energy is the **libido**, and the **death instinct** (Thanatos), which, turned outwards, appears as aggression and war. Investing psychic energy in an object is **cathexis**; the **id** is the reservoir of the instincts.

The ethologists **Konrad Lorenz** and **Nikolaas Tinbergen** (Nobel Prize 1973) described **fixed action patterns** released by a **sign stimulus**: a male stickleback attacks anything with a red belly. Yet as an account of human motives, instinct theory had already collapsed by the late

1920s: naming is not explaining (a "gregarious instinct" is circular), the lists ran into thousands, and **Zing-Yang Kuo** (1930) found that kittens reared with rats seldom killed them.

> EXAM TRAP
>
> The instinct theory of motivation is **McDougall's**, not James's, Freud's or Maslow's. Freud's "innate forces" are **instincts**, not needs (Maslow, Murray), drives (Hull) or the id (their reservoir). War and violence express the **death instinct**.

Drives, arousal and incentives ●●●

Drive theory grew from physiology. **Walter Bradford Cannon** (1871 to 1945) named **homeostasis** (1926; *The Wisdom of the Body*, 1932), the body's thermostat-like correction around a set point, building on the *milieu intérieur* of **Claude Bernard** (1813 to 1878). On it **Clark Leonard Hull** (1884 to 1952) built **drive-reduction theory** (1943): a need creates a drive, a state of tension, and any response that reduces the drive is reinforced. In Hull's equation (${}_{S}E_{R} = {}_{S}H_{R} \times D \times K \times V$), K is **incentive motivation** (reward size) and V **stimulus-intensity dynamism** (stronger stimuli evoke the learned response more readily). **Primary drives** arise from tissue needs; **secondary drives**, such as learned fear, are acquired.

> EXAM TRAP
>
> "Greater stimulus intensity, greater probability of the learned response" is **stimulus-intensity dynamism**, not incentive motivation, response amplitude (vigour) or reaction threshold (the minimum for any response). A deviation from homeostasis creates a **drive**, not equilibrium (the restored state), regulation (the correction) or dissonance (Festinger). Drive theory is about **reducing physiological needs**.

Yet **Harry Frederick Harlow**'s (1905 to 1981) monkeys took apart puzzles that hid no food (1950), rats worked for non-nutritive saccharin, and students paid to lie in sensory-deprivation cubicles (1954) quit within days. **Donald Olding Hebb**'s (1904 to 1985) **optimal arousal theory** (1955) says organisms seek a moderate level of arousal. In 1908 **Robert Mearns Yerkes** and **John Dillingham Dodson** had found that mice learned a brightness discrimination fastest under intermediate shock, and the harder the discrimination, the weaker the best shock. The **Yerkes-Dodson law**, as later taught (Figure 1.12), makes performance an **inverted-U** function of arousal or anxiety, with a lower optimum for complex tasks than for simple ones.

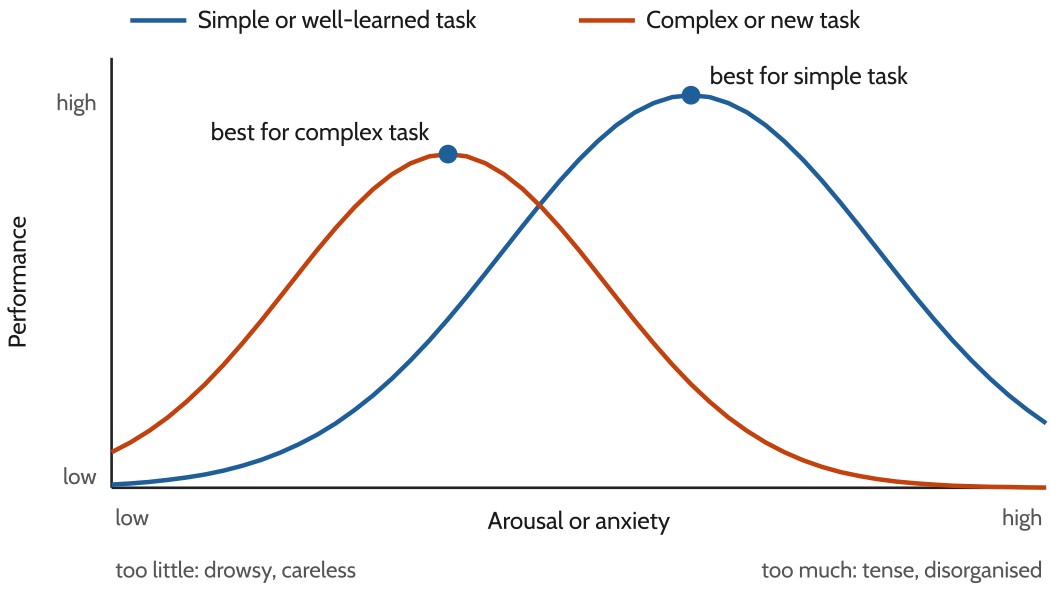

Figure 1.12 The Yerkes-Dodson law. Performance is an inverted-U function of arousal, and the peak for a complex or new task (dashed) lies at a lower arousal level than the peak for a simple or well-learned task (solid).

Robert Zajonc (1965) explained social facilitation the same way: an audience raises arousal, which strengthens the dominant response, helping easy, well-learned tasks and impairing novel or difficult ones. People differ too: **Marvin Zuckerman**'s (1928 to 2018) **sensation seeking** is the urge for varied, novel, intense experience despite risk.

> EXAM TRAP
>
> Yerkes-Dodson is an **inverted-U** relationship between performance and anxiety: more precise than "curvilinear", and never linear or M-shaped. A complex task is done best at **moderate arousal, with a lower optimum**, not at very high arousal or at a level independent of difficulty.

Drives push; incentives pull. The **incentive theory** holds that behaviour is driven mainly by external rewards: **Otto Tinklepaugh**'s monkeys (1928) refused lettuce after seeing a banana hidden. The **expectancy-value theory** multiplies the expectancy of success by the value of the outcome, so either at zero kills motivation. Its roots are the learned expectancies of **Edward Chace Tolman** (1886 to 1959) and the **social learning** theory of **Julian Bernard Rotter** (1954), in which behaviour depends on expectancy and reinforcement value; organisational texts give the social learning name to the framework built on expectancy, demand and incentive, which combines the **cognitive** framework (Tolman's expectancies) with the **behaviouristic** one (reinforcement); "supportive" is not a framework but a model of organisational behaviour (autocratic, custodial, supportive, collegial). **Richard Lester Solomon** and John Corbit's **opponent-process theory** (1974) explains acquired motives such as addiction: an opposite after-reaction grows with repetition, bringing tolerance and withdrawal.

Maslow's hierarchy and other lists of needs ●●●

Abraham Harold Maslow (1908 to 1970), with Carl Rogers a founder of humanistic psychology (Section 1.2), set out in 1943 ("A Theory of Human Motivation") and 1954 (*Motivation and Personality*) a **hierarchy of needs** (Figure 1.13): physiological, safety, belongingness and love (social), esteem, and self-actualisation, becoming all that one can be. By **prepotency**, unmet lower needs are strongest and must be largely satisfied before higher ones take charge: a starving man dreams of food, not fame. Physiological and safety needs are thus the **lower needs**. The four lower levels are **deficiency needs** (D-needs), which stop motivating once met; self-actualisation is a **growth** or being (B) need, which strengthens as it is satisfied.

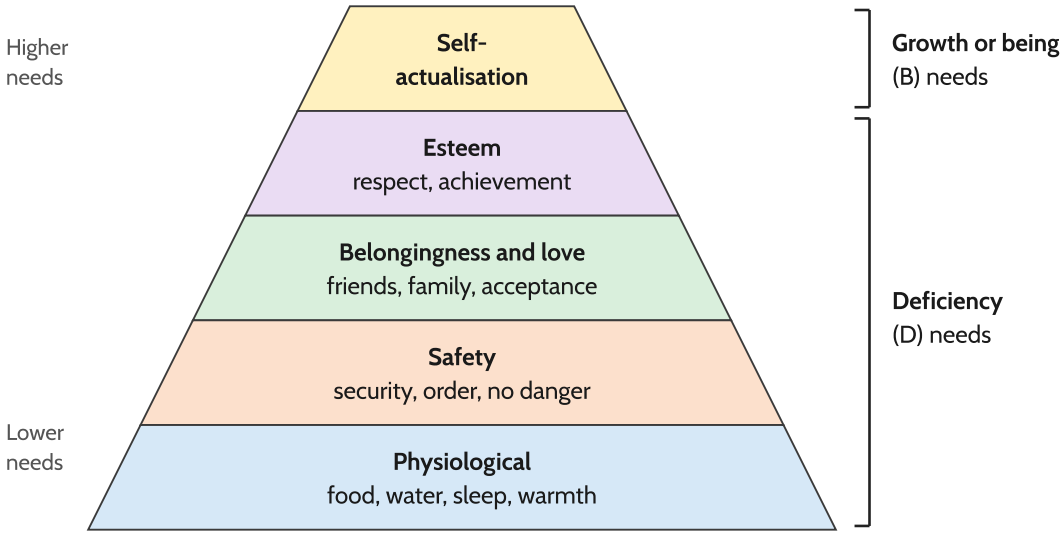

Figure 1.13 Maslow's hierarchy of needs (1943, 1954). The four lower levels are deficiency (D) needs; self-actualisation is a growth or being (B) need. Lower needs are prepotent: when unmet they dominate behaviour, and they must be largely met before higher needs take charge.

> **EXAM TRAP**
>
> Bottom to top: **physiological, safety, belongingness, esteem, self-actualisation**. "Which is NOT a level?" **Ego** (Freud). Physiological and safety needs are **lower needs**: "greatest potency" is half-true (they are prepotent), and "transformational" is a leadership term. The hierarchical theory is **A. H. Maslow's**, not McClelland's, Atkinson's or Mowrer's. RML 2025 keyed **physiological** as the need mostly met before esteem; the level immediately below esteem is **belongingness (social belonging)**, so choose that when offered.

Self-actualisers, by Maslow's estimate 1 per cent of people or fewer, show **metamotivation** towards metaneeds or B-values such as truth, beauty and justice; frustrated, these breed **metapathologies** such as cynicism and apathy. They report **peak experiences**, moments of ecstatic awe. Maslow later added cognitive and aesthetic needs and, beyond self-actualisation, **self-transcendence**. The term self-actualisation was coined by **Kurt Goldstein** (1878 to 1965) in 1939. Critics fault Maslow's hand-picked sample, weak evidence for the ranking and Western bias.

EXAM TRAP

"Striving for fulfilment of one's potential" is **self-actualisation**, not self-concept, ideal self (Rogers) or self-efficacy (Bandura). Self-actualisers show **metamotivation**, not D-motivation; "E-" and "F-motivation" are invented.

Clayton Paul Alderfer's **ERG theory** (1969) shrinks the ladder to existence, relatedness and growth, lets several needs act at once and adds **frustration-regression:** blocked at a higher need, a person falls back on a lower one (Table 1.21).

Table 1.21 Who proposed which list of needs.

Theorist	Year	The needs
Erich Fromm	1955	Relatedness, transcendence, rootedness, identity, frame of orientation
David McClelland	1953, 1961	Achievement, affiliation, power (learned)
Clayton Alderfer	1969	Existence, relatedness, growth
Deci and Ryan	1985	Autonomy, competence, relatedness (innate)
William Glasser	1965, 1998	Survival, love and belonging, power (achievement), freedom, fun

EXAM TRAP

One of **William Glasser**'s five basic genetic needs in reality therapy is **power or achievement**; self-actualisation (Maslow), the collective unconscious (Jung) and unconditional positive regard (Rogers) are baits. Viktor Frankl's **will to meaning** belongs to **logotherapy** (Section 1.2), not REBT, existential-humanistic therapy or ACT.

Murray, McClelland and achievement ●●●

Henry Alexander Murray (1893 to 1988), in *Explorations in Personality* (1938), listed about twenty psychogenic needs, among them n Achievement, n Affiliation (companionship), n Dominance (control of others) and n Exhibition (being seen and heard). A **press** is an environmental force that helps or hinders a need. **Christiana Morgan** and Murray's **Thematic Apperception Test** (TAT, 1935) reads needs and presses from stories told about ambiguous pictures (Assessment chapter).

EXAM TRAP

Murray's **need-press theory** underlies the **TAT**, a projective test whose stories reveal unconscious needs, conflicts and motives; not the Rorschach, Jung's word association test or sentence completion, and not a test of IQ or memory.

David Clarence McClelland (1917 to 1998) scored TAT stories for the **need for achievement** (nAch, 1953), a learned desire to meet standards of excellence, and later for **power** (nPow) and **affiliation** (nAff). High scorers prefer moderate risk, personal responsibility and quick feedback; businessmen he trained in Kakinada, Andhra Pradesh, in the 1960s went on to start more businesses. **John William Atkinson** (1923 to 2003) explained why in 1957: the tendency to approach success = motive to succeed × probability of success (Ps) × incentive (1 − Ps), and Ps × (1 − Ps) is 0.25 at Ps = 0.5 but only 0.09 at 0.1 or 0.9, so success-oriented people choose in-

termediate tasks, while those ruled by the **motive to avoid failure** pick very easy or very hard ones. **Matina Souretis Horner** (1972) proposed a **fear of success** in women, later read as cultural stereotyping.

Explanations matter too. Rotter's **locus of control** (1966) is internal when people credit their own effort, external when they credit luck or others (not low self-efficacy or Adler's inferiority complex). **Bernard Weiner** (born 1935) classified the causes of success and failure by **locus**, **stability** and **controllability**: ability is internal, stable and uncontrollable; effort internal, unstable and controllable. "I am useless at maths" breeds hopelessness; "I did not practise enough" breeds effort.

Intrinsic motivation and the self ●●●

Daniel Ellis Berlyne (1960) traced curiosity to novelty, complexity and incongruity, and **Robert White** (1959) united Harlow's manipulation drive, exploration and play as **effectance motivation**, the urge to deal effectively with the environment. That competence motive has two modern heirs.

The first is **self-efficacy**, introduced by **Albert Bandura** (1925 to 2021) in 1977: the belief that one can carry out the actions a particular task needs. It governs which tasks people attempt, how hard they try and how long they persist, and it grows mainly from mastery experiences, then from watching others, persuasion and bodily states. It is the core of his social cognitive theory (1986), built on **triadic reciprocal causation**: person, behaviour and environment influence one another. (Observational learning belongs to Section 1.6.)

> EXAM TRAP
>
> The reciprocal causation model belongs to **social cognitive theory**, not drive reduction or constructivism (learners building their own knowledge, after Piaget and Vygotsky), and that theory highlights **self-efficacy**, not external rewards only, biological drives or goal difficulty. "Two-factor theories" are Herzberg's (work), Mowrer's (avoidance), Schachter and Singer's (emotion) and Spearman's (intelligence).

> KEY STUDY: LEPPER, GREENE AND NISBETT'S MAGIC MARKERS (1973)
>
> **Mark Lepper**, **David Greene** and **Richard Nisbett** let nursery-school children who liked drawing with magic markers draw for an expected certificate, an unexpected one, or nothing. One to two weeks later, the expected-award children spent about half as much free time drawing as the others; surprise awards did no harm. **Criticism.** Verbal or informative rewards do not undermine interest either, so the effect is narrower than often claimed.

This **overjustification effect** occurs because the reward moves the reason for an enjoyed activity outside the person. **Edward L. Deci** and **Richard M. Ryan** built such findings into the second heir, **self-determination theory** (SDT, 1985), with three innate needs: **autonomy**, **competence** and **relatedness**. Its **cognitive evaluation theory** explains the magic markers: rewards felt as controlling lower intrinsic motivation, while informational ones, such as unexpected praise, raise it. Motivation runs from amotivation through increasingly internalised extrinsic regulation to intrinsic motivation, driven by internal reward and growth.

EXAM TRAP

SDT's needs are **autonomy, competence and relatedness**, not **affiliation** or achievement (McClelland), independence, superiority or excellence. Its motivation driven by internal reward and growth is **intrinsic**; amotivation is its absence, external regulation the least autonomous form.

Mihaly Csikszentmihalyi (1934 to 2021) called intrinsic motivation at its most absorbing **flow** (1975, 1990): action and awareness merge and time vanishes when **challenge and skill are balanced** (too hard brings anxiety, too easy boredom). It is not Maslow's peak experience, Rogers's fully functioning person or Frankl's will to meaning. Finally, **Leon Festinger**'s (1957) **cognitive dissonance**, the tension of acting against one's beliefs (Arvind, who finds smoking disgusting, smokes at a party), motivates its reduction by a change of attitude or behaviour; effort justification is one form of that reduction, not the state (Social Psychology chapter).

Conflict and frustration ●●

The blocking of goal-directed behaviour by obstacles, personal limits or conflicting motives is **frustration**. **Pressure** is the demand to perform or conform: Ramesh, made to work hard, fast and a lot in a call centre, is under pressure, not blocked (frustration), torn (conflict) or facing unpredictability.

Kurt Lewin (1890 to 1947) described the conflicts in 1935. **Approach-approach** (two attractive goals) is easiest to resolve; **avoidance-avoidance** (two evils) produces vacillation and attempts to leave the field; in an **approach-avoidance conflict** one goal both attracts and repels; and **double approach-avoidance**, several such goals, is the commonest in real life. **Neal Elgar Miller**'s (1909 to 2002) gradients (1944) explain the vacillation: both tendencies strengthen near the goal, but avoidance is steeper, so the person hovers where they cross.

Frustration may lead to **aggression**, direct or displaced; **regression**, as in Roger Barker, Tamara Dembo and Lewin's children (1941), who played like toddlers once attractive toys were screened off; **fixation**, as in **Norman Maier**'s (1949) rats, which kept jumping one way after the problem became soluble; withdrawal; or defence mechanisms (Personality chapter). The **frustration-aggression hypothesis** of **John Dollard**, Leonard Doob, Miller, **Orval Hobart Mowrer** and Robert Sears (1939) held that frustration always leads to aggression; **Leonard Berkowitz** (1989) recast it in terms of negative affect.

EXAM TRAP

An after-the-fact, socially acceptable reason for behaviour really driven by other motives is **rationalisation**; re-channelling sexual or aggressive impulses into valued outlets is **sublimation**; a safer target is **displacement**; attributing one's own impulse to others is **projection**; turning it into its exaggerated opposite is **reaction formation**; pushing a threatening thought out of awareness is **repression**; refusing to acknowledge an unpleasant reality is **denial**; compensation pushed to excess is **overcompensation** (the full list: Personality chapter).

Hunger, thirst and sex ●●●

In 1912 Walter Cannon and **A. L. Washburn** showed that stomach contractions coincide with hunger pangs, yet people without stomachs still feel hunger. **Glucostatic** (blood sugar) and **lipostatic** (body fat) set-point theories followed; John Pinel and Steven Barnes argue instead that we eat mainly for anticipated pleasure, with weight drifting to a **settling point**.

In 1940 Albert Hetherington and Stephen Ranson found that lesions of the **ventromedial hypothalamus** (VMH) made rats overeat and grow obese; in 1951 the Indian physiologist **Bal Krishan Anand** and John Brobeck found that lesions of the **lateral hypothalamus** (LH) caused aphagia (no eating) and adipsia (no drinking). Eliot Stellar's **dual-centre theory** (1954) made the LH the feeding centre and the VMH the satiety centre. Since revised (VMH lesions raise insulin; other nuclei matter), it now shares the stage with leptin from fat, insulin from the pancreas and ghrelin from the empty stomach.

> EXAM TRAP
>
> Satiety centre: **VMH**, lesion gives hyperphagia and obesity. Feeding centre: **LH**, lesion gives aphagia and weight loss (**L**ateral, **L**ose). "Ventromedial hippocampus" and "lateral thalamus" are invented; the cerebrum and basal ganglia are not hunger centres. The hypothalamus as a whole is the master homeostatic regulator (hunger, thirst, temperature; its suprachiasmatic nucleus is the circadian clock), not the cerebellum (movement) or pineal gland (melatonin).

Stanley Schachter's externality hypothesis (1968 to 1971) held that obese people respond more to external food cues, such as the clock, than to internal ones (eating disorders: Psychiatric Disorders chapter). Thirst is **osmotic** (salty body fluid draws water from cells) or **hypovolaemic** (blood volume falls, signalled through angiotensin II). **Alfred Charles Kinsey** surveyed the sexual behaviour of men (1948) and women (1953); **William Howell Masters** and **Virginia Eshelman Johnson** (1966) observed a four-phase **sexual response cycle** of excitement, plateau, orgasm and resolution; **Helen Singer Kaplan** (1979) put **desire** first (desire, excitement, orgasm). Hormones such as testosterone matter less in human sexuality than in other mammals. In Carl Warden's obstruction box (1931) the maternal drive outran thirst and hunger; pain avoidance (aroused by tissue damage) and sleep (Section 1.5) complete the primary motives.

> EXAM TRAP
>
> Masters and Johnson: **excitement, plateau, orgasm, resolution**. Sequences that insert desire (Kaplan), start with plateau or end in "refractory" (the male recovery period) are scrambles. The reports on sexual behaviour are **Kinsey's**, not Freud's, Anna Freud's or Horney's.

Motivation at work, school and clinic ●●●

Frederick Herzberg (1923 to 2000) asked about two hundred engineers and accountants when they had felt exceptionally good or bad at work. His **two-factor theory** (1959) separates **motivators** (Table 1.22), in the work itself, which create satisfaction, from **hygiene factors**, in its context, which cause dissatisfaction when poor but, when improved, merely remove it.

Table 1.22 Herzberg's two factors.

Motivators (job content)	Hygiene factors (job context)
Achievement, recognition	Company policy, supervision
The work itself, responsibility	Relationships, working conditions
Advancement, growth	Salary, status, job security

Victor Harold Vroom's **expectancy theory** (1964) makes motivational force = expectancy (effort leads to performance) × instrumentality (performance leads to outcomes) × valence (value of the outcomes). If Priya, a sales executive, has E = 0.8, I = 0.9 and V = 0.7, her force is about 0.50; if her firm is known not to pay bonuses, I drops to 0.2 and her force to about 0.11. **John Stacy Adams**'s **equity theory** (1963) says people compare their ratio of outcomes to inputs with others' and act to remove inequity. In the **goal-setting theory** of **Edwin Locke** and **Gary Latham** (from 1968), specific, difficult but attainable goals, with commitment and feedback, beat vague "do your best" goals.

EXAM TRAP

Hygiene factors **prevent dissatisfaction but do not by themselves motivate** (they are not self-actualisation needs, nor for high achievers only); job satisfaction depends on **work motivation and job environment**, not "or" or "only". Vroom's outcomes must be **valuable**; guaranteed rewards, effort unrelated to performance and unimportant rewards kill motivation. Goals should be **specific and challenging**, with feedback, not vague, easy or value-free.

In school, informational feedback, praise for effort and attribution retraining (Carol Dweck, 1975) sustain effort. In the clinic, **motivational interviewing** (William Richard Miller and Stephen Rollnick, 1991) treats resistance to change as ambivalence, an approach-avoidance conflict, and draws out the client's own reasons to change.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. A physiological deficiency is a _____; the tense state it produces is a _____.
2. For a complex task the optimal arousal is _____ than for a simple one.
3. The level immediately below esteem in Maslow's hierarchy is _____.
4. Murray's environmental force that helps or hinders a need is a _____.
5. Self-determination theory's three needs are autonomy, competence and _____.
6. In Herzberg's theory, salary is a _____ factor.

Answers: (1) need, drive, (2) lower, (3) belongingness and love, (4) press, (5) relatedness, (6) hygiene.

1.11 Emotion and Stress HIGH YIELD · 74 QS

Your heart pounds as the question paper is handed out. Is that pounding the fear itself, or something your mind has still to interpret? This section takes emotion apart, then turns to stress and coping.

What an emotion is ●●

An **emotion** is a brief, whole-person response to an event that matters to you, with three components: physiological arousal (a pounding heart, sweating palms), expressive behaviour (face, voice, posture, action) and conscious experience ("I am afraid"). The theories that follow quarrel mainly about the order of these three. A feeling is the subjective part of an emotion. An emotion lasts seconds to minutes and has an object (you are angry *at* someone); a **mood** lasts hours or days, is weaker and often has no object. **Affect** is the umbrella term for your present good or bad feeling state and, clinically, its observable expression, rated for quality, range and reactivity; persistence is what defines mood. Temperament is the inborn style of emotional reactivity (Alexander Thomas and Stella Chess's easy, difficult and slow-to-warm-up types are in the Developmental chapter).

Are there basic emotions, innate and universal, each with its own face? **Paul Ekman** (born 1934) and **Wallace V. Friesen** found six with universally recognised expressions: happiness, sadness, fear, anger, surprise and disgust, with contempt added in 1986. **Silvan Tomkins** (1962) listed nine innate affects; **Carroll Izard**'s differential emotions theory (1977) listed ten, including shame and guilt; and **Robert Plutchik**'s wheel (1980) set eight primaries opposite one another (joy and sadness, trust and disgust, fear and anger, surprise and anticipation), with neighbours blending into dyads such as love (joy plus trust). Dimensional models place emotions in a space instead: Wundt's three-dimensional theory of feeling (Section 1.1) and the circumplex of **James Russell** (1980), a circle of valence (pleasant to unpleasant) by arousal.

> EXAM TRAP
>
> **Jealousy, shame, guilt, pride and love are not among Ekman's six.** Shame and guilt are Izard's, trust and anticipation Plutchik's. "Our current mood and feelings" is **affect**, not mood.

Body first or label first ●●●

Common sense says you see a bear, feel afraid, and therefore tremble. **William James** (1842 to 1910) in 1884 and the Danish physician **Carl Georg Lange** (1834 to 1900) in 1885 reversed the order. The **James-Lange theory** holds that the stimulus first triggers bodily changes and that the emotion *is* our perception of them: "we feel sorry because we cry, angry because we strike, afraid because we tremble." The textbook sequence is stimulus, arousal, awareness ("interpretation") of the arousal, then emotion (Figure 1.14, top row), where the "interpretation" is just the brain noticing what the body is doing.

Walter Bradford Cannon (1871 to 1945) objected in 1927 that emotion survives when the viscera are cut off from the brain; that the same visceral changes accompany fear, anger, fever and exercise; that the viscera are too insensitive and slow for quick, fine distinctions; and that injected adrenaline produces symptoms without real emotion. In the **Cannon-Bard theory**, developed with **Philip Bard** (1898 to 1977), the thalamus signals the cortex and the body at once, so feeling and arousal occur simultaneously and independently (second row). If arousal is so similar across emotions, what tells fear from joy? In 1962 **Stanley Schachter** and **Jerome Singer** answered with the **two-factor theory**: emotion is physiological arousal, which sets its intensity, plus a cognitive label drawn from the situation, which names it.

KEY STUDY: SCHACHTER AND SINGER'S ADRENALINE EXPERIMENT (1962)

Students given adrenaline (epinephrine), described as a vitamin, were told its true effects, told nothing or misinformed; a placebo group got saline. Each then waited with a stooge acting euphoric or angry. The aroused but uninformed caught the stooge's mood most; the informed, who could blame the injection, least. **Criticism.** The effects were small, and in replications unexplained arousal simply felt anxious.

A label can be wrong. In **misattribution of arousal** people mistake the source of their arousal: in 1974 **Donald Dutton** and **Arthur Aron** found that men approached by an attractive woman on the swaying Capilano suspension bridge telephoned her far more often than men on a low, solid bridge. Fear had been read as attraction. **Dolf Zillmann**'s excitation transfer theory (1970s) adds that leftover arousal from one event intensifies the next emotion.

EXAM TRAP

Match theory to sequence. Common sense: emotion, then body (the view James rejected). James-Lange: body first, then emotion. Cannon-Bard: emotion and visceral reaction *simultaneous*, via the thalamus. Schachter-Singer: *cognition plus arousal*, and the only one that explains misattribution. No classic theory makes the somatic nervous system the seat of emotion, and "emotions are learned through reinforcement" is behaviourism.

Appraisal, faces and opposing processes

Richard S. Lazarus (1922 to 2002) put thought first. In his cognitive appraisal (cognitive-mediational) theory, **primary appraisal** asks "Does this matter to me?" (harm-loss, threat or challenge), **secondary appraisal** asks "What can I do about it?", and reappraisal revises both (Figure 1.14, bottom row). **Robert Zajonc** (1923 to 2008) argued for affective primacy ("preferences need no inferences", 1980; debated with Lazarus in 1984): by the mere exposure effect, repeated exposure raises liking even for shapes seen too briefly to be recognised, so affect can come first. The **facial feedback hypothesis**, rooted in Tomkins's work, says that our own expressions modulate our feelings; in the pen study of **Fritz Strack**, **Leonard Martin** and **Sabine Stepper** (1988), holding a pen in the teeth, which forces a smile, made cartoons seem funnier, an effect now judged real but small. **Charles Darwin**'s *The Expression of the Emotions in Man and Animals* (1872) founded the evolutionary view of expressions as inherited adaptations. **Richard Solomon**'s opponent-process theory (1974) holds that every strong emotion triggers a slow, opposite after-reaction that grows with repetition: a drug's high fades while withdrawal deepens.

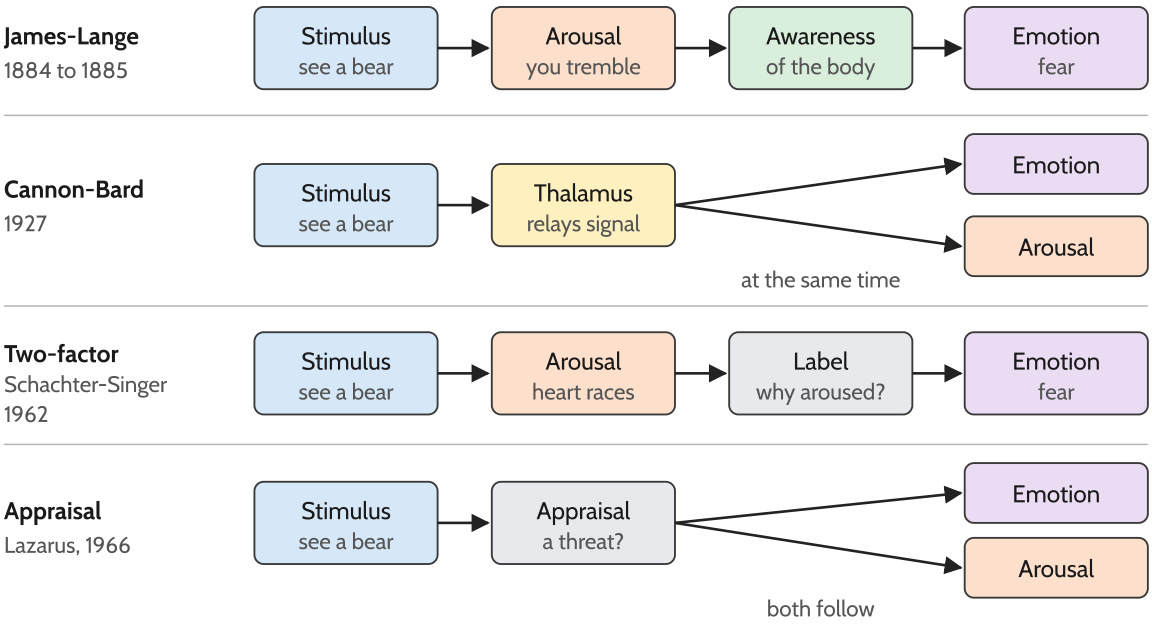

Figure 1.14 Four classic theories of emotion compared. Follow each row from the stimulus to the emotion. James-Lange: the body reacts first and the felt emotion is awareness of that reaction. Cannon-Bard: the thalamus sends signals that produce feeling and arousal at the same time. Two-factor: arousal plus a cognitive label. Appraisal (cognitive-mediational): an evaluation of the stimulus comes first, and both emotion and arousal follow from it.

The emotional brain and body ●●●

The body's side of emotion is run by the autonomic nervous system. Its sympathetic division prepares for fight or flight: it dilates the pupils, speeds the heart and breathing, dries the mouth, halts digestion and releases adrenaline from the adrenal medulla, and its sweating lowers skin resistance (the galvanic skin response). The parasympathetic division ("rest and digest") reverses each effect, constricting the pupils. Heartbeat and breathing themselves are paced by the medulla of the hindbrain; the hypothalamus only orchestrates them.

Bard's cats, with the cortex removed, showed **sham rage**, an undirected hissing fury that vanished only when the hypothalamus was removed too: the hypothalamus organises rage and the cortex restrains it. In 1937 **James Papez** (1883 to 1958) proposed an emotional loop, the **Papez circuit**: hippocampus, fornix, mammillary bodies, anterior thalamus, cingulate gyrus, entorhinal cortex and back to the hippocampus. **Paul MacLean** (1913 to 2007) added the amygdala and other structures in 1952 and named the whole the **limbic system**, the classic "seat of emotion", which regulates emotion, motivation and memory (Table 1.23).

Its key member is the **amygdala**, deep in each temporal lobe. When **Heinrich Klüver** and **Paul Bucy** removed both anterior temporal lobes of rhesus monkeys (1937, 1939), wild animals became tame. **Klüver-Bucy syndrome** is placidity (loss of fear, even of snakes), psychic blindness or visual agnosia, hyperorality, hyperphagia and hypersexuality; human cases often add aphasia and amnesia, because nearby temporal cortex is damaged too. The amygdala detects threat, drives fear conditioning and gives emotional events their extra memorability, an advantage that disappears when noradrenaline is blocked with propranolol (Cahill and McGaugh, 1994). **Joseph LeDoux** (born 1949) found a fast, crude low road from the thalamus straight to the amygdala and a slower, detailed high road through the cortex. Such subcorti-

cal routes probably explain affective blindsight, the recognition of emotional faces shown in a cortically blind field.

The prefrontal cortex regulates emotion (**Phineas Gage**'s rod injury of 1848; **Antonio Damasio**'s somatic markers, 1994). Damage to its control of the brainstem, after stroke, multiple sclerosis, ALS or brain injury, can cause **pseudobulbar affect**: sudden, involuntary laughing or crying out of proportion to the actual mood, which is not depression. The polygraph records arousal, not lies, so frightened innocent suspects can fail it; **David T. Lykken**'s guilty knowledge test (1959), which asks about details only the culprit knows, is more accurate.

Table 1.23 Brain structures in emotion.

Structure	Role	Classic evidence
Hypothalamus	Rage, stress hormones, hunger, thirst	Sham rage (Bard)
Limbic system	Emotion, motivation, memory	Papez (1937); MacLean (1952)
Amygdala	Fear, aggression, emotional memory	Klüver and Bucy; LeDoux
Hippocampus	Memory for facts	Amnesia, as in H.M.
Distractors	Reticular formation (arousal), pituitary, corpus callosum, cerebellum (coordination), caudate (movement), suprachiasmatic nucleus (body clock)	Not emotion centres

EXAM TRAP

Limbic system, not one of its parts or the broad forebrain, for "seat of emotion"; its function is emotion, while hunger and thirst are the hypothalamus's. **Amygdala**, not hippocampus, for fear, aggression and the emotional enhancement of memory. Amygdala removal gives tameness, not attack (sham rage), obesity (ventromedial hypothalamus) or amnesia (hippocampus); options with "hyperalertness" or "avolition, alogia" are wrong. Pupil constriction is the odd one out among sympathetic effects.

Expression and four emotions ●●

The Fore of Papua New Guinea, who had seen no films or magazines, matched Ekman and Friesen's faces to stories far above chance (1971), confusing only fear with surprise. Cultures differ instead in **display rules**, learned norms for when an emotion may be shown. The Facial Action Coding System (1978) codes faces as muscle action units, and a genuine Duchenne smile engages the muscles around the eyes. Posture, gesture and voice carry emotion too. Women are, on average, more facially expressive, largely through socialisation. Fear answers a present danger; anxiety is diffuse and future-oriented; a phobia is a persistent, irrational fear of a specific object or situation. The catharsis hypothesis, that venting anger drains it, fails: hitting a punching bag while thinking of an insulter increased aggression (**Brad Bushman**, 2002). Disgust guards against contamination (**Paul Rozin** and **April Fallon**, 1987).

Ed Diener (1946 to 2021) defined **subjective well-being** (1984) by three components: life satisfaction (cognitive), frequent positive affect and infrequent negative affect. Good fortune fades: by **Harry Helson**'s adaptation-level theory (1964) and the hedonic treadmill of **Philip**

Brickman and **Donald Campbell** (1971), lottery winners are no happier than others a year or so later, and relative deprivation (**Samuel Stouffer**, 1949) makes us judge ourselves against others. **Barbara Fredrickson**'s **broaden-and-build theory** (1998) says positive emotions widen thought and build lasting resources. **Savoring** (**Fred Bryant** and **Joseph Veroff**, 2007) is enhancing and prolonging the enjoyment of positive experiences.

> EXAM TRAP
>
> **Diener:** life satisfaction, not physical, spiritual or social wellness. **Fredrickson:** broaden-and-build; flow is Csikszentmihalyi, cognitive distortions Beck, psychodynamic theory Freud. **Savoring** is not avoiding negatives, relaxing or reflecting on trauma.

How emotions develop and are regulated ●●●

Katharine Banham Bridges (1932) saw the newborn's general excitement split first into *distress* and *delight*, and only then into anger, fear, elation and the rest. Primary emotions are present by about six months. The **self-conscious emotions** (embarrassment, pride, shame, guilt) need a sense of self, so they appear after 12 months, from about 18 months. **Edward Tronick**'s still-face procedure (1978), in which a mother suddenly freezes her face, shows that even young infants expect emotional give-and-take.

In the process model of **James Gross** (1998), emotion regulation has five families of strategies, in the order an emotion unfolds: situation selection, situation modification, attentional deployment (distraction, rumination), cognitive change, chiefly **cognitive reappraisal** ("my racing heart is my body getting ready"), and response modulation, which includes expressive suppression. **Alexithymia** (**Peter Sifneos**, 1973) is difficulty identifying and describing one's own feelings, not anhedonia (loss of pleasure). **Daniel Goleman**'s emotional intelligence (1995; Section 1.9) has five domains: knowing one's emotions, managing them, motivating oneself, recognising others' emotions and handling relationships (later relabelled self-awareness, self-regulation, motivation, empathy and social skills); empathy has emotional, cognitive and empathic-concern components.

> EXAM TRAP
>
> **Reappraisal is cognitive change**; suppression is response modulation. **First infant emotions:** distress and delight (Bridges); fear, rage and love are Watson's. **Goleman:** if the 1995 list (knowing and managing one's emotions, motivating oneself, recognising others' emotions, handling relationships) and the later self-awareness relabelling both appear, choose the 1995 wording.

Stress and the body's emergency systems ●●●

Stress has been defined as a stimulus (the stressor), as a response (Selye's "nonspecific response of the body to any demand") and as a transaction in which demands are appraised as exceeding one's resources. Cannon's **fight-or-flight response** (1915) is the fast route: within seconds the sympathetic nerves act and the adrenal medulla releases adrenaline.

What if the emergency never ends? **Hans Selye** (1907 to 1982) found that every noxious agent he gave rats, from tissue extracts to cold, produced the same triad: enlarged adrenal glands, shrunken immune tissue and stomach ulcers. In 1936 he proposed the **general adaptation**

syndrome (*The Stress of Life*, 1956), with three stages in fixed order. In the *alarm reaction* the body mobilises, as in fight or flight. In the *stage of resistance* cortisol stays high and the body adapts, at a cost. In the *stage of exhaustion* adaptive capacity runs out and diseases of adaptation such as ulcers and hypertension appear. He later (1974) separated **eustress**, beneficial stress, from distress.

Resistance runs on the slower **hypothalamic-pituitary-adrenal axis** (HPA axis): the hypothalamus releases corticotropin-releasing hormone (CRH), the anterior pituitary adrenocorticotropic hormone (ACTH), and the adrenal cortex cortisol, which feeds back on the hypothalamus and anterior pituitary to switch the system off. **Shelley E. Taylor** (2000) added tend-and-befriend: females especially meet stress by protecting offspring and seeking allies, a pattern linked to oxytocin.

> EXAM TRAP
>
> **Alarm, resistance, exhaustion.** Illness is not a stage, and the stages are Selye's, not life events theory, the health belief model or protection motivation theory. **The adrenal glands** are the stress glands; the pituitary only releases ACTH, and the posterior pituitary is not in the HPA axis. Insulin (pancreas), thyroxine (thyroid) and melatonin (pineal) are distractors.

Sources of stress and the transactional model ●●●

Thomas Holmes and **Richard Rahe**'s **Social Readjustment Rating Scale** (1967) scores 43 life events in life change units, from death of a spouse (100) through marriage (50) to minor violations of the law (11); because change itself is the stressor, even holidays count. **Allen Kanner**, Lazarus and colleagues (1981) showed that **daily hassles**, the irritations of everyday life, predict symptoms better than major events, while uplifts, the small pleasures that offset them, matter less. An acute stressor is brief (a pop quiz); a chronic one lasts (HIV disease, cancer, marital discord); a cataclysmic one, such as a flood or war, strikes many people at once. Frustration and conflict are covered under motivation (Section 1.10). At work, organisational stressors are task, role and interpersonal demands, structure and leadership, as distinct from environmental (political uncertainty) and individual (family, money) factors. Role conflict (incompatible expectations) and **Robert Karasek**'s high-demand, low-control jobs (1979) can end in **burnout** (**Herbert Freudenberger**, 1974), which **Christina Maslach**'s inventory (1981) measures as exhaustion, depersonalisation and reduced accomplishment.

In Lazarus and **Susan Folkman**'s **transactional model of stress** (1984), stress is a person-environment relationship appraised as taxing or exceeding one's resources: primary appraisal judges harm-loss, threat or challenge, secondary appraisal judges the resources for coping, and coping and reappraisal follow. In one exam hall, Ananya appraises the paper as a threat and her preparation as thin, and shakes; Kiran sees a challenge, trusts his practice, and feels keyed up but focused. The **diathesis-stress model** (**Paul Meehl**, 1962; **Joseph Zubin** and **Bonnie Spring**, 1977) explains why stress makes one person ill and spares another: a disorder appears when a pre-existing vulnerability, often biological, is triggered by stress; usually neither is enough alone.

EXAM TRAP

The SRRS measures life events, not hassles, physiological reactions or cognitive change. **Secondary appraisal** is the resources question; "coping resource assessment" is not the model's term. **Diathesis-stress** is not "genes only", "environment only", "unconscious conflicts only", "learned experiences alone" or supernatural; humanistic, sociocultural and cognitive models blame blocked growth, social forces and faulty thinking.

Coping, personality and health ●●●

Problem-focused coping acts on the stressor (a revision plan); **emotion-focused coping** acts on the feelings (reappraisal, support, acceptance, but also denial). The best choice fits controllability: plan while the exam can still be prepared for, reappraise once the result is out. Approach coping beats avoidance in the long run. Defence mechanisms (Personality chapter) differ in being unconscious: repression involuntarily blocks painful feelings (suppression is deliberate); displacement redirects anger to a safer target (anger at a husband shouted at a child); sublimation channels it into a valued outlet (kickboxing); projection attributes one's own feelings to others; isolation of affect strips the feeling from an idea (a gruesome fantasy told calmly), whereas denial rejects the reality; rationalisation offers a plausible excuse; and reaction formation turns a feeling into its opposite.

Social support protects both directly and as a buffer when stress is high (**Sheldon Cohen** and **Thomas Wills**, 1985). **Robert S. Weiss** (1973) separated **emotional isolation**, lacking a deep attachment to one specific person, from social isolation, lacking a circle of friends. **Suzanne Kobasa**'s **hardiness** (1979) is commitment, control and challenge; **Aaron Antonovsky**'s sense of coherence (1979) is comprehensibility, manageability and meaningfulness; resilience is bouncing back from adversity. Dispositional optimism (**Michael Scheier** and **Charles Carver**, 1985), a general expectancy of good outcomes, aids coping and recovery. **Martin Seligman** and **Steven Maier** (1967) found that dogs given inescapable shock stopped trying to escape, learned helplessness (Section 1.6), which Seligman later reversed into learned optimism.

In 1959 the cardiologists **Meyer Friedman** and **Ray Rosenman** described the **Type A behaviour pattern**: competitive striving, time urgency and hostility. In their Western Collaborative Group Study, Type A men developed coronary heart disease about twice as often as relaxed Type B men; later work found hostility to be the toxic component. A cancer-prone Type C, unassertive and suppressing negative emotion, has weak support. **Johan Denollet**'s **Type D personality** (mid-1990s) is negative affectivity plus social inhibition, and predicts a poor outlook in heart disease.

George Engel's **biopsychosocial model** (1977) explains illness through biological, psychological and social factors together. **Psychoneuroimmunology** began when **Robert Ader** and **Nicholas Cohen** (1975) conditioned immune suppression in rats by pairing saccharin with an immunosuppressive drug; chronic stress suppresses immunity and slows healing. Three models predict health behaviour: the health belief model (**Irwin Rosenstock** and colleagues, 1950s: perceived susceptibility, severity, benefits, barriers and cues to action), **Ronald Rogers**'s protection motivation theory (1975: threat appraisal and coping appraisal) and **Icek Ajzen**'s theory of planned behaviour (1985: attitude, subjective norm and perceived behavioural control, close to Albert Bandura's self-efficacy, belief in one's capability to act).

Stress management (Clinical chapter) includes **Edmund Jacobson**'s progressive muscle relaxation (1929), tensing and releasing muscle groups in turn; **Herbert Benson**'s relaxation response (1975); biofeedback, learning to control a bodily signal displayed on a monitor; meditation, as in **Jon Kabat-Zinn**'s mindfulness-based stress reduction (1979); regular aerobic exercise; and cognitive restructuring, replacing distorted thoughts with realistic ones, which **Donald Meichenbaum** combined with coping skills in stress inoculation training (1970s).

EXAM TRAP

Type A is competitiveness, impatience and hostility; options with "non-competitiveness" are wrong, and introversion or extraversion is no risk factor. Offered both Type A and hostility, RML 2024 keyed Type A; for the toxic *component*, answer hostility. **Type D** is not hostility, optimism or perfectionism. **Learned helplessness** is Seligman and Maier, not Beck, Sullivan or Kahneman. **Biopsychosocial** means social, not cultural. Emotional isolation is not incompetence or low self-esteem.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. "We feel afraid because we tremble" summarises the _____ theory.
2. MacLean's name for the classic "seat of emotion" is the _____ system.
3. Tameness, hyperorality and hypersexuality after bilateral amygdala damage make up _____ syndrome.
4. In Gross's process model, reappraisal belongs to _____ change.
5. Selye's general adaptation syndrome runs alarm, _____, exhaustion.
6. Denollet's Type D combines negative affectivity with social _____.

Answers: (1) James-Lange, (2) limbic, (3) Klüver-Bucy, (4) cognitive, (5) resistance, (6) inhibition.

The Case of Max Friedrich's Key, Revisited

Go back to the winter of 1879 and the student pressing his key. You can now take his act apart with the whole chapter. The stimulus had to cross a threshold (Section 1.3) and be organised into an object (Section 1.4); apperception was attention bringing it into focus (Section 1.5); practice shortened the times, which is learning (Section 1.6); the instruction was held in memory (Section 1.7); choosing when to press was a small decision (Section 1.8); observers differed in speed, one root of the study of individual differences and intelligence (Section 1.9); and the readiness and strain of waiting were motivation and emotion (Sections 1.10 and 1.11). Friedrich's numbers are forgotten, but his question, how long a mental act takes and what it is made of, runs through every section of this chapter.

Chapter summary

1.1 Psychology as a Science: Roots, Founders and the First Schools. Psychology, once the study of the soul, is now the scientific study of behaviour and covert mental processes. Its long past lies in philosophy (Plato's nativism, Aristotle's laws of association, Descartes' dualism, Locke's *tabula*

rasa, the associationists, Leibniz and Herbart on apperception, Kant) and in physiology (Muller, Helmholtz, localisation), with Darwin and Galton adding function and individual differences. Fechner's psychophysics (1860) and Donders' chronometry measured the mind. Wundt founded the first laboratory at Leipzig in 1879 and taught voluntarism through controlled introspection. Titchener's structuralism sought the elements of consciousness; James's functionalism asked what the mind is for, and Witmer opened the first clinic (1896).

1.2 Schools and Systems, the Cognitive Revolution and Indian Psychology. From about 1900 to 1960 psychology was a *preparadigmatic* field of rival schools. Watson founded behaviourism in 1913; Tolman, Hull, Guthrie and Skinner refined it. Gestalt psychology (Wertheimer's phi phenomenon, 1912; Kohler; Koffka) held that the whole differs from the sum of its parts, and Lewin defined the life space. Psychoanalysis was the first force, humanism (Maslow, Rogers) the third and transpersonal psychology the fourth, beside existential psychology and Seligman's positive psychology. The cognitive revolution (1956 onwards) restored the mind. Sengupta and Bose pioneered modern psychology at Calcutta; Indian psychology rests on Samkhya's purusha, prakriti and three gunas, Patanjali's eight limbs, the four states and the five koshas.

1.3 Sensation and Psychophysics. Sensation is the detection and transduction of physical energy; perception is its interpretation, so illusions distort perception. Muller's specific nerve energies and the all-or-none law explain how quality and intensity are coded, and every sense but smell relays through the thalamus. Fechner's psychophysics (1860) defines the absolute threshold as 50 percent detection and the JND as the smallest detectable change; Weber's law, Fechner's logarithmic law and Stevens' power law link stimulus to sensation, and signal detection theory separates sensitivity (d′) from response bias. Rods serve scotopic vision and foveal cones colour and acuity; colour is trichromatic then opponent; pitch is coded by place and timing; gate control explains pain.

1.4 Perception: Organising and Interpreting the World. Perception is the selection, organisation and interpretation of sensory signals. Constructivist theories (Helmholtz's unconscious inference, Gregory's hypotheses, the Bayesian union of priors and evidence) treat it as top-down inference; Gibson's direct, ecological theory, with its affordances, is the classic bottom-up account, and Neisser's perceptual cycle joins the two. The Gestalt laws, crowned by Prägnanz, organise the field into figure, ground and groups. Depth comes from two binocular cues (retinal disparity, convergence) and many monocular ones; the visual cliff (Gibson and Walk, 1960) showed depth perception by crawling age. Constancies keep objects stable, while illusions such as the Müller-Lyer reveal misapplied size scaling and differ from hallucinations and delusions.

1.5 Attention and Consciousness. Attention selects the little that reaches awareness; its span, studied by Jevons (1871) and with the tachistoscope, is about six to eight items. Cherry's dichotic listening (1953) led to Broadbent's early filter (1958), Treisman's attenuation model (1964) and late selection (Deutsch and Deutsch, 1963). In the Stroop effect (1935) automatic reading defeats selective attention, and vigilance declines over a watch (Mackworth). James described a stream of consciousness, Freud conscious, preconscious and unconscious levels. The suprachiasmatic nucleus times sleep, whose NREM and REM stages cycle every 90 minutes, four to six times a night. Dreams are explained by Freud, Jung and activation-synthesis; hypnosis rests on suggestibility.

1.6 Learning: Conditioning, Cognition and Observation. Learning is a relatively permanent change in behaviour, or its potential, resulting from experience (Kimble). In classical conditioning (Pavlov)

a neutral stimulus paired with a UCS becomes a CS; extinction, spontaneous recovery, generalisation and discrimination follow. Watson's Little Albert showed conditioned fear, and Jones and Wolpe reversed it through counterconditioning and systematic desensitisation (reciprocal inhibition). Rescorla, Kamin and Garcia showed that conditioning depends on prediction and preparedness. In operant conditioning (Thorndike's law of effect, Skinner) consequences change behaviour: reinforcement increases it, punishment decreases it, and schedules shape its pattern, variable ratio most strongly. Tolman's latent learning, Köhler's insight and Bandura's observational learning showed much learning without direct reinforcement.

1.7 Memory: Encoding, Storage, Retrieval and Forgetting. Memory is studied as encoding, storage and retrieval; Ebbinghaus (1885) measured forgetting with nonsense syllables and savings. Atkinson and Shiffrin's multi-store model (1968) separates sensory, short-term (seven plus or minus two chunks, mainly acoustic) and long-term memory, and Baddeley and Hitch (1974) recast short-term memory as working memory: central executive, phonological loop, visuospatial sketchpad and, from 2000, the episodic buffer. Depth of processing, Tulving's episodic and semantic memory, implicit memory, schemas and Bartlett's reconstruction, encoding specificity, decay, interference and repression explain remembering and forgetting, while H.M., Korsakoff's syndrome and Lashley's search for the engram reveal its biology.

1.8 Thinking, Problem Solving, Reasoning, Decision Making and Language. Thinking manipulates internal representations: images (Shepard and Metzler's mental rotation; Paivio's dual coding), concepts (Bruner's hypothesis testing for artificial concepts, Rosch's prototypes for natural ones) and propositions. Problems yield to algorithms, heuristics or Gestalt insight, and are blocked by functional fixedness (Duncker, 1945) and mental set (Luchins, 1942). Deduction runs from theory to data, induction from data to theory. Simon's bounded rationality and satisficing, and Tversky and Kahneman's heuristics and prospect theory, describe real decisions. Creativity is divergent production passing through Wallas's four stages. Language is built from phonemes and morphemes; Chomsky's LAD, the critical period, Whorf's linguistic relativity and Vygotsky's inner speech complete the picture.

1.9 Intelligence: Theories, Measurement, Heredity and Environment. Intelligence has been defined as judgement (Binet), *what the tests test* (Boring) and Wechsler's aggregate or global capacity (1944). Binet and Simon's 1905 scale introduced mental age; Stern coined the IQ (1912), Terman's Stanford-Binet (1916) made the ratio IQ famous, and Wechsler's deviation IQ (mean 100, SD 15) replaced it. The WAIS-IV has four indices; Raven's matrices test fluid intelligence, and Bhatia's Battery suits illiterate Indian children. Spearman's g, Thurstone's primary abilities, Guilford's cube, Cattell's fluid and crystallised intelligence, Gardner's multiple intelligences and Sternberg's triarchic theory offer rival structures. Heritability is a population statistic, and intellectual disability requires adaptive deficits as well as low IQ.

1.10 Motivation. Motivation energises and directs behaviour: a need produces a drive that pushes, while incentives pull. McDougall's instincts gave way to Hull's drive reduction (1943), which could not explain curiosity; Hebb proposed optimal arousal, and the Yerkes-Dodson law (1908) makes performance an inverted U of arousal with a lower optimum for complex tasks. Maslow's hierarchy of needs runs physiological, safety, belongingness, esteem, self-actualisation; Alderfer reduced it to ERG. Murray's needs and press underlie the TAT; McClelland studied achievement, power and affiliation. Self-efficacy, self-determination theory and flow explain intrinsic motivation. Lewin classified conflicts; the VMH is the satiety centre, the LH the feeding centre; Herzberg, Vroom and Locke explain work.

1.11 Emotion and Stress. An **emotion** joins arousal, expression and conscious experience; a mood is longer and diffuse, and affect is the current feeling state. Ekman named six universal basic emotions. **James-Lange** puts bodily change before feeling, **Cannon-Bard** makes them simultaneous via the thalamus, **Schachter and Singer** add a cognitive label (explaining misattribution), and Lazarus puts appraisal first. The **limbic system**, above all the **amygdala**, organises fear. Stress runs through Selye's alarm, resistance and exhaustion and the HPA axis; life events, hassles and appraisal shape it, and coping, support, hardiness and Type A or D personality link it to illness.

Who's who

Table 1.24 The people in this chapter.

Name	When	Examined for
Hermann Ebbinghaus	1850 to 1909	Memory measured with nonsense syllables and savings (On Memory, 1885); "Psychology has a long past, but only a short history" (1908)
Wilhelm Maximilian Wundt	1832 to 1920	First psychology laboratory, Leipzig (1879); Principles of Physiological Psychology (1873 to 1874); voluntarism; experimental introspection; Volkerpsychologie (folk psychology)
Aristotle	384 to 322 BCE	De Anima, the first treatise on the soul; empiricist; laws of association (similarity, contrast, contiguity)
William James	1842 to 1910	Father of American psychology; The Principles of Psychology (1890); stream of consciousness; habit; the self; inspiration of functionalism
William McDougall	1871 to 1938	Defined psychology as the science of conduct or behaviour (1905, 1908)
John Broadus Watson	1878 to 1958	Founder of behaviourism (1913); Angell's student at Chicago
Robert Sessions Woodworth	1869 to 1962	S-O-R formula; Dynamic Psychology (1918); psychology as the activities of the individual in relation to the environment
Norman L. Munn	Twentieth century	"The scientific study of behaviour", the definition credited to him in Indian textbooks
Plato	About 427 to 347 BCE	Nativist and rationalist; learning as recollection of inborn knowledge
Rene Descartes	1596 to 1650	Dualism; interaction through the pineal gland; innate ideas; the reflex
John Locke	1632 to 1704	Tabula rasa (1690); sensation and reflection; simple and complex ideas; primary and secondary qualities
David Hartley	1705 to 1757	Founder of associationism as a doctrine (1749)
James Mill	1773 to 1836	Mental mechanics: a complex idea is the sum of its parts
John Stuart Mill	1806 to 1873	Mental chemistry: ideas fuse into a new whole
Thomas Hobbes	1588 to 1679	Materialism; forerunner of British empiricism
George Berkeley	1685 to 1753	To be is to be perceived (esse est percipi)
David Hume	1711 to 1776	Laws of association: resemblance, contiguity, cause and effect
Alexander Bain	1818 to 1903	Associationism tied to the nervous system; founded the journal Mind (1876)
Gottfried Wilhelm Leibniz	1646 to 1716	Monads; petites perceptions; coined apperception; psychophysical parallelism
Johann Friedrich Herbart	1776 to 1841	Threshold of consciousness; apperceptive mass
Immanuel Kant	1724 to 1804	Innate categories; held that psychology could not be a science
Thomas Henry Huxley	1825 to 1895	Epiphenomenalism: mind as a by-product of the brain
Baruch Spinoza	1632 to 1677	Double-aspect theory of mind and body
Charles Bell	1774 to 1842	Bell-Magendie law (1811)
Francois Magendie	1783 to 1855	Bell-Magendie law (1822): dorsal roots sensory, ventral roots motor
Johannes Muller	1801 to 1858	Doctrine of specific nerve energies; Handbook of Human Physiology (1833 to 1840)

Name	When	Examined for
Hermann von Helmholtz	1821 to 1894	Speed of the nerve impulse (1850); unconscious inference; trichromatic theory of colour vision
Thomas Young	1773 to 1829	Suggested three colour receptors (1802): the Young-Helmholtz theory
Franz Joseph Gall	1758 to 1828	Founder of phrenology
Johann Gaspar Spurzheim	1776 to 1832	Popularised phrenology under that name
Pierre Flourens	1794 to 1867	Ablation studies of the brain (1820s)
Paul Broca	1824 to 1880	Speech production area in the left frontal lobe; patient Tan (1861)
Leborgne ("Tan")	1861	Broca's patient who understood speech but could say little more than "tan"
Carl Wernicke	1848 to 1905	Area for understanding speech in the left temporal lobe (1874)
Gustav Fritsch	1838 to 1927	Motor cortex mapped by electrical stimulation, with Hitzig (1870)
Eduard Hitzig	1838 to 1907	Motor cortex mapped by electrical stimulation, with Fritsch (1870)
Charles Darwin	1809 to 1882	Natural selection (1859); The Expression of the Emotions (1872); continuity, function and variation
Francis Galton	1822 to 1911	Individual differences; Hereditary Genius (1869); anthropometric laboratory (1884); correlation; questionnaire; word association; twin studies; eugenics
George John Romanes	1848 to 1894	Animal Intelligence (1882); anecdotal comparative psychology
Conwy Lloyd Morgan	1852 to 1936	Morgan's canon of parsimony (1894)
Ernst Heinrich Weber	1795 to 1878	Just noticeable difference as a constant fraction of the standard (Weber's law)
Gustav Theodor Fechner	1801 to 1887	Founder of psychophysics; insight of 22 October 1850; Elements of Psychophysics (1860); $S = k \log I$
Franciscus Cornelis Donders	1818 to 1889	Subtraction method (1868); mental chronometry
Emil Kraepelin	1856 to 1926	Worked in Wundt's laboratory; classification of psychiatric disorders
Charles Spearman	1863 to 1945	Leipzig doctorate; factor analysis
Vladimir Mikhailovich Bekhterev	1857 to 1927	Studied with Wundt; reflexology
Edward Bradford Titchener	1867 to 1927	Structuralism (1898) at Cornell; elements and attributes; stimulus error; context theory of meaning; the Experimentalists (1904)
Oswald Kulpe	1862 to 1915	Wurzburg school; imageless thought; Aufgabe and set
Narziss Ach	1871 to 1946	Wurzburg: the determining tendency and set (1905)
Franz Brentano	1838 to 1917	Act psychology; intentionality (1874)
Carl Stumpf	1848 to 1936	Tone psychology; teacher of the Gestalt founders
Georg Elias Muller	1850 to 1934	Gottingen laboratory; memory research
Carl Lange	1834 to 1900	Theory of emotion (1885), independently of James: the James-Lange theory
John Dewey	1859 to 1952	"The Reflex Arc Concept in Psychology" (1896); Chicago functionalism
James Rowland Angell	1869 to 1949	"The Province of Functional Psychology" (1906); taught Watson
Harvey A. Carr	1873 to 1954	The adaptive act; mature Chicago functionalism
James McKeen Cattell	1860 to 1944	Coined "mental test" (1890); reaction time and individual differences; Columbia
Edward Lee Thorndike	1874 to 1949	Animal learning at Columbia; law of effect
Granville Stanley Hall	1844 to 1924	First American PhD in psychology (1878); first American research laboratory (1883); American Journal of Psychology (1887); first APA president (1892); adolescence as storm and stress; recapitulation
Lightner Witmer	1867 to 1956	First psychological clinic (1896); first used "clinical psychology" in print (1907); father of clinical psychology
Hugo Munsterberg	1863 to 1916	Founder of forensic and industrial psychology; Wundt's student
Mary Whiton Calkins	1863 to 1930	First woman APA president (1905); refused a Harvard doctorate
Margaret Floy Washburn	1871 to 1939	First woman PhD in psychology (1894); The Animal Mind (1908)
Leta Stetter Hollingworth	1886 to 1939	Rejected claims of female inferiority with experimental evidence
Thomas Carlyle	1795 to 1881	Great-man (great-person) theory of history (1841)
Edwin Garrigues Boring	1886 to 1968	Historian of experimental psychology; champion of the Zeitgeist view
Alfred Russel Wallace	1823 to 1913	Natural selection, independently of Darwin (1858): a multiple discovery

Name	When	Examined for
Alfred Binet	1857 to 1911	First practical intelligence test (1905); father of intelligence testing
Narendra Nath Sengupta	1889 to 1944	First Indian laboratory (1915) and department (1916) at Calcutta; father of modern psychology in India
Girindrasekhar Bose	1887 to 1953	Indian Psychoanalytic Society (1922); father of psychoanalysis in India
M. V. Gopalaswamy	1924	Founded the psychology department at the University of Mysore
Sigmund Freud	1856 to 1939	Founder of psychoanalysis; free association; the unconscious mind
Carl Jung	1875 to 1961	Used word association as a clinical technique
Max Wertheimer	1880 to 1943	Founder of Gestalt psychology (1912)
Ivan Pavlov	1849 to 1936	Classical conditioning
Wilhelm Roentgen	1895	Physicist who discovered X-rays (1895); a distractor for folk psychology
Abraham Maslow	1908 to 1970	Humanistic psychology
Carl Rogers	1902 to 1987	Humanistic psychology
Burrhus Frederic Skinner	1904 to 1990	Operant conditioning; radical behaviourism
Jerome Bruner	1915 to 2016	"New Look" research on perception
Roy Schafer	1940s	"New Look" research on perception; a distractor for Titchener's definition of perception
Gregory Kimble	1961	Learning theorist (the standard definition of learning, 1961); a distractor for Titchener's definition of perception
Erik Erikson	1902 to 1994	Psychosocial stages; a distractor for "clinical psychology" in print
Karen Horney	1885 to 1952	Psychoanalyst; never APA president, a distractor for Calkins
Thomas Samuel Kuhn	1922 to 1996	Paradigm, paradigm shift and preparadigmatic science (The Structure of Scientific Revolutions, 1962)
Robert Irving Watson	1909 to 1980	Historian who named the prescriptions, the enduring issues of psychology (1967)
Ivan Petrovich Pavlov	1849 to 1936	Conditioned reflexes, a forerunner of behaviourism
Rosalie Rayner	1898 to 1935	Little Albert study with Watson (1920)
Percy Williams Bridgman	1882 to 1961	Operationism (1927), the basis of the operational definition
Edward Chace Tolman	1886 to 1959	Purposive behaviourism; molar behaviour; intervening variables; cognitive map; latent learning
Charles Honzik	1930 (study)	Latent learning experiment with Tolman
Clark Leonard Hull	1884 to 1952	Hypothetico-deductive system; drive reduction; reaction potential = habit strength × drive
John Dollard	1950 (with Miller)	Early social learning theory: Freud recast in Hull's terms
Neal Elgar Miller	1950 (with Dollard)	Early social learning theory with Dollard
Edwin Ray Guthrie	1886 to 1959	Contiguity; one-trial learning
Zing-Yang Kuo	1898 to 1970	Behaviourist who rejected the concept of instinct
Harry Harlow	1905 to 1981	Learning sets (a wrong option for the cognitive map)
James L. Gould	1986 (honeybee study)	Tested Tolman's cognitive map idea in honeybees; did not propose it
Joseph Wolpe	1915 to 1997	Systematic desensitisation (1958)
Norman Leslie Munn	Modern	Textbook author who popularised "the scientific study of behaviour"; keyed in one Panwar item as first to establish it (prefer Watson if offered)
Albert Bandura	1925 to 2021	Self-efficacy (1977); later social learning theory with Walters (1963)
Richard Walters	1963 (with Bandura)	Later social learning theory
Wolfgang Kohler	1887 to 1967	Insight in Sultan the chimpanzee on Tenerife (1913 to 1920); transposition; isomorphism
Kurt Koffka	1886 to 1941	Gestalt co-founder; geographical versus behavioural environment; Principles of Gestalt Psychology (1935)
Kurt Lewin	1890 to 1947	Field theory; life space; B = f(P, E); group dynamics; action research; father of modern social psychology
Bluma Zeigarnik	1900 to 1988	Better recall of interrupted tasks (1927)
Maria Ovsiankina	1898 to 1993	Spontaneous resumption of interrupted tasks (1928)

Name	When	Examined for
Frederick Salomon Perls	1893 to 1970	Fritz Perls: Gestalt therapy, here-and-now awareness, the empty chair
Laura Perls	1940s and 1950s	Co-developer of Gestalt therapy
Paul Goodman	1951 (Gestalt Therapy)	Co-developer of Gestalt therapy
Jean-Martin Charcot	1825 to 1893	Showed Freud hysterical symptoms produced and removed under hypnosis (Paris, 1885)
Pierre Janet	1859 to 1947	Explained hysteria as dissociation
Hippolyte Bernheim	1840 to 1919	Nancy school: hypnosis as suggestion
Josef Breuer	1842 to 1925	Anna O. and the cathartic method; Studies on Hysteria with Freud (1895)
Anna O. (Bertha Pappenheim)	Treated 1880 to 1882	Breuer's patient; the "talking cure"
Alfred Adler	1870 to 1937	Individual psychology
Carl Gustav Jung	1875 to 1961	Analytical psychology; the collective unconscious
Erich Fromm	1900 to 1980	Humanistic psychoanalysis
Harry Stack Sullivan	1892 to 1949	Interpersonal psychiatry
Erik Homburger Erikson	1902 to 1994	Psychosocial stages
Anna Freud	1895 to 1982	Ego psychology; defence mechanisms
Melanie Klein	1882 to 1960	Object relations
Abraham Harold Maslow	1908 to 1970	Named the third force; self-actualisation; foresaw the fourth force (1968)
Carl Ransom Rogers	1902 to 1987	Actualising tendency; unconditional positive regard
Charlotte Buhler	1893 to 1974	Humanistic founder; lifespan view
Rollo May	1909 to 1994	Brought existentialism to American psychology; normal versus neurotic anxiety (1950)
Edmund Husserl	1859 to 1938	Phenomenology; bracketing
Irvin David Yalom	Born 1931	Four ultimate concerns: death, freedom, isolation, meaninglessness (1980)
Soren Kierkegaard	1813 to 1855	Anxiety as the dizziness of freedom
Martin Heidegger	1889 to 1976	Dasein, being-in-the-world (1927)
Jean-Paul Sartre	1905 to 1980	Existence precedes essence
Ludwig Binswanger	1881 to 1966	Daseinsanalysis
Medard Boss	1903 to 1990	Daseinsanalysis
Viktor Emil Frankl	1905 to 1997	Logotherapy; will to meaning; existential vacuum; noogenic neurosis
Martin Elias Peter Seligman	Born 1942	Founded positive psychology as President of the American Psychological Association (1998)
Mihaly Csikszentmihalyi	1934 to 2021	Co-founder of positive psychology; flow
C. R. Snyder	Modern	Hope theory: the will and the ways to reach a goal
Howard Gardner	Born 1943	Multiple intelligences (a wrong option for the founder of positive psychology)
Steven Pinker	Born 1954	Language and evolutionary psychology (The Language Instinct, 1994); a wrong option for the founder of positive psychology
Karl Spencer Lashley	1890 to 1958	Serial order in behaviour needs central plans (1948)
Allen Newell	1927 to 1992	Logic Theorist with Simon (1956)
Herbert Alexander Simon	1916 to 2001	Logic Theorist with Newell (1956)
Noam Chomsky	Born 1928	Formal models of language (1956); review of Skinner's Verbal Behavior (1959)
George Armitage Miller	1920 to 2012	"The magical number seven, plus or minus two" (1956)
Donald Eric Broadbent	1926 to 1993	First major information-processing theory (1958)
Ulric Neisser	1928 to 2012	Father of cognitive psychology; Cognitive Psychology (1967)
Frederic Charles Bartlett	1886 to 1969	Schemas (1932); forerunner of cognitive psychology
Jean Piaget	1896 to 1980	Studied children's thinking from the 1920s; pioneer of cognitive psychology
David Rumelhart	1942 to 2011	Parallel distributed processing with McClelland (1986)
James McClelland	Born 1948	Parallel distributed processing with Rumelhart (1986)

Name	When	Examined for
Albert Ellis	1913 to 2007	Rational emotive behaviour therapy
Aaron Temkin Beck	1921 to 2021	Cognitive therapy; cognitive distortions
Anne Treisman	1935 to 2018	Attenuation work from about 1960 (a wrong option for the first information-processing theory)
Lawrence Kohlberg	1981 (The Philosophy of Moral Development)	Moral development (a wrong option for the pioneer of cognitive psychology)
George Libman Engel	1913 to 1999	Biopsychosocial model (1977)
Asutosh Mookerjee	1864 to 1924	Calcutta vice-chancellor (garbled as "Amitosh Mukherjee" in one item); not a founder of psychology
Durganand Sinha	1922 to 1998	Indigenisation of psychology in India
Uma Chowdhury	About 1960	Indian adaptation of the Thematic Apperception Test (also spelled Chaudhuri; probably the "Kamala Chaudhuri" of one item)
Jai B. P. Sinha	1980	Nurturant-task leader
Udai Pareek	1925 to 2010	Role stress
Sudhir Kakar	Born 1938	Psychoanalysis of Indian childhood and society
S. K. Mangal	Modern	Indian textbook author; did not found a laboratory
A. K. Singh	Modern	Indian textbook author; did not found a laboratory
G. S. Rathi	Modern	A wrong option for India's first laboratory
Matthijs Cornelissen	Early 2000s	Led the Indian Psychology movement; founded the Indian Psychology Institute
Koneru Ramakrishna Rao	Born 1932	Co-editor of the Handbook of Indian Psychology (2008); probably the "G. R. S. Rao" of one item
Anand C. Paranjpe	2008 (Handbook)	Co-editor of the Handbook of Indian Psychology
Ajit K. Dalal	2008 (Handbook)	Co-editor of the Handbook of Indian Psychology
Girishwar Misra	2011	Co-editor of Foundations of Indian Psychology
Suneet Varma	2011	Co-editor of Foundations of Indian Psychology
Kapila	Ancient (traditional)	Traditional founder of Samkhya
Ishvarakrishna	About fourth or fifth century CE	Samkhya Karika
Patanjali	Between about the second century BCE and the fourth century CE	Yoga Sutras; chitta-vritti-nirodha; the eight limbs
Adi Shankaracharya	Eighth century CE	Advaita Vedanta: atman is Brahman; the world is maya
Siddhartha Gautama (the Buddha)	Traditionally 563 to 483 BCE	Four Noble Truths; Eightfold Path; anatta
Sri Aurobindo	1872 to 1950	Integral yoga; the psychic being
Marsha Linehan	Born 1943	Dialectical behaviour therapy, with Zen mindfulness
Jerome Seymour Bruner	1915 to 2016	New Look in perception (Section 1.4); distractor for Titchener's definition
Johannes Peter Muller	1801 to 1858	Doctrine of specific nerve energies
Eugene Galanter	1962	Classic absolute threshold examples (a candle flame 48 km away, a watch tick at 6 m)
Stanley Smith Stevens	1906 to 1973	Power law of sensation (1957); magnitude estimation; cross-modality matching
Wilhelm Wundt	1832 to 1920	Father of experimental psychology; distractor for the founder of psychophysics
David M. Green	1966	Signal detection theory, with Swets (1966)
John A. Swets	1966	Signal detection theory, with Green (1966)
Jan Evangelista Purkyne (Purkinje)	1787 to 1869	Purkinje shift: reds darken before blues at dusk (1825)
Ernst Mach	1838 to 1916	Mach bands (1865), explained by lateral inhibition
Ewald Hering	1834 to 1918	Opponent-process theory of colour vision (1878)

Name	When	Examined for
Shinobu Ishihara	1879 to 1963	Ishihara colour vision plates (1917)
Georg von Bekesy	1899 to 1972	Travelling wave on the basilar membrane, place theory of pitch; Nobel Prize 1961
William Rutherford	1839 to 1899	Frequency (telephone) theory of pitch (1886)
Ernest Rutherford	1871 to 1937	Physicist wrongly credited with the frequency theory of pitch in some answer keys
Ernest Glen Wever	1902 to 1991	Volley principle of pitch (1930, with Bray)
Charles William Bray	born 1904	Volley principle of pitch (1930, with Wever)
Kikunae Ikeda	1864 to 1936	Named umami, the fifth basic taste (1908)
Hans Henning	1916	Smell prism (six primary odours) and taste tetrahedron (four tastes)
Ronald Melzack	1929 to 2019	Gate control theory of pain (1965, with Wall)
Patrick David Wall	1925 to 2001	Gate control theory of pain (1965, with Melzack)
Silas Weir Mitchell	1829 to 1914	Coined the term phantom limb (1871)
Vilayanur Subramanian Ramachandran	born 1951	Phantom limbs explained by cortical reorganisation (1990s)
Harry McGurk	1976	McGurk effect, with MacDonald (1976)
John MacDonald	1976	McGurk effect, with McGurk (1976)
Donald Olding Hebb	1904 to 1985	McGill sensory deprivation studies in his laboratory
William Bexton	1954	McGill sensory deprivation study (Bexton, Heron and Scott, 1954)
Woodburn Heron	1954	McGill sensory deprivation study (Bexton, Heron and Scott, 1954)
Thomas Scott	1954	McGill sensory deprivation study (Bexton, Heron and Scott, 1954)
Charles Bonnet	1720 to 1793	Charles Bonnet syndrome: visual hallucinations with insight in people with sight loss (1760)
Richard Langton Gregory	1923 to 2010	Constructivist approach and perceptual hypotheses (Eye and Brain, 1966); inappropriate constancy scaling (1963); studied S.B. (1963)
James Jerome Gibson	1904 to 1979	Direct (ecological) perception, the classic bottom-up theory; affordances; optic flow
Wolfgang Köhler	1887 to 1967	Gestalt co-founder
Edgar Rubin	1886 to 1951	Figure and ground (1915); Rubin's vase
Eleanor Jack Gibson	1910 to 2002	Visual cliff (with Walk, 1960)
Richard Walk	1920 to 1999	Visual cliff (with Eleanor Gibson, 1960)
Emil Emmert	1844 to 1911	Emmert's law of afterimage size (1881)
Franz Carl Müller-Lyer	1857 to 1916	Müller-Lyer illusion (1889)
Marshall Segall	1963, 1966	Carpentered-world hypothesis (with Campbell and Herskovits)
Donald Thomas Campbell	1916 to 1996	Carpentered-world hypothesis (with Segall and Herskovits)
Melville Jean Herskovits	1895 to 1963	Carpentered-world hypothesis (with Segall and Campbell)
Mario Ponzo	1882 to 1960	Ponzo illusion (1911)
Adelbert Ames Jr.	1880 to 1955	Ames room
Johann Karl Friedrich Zöllner	1834 to 1882	Zöllner illusion (1860)
Johann Christian Poggendorff	1796 to 1877	Poggendorff illusion (1860)
Louis Albert Necker	1786 to 1861	Necker cube, an ambiguous figure (1832)
Lionel Penrose	1898 to 1972	Penrose triangle, an impossible figure (1958, with his son Roger)
Roger Penrose	born 1931	Penrose triangle (1958, with his father Lionel)
Muzafer Sherif	1906 to 1988	Used the autokinetic effect to study norm formation (1935)
Oliver Selfridge	1926 to 2008	Pandemonium model of feature analysis (1959)
David Hubel	1926 to 2013	Feature detectors in cat visual cortex (with Wiesel, from 1959)
Torsten Wiesel	born 1924	Feature detectors in cat visual cortex (with Hubel, from 1959)
Eleanor Rosch	born 1938	Prototypes (1970s)
Irving Biederman	1987	Recognition-by-components theory; about 36 geons
David Marr	1945 to 1980	Computational theory: primal sketch, 2.5-D sketch, 3-D model
Gerald Reicher	1969	Word superiority effect
Leigh Minturn	1955	B or 13 context effect (with Bruner)

Name	When	Examined for
Cecile Goodman	1947	Coin study: economic deprivation affects perception (with Bruner)
Elliott McGinnies	1949	Perceptual defence for taboo words
George Malcolm Stratton	1865 to 1957	Inverting lenses (1896, 1897): perceptual adaptation
Jean Wallace	1963	Studied the restored-sight patient S.B. with Gregory
Herman Witkin	1916 to 1979	Field dependence-independence; rod-and-frame and embedded figures tests
James Vicary	1915 to 1977	Subliminal advertising hoax (1957, admitted 1962)
Anthony Marcel	1983	Masked semantic priming without awareness
Lawrence Weiskrantz	1926 to 2018	Blindsight (1974)
Joseph Banks Rhine	1895 to 1980	Parapsychology at Duke; Zener-card studies
Karl Zener	1903 to 1964	Zener cards
K. D. Broota	1970s	Indian work on cultural differences in perceptual selectivity (with Ganguli)
H. C. Ganguli	1970s	Indian work on cultural differences in perceptual selectivity (with Broota)
William Stanley Jevons	1835 to 1882; 1871	Bean experiment on the span of attention
George Miller	1956	Magical number seven, plus or minus two: memory span, not span of attention (Section 1.7)
Colin Cherry	1914 to 1979; 1953	Cocktail party problem; dichotic listening and shadowing
Neville Moray	1959	Own name heard in the unattended ear; no memory for words repeated 35 times
Donald Broadbent	1926 to 1993; 1958	Filter model (early selection); first major information-processing theory
J. Anthony Deutsch and Diana Deutsch	1963	Late selection model
Donald Norman	born 1935; 1968	Pertinence model of late selection
William Johnston and Steven Heinz	1978	Multimode (flexible) theory of selection
Nilli Lavie	1994 to 1995	Perceptual load theory
Michael Posner	born 1936	Spotlight; cueing task (1980); endogenous and exogenous attention; automaticity criteria (with Charles Snyder, 1975)
Charles Snyder	1975	Criteria for automatic processing (with Posner)
Garry Gelade	1980	Feature integration theory (with Treisman)
Daniel Simons and Christopher Chabris	1999	The invisible gorilla (inattentional blindness)
Daniel Kahneman	1934 to 2024; 1973	Capacity model (Attention and Effort)
Christopher Wickens	1984	Multiple resource theory
Richard Shiffrin and Walter Schneider	1977	Automatic versus controlled processing
Patricia Devine	1989	Stereotypes activated automatically; low-prejudiced people refute them by controlled processing
John Ridley Stroop	1897 to 1973; 1935	The Stroop effect
Norman Mackworth	1948	Clock test; vigilance decrement
Jerome L. Singer	1924 to 2010	Research on daydreaming
B. F. Skinner	1904 to 1990	Operant conditioning (Section 1.6); not the coiner of stream of consciousness
Anthony Greenwald and colleagues	1998	Implicit Association Test (unconscious stereotyping)
David Chalmers	born 1966; 1995	The hard problem of consciousness
Charles Tart	born 1937; 1969	Altered states of consciousness
Hans Berger	1873 to 1941; 1929	First human EEG
Eugene Aserinsky	1921 to 1998; 1953	Discovered REM sleep with Kleitman
Nathaniel Kleitman	1895 to 1999	REM discovery (1953); sleep cycle with Dement (1957)

Name	When	Examined for
William Charles Dement	1928 to 2020	Sleep cycle (1957, with Kleitman); REM deprivation and REM rebound (1960)
Howard Roffwarg and colleagues	1966	Sleep and REM across the lifespan
Viktor Frankl	1905 to 1997	Paradoxical intention (logotherapy), used for insomnia (Chapter 7)
J. Allan Hobson and Robert McCarley	1977	Activation-synthesis hypothesis of dreams
Antti Revonsuo	2000	Threat-simulation theory of dreams
Stephen LaBerge	early 1980s	Verified lucid dreaming with eye-movement signals
Franz Anton Mesmer	1734 to 1815	Animal magnetism (1770s)
James Braid	1795 to 1860	Coined hypnotism; Neurypnology (1843)
Ambroise-Auguste Liébeault	1823 to 1904	Founder of the Nancy school
André Weitzenhoffer	1959	Stanford Hypnotic Susceptibility Scales (with Hilgard)
Ernest Ropiequet Hilgard	1904 to 2001	Stanford scales (1959); neodissociation theory and the hidden observer
Theodore Xenophon Barber	1927 to 2005	Social-cognitive view of hypnosis
Nicholas Spanos	1942 to 1994	Social-cognitive theory of hypnosis
Maharishi Mahesh Yogi	1918 to 2008	Transcendental meditation (1950s)
Herbert Benson	1935 to 2022; 1975	The relaxation response
Nikolaas Tinbergen	1907 to 1988	Fixed action patterns triggered by sign stimuli (stickleback)
Eric Kandel	Born 1929	Habituation and sensitisation traced to synapses in Aplysia
Robert Fantz	1964	Habituation method for studying infant discrimination
Konrad Lorenz	1903 to 1989	Imprinting in greylag goslings (1935); critical period
Shenger-Krestovnikova	1921	Experimental neurosis (circle and ellipse), in Pavlov's laboratory
W. J. Brogden	1939	Sensory preconditioning
Mary Cover Jones	1897 to 1987	Peter and the rabbit (1924); counterconditioning
Edmund Jacobson	1888 to 1983	Progressive muscle relaxation
Thomas Stampfl	1960s	Implosive therapy
Joseph Cautela	1967	Covert sensitisation
Victor Meyer	1966	Exposure and response prevention for obsessive-compulsive disorder
Robert Rescorla	1940 to 2020	Contingency (1968); Rescorla-Wagner model (1972)
Leon Kamin	1927 to 2017	Blocking (1969)
Robert Lubow	1959	Latent inhibition (with Moore)
A. U. Moore	1959	Latent inhibition (with Lubow)
Allan Wagner	1934 to 2018	Rescorla-Wagner model (1972)
Wolfram Schultz	1990s	Dopamine neurons of the ventral tegmental area signal reward prediction error
John Garcia	1917 to 2012	Taste aversion; Garcia and Koelling (1966); equipotentiality refuted
Robert Koelling	1966	Bright, noisy, tasty water (with Garcia)
Martin Seligman	Born 1942	Preparedness (1970); learned helplessness (1967, with Maier)
Teodoro Ayllon	1960s; book 1968	Token economy at Anna State Hospital (with Azrin)
Nathan Azrin	1960s; book 1968	Token economy (with Ayllon)
Orval Hobart Mowrer	1907 to 1982	Two-factor theory of avoidance (1947); bell-and-pad for bed-wetting (1938)
William Kaye Estes	1919 to 2011	Punishment suppresses rather than erases a response (1944); stimulus sampling theory (1950)
Charles Ferster	1957	Schedules of reinforcement (with Skinner)
David Premack	1925 to 2015	Premack principle (1959)
Steven Maier	1967	Learned helplessness (with Seligman)
Keller Breland	1961	Instinctive drift (with Marian Breland)
Marian Breland	1961	Instinctive drift (with Keller Breland)
Richard Herrnstein	1930 to 1994	Matching law (1961)
Leo Crespi	1942	Contrast effects (the Crespi effect)

Name	When	Examined for
Neal Miller	1909 to 2002	Operant conditioning of autonomic responses and biofeedback; imitation (with Dollard, 1941)
Arnold Kegel	1948	Perineometer, an early form of biofeedback
Dorothea Ross	1961	Bobo doll studies (with Bandura and Sheila Ross)
Sheila Ross	1961	Bobo doll studies (with Bandura and Dorothea Ross)
Julian Rotter	1916 to 2014	Social learning theory (1954): behaviour potential, expectancy, reinforcement value
Robert Mills Gagné	1916 to 2002	Eight types of learning; The Conditions of Learning (1965)
William Lowe Bryan	1860 to 1955	Telegraphy learning curves and the plateau (with Harter, 1897, 1899)
Noble Harter	1897, 1899	The plateau in telegraphy learning (with Bryan)
William C. F. Krueger	1929	Overlearning improves retention
Edwin Locke	Born 1938	Goal setting: specific goals improve performance
Gary Latham	1970s onwards	Goal setting (with Locke)
Sidney Leavitt Pressey	1888 to 1979	The first teaching machine (1920s)
Clive Wearing	1985	British musician whose encephalitis (1985) left dense amnesia with spared musical skill
Raymond Nickerson	1979	Coin-drawing study showing encoding failure (with Marilyn Adams)
Marilyn Adams	1979	Coin-drawing study showing encoding failure (with Raymond Nickerson)
Richard C. Atkinson	1968	Multi-store (modal) model with Shiffrin; keyword method (1975)
Richard M. Shiffrin	1968	Multi-store (modal) model with Atkinson
George Sperling	born 1934	Partial-report experiments on iconic memory (1960)
Christopher Darwin	1972	Three-eared man study of echoic memory (with Turvey and Crowder)
Michael Turvey	1972	Three-eared man study of echoic memory (with Darwin and Crowder)
Robert Crowder	1972	Three-eared man study of echoic memory (with Darwin and Turvey)
Nelson Cowan	2001	Pure short-term memory capacity of about four chunks
John Brown	1958	Brown-Peterson technique for short-term memory duration
Lloyd Peterson	1959	Brown-Peterson technique: unrehearsed short-term memory lasts about 18 to 30 seconds
Margaret Peterson	1959	Brown-Peterson technique: unrehearsed short-term memory lasts about 18 to 30 seconds
Geoffrey Keppel	1962	Showed Brown-Peterson forgetting was interference, not decay (with Underwood)
Benton Underwood	1962	Showed Brown-Peterson forgetting was interference, not decay (with Keppel)
Reuben Conrad	1964	Acoustic confusions show acoustic coding in short-term memory
Alan Baddeley	born 1934	Acoustic versus semantic coding (1966); working memory model with Hitch (1974); episodic buffer (2000); divers study with Godden (1975)
Bennet Murdock	1962	Reliability of the serial position curve in free recall
Murray Glanzer	1966	A filled delay abolishes recency but spares primacy (with Cunitz)
Anita Cunitz	1966	A filled delay abolishes recency but spares primacy (with Glanzer)
Tim Shallice	1970	Patient K.F.: impaired short-term memory with normal long-term learning (with Warrington)
Elizabeth Warrington	1970	Patient K.F.: impaired short-term memory with normal long-term learning (with Shallice)
J. W. McCrary	1953	Keyed in RML 2024 (as McCray) for first observing the primacy-recency effect, with Hunter
W. S. Hunter	1953	Keyed in RML 2024 for first observing the primacy-recency effect, with McCrary
Graham Hitch	1974	Working memory model with Baddeley
Robert Logie	1995	Visual cache and inner scribe of the visuospatial sketchpad
Fergus Craik	1972	Levels of processing with Lockhart; Craik and Tulving (1975)
Robert Lockhart	1972	Levels of processing with Craik
T. B. Rogers	1977	Self-reference effect (with Kuiper and Kirker)
Nicholas Kuiper	1977	Self-reference effect (with Rogers and Kirker)

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W. S. Kirker	1977	Self-reference effect (with Rogers and Kuiper)
Donald Morris	1977	Transfer-appropriate processing (with Bransford and Franks)
John Bransford	1977	Transfer-appropriate processing (with Morris and Franks)
Jeffery Franks	1977	Transfer-appropriate processing (with Morris and Bransford)
Allan Paivio	1925 to 2016	Dual coding theory (1971): verbal and imagery codes
Endel Tulving	1927 to 2023	Episodic versus semantic memory (1972); encoding specificity with Thomson (1973); cue-dependent forgetting (1974); patient K.C.
Peter Graf	1985	Named explicit and implicit memory (with Schacter)
Daniel Schacter	1985	Named explicit and implicit memory (with Graf)
Larry Squire	1980	Declarative versus nondeclarative memory and their taxonomy
David Meyer	1971	Semantic priming in lexical decision, BREAD and BUTTER (with Schvaneveldt)
Roger Schvaneveldt	1971	Semantic priming in lexical decision, BREAD and BUTTER (with Meyer)
David Rubin	1986	Reminiscence bump (with colleagues); flashbulb versus everyday memories with Talarico (2003)
Roger Brown	1925 to 1997	Flashbulb memory with Kulik (1977); tip-of-the-tongue state with McNeill (1966)
James Kulik	1977	Flashbulb memory and the Now Print! mechanism (with Brown)
Nicole Harsch	1992	Challenger flashbulb memory study (with Neisser)
Jennifer Talarico	2003	9/11 flashbulb memories: confidence, not consistency (with Rubin)
John Flavell	1971	Coined metamemory (1971); metacognition
David McNeill	1966	Tip-of-the-tongue state (with Brown)
Allan Collins	1969	Hierarchical network model with Quillian (1969); spreading activation with Loftus (1975)
Ross Quillian	1969	Hierarchical network model and cognitive economy (with Collins)
Elizabeth Loftus	born 1944	Misinformation effect (Loftus and Palmer, 1974); spreading activation with Collins (1975); false memory implantation; critic in the recovered memory controversy
Roger Schank	1977	Scripts, such as the restaurant script (with Abelson)
Robert Abelson	1977	Scripts, such as the restaurant script (with Schank)
Gordon Allport	1947	Levelling and sharpening, from the study of rumour (with Postman)
Leo Postman	1947	Levelling and sharpening, from the study of rumour (with Allport)
John Palmer	1974	Smashed versus hit study of the misinformation effect (with Loftus)
Marcia Johnson	1993	Source monitoring framework (with colleagues)
Zena Pearlstone	1966	Category cues show available but inaccessible memories (with Tulving)
Donald Thomson	1973	Encoding specificity principle (with Tulving)
Duncan Godden	1975	Context-dependent memory in divers (with Baddeley)
Donald Goodwin	1969	State-dependent memory with alcohol
Gordon Bower	1932 to 2020	Mood-dependent and mood-congruent memory (1981)
John Jenkins	1924	Less forgetting after sleep than after waking (with Dallenbach)
Karl Dallenbach	1924	Less forgetting after sleep than after waking (with Jenkins)
Arthur Melton	1940	Two-factor theory of interference: response competition and unlearning (with Irwin)
J. M. Irwin	1940	Two-factor theory of interference (with Melton)
Michael Anderson	1994	Retrieval-induced forgetting (with Robert and Elizabeth Bjork)
Robert Bjork	1994	Retrieval-induced forgetting (with Anderson and Elizabeth Bjork)
Elizabeth Bjork	1994	Retrieval-induced forgetting (with Anderson and Robert Bjork)
Georg Elias Müller	1900	Consolidation (with Pilzecker)
Alfons Pilzecker	1900	Consolidation (with Müller)
Théodule Ribot	1881	Ribot's law: the new perishes before the old
Philip Ballard	1913	Reminiscence: recall rises after a delay without study
Hedwig von Restorff	1933	Von Restorff (isolation) effect: a distinctive item is remembered better
Arthur Gates	1917	Testing effect: recitation beats rereading

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William Beecher Scoville	1953	Removed H.M.'s medial temporal lobes; reported the case with Milner (1957)
Henry Gustav Molaison	1926 to 2008	Patient H.M.: profound anterograde amnesia after bilateral medial temporal lobe removal, with spared procedural learning
Brenda Milner	1957	Studied H.M. (Scoville and Milner, 1957); mirror-tracing learning without awareness
Sergei Korsakoff	late 1880s	Described Korsakoff's syndrome
Richard Semon	1904	Coined the term engram
Timothy Bliss	1973	Discovered long-term potentiation (with Lømo)
Terje Lømo	1973	Discovered long-term potentiation (with Bliss)
Simonides of Ceos	about 556 to 468 BCE	Credited with the method of loci
Francis P. Robinson	1946	SQ3R study method
Roger Newland Shepard	1929 to 2022	Mental rotation (1971, with Jacqueline Metzler)
Jacqueline Metzler	1971	Mental rotation, with Shepard
Stephen Michael Kosslyn	born 1948	Image scanning (1978): images as depictive representations
Zenon Walter Pylyshyn	1937 to 2022	Propositional theory and the imagery debate (1973)
Eugen Bleuler	1857 to 1939	Autistic (dereistic) thinking, 1911
Robert Ennis	20th century	A widely used definition of critical thinking
Jacqueline Goodnow	1956	A Study of Thinking, with Bruner and Austin
George Austin	1956	A Study of Thinking, with Bruner and Goodnow
Ludwig Wittgenstein	1889 to 1951	Family resemblance (1953)
Mary Gick	1980	Analogical problem solving: the radiation problem, with Holyoak
Keith Holyoak	1980	Analogical problem solving: the radiation problem, with Gick
Robert Jeffrey Sternberg	born 1949	Problem-solving cycle; investment theory of creativity (1991, with Lubart); analytical, creative and practical intelligence
Norman Maier	1900 to 1977	Nine-dot problem (1930)
Janet Metcalfe	1987	Warmth ratings show insight solutions arrive suddenly (with Wiebe)
David Wiebe	1987	Warmth ratings, with Metcalfe
Karl Duncker	1903 to 1940	Candle problem and functional fixedness (1945)
Abraham Luchins	1914 to 2005	Water jars; mental set (Einstellung), 1942
Charles Sanders Peirce	1839 to 1914	Abductive reasoning, inference to the best explanation
Karl Raimund Popper	1902 to 1994	Hypothetico-deductive method
Robert Woodworth	1935	Atmosphere effect in syllogisms, with Sells
Saul Sells	1935	Atmosphere effect in syllogisms, with Woodworth
Peter Cathcart Wason	1924 to 2003	2-4-6 task (1960), selection task (1966), confirmation bias
Philip Nicholas Johnson-Laird	born 1936	Mental models theory of reasoning (1983)
Gerd Gigerenzer	born 1947	Fast and frugal heuristics; recognition heuristic; ecological rationality
Amos Tversky	1937 to 1996	Heuristics and biases (1972 to 1974), prospect theory (1979), framing (1981), with Kahneman
Baruch Fischhoff	born 1946	Hindsight bias, the knew-it-all-along effect (1975)
Joy Paul Guilford	1897 to 1987	Convergent and divergent thinking; 1950 APA address; fluency, flexibility, originality, elaboration
Graham Wallas	1858 to 1932	The Art of Thought (1926): preparation, incubation, illumination, verification
Ellis Paul Torrance	1915 to 2003	Torrance Tests of Creative Thinking (1966)
Sarnoff Mednick	1928 to 2015	Remote Associates Test (1962)
Edward de Bono	1933 to 2021	Lateral thinking (1967)
Alex Faickney Osborn	1888 to 1966	Brainstorming; Applied Imagination (1953)
Todd Lubart	1991	Investment theory of creativity, with Sternberg
Charles Francis Hockett	1916 to 2000	Design features of language (1960)
Dan Isaac Slobin	born 1939	Language-making capacity (1985)
Eric Heinz Lenneberg	1921 to 1975	Critical period for language (1967)

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Genie	found 1970	Child isolated until 13: learned words but not grammar
Katherine Nelson	1973	Referential style of early word learning
Benjamin Lee Whorf	1897 to 1941	Linguistic relativity (the Sapir-Whorf hypothesis)
Edward Sapir	1884 to 1939	Whorf's teacher; the Sapir-Whorf hypothesis
Brent Berlin	1969	Basic colour terms in a near-universal order, with Kay
Paul Kay	1969	Basic colour terms, with Berlin
Lev Semyonovich Vygotsky	1896 to 1934	Thinking and Speech (1934): thought depends on language; private speech becomes inner speech; heaps, complexes, true concepts
David Wechsler	1896 to 1981	Aggregate or global capacity definition (1944); Wechsler-Bellevue (1939); deviation IQ; WAIS, WISC and WPPSI
Douglas Detterman	1986	With Sternberg, the 1986 expert symposium on defining intelligence
Carol Susan Dweck	Born 1946	Entity versus incremental theories of intelligence; fixed and growth mind-sets (Mindset, 2006); praise effort, not ability
Clark Wissler	1901	Showed that Cattell's mental tests did not predict college grades
Théodore Simon	1873 to 1961	Binet's collaborator on the 1905, 1908 and 1911 scales
Henry Herbert Goddard	1866 to 1957	Translated the Binet-Simon scale into English (1908); coined moron (1910)
William Stern	1871 to 1938	Coined the intelligence quotient (1912): mental age divided by chronological age
Lewis Madison Terman	1877 to 1956	Stanford-Binet (1916) and ratio IQ = MA ÷ CA × 100; gifted Termites study (from 1921)
Robert Mearns Yerkes	1876 to 1956	Army Alpha (verbal) and Army Beta (non-verbal) group tests (1917)
Gale Roid	2003	Stanford-Binet Fifth Edition (SB5): five factors, verbal and non-verbal, mean 100 and SD 15
Frank Schmidt and John Hunter	1998	General mental ability predicts job performance with a validity of about .5
John Carlyle Raven	1902 to 1970	Standard Progressive Matrices (1938): culture-fair test of fluid intelligence and the best single marker of g
Raymond Bernard Cattell	1905 to 1998	Fluid versus crystallised intelligence (1963); investment theory; Culture Fair Intelligence Test; the 16PF (personality)
V. V. Kamat	1930s	Binet-Kamat Test, a heavily verbal Indian adaptation of the Stanford-Binet
Arthur J. Malin	1969	Malin's Intelligence Scale for Indian Children (MISIC), a half-verbal adaptation of the WISC
C. M. Bhatia	1955	Bhatia's Battery of Performance Tests of Intelligence: five performance subtests, no reading, rural and urban norms
J. B. P. Sinha	1980	Nurturant-task leader (organisational psychology); a distractor, not the author of Bhatia's Battery
Charles Edward Spearman	1863 to 1945	Two-factor theory (1904): general factor g and specific factors s; positive manifold; first factor analysis
Louis Leon Thurstone	1887 to 1955	Seven primary mental abilities (1938)
Philip Ewart Vernon	1905 to 1987	Hierarchical model (1950): g, then the major group factors v:ed and k:m
John Bissell Carroll	1916 to 2003	Three-stratum theory (1993): g, eight broad and about 70 narrow abilities
Kevin McGrew	Late 1990s	Merged the Cattell-Horn and Carroll models into the CHC theory
John Leonard Horn	1928 to 2006	Developed the Gf-Gc theory with Cattell; extended it to about nine broad abilities
Godfrey Hilton Thomson	1881 to 1955	Sampling theory (1916): test correlations arise from overlapping samples of many small bonds
Arthur Robert Jensen	1923 to 2012	Level I (associative) and Level II (conceptual) abilities; 1969 claim that group IQ gaps are partly genetic
Jagannath Prasad Das	Born 1931	PASS model with Naglieri and Kirby (1994); Cognitive Assessment System
Jack A. Naglieri	1994	Co-author of the PASS model and the Cognitive Assessment System
John R. Kirby	1994	Co-author of the PASS model
Alexander Romanovich Luria	1902 to 1977	Functional units of the brain, the basis of the PASS model

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Stephen J. Ceci	1990	Bioecological view: cognitive potentials develop only in supporting contexts
Peter Salovey	1990	With Mayer, emotional intelligence (1990); four-branch ability model; MSCEIT
John D. Mayer	1990	With Salovey, emotional intelligence (1990); four-branch ability model; MSCEIT
Daniel Goleman	Born 1946	Popularised emotional intelligence (1995); five-part mixed model
Reuven Bar-On	1997	Emotional Quotient Inventory (EQ-i), a self-report mixed model; the term EQ
Danah Zohar	2000	Spiritual intelligence
Richard Haier	1988	PET study: high scorers used less brain glucose (neural efficiency)
Thomas J. Bouchard Jr.	1981; from 1979	Review of IQ correlations among relatives with McGue (1981); Minnesota Study of Twins Reared Apart
Matt McGue	1981	With Bouchard, the 1981 review of IQ correlations among relatives
Eric Turkheimer	2003	Heritability of IQ near .10 in the poorest families: gene-environment interaction
Richard Lewontin	1929 to 2021	Seed analogy (1970): heritability within groups says nothing about differences between groups
Irving Gottesman	1930 to 2016	Reaction range (1963): genes set limits, environment decides where within them a person lands
James Robert Flynn	1934 to 2020	The Flynn effect: average IQ scores rose about 3 points per decade (1984)
Richard J. Herrnstein	1930 to 1994	The Bell Curve (1994) with Murray
Charles Murray	Born 1943	The Bell Curve (1994) with Herrnstein
Claude Steele	1995	Stereotype threat, with Aronson (1995)
Joshua Aronson	1995	Stereotype threat, with Steele (1995)
Edgar Arnold Doll	1889 to 1968	Vineland Social Maturity Scale (1935): social age and social quotient
Joseph Renzulli	Born 1936	Three-ring conception of giftedness (1978): above-average ability, task commitment, creativity
Henry Alexander Murray	1893 to 1988	Needs and press (Explorations in Personality, 1938); viscerogenic and psychogenic needs; TAT with Morgan (1935)
Walter Bradford Cannon	1871 to 1945	Coined homeostasis (1926; The Wisdom of the Body, 1932); stomach contractions and hunger with Washburn (1912)
Claude Bernard	1813 to 1878	The milieu intérieur, the constant internal environment
Leon Festinger	1919 to 1989	Cognitive dissonance (1957)
Harry Frederick Harlow	1905 to 1981	Puzzle-solving monkeys and the manipulation drive (1950); Maslow's doctoral adviser
John Dillingham Dodson	1879 to 1955	Yerkes-Dodson law (1908)
Robert Zajonc	1923 to 2008	Drive theory of social facilitation (1965): arousal strengthens the dominant response
Marvin Zuckerman	1928 to 2018	Sensation seeking
Otto Tinklepaugh	1928	Monkeys expecting banana refused lettuce: incentive theory
Julian Bernard Rotter	1916 to 2014	Social learning theory: expectancy and reinforcement value (1954); locus of control (1966)
Richard Lester Solomon	1918 to 1995	Opponent-process theory of acquired motivation (with Corbit, 1974)
John Corbit	1974	Opponent-process theory with Solomon
Kurt Goldstein	1878 to 1965	Coined the term self-actualisation (1939)
Clayton Paul Alderfer	1940 to 2015	ERG theory (1969): existence, relatedness, growth; frustration-regression
David Clarence McClelland	1917 to 1998	Need for achievement (1953), power and affiliation; the Kakinada training studies
Edward L. Deci	born 1942	Self-determination theory with Ryan (1985); cognitive evaluation theory
Richard M. Ryan	born 1953	Self-determination theory with Deci (1985): autonomy, competence, relatedness
William Glasser	1925 to 2013	Reality therapy: five basic needs, including power or achievement
Christiana Morgan	1897 to 1967	Thematic Apperception Test with Murray (1935)

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John William Atkinson	1923 to 2003	Achievement motivation theory (1957): motive to succeed versus motive to avoid failure; preference for intermediate risk
Matina Souretis Horner	born 1939	Fear of success (1972)
Bernard Weiner	born 1935	Attribution theory of achievement: locus, stability, controllability
Daniel Ellis Berlyne	1924 to 1976	Curiosity: novelty, complexity, incongruity (1960)
Robert White	1904 to 2001	Effectance or competence motivation (1959)
Lev Vygotsky	1896 to 1934	Constructivism (a wrong option for reciprocal causation)
Frederick Herzberg	1923 to 2000	Two-factor (motivator-hygiene) theory (1959)
Stanley Schachter	1922 to 1997	Externality hypothesis of obesity (1968 to 1971); two-factor theory of emotion with Singer
Jerome Singer	1962	Two-factor theory of emotion with Schachter
Mark Lepper	1973	Overjustification effect: the magic-marker study with Greene and Nisbett
David Greene	1973	Magic-marker study with Lepper and Nisbett
Richard Nisbett	born 1941	Magic-marker study with Lepper and Greene (1973)
Roger Barker	1903 to 1990	Regression after frustration: toys behind a screen (with Dembo and Lewin, 1941)
Tamara Dembo	1941	Regression study with Barker and Lewin
Leonard Doob	1909 to 2000	Co-author of Frustration and Aggression (1939)
Robert Sears	1908 to 1989	Co-author of Frustration and Aggression (1939)
Leonard Berkowitz	1926 to 2016	Reformulated frustration-aggression hypothesis (1989): negative affect
A. L. Washburn	1912	Swallowed the balloon in Cannon's hunger study
John P. J. Pinel	2021 (Biopsychology, with Barnes)	Positive-incentive view of eating; settling point
Steven J. Barnes	2021 (Biopsychology, with Pinel)	Positive-incentive view of eating; settling point
Albert Hetherington	1940 (with Ranson)	VMH lesions cause hyperphagia and obesity
Stephen Walter Ranson	1880 to 1942	VMH lesions cause hyperphagia and obesity (with Hetherington, 1940)
Bal Krishan Anand	1951 (with Brobeck)	Lateral hypothalamic lesions cause aphagia: the feeding centre
John Brobeck	1951 (with Anand)	Lateral hypothalamic lesions cause aphagia
Eliot Stellar	1919 to 1993	Dual-centre theory of hunger (1954)
Alfred Charles Kinsey	1894 to 1956	Reports on male (1948) and female (1953) sexual behaviour
William Howell Masters	1915 to 2001	Sexual response cycle: excitement, plateau, orgasm, resolution (1966)
Virginia Eshelman Johnson	1925 to 2013	Sexual response cycle with Masters (1966)
Helen Singer Kaplan	1929 to 1995	Triphasic model: desire, excitement, orgasm (1979)
Carl Warden	1931	Columbia obstruction box: maternal drive strongest
Victor Harold Vroom	born 1932	Expectancy (VIE) theory of work motivation (1964)
John Stacy Adams	1963	Equity theory
Carol Dweck	born 1946	Attribution retraining of helpless children (1975)
William Richard Miller	born 1947	Motivational interviewing (with Rollnick, 1991)
Stephen Rollnick	1991 (with Miller)	Motivational interviewing
Paul Ekman	born 1934	Six universal basic emotions (contempt added 1986); Fore study (1971); Facial Action Coding System (1978)
Wallace V. Friesen	1971; 1978	With Ekman: Fore universality study and the Facial Action Coding System
Silvan Tomkins	1962	Affect theory: nine innate affects; the face as the seat of affect, root of the facial feedback hypothesis
Carroll Izard	1977	Differential emotions theory: ten basic emotions, including shame and guilt
Robert Plutchik	1980	Wheel of eight primary emotions and their dyads
James Russell	1980	Circumplex model: emotions on valence and arousal
Alexander Thomas		With Chess: easy, difficult and slow-to-warm-up temperaments (pointer)

Name	When	Examined for
Stella Chess		With Thomas: easy, difficult and slow-to-warm-up temperaments (pointer)
Carl Georg Lange	1834 to 1900	James-Lange theory of emotion (1885)
Philip Bard	1898 to 1977	Cannon-Bard theory; sham rage in decorticate cats
Donald Dutton	1974	Capilano bridge study: misattribution of arousal
Arthur Aron	1974	Capilano bridge study: misattribution of arousal
Dolf Zillmann	1970s	Excitation transfer theory
Richard S. Lazarus	1922 to 2002	Cognitive appraisal theory (primary and secondary appraisal); daily hassles (1981); transactional model of stress (1984)
Fritz Strack	1988	Pen study supporting the facial feedback hypothesis
Leonard Martin	1988	Pen study of facial feedback, with Strack and Stepper
Sabine Stepper	1988	Pen study of facial feedback, with Strack and Martin
Richard Solomon	1918 to 1995	Opponent-process theory of emotion (1974)
James Papez	1883 to 1958	Papez circuit of emotion (1937)
Paul MacLean	1913 to 2007	Named the limbic system (1952), the classic seat of emotion
Heinrich Klüver	1897 to 1979	Klüver-Bucy syndrome after bilateral temporal lobe removal in monkeys (1937, 1939)
Paul Bucy	1904 to 1992	Klüver-Bucy syndrome (1937, 1939)
Joseph LeDoux	born 1949	Low road and high road to the amygdala in fear
Larry Cahill	1994	With McGaugh: propranolol abolishes the emotional enhancement of memory
James McGaugh	1994	With Cahill: the amygdala and noradrenaline in emotional memory
H.M. (Henry Molaison)	operated on 1953; died 2008	Amnesic patient: the hippocampus, not the amygdala, stores facts
Phineas Gage	1823 to 1860	Prefrontal injury (1848) that changed his emotional and social conduct
Antonio Damasio	born 1944	Somatic marker hypothesis (1994)
David T. Lykken	1959	Guilty knowledge test, more accurate than the polygraph
Brad Bushman	2002	Venting experiment against the catharsis hypothesis
Paul Rozin	1987	With Fallon: disgust as protection against contamination
April Fallon	1987	With Rozin: disgust
Ed Diener	1946 to 2021	Subjective well-being, tripartite model (1984): life satisfaction, positive affect, negative affect
Harry Helson	1898 to 1977	Adaptation-level theory (1964)
Philip Brickman	1971	With Campbell: the hedonic treadmill
Donald Campbell	1971	With Brickman: the hedonic treadmill
Samuel Stouffer	1949	Relative deprivation (The American Soldier)
Barbara Fredrickson	1998	Broaden-and-build theory of positive emotions
Fred Bryant	2007	With Veroff: savoring
Joseph Veroff	2007	With Bryant: savoring
Aaron Beck	1967	Cognitive model of depression (1967) and cognitive distortions (a distractor in questions on Fredrickson and learned helplessness)
John B. Watson	1878 to 1958	Three innate emotions: fear, rage and love
Katharine Banham Bridges	1932	Differentiation of infant emotions: distress and delight first
Edward Tronick	1978	Still-face procedure: infants expect emotional give-and-take
James Gross	1998	Process model of emotion regulation: reappraisal is cognitive change, suppression is response modulation
Peter Sifneos	1920 to 2008	Coined alexithymia (1973)
Hans Selye	1907 to 1982	General adaptation syndrome (1936; The Stress of Life, 1956): alarm, resistance, exhaustion; eustress (1974)
Shelley E. Taylor	born 1946	Tend-and-befriend (2000)
Thomas Holmes	1967	With Rahe: Social Readjustment Rating Scale (life events)
Richard Rahe	1967	With Holmes: Social Readjustment Rating Scale (life events)
Allen Kanner	1981	With Lazarus and colleagues: daily hassles and uplifts

Name	When	Examined for
Robert Karasek	1979	Job demand-control model
Herbert Freudenberger	1926 to 1999	Introduced the term burnout (1974)
Christina Maslach	born 1946	Maslach Burnout Inventory (1981): exhaustion, depersonalisation, reduced accomplishment
Susan Folkman	1984	With Lazarus: transactional model of stress
Paul Meehl	1920 to 2003	Diathesis-stress model of schizophrenia (1962)
Joseph Zubin	1977	With Spring: vulnerability-stress model
Bonnie Spring	1977	With Zubin: vulnerability-stress model
Sheldon Cohen	1985	With Wills: direct and stress-buffering effects of social support
Thomas Wills	1985	With Cohen: stress-buffering effect of social support
Robert S. Weiss	1973	Emotional isolation versus social isolation
Suzanne Kobasa	1979	Hardiness: commitment, control, challenge
Aaron Antonovsky	1923 to 1994	Sense of coherence (1979): comprehensibility, manageability, meaningfulness
Michael Scheier	1985	With Carver: dispositional optimism
Charles Carver	1985	With Scheier: dispositional optimism
Meyer Friedman	1910 to 2001	With Rosenman: Type A behaviour pattern (1959); Western Collaborative Group Study
Ray Rosenman	1920 to 2013	With Friedman: Type A behaviour pattern (1959)
Johan Denollet	mid-1990s	Type D personality: negative affectivity plus social inhibition
George Engel	1913 to 1999	Biopsychosocial model (1977)
Robert Ader	1932 to 2011	With Nicholas Cohen: conditioned immunosuppression (1975), the founding study of psychoneuroimmunology
Nicholas Cohen	1975	With Ader: conditioned immunosuppression
Irwin Rosenstock	1950s	Health belief model
Ronald Rogers	1975	Protection motivation theory
Icek Ajzen	1985	Theory of planned behaviour: attitude, subjective norm, perceived behavioural control
Jon Kabat-Zinn	born 1944	Mindfulness-based stress reduction (1979)
Donald Meichenbaum	born 1940	Stress inoculation training (1970s)

Key terms

mental processes, clinical psychology, comparative psychology, evolutionary psychology, nativism, empiricism, rationalism, laws of association, dualism, innate ideas, tabula rasa, associationism, mental chemistry, apperceptive mass, psychophysical parallelism, Bell-Magendie law, psychophysics, mental chronometry, immediate experience, voluntarism, creative synthesis, tridimensional theory of feeling, introspection, Volkerpsychologie, structuralism, stimulus error, context theory of meaning, imageless thought, functionalism, stream of consciousness, reflex arc, mental test, psychological clinic, great-person theory, Zeitgeist, preparadigmatic science, prescriptions, behaviourism, peripheralism, operationism, cognitive map, latent learning, phi phenomenon, insight, transposition, life space, action research, Gestalt therapy, structural model, basic anxiety, phenomenology, Dasein, logotherapy, noogenic neurosis, transpersonal psychology, savouring, cognitive revolution, information-processing approach, connectionism, eclecticism, Indian psychology, theory of opposite wishes, indigenisation, nurturant-task leader, astika, purusha, prakriti, gunas, antahkarana, chitta-vritti-nirodha, nishkama karma, atman, turiya, panchakosha, psychic being, sensation, perception, transduction, doctrine of specific nerve energies, all-or-none law, sensory adaptation, absolute threshold, just noticeable difference, Weber's law, Fechner's law, magnitude estimation, Stevens' power law, method of limits, method of constant

stimuli, signal detection theory, d prime, response bias, receiver operating characteristic, accommodation, fovea, blind spot, scotopic vision, dark adaptation, Purkinje shift, saccades, Mach bands, lateral inhibition, trichromatic theory, opponent-process theory, place theory, frequency theory, volley principle, papillae, umami, gate control theory, phantom limb, semicircular canals, synaesthesia, Charles Bonnet syndrome, Bottom-up processing, top-down processing, unconscious inference, constructivist approach, Bayesian approach, direct perception, affordances, law of proximity, law of closure, Prägnanz, binocular cues, retinal disparity, convergence, monocular cues, texture gradient, Motion parallax, visual cliff, Perceptual constancy, size constancy, size-distance invariance, Emmert's law, illusion, Müller-Lyer illusion, inappropriate constancy scaling, carpentered-world hypothesis, hallucination, autokinetic effect, Pandemonium, prototype, recognition-by-components, perceptual set, field dependence-independence, Subliminal perception, Extrasensory perception, span of attention, cocktail party phenomenon, dichotic listening, filter model, early selection, attenuation model, late selection, feature integration theory, inattentional blindness, change blindness, automatic process, controlled processing, Stroop effect, sustained attention, Consciousness, topographic model, preconscious, unconscious, psychodynamic perspective, hard problem of consciousness, altered state of consciousness, circadian rhythm, suprachiasmatic nucleus, REM sleep, slow-wave sleep, sleep cycle, REM rebound, cataplexy, manifest content, latent content, dream work, collective unconscious, activation-synthesis hypothesis, Hypnosis, suggestibility, neodissociation theory, Concentrative meditation, mindfulness meditation, tolerance, learning, habituation, sensitisation, imprinting, classical conditioning, unconditioned stimulus, conditioned stimulus, extinction, spontaneous recovery, stimulus generalisation, stimulus discrimination, counterconditioning, reciprocal inhibition, systematic desensitisation, flooding, conditioned taste aversion, preparedness, law of effect, operant conditioning, negative reinforcement, secondary reinforcers, token economy, two-factor theory, response cost, shaping, schedule of reinforcement, Premack principle, learned helplessness, biofeedback, insight learning, observational learning, reciprocal determinism, self-efficacy, savings method, multi-store model, sensory memory, chunking, serial position effect, working memory, phonological loop, visuospatial sketchpad, central executive, episodic buffer, levels of processing, self-reference effect, episodic memory, semantic memory, procedural memory, implicit memory, flashbulb memory, tip-of-the-tongue state, schema, misinformation effect, source monitoring, encoding specificity principle, context-dependent memory, state-dependent memory, decay theory, proactive interference, retroactive interference, repression, anterograde amnesia, retrograde amnesia, Korsakoff's syndrome, confabulation, engram, long-term potentiation, Thinking, mental rotation, dual-coding theory, concept, family resemblance, problem space, algorithm, heuristic, Means-end analysis, hill climbing, functional fixedness, mental set, Deductive reasoning, Inductive reasoning, abductive reasoning, belief bias, confirmation bias, bounded rationality, satisficing, conjunction fallacy, Prospect theory, loss aversion, Hindsight bias, Creativity, phoneme, morpheme, pragmatics, prosody, deep structure, language acquisition device, poverty of the stimulus, critical period, holophrase, telegraphic speech, overregularisation, linguistic relativity, inner speech, Aphasia, operational definition, Achievement, aptitude, growth mindset, mental age, intelligence quotient (IQ), Stanford-Binet, ratio IQ, deviation IQ, concurrent validity, culture-fair test, Raven's Progressive Matrices, Factor analysis, positive manifold, general factor (g), primary mental abilities, structure of intellect, Fluid intelligence (Gf), Crystallised intelligence (Gc), multiple intelligences, triarchic theory, successful intelligence, PASS model, emotional intelligence, twin studies, Heritability, reaction range, Flynn effect, stereotype threat, Intellectual disability, adaptive functioning, Global developmental delay, Vineland Social Maturity Scale, Savant syndrome, motivation, need, drive, primary motives, secondary motives, intrinsic motivation, extrinsic motivation, instinct, hormic psychology, homeostasis, drive-reduction theory, optimal arousal theory, Yerkes-

Dodson law, incentive theory, expectancy-value theory, hierarchy of needs, prepotency, deficiency needs, metamotivation, ERG theory, press, need for achievement, triadic reciprocal causation, overjustification effect, self-determination theory, cognitive evaluation theory, flow, frustration, ventromedial hypothalamus, lateral hypothalamus, sexual response cycle, expectancy theory, goal-setting theory, emotion, mood, Affect, James-Lange theory, Cannon-Bard theory, misattribution of arousal, primary appraisal, secondary appraisal, facial feedback hypothesis, sham rage, Papez circuit, limbic system, amygdala, Klüver-Bucy syndrome, pseudobulbar affect, display rules, subjective well-being, broaden-and-build theory, Savoring, self-conscious emotions, cognitive reappraisal, Alexithymia, fight-or-flight response, general adaptation syndrome, eustress, hypothalamic-pituitary-adrenal axis, Social Readjustment Rating Scale, daily hassles, burnout, transactional model of stress, diathesis-stress model, Problem-focused coping, emotion-focused coping, emotional isolation, hardiness, Type A behaviour pattern, Type D personality, biopsychosocial model, Psychoneuroimmunology

From the past papers

1. Functionalism, an early school of thought in psychology, was most closely associated with which psychologist? (IHBAS 2024)

   (a) Edward Titchener (b) John Watson (c) William James (d) G. Stanley Hall

   Answer: (c) William James. James's *Principles of Psychology* (1890) inspired the functionalist question of what the mind is for, and Dewey and Angell made it a formal school at Chicago. Titchener founded the rival structuralism, Watson founded behaviourism, and Hall, though functionalist in spirit, is examined for institutions such as the first American research laboratory (1883) and the APA (1892).

2. The first person to use the term clinical psychology in print was (RML 2020)

   (a) Erik Erikson (b) Sigmund Freud (c) Lightner Witmer (d) Cattell

   Answer: (c) Lightner Witmer. Witmer opened the first psychological clinic at the University of Pennsylvania in 1896 and used the term in print in his journal *The Psychological Clinic* (1907). Erikson is known for the psychosocial stages, Freud for psychoanalysis, and James McKeen Cattell for coining "mental test".

3. Match List I with List II LIST I (Psychologist) LIST 2 (Theory) A. Dollard and Miller 1. Radical Behaviorism B. Skinner 2. Self-Actualization C. Bandura and Walters 3. Early Social Learning D. Maslow 4. Later Social Learning (IHBAS 2024)

   (a) A-1, B-2, C-3, D-4 (b) A-3, B-1, C-4, D-2 (c) A-2, B-1, C-3, D-4 (d) A-3, B-2, C-4, D-1

   Answer: (b) A-3, B-1, C-4, D-2. Dollard and Miller recast Freud in Hull's terms in 1950 (early social learning); Skinner's system is radical behaviourism; Bandura and Walters gave the later social learning theory (1963); Maslow is self-actualisation. Option (d) tempts only if you forget that Skinner, not Maslow, is the radical behaviourist.

4. Which is the correct order of the last four of Patanjali's eight limbs? (NIMHANS 2026 recall)

   (a) Dharana, pratyahara, dhyana, samadhi (b) Pratyahara, dharana, dhyana, samadhi (c) Pratyahara, dhyana, dharana, samadhi (d) Dhyana, dharana, pratyahara, samadhi

   Answer: (b) Pratyahara, dharana, dhyana, samadhi. The senses are first withdrawn inward (pratyahara), the mind is then fixed on one point (dharana), the fixing becomes an unbroken flow (dhyana) and finally absorption (samadhi). Each wrong option swaps two of these steps.

5. Which statement is INCORRECT? (Amity 2023)

   (a) Cones are most prevalent in the central retina (b) Cones are sensitive to moderate to high levels of light (c) Cones provide only monochromatic information (d) Cones provide excellent acuity

   Answer: (c) Cones provide only monochromatic information. Cones, with three pigments, give colour vision; it is the rods, with one pigment (rhodopsin), that give colourless scotopic vision. The other three statements are true: cones are packed in the fovea, work in moderate to bright light and give excellent acuity.

6. In signal detection theory, the ability to distinguish a stimulus from background noise is: (IHBAS 2025)

   (a) Sensitivity (d′) (b) Criterion (β) (c) ROC curve (d) JND

   Answer: (a) Sensitivity (d'). d′ is the separation between the noise and signal-plus-noise distributions and does not depend on the observer's willingness to say "yes". The criterion (β) is that willingness (response bias), the ROC curve plots hit rate against false alarm rate, and the JND belongs to classical psychophysics.

7. Among monocular cues for depth perception is (JEMAS 2025)

   (a) Convergence (b) Retinal disparity (c) Assimilation (d) Interposition

   Answer: (d) Interposition. An object that partly hides another is seen as nearer, and one eye is enough to see it. Convergence and retinal disparity are the only two binocular cues, and assimilation is not a depth cue at all (it is Piaget's term).

8. Illusions are (JEMAS 2025)

   (a) misinterpretation of stimuli arising from an external object (b) real perceptual object perceived in a distorted way (c) vivid mental images (d) perception without an adequate external stimulus

   Answer: (a) Misinterpretation of stimuli arising from an external object. In an illusion a real stimulus is misread. Option (b) is a sensory distortion such as micropsia, (c) is imagery, and (d) is the definition of a hallucination.

9. The Stroop effect demonstrates interference in (JEMAS 2025)

   (a) Long-term memory retrieval (b) Working memory (c) Selective attention (d) Procedural memory

   Answer: (c) Selective attention. Automatic word reading cannot be filtered out while naming the ink colour, so the task shows a failure of selective attention. The memory options are distractors; no stored memory is being retrieved or rehearsed.

10. According to Freud, the ___ mind includes accessible memories, which can be recalled at will (RML 2020)

    (a) Conscious (b) Preconscious (c) Unconscious (d) Psychic determinism

    Answer: (b) Preconscious. The conscious holds only what is in awareness now, and the unconscious cannot be recalled at will; psychic determinism is a principle (no mental event is accidental), not a level of mind.

11. Systematic desensitization technique is based upon the idea of: (IHBAS 2025)

    (a) Reciprocal inhibition (b) Law of effect (c) Habituation (d) Adaptation to stress

    Answer: (a) Reciprocal inhibition. Wolpe paired each step of an anxiety hierarchy with relaxation, a response incompatible with anxiety, so the anxiety bond weakens. The law of effect is Thorndike's operant principle, habituation underlies exposure and response prevention, and adaptation to stress belongs to stress theory.

12. Among the following schedules of reinforcement, which typically produces the highest rate of response and is most resistant to extinction? (Real PYQ 2024)

    (a) Fixed Ratio (b) Variable Ratio (c) Fixed Interval (d) Variable Interval

    Answer: (b) Variable Ratio. Ratio schedules reward fast responding, and unpredictability removes any reason to pause and makes it hard to tell that reinforcement has stopped, as with slot machines. Fixed ratio shows a pause after each reinforcer, fixed interval a scallop and variable interval only a moderate, steady rate.

13. The famous patient H.M. (Henry Molaison) underwent surgical removal of the medial temporal lobes, including the hippocampus, to treat severe epilepsy. While he retained most of his intellectual abilities and working memory, his primary and most profound deficit was in: (Real PYQ 2024)

    (a) Procedural memory (b) Semantic memory (c) Anterograde amnesia (d) Retrograde amnesia

    Answer: (c) Anterograde amnesia. After the 1953 operation H.M. could not form new declarative memories, although his intelligence, digit span and childhood memories survived. Procedural memory was spared (he learned mirror tracing), old semantic knowledge was intact, and his retrograde amnesia covered only the years before surgery.

14. Which of the following terms refers to difficulty recalling information due to interference from new information? (IHBAS 2024)

    (a) Retroactive interference (b) Proactive interference (c) Decay theory (d) Retrieval failure

    Answer: (a) Retroactive interference. New learning reaching back to disrupt older memories is retroactive (RETRO: new disrupts old); proactive interference is the reverse, old disrupting new. Decay blames the passage of time and retrieval failure a mismatch of cues, not competition from new learning.

15. In which method does the researcher gather data, look for patterns, and then generate a theory or hypothesis? (RML 2023)

    (a) Deductive reasoning (b) Hypothetico-deductive method (c) Inductive reasoning (d) Abductive reasoning

    Answer: (c) Inductive reasoning. Induction runs from specific observations to a probable generalisation, so data come first and theory last, as in most qualitative research. Deduction and the hypothetico-deductive method run the other way, from a theory to a tested prediction; abduction infers the best explanation of a surprising fact.

16. What is the primary difference between Piaget and Vygotsky's theories of language development? (RML 2021)

    (a) Piaget argued thought depends on language while Vygotsky argued language depends on thought (b) Vygotsky argued thought depends on language while Piaget argued language depends on thought (c) Vygotsky was concerned with early language development while Piaget was concerned with later language development. (d) None of the above

    Answer: (b). For Piaget, language expresses concepts already built through action; for Vygotsky, thought and speech merge at about two, after which thought becomes verbal and depends on language. Option (a) swaps the names, and (c) is false because both studied the same preschool years.

17. The term IQ was invented by (JEMAS 2025)

    (a) Lewis Termon (sic) (b) Gordon Allport (c) William Stern (d) John Watson

    Answer: (c) William Stern. Stern proposed the intelligence quotient, mental age divided by chronological age, in 1912. Terman is the tempting wrong answer: he multiplied the quotient by 100 and popularised it in the 1916 Stanford-Binet, but did not coin it; Allport (traits) and Watson (behaviourism) never worked on the IQ.

18. A 10-year-old child from a rural area, with no formal schooling, is brought for an intellectual assessment due to academic struggles. The child is illiterate and has had minimal exposure to formal testing materials. Which of the following tests would be the MOST appropriate to use to minimize cultural and educational bias? (Real PYQ 2024)

    (a) Bhatia's Battery of Performance Tests of Intelligence (b) Wechsler Intelligence Scale for Children (WISC-IV) (c) Stanford-Binet Intelligence Scale (5th ed.) (d) Malin's Intelligence Scale for Indian Children (MISIC)

    Answer: (a) Bhatia's Battery of Performance Tests of Intelligence. Its five performance subtests need no reading, and it has Indian rural norms. The WISC-IV and SB5 rest on school-built verbal knowledge and foreign norms; MISIC is the tempting Indian option, but half of it is verbal, adapted from the WISC.

19. The 'satiety centre' of the brain is located in the: (RML 2024)

    (a) Lateral hypothalamus (b) Ventromedial hippocampus (c) Lateral thalamus (d) Ventromedial hypothalamus

    Answer: (d) Ventromedial hypothalamus. Bilateral VMH lesions make rats overeat and grow obese (Hetherington and Ranson, 1940), so Stellar's dual-centre theory (1954) called the VMH the satiety centre. The lateral hypothalamus is the opposite, the feeding centre, whose destruction causes aphagia (Anand and Brobeck, 1951); "ventromedial hippocampus" and "lateral thalamus" are invented by changing one word.

20. According to Herzberg's Two-Factor Theory, which of the following factors are considered motivators that lead to job satisfaction? (RML 2023)

    (a) Salary and benefits (b) Job security and work conditions (c) Recognition and opportunities for advancement (d) Company policies and interpersonal relationships

    Answer: (c) Recognition and opportunities for advancement. Herzberg's motivators lie in the content of the work: achievement, recognition, the work itself, responsibility, advancement and growth. Salary, benefits, job security, working conditions, company policies and relationships are hygiene factors, which prevent dissatisfaction when good but do not by themselves motivate.

21. The seat of emotion is to be found in _____. (RML 2024)

    (a) Reticular formation (b) Hind brain (c) Limbic system (d) Forebrain

    Answer: (c) Limbic system. MacLean's limbic system (1952), built on Papez's circuit and including the amygdala, is the classic seat of emotion. The reticular formation governs arousal and wakefulness, the hindbrain paces heartbeat and breathing, and "forebrain" is too broad, although the limbic system lies within it.

22. 'Psychological disorders develop when a biological predisposition is set off by a stressful situation', is proposed by (JEMAS 2025)

    (a) Humanistics existential models. (b) Socio-cultural model. (c) Diathesis-stress model. (d) Cognitive model.

    Answer: (c) Diathesis-stress model. A pre-existing vulnerability (diathesis) set off by a stressful trigger is the model's definition (Meehl, 1962; Zubin and Spring, 1977). Humanistic-existential models blame blocked growth, sociocultural models social forces and cognitive models faulty thinking; none is defined by predisposition plus stress.

High-yield sheet

The facts this chapter's questions ask most, most-asked first. Last read before the exam.

1 **Systematic desensitisation** (Joseph Wolpe, *Psychotherapy by Reciprocal Inhibition*, 1958) rests on **reciprocal inhibition**: relaxation, a response incompatible with anxiety, is paired step by step with a graded anxiety hierarchy. ASKED 17 TIMES

2 **Working memory** (Alan Baddeley and Graham Hitch, 1974) has four components: the central executive (an attentional controller that stores nothing, loaded by random number generation), the phonological loop (auditory-verbal information), the visuospatial sketchpad (visual and spatial information, including mental rotation) and the episodic buffer, added in 2000, which binds information into coherent episodes. ASKED 11 TIMES

3 The **token economy** is operant conditioning with tokens as secondary reinforcers, developed by Teodoro Ayllon and Nathan Azrin (*The Token Economy*, 1968); it is the listed therapy not derived from classical conditioning. ASKED 10 TIMES

4 **Abraham Maslow**'s hierarchy of needs (1943, 1954), bottom to top: physiological, safety, belongingness and love (social), esteem, self-actualisation. Physiological and safety needs are the lower needs; ego is not a level. ASKED 10 TIMES

5 **Korsakoff's syndrome** follows thiamine (vitamin B1) deficiency in chronic alcoholism, often after Wernicke's encephalopathy (confusion, ataxia, nystagmus); it damages the mammillary bodies and mediodorsal thalamus and causes dense anterograde amnesia, extensive retrograde amnesia and confabulation, sincere false stories that fill memory gaps. ASKED 8 TIMES

6 The **empty chair** technique belongs to **Gestalt therapy** (Fritz Perls, with Laura Perls and Paul Goodman), an experiential here-and-now therapy that is not the Gestalt psychology of Wertheimer, Kohler and Koffka. ASKED 7 TIMES

7 **Variable ratio** reinforcement gives the highest, steadiest rate and the greatest resistance to extinction; fixed ratio gives high rates with a post-reinforcement pause, fixed interval the scallop, variable interval moderate steady rates, and continuous reinforcement the fastest learning and fastest extinction. ASKED 7 TIMES

8 **Extinction** is the fading of a conditioned response when the CS is repeatedly presented without the UCS, or of an operant response that is no longer reinforced; a parent who ignores tantrums is using extinction, not operant escape. ASKED 7 TIMES

9 Reinforcement always increases behaviour and punishment decreases it; positive adds and negative removes a stimulus, so switching off a shock is **negative reinforcement**, while time-out and response cost are **negative punishment**. ASKED 7 TIMES

10 **H.M.** (Henry Molaison) had both medial temporal lobes, including the hippocampus and amygdala, removed by William Beecher Scoville in 1953; Scoville and Brenda Milner (1957) reported a profound anterograde amnesia for declarative memory, with normal short-term memory, limited retrograde amnesia and spared procedural learning (mirror tracing). ASKED 7 TIMES

11 **Raymond Cattell** (with John Horn) split g into **fluid intelligence** (Gf), reasoning with novel problems independently of acquired knowledge and measured best by Raven's Progressive Matrices, and **crystallised intelligence** (Gc), the repertoire of information, cognitive skills and strategies, measured by vocabulary and information. ASKED 7 TIMES

12 **The limbic system** is the classic "seat of emotion": Paul MacLean named it in 1952, enlarging James Papez's 1937 circuit, and it regulates emotion, motivation and memory; hunger and thirst belong to the hypothalamus. ASKED 7 TIMES

13 **The amygdala** is the structure for fear and aggression: it detects threat, drives fear conditioning and gives emotional events their extra memorability, while the hippocampus stores the facts. ASKED 7 TIMES

14 The **hippocampus** is the structure damaged in anterograde amnesia and is critical for consolidating new declarative memories; it is not the store of remote memories. ASKED 6 TIMES

15 **Narendra Nath Sengupta** set up India's first psychology laboratory (1915) and first Department of Psychology (1916) at the University of Calcutta and is the father of modern psychology in India; keys naming Bose or Uma Chaudhuri for the first laboratory are wrong. ASKED 5 TIMES

16 **Humanistic psychology** (Maslow, Rogers) is the **third force**, after psychoanalysis (first) and behaviourism (second); **transpersonal psychology** is the fourth force. ASKED 5 TIMES

17 The **Stroop effect** (John Ridley Stroop, 1935): naming the ink colour of a conflicting colour word (RED printed in green) is slowed because automatic word reading interferes; it shows a limit of selective attention (interference in reaction time), not a memory effect. ASKED 5 TIMES

18 **Proactive interference**: earlier learning disrupts newer learning (dialling your old phone number); **retroactive interference**: newer learning disrupts older memories. RML 2024 keys English grammar intruding on newly learned Spanish as retroactive, but it is proactive. ASKED 5 TIMES

19 **Inductive reasoning** moves from specific observations to probable generalisations (gather data, find patterns, build theory: the logic of theory development and qualitative research); **deductive reasoning** starts from a theory and tests its predictions against observations. ASKED 5 TIMES

20 **Graham Wallas** (*The Art of Thought*, 1926): preparation, incubation, illumination, verification; incubation (setting the problem aside) is second and illumination (the insight) third. ASKED 5 TIMES

21 **Functional fixedness** (Karl Duncker's candle problem, 1945) is seeing an object only in its usual use; **mental set** (Abraham Luchins's water jars, 1942) is reusing a strategy that worked before, and was keyed as the most basic and common obstacle to problem solving. ASKED 5 TIMES

22 **Henry Murray**'s need-press theory (1938) underlies the **Thematic Apperception Test** (Morgan and Murray, 1935), a projective test whose stories reveal the teller's needs, conflicts and motives. ASKED 5 TIMES

23 **Wilhelm Wundt** opened the first psychology laboratory at **Leipzig in 1879** (not Vienna, Paris or Cambridge); Galton's London laboratory (1884) was anthropometric. ASKED 4 TIMES

24 Patanjali's **eight limbs**, in order: yama, niyama, asana, pranayama, pratyahara (withdrawal of the senses), dharana (concentration on one point), dhyana (unbroken meditation), samadhi (absorption); karma and jnana are not limbs. ASKED 4 TIMES

25 **Tamas** is the guna of inertia, lethargy, darkness, dullness and ignorance; a person who has stopped eating and cannot get herself to act shows tamas alone. ASKED 4 TIMES

26 **Every sense except smell relays through the thalamus** on its way to the cortex (vision via the lateral geniculate nucleus, hearing via the medial geniculate nucleus); the olfactory nerve (cranial nerve I) bypasses it. ASKED 4 TIMES

27 The **suprachiasmatic nucleus** (SCN) of the hypothalamus is the master circadian pacemaker, entrained by light through the retinohypothalamic tract; SCN lesions abolish circadian rhythms, not total sleep. ASKED 4 TIMES

28 **REM (paradoxical) sleep**: a desynchronised, fast, low-voltage EEG like waking, rapid eye movements, atonia of the core muscles, genital arousal and vivid dreams; EEG synchrony and delta waves belong to N3, spindles and K-complexes to N2. ASKED 4 TIMES

29 **Carl Jung** split the unconscious into a personal (individual) unconscious and the **collective unconscious**, an inherited layer of archetypes shared by all humanity, which he explained by the similar brain structure of all human races. ASKED 4 TIMES

30 **William Stern** coined the intelligence quotient in 1912; **Lewis Terman**'s Stanford-Binet (1916) multiplied it by 100, giving the ratio IQ = (MA ÷ CA) × 100, so a mental age of 8 at age 10 gives an IQ of 80. ASKED 4 TIMES

CHAPTER 2

Psychological Assessment

IN THIS CHAPTER

2.1 ● Psychological Testing: Nature, History, Types and Ethics: Define a psychological test and assessment, trace testing from Chinese civil-service examinations to the MMPI, classify tests (objective versus subjective, speed versus power, maximum versus typical performance), list the characteristics of a good test, recognise the errors of clinical judgement and apply the ethics of test use.

2.2 ● Building a Test: Items, Item Analysis, IRT and Response Bias: Write and format test items, run a classical item analysis (difficulty, discrimination, item-total correlations, distractors, ceiling and floor effects), explain item response theory and the item characteristic curve, and recognise response sets and rater errors such as acquiescence and halo.

2.3 ●●● Reliability: Consistency and the Standard Error of Measurement: Explain how classical test theory defines reliability, choose and compute the right estimate (test-retest, parallel forms, split-half with Spearman-Brown, Rulon, Flanagan, KR-20, KR-21, alpha, kappa) and put a standard-error band around one person's score.

2.4 ●● Validity and the Accuracy of Test Decisions: Define validity in its classical and unified senses, tell face, content, criterion-related (concurrent and predictive) and construct validity apart by their evidence, link validity to reliability through the validity ceiling, and judge test decisions with sensitivity, specificity, predictive values, base rates and ROC curves.

2.5 ●● Norms, Standard Scores and Interpreting a Score: Interpret any test score by placing it against norms: tell norm- from criterion-referenced scores, judge a norm sample, and convert between percentiles, z, T, stanine, sten, scaled scores and deviation IQs on the normal curve.

2.6 ●● Intelligence Tests: Binet, Wechsler, Non-verbal and Indian Scales: Identify and compare the major intelligence tests (the Binet scales, the Wechsler family, the Kaufman and Das-Naglieri batteries, non-verbal and culture-fair tests and the Indian scales), say what an IQ test can and cannot tell you, and choose the right test for a given person.

2.7 ● Assessing Development, Adaptive Behaviour and Disability: Explain how development, social maturity and adaptive behaviour are measured (DQ, SQ, Gesell, Bayley, Denver, VSMS, Vineland and the Indian scales), how intellectual disability and global developmental delay are assessed, and how intellectual disability, specific learning disability and autism are certified in India.

2.8 ● Aptitude, Achievement, Creativity, Interest and Attitude Measures: Tell achievement, aptitude and intelligence tests apart, name the main aptitude batteries, creativity tests and interest and value inventories, and explain how Thurstone, Likert, Guttman, Bogardus, semantic differential and implicit attitude measures are built and scored.

2.9 ●● Objective Personality Inventories: Distinguish objective inventories from projective tests, name Cattell's L-, Q- and T-data, match inventories to their construction strategy, interpret the MMPI's validity and clinical scales and its revisions, and identify the authors, items, scales and scores of the 16PF, EPQ, NEO-PI-R, MBTI, EPPS, CPI, MCMI, STAI, Rep test and Q-sort.

2.10 ●●● Projective Techniques: Explain the projective hypothesis, sort projective techniques by Lindzey's five categories, describe how the Rorschach and TAT are given, scored and interpreted, recognise the completion, expressive and choice techniques with their authors, and weigh the evidence on their reliability and validity.

2.11 ●● Neuropsychological Assessment and Clinical Rating Scales: Explain what neuropsychological assessment measures and how, recognise the main batteries and the memory, visual-motor, executive, attention and language tests and bedside screens, and match each clinical rating scale to its author, format, rater and purpose.

Weightage. 336 of the 3180 questions mapped in this book (10.6%) test this chapter, 136 of them from real past papers. 2 of its 11 sections are High yield.

London, 1884. Among the displays on diet, dress and sanitation at the International Health Exhibition in South Kensington stood a long, narrow room unlike anything else in the show. For threepence a visitor could walk along a counter of instruments while attendants measured height, arm span, weight, breathing power, strength of squeeze, keenness of sight and hearing, colour sense and speed of reaction. The figures went onto a card: the visitor kept one copy and the laboratory kept the other. By the time the exhibition closed, 9,337 people had passed through **Francis Galton**'s Anthropometric Laboratory.

Galton was not running a sideshow. He had argued in *Hereditary Genius* (1869) that ability is inherited, and he believed that an able mind showed itself in keen senses and quick reactions. Measure the senses accurately enough, in enough people, and you would be measuring intelligence. His data were superb: taken the same way for every visitor and from thousands of people, they showed how human traits spread around an average, and to analyse them Galton worked out the idea of correlation. At the University of Pennsylvania, **James McKeen Cattell** built the same kinds of measure into a battery and in 1890 gave it a name: the mental test.

Then, in 1901, Cattell's student **Clark Wissler** asked the obvious question. Did the tests predict anything that mattered? At Columbia University they barely related to students' grades. Careful, precise measurement had captured the wrong thing. That gap, between measuring something consistently and measuring what matters, runs through this whole chapter, and at its end we return to Galton's counter of instruments.

2.1 Psychological Testing: Nature, History, Types and Ethics

LOWER YIELD · 11 QS

Later sections each study one kind of test or one property of tests. This one builds their frame: what a test is, where testing came from, how tests are sorted and judged, how clinicians reason from scores, and testing ethics.

What a test is

Anne Anastasi (1908 to 2001), author of the standard text *Psychological Testing* (1954), defined a **psychological test** as "essentially an objective and standardized measure of a sample of behavior". *Objective*: the score does not depend on who gives or scores it. *Standardised*: everyone meets the same items, instructions, time and scoring. *Sample*: the test observes a small, chosen part of behaviour and infers the rest, as a doctor judges your blood from one drop. **Lee J. Cronbach** (1916 to 2001) stressed comparison: a test is "a systematic procedure for comparing the behavior of two or more persons".

Measurement assigns numbers to attributes by rules (Ravi scored 42); **evaluation** judges the number against a standard (42 is a pass); **testing** obtains a behaviour sample with a standardised instrument; and **assessment**, the widest, answers a referral question by integrating tests with interview, observation, history and records. A test yields a score; an assessment yields a reasoned conclusion about a person.

The **psychometric approach** measures individual differences with standardised tests, and its logic of fixed items, norms, reliability and validity serves intelligence, symptoms of psychopathology and neurological or cognitive deficits alike. Levels of measurement (nominal, ordinal, interval, ratio) are taught in Chapter 9.

From Chinese examinations to the MMPI

Testing is older than psychology: China examined candidates for public office under the Han dynasty (206 BC to AD 220), and under the T'ang dynasty written civil-service examinations became the rule. Modern testing has two nineteenth-century roots, one clinical: the French psychiatrist **Jean-Étienne Esquirol** (1772 to 1840) separated mental retardation from mental illness, grading retardation by the person's use of language, and **Édouard Séguin** (1812 to 1880), who trained retarded children through sensory and motor exercises, left us the **form board**, a performance task of fitting shapes into matching holes.

The other root is individual differences. **Francis Galton** (1822 to 1911) argued in *Hereditary Genius* (1869) that ability is inherited and, believing that all knowledge enters through the senses, measured sensory discrimination and reaction time in his London **Anthropometric Laboratory** of 1884 (the scene that opened this chapter). His idea of correlation became, in the hands of **Karl Pearson** (1857 to 1936), the product-moment correlation coefficient. Galton is called the father of mental testing and of psychometrics. **James McKeen Cattell** (1860 to 1944), who had worked with Wundt and Galton, coined the term **mental test** in his 1890 paper "Mental Tests and Measurements"; his tests, too, were sensory and motor. In 1901 **Clark Wissler** (1870 to 1947) found that they barely correlated with Columbia students' grades. Measuring the senses precisely was not measuring intelligence.

The breakthrough came in Paris, where **Alfred Binet** (1857 to 1911) and **Théodore Simon** tested judgement, comprehension and reasoning instead (Table 2.1). From the 1960s, the period of accountability, American courts heard challenges to tests as biased against minorities and women. Indian tests (Kamat 1934, Bhatia 1955, Malin 1969, the NIMHANS and PGIMER batteries) appear in Sections 2.6, 2.7 and 2.11.

Table 2.1 Milestones of the testing century.

Year	Who	Milestone
1905	Binet and Simon	Binet-Simon scale, the first practical intelligence test
1912	**William Stern**	Intelligence quotient: mental age divided by chronological age
1916	**Lewis Terman**	Stanford-Binet; IQ = MA/CA × 100
1917	**Robert Yerkes**, using **Arthur Otis**'s group items	Army Alpha (literate recruits) and Army Beta (non-verbal) group tests
1917	**Robert S. Woodworth**	Personal Data Sheet, the first personality inventory
1921	**Hermann Rorschach**	Inkblot test
1927	**Edward K. Strong**	Strong Vocational Interest Blank
1928 to 1929	**L. L. Thurstone**	Attitude scaling; later multiple factor analysis
1935	**Christiana Morgan** and **Henry Murray**	Thematic Apperception Test (TAT)
1938	**Oscar Buros**	First *Mental Measurements Yearbook* of test reviews

Year	Who	Milestone
1939	**David Wechsler**	Wechsler-Bellevue scale for adults
1943	**Starke Hathaway** and **J. Charnley McKinley**	Minnesota Multiphasic Personality Inventory (MMPI)

EXAM TRAP

"Mental tests" was coined by Cattell (1890), not Binet (first practical intelligence test, 1905), Terman (Stanford-Binet, 1916) or Wechsler (1939). Galton is the father of mental testing; Stern proposed the intelligence quotient and Terman multiplied it by 100. Woodworth's data sheet, built in 1917 to 1918, is dated 1919 or 1920 in some books.

Kinds of tests ●●

Four ways of sorting tests matter most; Table 2.2 gives the rest.

Scoring. On an **objective test** a fixed key leaves no room for the scorer's judgement, so any two scorers agree: multiple-choice items, rating scales and self-report inventories such as the MMPI are objective. On a **subjective test** (an essay, an interview rating, a projective test such as the Rorschach) the scorer judges each free response. "Projective" names a technique of ambiguous stimuli (Section 2.10); "introjective" is no kind of test (introjection is a defence mechanism, Chapter 4).

Time. A **speed test** has easy items of similar difficulty under a time limit so strict that few finish; the score is how many are done, as in a clerical checking test. A **power test** gives generous time, or none, but its items climb in difficulty until some are too hard for anyone; the score reflects the hardest level mastered (Figure 2.1). Most real tests are **speeded tests**, a blend of both.

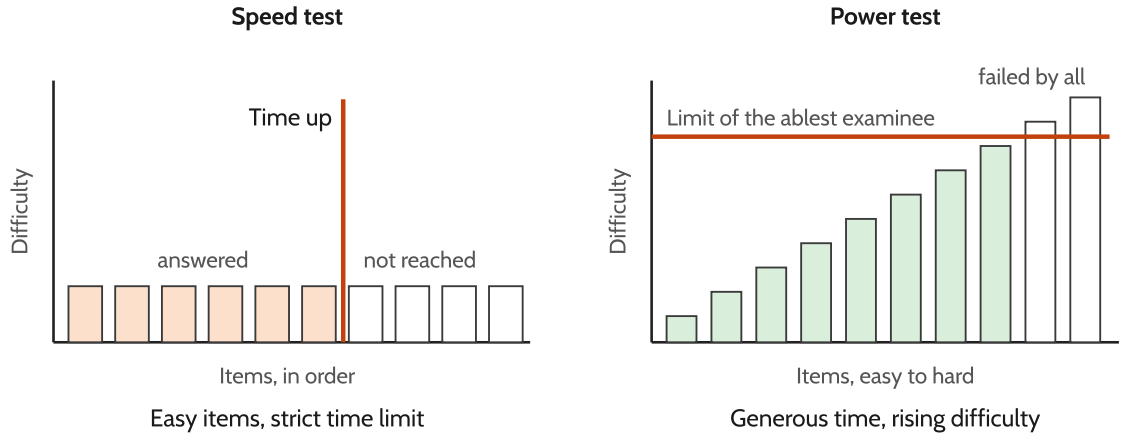


Figure 2.1 Speed versus power. Left: easy items, and time runs out first (white items are never reached). Right: time for everything, but difficulty rises until the white items defeat everyone.

What is asked. Cronbach, in *Essentials of Psychological Testing* (1949), separated tests of **maximum performance**, which ask you to do your best under standard conditions (intelligence, aptitude and achievement tests: the **ability tests**), from tests of **typical performance**, which

ask how you usually feel or act (personality, interest and attitude measures). An IQ test is an ability test.

Purpose. An **intelligence test** measures general mental ability; an **aptitude test**, the potential to learn a specific skill; an **achievement test**, what has already been learned; a **personality test**, traits, emotions, attitudes and motives; an interest inventory, liking for activities and occupations; a **neuropsychological test**, brain-behaviour functions in suspected brain damage; and a diagnostic test, specific weaknesses such as reading errors.

Table 2.2 Ways of classifying tests.

Basis	Types	The difference
Administration	Individual; group	One person, trained examiner (WAIS); many at once (Army Alpha)
Item content	Verbal; non-verbal; non-language	Words in items; pictures, words only in instructions (Raven's); gesture only (Army Beta)
Medium	Paper-and-pencil; performance	Written answers; handling blocks or form boards
Construction	Standardised; teacher-made	Fixed procedures and norms; classroom use, no norms
Reference	Norm-referenced; criterion-referenced	Against other people; against a mastery standard (Section 2.5)
Culture	Culture-fair	Minimal language and schooling (Section 2.6)
Delivery	Computerised; computer-adaptive	On screen; items chosen by earlier answers (Section 2.2)

EXAM TRAP

Speed versus power concerns time limit and item difficulty only, not reaction time and not individual versus group administration; and it is power items that rise in difficulty. "Maximum performance" means an **ability test**: speed and power tests are ability tests differing in timing, and a personality inventory measures typical performance.

Characteristics of a good test ●●

A good test has five characteristics:

- **Objectivity**: freedom from the examiner's judgement in administration, scoring and interpretation, so that different examiners agree.
- **Reliability**: consistency of scores over occasions, forms and scorers (Section 2.3). Consistency is a criterion of good measurement; high cost, low reliability and subjectivity mark poor measurement.
- **Validity**: measuring what the test claims to measure (Section 2.4). Reliability is necessary but not sufficient: head size can be measured consistently, yet it is a worthless intelligence test.
- **Norms**: a representative sample's performance, against which a raw score is read (Section 2.5).
- **Practicability** (usability): reasonable time, length, cost and ease of scoring.

Standardisation means fixing uniform procedures for administration and scoring (the same instructions, materials, time limits and key for everyone) and establishing norms, so that

scores are comparable across people and occasions. It is a precondition of validity, not a guarantee of it; it does not remove cultural bias; and it has nothing to do with computer administration.

> **EXAM TRAP**
>
> RML 2024 keyed **objectivity** as the most important characteristic of a psychological test, following Indian texts such as A. K. Singh's that list it first: without it the score reflects the examiner. Most textbooks, and the 2014 *Standards* ("the most fundamental consideration"), name **validity**, since an objective test of the wrong thing is useless. On RML papers expect objectivity; elsewhere choose validity. Subjectivity and rationality are not test characteristics.

Building, using and misusing tests

A test is built in the eight steps of Figure 2.2, which Sections 2.2 to 2.5 take in turn (A. K. Singh counts seven, folding item analysis into the try-out).

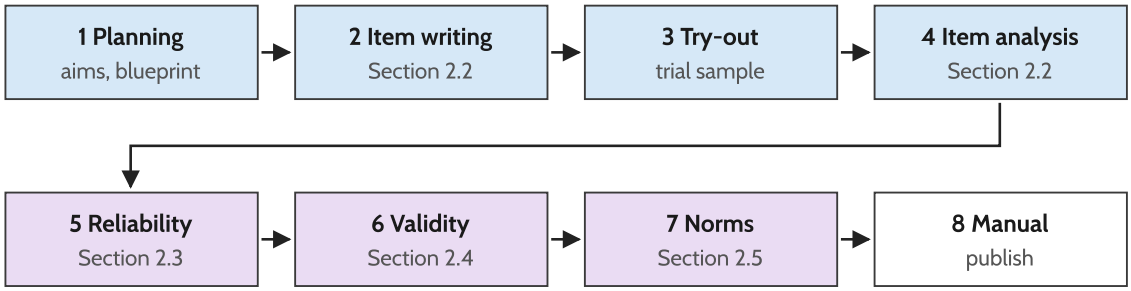


Figure 2.2 The steps of test construction. The top row builds and selects the items; the bottom row checks the scores, and the manual reports it all.

Tests serve selection (choosing applicants), classification and placement (assigning people to categories or programmes), diagnosis and treatment planning, self-knowledge and guidance, programme evaluation and research. Their limitations: they can invade privacy; a label such as "retarded" or "gifted" may stick after the person has changed; they sample only limited aspects of behaviour; test anxiety depresses some scores; and fixed keys penalise creative examinees whose unusual answers earn no credit.

Clinical judgement and its errors

Clinicians treat assessment information as samples, correlates or signs. A **sample** directly resembles the behaviour of interest: how a client tackles a frustrating puzzle samples how she meets frustration. A **correlate** has a statistically demonstrated link with something outside the test, as a high depression-scale score goes with clinicians' ratings of depression. A **sign** is read as an indicator of a hidden state, as when a Rorschach response is taken to signal disordered thinking.

Interpretation can be **nomothetic**, placing the person against norms for everyone (her anxiety is at the 95th percentile), or **idiographic**, tracing what is unique to her (what this worry means in her life); good assessment uses both. In *Clinical versus Statistical Prediction* (1954)

Paul E. Meehl (1920 to 2003) compared **clinical prediction**, combining the data in the clinician's head, with **actuarial prediction**, combining them by a formula derived from past cases. The formula usually equalled or beat the clinician, and later reviews agree.

Judgement also errs systematically. **Confirmatory bias** covers ignoring information that does not support one's hypothesis, ignoring information that does not fit one's stereotypes, and reading ambiguous information as support (confirmation bias in general: Chapter 1). **Illusory correlation** is perceiving a relation that is not there (Key study). **Overconfidence** grows with more data even when accuracy does not, and **neglect of base rates** over-diagnoses rare conditions (Section 2.4). The **Barnum effect** is accepting vague, universally true descriptions as uniquely accurate: in 1949 **Bertram Forer** (1914 to 2000) tested his students, then gave each the same "personal" sketch, largely from an astrology book ("You have a great need for other people to like and admire you"); they rated it 4.26 out of 5 for accuracy. Meehl named the effect in 1956 after the showman P. T. Barnum.

KEY STUDY: ILLUSORY CORRELATION (CHAPMAN AND CHAPMAN, 1967 AND 1969)

Clinicians believed Draw-a-Person signs, such as large eyes meaning suspiciousness, that research did not support. **Loren Chapman** and **Jean Chapman** showed naive students drawings, each paired at random with a symptom statement. The students "found" the very relations clinicians reported, though none existed, because the pairs were already linked in their minds (eyes go with suspicion). In 1969 they found the same for Rorschach signs: popular invalid signs were seen and valid ones missed.

The safeguards are **multimethod assessment** (tests, interview, observation, records), multiple informants (client, family, teachers), actuarial aids and a deliberate search for disconfirming evidence. Assessment ends in a written report (referral question, background, tests, behaviour, results, interpretation, recommendations) and feedback the client can understand. Interviewing, the mental status examination and behavioural assessment are in Chapter 7.

EXAM TRAP

All three statements describe confirmatory bias: answer "All of the above". Assessment information is conceived as **samples, correlates or signs**; the other triads offered (factual, hypothetical or imaginary; predictions, estimates or suggestions; examples, prototypes or categories) are invented.

Ethics and responsible test use

Tests can harm as well as help. The rules are the *Standards for Educational and Psychological Testing* of the American Educational Research Association, American Psychological Association and National Council on Measurement in Education (AERA, APA and NCME; 2014 edition), Section 9 (Assessment) of the APA Ethics Code and, in India, the code of the Rehabilitation Council of India (RCI), where clinical psychologists register (ethics and law in full: Chapter 7). The main duties:

- **Competence.** Use only tests you are trained for. Publishers sell by **test-user qualification levels**: Level A (achievement and proficiency tests) needs only the manual; Level B (group intelligence, aptitude and interest tests) needs training in test construction and

statistics; Level C (individual intelligence tests, clinical personality inventories, projective techniques) needs an advanced degree and supervised experience.

- **Informed consent.** Explain purpose, nature, fees, who sees results and the limits of confidentiality, unless law or a court orders the testing.
- **Confidentiality and release of test data.** Results go only to the client, those the client authorises, or as law requires; raw data may be withheld to prevent harm or misuse.
- **Test security.** Items, manuals and protocols are kept from the public; a leaked test measures nothing.
- **Current editions and relevant norms.** Obsolete tests and outdated or foreign norms mislead.
- **Fairness.** A test is **biased** when its scores mean different things for different groups (Section 2.4).
- **Disability. Accommodations** (Braille or large print, a sign-language interpreter, extra time, a scribe) remove barriers unrelated to what is measured; the report records them.
- **Translation.** A translated test is **back-translated** into the original language by an independent translator and the versions compared; reliability, validity and norms must then be re-established.
- **Explaining results and labelling.** Clients are owed an understandable explanation; labels are used sparingly, because they stigmatise and can fulfil themselves.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Anastasi called a test an objective, standardised measure of a _____ of behaviour.
2. The term "mental tests" was coined in 1890 by _____.
3. Easy items under a time limit so strict that few finish make a _____ test.
4. Intelligence, aptitude and achievement tests measure _____ performance.
5. Uniform administration and scoring procedures plus norms define _____.
6. Clinicians treat assessment information as samples, correlates or _____.
7. Accepting a vague, universal description as uniquely one's own is the _____ effect.

Answers: (1) sample, (2) James McKeen Cattell, (3) speed, (4) maximum, (5) standardisation, (6) signs, (7) Barnum.

2.2 Building a Test: Items, Item Analysis, IRT and Response Bias

LOWER YIELD · 11 QS

A test begins as a pile of draft items, and most will not survive. The constructor writes far more than needed, tries them out and keeps those of the right difficulty that separate people high and low on the trait. Then come the biases that test-takers and raters add.

Writing items

An **item** is the smallest scorable unit of a test, a question or task with a scoring rule. A good item measures only its trait, is unambiguous, gives no clues, is of suitable difficulty and discriminates.

Essay items (free-response items) tap organisation and evaluation but are scored slowly and subjectively. **Objective items**, once called *new-type* items, are scored alike by anyone. In **supply type** items the examinee produces the answer (short answer, completion); in **selection type** items he chooses it: true-false or alternative response (a 50 percent guessing chance), matching (more responses than stems, to block elimination), rearrangement (put steps in order) and the **multiple-choice item**, with a stem (the question), a key (the correct option) and **distractors**, wrong options that attract those who do not know.

The writing rules make the trait alone decide the answer: one idea per item ("I enjoy parties and sports" defeats someone who likes only one), clear wording, no double negatives, no clues such as a longer key or **specific determiners** (*always* and *never* suggest false), plausible distractors and, on personality scales, **balanced keying** (half the items keyed "agree", half "disagree").

Item formats, scoring and guessing

Psychological tests use the wider formats of Table 2.3.

Table 2.3 Item formats in psychological tests.

Format	The respondent
Dichotomous	Chooses between two answers (true or false)
Polytomous	Chooses among three or more
Likert format	Rates agreement, usually on five points (Section 2.8)
Category format	Rates on numbered categories, often 1 to 10
Visual analogue scale	Marks a point on an unbroken line
Adjective checklist	Ticks the adjectives that apply
Forced choice	Picks between options matched for desirability
Q-sort	Sorts statements from most to least like me (Section 2.9)

Answers are scored with a **stencil key** (overlay key: a card with holes at the correct answers, laid over the sheet) or a **strip key** (the answers in a column laid beside the test's). A **correction for guessing** removes lucky hits: S = R - W / (n - 1), with R right, W wrong (omissions excluded) and n options. A blind guesser on four options gets one right per three wrong, so 60 right and 30 wrong give S = 60 - 10 = 50. The formula assumes, wrongly, that every error is a blind guess.

Item difficulty, ceiling and floor

Item analysis uses tryout data to decide which items to keep, revise or discard, judging each by its difficulty, its discrimination and its distractors.

The **item difficulty index** (p) is the proportion passing the item (120 of 200 gives 0.60), so a high p means an easy item. An item everyone passes or everyone fails tells no one apart. The

variance of a pass-fail item, p × q, peaks at p = 0.50, which gives maximum variance and discrimination. Guessing raises the target: for multiple choice the optimum lies midway between chance and 1.00, about 0.62 for four options, 0.75 for true-false (texts aiming at maximum reliability give 0.74 and 0.85). Most tests keep p between 0.30 and 0.70.

A **ceiling effect** occurs when a test is too easy: most respondents score near the maximum, so it cannot discriminate among high performers or register further gains; scores are negatively skewed (Figure 2.3). A **floor effect** is the reverse: the test is too difficult, scores pile up near the minimum (positively skewed) and low performers cannot be told apart. Both shrink the spread of scores, lowering reliability and validity.

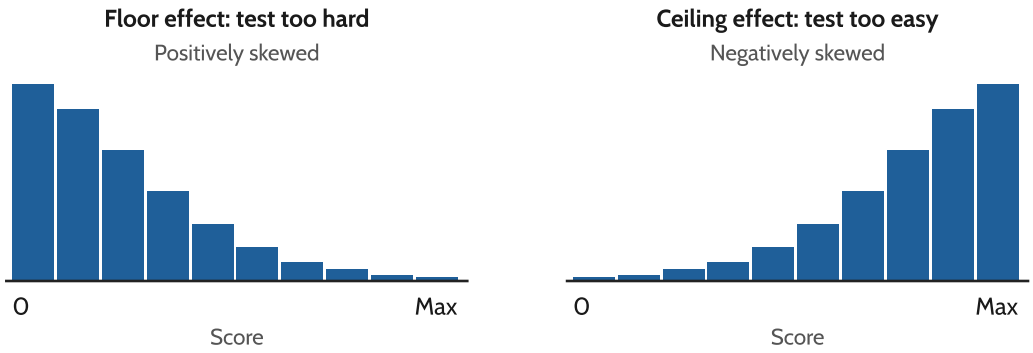


Figure 2.3 Floor and ceiling effects: scores pile up at the minimum (test too hard) or the maximum (too easy), and the piled-up people cannot be told apart.

EXAM TRAP

"The test is too difficult for everyone" is a **floor** effect. A ceiling effect does not mean "scores are normally distributed" (they are negatively skewed) or "perfect reliability": no reliability coefficient exceeds 1, and bunched scores lower it.

Item discrimination and distractors

The **item discrimination index** (D) asks whether high scorers on the whole test did better on this item. Take the top and bottom 27 percent by total score, the split that **Truman Lee Kelley** (1884 to 1961) showed in 1939 best balances extremeness against group size; then D = p_{upper} - p_{lower}, ranging from -1 to +1. If 40 of 50 upper and 15 of 50 lower examinees pass, D = 0.80 - 0.30 = 0.50 (and p = 0.55).

Robert L. Ebel's guide: 0.40 and above, very good; 0.30 to 0.39, reasonably good; 0.20 to 0.29, marginal; below 0.20, poor. A **negative D** means the lower group did better: the item is miskeyed, ambiguous (able examinees see a second defensible answer) or otherwise faulty, and must be revised or discarded.

EXAM TRAP

D = -0.20 does not mean "well-functioning", "too easy" or "at ideal difficulty": difficulty is p, not D. A negative D says the item works backwards; check the key first.

Discrimination can also be an **item-total correlation**: point-biserial r (pass-fail item against a continuous total), biserial r (the pass-fail split assumed to cut a normal ability), phi (two true dichotomies) or tetrachoric r (two artificial dichotomies). Since the item is part of the total, the correlation is inflated, most in short tests; the corrected form uses the other items. The item's correlation with an outside criterion, times its standard deviation, is the **item validity index**.

Distractor analysis checks each option: a working distractor draws more of the lower group; a non-functioning one draws almost nobody; one drawing the upper group signals ambiguity or miskeying. Plausible distractors make an item harder and usually more discriminating. Difficulty caps discrimination: at p = 1.00 or 0, D must be 0, and D can approach 1 only near p = 0.50, though an ambiguous item can sit at p = 0.50 with D near zero. These indices assume a **power test**; in a **speed test** the last items are reached only by fast workers, so their p and D reflect speed, not content.

Item response theory ●●

All this is **classical test theory** (CTT; X = T + E, Section 2.3), whose p and D depend on the sample: an item with p = 0.90 among postgraduates may have p = 0.30 in Class 8. **Item response theory** (IRT), or latent trait theory, models each item instead. It assumes an unobservable **latent trait**, θ (theta), and expresses the probability of a correct (or keyed) response as a function of the person's θ and the item's parameters.

Graphed, this is the **item characteristic curve** (ICC): the probability of a correct response plotted against the examinee's level on the latent trait, not against time taken, number of examinees or reliability. It is an S-shaped ogive (Figure 2.4). Its location on θ is the difficulty, **b** (where the probability reaches 0.50, or halfway between c and 1); its slope there is the discrimination, **a**; its lower asymptote is the pseudo-guessing parameter, **c**, the chance that a very low-θ examinee still succeeds.

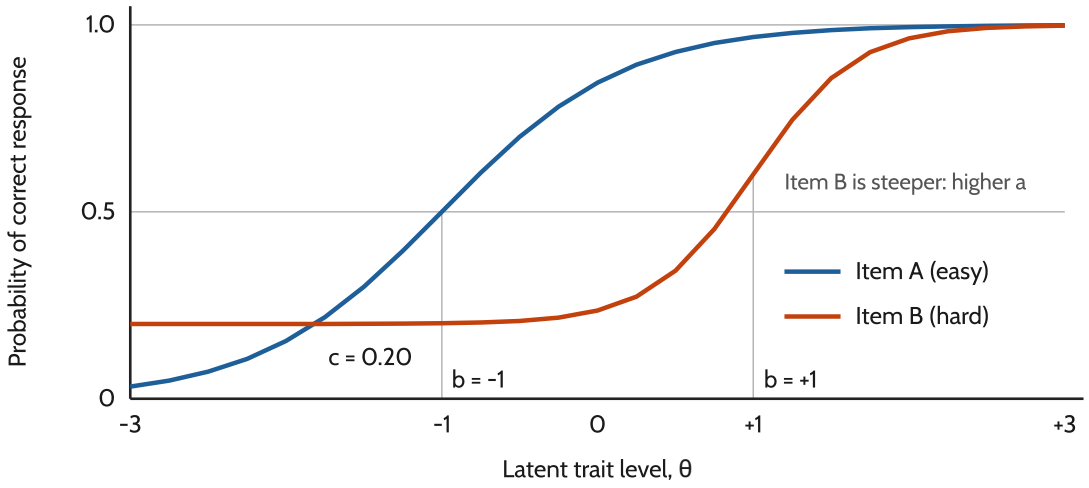


Figure 2.4 Item A (solid) lies left, so it is easy (b = -1). Item B (dashed) lies right (b = +1), rises more steeply (higher a) and never drops below its guessing floor, c = 0.20.

The **Rasch model**, or one-parameter logistic model (1PL), published in 1960 by the Danish mathematician **Georg Rasch** (1901 to 1980), estimates difficulty only. The two-parameter model adds discrimination and the three-parameter model adds guessing; **Allan Birnbaum** set out these logistic models in 1968, building on the work of **Frederic M. Lord** (1912 to 2000) from 1952. IRT assumes **unidimensionality** (one trait) and **local independence** (at a fixed θ, responses to different items are unrelated). An **item information function** shows how precisely an item measures at each θ, peaking near b; summed, they give the test information function.

When the model fits, item parameters are sample-invariant and θ does not depend on which items were given, since persons and items share one scale. This allows **computerised adaptive testing** (CAT), which picks each next item to suit the current estimate of θ and so needs far fewer items; item banking; test equating across forms; and detection of **differential item functioning** (DIF), or item bias, when equally able people from two groups have different ICCs for an item (Table 2.4).

Table 2.4 Classical test theory and item response theory.

Feature	Classical test theory	Item response theory
Model	X = T + E	Probability of a response given θ and item parameters
Level	Whole test	Each item
Item statistics	p and D, sample-dependent	b, a, c, sample-invariant
Person score	Number right, tied to the items given	θ, on the item difficulty scale
Precision	One standard error of measurement for all	Varies with θ

> **EXAM TRAP**
>
> IRT does not treat item statistics as sample-dependent (CTT does), is not limited to one parameter (only the Rasch model is), estimates item difficulty and a person parameter (θ), and makes adaptive testing possible. Group means and variances, test-retest stability, internal consistency and inter-rater agreement belong to classical reliability (Section 2.3); validity coefficients pooled across studies are validity generalisation (Section 2.4).

Response sets and response styles

Even a good item can be answered for the wrong reason. **Lee Joseph Cronbach** (1916 to 2001) showed in 1946 and 1950 that habitual ways of answering, **response sets**, shift scores regardless of content. A response set now usually means a temporary, situational tendency and a **response style** a stable, trait-like one; **Douglas N. Jackson** (1929 to 2004) and **Samuel Messick** (1931 to 1998) argued in 1958 that much variance on inventories reflects style, not content (Table 2.5).

Table 2.5 Response sets and their controls.

Set or style	The respondent	Control
Acquiescence (yea-saying), nay-saying	Agrees, or disagrees, regardless of content	Balanced keying
Social desirability, faking good	Gives answers that look good	Forced choice; L and K scales
Faking bad, malingering	Exaggerates problems	F scale

Set or style	The respondent	Control
Deviation	Gives atypical answers	Read as a sign in itself
Extreme or central-tendency responding	Uses only the ends, or only the middle	Clear anchors; no midpoint
Evasiveness, cautiousness	Marks "?" or omits items	Fewer "?" options; instructions
Guessing, working speedily	Guesses; rushes long tests	Correction formula; ample time

Acquiescence is agreeing regardless of content, so a yea-sayer scores high on an all-"agree" scale without having the trait; nay-saying is the opposite. **Social desirability** is choosing the answer that presents oneself well, often without intent; **Allen L. Edwards** showed, in work summed up in his 1957 book, that it strongly predicts which items are endorsed. Its deliberate forms are faking good and faking bad, and faking bad for external gain is malingering. **Irwin A. Berg**'s **deviation hypothesis** holds that atypical responding is general: people who answer deviantly even where content does not matter also deviate in behaviour.

A consistent style can raise reliability yet lower validity. Reverse-scored items make a yea-sayer's agreements cancel out; the Edwards Personal Preference Schedule pairs statements matched for desirability; the MMPI's L (lie), F (infrequency) and K (correction) scales are covered in Section 2.9; subtle items, rapport and honest-answering instructions also help.

> EXAM TRAP
>
> Agreeing *regardless of content* is **acquiescence**. Social desirability depends on content; central tendency is choosing the middle; halo is a rater's error, not a test-taker's.

Rating scales and rater errors ●●

When another person judges the trait, bias moves to the rater. Rating scales may be numerical (numbered points), graphic (a line to mark), descriptive (a phrase per level), percentage (a percentile band), standard or man-to-man (known people as anchors), cumulated points or checklist (ticked statements summed) or forced-choice (equally favourable statements, curbing leniency). The **behaviourally anchored rating scale** (BARS) of **Patricia Cain Smith** and **Lorne M. Kendall** (1963) anchors each point with a concrete behaviour. Ranking orders people; paired comparison compares every pair.

The **halo effect** (halo error), named by **Edward Lee Thorndike** (1874 to 1949) in 1920 from officers' ratings of soldiers, is judging a person on the basis of one attribute or a general impression that colours every other rating: a clinician who sees anxiety improve rates depression improved too. Table 2.6 lists the other errors, reduced by rater training, behavioural anchors, rating one trait across all persons at a time and pooling raters.

Table 2.6 Rater errors.

Error	The rater
Leniency (generosity)	Rates everyone too favourably
Severity	Rates everyone too harshly
Central tendency	Avoids both ends; rates everyone average
Logical error	Rates traits alike because he believes they go together

Error	The rater
Contrast error	Rates others as opposite to himself, or against the previous ratee
Proximity error	Rates traits placed close together on the form alike
Similarity ("similar to me")	Favours people who resemble himself

EXAM TRAP

Anxiety improvement spuriously improving depression ratings is **halo**, not a **ceiling effect** (a test too easy to show gains), the **Hawthorne effect** (people change because they are studied; Chapter 9) or a **therapeutic effect** (real change from treatment): the rating improved, not the depression.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The proportion of examinees passing an item is its _____ index.
2. Kelley's extreme groups are the top and bottom _____ percent.
3. When the lower group outscores the upper group, D is _____.
4. A test so easy that most score near the maximum shows a _____ effect.
5. The ICC plots the probability of a correct response against the _____.
6. The one-parameter logistic model is the _____ model.
7. Agreeing with items regardless of content is _____.
8. Judging a person on one attribute is the _____ effect.

Answers: (1) difficulty, (2) 27, (3) negative, (4) ceiling, (5) latent trait (θ), (6) Rasch, (7) acquiescence, (8) halo.

2.3 Reliability: Consistency and the Standard Error of Measurement

HIGH YIELD · 64 QS

Step on a bathroom scale three times in a minute, read 61, 64 and 58 kilograms, and you stop trusting it. A psychological test is a scale for something invisible, so the first question about any score is the same: measured again, would it come out the same? What follows separates score from error, estimates the error four ways, and puts an honest band around one person's score.

True scores, error and the meaning of reliability ●●●

In 1904 **Charles Spearman** proposed the model on which **classical test theory** rests: every observed score is a true score plus error, **X = T + E**. The **true score** is the average a person would earn over endless independent testings. The model assumes that errors are random, averaging zero; that they are uncorrelated with true scores; and that errors on one test are uncorrelated with errors on another. The variances therefore add: observed variance = true variance + error variance.

The **reliability coefficient** (r_{xx}) is the share of observed-score variance that is true-score variance: $r_{xx} = \sigma^2_T / \sigma^2_X = 1 - \sigma^2_E / \sigma^2_X$. If scores vary with variance 100, of which 80 is true and 20 error, reliability is .80. It runs from 0 (all error) to 1 (no error). The **domain-sampling model** sees a test's items as a sample from a large domain, such as all the words a spelling test could use; the score on the whole domain is the universe score. The more items sampled, the closer the test score comes to it, so longer tests are more reliable.

Only **random error** (a lapse of attention, a lucky guess, a noisy hall), which moves scores unpredictably, lowers reliability. A **systematic error** moves every score the same way, like a scale that always reads two kilograms heavy; being consistent, it leaves reliability intact but biases the scores and so threatens validity. Reliability therefore means **consistency**, **dependability** and **freedom from measurement error**. It is distinct from validity (measuring what the test claims) and sensitivity (detecting a condition or a change), both in Section 2.4, from generalizability (findings extending to other people and settings), and from fairness to groups (Section 2.1). A **standardised** test has uniform administration and scoring (Section 2.1); a **normed** test is read against a reference group (Section 2.5).

> EXAM TRAP
>
> Reliability is "freedom from measurement error, consistency and dependability": **all of the above**. A study is reliable when **its measures are stable on different occasions**. "Conducted by a reputable researcher" is trust in a person, "findings can be generalised to other settings" is external validity, and "methods stated clearly enough to be repeated" is replicability (Chapter 9).

Stability and equivalence: test-retest and parallel forms ●●●

In **test-retest reliability** the same test is given to the same people on two occasions and the scores are correlated, giving the **coefficient of stability**. Giving a new anxiety scale to students twice, two weeks apart, and correlating the scores is the classic case. It detects time-sampling error: changes in health, mood and conditions between sittings. Too short an interval lets memory and practice inflate the coefficient; too long lets real change lower it, so a fortnight to a few weeks is usual. The method suits traits expected to stay stable over the interval, such as intelligence and personality, not rapidly changing states such as current mood, where a low coefficient reflects true change.

> EXAM TRAP
>
> Test-retest reliability means the test **gives consistent results over time**. Correlating two halves is split-half reliability; two raters scoring one protocol is inter-rater reliability; correlating with a future criterion, or "measuring what it claims", is validity (Section 2.4); the client doing better the second time is a **practice effect**. It fits stable traits, not mood states, single-item measures or open-ended interviews.

A second method, **alternate-forms reliability** (**parallel-forms reliability**), uses two versions built to one blueprint: the same number, kind, content and format of items, equal difficulty, and equal means and variances. Given in one session, their correlation is a **coefficient of equivalence**, which detects content-sampling error; given weeks apart, it is a **coefficient of stability and equivalence**, the most demanding estimate. Parallel forms avoid remem-

bered answers and suit retesting and speed tests, but they are costly to write, rarely truly equivalent, and practice on the type of task still carries over.

Split-half reliability and the Spearman-Brown formula ●●●

Can reliability come from one sitting? Methods that do so estimate **internal consistency reliability**: how well the parts of a test agree with each other within one administration. In **split-half reliability** one test is scored as two halves, which are then correlated. First half against second half is a poor split, since warming up, fatigue and rising difficulty make the halves unequal; an odd-even split, or random halves matched for content and difficulty, is better. The major challenge is dividing the test into two nearly equivalent halves, and different splits give different answers.

The half correlation describes a test half as long, so it underestimates the whole. Spearman and **William Brown** published the correction independently in 1910. The **Spearman-Brown prophecy formula** steps the half correlation up: $r_{full} = 2r_{half} / (1 + r_{half})$, so halves correlating .60 give 1.20 / 1.60 = **.75**. Its general form predicts reliability when length changes by a factor k (new length over old): $\mathbf{r_{new} = kr / [1 + (k - 1)r]}$. Adding parallel items (k above 1) raises reliability, because true-score parts add up while random errors partly cancel; shortening (k below 1) lowers it. Rearranged, the length needed for a target is $k = r_{target}(1 - r) / [r(1 - r_{target})]$. Table 2.7 works four cases, and Figure 2.5 shows the diminishing returns.

Table 2.7 The Spearman-Brown formula, worked four ways.

Question	k	Working	Answer
Halves correlate .60; whole test?	2	1.20 / 1.60	.75
r = .50; test made three times as long?	3	1.50 / (1 + 2 × .50)	.75
r = .80; test cut to half length?	0.5	.40 / (1 - .5 × .80)	.67
r = .60; how much longer to reach .90?	?	(.90 × .40) / (.60 × .10)	6 times

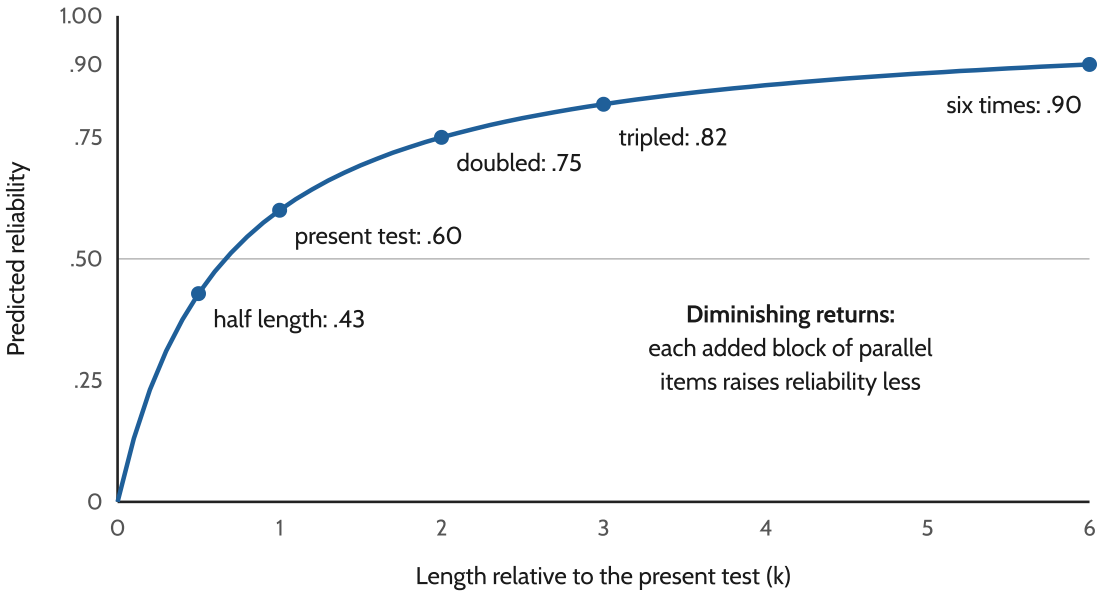


Figure 2.5 Spearman-Brown predictions for a test whose present reliability is .60. Doubling gains .15, tripling only .07 more, and .90 needs six times the items.

Spearman-Brown assumes parallel halves, with equal means and variances; when the half variances differ it overestimates. Two formulas need no step-up. **Phillip Rulon** (1939) treated the difference between a person's two half-scores as error: **Rulon's formula** is $r = 1 - \sigma^2_d / \sigma^2_X$, the variance of the differences over the variance of totals. **John Flanagan** (1937) used the half variances: $r = 2[1 - (\sigma^2_a + \sigma^2_b) / \sigma^2_X]$ (**Louis Guttman** gave the same formula in 1945). If each half has variance 6, totals 20 and differences 4, Rulon gives 1 - 4 / 20 = .80 and Flanagan 2(1 - 12 / 20) = .80; the halves correlate .67, which Spearman-Brown also steps up to .80.

> EXAM TRAP
>
> Spearman-Brown **estimates full-test reliability from a split-half correlation, or predicts reliability when length changes**; it raises reliability by adding parallel items, not by removing items, making them ambiguous or shrinking the sample. It does not estimate rater agreement, correct validity for unreliability, estimate a speed test's reliability from one sitting, or predict validity, effect sizes or factor structure. Rulon needs **variances**: SDs of 3 (differences) and 6 (totals) give 1 - 9 / 36 = .75.

Odd-even splitting suits a **power test**: graded items, generous time. On a pure **speed test** of easy items the score is how far a person gets, so whoever reaches item 60 has about 30 odd and 30 even right. The halves match by construction and the correlation nears 1.00 spuriously, as do KR-20 and alpha from one timed sitting. Speed tests need test-retest, parallel forms or **separately timed halves**, each given with half the time and stepped up with Spearman-Brown. Projective and many neuropsychological tests lack comparable, objectively scored items to split. Every internal consistency method shares one limitation: a single administration says nothing about stability over time.

EXAM TRAP

The limitation of internal consistency methods: they **measure consistency within a single administration only**. Needing several items and falling with mixed content are conditions of use, so a mock offering "all of the above" still keys the single administration.

Kuder-Richardson formulas and coefficient alpha ●●●

Why trust one arbitrary split? In 1937 **G. Frederic Kuder** and **Marion Richardson** derived an internal consistency coefficient from item statistics. **KR-20** suits items scored dichotomously (right or wrong, yes or no) that may vary in difficulty: **KR-20 = [k / (k - 1)] × [1 - Σpq / σ^2_X]**, where k is the number of items, p the proportion passing an item, q = 1 - p, and σ^2_X the variance of total scores.

The product pq is the item's variance: an item passed by 30 percent has pq = .30 × .70 = .21, and the maximum is .25 at p = .50. Items measuring one thing covary, so total variance far exceeds the summed item variances. For 20 items with Σpq = 3.2, mean 12 and variance 16: 3.2 / 16 = .20, 1 - .20 = .80, and × 20 / 19 gives **.84**. **KR-21** assumes equal item difficulty, replacing Σpq with M(k - M) / k, so it needs only k, mean and variance: KR-21 = [k / (k - 1)] × [1 - M(k - M) / (kσ^2)]. For the same test, 12 × 8 = 96, 96 / (20 × 16) = .30, 1 - .30 = .70, and × 20 / 19 gives **.74**. When item difficulties differ, KR-21 is lower than KR-20: a quick, conservative estimate.

Lee Cronbach (1951) generalised KR-20 to items with several score points, such as Likert ratings. The resulting **coefficient alpha** is α = [k / (k - 1)] × [1 - Σσ^2_i / σ^2_X]: five Likert items with variances summing to 6 and total variance 20 give 1.25 × .70 = **.875**. For 0 or 1 items σ^2_i = pq, so alpha equals KR-20. Cronbach showed that alpha is the mean of all possible split-half coefficients (each by Rulon's or Flanagan's formula); averaging over every split increases the accuracy of the estimate. **Cyril Hoyt** (1941) reached the same value by analysis of variance of persons by items: r = ($MS_{persons}$ - $MS_{residual}$) / $MS_{persons}$.

For alpha, as for any reliability coefficient, about .70 is acceptable for research and group comparisons, decisions about an individual patient need .80 to .90 or more, and .40 is poor: the items do not hang together. Alpha rises with test length, very high values can mean redundant items, and a high alpha does not prove that a test measures a single trait.

EXAM TRAP

Cronbach's alpha measures **internal consistency reliability**: how well a scale's items hang together. It is not validity of any kind (content, construct, concurrent, predictive or criterion), not test-retest, parallel-forms or inter-rater reliability, and not a variance or regression statistic. KR-20 differs from alpha only in being limited to dichotomous items, whatever their content.

Agreement between scorers ●●

Scorer or **inter-rater reliability** is the degree to which two or more clinicians agree when independently scoring or interpreting the same test responses. It matters wherever judgement enters, since scorer disagreement becomes error in the score: projective protocols, essays, behaviour observation and psychiatric diagnosis. Agreement of two tests on one construct is convergent validity, predicting future behaviour is predictive validity, and items relating to each other is internal consistency.

Simple percentage agreement flatters raters, since some agreement is chance. **Jacob Cohen** (1960) corrected for it with **Cohen's kappa** for categorical judgements: $\kappa = (P_o - P_e) / (1 - P_e)$. Two clinicians classify 100 patients: both say depressed for 40 and not depressed for 30, so observed agreement $P_o = .70$. Clinician A calls 50 depressed and B calls 60, so chance agreement $P_e = (.50 \times .60) + (.50 \times .40) = .50$, and $\kappa = .20 / .50 =$ **.40**. Seventy percent raw agreement is only fair beyond chance. **J. Richard Landis** and **Gary Koch** (1977) labelled kappa below 0 poor, .00 to .20 slight, .21 to .40 fair, .41 to .60 moderate, .61 to .80 substantial and .81 to 1.00 almost perfect, so .85 is almost perfect. Table 2.8 lists related statistics.

Table 2.8 Statistics of agreement between raters.

Statistic	Ratings	Raters	Note
Percentage agreement	Categories	Two	Inflated by chance
Cohen's kappa (1960)	Categories	Two	Corrected for chance
Weighted kappa (Cohen, 1968)	Ordered categories	Two	Near-misses earn partial credit
Fleiss' kappa (1971)	Categories	Three or more	After **Joseph Fleiss**
Intraclass correlation	Continuous scores	Two or more	Share of variance from true differences
Kendall's W	Rankings	Several	**Maurice Kendall**'s concordance, 0 to 1

> EXAM TRAP
>
> Cohen's kappa measures **inter-rater agreement on categorical data, corrected for chance**. It is not an effect size for ANOVA (that is eta squared, Chapter 9), and not internal consistency, test-retest reliability, or convergent, criterion or predictive validity.

The standard error of measurement ●●●

A reliability coefficient describes a group. A psychologist facing Ravi, whose IQ came out at 110, needs to know how far his true score might lie from 110. The **standard error of measurement** (SEM), the standard deviation of the errors one person's scores would show over many retestings, answers that: $\mathbf{SEM = SD \times \sqrt{(1 - r_{xx})}}$.

With SD 15 and reliability .91, SEM = 15 × √.09 = 15 × .30 = **4.5**; with reliability .80, SEM = 15 × √.20 = about **6.7**. At r = 1 the SEM is 0; at r = 0 it equals the SD. The true score lies within ±1 SEM of the observed score about 68 percent of the time and within ±1.96 SEM 95 percent of the time, so Ravi's true IQ is probably 105.5 to 114.5 (68 percent) or 101.2 to 118.8 (95 percent), as Figure 2.6 shows.

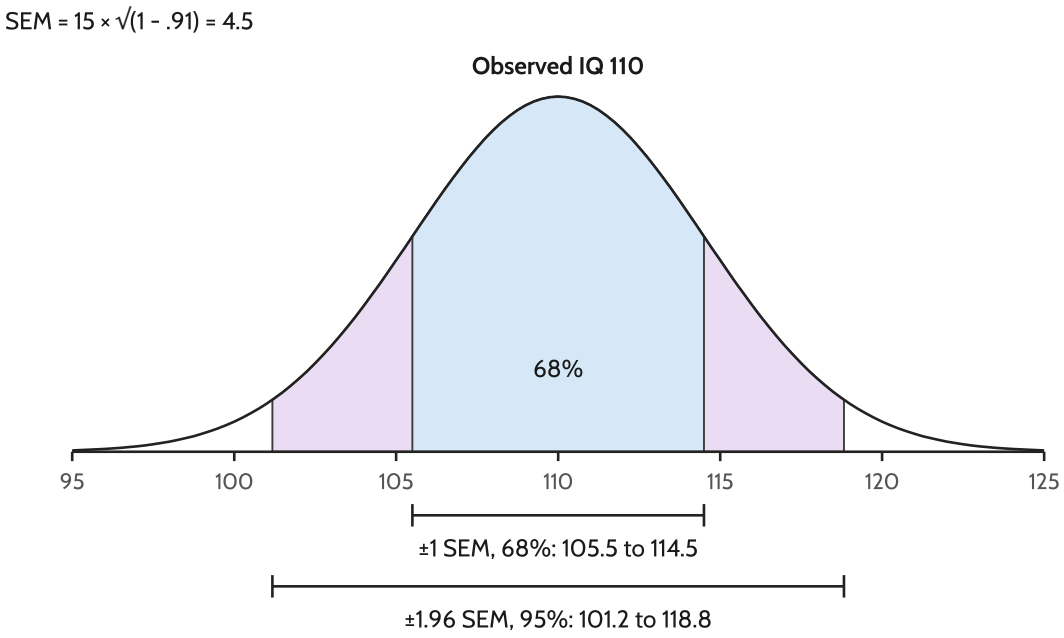


Figure 2.6 Confidence bands around an observed IQ of 110 (SD 15, reliability .91). The narrow band holds the true score about two times in three; the wide band, 19 times in 20.

EXAM TRAP

SEM = **SD × √(1 - r)**, used to **estimate the range within which a person's true score falls.** Wrong options: SD × √r, SD × r (15 × .91 = 13.65), SD × (1 - r) without the root (1.35), √(1 - r) alone (.30), SD / r, SD × (1 + r), reliability × validity, mean minus SD. It does not correlate two tests, index item difficulty (Section 2.2) or predict performance.

CLINICAL LINK

DSM-5 places the IQ criterion for intellectual disability at about 70 ± 5 (65 to 75), a margin for measurement error. Report bands, never a single number treated as exact.

Comparing two scores needs the **standard error of the difference**, $\sqrt{(SEM_1^2 + SEM_2^2)}$. For two index scores (SD 15) with reliabilities .91 and .84, the SEMs are 4.5 and 6.0 and the standard error of the difference is √56.25 = 7.5, so the scores must differ by about 1.96 × 7.5 ≈ 15 points to be trusted at 95 percent. A **difference score** is unreliable when its two tests correlate highly: $r_{dd} = [(r_{xx} + r_{yy}) / 2 - r_{xy}] / (1 - r_{xy})$, so reliabilities .90 and .80 with a correlation of .70 give .15 / .30 = .50, because the shared true score cancels and the errors remain. A **composite score** from several positively correlated tests is usually more reliable than its parts.

The **index of reliability**, $\sqrt{r_{xx}}$, is the correlation between observed and true scores (√.81 = .90) and caps validity (Section 2.4). The **estimated true score**, $M + r_{xx}(X - M)$, regresses towards the mean: an IQ of 130 with reliability .90 gives 100 + .90 × 30 = 127, because extreme scores contain some luck. Spearman's (1904) **correction for attenuation**, $r_{xy} / \sqrt{(r_{xx}\, r_{yy})}$, estimates the correlation between perfectly reliable measures: a predictor and criterion correlating .36, with reliabilities .81 and .64, give .36 / (.90 × .80) = .36 / .72 = **.50** (Section 2.4). Divide by the root; multiplying (.26) or dropping the root (.69) are the usual slips.

What raises or lowers reliability

Reliability belongs to scores from a particular group under particular conditions, not to a test for all time.

- **Length.** More parallel items raise it, as Spearman-Brown predicts.
- **Group heterogeneity.** A wide range of ability raises r; restriction of range (testing only admitted students) lowers it.
- **Item difficulty and discrimination.** Items near p = .50 with high item-total correlations add most; items everyone passes or fails add nothing (Section 2.2).
- **Guessing.** It adds random error, most on true-false items.
- **Speededness and interval.** Speed inflates single-session estimates; a longer retest interval lowers test-retest coefficients.
- **Scoring and conditions.** Objective keys raise reliability; subjective scoring, ambiguous items, noise, fatigue and low motivation lower it.

Cronbach and **Goldine Gleser** (1965) applied the **bandwidth-fidelity dilemma** from information theory to testing: a test covering many traits briefly (wide bandwidth) measures each with low precision (fidelity), while a narrow test measures one trait precisely but says little else.

Beyond classical theory

Classical theory lumps all error into one E; **generalizability theory**, from Cronbach, Gleser, **Harinder Nanda** and **Nageswari Rajaratnam** (1972), uses analysis of variance to split observed-score variance into persons and facets such as raters, items and occasions. A **G study** estimates these variance components; a **D study** (decision study) forecasts dependability under other designs, such as two raters instead of one. The result is a **generalizability coefficient**, of which classical reliability is a special case. Item response theory, which models each item's response curve, replaces one reliability for everyone with **test information**, which varies along the trait (Section 2.2). Table 2.9 gathers the methods.

Table 2.9 The methods of estimating reliability.

Method	Error it detects	Statistic	Best use
Test-retest	Time sampling	r (stability)	Stable traits; speed tests
Parallel forms	Content; time if delayed	r (equivalence)	Retesting; speed tests
Split-half	Content sampling	Spearman-Brown; Rulon; Flanagan	Power tests
KR-20, KR-21, alpha	Content sampling, heterogeneity	Internal consistency	Homogeneous scales
Inter-rater	Scorer differences	Kappa, ICC, W	Projectives, diagnosis
Generalizability	Several facets at once	G coefficient	Designing measurement

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Reliability is the ratio of _____ variance to observed-score variance.
2. The same test given twice to the same people yields the coefficient of _____.
3. Halves correlating .60 give a full-test reliability of _____ by Spearman-Brown.
4. An odd-even split from one timed sitting is least appropriate for a _____ test.
5. Cronbach's alpha measures _____ reliability.
6. KR-21 assumes all items are of equal _____.
7. Rater agreement on categories, corrected for chance, is Cohen's _____.
8. With SD 15 and reliability .91, the SEM is _____.

Answers: (1) true-score, (2) stability, (3) .75, (4) speed, (5) internal consistency, (6) difficulty, (7) kappa, (8) 4.5.

2.4 Validity and the Accuracy of Test Decisions MEDIUM YIELD · 32 QS

A bathroom scale that always reads two kilograms light is perfectly consistent and consistently wrong. Section 2.3 asked whether a test measures consistently; this section asks whether it measures the right thing, and then follows its scores into the decisions they drive.

What validity means

The classic definition, associated with **Henry Edward Garrett** (1894 to 1973), is that **validity** is the extent to which a test measures what it claims (purports) to measure. Validity is specific to a purpose and a population (a test valid in one culture need not be valid in another), it is a matter of degree, and every new use needs new evidence.

The American Psychological Association's *Technical Recommendations* of 1954 named content, predictive, concurrent and construct validity, and the 1966 *Standards* merged predictive and concurrent into criterion-related validity, giving the classical trinity of content, criterion-related and construct validity. In 1989 **Samuel Messick** replaced the trinity with a **unified view of validity**: one integrated judgement of how well evidence and theory support the interpretations and uses of scores, so validity belongs to an inference drawn from scores, not to the test itself. The 2014 *Standards for Educational and Psychological Testing* list five sources of evidence: test content, response processes, internal structure, relations with other variables, and the consequences of testing.

Face validity and content validity ●●

Face validity is the appearance of validity: on superficial inspection a test seems to measure what it claims, because its questions intuitively seem relevant, as when a depression questionnaire asks "I feel sad most of the day". It is judged by test takers and lay people, not by experts or by any statistic, so it is not validity in the technical sense. It still earns rapport and acceptance, but an obvious purpose makes faking good or faking bad easy, which is why some inventories use subtle items (Section 2.9).

Content validity is how adequately the items sample the whole domain: a depression scale should cover mood, interest, sleep, appetite, energy, guilt, concentration and suicidal think-

ing, not sadness alone. The developer drafts a **table of specifications** (test blueprint), a grid of content areas against objectives that fixes the number of items in each cell, and experts then judge every item. **C. H. Lawshe** quantified their judgement in 1975 with the **content validity ratio**, CVR = $(n_e - N/2) / (N/2)$, where n_e of N experts rate an item essential; it runs from −1 to +1. Item validity means each item belongs to the domain; sampling validity means the items together cover all of it. The two threats are **construct under-representation** (part of the domain left out) and **construct-irrelevant variance** (scores also reflecting something else, as when a mathematics test set in English partly measures reading). Content validity is the main evidence for achievement tests.

> EXAM TRAP
>
> Face validity has nothing to do with faces: "facial expressions used for diagnosis" and "a scale of emotional responding" are invented distractors, and "a hypothetical attribute that may not be directly observable" defines a construct (2.4.5). Face validity is judged by lay people; content validity by experts against a blueprint.

Criterion-related validity: concurrent and predictive ●●●

A criterion is an independent, external measure of the attribute, such as a psychiatrist's diagnosis, a gold-standard scale, grades or job ratings, and **criterion-related validity** is the correlation between test and criterion. Its two types differ only in timing.

In **concurrent validity** test and criterion are measured at about the same time. The classic design gives a new intelligence test together with an established one; likewise, a new 20-item depression questionnaire that correlates r = 0.85 with a gold-standard scale completed the same day shows concurrent validity. A high correlation with an established measure of the same attribute shows that the test reflects present status. In **predictive validity** the criterion is measured later, as when entrance scores are correlated with first-year marks. Selection tests need it, and it usually runs lower than concurrent validity because people change in the interval. A criterion recorded before the test gives postdictive validity.

The correlation is the validity coefficient; its square, the coefficient of determination, is the share of criterion variance the test explains (r = 0.60 explains 36 percent). The standard error of estimate, $SE_{est} = SD_y \sqrt{(1 - r^2)}$, is the error of an individual prediction: with r = 0.60 and a criterion SD of 10 marks it is 8 marks. It is not the standard error of measurement (Section 2.3).

> EXAM TRAP
>
> Concurrent and predictive are the two criterion-related types; face, content, factorial and construct validity are not criterion based. RML 2024 keyed "predictive validity" as the one NOT criterion based: that key is wrong. Of its options, face validity (appearance only) and factorial validity (construct evidence) are not criterion based; face validity is the safer mark. RML 2021 defined concurrent validity as "scores correlate highly with scores from tests that measure the same attribute"; "two tests done at the same time" is only the timing, "two clinicians agree" is inter-rater reliability, and split-half or retest correlations are reliability (Section 2.3).

Criteria, cut-offs and cross-validation ●

A coefficient is no better than its criterion, and finding a good one is the criterion problem. A good criterion is relevant, reliable (an unreliable one caps the coefficient, 2.4.6), free from bias and available at reasonable cost. In **criterion contamination** the criterion is influenced by knowledge of the predictor scores, or includes predictor-related content, so the correlation is inflated artificially: supervisors who know employees' selection-test scores tend to rate high scorers highly. The cure is blind ratings. Criterion deficiency is the opposite flaw, a criterion that omits important parts of performance, and plain random error in the criterion lowers rather than inflates the coefficient.

Restriction of range also lowers validity: grades exist only for students the entrance test admitted, and their narrow band of scores underestimates the test. Items chosen because they correlated with the criterion in one sample capitalise on chance, so the coefficient falls in a fresh sample. Re-checking it there is **cross-validation**, and the fall is shrinkage, largest with small samples and many items tried for few kept.

An expectancy table shows the chance of criterion success at each band of test scores. A cut-off score divides those selected or diagnosed from the rest, and since no test is perfectly valid every cut-off makes some errors (2.4.8). **Incremental validity** (**Lee Sechrest**, 1963) is the prediction a test adds beyond measures already available, such as school marks. Synthetic validity, a term of Lawshe's, infers validity for a job too small to study directly from known validities for its components.

Construct validity ●●

What is the criterion for anxiety? Many attributes have none. A **construct** is a hypothetical or inferred attribute, such as intelligence, anxiety or extraversion, that cannot be directly observed but is used to explain behaviour. **Construct validity**, introduced in the 1954 recommendations and set out by **Lee Joseph Cronbach** (1916 to 2001) and **Paul Everett Meehl** (1920 to 2003) in 1955, is the degree to which a test measures the theoretical construct it claims to measure. They placed each construct in a **nomological network**, the web of lawful relations the theory predicts with other constructs and with behaviour. Validation then tests whether scores behave as the network predicts.

Convergent validity is a high correlation with other measures of the same construct; **discriminant validity** (divergent validity) is a low correlation with measures of theoretically unrelated constructs. A new resilience scale should correlate highly with an established resilience measure and weakly with an unrelated trait. **Donald Thomas Campbell** (1916 to 1996) and **Donald W. Fiske** showed in 1959 how to test both at once.

KEY STUDY: CAMPBELL AND FISKE'S MULTITRAIT-MULTIMETHOD MATRIX (1959)

Does a score reflect the trait, or only the way it was measured? Campbell and Fiske measured two or more traits, each by two or more methods (self-report, peer rating, observation), and intercorrelated everything in a **multitrait-multimethod matrix**. The monotrait-heteromethod values (same trait, different methods) should be high, showing convergence, and higher than the heterotrait-monomethod values (different traits, same method). When different traits correlate highly merely because both were self-reported, the matrix has exposed method variance, which proved common in the published matrices they examined. Criticism: the criteria are judged by eye, without a single statistical test, so confirmatory factor analysis is now used to model the matrix.

Factorial validity is shown when items load on the expected factor in exploratory or confirmatory factor analysis, evidence of internal structure (the technique is in Chapter 9). In the **known-groups method** (contrasted groups) the test separates groups known to differ, such as depressed inpatients and community volunteers. Scores should also change with development as the theory expects and move with experimental interventions (anxiety scores falling after therapy). Internal consistency (Section 2.3) supports a construct claim but cannot establish it alone. **Consequential validity**, Messick's addition, weighs the social consequences of using the scores; **ecological validity** is how well test performance represents everyday functioning, a key question in neuropsychology. The validity of research designs (internal, external, statistical conclusion and the construct validity of a manipulation) belongs to Chapter 9.

EXAM TRAP

Construct validity was keyed as "the degree to which a test measures what it claims to measure" (RML 2025) and as "the level to which the theoretical construct is related to other similar measures" (RML 2021); in the 2021 paper "scores correlate highly with tests of the same attribute" defined concurrent validity. The 2025 wording is Garrett's general definition, which fits because the unified view treats all validity as construct validity. The distractors are reliability (consistency across administrations or among items), content validity (the whole domain) and inter-rater reliability (rater agreement). For discriminant validity choose the *weak* correlation with an unrelated construct; a high correlation with an established measure of the same construct is convergent, "covers all facets" is content and "predicts future coping" is predictive.

Reliability and validity together ●●

A test can be reliable without being valid: a scale with high internal consistency but low criterion validity consistently measures something, but not the intended construct. It cannot be valid without being reliable, because scores that are mostly random error correlate with nothing. Reliability is therefore necessary but not sufficient for validity; consistency does not make a test valid, and validity is more than consistency.

A validity coefficient cannot exceed $\sqrt{(r_{xx} \times r_{yy})}$, the square root of the product of the test and criterion reliabilities. With a perfectly reliable criterion this **validity ceiling** is $\sqrt{r_{xx}}$, the index of reliability: $r_{xx} = 0.81$ allows at most 0.90. Run in reverse, the logic gives the **correction for attenuation** of **Charles Edward Spearman** (1863 to 1945), published in 1904, which estimates the correlation if both measures were perfectly reliable: $r_c = r_{xy} / \sqrt{(r_{xx} \times r_{yy})}$. An observed 0.36 with reliabilities of 0.81 and 0.64 becomes 0.36 / 0.72 = 0.50.

One apparent conflict remains. Very high homogeneity maximises internal consistency but narrows coverage, so a broad criterion is often predicted better by a heterogeneous test with moderate inter-item correlations: the bandwidth-fidelity trade-off of Section 2.3 seen from the validity side.

> **EXAM TRAP**
>
> The TRUE statement (RML 2024) is "a test can be reliable without being valid". "If a test measures what it is intended to measure, then by definition it is reliable" is the near-miss: a valid test must be reliable, but not by definition. Validity need not come "first chronologically". The ceiling is the *square root* of reliability, not reliability itself, its square or reliability minus the SEM.

Sensitivity, specificity and predictive values ●●

Once a test is used to decide (depressed or not, select or reject), validity shows up as right and wrong calls. Crossing the test result with the true state, set by a gold-standard diagnosis, puts every person in one of four cells (Figure 2.7): a true positive, a **false positive** (positive without the condition), a false negative (the condition missed) or a true negative.

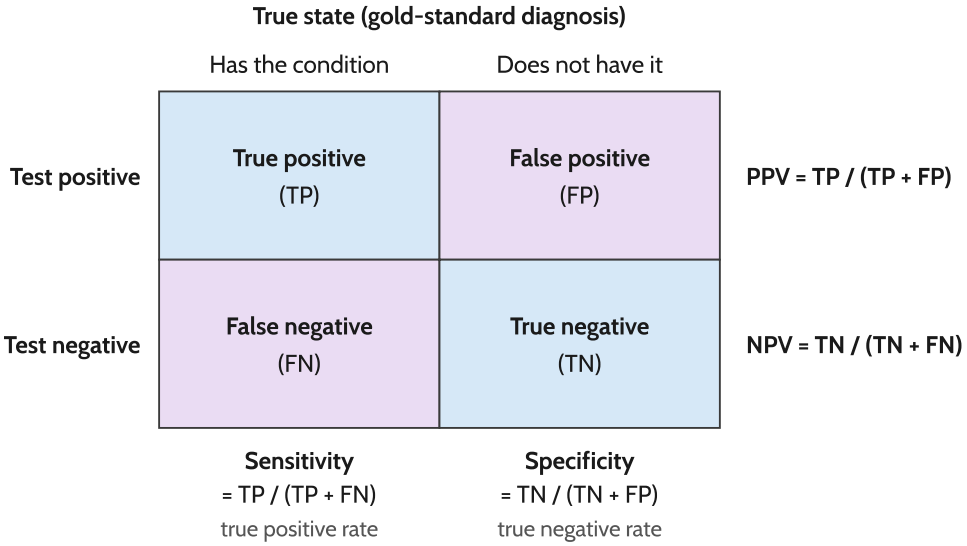


Figure 2.7 The four outcomes of a diagnostic decision. Sensitivity and specificity read down the columns; the predictive values read across the rows. Only the diagonal cells are correct decisions.

Sensitivity = TP / (TP + FN) is the true positive rate: the proportion of people who have the condition whom the test correctly identifies. **Specificity** = TN / (TN + FP) is the proportion of people without the condition whom the test correctly clears. Both describe the test itself at a given cut-off. A screen with sensitivity 90 percent and specificity 70 percent calls 90 percent of depressed people depressed and 70 percent of healthy people healthy. A high false positive rate means normal people are diagnosed as disordered; false negatives leave disordered people undetected. The proportion of all decisions that are correct is the hit rate.

The clinician, though, holds a result, not a diagnosis, and needs the rows. The **positive predictive value**, PPV = TP / (TP + FP), is the probability that a positive result is correct, and the negative predictive value, NPV = TN / (TN + FN), the probability that a negative one is. Both

depend heavily on the **base rate** (prevalence) of the condition in the people tested, as Meehl and **Albert Rosen** showed in 1955 (Table 2.10).

Table 2.10 One screen (sensitivity 90 percent, specificity 70 percent) given to 1000 people in two settings.

Outcome	Clinic: base rate 50 percent	Community: base rate 5 percent
Depressed: caught (TP) / missed (FN)	450 / 50	45 / 5
Healthy: cleared (TN) / flagged (FP)	350 / 150	665 / 285
PPV: positives truly depressed	450 / 600 = 75 percent	45 / 330 = 14 percent
NPV: negatives truly healthy	350 / 400 = 88 percent	665 / 670 = 99 percent
Hit rate: all correct calls	800 / 1000 = 80 percent	710 / 1000 = 71 percent

The test is unchanged, yet in the community six positive results in seven are false alarms, and a clinician who simply called everyone healthy would be right 95 percent of the time, beating the test's 71 percent. A sign is worth using only if it improves on the base rate.

> EXAM TRAP
>
> TP / (TP + FP) is the PPV, not sensitivity. "Probability that a positive test reflects true disease" is the PPV; "probability of disease prior to testing" is the base rate. Sensitivity of 90 percent does not mean 90 percent of positive results are correct. The PPV is most affected by the base rate, not by sensitivity alone, specificity alone or the number of items. An overstrict diagnosis produces false negatives, not false positives.

Cut-offs, ROC curves and selection

Sensitivity and specificity hold only for one cutting score. Lower the cut-off and sensitivity rises while specificity falls; raise it and the reverse happens. The choice weighs the costs of the two errors: missing a suicidal patient is worse than a false alarm, so screens use lenient cut-offs. The **receiver operating characteristic curve** (ROC curve), borrowed from signal detection theory, plots sensitivity (true positive rate) against 1 – specificity (false positive rate), one point for each possible cut-off (Figure 2.8). A useless test follows the diagonal; a better one bows toward the top-left corner. The area under the curve (AUC) summarises discrimination: 0.5 is chance, 1.0 is perfect, and it equals the probability that a randomly chosen case scores higher than a randomly chosen non-case.

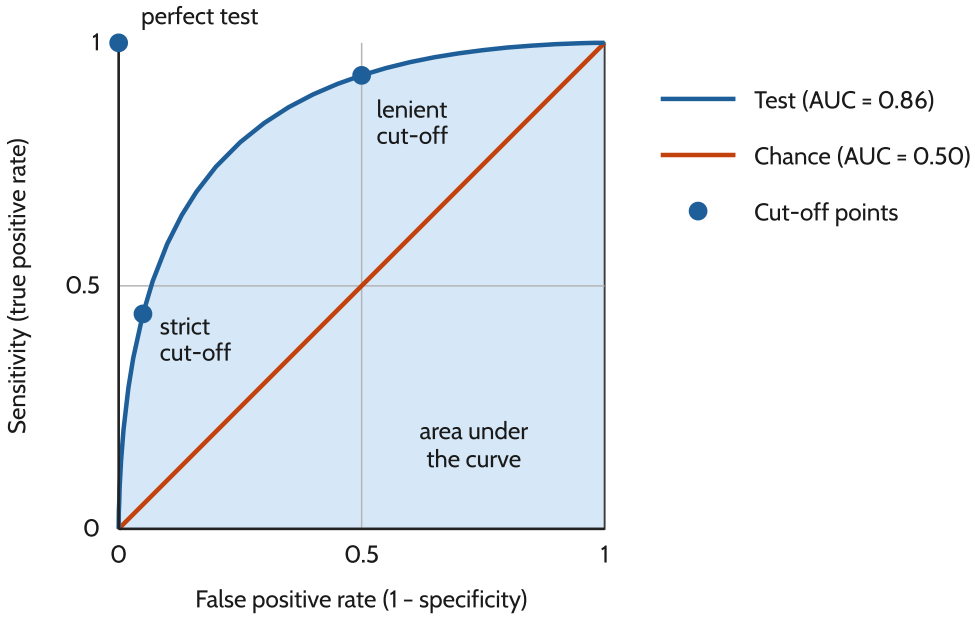


Figure 2.8 A receiver operating characteristic (ROC) curve. The strict cut-off flags few healthy people but misses cases; the lenient one catches nearly every case at the price of many false alarms. The shaded area is the AUC.

In selection, the **selection ratio** is the proportion of applicants who can be accepted (50 posts for 500 applicants gives 0.10). **H. C. Taylor** and **J. T. Russell** published the **Taylor-Russell tables** in 1939, giving the percentage of selectees who will succeed for a given validity coefficient, selection ratio and base rate. Even a modest validity helps a great deal when the selection ratio is low, and little when almost everyone must be accepted or the base rate is extreme. Utility analysis, developed by Cronbach and **Goldine Gleser** (1957), adds costs and gains: a test is worth using when better decisions repay the cost of testing.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. A test that merely looks relevant to the people taking it has _____ validity.
2. Correlating a new intelligence test with an established one given at the same time tests _____ validity.
3. Supervisors who know applicants' test scores and then rate their work cause criterion _____.
4. A weak correlation with a measure of an unrelated construct is evidence of _____ validity.
5. A test with reliability 0.64 can have a validity coefficient no higher than _____.
6. Sensitivity equals TP divided by _____.
7. As the base rate of a disorder falls, the positive predictive value of a screen _____.

Answers: (1) face, (2) concurrent, (3) contamination, (4) discriminant (divergent), (5) 0.80, (6) TP + FN, (7) falls.

2.5 Norms, Standard Scores and Interpreting a Score MEDIUM YIELD · 32 QS

A ten-year-old answers 34 reasoning items correctly. Is that good? Until you know how hard the items were and how other ten-year-olds do, the **raw score** (items passed or points earned) means nothing; it needs a frame of reference. There are two such frames, and the commoner one, comparison with other people, rests on norms, on derived scores and on the normal curve that ties them together.

Norm-referenced and criterion-referenced scores

In **norm-referenced interpretation** a score is compared with those of a **norm group** or **standardisation sample**, a representative sample of the people the test is meant for. "Higher than 90 per cent of her peers", "5th in his class" and "1 standard deviation (SD) above the mean" are norm-referenced: they rank a person among others, as selection requires. In **criterion-referenced interpretation** the score is compared with a fixed standard, whatever others do: "85 per cent of the addition problems correct, meeting the mastery cut-off". **Robert Glaser** drew the distinction in 1963. A criterion-referenced test asks what a person can do, so a whole class can pass it, or no one.

A **domain-referenced test** samples items from a defined domain (all two-digit additions, say), so the percentage correct estimates how much of the domain the person commands; a **mastery test** adds a cut-off and reports pass or fail, like a driving test. **Norms are not standards**: they describe how people do perform, not how they should, so matching the norm of a poorly taught school is not being good enough. An **ipsative score** compares a person with no one else: it expresses one trait relative to the same person's other traits, as in forced-choice inventories (Section 2.9), and cannot be compared across people.

> EXAM TRAP
>
> Of the usual options, only "85 per cent correct, meeting the mastery cut-off" is criterion-referenced; percentile standing, class rank and "1 SD above the mean" are norm-referenced. Norm-referencing compares with a representative standardisation sample, not with a cut-off, an absolute criterion or the person's own earlier scores.

Standardisation and the norm sample

The process of **standardisation** means giving and scoring a test identically for everyone (same instructions, materials, time limits and scoring rules) and giving it so to a large sample whose scores become the **norms**. The test maker defines the **target population**, the people the test is meant for, and samples it: simple **random sampling** is ideal but rarely practical, so **stratified sampling** (quotas matching the census on age, sex, region, residence and socio-economic status) or **cluster sampling** (whole schools chosen at random) is used (Section 9.3). Good norms are large, **representative**, **recent** (the Flynn effect, a rise of about 3 intelligence quotient (IQ) points a decade, makes old norms too easy; Section 1.9), **relevant** to the person tested and fully described in the manual. Hence **national norms**, **local norms** (one region or school) and **subgroup norms** (by age, sex, grade, language or occupation). American norms compare an Indian child with American children, so Indian versions of Western tests were renormed (Section 2.6).

Standardisation lets the clinician do all three of the following: see where a client's score falls on the test's distribution of scores; estimate from it whether the client might meet the diagnostic criteria for a disorder (an IQ about 2 SD below the mean is one criterion for intellectual disability, Section 2.7); and estimate how probable that diagnosis is. Computers now score tests against the norm tables and write narrative reports (as for the Minnesota Multiphasic Personality Inventory-2, MMPI-2), saving time and errors, though not replacing clinical judgement.

Age and grade norms; ratio and deviation IQ

The oldest derived scores express performance as the age or school grade at which it is typical. An **age-equivalent norm** is the average performance at each age; **Alfred Binet**'s 1908 scale grouped items by age, and the age level a child reached was the **mental age** (MA). A **grade-equivalent norm** does the same for school grades, interpolating in-between values. Both mislead. Growth is uneven (faster at 2 than at 14), so a year is not a fixed unit; traits such as personality do not grow with age; grade equivalents in different subjects are not comparable; and a Class 4 child with an arithmetic grade equivalent of 7.2 has not mastered Class 7 work, only scored on a Class 4 test as an average Class 7 pupil is estimated to.

William Stern divided MA by chronological age (CA) in 1912, and **Lewis Madison Terman** multiplied by 100 in the 1916 Stanford-Binet: the **ratio IQ** = (MA ÷ CA) × 100, so MA 8 and CA 10 give 80. It fails for adults, because mental age stops rising in the late teens (a 40-year-old with an MA of 16 would score 40). **David Wechsler**'s Wechsler-Bellevue scale (1939) introduced the **deviation IQ**, a standard score comparing a person with same-age peers, with a mean of 100 and an SD of 15 at every age (Sections 1.9 and 2.6).

> EXAM TRAP
>
> MA 8 and CA 10 give 80; 125 inverts the ratio (CA ÷ MA). The deviation IQ is not "only for adults", is not less reliable and is not built from unstandardised raw scores: it is a standard score against age peers.

Percentile ranks

The **percentile rank** (PR) is the percentage of the norm group scoring below a given raw score (some books say "at or below"). If 70 per cent of the sample scored below Meera's 65, her PR is 70 and her score is the 70th percentile (P_{70}). It is not the **percentage correct**, a raw score: 70 per cent correct could be the 30th percentile on an easy test and the 99th on a hard one. The **quartiles** cut the distribution into four ($Q_1 = P_{25}$; $Q_2 = P_{50}$, the median; $Q_3 = P_{75}$) and **deciles** into ten ($D_5 = P_{50}$).

Percentiles are **ordinal**, with unequal units. Scores crowd around the middle of a normal distribution, so a small raw difference there jumps many percentile points while a large one in the tails moves few: the 50th to the 60th percentile is a quarter of an SD (IQ 100 to 104), the 90th to the 99th more than a whole SD (IQ 119 to 135). So percentiles exaggerate differences near the middle, hide them in the tails and should not be averaged; equal-unit standard scores solve this.

The z score ●●

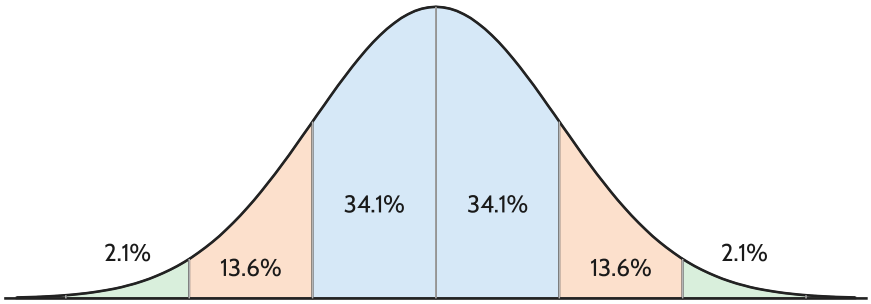
A **standard score** states how far a score lies from the mean in SD units. The basic one is the **z score**: z = (X − M) ÷ SD, with X the raw score and M and SD those of the norm group. z scores have a mean of 0 and an SD of 1; z = +1.0 is exactly one SD above the mean, and negative values lie below it. Their great use is comparing results from different distributions. Arjun scores 60 in psychology (mean 50, SD 5) and 70 in statistics (mean 60, SD 10). The raw scores favour statistics, but psychology is z = +2.0 and statistics z = +1.0, so psychology was his better paper.

Look-alikes answer other questions (Sections 9.8 and 9.9). The **variance** (SD squared) and the **semi-interquartile range**, $(Q_3 - Q_1) \div 2$, measure the spread of a whole distribution; the **coefficient of variation**, (SD ÷ M) × 100, compares spread across distributions in different units; the **standard error** is the SD of a sampling distribution, such as that of sample means (the standard error of measurement is in Section 2.3); the median, not z, describes the centre of a skewed distribution. Only z places one person within a distribution.

> **EXAM TRAP**
>
> The measure of how far a data point lies from the mean in SD units, which lets scores from different distributions be compared, is the z score. It is not a measure of variability or of the central tendency of a skewed distribution.

The normal curve: from z to percentile ●●●

Why does z = +1 always mean the 84th percentile? Because test scales are built on the **normal curve**, the symmetrical bell in which mean, median and mode coincide, and its areas are fixed: about 34.1 per cent of cases lie between the mean and 1 SD on each side, 13.6 per cent between 1 and 2 SD and 2.1 per cent between 2 and 3 SD. So about 68 per cent fall within ±1 SD, 95 per cent within ±2 SD and 99.7 per cent within ±3 SD (Figure 2.9). The **z table** of areas under the curve is derived from the normal distribution and gives the proportion of cases below any z.



z score	−3	−2	−1	0	+1	+2	+3
Percentile	0.1	2	16	50	84	98	99.9
T score	20	30	40	50	60	70	80
IQ (SD 15)	55	70	85	100	115	130	145
Scaled score	1	4	7	10	13	16	19

Stanine	1	2	3	4	5	6	7	8	9
% of cases	4	7	12	17	20	17	12	7	4

Figure 2.9 The normal curve and the scales built on it. Read straight down from any point to convert between z, percentile, T score, IQ (SD 15), scaled score and stanine; stanine bands are half an SD wide except the open-ended 1 and 9.

The landmarks: z = 0 is the 50th percentile; +1 the 84th and −1 the 16th; +0.5 about the 69th; +1.5 about the 93rd; +2 about the 98th (97.7) and −2 the 2nd; +3 the 99.9th. With IQ (mean 100, SD 15), 100 is the 50th percentile, 85 the 16th, and 130, two SDs above the mean, about the 98th, so only about 2 in 100 score higher. On a Binet-type scale with an SD of 16, an IQ of 136 is z = 36 ÷ 16 = 2.25; the z table gives .9878 below it, so about 98.8 per cent score lower (98.7 if the entry is truncated).

> **EXAM TRAP**
>
> ±2 SD holds about 95 per cent (exactly 95 within ±1.96, so 97.5 per cent lies below +1.96); ±1 SD holds 68 and ±3 SD 99.7. The z table comes from the normal distribution, not from the SD, t scores or random sampling. The 84th percentile is z = +1, not +1.5 (93rd) or +2 (98th).

Linear and normalised scores; the T score ●●

z scores carry signs and decimals, so other standard scores rescale them: new score = new mean + (new SD × z). This is a **linear transformation** (new = a + b × old), as are z itself and the **deviation score** X − M; it keeps the raw distribution's shape, so skewed raw scores stay skewed. A **normalised standard score** (an area, or non-linear, transformation) instead turns each raw score into its percentile rank, looks up the z that cuts off that percentage of the normal curve and rescales it, forcing a normal shape. Stanines, stens, the C scale, normalised T scores and most modern intelligence-test scales are normalised, which is justified only with large samples and when the skew lies in the test, not the trait.

The **T score** of **William A. McCall** (1922) has a mean of 50 and an SD of 10: T = 50 + 10z, usually 20 to 80 (±3 SD). Computed from z it is linear; McCall's own scale was normalised. T scores

are the metric of the MMPI and most personality inventories, including the Revised NEO Personality Inventory (NEO-PI-R; Section 2.9). T 70, 2 SD above the mean (about the 98th percentile), was the original MMPI's clinical cut-off; the MMPI-2 uses T 65, 1.5 SD above (about the 93rd).

> EXAM TRAP
>
> T scores have a mean of 50 and an SD of 10, not 100 and 15 (IQ), 0 and 1 (z) or 500 and 100 (CEEB). T 70 = +2 SD, T 65 = +1.5 SD, T 60 = +1 SD.

Stanines, stens and the C scale ●●

The **stanine** ("standard nine"), developed by the United States Army Air Forces in World War II, runs from 1 to 9 with a mean of 5 and an SD of about 2. Stanines are assigned by rank: the lowest 4 per cent of the norm group get 1, the next 7 per cent 2, and so on in the proportions 4, 7, 12, 17, 20, 17, 12, 7 and 4 per cent. Each band but the open-ended 1 and 9 is half an SD wide. Stanine 5 spans z = −0.25 to +0.25, the 40th to 60th percentile, and holds 20 per cent of cases; stanine 4 covers about the 23rd to 40th percentile, and 9 begins at the 96th (Figure 2.9). Built from percentile ranks, stanines are a normalised, non-linear transformation. Stanines 1 to 3 are below average, 4 to 6 average and 7 to 9 above average; single digits are easy to record but coarse.

The **sten** ("standard ten") runs from 1 to 10 with a mean of 5.5 and an SD of 2. It has no middle band: stens 5 and 6 lie just below and above the mean, and 4 to 7 cover the middle 68 per cent. Stens are the scale of **Raymond Bernard Cattell**'s Sixteen Personality Factor Questionnaire (16PF, 1949), not of the Wechsler scales, the Stanford-Binet or the Rorschach. **J. P. Guilford**'s **C scale** (1954) extends the stanine to 11 units (0 to 10; mean 5, SD 2) by splitting off about the extreme 1 per cent at each end.

> EXAM TRAP
>
> Stanine 5 holds 20 per cent, not 17 (stanines 4 and 6). Of z, 50 + 10z, X − M and the stanine, only the stanine is a non-linear transformation.

Scaled scores, deviation IQs and CEEB scores ●●

Each Wechsler subtest is reported as a **scaled score** (mean 10, SD 3, range 1 to 19), and the index scores and Full Scale IQ as deviation IQs (mean 100, SD 15). A scaled score of 10 is exactly average, the 50th percentile; 7 is 1 SD below (about the 16th percentile, like an IQ of 85), which the Wechsler Adult Intelligence Scale, fourth edition (WAIS-IV), calls low average; 13 is 1 SD above (84th); 4 is 2 SD below (2nd), the borderline region. The Stanford-Binet used an SD of 16 from Form L-M (1960) through the fourth edition (1986), returning to 15 in the fifth (2003), and Cattell's Culture Fair Intelligence Test also uses 16. So IQ 130 is z = +2.0 (97.7th percentile) on a Wechsler scale but +1.875 (97th) on an SD-16 scale: compare tests through z. A **CEEB score** (College Entrance Examination Board) has a mean of 500 and an SD of 100, as on the older SAT and Graduate Record Examinations (GRE). Tables 2.11 and 2.12 summarise them all.

Table 2.11 The standard-score systems.

Score	Mean	SD	Range	Type; main use

Score	Mean	SD	Range	Type; main use
z score	0	1	about −3 to +3	Linear; comparing distributions
T score	50	10	20 to 80	Linear or normalised; MMPI, NEO-PI-R
Deviation IQ	100	15	40 to 160	Normalised; Wechsler, SB5
Older Binet IQ	100	16	varies	Normalised; SB L-M to SB4, Culture Fair
Scaled score	10	3	1 to 19	Normalised; Wechsler subtests
Stanine	5	about 2	1 to 9	Normalised; school tests
Sten	5.5	2	1 to 10	Normalised; 16PF
C scale	5	2	0 to 10	Normalised; Guilford
CEEB score	500	100	200 to 800	Older SAT, GRE

Table 2.12 Converting between scores (IQ in brackets for an SD of 16).

z (percentile)	T score	IQ, SD 15 (16)	Scaled score	Stanine
−3.0 (0.1)	20	55 (52)	1	1
−2.0 (2)	30	70 (68)	4	1
−1.5 (7)	35	77.5 (76)	5.5	2
−1.0 (16)	40	85 (84)	7	3
−0.5 (31)	45	92.5 (92)	8.5	4
0 (50)	50	100 (100)	10	5
+0.5 (69)	55	107.5 (108)	11.5	6
+1.0 (84)	60	115 (116)	13	7
+1.5 (93)	65	122.5 (124)	14.5	8
+2.0 (98)	70	130 (132)	16	9
+3.0 (99.9)	80	145 (148)	19	9

EXAM TRAP

Mean 10 and SD 3 identify the scaled score, and 10 is the 50th percentile. One mock keys a scaled score of 7 as "within average range (1 SD below mean)": the best of its options, since 7 is neither impaired nor borderline (4 is), but the WAIS-IV calls it low average.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. "85 per cent correct, meeting the mastery cut-off" is a _____-referenced interpretation.
2. A mental age of 8 and a chronological age of 10 give a ratio IQ of _____.
3. About _____ per cent of cases fall within ±2 SD of the mean.
4. A percentile rank of 84 corresponds to z = _____.
5. A T score of 65 lies _____ SD above the mean.
6. Stanine 5 holds _____ per cent of cases.
7. Stens (mean 5.5) are the scale of Cattell's _____.

Answers: (1) criterion, (2) 80, (3) 95, (4) +1, (5) 1.5, (6) 20, (7) 16PF.

2.6 Intelligence Tests: Binet, Wechsler, Non-verbal and Indian Scales

MEDIUM YIELD · 35 QS

In 1904 the French education ministry asked which Paris schoolchildren could not keep up without special help. The answer, a list of graded tasks, founded psychology's most used family of tests. This section follows it through the Binet, Wechsler and Kaufman scales, the non-verbal tests and the Indian adaptations; theories of intelligence and the IQ itself were introduced in Section 1.9.

What an intelligence test can and cannot do ●●

An **individual test** is given by one examiner to one person, who can be watched and questioned (the Stanford-Binet, the Wechsler scales); a **group test** is given to many people at once. A **verbal test** uses language in its items; a **non-verbal test** keeps language out of the items; a **performance test** asks the person to handle materials, such as blocks or a form board, usually against the clock.

What does a good IQ test offer? High test-retest reliability, high internal consistency (both above .90 for the major full-scale IQs) and good validity for predicting school achievement (it correlates about .5 with grades). It does not offer a good prediction of behaviour in general: honesty, persistence and social skill depend on personality, motivation and circumstance.

Intelligence tests diagnose intellectual disability and giftedness, guide educational placement, set a neuropsychological baseline and document eligibility for a disability allowance, but not for housing benefit (an income question), suicide risk (a clinical risk assessment, Chapter 7) or, with an adult scale, a child's special educational needs. The Wechsler Adult Intelligence Scale (WAIS) is an intelligence test, not a personality, aptitude (predicting future success), attitude or achievement (measuring past learning) test (Sections 2.8 and 2.9), and it does not look for neurological soft signs, subtle non-localising signs such as clumsiness.

EXAM TRAP

"An IQ test does NOT provide ..." is answered by "a good predictor of behaviour". High test-retest reliability, high internal consistency and good validity (for school achievement) are exactly what it does provide.

The Binet scales ●●

Alfred Binet (1857 to 1911) and the physician **Theodore Simon** (1872 to 1961) published the first practical intelligence test in 1905: 30 items in order of difficulty. In 1908 they grouped the items by the age at which most children pass them, creating the first **age scale**; the highest level a child reaches is the **mental age**. The 1911 revision, the year Binet died, extended it to adults.

Lewis Madison Terman (1877 to 1956) standardised it at Stanford University as the Stanford-Binet (1916), using William Stern's ratio IQ (mental age ÷ chronological age × 100): the original Stanford-Binet used the ratio IQ, not a deviation IQ, percentile rank or T-score (Section 2.5). Later editions (Table 2.13) moved away from mental age: parallel Forms L and M with **Maud Merrill** (1937); Form L-M (1960), with a deviation IQ of SD 16; and the SB4 (1986) of **Robert L. Thorndike**, **Elizabeth Hagen** and **Jerome M. Sattler**, a **point scale** (each subtest scored by items passed, whatever their age level) reporting Standard Age Scores, SD 16.

The Stanford-Binet Fifth Edition (SB5, 2003) of **Gale H. Roid** covers ages 2 to 85 and over, with mean 100 and SD 15. Its five factors, Fluid Reasoning, Knowledge, Quantitative Reasoning, Visual-Spatial Processing and Working Memory, are each tested in a verbal and a non-verbal domain: 10 subtests yielding Full Scale, Verbal and Nonverbal IQs. Two **routing subtests** (Object Series/Matrices and Vocabulary) set the starting level for the rest. Within a subtest the examiner finds a **basal level**, below which all items are assumed passed, and a **ceiling level**, where failures stop testing.

Table 2.13 The Binet line, from mental age to deviation IQ.

Edition	Year	Authors	Scale and score
Binet-Simon	1905, 1908, 1911	Binet, Simon	30 items; age scale, mental age (1908)
Stanford-Binet	1916	Terman	Ratio IQ
Forms L and M	1937	Terman, Merrill	Parallel forms; ratio IQ
Form L-M	1960	Terman, Merrill	Deviation IQ, SD 16
SB4	1986	Thorndike, Hagen, Sattler	Point scale; Standard Age Scores, SD 16
SB5	2003	Roid	Ages 2 to 85+; 5 factors, 10 subtests; SD 15

EXAM TRAP

The Stanford-Binet grew from Binet and Simon, revised by Terman at Stanford; not Wechsler and Terman, Spearman and Thurstone (factor theorists, Section 1.9) or Galton and Cattell (sensory tests, Section 2.1). The SB5's factors are not "sensory, motor, memory, attention, language" (a neuropsychological checklist) or "Verbal and Performance IQ only" (old Wechsler).

Wechsler, the deviation IQ and the WAIS-IV ●●●

David Wechsler (1896 to 1981), chief psychologist at Bellevue Hospital in New York, found the Stanford-Binet unsuited to the adults he tested. It had little appeal for adults; it put too much emphasis on word manipulation; it stressed speed over accuracy; and it relied on mental age, which stops growing in the late teens, so an adult's ratio IQ falls with every birthday. His Wechsler-Bellevue Intelligence Scale (1939) answered each flaw: a point scale with a Performance scale of non-verbal tasks beside the Verbal scale, scored by a new **deviation IQ**, also called the standard IQ because it is a standard score.

The deviation IQ compares the actual score with the expected mean score for the person's age group: IQ = 100 + 15 × (score minus age-group mean) ÷ age-group SD. Scoring at your age group's mean gives 100 at any age; one SD above gives 115. Wechsler defined intelligence (1944) as "the aggregate or global capacity of the individual to act purposefully, to think rationally, and to deal effectively with his environment".

> EXAM TRAP
>
> Wechsler's new method was the deviation ("standard") IQ, not a mental-age, physical-age or age-relative formula IQ; mocks write it as actual score ÷ expected mean score for age. His target flaws were no appeal for adults, word manipulation, speed over accuracy and mental age, never "children", "physical age" or "accuracy over speed".

The adult line runs from the WAIS (1955; 6 verbal and 5 performance subtests) through the WAIS-R (1981) and WAIS-III (1997), which still gave Verbal, Performance and Full Scale IQs, to the WAIS-IV (2008; ages 16:0 to 90:11), which dropped the first two. It yields a **Full Scale IQ** (FSIQ) from 10 **core subtests** and four **index scores**, composites for narrower abilities: Verbal Comprehension (VCI), Perceptual Reasoning (PRI), Working Memory (WMI) and Processing Speed (PSI) (Figure 2.10). Five **supplemental subtests** can replace a spoiled core subtest. The VCI reflects verbal reasoning and acquired knowledge; the PRI, reasoning with visual, non-verbal material. The **Working Memory Index**, built on holding and manipulating digits and mental sums, reflects attention, concentration and mental control; the **Processing Speed Index**, how quickly simple visual material is scanned and acted on. The optional **General Ability Index** (GAI, VCI plus PRI) estimates reasoning with less weight on memory and speed; the **Cognitive Proficiency Index** (CPI) pairs WMI and PSI. The WAIS-5 (2024) splits perceptual reasoning into Visual Spatial and Fluid Reasoning: five primary indices.

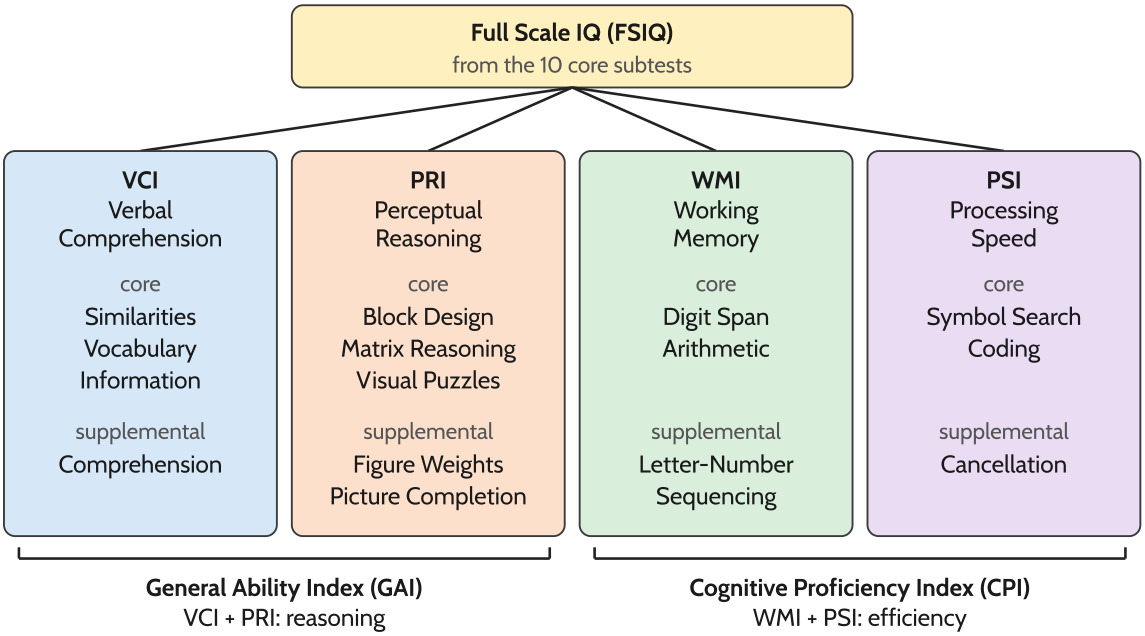


Figure 2.10 The WAIS-IV. Count the core subtests under each index: three verbal, three perceptual, two working memory, two processing speed. The optional GAI and CPI pair the indices.

EXAM TRAP

The WAIS-IV gives four indices (VCI, PRI, WMI, PSI) beside the FSIQ, not two, three or five; the GAI is an optional summary. Core subtests: 3 verbal, 3 perceptual. "VIQ, PIQ, FSIQ and Memory Quotient" mixes the WAIS-III with the Wechsler Memory Scale (Section 2.11); "crystallised, fluid, visual, auditory (or quantitative)", "VIQ, PIQ, memory, attention" and "cognitive, behavioural, emotional, adaptive" are not WAIS-IV indices.

WISC, WPPSI and reading a Wechsler profile ●●

The Wechsler Intelligence Scale for Children (WISC, 1949) and the Wechsler Preschool and Primary Scale of Intelligence (WPPSI, 1967) extend the design downwards (Table 2.14). The WISC-V (2014; ages 6:0 to 16:11) has five primary indices, Verbal Comprehension, Visual Spatial, Fluid Reasoning, Working Memory and Processing Speed, from 10 primary subtests, and takes the FSIQ from seven; Information, Arithmetic and Letter-Number Sequencing are among its secondary subtests.

Table 2.14 The Wechsler family today.

Scale	Editions	Ages	Structure
WAIS-IV	1955, 1981, 1997, 2008	16:0 to 90:11	FSIQ + 4 indices; 10 core, 5 supplemental subtests
WAIS-5	2024	16:0 to 90:11	5 primary indices
WISC-V	1949, 1974, 1991, 2003, 2014	6:0 to 16:11	5 primary indices; FSIQ from 7 subtests
WPPSI-IV	1967 to 2012	2:6 to 7:7	Preschool and early school

> **EXAM TRAP**
>
> Age ranges: WISC-V 6:0 to 16:11; WPPSI-IV 2:6 to 7:7; WAIS-IV 16:0 to 90:11; Bayley scales to 42 months (Section 2.7). A 2024 paper keys Letter-Number Sequencing as "core in the WAIS-IV but not the WISC-V". In fact it is supplemental in the WAIS-IV and secondary in the WISC-V (core in the WISC-IV). Digit Span, Symbol Search and Matrix Reasoning are core or primary in both, so mark Letter-Number Sequencing, but Information and Arithmetic truly fit the stem.

Which ability does each subtest tap? Vocabulary and Information measure crystallised intelligence, knowledge acquired through schooling and culture. Matrix Reasoning (completing a visual pattern) measures fluid reasoning, solving novel problems without learned knowledge; Block Design (copying a design with coloured cubes) measures visuospatial analysis and construction; Coding (formerly Digit Symbol: copying number-paired symbols at speed) measures processing speed.

The FSIQ is placed in a descriptive band (Table 2.15). Below 70 is extremely low, but intellectual disability also requires deficits in adaptive behaviour (Section 2.7). Scores of 70 to 84 straddle borderline (70 to 79) and low average (80 to 89); older systems called 71 to 84 borderline intellectual functioning. The examiner then studies **index discrepancies**, differences between index scores, and **scatter**, unevenness among subtest scores. A VCI of 115 beside a PSI of 82 is interpreted only if the gap is statistically significant and rare in the norm sample; the FSIQ then averages over a real split, and the separate indices or the GAI tell more.

Table 2.15 Wechsler descriptive classification of IQ (WAIS-IV labels).

IQ	Classification
130 and above	Very superior
120 to 129	Superior
110 to 119	High average
90 to 109	Average
80 to 89	Low average
70 to 79	Borderline
69 and below	Extremely low

Kaufman, Das-Naglieri and Woodcock-Johnson

Alan S. Kaufman and **Nadeen L. Kaufman** built their tests on theory. The Kaufman Assessment Battery for Children (K-ABC, 1983) used Alexander Luria's sequential processing (handling items in serial order) and simultaneous processing (integrating many at once), and scored achievement apart from ability. The KABC-II (2004, ages 3 to 18) lets the examiner choose a Luria model or a Cattell-Horn-Carroll (CHC) model. The Kaufman Brief Intelligence Test (K-BIT, 1990; KBIT-2, 2004, ages 4 to 90) is a quick screen with a Verbal or Crystallised scale and a Nonverbal or Fluid scale (matrices). Reading, mathematics, quantitative and analytical scales belong to achievement and scholastic aptitude tests, such as the ACT, SAT and GRE admission tests (Section 2.8); an "EQ scale" belongs to emotional intelligence (Section 1.9).

The Indian-born **J. P. Das** and **Jack A. Naglieri** turned Luria's three functional units of the brain into the **PASS theory** (Planning, Attention, Simultaneous and Successive processing), measured by their Cognitive Assessment System (1997). The Woodcock-Johnson Tests of Cognitive Abilities (**Richard W. Woodcock** and **Mary Bonner Johnson**, 1977) are, in current editions, built on CHC theory.

Non-verbal and culture-fair tests ●●

Vocabulary items reward schooling and language as much as intelligence. A **culture-free test** would leave culture out entirely; none exists, since even a pattern on paper assumes familiarity with pictures and timing. A **culture-fair test** merely minimises language, reading and specific knowledge, so that people from different backgrounds start more nearly level.

The classic is Raven's Progressive Matrices by **John Carlyle Raven** (1902 to 1970). Each item is a pattern with a piece missing. The Standard Progressive Matrices (1938) has 60 items in five sets, A to E, growing progressively harder; the Coloured version (1947) serves children of 5 to 11, elderly people and disabled people; the Advanced version serves able adults. Raven aimed at Spearman's g through **eductive ability**, seeing relations in confusion, as against recalling learned information. The test measures fluid, non-verbal reasoning by abstract pattern completion with minimal language, and is one of the strongest single markers of g.

> EXAM TRAP
>
> Raven's measures fluid intelligence; it is culture-fair, not culture-free, and does not measure crystallised, verbal or social intelligence, reading comprehension or memory alone. Offered "general intelligence" beside "fluid", choose fluid: Raven's marks g, but its content is fluid and non-verbal. "Both" is wrong.

Raymond Bernard Cattell (1905 to 1998) named his test the Culture Free Intelligence Test (1940), then renamed it Culture Fair. Its three Scales are graded by age and ability; Scales 2 and 3 use four figural subtests: series, classification, matrices and conditions. Other tests remove speech from the task: the Leiter International Performance Scale of **Russell Graydon Leiter** (1929; Leiter-3, 2013) gives its instructions by pantomime, for deaf or autistic children; the Columbia Mental Maturity Scale needs only pointing to the drawing that does not belong, suiting children with cerebral palsy; and the Peabody Picture Vocabulary Test of **Lloyd M. Dunn** (1959) measures receptive vocabulary as the person points to the picture the examiner names.

The simplest non-verbal test is a pencil and the request "draw a man". **Florence Laura Goodenough** (1886 to 1959) noticed that children add details to drawings in a regular developmental order. Her **Draw-a-Man test** (1926) credits up to 51 details as a non-verbal estimate of intellectual maturity and visual-motor ability; **Dale B. Harris** revised it in 1963. Because it asks so little, it works with a client too disorganised for verbal or performance subtests. **Karen Machover**'s Draw-a-Person uses the same request projectively, to read personality (Section 2.10).

> EXAM TRAP
>
> A disorganised client who cannot manage verbal or performance tasks is asked to draw a person: the domain targeted is visual-motor integration and cognitive estimation (Goodenough), not projective personality assessment (Machover), verbal comprehension or processing speed.

Group testing arrived with the First World War. In 1917 a committee led by **Robert Mearns Yerkes** (1876 to 1956) built the Army Alpha, a verbal test for literate recruits, and the Army Beta, a non-verbal one for illiterate and non-English-speaking recruits. **Arthur Otis**, Terman's student, devised the multiple-choice group format the Alpha borrowed (today's Otis-Lennon School Ability Test). The Primary Mental Abilities tests of **Louis Leon Thurstone** (1887 to 1955) score seven primary abilities separately instead of giving one IQ.

Performance tests and Indian scales ●●

India needed tests that did not lean on literacy or on one language, so its psychologists normed performance tests and adapted Western scales (Table 2.16). In the Kohs Block Design test (**Samuel C. Kohs**, 1920; book 1923) the person arranges coloured cubes to copy a printed design; Wechsler borrowed it as Block Design. In the Alexander Pass-Along test (**W. P. Alexander**, 1932) blocks are slid along a tray, never lifted, to reach a target pattern. The Seguin Form Board, a teaching device of **Edouard Seguin** (1812 to 1880) for children with intellectual disability, became a test in **Henry Herbert Goddard**'s version: a child replaces 10 shapes over three trials, and the shortest time converts to a mental age. In the Gesell Drawing Test of **Arnold Gesell** (1880 to 1961) a child copies geometric figures mastered at typical ages.

C. M. Bhatia assembled **Bhatia's Battery of Performance Tests of Intelligence** (1955), standardised on Indian children. Its five subtests are Kohs Block Design, Alexander Pass-Along, Pattern Drawing, Immediate Memory (digits, the one non-performance task) and Picture Construction; the Seguin Form Board is not among them. Needing no reading or writing, it is the test of choice for an illiterate child or adult from a rural area, because it minimises cultural and educational bias; the WISC-IV and SB5 carry Western verbal content, and MISIC is half verbal and assumes schooling. Being visual, it cannot serve a patient with cortical blindness.

> EXAM TRAP
>
> For an illiterate rural 10-year-old, choose Bhatia's Battery over the WISC-IV, SB5 or MISIC. Its population is illiterate or rural Indians, adolescents and adults included, not urban English speakers or preschoolers only.

The adaptations follow their parents. The **Binet-Kamat Test** (**V. V. Kamat**, 1934; revised 1967) adapted the 1916 Stanford-Binet; it is a heavily verbal age scale giving a mental age and ratio IQ. **Malin's Intelligence Scale for Indian Children** (MISIC; **Arthur J. Malin**, Nagpur, 1969) adapted the WISC for ages 6 to 15, so it can be given from 6 years, and yields Verbal and Performance Quotients and a Full Scale IQ. The **Wechsler Adult Performance Intelligence Scale** (WAPIS; **P. Ramalingaswamy**, 1974) adapted the WAIS performance subtests for Indian adults. **Pramila Phatak** of Baroda produced Indian norms for the Draw-a-Man test, and

Raven's matrices also have Indian norms. Pearson's Indian editions of the WISC-IV and WAIS-IV are newer and poorly documented.

> **EXAM TRAP**
>
> A mock keys "WAIS-IV India edition" as the Indian WAIS common in clinics. Learn WAPIS (Ramalingaswamy) as the classic Indian adult adaptation; the BKT and MISIC are children's tests, so "all of the above" fails.

Table 2.16 Performance and Indian tests compared.

Test	Author, year	Ages	Key fact
Raven's SPM	J. C. Raven, 1938	About 6 to adult	60 items; Indian norms exist
Draw-a-Man	Goodenough, 1926	About 3 to 15	51 points; Indian norms, Phatak
Seguin Form Board	Seguin; Goddard	Young children	10 shapes, 3 trials
Bhatia's Battery	C. M. Bhatia, 1955	Normed 11 to 16	5 subtests; no reading
Binet-Kamat	V. V. Kamat, 1934	3 to 22	From 1916 Stanford-Binet; ratio IQ
MISIC	A. J. Malin, 1969	6 to 15	From WISC; VQ, PQ, FSIQ
WAPIS	P. Ramalingaswamy, 1974	Adults	From WAIS performance scale

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. An IQ test is reliable and predicts school achievement, but it is not a good predictor of _____ in general.
2. The original 1916 Stanford-Binet reported the _____ IQ.
3. The WAIS-IV Working Memory Index has two core subtests, Digit Span and _____.
4. The WISC-V is normed for ages 6:0 to _____.
5. The SB5 subtests that set the starting level for the rest are the _____ subtests.
6. Raven's Progressive Matrices is culture-fair, not culture-_____.
7. Arthur J. Malin's MISIC is the Indian adaptation of the _____.

Answers: (1) behaviour, (2) ratio, (3) Arithmetic, (4) 16:11, (5) routing, (6) free, (7) WISC.

2.7 Assessing Development, Adaptive Behaviour and Disability

LOWER YIELD · 21 QS

An intelligence test asks how well a person reasons when tested. Two other questions matter as much. Has this infant reached the milestones expected for her age? And how well does a child or adult actually manage everyday life? Their answers decide who is diagnosed with intellectual disability and who in India is certified as disabled.

Beyond IQ: quotients, adaptive behaviour and informants

An infant cannot follow test instructions, so you measure how far she has come along the expected path. A **developmental scale** lists milestones at the ages most children reach them. The level reached is the **developmental age** (DA), and the **developmental quotient** is DQ = DA / CA × 100, CA being chronological age. A social maturity scale gives a **social age** (SA) and a **social quotient**, SQ = SA / CA × 100: if 6-year-old Ravi (72 months) manages daily life like a typical child of 54 months, his SQ is 75. Modern scales replace these crude ratios with deviation scores, mean 100 and SD 15 (Section 2.5).

Why measure everyday competence apart from IQ? Because one child scoring 65 dresses, shops and takes the bus alone, while another scoring 75 cannot look after herself. The American Association on Intellectual and Developmental Disabilities (AAIDD, formerly the AAMD) defines **adaptive behaviour** as the conceptual (language, money, number), social (relationships, rules, avoiding victimisation) and practical (self-care, safety, travel, work) skills that people learn and perform in everyday life. Since it is what a person habitually does, not what he can do at his best, it is usually rated by an **informant**, a parent, caregiver or teacher, which also suits infants and the severely disabled.

A **screening test** is short, is given to many children and only flags those needing a closer look; a **diagnostic test** is longer and individual, and yields a level on which a diagnosis can rest. And because symptoms vary across contexts (the boy restless in a noisy classroom may be calm at home), clinicians use **multi-informant assessment**, combining parents, teachers, the child and clinic observation.

KEY STUDY: ACHENBACH'S CROSS-INFORMANT CORRELATIONS (1987)

Thomas M. Achenbach and colleagues pooled studies in which several people rated the same children. Similar informants, such as two parents, agreed moderately (mean correlation about 0.60); different kinds, such as parent and teacher, only weakly (about 0.28). Disagreement, they argued, reflects real differences between settings, so each informant adds valid information: the logic of Achenbach's parallel parent Child Behavior Checklist, Teacher's Report Form and Youth Self-Report. Criticism: it does not say whose report should decide.

EXAM TRAP

Multi-informant assessment is recommended because **symptoms may vary across contexts**, not because parents over-report, teachers add nothing or clinic observation alone is insufficient.

Infant and preschool developmental scales

Arnold Gesell (1880 to 1961) of the Yale Clinic of Child Development filmed thousands of infants and turned the timing of their behaviour into norms (1925, 1940). His **Gesell Developmental Schedules** place observed behaviour at an age level in four fields: motor, adaptive (solving problems with objects, not adaptive behaviour in the Vineland sense), language and personal-social, each giving a DQ.

Nancy Bayley (1899 to 1994) built the leading diagnostic infant test, the **Bayley Scales of Infant Development** (BSID, 1969). The BSID-II (1993) gave a Mental Development Index

(MDI) and a Psychomotor Development Index (PDI). The Bayley-III (2006), renamed Infant and Toddler Development, and the Bayley-4 (2019) have five scales: cognitive, language and motor, tested with the child, and social-emotional and adaptive behaviour, reported by the caregiver. Its range since 1993, the first month to 42 months, is its signature.

For screening, the **Denver Developmental Screening Test** of **William K. Frankenburg** and **Josiah B. Dodds** (1967; Denver II, 1990) covers birth to 6 years in four sectors: personal-social, fine motor-adaptive, language and gross motor. The result is normal, suspect or untestable, never a quotient. The **McCarthy Scales of Children's Abilities** (**Dorothea McCarthy**, 1972; 2½ to 8½ years) yield a General Cognitive Index. **Virginia Apgar**'s newborn score (1953) and **T. Berry Brazelton**'s Neonatal Behavioral Assessment Scale (1973) are taught in Chapter 6.

The Indian **Developmental Screening Test** (DST) of **J. Bharath Raj** (Mysore, 1977, 1983) places 88 milestones on Indian norms from birth to 15 years and gives a DQ. The **Developmental Assessment Scales for Indian Infants** (DASII) of **Pramila Phatak** and colleagues (Baroda) restandardised the Bayley on Indian infants from birth to 30 months, giving motor and mental DQs. The **Trivandrum Developmental Screening Chart** of **M. K. C. Nair** and colleagues (1991), built from the Baroda norms, screens children under 2 in minutes.

> EXAM TRAP
>
> The scale for **birth to 42 months** across cognitive, language and motor domains is the **Bayley**, not the WPPSI (from 2½ years), the Stanford-Binet (from 2; both Section 2.6) or the McCarthy (from 2½). Do not confuse Bharath Raj's Indian DST with the American Denver, or the DASII (the Indian Bayley) with the TDSC (a Kerala screening chart).

Social maturity: the Vineland Social Maturity Scale ●●

Intelligence tests missed something institution staff saw daily: residents of the same mental age differed hugely in self-care. **Edgar Arnold Doll** (1889 to 1968), director of research at the Vineland Training School in New Jersey, measured that difference with the **Vineland Social Maturity Scale** (VSMS, 1935), the first standardised measure of social competence, or adaptive behaviour.

Its 117 items run in the order in which they are usually achieved, grouped year by year through childhood and in wider bands up to "25 and above", so the scale spans birth to adulthood. They fall into eight categories: self-help general, self-help eating, self-help dressing, self-direction, occupation, communication, locomotion and socialisation. The scale is not given to the person: an informant who knows him well is interviewed, habitually performed items are credited under standardised scoring rules, and the total converts to a social age and an SQ.

The Indian version, **Malin's VSMS**, is by **Arthur J. Malin** of Nagpur, who also adapted the WISC as MISIC (Section 2.6). It keeps the eight categories but has 89 items for birth to 15 years on Indian norms, and its SQ is used with IQ in Indian disability certification.

EXAM TRAP

The VSMS was devised by **Doll**, not Malin (the Indian adaptation), Bharath Raj (the DST), C. M. Bhatia (a performance battery), Terman, Wechsler or Binet. It yields **social age and social quotient**, not mental age and IQ, developmental age and DQ, or emotional age and EQ. Its items run past 25 years, so "applicable from 0 to 25 years only" is false; age-graded items set out in age-level groups (the "separate forms for different age groups" of the options) and standardised scoring are genuine features. Malin VSMS: 89 items, SQ; Bharath Raj DST: 88 items, DQ; both 0 to 15 years.

Adaptive behaviour scales and structured interviews ●●

Sara S. Sparrow, **David A. Balla** and **Domenic V. Cicchetti** rebuilt Doll's scale as the **Vineland Adaptive Behavior Scales** (1984; Vineland-II, 2005; Vineland-3, 2016); the current edition covers birth to 90 years. They measure adaptive functioning in daily life in four domains, communication, daily living skills, socialisation and motor skills (for young children), plus an optional maladaptive behaviour domain; the first three form the **Adaptive Behavior Composite** (mean 100, SD 15). Given as a caregiver interview or parent and teacher forms, the Vineland supplies the adaptive half of an intellectual disability diagnosis. The older **AAMD Adaptive Behavior Scale** (**Kazuo Nihira** and colleagues, 1969) rates adaptive skills in Part One and maladaptive behaviour in Part Two.

The Vineland belongs to a family of clinician-administered instruments with the **K-SADS** (Kiddie Schedule for Affective Disorders and Schizophrenia; **Joaquim Puig-Antich**, 1978), a semi-structured psychiatric interview of parent and child, and the ADI-R and ADOS-2 for autism (below). Table 2.17 compares this section's scales.

EXAM TRAP

The Vineland measures **adaptive behaviour**, not IQ or verbal IQ, personality, psychopathology, psychotic symptoms, neurological soft signs, memory deficits or suicidal ideation. The Vineland-3, K-SADS, ADI-R and ADOS-2 are **interviews and observation schedules**, not EEG paradigms, projective techniques or self-report cognitive tests.

Table 2.17 Developmental, social maturity and adaptive behaviour scales compared.

Scale	Author, year	Items or parts	Ages	Score
Gesell Schedules	Arnold Gesell, 1925, 1940	4 fields	Infancy to 5 or 6 years	DQ
Bayley	Nancy Bayley, 1969 to 2019	5 scales from 2006	1 to 42 months	MDI, PDI (BSID-II); composites
Denver II	Frankenburg, Dodds, 1967, 1990	4 sectors	0 to 6 years	Normal, suspect, untestable
McCarthy Scales	Dorothea McCarthy, 1972	18 subtests	2½ to 8½ years	General Cognitive Index
DST (India)	J. Bharath Raj, 1977, 1983	88 items	0 to 15 years	DQ
DASII (India)	Pramila Phatak, Baroda	Motor, mental	0 to 30 months	Motor and mental DQ
TDSC (India)	M. K. C. Nair, 1991	Short chart	0 to 2 years	Delay or not

Scale	Author, year	Items or parts	Ages	Score
VSMS	Edgar A. Doll, 1935	117 items, 8 categories	Birth to 25 and above	SA, SQ
Malin's VSMS	A. J. Malin, Nagpur	89 items, 8 categories	0 to 15 years	SA, SQ
Vineland ABS	Sparrow, Balla, Cicchetti, 1984	4 domains	Birth to 90 (since 2005)	ABC, mean 100
AAMD ABS	Kazuo Nihira, 1969	2 parts	Children, adults	Domain scores

Assessing intellectual disability ●●

DSM-5 (2013) diagnoses **intellectual disability** (formerly mental retardation) on three criteria. (A) Deficits in intellectual functions such as reasoning and learning, confirmed by clinical assessment and individual standardised testing: about 2 SD below the mean, an IQ of about 65 to 75 allowing for measurement error (Section 2.3). (B) Deficits in adaptive functioning in conceptual, social and practical domains. (C) Onset in the developmental period. The disorder itself is in Chapter 8.

Severity (mild, moderate, severe, profound) is graded by **adaptive functioning**, not IQ, since that decides the support needed. Older systems used IQ bands (Table 2.18): ICD-10's mild 50 to 69, moderate 35 to 49, severe 20 to 34, profound below 20, and DSM-IV-TR's overlapping bands. ICD-11 (in force 2022) renamed the condition **disorders of intellectual development**, adding a provisional level.

Old educational labels set expectations: mild was **educable** (academics to about sixth grade by the late teens), moderate trainable (about second grade), severe and profound custodial. So an IQ of about 50, at the mild-moderate border, legitimately predicts learning to about fifth grade, not lifelong institutional care, constant supervision or high-school graduation. And **borderline intellectual functioning** (IQ about 71 to 84) is not intellectual disability, and an FSIQ of 70 without adaptive deficits is no diagnosis.

Table 2.18 Levels of intellectual disability by IQ band (DSM-5 grades by adaptive functioning instead). Indian percentages are those widely quoted from the earlier certification guidelines.

Level	ICD-10 IQ (adult mental age)	DSM-IV-TR IQ	Old label	Indian disability
Borderline (not ID)	Not a category	71 to 84	None	25 percent (IQ 70 to 79)
Mild	50 to 69 (9 to under 12 years)	50 to 55 up to about 70	Educable	50 percent
Moderate	35 to 49 (6 to under 9)	35 to 40 up to 50 to 55	Trainable	75 percent
Severe	20 to 34 (3 to under 6)	20 to 25 up to 35 to 40	Custodial	90 percent
Profound	Below 20 (under 3)	Below 20 or 25	Custodial	100 percent

> **EXAM TRAP**
>
> An RML 2020 paper keys severe intellectual disability as IQ 21 to 34; the standard ICD-10 band is **20 to 34**, so pick 21 to 34 only when 20 to 34 is absent. An FSIQ of 70 is never "mild ID solely on score": without adaptive deficits the keyed answer is borderline intellectual functioning, because ID needs both criteria.

Global developmental delay or intellectual disability? ●●

What if the child is 3 and cannot sit through an IQ test? DSM-5 reserves **global developmental delay** for children under 5 years who miss milestones in several areas but are too young for reliable standardised testing; a DQ describes them until they are reassessed. From 5 years the diagnosis is intellectual disability. The key difference is the **age of assessment**. So the label is inappropriate for a 7-year-old with an FSIQ of 65 and severe adaptive impairment, who meets criteria for intellectual disability, but fits a child too young to test or one whose test or adaptive data are incomplete (Figure 2.11).

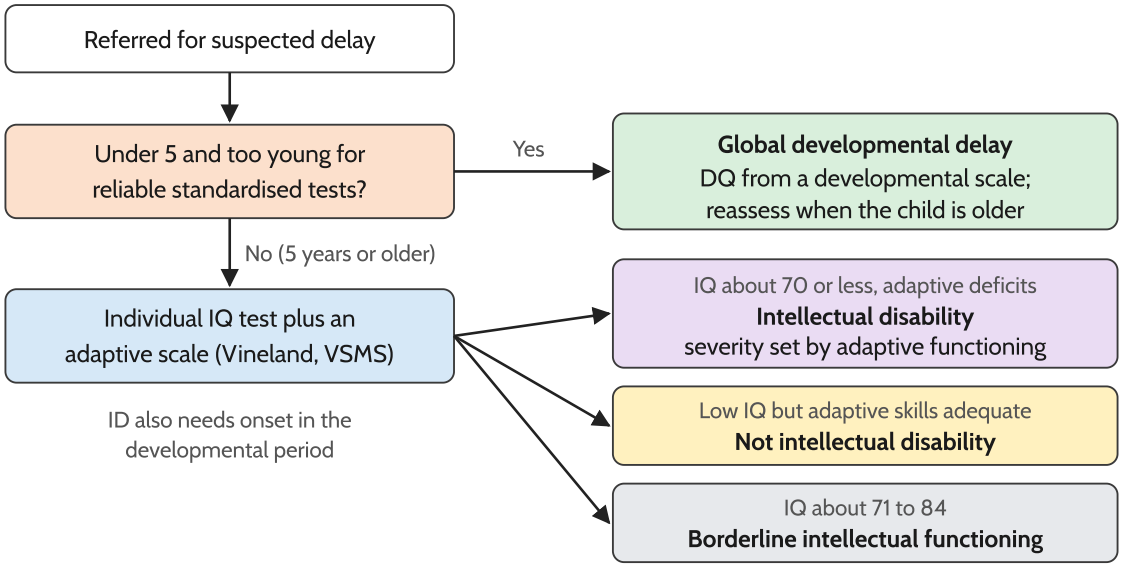


Figure 2.11 Age decides between global developmental delay and intellectual disability; after 5, both IQ and adaptive deficits are needed.

> **EXAM TRAP**
>
> Global developmental delay versus intellectual disability turns on the **age of assessment** (under 5 versus 5 and above), not on social deficits, language delay alone or motor impairment.

Disability certification in India ●

The **Rights of Persons with Disabilities Act 2016** (RPWD Act) recognises 21 disabilities, up from 7 under the 1995 Act, among them intellectual disability, specific learning disabilities, autism spectrum disorder and mental illness. Forty percent or more of a specified disability is a **benchmark disability**, which opens reservations and other benefits. Guidelines notified under the Act in 2018 set out how each disability is assessed.

For intellectual disability, IQ comes from Indian or Indian-normed tests (Binet-Kamat, MISIC, WAPIS, Bhatia's battery, the Seguin Form Board, Raven's matrices: Section 2.6), the social quotient from the VSMS, and a DQ for children under 5, who are certified with global developmental delay. A medical board that includes a clinical psychologist certifies. The percentages widely quoted for each IQ band (Table 2.18) come from earlier guidelines; check them against the current notification.

A **specific learning disability** (dyslexia, dysgraphia, dyscalculia; Chapter 8) is assessed and certified only from 8 years of age, since before then ordinary variation in learning to read, write and calculate cannot be told apart from a disorder. An intelligence test shows that general ability is not the problem, and the **NIMHANS Index for Specific Learning Disabilities**, with curriculum-based tests, measures the deficits. Disability from mental illness is rated with IDEAS, the Indian Disability Evaluation and Assessment Scale (Section 2.11).

> EXAM TRAP
>
> The minimum age for assessing specific learning disability under the RPWD guidelines is **8 years**, not 7, 10 or 14.

Assessing autism: ISAA, CARS and the diagnostic pair ●●

Two rating scales dominate Indian autism assessment (the disorder is in Chapter 8). In the **Childhood Autism Rating Scale** (CARS) of **Eric Schopler**, **Robert J. Reichler** and **Barbara Rochen Renner** (North Carolina; 1980, manual 1988; CARS-2, 2010), the clinician observes the child, hears the parents and rates 15 areas, such as relating to people and imitation, from 1 (normal for age) to 4 (severely abnormal). Totals run from 15 to 60; 30 and above indicates autism.

The **Indian Scale for Assessment of Autism** (ISAA) was developed in 2009 by the National Institute for the Mentally Handicapped, Secunderabad, partly from CARS items. Its 40 items cover six domains (social relationship and reciprocity, emotional responsiveness, speech-language and communication, behaviour patterns, sensory aspects, cognitive component), each rated 1 (rarely) to 5 (always) from observation and parent interview. Totals run from 40 to 200: below 70 is not autism, and higher scores grade mild, moderate or severe autism with a disability percentage. ISAA is the instrument prescribed for certifying autism in India.

Diagnosis pairs the **Autism Diagnostic Interview-Revised** (ADI-R; **Michael Rutter**, **Ann Le Couteur** and **Catherine Lord**, 2003), a 93-item parent interview, with the **Autism Diagnostic Observation Schedule** (ADOS-2; Lord and colleagues, 2012), structured play and conversation in one of five modules. The M-CHAT-R/F (**Diana Robins** and colleagues, 2009) screens toddlers of 16 to 30 months with 20 parent questions. Not autism tools: the **Children's Global Assessment Scale** (CGAS; **David Shaffer**, 1983), a single 1 to 100 rating of overall functioning, and the Child Behavior Checklist (Section 2.11).

> EXAM TRAP
>
> IHBAS papers of 2020 and 2024 key **CARS** for severity and disability in autism, even for "the Indian tool" with ISAA offered. CARS is American; the Indian instrument prescribed for certifying autism disability is **ISAA**, which drew on CARS. Choose ISAA when offered. DAAS, PAAS, CGAS and the CBCL are distractors.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. DQ is developmental age divided by _____ age, multiplied by 100.
2. The VSMS was devised in 1935 by _____ at the Vineland Training School.
3. The infant scale for the first month to 42 months is the _____.
4. Malin's Indian VSMS has _____ items; Bharath Raj's DST has 88.
5. DSM-5 grades the severity of intellectual disability by _____ functioning.
6. Global developmental delay is diagnosed only under the age of _____ years.
7. In India, specific learning disability is certified only from _____ years, and autism with the _____.

Answers: (1) chronological, (2) Edgar A. Doll, (3) Bayley Scales, (4) 89, (5) adaptive, (6) 5, (7) 8; ISAA.

2.8 Aptitude, Achievement, Creativity, Interest and Attitude Measures

LOWER YIELD · 8 QS

Two Class 10 students in Mysuru both score 85 percent in mathematics. The marks show what each has learnt, but not what each could still learn, how original each is, what each enjoys or how each feels about a career in engineering. Each of those questions has its own family of tests.

Aptitude, achievement and intelligence

An **achievement test** measures what a person has already learnt: the knowledge and understanding gained in a specific academic area, usually through instruction. An **aptitude test** looks forward, estimating how well a person will learn or succeed in a specific area in future, such as mechanical, musical or clerical work. An intelligence test samples general ability and gives one broad score (Section 2.6). Achievement looks back and aptitude looks ahead, so achievement tests rest mainly on content validity and aptitude tests on predictive validity (Section 2.4).

Anne Anastasi (1908 to 2001) stressed that the difference is one of degree: all ability tests measure developed ability, and they lie on a continuum by how specific and how controlled the underlying learning is. Course-based achievement tests (a term of algebra) sit at one end; broad achievement tests, verbal aptitude and intelligence tests, and non-verbal tests follow; culture-fair tests, drawing on the least controlled everyday experience, sit at the other.

A teacher-made test serves one class and has no norms; a standardised achievement test has tried-out items, fixed administration and norms. By purpose, a **survey test** gives an overall level of attainment, a **diagnostic test** pinpoints specific weaknesses (exactly which kinds of subtraction a child gets wrong), and a **prognostic test** forecasts success in a later course.

Clinicians use individual batteries: the Wide Range Achievement Test (**Joseph Jastak**, 1946; reading, spelling and arithmetic), the Peabody Individual Achievement Test (**Lloyd Dunn** and **Frederick Markwardt**, 1970; many items answered by pointing), the Wechsler Individual Achievement Test (1992) and the Woodcock-Johnson Tests of Achievement (**Richard Woodcock** and **Mary Bonner Johnson**, 1977). The last two are co-normed with

ability tests to assess the **ability-achievement discrepancy**: achievement well below what measured intelligence predicts has traditionally signalled a specific learning disability (Section 2.7).

> EXAM TRAP
>
> The primary purpose of an achievement test is to measure understanding and knowledge in specific academic areas. "Future success potential" is aptitude, "cognitive abilities related to brain function" is neuropsychological testing (Section 2.11), and "unconscious conflicts" belong to projective techniques (Section 2.10).

Aptitude batteries and special aptitude tests

One intelligence score told a careers counsellor little. After **Louis Leon Thurstone** (1887 to 1955) split intelligence into primary mental abilities (Section 1.9), the 1940s produced the **multiple aptitude battery**: separately normed subtests that give a profile instead of one number. The **Differential Aptitude Tests** (DAT) of **George K. Bennett**, **Harold G. Seashore** and **Alexander G. Wesman** (1947), built to guide secondary school students, has eight subtests: verbal reasoning, numerical ability, abstract reasoning, clerical speed and accuracy, mechanical reasoning, space relations, spelling and language usage. Verbal reasoning plus numerical ability indexes scholastic aptitude, and **J. M. Ojha** adapted the DAT for India. The **General Aptitude Test Battery** (GATB) of the United States Employment Service (1947) measures nine aptitudes, from general learning ability to finger and manual dexterity, to match people to occupations, and the Armed Services Vocational Aptitude Battery (1968) serves US military recruitment.

A special aptitude test measures one narrow talent; an early classic is the Measures of Musical Talents of **Carl Seashore** (1866 to 1949), published in 1919 (Table 2.19). Admission tests such as the SAT, ACT and GRE are scholastic aptitude tests predicting success in college or graduate study.

Table 2.19 Classic special aptitude tests.

Test	Aptitude	The examinee
Seashore Measures of Musical Talents (1919)	Musical	Compares tones for pitch, loudness and time; recalls tone sequences
Meier Art Judgment Test	Artistic	Picks the better of an original and an altered artwork
Minnesota Clerical Test	Clerical	Checks pairs of numbers and names, at speed
Bennett Mechanical Comprehension Test	Mechanical	Reasons about pictured gears, pulleys and levers
Minnesota Paper Form Board	Spatial	Picks the figure that loose pieces would form

> EXAM TRAP
>
> Carl Seashore (music, 1919) is not Harold Seashore (DAT, 1947). The DAT is a multiple aptitude battery of eight subtests, not an intelligence test; the GATB has nine aptitudes.

Tests in the guidance programme

These batteries serve guidance, the organised help a school gives students in educational and vocational choices. A guidance programme's tools are the case study (an in-depth picture of one pupil from all sources), the interview (a face-to-face conversation to gather information and advise), the **cumulative record** (a running file of marks, test scores, health and activities kept over the years), observation, and tests of intelligence, aptitude, achievement and interest. Counselling then draws on them all.

> EXAM TRAP
>
> Guidance textbooks call *counselling* "the heart of the guidance programme", so choose counselling whenever it is offered. RML 2022 offered only case study, interview, cumulative record and aptitude test, and keyed the aptitude test, which no standard text supports: an aptitude test only predicts trainability. With exactly those four options, choose the interview, counselling's chief instrument and itself called the heart of the counselling process (Section 7.10), and expect some keys to differ.

Creativity and divergent thinking

Intelligence tests reward **convergent thinking**, narrowing information down to one correct answer. In his presidential address to the American Psychological Association in 1950, **Joy Paul Guilford** (1897 to 1987) tied creativity to **divergent thinking**, producing many varied answers to an open problem, the operation he called divergent production in his Structure of Intellect model (Section 1.9). His Alternate Uses task asks for uses of an object such as a brick. If Kavya writes wall, house, doorstop, paperweight and red powder for rangoli, her **fluency** (number of ideas) is five; her **flexibility** (number of categories: building, weighting, pigment) is three; her **originality** is the statistical rarity of each idea (rangoli powder scores, a wall does not); and **elaboration** is the detail she adds.

Ellis Paul Torrance (1915 to 2003) standardised such tasks in the **Torrance Tests of Creative Thinking** (1966), with verbal and figural (picture-completing) forms. **Sarnoff Mednick** (1962), in his associative theory, held that creative people reach remote associations easily. His **Remote Associates Test** (RAT) is a timed paper-and-pencil test of about 40 minutes in which the examinee finds one word linking three cue words: cottage, Swiss, cake gives "cheese". **Michael Wallach** and **Nathan Kogan** (1965), testing children in relaxed, game-like conditions, found creativity largely independent of intelligence. The Indian tests are the verbal and non-verbal Tests of Creative Thinking of **Baqer Mehdi** (1973) and the six Passi Tests of Creativity (**B. K. Passi**, 1979).

> EXAM TRAP
>
> "A timed, 40-minute paper-and-pencil test" is the RAT. Each item has one right answer, so its format is convergent, unlike the TTCT. The Alternate Uses task is Guilford's, not Torrance's.

Interest inventories and value scales

An **interest inventory** asks which activities, subjects and occupations a person likes. On the empirical route an item joins an occupational scale if people in that occupation answer it dif-

ferently from people in general, whatever its content; on the homogeneous route items are grouped by content into interest areas.

Edward Kellogg Strong Jr. (1884 to 1963) took the empirical route in the **Strong Vocational Interest Blank** (1927), revised by **David Campbell** as the Strong-Campbell Interest Inventory (1974) and now the Strong Interest Inventory. **G. Frederic Kuder** (1903 to 2000) took the homogeneous route in the Kuder Preference Record (1939): the person marks the most and least liked activity in each forced-choice triad, and the answers score ten areas (outdoor, mechanical, computational, scientific, persuasive, artistic, literary, musical, social service, clerical). As each choice raises one area by lowering another, the scores are **ipsative**: they show relative strengths within the person, not standing against others (Section 2.5). The later Kuder Occupational Interest Survey compares people with occupational groups.

John L. Holland (1919 to 2008), in a theory first published in 1959 and set out fully in 1973, sorted people and work environments into six **RIASEC** types (Realistic, Investigative, Artistic, Social, Enterprising, Conventional) arranged on a hexagon: adjacent types are alike, opposite ones (Realistic and Social) least alike. His self-scored **Self-Directed Search** (1971) gives a three-letter code matched to occupations.

For values, the Study of Values of **Gordon Allport** (1897 to 1967) and **Philip Vernon** (1931; with **Gardner Lindzey** from 1951; revised 1960) measures six values from the types of **Eduard Spranger** (1882 to 1963): theoretical, economic, aesthetic, social, political and religious. **Milton Rokeach** (1973) had people rank 18 terminal values (end states such as freedom) and 18 instrumental values (conduct such as honesty).

Thurstone and Likert scales ●●

An attitude is an evaluative disposition towards an object such as a group or an idea (its components and theories are in Chapter 3). An attitude scale infers it from responses to statements, combined into one score from unfavourable to favourable. The classic scales differ in how the statements are chosen and how the score is formed.

Thurstone and **Ernest J. Chave** (1929), measuring attitude towards the church, devised the **method of equal-appearing intervals**. Judges sorted over a hundred statements into 11 piles from most unfavourable to most favourable, without giving their own opinion. A statement's **scale value** is the median pile; its ambiguity is Q, the interquartile range of the placements, and high-Q statements are dropped. About 20 statements with evenly spread values form the scale. The respondent ticks those they agree with and scores the median (or mean) of their values: agreeing with statements valued 7.2, 8.1 and 8.4 gives 8.1, a favourable attitude.

This is laborious. **Rensis Likert** (1903 to 1981) offered a short cut in 1932, the **method of summated ratings**. A **Likert scale** is a set of closed-ended statements, each with a fixed number of ordered alternatives, classically five from strongly agree to strongly disagree, scored 5 to 1 (reversed for unfavourable statements). Respondents rate their own agreement on this symmetric continuum, and item scores are summed with equal weights. No judges are used: items are kept by item analysis if high and low total scorers answer them differently (Section 2.2). Easier to build and usually as reliable, Likert scales dominate questionnaires; strictly ordinal, their scores are usually treated as interval. Figure 2.12 contrasts the two methods.

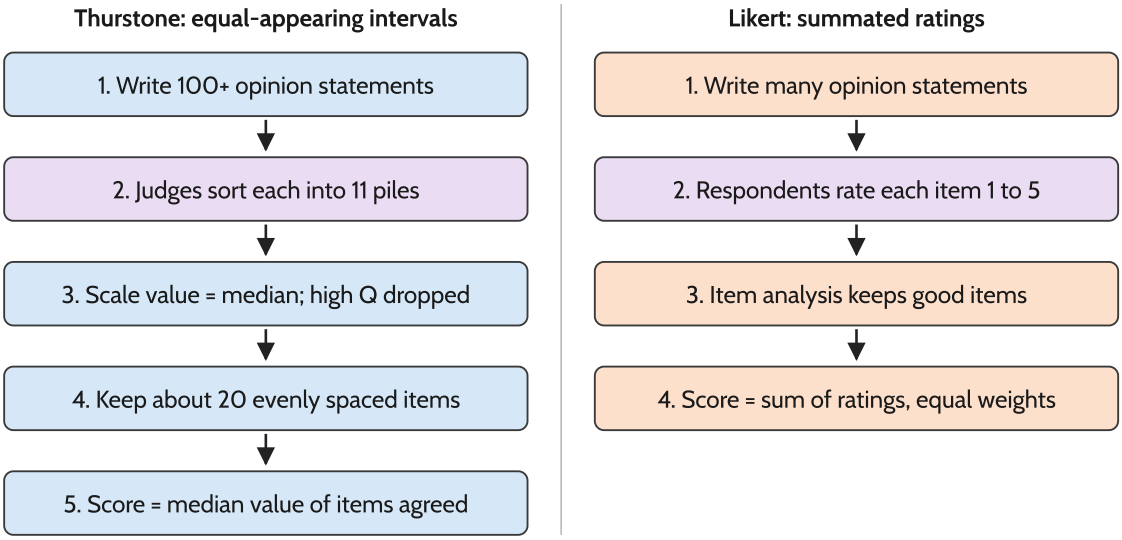


Figure 2.12 The step shaded alike is the key difference: Thurstone's judges pre-scale the statements, whereas Likert's respondents rate their own agreement and the ratings are summed.

EXAM TRAP

A summated rating scale is a Likert scale, never a Thurstone, Guttman or equal-appearing interval scale. Likert items are closed-ended with a fixed number of ordered alternatives, neither open-ended nor dichotomous. Judges pre-scaling statements is Thurstone; respondents rating their agreement, summed with equal weights, is Likert; a sentence-completion stimulus belongs to projective techniques (Section 2.10). Likert is a type of attitude scale, whereas nominal, ordinal and interval are levels of measurement (Chapter 9).

Guttman, Bogardus and the semantic differential

Louis Guttman (1916 to 1987) built a one-dimensional **cumulative scale** by scalogram analysis (1944). Statements are ordered so that endorsing a stronger one implies endorsing every weaker one ("I would accept a de-addiction centre in my city", "in my locality", "on my street"), so the total score reveals the exact pattern of answers. Fit is judged by the **coefficient of reproducibility**, which should be 0.90 or more.

The **social distance scale** of **Emory Bogardus** (1882 to 1973), published in 1925, is generally regarded as the oldest attitude scale. For each group the respondent marks the closest relation they would accept, on seven steps: close kinship by marriage, club chums, neighbours, co-workers, fellow citizens, visitors only, and exclusion from the country. In the **semantic differential** of **Charles Osgood**, **George Suci** and **Percy Tannenbaum** (1957), a concept such as "the police" is rated on seven-point bipolar adjective pairs (good to bad, strong to weak, fast to slow), which factor into evaluation, potency and activity; evaluation is the attitude proper.

In paired comparisons each object is paired with every other and one of each pair is chosen; Thurstone's **law of comparative judgement** (1927) converts the choices into scale values, and rank ordering is a quicker variant. **Allen Edwards** and **Franklin Kilpatrick** (1948) combined Thurstone judging, Likert item analysis and Guttman scaling in the **scale-discrimination technique**. Table 2.20 compares the main scales.

Table 2.20 The classic attitude scales compared.

Scale	Author, year	Respondent	Score	Hallmark
Equal-appearing intervals	Thurstone and Chave, 1929	Ticks agreed statements	Median scale value	Judges, 11 piles, Q
Summated ratings	Likert, 1932	Rates every item, usually 5 points	Sum of ratings	Item analysis, no judges
Cumulative	Guttman, 1944	Agrees or not with graded items	Number endorsed	Reproducibility 0.90
Social distance	Bogardus, 1925	Marks closest relation accepted	Step, 1 to 7	Oldest attitude scale
Semantic differential	Osgood, Suci and Tannenbaum, 1957	Rates a concept on bipolar adjectives	Factor means	Evaluation, potency, activity

Implicit and indirect measures

Direct scales invite socially desirable answers, and some bias lies outside awareness. In the **Implicit Association Test** (IAT) of **Anthony Greenwald**, **Debbie McGhee** and **Jordan Schwartz** (1998), words or pictures are sorted on a computer with two keys into paired categories, such as "flower or pleasant" versus "insect or unpleasant", and then with the pairing reversed. Sorting is faster when concepts sharing a key are already linked in memory, so reaction times index implicit, often unconscious attitudes and stereotypes. In the **bogus pipeline** of **Edward Jones** and **Harold Sigall** (1971), people answer more honestly when wired to a machine they believe detects true feelings. Physiological indices (skin conductance, pupil size, facial muscle activity) show how strongly a person reacts, rarely in which direction.

EXAM TRAP

The IAT measures implicit, unconscious attitudes and stereotypes through reaction times. It does not measure verbal ability, hidden mathematical ability or dream content.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. A test of knowledge already gained in a specific academic area is an _____ test.
2. The number of different categories of response in a divergent task is _____.
3. The RAT asks for one word linking _____ cue words.
4. Kuder's forced-choice triads yield _____ scores.
5. A summated rating scale is a _____ scale.
6. A Guttman scale needs a coefficient of reproducibility of at least _____.

Answers: (1) achievement, (2) flexibility, (3) three, (4) ipsative, (5) Likert, (6) 0.90.

2.9 Objective Personality Inventories

MEDIUM YIELD · 38 QS

In 1917, as the United States entered World War I, its army needed a quick way to spot recruits likely to break down under fire, and there were far too few psychiatrists to interview them

all. **Robert Sessions Woodworth** (1869 to 1962) put the psychiatrist's screening questions on paper, and the personality inventory was born. Ever since, psychologists have argued over how to choose the questions and how far to trust the answers.

Objective tests and Cattell's three kinds of data ●●

An **objective personality test** fixes the items, the answer format (true or false, yes or no, a rating) and the scoring key, so any two scorers get the same result. Rating scales and **personality inventories** (long self-report questionnaires scored on several scales) are objective; projective tests (Section 2.10) use ambiguous stimuli, free responses and interpretive scoring.

Raymond Bernard Cattell (1905 to 1998), whose trait theory is taught in Chapter 4, accepted a trait only if it appeared in three kinds of data:

- **L-data** (life record data): actual behaviour in everyday life, from school performance, work and criminal records, and ratings by observers who know the person.
- **Q-data** (questionnaire data): self-report ratings of one's own characteristics, attributes and interests.
- **T-data** (objective test data): behaviour in a controlled test situation whose purpose the person cannot see, and so cannot easily fake, such as a laboratory task scored for a trait he does not know is being measured.

> EXAM TRAP
>
> Cattell's scheme has exactly three letters, **L, Q and T**: there is no P-data (projective) or Z-data. School and criminal records are L-data; self-report questionnaires are Q-data; projective-test performance and Q-sort responses are not L-data. Q-data is not the Q-sort (2.9.7).

Four ways to build an inventory ●

The **logical-content** or **rational strategy** reasons out how people with a trait (anxiety, say) describe themselves and writes items whose content plainly reflects it, trusting face validity. Woodworth's **Personal Data Sheet** (built in 1917, published in 1919 and often dated 1920), the first personality inventory, asked 116 yes-no questions about symptoms and gave one score of emotional instability; **Hugh M. Bell**'s Adjustment Inventory (1934) scored home, health, social and emotional adjustment. Its weakness: respondents see what each item is after, and the method assumes insight and honesty.

The **theoretical strategy** writes items for the constructs of a personality theory: the Edwards Personal Preference Schedule (Henry Murray's needs), the Myers-Briggs Type Indicator (Carl Jung's types) and the Millon Clinical Multiaxial Inventory (Theodore Millon's theory).

Empirical criterion-group keying ignores content. A large item pool goes to a criterion group (say, patients diagnosed with depression) and a normal control group; only the items the two groups answer differently survive, whatever they say, and the scale is cross-validated on fresh groups. If depressed patients happened to answer "true" to "I like mechanics magazines" more often than controls, that item would join the depression scale. The MMPI, the California Psychological Inventory and the Strong Vocational Interest Blank (Section 2.8) were built this way.

The **factor-analytic strategy** factor-analyses the correlations among many items, keeps the items that load on each factor and names the factor afterwards: the 16PF, the Eysenck inventories, the NEO-PI-R, the **Guilford-Zimmerman Temperament Survey** (**J. P. Guilford** and Wayne Zimmerman, 1949; 300 items, ten traits such as sociability and restraint) and the **Comrey Personality Scales** (**Andrew L. Comrey**, 1970; 180 items, eight factors such as trust and orderliness). Modern inventories, such as the NEO-PI-R and the MCMI, use **combined strategies**.

Against response sets (Section 2.2) and deliberate faking, test makers use rapport and confidentiality, **forced-choice** items pairing statements of equal social desirability, disguised items, and **validity scales** and correction keys.

> EXAM TRAP
>
> The MMPI was derived by finding items that discriminate between criterion and control groups, **not** by the rational method or by assuming traits and writing questions for them. Comrey's 180-item test is **factor-analytic**. "Correlational" and "experimental" are research designs (Chapter 9), and criterion-referencing is a way of interpreting scores (Section 2.5), not of choosing items.

The MMPI and its validity scales ●●●

Around 1940, at the University of Minnesota Hospitals, the psychologist **Starke R. Hathaway** (1903 to 1984) and the neuropsychiatrist **J. Charnley McKinley** (1891 to 1950) set out to aid psychiatric diagnosis. They gave over a thousand statements to patient groups with known diagnoses and to about 700 "Minnesota normals", mostly hospital visitors, and kept for each scale only the items that separated that patient group from the normals. The **Minnesota Multiphasic Personality Inventory** (MMPI, 1943) had 550 different statements (566 printed, 16 repeated), answered true, false or cannot say. It is "multiphasic" because its items cover physical health, mood, family, sexual, religious and social attitudes, fears and unusual experiences.

The MMPI is therefore an **objective personality inventory** of **personality and psychopathology**, used mainly for psychiatric diagnosis. It is not a projective test, intelligence test, aptitude battery, interest inventory or neuropsychological screening battery, nor a tool for educational, career or physical-health assessment, and it does not measure unconscious drives (projective tests), the Big Five (the NEO) or leadership potential (the CPI). Because self-reports can mislead, it carries validity scales:

- **Cannot Say (?)**: items left blank or marked both ways; 30 or more make the record doubtful.
- **L (Lie) scale**: small faults nearly everyone admits, of the kind "I get angry sometimes". Answering "false" to them asserts a degree of personal virtue rarely seen in our culture, so a high L marks a *naive* attempt to look good by denying common minor faults.
- **F (Infrequency) scale**: items fewer than one normal person in ten endorses. A high F flags random or careless responding, over-reporting of symptoms (faking bad) or severe psychopathology; on the MMPI-2 an F above T 100 suggests random responding or gross exaggeration.

- **K (Correction) scale**: added by **Paul E. Meehl** (1920 to 2003) and Hathaway in 1946 to catch *subtle* defensiveness. A fraction of K is added to clinical scales 1, 4, 7, 8 and 9 (the **K-correction**) to compensate for defensive responding.

The MMPI-2 added **VRIN** (Variable Response Inconsistency: contradictory answers to similar item pairs, a sign of random responding), **TRIN** (True Response Inconsistency: answering "true", or "false", regardless of content), **F(B)** (Back F, for later items), **F(p)** (Infrequency-Psychopathology, items rare even among psychiatric inpatients, which separates faking bad from real disorder) and **S** (Superlative Self-Presentation).

> EXAM TRAP
>
> The three classic validity scales are **L, F and K** (four if Cannot Say is counted, as some texts do); VRIN, TRIN and S came later. L is naive faking good, K is subtle defensiveness plus a correction, and F is rare responding (random answers, faking bad or severe pathology). A very high F never means faking good, defensiveness or an excellent profile.

Clinical scales, profiles and the MMPI family ●●

The MMPI has **ten clinical scales** (Table 2.21): not 4, 13 or 16 (the 13 "basic scales" are the ten plus L, F and K). Eight were keyed against psychiatric diagnostic groups; scales 5 and 0 were not. Scales 1, 2 and 3 form the **neurotic triad**; 6, 7, 8 and 9 the **psychotic tetrad**.

Table 2.21 The ten MMPI clinical scales.

Scale	Name	High scores suggest
1 Hs	Hypochondriasis	Bodily preoccupation, vague physical complaints
2 D	Depression	Low mood, hopelessness, slowed activity
3 Hy	Hysteria	Physical symptoms under stress, denial of problems
4 Pd	Psychopathic Deviate	Antisocial behaviour, conflict with family and authority, impulsivity, alienation
5 Mf	Masculinity-Femininity	Interests atypical for one's sex (not pathology)
6 Pa	Paranoia	Suspiciousness, ideas of persecution
7 Pt	Psychasthenia	Anxiety, worry, obsessions and compulsions
8 Sc	Schizophrenia	Odd thinking, alienation, unusual experiences
9 Ma	Hypomania	Overactivity, excitement, flight of ideas
0 Si	Social Introversion	Shyness, social discomfort

Raw scores become T scores (mean 50, SD 10; Section 2.5): T 70 or more was elevated on the MMPI, and the MMPI-2 uses uniform T scores with a cut-off of 65. Clinicians read the **code type**, the two highest scales: **2-7** suggests anxious, worried depression; **4-9** impulsive, antisocial acting out; and the **conversion V** (1 and 3 high, 2 clearly lower) a person who turns stress into physical complaints without feeling much distress (Figure 2.13).

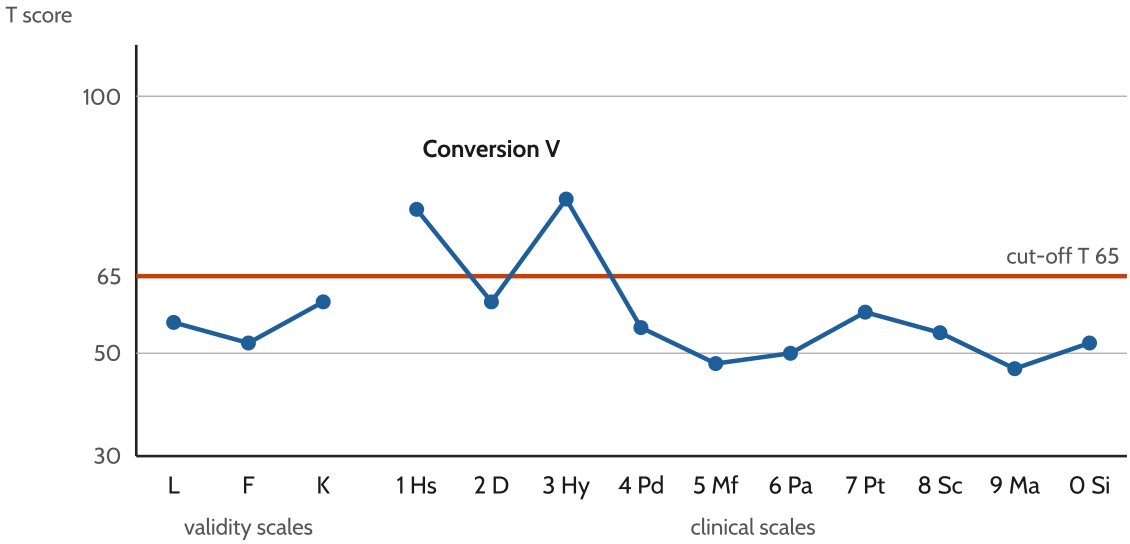


Figure 2.13 A valid MMPI-2 profile (L, F and K near 50) with a conversion V: scales 1 and 3 rise above the dashed cut-off while scale 2 stays lower.

The **MMPI-2** (**James N. Butcher** and colleagues, 1989) has 567 items, rewrote outdated ones, was normed on 2,600 US adults and added content and supplementary scales (such as health concerns and the MacAndrew alcoholism scale). The old clinical scales intercorrelate because all share a general distress factor, **demoralisation**. **Auke Tellegen** and colleagues (2003) extracted it first as **RCd** and rebuilt the **Restructured Clinical (RC) scales** (such as RC2 low positive emotions and RC4 antisocial behaviour) so that each measures a distinctive dimension. RC scales are the core of the **MMPI-2-RF** (**Yossef S. Ben-Porath** and Tellegen, 2008; 338 items) and the **MMPI-3** (2020; 335 items). The **MMPI-A** (1992; 478 items) serves adolescents aged 14 to 18.

INDIAN CONTEXT

H. N. Murthy's **Multiphasic Questionnaire** (MPQ; 1964) is a short (100-item) Indian adult inventory built from MMPI-type items, with a Lie scale and seven clinical scales such as hysteria, depression, anxiety, paranoia and schizophrenia. An elevated MPQ Lie scale indicates **all** of these: a tendency to fake and manipulate, social desirability, and mistrust (a guarded person who admits no flaw to the examiner).

Factor-analytic inventories: 16PF, Eysenck and NEO ●●●

Cattell turned his source traits into the **Sixteen Personality Factor Questionnaire** (16PF), first published in 1949; the fifth edition (1993) has 185 three-choice items for ages 16 and over. It scores **16 primary factors** found by factor analysis and labelled by letter (A warmth, B reasoning, C emotional stability, E dominance, and so on to Q4 tension), plus **five global, second-order factors** (extraversion, anxiety, tough-mindedness, independence, self-control) that approximate the Big Five. Scores are **stens** (standard tens: 1 to 10, mean 5.5, SD 2). Factor B is a brief reasoning scale, but the 16PF measures personality traits, not fluid, crystallised or general intelligence (Section 2.6).

> EXAM TRAP
>
> The 16PF is **Raymond B. Cattell's**: Kelly is to the Rep test as Cattell is to the 16PF. Hans Eysenck built the EPQ, Costa and McCrae the NEO-PI-R, Henry Murray the TAT, Gordon Allport the Study of Values, William Sheldon a body-type scheme, and Freud no inventory.

Hans Jurgen Eysenck (1916 to 1997), at the Maudsley Hospital, London, built three inventories. The **Maudsley Personality Inventory** (1959; 48 items) scored extraversion (E) and neuroticism (N); the **Eysenck Personality Inventory** (1964, with **Sybil B. G. Eysenck**; 57 items) added a Lie scale; and the **Eysenck Personality Questionnaire** (EPQ, 1975; 90 items) added psychoticism (P), giving PEN plus L. The EPQ-R followed in 1985.

Paul T. Costa Jr. and **Robert R. McCrae** built the **NEO Personality Inventory** on the Five-Factor Model (Chapter 4). The **NEO-PI-R** (1992) has 240 items on a five-point agree-disagree scale and scores five domains, **Openness, Conscientiousness, Extraversion, Agreeableness and Neuroticism** (OCEAN), each with **six facets** (30 in all), as T scores. The 60-item **NEO-FFI** scores domains only; the **NEO-PI-3** (2005) simplified the wording. The NEO is not based on Cattell's 16 factors, Eysenck's PEN, Murray's needs or MBTI types (personality-disorder scales belong to the MCMI), and OCEAN is neither the four humours (sanguine, choleric, melancholic, phlegmatic; Chapter 4) nor achievement, affiliation and power motives (Chapter 1). Shorter Big Five measures include the 44-item **Big Five Inventory** (Oliver John and colleagues, 1991); the **HEXACO** inventory of Kibeom Lee and Michael Ashton adds a sixth factor, Honesty-Humility.

Theory-based, clinical and Indian inventories

The **Myers-Briggs Type Indicator** (MBTI) of **Katharine Cook Briggs** (1875 to 1968) and her daughter **Isabel Briggs Myers** (1897 to 1980), first issued in 1943, is directly based on **Carl Gustav Jung**'s theory of psychological types. Forced-choice items place people on four dichotomies, extraversion-introversion, sensing-intuition, thinking-feeling and judging-perceiving, giving 16 types such as INTP. It is popular in counselling and team building but psychometrically weak: continuous scores that bunch near the middle are forced into types, and many people get a different type on retest weeks later.

The **Edwards Personal Preference Schedule** (EPPS) of **Allen L. Edwards** (1953, manual 1954; revised 1959) measures 15 needs from **Henry A. Murray**'s list, such as achievement, order, affiliation, dominance and aggression. To control social desirability, every item is a **forced choice** between two statements matched for desirability (225 pairs: 210 different and 15 repeated for a consistency score). The scores are therefore **ipsative**: each need is strong only relative to the person's other needs, so people cannot be compared directly.

The **California Psychological Inventory** (CPI) of **Harrison G. Gough** (1921 to 2010), published in 1956 (manual 1957), is the normal person's MMPI: largely criterion-keyed, with many MMPI items, but scoring **folk concepts**, everyday trait ideas found in all cultures, such as dominance, sociability, responsibility and self-control. It began with 480 items (now 434) and predicts achievement, leadership and delinquency. The **Millon Clinical Multiaxial Inventory** (MCMI) of **Theodore Millon** (1928 to 2014), first published in 1977, follows Millon's theory and the DSM, measuring personality disorders and clinical syndromes in patients (175 items; 195 in the MCMI-IV of 2015). It reports **base rate scores**, anchored to how common each disorder is in clinical samples rather than to normal T scores.

The **State-Trait Anxiety Inventory** (STAI) of **Charles D. Spielberger** (1927 to 2013) and colleagues (1970; Form Y 1983) has 40 items: 20 ask how you feel right now (state anxiety, passing) and 20 how you generally feel (trait anxiety, a stable proneness). Its purpose is to differentiate state from trait anxiety. Measuring psychoticism, extraversion and neuroticism is the EPQ's job; "normal and abnormal personality" fits broad inventories such as the MMPI and CPI; checking response validity is the job of validity scales. The older **Taylor Manifest Anxiety Scale** of **Janet A. Taylor** (1953) used 50 MMPI items judged to express overt anxiety.

INDIAN CONTEXT

Common Indian tools include **Sinha's Comprehensive Anxiety Test** (A. K. P. Sinha and L. N. K. Sinha), Hindi adaptations of Bell's Adjustment Inventory (Saxena, 1959; Mohsin and Hussain, 1981), the **PGI Health Questionnaire N-1** (N. N. Wig and S. K. Verma, PGIMER Chandigarh), a neuroticism screen derived from the Cornell Medical Index, and Hindi versions of the MPI and EPQ.

Idiographic and situational methods

Some methods study how one person construes the world. In the **Role Construct Repertory Test** (Rep test, 1955) of **George A. Kelly** (1905 to 1967), the person names people who fill roles in his life (mother, a liked teacher, someone disliked). Shown three at a time, he says how two are alike and differ from the third; each answer ("generous versus selfish") is a bipolar **personal construct**, and a **repertory grid** of figures by constructs maps his construct system (theory in Chapter 4).

In the **Q-sort**, set out by **William Stephenson** (1902 to 1989) in 1953, a person sorts statements, such as self-descriptions, into piles from least to most characteristic, with a fixed number per pile forming a forced, quasi-normal distribution (Figure 2.14). Since every sort has the same mean and spread, it shows which statements are more characteristic *of this person*: within-person, not between-person, variation. **Carl Rogers** (1902 to 1987) had clients sort for the real self and the ideal self; Butler and Haigh (1954) found the mean real-ideal correlation rose from about zero before client-centred therapy to about .3 after it (.34 at the end, .31 at follow-up). **Jack Block** (1924 to 2010) built the 100-item **California Q-set** for observers. The Q-sort complements inventories; it never replaced the MMPI.

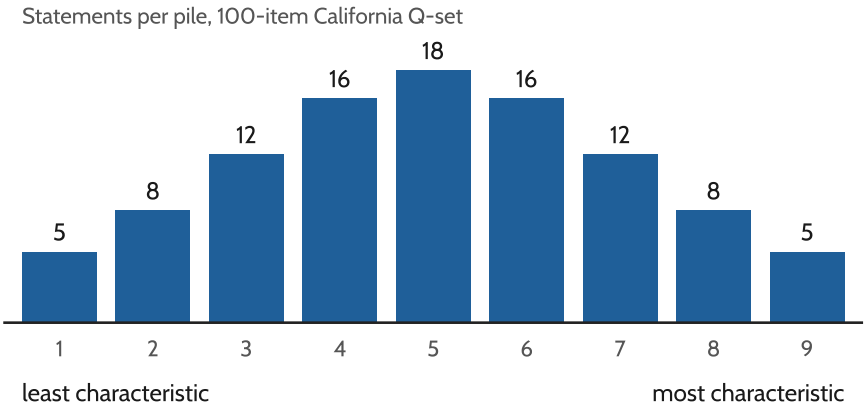


Figure 2.14 A Q-sort forces statements into a fixed quasi-normal shape: only five may go in each extreme pile, eighteen in the neutral middle pile.

On an **adjective checklist**, such as Gough's Adjective Check List of 300 adjectives, the person ticks every word that applies. **Situational tests** watch behaviour in staged real-life tasks. In the Character Education Inquiry, **Hugh Hartshorne** and **Mark A. May** (1928) gave school-children natural-seeming chances to cheat, lie and steal; honesty in one situation predicted honesty in another only weakly. The US Office of Strategic Services (OSS) selected World War II agents through stressful staged tasks; in the **leaderless group discussion** candidates tackle a problem with no appointed leader while assessors rate who leads, as India's Services Selection Board still does.

Judging inventories

Inventories offer objective scoring, large norms, good reliability and economy (one psychologist can test a hundred people at once). Their weaknesses are those of self-report: response sets and faking, limited self-insight, the reading level they demand, and cultural bias when items and norms from one country (rural Minnesota in 1940) travel to another. **Walter Mischel** (1968) added that trait scores predict behaviour in any one situation only weakly, a debate for Chapter 4. Table 2.22 compares the main inventories.

Table 2.22 The main objective personality inventories compared.

Inventory	Author, year	Items	Strategy	Scales and scores
MMPI	Hathaway and McKinley, 1943	550	Empirical keying	?, L, F, K; 10 clinical; T
MMPI-2	Butcher and colleagues, 1989	567	Empirical keying	Adds content scales; uniform T
MMPI-2-RF	Ben-Porath and Tellegen, 2008	338	Restructured	RC scales; T
CPI	Gough, 1956 (manual 1957)	480, now 434	Empirical keying	Folk concepts; T
16PF	Cattell, 1949 (5th edition 1993)	185	Factor-analytic	16 primary, 5 global; stens
EPQ	H. J. and S. B. G. Eysenck, 1975	90	Factor-analytic	P, E, N, L
NEO-PI-R	Costa and McCrae, 1992	240	Factor-analytic, combined	5 domains, 30 facets; T
GZTS	Guilford and Zimmerman, 1949	300	Factor-analytic	10 traits
Comrey Scales	Comrey, 1970	180	Factor-analytic	8 factors
EPPS	Edwards, 1953	225 pairs	Theoretical (Murray)	15 needs; ipsative
MBTI	Briggs and Myers, 1943	Forced choice	Theoretical (Jung)	4 dichotomies, 16 types
MCMI	Millon, 1977	175 to 195	Theoretical, combined	Disorders, syndromes; base rates

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. School performance and criminal records are Cattell's _____-data.
2. The MMPI kept items that discriminated between _____ groups and control groups.
3. Denying common minor faults raises the MMPI _____ scale.
4. A fraction of the _____ scale is added to clinical scales 1, 4, 7, 8 and 9.
5. The RC scales first remove the common factor called _____.
6. The NEO-PI-R scores five domains with _____ facets each.
7. Forced-choice pairs make EPPS scores _____.

Answers: (1) L, (2) criterion, (3) L (Lie), (4) K (Correction), (5) demoralisation (RCd), (6) six, (7) ipsative.

2.10 Projective Techniques HIGH YIELD · 44 QS

Hand someone a symmetrical smudge of ink and ask, "What might this be?" One person sees two dancers, another a crushed bat, a third "blood, because it is red". The blot is the same, so the differences must come from the viewers. That idea produced clinical psychology's most famous and most disputed tests. This section explains it, sorts the tests into five families, works through the Rorschach and the Thematic Apperception Test, then the smaller techniques and the evidence on their worth.

The projective hypothesis ●●●

The objective inventories of Section 2.9 ask clear questions and score fixed answers. A **projective technique** reverses that logic. It presents an ambiguous, unstructured stimulus (an inkblot, a vague picture, half a sentence, a blank page), gives minimal instructions and accepts any answer. Because the stimulus supplies so little structure, whatever structure appears in the response is assumed to come from the person. This is the **projective hypothesis**: faced with an ambiguous stimulus, a person projects needs, conflicts, unconscious processes and a characteristic perceptual style into the response, usually without knowing it.

The word comes from **Sigmund Freud** (1856 to 1939), for whom **projection** was a defence that attributes one's own unacceptable impulse to someone else (Section 4.3). Testing borrowed a broader sense: everyone reads their inner world into an unclear outer one. **Lawrence K. Frank** coined the term **projective methods** in 1939, likening them to an X-ray that shows inner structure without opening the body.

Projective tests share four features: an **ambiguous stimulus**; an **open response**, with almost unlimited possible answers; a **disguised purpose**, so the person cannot tell how answers will be read; and **global interpretation** of the whole personality rather than one trait. They therefore claim to reveal **unconscious processes**, not intellectual attainment, parenting style or career aptitude. Their main advantage follows from the disguise: a person who cannot tell what a "good" answer is cannot easily fake one, so the tests avoid the self-report biases of inventories (social desirability, faking, defensiveness).

EXAM TRAP

The advantage is **avoiding self-report bias.** "They let people use the defence of projection" twists the theory; superior retest reliability and numerical data are strengths of objective inventories. Beside the EPQ, 16PF, MMPI-2, NEO-PI-R, MBTI or MCMI-III, only the **Rorschach and TAT** are projective: the rest are objective self-report inventories (Section 2.9). Intelligence tests and direct observation are not projective either.

Lindzey's five families ●●

Gardner Lindzey (1959) sorted the many techniques by the kind of response they demand (Figure 2.15). In **association techniques** the person reports the first word, image or percept the stimulus calls up. In **construction techniques** the person builds a product, usually a story with a past, present and future. In **completion techniques** the person finishes an incomplete stimulus. In **choice or ordering techniques** the person picks or ranks items by preference, a step towards objective testing. In **expression techniques** the person produces something freely (a drawing, a painting, play), and the manner of production counts as much as the product.

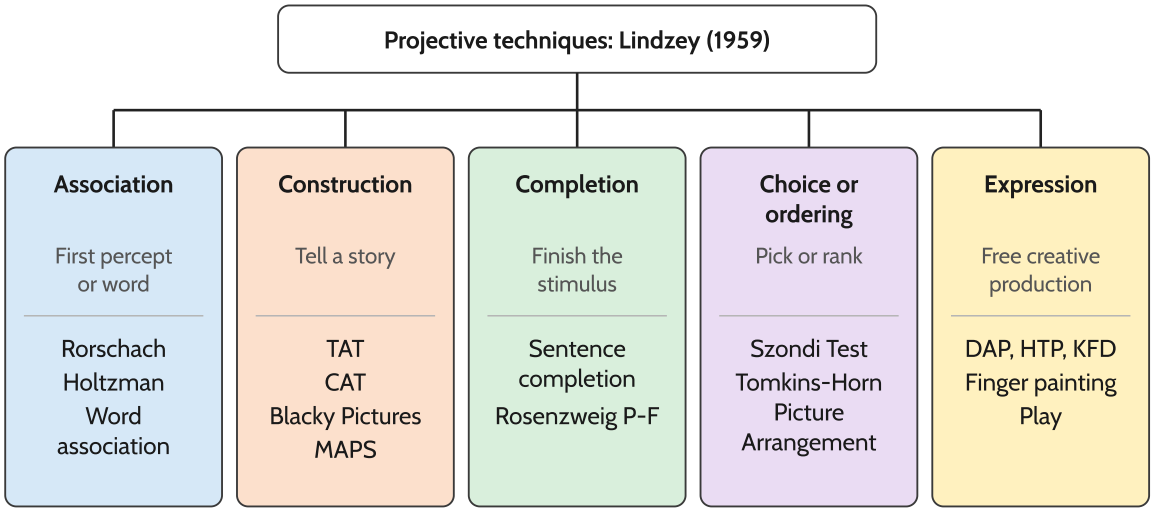


Figure 2.15 Lindzey's five families, named by the response each demands. Anchor one test per family: Rorschach, TAT, sentence completion, Szondi, Draw-A-Person.

Frank (1939) had grouped methods by what the person does with the material: **constitutive** (structuring formless material, as in finger painting), **constructive** (building a directed product), **interpretive** (giving material personal meaning, as in the TAT), **cathartic** (releasing emotion, as in play) and **refractive** (personality refracted through a conventional medium, as in handwriting). The categories overlapped, and Lindzey's replaced them.

EXAM TRAP

The TAT is **construction**, the Rorschach **association**, sentence completion completion and Draw-A-Person expression. RML 2021 keyed the **word association test** as "not a projective test". That key is wrong: Lindzey lists word association as an association technique, and Galton's word association (1879) is usually called the first projective technique. By the standard classification all four options (word association, Rorschach, TAT, sentence completion) are projective, so no option is correct. Learn word association as projective; if the identical item returns, the paper's key was word association.

The Rorschach: ten cards and how they are given ●●●

Hermann Rorschach (1884 to 1922), a Swiss psychiatrist, published the test in *Psychodiagnostik* (1921). From many blots he kept the ten that best separated psychiatric groups, then died the next year, aged 37. The **Rorschach Inkblot Test** has **ten** bilaterally symmetrical cards (not 12, 15 or 20), always shown in the same order. Five are achromatic, black and grey (I, IV, V, VI, VII); two are black and red (II, III); three are multicoloured (VIII, IX, X), as Figure 2.16 shows. Hence the textbook phrase "five black-and-white and five coloured".

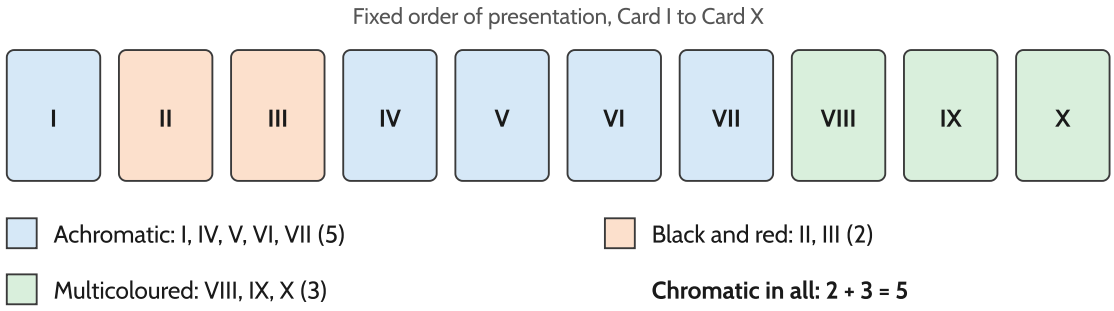


Figure 2.16 The ten Rorschach cards. Five are achromatic; the other five carry colour, two only as red patches on black and three as full pastel colour.

Administration has three phases. In the **response phase** (free association, or performance proper) the examiner asks "What might this be?", records every response verbatim with its reaction time, and does not prompt. In the **inquiry phase**, after all ten cards, the examiner goes back through each response to clarify where on the blot it was seen (location) and what made it look that way (determinants), using non-leading questions ("What made it look like that?"). Optional **testing the limits** follows: the examiner asks directly, for example, whether the person can see a common percept they missed. A longer reaction time, rejection or confusion on the coloured cards than on the achromatic ones is **colour shock**, read as a sign of emotional disturbance.

The Rorschach assesses personality, cognition and perception: how a person organises an ambiguous field, how they think, and whether thinking is disordered. Rorschach built it as a diagnostic aid, so it is used to detect thought disorder, help identify mental illness and infer unconscious intentions. It does not interpret dreams, record conscious desires, measure IQ, memory or brain size, or diagnose physical illness.

EXAM TRAP

The inquiry clarifies each response **without leading questions**; cards are never shuffled, responses are recorded from the first card, and inquiry is not "scoring only". Ten cards, Hermann Rorschach, 1921: not 20 by Henry Murray, 15 by John Exner or 5 by Carl Jung, and no blank card (that is TAT card 16). Five chromatic and five achromatic, not three and seven. RML 2025 keyed "five black and white and five multi-colored cards": the five-five split is right but the wording is loose, since the achromatic cards are black and grey and only VIII, IX and X are fully multicoloured (II and III are black and red). It is still the only correct option; "only black and white", "only coloured" and "ten multicoloured" are wrong.

Scoring and reading a Rorschach record ●●●

Each response is coded first for its **location**, the part of the blot used: **W** the whole blot, **D** a common detail, **Dd** an unusual detail, and **S** white space (always coded with another location, as WS or DS). In Exner's system many W responses, especially well-organised ones, reflect **organisational and integrative effort**, the drive to pull a complex field into one meaningful whole. **Developmental quality** grades that integration: synthesised (+), ordinary (o) or vague (v). The **determinants** record what feature of the blot shaped the percept (Table 2.23); the colour determinants index emotional reactivity and impulse control.

Table 2.23 Rorschach determinants, with an example response and the usual reading.

Code	Determinant	Example	Usually read as
F	Pure form: shape alone	"A bat" (outline only)	Control, reasoning
M	Human movement	"Two people dancing"	Imagination, inner life
FM	Animal movement	"Two bears fighting"	Basic urges, need states
m	Inanimate movement	"A rocket exploding"	Situational stress
FC	Form first, colour adds	"A red butterfly"	Controlled emotion
CF	Colour first, vague form	"Fire"	Weaker emotional control
C	Pure chromatic colour	"Blood, because it is red"	Impulsive affect
Cn	Colour naming	"That is pink"	Serious disorganisation
C'	Achromatic colour	"A black cloud"	Held-in, painful feeling
T, V, Y	Shading: texture, vista (depth), diffuse	"Fur"; "a deep well"; "smoke"	Need for closeness; self-criticism; helplessness
FD	Form dimension: depth from size	"A giant seen from below"	Introspection
Fr, (2)	Reflection; pair	"A girl in a mirror"; "two dogs"	Self-focus

The **form quality** rating shows how well the percept fits the blot's actual shape, from good or ordinary to minus (distorted). It is the main index of **reality testing**: many minus responses mean perception driven by inner preoccupation rather than by the stimulus, as in psychosis. **Content** codes what was seen: H (whole human), Hd (human detail), A (animal), An (anatomy), Bl (blood) and others. A **popular response** (P) is one given very often, such as a bat or butterfly on Card V, and shows conventional perception; an **original response** is rare, a sign of creativity when well formed and of bizarre thinking when not.

Two further signs matter. An **absence of human responses** suggests **poor interpersonal orientation**, little interest in or understanding of people (not apathy or lack of empathy as such). A **confabulatory response** (DW) generalises from one detail to the whole blot ("a crab, because of these claws, so the whole thing is a crab"), and indicates **impaired reality testing**, not a regressed ego, cognitive immaturity or lack of empathy.

> EXAM TRAP
>
> Shape alone is **F**; "blood because it is red", with no form, is pure **C**. FC and CF mix form and colour; Cn only names the colour. **M** is human movement only (FM animal, m inanimate); C' is achromatic.

Scoring systems: Exner, R-PAS and Holtzman ●●

Rorschach left no full scoring system, and five American ones grew up in parallel: those of **Samuel Beck**, **Bruno Klopfer**, **Marguerite Hertz**, **Zygmunt Piotrowski**, and **David Rapaport** with **Roy Schafer**. Their codes differed, so research could not be compared. **John E. Exner** (1928 to 2006) combined their best-supported elements in his **Comprehensive System** (1974). It standardised administration, required at least 14 responses for a valid record, and tallied the codes into a **Structural Summary** of ratios and indices compared with norms. **Lambda** compares pure F responses with all others; a high Lambda suggests a simplifying, avoidant style. **Experience Balance** (EB) weighs human movement against colour: an **introversive** person (more M) works things out internally, an **extratensive** person (more colour) through feeling and action. Constellations flag specific risks: PTI (Perceptual-Thinking Index, formerly the Schizophrenia Index, SCZI), DEPI (depression), S-CON (suicide), CDI (coping deficit) and HVI (hypervigilance). The system gives **no IQ score**.

After Exner's death, **Gregory Meyer** and colleagues published the **Rorschach Performance Assessment System** (R-PAS, 2011). It keeps only research-supported variables, asks for two or three responses per card ("prompt and pull") so records are comparable in length, and reports standard scores against international norms. **Wayne Holtzman** had earlier solved the length problem differently: the **Holtzman Inkblot Technique** (1961) has two parallel forms of 45 cards each and allows one response per card, so every record is the same length and alternate-form reliability can be computed.

> EXAM TRAP
>
> There is **no "Murray system"** of Rorschach scoring: Henry Murray gave us the TAT. The Comprehensive System is Exner's (not Beck's, Klopfer's or Piotrowski's) and scores the Rorschach, not the MMPI, TAT or Wechsler scales.

The Thematic Apperception Test ●●●

The **Thematic Apperception Test** (TAT) was introduced in 1935 by **Christiana Morgan** (1897 to 1967) and **Henry Alexander Murray** (1893 to 1988) at the Harvard Psychological Clinic; Murray's manual followed in 1943. To apperceive is to perceive in the light of past experience. The theory is Murray's **need-press theory** of psychogenic needs (*Explorations in Personality*, 1938; Section 4.6), not personal constructs (George Kelly), operant conditioning (B. F. Skinner) or cognitive distortions (Aaron Beck). For each picture of people in an ambiguous

social scene, the person tells a dramatic story: what led up to it, what is happening, what the characters think and feel, and how it ends. The full set has **31 cards**: 30 pictures plus **blank card 16**, on which the person imagines a scene. Classically 20 cards were given over two sessions; in practice a clinician selects about **5 to 12** per session.

The interpreter assumes the storyteller identifies with the **hero** and lends the hero their own **needs**, such as need for achievement (n Ach), affiliation (n Aff) and aggression (n Agg). The **press** is the force of the environment on the hero: **alpha press** as it really is, **beta press** as the person perceives it. A recurring need-press combination is a **thema**, and the **outcome** (success, failure, rescue) shows the person's expectations. **Leopold Bellak** later systematised this scoring, and **David McClelland** scored stories for the achievement motive (1953). The TAT therefore reveals unconscious thoughts, conflicts and motives, and a client's characteristic ways of thinking, feeling and behaving in close relationships; when stories about ambiguous pictures are wanted, it is the test of choice. It is a projective personality test, not an intelligence, achievement or neuropsychological test, and it does not measure vocabulary or memory.

Unlike the Rorschach, the TAT uses ambiguous social scenes with characters, not symmetrical inkblots, and asks for stories, not percepts; it shows the content of motives and relationships more than the structure of thought. Its descendants adapt the pictures to the teller. The **Children's Apperception Test** (CAT; Leopold and **Sonya Sorel Bellak**, 1949) has 10 cards of animals in human situations for children of 3 to 10, who identify readily with animals; the CAT-H (1965) uses human figures and the CAT-S is a supplement. The Bellaks' Senior Apperception Technique (1973) serves older adults, and the Roberts Apperception Test for Children (1982) adds normed scoring. **Gerald Blum**'s Blacky Pictures (1950) show a dog family in scenes built around Freud's psychosexual stages, and **Edwin Shneidman**'s Make-A-Picture-Story test (MAPS, 1947) lets the person place cut-out figures on backgrounds before telling the story.

> EXAM TRAP
>
> TAT: **Murray and Morgan, 1930s**, a **construction** technique. Not Freud, Jung, Skinner, Rorschach, Exner or Bellak (Bellak made the CAT). The CAT differs mainly in its **animal figures designed for children**; it is not objective, not an intelligence test and uses no inkblots.

> INDIAN CONTEXT
>
> **Uma Chowdhury** adapted the TAT (1960) and the CAT for India, with Indian figures, dress and settings. **Udai Pareek** adapted the Rosenzweig Picture-Frustration Study (below); do not swap them.

Completion and verbal techniques

The oldest projective method is the **word association test**. **Francis Galton** tried it on himself in 1879. **Carl Gustav Jung** made it clinical (1910) with 100 stimulus words, timing each reply and watching for **complex indicators** (a delayed reaction, no reply, repeating the stimulus word, an odd answer) that betray an emotionally charged complex (Section 4.4). **Grace Kent** and **Aaron Rosanoff** published a 100-word list in 1910 with frequency norms from 1,000 normal adults, so a reply could be scored as common or idiosyncratic.

The **sentence completion** method grew from **Hermann Ebbinghaus**'s completion method: the person finishes stems such as "My father..." or "I regret...". The **Rotter Incomplete Sentences Blank** (**Julian Rotter** and **Janet Rafferty**, 1950) has 40 stems, each rated for conflict or positive adjustment and summed into one adjustment score. The Sacks Sentence Completion Test (**Joseph Sacks** and **Sidney Levy**, 1950) has 60 items covering 15 attitude areas in four groups: family, sex, interpersonal relationships and self-concept.

The **Rosenzweig Picture-Frustration Study** (**Saul Rosenzweig**, 1945; some Indian texts give later dates) shows 24 cartoon-style frames of everyday frustrations; the respondent writes what the frustrated character says in an empty speech balloon. Replies are scored for **direction of aggression**: **extrapunitive** (blame turned outward), **intropunitive** (blame turned on oneself) or **impunitive** (blame avoided, the frustration glossed over); and for **type of reaction**: obstacle-dominance (the obstacle stands out), ego-defence (protecting the self) or need-persistence (seeking a solution). The **Group Conformity Rating** shows how often the person gives the commonest reply.

Expressive and choice techniques

In **Karen Machover**'s **Draw-A-Person** test (1949) the person draws a person, then one of the other sex. Her **body-image hypothesis** holds that the drawing reflects the drawer's self-concept and unconscious attitudes, so size, placement, erasures and omitted parts are read as signs; it is not a test of fine motor skill alone, academic achievement or brain lesions. Contrast **Florence Goodenough**'s Draw-a-Man, which counts details as an index of intellectual maturity (Section 2.6). **John Buck**'s **House-Tree-Person** test (1948) asks for a house (home and family), a tree (deeper feelings about the self) and a person (self-concept and relationships), first in pencil (achromatic phase), then in crayon (chromatic phase), each followed by questioning. In the **Kinetic Family Drawing** (**Robert Burns** and **S. Harvard Kaufman**, 1970) the child draws everyone in the family doing something, revealing perceived closeness and conflict. **Elizabeth Koppitz** listed emotional indicators in children's figure drawings, such as a tiny figure or gross asymmetry. **Max Hutt** read the copying of the Bender designs projectively (Section 2.11).

The **Szondi Test** (**Leopold Szondi**, 1947) is the classic choice technique: from 48 photographs of psychiatric patients, in six sets of eight, the person picks the faces they like and dislike most; its genetic theory is unsupported. **Margaret Lowenfeld**'s World Technique (a world built in a sand tray) and Mosaic Test (patterns from coloured pieces), the **Hand Test** (**Edward Wagner**, 1962: "What might this hand be doing?", read for acting-out tendencies), play, finger painting and **graphology** (handwriting analysis, with little validity) complete the expressive family.

> EXAM TRAP
>
> House-Tree-Person is **John Buck**'s, not Aaron T. Beck's (cognitive therapy), David D. Burns's (author of *Feeling Good*) or Karen Machover's (Draw-A-Person). Kinetic Family Drawing is Robert Burns's, a different Burns.

How good are projective tests?

The case against them is psychometric. Many techniques have competing scoring systems, so two clinicians may not be giving the same test. Inter-scorer reliability is acceptable only with

structured systems such as Exner's or Rotter's; retest reliability is weak, because stories and percepts change between occasions. Global, intuitive interpretation has poor validity, and even scored variables show doubtful **incremental validity**: they add little to an interview and an inventory. Defenders reply that some Rorschach indices, notably of thought disorder and perceptual accuracy, hold up, and that the tests generate rich clinical hypotheses; they remain in clinical training and practice in India. Table 2.24 gathers the main tests.

KEY STUDY: ILLUSORY CORRELATION (CHAPMAN AND CHAPMAN, 1967, 1969)

Loren Chapman and **Jean Chapman** asked why clinicians trusted signs that research did not support. They paired figure drawings with symptom statements at random and showed them to students, who "discovered" the links clinicians reported, such as odd eyes with suspiciousness, though the real correlation was zero. In 1969 the same happened with invalid Rorschach signs, while valid signs went unnoticed. This **illusory correlation** comes from expectation, not data. The task was artificial, but practising clinicians reported the same invalid signs.

Table 2.24 The main projective tests compared.

Test	Author, year	Stimulus	Lindzey	Key fact
Rorschach	Rorschach, 1921	10 inkblots	Association	5 achromatic, 5 chromatic
Holtzman (HIT)	Holtzman, 1961	2 forms of 45 inkblots	Association	One response per card
Word association	Galton 1879; Jung 1910	100 words (Jung)	Association	Complex indicators
TAT	Morgan and Murray, 1935	31 cards, card 16 blank	Construction	Need and press
CAT	Bellak and Bellak, 1949	10 animal pictures	Construction	Ages 3 to 10
Rotter ISB	Rotter and Rafferty, 1950	40 stems	Completion	Adjustment score
Rosenzweig P-F	Rosenzweig, 1945	24 cartoons	Completion	Extra-, intro-, impunitive
Szondi Test	Szondi, 1947	48 patient photographs	Choice	Liked and disliked faces
Draw-A-Person	Machover, 1949	Blank page	Expression	Body-image hypothesis
House-Tree-Person	Buck, 1948	Blank pages	Expression	Achromatic then chromatic

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The term "projective methods" was coined by _____ in 1939.
2. Of the ten Rorschach cards, cards II and III are black and _____.
3. The Rorschach phase in which each response is clarified without leading questions is the _____ phase.
4. A Rorschach response based on the shape of the blot alone is scored _____.
5. The Rorschach Comprehensive System (1974) was developed by _____.
6. In Lindzey's classification the TAT is a _____ technique.
7. The TAT rests on Murray's theory of needs and _____.
8. The House-Tree-Person test was developed by _____ in 1948.

Answers: (1) Lawrence K. Frank, (2) red, (3) inquiry, (4) F (pure form), (5) John Exner, (6) construction, (7) press, (8) John Buck.

2.11 Neuropsychological Assessment and Clinical Rating Scales

MEDIUM YIELD · 40 QS

A retired engineer forgets appointments and loses words mid-sentence. Is it depression, early dementia or a small stroke, and which abilities are gone? Neuropsychological tests profile what the brain can still do; clinical rating scales measure how severe symptoms are and whether treatment helps.

What neuropsychological assessment measures ●●

Neuropsychological assessment infers how the brain is working from behaviour: each standardised task samples one function, and a characteristic pattern of failures points to the damaged system. Its domains are orientation and consciousness, attention and concentration, processing speed, memory, language, visuospatial and constructional skill, **executive functions** (planning, shifting, inhibiting and monitoring goal-directed behaviour) and motor and sensory function; mood is checked too, because it colours every score. Opinions, attitudes and reactions are not neuropsychological domains. The purposes are diagnosis, localisation, a baseline for measuring change, rehabilitation planning and judgments of capacity. Localisation itself is in Section 5.11.

A **fixed battery** gives every patient the same tests scored against norms (Halstead-Reitan, Luria-Nebraska); a **flexible battery** adds tests to a core to fit the referral question. The **process approach** of **Edith Kaplan** (1924 to 2009) in Boston holds that how a patient fails matters as much as the score (a piecemeal copy and an incomplete one may score alike). Its roots lie in the qualitative method of **Alexander Romanovich Luria** (1902 to 1977), who used no norms but watched exactly where a function broke down.

A score needs a standard. The **normative comparison** (nomothetic) standard compares the patient with healthy people of the same age and education. The **individual comparison** standard compares present performance with the person's own estimated **premorbid level**, which one Indian paper calls the 'ideometric' approach: an engineer who now scores average has declined. **David Wechsler** (1896 to 1981) noticed that some subtests 'hold' with age and ill-

ness (Vocabulary, Information, Picture Completion, Object Assembly) while others 'don't hold' (Digit Span, Similarities, Digit Symbol, Block Design). His **deterioration index** or quotient, (hold minus don't hold) divided by hold, estimates how far brain or psychiatric illness has eroded cognitive and intellectual ability. The **National Adult Reading Test** (**Hazel Nelson**, 1982) predicts premorbid IQ from reading aloud 50 irregular words such as 'debt', a skill that survives early dementia. Such profiles replaced the old screening question 'organic or functional?', once answered with a single test such as the Bender-Gestalt.

> EXAM TRAP
>
> Estimating the impact of brain or psychiatric illness on intellectual ability means the **deterioration quotient**. Distractors: a mediation index belongs to statistics (Chapter 9), sense of coherence to **Aaron Antonovsky**'s salutogenesis (Section 7.9), theory of mind to social cognition (Chapter 6). The 'ideometric' approach weighs premorbid functioning against current status; comparison with population norms is the normative approach, and neither quantitative scoring nor emotional well-being is meant.

Neuropsychological batteries ●●

The classic fixed battery began with **Ward Halstead**'s studies of frontal-lobe patients (1947) and was validated by his student **Ralph Reitan** (1922 to 2014). The **Halstead-Reitan Neuropsychological Battery** is used for comprehensive neuropsychological assessment: to detect and localise brain dysfunction and gauge its severity. Its tests are the **Category Test** (abstract reasoning and concept formation); the **Tactual Performance Test** (the blindfolded patient fits blocks into a form board, then draws it from memory, giving time, memory and localisation scores); the **Seashore Rhythm Test**; the **Speech-Sounds Perception Test**; the **Finger Tapping** (Finger Oscillation) **Test** of motor speed; and the **Trail Making Test**, Parts A and B. Reitan added the Reitan-Indiana Aphasia Screening Test, a sensory-perceptual examination, grip strength and lateral dominance. A **Halstead Impairment Index** (the proportion of key scores in the impaired range) of 0.5 or more suggests brain damage.

The **Luria-Nebraska Neuropsychological Battery** gave Luria's methods American norms: **Anne-Lise Christensen** compiled them (1975), and **Charles Golden**, **Thomas Hammeke** and **Arnold Purisch** standardised them (1978; manual 1980). Form I has 269 items scored 0 to 2 in 11 clinical scales, from motor and rhythm to memory and intellectual processes; high T scores mean impairment.

The Indian **NIMHANS Neuropsychological Battery** (Bangalore; adult manual by **Shobini Rao**, D. K. Subbakrishna and K. Gopukumar, 2004, plus a children's battery) is a set of individually normed tests with Indian norms by age, sex and education: colour trails, digit vigilance, N-back, WCST, Tower of London, Stroop, verbal fluency, auditory verbal learning and a complex figure. The **PGI Battery of Brain Dysfunction** (PGIMER, Chandigarh; **Dwarka Pershad** and colleagues) combines the PGI Memory Scale, verbal and performance intelligence tests (the latter built on **C. M. Bhatia**'s, Section 2.6), the Nahor-Benson perceptual test and the Bender-Gestalt. The **PGI Memory Scale** (Pershad and **N. N. Wig**, 1977) has 10 subtests: remote and recent memory, mental balance, attention and concentration, delayed and immediate recall, retention for similar and for dissimilar pairs, visual retention and recognition.

EXAM TRAP

The Halstead-Reitan assesses brain function, not personality, vocational interest, attachment, self-reported depression or unconscious conflict, and it is not a projective test. The Rorschach is never part of it (the WAIS and MMPI are often given alongside). Luria did not write the Luria-Nebraska. PGI means Chandigarh; NIMHANS means Bangalore.

Memory and visual-motor tests ●●●

The **Wechsler Memory Scale** (Wechsler, 1945) began as seven brief subtests summed into a **memory quotient**. The WMS-IV (2009) gives five indices: **Auditory Memory, Visual Memory, Visual Working Memory, Immediate Memory and Delayed Memory**. Verbal and Performance IQ, like crystallised and fluid scores, belong to intelligence tests (Section 2.6); Mechanical Reasoning is a Differential Aptitude Tests subtest (Section 2.8).

In the **Rey Auditory Verbal Learning Test** (**André Rey**, Geneva) the examiner reads 15 words over five trials, then an interference list, then asks for the first list again immediately and after a delay; a flat learning curve or heavy delayed loss signals a memory disorder. The California Verbal Learning Test (1987) uses a 16-word 'shopping list' of four categories, showing whether the person groups words by meaning. Visual memory is tested by the **Rey-Osterrieth Complex Figure**, a detailed drawing first copied (planning and construction) and later drawn from memory, and by the **Benton Visual Retention Test** (**Arthur Benton**, 1946), in which 10 designs are each shown briefly and drawn from memory.

The classic visual-motor test is the **Bender Visual-Motor Gestalt Test** (1938) of **Lauretta Bender** (1897 to 1987). She adapted nine figures (A and 1 to 8) from designs the Gestalt psychologist **Max Wertheimer** (1880 to 1943) had used to study perceptual organisation. Copying them requires perceiving each design as a whole and reproducing it by hand, so the test assesses **visual-motor integration**. Its errors screen for organic brain damage and neurological dysfunction in adults and for developmental immaturity in children, scored by **Elizabeth Koppitz**'s developmental system (1963). The Bender-Gestalt II (2003) has 16 designs and a recall phase. **Max Hutt** also read the drawings projectively, a secondary use.

Simplest of all is **digit span**. Repeating digits in the order heard (forward span, about seven in adults) measures attention span. Repeating them in reverse (backward span) forces the person to hold and rearrange the digits at once, so it measures **working memory** (Section 1.7), not long-term, sensory or procedural memory.

EXAM TRAP

The Bender does not test verbal intelligence, vocational interest, personality, unconscious conflict, depression, schizophrenia, ADHD, reading, auditory or working memory, verbal abstraction or premorbid intelligence (that is the NART). Bender (1938) means copying nine figures; Benton (1946) means drawing ten designs from memory.

Executive function, attention, thinking and language ●●●

The most widely used test of executive function is the **Wisconsin Card Sorting Test** (WCST) of **David Grant** and **Esta Berg** (1948), with a clinical manual by **Robert Heaton** (1981). The patient sorts 128 response cards under four stimulus cards by colour, form or number, told only 'right' or 'wrong', and after ten correct sorts the rule changes without warning (Figure 2.17). A

healthy person notices the run of 'wrong' and shifts; a frontal patient keeps sorting by the old rule. The scores are **perseverative errors**, categories achieved and failure to maintain set. The WCST measures **set-shifting** and cognitive flexibility and is most sensitive to dysfunction of the **dorsolateral prefrontal cortex**.

Figure 2.17 The Wisconsin Card Sorting Test. Each response card fits more than one rule; clinging to a rule after it has changed is perseveration.

KEY STUDY: MILNER AND THE FRONTAL LOBES (1963)

In Montreal **Brenda Milner** gave the WCST to patients having brain surgery for epilepsy. Those with dorsolateral frontal excisions achieved few categories and made many perseverative errors, some while saying the rule had changed; temporal and posterior cases did not. **Criticism:** perseveration also follows damage elsewhere and occurs in schizophrenia, so the test is sensitive to frontal damage but not specific to it.

EXAM TRAP

The 'Minnesota Number Ordering Test' and 'Manhattan Card Sorting Test' do not exist. The WCST does not measure memory, language, reading, visual acuity, mood or manual dexterity, and its region is the dorsolateral prefrontal cortex, not the hippocampus, primary motor cortex or cerebellum.

The **Trail Making Test**, from the Army Individual Test Battery (1944), has two parts: Part A joins 25 numbered circles in order (attention and speed), and Part B alternates numbers and letters (1, A, 2, B and so on), adding set-shifting. In children with ADHD it assesses inattention; impulsivity, over-activity and social skills are better judged from continuous performance errors and rating scales. The **Stroop Colour-Word Test** (**John Ridley Stroop**, 1935) asks for the ink colour of conflicting colour words ('RED' printed in green), measuring **response inhibition**. The **Tower of London** (**Tim Shallice**, 1982), moving beads on three pegs to a goal in the fewest moves, measures planning, as does the Tower of Hanoi. **Verbal fluency** asks for words beginning with a letter in a minute (the Controlled Oral Word Association or FAS test) or from a category such as animals.

Sustained attention is tested by the **Continuous Performance Test** (**H. Enger Rosvold** and colleagues, 1956): in a long stream of letters the person responds only to a target, so omissions reflect inattention and commissions impulsivity. The Test of Everyday Attention measures sustained, selective and divided attention through everyday tasks such as map searching. Neither shows which side of the body is affected; Benton's **Finger Localization Test** does. The patient, without looking, identifies fingers the examiner touches, and consistently more errors on one hand point to the opposite hemisphere.

Concept formation, and with it **thought disorder**, is assessed by sorting. In the **Object Sorting Test** of **Kurt Goldstein** (1878 to 1965) and **Martin Scheerer** (1941) the patient groups everyday objects and explains why they belong together; concrete, bizarre or loose groupings expose disordered thinking directly. The Rorschach reveals it only indirectly (Section 2.10), the TAT taps needs and conflicts, and the Bender visual-motor skill. Proverbs and similarities test abstract against **concrete thinking** in the same way.

Language tests include the **Boston Naming Test** (60 line drawings), the **Token Test** of **Ennio De Renzi** and **Luigi Vignolo** (1962), which checks comprehension with commands such as 'touch the small red circle', and the Boston Diagnostic Aphasia Examination and Western Aphasia Battery. The aphasias are in Section 5.10. Table 2.25 gathers the tests by domain.

Table 2.25 Which neuropsychological test for which domain.

Domain	Classic tests
Consciousness, orientation	Glasgow Coma Scale; MMSE orientation
Attention and speed	Digit span forward, Trail Making A, CPT, Test of Everyday Attention
Working memory	Digit span backward, N-back
Verbal memory	WMS-IV, Rey AVLT, CVLT, PGI Memory Scale
Visual memory	Benton Visual Retention Test, Rey figure recall
Visuoconstruction	Bender-Gestalt, Rey figure copy, clock drawing
Executive function	WCST, Trail Making B, Stroop, Tower of London, fluency, Category Test
Concept formation	Object Sorting Test, proverbs, similarities
Language	Boston Naming Test, Token Test, BDAE, WAB
Motor and sensory	Finger Tapping, grip strength, Finger Localization

Bedside screening and staging

A short screen settles whether cognition is impaired at all. The **Mini-Mental State Examination** (MMSE) of **Marshal Folstein**, **Susan Folstein** and **Paul McHugh** (1975) takes about ten minutes and is scored out of 30: orientation (10), registration (3), attention by serial sevens (5), recall (3), and language and copying (9). A score of 23 or below (cut-off 23/24) suggests impairment. It screens for and tracks global cognitive impairment and dementia, not personality, depression or IQ. Because it is verbal and education-bound, **Mary Ganguli** built the Hindi Mental State Examination (1995) for rural, largely illiterate older Indians.

The **Montreal Cognitive Assessment** (MoCA) of **Ziad Nasreddine** (2005) is also out of 30 and takes about ten minutes, but adds executive tasks (a short trail, fluency, abstraction), cube copying and **clock drawing**; 26 and above is normal. It is therefore more sensitive than the

MMSE to mild cognitive impairment. Addenbrooke's Cognitive Examination III extends the idea to 100 points.

Staging scales grade severity. The **Glasgow Coma Scale** of **Graham Teasdale** and **Bryan Jennett** (1974) rates level of consciousness from the best eye opening (1 to 4), verbal response (1 to 5) and motor response (1 to 6), totalling 3 to 15; 8 or below means severe injury or coma. In dementia the Clinical Dementia Rating grades six domains of cognition and daily function from 0 to 3, the Global Deterioration Scale (**Barry Reisberg**, 1982) describes seven stages, and the Hachinski Ischemic Score separates vascular from Alzheimer-type dementia (Section 8.10).

> EXAM TRAP
>
> The Glasgow Coma Scale measures consciousness, not attention, memory or intelligence, and its minimum is 3, not 0. The MoCA's advantage over the MMSE is sensitivity to mild impairment, not brevity, and neither is reserved for neurologists.

Rating scales for depression and anxiety ●●

Rating scales grade symptom severity and track change. What matters first is who completes them. In a **self-report inventory** the patient rates himself; in a **clinician-rated scale** a trained interviewer rates the patient from interview and observation. Structured diagnostic interviews (SCID, PSE) are in Section 7.2, and life-event scales such as Holmes and Rahe's in Section 1.11.

The **Beck Depression Inventory** of **Aaron Temkin Beck** (1921 to 2021), first published in 1961, is the most used self-report depression measure. The BDI-II (1996) has 21 items, each rated 0 to 3 for the past two weeks, so totals run from 0 to 63: **0 to 13 minimal, 14 to 19 mild, 20 to 28 moderate, 29 to 63 severe.** For a patient with two months of sadness, anhedonia and worthlessness, the BDI is the instrument for further evaluation, not the Rorschach (projective), the MMPI (broad personality, Section 2.9) or the WAIS (IQ). Clinician-rated alternatives are the **Hamilton Rating Scale for Depression** (**Max Hamilton**, 1960) and the Montgomery-Asberg scale, which is sensitive to treatment change; the Zung scale (1965) and PHQ-9 are brief self-report screens, and Beck's Hopelessness Scale and Scale for Suicide Ideation assess suicide risk. For anxiety, the clinician-rated **Hamilton Anxiety Rating Scale** pairs with the self-report Beck Anxiety Inventory and GAD-7. The Fear Survey Schedule of **Joseph Wolpe** and **Peter Lang** (1964) lists feared situations; the SUDS rating used in exposure is in Section 7.4.

> EXAM TRAP
>
> RML 2025 keyed the BDI as 'typically used for neuropsychological assessment' against a Likert scale (an attitude-scaling format, Section 2.8), the 16PF (a personality inventory, Section 2.9) and the Peabody Individual Achievement Test (an achievement test, Section 2.8). The BDI is a self-report depression inventory, not a neuropsychological test; neuropsychologists add it to see whether low mood is depressing test scores. Choose it only because the others are further off.

Scales for OCD, psychosis, substances, children and forensic work ●●

The **Yale-Brown Obsessive Compulsive Scale** (Y-BOCS) of **Wayne Goodman** and colleagues (1989) is the standard clinician-rated measure of OCD severity. After a symptom checklist, ten core items rate obsessions (five) and compulsions (five) on the same five dimensions, time, interference, distress, resistance and control, each 0 to 4, for a total of 0 to 40. It does not rate depression, bipolar disorder, schizophrenia, PTSD or suicidal ideation.

For psychosis the **Brief Psychiatric Rating Scale** (**John Overall** and **Donald Gorham**, 1962) covers general psychopathology, and the **Positive and Negative Syndrome Scale** (PANSS) of **Stanley Kay**, **Abraham Fiszbein** and **Lewis Opler** (1987) measures symptom severity in schizophrenia with 30 items: 7 positive, 7 negative and 16 general psychopathology. **Nancy Andreasen**'s SANS and SAPS rate negative and positive symptoms separately; the Young Mania Rating Scale rates mania. General screens such as the **General Health Questionnaire** (**David Goldberg**, 1972) and **Leonard Derogatis**'s SCL-90-R detect distress without making a diagnosis. For substance use there are the WHO's Alcohol Use Disorders Identification Test (AUDIT), CAGE (Cut down, Annoyed, Guilty, Eye-opener), the Michigan Alcoholism Screening Test (MAST), the CIWA-Ar for alcohol withdrawal, the SADQ for dependence severity and the Fagerstrom Test for nicotine dependence.

The Clinical Global Impression rates severity and improvement from 1 to 7; the Global Assessment of Functioning, dropped in DSM-5, gave way to the WHO's WHODAS 2.0. In India, disability from mental illness is certified with **IDEAS**, the Indian Disability Evaluation and Assessment Scale (Indian Psychiatric Society, 2002). Children are rated with the Conners Rating Scales (ADHD), **Thomas Achenbach**'s **Child Behavior Checklist** and its Indian adaptation, **Savita Malhotra**'s Childhood Psychopathology Measurement Schedule (PGIMER); the Edinburgh Postnatal Depression Scale (1987) screens new mothers. In forensic work the **Psychopathy Checklist-Revised** of **Robert Hare** (1980, revised 1991) rates 20 items 0 to 2 from interview and case records. Hare built it on **Hervey Cleckley**'s portrait of the psychopath in *The Mask of Sanity* (1941), but Cleckley devised no checklist. Table 2.26 compares the main scales.

Table 2.26 Clinical rating scales compared.

Scale	Author, year	Items	Rater	Measures
BDI-II	Beck 1961; 1996	21 (0 to 3)	Self	Depression severity
HAM-D	Hamilton 1960	17	Clinician	Depression severity
MADRS	Montgomery, Asberg 1979	10	Clinician	Depression, change
HAM-A	Hamilton 1959	14	Clinician	Anxiety
BAI	Beck 1988	21	Self	Anxiety
Y-BOCS	Goodman 1989	10 (0 to 4)	Clinician	OCD severity
BPRS	Overall, Gorham 1962	18 (16 at first)	Clinician	General psychopathology
PANSS	Kay, Fiszbein, Opler 1987	30	Clinician	Schizophrenia severity
YMRS	Young 1978	11	Clinician	Mania
AUDIT / CAGE	WHO / Ewing	10 / 4	Self or interviewer	Problem drinking
EPDS	Cox 1987	10	Self	Postnatal depression

Scale	Author, year	Items	Rater	Measures
PCL-R	Hare 1980, 1991	20 (0 to 2)	Clinician	Psychopathy

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Comparing current performance with the person's own premorbid level is the _____ comparison standard.
2. Wechsler's index contrasting hold and don't-hold subtests is the _____ quotient.
3. A Halstead Impairment Index of _____ or more suggests brain damage.
4. The Bender-Gestalt assesses _____ integration.
5. Backward digit span measures _____ memory.
6. The WCST is most sensitive to dysfunction of the _____ prefrontal cortex.
7. A BDI-II score of 24 falls in the _____ band.

Answers: (1) individual, (2) deterioration, (3) 0.5, (4) visual-motor, (5) working, (6) dorsolateral, (7) moderate.

Galton's Anthropometric Laboratory, Revisited

Look again at Galton's counter with this chapter behind you. Every visitor met the same instruments, instructions and recording, so his procedures were standardised and objective (Section 2.1). Measures such as height and grip strength are highly reliable (Section 2.3), yet Wissler showed that sensory and motor tests of this kind had almost no criterion validity for academic success (Section 2.4): consistency is necessary, not sufficient. Galton's thousands of cards were among the first large norms, and his spread of scores around an average foreshadowed standard scores (Section 2.5). Binet's turn to judgement and reasoning founded the intelligence tests of Section 2.6, and every later test has had to prove what Galton assumed. His belief in inherited ability, which fed eugenics, is one reason testing now carries rules of fairness.

Chapter summary

2.1 Psychological Testing: Nature, History, Types and Ethics. A **psychological test** is an objective, standardised measure of a sample of behaviour (Anastasi); assessment integrates tests with interview, observation and records. Testing runs from Chinese examinations through Galton's laboratory (1884), Cattell's 'mental tests' (1890) and Wissler's disproof (1901) to Binet and Simon (1905), the Army Alpha and Beta, Woodworth's inventory and the MMPI (1943). Tests are objective or subjective, speed or power, of maximum or typical performance. A good test is objective, reliable, valid, normed and practicable; standardisation fixes procedures and norms. Clinicians read data as samples, correlates or signs; actuarial prediction usually beats clinical judgement, which suffers confirmatory bias, illusory correlation and the Barnum effect.

2.2 Building a Test: Items, Item Analysis, IRT and Response Bias. Items are **essay** or **objective** (supply or selection type), and guessing is corrected by $S = R - W/(n - 1)$. **Item analysis** gives difficulty p (pro-

portion passing; ideal 0.50, about 0.62 for four options) and discrimination D = p upper - p lower on Kelley's 27 percent groups; a negative D flags a miskeyed item. Ceiling and floor effects stop a test discriminating at the top or bottom. In IRT the item characteristic curve plots success against the latent trait θ (b, a, c; Rasch 1PL to 3PL), and sample-invariant parameters allow adaptive testing. Acquiescence, social desirability and halo (Thorndike, 1920) bias scores.

2.3 Reliability: Consistency and the Standard Error of Measurement. Reliability is the consistency of a measure: dependability and freedom from measurement error. In Spearman's classical test theory (1904) every observed score is a true score plus random error, and reliability is the ratio of true-score to observed-score variance. Test-retest correlation gives the coefficient of stability, for stable traits; parallel forms give the coefficient of equivalence. Single-sitting internal consistency methods are split-half (stepped up by the Spearman-Brown formula, so .60 becomes .75), Rulon and Flanagan, KR-20 and KR-21 for right-wrong items, and Cronbach's alpha (1951). Cohen's kappa measures rater agreement beyond chance. The standard error of measurement, SD × √(1 - r), bands one person's score.

2.4 Validity and the Accuracy of Test Decisions. Validity is the extent to which a test measures what it claims to measure; since Messick (1989) it is a unified judgement about the interpretations and uses of scores. Face validity is mere appearance; content validity is adequate sampling of the domain (blueprint, Lawshe's CVR). Criterion-related validity is concurrent (criterion measured now) or predictive (criterion measured later). Construct validity (Cronbach and Meehl, 1955) rests on convergent and discriminant evidence and the multitrait-multimethod matrix (Campbell and Fiske, 1959). A test can be reliable without being valid, and validity cannot exceed the square root of reliability. Sensitivity and specificity describe the test; predictive values depend on the base rate.

2.5 Norms, Standard Scores and Interpreting a Score. A raw score means nothing without a frame of reference. Norm-referenced interpretation compares a person with a representative standardisation sample; criterion-referenced interpretation (Glaser, 1963) compares with a fixed mastery standard. Good norms are large, representative, recent and relevant. The ratio IQ (MA ÷ CA × 100) gave way to Wechsler's deviation IQ (mean 100, SD 15). Percentile ranks are ordinal. The z score (mean 0, SD 1) and the normal curve (68, 95, 99.7 per cent within 1, 2, 3 SD) link T scores (50, 10), stanines (5, 2), stens (5.5, 2), scaled scores (10, 3) and CEEB scores (500, 100).

2.6 Intelligence Tests: Binet, Wechsler, Non-verbal and Indian Scales. Intelligence tests are reliable and predict school achievement but not behaviour in general. Binet and Simon (1905) built the first test; Terman's Stanford-Binet (1916) used the ratio IQ, and the SB5 (Roid, 2003) measures five factors in verbal and non-verbal domains. Wechsler's Bellevue scale (1939) introduced the deviation IQ; the WAIS-IV (2008; 16:0 to 90:11) gives an FSIQ and four indices (VCI, PRI, WMI, PSI) from 10 core subtests, and the WISC-V covers 6:0 to 16:11. Raven's Progressive Matrices measures fluid reasoning and is culture-fair. In India, Bhatia's performance battery suits illiterate people, while the Binet-Kamat, MISIC and WAPIS adapt the Stanford-Binet, WISC and WAIS.

2.7 Assessing Development, Adaptive Behaviour and Disability. Development and everyday competence are scored apart from IQ as DQ (developmental age / CA × 100) and SQ (social age / CA × 100), from several informants because symptoms vary across contexts. Infant scales include the Bayley (1 to 42 months), Gesell's schedules, Denver II and the Indian DST, DASII and TDSC.

Doll's **VSMS** (1935; 117 items; SQ) and Malin's Indian version (89 items) measure social maturity; the **Vineland Adaptive Behavior Scales**, adaptive functioning. DSM-5 **intellectual disability** needs IQ and adaptive deficits, with severity graded adaptively; **global developmental delay** is for under-5s. Under the RPWD Act 2016, learning disability is certified from 8 years, autism with **ISAA**.

2.8 Aptitude, Achievement, Creativity, Interest and Attitude Measures. An **achievement test** measures what has been learnt in a specific academic area; an **aptitude test** predicts future learning in a specific area, and Anne Anastasi placed both on one continuum. Multiple aptitude batteries (the DAT with eight subtests, 1947; the GATB with nine aptitudes) serve guidance. Guilford's **divergent thinking** is scored for fluency, flexibility, originality and elaboration (TTCT; Mednick's RAT). Strong keyed interests empirically, Kuder used ipsative forced-choice triads and Holland proposed RIASEC. Thurstone's **equal-appearing intervals** use judges and median scale values; Likert's **summated ratings** sum respondents' agreement ratings; Guttman's cumulative scale needs reproducibility of 0.90; Bogardus measured social distance; the IAT measures implicit attitudes by reaction time.

2.9 Objective Personality Inventories. An **objective personality test** fixes items, answer format and scoring. Cattell accepted three data sources: **L-data** (life records), **Q-data** (questionnaires) and **T-data** (objective tests). Inventories are built rationally (Woodworth's Personal Data Sheet, 1917), theoretically (EPPS, MBTI, MCMI), by **empirical criterion keying** (MMPI, CPI) or by factor analysis (16PF, EPQ, NEO-PI-R, GZTS, Comrey). The **MMPI** (Hathaway and McKinley, 1943) assesses personality and psychopathology with validity scales **L, F and K** and ten clinical scales; the MMPI-2, MMPI-2-RF (RC scales) and MMPI-3 followed. The Rep test and Q-sort study one person at a time.

2.10 Projective Techniques. **Projective techniques** rest on the **projective hypothesis** (term coined by Frank, 1939): faced with an ambiguous stimulus, people project needs, conflicts and unconscious processes into their answers, which resists self-report bias. **Lindzey** (1959) grouped them as association, construction, completion, choice or ordering and expression. The **Rorschach** (1921) has 10 cards (5 achromatic, 2 black-red, 3 multicoloured), a response and inquiry phase, and codes for location, determinants and form quality; Exner's Comprehensive System (1974) and R-PAS (2011) standardised it. The **TAT** (Morgan and Murray, 1935; 31 cards) is read for needs and press. Sentence completion, the Rosenzweig P-F, figure drawings and the Szondi complete the family, all psychometrically weak.

2.11 Neuropsychological Assessment and Clinical Rating Scales. **Neuropsychological assessment** infers brain function from behaviour across attention, memory, language, visuospatial, executive, motor and sensory domains, using fixed batteries (**Halstead-Reitan**, Luria-Nebraska), flexible batteries or Kaplan's process approach, and comparing scores with norms or with the person's own premorbid level (Wechsler's **deterioration quotient**, the NART). India has the NIMHANS and PGI batteries. Key tests are the WMS-IV, Rey AVLT, the **Bender-Gestalt** (visual-motor integration), backward digit span (working memory) and the **WCST** (set-shifting, dorsolateral prefrontal cortex). The MMSE, MoCA and Glasgow Coma Scale screen and stage. Rating scales are self-report (BDI) or clinician-rated (Hamilton scales, Y-BOCS, PANSS, PCL-R).

Who's who

Table 2.27 The people in this chapter.

Name	When	Examined for
Anne Anastasi	1908 to 2001	Defined a test as an objective and standardised measure of a sample of behaviour; Psychological Testing (1954)
Lee J. Cronbach	1916 to 2001	A test as a systematic procedure for comparing persons; maximum versus typical performance (1949)
A. K. Singh	Indian textbook author	Lists objectivity first among the five characteristics of a good test
Jean-Étienne Esquirol	1772 to 1840	Separated mental retardation from mental illness; graded it by use of language
Édouard Séguin	1812 to 1880	Sensory-motor training of retarded children; the Séguin form board
Francis Galton	1822 to 1911	Hereditary Genius (1869); Anthropometric Laboratory, London (1884); father of mental testing and psychometrics
Karl Pearson	1857 to 1936	Product-moment correlation coefficient
Wilhelm Wundt	1832 to 1920	Cattell trained in his Leipzig laboratory (Chapter 1)
James McKeen Cattell	1860 to 1944	Coined the term 'mental tests' (1890, Mental Tests and Measurements)
Clark Wissler	1870 to 1947	Showed in 1901 that Cattell's sensory tests did not predict college grades
Alfred Binet	1857 to 1911	With Simon, the first practical intelligence test (1905)
Théodore Simon	1873 to 1961	Co-author of the Binet-Simon scale
William Stern	1871 to 1938	Proposed the intelligence quotient, mental age divided by chronological age (1912)
Lewis Terman	1877 to 1956	Stanford-Binet (1916); IQ = MA/CA x 100
Arthur Otis	1917	Group test items used in the Army Alpha
Robert Yerkes	1876 to 1956	Led the Army Alpha (literate) and Army Beta (non-verbal) group tests (1917)
Robert S. Woodworth	1869 to 1962	Personal Data Sheet (1917 to 1918), the first personality inventory
Hermann Rorschach	1884 to 1922	Inkblot test (1921)
Edward K. Strong	1927	Strong Vocational Interest Blank
L. L. Thurstone	1887 to 1955	Attitude scaling (1928 to 1929); multiple factor analysis
Christiana Morgan	1897 to 1967	Thematic Apperception Test with Murray (1935)
Henry Murray	1893 to 1988	Thematic Apperception Test with Morgan (1935)
Oscar Buros	1905 to 1978	Mental Measurements Yearbook (from 1938)
David Wechsler	1896 to 1981	Wechsler-Bellevue scale (1939)
Starke Hathaway	1903 to 1984	MMPI with McKinley (1943)
J. Charnley McKinley	1891 to 1950	MMPI with Hathaway (1943)
V. V. Kamat	1934	Indian revision of the Binet scale (Section 2.6)
C. M. Bhatia	1955	Battery of performance tests of intelligence (Section 2.6)
A. J. Malin	1969	Malin's Intelligence Scale for Indian Children, an Indian WISC (Section 2.6)
Paul E. Meehl	1920 to 2003	Clinical versus Statistical Prediction (1954); named the Barnum effect (1956)
Bertram Forer	1914 to 2000	Demonstrated the Barnum (Forer) effect with a universal personality sketch (1949)
P. T. Barnum	1810 to 1891	Showman after whom Meehl named the Barnum effect
Loren Chapman	1967 and 1969	Illusory correlation in Draw-a-Person and Rorschach signs, with Jean Chapman
Jean Chapman	1967 and 1969	Illusory correlation in Draw-a-Person and Rorschach signs, with Loren Chapman
Truman Lee Kelley	1884 to 1961	Upper and lower 27 percent groups for item analysis (1939)
Robert L. Ebel	1965 (definition of item discrimination)	Rule of thumb for grading the discrimination index D (0.40 and above very good, below 0.20 poor)

Name	When	Examined for
Georg Rasch	1901 to 1980	One-parameter logistic (Rasch) model of item response theory (1960)
Allan Birnbaum	1968 (logistic IRT models)	Two- and three-parameter logistic models of item response theory
Frederic M. Lord	1912 to 2000	Laid the foundations of item response theory (from 1952)
Lee Joseph Cronbach	1916 to 2001	Response sets (1946, 1950)
Douglas N. Jackson	1929 to 2004	With Messick, content versus style in personality assessment (1958): response styles
Samuel Messick	1931 to 1998	With Jackson, content versus style in personality assessment (1958): response styles
Allen L. Edwards	1957 (The Social Desirability Variable)	Social desirability response set; forced-choice Edwards Personal Preference Schedule
Irwin A. Berg	1950s	Deviation hypothesis of response sets
Patricia Cain Smith	1963	Behaviourally anchored rating scales (BARS), with Lorne Kendall
Lorne M. Kendall	1963	Behaviourally anchored rating scales (BARS), with Patricia Cain Smith
Edward Lee Thorndike	1874 to 1949	Named the halo effect (1920), from officers' ratings of soldiers
Charles Spearman	1863 to 1945	True-score model X = T + E and correction for attenuation (1904); Spearman-Brown formula (1910)
William Brown	1881 to 1952	Spearman-Brown prophecy formula, published independently of Spearman in 1910
Phillip Rulon	1939	Rulon's split-half formula: 1 minus the variance of half-score differences over total-score variance
John Flanagan	1906 to 1996	Flanagan's split-half formula from the two half variances (1937)
Louis Guttman	1916 to 1987	Gave the same split-half formula as Flanagan (1945)
G. Frederic Kuder	1903 to 2000	KR-20 and KR-21 with Marion Richardson (1937)
Marion Richardson	1937	KR-20 and KR-21 with G. Frederic Kuder
Lee Cronbach	1916 to 2001	Coefficient alpha (1951); bandwidth-fidelity dilemma with Gleser (1965); generalizability theory (1972)
Cyril Hoyt	1941	Reliability by analysis of variance of persons by items, equal to alpha
Jacob Cohen	1923 to 1998	Cohen's kappa for chance-corrected rater agreement (1960); weighted kappa (1968)
J. Richard Landis	1977	Kappa benchmarks with Gary Koch: .81 to 1.00 almost perfect, .21 to .40 fair
Gary Koch	1977	Kappa benchmarks with J. Richard Landis
Joseph Fleiss	1937 to 2003	Fleiss' kappa for three or more raters (1971)
Maurice Kendall	1907 to 1983	Kendall's coefficient of concordance W for agreement among rankings
Goldine Gleser	1915 to 2004	Bandwidth-fidelity dilemma with Cronbach (1965); generalizability theory (1972)
Harinder Nanda	1972	Co-author of generalizability theory with Cronbach, Gleser and Rajaratnam
Nageswari Rajaratnam	1972	Co-author of generalizability theory with Cronbach, Gleser and Nanda
Henry Edward Garrett	1894 to 1973	Classic definition: validity is the extent to which a test measures what it claims (purports) to measure
C. H. Lawshe	1975	Content validity ratio (1975); the term synthetic validity
Lee Sechrest	1963	Incremental validity: the prediction a test adds beyond existing measures
Paul Everett Meehl	1920 to 2003	Construct validity with Cronbach (1955); base rates and predictive values with Rosen (1955)
Donald Thomas Campbell	1916 to 1996	Convergent and discriminant validity; multitrait-multimethod matrix with Fiske (1959)
Donald W. Fiske	1959	Multitrait-multimethod matrix with Campbell (1959)
Charles Edward Spearman	1863 to 1945	Correction for attenuation (1904)
Albert Rosen	1955	With Meehl, showed that the accuracy of a test sign or cutting score depends on the base rate (1955)
H. C. Taylor	1939	Taylor-Russell tables: success rate from validity, selection ratio and base rate (1939)

Name	When	Examined for
J. T. Russell	1939	Taylor-Russell tables, with Taylor (1939)
Robert Glaser	1921 to 2012	Distinguished criterion-referenced from norm-referenced measurement (1963)
Lewis Madison Terman	1877 to 1956	Stanford-Binet (1916); ratio IQ = (MA / CA) x 100
William A. McCall	1922	T score: mean 50, SD 10 (1922)
Raymond Bernard Cattell	1905 to 1998	16PF (1949), scored in stens; Culture Fair Intelligence Test (SD 16)
J. P. Guilford	1897 to 1987	C scale: 11 units, mean 5, SD 2 (1954)
James Robert Flynn	1934 to 2020	Flynn effect: IQ scores rising about 3 points a decade, so old norms grow too easy
Theodore Simon	1872 to 1961	Co-author of the Binet-Simon scale (1905, 1908, 1911)
Maud Merrill	1937	Co-author with Terman of Stanford-Binet Forms L and M (1937) and Form L-M (1960)
Robert L. Thorndike	1986	Senior author of the Stanford-Binet Fourth Edition (SB4), a point scale
Elizabeth Hagen	1986	Co-author of the SB4
Jerome M. Sattler	1986	Co-author of the SB4
Gale H. Roid	2003	Stanford-Binet Fifth Edition (SB5): five factors, verbal and non-verbal domains, SD 15; co-author of the Leiter-3
Louis Leon Thurstone	1887 to 1955	Primary Mental Abilities tests: seven abilities scored separately
Alan S. Kaufman	1983	K-ABC (1983), KABC-II (2004) and the Kaufman Brief Intelligence Test (K-BIT 1990, KBIT-2 2004)
Nadeen L. Kaufman	1983	Co-author of the K-ABC, KABC-II and K-BIT
Alexander Luria	1902 to 1977	Sequential and simultaneous processing and three functional units, the basis of the K-ABC and the PASS theory
J. P. Das	1997	Indian-born co-author of the PASS theory and the Das-Naglieri Cognitive Assessment System
Jack A. Naglieri	1997	Co-author of the Cognitive Assessment System (PASS)
Richard W. Woodcock	1977	Woodcock-Johnson Tests of Cognitive Abilities, now built on CHC theory
Mary Bonner Johnson	1977	Co-author of the Woodcock-Johnson tests
John Carlyle Raven	1902 to 1970	Raven's Progressive Matrices: Standard (1938), Coloured (1947), Advanced; fluid reasoning, culture-fair
Russell Graydon Leiter	1929	Leiter International Performance Scale, instructions by pantomime (Leiter-3, 2013)
Lloyd M. Dunn	1959	Peabody Picture Vocabulary Test: receptive vocabulary by pointing to pictures
Florence Laura Goodenough	1886 to 1959	Draw-a-Man test (1926), 51 scoring points, intellectual maturity
Dale B. Harris	1963	Revised the Draw-a-Man as the Goodenough-Harris Drawing Test
Karen Machover	1949	Projective Draw-a-Person test (Section 2.10)
Robert Mearns Yerkes	1876 to 1956	Army Alpha (verbal) and Army Beta (non-verbal) group tests, 1917
Samuel C. Kohs	1920 (book 1923)	Kohs Block Design test, borrowed by Wechsler and by Bhatia
W. P. Alexander	1932	Alexander Pass-Along test, part of Bhatia's Battery
Edouard Seguin	1812 to 1880	Seguin Form Board, a teaching device for children with intellectual disability
Henry Herbert Goddard	1866 to 1957	Goddard's version of the Seguin Form Board: 10 shapes, three trials, shortest time
Arnold Gesell	1880 to 1961	Gesell Drawing Test: copying geometric figures as a developmental measure
Arthur J. Malin	1969	Malin's Intelligence Scale for Indian Children (MISIC), Nagpur adaptation of the WISC, ages 6 to 15
P. Ramalingaswamy	1974	Wechsler Adult Performance Intelligence Scale (WAPIS), Indian adaptation of the WAIS performance scale
Pramila Phatak	Baroda	Indian norms for the Draw-a-Man test
Thomas M. Achenbach	1987	Cross-informant approach: parent, teacher and child ratings agree only modestly; Child Behavior Checklist, Teacher's Report Form, Youth Self-Report

Name	When	Examined for
Nancy Bayley	1899 to 1994	Bayley Scales of Infant Development (1969); BSID-II (1993) MDI and PDI; Bayley-III (2006), Bayley-4 (2019), 1 to 42 months
William K. Frankenburg	1967	Denver Developmental Screening Test with Josiah Dodds; Denver II (1990), 0 to 6 years
Josiah B. Dodds	1967	Co-author of the Denver Developmental Screening Test
Dorothea McCarthy	1972	McCarthy Scales of Children's Abilities (2½ to 8½ years); General Cognitive Index
Virginia Apgar	1953	Apgar score for newborns (Chapter 6)
T. Berry Brazelton	1973	Neonatal Behavioral Assessment Scale (Chapter 6)
J. Bharath Raj	1977	Developmental Screening Test, Mysore (88 items, 0 to 15 years, DQ)
M. K. C. Nair	1991	Trivandrum Developmental Screening Chart (0 to 2 years)
Edgar Arnold Doll	1889 to 1968	Vineland Social Maturity Scale (1935), Vineland Training School, New Jersey; social age and social quotient
Sara S. Sparrow	1984	Vineland Adaptive Behavior Scales with Balla and Cicchetti; Vineland-II (2005), Vineland-3 (2016)
David A. Balla	1984	Co-author of the Vineland Adaptive Behavior Scales
Domenic V. Cicchetti	1984	Co-author of the Vineland Adaptive Behavior Scales
Kazuo Nihira	1969	AAMD Adaptive Behavior Scale (Part One adaptive, Part Two maladaptive)
Joaquim Puig-Antich	1978	K-SADS, semi-structured child psychiatric interview
Eric Schopler	1980	Childhood Autism Rating Scale (CARS) with Reichler and Renner; TEACCH
Robert J. Reichler	1980	Co-author of the Childhood Autism Rating Scale
Barbara Rochen Renner	1988	Co-author of the CARS manual
Michael Rutter	2003	Autism Diagnostic Interview-Revised (ADI-R) with Le Couteur and Lord
Ann Le Couteur	2003	Co-author of the ADI-R
Catherine Lord	2012	ADOS-2 (autism observation schedule); co-author of the ADI-R
Diana Robins	2009	M-CHAT-R/F toddler autism screen (16 to 30 months)
David Shaffer	1983	Children's Global Assessment Scale (CGAS), a 1 to 100 rating of overall functioning
Joseph Jastak	1946	Wide Range Achievement Test (reading, spelling, arithmetic)
Lloyd Dunn	1970	Peabody Individual Achievement Test (with Frederick Markwardt)
Frederick Markwardt	1970	Peabody Individual Achievement Test (with Lloyd Dunn)
Richard Woodcock	1977	Woodcock-Johnson Tests of Achievement (with Mary Bonner Johnson)
George K. Bennett	1947	Differential Aptitude Tests (with Seashore and Wesman); Bennett Mechanical Comprehension Test
Harold G. Seashore	1947	Co-author of the Differential Aptitude Tests
Alexander G. Wesman	1947	Co-author of the Differential Aptitude Tests
J. M. Ojha	Indian adaptation	Indian adaptation of the Differential Aptitude Tests
Carl Seashore	1866 to 1949	Seashore Measures of Musical Talents (1919), an early special aptitude test
Joy Paul Guilford	1897 to 1987	Convergent versus divergent thinking (APA presidential address, 1950); Alternate Uses task; Structure of Intellect
Ellis Paul Torrance	1915 to 2003	Torrance Tests of Creative Thinking (1966), verbal and figural forms
Sarnoff Mednick	1962	Associative theory of creativity; Remote Associates Test (timed, about 40 minutes)
Michael Wallach	1965	With Kogan: creativity largely independent of intelligence when children are tested in game-like conditions
Nathan Kogan	1965	With Wallach: creativity largely independent of intelligence
Baqer Mehdi	1973	Verbal and Non-verbal Tests of Creative Thinking (India)
B. K. Passi	1979	Passi Tests of Creativity, six tests (India)
Edward Kellogg Strong Jr.	1884 to 1963	Strong Vocational Interest Blank (1927), empirical occupational keying
David Campbell	1974	Strong-Campbell Interest Inventory
John L. Holland	1919 to 2008	Theory of vocational choice (1959; set out fully 1973); RIASEC hexagon; Self-Directed Search (1971)

Name	When	Examined for
Gordon Allport	1897 to 1967	Study of Values (with Philip Vernon, 1931; Gardner Lindzey from 1951)
Philip Vernon	1931	Co-author of the Study of Values
Gardner Lindzey	1951	Joined the Study of Values (revised 1960)
Eduard Spranger	1882 to 1963	Six value types: theoretical, economic, aesthetic, social, political, religious
Milton Rokeach	1973	Rokeach Value Survey: 18 terminal and 18 instrumental values ranked
Ernest J. Chave	1929	Method of equal-appearing intervals with Thurstone (attitude towards the church)
Rensis Likert	1903 to 1981	Method of summated ratings (1932): the Likert scale
Emory Bogardus	1882 to 1973	Social distance scale (1925), seven steps; generally regarded as the oldest attitude scale
Charles Osgood	1916 to 1991	Semantic differential (1957, with Suci and Tannenbaum): evaluation, potency, activity
George Suci	1957	Co-author of the semantic differential
Percy Tannenbaum	1957	Co-author of the semantic differential
Allen Edwards	1948	Scale-discrimination technique (with Kilpatrick)
Franklin Kilpatrick	1948	Scale-discrimination technique (with Edwards)
Anthony Greenwald	1998	Implicit Association Test (with McGhee and Schwartz)
Debbie McGhee	1998	Co-author of the Implicit Association Test
Jordan Schwartz	1998	Co-author of the Implicit Association Test
Edward Jones	1971	Bogus pipeline (with Harold Sigall)
Harold Sigall	1971	Bogus pipeline (with Edward Jones)
Robert Sessions Woodworth	1869 to 1962	Personal Data Sheet (built 1917, published 1919, often dated 1920), the first personality inventory, rational strategy
Hugh M. Bell	1934	Bell Adjustment Inventory (home, health, social, emotional adjustment)
Wayne Zimmerman	1949	Co-author of the Guilford-Zimmerman Temperament Survey
Andrew L. Comrey	1970	Comrey Personality Scales: factor-analytic, 180 items, eight factors
Starke R. Hathaway	1903 to 1984	MMPI (1943) with McKinley; empirical criterion keying
James N. Butcher	1989	MMPI-2 (567 items) with colleagues
Auke Tellegen	2003	Restructured Clinical (RC) scales; MMPI-2-RF (2008) with Ben-Porath
Yossef S. Ben-Porath	2008	MMPI-2-RF (338 items) and MMPI-3 (2020) with Tellegen
H. N. Murthy	1964	Multiphasic Questionnaire (MPQ), 100-item Indian MMPI-type inventory with a Lie scale
Hans Jurgen Eysenck	1916 to 1997	MPI (1959), EPI (1964), EPQ (1975): psychoticism, extraversion, neuroticism
Sybil B. G. Eysenck	1964 and 1975	Co-author of the EPI and EPQ
Paul T. Costa Jr.	1992	NEO-PI-R with McCrae: Five-Factor Model, 240 items
Robert R. McCrae	1992	NEO-PI-R with Costa
Oliver John	1991	Big Five Inventory (44 items), with colleagues
Kibeom Lee	2004	HEXACO inventory with Ashton (adds Honesty-Humility)
Michael Ashton	2004	HEXACO inventory with Lee
Katharine Cook Briggs	1875 to 1968	Myers-Briggs Type Indicator (first form 1943), from Jung's types
Isabel Briggs Myers	1897 to 1980	Myers-Briggs Type Indicator with her mother
Carl Gustav Jung	1875 to 1961	Psychological types, the basis of the MBTI
Henry A. Murray	1893 to 1988	Theory of needs behind the EPPS; the TAT
Harrison G. Gough	1921 to 2010	California Psychological Inventory (1956, manual 1957; folk concepts); Adjective Check List
Theodore Millon	1928 to 2014	Millon Clinical Multiaxial Inventory (1977; MCMI-IV 2015); base rate scores
Charles D. Spielberger	1927 to 2013	State-Trait Anxiety Inventory (1970; Form Y 1983)
Janet A. Taylor	1953	Taylor Manifest Anxiety Scale (50 MMPI items)
A. K. P. Sinha and L. N. K. Sinha	Indian	Sinha's Comprehensive Anxiety Test
Saxena	1959	Hindi adaptation of the Bell Adjustment Inventory
Mohsin and Hussain	1981	Hindi adaptation of the Bell Adjustment Inventory (students' form)

Name	When	Examined for
N. N. Wig and S. K. Verma	PGIMER Chandigarh	PGI Health Questionnaire N-1, from the Cornell Medical Index
George A. Kelly	1905 to 1967	Role Construct Repertory Test (Rep test, 1955); repertory grid
William Stephenson	1902 to 1989	Q-sort (Q-methodology, 1953)
Carl Rogers	1902 to 1987	Real-self and ideal-self Q-sorts
Butler and Haigh	1954	Real-ideal Q-sort correlation rose from about zero to about .3 after client-centred therapy (.34 at the end, .31 at follow-up)
Jack Block	1924 to 2010	California Q-set (100 items)
Hugh Hartshorne	1928	Character Education Inquiry with May: honesty is situation-specific
Mark A. May	1928	Character Education Inquiry with Hartshorne
Walter Mischel	1968	Critique of trait consistency (taught in Chapter 4)
William Sheldon	1898 to 1977	Body types; distractor for the 16PF
Sigmund Freud	1856 to 1939	Psychoanalysis; built no inventory (distractor)
Lawrence K. Frank	1939	Coined 'projective methods'; constitutive, constructive, interpretive, cathartic and refractive categories
Samuel Beck	1944 (Rorschach's Test)	One of the five American Rorschach scoring systems
Bruno Klopfer	1942 (The Rorschach Technique)	One of the five American Rorschach scoring systems
Marguerite Hertz	Mid-twentieth century	One of the five American Rorschach scoring systems
Zygmunt Piotrowski	1957 (Perceptanalysis)	One of the five American Rorschach scoring systems
David Rapaport	1945 to 1946 (Diagnostic Psychological Testing)	Rorschach scoring system with Roy Schafer
Roy Schafer	1945 to 1946	Rorschach scoring system with David Rapaport
John E. Exner	1928 to 2006	Rorschach Comprehensive System (1974); Structural Summary, Lambda, EB, constellations
Gregory Meyer	2011	Rorschach Performance Assessment System (R-PAS), with colleagues
Wayne Holtzman	1961	Holtzman Inkblot Technique: two forms of 45 cards, one response per card
Henry Alexander Murray	1893 to 1988	TAT with Morgan (1935; manual 1943); need-press theory, Explorations in Personality (1938)
George Kelly	1955	Personal construct theory; not the basis of the TAT
B. F. Skinner	1904 to 1990	Operant conditioning; distractor for the TAT's theory and author
Aaron T. Beck	1921 to 2021	Cognitive therapy and cognitive distortions; not the author of House-Tree-Person
Leopold Bellak	1949	Children's Apperception Test with Sonya Sorel Bellak; TAT scoring; Senior Apperception Technique (1973)
Sonya Sorel Bellak	1949	Children's Apperception Test with Leopold Bellak
David McClelland	1953	Achievement motive (n Ach) scoring of TAT stories
Gerald Blum	1950	Blacky Pictures: dog family, psychosexual themes
Edwin Shneidman	1947	Make-A-Picture-Story test (MAPS)
Uma Chowdhury	1960	Indian adaptation of the TAT; Indian CAT
Udai Pareek	1925 to 2010	Indian adaptation of the Rosenzweig Picture-Frustration Study
Grace Kent	1910	Kent-Rosanoff word list with frequency norms
Aaron Rosanoff	1910	Kent-Rosanoff word list with frequency norms
Hermann Ebbinghaus	1850 to 1909	Completion method, forerunner of sentence completion tests
Julian Rotter	1950	Rotter Incomplete Sentences Blank (40 stems, adjustment score), with Janet Rafferty
Janet Rafferty	1950	Rotter Incomplete Sentences Blank, with Julian Rotter
Joseph Sacks	1950	Sacks Sentence Completion Test (60 items), with Sidney Levy

Name	When	Examined for
Sidney Levy	1950	Sacks Sentence Completion Test, with Joseph Sacks
Saul Rosenzweig	1945	Picture-Frustration Study: 24 cartoons; extrapunitive, intropunitive, impunitive
Florence Goodenough	1926	Draw-a-Man as a measure of intellectual maturity (contrast with Machover)
John Buck	1948	House-Tree-Person; achromatic then chromatic phases
Robert Burns	1970	Kinetic Family Drawing, with S. Harvard Kaufman
S. Harvard Kaufman	1970	Kinetic Family Drawing, with Robert Burns
David D. Burns	1980 (Feeling Good)	Cognitive therapy writer; distractor for House-Tree-Person
Elizabeth Koppitz	1968	Emotional indicators in children's human figure drawings
Max Hutt	From the 1940s	Projective use of the Bender-Gestalt designs
Leopold Szondi	1947	Szondi Test: 48 photographs of patients, liked and disliked faces
Margaret Lowenfeld	From 1929	World Technique (sand tray) and Mosaic Test
Edward Wagner	1962	Hand Test: 'What might this hand be doing?'; acting out
Edith Kaplan	1924 to 2009	Boston process approach to neuropsychological assessment; co-author of the CVLT and Boston Naming Test
Alexander Romanovich Luria	1902 to 1977	Qualitative, syndrome-analysis approach to neuropsychological assessment
Hazel Nelson	1982	National Adult Reading Test for estimating premorbid IQ
Aaron Antonovsky	1923 to 1994	Sense of coherence (salutogenesis), a distractor for the deterioration quotient
Ward Halstead	1947	Founded the Halstead-Reitan battery (Brain and Intelligence, 1947)
Ralph Reitan	1922 to 2014	Validated and extended the Halstead-Reitan Neuropsychological Battery
Anne-Lise Christensen	1975	Compiled Luria's methods, the basis of the Luria-Nebraska battery
Charles Golden	1978, 1980	Luria-Nebraska Neuropsychological Battery (with Hammeke and Purisch)
Thomas Hammeke	1978, 1980	Co-author of the Luria-Nebraska Neuropsychological Battery
Arnold Purisch	1978, 1980	Co-author of the Luria-Nebraska Neuropsychological Battery
Shobini Rao	2004	NIMHANS Neuropsychological Battery (with Subbakrishna and Gopukumar)
Dwarka Pershad	1977	PGI Memory Scale (with Wig) and the PGI Battery of Brain Dysfunction
N. N. Wig	1977	PGI Memory Scale (with Pershad)
Andre Rey	1941	Rey Auditory Verbal Learning Test and the Rey complex figure
Paul-Alexandre Osterrieth	1944	Standardised the Rey-Osterrieth Complex Figure
Arthur Benton	1909 to 2006	Benton Visual Retention Test (1946); Finger Localization Test
Lauretta Bender	1897 to 1987	Bender Visual-Motor Gestalt Test (1938)
Max Wertheimer	1880 to 1943	Gestalt designs adapted for the Bender-Gestalt
David Grant	1948	Wisconsin Card Sorting Test (with Esta Berg)
Esta Berg	1948	Wisconsin Card Sorting Test (with David Grant)
Robert Heaton	1981	Clinical manual of the Wisconsin Card Sorting Test
Brenda Milner	1963	Showed perseveration on the WCST after dorsolateral frontal excisions
John Ridley Stroop	1935	Stroop Colour-Word Test of response inhibition
Tim Shallice	1982	Tower of London test of planning
H. Enger Rosvold	1956	Continuous Performance Test of sustained attention
Kurt Goldstein	1878 to 1965	Object Sorting Test (with Scheerer, 1941); abstract versus concrete attitude
Martin Scheerer	1941	Object Sorting Test (with Goldstein)
Ennio De Renzi	1962	Token Test of language comprehension (with Vignolo)
Luigi Vignolo	1962	Token Test (with De Renzi)
Marshal Folstein	1975	Mini-Mental State Examination (with Susan Folstein and Paul McHugh)
Susan Folstein	1975	Mini-Mental State Examination
Paul McHugh	1975	Mini-Mental State Examination
Mary Ganguli	1995	Hindi Mental State Examination

Name	When	Examined for
Ziad Nasreddine	2005	Montreal Cognitive Assessment
Graham Teasdale	1974	Glasgow Coma Scale (with Bryan Jennett)
Bryan Jennett	1974	Glasgow Coma Scale (with Graham Teasdale)
Barry Reisberg	1982	Global Deterioration Scale (seven stages)
Vladimir Hachinski	1975	Hachinski Ischemic Score (vascular versus Alzheimer-type dementia)
Aaron Temkin Beck	1921 to 2021	Beck Depression Inventory (1961; BDI-II 1996), Beck Anxiety Inventory, Hopelessness Scale
Max Hamilton	1959, 1960	Clinician-rated Hamilton Anxiety (1959) and Depression (1960) scales
William Zung	1965	Self-Rating Depression Scale
Stuart Montgomery and Marie Asberg	1979	Montgomery-Asberg Depression Rating Scale (10 items)
Joseph Wolpe	1964	Fear Survey Schedule (with Peter Lang)
Peter Lang	1964	Fear Survey Schedule (with Joseph Wolpe)
Wayne Goodman	1989	Yale-Brown Obsessive Compulsive Scale
John Overall	1962	Brief Psychiatric Rating Scale (with Donald Gorham)
Donald Gorham	1962	Brief Psychiatric Rating Scale (with John Overall)
Stanley Kay	1987	Positive and Negative Syndrome Scale (with Fiszbein and Opler)
Abraham Fiszbein	1987	Positive and Negative Syndrome Scale
Lewis Opler	1987	Positive and Negative Syndrome Scale
Nancy Andreasen	1982, 1984	Scales for the Assessment of Negative (SANS) and Positive (SAPS) Symptoms
Robert Young	1978	Young Mania Rating Scale (11 items)
David Goldberg	1972	General Health Questionnaire
Leonard Derogatis	1970s	Symptom Checklist-90-Revised
John Ewing	1984	CAGE alcohol screening questionnaire
Thomas Achenbach	1983	Child Behavior Checklist
Savita Malhotra	1980s	Childhood Psychopathology Measurement Schedule (PGIMER), Indian adaptation of the CBCL
John Cox	1987	Edinburgh Postnatal Depression Scale
Robert Hare	1980, 1991	Psychopathy Checklist-Revised (20 items scored 0 to 2)
Hervey Cleckley	1903 to 1984	The Mask of Sanity (1941), the clinical basis of the PCL-R
Thomas Holmes and Richard Rahe	1967	Life-event (social readjustment) rating scale; see Section 1.11

Key terms

psychological test, Measurement, evaluation, assessment, psychometric approach, form board, mental test, objective test, subjective test, speed test, power test, maximum performance, ability tests, typical performance, aptitude test, achievement test, personality test, neuropsychological test, Objectivity, Reliability, Validity, Norms, Practicability, Standardisation, sample, correlate, sign, nomothetic, idiographic, actuarial prediction, Confirmatory bias, Illusory correlation, Barnum effect, test-user qualification levels, back-translated, item, Essay items, Objective items, supply type, selection type, multiple-choice item, distractors, specific determiners, stencil key, correction for guessing, Item analysis, item difficulty index, ceiling effect, floor effect, item discrimination index, item-total correlation, Distractor analysis, classical test theory, Item response theory, latent trait, item characteristic curve, Rasch model, unidimensionality, local independence, item information function, computerised adaptive testing, differential item functioning, response sets, response style, Acquiescence, Social desirability, deviation hypothesis, behaviourally anchored rating scale, halo effect, true score, reliability coefficient, domain-sampling model, random error, systematic error, test-retest reliability, coefficient of stability, alternate-forms relia-

bility, parallel-forms reliability, coefficient of equivalence, coefficient of stability and equivalence, internal consistency reliability, split-half reliability, Spearman-Brown prophecy formula, Rulon's formula, KR-20, KR-21, coefficient alpha, inter-rater reliability, Cohen's kappa, standard error of measurement, standard error of the difference, difference score, composite score, index of reliability, estimated true score, correction for attenuation, bandwidth-fidelity dilemma, generalizability theory, G study, D study, generalizability coefficient, test information, unified view of validity, Face validity, Content validity, table of specifications, content validity ratio, construct under-representation, construct-irrelevant variance, criterion-related validity, concurrent validity, predictive validity, criterion contamination, Restriction of range, cross-validation, Incremental validity, construct, Construct validity, nomological network, Convergent validity, discriminant validity, multitrait-multimethod matrix, Factorial validity, known-groups method, Consequential validity, ecological validity, validity ceiling, false positive, Sensitivity, Specificity, positive predictive value, base rate, receiver operating characteristic curve, selection ratio, Taylor-Russell tables, raw score, norm-referenced interpretation, norm group, standardisation sample, criterion-referenced interpretation, domain-referenced test, mastery test, ipsative score, national norms, local norms, subgroup norms, age-equivalent norm, mental age, grade-equivalent norm, ratio IQ, deviation IQ, percentile rank, quartiles, deciles, standard score, z score, normal curve, z table, linear transformation, deviation score, normalised standard score, T score, stanine, sten, C scale, scaled score, CEEB score, individual test, group test, verbal test, non-verbal test, performance test, age scale, point scale, routing subtests, basal level, ceiling level, Full Scale IQ, core subtests, index scores, supplemental subtests, Working Memory Index, Processing Speed Index, General Ability Index, Cognitive Proficiency Index, index discrepancies, scatter, PASS theory, culture-free test, culture-fair test, eductive ability, Draw-a-Man test, Bhatia's Battery of Performance Tests of Intelligence, Binet-Kamat Test, Malin's Intelligence Scale for Indian Children, Wechsler Adult Performance Intelligence Scale, developmental scale, developmental age, developmental quotient, social age, social quotient, adaptive behaviour, informant, screening test, diagnostic test, multi-informant assessment, Gesell Developmental Schedules, Bayley Scales of Infant Development, Denver Developmental Screening Test, McCarthy Scales of Children's Abilities, Developmental Screening Test, Developmental Assessment Scales for Indian Infants, Trivandrum Developmental Screening Chart, Vineland Social Maturity Scale, Malin's VSMS, Vineland Adaptive Behavior Scales, Adaptive Behavior Composite, AAMD Adaptive Behavior Scale, K-SADS, intellectual disability, disorders of intellectual development, educable, borderline intellectual functioning, global developmental delay, Rights of Persons with Disabilities Act 2016, benchmark disability, specific learning disability, NIMHANS Index for Specific Learning Disabilities, Childhood Autism Rating Scale, Indian Scale for Assessment of Autism, Autism Diagnostic Interview-Revised, Autism Diagnostic Observation Schedule, Children's Global Assessment Scale, survey test, prognostic test, ability-achievement discrepancy, multiple aptitude battery, Differential Aptitude Tests, General Aptitude Test Battery, cumulative record, convergent thinking, divergent thinking, fluency, flexibility, originality, elaboration, Torrance Tests of Creative Thinking, Remote Associates Test, interest inventory, Strong Vocational Interest Blank, ipsative, RIASEC, Self-Directed Search, method of equal-appearing intervals, scale value, method of summated ratings, Likert scale, cumulative scale, coefficient of reproducibility, social distance scale, semantic differential, law of comparative judgement, scale-discrimination technique, Implicit Association Test, bogus pipeline, objective personality test, L-data, Q-data, T-data, logical-content, Personal Data Sheet, theoretical strategy, Empirical criterion-group keying, factor-analytic strategy, forced-choice, validity scales, Minnesota Multiphasic Personality Inventory, L (Lie) scale, F (Infrequency) scale, K (Correction) scale, neurotic triad, psychotic tetrad, conversion V, demoralisation, Restructured Clinical (RC) scales, Sixteen Personality Factor Questionnaire, stens, Eysenck Personality Questionnaire, NEO Personality Inventory,

Myers-Briggs Type Indicator, Edwards Personal Preference Schedule, California Psychological Inventory, folk concepts, Millon Clinical Multiaxial Inventory, base rate scores, State-Trait Anxiety Inventory, Role Construct Repertory Test, Q-sort, Situational tests, projective technique, projective hypothesis, projection, association techniques, construction techniques, completion techniques, choice or ordering techniques, expression techniques, Rorschach Inkblot Test, inquiry phase, colour shock, location, determinants, form quality, reality testing, popular response, confabulatory response, Comprehensive System, Lambda, Experience Balance, Rorschach Performance Assessment System, Holtzman Inkblot Technique, Thematic Apperception Test, need-press theory, press, thema, Children's Apperception Test, word association test, complex indicators, sentence completion, Rosenzweig Picture-Frustration Study, Draw-A-Person, body-image hypothesis, House-Tree-Person, Neuropsychological assessment, executive functions, fixed battery, flexible battery, process approach, normative comparison, individual comparison, premorbid level, deterioration index, National Adult Reading Test, Halstead-Reitan Neuropsychological Battery, Halstead Impairment Index, Luria-Nebraska Neuropsychological Battery, NIMHANS Neuropsychological Battery, PGI Battery of Brain Dysfunction, Wechsler Memory Scale, Bender Visual-Motor Gestalt Test, visual-motor integration, working memory, Wisconsin Card Sorting Test, perseverative errors, set-shifting, Trail Making Test, response inhibition, thought disorder, Object Sorting Test, Mini-Mental State Examination, Montreal Cognitive Assessment, Glasgow Coma Scale, self-report inventory, clinician-rated scale, Beck Depression Inventory, Yale-Brown Obsessive Compulsive Scale, Positive and Negative Syndrome Scale, Psychopathy Checklist-Revised

From the past papers

1. The primary difference between a "speed test" and a "power test" in psychological assessment is that a speed test: (Real PYQ 2024)

   (a) Has items of increasing difficulty, while a power test has items of uniform difficulty. (b) Has a strict time limit that prevents most test-takers from completing all items, while a power test has items so difficult that no one can answer them all correctly. (c) Measures reaction time, while a power measures cognitive ability. (d) Is administered individually, while a power test can be administered in groups.

   Answer: (b). A speed test uses easy items under a time limit so strict that few finish; a power test allows generous time, but its items grow so hard that no one passes them all. Option (a) reverses the two, (c) confuses a timed test with reaction time, and (d) confuses timing with individual versus group administration.

2. Which of the following is the most important characteristic of a psychological test? (RML 2024)

   (a) Objectivity (b) Subjectivity (c) Validity (d) Rationality

   Answer (keyed): (a) Objectivity. RML follows Indian texts such as A. K. Singh, which list objectivity first: a score must not depend on who gives or scores the test. Most textbooks and the 2014 *Standards* call validity the most fundamental property, so on other papers choose (c). Subjectivity is a defect, and rationality is not a test characteristic.

3. Which of the following refers to the notion of the confirmatory bias? (RML 2021)

   (a) Clinicians ignore information that does not support their hypothesis (b) Clinicians ignore information that does not support their stereotypes. (c) Clinicians interpret ambiguous information as supporting their hypothesis. (d) All of the above

   Answer: (d) All of the above. Confirmatory bias is the tendency to notice, weigh and interpret information so that it confirms what one already believes; ignoring contrary evidence, ignoring evidence against one's stereotypes and reading ambiguity as support are all forms of it.

4. The improvement in anxiety may sometimes spuriously improve depression. This is called: (RML 2024)

   (a) Halo effect (b) Ceiling effect (c) Hawthorne effect (d) Therapeutic effect

   Answer: (a) Halo effect. One attribute (the improved anxiety) colours the rating of another, so the depression rating improves although the depression has not. A ceiling effect is a test too easy to register further gains, the Hawthorne effect is change because people know they are being studied, and a therapeutic effect is real improvement from treatment.

5. A response style or bias, in which there is tendency to agree with test items regardless of their contents is called: (RML 2024)

   (a) Halo effect (b) Central tendency effect (c) Social desirability effect (d) Acquiescence effect

   Answer: (d) Acquiescence effect. Acquiescence (yea-saying) is agreeing regardless of content and is controlled by balanced keying. Social desirability depends on content (the answer that looks good), central tendency is choosing the middle of the scale, and halo is a rater's error, not a test-taker's.

6. The tendency of the rater/interviewer to judge an individual on the basis of one single attribute is referred to as : (IHBAS 2024)

   (a) Similarity error (b) Central tendency error (c) Contrast error (d) Halo error

   Answer: (d) Halo error. Named by Thorndike (1920), halo lets one attribute or a general impression colour every other rating. Similarity error favours people like the rater, central tendency error rates everyone average, and contrast error rates others as opposite to the rater.

7. Which of the following is assessed by Cronbach's alpha? (RML 2021)

   (a) Concurrent validity (b) Inter-rater reliability (c) Test retest reliability (d) Internal consistency

   Answer: (d) Internal consistency. Alpha (Lee Cronbach, 1951) estimates how well a scale's items hang together within one administration. Concurrent validity needs an outside criterion, inter-rater reliability needs two scorers (kappa or the intraclass correlation), and test-retest reliability needs two occasions.

8. A test developer administers a new anxiety scale to a group of college students and, two weeks later, administers the same scale again to the same group. She then correlates the two sets of scores. This process is used to establish: (Real PYQ 2024)

   (a) Split-half reliability (b) Inter-rater reliability (c) Test-retest reliability (d) Parallel forms reliability

   Answer: (c) Test-retest reliability. Same scale, same people, two occasions, scores correlated: the coefficient of stability. Split-half correlates two halves of one sitting, inter-rater reliability compares scorers, and parallel forms would need a second, equivalent version of the scale.

9. Odd-even method of splitting the test to estimate split-half reliability is MOST appropriate for : (RML 2022)

   (a) Projective tests (b) Neuropsychological tests (c) Power tests (d) Speed tests

   Answer: (c) Power tests. With graded items and generous time, odd and even halves are fair parallel samples. On a speed test the halves match by construction and the coefficient is spuriously high, and projective and many neuropsychological tests lack comparable, objectively scored items to split.

10. To determine whether a newly developed intelligence test truly accurately measures intelligence or not, it is necessary to administer the test with one that has been in the field for some time. Which of the following needs to be looked for? (JEMAS 2025)

    (a) Construct validity (b) Concurrent validity (c) Content validity (d) Predictive validity

    Answer: (b) Concurrent validity. Correlating a new test with an established test given at about the same time is the concurrent type of criterion-related validity. Predictive validity needs a criterion measured later, content validity concerns sampling of the domain, and construct validity rests on a network of evidence rather than one established test.

11. Which of the following statements is TRUE? (RML 2024)

    (a) A test can be valid without being reliable (b) A test can be reliable without being valid (c) If a test measures what it is intended to measure, then by definition it is reliable (d) If a test gives consistent scores when administered repeatedly then it is valid

    Answer: (b) A test can be reliable without being valid. A test can consistently measure the wrong thing. It cannot be valid without being reliable, but option (c) overstates this as true by definition, and consistency alone (d) never proves validity.

12. A high false positive rate in diagnosis means : (IHBAS 2025)

    (a) Diagnosis is too strict (b) Normal people are diagnosed as disordered (c) Diagnosis has low reliability (d) Disordered people go undetected

    Answer: (b) Normal people are diagnosed as disordered. A false positive is a positive call on someone without the condition, the mark of low specificity. Undetected disordered people (d) are false negatives, which an over-strict diagnosis (a) produces; reliability (c) is a different property.

13. Which statistical measure tells us how far individual data points are dispersed from the mean expressed in standard deviation units? (Real PYQ 2025)

    (a) Variance (b) Z-score (Standard score) (c) Coefficient of variation (d) Standard error

    Answer: (b) Z-score (Standard score). $z = (X - M) \div SD$ states how many SDs one score lies from the mean, so scores from different distributions can be compared. Variance (SD squared) describes the spread of a whole distribution, the coefficient of variation compares spread across distributions, and the standard error is the SD of a sampling distribution.

14. IQ if any body has a mental age of 8 and a chronological age of 10 is: (RML 2020)

    (a) 108 (b) 125 (c) 80 (d) 92

    Answer: (c) 80. Ratio IQ = $(MA \div CA) \times 100 = (8 \div 10) \times 100 = 80$. The tempting 125 inverts the ratio ($CA \div MA$); a mental age below the chronological age must give an IQ below 100.

15. Which of the following is a non-linear transformation of raw scores? (Recall set)
(a) z-score (b) 50 + 10z (c) Stanine (d) Deviation score (X – M)

Answer: (c) Stanine. Stanines are assigned from percentile ranks in fixed proportions (4, 7, 12, 17, 20, 17, 12, 7, 4 per cent), a normalising, area transformation that forces a normal shape. z, the T score 50 + 10z and the deviation score X – M all apply one multiply-and-add rule to every score, so they are linear and keep the raw distribution's shape.

16. In the WAIS-IV (Wechsler Adult Intelligence Scale - Fourth Edition), how many core subtests are there in the Verbal Comprehension Index and the Perceptual Reasoning Index respectively? (Real PYQ 2025)
(a) 3 verbal, 4 performance (b) 3 verbal, 3 performance (c) 4 verbal, 3 performance (d) 2 verbal, 4 performance

Answer: (b) 3 verbal, 3 performance. The VCI core is Similarities, Vocabulary and Information (Comprehension is supplemental); the PRI core is Block Design, Matrix Reasoning and Visual Puzzles (Figure Weights and Picture Completion are supplemental). Counting a supplemental subtest as core produces the tempting 4.

17. A 10-year-old child from a rural area, with no formal schooling, is brought for an intellectual assessment due to academic struggles. The child is illiterate and has had minimal exposure to formal testing materials. Which of the following tests would be the MOST appropriate to use to minimize cultural and educational bias? (Real PYQ 2024)
(a) Bhatia's Battery of Performance Tests of Intelligence (b) Wechsler Intelligence Scale for Children (WISC-IV) (c) Stanford-Binet Intelligence Scale (5th ed.) (d) Malin's Intelligence Scale for Indian Children (MISIC)

Answer: (a) Bhatia's Battery. Its five performance subtests need no reading and carry Indian norms. The WISC-IV and SB5 rest on Western verbal content and norms, and MISIC, though Indian, is half verbal and assumes schooling.

18. An IQ test does NOT provide which of the following? (RML 2021)
(a) High test retest reliability (b) Good predictor of behaviour (c) High internal consistency (d) Good validity

Answer: (b) Good predictor of behaviour. IQ tests have high test-retest reliability, high internal consistency and good validity for predicting school achievement, but how a person behaves in general depends on personality, motivation and circumstance.

19. Vineland Social Maturity Scale (VSMS) is a measure of : (RML 2022)
(a) Mental age and Intelligence quotient (b) Social age and social quotient (c) Developmental age and developmental quotient (d) Emotional age and emotional quotient

Answer: (b) Social age and social quotient. Doll's VSMS converts the items an informant credits into a social age, and SQ = SA / CA × 100. Mental age and IQ come from intelligence tests, developmental age and DQ from developmental scales such as Gesell's or the DST.

20. A key differentiation between global developmental delay and intellectual disability is (JEMAS 2025)
(a) Age of assessment (under 5 years for GDD, 5 years and above for ID) (b) Presence of social deficits (c) Language delay alone (d) Motor impairment

Answer: (a) Age of assessment. DSM-5 reserves global developmental delay for children under 5 who are too young for reliable standardised testing and must be reassessed; from 5 years the diagnosis is intellectual disability. The deficits are of the same kind, so social, language or motor problems do not separate them.

21. Name of the Indian tool to assess the severity and disability of children with autism : (IHBAS 2024)

    (a) CGAS (b) Childhood Autism Rating Scale (c) CBCL (d) ISAA

    Answer: (d) ISAA (the paper keys (b) CARS). The Indian Scale for Assessment of Autism (NIMH Secunderabad, 2009; 40 items) is the Indian instrument prescribed for certifying autism and its disability percentage; CARS is the American scale of Schopler and colleagues on which ISAA partly drew. CGAS rates global functioning and the CBCL general behaviour problems.

22. Summated rating scale is also known as : (RML 2022)

    (a) Likert scale (b) Thurstone scale (c) Guttmann scale (d) Equal appearing interval scale

    Answer: (a) Likert scale. Likert (1932) had respondents rate their agreement with each statement and summed the ratings, hence summated ratings. The Thurstone scale and the equal-appearing interval scale are one and the same judge-scaled method, and the Guttman scale is cumulative.

23. The Implicit Association Test is designed to measure: (RML 2022)

    (a) Verbal ability (b) Unconscious stereotyping (c) Hidden mathematical ability (d) Dream content

    Answer: (b) Unconscious stereotyping. Greenwald, McGhee and Schwartz (1998) built the IAT to reveal implicit attitudes and stereotypes from reaction times in a sorting task. It is not an ability test of any kind, and dream content is not its target.

24. What is the primary purpose of achievement tests in psychological assessment? (RML 2025)

    (a) To measure future success potential (b) To assess cognitive abilities related to brain function (c) To measure understanding and knowledge in specific academic areas (d) To evaluate unconscious conflicts and emotions

    Answer: (c) To measure understanding and knowledge in specific academic areas. Achievement tests look back at what has been learnt. Future potential is the aim of aptitude tests, brain-related abilities of neuropsychological tests, and unconscious conflicts of projective techniques.

25. Which of the following statements is TRUE with respect to Minnesota Multiphasic Personality Inventory (MMPI)? (RML 2022)

    (a) It utilizes rational method of test construction (b) It is constructed by assuming specific personality traits and formulating questions to answer them (c) It is derived by finding items that discriminate between criterion and control groups (d) It is true with respect to both a and b

    Answer: (c) It is derived by finding items that discriminate between criterion and control groups. Hathaway and McKinley kept only items that separated patient groups from normal controls, whatever their content (empirical criterion keying). Options (a) and (b) describe the rational, logical-content strategy of Woodworth's Personal Data Sheet.

26. According to Raymond Cattell's classification of personality data sources, which of the following is NOT recognized as a legitimate data type? (Real PYQ 2025)

    (a) Q-data (Questionnaire data) (b) L-data (Life record data) (c) T-data (Objective test data) (d) P-data (Projective data)

    Answer: (d) P-data. Cattell recognised only three sources: L-data (life records such as school and criminal records, and observer ratings), Q-data (self-report questionnaires) and T-data (objective tests in controlled situations whose purpose is hidden).

27. Elevated Lie scale on Multiphasic Questionnaire (MPQ) is indicative of: (IHBAS 2020)
(a) Tendency to fake and manipulate (b) Social desirability (c) Mistrust (d) All the above
Answer: (d) All the above. A high Lie score means the person denies common minor faults: an attempt to fake a good impression, a pull towards social desirability, and a guarded mistrust that admits no flaw to the examiner.

28. How many Rorschach cards are coloured? (RML 2022)
(a) 10 (b) 5 (c) 3 (d) 4
Answer: (b) 5. Two cards (II, III) are black with red and three (VIII, IX, X) are multicoloured, so five carry colour and five (I, IV, V, VI, VII) are achromatic. Three counts only the fully coloured cards; ten is the whole set.

29. Thematic Apperception Test (TAT) is based on which technique of psychological testing? (RML 2022)
(a) Expressive technique (b) Construction technique (c) Association technique (d) Completion technique
Answer: (b) Construction technique. In Lindzey's scheme the person constructs a story about each picture. Association is the Rorschach and word association, completion is sentence completion and the Rosenzweig P-F, and expression is drawing and play.

30. Absence of 'Human' Response on Rorschach Psychodiagnostics Test indicates: (IHBAS 2020)
(a) Apathy (b) Poor interpersonal orientation (c) Lack of empathy (d) All the above
Answer: (b) Poor interpersonal orientation. Human content reflects interest in and understanding of people, so its absence points to poor interpersonal orientation. Apathy and lack of empathy are not the standard reading, so 'All the above' is wrong as well.

31. One of the most widely used tests to assess executive functioning: (Amity 2023)
(a) Wisconsin Card Sorting Test (b) Minnesota Number Ordering Test (c) Manhattan Card Sorting Test (d) None of the above
Answer: (a) Wisconsin Card Sorting Test. Grant and Berg's WCST (1948) measures set-shifting and cognitive flexibility through perseverative errors when the sorting rule changes unannounced, and is most sensitive to dorsolateral prefrontal dysfunction. The Minnesota and Manhattan tests are invented distractors, so 'none of the above' is wrong too.

32. Which of the following is a clinician-rated questionnaire? (RML 2022)
(a) Beck Depression Inventory (b) Beck Anxiety Inventory (c) Hamilton Anxiety Rating Scale (d) Both b and c
Answer: (c) Hamilton Anxiety Rating Scale. Max Hamilton's 14-item anxiety scale (1959) is completed by a trained clinician from interview and observation. Both Beck inventories are self-report: the patient rates his own symptoms, so (a), (b) and (d) are wrong.

33. Impact of psychiatric illness on one's cognitive and intellectual abilities can be estimated on the basis of: (IHBAS 2020)
(a) Deterioration quotient (b) Mediation index (c) Sense of coherence index (d) Theory of Mind index
Answer: (a) Deterioration quotient. Wechsler compared 'hold' subtests (Vocabulary, Information, Picture Completion, Object Assembly) with 'don't hold' subtests (Digit Span, Similarities, Digit Symbol, Block Design) to estimate loss from illness. Mediation is a statistical concept, sense of coherence is Antonovsky's health concept, and theory of mind concerns understanding others' minds.

High-yield sheet

The facts this chapter's questions ask most, most-asked first. Last read before the exam.

1 **Cronbach's alpha** (Lee Cronbach, 1951) measures **internal consistency reliability**: not validity of any kind, not test-retest and not inter-rater reliability. ASKED 10 TIMES

2 The **Spearman-Brown prophecy formula** (Spearman and Brown, 1910) steps a split-half correlation up to full length, 2r / (1 + r), so .60 becomes .75; in general form kr / [1 + (k - 1)r] it shows that adding parallel items raises reliability. ASKED 9 TIMES

3 **Test-retest reliability** correlates the same test given to the same people on two occasions (the coefficient of stability); it suits stable traits such as intelligence and personality, not changing states such as current mood. ASKED 9 TIMES

4 **Criterion-related validity** has two types: **concurrent** (test and criterion measured at about the same time, such as a new intelligence test given with an established one) and **predictive** (criterion measured later, such as entrance scores and later grades). ASKED 9 TIMES

5 The **Thematic Apperception Test** was developed by Christiana Morgan and Henry Murray (1935, Harvard Psychological Clinic), rests on Murray's need-press theory, and is a projective **construction** technique in which the person tells stories about ambiguous pictures of social scenes. ASKED 9 TIMES

6 The three classic MMPI validity scales are **L** (Lie: naive denial of common minor faults), **F** (Infrequency: random responding, faking bad or severe pathology; T above 100 suggests random responding or gross exaggeration) and **K** (Correction: subtle defensiveness; a fraction of K is added to scales 1, 4, 7, 8 and 9). ASKED 8 TIMES

7 **KR-20** (Kuder and Richardson, 1937) is for dichotomously scored items of varying difficulty; KR-21 assumes equal difficulty and gives a lower estimate when difficulties differ; alpha equals KR-20 when every item is scored 0 or 1. ASKED 7 TIMES

8 On the normal curve z = 0 is the 50th percentile, z = +1 the 84th (−1 the 16th), z = +1.5 about the 93rd and z = +2 about the 98th; so IQ 100, 85 and 130 (SD 15) are the 50th, 16th and about 98th percentiles, and IQ 136 on an SD-16 scale (z = 2.25) has about 98.8 per cent below it. ASKED 7 TIMES

9 The **WAIS-IV** (2008; ages 16:0 to 90:11) yields a Full Scale IQ and four index scores, Verbal Comprehension, Perceptual Reasoning, Working Memory and Processing Speed, from 10 core subtests: 3 verbal (Similarities, Vocabulary, Information), 3 perceptual (Block Design, Matrix Reasoning, Visual Puzzles), 2 working memory (Digit Span, Arithmetic) and 2 processing speed (Symbol Search, Coding). ASKED 7 TIMES

10 The **16PF** is Raymond B. Cattell's (1949; fifth edition 1993, 185 items): 16 primary source traits found by factor analysis and five global factors, scored in stens; it measures personality traits, not intelligence. ASKED 7 TIMES

11 **Reliability** means consistency, dependability and freedom from measurement error (all of these): a reliable measure gives stable results on different occasions. ASKED 6 TIMES

12 **SEM = SD × √(1 - r)**, which estimates the range within which a person's true score probably falls: with SD 15 and r = .91 the SEM is 4.5 (±1 SEM 68 percent, ±1.96 SEM 95 percent). ASKED 6 TIMES

13 The **Vineland Adaptive Behavior Scales** (Sparrow, Balla and Cicchetti, 1984; Vineland-3, 2016) measure adaptive functioning in daily life (communication, daily living skills, socialisation, motor skills), not IQ, personality or psychopathology. ASKED 6 TIMES

14 The **MMPI** (Hathaway and McKinley, 1943; 550 items) is an objective personality inventory that assesses personality and psychopathology for psychiatric diagnosis, covering mood, physical concerns and social attitudes; it is not projective, an intelligence test or an aptitude battery. ASKED 6 TIMES

15 Projective tests present ambiguous stimuli to reveal **unconscious processes**; their main advantage is that they avoid self-report biases and are hard to fake. ASKED 6 TIMES

16 John Exner's **Comprehensive System** (1974) is the standard Rorschach scoring system: it codes location, determinants, form quality and content, gives no IQ, and there is no 'Murray system' of Rorschach scoring. ASKED 6 TIMES

17 The **Wisconsin Card Sorting Test** (David Grant and Esta Berg, 1948) is the most widely used test of executive function: it measures set-shifting and cognitive flexibility through perseverative errors and is most sensitive to **dorsolateral prefrontal cortex** dysfunction. ASKED 6 TIMES

18 A test can be **reliable without being valid** but cannot be valid without being reliable: reliability is necessary but not sufficient for validity. ASKED 5 TIMES

19 Wechsler **scaled scores** have a mean of 10 and an SD of 3 (range 1 to 19): 10 is the 50th percentile, 7 is 1 SD below (about the 16th, low average), 13 is 1 SD above and 4 is 2 SD below. ASKED 5 TIMES

20 The **Rorschach** (Hermann Rorschach, *Psychodiagnostik*, 1921) has **10** symmetrical inkblot cards: 5 achromatic (I, IV, V, VI, VII), 2 black and red (II, III) and 3 multicoloured (VIII, IX, X), so five achromatic and five chromatic. ASKED 5 TIMES

21 The **Bender Visual-Motor Gestalt Test** (Lauretta Bender, 1938; nine figures adapted from Wertheimer, 16 in Bender-Gestalt II, 2003) assesses **visual-motor integration** and screens for organic brain damage and developmental immaturity; it is not a personality test. ASKED 5 TIMES

22 The **Halstead-Reitan Neuropsychological Battery** (Halstead 1947; Reitan) is a fixed battery for comprehensive neuropsychological assessment (Category Test, Tactual Performance, Seashore Rhythm, Speech-Sounds Perception, Finger Tapping, Trail Making); an Impairment Index of 0.5 or more suggests brain damage, and the Rorschach is not part of it. ASKED 5 TIMES

23 The **BDI-II** (Aaron Beck, 1996) is a self-report inventory of 21 items rated 0 to 3 for the past two weeks: 0 to 13 minimal, 14 to 19 mild, 20 to 28 moderate, 29 to 63 severe. ASKED 5 TIMES

24 **Sensitivity** = TP / (TP + FN), the proportion of people with the condition whom the test correctly identifies; **specificity** = TN / (TN + FP), the proportion without it whom the test correctly clears. ASKED 4 TIMES

25 **Construct validity** (Cronbach and Meehl, 1955) is the degree to which a test measures the theoretical construct it claims to measure; **convergent validity** is a high correlation with measures of the same construct and **discriminant validity** a low correlation with measures of unrelated constructs. ASKED 4 TIMES

26 **T scores** (McCall, 1922) have a mean of 50 and an SD of 10 ($T = 50 + 10z$) and are used on the MMPI and NEO-PI-R; T 70 is 2 SD above the mean (original MMPI cut-off) and T 65 is 1.5 SD above (MMPI-2 cut-off). ASKED 4 TIMES

27 The **z score**, $z = (X - M) \div SD$, has a mean of 0 and an SD of 1, expresses a score's distance from the mean in SD units and lets results from different distributions be compared. ASKED 4 TIMES

28 **Stanines** run from 1 to 9 (mean 5, SD about 2) in the proportions 4, 7, 12, 17, 20, 17, 12, 7, 4 per cent; stanine 5 holds 20 per cent, the 40th to 60th percentile (mean ± 0.25 SD), and is a normalised, nonlinear transformation. ASKED 4 TIMES

29 The **Vineland Social Maturity Scale** was devised by **Edgar A. Doll** (1935, Vineland Training School): 117 items from birth to adulthood (25 and above), given by informant interview, yielding **social age and social quotient** (SQ = SA / CA × 100). ASKED 4 TIMES

30 A **Likert scale** is a **summated rating scale** (Rensis Likert, 1932): closed-ended statements with a fixed number of ordered alternatives, usually five from strongly agree to strongly disagree, rated by respondents and summed with equal weights. It is a type of attitude scale, not a level of measurement. ASKED 4 TIMES

CHAPTER 3

Social Psychology

IN THIS CHAPTER

3.1 • Social Psychology: History, Theories, Concepts and Methods: Define social psychology and set it apart from its neighbours, trace its history from Le Bon, Triplett and Lewin to the replication crisis, use the sociological ideas of anomie, socialisation, status and role, match the main theories to their founders, and outline the field's methods and research ethics.

3.2 • Social Cognition and First Impressions: Explain how people think about the social world (cognitive misers, dual processes, schemas, priming, heuristics and the errors of social thought), recognise the self-fulfilling prophecy and the halo effect, and describe how first impressions form, how accurate thin slices are and how people manage the impressions they make.

3.3 • Communication, Nonverbal Cues and Attribution: Trace a message through the communication process and its classic models, read nonverbal cues and personal space, explain behaviour with the attribution theories of Heider, Jones and Davis, Kelley and Weiner, and recognise each attributional bias from its description.

3.4 • Attitudes, Persuasion and Cognitive Dissonance: Define attitudes, their components, strength and functions, explain how they form and when they predict behaviour (reasoned action, planned behaviour, MODE), analyse persuasion through the Yale factors and the elaboration likelihood model, explain resistance (reactance, forewarning, inoculation), and apply balance and cognitive dissonance theory with its classic paradigms and rivals.

3.5 • Stereotypes, Prejudice and Intergroup Relations: Distinguish stereotype, prejudice and discrimination, explain where prejudice comes from (personality, frustration, learning, realistic conflict and social identity), describe the processes that sustain stereotypes and the barriers they create at work, and state when contact reduces prejudice.

3.6 • Affiliation, Attraction and Close Relationships: Explain why people affiliate and compare themselves with others, what draws them together, how exchange, equity and investment theories account for staying or leaving, and how self-disclosure, love, break-up and loneliness are described and measured.

3.7 • Social Influence: Conformity, Compliance, Obedience and Power: Tell apart conformity, compliance and obedience, explain why people conform or resist, and recall the classic studies of Sherif, Asch, Milgram and Zimbardo, Cialdini's compliance principles and tactics, and French and Raven's bases of power.

3.8 • Prosocial Behaviour: Why and When People Help: Explain why and when people help: the empathy-altruism, egoistic, evolutionary and normative accounts, the bystander effect with its five-step model and its two classic explanations, and the situations, personalities and recipient reactions that raise or lower helping.

3.9 • Aggression: Its Roots, Causes and Control: Define aggression and its forms, compare the instinct, biological, frustration-aggression, learning and General Aggression Model explanations, link each major cause of aggression to its researchers, and judge which methods of control work and why catharsis fails.

3.10 •• Groups: Structure, Performance, Decisions and Leadership: Define a group and its structure, map a group with sociometry, order Tuckman's stages, predict when the presence of others raises or lowers effort and performance, diagnose and prevent group polarisation and groupthink, and compare the main theories of leadership.

3.11 • Applied Social Psychology: Law, Culture and Social Work: Apply social psychology to health, environment, eyewitnesses, juries and offenders, and name the roots, methods, models, principles and pioneers of social work, from case work and group work to culturally sensitive practice, child welfare and the first Indian school.

Weightage. 136 of the 3180 questions mapped in this book (4.3%) test this chapter, 93 of them from real past papers. 0 of its 11 sections are High yield.

In the early hours of 13 March 1964, Catherine "Kitty" Genovese, a 28-year-old bar manager, parked her car near her flat in Kew Gardens, a quiet neighbourhood of Queens, New York, and started the short walk home. A stranger, Winston Moseley, chased her and stabbed her. A neighbour shouted from a window and Moseley fled, but about ten minutes later he came back, found her lying wounded in a hallway at the back of her building, and attacked her again. She died on the way to hospital.

Two weeks later *The New York Times* put the killing on its front page. For more than half an hour, it reported, 38 respectable, law-abiding citizens had watched a killer stalk and stab a woman, and not one had telephoned the police during the attack. The story went round the world as proof of big-city apathy. Two young psychologists in New York, John Darley and Bibb Latane, suspected another explanation: perhaps each witness, knowing that others could also hear, assumed that someone else would act. They staged an emergency in the laboratory, reported in 1968, in which students talking over an intercom heard another student apparently have a seizure. The more listeners a student believed were present, the less likely that student was to go for help.

The finding became one of the most famous in psychology, yet the newspaper story behind it was badly wrong. Later investigations, among them a 2007 review by the psychologists Rachel Manning, Mark Levine and Alan Collins, found no 38 watching witnesses: few neighbours saw any part of the attacks, most heard only a cry in the night, and one, Sophia Farrar, went down and held Kitty in her arms as she lay dying. In 2016 the *Times* itself conceded that its report had exaggerated the number of witnesses and what they saw. One night in Queens thus holds social psychology in small: how the presence of others changes what each of us does, and how a whole society can come to share a story that is not quite true.

3.1 Social Psychology: History, Theories, Concepts and Methods

LOWER YIELD · 6 QS

Why do you drive more carefully with your father beside you? Social psychology asks how other people, present or imagined, shape what we think, feel and do. This section defines the field, traces its history, and covers its sociological roots, theories and methods.

What social psychology studies

The classic definition is that of **Gordon Willard Allport** (1897 to 1967), from 1954: **social psychology** studies how the thoughts, feelings and behaviour of individuals are influenced by the actual, imagined or implied presence of others: an audience in the hall, a mother whose frown you picture, a "No smoking" sign left by strangers. **Robert A. Baron** and **Nyla R. Branscombe** call it the scientific study of the nature and causes of individual behaviour, feeling and thought in social situations: *scientific* because evidence settles questions, *individual* because the unit is the person, even in a crowd.

Behaviour reflects the **person-situation interaction**: a student silent in class talks freely among friends. **Psychological social psychology** studies the individual's mental processes, mostly by experiment; **sociological social psychology** studies how groups, roles and social

structure shape people, mostly by surveys and observation, drawing on symbolic interactionism (below). Sociology proper studies groups and societies, and personality psychology stable differences between people (Chapter 4); social psychology asks how situations make different people act alike. Common sense has a proverb for every outcome ("birds of a feather flock together", yet "opposites attract"), and hindsight bias (Section 1.8) makes findings look obvious afterwards; only testing shows which holds when.

> **EXAM TRAP**
> "Actual, imagined or implied presence of others" is **Gordon Allport** (1954), though Baron and Branscombe credit it to his brother Floyd (1924). "Nature and causes of individual behaviour ... in social situations" is **Baron and Branscombe**.

Founders and landmarks

Two roots, crowds and experiments, fed the field (Table 3.1). Le Bon's crowd mind, Ringelmann's rope pullers (each pulled less hard as the team grew) and Triplett's children, who wound fishing reels faster in pairs than alone in what is usually called the first social psychology experiment, are taught in Section 3.10. The first two textbooks (1908) pulled opposite ways: the British psychologist **William McDougall** traced social behaviour to inborn instincts, the American sociologist **Edward Alsworth Ross** to imitation and suggestion in crowds. **Floyd Henry Allport** (1924) made the field experimental and individual-centred, and **Muzafer Sherif** (1935, 1936) watched group norms form as people judged the apparent movement of a light in the dark (Section 3.7).

Table 3.1 Landmarks in the history of social psychology.

Year	Who	Landmark
1895	**Gustave Le Bon**	*The Crowd*: contagion, the crowd mind
1898	**Norman Triplett**	First experiment: others speed us up
1908	McDougall; Ross	First textbooks: instinct; imitation
1913	**Max Ringelmann**	Rope pulling: effort falls in groups
1924	Floyd Allport	Experimental, individual-centred text
1935, 1936	Muzafer Sherif	Norm formation (autokinetic effect)
1939	Lewin, **Ronald Lippitt**, **Ralph K. White**	Leadership climates
1945 to 1947	Kurt Lewin	Group dynamics, action research, T-groups
1950	**Theodor W. Adorno** and colleagues	*The Authoritarian Personality*
1951	**Solomon Asch**	Conformity to a wrong majority
1953	**Carl Hovland** and colleagues	Yale persuasion research
1954, 1957	**Leon Festinger**	Social comparison; cognitive dissonance
1958	**Fritz Heider**	*The Psychology of Interpersonal Relations*
1961	Serge Moscovici	Social representations
1963	**Stanley Milgram**	Obedience
1971	**Philip Zimbardo**	Stanford prison study

Year	Who	Landmark
1973	Kenneth Gergen	"Social psychology as history"
2010s	Many teams	Replication crisis

Kurt Lewin ●●

Kurt Lewin (1890 to 1947), a Gestalt-trained refugee from Nazi Germany, is called the father of modern experimental social psychology. His field theory (Section 1.2) states B = f(P, E): behaviour is a function of the person and the environment, together the **life space**, the person plus the *psychological* environment as it exists for that person at that moment, not society, the group or the physical setting.

Lewin wanted research to change real life. In 1945 he founded the Research Center for Group Dynamics at the Massachusetts Institute of Technology (MIT); **group dynamics** is the study of forces within groups. In 1946 he described **action research**, done with the people affected, as a spiral of steps, each a circle of planning, action and fact-finding about the results (Figure 3.1). A 1946 workshop produced the **T-group** (training group), whose members learn from frank feedback on their behaviour in the group, spread by the National Training Laboratories (1947) as sensitivity training. His motto: "There is nothing so practical as a good theory."

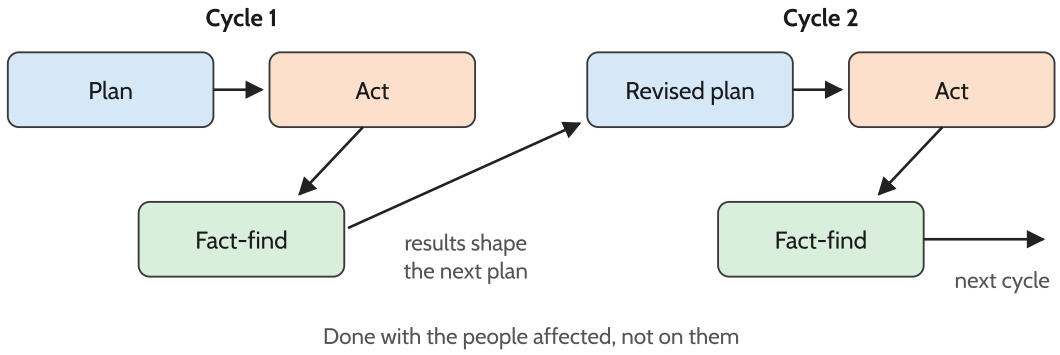


Figure 3.1 Lewin's action research (1946): fact-finding about each action feeds a revised plan.

EXAM TRAP

Life space = person + psychological environment, not society, group or physical environment. Lewin founded **action research**, not psychoanalysis (Freud), psychophysics (Fechner) or functionalism (James).

Crisis and new directions

After 1945 the field grew around the war's questions: persuasion, prejudice, conformity and obedience (Table 3.1). By the late 1960s a crisis had set in: laboratory experiments on students looked artificial, ethically doubtful and culture-bound, and **Kenneth J. Gergen** argued in "Social psychology as history" (1973) that its findings describe one society at one moment, not timeless laws. Europeans made the field more social: **Henri Tajfel** (1919 to 1982) studied group identity (Section 3.5), and Serge Moscovici minorities and shared knowledge. Then came the social cognition era of the 1970s and 1980s (Section 3.2), social neuroscience (the brain activi-

ty behind social thought), implicit processes (automatic attitudes outside awareness, measured by the Implicit Association Test, 1998) and cultural psychology (how culture shapes social thought and behaviour). In the 2010s the **replication crisis** broke: when 100 published psychology studies were repeated in 2015, only about a third gave the original result.

Sociological roots: Durkheim to Mead

Emile Durkheim (1858 to 1917), founder of French sociology, treated society as a reality in its own right. **Social facts** are ways of acting, thinking and feeling that exist outside individuals and constrain them, such as laws and customs; **collective representations** are the ideas and symbols a society shares. His most examined idea is **anomie**: normlessness, a state in which social norms are unclear, weak or no longer apply, so people no longer know what is expected of them. He introduced it in *The Division of Labour in Society* (1893); in *Suicide* (1897), anomic suicide rises when sudden change, a crash or a windfall, unsettles the norms. **Robert King Merton** (1910 to 2003) recast anomie as strain (1938), a gap between cultural goals such as wealth and the legitimate means to them, met by conformity, innovation (illegitimate means such as fraud), ritualism (rules kept, goal dropped), retreatism (both rejected) or rebellion (both replaced).

> EXAM TRAP
>
> **Anomie** = normlessness. The progression from traditional to modern society is *modernisation*: Durkheim's mechanical solidarity (bonds of sameness) giving way to organic solidarity (interdependent specialists), or **Ferdinand Tonnies**'s (1887) *Gemeinschaft* (community) giving way to *Gesellschaft* (society). Adapting to a dominant culture's values is *assimilation* or *acculturation*; internalising society's norms is *socialisation*.

A second root is **symbolic interactionism**: people act towards things according to the meanings those things have for them, and meanings, and the self, are built in interaction through symbols, above all language. **George Herbert Mead** (1863 to 1931) founded it; students published his lectures as *Mind, Self and Society* (1934), and **Herbert Blumer** named it in 1937. **Charles Horton Cooley** (1864 to 1929) added the looking-glass self (1902: we imagine how we appear to others, imagine their judgement and feel pride or shame; Chapter 4) and the primary group (1909), such as the family.

Socialisation and resocialisation

Socialisation is the lifelong learning of a society's norms, values and roles until they become one's own: *primary* in childhood within the family, *secondary* through school, peers, work and the media, the other agents of socialisation (gender socialisation is in Chapter 6). Merton's **anticipatory socialisation** is adopting a group's norms before joining it, as a medical aspirant already talks like a doctor.

Resocialisation is unlearning old norms, values and roles and learning new ones after a major transition: college, migration, marriage, the army or prison. A student's first weeks at college, before she finds friends and fits in, are a mild case. It is most intense in the **total institutions** described by **Erving Goffman** (1922 to 1982) in *Asylums* (1961), such as prisons, mental hospitals and barracks, which cut people off from society, manage all of daily life and strip away the old identity.

CLINICAL LINK

Long-stay inmates may develop **institutionalisation**: apathy, lost initiative, submissiveness and dependence on routine. **Russell Barton** called it institutional neurosis (1959), a product of the institution rather than the illness (Chapter 7).

EXAM TRAP

A newcomer adapting to college, a new country or the army shows **resocialisation**, not the *looking-glass self* (a self-image built from imagined judgements), *self-awareness* (attention turned on oneself) or *institutionalisation* (the dependent apathy of long-stay inmates).

Status and roles

A **status** is a position in a group: *ascribed* if given at birth (sex, caste), *achieved* if earned (doctor), in **Ralph Linton**'s distinction (1936). A **role** is the behaviour expected of whoever holds a status. Table 3.2 separates the easily confused terms; Zimbardo's prison study shows the power of roles (Section 3.7).

Table 3.2 The vocabulary of roles.

Term	Meaning	Example
Role expectation	What others expect of the person in a position	Patients expect a doctor to listen
Role taking (Mead)	Seeing a situation from another's viewpoint: one person at a time (play stage), all the players at once (game stage), then the community, the *generalised other*	A fielder reading the bowler and wicketkeeper
Playing at a role	A young child's imitative make-believe	Playing "teacher" to dolls
Role playing (role enactment)	The behaviour a person actually engages in while enacting a role	A new teacher taking her first class
Role conflict	Clashing demands: inter-role (between roles) or intra-role (within one)	Mother and employee on a sick child's day
Role strain (**William J. Goode**, 1960)	Felt difficulty in meeting one role's obligations	A student with too many deadlines
Role set (Merton, 1957)	All the role partners of one status	A student's teachers, classmates, parents
Role ambiguity	Unclear expectations	A job with no job description

EXAM TRAP

"The behaviour an individual engages in while enacting a role" is **role playing**. *Role taking* is mental (seeing through another's eyes), *playing at a role* is a child's make-believe, and *role expectation* lies in other people, not in the performer.

Theoretical perspectives

The field works with several theories (Table 3.3), most taught where they are used.

Table 3.3 Theoretical perspectives in social psychology.

Theory	Theorist, year	Core idea
Field theory	Kurt Lewin	B = f(P, E)
Cognitive consistency	Heider (1958); Festinger (1957)	We strive for consistent beliefs
Social exchange	**George C. Homans** (1961); **John Thibaut** and **Harold Kelley** (1959); **Peter Blau** (1964)	Relationships weighed by rewards and costs
Social learning, social cognitive	**Albert Bandura**	Learning by observing models
Psychoanalytic group psychology	**Sigmund Freud** (1921)	Members idealise the leader and identify with one another
Sociobiology, evolutionary	**Edward O. Wilson** (1975)	Natural selection shapes social behaviour; kin selection
Social representations	Serge Moscovici (1961)	Shared common-sense theories
Social identity	Tajfel and **John Turner** (1979)	Self-esteem from group membership
Uncertainty-identity	**Michael A. Hogg**	Group identity reduces self-uncertainty
Need to belong	**Roy F. Baumeister** and **Mark R. Leary** (1995)	A basic drive for lasting bonds
Sociometer	Mark R. Leary	Self-esteem gauges acceptance
Role theory	Mead; Ralph Linton (1936)	Behaviour follows the roles people occupy
Social constructionism	Kenneth Gergen (1985)	Knowledge is built socially, not discovered
Dual-process models	Petty and Cacioppo, others	Fast automatic versus slow deliberate thinking
Liberation psychology	**Ignacio Martin-Baro**	Psychology from the viewpoint of the oppressed

A **social representation** is a shared, changing system of values, ideas and practices, a common-sense theory through which a group makes the unfamiliar familiar: a collective successor to Durkheim's collective representations.

KEY STUDY: MOSCOVICI AND PSYCHOANALYSIS IN FRANCE (1961)

Serge Moscovici (1925 to 2014), a Romanian-born French psychologist, asked in *La Psychanalyse, son image et son public* (1961) how a scientific theory becomes common sense. Surveying Parisians and analysing the Catholic, Communist and popular press, he found each group remaking psychoanalysis by **anchoring**, fitting the unfamiliar into a familiar category (analysis seen as a kind of confession), and **objectification**, turning an abstract idea into a concrete thing (a "complex" became something a person *has*, like a cold). Critics find the concept too broad to test precisely.

EXAM TRAP

Social representations = **Serge Moscovici**. Michael A. Hogg is social identity and uncertainty-identity theory; Baumeister and Leary proposed the need to belong, and Leary the sociometer.

Methods and ethics

Social psychology uses the methods of Chapter 9 (Table 3.4). A theory predicts; results support or revise it, never prove it.

Table 3.4 Research methods in social psychology.

Method	What it does, and its catch
Systematic observation	Records behaviour, often in natural settings (naturalistic); participant observers join the group, and may alter it
Survey	Questions large samples; self-report bias
Correlational	Measures how two variables co-vary; correlation is not causation
Laboratory experiment	Manipulates an independent variable, measures a dependent variable; shows cause, can be artificial
Field experiment	Experiment in a natural setting; realistic, less control
Quasi-experiment	Compares existing groups without random assignment; cause uncertain
Archival	Analyses existing records; no reactivity
Cross-cultural	Compares cultures; tests generality
Meta-analysis	Statistically combines many studies into one overall effect size

Random assignment, giving every participant an equal chance of any condition, lets an experiment show cause. **Elliot Aronson** and **J. Merrill Carlsmith** (1968) separated **experimental realism**, how far a study absorbs participants, from **mundane realism**, how far it resembles everyday life; Milgram's study had much of the first, little of the second. External validity is how far results generalise. A **mediating variable** explains *how* or *why* one variable affects another (a violent film raises aggressive thoughts, which raise aggression); a **moderating variable** changes *when* or *for whom* it does (more in people already angry).

Deception, misleading participants about a study's purpose so they behave naturally, needs two safeguards: informed consent beforehand (what they will do, and that they may leave at any time) and **debriefing** afterwards, by *dehoaxing* (revealing the deception) and *desensitising* (removing distress, such as the belief that one is cruel). The Milgram and Zimbardo controversies made both standard (Section 3.7). Participants also respond to **demand characteristics**, cues revealing what the experimenter hopes to find (**Martin Orne**, 1962), and experimenters bias results through expectancy effects (**Robert Rosenthal**, 1966); double-blind procedures guard against both.

INDIAN CONTEXT

Gardner Murphy led a UNESCO study of social tensions in India, *In the Minds of Men* (1953). **Durganand Sinha** (1922 to 1998) urged indigenising psychology (Section 1.2), **Jai B. P. Sinha** proposed the nurturant-task leader (Section 3.10) and **Udai Pareek** (1925 to 2010) studied role stress. Research centres on caste, communal relations and poverty.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Allport: the actual, imagined or _____ presence of others.
2. Lewin's life space is the person plus the _____ environment.
3. Lewin founded _____ research: planning, action and fact-finding.
4. Durkheim's term for normlessness is _____.
5. Unlearning old norms and learning new ones at college is _____.
6. Behaviour performed while enacting a role is role _____.
7. Social representations theory (1961) is _____'s.
8. Cues revealing the experimenter's hypothesis are _____ characteristics.

Answers: (1) implied, (2) psychological, (3) action, (4) anomie, (5) resocialisation, (6) playing, (7) Serge Moscovici, (8) demand

3.2 Social Cognition and First Impressions LOWER YIELD · 7 QS

A stranger sits down at your hostel mess table, and within seconds you have judged him friendly, clever or arrogant; from then on you notice what fits. This section follows social thought from the frameworks that store it, through the shortcuts and tilts that bend it, to the impressions we form of others and manage in ourselves.

The social thinker

Social cognition is how we interpret, analyse, remember and use information about the social world. **Susan Fiske** and **Shelley Taylor** (1984) called people **cognitive misers**: with limited capacity and a flood of social information, we save effort with shortcuts and settle for good-enough answers. In their 1991 revision the thinker became a **motivated tactician**, choosing shortcuts or careful thought according to goals and stakes: you skim a passing stranger but study an interviewer's face.

That choice is **dual-process** thinking. **Automatic processing** is fast, effortless, unintended and often unconscious; **controlled processing** is slow, effortful, deliberate and conscious (System 1 and System 2, Section 1.8). Fatigue and time pressure push us onto the automatic track. The dominant account of how people organise social information in memory and use it to understand behaviour is the **schema approach**.

EXAM TRAP

"How people organise information in memory and use it to understand behaviour" = the **schema approach**. A *motivational approach* explains social thought by needs and goals; a *perceptual approach* treats knowing people like perceiving objects (Asch's Gestalt view, below); "organisational approach" merely echoes the stem's verb.

Schemas, priming and perseverance

A **schema** is an organised mental framework, built from experience, that holds our knowledge about a person, group, role or event and guides how we take in new information (Bartlett's idea, Section 1.7). Table 3.5 lists the kinds.

Table 3.5 Kinds of social schema.

Schema	What it holds	Example
Person schema	Traits and goals of a person or type	Your cautious uncle
Self-schema	Beliefs about oneself (**Hazel Markus**, 1977; Chapter 4)	"I am a good debater"
Role schema	Expected behaviour in a role	What a professor does
Event schema or **script**	The usual sequence of an event (**Roger Schank** and **Robert Abelson**, 1977)	Eating at a restaurant
Group schema or stereotype	Traits assumed of a group (Section 3.5)	"Engineers are logical"

Schemas steer **attention** towards what fits, shape **encoding** (consistent information is stored readily, though a sharply inconsistent event stands out) and bias **retrieval** towards schema-consistent details. Recently activated schemas dominate. **Priming** is a temporary rise in a schema's availability: after a violent film, a driver who takes your parking spot looks aggressive. **Automatic priming** works below awareness: **John Bargh**, Mark Chen and Lara Burrows (1996) had students unscramble sentences containing words linked to old age (bingo, Florida), and they then walked down the corridor more slowly. It failed to replicate: Stéphane Doyen and colleagues (2012), timing walkers with infrared sensors, found slowing only when experimenters expected it. A primed schema fades by **unpriming** once it is expressed in thought or action (Betsy Sparrow and Daniel Wegner, 2006).

Schemas also resist disconfirmation, the **perseverance effect**. In the classic study of **belief perseverance**, **Lee Ross**, Mark Lepper and Michael Hubbard (1975) gave students false feedback that they were good or poor at telling real suicide notes from fakes; told later that the feedback was invented, they still rated their ability in line with it. Confirmation bias (Section 1.8) keeps schemas alive. Even figures of speech prime schemas: in **reasoning by metaphor** we grasp an abstract idea through a concrete one ("our relationship is on the right track"), and the metaphor shapes the judgement. Worst of all, a schema can make itself come true.

The self-fulfilling prophecy

The sociologist **Robert King Merton** (1910 to 2003) named and formalised the **self-fulfilling prophecy** in 1948: "a false definition of the situation evoking a new behaviour which makes the originally false conception come true." A rumour that a sound bank is failing starts a run

on it, and it fails. In social psychology the term covers any expectation, about oneself or another, that leads us to act in ways that make the expected event or behaviour more likely (Figure 3.2).

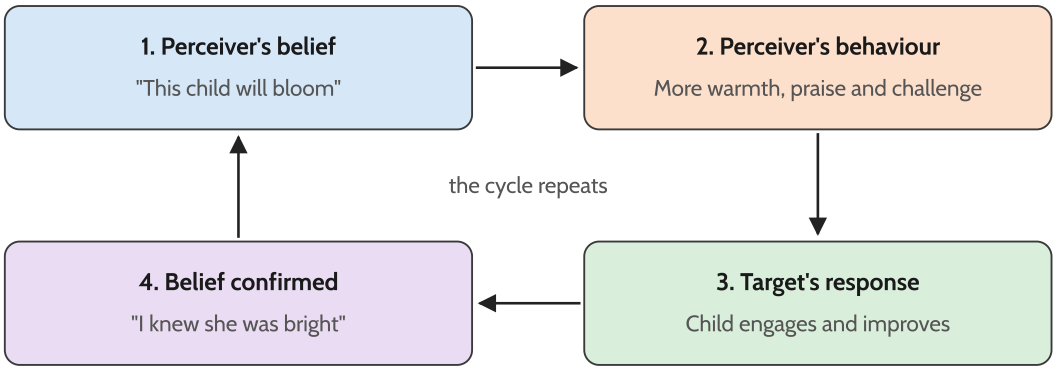


Figure 3.2 The self-fulfilling prophecy: the perceiver's expectation draws matching behaviour from the target, which seems to prove it.

KEY STUDY: PYGMALION IN THE CLASSROOM (1968)

Robert Rosenthal (born 1933) and **Lenore Jacobson** tested every child in a Californian elementary school and told teachers that about 20 percent, in fact chosen at random, would soon "bloom" academically. Eight months later these children had gained more IQ points than their classmates, most clearly in the first two grades; teachers had given them more warmth and challenge. Critics questioned the IQ measures; later reviews find expectancy effects real but usually small.

A high expectation that raises performance is the **Pygmalion effect** (Rosenthal effect), after the sculptor whose statue came to life; a low one that depresses it is the **Golem effect** (Elisha Babad, Jacinto Inbar and Rosenthal, 1982). A perceiver's expectation drawing matching behaviour from the target is **behavioural confirmation**. **Mark Snyder**, Elizabeth Tanke and Ellen Berscheid (1977) gave men a photograph, attractive or not, of the woman they would telephone (it was not her); judges hearing only the women's side rated the "attractive" women warmer and more sociable. Carl Word, Mark Zanna and Joel Cooper (1974) found that white interviewers sat farther away and ended interviews sooner with Black applicants; white applicants treated this coolly then performed worse.

EXAM TRAP

Acting on an expectation until it comes true = **self-fulfilling prophecy**. **Confirmation bias** only seeks confirming evidence and changes nothing in the world; the **base-rate fallacy** ignores how common something is (Section 1.8); "motivational reaction" is not a term. Merton formalised it, not Erving Goffman (dramaturgy, below; stigma, 1963; total institutions), **Charles Horton Cooley** (1864 to 1929; looking-glass self, 1902; primary group) or **Émile Durkheim** (1858 to 1917; anomie, a state of normlessness).

Heuristics in social judgement

Why shortcuts? Because of **information overload**: demands exceed our capacity, so we use **heuristics**, simple rules giving quick, usually adequate judgements (**Amos Tversky** and **Daniel Kahneman**, 1974; Section 1.8). **Representativeness** judges a quiet, bookish neighbour a librarian, ignoring that managers far outnumber librarians (base-rate neglect). **Availability** judges by ease of recall: people who recalled six of their own assertive acts rated themselves more assertive than those asked for a harder twelve (**Norbert Schwarz** and colleagues, 1991). **Anchoring and adjustment** starts from a value, such as an asking price, and adjusts too little. The **status quo heuristic** ("what is, is good") favours the established: the same chocolate tasted better when said to be on sale since 1937 rather than 2003. The **affect heuristic** (Paul Slovic) lets a feeling of liking or dread stand in for judging risk and benefit.

Errors and tilts in social thought

Other tilts save effort or protect our feelings at the cost of accuracy (Table 3.6).

Table 3.6 Errors and tilts in social thought.

Tilt	Shown or named by	What happens
Negativity bias	Paul Rozin and Edward Royzman (2001)	Bad information outweighs equally strong good information
Optimistic bias (unrealistic optimism)	**Neil Weinstein** (1980)	Good events seem likelier, bad ones less likely, for me than for others
Overconfidence	Section 1.8	Confidence exceeds accuracy
Planning fallacy	Kahneman and Tversky (1979); Roger Buehler, Dale Griffin and Ross (1994)	Underestimating task time: thesis students predicted 33.9 days, took 55.5
Counterfactual thinking	Victoria Medvec, Scott Madey and **Thomas Gilovich** (1995)	Imagining what might have been: upward (better) breeds regret, downward (worse) relief
Magical thinking	**Paul Rozin** (1990)	Law of contagion (contact transfers essence); law of similarity (alike in look, alike in nature)
False consensus effect	Ross, David Greene and Pamela House (1977)	Overestimating how many share our opinions and choices
False uniqueness effect		Underestimating how many share our desirable traits and abilities
Illusion of control	**Ellen Langer** (1975)	Feeling control over chance: a self-chosen lottery ticket is valued more
Ironic rebound after thought suppression	**Daniel Wegner** (1987)	Trying not to think of a white bear makes it return more often

On film of the 1992 Olympics, bronze medallists looked happier than silver medallists: silver thinks upward ("I nearly had gold"), bronze downward ("I nearly got nothing"). Ironic rebound arises because an automatic monitoring process keeps checking for the forbidden thought, and so keeps it primed. **Terror management theory** (Jeff Greenberg, Sheldon Solomon and Tom Pyszczynski, 1986) holds that awareness of certain death creates anxiety we buffer with cultural world-views, self-esteem and supernatural beliefs.

Feeling also steers thought. In **mood-congruent judgement** we notice, recall and judge in line with our mood (Section 1.7). Using **affect as information** we ask "how do I feel about it?":

people telephoned on sunny days reported more life satisfaction than on rainy days, unless first asked about the weather, which let them discount their mood (Schwarz and Gerald Clore, 1983). **Joseph Forgas**'s **affect infusion model** (1995) adds that mood seeps into judgements we must construct, but hardly into ready-made or goal-driven ones.

Forming impressions ●●

Impression formation is the combining of information about a person into an overall judgement. Its founding experiment used only adjectives.

KEY STUDY: ASCH'S WARM AND COLD STRANGER (1946)

Solomon Asch (1907 to 1996), a Gestalt thinker, read participants a stranger's traits: intelligent, skilful, industrious, *warm*, determined, practical, cautious. Others heard *cold* instead. The warm stranger was judged far more generous, happy and sociable; swapping *polite* for *blunt* changed little. Warm-cold were **central traits**, which reorganise the whole impression; polite-blunt were **peripheral traits**. Traits, Asch concluded, form an integrated whole rather than a sum. Critics later showed that centrality depends on context.

Asch also showed a **primacy effect**: a person described as "intelligent, industrious, impulsive, critical, stubborn, envious" was liked more than one given the same words in reverse, because early traits set the meaning of later ones. **Harold Kelley** (1950) told students a guest lecturer was "rather cold" or "very warm"; after the same class, the warm group rated him more considerate and sociable, and more of them joined the discussion.

Such leaps rest on **implicit personality theories**, unspoken beliefs about which traits go together ("warm people are generous"), a term of **Jerome Bruner** and **Renato Tagiuri** (1954). Their best-known product is the **halo effect**: an overall impression, or one salient positive trait, colours ratings of other, unrelated traits. **Edward Lee Thorndike** named it in 1920: army officers who rated a soldier high on physique also rated him high on intelligence, loyalty and leadership. The negative version is the **horns effect**. "What is beautiful is good" is a halo from attractiveness (Section 3.6); in rating scales the halo is a rater error (Chapter 2).

EXAM TRAP

Halo effect = an overall impression or one positive trait influencing ratings of *other, specific* traits. Not judging by group membership (stereotyping, Section 3.5); not initial impressions shaping later judgements (primacy, or anchoring); not overlooking a liked person's faults (a consequence of a halo, not its definition); not crediting oneself for success and circumstances for failure (self-serving bias, Section 3.3); not overestimating how far others share our opinions (false consensus, Table 3.6).

In **Norman Anderson**'s **information integration theory** (1965) each trait has a value and a weight. An **additive model** sums them, so adding "punctual" to "brilliant, generous" should help; a **weighted averaging model**, which the evidence mostly favours, averages them, so the impression dips. By the **negativity effect**, negative traits weigh more because bad behaviour seems more revealing.

Thin slices and first impressions

How fast and how good is a first impression? **Nalini Ambady** and Rosenthal (1992) named **thin slicing**: accurate inferences about people from very brief samples of behaviour. In 1993 they showed students silent clips of college teachers lasting 30 seconds or less (later as little as six); ratings of the clips predicted the teachers' end-of-semester student evaluations. **Malcolm Gladwell** popularised the idea in *Blink* (2005). Janine Willis and Alexander Todorov (2006) found that a 100-millisecond look at a face yields trait judgements, trustworthiness above all, that correlate highly with unhurried ones; longer looks mainly raise confidence. Thin slices beat chance but are not infallible.

> EXAM TRAP
>
> A teacher who watches two students for under a minute and correctly picks the class leader shows **thin slicing**. Not **social facilitation** (better performance of well-learned tasks before others, Section 3.10), not a **self-fulfilling prophecy** (her expectation did not create the leadership), not the **fundamental attribution error** (over-explaining others' behaviour by disposition, Section 3.3).

Managing the impressions we make

We also try to control the impressions others form of us: **impression management**, or **self-presentation**. **Erving Goffman** (1922 to 1982) framed it as theatre in *The Presentation of Self in Everyday Life* (1959). In his **dramaturgy**, on the **front stage** we play a role before an audience (the doctor's calm on the ward); **back stage** we drop it (doctors joking in the staff room). **Face-work** maintains one's own and others' face, the positive social value claimed in an encounter.

Tactics are **self-enhancement**, raising one's own appeal (grooming, recounting achievements), or **other-enhancement**, making the target feel good, chiefly **ingratiation**, winning liking through flattery and agreement, analysed by **Edward E. Jones** (1964). Jones and Thane Pittman (1982) classified five strategies by the impression sought (Table 3.7).

Table 3.7 Jones and Pittman's strategies of self-presentation (1982).

Strategy	Wants to seem	Typical behaviour
Ingratiation	Likeable	Flattery, agreement, favours
Self-promotion	Competent	Displaying achievements
Intimidation	Dangerous	Threats, shows of anger
Exemplification	Morally worthy	Self-sacrifice, staying late
Supplication	Helpless	Pleading need, so others help

Self-monitoring (Snyder, 1974) is the tendency to read situational cues and adjust one's behaviour to fit: high self-monitors are social chameleons, low self-monitors act from inner attitudes. In **self-handicapping** people create an obstacle in advance to excuse possible failure: Steven Berglas and Jones's (1978) students, given unearned success on unsolvable problems, chose a drug said to impair performance before a retest. By the **spotlight effect** we overestimate how much others notice us: students in an embarrassing T-shirt guessed half the room noticed, when about a quarter did (Gilovich, Medvec and Kenneth Savitsky, 2000). The **illu-**

sion of transparency overestimates how well others read our inner states, such as nervousness.

Does it work? Usually: applicants who use it get higher interview ratings. Overdone, it backfires: **Roos Vonk** (1998) described the **slime effect**, strong dislike of people who flatter superiors and scorn subordinates. The self-concept and self-esteem are in Chapter 4.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Fiske and Taylor's (1984) term for people who save mental effort with shortcuts is the _____.
2. Knowledge of the usual sequence of an event, such as eating at a restaurant, is a _____.
3. A false belief that evokes behaviour which makes it come true is a _____, formalised by _____ in 1948.
4. Bronze medallists are happier than silver medallists because of _____ counterfactual thinking.
5. In Asch's study, warm and cold were _____ traits.
6. One positive trait colouring ratings of unrelated traits is the _____ effect, named by _____ in 1920.
7. Accurate judgements of teachers from silent clips of 30 seconds or less illustrate _____.

Answers: (1) cognitive miser, (2) script (event schema), (3) self-fulfilling prophecy; Robert K. Merton, (4) downward, (5) central, (6) halo; Edward Thorndike, (7) thin slicing.

3.3 Communication, Nonverbal Cues and Attribution

LOWER YIELD · 12 QS

A friend answers your long message with one word: "ok." Is she busy, annoyed, or simply brief? Two processes are at work: communication, which carries meaning from one head to another through words, faces, gestures and distances, and attribution, by which you decide why someone acted as they did. This section follows the message first, then turns to the rules, and the biases, by which we explain behaviour.

The communication process and its models

Communication is the process by which a person or group transmits information, ideas and feelings to another through shared symbols. Every exchange has the same parts (Figure 3.3). The sender, or source, first performs **encoding**, turning an idea into symbols such as words or a gesture. The message travels through a **channel**, the medium that carries it (speech, print, a phone screen, the face). The receiver, or recipient, performs **decoding**, turning the symbols back into meaning, and replies with **feedback**. Anything that distorts the message on the way is **noise**.

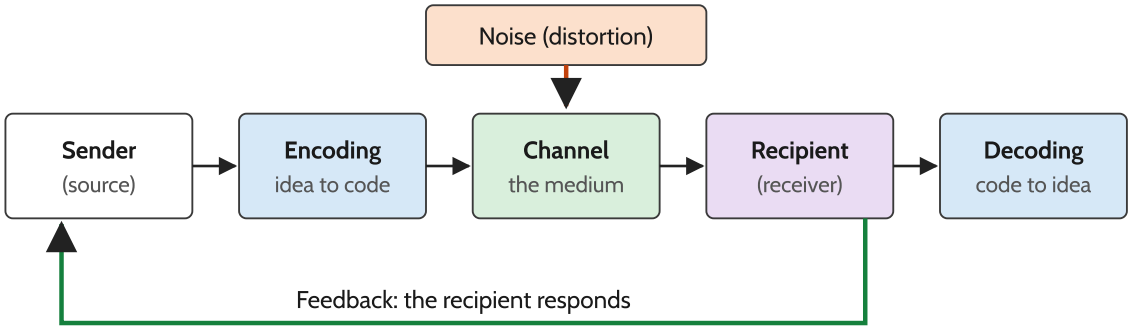


Figure 3.3 The communication process: encoding, channel, recipient, decoding. Noise (dashed arrow) distorts the message in the channel; feedback (dotted line) makes the exchange two-way.

EXAM TRAP

The sequence is **encoding, channel, recipient, decoding**: a message must be encoded before it can enter a channel, and received before it can be decoded.

Table 3.8 compares four classic models: those of **Harold Lasswell** (1902 to 1978); **Claude Shannon** (1916 to 2001) and **Warren Weaver** (1894 to 1978); **Wilbur Schramm** (1907 to 1987), for whom a message is understood only where the parties' fields of experience (language, culture, knowledge) overlap; and **David Berlo** (1929 to 1996), whose **SMCR model** adds the source's and receiver's skills, attitudes, knowledge, social system and culture.

Table 3.8 Four classic models of communication.

Model	Year	Elements	Remembered for
Lasswell	1948	Who says what, in which channel, to whom, with what effect	Verbal, linear; no feedback
Shannon and Weaver	1949	Source, transmitter, channel, noise, receiver, destination	Linear; introduced noise
Schramm	1954	Each party encodes, interprets and decodes	Circular; field of experience
Berlo (SMCR)	1960	Source, message, channel, receiver	Skills, attitudes, knowledge, culture

Verbal communication uses words; nonverbal communication uses everything else. In one-way communication (a lecture, a notice) the receiver cannot reply; in two-way communication feedback flows back, which is slower but more accurate. Barriers are **semantic** (words or jargon understood differently), **physical** (noise, distance, a poor network), **psychological** (anxiety, anger, prejudice, inattention) and **cultural** (different norms for gestures, eye contact and directness).

In the **communication networks** studied by **Alex Bavelas** (1913 to 1993) in 1950 and **Harold Leavitt** (1922 to 2007) in 1951, small groups solved problems by passing notes along fixed links. The centralised **wheel**, everyone linked only to a hub, was fastest and most accurate on simple tasks, and the hub became leader, but the others were least satisfied. The **chain** was intermediate. The decentralised **circle** was slower, with no clear leader but the highest morale,

and in the later **all-channel** network everyone talks to everyone, which suits complex problems (Section 3.10).

Nonverbal channels and personal space

Nonverbal communication is the exchange of information through facial expression, gaze, body movement, posture, touch, voice and even scent, apart from the meaning of words. Such cues are hard to suppress, so hidden feelings leak through them. As Section 1.11 showed, **Paul Ekman** (born 1934) and **Wallace Friesen** found six universally recognised expressions (happiness, sadness, fear, anger, surprise, disgust), with contempt added in 1986, while cultures differ in **display rules** about when and how strongly an emotion may be shown. The facial feedback hypothesis is also in Section 1.11.

Eye contact usually signals liking, and avoided eyes suggest dislike or shyness, but **staring**, gaze held regardless of what the other person does, signals hostility and makes people leave. Body language includes posture (an open posture signals ease, frequent self-touching nervousness) and gestures, the most specific being **emblems** with a fixed meaning in one culture, such as the namaste or the Indian head wobble meaning "all right". **Ray Birdwhistell** (1918 to 1994) founded **kinesics**, the study of body movement as communication (*Introduction to Kinesics*, 1952). Touch conveys affection, care or dominance depending on who touches whom and where. **Paralanguage** is the voice apart from the words (pitch, tone, loudness, rate, pauses), which is why "fine" can mean its opposite. Even scent is a channel: subtle changes in body odour, such as those across the menstrual cycle, can affect others who do not consciously notice them.

Edward T. Hall (1914 to 2009) coined **proxemics** (1963), the study of how people use interpersonal distance, and in *The Hidden Dimension* (1966) described four zones (Figure 3.4). He measured them among Americans; contact cultures (Arab, Latin American) stand closer. In *Beyond Culture* (1976) he contrasted **high-context cultures**, where meaning lies largely in the setting, the relationship and nonverbal cues, so speech is indirect (India, Japan, China), with low-context cultures, where explicit words carry it (the United States, Germany). An Indian "I will try to come" may be a polite no.

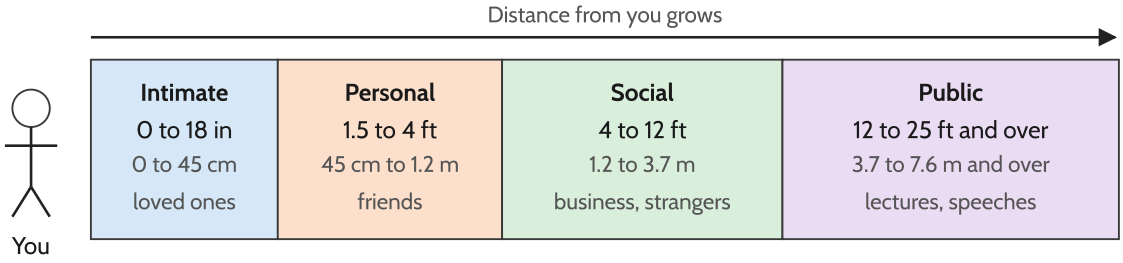


Figure 3.4 Hall's (1966) zones of personal space, not drawn to scale. Learn the boundaries: 18 inches, 4 feet, 12 feet and 25 feet.

Detecting deception

Can nonverbal cues expose a lie? Four help. **Microexpressions** are fleeting facial expressions, lasting a fraction of a second, that leak a concealed emotion before it is masked. **Interchannel discrepancies** are mismatches between channels: a liar may control the face

but not the voice or hands. Eye contact changes: liars blink more, have more dilated pupils, and show too little or unusually much eye contact. Exaggerated expressions, an overdone smile or excessive regret, are the fourth sign, and vocal pitch tends to rise. Yet a meta-analysis of more than 200 studies by **Charles Bond** and **Bella DePaulo** (2006) found accuracy of only about 54 percent, barely above chance, and truths were spotted more often than lies because people assume honesty.

> EXAM TRAP
> Averted eyes do not reliably mark a lie: liars may stare to look honest. No single cue betrays a lie, and most judges stay near chance.

Heider, Jones and Davis: from acts to causes ●●

Attribution is the process by which people infer the causes of their own and others' behaviour. **Fritz Heider** (1896 to 1988), in *The Psychology of Interpersonal Relations* (1958), described people as amateur scientists; his **naive psychology** (common-sense psychology) concerns the implicit rules they use to infer causes. His key distinction: a **dispositional attribution** (internal) locates the cause in the person (traits, abilities, attitudes, effort), while a **situational attribution** (external) locates it outside (the situation, other people, the environment, luck, roles, culture). "He failed because he is lazy" is dispositional; "the paper was unusually hard" is situational.

> EXAM TRAP
> Causes resting on the actor's personality traits are **dispositional**; situational, environmental and cultural causes are all external. Matching: Heider = implicit rules of attribution; **Fred Fiedler** (1922 to 2017) = contingency theory of leadership (Section 3.10); **Sigmund Freud** (1856 to 1939) = unconscious conflict; **Albert Bandura** (1925 to 2021) = social learning theory.

Edward E. Jones and **Keith Davis** asked when an act reveals a trait. By their **correspondent inference theory** (1965), we infer that behaviour corresponds to a stable disposition when it is (1) freely chosen; (2) produces **non-common effects**, outcomes that only the chosen option yields, which pinpoint the motive; and (3) low in social desirability, since out-of-role behaviour is more revealing: a guest who calls your mother's biryani too salty reveals more than one who praises it. **Hedonic relevance** (the act affects us) and **personalism** (it seems aimed at us) strengthen the inference.

Kelley's covariation model and causal schemas

Harold Kelley (1921 to 2003) asked whether a cause lies in the person, the stimulus or the circumstances. His **covariation model** (1967) says that, observing behaviour over several occasions, we attribute it to the cause it covaries with, using three kinds of information:

- **Consensus**: do other people react to this stimulus in the same way?
- **Consistency**: does this person react this way to the stimulus on other occasions?
- **Distinctiveness**: does this person react differently to other stimuli?

Ananya laughs at a comedian. If few others laugh, she laughs at him every time and she laughs at every comedian (low, high, low), the cause is Ananya. If everyone laughs, she always does and she laughs at no other comedian (all high), the comedian is funny. If she has never laughed at him before (low consistency), a passing circumstance is responsible. Consistency is high in both main patterns, so only consensus and distinctiveness separate internal from external (Table 3.9).

Table 3.9 Kelley's covariation model.

Consensus	Consistency	Distinctiveness	Attribution
Low	High	Low	Internal: the person
High	High	High	External: the stimulus
Low	High	High	Person and stimulus together
Any	Low	Any	A temporary circumstance

For a single observation, Kelley (1972) proposed causal schemas. By the **discounting principle**, a cause gets less weight when other plausible causes exist (a film star is paid to praise a brand); by the **augmenting principle**, it gets more weight when the effect occurs despite an obstacle (a student tops the class while working nights).

Weiner's three dimensions

Bernard Weiner (born 1935), studying achievement from the 1970s, classified causes by **locus** (internal or external), **stability** and **controllability**. Ability is internal, stable and uncontrollable; effort is internal, unstable and controllable; task difficulty is external and stable; luck is external and unstable. Stability shapes expectations, and the dimensions shape emotion: success put down to internal causes brings pride; failure from low ability brings shame, from low effort guilt. Another person's misfortune seen as controllable ("he drank") provokes anger, and an uncontrollable one ("a lay-off") pity and helping (Section 3.8).

CLINICAL LINK

In the attributional reformulation of learned helplessness, **Lyn Abramson**, **Martin Seligman** (born 1942) and **John Teasdale** (1978) proposed that people who explain bad events by **internal** ("it is me"), **stable** ("it will last") and **global** ("it spoils everything") causes are most vulnerable to depression (Section 1.6; Chapter 8).

Attributional biases ●●●

Our attributions tilt in predictable directions (Table 3.10). The strongest tilt is the **fundamental attribution error** (FAE), named by **Lee Ross** (1942 to 2021) in 1977: overestimating dispositional and underestimating situational causes of other people's behaviour. A man arrives late and spills his coffee; we call him clumsy, not delayed and hurried. The **correspondence bias**, inferring dispositions that correspond to behaviour even when situational causes are present, is usually treated as a synonym (reviewed in 1995 by **Daniel Gilbert**, born 1957, and **Patrick Malone**).

KEY STUDY: JONES AND HARRIS, THE CASTRO ESSAYS (1967)

Method. Jones and **Victor Harris** had participants read essays for or against Fidel Castro, telling some that the writer chose the side and others that it was assigned. **Result.** Even with assigned sides, pro-Castro writers were judged fairly pro-Castro. **Conclusion.** People underweight a situational constraint they know about. **Criticism.** Readers may assume nobody argues well for a view he rejects, but the bias replicates across many tasks.

Why does it happen? The actor is perceptually salient while the situation fades into the background. In Gilbert's two-step model we first make an automatic dispositional inference, then effortfully correct it for the situation; under cognitive load the correction fails.

For our own behaviour the tilt reverses. The **actor-observer effect** (Jones and **Richard Nisbett**, 1971) is attributing our own behaviour to situations and others' to dispositions: he trips because he is clumsy, I trip because the floor was wet. The actor sees the situation; the observer sees the actor. **Michael Storms** (1973) reversed the effect by showing actors and observers videotapes shot from the opposite viewpoint. The **self-serving bias** concerns our own outcomes: successes go to internal causes, failures to external ones (an unfair paper, bad luck); it protects self-esteem and is weaker in collectivist cultures.

The **just-world hypothesis** of **Melvin Lerner** (1965, 1980) holds that people need to believe the world is orderly, predictable and fair, a place where people get what they deserve. An innocent person's suffering threatens the belief, which is restored by deciding the victim deserved it: victim blaming. In Lerner and **Carolyn Simmons**'s experiment (1966), women who watched a fellow student receive apparent electric shocks, and could not end or compensate her suffering, derogated her. **Zick Rubin** and **Letitia Anne Peplau** measured the belief with their Just World Scale (1975).

EXAM TRAP

Just-world victim blaming comes from the **need to believe that the world is orderly, predictable and fair**, not from sadism, not from the FAE itself (it looks dispositional, but the motive is protecting the belief), and not from downward social comparison with someone worse off (Section 3.6).

Table 3.10 Attributional biases compared.

Bias	Who, when	What happens	Example
Fundamental attribution error (correspondence bias)	Ross 1977	Others' acts put down to disposition	"He snapped at the waiter: he is rude."
Actor-observer effect	Jones and Nisbett 1971	My acts to situations, yours to disposition	"I was late because of traffic; you are careless."
Self-serving bias	**Dale Miller** and **Michael Ross** 1975	My successes internal, my failures external	"I topped through talent; I failed because the paper was unfair."
False consensus effect	Ross 1977 (Section 3.2)	Overestimating how many share my views	"Everyone finds this film boring."
Just-world hypothesis	Lerner 1965, 1980	Victims seen as deserving their fate	"She must have asked for it."

Bias	Who, when	What happens	Example
Ultimate attribution error	**Thomas Pettigrew** 1979 (Section 3.5)	FAE for groups: out-group's bad acts dispositional, good acts luck	"They cheat by nature; he is an exception."
Hostile attribution bias	**Kenneth Dodge** 1980 (Section 3.9)	Ambiguous acts read as hostile; marks aggressive children	"He bumped me on purpose."
Defensive attribution	**Kelly Shaver** 1970	More blame for severe accidents, less if the person resembles us	"He was careless; it could not happen to me."

EXAM TRAP

Ask whose behaviour and whose outcome. FAE: **other people's** behaviour, disposition overestimated (the reversed wording is a distractor). Actor-observer: my behaviour versus yours. Self-serving: my own successes and failures. Non-attribution distractors: **confirmation bias**, favouring evidence for what we already believe (Section 1.8); **belief bias**, judging an argument valid because its conclusion is believable; the **halo effect**, one good trait colouring the whole impression (Section 3.2); **cognitive dissonance**, discomfort from inconsistent cognitions (Section 3.4); **stereotyping**, a generalised belief about a group (Section 3.5). The optimism bias is in Section 3.2.

Culture and applications

Is the FAE truly fundamental? It is strongest in individualist cultures, which see the person as an independent agent, and weaker in collectivist ones, which see behaviour as embedded in roles and relationships.

INDIAN CONTEXT

Joan Miller (1984) compared Hindu Indians in Mysore with Americans, aged 8, 11, 15 and adult, asking them to explain good and bad acts of people they knew. American adults gave far more dispositional explanations; Hindu adults gave more situational ones (roles, relationships, circumstances). At age 8 the groups barely differed and the gap grew with age: attributional style is learned from culture.

Similarly, **Michael Morris** and **Kaiping Peng** (1994) found that *The New York Times* explained two 1991 American mass murders by the killers' personalities, while a Chinese-language newspaper stressed their situations (isolation, conflicts at work, easy access to guns).

Because attributions shape feelings and effort, changing them helps. In **attribution retraining**, **Timothy Wilson** and **Patricia Linville** (1982) showed worried first-year students evidence that grades usually improve after the first year, inviting an unstable explanation for poor grades; they later earned better grades and dropped out less than controls. Attributional therapy for depression teaches people to credit themselves for success and to stop blaming lasting flaws for failure. Happy couples credit a partner's kindness to disposition and excuse lapses as situational; distressed couples do the reverse (Section 3.6). And computer-mediated communication, stripped of faces and tone, gives the correspondence bias free rein: a terse "ok." reads as rudeness, not haste.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. In the communication process, turning an idea into symbols is _____.
2. Lasswell's model asks who says what, in which _____, to whom, with what effect.
3. Hall's personal distance runs from 1.5 to _____ feet.
4. Fleeting facial expressions that leak a concealed emotion are _____.
5. Low consensus, high consistency and low distinctiveness lead to an _____ attribution.
6. Attributing your own behaviour to the situation but others' to their dispositions is the _____ effect.
7. Victim blaming protects the belief that the world is orderly, predictable and _____.

Answers: (1) encoding, (2) channel, (3) 4, (4) microexpressions, (5) internal (dispositional), (6) actor-observer, (7) fair.

3.4 Attitudes, Persuasion and Cognitive Dissonance

LOWER YIELD · 15 QS

Advertisements, election rallies and anti-smoking posters all bet that changing attitudes changes behaviour. This section follows attitudes from their origins to behaviour, persuasion, resistance and dissonance.

What an attitude is

In 1935 **Gordon Willard Allport** (1897 to 1967) called the attitude "the most distinctive and indispensable concept in contemporary American social psychology". An **attitude** is a learned, fairly lasting evaluation of an **attitude object** (a person, group, thing or idea) along a like-dislike dimension. The **tricomponent model** (ABC model) of **Milton J. Rosenberg** and Carl Hovland (1960) gives it three parts: *affect* (feelings: "chai cheers me up"), *behaviour* or conation (action tendencies: "I stop at the stall daily") and *cognition* (beliefs: "tea is good for me").

Attitude strength combines extremity, certainty, accessibility (how fast the attitude comes to mind) and importance. Ambivalence, holding positive and negative evaluations of one object at once, weakens an attitude's hold on behaviour. Explicit attitudes are conscious and self-reported; **implicit attitudes** are automatic evaluations measured from reaction times, as in the Implicit Association Test (Section 3.5), and the two can disagree.

Daniel Katz (1903 to 1998) named four **functions of attitudes** in 1960: *utilitarian* or adjustive (gaining rewards, avoiding punishments), *knowledge* (organising a confusing world), *ego-defensive* (protecting self-esteem from threatening truths, as when scorn for an outgroup props up a shaky self) and *value-expressive* (displaying central values). In 1956 **M. Brewster Smith**, **Jerome Seymour Bruner** and **Robert W. White** had named *object appraisal* (sizing up objects for one's interests), *social adjustment* (easing relations with valued people) and *externalisation* (projecting inner conflicts onto outside objects). Measurement belongs to Section 2.8: Bogardus's social distance scale (1925), Thurstone's equal-appearing intervals (1928), Likert's

summated ratings (1932), Guttman's cumulative scale and Osgood's semantic differential (1957).

How attitudes form

Most attitudes are learned (Section 1.6). In classical conditioning a neutral object takes on the feeling of what accompanies it: **Arthur W. Staats** and **Carolyn K. Staats** (1958) paired nationality names such as "Dutch" and "Swedish" with pleasant or unpleasant words, and the pleasantly paired names were later rated more favourably. In **subliminal conditioning** the pairing works without awareness: **Jon A. Krosnick** and colleagues (1992) flashed pleasant or unpleasant pictures too briefly to be noticed before photographs of a stranger, who was then liked more after the pleasant ones.

In instrumental conditioning, views that are praised or agreed with are strengthened, which is why children echo their parents' opinions until adolescence. We also copy the attitudes of models (observational learning) and adopt those of reference groups, the groups whose standards we judge ourselves by, through social comparison, which Leon Festinger (1954) described as evaluating our own opinions and abilities against other people's. Mere exposure is taught in Section 3.6. Identical twins share more attitudes than fraternal twins, and **Abraham Tesser** (1993) found highly heritable attitudes more accessible and harder to change. Online word of mouth (reviews, ratings, posts) now carries the weight of peers.

> EXAM TRAP
>
> Praise or agreement from a friend **strengthens** an expressed attitude, never weakens it: positive reinforcement raises the likelihood of whatever it follows.

When attitudes predict behaviour

> KEY STUDY: LAPIERE'S TRAVELS (1934)
>
> In the early 1930s, a time of open anti-Asian prejudice, **Richard Tracy LaPiere** (1899 to 1986) toured the United States with a young Chinese couple. Of about 250 hotels and restaurants, one refused them. Afterwards he wrote to the same places asking whether they would serve Chinese guests, and over 90 percent of those replying said no. Critics note that different staff may have answered the letters.

In 1969 **Allan W. Wicker**'s review found attitude-behaviour correlations rarely above .30. Research then asked when attitudes do predict behaviour.

Strong attitudes predict best. Extremity often reflects **vested interest**: John Sivacek and **William D. Crano** (1982) found that 47 percent of students who would be barred by a higher drinking age agreed to campaign against it, against 12 percent of unaffected students. Certainty (clarity plus a sense of correctness), personal relevance and knowledge about the object also help. So does direct experience. During a Cornell University housing shortage, **Dennis T. Regan** and **Russell H. Fazio** (1977) found that students forced to sleep on lounge cots acted on their attitudes (petitions, committees) far more consistently than students who had only heard about it, and Fazio and **Mark P. Zanna** (1981) showed that direct experience makes attitudes clearer and more accessible. Attitudes formed by indirect experience predict worst.

Specificity must also match. The **principle of compatibility** of **Icek Ajzen** and **Martin Fishbein** (1977) requires attitude and behaviour to correspond in action, target, context and time. In Andrew Davidson and James Jaccard's (1979) study, general attitude towards birth control correlated .08 with women's pill use over the next two years, but attitude towards using the pill in that period correlated .57. Finally, situational constraints and perceived norms intervene: **Deborah A. Prentice** and **Dale T. Miller** (1993) found Princeton students privately less comfortable with heavy drinking than they believed their peers were, and the men's attitudes drifted towards that imagined norm.

> EXAM TRAP
>
> Behaviour is hardest to predict from attitudes formed by **indirect experience** (hearsay, media). Direct experience, personal relevance, knowledge of the object, certainty and matching specificity all strengthen the link.

Reasoned action, planned behaviour and MODE

For deliberate behaviour, the **theory of reasoned action** (Fishbein and Ajzen, 1975) makes behavioural intention the immediate predictor of behaviour. Intention rests on the *attitude towards the behaviour* (beliefs about its outcomes, weighted by their value) and the **subjective norm** (what important others want, weighted by the motivation to comply). Ajzen (1985, 1991) added **perceived behavioural control**, the belief that one can perform the act, close to Albert Bandura's self-efficacy, making the **theory of planned behaviour**; control affects intention and also behaviour directly. It is the standard model of health behaviours such as condom use and quitting smoking; the health belief model and protection motivation theory are in Section 7.9.

Spontaneous behaviour works differently. In Fazio's **attitude-to-behaviour process model** (1986), an event automatically activates an accessible attitude, which together with knowledge of norms shapes how the situation is defined, and that definition drives behaviour. His **MODE model** (1990: Motivation and Opportunity as DEterminants) unites the accounts: people deliberate only when motivated and given the opportunity (time, capacity); otherwise accessible attitudes act automatically.

> EXAM TRAP
>
> Planned behaviour's predictors of intention are attitude, subjective norms and perceived behavioural control, often worded as self-efficacy (RML 2023). Genetics, personality, socioeconomic status, cultural beliefs, social support, health literacy, physical environment, emotional intelligence and coping are not in the model.

Persuasion: who says what to whom

Persuasion is attitude change produced by messages. At Yale University **Carl Iver Hovland** (1912 to 1961), **Irving Lester Janis** (1918 to 1990) and **Harold H. Kelley** (1921 to 2003) summarised their programme in *Communication and Persuasion* (1953). The **Yale communication approach** asks who says what to whom (Section 3.3) and varies source, message, audience and channel (Table 3.11). Credibility means expertise plus trustworthiness.

Table 3.11 What makes a message persuasive: the Yale findings.

Variable	What persuades more
Source	Credible, attractive, likeable or similar; fast speech; not seeming to persuade (overheard, arguing against own interest)
One- or two-sided message	One-sided if the audience agrees or is less educated; two-sided (refuting the other side) if it is opposed or better educated (Hovland, Lumsdaine and Sheffield 1949)
Fear appeal	Moderate fear plus clear, doable advice; intense fear invites denial (Janis and Feshbach 1953)
Discrepancy, repetition	Moderate on both; credible sources can push bigger gaps
Order	Primacy if two messages come back to back and judgement is delayed; recency if time separates them and judgement is immediate (Miller and Campbell 1959)
Audience	Moderate self-esteem and intelligence; low need for cognition (for cue-based appeals); impressionable years (late adolescence, early adulthood); good mood; initial view near the message; no forewarning
Channel	Face to face beats mass media; print suits complex messages

Audience factors are **receiver characteristics**, the factors in the audience that influence persuasion. **Need for cognition**, the enjoyment of effortful thinking (**John T. Cacioppo** and **Richard E. Petty**, 1982), makes people weigh arguments rather than cues. **William J. McGuire** (1925 to 2007) explained why moderate self-esteem and intelligence persuade best with his information-processing chain of attention, comprehension, yielding, retention and behaviour: able, confident people comprehend more but yield less.

In the **sleeper effect**, a low-credibility message gains impact over time because the discounting cue (its source) is forgotten while the content is remembered. Hovland and **Walter Weiss** (1951) found that a message credited to the physicist Robert Oppenheimer persuaded more than the same message credited to the Soviet paper *Pravda*, but four weeks later the difference had almost vanished.

> EXAM TRAP
>
> "Receiver characteristics" (printed "receiver-operant characteristics" in IHBAS 2025) are audience factors influencing persuasion, not sensory thresholds or the receiver operating characteristic, the signal-detection curve of hits against false alarms (Section 1.3), not reinforcement controlling recall, and not biases in interpreting messages (biased assimilation, 3.4.7).

Two routes to persuasion

Why do these factors work? In the **elaboration likelihood model** (ELM) of Petty and Cacioppo (1981, 1986), elaboration is thinking about a message's arguments, and it depends on *motivation* (personal relevance, need for cognition) and *ability* (time, knowledge, no distraction). With both, people take the **central route**: systematic processing with careful attention to argument quality, giving durable change that resists attack and predicts behaviour. Without them they take the **peripheral route**, using mental shortcuts triggered by cues such as an attractive source, the number of arguments or a good mood, and the change is temporary (Figure 3.5). In a 1981 study by Petty, Cacioppo and Goldman, argument quality decided students' views on a comprehensive-examination proposal due next year, but source expertise decided them when it was a decade away.

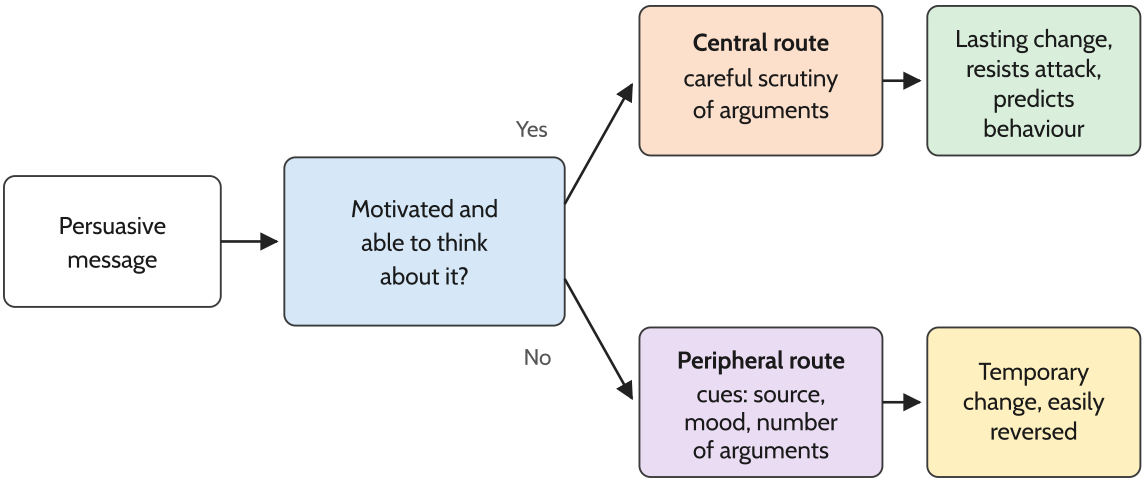


Figure 3.5 The elaboration likelihood model (Petty and Cacioppo, 1986). Motivation and ability decide the route; only central-route change lasts.

Shelly Chaiken's **heuristic-systematic model** (1980) contrasts systematic processing with heuristic processing by rules of thumb ("experts are right"), and lets both run at once.

> EXAM TRAP
>
> Systematic processing is the central route and involves careful attention; heuristic processing is the peripheral route and uses mental shortcuts. The ELM has exactly two routes: there is no "emotional" or "situational" route (IHBAS 2020).

Resisting persuasion

Audiences also push back. **Jack Williams Brehm** (1928 to 2009) defined **reactance** in 1966 as the unpleasant state, and motivation, to restore a freedom that is threatened, often by wanting or doing the forbidden thing more; hard selling may even produce a *boomerang effect*, change opposite to the one advocated. Reactance is not compliance (changing behaviour at a direct request), conformity (following group norms) or obedience (following authority), all in Section 3.7, nor social comparison (3.4.2).

Forewarning of persuasive intent aids resistance: **Jonathan L. Freedman** and **David O. Sears** (1965) found teenagers warned ten minutes ahead of a talk against teenage driving were persuaded less than those warned two minutes ahead or not at all, because they had time for counterarguing, generating thoughts that refute the message. People also practise selective avoidance of challenging information and selective exposure to agreeable information. In **inoculation** (McGuire, 1961), a weak attack plus its refutation (refutational defence) protects a belief better than supportive arguments alone, as a vaccine does. **Charles G. Lord**, **Lee Ross** and **Mark R. Lepper** (1979) showed biased assimilation: supporters and opponents of capital punishment read the same mixed evidence, each rated the agreeing study as stronger, and both became more extreme (**attitude polarisation**). Ego depletion, tiredness after a demanding task, weakens resistance.

EXAM TRAP

Reactance (Brehm, 1966) is the motivation to restore a threatened freedom, so the restricted option grows more attractive. It is not complying with requests or authority, following norms or the majority, the effect of persuasive messages, or comparing one's abilities with others.

CLINICAL LINK

Lecturing a client to stop drinking can provoke reactance; motivational interviewing (Section 7.6) avoids it by letting the client voice the reasons for change.

Consistency theories and cognitive dissonance ●●●

Much resistance reflects a drive for consistency. In the **balance theory** of **Fritz Heider** (1896 to 1988; 1946, 1958), a person (P), another person (O) and an object (X) form a triad of positive or negative relations. It is balanced when the product of the three signs is positive, and imbalanced, motivating change, when it is negative (Figure 3.6). In the **congruity theory** of **Charles E. Osgood** and **Percy H. Tannenbaum** (1955), when a source endorses an object we rate differently, as when an admired cricketer promotes a disliked brand, both evaluations move towards each other.

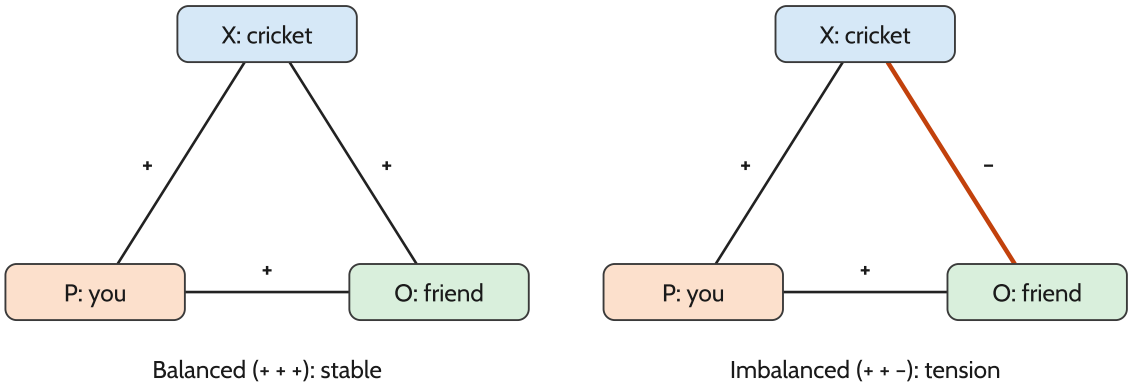


Figure 3.6 Heider's balance theory. If your friend hates the cricket you love (dashed line), you may like cricket less, like the friend less, or win the friend over.

Leon Festinger (1919 to 1989) proposed **cognitive dissonance** theory in 1957. Dissonance is the aversive tension of holding inconsistent cognitions, especially of acting against one's own attitude; it grows with the importance of the cognitions and with the ratio of dissonant to consonant ones. People reduce it by:

- changing the attitude, the usual outcome when the behaviour cannot be undone;
- changing the behaviour;
- adding consonant cognitions and seeking supportive, not conflicting, information;
- **trivialisation**, deciding the inconsistency does not matter (**Linda Simon**, **Jeff Greenberg** and Brehm, 1995);

- **self-affirmation**, bolstering self-worth through other values while the inconsistency stays (**Claude Mason Steele**, 1988);
- denying responsibility.

Take a case. Arvind detests smoking; his grandfather died of lung cancer. At a party his friends smoke, and so does he. "Smoking is deadly" and "I smoked" clash, both matter to him, and he feels dissonance. He cannot unsmoke the cigarette, so he may soften his attitude ("one is harmless"), add a cognition ("I did it for my friends"), blame the pressure, or vow never again. This is not effort justification (valuing what cost effort), an increased drive to smoke, or drive reduction (Hull's need-reduction principle, Section 1.10).

> EXAM TRAP
>
> Cognitive dissonance is Festinger's theory, not Milgram's (obedience), Asch's (conformity) or Zimbardo's (prison simulation), and not operant conditioning, classical conditioning or social learning theory (Section 1.6). Trivialisation and self-affirmation reduce dissonance, not compliance, reactance or self-regulation (control of one's own behaviour towards goals). Dissonance is likeliest when one must decide or act against one's beliefs, not when thoughts and behaviour agree, after praise or during happy memories.

Dissonance paradigms and rival explanations

> KEY STUDY: FESTINGER AND CARLSMITH (1959)
>
> Festinger and **James Merrill Carlsmith** had students do an hour of dull tasks (turning pegs, loading spools), then paid them $1 or $20 to tell the next participant the tasks were interesting. The $20 group, like controls who told no lie, later rated the tasks dull; the $1 group rated them more enjoyable. Lacking adequate external justification, they reduced dissonance by changing their attitude. This is **induced compliance** (forced compliance) with insufficient justification, and its result is the *less-leads-to-more effect*.

Table 3.12 sets out the other classic paradigms.

Table 3.12 The classic dissonance paradigms.

Paradigm	Classic study	Finding
Effort justification	Aronson and Judson Mills 1959	A severe, embarrassing initiation made a dull group discussion seem more interesting
Free choice (post-decision)	Brehm 1956	After choosing between similar appliances, the chosen was rated higher, the rejected lower (spreading of alternatives)
Belief disconfirmation	Festinger, Henry Riecken and Stanley Schachter 1956, *When Prophecy Fails*	When a cult's prophecy failed, committed members recruited more
Insufficient deterrence	Aronson and Carlsmith 1963	Children mildly threatened against playing with a toy later liked it less than severely threatened children
Hypocrisy induction	Jeff Stone, Aronson and colleagues 1994	Students who urged condom use, then recalled their own lapses, bought more condoms

What makes inconsistency hurt? **Elliot Aronson** (born 1932) proposed self-consistency: behaviour that threatens a decent, competent self-concept. The new look of **Joel Cooper** and Fazio (1984) required personal responsibility for foreseeable aversive consequences. **Daryl J. Bem**'s **self-perception theory** (1967, 1972) denied any tension: when attitudes are vague we infer them from our own behaviour, as an observer would ("I praised it for a dollar, so I must like it"). Yet when Zanna and Cooper (1974) let people blame their tension on a placebo pill, attitude change vanished, so arousal matters; self-perception fits vague attitudes, dissonance clear ones. Impression management theory (**James T. Tedeschi** and colleagues, 1971) held that people merely present consistent attitudes to others. Culture matters too: **Shinobu Kitayama** and colleagues (2004) found that Japanese participants showed post-decision dissonance when others' views were made salient, Americans when choosing for themselves.

> **EXAM TRAP**
>
> Dissonance and self-perception both predict that attitudes follow behaviour. Choose Festinger when the stem mentions tension or a clear attitude contradicted; choose Bem when a vague attitude is calmly inferred. Effort justification is one dissonance paradigm, not a synonym for dissonance. The theory dates from 1957; the $1 versus $20 study was published in 1959, though some lists give 1957.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Katz's function that protects self-esteem from threatening truths is the _____ function.
2. Attitudes formed through _____ experience predict behaviour best.
3. Planned behaviour added perceived behavioural _____ to reasoned action.
4. Careful attention to argument quality marks the _____ route.
5. A low-credibility message gaining impact over time shows the _____ effect.
6. Brehm's motivation to restore a threatened freedom is _____.
7. Festinger and Carlsmith's students paid _____ came to like the dull task more.
8. Bem's tension-free alternative to dissonance is _____ theory.

Answers: (1) ego-defensive, (2) direct, (3) control, (4) central, (5) sleeper, (6) reactance, (7) one dollar, (8) self-perception.

3.5 Stereotypes, Prejudice and Intergroup Relations LOWER YIELD · 13 QS

In 1954 two groups of eleven-year-old boys at an Oklahoma camp came to loathe each other within days, though nothing divided them but a name and a tournament. How do ordinary people come to think less of whole groups, and what undoes it?

Stereotype, prejudice and discrimination

Bias has three components, matching the ABC of attitudes (Section 3.4). A **stereotype** is cognitive: a belief about the traits typical of a group's members, what the journalist **Walter Lippmann** called "pictures in our heads" (*Public Opinion*, 1922). **Daniel Katz** and **Kenneth**

Braly (1933) measured them with an adjective checklist: Princeton students agreed strongly on the traits of ten ethnic groups, including groups they had seldom or never met. **Prejudice** is affective: a negative attitude toward people because they belong to a group, in **Gordon Willard Allport**'s words (*The Nature of Prejudice*, 1954) "an antipathy based upon a faulty and inflexible generalization". **Discrimination** is behavioural: unfair negative action toward a group's members.

William Graham Sumner (*Folkways*, 1906) named the **in-group** ("we") and the **out-group** ("they") and defined **ethnocentrism**, judging other groups by the standards of one's own, placed at the centre and above the rest. Allport ranked prejudice in action in five steps of rising harm (**Allport's scale**): **antilocution** (hostile talk), **avoidance**, **discrimination**, **physical attack** and **extermination**. Prejudice is named by its target: **racism**, **sexism** and **ageism** (against the old; coined by **Robert Neil Butler** in 1969). In India, **casteism** is prejudice and discrimination based on caste, a hereditary, endogamous ranking (untouchability is abolished by Article 17 of the Constitution), and **communalism** joins loyalty to one's own religious community with hostility toward others.

> EXAM TRAP
>
> Stereotype = belief, prejudice = feeling, discrimination = action. "Pictures in our heads" is Lippmann, not Allport. Allport's scale starts with antilocution, not avoidance. Ethnocentrism is Sumner (1906).

Where prejudice comes from

Early explanations looked inside the prejudiced person. **Theodor Adorno**, **Else Frenkel-Brunswik**, **Daniel Levinson** and **Nevitt Sanford** (*The Authoritarian Personality*, 1950) described an **authoritarian personality**: rigid, conventional, submissive to authority and hostile to minorities, shaped by harsh, punitive parenting, and measured by the **F-scale** (F for fascism). Its modern heirs are **Bob Altemeyer**'s **right-wing authoritarianism** (1981: authoritarian submission, authoritarian aggression, conventionalism) and **Jim Sidanius** and **Felicia Pratto**'s **social dominance orientation**, a preference for a hierarchy in which some groups rank above others.

Scapegoat theory holds that frustration is displaced onto a weak, visible minority (Section 3.9): **Carl Hovland** and **Robert Sears** (1940) found that lynchings in the American South rose as cotton prices fell. In the **social learning view**, children absorb prejudice from parents, peers and media and are rewarded with approval for voicing it.

At the group level, **realistic conflict theory** (named by **Donald Thomas Campbell** in 1965, argued by **Muzafer Sherif** in 1966) holds that competition for scarce resources breeds hostility. Often what hurts is **relative deprivation**, having less than one feels entitled to compared with others: in *The American Soldier* (1949), **Samuel Stouffer** and colleagues found fast-promoted Air Corps men less satisfied with promotion than slow-promoted military police. **Walter Garrison Runciman** (1966) separated **egoistic** (personal) from **fraternal** (group) deprivation; the fraternal kind fuels prejudice and **collective action** such as protest. **Ted Robert Gurr** (*Why Men Rebel*, 1970) traced rebellion to the gap between what people expect and what they get.

KEY STUDY: THE ROBBERS CAVE EXPERIMENT (1954)

Question. Does competition create intergroup hostility, and what removes it? **Method.** Sherif and colleagues took 22 unacquainted boys aged about 11 to 12, in two matched groups, to a summer camp at Robbers Cave State Park, Oklahoma. Phase 1, **in-group formation**: each group lived apart and named itself (the Eagles, the Rattlers). Phase 2, **competition and friction**: a tournament with prizes for the winners. Phase 3, **reduction of conflict**. **Result.** Competition brought name-calling, a burned flag and cabin raids. Mere contact (shared meals, films) failed and even made things worse: meals became food fights. Hostility fell only with **superordinate goals**, which both groups wanted but neither could reach alone: restoring the water supply, pooling money to rent a film, pulling a stalled truck. **Criticism.** The boys were all white, Protestant and middle class, and staff stage-managed events. The report appeared in 1961.

EXAM TRAP

Superordinate goals requiring cooperation reduced the hostility. Mere contact failed, more competition is what created the hostility, and no dominant leader was assigned. Robbers Cave is not Sherif's autokinetic study of norms (Section 3.7).

Social identity theory ●●●

Sherif's boys traded insults as soon as they learned of each other, before any contest. Is competition even needed? **Henri Tajfel** (1919 to 1982), a Polish Jew who lost most of his family in the Holocaust, looked for the minimal conditions for discrimination. In the **minimal group paradigm** (Tajfel, Billig, Bundy and Flament, 1971), Bristol schoolboys were told they had been grouped by a preference for Paul Klee or Wassily Kandinsky (in truth, at random), then divided points worth money between two anonymous others. Groups with no history, contact or competition still produced in-group favouritism, and many boys chose to **maximise the difference** between groups even at a cost to their own group's absolute gain.

Tajfel and **John Turner** (1947 to 2011) explained this with **social identity theory** (1979, 1986): a large part of our self-concept and self-esteem comes from the groups we belong to. Besides a **personal identity** (one's own traits, achievements and preferences) we have a **social identity** derived from group memberships (Indian, Kannadiga, psychology student). Three processes link membership to bias (Figure 3.7): **social categorisation** into in-groups and out-groups, **social identification** with the in-group, and **social comparison** with out-groups. Since self-esteem rides on the comparison, we seek **positive distinctiveness**: our group must come out ahead. Prejudice thus arises from categorisation itself, with no clash of interests required.

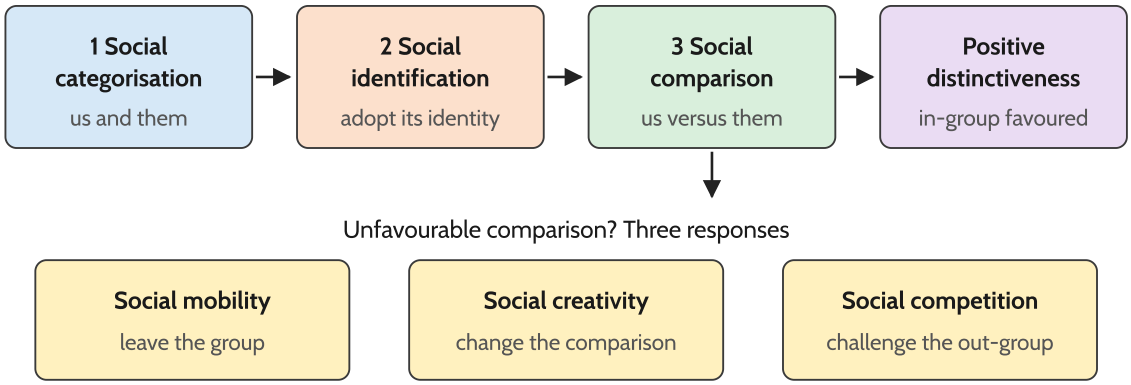


Figure 3.7 Social identity theory (Tajfel and Turner 1979): three processes, and three responses when the in-group compares badly.

When the comparison is unflattering, members may leave for a higher group (**social mobility**, if boundaries are permeable), change the comparison by picking a new dimension or a lower group (**social creativity**: "poorer but kinder"), or challenge the higher group directly (**social competition**). The **self-categorisation theory** of Turner and colleagues (1987) explains when we see ourselves as individuals and when as interchangeable group members: the category that best fits the situation becomes salient. **Marilynn Brewer**'s **optimal distinctiveness theory** (1991) holds that we balance a need to belong against a need to be different, so we identify most with groups that are inclusive yet distinctive.

> EXAM TRAP
>
> In social identity theory, identity and prejudice come from **group membership and categorisation into in-groups and out-groups**, driven by the need for a positive social identity. Wrong options: competition for scarce resources (realistic conflict); birthplace, prejudice experienced, economic status, personal achievements, preferences or attributes (personal identity); internal or external attributions (Section 3.3); cognitive dissonance (Section 3.4); social facilitation (Section 3.10); genetic kin selection (Section 3.8); operant reinforcement (Chapter 1).

How stereotypes work in the mind

Categorisation also distorts perception. In the **out-group homogeneity effect** "they are all alike": **George Quattrone** and **Edward Ellsworth Jones** (1980) found Princeton and Rutgers students generalised more from one rival student's choice to his whole university than from an in-group member's; the reverse is **in-group differentiation**. An **illusory correlation** is a perceived group-trait link the data do not support: **David Hamilton** and **Robert Gifford** (1976) gave a large and a small group the same proportion of undesirable acts, yet people paired the two distinctive things, minority and rare bad acts, and judged the small group worse. **Thomas Pettigrew**'s **ultimate attribution error** (1979) extends the fundamental attribution error (Section 3.3) to groups: an out-group's failings are put down to its nature, its successes to luck.

Stereotypes persist because they save effort: **C. Neil Macrae**, **Alan Milne** and **Galen Bodenhausen** (1994) found that people given stereotype labels recalled more traits and did

better on a simultaneous second task, so stereotypes are **energy-saving devices**. They also resist disconfirmation. **Renee Weber** and **Jennifer Crocker** (1983) compared **bookkeeping** (each exception nudges the stereotype), **conversion** (one dramatic case overturns it) and **subtyping** (exceptions are fenced off as atypical, leaving it intact); disconfirmations spread over many typical members changed it most.

The **stereotype content model** of **Susan Fiske**, **Amy Cuddy**, **Peter Glick** and **Jun Xu** (2002) places groups on **warmth** and **competence**, each combination evoking its own emotion (Figure 3.8). Stereotypes can also impair performance through **stereotype threat**, the fear of confirming a negative stereotype about one's group. **Claude Steele** and **Joshua Aronson** (1995) found that Black Stanford students scored below equally able White students on hard verbal items only when these were presented as a test of ability; **Steven Spencer**, Steele and **Diane Quinn** (1999) found the same for women on a hard mathematics test.

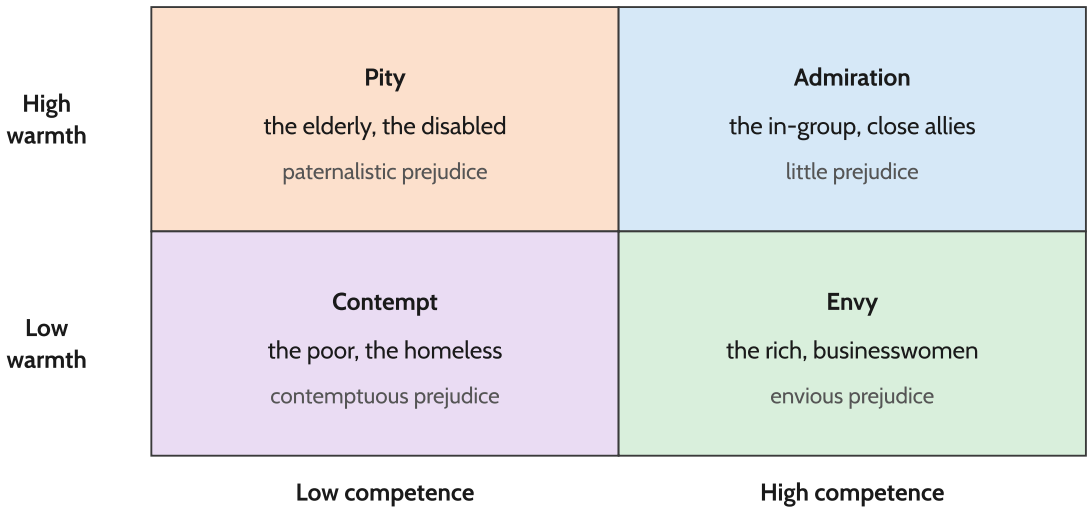


Figure 3.8 The stereotype content model (Fiske and colleagues 2002): warmth and competence decide the emotion a group evokes, so much prejudice is mixed.

Automatic and controlled processes ●●

Nearly everyone knows the cultural stereotypes; is everyone then prejudiced? **Patricia Devine**'s (1989) **dissociation model** says no. High- and low-prejudiced White Americans knew the stereotype of Black Americans equally well, and after stereotype-related words were flashed too fast to see, both judged an ambiguous man as more hostile: activation by **automatic processing** (fast, unintended, outside awareness) happens in everyone. Given time to list their thoughts, however, low-prejudiced people wrote more egalitarian ones. Devine concluded that low-prejudiced people use **controlled processing** (slow, deliberate, effortful) to inhibit the activated stereotype and replace it with egalitarian beliefs, much like breaking a bad habit.

Automatic associations can be measured without asking. The **Implicit Association Test** (IAT) of **Anthony Greenwald**, **Debbie McGhee** and **Jordan Schwartz** (1998) times how fast people sort faces and words with two response keys. Being faster when one group shares a key with "good" words than with "bad" words reveals an implicit, unconscious association (**implicit prejudice** or stereotyping) that people may be unwilling or unable to report. In the

bona fide pipeline of **Russell Fazio** and colleagues (1995), a briefly shown Black or White face primes a good-or-bad judgement of adjectives; faster responses to negative words after Black faces reveal an automatic negative attitude.

> EXAM TRAP
>
> Low-prejudiced people show **controlled processing that refutes the stereotype.** Automatic processing with involuntary stereotyping is the first stage, shared by all and left unchecked by high-prejudiced people; "automatic, voluntary" is self-contradictory; controlled processing prevents involuntary stereotyping rather than causing it. The IAT measures implicit associations by reaction time: not verbal or hidden mathematical ability, and not dream content (psychoanalysis).

Modern and subtle prejudice

Open bigotry has declined in surveys, but prejudice survives in forms that let people keep a fair self-image (Table 3.13).

Table 3.13 Modern and subtle forms of prejudice.

Form	Proposed by	What it is
Modern racism	**John McConahay** (1986)	Denying that discrimination persists; resenting minorities' "special favours"
Symbolic racism	**David Sears**	Hostility voiced as defence of values such as self-reliance
Aversive racism	**Samuel Gaertner** and **John Dovidio** (1986)	Sincere egalitarians with hidden negative feelings discriminate when a non-racial excuse exists
Tokenism	(general term)	A trivial favour or a few minority hires used to deflect charges of bias
Microaggressions	Term: **Chester Pierce** (1970); **Derald Wing Sue** and colleagues (2007)	Brief everyday slights, often unintended
Ambivalent sexism	Glick and Fiske (1996)	**Hostile** sexism (resentment) plus **benevolent** sexism (protective, but casting women as weak)

Gender roles and the workplace ●●

Descriptive gender stereotypes say what women and men are like; **prescriptive** ones say what they should be like, so a forceful woman leader meets backlash. **Alice Eagly**'s **social role theory** (1987; Eagly and **Wendy Wood**, 1999) traces sex differences and gender stereotypes to the division of labour: because men held provider and leader roles and women homemaker and caregiver roles, people infer that each sex has the traits its roles demand, and each sex learns the skills those roles reward. **David Buss**'s evolutionary rival (37 cultures, 1989) found men valuing a mate's youth and looks and women valuing earning prospects, which he traced to reproductive pressures in evolution; Eagly and Wood found these differences smaller in more gender-equal nations.

EXAM TRAP

Social role theory of gender = **Alice Eagly**. Buss is the evolutionary rival; Clea Mehta and JoNell Strough (2009) reviewed sex segregation in friendships across the life span. Anderson is only a distractor name: no Anderson is linked to social role theory (Craig Anderson, with Brad Bushman, built the General Aggression Model, Section 3.9).

At work, stereotypes become named barriers (Table 3.14). The best known, the **glass ceiling**, is an invisible barrier of attitudes and organisational bias that stops qualified women and minorities from rising to top positions; a 1986 *Wall Street Journal* article popularised the term. The halo effect (one good trait colouring every judgement) and the Pygmalion effect (expectations fulfilling themselves) are impression effects (Section 3.2).

Table 3.14 Named barriers and biases at work.

Term	Source	Meaning
Glass cliff	**Michelle Ryan** and **Alexander Haslam** (2005)	Women more often appointed to lead in a crisis, to precarious posts
Sticky floor	**Catherine White Berheide** (1992)	Women kept in the lowest-paid, least mobile jobs at the bottom
Maternal wall	**Joan Williams**	Bias against mothers, presumed less committed
Glass escalator	**Christine Williams** (1992)	Men in female-dominated jobs such as nursing rise fast
Matthew effect	**Robert King Merton** (1968)	Accumulated advantage: the already recognised get still more credit

EXAM TRAP

Glass ceiling = blocked at the **top**; sticky floor = stuck at the **bottom**; glass cliff = promoted onto a precarious post in a crisis; maternal wall = blocked by motherhood. The Matthew effect is accumulated advantage, not a barrier.

Reducing prejudice

Allport's **contact hypothesis** (1954) holds that contact reduces prejudice only under four conditions: **equal status**, **common goals**, **intergroup cooperation** and **support of authorities, law or custom**. That is why mere contact during conflict failed at Robbers Cave. Pettigrew and **Linda Tropp**'s (2006) meta-analysis of over 500 studies (515, with more than 250,000 participants) confirmed that contact reduces prejudice, most strongly when the conditions hold. In the **extended contact effect** (**Stephen Wright** and colleagues, 1997), merely knowing that an in-group member has a close out-group friend lowers prejudice. **Segregation**, **minimal contact** and **competition** keep prejudice alive.

In **Elliot Aronson**'s **jigsaw classroom** (first used in 1971 in newly desegregated schools in Austin, Texas; book *The Jigsaw Classroom*, 1978), each child in a mixed group learns one piece of the lesson and must teach it, so every child needs every other; liking and self-esteem rose. Other strategies change categories or retrain the mind (Table 3.15).

Table 3.15 Further ways to reduce prejudice.

Strategy	Source	How it works
Recategorisation: common in-group identity model	Gaertner and Dovidio (2000)	"Us" and "them" become one inclusive "we"
Decategorisation	Brewer and **Norman Miller** (1984)	Meet out-group members as individuals
Mutual intergroup differentiation	**Miles Hewstone** and **Rupert Brown** (1986)	Keep both identities salient while cooperating, so liking generalises
Learning not to hate	Social learning view	Parents and teachers model tolerance
Collective guilt	Powell, **Nyla Branscombe** and Schmitt (2005)	Seeing inequality as one's own group's advantage lowers racism
Saying no to stereotypes	**Kerry Kawakami** and colleagues (2000)	Repeatedly negating stereotypic pairings weakens automatic activation
Social influence	**Charles Stangor** and colleagues (2001)	Learning that peers are less prejudiced reduces one's own stereotyping

EXAM TRAP

Of segregation, minimal contact, competition and intergroup contact with cooperation, only the last reduces prejudice, and only when Allport's conditions hold.

INDIAN CONTEXT

In rural Uttar Pradesh, **Karla Hoff** and **Priyanka Pandey** (2006) found that low-caste boys solved mazes as well as high-caste boys until each boy's caste was announced publicly, an effect resembling stereotype threat. Cooperative contact helps: in **Matt Lowe**'s (2021) field experiment in Uttar Pradesh, young men randomly placed on caste-mixed cricket teams formed more cross-caste friendships, whereas merely playing against other castes did not. **Ashutosh Varshney** (2002) found that cities with strong Hindu-Muslim civic associations stayed peaceful when communal riots struck elsewhere.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. A belief about the traits of a group's members is a _____.
2. Allport's five-step scale begins with _____.
3. At Robbers Cave, hostility fell only with _____ goals.
4. Boys grouped by a supposed preference for Klee or Kandinsky favoured their own group in Tajfel's _____ group paradigm.
5. In social identity theory, the need for positive _____ drives favourable comparison of the in-group.
6. Devine found that low-prejudiced people use _____ processing to inhibit automatically activated stereotypes.
7. Women appointed to lead in a crisis stand on a glass _____; women kept in the lowest jobs face a sticky _____.
8. In Aronson's _____ classroom, each child must teach one piece of the lesson.

Answers: (1) stereotype, (2) antilocution, (3) superordinate, (4) minimal, (5) distinctiveness, (6) controlled, (7) cliff, floor, (8) jigsaw.

3.6 Affiliation, Attraction and Close Relationships

LOWER YIELD · 7 QS

Think back to your first weeks in a hostel: most of the friends you made were probably the people whose rooms were nearest yours. Why do we seek company at all, why these people rather than others, and what keeps a relationship alive or ends it? This section moves from the need to affiliate, through attraction and the theories of close relationships, to love, break-up and loneliness.

The need to affiliate ●●

Roy Baumeister and **Mark Leary** argued in 1995 that the **need to belong**, a drive to form and keep a few lasting, positive relationships, is a fundamental human motive. People differ in its strength; the stable trait is the **need for affiliation** (nAff), listed by **Henry Murray** and measured by **David McClelland** (Section 1.10). The older word is **gregariousness**, the urge to be with others and live in groups. **William McDougall** listed a gregarious instinct in 1908, but gregariousness is now classed as a **social motive**: a secondary, learned or psychosocial motive, like affiliation, achievement and power, that arises and is satisfied in interaction with others. Biological (primary) motives such as hunger and thirst serve bodily survival instead.

EXAM TRAP

Gregariousness has a **social** motive. "Biological" tempts because McDougall called it an instinct, but instinct theory collapsed (Section 1.10). "Psychological" is not a standard separate class, and personal motives are an individual's own habits, interests and aspirations.

KEY STUDY: SCHACHTER'S ANXIETY AND AFFILIATION EXPERIMENT (1959)

Stanley Schachter (1922 to 1997) had women students met by a fictitious "Dr Gregor Zilstein", who warned of electric shocks that would be painful (high anxiety) or feel like a tickle (low anxiety). Offered a wait alone or with others, about two-thirds of the high-anxiety women chose company, against about one-third of the low-anxiety women, and they wanted only others facing the same shocks: "misery doesn't just love any kind of company, it loves only miserable company". No shocks were given.

Why do frightened people want each other? All three usual answers hold: we affiliate **to obtain information** about what is happening, **to reduce anxiety**, and **to compare ourselves with similar others**. **Craig Hill** (1987) found four affiliation motives: **positive stimulation**, **emotional support**, **attention** and **social comparison**.

The need shows most sharply when thwarted. **Ostracism**, being ignored and excluded, was studied by **Kipling Williams** with **Cyberball** (2000), an online ball-tossing game in which the other "players" soon stop throwing to the participant; minutes of exclusion by strangers lower mood and threaten belonging, self-esteem, control and meaningful existence. Feelings shape liking too: in the **reinforcement-affect model** of **Donn Byrne** and **Gerald Clore** (1970) we like people who make us feel good and, by classical conditioning, even strangers merely present when something pleasant happens.

Social comparison

Leon Festinger (1919 to 1989) proposed **social comparison theory** in 1954: people are driven to evaluate their opinions and abilities, and when no objective standard exists they compare themselves with others, preferably similar others. A mark of 85 percent means little until you know how your classmates scored. **Upward social comparison**, with someone better off, can inspire but often brings envy; **downward social comparison**, with someone worse off, was described by **Thomas Wills** (1981) as a route to self-enhancement when self-esteem is threatened. **Abraham Tesser**'s **self-evaluation maintenance model** (1988) adds closeness and relevance: if your cousin tops the medical entrance and medicine matters to you, the **comparison process** stings and you may distance yourself; if she wins a dance contest you do not care about, the **reflection process** lets you bask in her glory.

Social media supply an endless upward stream of other people's best moments. **Andrew Przybylski** and colleagues (2013) defined the **fear of missing out** (FOMO) as a pervasive apprehension that others might be having rewarding experiences from which one is absent; it goes with heavier social media use and lower mood. FOMO is a social comparison phenomenon.

EXAM TRAP

FOMO is linked to **social comparison**, not classical conditioning (learning by pairing stimuli), operant extinction (a response fading when reinforcement stops) or "cognitive dissonance only" (discomfort from inconsistent cognitions, Section 3.4).

Nearness and familiarity

The plainest force of attraction is nearness. In 1950 Festinger, Schachter and **Kurt Back** studied Westgate West, married-student housing at the Massachusetts Institute of Technology where flats were assigned at random. Of next-door neighbours, 41 percent were named as close friends, against 22 percent of those two doors away and 10 percent of those at the far end of the corridor. **Functional distance**, how often architecture brings paths together, mattered too: residents beside the stairs and mailboxes made more friends upstairs. **Mady Segal** (1974) found that police trainees seated alphabetically befriended those with neighbouring surnames. This is the **propinquity effect**: the more we see and interact with a person, the more likely he or she is to become our friend or partner.

The mechanism is largely the **mere exposure effect** of **Robert Zajonc** (1968): repeated exposure to a stimulus, without interaction or reward, increases liking for it, whether nonsense words, Chinese-like characters or faces. **Richard Moreland** and **Scott Beach** (1992) had four women attend a class 0, 5, 10 or 15 times without speaking; students liked them more the more often they came. **Theodore Mita** and colleagues (1977) found that women preferred mirror images of their own faces, the version they see daily, while friends preferred the true image. A stimulus disliked at first, however, can become more disliked.

> EXAM TRAP
>
> "The more we see and interact with someone, the more likely they become our friend or partner" is the **propinquity effect**. The **exposure effect** is narrower (mere exposure to any stimulus raises liking) and is its mechanism; **matching** and **reciprocity** are taught below. For mere exposure, "performing better when observed" is social facilitation, "attitude change after inconsistent behaviour" is cognitive dissonance and "riskier group decisions" is the risky shift (Sections 3.4 and 3.10).

Looks, similarity and reciprocity

Appearance acts first. **Karen Dion**, **Ellen Berscheid** and **Elaine Hatfield** (then Walster) showed in 1972 that attractive people are judged kinder, more sociable and more successful: the **what is beautiful is good** stereotype. At a 1966 computer dance at the University of Minnesota, Walster and colleagues paired students at random; the only predictor of wanting another date was the partner's attractiveness. **Judith Langlois** and **Lori Roggman** (1990) found computer-averaged faces more attractive than most faces composing them (averageness); symmetry also attracts. Yet most people must settle: the **matching hypothesis** (Walster and colleagues 1966; tested by Berscheid and colleagues 1971) holds that we pair with partners of roughly our own attractiveness to avoid rejection, and **Bernard Murstein** (1972) found real couples closer in attractiveness than chance.

Similarity matters more over time. **Theodore Newcomb** measured transfer students' attitudes by post, housed them together, and found that those initially similar became friends (*The Acquaintance Process*, 1961). Byrne's "bogus stranger" experiments gave his **law of attraction** (1971): attraction rises with the *proportion* of similar attitudes. **Milton Rosenbaum**'s **repulsion hypothesis** (1986), that dissimilarity repels while similarity adds nothing, overstates the case, though dissimilarity does weigh slightly more. **Robert Winch**'s (1958) **complementarity of needs**, opposites attracting (dominant with submissive), has found little support.

We also like those who like us (**reciprocity of liking**). In the **gain-loss effect** of **Elliot Aronson** and **Darwyn Linder** (1965), participants who overheard a confederate's evaluations liked her most when they turned from negative to positive, more than when consistently positive, and least when they turned from positive to negative.

Exchange, equity and investment

Why do people stay in some relationships and leave others? **Social exchange theory**, from **John Thibaut** and **Harold Kelley** (1959) and **George Homans** (1961), treats a relationship as a trade: rewards minus costs. The outcome is judged against the **comparison level** (CL), what one feels one deserves from past relationships, which sets satisfaction, and the **comparison level for alternatives** (CLalt), the best outcome available elsewhere, which decides staying. Meera may be dissatisfied with Arjun (below her CL) yet stay because every alternative looks worse (above her CLalt). Table 3.16 sets this beside three relatives.

Table 3.16 Four theories of close relationships.

Theory	Proposed by	Core idea
Social exchange	Thibaut and Kelley 1959; Homans 1961	Rewards minus costs, judged against CL and CLalt
Equity theory	Walster (Hatfield), Berscheid and Walster 1978	Each partner's outcomes in proportion to inputs
Investment model	Caryl Rusbult 1980	Satisfaction, poor alternatives and large investments build commitment
Communal versus exchange	Clark and Mills 1979	Help given to meet needs versus benefits repaid in kind

In **equity theory** (Walster, Berscheid and **G. William Walster** 1978) both the over-benefited, who feel guilty, and the under-benefited, who feel angry, are distressed. **Caryl Rusbult**'s **investment model** explains staying when unhappy: years together, children or a shared home cannot be recovered, so commitment stays high, as in women who return to abusive partners. **Margaret Clark** and **Judson Mills** showed that tit for tat suits **exchange relationships** with strangers; in **communal relationships** with family and close friends, prompt repayment can even reduce liking.

Self-disclosure

Relationships deepen through **self-disclosure**, revealing personal information about oneself to another. **Sidney Jourard** (1926 to 1974), in *The Transparent Self* (1964), saw being fully known by someone as a mark of healthy personality; his **Jourard Self-Disclosure Questionnaire** asked how much people had told mother, father, friends and spouse about areas such as tastes, work, money and body. **Irwin Altman** and **Dalmas Taylor**'s **social penetration theory** (1973) pictures the self as an onion: relationships grow in **breadth** (range of topics) and **depth** (intimacy), moving from outer layers to the core. Disclosure invites matching disclosure (**disclosure reciprocity**), and we like those who confide in us.

T. L. Morton (1978), followed by the textbook of **Shelley Taylor**, **Letitia Anne Peplau** and **David Sears**, separated **descriptive self-disclosure**, revealing facts and information about oneself (where you grew up), from **evaluative self-disclosure**, revealing one's personal feel-

ings, opinions and judgements (that you feel lonely). Evaluative disclosure is more intimate and comes later.

> **EXAM TRAP**
>
> IHBAS 2020 keyed "descriptive self-disclosure" for communicating one's feelings. By the standard definitions that is **evaluative**; descriptive disclosure is about facts. Learn the standard pair, and know that this paper's key reversed it. "Prescriptive" and "non-evaluative" self-disclosure are not standard types.

Love

Zick Rubin (1970) built separate **Liking** (respect, affection) and **Loving Scales** (attachment, caring, intimacy); high scorers on loving gazed at each other longer. Berscheid and Hatfield separated **passionate love**, an intense, aroused longing for union with the other (measured by Hatfield and **Susan Sprecher**'s Passionate Love Scale, 1986), from **companionate love**, the calmer affection we feel for those with whom our lives are deeply entwined. **Robert Sternberg**'s **triangular theory of love** (1986) builds love from **intimacy** (closeness), **passion** (physical attraction and arousal) and **commitment** (the decision to maintain the love); their combinations give eight kinds (Table 3.17).

Table 3.17 Sternberg's eight kinds of love.

Kind	Intimacy	Passion	Commitment	For example
Non-love	no	no	no	casual acquaintances
Liking	yes	no	no	true friendship
Infatuation	no	yes	no	love at first sight
Empty love	no	no	yes	a stale marriage
Romantic love	yes	yes	no	a holiday romance
Companionate love	yes	no	yes	a long, settled marriage
Fatuous love	no	yes	yes	a whirlwind wedding
Consummate love	yes	yes	yes	complete love

The sociologist **John Alan Lee** (1973) described six love styles: eros (passionate, beauty-led), ludus (playful, uncommitted), storge (friendship-based), pragma (practical, a checklist partner), mania (possessive, jealous) and agape (selfless). Berscheid and Walster's **two-factor theory of love** (1974) applies Schachter's two-factor theory of emotion: passionate love is physiological arousal labelled "love" in the presence of a suitable person, so arousal from elsewhere can be misattributed to attraction, as in **Donald Dutton** and **Arthur Aron**'s 1974 bridge study (Section 1.11).

Love also carries the stamp of early attachment (infants in Chapter 6). **Cindy Hazan** and **Phillip Shaver** (1987) classified adults as **secure**, **avoidant** (uneasy with closeness) or **anxious-ambivalent** (craving closeness, fearing abandonment), just over half secure, much as in infants. **Kim Bartholomew** and **Leonard Horowitz** (1991) crossed the model of self with the model of others to get four **adult attachment styles** (Figure 3.9).

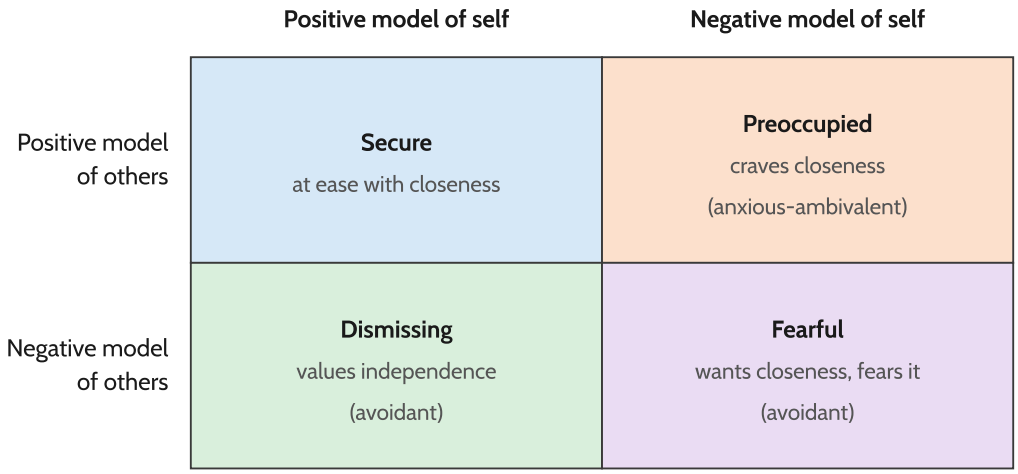


Figure 3.9 Bartholomew and Horowitz's four styles; brackets give Hazan and Shaver's nearest category.

Anxious attachment also feeds **jealousy**, the fear, anger and hurt felt when a rival seems to threaten a valued relationship. **David Buss** and colleagues (1992) reported men more upset by sexual and women by emotional infidelity, an evolutionary reading that critics say depends on question wording.

INDIAN CONTEXT

In Jaipur, **Usha Gupta** and **Pushpa Singh** (1982) found that couples who married for love reported less love after about five years, while those in arranged marriages reported more love once past the newlywed years.

When relationships fail

Rusbult classified responses to dissatisfaction as active or passive and constructive or destructive: **exit** (active, destructive: leaving), **voice** (active, constructive: discussing problems), **loyalty** (passive, constructive: waiting faithfully) and **neglect** (passive, destructive: letting things decay). **Steve Duck** (1982) traced dissolution through four phases: **intrapsychic** (brooding privately on the partner's faults), **dyadic** (confronting the partner), **social** (telling friends and family) and **grave-dressing** (building an acceptable story of the break-up). **John Gottman**'s "four horsemen" are taught with couple therapy in Chapter 7.

Loneliness

Peplau and **Daniel Perlman** (1982) defined **loneliness** as the unpleasant experience that occurs when a person's network of social relationships is deficient in quantity or quality. It is subjective: one can be lonely in a crowd. **Robert Weiss** (1973) separated two kinds. **Emotional isolation** (emotional loneliness) is the lack of a close, intimate attachment to one specific person, felt by a widow though friends surround her. **Social isolation** (social loneliness) is the lack of a wider network of friends and associates, felt by a newcomer to a city even in a happy marriage.

> **EXAM TRAP**
>
> Emotional isolation means no deep attachment to **one specific person**; social isolation means too few **friends or associates**. Incompetence, inadequacy and low self-esteem are correlates of loneliness, not Weiss's types.

The **UCLA Loneliness Scale** of **Daniel Russell**, Peplau and colleagues (1978, revised 1980) has 20 statements such as "I lack companionship". Loneliness goes with depression, poorer health and shorter life (**John Cacioppo** and colleagues), and lonely people often benefit from **social skills training** (starting conversations, disclosing appropriately) and from challenging the negative expectations that keep them withdrawn.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Schachter's anxious participants preferred to wait with others facing the _____ situation.
2. Comparing yourself with someone worse off is _____ social comparison.
3. FOMO is best explained as a _____ phenomenon.
4. Friendship growing from frequent contact with neighbours is the _____ effect.
5. In exchange theory, the standard that decides whether a person stays is the _____.
6. Revealing feelings and opinions is _____ self-disclosure.
7. Passion plus commitment without intimacy is _____ love.
8. Lacking a close attachment to one specific person is Weiss's _____.

Answers: (1) same, (2) downward, (3) social comparison, (4) propinquity, (5) comparison level for alternatives, (6) evaluative, (7) fatuous, (8) emotional isolation.

3.7 Social Influence: Conformity, Compliance, Obedience and Power

LOWER YIELD · 18 QS

Other people change what you do all day, often without a word. You join the queue at a chai stall, lend a friend your notes, fall silent when the invigilator orders it. This section follows influence from the unspoken norm through direct requests to outright commands, then turns to the roles and the bases of power behind them.

Kinds of social influence ●●

Social influence is any effort by one or more people to change the behaviour, attitudes or feelings of others (persuasion, changing attitudes by messages, is in Section 3.4). Three forms differ in how directly pressure is applied (Table 3.18). **Conformity** is changing your behaviour or beliefs to match a group or its norms, although nobody has asked you to. **Compliance** is changing in response to a direct request, which you may accept or refuse. **Obedience** is changing in response to a command from an authority. Textbooks add **symbolic social influence**: absent people still steer us through our mental image of them.

Table 3.18 The forms of social influence.

Form	Source of pressure	Example
Conformity	Group norms; no request	Standing for the national anthem
Compliance	A direct request	A friend asks to borrow your notes
Obedience	A command from an authority	A doctor orders a nurse to give a drug
Symbolic influence	Mental image of absent others	Studying because you picture a parent's disappointment

Herbert Kelman (1958) graded how deep the change goes. In compliance (his sense) a person goes along outwardly to gain reward or avoid punishment, and only while watched. In **identification** he adopts the behaviour to be like an admired person or group. In **internalisation** he adopts it because it fits his own values, and it lasts. The same contrast appears as **public compliance** (going along outwardly while privately disagreeing) versus **private acceptance** (truly coming to believe what the group believes).

> EXAM TRAP
>
> "Changing your behaviour to match others in a group" is **conformity**. *Norming* is a stage of **Bruce Tuckman**'s group development (forming, storming, norming, performing; Section 3.10); *standardisation* means uniform test procedures (Chapter 2). Conformity is change under group pressure with nobody asking; compliance answers a direct request. Conformity is not "always involuntary", and neither term means obedience.

Social norms

Social norms are rules indicating how people are expected to behave in a situation. **Explicit norms** are written or stated ("No Parking"); **implicit norms** are unstated yet known to all (do not talk during a concert). **Robert Cialdini**, **Raymond Reno** and **Carl Kallgren** (1990) distinguished **descriptive norms**, what most people actually do, from **injunctive norms**, what is approved or disapproved, what ought to be done. In their littering study, people found a handbill on their car windscreen. More dropped it where the ground was already littered; fewest did where the ground was clean and a confederate conspicuously dropped one, because the lone litterbug highlighted the descriptive norm that nobody litters here. Making the injunctive norm salient (a passer-by picking up litter, in a follow-up) cut littering even in a littered setting. Their **focus theory of normative conduct** holds that a norm guides behaviour only when it is focal, salient at the moment of acting.

Prescriptive norms tell people what they ought to do (greet elders); **proscriptive norms** tell them what not to do (do not cheat). A **perceived norm** is a person's belief about what others do or approve, which may be wrong: students who overestimate their classmates' drinking drink more. Societies whose norms break down suffer anomie (Section 3.1).

> EXAM TRAP
>
> Norms telling people how they *ought* to behave are **prescriptive** (injunctive) norms. "Idealistic norms" is not a standard type; descriptive norms describe what is done; perceived norms are beliefs about norms.

Sherif and Asch: the classic studies ●●

How does a norm form? **Muzafer Sherif** (1935, 1936) used the **autokinetic effect**: in a dark room a stationary point of light seems to move, because there is no frame of reference. Alone, people gave widely different estimates of the movement. In small groups over several sessions their estimates converged on a shared group norm, which each person kept when later tested alone. In an ambiguous situation people used one another as information and privately accepted the answer. **Robert Jacobs** and **Donald Campbell** (1961) had confederates set an exaggerated norm, then replaced members one at a time; the arbitrary norm was handed down for several "generations" after the confederates had gone. **Solomon Asch** (1951, 1956) asked the harder question: will people conform when the answer is obvious?

KEY STUDY: ASCH'S LINE JUDGEMENTS (1951, 1956)

Method: one real participant sat among about seven confederates and said aloud, near the end of the row, which of three comparison lines matched a standard line (Figure 3.10). On 12 of 18 trials, the **critical trials**, every confederate gave the same wrong answer. **Result:** alone, people almost never erred (under 1 percent of judgements); in the group, conformity occurred on about 37 percent of critical trials. About three quarters (76 percent in Baron's account) conformed at least once; about a quarter never did. **Conclusion:** the answer was obvious, so people yielded to avoid standing out: normative influence towards an incorrect majority, mostly public compliance. **Criticism:** only male American students were tested, and such conformity has since declined.

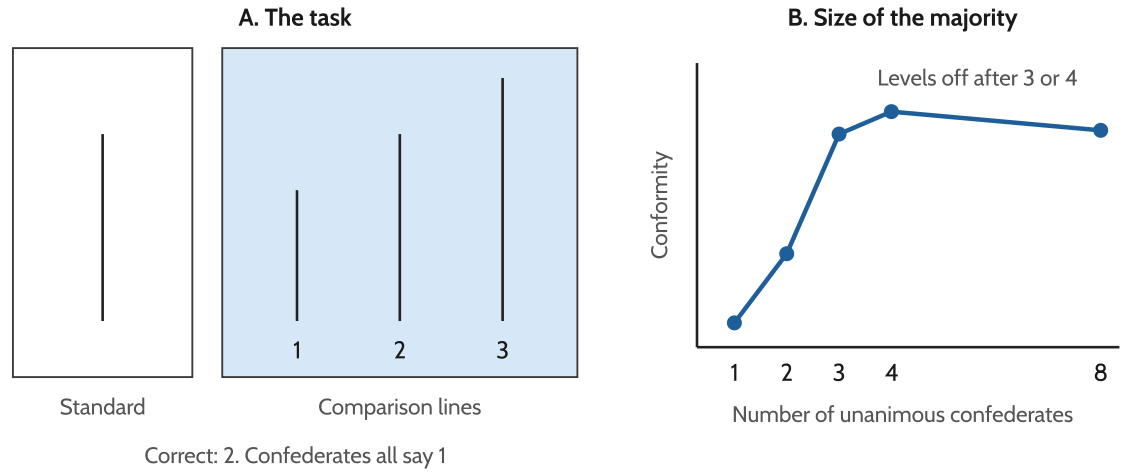


Figure 3.10 Asch's conformity task. (A) The match is obvious, yet on critical trials every confederate named a wrong line. (B) Conformity rose steeply as the unanimous majority grew to three, then levelled off (schematic, after Asch's group-size variation).

Asch's variations show what drives the effect. Conformity rose with the size of the majority up to three or four, then levelled off. A single dissenting partner cut it sharply, even one giving a different wrong answer: unanimity is what matters. Written, private answers cut it steeply too. **Richard Crutchfield** (1955) mechanised the method: people in separate booths saw others' supposed answers as lights on a panel, so no confederates were needed. **Rod Bond** and **Peter Smith**'s (1996) meta-analysis of 133 Asch-type studies in 17 countries found more conformity in collectivist cultures and a decline in the United States since the 1950s.

EXAM TRAP

Asch tested the **limits of conformity** (not social perception, leadership or coercive power) and showed **normative influence towards an incorrect majority**: not obedience (nobody gave orders), the bystander effect or diffusion of responsibility (Section 3.8), or cognitive dissonance (Section 3.4). Private answers *lowered* conformity. Sherif's dot of light: ambiguity, informational influence, norm formation (not obedience or implicit bias, Section 3.5).

Why people conform

Morton Deutsch and **Harold Gerard** (1955) named two motives. **Normative social influence** rests on the desire to be liked and accepted; it produces public compliance and is strongest when a valued group is watching, as in Asch's study. **Informational social influence** rests on the desire to be correct; it produces private acceptance and is strongest when the situation is ambiguous or we lack expertise, as in Sherif's study.

Conformity rises with the group's **cohesiveness** (how much we are attracted to it and want to belong), its size up to a point, its unanimity, its members' status or expertise, public responding, a difficult or ambiguous task, and a collectivist culture. Gender differences are small: **Alice Eagly** and **Linda Carli**'s (1981) review found women slightly more influenceable, mainly under face-to-face group pressure. Conformity keeps traffic and queues orderly, but its downside is that it can lead ordinary people into cruelty (see the power of roles).

EXAM TRAP

Normative influence is the desire to be **accepted and liked**; informational influence, the desire to be **correct**. "Personal beliefs and values" describe internalisation; "power and authority" describe obedience and the bases of power.

Resisting the majority; minority influence

Conformity pressure can be resisted. **C. R. Snyder** and **Howard Fromkin** (1980) proposed a **need for uniqueness**; later studies found that people told they are "exactly average" resist the majority more. The related desire for **individuation**, to stand out as a distinct person, works the same way. People reminded of a time they held power over others conform less, and pressure that threatens freedom can provoke **reactance** (Section 3.4).

Can a minority sway a majority? **Serge Moscovici**, **Elisabeth Lage** and **Martine Naffrechoux** (1969) showed groups of four participants and two confederates blue slides. When the confederates consistently called every slide "green", about 8 percent of the majority's judgements became "green", and in a later private test participants labelled more blue-green discs green; an inconsistent minority had almost no effect. In his **conversion theory** (1980) Moscovici argued that majorities win quick public compliance, whereas a consistent minority makes people think and produces slower, genuine private conversion. **Edwin Hollander** (1958) explained safe dissent: by conforming early a member earns **idiosyncrasy credits**, goodwill he can later spend on deviation. The most basic influence may be **emotional contagion**, the automatic catching of another's mood, which **Elaine Hatfield**, **John Cacioppo** and **Richard Rapson** (1994) traced to our mimicry of others' faces and postures.

Compliance tactics and cults ●●

Cialdini worked undercover among "compliance professionals" (salespeople, fund-raisers, advertisers) and in *Influence* (1984) traced their tactics to six principles: **reciprocity** (we repay favours and concessions), **commitment and consistency** (we act in line with what we have agreed to), **social proof** or social validation (we copy similar others), **liking** or friendship, **authority** (we defer to real or apparent experts) and **scarcity** (we want what is rare or running out). Table 3.19 matches tactics to principles.

Table 3.19 Compliance tactics and the principle each exploits.

Tactic	What happens	Principle
Ingratiation (**Edward E. Jones**, 1964)	Flattery, agreement or favours make the target like you first	Liking
Foot-in-the-door (1966)	A small request nobody refuses, then the larger real one	Consistency
Low-ball (Cialdini and colleagues, 1978)	A good deal is accepted, then made worse	Commitment
Door-in-the-face (Cialdini and colleagues, 1975)	A huge request is refused, then the smaller real one follows	Reciprocity
That's-not-all (**Jerry Burger**, 1986)	Before any answer, the offer is sweetened	Reciprocity
Playing hard to get	You or your product seem much in demand	Scarcity
Fast-approaching deadline	"Offer ends Sunday"	Scarcity
Pique (Santos, Leve and Pratkanis, 1994)	An odd request ("spare 17 cents?") breaks automatic refusal	Arousing attention
Foot-in-the-mouth (**Daniel Howard**, 1990)	The target first answers "How are you today?"	Commitment to a friendly exchange

KEY STUDY: THE FOOT-IN-THE-DOOR (1966)

Jonathan Freedman and **Scott Fraser** demonstrated the **foot-in-the-door technique** by asking Californian homeowners to put a large, ugly "Drive Carefully" sign on their lawns. About 17 percent of those approached cold agreed, but about 76 percent of those who, two weeks earlier, had agreed to show a small "Be a safe driver" sign. Two explanations: **consistency** (refusing now feels inconsistent) and **self-perception** (watching yourself agree, you infer "I support good causes"; Section 3.4).

The **door-in-the-face technique** reverses the order. Cialdini and his colleagues (1975) asked students to counsel juvenile delinquents two hours a week for two years; all refused. Asked next to take delinquents on a two-hour zoo trip, about 50 percent agreed, against under 17 percent asked only the small favour. The retreat looks like a concession, and reciprocity demands one back (reciprocal concessions). In the **low-ball technique** the initial commitment does the work: a car buyer who agreed to a bargain still buys when the "manager" cancels the discount.

EXAM TRAP

Foot-in-the-door: **small, then large**. Door-in-the-face: **large, refused, then small**. Low-ball: yes first, then the deal worsens. That's-not-all: the deal improves before any yes. "Re-ward" is **not** one of Cialdini's six principles, and "conformity technique" is not a compliance tactic: conformity involves no request.

Cults use these principles at full strength. Recruiters begin with **love-bombing**, flooding a newcomer with affection and attention; they escalate commitment step by step (a free dinner, a retreat, a donation, then residence); they isolate recruits from family and former friends, control their information and build an us-versus-them identity. **Robert Jay Lifton** (1961), studying Chinese Communist "thought reform", described such total environments, including milieu control (control of all communication). At Jonestown, Guyana (1978), more than 900 followers of **Jim Jones** died in a mass murder and suicide he ordered. Talking with recruits' families is exactly what cults avoid.

Obedience: Milgram's experiments

Would you shock an innocent stranger because a man in a laboratory coat told you to? **Stanley Milgram** (1933 to 1984) asked this partly to understand how ordinary people carried out the Holocaust.

KEY STUDY: MILGRAM'S OBEDIENCE STUDY (1963)

Method: at Yale, 40 men aged 20 to 50 joined a supposed study of punishment and memory. A rigged draw made each the **teacher**; the **learner**, a likeable confederate, sat strapped in next door. For each error the teacher moved one switch up a generator from 15 to 450 volts in 15-volt steps, labelled up to "Danger: severe shock" and "XXX". No real shocks were given. At 300 volts the learner pounded on the wall, then fell silent. When teachers hesitated, the experimenter used four standard prods, from "Please continue" to "You have no other choice, you must go on." **Result:** psychiatrists had predicted that about one in a thousand would go to the end. All 40 reached 300 volts and 26 (65 percent) went to 450 volts, many sweating, trembling and protesting. **Conclusion:** most ordinary people obey a legitimate authority even to the point of apparently harming another person.

His variations (*Obedience to Authority*, 1974) show what raises and lowers obedience (Table 3.20).

Table 3.20 Milgram's variations: percentage who went to 450 volts.

Condition	Full obedience
Remote: learner unseen, pounds on wall (baseline)	65%
Voice feedback: protests heard	62.5%
Proximity: learner in the same room	40%
Touch-proximity: teacher presses learner's hand onto plate	30%
Orders given by telephone	About 20%
Run-down Bridgeport office, no link to Yale	About 48%
An ordinary man gives the orders	20%

Condition	Full obedience
Two peer teachers rebel	10%
Women as teachers	65%

Milgram's explanation was the **agentic state**: within a hierarchy a person stops acting autonomously and becomes the agent of another's wishes, transferring responsibility to the authority. The authority seemed legitimate (Yale, the coat, science); demands escalated gradually, each step barely worse than the last; and events moved too fast for reflection. Obedience falls when people are reminded that they are responsible, see disobedient models, question the authority's expertise and motives, and know about this research.

Diana Baumrind (1964) attacked the study's ethics: participants were deceived and severely stressed. Milgram replied that all were debriefed and 84 percent were glad to have taken part. Burger (2009) ran a safer partial replication, stopping at 150 volts (beyond which nearly all of Milgram's obedient men went to the end): about 70 percent were willing to continue past it. **Hannah Arendt** (1963), reporting the trial of the Nazi official **Adolf Eichmann**, described the **banality of evil**: monstrous crimes done by ordinary people doing their jobs.

CLINICAL LINK

Charles Hofling and colleagues (1966) had an unknown "doctor" telephone hospital nurses and order 20 milligrams of an unfamiliar drug, "Astroten", twice the maximum dose on its label. Of 22 nurses, 21 began to give it before being stopped.

EXAM TRAP

Milgram's baseline: **65 percent** (26 of 40) to 450 volts, all 40 to 300 volts; not 10, 35 or 100 percent. The lesson is the power of the situation, not personality, and it concerns obedience to authority, not conformity when an answer is ambiguous.

The power of roles

KEY STUDY: THE STANFORD PRISON EXPERIMENT (1971)

Method: Philip Zimbardo, with **Craig Haney** and **Curtis Banks**, randomly made 24 psychologically healthy male students guards or prisoners in a mock prison in Stanford's psychology basement. Prisoners were "arrested" at home and known only by numbers; guards wore uniforms and mirrored sunglasses. **Result:** the guards became abusive, and the prisoners, after a brief rebellion, passive and distressed. Planned for two weeks, the study was stopped after six days. **Conclusion:** social roles and situational forces can turn ordinary people cruel. **Criticism:** demand characteristics (guards may have acted as they thought was wanted), Zimbardo's double role as researcher and prison superintendent, and ethical harm.

In the BBC Prison Study (2006), **Stephen Reicher** and **Alexander Haslam** found that guards who did not identify with their role did not turn tyrannical, and prisoners who shared a group identity challenged them: roles shape behaviour when people identify with them. In

The Lucifer Effect (2007) Zimbardo applied his analysis to American soldiers' abuse of Iraqi prisoners at Abu Ghraib.

> EXAM TRAP
>
> "The power of roles" is **Zimbardo's prison experiment**, showing situational forces, not genetics, conditioning or reinforcement schedules (Chapter 1). Not the *Robbers Cave experiment* (Sherif's intergroup conflict study, Section 3.5); not Milgram's *message experiment*, the small-world study (1967), whose letters reached a stranger through about six acquaintances ("six degrees of separation"), nor his lost-letter technique; not "Zilstein's shock research", the fictitious Dr Gregor Zilstein of **Stanley Schachter**'s (1959) affiliation study (Section 3.6).

Bases of social power

Social power is the capacity to influence others. **John French** and **Bertram Raven** (1959) described five bases; Raven (1965) added informational power (Table 3.21). Raven later grouped tactics as **harsh** (coercion, impersonal reward, insisting on one's position), which breed resentment, or **soft** (expert, referent and informational), which win lasting acceptance.

Table 3.21 The six bases of social power.

Base	Rests on	Example
Reward	Ability to give rewards	A manager who grants bonuses
Coercive	Ability to punish	A warden who can confine
Legitimate	An accepted right to give orders	A police officer; Milgram's experimenter
Expert	Superior knowledge or skill	A doctor's advice
Referent	Identification with an admired agent	A cricket star's endorsement
Informational	The content of the message itself	A convincing argument

> EXAM TRAP
>
> **Reciprocity** is a Cialdini compliance principle, not one of Raven's six bases of power. Coercion, expertise and information are bases.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Changing in response to a direct request is _____.
2. Norms that say what most people actually do are _____ norms.
3. Sherif used the _____ effect to show how group norms form.
4. Asch's participants conformed on about _____ percent of critical trials.
5. Conforming from a desire to be liked and accepted is _____ social influence.
6. A large request refused, then a smaller one, is the _____ technique.
7. In Milgram's baseline study, _____ percent went to 450 volts.
8. Raven's sixth base of power is _____ power.

Answers: (1) compliance, (2) descriptive, (3) autokinetic, (4) 37, (5) normative, (6) door-in-the-face, (7) 65, (8) informational.

3.8 Prosocial Behaviour: Why and When People Help

LOWER YIELD · 15 QS

In March 1964 a young woman was stabbed to death in New York while, the newspapers said, 38 neighbours did nothing. Was this proof that city people no longer cared? Two young psychologists suspected that the number of witnesses was itself the problem. Their answer, the bystander effect, is the heart of this section. First, though: why does anyone help at all?

Prosocial behaviour, helping and altruism

Any action that benefits others, such as sharing, comforting, donating or rescuing, is **prosocial behaviour**. Prosocial behaviour that intentionally benefits a particular person is **helping**. Defined by motive rather than act, **altruism** is helping with no expected reward, often at a cost to oneself, purely for the other's welfare. In **cooperation**, people work together towards a goal that benefits all of them, so nobody need be the giver. A student who lends her notes is helping; if she expects nothing back and loses revision time, she may be acting altruistically; if two friends split the syllabus and swap notes, they are cooperating.

Does true altruism exist? The doctrine of **psychological egoism** says no: every helpful act ultimately serves the helper, by winning praise, avoiding guilt or feeling good. This **egoism versus altruism debate** was fought hardest over empathy.

Empathy and the egoism versus altruism debate

The act of perceiving, understanding and responding to another person's emotional state and ideas is **empathy**. Its emotional side, **affective empathy**, is feeling with the person; **cognitive empathy**, or perspective taking, is accurately understanding what they think and feel; **empathic concern** is the warm, other-oriented compassion that follows. **Mark H. Davis** measured these with his Interpersonal Reactivity Index (1980), whose four scales are perspective taking, fantasy, empathic concern and personal distress (one's own anxiety at another's suffering). Feeling concern or pity *for* someone, rather than feeling *with* them, is **sympathy**. Empathy is a foundation of helping, and a core condition a counsellor offers a client (Chapter 7).

EXAM TRAP

"Perceiving, understanding and responding to the emotional state and ideas of another person" is **empathy**, not sympathy. **Transference** (a client redirecting feelings about past figures onto the therapist) is a therapy term (Chapter 7); **displacement** (shifting an impulse onto a safer target) is a defence mechanism (Chapter 4).

C. Daniel Batson's **empathy-altruism hypothesis** (1981; *The Altruism Question*, 1991) holds that empathic concern produces truly altruistic motivation, a wish to relieve the other's suffering for their sake, whereas personal distress produces only an egoistic wish to escape. To tell the two apart, Batson made escape easy.

KEY STUDY: ELAINE AND THE SHOCKS (BATSON AND COLLEAGUES 1981)

Women students watched "Elaine" react badly to electric shocks and were asked whether they would take her remaining shocks. Some were led to feel high empathy (Elaine shared their values). With **easy escape** they could leave after watching two of ten trials; with **difficult escape** they had to watch all ten. Low-empathy participants helped when escape was hard but mostly left when it was easy, as egoists would. High-empathy participants helped at high rates even when they could walk away. **Criticism:** they might still be avoiding guilt.

Robert Cialdini and colleagues replied with the egoistic **negative-state relief model** (1981, 1987): seeing suffering, or being sad for any reason, makes us feel bad, and we help to lift our own mood. Empathic people helped less when told that a drug had "fixed" their mood, so helping could not make them feel better. The **empathic joy hypothesis** (**K. D. Smith**, **J. P. Keating** and **Ezra Stotland** 1989) says we help for the pleasure of seeing the person relieved: empathic participants helped only if they expected to learn how their help had worked. Batson kept finding helping that these motives cannot explain, and most psychologists now accept that both altruistic and egoistic motives exist.

Genes, status and norms

How could a costly urge to help survive natural selection? **William D. Hamilton**'s **kin selection** theory (1964) answers that helping a relative helps copies of your own genes, so helping is favoured when relatedness multiplied by the benefit exceeds the cost. **Eugene Burnstein**, **Christian Crandall** and **Shinobu Kitayama** (1994) found that people would help close kin more than distant kin or strangers, and young relatives more than old ones, most clearly when life was at stake. For non-relatives, **Robert Trivers** proposed **reciprocal altruism** (1971): helping pays when the favour is likely to be returned.

Two accounts are frankly strategic. According to **competitive altruism** (**Charlie Hardy** and **Mark Van Vugt** 2006), people help to gain status: players gave more when giving was public, and the most generous were preferred as partners. Helping an out-group to reduce its threat to one's in-group is **defensive helping** (**Arie Nadler** and colleagues 2009): students offered more help to a rival school said to outscore them, as if to keep it dependent.

Finally, norms. The **norm of reciprocity** (**Alvin Gouldner** 1960) says we should help those who helped us; the **social responsibility norm** (**Leonard Berkowitz** and L. R. Daniels 1963) says we should help those who depend on us, whether or not they can repay. A **personal**

norm, one's own felt moral obligation to act, predicts helping better than vague social norms.

The bystander effect ●●

In the early hours of 13 March 1964, **Kitty Genovese** (1935 to 1964), aged 28, was stabbed near her apartment building in Kew Gardens, Queens, New York. *The New York Times* reported that 38 neighbours had seen or heard the attack without calling the police. The figure was later shown to be exaggerated (few saw the attack clearly, and some did call or go to her), but the story shocked the country, and commentators blamed big-city apathy.

Bibb Latané (born 1937) and **John Darley** (1938 to 2018) suspected that the witnesses failed to act because there were so many of them. Their experiments (1968) and book *The Unresponsive Bystander* (1970) established the **bystander effect**: the more bystanders present at an emergency, the less likely any one of them is to help, and the slower the help that comes. People are more likely to help when alone.

KEY STUDY: A SEIZURE OVER THE INTERCOM (DARLEY AND LATANÉ 1968)

New York University students talked over an intercom from separate booths. One voice, a tape, began choking and begging for help as if having a seizure. Participants believed that only they, or they and one other, or they and four others could hear. Before the fit ended, 85 percent of those who thought they were alone left the booth to help, 62 percent with one other listener, and 31 percent with four others (Figure 3.11). Non-helpers were not apathetic; many were trembling. **Criticism:** a distressing deception.

In the **smoke-filled room** (Latané and Darley 1968), students filling in questionnaires saw smoke pour through a vent. Alone, 75 percent reported it; beside two passive confederates who kept writing, 10 percent did; in groups of three naive students, someone reported it in 38 percent of groups. In the **lady in distress** study (Latané and **Judith Rodin** 1969), a woman was heard falling next door: 70 percent of people alone went to help, but only 7 percent of those sitting with an unresponsive confederate.

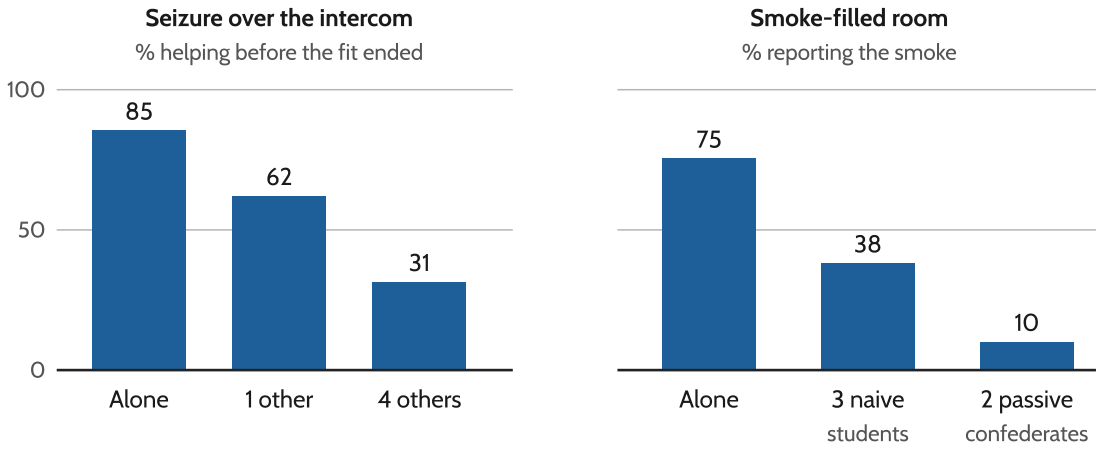


Figure 3.11 The bystander effect in two classic 1968 studies by Darley and Latané. The more people present, the less likely any one person is to act.

EXAM TRAP

The research followed the murder of **Kitty Genovese**, not Martin Luther King Jr. (assassinated 1968), Emmett Till (a Black teenager lynched in 1955) or Malala Yousafzai (shot by the Taliban in 2012). People help more when **alone**; seeing others not help makes intervention **less** likely. Groups did notice the smoke a little later, but the effect is about helping, not noticing.

Five decisions, and why bystanders fail ●●●

Latané and Darley (1970) traced the failure to five decisions, each of which must be answered yes before help is given (Table 3.22).

Table 3.22 Latané and Darley's decision model of helping.

Step	The bystander must	Obstacle at this step
1	Notice the event	Distraction, hurry, overload
2	Interpret it as an emergency	Ambiguity; pluralistic ignorance
3	Assume responsibility	Diffusion of responsibility
4	Know how to help	Lack of competence
5	Decide to help	Audience inhibition; costs

Two obstacles carry the bystander effect. The first, **diffusion of responsibility** (step 3), is the spreading of responsibility across all the bystanders, so that each feels less personally obliged and assumes someone else will act. In the intercom study nobody could see anyone else, so diffusion alone explains the result. The second, **pluralistic ignorance** (step 2), occurs when bystanders use the behaviour of others to decide whether help is needed and mistakenly conclude that it is not. Among strangers, nobody knows for sure what is happening; each stays calm to avoid looking foolish, reads the others' calm as evidence that all is well, and the whole group hesitates and does nothing, as in the smoke-filled room. A third obstacle, **audience inhibition** (evaluation apprehension), is the fear of being judged for overreacting or helping clumsily.

Push a familiar scene through the model. A woman is beaten in a busy street while a crowd films it on phones and nobody calls for help. Each sees many others who could act (diffusion), sees nobody stepping in and half concludes it is a private quarrel (pluralistic ignorance), and fears looking foolish (audience inhibition). That is the bystander effect, not social loafing or group polarisation.

EXAM TRAP

The **two classic explanations** are diffusion of responsibility and pluralistic ignorance; asked for one, the key is diffusion of responsibility. Pluralistic ignorance is not ignoring others or eyewitnesses, nor being suggestible about the cause: it is relying on others and wrongly deciding, from their calm, that help is unnecessary. "Someone else will do it" is diffusion; "nobody looks worried, so it must be nothing" is pluralistic ignorance.

Table 3.23 defines the terms most often offered in their place.

Table 3.23 Terms confused with the bystander effect and its causes.

Term	What it actually means	Section
Deindividuation	Loss of self-awareness and restraint in an anonymous crowd	3.10
Social loafing	Less individual effort on a pooled group task	3.10
Social facilitation	Others' presence improves easy, impairs hard tasks	3.10
Group polarisation	Discussion makes a group's view more extreme	3.10
Groupthink	A cohesive group suppresses dissent and decides badly	3.10
Crowd behaviour	Collective action of a large, temporary gathering	3.1, 3.10
Conformity; compliance	Yielding to group pressure; to a direct request	3.7
Normative influence	Conforming to be liked or accepted	3.7
Social comparison	Judging oneself against others	3.6
Mere exposure	Repeated exposure increases liking	3.6
Reactance	Doing the opposite when freedom is threatened	3.4
Cognitive dissonance	Tension from inconsistent beliefs and actions	3.4
Just-world phenomenon	Belief that people get what they deserve	3.3
Halo effect	One good trait colours the whole impression	3.2
Amotivation	No intention to act at all	Ch. 1

The effect weakens when bystanders are friends who can talk (in the lady in distress study, 70 percent of pairs of friends helped against 40 percent of pairs of strangers) and in small towns where people know one another. A meta-analysis by **Peter Fischer** and colleagues (2011) found it much weaker in dangerous emergencies, which are quickly recognised as real and in which other bystanders become physical support.

Weighing costs: the arousal: cost-reward model

Why do bystanders sometimes help at once? In 1969 **Irving Piliavin**, Judith Rodin and **Jane Allyn Piliavin** staged collapses on New York subway trains. A man with a cane who seemed ill was helped in 95 percent of trials, usually within seconds, and larger groups did not help less; a man who smelled of alcohol and carried a bottle was helped in only about half. Their **arousal: cost-reward model** (developed fully in 1981) says that seeing someone in distress causes unpleasant arousal, which bystanders reduce by whichever response costs least. The costs of helping (effort, danger, embarrassment) are weighed against the costs of not helping (guilt, blame, the victim's suffering). The drunk man raised the costs of helping and, seeming responsible for his plight, lowered the guilt of walking away (Figure 3.12).

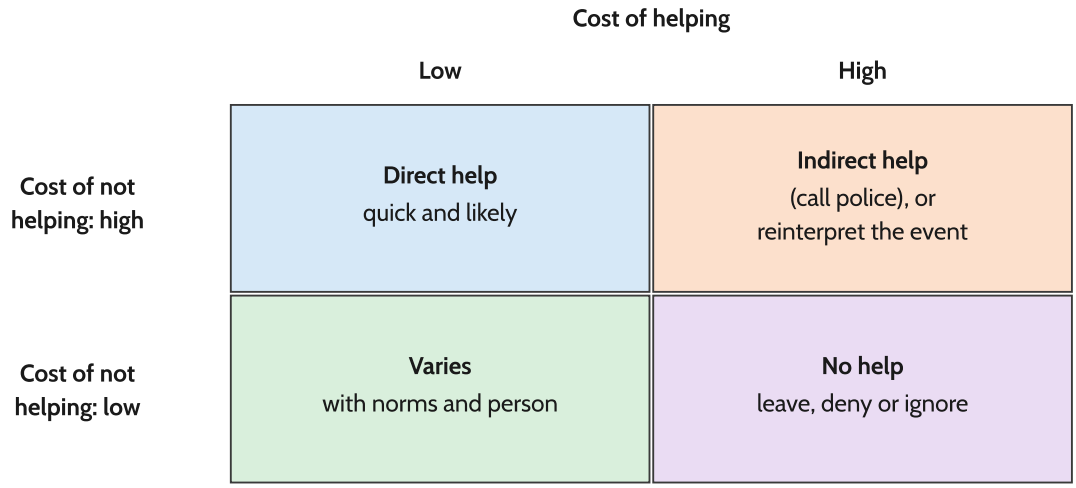


Figure 3.12 Predictions of the arousal: cost-reward model. Help is most likely when helping is cheap and not helping is costly.

Situations that raise or lower helping

We help people who are **similar** to us, partly because similarity raises empathic concern. We help less when victims seem **responsible for their plight**: in **Bernard Weiner**'s attribution analysis (Section 3.3), a controllable cause such as drinking invites anger, an uncontrollable one such as illness invites sympathy and help.

KEY STUDY: THE GOOD SAMARITAN (DARLEY AND BATSON 1973)

Princeton Theological Seminary students, some preparing a talk on the parable of the Good Samaritan, were told they had time to spare, were on time, or were already late. On the way they passed a man slumped and groaning in a doorway. Of those in no hurry, 63 percent helped; moderate hurry, 45 percent; great hurry, 10 percent. The topic of the talk made no reliable difference. **Being in a hurry**, not character, decided who stopped.

Stanley Milgram (1970) proposed **urban overload**: city dwellers face so much stimulation that they screen much of it out, people in need included, and helping is lower in big cities than in small towns.

Mood matters. **Alice Isen** and **Paula Levin** (1972) found that students given a cookie were more willing to help, and that people who had just found a dime in a public telephone were far more likely to help a stranger pick up dropped papers (about 88 against 4 percent). This **feel good, do good effect** fades if helping would spoil the mood. **Guilt** also raises helping, as people repair their self-image.

Goodness is contagious. Motorists stopped more often for a woman with a flat tyre if they had just passed another woman being helped (**James Bryan** and **Mary Ann Test** 1967): a **prosocial model** shows what to do and that helping is the norm. After playing a prosocial video game such as Lemmings, 57 percent helped pick up spilt pencils, against 33 percent after a neutral game (**Tobias Greitemeyer** and **Silvia Osswald** 2010). Watching an act of kindness produces **elevation**, an uplifted feeling described by **Jonathan Haidt** (2000) that makes the watcher want to help.

Gratitude raises future helping. **Adam Grant** and **Francesca Gino** (2010) had students comment on a stranger's job application letter; when he thanked them, about twice as many helped again. Being thanked raised the helpers' self-worth (feeling valued by others) and self-efficacy (feeling competent), but only self-worth carried the rise in helping.

> EXAM TRAP
>
> Gratitude raises helping by enhancing helpers' **self-worth and self-efficacy**; pairs built on self-esteem or self-confidence are wrong. If asked which one did the work, it is self-worth.

Helping falls after **social exclusion**, which drains empathy (people told they would end up alone gave less to a student fund; **Jean Twenge** and colleagues 2007); in **darkness** and other conditions of **anonymity**, which remove the approval helping earns (**Chen-Bo Zhong** and colleagues 2010); and when people think of their **time in money terms** (**Sanford DeVoe** and **Jeffrey Pfeffer** 2010).

Who helps, and how help is received

The **altruistic personality** is a consistent tendency to help others across situations, rather than helping only when one feels like it, calculating costs and benefits, or being ready only in emergencies. **Hans Werner Bierhoff**, **Renate Klein** and **Peter Kramp** (1991) compared people who had given first aid at road accidents with people who had not, and found five components in the helpers: empathy, belief in a just world, social responsibility, internal locus of control and low egocentrism. **Samuel Oliner** and **Pearl Oliner** (1988) interviewed more than 400 rescuers of Jews in Nazi Europe: compared with non-rescuers, their parents had modelled caring and disciplined by reasoning, and they felt responsible for people beyond their own group. In volunteering, **Allen Omoto** and **Mark Snyder** (1995) found five motives among AIDS volunteers (values, understanding, personal development, community concern, esteem enhancement); the self-oriented ones best predicted who kept serving.

Help is not always welcome. The **threat-to-self-esteem model** (Nadler and **Jeffrey Fisher** 1986) holds that aid carries both support (someone cares) and threat (you are inferior and dependent). It threatens most when the helper is a similar peer, the task is central to one's self-image and the help cannot be repaid; then people resent help or avoid seeking it. Nadler also contrasted **dependency-oriented help**, which hands over the full solution, with **autonomy-oriented help**, which gives the tools to solve the problem oneself. The helper's perceived motive matters too: help that seems freely chosen and caring is welcomed; help that seems forced by a role or given for show is resented.

Are helping and aggression, then, opposites? Not quite. Their motives overlap (people help to gain status or keep rivals dependent), their acts can look alike (a surgeon's knife), and unwanted help can harm. Section 3.9 takes up aggression.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Helping with no expected reward, at a cost to oneself, is _____.
2. Batson's _____ hypothesis says empathic concern produces truly altruistic motivation.
3. Bystander research was prompted by the 1964 murder of _____.
4. In the seizure study, 85 percent of those who thought they were alone helped, against _____ percent with four others.
5. Bystanders who read one another's calm as a sign that help is not needed show _____.
6. Darley and Batson's seminarians helped least when they were in a _____.
7. Gratitude raises helping mainly through the helper's sense of _____.
8. Nadler and Fisher's _____ model explains why help can be resented.

Answers: (1) altruism, (2) empathy-altruism, (3) Kitty Genovese, (4) 31, (5) pluralistic ignorance, (6) hurry, (7) self-worth, (8) threat-to-self-esteem.

3.9 Aggression: Its Roots, Causes and Control LOWER YIELD · 3 QS

A batsman given out wrongly smashes his bat against the boundary board; a schoolgirl spreads a cruel rumour; a hired killer shoots a stranger without anger. All three are aggression, yet their roots differ. This section traces those roots, the triggers, and the methods that actually reduce aggression.

What counts as aggression

Robert A. Baron and **Deborah R. Richardson** (1994) define **aggression** as behaviour directed toward the goal of harming another living being who is motivated to avoid such treatment. So hateful thoughts alone are not aggression (it is behaviour); accidental harm is not, though a punch that misses is (it is intended); and kicking a door or a wanted surgical cut is not (the target must be living and unwilling). Anger is an emotion, aggression a behaviour: you can be furious and do nothing, and the hired killer aggresses without anger.

The main distinction is by goal. In **hostile aggression** (reactive or affective aggression) harm is the end in itself: hot, impulsive, a reply to provocation. In **instrumental aggression** (proactive aggression) harm is a means to another end, such as money or status: cold and planned. Table 3.24 adds the other cuts. The term **relational aggression** comes from **Nicki R. Crick** and **Jennifer K. Grotpeter** (1995), who found that girls used it more than boys; studies that counted only hitting had made girls look unaggressive.

Table 3.24 Further forms of aggression.

Form	What it is	Example
Physical	Harm by bodily force or weapons	A slap
Verbal	Harm by words	Insults, threats
Direct	Face to face; the aggressor is known	A shove

Form	What it is	Example
Indirect	Through others or in secret	An anonymous complaint
Relational	Damages the victim's friendships and status	Exclusion, gossip

Instinct, evolution and biology

The oldest answer is that aggression is built in. **Sigmund Freud** (1856 to 1939) proposed in 1920 a **death instinct**, **thanatos**, opposing the life instinct (eros): a drive toward self-destruction that is turned outward as aggression (Chapter 4). The ethologist **Konrad Lorenz** (1903 to 1989), in *On Aggression* (1966), saw an inherited fighting instinct that spaced animals out and let the strongest males breed. His **hydraulic model** pictured aggressive energy building like water in a tank until a releasing stimulus, or eventually none at all, lets it out. Well-armed animals evolved **ritualised fighting** (stags shove with locked antlers rather than gore) and **appeasement gestures** (a beaten wolf bares its throat) that stop the winner's attack; weakly armed humans evolved weak inhibitions, and our weapons now kill beyond the reach of appeasement. Critics note that aggression varies too much across cultures to be one pressure-cooker drive. Both views fed the catharsis idea (3.9.8).

Evolutionary psychology recasts instinct as inherited potential: aggression that won mates and resources was passed on. It predicts that young men commit most violence, mostly against rival men (**male competition**), and that **sexual jealousy** is a leading cause of violence between partners.

The clearest chemical finding is that low **serotonin** activity, shown by low levels of its breakdown product 5-HIAA in cerebrospinal fluid, is linked to impulsive aggression; serotonin helps restrain impulses. **Testosterone** correlates modestly with dominance and aggression. Table 3.25 sets out the usual distractors. The amygdala fuels anger, the hypothalamus organises attack and the prefrontal cortex applies the brakes (Chapter 5). Genes act through environments: in the New Zealand cohort of **Avshalom Caspi** and colleagues (2002), maltreated boys became antisocial adults far more often if they carried the low-activity form of the **MAOA** gene, which codes the enzyme that breaks down serotonin, norepinephrine and dopamine; without maltreatment the variant carried little risk.

Table 3.25 Chemicals and aggression.

Chemical	Main job	Link to aggression
Serotonin	Mood, impulse control	Low: impulsive aggression
Dopamine	Reward, motivation, movement	Not the aggression marker
Epinephrine (adrenaline)	Adrenal fight-or-flight hormone	Rises in anger and fear alike
Norepinephrine	Alertness, arousal	Arousal in any strong emotion
Testosterone	Male sex hormone	High: modestly more aggression

> **EXAM TRAP**
>
> Low **serotonin**, not low dopamine, epinephrine or norepinephrine, is the impulsive-aggression finding. Testosterone is a hormone, not a neurotransmitter.

Frustration-aggression and its reformulation

Social psychologists moved the drive's source outside the person. The **frustration-aggression hypothesis**, put forward at Yale in 1939 by **John Dollard**, **Leonard W. Doob**, **Neal E. Miller**, **O. H. Mowrer** and **Robert R. Sears**, claimed that frustration, the blocking of a goal (Section 1.10), always leads to aggression, and that aggression always comes from frustration. Neither holds: frustration often brings sadness or problem solving, and the hired killer is not frustrated. Miller (1941) soon withdrew the first claim: aggression is only one of several responses to frustration. The group also explained the target. When the frustrator is powerful or absent, aggression shifts to a safer target (**displacement**: scolded by his boss, a clerk shouts at his children); when that target is a weak minority blamed for the trouble, displacement becomes **scapegoating** (Section 3.5).

Leonard Berkowitz (1926 to 2016) reformulated the hypothesis in 1989, in what became his **cognitive neoassociationist model**. Frustration causes aggression only to the extent that it produces **negative affect**, so unexpected or unfair frustrations provoke most. And any aversive event (pain, heat, foul smells, noise) does the same. Negative affect automatically primes both **fight** (rudimentary anger) and **flight** (rudimentary fear); the situation, past learning and later appraisal decide which wins (Figure 3.13). Stimuli associated with violence act as **aggressive cues**, tipping an angry person toward fight.

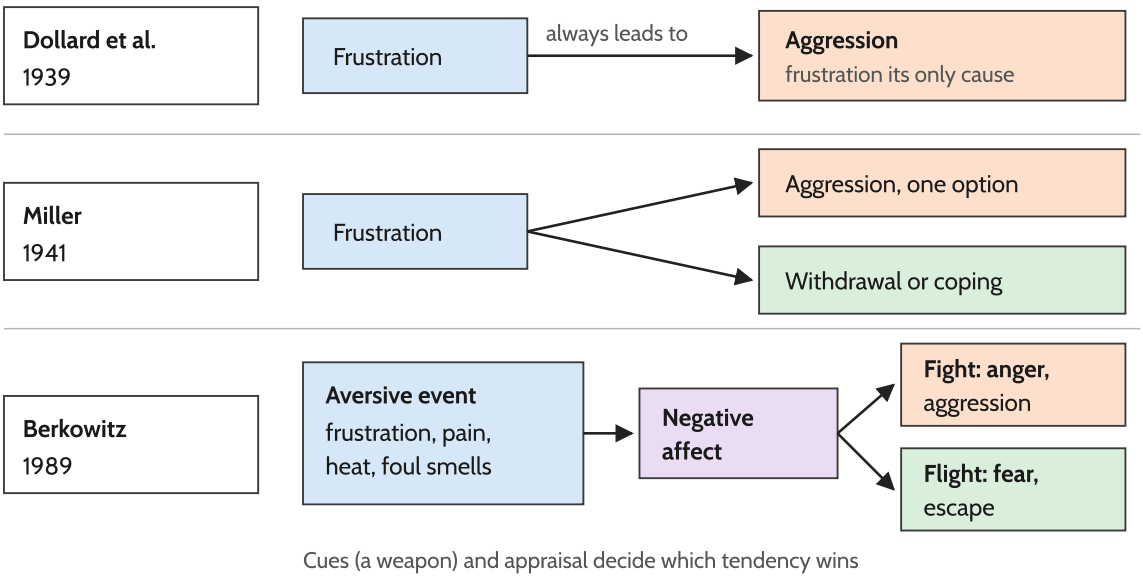


Figure 3.13 From certain cause (1939) to one possible response (1941) to negative affect from any aversive event (1989).

KEY STUDY: THE WEAPONS EFFECT (BERKOWITZ AND LEPAGE, 1967)

In the study by Berkowitz and **Anthony LePage**, male students, angered or not by a partner's electric shocks, could shock him back while a shotgun and a revolver, or badminton racquets, or nothing lay nearby. Angered men beside the guns gave the most shocks. "The finger pulls the trigger, but the trigger may also be pulling the finger." Some replications failed, but meta-analyses support a modest **weapons effect**.

Arousal from elsewhere also feeds aggression. In **Dolf Zillmann**'s **excitation transfer theory** (1971), arousal fades slowly, so arousal left over from one event (a near miss on the Mysuru ring road, a gym session) is misattributed to the next, and a minor annoyance produces rage, especially when the leftover arousal goes unnoticed.

> EXAM TRAP
>
> The **reformulation** is **Berkowitz's** (1989), not Dollard's (the 1939 original), Miller's (the 1941 softening) or Lazarus's (Richard Lazarus: appraisal and stress, Section 1.11). Weapons effect: Berkowitz and LePage; leftover arousal: Zillmann.

Learning, scripts and the General Aggression Model

Learning theories ask where the particular ways of aggressing come from. **Albert Bandura** (1925 to 2021) answered: largely from watching others.

> KEY STUDY: THE BOBO DOLL EXPERIMENTS (1961 TO 1965)
>
> Bandura, **Dorothea Ross** and **Sheila A. Ross** (1961) let 72 nursery-school children watch an adult punch, kick and shout at an inflatable Bobo doll, or play quietly. Left with the doll, those who had seen aggression copied the model's specific acts and words, unrewarded. Filmed and cartoon models worked almost as well (1963). In 1965, children who saw the model punished (vicarious punishment) imitated less than those who saw it rewarded (vicarious reinforcement), yet a reward for reproducing the acts made all groups equal: consequences to the model affect performance, not learning. **Criticism.** Bobo is built to be hit.

This is **observational learning** through **modelling**, with no direct reinforcement of the learner (Bandura's four processes are in Section 1.6).

> EXAM TRAP
>
> Bobo doll = **observational learning of aggression through modelling**. Not operant conditioning (no child was reinforced for hitting), not classical conditioning of fear (Little Albert, Section 1.6), not sensory deprivation (the McGill isolation studies, Section 1.5).

Repeated observation builds **aggressive scripts** (**L. Rowell Huesmann**): stored programmes for how a conflict unfolds and how to respond ("if disrespected, hit back"), retrieved fast when provoked. The **General Aggression Model** of **Craig A. Anderson** and **Brad J. Bushman** (2002) integrates these theories (Figure 3.14): person and situation inputs shape the present internal state (affect, cognition, arousal); a quick appraisal yields impulsive action, a considered one thoughtful action; and each outcome feeds back, making the next aggressive episode easier.

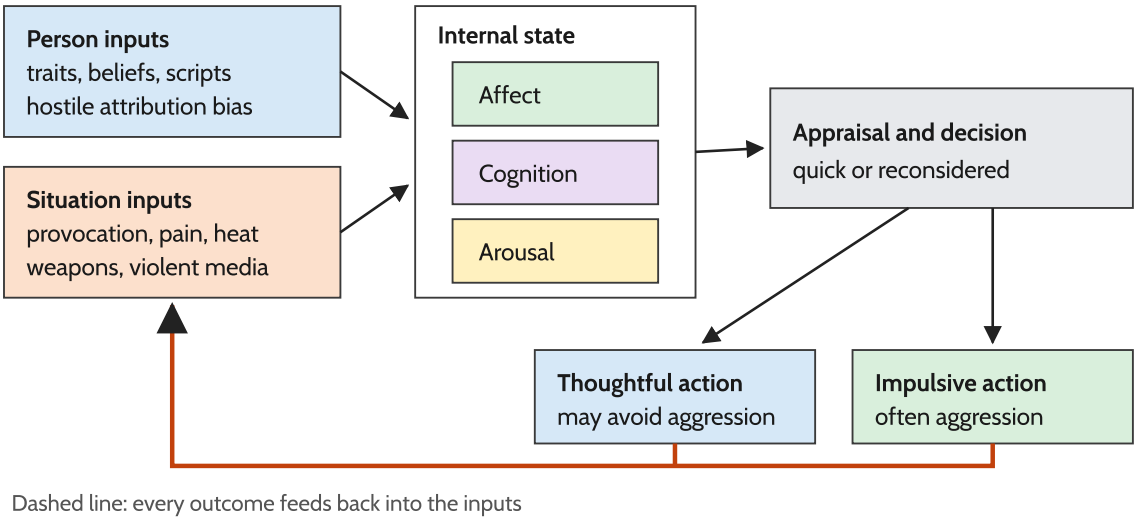


Figure 3.14 The General Aggression Model: inputs act through feelings, thoughts and arousal, and appraisal decides the act.

Social, cultural and personal causes

The strongest trigger is direct **provocation** (insults, unfair criticism, condescension, teasing seen as meant to hurt): people return what they receive, or a little more. Displaced aggression (3.9.3) and leftover arousal add fuel. Table 3.26 gathers the other causes.

Table 3.26 Further causes of aggression.

Cause	Who	Key finding
Social exclusion	**Jean M. Twenge** and colleagues (2001)	Rejected people blasted more noise at others, even at a neutral stranger
Media violence	**Leonard D. Eron** and Huesmann; Anderson (video games)	Boys' TV violence at 8 predicted aggression at 19 and serious crime by 30; violent games raise aggression and lower empathy
The culture of honour	**Richard E. Nisbett** and **Dov Cohen** (1996)	Herding regions (the US South) demand violent answers to insults and infidelity; insulted Southerners showed more cortisol, testosterone and aggression
Gender role: precarious manhood	**Joseph A. Vandello** and **Jennifer K. Bosson** (2008)	Manhood is hard won and easily lost; in a 2009 follow-up, threatened men punched a bag harder
A hostile attribution bias	**Kenneth A. Dodge** (1980)	Aggressive children read ambiguous acts as deliberate, then retaliate
Narcissism: threatened egotism	Bushman and **Roy F. Baumeister** (1998)	Insulted narcissists aggressed most; low self-esteem did not predict aggression
Type A pattern	Section 1.11	More hostile, not instrumental, aggression

Media violence works by **modelling**, by **priming** aggressive thoughts and scripts, and by **desensitisation**, the blunting of emotional reactions to real violence. As for sex differences, men are more physically and directly aggressive; women equal or exceed men in relational and indirect aggression; and the gap shrinks under strong provocation.

Heat, alcohol and other aversive conditions

Aversive conditions act as Berkowitz predicted. The **hot temperature effect** is real: across 50 US cities from 1950 to 1995, hotter years brought more murders and assaults but not more property crime (Anderson and colleagues). Yet in Baron's laboratory, very hot, angry participants became *less* aggressive. The **negative affect escape model** of Baron and **Paul A. Bell** explains this: discomfort raises aggression up to a moderate level, beyond which escape takes priority, an inverted U. Alcohol raises aggression in both sexes. In the **alcohol myopia** account of **Claude M. Steele** and **Robert A. Josephs** (1990), alcohol narrows attention to the salient cue (the insult) and pushes out distant ones (consequences, norms). Crowding, uncontrollable noise and pain act likewise, through negative affect.

Bullying, cyberbullying and workplace aggression

Dan Olweus, pioneer of school research in Norway (*Bullying at School*, 1993), defined **bullying** as intentional, repeated aggression against a victim with less power. He distinguished bullies, victims (often anxious and friendless) and bully-victims (provocative victims who also bully, and fare worst); his whole-school programme of supervision, clear rules and consistent action roughly halved bullying. Online, **cyberbullying** follows victims home and reaches vast audiences; anonymity and the unseen victim produce **online disinhibition**, a cousin of deindividuation (Section 3.10). Workplace aggression is mostly verbal, indirect and covert, and perceived injustice often triggers it.

Preventing and controlling aggression

Punishment deters only when it is prompt, certain, strong and seen as justified, conditions courts rarely meet; unfair punishment acts as fresh provocation, and physical punishment models aggression. The **catharsis hypothesis**, rooted in Freud, adopted by Dollard's group (1939) and revived by Lorenz, holds that venting anger safely (punching a pillow, shouting) drains the drive and reduces later aggression. It fails.

KEY STUDY: VENTING FEEDS THE FLAME (BUSHMAN, 2002)

Bushman angered about 600 students with harsh feedback on their essays. One group then hit a punching bag while thinking about the critic, a second while thinking about fitness, and a third sat quietly. When all could blast the critic with noise, the rumination group was angriest and most aggressive; doing nothing produced the least aggression.

What works? **Cognitive interventions**: a sincere **apology**; **pre-attribution**, a warning that someone is ill or stressed, so later rudeness is blamed on circumstances; and a **delay** that lets arousal fade. Exposure to **non-aggressive models** who stay calm under provocation. **Social skills training** in negotiating and complaining without attack. Baron's **incompatible response technique**, which induces empathy or humour, feelings that cannot coexist with anger. Strengthened **self-regulation**, which fatigue, stress and alcohol deplete (Baumeister). And **forgiveness**, which ends the cycle of revenge.

EXAM TRAP

Which method is **not** supported? **Catharsis**: venting raises aggression. "Severe punishment always deters" is also false.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Aggression harms a living being who is motivated to _____ such treatment.
2. Gossip and exclusion that damage friendships are _____ aggression.
3. Low levels of _____ are linked to impulsive aggression.
4. The frustration-aggression hypothesis was reformulated by _____ in terms of negative affect.
5. Angered men gave more shocks when guns lay nearby: the _____ effect.
6. Leftover arousal intensifying anger later is Zillmann's _____ theory.
7. The Bobo doll studies show observational learning through _____.
8. Bushman's punching-bag study contradicted the _____ hypothesis.

Answers: (1) avoid, (2) relational, (3) serotonin, (4) Leonard Berkowitz, (5) weapons, (6) excitation transfer, (7) modelling, (8) catharsis.

3.10 Groups: Structure, Performance, Decisions and Leadership

MEDIUM YIELD · 25 QS

Your family, your entrance batch and your cricket team all shape how hard you try, what you decide and whom you follow. What turns people into a group, how do groups grow, how do others change our effort, why do groups decide badly, and what makes a leader effective?

What makes a group

A **group** is two or more people who interact, depend on one another for their goals, see themselves as belonging together and share a structure of roles and norms. People at a bus stop share a place, not a group. **Entitativity**, **Donald T. Campbell**'s (1958) term, is how far a collection of people is seen as one coherent unit; frequent interaction, shared goals and similarity raise it, and highly entitative groups such as families matter most to their members. Table 3.27 sorts the kinds of groups.

Table 3.27 Kinds of groups.

Distinction	Who	Meaning
Primary versus secondary	**Charles Horton Cooley** (1909)	Primary: small, intimate, face-to-face (family). Secondary, the later contrast: large, impersonal, goal-based (a company)
Formal versus informal		Set up with assigned roles (a ward team) versus arising on its own (lunch friends)
Membership versus reference	**Herbert Hyman** (1942)	Groups you belong to versus groups whose standards you judge yourself by
In-group versus out-group	**William Graham Sumner** (1906)	"Us" versus "them"; root of ethnocentrism (Section 3.5)
Intimacy groups, task groups, social categories, loose associations	**Brian Lickel** and colleagues (2000)	Most to least entitative: a family; a work team; "women"; a bus queue

A reference group can remake attitudes. In **Theodore Newcomb**'s Bennington College study (late 1930s, published 1943), women from conservative homes who adopted the liberal college as their reference group became more liberal, and most stayed so 25 years later. Formed versus natural groups and social group work are in Section 3.11.

Group structure and membership ●

Every lasting group develops a structure with four parts. **Roles** are behaviours expected of a position. **Kenneth Benne** and **Paul Sheats** (1948) distinguished **task roles** that move the work on (initiator, information seeker, coordinator, evaluator-critic), **maintenance roles** or socio-emotional roles that hold members together (encourager, harmoniser, compromiser, gatekeeper) and **self-oriented roles** that serve the individual at the group's cost (aggressor, blocker, recognition seeker, dominator). **Role differentiation** follows as members specialise, often one leading the task while another keeps the peace. **Status** is rank within the group; high-status members talk more and may deviate more. **Norms** are the group's rules, often unspoken. **Cohesiveness** is the sum of forces holding members in: liking, commitment to the task and pride. Cohesive groups demand more conformity (Section 3.7) and satisfy members more, but perform better only when their norms favour effort, and very high cohesion feeds groupthink.

People join for belonging, goals, self-knowledge, status and identity (Section 3.5), and pay in freedom, time and effort. A high price even raises a group's value: women who passed an embarrassing initiation liked a dull group more (**Elliot Aronson** and **Judson Mills**, 1959), a dissonance effect (Section 3.4).

Richard Moreland and **John Levine** (1982) described **group socialisation** as five phases in which group and person evaluate each other: **investigation** (prospective member), **socialisation** (new member), **maintenance** (full member), **resocialisation** (marginal member, as commitment falls) and **remembrance** (ex-member). Entry, acceptance, divergence and exit mark the transitions. Sometimes the whole group breaks. A **schism** is the splintering of a group into distinct factions. **Fabio Sani** (2005) found that clergy who left the Church of England after it first ordained women in 1994 felt the change had subverted its identity, and that their protests would go unheard.

> EXAM TRAP
>
> Splintering into factions is a **schism**. A *chapter* is a local branch, *division* an everyday word and *deviation* one member breaking a norm.

Sociometry: mapping who chooses whom ●●

A hostel's room list shows its official structure; how would you find the real one? **Jacob Levy Moreno** (1889 to 1974) answered with **sociometry**, a method of measuring an individual's social behaviour and relations in a group from the attractions and rejections among its members. He set it out in *Who Shall Survive?* (1934), with **Helen Hall Jennings**, after regrouping a girls' training school by the residents' own choices.

In the **sociometric test** each member privately names whom he or she would choose, and sometimes reject, for working, living or playing together. Answers fill a **sociometric matrix**, choosers down the side and the chosen across the top; a column total divided by the number

of other members (N minus 1) is a **sociometric index** of choice status. Drawn out, the matrix becomes a **sociogram**, a diagram of choices and rejections that shows attraction, social distance and patterns of choice within a group (Figure 3.15). A **star** receives many choices, an **isolate** none; a **neglectee** receives few but is not rejected; a **rejectee** draws rejections. A mutual pair choose each other, a chain runs A to B to C, and a **clique** chooses within itself. The sociogram thus captures implicit and explicit communication and interaction patterns; teachers use it to find isolated children. Moreno also founded psychodrama (Chapter 7).

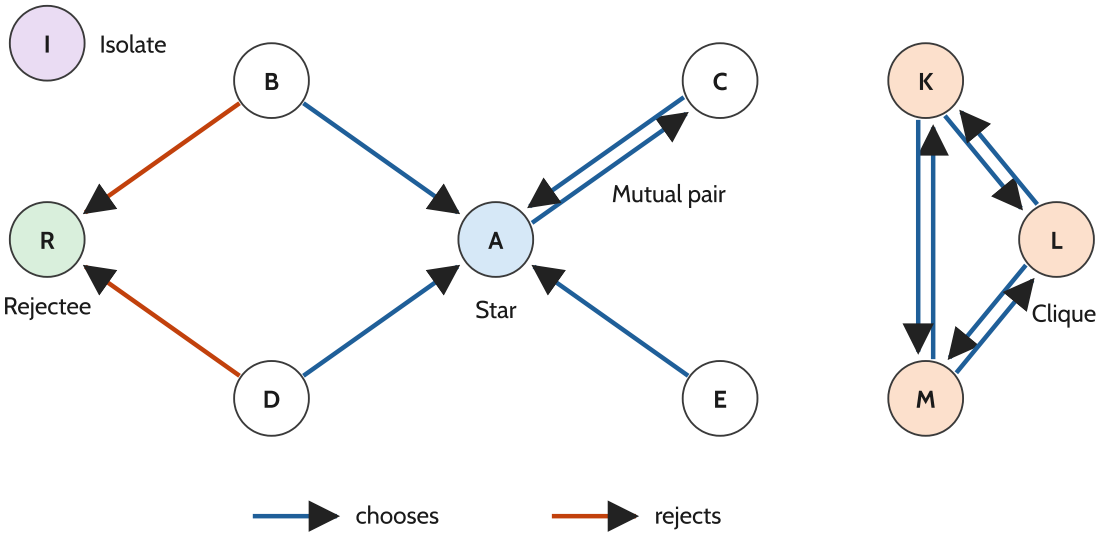


Figure 3.15 A sociogram. A is the star, A and C a mutual pair, K, L and M a clique; R is rejected and nobody chooses I.

EXAM TRAP

Sociometry, not the sample method (Chapter 9), the interview method or the *Verstehen* method (interpretive understanding of social action, **Wilhelm Dilthey**, **Max Weber**), measures social behaviour in a group. A sociogram does not measure IQ, a leader's personality or economic status. A **genogram** is a three-generation family tree (Chapter 7), an **ecomap** a family's links with its environment (Section 3.11), a **cognitive map** Tolman's mental map (Section 1.6).

How groups develop ●●●

In 1965 **Bruce Tuckman** reviewed studies of therapy, training and laboratory groups and proposed four stages; with **Mary Ann Jensen** in 1977 he added a fifth (Figure 3.16). In **forming**, members orient themselves, are polite and unsure, and depend on the leader. In **storming**, conflict and resistance surface as members struggle over leadership, roles and the group's definition. In **norming**, cohesion grows, norms and roles are agreed, and the group clarifies its purpose and strategy for reaching its goals. In **performing**, the work gets done. In **adjourning**, the group disbands, often with a sense of loss.

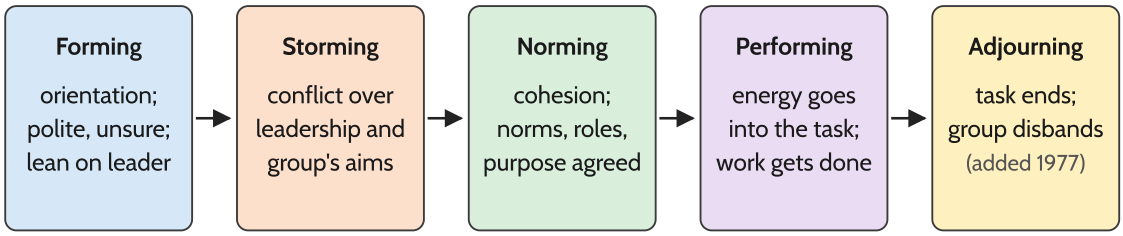


Figure 3.16 Tuckman's stages of group development, in their fixed order.

Real groups do not always march in step. **Connie Gersick** (1988) found that project teams with deadlines show **punctuated equilibrium**: inertia, a burst of change near the midpoint of their time, then focused work. The Boston model used in social group work is in Section 3.11.

> EXAM TRAP
>
> **Proximity** is not a stage of group formation; it is a cause of attraction (Section 3.6). "Clarified its purpose and strategy" is **norming**; "struggle over leadership and group definition" is **storming**. The order is forming, storming, norming, performing, adjourning.

Performing in front of others ●●

Does an audience help you or hurt you? The question is as old as experimental social psychology.

> KEY STUDY: TRIPLETT'S FISHING REELS (1898)
>
> **Norman Triplett** noticed in cycling records that riders were faster when paced or racing a rival than when riding alone against the clock. He then had children wind a fishing reel as fast as possible, alone or beside another child doing the same; many wound faster in company, though some did worse. Published in 1898, it is often called the first social psychology experiment. Criticism: the gains were small and uneven.

Floyd Henry Allport named the effect **social facilitation** (1920, 1924), better performance when others are present, and called working side by side on the same task coaction. Yet others sometimes made people worse, called social inhibition or impairment. **Robert Zajonc** resolved the puzzle in 1965 with **drive theory**: the mere presence of others raises arousal, and arousal strengthens the **dominant response**, the one most likely in that situation. On a simple or well-learned task the dominant response is correct, so performance improves; on a complex or new task it is likely wrong, so performance worsens (Figure 3.17).

Push a case through it. Tim has trained his voice for years but has hardly danced, and his school musical needs both. The audience raises his arousal: his singing improves and his dancing deteriorates. An audience helps most on easy, well-learned tasks, not novel, difficult, creative or first-time ones. Zajonc, **Alexander Heingartner** and **Edward Herman** (1969) found it even in cockroaches: watched by other roaches, they ran a straight runway faster but a maze more slowly.

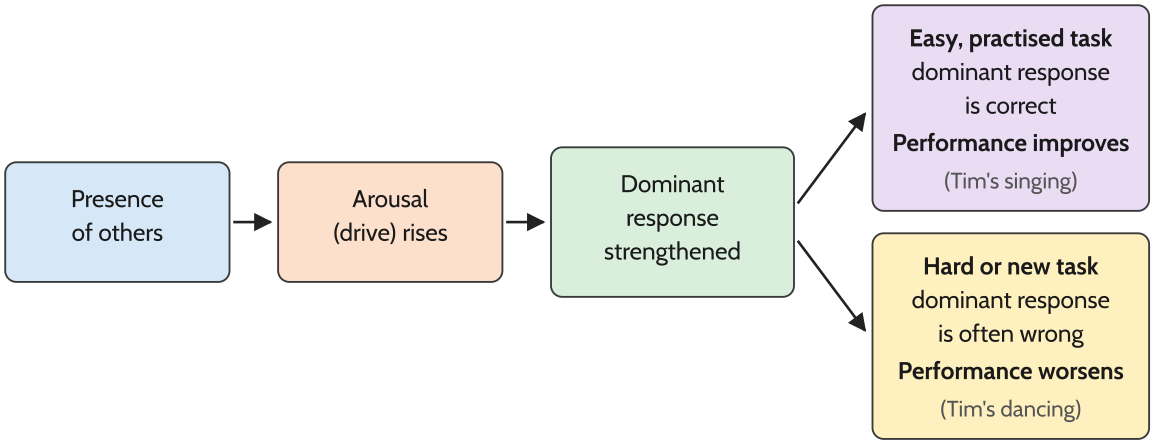


Figure 3.17 Zajonc's drive theory (1965): one rise in arousal, two opposite outcomes.

Two rivals refine the account. In **evaluation apprehension** (**Nickolas Cottrell** and colleagues, 1968), concern about being judged does the arousing: a blindfolded audience produced no facilitation. In **distraction-conflict theory** (**Robert S. Baron**, 1986), attending to both task and onlookers creates conflict that arouses and narrows attention. The arousal-performance curve is the Yerkes-Dodson law (Section 1.10).

> EXAM TRAP
>
> Triplett used **fishing reels**. Rope pulling is Ringelmann, the shock generator Milgram (Section 3.7), a maze with food reward animal learning (Section 1.6). Diffusion of responsibility explains bystanders not helping (Section 3.8).

Social loafing and deindividuation ●●●

Now pull a rope with seven others: nobody can tell how hard you pull. The French engineer **Max Ringelmann** found that the bigger the team, the less each member pulled (published 1913). Part of this **Ringelmann effect** is coordination loss, but **Alan Ingham** and colleagues (1974) showed that blindfolded people who merely believed others were pulling also pulled less: motivation falls too.

That loss is **social loafing**: less effort when individual efforts are pooled than when working alone. **Bibb Latané**, **Kipling Williams** and **Stephen Harkins** (1979) had blindfolded students in headphones clap and shout alone or in supposed groups of two, four or six: each person's output fell as the group grew. **Steven Karau** and Williams's collective effort model (1993) explains why: people work hard only when they expect their effort to matter for an outcome they value. Loafing therefore falls when each contribution is identifiable and evaluated, the task is challenging, appealing or involving, members are friends or committed, effort is believed to bring valued rewards, and the group is small. A teacher such as Miss Pooja should grade both the group and each member; group grades alone, random assignment or optional presentations leave individual effort hidden.

> **EXAM TRAP**
>
> "Groups loaf less when evaluation apprehension is absent" is **false**: removing the fear of judgement lets people loaf. Social loafing's other name is the **Ringelmann effect**; the free-rider effect is a related but separate term (Table 3.28). The Triplett effect is social facilitation, the McGurk effect is lips changing the speech sound heard (Section 1.3), and the sleeper effect is delayed persuasion (Section 3.4).

Loafing is not universal. **P. Christopher Earley** (1989, 1993) found that Chinese management trainees worked harder in a group than alone, where Americans loafed: **social striving**, typical of collectivist cultures. Table 3.28 separates the look-alike terms.

Table 3.28 Effort and restraint in groups: look-alike terms.

Term	Who	What happens
Social loafing (Ringelmann effect)	Ringelmann 1913; Latané and colleagues 1979	Less effort when contributions are pooled
Free-rider effect	**Norbert Kerr** and **Steven Bruun** 1983	Letting others carry the load because one's input seems dispensable
Sucker effect	Kerr 1983	Cutting effort to avoid being exploited by loafers
Social compensation	Williams and Karau 1991	Working harder to cover weak teammates on an important task
Social striving	Earley 1989, 1993	Working harder in a group than alone
Deindividuation	Festinger and colleagues 1952	Lost self-awareness and restraint under anonymity
Bystander effect	Darley and Latané 1968	Less helping when others are present (Section 3.8)

Gustave Le Bon, in *The Crowd* (1895), argued that people in a mob lose their minds to a contagious group mind. **Leon Festinger**, **Albert Pepitone** and Newcomb (1952) named the state **deindividuation**: anonymity and shared accountability lower self-awareness, so restraints weaken. **Philip Zimbardo** (1969) found that hooded women gave longer shocks than name-tagged women, and **Edward Diener** and colleagues (1976) saw anonymous Halloween trick-or-treaters in groups take extra sweets far more often than named children alone. The **SIDE model** (social identity model of deindividuation effects; **Stephen Reicher**, **Russell Spears** and **Tom Postmes**, 1995) adds that anonymity strengthens a salient group identity, so people follow group norms more, violent or peaceful. Cyberbullying and online disinhibition, cruelty behind a screen name, are deindividuation, not social facilitation, polarisation or the bystander effect (bullying is in Section 3.9).

Group decision making ●●

Do several heads beat one? **James H. Davis** (1973) showed that group decisions follow **social decision schemes**: **majority wins** on opinions, **truth wins** when one member can demonstrate a correct answer, and **first shift**, where the group follows the first member to change position, as in deadlocked juries.

Groups are often more extreme, not more moderate. In 1961 **James Stoner** found that groups recommended riskier courses than their members had alone, the **risky shift**, but later studies found that groups that began cautious became more cautious. **Serge Moscovici** and

Marisa Zavalloni (1969) showed the general rule: French students' liking for de Gaulle and dislike of Americans both grew after discussion. This is **group polarisation**: discussion pushes members towards a more extreme version of their initial leaning, through **persuasive arguments** (more arguments heard for the favoured side), **social comparison** (wanting to look better than average on the valued side) and **social identity** (converging on the group's prototype).

> EXAM TRAP
>
> More extreme group decisions are **group polarisation**. "Extremity" is not a technical term, groupthink is cohesive suppression of dissent, and conformity is yielding to a norm (Section 3.7).

Irving Janis, in *Victims of Groupthink* (1972; revised 1982), analysed fiascoes such as Kennedy's Bay of Pigs invasion (1961) and the ignored warnings before Pearl Harbor (1941); later analysts added the 1986 *Challenger* launch. **Groupthink** is a mode of thinking in highly cohesive groups in which the drive for unanimity overrides realistic appraisal of alternatives: collective decision making that discourages dissent and leads to poor decisions. Antecedents are high cohesion, insulation, a directive leader, high stress and no methodical procedures. The eight symptoms are an illusion of invulnerability (excess optimism and risk), collective rationalisation (warnings explained away), belief in the group's morality, stereotyped views of out-groups, direct pressure on dissenters, self-censorship of doubts, an illusion of unanimity (silence read as agreement) and self-appointed mindguards who screen out contrary information. Remedies: the leader stays impartial and withholds his or her view early, appoints a devil's advocate or a subgroup to critique the plan, invites outside experts, uses subgroups and holds a second-chance meeting.

> EXAM TRAP
>
> To curb groupthink, the leader should **ask a subset of the group to critique** the plan. Team-building raises cohesion, stating his view first steers the group, and forbidding argument silences dissent: all make it worse. Groupthink is not individual thinking, brainstorming or mere peer pressure, and it yields poor decisions.

Groups also waste what members know. **Garold Stasser** and **William Titus** (1985) spread facts so that the best student-body candidate stood out only if members pooled unshared facts, a **hidden profile**; groups discussed mainly shared facts (biased information sampling) and chose worse. In **brainstorming**, popularised by **Alex Osborn**, members produce many ideas with criticism postponed, yet as many people working alone produce more, because of **production blocking**: one person speaks at a time and ideas fade while waiting. The **nominal group technique** has members write ideas silently, share them in turn and vote privately; the **Delphi technique**, from the RAND Corporation, polls experts anonymously over rounds with feedback until opinion converges. **Escalation of commitment** is continuing to invest in a failing course to justify past costs.

Cooperation, conflict and fairness

In a **social dilemma**, each person gains by acting selfishly but all lose if everyone does. In the **prisoner's dilemma**, two suspects questioned apart each do best by betraying the other, yet both do worse if both betray than if both stay silent. **Garrett Hardin**'s **tragedy of the commons** (1968) is the many-person version: each herder adds cows to the shared pasture until it is ruined, as with groundwater or fish stocks today. Communication, trust, small groups and sanctions raise cooperation.

Conflict yields to **bargaining**, best ending in an **integrative agreement** that gives both sides more than a split by trading on issues each values differently; to **superordinate goals** that neither side can reach alone (Sherif's Robbers Cave, Section 3.5); and to **Charles Osgood**'s **GRIT** (graduated and reciprocated initiatives in tension reduction, 1962), small announced concessions that invite reciprocation. Conflict is often about fairness: **distributive justice** is fair outcomes, judged in **J. Stacy Adams**'s equity theory (1965) by comparing our outcome-to-input ratio with others'; **procedural justice** is fair procedures; **transactional** or interactional **justice** is courteous treatment and honest explanation.

Leadership •

Leadership is influencing group members towards shared goals. The **great-person theory** held that leaders are born with special traits, but **Ralph Stogdill**'s 1948 review found traits only weakly linked to leadership: a leader in one situation may not lead in another. Attention turned to what leaders do.

> **KEY STUDY: LEWIN, LIPPITT AND WHITE (1939)**
>
> **Kurt Lewin**, **Ronald Lippitt** and **Ralph White** had adults lead clubs of ten-year-old boys in three styles. Under **autocratic** leaders, who made every decision, boys worked while watched but stopped when the leader left, and grew hostile or apathetic. Under **democratic** leaders, who guided group decisions, work was more original, continued without the leader, and the boys liked this style best. Under **laissez-faire** leaders, who left the boys alone, little was done. Criticism: small American samples.

The Ohio State studies, using the Leader Behaviour Description Questionnaire (LBDQ), found two independent dimensions: initiating structure (organising work and roles) and consideration (warmth and concern for subordinates); one leader can be high on both. The Michigan studies contrasted job-centred with employee-centred supervisors. **Robert Blake** and **Jane Mouton**'s managerial grid (1964) crosses concern for production with concern for people, from impoverished (1,1), country club (1,9), authority-compliance (9,1) and middle of the road (5,5) to the ideal, team management (9,9).

Contingency theories match leader to situation. In **Fred Fiedler**'s **contingency model** (1967), leaders rate their **least preferred co-worker** (LPC): rating even that person kindly (high LPC) marks a relationship-oriented leader, rating harshly (low LPC) a task-oriented one. Situational favourableness comes from three dimensions: **leader-member relations**, **task structure** and **position power** (the leader's formal power and authority to reward and punish); good relations count most. Task-oriented leaders do best in very favourable or very unfavourable situations, relationship-oriented leaders in between; since style is fixed, Fiedler advised changing the situation. Table 3.29 lists later theories.

EXAM TRAP

Employee motivation is not one of Fiedler's dimensions. The three are leader-member relations, task structure and position power ("intrinsic power and authority" in some papers).

Table 3.29 Later theories of leadership.

Theory	Who	Core idea
Situational leadership	**Paul Hersey** and **Kenneth Blanchard** (1969)	Fit style to followers' readiness: telling, selling, participating, delegating
Path-goal theory	**Robert House** (1971)	Clear the path to goals: directive, supportive, participative or achievement-oriented
Normative decision model	**Victor Vroom** and **Philip Yetton** (1973)	A decision tree sets how far to involve subordinates
Idiosyncrasy credits	**Edwin Hollander** (1958)	Conformity and competence earn room to innovate (Section 3.7)
Leader-member exchange	**George Graen** (1975)	Close ties with an in-group of followers, formal ones with an out-group
Transactional leadership	**James MacGregor Burns** (1978)	Rewards exchanged for performance
Transformational leadership	Burns (1978); **Bernard Bass** (1985)	Idealised influence, inspirational motivation, intellectual stimulation, individualised consideration
Charismatic leadership	Weber; House (1977)	Devotion to a leader seen as extraordinary
Servant leadership	**Robert Greenleaf** (1970)	The leader serves followers' growth first

INDIAN CONTEXT

J. B. P. Sinha's **nurturant-task leader** (1980) is firm about work but caring like a parent, with nurturance conditional on the task being done; it suits hierarchical Indian workplaces and should grow into participative leadership. Women and leadership are in Section 3.5; Herzberg's work motivation is in Section 1.10.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. A diagram of choices and rejections within a group is a _____.
2. Tuckman's stage of struggle over leadership is _____.
3. Zajonc: others' presence strengthens the _____ response.
4. Social loafing is also called the _____ effect.
5. Working harder in a group, seen in collectivist cultures, is social _____.
6. Discussion making views more extreme is group _____.
7. Fiedler's dimensions: leader-member relations, task structure and _____.
8. Splintering of a group into factions is a _____.

Answers: (1) sociogram, (2) storming, (3) dominant, (4) Ringelmann, (5) striving, (6) polarisation, (7) position power, (8) schism.

3.11 Applied Social Psychology: Law, Culture and Social Work

LOWER YIELD · 15 QS

Kurt Lewin wanted social psychology to change lives, through action research that studies a problem while trying to solve it (Section 3.1). This closing section follows that wish into hospitals and crowded cities, police stations and courtrooms, and then into social work, the profession that helps individuals, groups and communities.

Health, environment, work and happiness

Why does the same stress make one person ill and leave another well? Partly because of other people. **Social support** is the perception or experience of being loved, cared for and part of a network of mutual help. **Sheldon Cohen** and **Thomas Ashby Wills** (1985) contrasted two models. In the main-effect model, social integration benefits health at any level of stress. In the **stress-buffering hypothesis**, support protects chiefly under high stress, by making the stressor seem less threatening and by softening the reaction to it. Loneliness is in Section 3.6, and health behaviour models in Chapter 7.

Places matter too (personal space is in Section 3.3). **Daniel Stokols** (1972) separated density, a physical count of people per unit of space, from **crowding**, the unpleasant feeling that there are too many people, with lost control and privacy: a festival is dense yet enjoyable, while a lift with three strangers feels crowded. In 1962 **John B. Calhoun** let rats breed with ample food but limited space. Their society collapsed into a **behavioural sink** of neglected young, soaring infant mortality, aggression and withdrawal. In people, density harms mainly when it removes control. **Irwin Altman** (1975) described **territoriality**, the marking and defending of places: primary territories are owned and central to life (a home), secondary territories are used regularly but shared (a canteen table) and public territories are open to anyone for a while (a park bench). Noise, likewise, harms most when it is unpredictable and uncontrollable.

At work, social psychology informs motivation (Herzberg, Section 1.10), the Hawthorne effect, in which people change their behaviour because they know they are being studied (Chapter 9), and leadership (Section 3.10). **Ed Diener** (1946 to 2021) defined **subjective well-being** as a person's own evaluation of life: high life satisfaction, frequent positive feelings and infrequent negative ones.

Eyewitness testimony

Why do defence lawyers so often protest the use of eyewitness testimony? Witnesses are not especially forgetful, and poor eyesight is rarely the issue; the trouble is that a memory can be vivid, confident and false. **Source monitoring** is the process of judging where a memory came from: seen, heard about or imagined (**Marcia Johnson** and colleagues, 1993). A source monitoring error attaches a genuine memory to the wrong origin, so a face seen in a mugshot album, on the news or as an innocent bystander is taken for the culprit's. In one celebrated case a rape victim identified the psychologist **Donald Thomson**, who had been on the live television programme she was watching when she was attacked: his alibi was the source of her memory. Misleading questions act the same way. In the 'smashed' versus 'hit' study of **Elizabeth Loftus** and **John Palmer** (1974; Section 1.7), the wording of a question changed what witnesses remembered, down to broken glass that was never there.

A confident witness is not usually right: across studies, confidence predicts accuracy only weakly, and it grows after the event. In the **post-identification feedback effect** (**Gary Wells** and colleagues, 1998), witnesses told 'Good, you identified the suspect' later claimed a better view and more certainty, even when they had picked an innocent person. Some sources of error the legal system cannot control. In **weapon focus** a visible weapon draws attention away from the face; in **own-race bias** people recognise faces of their own race more accurately than faces of other races; and high stress during the crime usually impairs identification. Other sources it can control, chiefly the lineup (Table 3.30).

Table 3.30 Lineup reforms that reduce false identifications.

Reform	What is done	Why it helps
Double-blind	The officer running it does not know the suspect	No unintended hints or feedback
Sequential lineup	Faces shown one at a time, each judged yes or no	Stops choosing the closest match
Unbiased instructions	Witness told the culprit may not be present	Removes pressure to choose someone
Fair fillers	Others match the witness's description	The suspect does not stand out
Confidence recorded at once	Certainty noted in the witness's words	Before feedback inflates it

The **cognitive interview** of **Ronald Fisher** and **Edward Geiselman** (first tested in 1984) gives four instructions: **reinstate the context** (mentally return to the scene and your feelings), **report everything**, however trivial, **change the order** (recall backwards as well as forwards) and **change the perspective** (describe what another person present would have seen). Because a memory has several retrieval routes, it draws out much more correct detail than a standard police interview, with few added errors.

> EXAM TRAP
>
> Defence lawyers protest eyewitness testimony because **source errors can cause a false identification**, not because witnesses forget more, see poorly or are usually right when confident. The Ronald Fisher of the cognitive interview is not Ronald Aylmer Fisher, the statistician.

Juries, profiling and rehabilitation

In the **story model** of **Nancy Pennington** and **Reid Hastie** (1986), jurors faced with conflicting evidence build a narrative of what happened from the evidence and their own knowledge of how such events go, then choose the verdict that fits the most coherent story; evidence presented in story order persuades best. Pretrial publicity against the accused raises guilty verdicts, and instructions to ignore it help little. Attractive defendants are more often acquitted and given lighter sentences, unless their looks aided the crime, as in a swindle. Stereotypes about race, sex and class bias verdicts (Section 3.5), though careful deliberation weakens them.

Offender profiling infers the characteristics of an unknown offender from the crime. FBI agents such as **Robert Ressler** and **John Douglas** interviewed convicted murderers and built a top-down typology. **Organised** offenders plan, control the scene, bring and remove a weapon and are socially competent; **disorganised** offenders act on impulse, leave a chaotic

scene and a weapon of opportunity, and are socially inadequate. **David Canter**'s **investigative psychology** works bottom-up, from statistical patterns across many solved cases; his 1986 profile helped catch England's 'railway rapist'. Support for the organised-disorganised split is weak.

Profiling and the cognitive interview help to catch offenders; rehabilitation tries to change them. After **Robert Martinson**'s 'nothing works' (Key study), meta-analyses from the late 1980s built the 'what works' evidence. Successful programmes are **community-based** rather than institutional, keeping offenders tied to work, family and prosocial peers; they are **cognitive-behavioural**, structured and multimodal. They follow the **risk-need-responsivity** (RNR) principles of **Don Andrews** and **James Bonta** (1990): match treatment intensity to the risk of reoffending (**risk**); target criminogenic needs, the changeable causes of crime such as antisocial attitudes, antisocial peers and substance misuse (**need**); and teach in the way the offender learns best (**responsivity**). Punishment alone does not reduce reoffending.

KEY STUDY: MARTINSON AND 'NOTHING WORKS' (1974)

Question: does rehabilitation reduce reoffending? **Method:** Martinson and colleagues reviewed 231 controlled evaluations of correctional programmes. **Result:** 'with few and isolated exceptions', rehabilitation had had no appreciable effect on recidivism; the press shortened this to 'nothing works', and policy turned to punishment. **Criticism:** many programmes were weak or badly run, and Martinson himself withdrew the strong claim in 1979.

EXAM TRAP

A quality of successful offender rehabilitation is **being community-based**. Cognitive interviews and offender profiling are investigative tools, not treatment, and a focus on the relationship with parents is not a defining quality.

Social work: definition, roots and methods ●●

The 2014 global definition of the International Federation of Social Workers (IFSW) and the International Association of Schools of Social Work (IASSW) calls social work a practice-based profession and academic discipline that promotes social change and development, social cohesion, and the empowerment and liberation of people. Its lens is the **person-in-environment**: problems arise between people and their families, neighbourhoods and institutions, so help may target either side.

It has two roots. The **Charity Organisation Society** (London, 1869) sent 'friendly visitors' to investigate applicants and coordinate relief, the seed of case work. The **settlement houses** placed educated volunteers among the poor to press for reform: **Toynbee Hall** (London, 1884, under **Samuel Barnett**) and **Hull House** (Chicago, 1889, founded by **Jane Addams** and **Ellen Gates Starr**), the seed of group work and community organisation.

The **primary methods**, or direct methods, are social case work, social group work and community organisation; the secondary or auxiliary methods are social welfare administration, social action and social work research. **Direct practice**, face-to-face work with clients, has historically been conceptualised as case work and case management, as individual psy-

chotherapy (clinical social work) and as generalist practice. Policy practice, shaping laws and agency rules, is indirect or macro practice.

India's first school of social work, and Asia's, was the **Sir Dorabji Tata Graduate School of Social Work**, opened in Bombay in 1936 under the American missionary **Clifford Manshardt**; in 1944 it was renamed the **Tata Institute of Social Sciences**. The Delhi School of Social Work followed in 1946, and psychiatric social work training grew up at Ranchi and at NIMHANS, Bengaluru.

> EXAM TRAP
>
> 'Tata Institute of Social Profession' and 'Tata Institute of Social Service' are invented, and the Tata Institute of Social Sciences is the 1944 name; the first school was the **Sir Dorabji Tata Graduate School of Social Work**. Policy practice is not a form of direct practice.

Social case work ●●

Social case work, the oldest method, helps one person or family at a time with problems of social functioning, through a professional relationship and community resources. Its founder, **Mary Ellen Richmond** (1861 to 1928), gave it a method in *Social Diagnosis* (1917), which taught workers to gather evidence systematically before judging, and defined it in *What Is Social Case Work?* (1922) as the processes that develop personality through adjustments consciously effected, individual by individual, between people and their social environment.

The process runs in phases (Figure 3.18): **intake** (can this agency help?), **study** (gathering the facts), **assessment** or **social diagnosis** (defining the problem and its causes), **intervention** or treatment, **evaluation**, and **termination** with **follow-up**. Content analysis, the coding of themes in texts, is a research technique, not a phase. The worker's techniques are **interviewing** (the central one), **observation**, **counselling**, **supportive techniques** (reassurance, encouragement), **home visits**, **collateral contacts** (speaking, with consent, to relatives, teachers or employers), **environmental modification** (changing the client's surroundings, such as arranging a pension) and **referral**. Lobbying legislators belongs to social action, not case work.

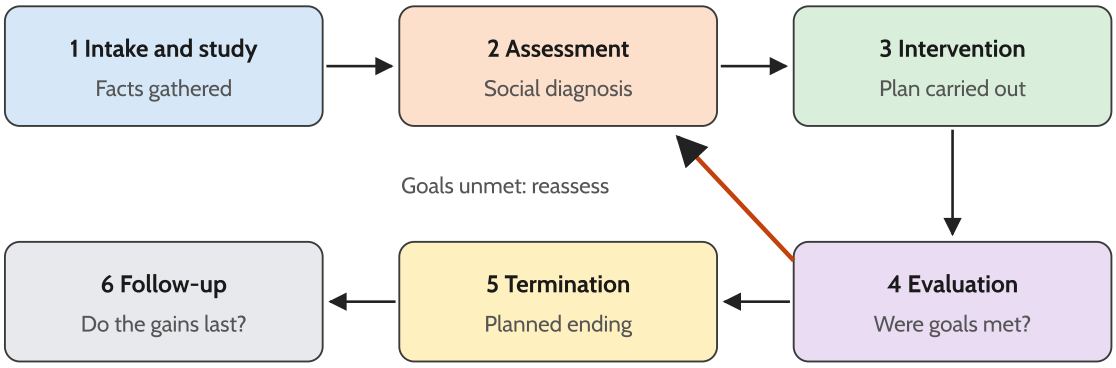


Figure 3.18 The social case work process. When evaluation finds goals unmet, the worker returns to assessment (dashed arrow).

By the 1930s case work had split into the **diagnostic school**, built on Freud, and the **functional school**, built on Otto Rank's will therapy (Table 3.31). **Florence Hollis** developed the diagnostic school into the psychosocial approach, and **Charlotte Towle**'s *Common Human Needs* (1945) taught that every welfare applicant has the same basic needs as anyone else.

Table 3.31 Schools and models of social case work.

Approach	Key names	Central idea
Diagnostic (psychosocial)	**Gordon Hamilton** (1940); Hollis (1964)	Understand the personality, diagnose, then treat; the client as person-in-situation
Functional	**Jessie Taft**, **Virginia Robinson** (Pennsylvania)	The client's will drives change, within the agency's function and time limits
Problem-solving	**Helen Harris Perlman** (1957)	Four Ps: a person with a problem comes to a place for a process of help
Task-centred	**William Reid** and **Laura Epstein** (1972)	Brief, time-limited work on agreed problems through concrete tasks

Whatever the school, the relationship carries the help. **Felix Biestek** (1957) set out seven principles of the casework relationship: **individualisation** (each client is unique), **purposeful expression of feelings**, **controlled emotional involvement** (responding with feeling without being swept away), **acceptance**, a **non-judgemental attitude**, **client self-determination** (the right to make one's own choices) and **confidentiality**.

KEY STUDY: FISCHER, 'IS CASEWORK EFFECTIVE?' (1973)

Question: does professional case work help clients? **Method: Joel Fischer** reviewed the controlled studies comparing clients who received professional case work with control groups who did not. **Result:** there was no good evidence that case work worked, and in some studies treated clients did worse. **Impact:** the review shook the profession and pushed it toward empirical, evidence-based practice. **Criticism:** the studies often had vague goals and crude outcome measures.

EXAM TRAP

Richmond founded case work; **Joel Fischer** studied its effectiveness. Irwin Epstein wrote on social work research, John Goyder on survey nonresponse, and Jacob Cohen was the statistician of kappa, effect size and power (Chapter 9). Not part of the case work process: **content analysis**. Not a case work technique: **lobbying**.

Social group work ●●●

Social group work uses the group itself as a helper: members gain from one another as well as from the worker. **Grace Longwell Coyle** (1892 to 1962), whose *Social Process in Organized Groups* appeared in 1930, established group work as a method within social work. **Gisela Konopka** (1910 to 2003) later carried it into treatment settings such as institutions and child guidance clinics. **Harleigh B. Trecker** listed ten principles: planned group formation, specific objectives, a purposeful worker-group relationship, continuous individualisation, guided group interaction, democratic group self-determination, flexible functional organisation, progressive programme experiences, resource utilisation and evaluation.

Three models dominate. The social goals model, from the settlements, builds social consciousness and responsible citizenship. The remedial model of **Robert Vinter** uses the group to treat individual members, with the worker as change agent. The reciprocal, or mediating, model of **William Schwartz** makes the group a mutual-aid system, with the worker mediating between members and between the group and society.

Ronald Toseland and **Robert Rivas** distinguish **formed groups**, deliberately convened by a worker or agency for an identified purpose, usually with a professionally trained leader and an agency auspice (a support group for caregivers), from **natural groups**, which come together spontaneously out of friendship, shared interest or circumstance (a circle of friends, a street gang). All groups share certain properties: a purpose, stages of development and criteria for their composition, along with structure, norms, roles and cohesion (Section 3.10). No group needs an 'analytic base'.

Some members serve the group in hidden ways. **Lawrence Shulman** described the scapegoat, blamed for what the group dislikes in itself; the deviant member, whose odd behaviour often voices what others feel; the **gatekeeper**, who blocks or diverts the group, by joking or changing the subject, whenever talk nears a painful topic; the internal leader, who takes initiative from within; the defensive member, who denies having a problem; the quiet member; and the monopoliser, who dominates the talk. Groups also change over time. The **Boston model** of **James Garland**, **Hubert Jones** and **Ralph Kolodny** (1965) has five stages: pre-affiliation (approach and avoidance), power and control, intimacy, differentiation and separation. Tuckman's forming to adjourning sequence is in Section 3.10.

> EXAM TRAP
>
> Group work as a method: **Grace Coyle**, not Mary Richmond (case work), Charlotte Towle (*Common Human Needs*) or Gisela Konopka (group work in treatment settings). 'Focusing on change and possibilities' is not one of Trecker's principles. Formed groups do not come together spontaneously.

Community organisation

Community organisation works with whole communities. **Murray G. Ross** (1955) defined it as a process by which a community identifies and ranks its needs, develops the confidence and will to work on them, finds resources, takes action, and in doing so develops cooperative attitudes and practices. **Jack Rothman** (1968) distinguished three models (Table 3.32). Social action is linked above all to **Saul Alinsky** (1909 to 1972), who organised Chicago's Back of the Yards neighbourhood and wrote *Rules for Radicals* (1971).

Table 3.32 Rothman's three models of community organisation.

Model	Aim	Worker's role
Locality development	Self-help and consensus through broad local participation	Enabler
Social planning	Expert, data-based solving of problems such as housing or health	Planner, fact-gatherer
Social action	Shifting power and resources to a disadvantaged group, through conflict if needed	Activist, advocate

Culturally sensitive practice and child welfare ●●

Cultures differ in measurable ways. Surveying IBM employees in 40 countries, **Geert Hofstede** (1980) found four dimensions: power distance (acceptance of unequal power), **individualism-collectivism** (a self defined as an independent 'I' or by the 'we' of family and group), masculinity-femininity (competition and achievement versus care) and uncertainty avoidance (discomfort with ambiguity, eased by rules). Long-term orientation and indulgence-restraint were added later. Collectivism brings more conformity (Section 3.7), a weaker fundamental attribution error (Section 3.3) and social striving (Section 3.10).

Ethnic-sensitive practice (**Wynetta Devore** and **Elfriede Schlesinger**, 1981) uses descriptive knowledge specific to different racial, cultural and ethnic groups in every phase of practice, so that the worker grasps the client's 'ethnic reality', the outlook shaped jointly by ethnic group and social class. Three neighbouring terms differ: ethno-psychotherapy (**Georges Devereux**) joins psychoanalysis and anthropology to treat clients from other cultures; sensitivity training is the T-group method (Section 3.1); and matching assigns a worker of the client's own ethnicity. **Cultural competence** (**Derald Wing Sue** and colleagues) combines awareness of one's own biases, knowledge of the client's worldview and culturally appropriate skills.

How does a worker learn a world not his own? The **ethnographic interview** of the anthropologist **James Spradley** (1979), adapted for social work by **James Leigh** (1998), treats the client as a cultural guide who explains what things mean in their world: 'Can you tell me why this band is so popular among your age group?' A psychodynamic inquiry ('What does that remind you of from childhood?') seeks unconscious conflict; a task-focused inquiry ('What could you do before next week?') seeks concrete steps; a strengths-based inquiry ('How have you coped so far?'), from the **strengths perspective** of **Dennis Saleebey** (1992), seeks resources rather than deficits. The **ecomap** of **Ann Hartman** (1978) draws the family in a circle linked to school, work, services and friends by lines marking strong, weak or stressful ties; the **genogram** charts three or more generations of a family.

> EXAM TRAP
>
> Descriptive knowledge specific to racial, cultural and ethnic groups is **ethnic-sensitive practice**, not ethno-psychotherapy, sensitivity training or matching. Asking a client to explain what something means in their world is **ethnographic** inquiry.

Child maltreatment takes four forms: physical abuse, sexual abuse, emotional abuse and **neglect**, the failure to provide for a child's basic needs. Neglect has several categories: **physical neglect** (inadequate food, housing, clothing or hygiene); **health or medical neglect** (failing to immunise, to seek medical care or to follow medical directives); **inadequate supervision** (children left unattended); **educational neglect**; and **emotional neglect**. Using licensed day care is responsible substitute care, not neglect. Sexual offences against children in India fall under the POCSO Act 2012 (Chapter 7).

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Support that protects mainly under high stress fits the _____ hypothesis.
2. Attaching a genuine memory to the wrong origin is a _____ error.
3. The cognitive interview's fourth instruction is to change the _____.
4. Andrews and Bonta's principles are risk, need and _____.
5. The founder of social case work, author of *Social Diagnosis* (1917), was _____.
6. Group work was established as a method within social work by _____.
7. In Shulman's terms, the member who blocks painful topics is the _____.
8. India's first school of social work (Bombay, 1936) was the _____.

Answers: (1) stress-buffering, (2) source monitoring, (3) perspective, (4) responsivity, (5) Mary Ellen Richmond, (6) Grace Coyle, (7) gatekeeper, (8) Sir Dorabji Tata Graduate School of Social Work.

The Murder of Kitty Genovese, Revisited

Return to Kew Gardens with the whole chapter behind you. Allport's definition (Section 3.1) fits that night exactly: the presence of other neighbours, actual and imagined, changed what each one did. The public's verdict of apathy blamed character and ignored the situation, the fundamental attribution error (Section 3.3); diffusion of responsibility and the bystander research it launched are Section 3.8. Neighbours who took the cries for a drunken quarrel show how we read ambiguous scenes (Section 3.2); the persuasive power of a front-page story is Section 3.4; Moseley's violence belongs to Section 3.9. And the 38-witnesses legend, repeated in textbooks for some forty years before psychologists checked it, carries the lesson of the replication crisis: in social psychology even the stories need testing.

Chapter summary

3.1 Social Psychology: History, Theories, Concepts and Methods. Social psychology studies how individuals' thoughts, feelings and behaviour are influenced by the actual, imagined or implied presence of others (Gordon Allport, 1954). It grew from Le Bon's crowd (1895), Triplett's first experiment (1898) and the 1908 texts of McDougall and Ross; Kurt Lewin, its modern father, added the life space (person plus psychological environment), group dynamics and action research. A crisis around 1970, European work and the social cognition era followed. From sociology came Durkheim's anomie (normlessness), symbolic interactionism, socialisation and resocialisation, status and role. Its theories include Moscovici's social representations (1961); its methods run from observation to experiment, with debriefing after deception.

3.2 Social Cognition and First Impressions. Social cognition is how we interpret, remember and use social information. As cognitive misers (Fiske and Taylor, 1984) we lean on automatic processing, schemas (scripts, self-schemas, stereotypes), priming and heuristics. Schemas persist (belief perseverance) and can fulfil themselves: Merton's self-fulfilling prophecy (1948), the Pygmalion effect (Rosenthal and Jacobson, 1968) and the Golem effect. Social thought tilts towards negativity, optimism, the planning fallacy, counterfactuals, magical thinking, false consensus and

the illusion of control. Asch's central traits (warm-cold) and the halo effect (Thorndike, 1920) shape impressions; thin slices (Ambady and Rosenthal) are surprisingly accurate; impression management follows Goffman's dramaturgy and Jones and Pittman's strategies.

3.3 Communication, Nonverbal Cues and Attribution. Communication runs encoding, channel, recipient, decoding, with feedback and noise; the classic models are Lasswell (1948), Shannon and Weaver (1949), Schramm (1954) and Berlo's SMCR (1960). Nonverbal channels include face, gaze, body language, touch and paralanguage; Hall's proxemics sets intimate, personal, social and public zones, and lies are detected with only about 54 percent accuracy. Attribution theory began with Heider (1958): dispositional versus situational causes. Jones and Davis explained correspondent inferences, Kelley used consensus, consistency and distinctiveness, and Weiner added stability and controllability. The fundamental attribution error (Ross 1977), actor-observer effect, self-serving bias and just-world hypothesis tilt our explanations, more so in individualist cultures.

3.4 Attitudes, Persuasion and Cognitive Dissonance. An attitude is a learned evaluation of an object with affective, behavioural and cognitive parts (Rosenberg and Hovland, 1960), serving Katz's utilitarian, knowledge, ego-defensive and value-expressive functions. LaPiere (1934) exposed the attitude-behaviour gap; strong, specific attitudes formed by direct experience predict best, and the theory of planned behaviour (Ajzen) adds perceived control to attitude and subjective norm. Persuasion depends on source, message and audience (the Yale approach) and runs by the central or peripheral route (Petty and Cacioppo). Reactance (Brehm, 1966), forewarning and inoculation aid resistance. Festinger's cognitive dissonance (1957) explains why $1 beat $20; Bem's self-perception theory is its main rival.

3.5 Stereotypes, Prejudice and Intergroup Relations. A stereotype is a belief about a group, prejudice a negative attitude (Allport, 1954) and discrimination unfair action. Prejudice has been traced to the authoritarian personality, displaced frustration, social learning, realistic conflict (at Robbers Cave only superordinate goals ended hostility) and relative deprivation. Tajfel's minimal groups showed that categorisation alone breeds in-group favouritism; social identity theory (Tajfel and Turner, 1979) says much of our self-esteem comes from our groups. Low-prejudiced people inhibit automatically activated stereotypes by controlled processing (Devine, 1989); the IAT measures implicit bias. Social roles (Eagly) shape gender stereotypes, which build glass ceilings and sticky floors. Contact reduces prejudice under Allport's conditions.

3.6 Affiliation, Attraction and Close Relationships. The need to belong (Baumeister and Leary 1995) is fundamental; gregariousness is a social motive. Schachter (1959) showed that anxiety raises affiliation with others in the same plight, and we affiliate for information, anxiety reduction and social comparison (Festinger 1954), which also explains FOMO. Attraction grows from propinquity and mere exposure (Zajonc 1968), looks, matching, similarity and reciprocity. Exchange, equity and investment theories explain staying or leaving. Self-disclosure deepens relationships (descriptive facts, evaluative feelings). Love is passionate or companionate, triangular (Sternberg 1986) and shaped by attachment style. Weiss separated emotional from social loneliness.

3.7 Social Influence: Conformity, Compliance, Obedience and Power. Social influence takes three forms: conformity (matching a group with no request), compliance (yielding to a direct request) and obedience (following an authority's command). Norms may be descriptive or injunctive, prescriptive or proscriptive. Sherif's autokinetic study showed norms forming through infor-

mational influence; Asch's line studies showed normative conformity to an obviously wrong majority (about 37 percent of critical trials). Consistent minorities can convert majorities (Moscovici). Cialdini traced compliance tactics such as the foot-in-the-door and door-in-the-face to six principles. In Milgram's study 65 percent obeyed to 450 volts; Zimbardo's prison study showed the power of roles. French and Raven named six bases of power.

3.8 Prosocial Behaviour: Why and When People Help. **Altruism** is helping with no expected reward, at a cost to oneself. Batson's **empathy-altruism hypothesis** (the Elaine study) claims true altruism; Cialdini's **negative-state relief model** and the **empathic joy hypothesis** are egoistic, while kin selection (Hamilton), reciprocal altruism (Trivers), competitive altruism, defensive helping and norms add other motives. After the 1964 murder of **Kitty Genovese**, Latané and Darley showed the **bystander effect**: people help more when alone (seizure study, 85 versus 31 percent). Their five-step model locates its two classic causes, **diffusion of responsibility** and **pluralistic ignorance**. Costs (Piliavin), hurry, mood, models, gratitude, the altruistic personality and the threat-to-self-esteem model complete the picture.

3.9 Aggression: Its Roots, Causes and Control. **Aggression** is behaviour aimed at harming a living being who wants to avoid it (Baron and Richardson, 1994): **hostile** or **instrumental**, physical or verbal, direct, indirect or **relational**. Freud's thanatos and Lorenz's hydraulic instinct gave way to biology (low **serotonin**, MAOA with maltreatment) and to the **frustration-aggression hypothesis** (Dollard and colleagues, 1939), softened by Miller (1941) and reformulated by **Berkowitz** (1989) as negative affect plus aggressive cues (the weapons effect). Bandura's Bobo doll studies showed observational learning; the General Aggression Model integrates these. Provocation, exclusion, media violence, honour cultures, heat and alcohol raise aggression; **catharsis** fails, while apology, incompatible responses and self-regulation help.

3.10 Groups: Structure, Performance, Decisions and Leadership. A **group** is two or more interacting, interdependent people with a shared identity and a structure of roles, status, norms and cohesiveness. Moreno's **sociometry** (1934) maps choices in a **sociogram** of stars, isolates and cliques. Groups pass through Tuckman's **forming, storming, norming, performing** and adjourning. Others' presence brings **social facilitation** on easy tasks and impairment on hard ones (Triplett 1898; Zajonc 1965); pooled effort brings **social loafing** (Ringelmann), reversed as social striving in collectivist cultures; anonymity brings **deindividuation**. Discussion causes **group polarisation**, and cohesion with insulation causes Janis's **groupthink**. Leadership theory runs from traits and styles to Fiedler's contingency model and transformational leadership.

3.11 Applied Social Psychology: Law, Culture and Social Work. Social support protects health, chiefly as a **stress buffer** (Cohen and Wills, 1985); crowding is felt, density is counted (Stokols). Defence lawyers challenge eyewitnesses because **source monitoring errors** cause false identifications and confidence predicts accuracy weakly; double-blind sequential lineups and the **cognitive interview** help. Jurors build stories; effective rehabilitation is community-based, cognitive-behavioural and follows **risk-need-responsivity**. Social work grew from the Charity Organisation Society and settlements. Mary Richmond founded **case work** (intake to follow-up); Fischer (1973) doubted its effectiveness. Grace Coyle established **group work**; Trecker gave its principles and Shulman its member roles. India's first school was the **Sir Dorabji Tata Graduate School of Social Work** (1936).

Who's who

Table 3.33 The people in this chapter.

Name	When	Examined for
Gordon Willard Allport	1897 to 1967	Classic definition of social psychology (1954): the actual, imagined or implied presence of others
Robert A. Baron	Modern textbook	With Branscombe: the scientific study of the nature and causes of individual behaviour, feeling and thought in social situations
Nyla R. Branscombe	Modern textbook	With Baron: definition of social psychology
Gustave Le Bon	1895	The Crowd: the crowd mind, contagion and suggestion
Norman Triplett	1898	Usually called the first social psychology experiment (cyclists, fishing reels; social facilitation)
Max Ringelmann	Published 1913	Rope pulling: effort per person falls as the group grows (root of social loafing)
William McDougall	1908	One of the first two social psychology textbooks; instinct based
Edward Alsworth Ross	1908	Social Psychology; sociological, imitation and suggestion in crowds
Floyd Henry Allport	1924	Social Psychology: experimental and individual centred
Muzafer Sherif	1935, 1936	Group norm formation with the autokinetic effect; The Psychology of Social Norms
Kurt Lewin	1890 to 1947	Father of modern experimental social psychology; B = f(P, E) and the life space; action research (1946); group dynamics; Research Center for Group Dynamics at MIT (1945); T-groups
Ronald Lippitt	1939	With Lewin and White: autocratic, democratic and laissez-faire leadership climates
Ralph K. White	1939	With Lewin and Lippitt: leadership climates
Theodor W. Adorno	1950	The Authoritarian Personality (with colleagues)
Solomon Asch	1951	Conformity to a wrong majority
Carl Hovland	1953	Yale persuasion research
Leon Festinger	1954, 1957	Social comparison theory; cognitive dissonance theory
Fritz Heider	1958	The Psychology of Interpersonal Relations; cognitive consistency
Stanley Milgram	1963	Obedience studies; ethics controversy; high experimental, low mundane realism
Philip Zimbardo	1971	Stanford prison study; the power of roles; ethics controversy
Kenneth J. Gergen	1973, 1985	"Social psychology as history" (1973); social constructionism
Henri Tajfel	1919 to 1982	European social psychology; social identity theory (with Turner, 1979)
John Turner	1979	With Tajfel: social identity theory
Serge Moscovici	1925 to 2014	Social representations theory (1961): anchoring and objectification; minority influence
Emile Durkheim	1858 to 1917	Social facts, collective representations, anomie (The Division of Labour in Society 1893; Suicide 1897)
Robert King Merton	1910 to 2003	Strain theory of anomie (1938); anticipatory socialisation; role set (1957)
Ferdinand Tonnies	1887	Gemeinschaft (community) and Gesellschaft (society)
George Herbert Mead	1863 to 1931	Symbolic interactionism; Mind, Self and Society (1934); role taking, play and game stages, generalised other
Herbert Blumer	1937	Named symbolic interactionism
Charles Horton Cooley	1864 to 1929	Looking-glass self (1902); primary group (1909)
Erving Goffman	1922 to 1982	Total institutions, Asylums (1961)
Russell Barton	1959	Institutional neurosis
Ralph Linton	1936	Ascribed versus achieved status
William J. Goode	1960	Role strain
George C. Homans	1961	Social exchange theory
John Thibaut	1959	With Kelley: social exchange, The Social Psychology of Groups
Harold Kelley	1959	With Thibaut: social exchange
Peter Blau	1964	Social exchange: Exchange and Power in Social Life

Name	When	Examined for
Albert Bandura	1977, 1986	Social learning and social cognitive theory
Sigmund Freud	1921	Group Psychology and the Analysis of the Ego; not the founder of action research
Edward O. Wilson	1975	Sociobiology
Michael A. Hogg	2000s	Uncertainty-identity theory; social identity
Roy F. Baumeister	1995	With Leary: the need to belong
Mark R. Leary	1995	Need to belong (with Baumeister); sociometer theory of self-esteem
Richard E. Petty	1986	With Cacioppo: dual-process (elaboration likelihood) model
John T. Cacioppo	1986	With Petty: dual-process model
Ignacio Martin-Baro	1942 to 1989	Liberation psychology
Gustav Theodor Fechner	1860	Psychophysics (distractor for Lewin; Chapter 1)
William James	1890	Functionalism (distractor for Lewin; Chapter 1)
Elliot Aronson	1968	With Carlsmith: experimental versus mundane realism
J. Merrill Carlsmith	1968	With Aronson: experimental versus mundane realism
Martin Orne	1962	Demand characteristics
Robert Rosenthal	1966	Experimenter expectancy effects
Gardner Murphy	1953	UNESCO study of social tensions in India, In the Minds of Men
Durganand Sinha	1922 to 1998	Indigenisation of psychology in India
Jai B. P. Sinha	1980	Nurturant-task leader
Udai Pareek	1925 to 2010	Organisational role stress
Susan Fiske	1984	The cognitive miser (with Shelley Taylor); later the motivated tactician (1991)
Shelley Taylor	1984	The cognitive miser, with Susan Fiske
Frederic Charles Bartlett	1886 to 1969	Origin of the schema concept in reconstructive memory (1932)
Hazel Markus	1977	Self-schemas
Roger Schank	1977	Scripts (event schemas), with Robert Abelson
Robert Abelson	1977	Scripts (event schemas), with Roger Schank
John Bargh	1996	Automatic priming: the elderly prime slowed walking (with Mark Chen and Lara Burrows)
Mark Chen	1996	Co-author of the elderly-priming study
Lara Burrows	1996	Co-author of the elderly-priming study
Stéphane Doyen	2012	Failed replication of the elderly-priming study (slowing only when experimenters expected it)
Betsy Sparrow	2006	Unpriming, with Daniel Wegner
Daniel Wegner	1948 to 2013	Thought suppression and ironic rebound, the white bear (1987); unpriming (2006)
Lee Ross	1942 to 2021	Belief perseverance (1975); false consensus effect (1977); planning fallacy study (1994)
Mark Lepper	1975	Belief perseverance, with Ross and Hubbard
Michael Hubbard	1975	Belief perseverance, with Ross and Lepper
Lenore Jacobson	1968	Pygmalion in the Classroom, with Robert Rosenthal
Elisha Babad	1982	Golem effect, with Jacinto Inbar and Robert Rosenthal
Jacinto Inbar	1982	Golem effect, with Babad and Rosenthal
Mark Snyder	1974, 1977	Self-monitoring (1974); behavioural confirmation telephone study (1977)
Elizabeth Tanke	1977	Behavioural confirmation telephone study, with Snyder and Berscheid
Ellen Berscheid	1977	Behavioural confirmation telephone study, with Snyder and Tanke
Carl Word	1974	Interviewer study of the self-fulfilling prophecy, with Zanna and Cooper
Mark Zanna	1974	Interviewer study, with Word and Cooper
Joel Cooper	1974	Interviewer study, with Word and Zanna
Émile Durkheim	1858 to 1917	Anomie
Amos Tversky	1937 to 1996	Heuristics of judgement (1974); planning fallacy (1979)
Daniel Kahneman	1934 to 2024	Heuristics of judgement (1974); planning fallacy (1979)

Name	When	Examined for
Norbert Schwarz	1983, 1991	Ease of retrieval in availability (1991); affect as information, with Gerald Clore (1983)
Paul Slovic	2000s	Affect heuristic
Paul Rozin	1990, 2001	Magical thinking: laws of contagion and similarity; negativity bias (with Edward Royzman, 2001)
Edward Royzman	2001	Negativity bias, with Paul Rozin
Neil Weinstein	1980	Optimistic bias (unrealistic optimism)
Roger Buehler	1994	Planning fallacy study, with Dale Griffin and Lee Ross
Dale Griffin	1994	Planning fallacy study
Victoria Medvec	1995, 2000	Olympic medallists and counterfactual thinking (1995); spotlight effect (2000)
Scott Madey	1995	Olympic medallists and counterfactual thinking
Thomas Gilovich	1995, 2000	Bronze happier than silver medallists (1995); spotlight effect (2000)
David Greene	1977	False consensus effect, with Ross and House
Pamela House	1977	False consensus effect, with Ross and Greene
Ellen Langer	born 1947	Illusion of control, the lottery-ticket study (1975)
Jeff Greenberg	1986	Terror management theory, with Solomon and Pyszczynski
Sheldon Solomon	1986	Terror management theory
Tom Pyszczynski	1986	Terror management theory
Gerald Clore	1983	Affect as information, the sunny and rainy day study, with Norbert Schwarz
Joseph Forgas	1995	Affect infusion model
Jerome Bruner	1915 to 2016	Implicit personality theory, with Renato Tagiuri (1954)
Renato Tagiuri	1954	Implicit personality theory, with Jerome Bruner
Edward Lee Thorndike	1874 to 1949	Named the halo effect (1920) from officers' ratings of soldiers
Norman Anderson	1965	Information integration theory: weighted averaging versus additive models of impression formation
Nalini Ambady	1959 to 2013	Thin slicing, with Robert Rosenthal (1992, 1993)
Malcolm Gladwell	2005	Popularised thin slicing in Blink
Janine Willis	2006	100-millisecond face judgements, with Alexander Todorov
Alexander Todorov	2006	100-millisecond face judgements, with Janine Willis
Edward E. Jones	1926 to 1993	Ingratiation (1964); self-presentation strategies with Pittman (1982); self-handicapping with Berglas (1978)
Thane Pittman	1982	Five strategies of self-presentation, with Edward E. Jones
Steven Berglas	1978	Self-handicapping, with Edward E. Jones
Kenneth Savitsky	2000	Spotlight effect, with Gilovich and Medvec
Roos Vonk	1998	Slime effect
Harold Lasswell	1902 to 1978	Communication model (1948): who says what, in which channel, to whom, with what effect
Claude Shannon	1916 to 2001	Shannon and Weaver model of communication (1949); introduced noise
Warren Weaver	1894 to 1978	Shannon and Weaver model (1949): source, transmitter, channel, noise, receiver, destination
Wilbur Schramm	1907 to 1987	Interactive, circular model of communication (1954); field of experience
David Berlo	1929 to 1996	SMCR model of communication (1960): source, message, channel, receiver
Alex Bavelas	1913 to 1993	Communication networks in small groups (1950)
Harold Leavitt	1922 to 2007	Communication networks (1951): wheel fastest and most accurate on simple tasks, circle highest morale
Paul Ekman	born 1934	Six universal facial expressions plus contempt; display rules; microexpressions
Wallace Friesen	1969 onwards	With Ekman: universal facial expressions and display rules
Ray Birdwhistell	1918 to 1994	Founded kinesics, the study of body movement as communication (Introduction to Kinesics, 1952)

Name	When	Examined for
Edward T. Hall	1914 to 2009	Proxemics (1963); four zones of personal space in The Hidden Dimension (1966); high- and low-context cultures (1976)
Charles Bond	2006	Meta-analysis with DePaulo: lie detection accuracy about 54 percent
Bella DePaulo	2006	Meta-analysis with Bond: lie detection accuracy about 54 percent
Fred Fiedler	1922 to 2017	Contingency theory of leadership
Keith Davis	1965	Correspondent inference theory with Jones: free choice, non-common effects, low social desirability
Bernard Weiner	born 1935	Attribution theory of achievement and emotion: locus, stability, controllability
Lyn Abramson	1978	Attributional reformulation of learned helplessness: internal, stable, global causes
Martin Seligman	born 1942	Learned helplessness; its attributional reformulation (1978)
John Teasdale	1978	Attributional reformulation of learned helplessness
Daniel Gilbert	born 1957	Two-step model of attribution; review of the correspondence bias with Malone (1995)
Patrick Malone	1995	Review of the correspondence bias with Gilbert
Victor Harris	1967	Castro essay study of the fundamental attribution error, with Jones
Richard Nisbett	born 1941	Actor-observer effect with Jones (1971)
Michael Storms	1973	Videotape study that reversed the actor-observer effect
Dale Miller	1975	Review of the self-serving bias with Michael Ross
Michael Ross	1975	Review of the self-serving bias with Dale Miller
Melvin Lerner	1965, 1980	Just-world hypothesis; victim blaming
Carolyn Simmons	1966	Lerner and Simmons experiment: observers derogated an innocent victim they could not help
Zick Rubin	1975	Just World Scale, with Peplau
Letitia Anne Peplau	1975	Just World Scale, with Rubin
Thomas Pettigrew	1979	Ultimate attribution error
Kenneth Dodge	1980	Hostile attribution bias in aggressive children
Kelly Shaver	1970	Defensive attribution
Joan Miller	1984	Hindu Indians in Mysore gave more situational and fewer dispositional explanations than Americans; the gap grew with age
Michael Morris	1994	With Peng: English-language newspapers explained murders dispositionally, a Chinese-language newspaper situationally
Kaiping Peng	1994	With Morris: culture and attribution in newspaper reports of murders
Timothy Wilson	1982	Attribution retraining of worried first-year students, with Linville
Patricia Linville	1982	Attribution retraining of worried first-year students, with Wilson
Milton J. Rosenberg	1960	Tricomponent (ABC) model of attitudes, with Carl Hovland
Carl Iver Hovland	1912 to 1961	Yale communication approach, Communication and Persuasion (1953); sleeper effect with Weiss (1951); one- versus two-sided messages (1949); ABC model (1960)
Irving Lester Janis	1918 to 1990	Yale communication approach (1953); fear appeals with Feshbach (1953)
Harold H. Kelley	1921 to 2003	Co-author of Communication and Persuasion (1953), the Yale approach
Daniel Katz	1903 to 1998	Four functions of attitudes: utilitarian, knowledge, ego-defensive, value-expressive (1960)
M. Brewster Smith	1956	Functions of attitudes: object appraisal, social adjustment, externalisation (with Bruner and White)
Jerome Seymour Bruner	1956	Functions of attitudes with Smith and White
Robert W. White	1956	Functions of attitudes with Smith and Bruner
Emory S. Bogardus	1925	Social distance scale (Section 2.8)
Louis Leon Thurstone	1928	Equal-appearing intervals attitude scale (Section 2.8)
Rensis Likert	1932	Summated ratings attitude scale (Section 2.8)
Louis Guttman	1944	Cumulative (scalogram) attitude scale (Section 2.8)
Charles E. Osgood	1916 to 1991	Congruity theory with Tannenbaum (1955); semantic differential (1957)
Percy H. Tannenbaum	1955	Congruity theory with Osgood

Name	When	Examined for
Arthur W. Staats	1958	Classical conditioning of attitudes: nationality names paired with pleasant or unpleasant words (with Carolyn Staats)
Carolyn K. Staats	1958	Classical conditioning of attitudes (with Arthur Staats)
Jon A. Krosnick	1992	Subliminal conditioning of attitudes towards a stranger
Abraham Tesser	1993	Heritable attitudes are more accessible and resistant to change
Richard Tracy LaPiere	1899 to 1986	Attitude-behaviour gap: Chinese couple refused once in about 250 places, yet over 90 percent of letters refused (1934)
Allan W. Wicker	1969	Review: attitude-behaviour correlations rarely above .30
John Sivacek	1982	Vested interest and the drinking-age campaign (with Crano)
William D. Crano	1982	Vested interest strengthens the attitude-behaviour link
Dennis T. Regan	1977	Cornell housing shortage: direct experience strengthens attitude-behaviour consistency (with Fazio)
Russell H. Fazio	1977, 1986, 1990	Direct experience (1977, 1981); attitude-to-behaviour process model (1986); MODE model (1990); new look of dissonance with Cooper (1984)
Mark P. Zanna	1974, 1981	Direct experience and attitude accessibility (with Fazio, 1981); placebo misattribution study of dissonance arousal (with Cooper, 1974)
Icek Ajzen	born 1942	Theory of reasoned action (1975) and planned behaviour (1985, 1991); principle of compatibility (1977)
Martin Fishbein	1936 to 2009	Theory of reasoned action with Ajzen (1975); principle of compatibility (1977)
Andrew R. Davidson	1979	Birth-control pill study of attitude-behaviour compatibility (with Jaccard)
James J. Jaccard	1979	Compatibility study with Davidson: specific attitudes predict specific acts
Deborah A. Prentice	1993	Perceived campus drinking norms and pluralistic ignorance (with Miller)
Dale T. Miller	1993	Perceived campus drinking norms (with Prentice)
Arthur A. Lumsdaine	1949	One-sided versus two-sided messages (with Hovland and Sheffield)
Fred D. Sheffield	1949	One-sided versus two-sided messages (with Hovland and Lumsdaine)
Seymour Feshbach	1953	Fear appeals with Janis: mild fear changed tooth-brushing most
Norman Miller	1959	Primacy versus recency in persuasion (with Campbell)
Donald T. Campbell	1959	Primacy versus recency in persuasion (with Miller)
R. Goldman	1981	Involvement study of the elaboration likelihood model (with Petty and Cacioppo)
William J. McGuire	1925 to 2007	Information-processing chain of persuasion; inoculation theory (1961)
Walter Weiss	1951	Source credibility and the sleeper effect (with Hovland)
Robert Oppenheimer	1904 to 1967	Physicist used as the credible source in Hovland and Weiss (1951)
Shelly Chaiken	1980	Heuristic-systematic model of persuasion
Jack Williams Brehm	1928 to 2009	Reactance theory (1966); free-choice dissonance and spreading of alternatives (1956); trivialisation (1995)
Jonathan L. Freedman	1965	Forewarning increases resistance to persuasion (with Sears)
David O. Sears	1965	Forewarning study with Freedman
Charles G. Lord	1979	Biased assimilation and attitude polarisation (with Ross and Lepper)
Mark R. Lepper	1979	Biased assimilation (with Lord and Ross)
Linda Simon	1995	Trivialisation as dissonance reduction (with Greenberg and Brehm)
Claude Mason Steele	1988	Self-affirmation reduces dissonance
Clark Leonard Hull	1884 to 1952	Drive-reduction theory (Section 1.10)
James Merrill Carlsmith	1936 to 1984	$1 versus $20 induced-compliance study with Festinger (1959); forbidden toy with Aronson (1963)
Judson Mills	1959	Severe initiation study of effort justification (with Aronson)
Henry W. Riecken	1956	When Prophecy Fails, belief disconfirmation (with Festinger and Schachter)
Stanley Schachter	1922 to 1997	When Prophecy Fails (1956) with Festinger and Riecken
Jeff Stone	1994	Hypocrisy induction and condom buying (with Aronson and colleagues)
Daryl J. Bem	born 1938	Self-perception theory (1967, 1972), rival to dissonance
James T. Tedeschi	1971	Impression management theory of dissonance effects

Name	When	Examined for
Shinobu Kitayama	2004	Culture and dissonance: Japanese show post-decision dissonance when others' views are salient
Walter Lippmann	1889 to 1974	Public Opinion (1922): stereotypes as pictures in our heads
Kenneth Braly	1933	Katz and Braly checklist study of Princeton students
William Graham Sumner	1840 to 1910	Folkways (1906): ethnocentrism, in-group and out-group
Robert Neil Butler	1927 to 2010	Coined ageism (1969)
Theodor Adorno	1903 to 1969	The Authoritarian Personality (1950); F-scale
Else Frenkel-Brunswik	1908 to 1958	Co-author of The Authoritarian Personality (1950)
Daniel Levinson	1920 to 1994	Co-author of The Authoritarian Personality (1950)
Nevitt Sanford	1909 to 1995	Co-author of The Authoritarian Personality (1950)
Bob Altemeyer	1981	Right-wing authoritarianism: submission, aggression, conventionalism
Jim Sidanius	1945 to 2021	Social dominance orientation (with Felicia Pratto)
Felicia Pratto	1990s	Social dominance orientation (with Jim Sidanius)
Robert Sears	1908 to 1989	Hovland and Sears (1940) cotton prices and lynchings
Donald Thomas Campbell	1916 to 1996	Named realistic conflict theory (1965)
Samuel Stouffer	1900 to 1960	The American Soldier (1949): relative deprivation
Walter Garrison Runciman	1934 to 2020	Egoistic versus fraternal relative deprivation (1966)
Ted Robert Gurr	1936 to 2017	Why Men Rebel (1970): expectations versus attainments
Michael Billig, R. P. Bundy and Claude Flament	1971	Co-authors of the minimal group experiments with Tajfel
Paul Klee and Wassily Kandinsky	Painters	Their paintings were the basis of the minimal group assignment
Marilynn Brewer	1991	Optimal distinctiveness theory; decategorisation (1984, with Miller)
George Quattrone	1980	Out-group homogeneity effect (with Jones)
Edward Ellsworth Jones	1926 to 1993	Out-group homogeneity effect (Quattrone and Jones, 1980)
David Hamilton	1976	Distinctiveness-based illusory correlation (with Gifford)
Robert Gifford	1976	Illusory correlation (Hamilton and Gifford)
C. Neil Macrae	1994	Stereotypes as energy-saving devices (with Milne and Bodenhausen)
Alan Milne	1994	Stereotypes as energy-saving devices
Galen Bodenhausen	1994	Stereotypes as energy-saving devices
Renee Weber	1983	Bookkeeping, conversion and subtyping models (with Crocker)
Jennifer Crocker	1983	Models of stereotype change (with Weber)
Amy Cuddy	2002	Stereotype content model
Peter Glick	1996	Ambivalent sexism: hostile and benevolent (with Fiske); stereotype content model (2002)
Jun Xu	2002	Stereotype content model
Claude Steele	1995	Stereotype threat (with Joshua Aronson)
Joshua Aronson	1995	Stereotype threat (with Steele)
Steven Spencer	1999	Stereotype threat in women's mathematics performance (with Steele and Quinn)
Diane Quinn	1999	Stereotype threat and women's mathematics performance
Patricia Devine	1989	Dissociation model: automatic activation, controlled inhibition by low-prejudiced people
Anthony Greenwald	1998	Implicit Association Test (with McGhee and Schwartz)
Debbie McGhee	1998	Implicit Association Test
Jordan Schwartz	1998	Implicit Association Test
Russell Fazio	1995	Bona fide pipeline
John McConahay	1986	Modern racism and the Modern Racism Scale
David Sears	1980s	Symbolic racism
Samuel Gaertner	1986	Aversive racism (with Dovidio); common in-group identity model (2000)
John Dovidio	1986	Aversive racism; common in-group identity model (2000, with Gaertner)
Derald Wing Sue	2007	Microaggressions: taxonomy of everyday slights (with colleagues)
Chester Pierce	1970	Coined the term microaggression

Name	When	Examined for
Alice Eagly	1987	Social role theory of gender
Wendy Wood	1999	Social role theory with Eagly; reanalysis of Buss's mate preference data
David Buss	1989	Evolutionary account of sex differences in mate preferences (37 cultures)
Clea Mehta and JoNell Strough	2009	Sex segregation in friendships across the life span (a distractor for social role theory)
Craig Anderson and Brad Bushman	2002	General Aggression Model (Section 3.9); not linked to social role theory
Michelle Ryan	2005	Glass cliff (with Haslam)
Alexander Haslam	2005	Glass cliff (with Ryan)
Catherine White Berheide	1992	Sticky floor
Joan Williams	2000s	Maternal wall: bias against mothers
Christine Williams	1992	Glass escalator
Linda Tropp	2006	Meta-analysis of over 500 contact studies (with Pettigrew)
Stephen Wright	1997	Extended contact effect
Miles Hewstone	1986	Mutual intergroup differentiation (with Brown)
Rupert Brown	1986	Mutual intergroup differentiation (with Hewstone)
Adam Powell, Nyla Branscombe and Michael Schmitt	2005	Collective guilt framed as in-group advantage lowers racism
Kerry Kawakami	2000	Training to say no to stereotypes
Charles Stangor	2001	Social influence: learning peers are less prejudiced reduces stereotyping
Karla Hoff and Priyanka Pandey	2006	Caste identity made public lowered low-caste boys' performance (Uttar Pradesh)
Matt Lowe	2021	Caste-mixed cricket teams increased cross-caste friendships (Uttar Pradesh)
Ashutosh Varshney	2002	Hindu-Muslim civic associations and communal peace in Indian cities
Roy Baumeister	born 1953	The need to belong as a fundamental motive (with Mark Leary, 1995)
Mark Leary	1995	The need to belong (with Roy Baumeister)
Henry Murray	1893 to 1988	Listed the need for affiliation among human needs
David McClelland	1917 to 1998	Measured achievement, power and affiliation needs from TAT stories
Craig Hill	1987	Four affiliation motives: positive stimulation, emotional support, attention, social comparison
Kipling Williams	2000	Ostracism research; the Cyberball game
Donn Byrne	1970, 1971	Reinforcement-affect model (with Gerald Clore); law of attraction (proportion of similar attitudes)
Thomas Wills	1981	Downward social comparison as self-enhancement
Andrew Przybylski	2013	Defined and measured the fear of missing out (FOMO)
Kurt Back	1950	Westgate West housing study (with Festinger and Schachter)
Mady Segal	1974	Police trainees seated alphabetically befriended alphabetical neighbours
Robert Zajonc	1923 to 2008	Mere exposure effect (1968)
Richard Moreland	1992	Classroom exposure study (with Scott Beach)
Scott Beach	1992	Classroom exposure study (with Richard Moreland)
Theodore Mita	1977	Mirror-image face preference (with Dermer and Knight)
Karen Dion	1972	What is beautiful is good (with Berscheid and Walster)
Elaine Hatfield (Walster)	born 1937	Computer dance and matching hypothesis (1966); equity theory (1978); passionate and companionate love; Passionate Love Scale (1986)
Judith Langlois	1990	Averaged faces are attractive (with Lori Roggman)
Lori Roggman	1990	Averaged faces are attractive (with Judith Langlois)
Bernard Murstein	1972	Couples matched in physical attractiveness
Theodore Newcomb	1903 to 1984	Acquaintance study of transfer students; similarity breeds friendship (1961)
Milton Rosenbaum	1986	Repulsion hypothesis
Robert Winch	1958	Complementarity of needs in mate selection
Darwyn Linder	1965	Gain-loss effect (with Elliot Aronson)
George Homans	1910 to 1989	Social exchange theory (1961)

Name	When	Examined for
G. William Walster	1978	Equity theory (with Elaine Walster and Ellen Berscheid)
Caryl Rusbult	1952 to 2010	Investment model of commitment (1980); exit, voice, loyalty, neglect
Margaret Clark	1979	Communal versus exchange relationships (with Judson Mills)
Sidney Jourard	1926 to 1974	Self-disclosure; The Transparent Self (1964); Jourard Self-Disclosure Questionnaire
Irwin Altman	1973	Social penetration theory: breadth and depth (with Dalmas Taylor)
Dalmas Taylor	1973	Social penetration theory (with Irwin Altman)
T. L. Morton	1978	Descriptive versus evaluative self-disclosure
Susan Sprecher	1986	Passionate Love Scale (with Elaine Hatfield)
Robert Sternberg	born 1949	Triangular theory of love (1986): intimacy, passion, commitment
John Alan Lee	1973	Six love styles: eros, ludus, storge, pragma, mania, agape
Donald Dutton	1974	Suspension-bridge study of misattributed arousal (with Arthur Aron)
Arthur Aron	1974	Suspension-bridge study (with Donald Dutton)
Cindy Hazan	1987	Adult attachment styles: secure, avoidant, anxious-ambivalent (with Phillip Shaver)
Phillip Shaver	1987	Adult attachment styles (with Cindy Hazan)
Kim Bartholomew	1991	Four-category model: secure, preoccupied, dismissing, fearful (with Leonard Horowitz)
Leonard Horowitz	1991	Four-category attachment model (with Kim Bartholomew)
Usha Gupta	1982	Love in love versus arranged marriages in Jaipur (with Pushpa Singh)
Pushpa Singh	1982	Love in love versus arranged marriages in Jaipur (with Usha Gupta)
Steve Duck	1982	Phases of relationship dissolution: intrapsychic, dyadic, social, grave-dressing
John Gottman	born 1942	Four horsemen that predict divorce (Chapter 7)
Daniel Perlman	1982	Definition of loneliness (with Letitia Anne Peplau)
Robert Weiss	1973	Emotional versus social isolation (loneliness)
Daniel Russell	1978, 1980	UCLA Loneliness Scale
John Cacioppo	1951 to 2018	Loneliness and health
Herbert Kelman	1958	Three processes of influence: compliance, identification and internalisation
Bruce Tuckman	1965	Stages of group development (forming, storming, norming, performing); norming is a distractor for conformity
Robert Cialdini	1975 to 1990	Six principles of compliance in Influence (1984); door-in-the-face (1975) and low-ball (1978); descriptive and injunctive norms and the littering study (1990)
Raymond Reno	1990	Descriptive versus injunctive norms and focus theory, with Cialdini and Kallgren
Carl Kallgren	1990	Descriptive versus injunctive norms and focus theory, with Cialdini and Reno
Robert Jacobs	1961	Arbitrary autokinetic norm passed across generations, with Donald Campbell
Donald Campbell	1916 to 1996	Arbitrary autokinetic norm passed across generations, with Robert Jacobs (1961)
Richard Crutchfield	1955	Booth and light-panel apparatus for testing conformity without confederates
Rod Bond	1996	Meta-analysis of Asch-type studies with Peter Smith: more conformity in collectivist cultures, decline over time
Peter Smith	1996	Meta-analysis of Asch-type studies with Rod Bond
Morton Deutsch	1920 to 2017	Normative versus informational social influence, with Harold Gerard (1955)
Harold Gerard	1955	Normative versus informational social influence, with Morton Deutsch
Linda Carli	1981	Review with Alice Eagly on gender and influenceability
C. R. Snyder	1980	Need for uniqueness, with Howard Fromkin
Howard Fromkin	1980	Need for uniqueness, with C. R. Snyder

Name	When	Examined for
Elisabeth Lage	1969	Blue-green slide study of minority influence with Moscovici
Martine Naffrechoux	1969	Blue-green slide study of minority influence with Moscovici
Edwin Hollander	1958	Idiosyncrasy credits
Elaine Hatfield	1994	Emotional contagion, with Cacioppo and Rapson
Richard Rapson	1994	Emotional contagion, with Hatfield and Cacioppo
Jerry Burger	1986; 2009	That's-not-all technique (1986); partial replication of Milgram (2009)
Daniel Howard	1990	Foot-in-the-mouth technique
Michael Santos, Craig Leve and Anthony Pratkanis	1994	Pique technique
Jonathan Freedman	1966	Foot-in-the-door technique, with Scott Fraser ('Be a safe driver' sign)
Scott Fraser	1966	Foot-in-the-door technique, with Jonathan Freedman
Robert Jay Lifton	1961	Thought reform and the psychology of totalism; milieu control
Jim Jones	1931 to 1978	Cult leader of the Jonestown mass murder and suicide (1978)
Diana Baumrind	1964	Ethical critique of Milgram's obedience study
Hannah Arendt	1906 to 1975	Banality of evil (1963), from the Eichmann trial
Adolf Eichmann	1906 to 1962	Nazi official whose trial Arendt reported
Charles Hofling	1966	Hospital nurses study: 21 of 22 began to give an overdose ordered by phone
Craig Haney	1971	Co-researcher of the Stanford Prison Experiment
Curtis Banks	1971	Co-researcher of the Stanford Prison Experiment
Stephen Reicher	2006	BBC Prison Study with Alexander Haslam
John French	1959	Five bases of social power, with Bertram Raven
Bertram Raven	1959; 1965	Bases of social power with John French; added informational power (1965)
Mark H. Davis	1980	Interpersonal Reactivity Index: perspective taking, fantasy, empathic concern, personal distress
C. Daniel Batson	1981, 1991	Empathy-altruism hypothesis; the Elaine shock study with easy versus difficult escape; Good Samaritan study with Darley (1973)
K. D. Smith, J. P. Keating and Ezra Stotland	1989	Empathic joy hypothesis: helping for the pleasure of the victim's relief, which needs feedback
William D. Hamilton	1936 to 2000	Kin selection theory (1964)
Eugene Burnstein, Christian Crandall and Shinobu Kitayama	1994	More help to closer and younger kin, most clearly when life is at stake
Robert Trivers	born 1943	Reciprocal altruism (1971)
Charlie Hardy and Mark Van Vugt	2006	Competitive altruism: helping to gain status
Arie Nadler	1986, 2009	Defensive helping; threat-to-self-esteem model (with Fisher); dependency- versus autonomy-oriented help
Jeffrey Fisher	1986	Threat-to-self-esteem model of receiving help (with Nadler)
Alvin Gouldner	1920 to 1980	Norm of reciprocity (1960)
Leonard Berkowitz	1926 to 2016	Social responsibility norm (with Daniels, 1963)
L. R. Daniels	1963	Social responsibility norm (with Berkowitz)
Kitty Genovese	1935 to 1964	Murdered in Kew Gardens, Queens, on 13 March 1964; her case prompted bystander research
Bibb Latané	born 1937	Bystander effect with Darley: seizure and smoke-filled room studies (1968), The Unresponsive Bystander (1970), five-step decision model
John Darley	1938 to 2018	Bystander effect with Latané (1968, 1970); Good Samaritan study with Batson (1973)
Judith Rodin	born 1944	Lady in distress study with Latané (1969); New York subway study with the Piliavins (1969)
Martin Luther King Jr.	1929 to 1968	Wrong option: his assassination did not prompt bystander research
Emmett Till	1941 to 1955	Wrong option: his lynching did not prompt bystander research
Malala Yousafzai	born 1997	Wrong option: her shooting (2012) did not prompt bystander research
Peter Fischer	2011	Meta-analysis: the bystander effect is weaker in dangerous emergencies
Irving Piliavin	1969	New York subway study; arousal: cost-reward model

Name	When	Examined for
Jane Allyn Piliavin	1969, 1981	New York subway study; arousal: cost-reward model
Alice Isen	1942 to 2012	Feel good, do good: cookie and dime studies with Levin (1972)
Paula Levin	1972	Cookie and dime studies with Isen
James Bryan and Mary Ann Test	1967	Prosocial models: the flat tyre study
Tobias Greitemeyer and Silvia Osswald	2010	Prosocial video games increase helping
Jonathan Haidt	born 1963	Elevation: the uplift felt on witnessing kindness (2000)
Adam Grant and Francesca Gino	2010	Gratitude raises helpers' self-worth and self-efficacy; only self-worth carries later helping
Jean Twenge	2007	Social exclusion reduces helping
Chen-Bo Zhong	2010	Darkness and anonymity reduce prosocial behaviour
Sanford DeVoe and Jeffrey Pfeffer	2010	Thinking of time in money terms reduces volunteering
Hans Werner Bierhoff, Renate Klein and Peter Kramp	1991	Components of the altruistic personality: empathy, belief in a just world, social responsibility, internal locus of control, low egocentrism
Samuel Oliner and Pearl Oliner	1988	Rescuers of Jews in Nazi Europe; The Altruistic Personality
Allen Omoto and Mark Snyder	1995	Volunteer motives among AIDS volunteers
Deborah R. Richardson	1994	Definition of aggression with Baron
Nicki R. Crick	1995	Relational aggression (with Grotpeter); more used by girls
Jennifer K. Grotpeter	1995	Relational aggression with Crick
Konrad Lorenz	1903 to 1989	On Aggression (1966): fighting instinct, hydraulic model, ritualised fighting, appeasement gestures
Avshalom Caspi	2002	Low-activity MAOA gene plus childhood maltreatment predicts antisocial behaviour
John Dollard	1900 to 1980	Frustration-aggression hypothesis (1939), with Doob, Miller, Mowrer and Sears
Leonard W. Doob	1909 to 2000	Co-author of the frustration-aggression hypothesis (1939)
Neal E. Miller	1909 to 2002	Co-author of the frustration-aggression hypothesis (1939); revision (1941): aggression is only one response to frustration
O. H. Mowrer	1907 to 1982	Co-author of the frustration-aggression hypothesis (1939)
Robert R. Sears	1908 to 1989	Co-author of the frustration-aggression hypothesis (1939)
Anthony LePage	1967	Weapons effect study with Berkowitz
Dolf Zillmann	1971	Excitation transfer theory: leftover arousal intensifies later anger
Richard Lazarus	1922 to 2002	Appraisal and stress (Section 1.11); wrong option for the reformulation of the frustration-aggression hypothesis
Dorothea Ross	1961	Bobo doll study with Bandura and Sheila Ross
Sheila A. Ross	1961	Bobo doll study with Bandura and Dorothea Ross
L. Rowell Huesmann	1984 and 1986	Aggressive scripts; longitudinal media violence research with Eron
Craig A. Anderson	2002	General Aggression Model (with Bushman); violent video games; heat and violent crime
Brad J. Bushman	2002	General Aggression Model (with Anderson); venting study showing catharsis fails (2002); threatened egotism (1998)
Jean M. Twenge	2001	Social exclusion increases aggression
Leonard D. Eron	1972	Longitudinal study: TV violence viewed at 8 predicted aggression at 19 and serious crime by 30
Richard E. Nisbett	1996	Culture of honour (with Cohen)
Dov Cohen	1996	Culture of honour (with Nisbett)
Joseph A. Vandello	2008	Precarious manhood (with Bosson)
Jennifer K. Bosson	2008	Precarious manhood (with Vandello); threatened men punched harder (2009)
Kenneth A. Dodge	1980	Hostile attribution bias in aggressive children
Paul A. Bell	1992	Negative affect escape model of heat and aggression (with Baron)
Claude M. Steele	1990	Alcohol myopia (with Josephs)
Robert A. Josephs	1990	Alcohol myopia (with Steele)

Name	When	Examined for
Dan Olweus	1993	Bullying (Bullying at School): repeated aggression against a weaker victim; bullies, victims, bully-victims
Herbert Hyman	1942	Coined the reference group
Brian Lickel	2000	Intimacy groups, task groups, social categories and loose associations, ranked by entitativity
Kenneth Benne	1948	Task, maintenance and self-oriented roles in groups, with Paul Sheats
Paul Sheats	1948	Task, maintenance and self-oriented roles in groups, with Kenneth Benne
John Levine	1982	Group socialisation model with Richard Moreland
Fabio Sani	2005	Schism: groups splinter when members see a change as subverting the group's identity
Jacob Levy Moreno	1889 to 1974	Founded sociometry and the sociogram (Who Shall Survive? 1934); founded psychodrama
Helen Hall Jennings	1934	Moreno's collaborator on Who Shall Survive? and sociometry
Wilhelm Dilthey	1833 to 1911	Verstehen, interpretive understanding (a distractor for sociometry)
Max Weber	1864 to 1920	Verstehen method; charismatic authority
Mary Ann Jensen	1977	Added adjourning to Tuckman's stages
Connie Gersick	1988	Punctuated equilibrium model of group development
Alexander Heingartner	1969	Cockroach social facilitation study with Zajonc and Herman
Edward Herman	1969	Cockroach social facilitation study with Zajonc and Heingartner
Nickolas Cottrell	1968	Evaluation apprehension explanation of social facilitation
Robert S. Baron	1986	Distraction-conflict theory of social facilitation
Alan Ingham	1974	Pseudo-group rope pulling showed motivation loss, not only coordination loss
Stephen Harkins	1979	Social loafing clapping and shouting study
Steven Karau	1993	Collective effort model with Kipling Williams; social compensation (1991)
P. Christopher Earley	1989, 1993	Social striving: Chinese trainees worked harder in groups
Norbert Kerr	1983	Sucker effect; free-rider effect with Steven Bruun
Steven Bruun	1983	Free-rider effect with Norbert Kerr
Albert Pepitone	1952	Named deindividuation with Festinger and Newcomb
Edward Diener	1946 to 2021	Halloween trick-or-treater study of deindividuation (1976)
Russell Spears	1995	SIDE model of deindividuation effects
Tom Postmes	1995	SIDE model of deindividuation effects
James H. Davis	1973	Social decision schemes: majority wins, truth wins, first shift
James Stoner	1961	Discovered the risky shift
Marisa Zavalloni	1969	Group polarisation with Serge Moscovici
Irving Janis	1918 to 1990	Groupthink (Victims of Groupthink, 1972; revised 1982): eight symptoms and remedies
Garold Stasser	1985	Hidden profiles and biased information sampling, with William Titus
William Titus	1985	Hidden profile study with Garold Stasser
Alex Osborn	1888 to 1966	Popularised brainstorming
Garrett Hardin	1915 to 2003	Tragedy of the commons (1968)
Charles Osgood	1916 to 1991	GRIT, graduated and reciprocated initiatives in tension reduction (1962)
J. Stacy Adams	1965	Equity theory of distributive justice
Ralph Stogdill	1948	Review showing traits weakly predict leadership
Ralph White	1939	Leadership styles study with Lewin and Lippitt
Robert Blake	1964	Managerial grid with Jane Mouton
Jane Mouton	1964	Managerial grid with Robert Blake
Paul Hersey	1969	Situational leadership with Kenneth Blanchard: telling, selling, participating, delegating
Kenneth Blanchard	1969	Situational leadership with Paul Hersey
Robert House	1971, 1977	Path-goal theory (1971); charismatic leadership theory (1977)
Victor Vroom	1973	Normative decision model with Philip Yetton

Name	When	Examined for
Philip Yetton	1973	Normative decision model with Victor Vroom
George Graen	1975	Leader-member exchange (LMX)
James MacGregor Burns	1918 to 2014	Transactional versus transforming leadership (Leadership, 1978)
Bernard Bass	1925 to 2007	Transformational leadership and its four components (1985)
Robert Greenleaf	1904 to 1990	Servant leadership (1970)
J. B. P. Sinha	1980	Nurturant-task leader, an Indian leadership model
Edward Chace Tolman	1886 to 1959	Cognitive map (distractor for the sociogram; Section 1.6)
Frederick Herzberg	1923 to 2000	Work motivation (Section 1.10)
Sheldon Cohen	1985	Stress-buffering hypothesis of social support (with Wills)
Thomas Ashby Wills	1985	Stress-buffering versus main-effect models of social support (with Cohen)
Daniel Stokols	1972	Density (physical) versus crowding (psychological)
John B. Calhoun	1917 to 1995	Crowded rat colonies and the behavioural sink (1962)
Ed Diener	1946 to 2021	Subjective well-being: life satisfaction, frequent positive and infrequent negative feelings
Marcia Johnson	1993	Source monitoring framework (with colleagues)
Donald Thomson	Twentieth century	Australian psychologist wrongly identified by a victim who had seen him on live television: a source error
Elizabeth Loftus	born 1944	Misinformation effect ('smashed' versus 'hit', with Palmer, 1974); weapon focus
John Palmer	1974	Loftus and Palmer study of leading questions and memory
Gary Wells	1998	Post-identification feedback effect; lineup reform
Ronald Fisher	1984	Cognitive interview (with Geiselman); not the statistician
Edward Geiselman	1984	Cognitive interview (with Fisher)
Ronald Aylmer Fisher	1890 to 1962	Statistician; a name trap for the cognitive interview
Nancy Pennington	1986	Story model of juror decision making (with Hastie)
Reid Hastie	1986	Story model of juror decision making (with Pennington)
Robert Ressler	1980s	FBI organised versus disorganised offender typology
John Douglas	1980s	FBI offender profiling (with Ressler)
David Canter	1986	Investigative psychology; bottom-up, statistical offender profiling
Robert Martinson	1927 to 1979	'What works?' (1974), read as 'nothing works' in offender rehabilitation
Don Andrews	1990	Risk-need-responsivity principles of offender rehabilitation (with Bonta)
James Bonta	1990	Risk-need-responsivity principles (with Andrews)
Samuel Barnett	1844 to 1913	Toynbee Hall, London, the first settlement house (1884)
Jane Addams	1860 to 1935	Hull House, Chicago (1889, with Ellen Gates Starr)
Ellen Gates Starr	1859 to 1940	Co-founder of Hull House (1889)
Clifford Manshardt	1936	First director of the Sir Dorabji Tata Graduate School of Social Work, Bombay
Mary Ellen Richmond	1861 to 1928	Founder of social case work: Social Diagnosis (1917), What Is Social Case Work? (1922)
Otto Rank	1884 to 1939	Will therapy, the basis of the functional school of case work
Gordon Hamilton	1940	Diagnostic school of case work (Theory and Practice of Social Case Work)
Florence Hollis	1964	Psychosocial approach; the person-in-situation
Charlotte Towle	1945	Common Human Needs
Jessie Taft	1930s	Functional (Rankian) school of case work, Pennsylvania
Virginia Robinson	1930s	Functional (Rankian) school of case work, Pennsylvania
Helen Harris Perlman	1957	Problem-solving model of case work; the four Ps
William Reid	1972	Task-centred casework (with Laura Epstein)
Laura Epstein	1972	Task-centred casework (with William Reid)
Felix Biestek	1957	Seven principles of the casework relationship
Joel Fischer	1973	'Is casework effective?': no good evidence that case work worked
Irwin Epstein	Contemporary	Social work research (a distractor for Fischer)

Name	When	Examined for
John Goyder	Contemporary	Survey nonresponse research (a distractor for Fischer)
Jacob Cohen	1923 to 1998	Statistician of kappa, effect size and power (a distractor for Fischer)
Grace Longwell Coyle	1892 to 1962	Established group work as a method in social work; Social Process in Organized Groups (1930)
Gisela Konopka	1910 to 2003	Group work in treatment settings
Harleigh B. Trecker	Mid-twentieth century	Ten principles of social group work
Robert Vinter	1960s	Remedial model of group work
William Schwartz	1960s	Reciprocal (mediating) model of group work
Ronald Toseland	Contemporary	Formed versus natural groups (with Rivas)
Robert Rivas	Contemporary	Formed versus natural groups (with Toseland)
Lawrence Shulman	Contemporary	Member roles in groups: scapegoat, deviant, gatekeeper, internal leader, defensive, quiet, monopoliser
James Garland	1965	Boston model of group development (with Jones and Kolodny)
Hubert Jones	1965	Boston model of group development
Ralph Kolodny	1965	Boston model of group development
Murray G. Ross	1955	Classic definition of community organisation
Jack Rothman	1968	Three models of community organisation: locality development, social planning, social action
Saul Alinsky	1909 to 1972	Social action organising; Rules for Radicals (1971)
Geert Hofstede	1928 to 2020	Cultural dimensions (1980): power distance, individualism-collectivism, masculinity-femininity, uncertainty avoidance
Wynetta Devore	1981	Ethnic-sensitive practice (with Schlesinger)
Elfriede Schlesinger	1981	Ethnic-sensitive practice (with Devore)
Georges Devereux	1908 to 1985	Ethno-psychotherapy (ethnopsychiatry)
James Spradley	1979	The ethnographic interview
James Leigh	1998	Ethnographic interviewing in social work: the client as cultural guide
Dennis Saleebey	1992	The strengths perspective in social work
Ann Hartman	1978	The ecomap

Key terms

social psychology, person-situation interaction, Psychological social psychology, sociological social psychology, life space, group dynamics, action research, T-group, replication crisis, Social facts, collective representations, anomie, symbolic interactionism, Socialisation, anticipatory socialisation, Resocialisation, total institutions, institutionalisation, status, role, Role taking, Role playing, Role conflict, Role strain, Role set, social representation, anchoring, objectification, experimental realism, mundane realism, mediating variable, moderating variable, debriefing, demand characteristics, Social cognition, cognitive misers, motivated tactician, Automatic processing, controlled processing, schema approach, schema, script, Priming, unpriming, belief perseverance, self-fulfilling prophecy, Pygmalion effect, Golem effect, behavioural confirmation, status quo heuristic, Negativity bias, Optimistic bias, Planning fallacy, Counterfactual thinking, Magical thinking, False consensus effect, Illusion of control, Terror management theory, affect infusion model, central traits, implicit personality theories, halo effect, thin slicing, impression management, dramaturgy, Self-monitoring, self-handicapping, spotlight effect, slime effect, Communication, encoding, decoding, feedback, noise, SMCR model, communication networks, Nonverbal communication, display rules, kinesics, Paralanguage, proxemics, high-context cultures, Microexpressions, Interchannel discrepancies, Attribution, naive psychology, dispositional attribution, situational attribution, correspondent inference theory, non-common effects, covariation model, Consensus,

Consistency, Distinctiveness, discounting principle, augmenting principle, fundamental attribution error, correspondence bias, actor-observer effect, self-serving bias, just-world hypothesis, Ultimate attribution error, Hostile attribution bias, attribution retraining, attitude, attitude object, tricomponent model, Attitude strength, implicit attitudes, functions of attitudes, subliminal conditioning, vested interest, principle of compatibility, theory of reasoned action, subjective norm, perceived behavioural control, theory of planned behaviour, attitude-to-behaviour process model, MODE model, Persuasion, Yale communication approach, receiver characteristics, Need for cognition, sleeper effect, elaboration likelihood model, central route, peripheral route, heuristic-systematic model, reactance, Forewarning, inoculation, attitude polarisation, balance theory, congruity theory, cognitive dissonance, trivialisation, self-affirmation, induced compliance, self-perception theory, stereotype, Prejudice, Discrimination, in-group, ethnocentrism, Allport's scale, casteism, authoritarian personality, Scapegoat theory, realistic conflict theory, relative deprivation, superordinate goals, minimal group paradigm, social identity theory, positive distinctiveness, self-categorisation theory, optimal distinctiveness theory, out-group homogeneity effect, illusory correlation, subtyping, stereotype content model, stereotype threat, Implicit Association Test, bona fide pipeline, Modern racism, Aversive racism, Ambivalent sexism, social role theory, glass ceiling, Glass cliff, Sticky floor, contact hypothesis, jigsaw classroom, need to belong, need for affiliation, gregariousness, Ostracism, reinforcement-affect model, social comparison theory, Upward social comparison, downward social comparison, self-evaluation maintenance model, fear of missing out, propinquity effect, mere exposure effect, what is beautiful is good, matching hypothesis, law of attraction, complementarity of needs, gain-loss effect, Social exchange theory, comparison level, comparison level for alternatives, equity theory, investment model, communal relationships, self-disclosure, social penetration theory, descriptive self-disclosure, evaluative self-disclosure, passionate love, companionate love, triangular theory of love, two-factor theory of love, loneliness, Emotional isolation, Social isolation, UCLA Loneliness Scale, Social influence, Conformity, Compliance, Obedience, symbolic social influence, identification, internalisation, public compliance, private acceptance, Social norms, descriptive norms, injunctive norms, focus theory of normative conduct, Prescriptive norms, proscriptive norms, perceived norm, autokinetic effect, Normative social influence, Informational social influence, cohesiveness, need for uniqueness, conversion theory, idiosyncrasy credits, emotional contagion, reciprocity, foot-in-the-door technique, door-in-the-face technique, low-ball technique, love-bombing, agentic state, banality of evil, Social power, prosocial behaviour, helping, altruism, cooperation, psychological egoism, empathy, affective empathy, cognitive empathy, empathic concern, sympathy, empathy-altruism hypothesis, negative-state relief model, empathic joy hypothesis, kin selection, reciprocal altruism, competitive altruism, defensive helping, norm of reciprocity, social responsibility norm, personal norm, bystander effect, diffusion of responsibility, pluralistic ignorance, audience inhibition, arousal: cost-reward model, urban overload, feel good, do good effect, prosocial model, elevation, social exclusion, altruistic personality, threat-to-self-esteem model, dependency-oriented help, autonomy-oriented help, aggression, hostile aggression, instrumental aggression, relational aggression, death instinct, hydraulic model, ritualised fighting, appeasement gestures, serotonin, MAOA, frustration-aggression hypothesis, displacement, scapegoating, cognitive neoassociationist model, aggressive cues, weapons effect, excitation transfer theory, observational learning, modelling, aggressive scripts, General Aggression Model, provocation, culture of honour, precarious manhood, threatened egotism, desensitisation, hot temperature effect, negative affect escape model, alcohol myopia, bullying, cyberbullying, catharsis hypothesis, incompatible response technique, self-regulation, group, Entitativity, group socialisation, schism, sociometry, sociometric test, sociogram, forming, storming, norming, performing, adjourning, punctuated equilibrium, social facilitation, drive theory, evaluation apprehension, Ringelmann effect, social loafing, social

striving, deindividuation, SIDE model, risky shift, group polarisation, Groupthink, hidden profile, production blocking, Delphi technique, tragedy of the commons, GRIT, great-person theory, contingency model, least preferred co-worker, Transformational leadership, nurturant-task leader, Social support, stress-buffering hypothesis, crowding, behavioural sink, territoriality, subjective well-being, Source monitoring, post-identification feedback effect, weapon focus, own-race bias, cognitive interview, story model, Offender profiling, investigative psychology, risk-need-responsivity, person-in-environment, Charity Organisation Society, Direct practice, Social case work, social diagnosis, diagnostic school, functional school, formed groups, natural groups, gatekeeper, Boston model, Community organisation, individualism-collectivism, Ethnic-sensitive practice, Cultural competence, ethnographic interview, strengths perspective, ecomap, genogram, neglect

From the past papers

1. Who is credited with elaborating the "social representation theory"? (RML 2022)
   (a) Serge Moscovici (b) Michael A. Hogg (c) Roy F. Baumeister (d) Mark E. Leary
   Answer: (a) Serge Moscovici. Moscovici launched social representations theory in 1961 with his study of how psychoanalysis became French common sense, through anchoring and objectification. Hogg is known for social identity and uncertainty-identity theory; Baumeister and Mark R. Leary for the need to belong (1995), and Leary for the sociometer theory of self-esteem.

2. Which one refers to the behaviour that an individual engages in while enacting a role? (RML 2020)
   (a) Role taking (b) Role playing (c) Playing at a role (d) Role expectation
   Answer: (b) Role playing. Role playing (role enactment) is the actual behaviour of the person performing a role. Role taking is Mead's mental act of seeing a situation from another's viewpoint, playing at a role is a young child's make-believe, and role expectation is what other people expect of the occupant.

3. Some students experience an intense period of adaptation when they leave their old school environment and start college, especially during those initial days before they find companions, start to fit in and feel comfortable. This process is best described as: (RML 2021)
   (a) Looking-glass self (b) Self-awareness (c) Institutionalisation (d) Re-socialisation
   Answer: (d) Re-socialisation. Resocialisation is unlearning old norms and roles and learning new ones after a major transition such as starting college. The looking-glass self is Cooley's self-image built from others' imagined judgements, self-awareness is attention turned on oneself, and institutionalisation is the apathetic dependence of long-stay inmates.

4. A tendency to act in a way that is consistent with one's own belief, expectation, or cognition about an event or behavior, thereby increasing the likelihood that the event or behavior will occur, is termed as: (IHBAS 2020)
   (a) Self-fulfilling prophecy (b) Confirmation bias (c) Motivational reaction (d) Base-rate fallacy
   Answer: (a) Self-fulfilling prophecy. Acting on an expectation in ways that make it come true is Merton's self-fulfilling prophecy (1948). Confirmation bias only seeks evidence that fits a belief and changes nothing in the world, the base-rate fallacy ignores how common something is, and 'motivational reaction' is not a term.

5. What is the halo effect? (IHBAS 2024)

   (a) The tendency to judge others based on their group membership (b) The influence of a positive trait on the perception of other traits (c) The tendency to overlook negative behaviors in individuals (d) The effect of initial impressions on subsequent judgments

   Answer: (b) The influence of a positive trait on the perception of other traits. Thorndike (1920) named the halo effect: one salient positive trait or an overall impression colours ratings of unrelated traits. Judging by group membership is stereotyping, the effect of initial impressions is the primacy effect (or anchoring), and overlooking faults is a consequence of a halo, not its definition.

6. A teacher watches two students interact for less than a minute and correctly infers who is the class leader. This example best illustrates: (IHBAS 2025)

   (a) Social facilitation (b) Self-fulfilling prophecy (c) Thin slicing (d) Fundamental attribution error

   Answer: (c) Thin slicing. Ambady and Rosenthal showed that very brief samples of behaviour (silent clips of 30 seconds or less) yield accurate judgements. Social facilitation is better performance of well-learned tasks before others, the teacher's expectation did not create the leadership, and the fundamental attribution error over-explains behaviour by disposition.

7. Arrange the process of communication in sequence (RML 2020)

   (a) coding, decoding, recipient, channel (b) channel, encoding, recipient, decoding (c) channel, decoding, recipient, encoding (d) encoding, channel, recipient, decoding

   Answer: (d) encoding, channel, recipient, decoding. The sender first encodes the idea into symbols, the message travels through a channel to the recipient, who then decodes it. The other options send a message through the channel before it is encoded or decode it before it is received.

8. The tendency of people to attribute their own behavior to external causes and other's behavior to internal causes is referred to as : (IHBAS 2024)

   (a) Self-serving bias (b) Fundamental attribution error (c) Actor-Observer effect (d) Stereotyping

   Answer: (c) Actor-Observer effect. Jones and Nisbett (1971): as actors we explain our own behaviour by the situation, as observers we explain others' by their dispositions. The fundamental attribution error concerns only other people's behaviour, the self-serving bias concerns our own successes and failures, and stereotyping is a generalised belief about a group.

9. The "just-world hypothesis" explains victim-blaming as a result of: (Real PYQ 2024)

   (a) People's inherent sadism and desire to see others suffer. (b) The need to believe that the world is orderly, predictable, and fair. (c) A fundamental error in attributing behavior to dispositional factors. (d) The process of social comparison with a less fortunate other.

   Answer: (b) The need to believe that the world is orderly, predictable, and fair. For Melvin Lerner, an innocent person's suffering threatens this belief, so observers restore it by deciding the victim deserved it. Observers are distressed rather than sadistic, the dispositional blame is the result rather than the cause, and downward social comparison is a different process.

10. Arvind is a college student who has strong negative feelings about smoking cigarettes. His grandfather died of lung cancer and he believes smoking is disgusting. One night at a party, Arvind's friends all go outside for a cigarette and he decides to smoke as well. What will he most likely experience as a result? (RML 2021)

    (a) Justification of efforts (b) Increased drive to smoke (c) Cognitive Dissonance (d) Drive Reduction

    Answer: (c) Cognitive dissonance. Acting against a strong, important attitude creates Festinger's aversive tension. Effort justification is a different paradigm (valuing what cost effort), nothing suggests a stronger urge to smoke, and drive reduction is Hull's motivation theory.

11. In which of the following scenarios would it be most difficult to predict behaviour from attitudes? (RML 2021)

    (a) When the attitude has a great deal of personal relevance for the individual (b) When the attitude is formed by indirect experience (c) When the individual has a lot of knowledge about the attitude object (d) When the attitude is formed by the individual's direct experience

    Answer: (b) When the attitude is formed by indirect experience. Hearsay-based attitudes are weak and less accessible. Personal relevance, knowledge of the object and direct experience (Regan and Fazio, 1977) all strengthen the attitude-behaviour link.

12. In Social Psychology, reactance is : (IHBAS 2024)

    (a) The tendency to comply with direct requests (b) The motivation to restore freedom when it is threatened (c) The effect of persuasive messages on attitude change (d) The tendency to follow social norms without question

    Answer: (b) The motivation to restore freedom when it is threatened. Brehm (1966) defined reactance this way; it often makes the forbidden option more attractive. Complying with requests is compliance, following norms is conformity, and message-driven attitude change is persuasion.

13. According to social identity theory, prejudice arises primarily due to : (IHBAS 2025)

    (a) Competition for scarce resources (b) Cognitive dissonance (c) Categorization into in-groups and out-groups (d) Social facilitation

    Answer: (c) Categorization into in-groups and out-groups. Tajfel's minimal groups showed favouritism with no resources at stake, so categorisation plus the need for positive distinctiveness is enough. Competition for scarce resources is Sherif's realistic conflict theory; cognitive dissonance (Festinger) and social facilitation (performance in the presence of others) are unrelated.

14. Low-prejudiced people engage in: (IHBAS 2020)

    (a) Automatic information processing resulting in involuntary stereotyping (b) Controlled information processing resulting in refuting stereotype (c) Controlled information processing resulting in involuntary stereotyping (d) Automatic information processing resulting in voluntary stereotyping

    Answer: (b) Controlled information processing resulting in refuting stereotype. Devine (1989) found stereotypes are activated automatically in high- and low-prejudiced people alike; low-prejudiced people then use controlled processing to inhibit them. Option (a) is the first stage common to all; (c) and (d) pair the processes wrongly.

15. The phenomenon where a qualified individual is prevented from advancing to higher positions in their career due to an invisible barrier based on gender or minority status is called: (Real PYQ 2024)

    (a) The Matthew Effect (b) The Glass Ceiling Effect (c) The Halo Effect (d) The Pygmalion Effect

    Answer: (b) The Glass Ceiling Effect. The glass ceiling is an invisible barrier of attitudes and organisational bias near the top, popularised by a 1986 Wall Street Journal article. The Matthew effect is accumulated advantage (Merton, 1968); halo and Pygmalion effects concern impressions and expectations, not career barriers.

16. 'The more we see and interact with a person, the more likely he or she is to become our friend or sexual partner.' This denotes: (IHBAS 2020)

    (a) Propinquity effect (b) Exposure effect (c) Matching hypothesis (d) Reciprocity

    Answer: (a) Propinquity effect. Nearness and frequent interaction breed friendship (Festinger, Schachter and Back 1950). The exposure effect is narrower (mere repeated exposure to any stimulus raises liking) and is its mechanism; matching is choosing partners of similar attractiveness; reciprocity is liking those who like us.

17. Emotional isolation means: (IHBAS 2020)

    (a) Feeling a lack of deep emotional attachment to one specific person (b) Suffering from a lack of friends/associates (c) Suffering from feelings of incompetence (d) Feeling a sense of inadequacy and low self-esteem

    Answer: (a) Lack of deep attachment to one specific person. Robert Weiss (1973) separated emotional isolation from social isolation, the lack of a wider network of friends and associates (option b). Incompetence and low self-esteem are correlates of loneliness, not its types.

18. Which concept is closely related to FOMO (Fear of Missing Out) in social media research? (JEMAS 2025)

    (a) Social comparison (b) Classical conditioning (c) Operant extinction (d) Cognitive dissonance only

    Answer: (a) Social comparison. FOMO (Przybylski and colleagues 2013) arises from constant upward comparison with others' curated experiences, in the tradition of Festinger's social comparison theory (1954). Conditioning, extinction and dissonance are learning and consistency processes, not comparison.

19. Which of the following is an example of the power of roles? (IHBAS 2024)

    (a) Robber's Cave Experiment (b) Milgram's message experiment (c) Zilstein's shock research (d) Zimbardo's prison experiment

    Answer: (d) Zimbardo's prison experiment. Students randomly made guards or prisoners took on their roles so fully that the study was stopped after six days. Robbers Cave studied intergroup conflict, Milgram's message experiment was the small-world study, and Dr Zilstein was the fictitious experimenter of Schachter's affiliation study.

20. What is normative social influence? (IHBAS 2024)

    (a) Influence based on the desire to be correct and accurate (b) Influence based on the desire to be accepted and liked by others (c) Influence based on personal beliefs and values (d) Influence based on perceived power and authority

    Answer: (b) The desire to be accepted and liked. Deutsch and Gerard (1955) contrasted it with informational influence, the desire to be correct (option a). Personal values describe internalisation, and power and authority describe obedience.

21. According to Cialdini, techniques for gaining compliance rest to a large extent on six basic principles. These do NOT include: (RML 2022)

    (a) Scarcity (b) Social validation (c) Friendship/liking (d) Reward

    Answer: (d) Reward. The six principles are reciprocity, commitment and consistency, social validation (social proof), friendship or liking, authority and scarcity.

22. The bystander effect, where the presence of others inhibits helping, is primarily explained by two factors: (Real PYQ 2024)

    (a) Social loafing and deindividuation (b) Diffusion of responsibility and pluralistic ignorance (c) Social comparison and normative influence (d) Group polarization and groupthink

    Answer: (b) Diffusion of responsibility and pluralistic ignorance. These are Latané and Darley's two classic mechanisms: each bystander assumes someone else will act, and each reads the others' calm as a sign that nothing is wrong. The other pairs are group phenomena of other kinds (Sections 3.6, 3.7 and 3.10).

23. A state that occurs when bystanders in an emergency situation use the behaviour of others to determine whether help is actually required is called: (IHBAS 2020)

    (a) Pluralistic ignorance (b) Group conformity (c) Group compliance (d) Crowd behavior

    Answer: (a) Pluralistic ignorance. Bystanders look to one another to decide whether help is needed and, seeing calm faces, wrongly conclude it is not. Conformity and compliance are yielding to group pressure or to a direct request, and crowd behaviour is the collective action of a large gathering.

24. Altruistic Personality is characterized by: (IHBAS 2020)

    (a) Offering help only when one feels like (b) Consistently helping others (c) Calculating the cost and benefits of helping (d) Readiness to help in an emergency

    Answer: (b) Consistently helping others. The altruistic personality is a stable tendency to help across situations; Bierhoff, Klein and Kramp (1991) found its components to be empathy, belief in a just world, social responsibility, internal locus of control and low egocentrism. Helping only when in the mood, cost-benefit calculation and readiness only in emergencies describe situational helping.

25. Who reformulated the "frustration-aggression hypothesis"? (RML 2022)

    (a) Dollard (b) Miller (c) Berkowitz (d) Lazarus

    Answer: (c) Berkowitz. Leonard Berkowitz (1989) recast it: frustration leads to aggression only to the extent that it produces negative affect, with aggressive cues tipping the balance. Dollard and colleagues proposed the original (1939), Miller only softened it (1941), and Richard Lazarus is known for appraisal and stress.

26. Albert Bandura's Bobo doll experiment is a classic demonstration of: (RML Practice 1)

    (a) Operant conditioning through direct reinforcement (b) Observational learning of aggression through modeling (c) Classical conditioning of fear (d) The effects of sensory deprivation

    Answer: (b) Observational learning of aggression through modelling. Children copied an aggressive adult's specific acts without ever being reinforced, so it is not operant conditioning; classical conditioning of fear is Little Albert, and sensory deprivation is the McGill isolation research.

27. Low levels of which neurotransmitter leads to aggression (Entrance Mock 1)

    (a) Serotonin (b) Dopamine (c) Epinephrine (d) Norepinephrine

    Answer: (a) Serotonin. Low serotonin activity (low 5-HIAA in cerebrospinal fluid) is linked to impulsive aggression. Dopamine concerns reward and movement, while epinephrine and norepinephrine mark arousal in any strong emotion.

28. Which of the following is not a stage of group formation? (JEMAS 2025)

    (a) Storming (b) Proximity (c) Norming (d) Performing

    Answer: (b) Proximity. Tuckman's stages are forming, storming, norming, performing and adjourning. Proximity is a cause of interpersonal attraction, not a stage of group development.

29. Tim would like to participate in his high school's musical. He has been taking vocal lessons for years and is an excellent singer, but does not have much experience as a dancer. The director gives him a part where he has to sing and dance. Based on the theory of social facilitation, how will the presence of an audience affect Tim's performance? (RML 2021)

    (a) His dancing will improve, but his singing will be impaired (b) Both his singing and dancing will improve (c) His singing will improve, but his dancing will be impaired (d) Both his singing and dancing will be impaired

    Answer: (c) His singing will improve, but his dancing will be impaired. By Zajonc's drive theory, an audience raises arousal, which strengthens the dominant response: correct on a well-learned task (singing), often wrong on an unpractised one (dancing). Options (b) and (d) ignore that the effect depends on how well learned the task is.

30. A group of scientists is deciding on the use of live animals in research; most of the group has used live animals. How might the leader reduce groupthink? (RML 2021)

    (a) Make a cohesive team-building exercise a mandatory part of the meeting (b) Relate his opinion at the beginning of the discussion (c) Ask a subset of individuals in the group to critique animal research (d) Ask the group members not to argue with one another's opinions

    Answer: (c) Ask a subset to critique animal research. Janis's remedies include a devil's advocate or critical subgroup. Team-building raises cohesion, a leader stating views first steers the group, and forbidding argument suppresses dissent, so (a), (b) and (d) all worsen groupthink.

31. Defense attorneys often protest the prosecutor's use of eyewitness testimony because: (Amity 2023)

    (a) Eyewitnesses are more prone to forgetting (b) A confident witness is usually right (c) Source errors may cause an eyewitness to make a false identification (d) Poor eyesight of eyewitnesses tends to invalidate their testimony

    Answer: (c) Source errors may cause an eyewitness to make a false identification. A face met elsewhere (a mugshot, the news, a bystander) is misattributed to the crime. Witnesses are not unusually forgetful, confidence predicts accuracy only weakly and grows after feedback, and eyesight is rarely the issue.

32. Which of the following is not a part of the social case work process? (RML 2020)

    (a) Content analysis (b) Treatment (c) Assessment (d) Evaluation

    Answer: (a) Content analysis. Case work runs intake and study, assessment (social diagnosis), treatment or intervention, evaluation, termination and follow-up. Content analysis, coding themes in texts, is a research technique.

33. Who among the following established group work as a method within the profession of social work? (RML 2020)

    (a) Mary Richmond (b) Charlotte Towle (c) Grace Coyle (d) Geisla Konopka

    Answer: (c) Grace Coyle. Her *Social Process in Organized Groups* (1930) founded group work as a method. Richmond founded case work, Towle wrote *Common Human Needs*, and Konopka carried group work into treatment settings later.

High-yield sheet

The facts this chapter's questions ask most, most-asked first. Last read before the exam.

1 The **fundamental attribution error** (named by Lee Ross, 1977), also called the **correspondence bias**, is the tendency to overestimate dispositional and underestimate situational causes of *other people's* behaviour. ASKED 6 TIMES

2 **Leon Festinger** proposed **cognitive dissonance** theory (1957): holding inconsistent cognitions, or acting against one's attitude, produces aversive tension that people reduce, usually by changing the attitude when the behaviour cannot be undone. ASKED 6 TIMES

3 **Diffusion of responsibility** is the main explanation of the bystander effect: responsibility spreads across all the bystanders, so each feels less personally obliged and assumes someone else will act (Latané and Darley). ASKED 6 TIMES

4 **Social loafing**, also called the Ringelmann effect, is reduced effort when individual efforts are pooled (Latané, Williams and Harkins 1979); it falls when each contribution is identifiable and evaluated, so a teacher should grade both the group and each member. Its collectivist reversal is **social striving** (Earley). ASKED 6 TIMES

5 **Social identity theory** (Henri Tajfel and John Turner, 1979): a large part of our self-concept and self-esteem comes from the groups we belong to, and prejudice arises from categorisation into in-groups and out-groups and the need for positive distinctiveness, not from competition for resources. ASKED 5 TIMES

6 **Solomon Asch** (1951, 1956): among about seven confederates unanimously giving an obviously wrong line judgement, participants conformed on about 37 percent of critical trials and about three quarters conformed at least once; this is normative influence towards an incorrect majority, not obedience. ASKED 5 TIMES

7 **Pluralistic ignorance**: in an ambiguous emergency each bystander uses the calm behaviour of the others to decide whether help is needed, and mistakenly concludes that it is not (the smoke-filled room). ASKED 5 TIMES

8 **Bruce Tuckman** (1965) ordered group development as forming, storming (conflict over leadership and the group's definition), norming (cohesion; purpose and strategy clarified) and performing; adjourning was added with Mary Ann Jensen in 1977. Proximity is not a stage. ASKED 5 TIMES

9 **Social facilitation** began with Norman Triplett's 1898 fishing-reel study; Zajonc's drive theory (1965) says the presence of others raises arousal and strengthens the dominant response, improving easy, well-learned tasks and impairing new or difficult ones. ASKED 4 TIMES

10 The **halo effect** (Edward Thorndike, 1920): an overall impression or one positive trait influences ratings of a person's other, unrelated traits; the negative version is the horns effect. It is not primacy, stereotyping, self-serving bias or false consensus. ASKED 3 TIMES

11 **Normative social influence** rests on the desire to be liked and accepted; **informational social influence** on the desire to be correct (Deutsch and Gerard, 1955). Sherif's autokinetic study shows informational influence and the formation of group norms. ASKED 3 TIMES

12 The **bystander effect**: the more bystanders present, the less likely any one is to help and the slower the help; people help more when alone (seizure study: 85 percent alone versus 31 percent with four others). ASKED 3 TIMES

13 A **sociogram**, the tool of Jacob Moreno's sociometry (1934), maps choices and rejections to show social distance and patterns of choice in a group (stars, isolates, cliques); it captures implicit and explicit interaction patterns and does not measure IQ, personality or economic status. ASKED 3 TIMES

14 **Groupthink** (Irving Janis 1972) is a mode of thinking in highly cohesive groups where the drive for unanimity overrides realistic appraisal, discouraging dissent and producing poor decisions; a devil's advocate or critical subgroup and an impartial leader reduce it. ASKED 3 TIMES

15 **Kurt Lewin** (1890 to 1947), father of modern social psychology: in B = f(P, E) the **life space** is the person plus the psychological environment, and he founded **action research** (1946) and group dynamics. ASKED 2 TIMES

16 The **self-fulfilling prophecy** was named and formalised by the sociologist **Robert K. Merton** (1948): a false definition of a situation evokes behaviour that makes it come true. It is not confirmation bias or the base-rate fallacy. ASKED 2 TIMES

17 Melvin Lerner's **just-world hypothesis** explains victim blaming by the need to believe that the world is orderly, predictable and fair. ASKED 2 TIMES

18 In the **elaboration likelihood model** (Petty and Cacioppo), systematic processing with careful attention to argument quality is the **central route**; reliance on mental shortcuts and cues is the **peripheral route**. There is no emotional or situational route. ASKED 2 TIMES

19 **Reactance** (Jack Brehm, 1966) is the motivation to restore a threatened freedom, often by wanting the forbidden behaviour more. ASKED 2 TIMES

20 The **glass ceiling** is an invisible barrier that stops qualified women and minorities from rising to top positions; the term was popularised by a 1986 Wall Street Journal article. ASKED 2 TIMES

21 Intergroup hostility falls with cooperation: at Robbers Cave (Sherif, 1954) only **superordinate goals** ended it, and Allport's **contact hypothesis** (1954) requires equal status, common goals, cooperation and support of authorities, law or custom. ASKED 2 TIMES

22 The **propinquity effect**: the more we see and interact with a person, the more likely he or she becomes our friend or partner (Festinger, Schachter and Back 1950); its mechanism is Robert Zajonc's **mere exposure effect** (1968), in which repeated exposure to a stimulus increases liking for it. ASKED 2 TIMES

23 **Conformity** is changing to match a group with no direct request; **compliance** is changing in response to a direct request; **obedience** is following a command from an authority. ASKED 2 TIMES

24 In **Milgram**'s 1963 baseline study, 26 of 40 men (65 percent) obeyed to the maximum 450 volts, and all went to 300 volts. ASKED 2 TIMES

25 **Zimbardo**'s Stanford Prison Experiment (1971) demonstrates the power of social roles and situational forces; planned for two weeks, it was stopped after six days. ASKED 2 TIMES

26 **Cialdini**'s six compliance principles are reciprocity, commitment and consistency, social proof, liking, authority and scarcity (not reward); the foot-in-the-door (Freedman and Fraser, 1966) starts with a small request, then a larger one. ASKED 2 TIMES

27 The **two classic explanations** of the bystander effect are diffusion of responsibility and pluralistic ignorance, not social loafing, deindividuation, group polarisation or groupthink. ASKED 2 TIMES

28 **Social case work** was founded by Mary Ellen Richmond (*Social Diagnosis*, 1917); its process is intake and study, assessment (social diagnosis), intervention, evaluation, termination and follow-up, and its techniques include interviewing, observation, counselling, home visits, collateral contacts and referral, but not content analysis or lobbying. ASKED 2 TIMES

29 **Grace Coyle** established group work as a method within social work; **formed groups** are convened for an identified purpose with a trained leader and an agency auspice, whereas natural groups come together spontaneously. ASKED 2 TIMES

30 In Lawrence Shulman's terms the **gatekeeper** blocks or diverts the group from painful topics; Trecker's principles include democratic group self-determination, flexible functional organisation and progressive programme experiences. ASKED 2 TIMES

CHAPTER 4

Personality

IN THIS CHAPTER

4.1 ● What Personality Is: Definitions, Approaches and Early Type Theories: Define personality and its neighbouring terms, map the approaches and the six questions on which they differ, name the determinants and research methods, and describe the early type theories from the four humours to Sheldon's somatotypes.

4.2 ●● Freud's Psychoanalysis: Instincts, Structure and Psychosexual Stages: Trace how Freud built psychoanalysis from the study of hysteria, and use his methods, instincts, topographic and structural models, types of anxiety and psychosexual stages to explain personality, judging which of his ideas research supports.

4.3 ●●● Defence Mechanisms: How the Ego Protects Itself: Define defence mechanisms and their shared features, tell repression from its look-alikes, name the classic mechanisms from everyday vignettes, place them in Vaillant's four-level hierarchy, and match characteristic defences to particular disorders.

4.4 ●● Jung's Analytical Psychology and Adler's Individual Psychology: Explain how Jung's analytical psychology and Adler's individual psychology broke with Freud, and attribute each of their concepts (collective unconscious, archetypes, attitudes and functions, inferiority complex, fictional finalism, style of life, birth order) correctly.

4.5 ● Horney, Fromm, Sullivan and the Object-Relations Theorists: Explain how Horney, Fromm and Sullivan rooted personality in culture and relationships, and how ego psychology, object relations (Klein, Winnicott, Fairbairn, Mahler, Kernberg), Kohut's self psychology and Lacan revised Freud.

4.6 ● Traits I: Allport, Cattell and the Lexical Search: Explain the trait approach and the lexical hypothesis, Allport's personal dispositions, idiographic method, functional autonomy and proprium, and Cattell's surface and source traits, ergs and sentiments, L-, Q- and T-data and the 16PF.

4.7 ● Traits II: Eysenck, the Big Five and the Biology of Personality: Explain Eysenck's PEN model and its biology, the other biological temperament models, the Big Five with its facets and NEO inventories, HEXACO and the Dark Triad, what behaviour genetics shows about personality, and how the person-situation debate was resolved.

4.8 ● Humanistic and Existential Theories: Rogers, Maslow and May: Explain the humanistic image of the person through Rogers's self theory and Maslow's account of self-actualisation, and the existential view of Dasein, anxiety and meaning in Binswanger, May and Frankl.

4.9 • Behavioural, Social-Cognitive and Cognitive Theories: Explain personality through learning and cognition: Skinner's radical behaviourism, Dollard and Miller's early social learning, Bandura's reciprocal determinism, self-regulation and self-efficacy, Rotter's expectancy-value theory and locus of control, Mischel's CAPS, Kelly's personal constructs, Rep test and fixed-role therapy, and Witkin's and Kagan's cognitive styles.

4.10 • The Self, Positive Traits and Indian Approaches to Personality: Explain the classic theories of the self, separate self-concept from self-esteem, describe sensation seeking, explanatory style, well-being and character strengths, and identify the three gunas and the doshas of Indian approaches to personality.

Weightage. 202 of the 3180 questions mapped in this book (6.4%) test this chapter, 103 of them from real past papers. 1 of its 10 sections is High yield.

In 1920 a 22-year-old American named Gordon Allport was travelling home after a year teaching at Robert College in Istanbul. He stopped in Vienna to see one of his brothers and, on an impulse, sent a note to Sigmund Freud asking whether he might call. Freud, then in his sixties and already world famous, invited him over. When Allport entered the study, Freud said nothing at all. He simply sat and waited for the young visitor to explain why he had come.

The silence stretched until Allport, desperate for something to say, described an incident he had just seen on the tram. A small boy had seemed terrified of dirt: everything looked dirty to him, he changed his seat, and he told his mother not to let a dirty man sit beside him. Freud studied the neat, carefully groomed young man in front of him and asked, "Was that little boy you?"

Allport was shaken. He had told the story only to fill a silence; Freud had heard it as a disguised confession of his visitor's own unconscious fears. Allport denied for the rest of his life that he was the little boy, yet 47 years later he still wrote that his "single encounter with Freud was traumatic". The meeting convinced him that psychoanalysis dug too deep, and that psychology should first take people's conscious motives at face value. In 1922 he finished a Harvard dissertation on personality traits, the first research on traits in the United States, and went on to found the trait approach. Two great personality theorists had heard the same small story and seen entirely different things in it. Which of them was right is a question this whole chapter keeps asking.

4.1 What Personality Is: Definitions, Approaches and Early Type Theories

LOWER YIELD · 4 QS

You call a cousin moody and a friend a born leader, and you use those labels to predict what each will do next. Personality psychology does the same job with evidence. This section defines personality and its neighbouring terms, maps the approaches that fill this chapter, and ends with the oldest answers, which sorted people into types by body fluids, faces and physiques.

Persona and Allport's definition

The word personality comes from the Latin *persona*, the mask a Roman actor wore to show which character he was playing. The origin points to the public face we show others, and it raises the question every theory in this chapter must answer: is personality the mask, or what lies behind it?

The most quoted definition is that of **Gordon Willard Allport** (1897 to 1967), who weighed some 50 earlier definitions and offered his own in 1937: personality is "the dynamic organization within the individual of those psychophysical systems that determine his unique adjustments to his environment". In 1961 he replaced the last phrase with "characteristic behavior and thought". *Dynamic organisation* means personality grows, but in an organised way; *psychophysical* means mind and body work together; *determine* means it actively directs behaviour.

A working definition: **personality** is the relatively enduring, distinctive pattern of thoughts, feelings and behaviour that characterises a person, with internal aspects (motives, emotions) and external aspects (what others see), consistent across situations and over time. "Relatively" matters: a calm friend may still panic in an exam hall.

Temperament, character, trait and type

Temperament is the biologically based, early-appearing and relatively stable style of emotional reacting: how easily a person is aroused, how strongly they respond and what their usual mood is. Allport counted it, with physique and intelligence, as the inherited raw material of personality; infant temperament (Alexander Thomas and Stella Chess's easy, difficult and slow-to-warm-up babies) is in Chapter 6. **Character** is personality judged against moral standards; in Allport's phrase, "character is personality evaluated, and personality is character devaluated". A **trait** is a stable dimension along which people differ consistently, such as sociability; a **type** is a discrete category into which a person is placed, such as "melancholic". Table 4.1 adds three more terms.

Table 4.1 Personality and its neighbouring terms.

Term	Meaning	Example
Temperament	Biologically based, early-appearing emotional style	A baby who cries at every new face
Character	Personality evaluated morally; learned values	"She is honest and loyal"
Trait	Continuous dimension: more or less of it	7 out of 10 on sociability
Type	Discrete category: in or out	"He is an introvert"
Disposition	General tendency to act in a certain way; Allport's personal dispositions (Section 4.6)	A tendency to be generous
Habit	Specific learned, automatic response to a specific situation	Touching elders' feet on arrival
Attitude	Learned evaluation of a particular object, person or idea (Chapter 3)	Disliking fast food

Type versus trait is boxes versus a ruler: type theories sort people into a few categories; trait theories measure how much of each quality a person has, and find most people near the middle. Modern research favours traits; the early theories below are all type theories.

> EXAM TRAP
>
> "Biologically based", "inborn" or "visible from infancy" in a stem means **temperament**, not disposition (too loose a word), character (learned and moral) or trait (a descriptive dimension, not defined by biological origin). "Personality evaluated" is **character**.

What theories do: a map of the approaches

A **personality theory** is a set of related ideas that accounts for personality. A good one *describes* how people differ, *explains* why, *predicts* what a person will do and *generates research* that can test it. No theory does all four equally well, so the field is a family of approaches (Table 4.2, the map of this chapter).

Table 4.2 The approaches to personality and where they are taught.

Approach	Key names	Central idea	Where
Psychoanalytic	Freud	Unconscious conflict rooted in childhood	4.2, 4.3
Neo-psychoanalytic	Jung, Adler, Horney, Fromm, Sullivan	Social and cultural forces; less sex	4.4, 4.5
Ego psychology, object relations	Anna Freud, Melanie Klein	The ego's defences; early relationships	4.3, 4.5
Life-span	Erikson	Psychosocial stages across life	Chapter 6
Trait and type	Allport, Cattell, Eysenck, Costa and McCrae	Measurable stable dimensions	4.6, 4.7
Biological, evolutionary	Eysenck; twin researchers	Genes, brain, evolved tendencies	4.7
Humanistic, existential	Maslow, Rogers, May	Growth, choice, meaning	4.8
Behavioural	Skinner	Reinforcement shapes behaviour	4.9
Social-cognitive	Bandura, Rotter, Mischel	Learning plus thought, in situations	4.9
Cognitive	Kelly	Personal constructs	4.9
Limited-domain	Rotter, Zuckerman, Seligman	One facet at a time	4.9, 4.10
Indian	Samkhya, Ayurveda	Gunas and doshas	4.10

Six questions about human nature

Behind every theory lies an image of human nature. **Duane Schultz** and **Sydney Ellen Schultz** compare theorists on six questions: *free will versus determinism* (do we direct our lives?), *nature versus nurture* (inheritance or upbringing?), *past versus present* (is the adult fixed by childhood, the view called **historical determinism**?), *uniqueness versus universality*, *equilibrium versus growth* (do we only reduce tension, or strive to grow?) and *optimism versus pessimism* (are people basically good?).

Freud gave the darkest answers (Figure 4.1): behaviour determined by instincts and the first five years, life a struggle to reduce tension, people seen pessimistically; on nature-nurture and uniqueness he took a middle position. The humanists stand near the opposite pole: Carl Rogers saw people as free, shaped more by the present than by childhood, growing and basically good.

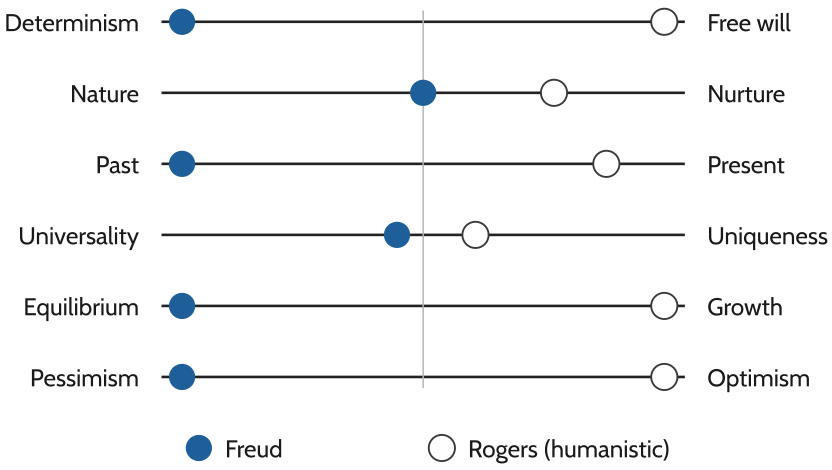


Figure 4.1 Schultz's six questions about human nature. Freud (filled) sits at the gloomy pole on four questions and near the middle on two; Rogers (open) sits near the opposite pole on most.

Determinants of personality

Heredity supplies temperament and more: twin studies (identical versus fraternal twins) and adoption studies (children compared with biological and adoptive parents) suggest genes account for up to about half the variation in traits (Section 4.7). **Environment** shapes the rest: the family (parenting style, parental models, and birth order, which Alfred Adler stressed; Section 4.4), culture (individualist and collectivist societies reward different traits) and social class (opportunities, stresses and values). Surprisingly, the environment siblings share matters less than each child's own experiences. The **situation** matters too: the student talkative in the canteen is silent in the viva. Walter Mischel's 1968 critique began a twenty-year person-situation debate (Section 4.9) that ended in **interactionism**: behaviour depends on the person, the situation and their interaction.

How personality is studied and assessed

The **clinical method** builds theory from the **case study**, a detailed history of one person from many sources; Freud, Jung and Adler built their theories from patients this way. It gives depth, but its data are subjective and open to the therapist's bias. The **experimental method** manipulates an independent variable and measures its effect on a dependent variable with all else held constant, so it alone shows cause; Skinner and Bandura worked this way, at the price of studying narrow slices of behaviour. The **correlational method** measures how strongly two variables go together, such as anxiety and exam marks, so it cannot prove cause. Its extension, **factor analysis**, finds the few factors underlying many intercorrelated measures; Raymond Cattell and Hans Eysenck built their trait theories with it.

The German philosopher **Wilhelm Windelband** (1848 to 1915) named a second divide in 1894. **Nomothetic** research (Greek *nomos*, law) seeks general laws from groups; **idiographic** research (Greek *idios*, one's own) studies the single individual in depth, the approach Allport championed (Section 4.6).

Assessment uses self-report inventories (MMPI-2, 16PF, NEO-PI-R, EPQ), projective techniques that ask for interpretations of ambiguous material (Rorschach inkblots, the Thematic

Apperception Test or TAT), interviews, behavioural observation and life records such as diaries and letters. Each is useful only if reliable (consistent) and valid (measures what it claims); the tests are in Chapter 2.

Humours and the four temperaments

The Greek physician **Hippocrates** (about 460 to 377 BC) taught that health depends on the balance of four bodily fluids, or **humours**: blood, yellow bile, black bile and phlegm. **Galen** (AD 129 to about 216), a Greek physician in Rome, is credited with turning this into a theory of temperament: the dominant humour sets a person's emotional style (Table 4.3). The chemistry is wrong, but the idea of a biological temperament was ahead of its time, and Hans Eysenck later fitted the four temperaments into the quadrants of his extraversion and neuroticism dimensions (Section 4.7).

Table 4.3 The four humours and temperaments, with Eysenck's later mapping.

Humour	Temperament	Character	Eysenck quadrant
Blood	Sanguine	Cheerful, hopeful, sociable	Stable extravert
Yellow bile (choler)	Choleric	Irritable, hot-tempered	Unstable extravert
Black bile	Melancholic	Sad, gloomy, anxious	Unstable introvert
Phlegm	Phlegmatic	Calm, slow, unexcitable	Stable introvert

Physiognomy and its look-alikes

Physiognomy judges character from facial features and outward appearance. The Greek *Physiognomonica*, attributed to **Aristotle** (384 to 322 BC) but probably by his followers, claimed that people who resemble certain animals share their characters. The Swiss pastor **Johann Kaspar Lavater** (1741 to 1801) made it a European fashion with his illustrated essays of the 1770s, which read each feature of the face as a sign of character. The Italian physician **Cesare Lombroso** (1835 to 1909) applied it to crime: in *Criminal Man* (1876) he claimed that "born criminals" carry **criminal stigmata**, such as a sloping forehead and large jaws, marks of a throwback to primitive humans. Later comparisons of prisoners with other citizens found no such type.

Phrenology, founded by **Franz Joseph Gall** (1758 to 1828) around 1800 and popularised under that name by **Johann Gaspar Spurzheim** (1776 to 1832), read the bumps of the skull, assuming each mental faculty has its own brain "organ" whose size pushes out the skull above it (Section 1.1): wrong about bumps, right that functions are localised. **Graphology** infers personality from handwriting style and shows no useful validity in controlled studies. Interpreting dreams for unconscious motives is psychoanalysis (Section 4.2).

> EXAM TRAP
>
> Face as a sign of character = **physiognomy**. Skull bumps and brain localisation of functions = **phrenology** (Gall). Handwriting = **graphology**. Dreams and unconscious motives = **psychoanalysis**. Body build = Kretschmer and Sheldon, below.

Body types: Kretschmer, Sheldon and later typologies ●●

The German psychiatrist **Ernst Kretschmer** (1888 to 1964) thought his patients' illnesses went with their build. In *Physique and Character* (1921) he described the **pyknic** type, short and round, with a **cyclothymic** temperament swinging between cheer and gloom and a liability to manic-depressive illness; the **asthenic** or leptosomic type, tall and thin, with a reserved, withdrawn **schizothymic** temperament and a liability to schizophrenia; the **athletic** type, muscular; and the **dysplastic** type, a disproportioned mix. One flaw: his manic-depressive patients were older, and people grow rounder with age.

The American physician **William Herbert Sheldon** (1898 to 1977) replaced boxes with dimensions. In *The Varieties of Human Physique* (1940) he rated thousands of standardised photographs of male college students on three components, named after the embryonic layers that form the dominant tissues: **endomorphy** (soft and round, digestive organs prominent), **mesomorphy** (muscle and bone) and **ectomorphy** (thin and fragile, a large skin surface for the body's mass). Each is rated from 1 to 7, so a **somatotype** is three numbers: 4-4-4 is the average build, 7-1-1 the extreme endomorph, 1-7-1 the extreme mesomorph and 1-1-7 the extreme ectomorph. In *The Varieties of Temperament* (1942) he matched each component to a temperament (Table 4.4).

Table 4.4 Sheldon's somatotypes and temperaments.

Physique	Build	Temperament	Described as
Endomorph	Round, soft	Viscerotonia	Relaxed, sociable, even-tempered; loves comfort, food and company
Mesomorph	Muscular, broad	Somatotonia	Assertive, energetic, noisy, callous; fond of physical activity and risk
Ectomorph	Thin, fragile	Cerebrotonia	Restrained, inhibited, private, self-conscious, introverted

KEY STUDY: SHELDON'S SOMATOTYPES (1940, 1942)

Does body build predict temperament? Sheldon rated young men on dozens of temperament traits and compared the ratings with their somatotypes, reporting correlations of about .8, far above almost anything in personality research. The flaw: he rated both physique and temperament himself, knowing each man's build, so his expectations and the stereotypes of the jolly fat man and the nervous thin one could create the correlation. Independent studies found only weak links. The theory is regarded as discredited.

EXAM TRAP

Sheldon's theory has been **discredited**: not supported by research, and not "accurate for thin people" or "for women only" (his subjects were men). Noisy, callous, fond of physical activity = **mesomorph**; relaxed and food-loving = endomorph; restrained and private = ectomorph. Pyknic and asthenic with psychosis = Kretschmer.

Type thinking outlived the body theories. The German philosopher **Eduard Spranger** (1882 to 1963) sorted people by dominant value into six types: theoretical (truth), economic (usefulness), aesthetic (beauty), social (love of people), political (power) and religious (unity); the Allport-Vernon-Lindzey *Study of Values* measures them (Section 4.6). Carl Jung's types are in Section 4.4; Type A and B (Meyer Friedman and Ray Rosenman), Type C and Johan Denollet's Type D in Section 1.11; the gunas and doshas in Section 4.10.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The word personality comes from the Latin _____, the mask of the Roman actor.
2. Allport (1937): the dynamic organisation within the individual of those _____ systems that determine his unique adjustments to his environment.
3. The biologically based, early-appearing style of emotional reacting is _____.
4. The view that adult personality is fixed in the early years is called _____ determinism.
5. The terms nomothetic and idiographic were introduced by _____.
6. In humoral theory, an excess of black bile produces the _____ temperament.
7. Judging character from facial features is _____; reading skull bumps is _____.
8. Kretschmer linked the round _____ build to manic-depressive illness, and Sheldon linked the mesomorph to _____.

Answers: (1) persona, (2) psychophysical, (3) temperament, (4) historical, (5) Wilhelm Windelband, (6) melancholic, (7) physiognomy; phrenology, (8) pyknic; somatotonia.

4.2 Freud's Psychoanalysis: Instincts, Structure and Psychosexual Stages

MEDIUM YIELD · 42 QS

You mean to thank a teacher and hear yourself use the name of one you disliked; you "forget" a dental appointment you were dreading. Most of us call these accidents. Sigmund Freud did not. To him every slip, dream and symptom had a hidden cause, and personality was an uneasy truce between inborn urges, the outside world and the morality we absorb as children. His system, **psychoanalysis**, was the first formal theory of personality and is still the best known: at once a theory of personality and a method of treating mental disorder. We follow it from hysteria to the verdict of research.

Freud's life and the birth of psychoanalysis ●●●

Sigmund Freud (1856 to 1939) was born in Freiberg, Moravia (now Příbor in the Czech Republic), and from the age of four lived in Vienna for almost 80 years. He qualified in medicine at Vienna in 1881 and began his career as a **neurologist**, not a psychiatrist or philosopher; "psychoanalyst" was a profession he had yet to invent. Neurologists were sent the patients nobody could explain, those with **hysteria**: genuine physical symptoms such as paralysis, blindness or loss of feeling (anaesthesia) in a limb, with no damage to the nerves. Freud came to see these as bodily reactions to emotional trauma, produced by painful memories kept out of awareness.

Three teachers pointed the way. In Paris (1885 to 1886) the neurologist **Jean-Martin Charcot** (1825 to 1893) showed him hysterical symptoms produced and removed by hypnosis, and hinted that their roots were sexual. At Nancy, which Freud visited in 1889, **Hippolyte Bernheim** (1840 to 1919) showed that hypnosis worked through suggestion. In Vienna the physician **Josef Breuer** (1842 to 1925) had found that a patient lost her symptoms when she talked through their origins under hypnosis. Breuer and Freud reported this in "On the Psychical

Mechanism of Hysterical Phenomena: Preliminary Communication" (1893), expanded into *Studies on Hysteria* (1895): hysterics, they wrote, suffer mainly from reminiscences.

KEY STUDY: ANNA O. AND THE TALKING CURE (BREUER AND FREUD, 1895)

Anna O., really **Bertha Pappenheim** (1859 to 1936), was a bright young Viennese woman who fell ill while nursing her sick father: blurred vision, weakness of her right arm and both legs, and trouble speaking. Breuer took one symptom at a time; under hypnosis she traced it back to the moment it began, relived the feeling, and the symptom faded. She called this the "talking cure". Breuer and Freud named the release of pent-up emotion **catharsis** (Greek for purification) and concluded that symptoms carry hidden meaning. Criticism: she was Breuer's patient, not Freud's, and she was not cured when treatment ended in 1882; she spent months in a Swiss sanatorium and took years to recover before becoming a pioneering social worker.

Freud went further than Breuer. By the mid-1890s he held that the buried trauma was always sexual: his **seduction theory** traced neurosis to sexual abuse in childhood, usually by an older male relative. In 1897 he abandoned it, deciding that many such memories were fantasies and that the wishes behind them mattered more than events. The same year he began a lifelong **self-analysis** through his own dreams, recovering childhood hostility to his father and attachment to his mother, the seed of the Oedipus complex. The major works followed (Table 4.5). He fled the Nazis to London in 1938 and died there in September 1939, after sixteen years of cancer of the mouth.

Table 4.5 Freud's path to psychoanalysis: the dates that are examined.

Year	Work or event	Why it matters
1893	"Preliminary Communication" (with Breuer)	Hysteria from traumatic memories
1895	*Studies on Hysteria* (with Breuer)	Anna O.; catharsis; the founding text
1900	*The Interpretation of Dreams*	Dreams as wish fulfilment; topographic model (appeared November 1899, dated 1900)
1901	*The Psychopathology of Everyday Life*	Parapraxes (slips)
1905	*Three Essays on the Theory of Sexuality*	Infantile sexuality; psychosexual stages
1909	Lectures at Clark University, Worcester, Massachusetts	Invited by G. Stanley Hall; American fame
1920	*Beyond the Pleasure Principle*	The death instinct
1923	*The Ego and the Id*	Structural model: id, ego, superego
1926	*Inhibitions, Symptoms and Anxiety*	Anxiety as the ego's danger signal
1930	*Civilization and Its Discontents*	Instinct versus civilisation; guilt

Freud's theory is **psychoanalytic**, not behaviouristic (Watson and Skinner: behaviour as learning), humanistic (Maslow and Rogers: growth and choice) or psychosocial (Erikson's social stages; Chapter 6). The broader family that grew from it is the **psychodynamic perspective**: many mental disorders can be traced to unconscious urges and to the struggle over their expression in the hidden depths of personality. Gestalt psychology studies perception as or-

ganised wholes (Chapter 1), and social learning theory explains behaviour by observation and reinforcement (Section 4.9).

CLINICAL LINK

Freud's hysteria survives as **conversion disorder**, called functional neurological symptom disorder in DSM-5-TR: real loss of movement or sensation that fits no neurological disease (Chapter 8).

EXAM TRAP

The 1893 paper and *Studies on Hysteria* are **Freud and Breuer**, never Freud with Charcot, Adler or Jung (Adler and Jung were later followers who broke away; Section 4.4). Freud began as a **neurologist**. To Freud, hysteria meant **physical symptoms produced by emotional trauma**; a phobia of horses is the case of Little Hans, and an unresolved Oedipus complex or oral behaviour are unrelated ideas. The father of psychoanalysis is Freud, not Bernheim, Adler or Jung.

Freud's methods: free association, dreams and slips ●●

Many patients could not be hypnotised, so Freud had them lie on a couch, with himself out of sight, and gave them one **fundamental rule**: say whatever comes to mind, however trivial, embarrassing or painful, without censoring or rearranging it. This is **free association**, a kind of daydreaming aloud, and it became the central technique of psychoanalysis. Because of psychic determinism (below), nothing in the chain was random; each idea led closer to the buried conflict.

The chain often stalls: the patient falls silent or changes the subject. This is **resistance**, the unconscious refusal to let painful material surface, which Freud took as a sign that the analysis was near the source. Patients also come to feel toward the analyst as they once felt toward their parents: **transference**. Both are worked through in therapy (Chapter 7).

Dream analysis was the second road in; Freud called the interpretation of dreams "the royal road to a knowledge of the unconscious". What you remember is the **manifest content**; its hidden meaning, a disguised wish, is the **latent content**. The dream-work that turns one into the other is taught in Section 1.5.

The third window was everyday mistakes. A **parapraxis** (plural parapraxes), popularly a **Freudian slip**, is a slip of the tongue or pen, a forgotten name or appointment, or a misplaced object that betrays an unconscious thought or wish. A student who writes to his professor "I look forward to missing your lecture" has, in Freud's eyes, said what he meant. Parapraxes are mistakes that reveal the unconscious; they need not be sexual. Their meaning is found as a symptom's is, through free association.

Freud reported his work as **case studies**, detailed histories of single patients: Anna O. (hysteria, Breuer's patient), Dora (hysteria, 1905), Little Hans (1909, a small boy's phobia of horses, read as Oedipal fear of his father), the Rat Man (obsessional neurosis, 1909), Schreber (paranoia, 1911, analysed only from his published memoir) and the Wolf Man (1918, a childhood dream of wolves). The method is rich but weak as evidence, as the evaluation shows.

EXAM TRAP

Lying on the couch saying whatever comes to mind is **free association**, not dreaming, narration or storytelling, and it is the central technique of **psychoanalysis**, not of behaviour, person-centred or cognitive therapy (Chapter 7). Freudian slips are analysed through **free association**: case studies are how Freud reported patients, behaviour analysis records observable acts and their consequences, and a sleep study records sleep stages, not meanings. A slip is not a **defence mechanism** (an unconscious strategy against anxiety; Section 4.3), a **cognitive distortion** (Aaron Beck's errors of thinking, such as overgeneralisation) or a **speech disorder** (such as stuttering).

Instincts and psychic energy ●●

What powers the mind? Freud's answer was the **instincts** (German *Triebe*, sometimes translated "drives"): the powerful innate forces that motivate all behaviour and set its direction. An instinct is the mental representation of a bodily need, the wish that hunger becomes in the mind. In "Instincts and Their Vicissitudes" (1915) Freud gave every instinct a **source** (the bodily need), an **aim** (to remove the excitation, that is, satisfaction), an **object** (whatever achieves the aim, the most changeable part) and an **impetus** or pressure (its strength). The aim is always tension reduction, so the model is homeostatic.

Freud grouped the instincts into two classes. The **life instincts**, or **Eros**, serve survival of the individual and the species: hunger, thirst and above all sex, which Freud defined broadly to include almost any bodily pleasure and affection. Their energy is the **libido**. In *Beyond the Pleasure Principle* (1920), shaken by the slaughter of the First World War, he added the **death instinct**, an unconscious drive toward decay and a return to the inanimate state, which his followers named **Thanatos**, a word Freud himself did not use in his writings. Turned inward it is self-destruction; turned outward it becomes the **aggressive drive** to destroy, conquer and kill, so wars and violence are, for Freud, evidence of the death instinct. Sex and aggression are what the id wants; the pleasure and reality principles (below) are the rules by which the id and the ego pursue them.

Libido can be attached to anything. **Cathexis** is the investment of psychic energy in the mental representation of an object, person or idea: if you adore a cricketer, your libido is cathected to him. An **anticathexis** is the opposing investment by which the ego or superego blocks an id cathexis; keeping a wish repressed costs energy. The cost matters because Freud saw the mind as a **closed energy system**, a fixed quantity of psychic energy that can be shifted but never added to, so whatever the ego spends holding back the id is lost to work and love. Beside it stands **psychic determinism**, the principle that every mental event has a cause, so no slip, dream or symptom is an accident.

EXAM TRAP

Freud's innate motivating forces are **instincts**; **needs** are Maslow's and Henry Murray's term, **drives** Clark Hull's, and the **id** is the reservoir that holds the instincts. Investing energy in an object is **cathexis**. Its neighbours: **libido** is the energy itself, **fixation** is libido left stuck at a childhood stage, **sublimation** channels it into socially valued work, **displacement** shifts an impulse to a safer target and **repression** keeps threatening material out of awareness (the last three are defences; Section 4.3).

The levels of the mind: the topographic model ●

Freud's first map, the **topographic model** (1900), sorts mental contents by how accessible they are. The **conscious** holds whatever you are aware of at this moment. The **preconscious** holds accessible memories, thoughts and perceptions that are not in awareness now but can be recalled at will: your mobile number, what you ate last night. The **unconscious**, by far the largest, holds instincts, repressed wishes and painful memories that cannot be recalled at will and show themselves only indirectly, in dreams, slips and symptoms. Freud likened the mind to an iceberg: the conscious is the small tip above the water, the preconscious lies just below the waterline, and the unconscious is the vast hidden mass, the real driving power of behaviour (Figure 4.2).

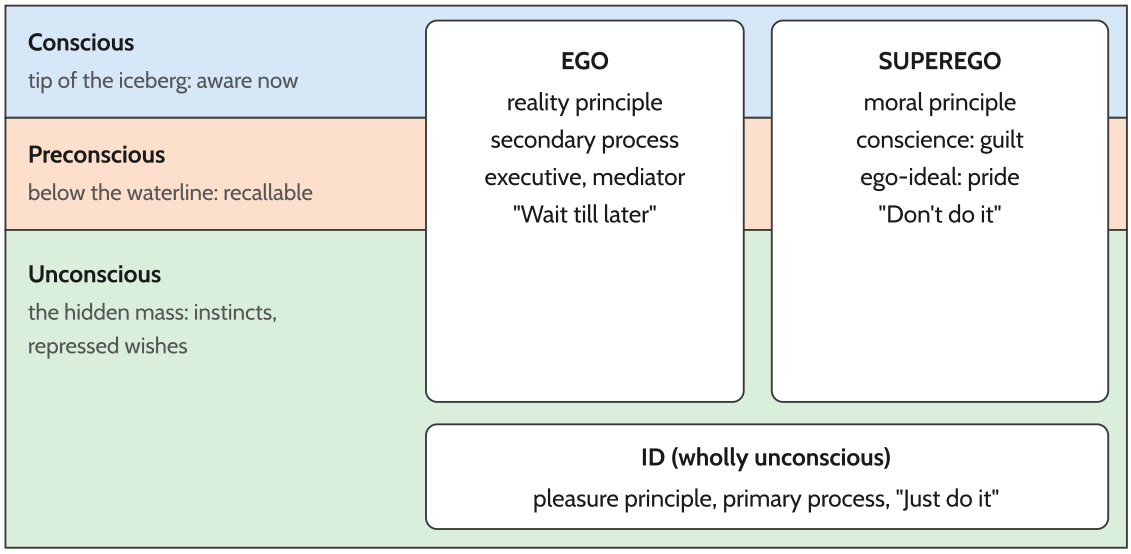


Figure 4.2 The two maps together. The shaded bands are the topographic levels of the iceberg; the boxes are the structures of 1923. The ego and superego reach into all three levels, while the id lies wholly below.

> EXAM TRAP
>
> Accessible memories recalled at will are **preconscious**, not conscious (in awareness now) or unconscious (not recallable at will). **Psychic determinism** is a principle, not a level of the mind.

Id, ego and superego: the structural model ●●●

In *The Ego and the Id* (1923) Freud replaced the three levels with three interacting structures, the **structural model**. The levels became properties of the structures (Figure 4.2).

The **id** is the reservoir of unorganised instinctual drives, present at birth, wholly unconscious and the source of all psychic energy. It operates on the **pleasure principle**: seek pleasure and avoid pain, now. Impulsive, irrational, selfish and amoral, it demands the immediate gratification of primitive needs and tolerates no delay ("Just do it"). Having no contact with reality, it can satisfy a need only by reflex or by **primary process thought**, forming a wish-fulfilling image of the object it lacks, as a hungry infant pictures the breast. This **wish fulfilment** also drives dreams.

Images feed no one, so from the id grows the **ego**, the rational part of personality, which works on the **reality principle**. Using **secondary process thought** (perception, memory, judgement and planning) and **reality testing**, it decides when, where and how the id's wishes can safely be met. It does not block the id; it postpones and redirects ("Wait till later"). Freud compared it to a rider guiding a stronger horse. It is called the **executive** of personality because it directs the energies supplied by the id and mediates between the id's desires, the superego and reality or social rules. The ego is partly conscious, partly preconscious and partly unconscious.

The **superego** is the moral arm, working on the moral principle. It forms at about five or six, when the Oedipus complex is resolved and the child internalises parental and social standards, so that parental control becomes self-control. It has two parts. The **conscience** is built from behaviours for which the child was punished and enforces them with guilt; the **ego-ideal** is built from good, correct behaviours for which the child was praised and rewards with pride. Like the id it admits no compromise: it seeks not to delay forbidden impulses but to stop them ("Don't do it"), and it strives for perfection. The ego thus serves three harsh masters: the id, reality and the superego (Table 4.6).

Table 4.6 The three structures of personality compared.

Feature	Id	Ego	Superego
Principle	Pleasure	Reality	Moral (perfection)
Thinking	Primary process; wish fulfilment	Secondary process; reality testing	Moral judgement
Awareness	Wholly unconscious	All three levels	All three levels, mostly unconscious
Appears	At birth	Infancy, out of the id	About 5 or 6, at Oedipal resolution
Role	Reservoir of instincts; all energy	Executive; mediator	Conscience (guilt) and ego-ideal (pride)
Motto	"Just do it"	"Wait till later"	"Don't do it"

> **EXAM TRAP**
>
> Gratification, impulsiveness and irrationality point to the **id**, and **only** the id seeks immediate gratification of primitive needs. The ego is the **executive**; "superordinate" and "subordinate" are not Freud's labels, and **congruence** is Carl Rogers's term for a match between self and experience (Section 4.8). The conscious and unconscious mind are levels, not principles. The **self** and the **libido** are not Freudian structures, and the superego's two parts are the **conscience and the ego-ideal**, not the id and the ego. "Do your own thing" and "Oh, sit on it" are not Freudian mottoes.

Anxiety: the ego's danger signal

Squeezed by three masters, the ego feels **anxiety**, fear without a clear object. In *Inhibitions, Symptoms and Anxiety* (1926) Freud made anxiety a **signal** that warns the ego of danger, with the overwhelming stimulation of birth as its prototype. He named three kinds by their source. **Reality anxiety** (objective anxiety) is fear of a real danger in the outside world, such as a speeding lorry; it is useful and fades when the danger passes. **Neurotic anxiety**, a conflict between id and ego, is the unconscious fear that id impulses will break through and

bring punishment. **Moral anxiety**, a conflict between id and superego, is fear of one's own conscience, felt as guilt or shame. When realistic action cannot remove the threat, the ego turns to unconscious defence mechanisms, the subject of Section 4.3.

The psychosexual stages ●●●

Freud believed adult personality was largely set by about the age of five, and his **psychosexual stages** (*Three Essays*, 1905) explain how. At each stage the libido centres on a different **erogenous zone**, a part of the body that gives pleasure, and each stage brings a conflict between the child's pleasure and the parents' demands. If a stage goes badly, part of the libido stays invested there: **fixation**. It results from **frustration** (needs unmet) or **overindulgence** (needs so richly met that the child will not move on), and it stamps a recognisable adult character. Under stress a person may show **regression**, a return to the behaviour of an earlier stage, usually the one where libido is fixated. Table 4.7 gives the sequence; the ages are approximate and differ between textbooks (some give oral birth to 1 year, anal 1 to 3, phallic 4 to 5 and latency 5 to puberty).

In the **oral stage** (birth to about 18 months) pleasure comes from the mouth, through sucking, biting and swallowing, and the central conflict is **feeding and weaning**. Its first mode, **oral-incorporative** (taking in), produces adults preoccupied with eating, drinking, smoking, kissing and nail-biting, who are dependent and gullible ("swallowing" whatever they are told). The later **oral-aggressive** or oral-sadistic mode, which comes with painful teething, leaves adults who are sarcastic ("biting" remarks), critical, hostile and argumentative. Freud linked alcohol dependence to oral fixation.

In the **anal stage** (about 18 months to 3 years) pleasure comes from holding back and expelling faeces, and **toilet training** is the first time society regulates an instinct. A child who defies it by soiling may become an **anal-expulsive** (anal-aggressive) adult: messy, wasteful, disorganised, even cruel. A child who resists by holding back may become **anal-retentive**: orderly, stingy, obstinate and compulsively neat.

The **phallic stage** (about 3 to 5 or 6 years) centres on the genitals and on incestuous fantasies. The boy faces the **Oedipus complex**, named after the king in Sophocles' play who unknowingly kills his father and marries his mother: he desires his mother and sees his father as a rival. Fearing that the father will cut off his penis (**castration anxiety**), he represses the desire and achieves **identification** with the same-sex parent, adopting his father's manner and standards; this gives him his sex role and creates the superego. The girl's first love is also her mother, but on finding she lacks a penis she feels **penis envy**, blames her mother and turns to her father; she resolves the conflict, less completely in Freud's view, by identifying with her mother. Freud called this the female Oedipus complex. **Carl Gustav Jung** (1875 to 1961) named it the **Electra complex** in 1913, after the Greek heroine who persuades her brother to kill their mother. Freud rejected the term, yet exams pair Electra with girls and Oedipus with boys.

In the **latency period** (about 6 to puberty) the sexual instinct is dormant, sublimated into school, sport, hobbies and friendships with children of the same sex; Freud did not count it as a true psychosexual stage, since no new zone appears. The **genital stage** (puberty onwards) reawakens sexuality, now directed to mature love. Its achievements are sex-role identity, adult social relationships and the capacity for love and work.

Table 4.7 Freud's psychosexual stages.

Stage and age	Erogenous zone	Central conflict	Adult fixation traits
Oral, birth to about 18 months	Mouth	Feeding and weaning	Incorporative: dependent, gullible, overeating, smoking, drinking, nail-biting. Aggressive: sarcastic, critical, hostile
Anal, about 18 months to 3 years	Anus	Toilet training	Retentive: orderly, stingy, obstinate. Expulsive: messy, wasteful, cruel
Phallic, about 3 to 5 or 6	Genitals	Oedipus complex; castration anxiety or penis envy; superego forms	Vain, brash, narcissistic
Latency, about 6 to puberty	None (dormant)	Sublimation into school, sport, same-sex friends	Not a fixation point
Genital, puberty onwards	Genitals (mature)	Mature love and work; sex-role identity; adult relationships	The healthy goal

EXAM TRAP

Smoking, drinking, overeating, nail-biting, sarcasm and criticalness all mean **oral** fixation, which comes from problems of **feeding and weaning**; toilet training is anal, Oedipal resolution phallic, and adolescent identity formation is Erik Erikson's psychosocial crisis (Chapter 6). Oral aggression is oral, not anal, and "genital identity confusion" is not a Freudian concept. Incestuous fantasies, the Oedipus complex and the emergence of the superego belong to the **phallic** stage; sublimation of the sex instinct is **latency**; sex-role identity and adult relationships are **genital**.

Evaluating Freud

Which parts of the theory survive testing? The strongest support is for an unconscious mind. In **subliminal perception** studies, stimuli flashed too briefly to be seen still shape thought: **Howard Shevrin** (1977) showed people a picture of a bee they could not report, and their free associations then included "sting" and "honey". **Implicit memory** (experience shaping performance without recollection) and **priming** point the same way. Some defences, such as repression, projection and displacement, have experimental support (Section 4.3), and **Michael Motley** (1987) found that men made anxious about an electric shock said "damn shock" for "sham dock", a laboratory Freudian slip. Reviewing some 2,500 studies, **Seymour Fisher** and **Roger Greenberg** (1977, 1996) judged the id, ego, superego, libido and death instinct untestable, found weak support for oral and anal character types and some Oedipal ideas, and found no support for the psychosexual stages or for dreams as disguised wishes.

The criticisms are heavy. Many concepts are **unfalsifiable**: if a man loves his mother it is Oedipal, and if he hates her he is defending against Oedipal love, so no result can count against the theory. The data came from Freud himself and a small, biased sample, a dozen or so published cases, mostly young, educated, well-off Viennese women. His notes were written from memory hours later, and patients' recollections were not corroborated. His account of women (penis envy, a weaker superego) is sexist, a charge led by Karen Horney (Section 4.5). His view of human nature is pessimistic and over-sexualised, and personality plainly goes on developing after five.

Still, Freud gave psychology its first comprehensive theory of personality, the ideas of unconscious motivation, defence and the weight of childhood, and the talking cure from which every psychotherapy descends. On the six questions about human nature (Section 4.1) Freud is a determinist (though analysis widens choice), takes a middle ground on nature and nurture (inborn id and stages, learned ego and superego), stresses the past (the first five years), sees both universality (shared instincts and stages) and uniqueness (the content of ego and superego), makes equilibrium (tension reduction) life's goal, and is a pessimist. In India, Girindrasekhar Bose, founder of the Indian Psychoanalytical Society (1922), corresponded with Freud and proposed his own theory of opposite wishes (Section 1.2).

EXAM TRAP

Research has begun to support the **unconscious mind**, not the existence of an id, ego and superego (untestable), the order of the psychosexual stages or a universal Oedipus complex.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Freud and _____ published *Studies on Hysteria* in 1895.
2. Saying whatever comes to mind on the couch, without censoring it, is _____.
3. The investment of psychic energy in an object, person or idea is _____.
4. Memories not in awareness but recallable at will lie in the _____.
5. The id works on the pleasure principle; the ego works on the _____ principle.
6. The two parts of the superego are the conscience and the _____.
7. Fear of one's own conscience, felt as guilt, is _____ anxiety.
8. Smoking, overeating, nail-biting and sarcasm suggest fixation at the _____ stage.

Answers: (1) Breuer, (2) free association, (3) cathexis, (4) preconscious, (5) reality, (6) ego-ideal, (7) moral, (8) oral.

4.3 Defence Mechanisms: How the Ego Protects Itself HIGH YIELD · 38 QS

Arjun has just failed his statistics paper. Within an hour he has told his friends that statistics is useless for a clinician anyway, and he has snapped at his younger brother over nothing. He is not lying, at least not knowingly. In the terms of Section 4.2, his ego is under threat: anxiety is signalling danger, and when the ego cannot remove a danger by realistic means it falls back on unconscious tricks that disguise it. This section defines those tricks, studies the most basic of them, repression, walks through the full catalogue, ranks them from psychotic to mature, and shows which ones mark particular disorders.

What defences are and what they share •

Sigmund Freud (1856 to 1939) wrote of defence from his first papers on hysteria in the 1890s, although for years he spoke almost only of repression. In *Inhibitions, Symptoms and Anxiety* (1926) he returned to the wider term: anxiety is a signal of danger to the ego, and defence is the name for every technique the ego uses to keep that danger out of awareness. Formally,

defence mechanisms are unconscious strategies the ego uses to protect itself against anxiety by denying or distorting reality. It was his daughter **Anna Freud** (1895 to 1982) who turned the idea into a catalogue. In *The Ego and the Mechanisms of Defence* (1936) she listed ten mechanisms (regression, repression, reaction formation, isolation, undoing, projection, introjection, turning against the self, reversal and sublimation), described others such as identification with the aggressor, and moved the ego to the centre of psychoanalysis. The standard list you learn today owes its full articulation to her.

The mechanisms differ in detail, but they share six features:

- They operate unconsciously. The person neither chooses a defence nor knows that one is working; if we knew we were lying to ourselves, the lie would no longer protect us.
- They deny or distort reality, either outer reality (a diagnosis, a loss) or inner reality (a wish, a feeling).
- They protect the ego and reduce anxiety. That is their whole function, and we usually use several at once.
- They have adaptive value in moderation. Freud held that everyone defends all the time and that we could not survive long without defences.
- They become pathological when they are rigid, overused or used where realistic action is needed; then they produce neurotic or even psychotic symptoms.
- They are not rational. A defence does not solve the problem; it changes how the problem is seen.

> **EXAM TRAP**
>
> Asked which statement does NOT describe defence mechanisms, the answer is "rational unconscious protective measures". "Having adaptive value", "pushing painful experiences out of consciousness" and "resulting in a distorted view of reality" are all true. The single word *rational* makes the option false.

Defence should not be confused with **coping**, the conscious, deliberate, flexible and reality-oriented effort to manage stress (problem-focused and emotion-focused coping are taught in Section 1.11). If Arjun draws up a revision timetable he is coping; if he decides the subject is useless he is defending. The line blurs only at the top of the maturity ladder (see below), where suppression and anticipation look much like good coping.

Repression, the primary defence ●●●

Freud called the theory of repression "the corner-stone on which the whole structure of psycho-analysis rests" (1914). In full, **repression** is the involuntary, unconscious blocking of threatening or anxiety-provoking memories, wishes and feelings from awareness, or the barring of them from ever reaching it. Freud separated two forms: **primal repression**, which keeps instinctual ideas from ever becoming conscious, and **repression proper** (after-pressure), which pushes out material that was once conscious and is now linked with the forbidden. Repression is the primary defence against anxiety: it is the most basic and most frequently used, the other mechanisms are often described as back-ups when it fails, and Freud saw it in all neurotic behaviour. It can act on memories, on perception (we may fail to see a disturbing event in front of us) and even on bodily function, as when a man represses his sex-

ual drive so strongly that he becomes impotent. What is repressed is not destroyed; it stays active in the unconscious and returns in disguise as dreams, slips and symptoms.

In memory terms, then, repression is an unconscious process that removes unpleasant memories from awareness. It is not a conscious effort to forget, not a strengthening of memory traces and not the deliberate recall of traumatic events. Several neighbouring terms are easily confused with it, and Table 4.8 sets them side by side.

Table 4.8 Repression and its look-alikes.

Term	Conscious?	What it is	Example
Repression	No, involuntary	Blocks threatening material from awareness	No memory of a childhood humiliation
Suppression	Yes, deliberate	Puts a worry aside to face later; mature in Vaillant's scheme	"I will think about the result after the exam"
Inhibition	No	Gives up an ego function (work, eating, sex) to avoid anxiety (Freud, 1926)	Cannot bring himself to start any work
Denial	No	Refuses to accept an outer reality that is perceived	"The biopsy report must be wrong"
Motivated forgetting	Can be either	The broad memory term: any forgetting driven by a motive (Chapter 1)	Includes both repression and suppression
Depression	Not a defence	A mood state and disorder (Chapter 8)	Persistent sadness, loss of interest

EXAM TRAP

Stems that say "involuntary", "unconscious" or "unable to gain access to conscious awareness" (Deepika cannot bring the threatening material into awareness) want **repression**. "Decides to put it out of mind" is **suppression**. Depression sounds similar but is a mood, not a mechanism, and "motivated forgetting" is the wider memory label, not the psychoanalytic defence.

The recovered memory controversy. From the late 1980s, adults in therapy began to "recover" memories of childhood sexual abuse, and some accused relatives were tried and imprisoned on that evidence. **Elizabeth Loftus** (born 1944) and others showed how easily suggestion, hypnosis and dream interpretation can implant vivid memories of events that never happened. Abuse is real and can go unrecalled for years, but Loftus, reviewing the evidence in 2006, concluded that there is little support for the idea that trauma is commonly banished from awareness and later reliably recovered. Repression remains a clinical concept rather than a proven memory mechanism (misinformation and false memory are taught in Chapter 1).

Repression, hysteria and dissociation. The concept was born in *Studies on Hysteria* (1895), written by Freud with **Josef Breuer** (1842 to 1925). Hysterical symptoms, they argued, came from painful memories pushed out of awareness that found expression through the body. For the same reason most psychodynamic theorists hold that dissociative symptoms (amnesia, fugue, multiple identities) are produced by repression: a whole segment of experience is walled off from consciousness. Conflict is what sets defence going, not the mechanism itself;

"ego states" describe the split-off parts rather than the process that splits them; and gender identity has no bearing on dissociation.

Five defences that redirect or disguise ●●●

Most questions on defences give a short story and ask you to name the mechanism. Five defences carry most of these stories. Three of them move an impulse somewhere, so the key question is where it goes; Figure 4.3 follows one impulse through all five.

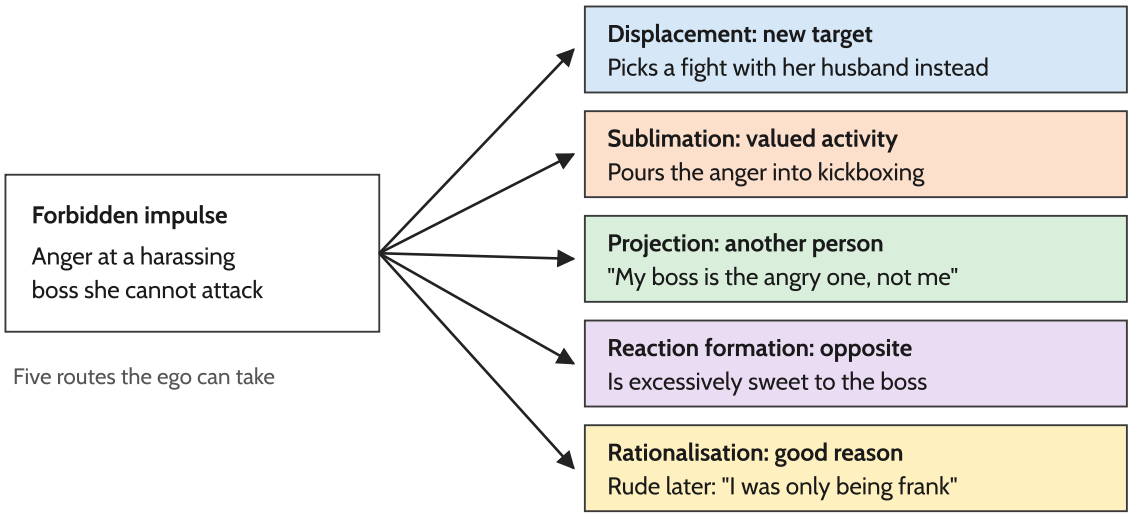


Figure 4.3 One impulse, five disguises. Notice what changes in each: the target (displacement), the activity (sublimation), the owner (projection), the direction (reaction formation) or the stated reason (rationalisation).

In **displacement** a feeling, usually anger, is redirected from a threatening target to a safer one that cannot hit back. The impulse stays the same; only its object changes. The woman angry with her husband shouts at her child; the teenager furious with a teacher goes home and shouts at his younger sibling; the man upset with his girlfriend yells at his dog; the woman harassed by her boss picks a fight with her husband. Because the substitute never discharges the tension as fully as the real target would, undischarged tension builds up. Displacement of fear, rather than anger, is the classic mechanism of phobias (see below).

In **projection** a person attributes his or her own unacceptable thoughts and impulses to someone else. The person says, in effect, "I don't hate him; he hates me." A husband who struggles with his own wish to stray accuses his faithful wife of flirting. The impulse is still expressed, but as somebody else's, which feels far less threatening.

In **sublimation**, sexual or aggressive energy is rechannelled into a socially valued activity. Roshan, who is full of anger and frustration, enrols in a kickboxing class; aggressive drives go into competitive sport, and frustrated sexual energy goes into painting or poetry. Displacement changes only the target, whereas sublimation transforms the impulse's aim into something society admires. Freud saw art, science and civilisation itself as built on sublimation and treated it as the healthiest defence; Vaillant later placed it among the mature ones.

The term **rationalisation**, introduced by Freud's follower **Ernest Jones** (1879 to 1958) in 1908, means a logical, socially acceptable reason invented after the act for behaviour that was really driven by unconscious or unacceptable motives. Clare complains about her duties, rarely finishes her tasks and is regularly late; when she is fired she says a co-worker badmouthed her. Arjun's "statistics is useless anyway" is the "sour grapes" form, after the fox in the fable who decides that the grapes he cannot reach are sour. Unlike a lie, the excuse is believed by the person who gives it.

In **reaction formation** the opposite of a threatening impulse is expressed: someone troubled by sexual longings becomes a crusader against pornography, and a child who resents a new baby smothers it with exaggerated affection. The giveaway is the excessive, rigid quality of the displayed feeling.

> EXAM TRAP
>
> Read the stem for its landing place: a **safer target** is displacement, a **socially valued activity** is sublimation, **another person's mind** is projection. Clare blames a co-worker, which tempts you towards projection or displacement, and she plays down her failings, which tempts you towards denial. But she does not deny the dismissal, and she moves no anger onto a safer target; she supplies an after-the-fact excuse that protects her self-esteem: rationalisation. Do not confuse rationalisation (an excuse) with intellectualisation (abstract thinking that keeps feeling away).

The wider catalogue

Beyond the big five lies a long list drawn from Freud, Anna Freud, **Alfred Adler** (1870 to 1937) and the object-relations school. Table 4.9 gives each one a definition and an everyday example; the defences found only in Vaillant's ladder are defined with it (below).

Table 4.9 The wider catalogue of defences.

Defence	What the person does	Everyday example
Denial	Refuses to acknowledge a painful outer reality	Parents keep a dead child's room unchanged
Regression	Retreats to the behaviour of an earlier, safer stage	A six-year-old wets the bed again after a sibling's birth
Fixation	Part of the libido stays tied to an earlier psychosexual stage (Section 4.2)	Lifelong oral habits such as nail-biting
Intellectualisation	Uses abstract, technical thinking to avoid feeling	Discusses her own cancer only as survival statistics
Isolation of affect	Keeps the thought but strips away its feeling	Describes a gruesome fantasy with no anxiety or disgust
Undoing	Performs an act that symbolically cancels a forbidden thought or act	Washes hands repeatedly after a hostile thought
Identification	Takes on another's qualities to feel safer or stronger	A boy copies his father's manner
Identification with the aggressor	Becomes like the person one fears (Anna Freud)	A child frightened at the dentist plays at being the dentist
Introjection	Takes another's values or qualities into the self; the reverse of projection	Parents' rules become one's own conscience

Defence	What the person does	Everyday example
Turning against the self	Redirects hostility from another person onto oneself	Bitter self-blame after a parent's death
Compensation and overcompensation	Makes up for a weakness by excelling elsewhere; or by over-striving in the very area of weakness (Adler, Section 4.4)	A frail boy becomes a chess champion; a stammerer becomes an orator
Conversion	Turns psychic conflict into a bodily symptom with symbolic meaning	An arm is paralysed in a man who wished to strike his father
Somatisation	Expresses distress as bodily complaints	Stomach aches every school morning
Dissociation	Drastically and temporarily alters memory, awareness or identity to escape distress	No memory of a terrifying accident
Splitting	Sees self or others as all good or all bad, never both	A doctor is "perfect" on Monday and "a monster" by Friday
Projective identification	Projects a disowned feeling into another, then behaves so that the other comes to feel it (Melanie Klein)	A patient provokes the therapist until the therapist feels the patient's rage
Acting out	Expresses an unconscious impulse in action so as not to feel it	A rejected teenager gets into fights instead of feeling hurt
Passive aggression	Expresses hostility indirectly, through passivity	Keeps "forgetting" a colleague's request

Several entries are worth separating. Three belong together because all appear in obsessional neurosis: **isolation of affect** keeps the idea in awareness but removes its emotion, **intellectualisation** goes further and buries the emotion under abstract analysis, and **undoing** adds an act that magically "cancels" the thought. Plain **identification** copies another's behaviour, and **identification with the aggressor**, Anna Freud's special case, copies the very person one fears; **introjection** goes deeper and swallows another's values whole, so that they feel like one's own. Note too that **regression**, a retreat to the behaviour of an earlier stage, is a general response to stress rather than the mark of any one disorder, and **dissociation**, though dramatic, is ranked neurotic rather than immature by Vaillant. In the same way, **conversion** (Freud's term from his work on hysteria) produces a single symbolic symptom such as paralysis or blindness, whereas somatisation is a general tendency to express distress through the body. Finally, **splitting**, described by **Melanie Klein** (1882 to 1960) in the infant's paranoid-schizoid position (Section 4.5), keeps good and bad images apart so that the bad cannot spoil the good; **projective identification**, also Klein's, adds the push that makes the other person actually carry the projected feeling. Adler's compensation is treated fully in Section 4.4.

Projection, introjection, empathy and denial apart

Projection has a broad sense as well as the narrow defensive one. Freud also used the word for any externalising of inner states, and in this broad sense projection is any distorting or selective effect of a person's inner world on his perception or conception of the outer world. This broad sense underlies the **projective hypothesis**: given an ambiguous stimulus, people read their own needs and conflicts into it. **Lawrence K. Frank** named such tests projective methods in 1939 (they are taught in Chapter 2).

The neighbouring terms run in other directions. Introjection moves something from outside to inside, the reverse of projection. By contrast, **empathy** is perceiving and sharing another person's feelings from that person's point of view; it is an accurate act of understanding, not a defence, and it distorts nothing. And **denial** blocks a particular, unwelcome fact of outer reality ("my son cannot be dead"); it does not colour perception in general.

> EXAM TRAP
>
> RML 2024 asked for "any distorting or selective effect of a person's inner world on his perception or conception of the outer world" and keyed **denial**. That wording is the broad definition of **projection**, the basis of the projective hypothesis, so the key is doubtful. Denial refuses a specific reality; introjection takes the world in; empathy reads others accurately. If the same stem appears, projection is the defensible answer.

Vaillant's hierarchy of maturity ●●●

Are some defences healthier than others? **George Eman Vaillant** (born 1934), an American psychiatrist, answered with decades of life-history data (see the Key study box) and arranged the defences in four levels of maturity (Table 4.10).

Table 4.10 Vaillant's four levels of defence (1977).

Level	Defences	Hallmark
I Psychotic	Delusional projection, psychotic denial, distortion	Reality testing lost; normal in dreams and early childhood
II Immature	Projection, schizoid fantasy, hypochondriasis, passive aggression, acting out	Normal in adolescence; in adults, common in personality disorders
III Neurotic (intermediate)	Intellectualisation, isolation, repression, displacement, reaction formation, dissociation	Common in healthy adults under stress; private, inner distortions
IV Mature	Altruism, anticipation, humour, sublimation, suppression	Integrate reality, feelings and relationships; most adaptive

Several names appear only in this ladder. At Level I, **delusional projection** hardens projection into a frank, usually persecutory delusion ("the neighbours are poisoning my water"); **psychotic denial** denies outer reality itself, as when a widow insists her husband is alive; **distortion** grossly reshapes reality to fit inner needs, as in a belief in one's own divine powers. At Level II, **schizoid fantasy** is retreat into private daydreams to avoid conflict and people, and **hypochondriasis** turns reproach towards others (after bereavement or loneliness) into complaints of illness. At Level IV, **altruism** gains satisfaction from constructive service to others, **humour** expresses painful feelings openly without hurting anyone, **anticipation** is realistic planning for future discomfort, and **suppression** is the conscious decision to postpone attention to a worry.

KEY STUDY: VAILLANT AND THE GRANT STUDY MEN (1977)

Question: do the defences a person habitually uses predict how well his life turns out? **Method:** the Grant Study of Adult Development has followed 268 Harvard men since 1938 through interviews, questionnaires and medical examinations. For *Adaptation to Life* (1977), judges rated the defences the men showed in crises over more than three decades. **Result:** the maturity of defences predicted adult adjustment: men who relied on mature defences had better marriages, careers, and physical and mental health, and defences tended to mature with age. **Criticism:** the sample was all male, mostly white and privileged, and the ratings rest on judges' inferences from life histories.

Some psychiatry textbooks extend the lists, calling Level I narcissistic and adding, for example, rationalisation and inhibition to the neurotic group and regression and somatisation to the immature group. The DSM-IV Defensive Functioning Scale (1994) later spread the defences over seven levels, from the high adaptive level down to defensive dysregulation, placing splitting and projective identification at the major image-distorting level.

EXAM TRAP

Repression, projection and reaction formation are **not** mature; only altruism, anticipation, humour, sublimation and suppression are. Splitting (primitive), sublimation (mature) and projection (immature) are **not** neurotic, so intellectualisation is the neurotic choice. **Dissociation is neurotic**, not immature. Denial, distortion and delusional projection together mark the psychotic level. No defence is "conscious" as a class; suppression is the one deliberate exception, and it sits in the mature group.

Defences typical of particular disorders ●●●

Psychoanalysts noticed that each neurosis seemed to favour its own defences, and Table 4.11 lists the classic pairings (the disorders themselves are taught in Chapter 8).

Table 4.11 Characteristic defences of particular disorders.

Disorder	Characteristic defences	Note
Obsessive-compulsive disorder	Isolation of affect, undoing, reaction formation, intellectualisation	Repression is the one not typical
Phobias	Displacement, with projection and avoidance	Little Hans (1909)
Hysteria, conversion and dissociative disorders	Repression, conversion, dissociation	*Studies on Hysteria* (1895)
Paranoia	Projection	Klein's splitting in the paranoid-schizoid position
Borderline personality disorder	Splitting, projective identification	Otto Kernberg
Depression	Introjection, turning against the self	Anger at a lost person turned inward
Mania	Denial	Denial of depressive pain
Schizophrenia	Psychotic-level defences	Delusional projection, distortion

KEY STUDY: LITTLE HANS (1909)

In *Analysis of a Phobia in a Five-Year-Old Boy* (1909), Freud described **Herbert Graf**, a Viennese boy who was terrified that a horse would bite him and would hardly go out into the street. The case was conducted mainly through the boy's father, a follower of Freud, who reported to him by letter. Freud concluded that Hans's Oedipal fear of his father had been displaced onto horses, whose blinkers and dark muzzles recalled the father's spectacles and moustache; a horse, unlike a father, can simply be avoided. **Criticism:** the evidence came from a sympathetic father asking leading questions, and **Joseph Wolpe** and **Stanley Rachman** (1960) argued that the phobia was conditioned after Hans saw a horse fall in the street.

The logic behind the table is worth knowing. In obsessional neurosis the disturbing idea stays fully conscious (that is what an obsession is), but its feeling is isolated, compulsions undo it, and excessive cleanliness or kindness stands in reaction to it; hysteria, by contrast, represses the idea itself. In phobia, a fear is displaced onto an object that can be avoided, as in Little Hans. In paranoia, the forbidden feeling is projected, so that hatred within becomes persecution from without. **Otto Kernberg** (born 1928) made splitting and projective identification the core of borderline personality organisation, which is why the person with borderline personality disorder swings between idealising and devaluing the same person; regression, undoing and sublimation are not characteristic. In depression, following Freud's *Mourning and Melancholia* (1917), the lost person is introjected and the anger meant for them is turned against the self.

EXAM TRAP

"Which defence is used in OCD: isolation, undoing, reaction formation, displacement?" Three options are true. When the key is **undoing** (RML 2024), take it: undoing is the defence most directly tied to compulsive rituals. When the stem asks which is NOT typical of OCD, the answer is **repression**. Asked where displacement is typically seen, answer **phobias** (IHBAS 2020), not hysteria, OCD or depression.

Development, research and clinical use

Defences have a developmental order. **Phebe Cramer**, coding children's Thematic Apperception Test stories (1987), found that denial dominates in early childhood, projection rises in later childhood, and identification takes over in adolescence, as Figure 4.4 shows. Each step demands more cognitive skill: denial needs only a refusal, projection needs a sense of other minds, identification needs a stable self that can borrow from others. A later study of pupils from Grade 2 to college found the same pattern: denial and projection fell with age while identification rose.

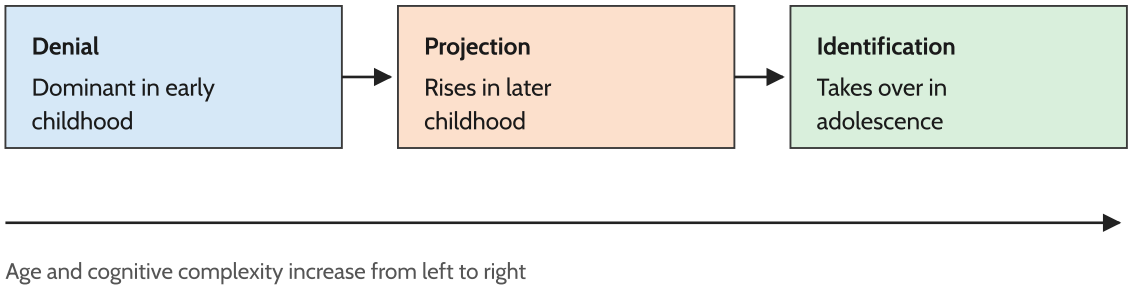


Figure 4.4 Cramer's developmental sequence. Simpler defences come first and give way to ones that need more cognitive skill.

Laboratory evidence for repression is debated. **Sam Glucksberg** and L. J. King (1967) paired some words with electric shock and found that people later forgot those words and related ones, which looks like repression. Others have found that habitual "repressors", people low in reported anxiety but high in defensiveness, recall fewer negative childhood memories. Yet **David S. Holmes** (1990), reviewing sixty years of studies, found no convincing controlled evidence that repression works as Freud described, and the recovered memory debate has kept the question open.

In everyday life defences are neither rare nor shameful: they soften losses, protect self-esteem and buy time. Clinically, they matter in two ways. The defences a person habitually uses help a clinician gauge the level of personality organisation, and in psychodynamic therapy defences show up as resistance, which the therapist gently interprets so that the patient can face what is being avoided and replace rigid, immature defences with mature ones (therapy is taught in Chapter 7).

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Anna Freud catalogued the defences in *The Ego and the Mechanisms of Defence*, published in _____.
2. The involuntary, unconscious blocking of threatening material from awareness is _____; its deliberate, conscious counterpart is _____.
3. A man upset with his girlfriend who yells at his dog is using _____.
4. Rechannelling aggression into kickboxing or competitive sport is _____.
5. Describing a gruesome fantasy without any anxiety or disgust shows _____ of affect.
6. In Vaillant's 1977 hierarchy, dissociation belongs to the _____ level, and altruism to the _____ level.
7. The primary defence associated with borderline personality disorder is _____.
8. Cramer's developmental sequence runs from denial to projection to _____.

Answers: (1) 1936, (2) repression; suppression, (3) displacement, (4) sublimation, (5) isolation, (6) neurotic; mature, (7) splitting, (8) identification.

4.4 Jung's Analytical Psychology and Adler's Individual Psychology

MEDIUM YIELD · 25 QS

In March 1907 a 31-year-old Swiss psychiatrist, Carl Jung, met Sigmund Freud in Vienna, and the two talked for almost thirteen hours. Six years later they were no longer speaking, and Alfred Adler had already walked out of Freud's circle. Each rebel kept part of Freud's framework but rebuilt the theory of personality at the point where he thought Freud had gone wrong.

The neo-Freudians and the break with Freud

The **neo-Freudians** began inside psychoanalysis, kept its belief in unconscious processes and in the weight of childhood, and then rebuilt the theory. The usual list is Carl Jung, Alfred Adler, Karen Horney, Erich Fromm, Harry Stack Sullivan and Erik Erikson (Horney, Fromm and Sullivan are in Section 4.5, Erikson in Chapter 6). Their changes ran one way: less sex and instinct; more social, cultural and conscious forces; and a person pulled toward goals, not only pushed by the past.

Adler left first. A member of Freud's circle from 1902 and president of the Vienna Psychoanalytic Society, he argued that inferiority and social striving, not sexuality, lay behind neurosis. In 1911 he resigned with 9 of the 23 members, and in 1912 he founded the Society for Individual Psychology. Jung was Freud's chosen successor. Freud called him his "crown prince", took him on the 1909 lecture trip to Clark University in Massachusetts, and saw him become first president of the International Psychoanalytic Association in 1910. But Jung would not accept that libido was purely sexual, and they parted in 1913 (Table 4.12).

Table 4.12 Freud, Jung and Adler compared.

Issue	Freud	Jung	Adler
School	Psychoanalysis	Analytical psychology	Individual psychology
Main motive	Sexual and aggressive instincts	General psychic energy	Striving for superiority; social interest
Unconscious	Personal; central	Personal and collective; even more central	Played down
Direction	Pushed by the past	Past and future	Pulled by future goals
Key period	First five years	Middle age	First four or five years
Left Freud	(founder)	1913	1911

> EXAM TRAP
>
> Lists of neo-Freudians hide outsiders: Hans Eysenck (traits, Section 4.7), Albert Bandura (social-cognitive, Section 4.9) and Carl Rogers (humanist, Section 4.8). Freud is the founder, not a neo-Freudian. Jung, Horney and Adler is the safe triple.

Jung's psyche: energy, ego and complexes

Carl Gustav Jung (1875 to 1961), a Swiss pastor's son, worked from 1900 at the Burghölzli hospital in Zurich under Eugen Bleuler. His **analytical psychology** begins with energy. Jung

widened libido from sexual energy to general psychic energy, the energy of all perceiving, thinking, feeling and wishing; sex is only one of its uses. It obeys three principles borrowed from physics. By the **principle of opposites**, conflict between polarities, such as love and hate, generates energy. By the **principle of equivalence**, energy withdrawn from one interest reappears in another of equal value, as when a retired cricketer's passion flows into coaching. By the **principle of entropy**, energy flows from stronger to weaker parts of the personality, tending toward a balance that is never reached.

The psyche has three levels (Figure 4.5). The ego is the conscious mind, the centre of awareness. Below it lies the **personal unconscious**, material once conscious but forgotten or suppressed, much like Freud's preconscious. There related memories and feelings gather into complexes. A **complex** is an emotionally charged cluster of ideas organised around a theme, such as a mother complex or a power complex; its owner rarely notices it, though others do. To detect complexes Jung developed, from 1904 with **Franz Riklin**, the **word association test**.

KEY STUDY: JUNG'S WORD ASSOCIATION TEST (1904 ONWARDS)

Question: can a hidden complex be detected objectively? **Method:** Jung read out 100 stimulus words (head, green, water, death), the patient answered each with the first word that came to mind, and Jung timed every reply and recorded physiological signs such as breathing and skin resistance. **Result:** certain words produced complex indicators: a long delay, no answer, repeating the stimulus word, the same reply to different words, a slip, or a bodily reaction. These marked the theme of a complex. **Legacy and criticism:** the test fed projective testing and lie detection, but its interpretation stays subjective. Francis Galton and Wilhelm Wundt had studied word association earlier; as a clinical tool for complexes it is Jung's.

EXAM TRAP

Word association test: Jung. Not Henry Murray (the Thematic Apperception Test, Chapter 2), not Freud (free association, talking freely with no stimulus words) and not Donald Winnicott (object relations: the transitional object and the good enough mother, Section 4.5).

The collective unconscious and the archetypes ●●●

Freud's unconscious was essentially personal. Jung held that the unconscious exists on two levels, the personal (individual) and the collective. The **collective unconscious** is the deepest layer of the psyche: inherited, universal, the same in every human being, the deposit of the experience of our human and pre-human ancestors. Jung accounted for it through evolution: the brain has the same structure in all human races, so it carries the same inherited predispositions. His evidence was that the same symbols (the dragon-slaying hero, the great flood, the sacred circle) recur in the myths of cultures that never met and in the dreams of patients who had never read those myths. We inherit readiness, not memories: not a fear of snakes, but the potential to fear them.

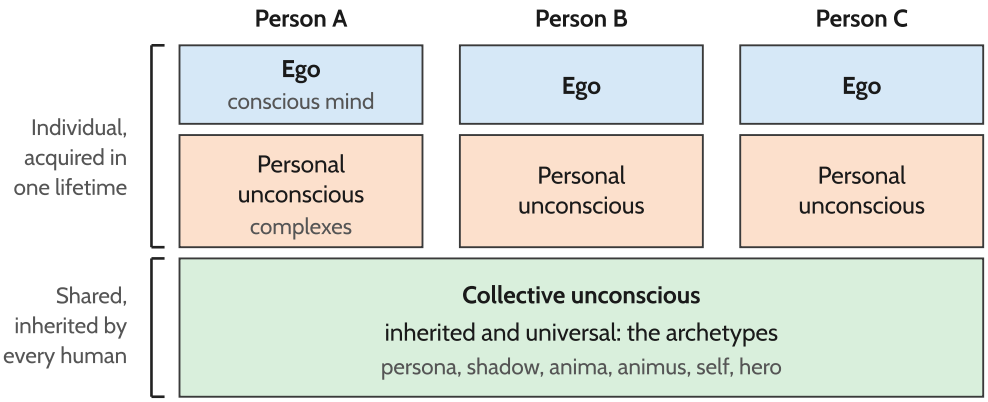


Figure 4.5 Jung's model of the psyche. Each person has an ego and a personal unconscious of his or her own, but all rest on the same collective unconscious, the inherited store of archetypes.

The contents of the collective unconscious are **archetypes**, which Jung also called primordial images: universal patterns that shape how we respond to life's recurring experiences, such as mother, birth and death (Table 4.13). The **persona** is the social mask; trouble starts when the ego mistakes it for the whole person. The **shadow**, the most powerful archetype, is the disowned animal side, the source of evil but also of vitality and creativity. The **anima** is the feminine side of a man and the **animus** the unconscious masculine side of a woman. The **self**, the archetype of wholeness and unity, is the centre of the whole psyche (the ego is only the centre of consciousness), symbolised by the **mandala**, a circle usually divided into four.

Table 4.13 Jung's main archetypes.

Archetype	What it is	Typical image
Persona	Social mask, public role	Uniform, stage mask
Shadow	Dark, animal, disowned side	Devil, villain
Anima	Feminine side of a man	Muse, idealised woman
Animus	Masculine side of a woman	Idealised man, rescuer
Self	Wholeness and unity	Mandala, circle
Hero	Triumph over great odds	Dragon slayer
Great mother	Nurturing source of life	Earth mother
Wise old man	Wisdom and guidance	Sage, guru
Trickster	Mischief that upsets order	Clown, cunning fox
Child	New beginnings, potential	Divine child

> **REMEMBER**
>
> *Anima* carries the feminine Latin ending -a (as in *femina*, woman): the woman within a man. *Animus* carries the masculine -us: the man within a woman.

> **EXAM TRAP**
>
> The collective unconscious is not repressed personal memories (the personal unconscious), not socially learned attitudes and not ruled by the reality principle (Freud's ego, Section 4.2). It is inherited, universal and archetypal, and it is Jung's, not Freud's, Adler's or Horney's. Asked for the two levels of the unconscious, answer collective and individual: anima and animus are archetypes, extraversion and introversion are attitudes, and intuitive and perceptive are type labels.

Attitudes, functions and the eight types

In *Psychological Types* (1921) Jung asked why people meet the same world so differently. His first answer was two attitudes, his own terms for the direction of psychic energy. In **extraversion** it flows outward, to people and things: the extravert is sociable and assertive. In **introversion** it flows inward, to one's own thoughts and feelings: the introvert is reserved and reflective. Everyone has both; one dominates consciousness, while the other sinks into the personal unconscious and still leaks out, as when a shy student is drawn to a loud friend.

His second answer was four functions in two opposed pairs (Figure 4.6). Thinking judges true or false; feeling judges pleasant or unpleasant. Because both evaluate, they are the **rational functions**. Sensing registers what is there, like a camera; intuiting grasps possibilities and hunches beyond the evidence. Because both perceive without evaluating, they are the **irrational functions** (also called nonrational). One function dominates; the other member of its pair, the **inferior function**, stays undeveloped and largely unconscious, so a strong thinker may be clumsy with feelings. Two attitudes times four functions give Jung's eight psychological types, from the extraverted thinking type (logical, rule-bound) to the introverted intuiting type (the visionary daydreamer).

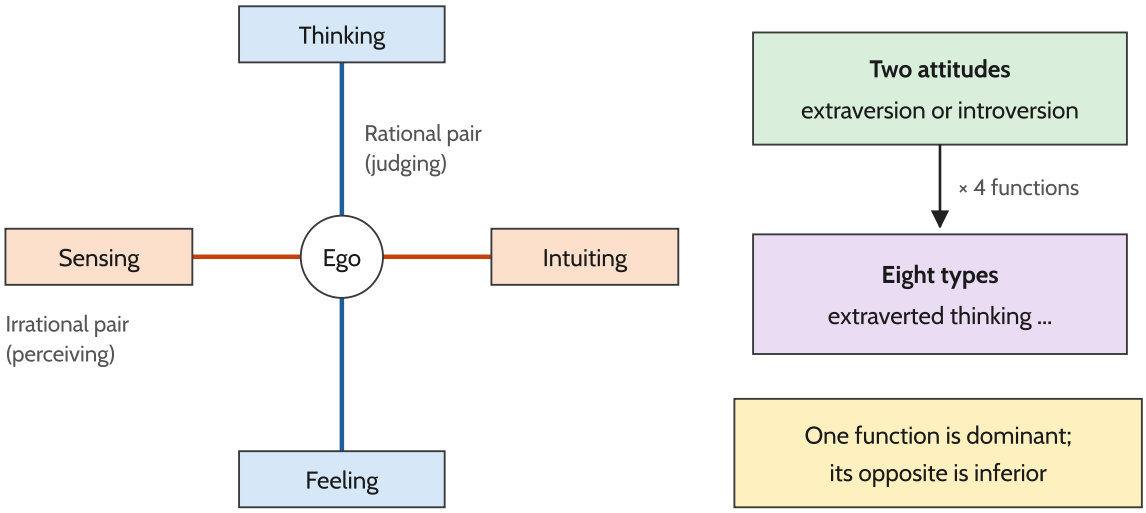


Figure 4.6 Jung's four functions form two opposed pairs: thinking and feeling judge (rational, vertical axis); sensing and intuiting perceive without judging (irrational, horizontal axis). Crossed with the two attitudes they give eight types.

Katharine Cook Briggs (1875 to 1968), who read Jung's book in 1923, and her daughter **Isabel Briggs Myers** (1897 to 1980), neither trained in psychology, turned the types into the **Myers-Briggs Type Indicator** (MBTI), whose first form appeared in 1943. It scores four di-

chotomies, extraversion-introversion (E-I), sensing-intuition (S-N), thinking-feeling (T-F) and judging-perceiving (J-P, their own addition), giving 16 types such as INTP. Hugely popular in workplaces, it is psychometrically weak: it forces continuous traits into either-or boxes, and its test-retest reliability is poor, with many people (in some studies about half) typed differently when retested a few weeks later. Researchers prefer trait inventories (Section 4.7; Chapter 2).

> EXAM TRAP
>
> Jung's two attitudes are extraversion and introversion; thinking, feeling, sensing and intuiting are functions; rational and irrational label the pairs of functions. The inventory based directly on Jung's types is the MBTI, not the MMPI (an empirically keyed clinical inventory), the 16PF (Raymond Cattell's factors, Section 4.6) or the NEO-PI-R (Paul Costa and Robert McCrae's Big Five, Section 4.7).

Individuation, method and evaluation of Jung

Freud thought personality was largely settled by five; Jung's crucial period was middle age, the time of his own breakdown after 1913. In the first half of life energy goes outward, to education, career and family, under consciousness and extraversion. Between about 35 and 40 many successful people feel empty. Jung saw this midlife crisis as natural, a signal to turn inward, from the material to the spiritual and from extraversion toward introversion. The goal of the second half of life is **individuation**, becoming a whole, undivided individual by integrating conscious and unconscious: dethrone the persona, accept the shadow, express the anima or animus, and transcend, so that the self, not the ego, becomes the centre. Such self-realisation is rarely complete. **Synchronicity**, a meaningful coincidence with no causal link, belongs to the same mystical side of Jung.

Jung assessed patients by word association, symptom analysis (free associations to the symptoms) and dream analysis. He studied series of dreams, saw them as prospective (preparing for the future) and compensatory (balancing a one-sided conscious attitude), and used **amplification**: circling back to one dream image and enlarging it with repeated associations and parallels from myth, instead of following free associations away from it. Critics call his evidence (myth, alchemy, the occult, his own visions) mystical, his writing obscure and the collective unconscious unfalsifiable. Yet introversion-extraversion (Section 4.7), complexes, word association, the midlife transition and individuation, a forerunner of Abraham Maslow's self-actualisation, all lasted.

Adler: the individual as a goal-seeking whole ●●

Alfred Adler (1870 to 1937), born near Vienna, was a sickly second son with rickets who nearly died of pneumonia at four and envied a healthy older brother. A physician, he joined Freud's circle as a colleague, never a pupil, and after the break founded **individual psychology**; he died on a lecture tour in Scotland. Individual here means indivisible: each person is a unique, unified whole, not an id, ego and superego at war, and the whole person moves toward a goal. The basic assumption of individual psychology is that human behaviour is purposeful and goal-directed: Freud explained behaviour by past causes, Adler by future goals that pull us forward. The goals are social. In place of Freud's sexual instinct, Adler held that people are motivated primarily by social urges, the need to belong and to count among oth-

ers. He gave the conscious more weight than the unconscious and saw people hopefully, as free to shape themselves.

> **EXAM TRAP**
>
> The basic assumption is purposeful, goal-directed behaviour. Optimism is a feature of Adler's view, not its basic assumption; "social beings understood through their relationships" describes his social emphasis but fits Sullivan's interpersonal theory better (Section 4.5); lifelong development with ego identity at the centre is Erik Erikson (Chapter 6). The motive is social urges: not conscious drives (drive is Freud's language), not neurotic needs (Karen Horney, Section 4.5) and not empathic understanding (a therapist condition of Carl Rogers, Section 4.8). Individual psychology is Adler's, not Horney's, Jung's or Erikson's.

Inferiority, compensation and striving for superiority ●●●

Adler's *Study of Organ Inferiority and Its Psychical Compensation* (1907) argued that a weak organ or body part drives the person to make up for it, sometimes overshooting into strength, as the Greek orator Demosthenes overcame a stutter. This idea of **organ inferiority** grew into a theory of all motivation. Every child starts small and dependent among powerful adults, so inferiority feelings are universal and normal, and they are the source of all human striving. The effort to overcome them is **compensation**; going beyond the deficit, as the frail boy who becomes an athlete does, is overcompensation (Figure 4.7).

When compensation fails, the feelings harden into an **inferiority complex**, Adler's term: a lasting lack of self-worth, with helplessness, hopelessness and avoidance of challenge. It has three childhood sources: organ inferiority, pampering (the spoilt child crumbles when school does not revolve around him) and neglect (the unwanted child learns that others are hostile). Children with organic defects or born into minority groups are therefore at risk. Overcompensation can instead produce a **superiority complex**, an exaggerated self-importance shown in boasting, vanity and belittling others.

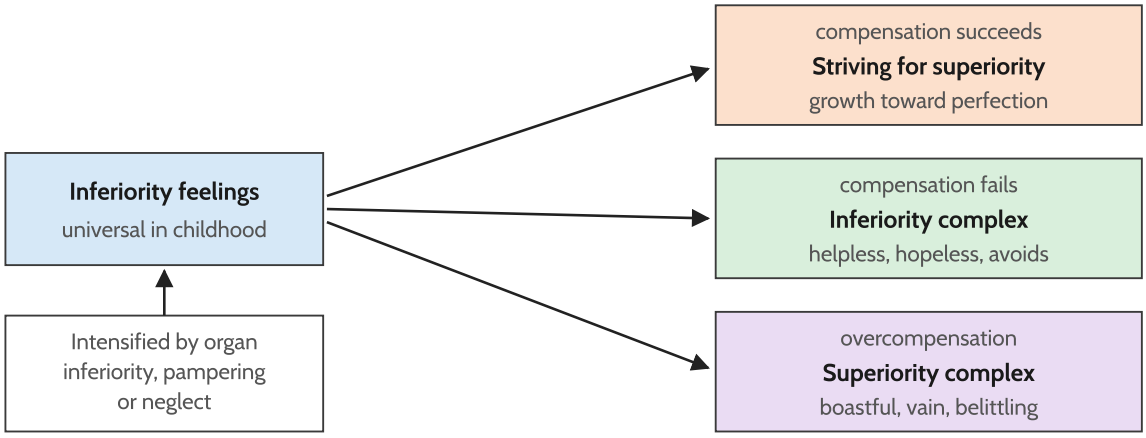


Figure 4.7 Adler's account of inferiority. Universal inferiority feelings drive all striving; successful compensation becomes striving for superiority, failure an inferiority complex, and overcompensation a superiority complex.

Neither complex is **striving for superiority**, the healthy, innate drive toward perfection (completion) that Adler called the fundamental fact of life: an effort to perfect oneself and

one's community, not to dominate others.

EXAM TRAP

The inferiority complex and the superiority drive are Adler's, not Jung's, Viktor Frankl's (logotherapy, Chapter 7) or Erikson's. Adler coined the term, not Franz Alexander (psychosomatic medicine), Gordon Allport (traits, Section 4.6) or Jung; obituaries wrongly credited Freud and Jung with it. Organ inferiority is Adler's too, not Sullivan's, Melanie Klein's (both Section 4.5) or Anna Freud's (Section 4.3). Overcompensation, lack of self-worth and hopelessness go with the complex; certainty in decision making is its opposite.

Fictional finalism, style of life and social interest

What goals pull us? Adler borrowed from the philosopher **Hans Vaihinger** (1852 to 1933), whose *The Philosophy of "As If"* (1911) argued that people live by useful fictions, unprovable ideas acted on as if they were true, such as "all people are created equal". Adler's **fictional finalism** is the idea that each person is guided by an imagined final goal, a subjective ideal of success or completeness; he preferred the terms subjective final goal or guiding self-ideal.

Each person pursues the goal through a unique **style of life**, the characteristic pattern of traits, habits and attitudes by which one strives for superiority. It grows from the child's compensations, birth order and parenting, and is largely fixed by four or five. Yet it is not fate: through the **creative power of the self** each child interprets heredity and environment and builds its own style.

The test of a healthy style is **social interest** (*Gemeinschaftsgefühl*, community feeling), an innate potential to cooperate with others for personal and social goals, which the mother must nurture. It is needed for the three life tasks of work, friendship and love, and its absence underlies neurosis and crime. Crossing social interest with degree of activity, Adler described four styles of life (Table 4.14).

Table 4.14 Adler's four styles of life.

Type	Social interest	Activity	Pattern
Ruling or dominant	Low	High	Dominates others; extreme forms turn delinquent
Getting or leaning	Low	Low	Expects to receive; dependent; the commonest type
Avoiding	Low	Lowest	Sidesteps problems to avoid failure
Socially useful	High	High	Cooperates and contributes; healthy

EXAM TRAP

Fictional finalism is an imagined goal that guides behaviour: not parental expectations, a repressed unconscious conflict or a biologically determined trait. In matching items, Jung with extraversion-introversion, Adler with the ruling dominant type and Albert Ellis with rational emotive therapy (Chapter 7) are correct pairs; Kohlberg with formal operations is wrong, because formal operations is Jean Piaget's stage and Lawrence Kohlberg's theory is of moral reasoning (Chapter 6).

Birth order, early recollections and Adler's legacy

For Adler, siblings raised by the same parents do not grow up in the same family, because each birth position is a different situation (Table 4.15). The first-born rules alone until a new baby dethrones it, and so knows power and wants it back; the second has a pacesetter ahead and, Adler said, behaves as if in a race. Large modern studies, however, find birth-order effects on personality traits weak or near zero.

Table 4.15 Adler's birth-order positions.

Position	Situation	Likely style
First-born	Dethroned by the next baby	Power-minded, past-oriented, conservative, responsible; may turn insecure
Second or middle	A pacesetter always ahead	Ambitious, competitive, rebellious, always in a race; may give up
Youngest	Never dethroned; family pet	Pampered; may stay spoilt and dependent or become highly ambitious
Only child	Never loses the centre	Matures early among adults; pampered; dislikes sharing the stage

> **EXAM TRAP**
>
> For Adler the middle (second) child is competitive and ambitious, racing to overtake the one ahead. "Overachieving" is too narrow (the second child may instead give up), "resentful of the older child's freedom" is not his description, and inferiority feelings are universal, not the mark of the middle child.

Adler's entrance gates to mental life were birth order, dreams and **early recollections**, a person's earliest memories which, true or imagined, reveal the style of life, because we remember what fits our present view of ourselves. Dreams, for him, rehearse solutions to present problems. His counselling methods are in Chapter 7. Critics call Adler commonsensical and unsystematic, but Abraham Maslow, Julian Rotter (social learning theory) and Albert Ellis (rational emotive therapy) built on his stress on goals and beliefs, and Karen Horney (Section 4.5) on his social emphasis.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Jung widened libido from sexual energy to general _____ energy.
2. Jung detected hidden complexes with the _____ test, timing each reply.
3. The deepest, inherited layer of the psyche, shared by all humanity, is the _____.
4. The feminine side of a man's psyche is the _____; the masculine side of a woman's is the _____.
5. Thinking and feeling are Jung's _____ functions; sensing and intuiting are his _____ functions.
6. Adler's 1907 book on _____ began his theory of compensation.
7. An imagined goal that guides behaviour is Adler's _____.
8. Adler said the _____ child behaves as if in a race.

Answers: (1) psychic, (2) word association, (3) collective unconscious, (4) anima; animus, (5) rational; irrational, (6) organ inferiority, (7) fictional finalism, (8) second (middle).

4.5 Horney, Fromm, Sullivan and the Object-Relations Theorists

LOWER YIELD · 10 QS

Freud's followers kept his couch but argued about what drives the person lying on it. Karen Horney, Erich Fromm and Harry Stack Sullivan moved the source of personality from instinct to culture and relationships. Others kept more of Freud and asked how the ego adapts, how loved people become figures inside the mind, how a cohesive self is built and how language shapes the unconscious. Jung and Adler are in Section 4.4.

Horney: basic anxiety and the neurotic trends ●●

Karen Horney (1885 to 1952), a Berlin-trained analyst who taught in the United States from 1932, argued in *The Neurotic Personality of Our Time* (1937) that personality is shaped primarily by social and cultural factors, above all the child's relationships with its parents, not by biological instincts, unconscious sexual conflicts or classical conditioning.

The helpless infant's overriding need is the childhood **need for safety and security**: to feel wanted and protected. What damages a child is **parental indifference**, a lack of genuine warmth shown through favouritism, unfair punishment, broken promises or ridicule. The child feels **basic hostility** towards such parents but must repress it, out of helplessness, fear, the wish to keep even counterfeit love, or guilt. The buried hostility returns as **basic anxiety**, in Horney's words an "insidiously increasing, all-pervading feeling of being lonely and helpless in a hostile world". It is the foundation on which every later neurosis is built.

EXAM TRAP

Pervasive loneliness and helplessness, the foundation of neurosis, is **basic anxiety**. Neurotic anxiety is Freud's fear that id impulses will break through (Section 4.2); existential anxiety is dread of death, freedom and meaninglessness (Section 4.8); "innate anxiety" is not a technical term. Instincts, sexual conflicts or conditioning are never Horney's answer.

To quiet basic anxiety the child seizes a strategy (winning affection, submitting, gaining power or withdrawing), and used rigidly a strategy hardens into a need. Horney listed ten **neurotic needs**, irrational defences against anxiety that become permanent parts of the personality: affection and approval, a dominant partner, power, exploitation, prestige, admiration, achievement, self-sufficiency and independence, perfection, and narrow limits to life. Everyone feels them at times; they are neurotic when pursued compulsively as the only route to safety. In *Our Inner Conflicts* (1945) she grouped them into three **neurotic trends** (Figure 4.8). The **compliant** person moves toward people, seeking love and leaning on others; the **aggressive** person moves against people, treating life as a struggle; the **detached** person moves away from people, keeping an emotional distance. A healthy person uses all three flexibly; the neurotic is locked into one, and the clash among them is the core of neurosis.

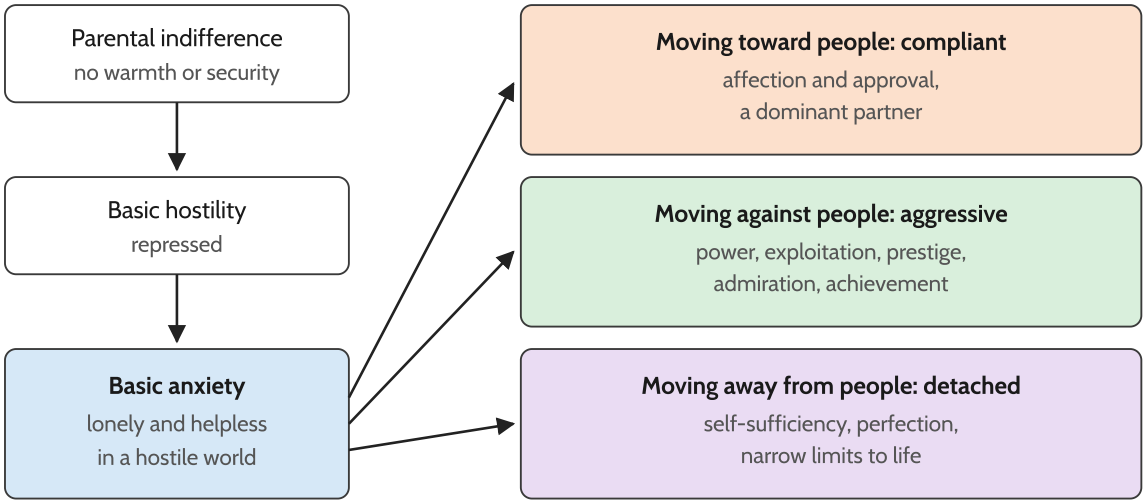


Figure 4.8 Horney's account of neurosis. Repressed hostility becomes basic anxiety, managed by one of three neurotic trends; the small text lists the neurotic needs in each.

INDIAN CONTEXT

For basic anxiety "in the Indian context" the key is **Sudhir Kakar** (born 1938), the psychoanalyst of Indian childhood and society (*The Inner World*, 1978; Section 1.2). He wins as the only psychoanalyst offered: Anand C. Paranjpe co-edited the *Handbook of Indian Psychology* (2008), Durganand Sinha (1922 to 1998) urged the indigenisation of psychology, and "Kamala Chaudhuri" is a garbled name.

Horney on the idealised self, women and "momism"

In *Neurosis and Human Growth* (1950) Horney turned to the self. A healthy self-image rests on a realistic appraisal of strengths and weaknesses. The neurotic, torn between incompatible trends, builds an **idealised self-image** of flawless virtue and tries to become it. The result is the **tyranny of the shoulds** ("I should be the perfect student, never tired, never angry"), demands no one can meet, which end in self-hatred.

Horney also challenged Freud's view of women. From 1922 she rejected Freud's penis envy as an idea drawn from neurotic women and seen through male eyes. Men, she countered, may feel **womb envy**, envy of women's power to bear children, which they sublimate into

achievement. In her **feminine psychology**, women's inferiority feelings have cultural causes, such as economic dependence and narrow roles, not anatomical ones. Her *Self-Analysis* (1942), drawn from her own lifelong practice, argued that people can ease many of their own neurotic conflicts without an analyst.

A fashion of the 1940s blamed mothers instead. **Momism** is the blaming of over-protective, idealised mothers for their sons' weakness. The writer **Philip Wylie** coined it in *Generation of Vipers* (1942), and the psychiatrist **Edward Strecker** took it up in *Their Mothers' Sons* (1946), blaming "moms" for the young men found unfit for military service in the Second World War.

> EXAM TRAP
>
> "Mom-ism" was coined by Wylie; when Wylie is not an option, the key is **Edward Strecker**, not Freud, Jung or Horney.

Fromm: escape from freedom and the existential needs

Erich Fromm (1900 to 1980), a German analyst trained also in sociology, called his approach humanistic psychoanalysis and studied whole societies. In *Escape from Freedom* (1941), written as fascism spread, he argued that modern people have won freedom from the old ties of church, clan and guild but pay for it with isolation. Many flee it through three **mechanisms of escape**: **authoritarianism** (submitting to a leader or dominating others), **destructiveness** (attacking the threatening world) and **automaton conformity** (becoming exactly what society expects). The healthy answer is positive freedom through spontaneous love and productive work.

In *The Sane Society* (1955) Fromm named five distinctly human **existential needs**, arising from our awareness of being separate from nature (Table 4.16).

Table 4.16 Fromm's five existential needs.

Need	What it is	Healthy versus unhealthy answer
Relatedness	Union with others	Productive love versus narcissism
Transcendence	Rising above passive existence	Creating versus destroying
Rootedness	Feeling at home in the world	Brotherliness versus clinging to mother and clan
Sense of identity	Knowing oneself as a separate being	Individuality versus herd conformity
Frame of orientation (consistent orientation)	A stable way of making sense of the world	Reason versus irrational belief

> EXAM TRAP
>
> Safety, love and belonging, esteem and self-actualisation are Maslow's (Section 1.10 compares the lists); competence, autonomy and relatedness are Deci and Ryan's self-determination theory; hunger, thirst and shelter are physiological needs.

Fromm's **character orientations** (*Man for Himself*, 1947) describe how people relate to the world. Four are nonproductive: the **receptive** person waits to be given all good things; the **exploitative** person takes them by force or cunning; the **hoarding** person holds on to possessions, feelings and thoughts; and the **marketing** person packages the self as a commodity for sale. The **productive** orientation is the healthy ideal: using one's own powers in love, reason and work.

Sullivan's interpersonal theory

The American psychiatrist **Harry Stack Sullivan** (1892 to 1949) went furthest. His **interpersonal theory** (*The Interpersonal Theory of Psychiatry*, published in 1953, after his death) holds that personality exists only in relations with other people, real or imagined: it is "the relatively enduring pattern of recurrent interpersonal situations which characterize a human life".

Anxiety, again, is the engine. It passes by empathy from the "mothering one" to the infant: a tense mother makes a tense baby. To avoid it the child builds a **self-system**, the organisation of experience that guards security, using **security operations** such as selective inattention, which ease anxiety but block learning. It forms **personifications**, images of self and others: the **good-me** (experiences met with tenderness), the **bad-me** (experiences met with the mother's anxiety) and the **not-me** (experiences so dreadful that they are dissociated, surfacing in nightmares and schizophrenia).

Experience matures through three modes. The **prototaxic** mode is the infant's undifferentiated flow of sensations. The **parataxic** mode links events that occur together as cause and effect, the prelogical thinking behind superstition. The **syntaxic** mode is logical and consensually validated through shared language. Sullivan's developmental epochs run from infancy and childhood through the juvenile era (the need for peers), preadolescence, and early and late adolescence to adulthood. Preadolescence brings the **chum**, a same-sex friend whose welfare matters as much as one's own: the first intimacy.

> **EXAM TRAP**
>
> A practice-mock key for "Sullivan's interpersonal theory" is "personality as an open system interacting with the environment". Accept it by elimination, but "open system" is Allport's phrase (*The Open System in Personality Theory*, 1960; Section 4.6). "Unconscious processes" is Freud, "observable behaviour" is behaviourism, and "social structures and institutions" is sociology, nearer Fromm. The eight stages from oral-sensory to maturity are Erikson's (Chapter 6), never Sullivan's.

Ego psychology: Anna Freud, Hartmann and Erikson

Ego psychology kept Freud's structural model but gave the ego more power. **Anna Freud** (1895 to 1982), Freud's youngest daughter, catalogued the defences in *The Ego and the Mechanisms of Defence* (1936; Section 4.3) and pioneered child analysis. **Heinz Hartmann** (1894 to 1970), the father of ego psychology, argued in *Ego Psychology and the Problem of Adaptation* (1939) that perception, memory, motor skill, language and thinking develop in a **conflict-free ego sphere**, not out of conflict with the id. These functions give the ego autonomy, and its task is adaptation to an "average expectable environment".

Erik Erikson (1902 to 1994), analysed by Anna Freud, carried the ego across the life span. His **epigenetic principle** holds that development unfolds by a ground plan, each part having its

own time of special ascendancy, as organs form in an embryo. His stages are psychosocial, not psychosexual: each is a crisis between the growing person and society's demands, not a shift of libido to a new body zone, and they continue into old age. The adolescent **identity crisis** is his term; the eight stages are in Chapter 6.

Object relations and Melanie Klein ●●

In psychoanalysis an **object** is the target of a drive or feeling: almost always a person, or the mental image of one, beginning with the mother. **Object relations** are the internalised mental representations of self and others (objects) and the relationships between them. A child does not just live with a mother; it carries a mother inside, coloured by love and hate, and later relationships are lived through these inner figures. The term does not mean the relationship among id, ego and superego, the use of defences, or the genital stage (Sections 4.2 and 4.3). The school's founder was **Melanie Klein** (1882 to 1960), who worked in Berlin and, from 1926, London.

Klein analysed very young children through her **play technique**, treating play with small toys as free association. At first, she concluded, the baby relates to **part objects**: the gratifying "good breast" and the frustrating "bad breast", felt as two different beings. In the **paranoid-schizoid position** of the first months, the infant protects the good by **splitting** it off from the bad, keeping love and hate apart, and projects its rage into the bad object, which then seems to persecute it. From about the middle of the first year, as the infant grasps that the loved and hated mother are one person, it enters the **depressive position**: it feels guilt over the harm its rage may have done and makes **reparation** through loving acts. Klein spoke of positions, not stages, because adults move between them all their lives (Table 4.17). In **projective identification** a person puts unwanted parts of the self into someone else and then treats that person as if they had them. **Envy** (*Envy and Gratitude*, 1957) is the infant's wish to spoil the good breast because it has what the infant lacks.

Table 4.17 Klein's two positions.

Feature	Paranoid-schizoid position	Depressive position
Begins	First months of life	About the middle of the first year
Object	Part objects: good breast, bad breast	Whole object: one mother, good and bad
Anxiety	Persecutory: fear of being attacked	Depressive: fear of having harmed the loved one
Main processes	Splitting, projection, projective identification	Guilt, mourning, reparation

> **EXAM TRAP**
>
> Klein's defence in paranoia is **splitting**: the paranoid-schizoid position rests on it, and projection works on what splitting has already separated. Projection is Freud's classic mechanism of paranoia (the Schreber case, 1911). Reaction formation (turning a feeling into its opposite) and suppression (conscious putting aside) are in Section 4.3.

Winnicott and the later object-relations theorists

The English paediatrician **Donald Winnicott** (1896 to 1971) held that the infant needs not a perfect mother but a **good-enough mother**, who meets its needs almost completely at first and then fails it in small, bearable ways, so that it learns to tolerate frustration. She provides

a **holding environment** of physical and emotional security. As the infant learns that the mother is separate, it adopts a **transitional object**, the teddy or blanket that is neither wholly "me" nor "not-me" and comforts it when she is away. Good-enough care lets a spontaneous **true self** develop; without it the infant builds a **false self**, a compliant shell that gives others what they expect.

Ronald Fairbairn (1889 to 1964) made the cleanest break with Freud: libido is object-seeking, not pleasure-seeking. **Margaret Mahler** (1897 to 1985) traced the infant's psychological birth through **separation-individuation**, from fusion with the mother to a separate self (Chapter 6). **Otto Kernberg** (born 1928) described **borderline personality organisation**: intense early frustration and aggression leave the person unable to unite good and bad images, so splitting persists into adulthood, identity is diffuse and others are all good or all bad, though reality testing is largely kept. **John Bowlby** (1907 to 1990), trained in this tradition, turned the infant's tie to its mother into attachment theory, an offshoot taught in Chapter 6.

Kohut's self psychology

Heinz Kohut (1913 to 1981) founded **self psychology**, which drew on object relations, in *The Analysis of the Self* (1971) and *The Restoration of the Self* (1977). Where Freud saw narcissism as love withdrawn from others, Kohut saw healthy narcissism as a normal line of development. The self grows through **selfobjects**, people experienced as part of the self because they perform functions the child cannot yet perform alone. Three selfobject needs must be met: **mirroring**, the gleam in the mother's eye that confirms the child's "Look at me!" and so sustains the **grandiose self**; **idealising**, merging with a calm, strong parent (the idealised parent imago); and **twinship** (alter ego), added later, feeling essentially alike with others. When parents respond well but imperfectly, each bearable failure leads the child to take over the selfobject's soothing function; this **transmuting internalisation** builds a cohesive self. When mirroring and idealising fail chronically, the self stays fragmented and enfeebled: empty and dependent on others to hold it together. Failed mirroring above all leaves the archaic grandiose self unintegrated, and it returns in adulthood as **narcissistic pathology**: grandiosity, a constant hunger for admiration and little empathy, as the person still seeks the mirroring that never came.

> EXAM TRAP
>
> Grandiosity, a need for constant admiration and a lack of empathy trace to failed parental **mirroring**. Failed idealising leaves a person searching for someone perfect to admire; twinship (alter ego) is the need to feel alike; "confrontation" is a therapist's technique, not a selfobject need.

> CLINICAL LINK
>
> A client says she is "no one", feels "empty inside", dreads abandonment and has intense, unstable relationships. Self psychology reads this as a fragmented self from failed parental mirroring and idealising. It is not oral fixation or an unresolved Oedipus complex with castration anxiety (Freud's psychosexual explanations, Section 4.2), nor guilt from a harsh superego, a neurotic conflict rather than a deficit of self. The borderline disorder itself is in Chapter 8.

Lacan, the "French Freud"

Jacques Lacan (1901 to 1981), the Paris analyst called the "French Freud", led a return to Freud, a close rereading of Freud's own texts against ego psychology's taming of them. In the **mirror stage**, between about 6 and 18 months, the infant delights in its unified mirror image; the ego is born from this identification with an outside image, and so is partly an illusion. Drawing on structural linguistics, Lacan taught that the unconscious is structured like a language. Experience moves in three registers: the **Imaginary** (images and identifications), the **Symbolic** (language, law and social order) and the **Real** (whatever resists being put into words).

EXAM TRAP

The other options are earlier French psychiatrists. Jean-Étienne Esquirol gave hallucination its modern meaning and described monomania. Jules Falret, with Charles Lasègue (1877), described folie à deux, a delusion shared by two people. Paul Briquet (1859) described the chronic, many-symptom hysteria called Briquet's syndrome, now somatisation (Chapter 8).

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Horney's all-pervading feeling of being lonely and helpless in a hostile world is _____ anxiety.
2. Horney's detached type moves _____ from people.
3. Fromm's escape by becoming exactly what society expects is _____ conformity.
4. Sullivan's mode that treats events occurring together as cause and effect is the _____ mode.
5. Hartmann's ego functions that develop free of conflict lie in the _____ ego sphere.
6. In Klein's _____ position the infant splits the good breast from the bad.
7. Winnicott's comforting teddy or blanket is a _____ object.
8. Kohut's people who perform functions the child's self cannot yet perform are _____.

Answers: (1) basic, (2) away, (3) automaton, (4) parataxic, (5) conflict-free, (6) paranoid-schizoid, (7) transitional, (8) selfobjects.

4.6 Traits I: Allport, Cattell and the Lexical Search

LOWER YIELD · 17 QS

Ask a friend to describe you and you will hear adjectives: friendly, stubborn, punctual, moody. Each one claims that you behave in a characteristic way across situations and across years. Trait theorists set out to make a science of this everyday habit. This section follows two of them, who began from the same dictionary and went opposite ways: Gordon Allport, who studied the unique individual, and Raymond Cattell, who used statistics to find the dimensions on which everyone differs. Eysenck and the Big Five follow in Section 4.7.

The trait approach and the lexical hypothesis

The **trait approach** describes personality through specific psychological attributes along which individuals differ in consistent and stable ways. A **trait** is one such attribute: an enduring dimension, such as sociability, on which everyone has a position. The approach assumes *stability* over time (the talkative child becomes a talkative adult), *consistency* across situations (sociable at a wedding and in the canteen alike) and *individual differences* (people sit at different points on the same dimension, so they can be compared and their behaviour predicted).

Three neighbouring ideas must be kept apart. A **type** sorts people into broad, discrete categories, such as Sheldon's body types (Section 4.1), and so describes personality through broad patterns; typing shades into stereotyping, a set of expected behaviours based on similarities among members of a group. **Situationism** holds that characteristics of the situation, not of the person, largely determine behaviour (Walter Mischel's challenge, Section 4.7). A state is a temporary condition: **Charles Spielberger** (1927 to 2013) separated state anxiety, a passing tension, from trait anxiety, a stable proneness to anxiety, and measured both with his State-Trait Anxiety Inventory (1970).

Where should one look for the traits that matter? The **lexical hypothesis** answers: in language. Differences between people that matter in daily life get noticed and named, so the important traits are already encoded as words, and the more important a trait, the more words a language has for it. In 1936 Gordon Allport and **Henry Odbert** searched an unabridged English dictionary and culled 17,953 words that distinguish one person's behaviour from another's. Only about 4,500 named stable traits (honest, sociable); the rest described temporary states (flustered), evaluations (admirable), or physical features and abilities. This list fed the whole factor-analytic tradition (Figure 4.9).

Such a list can be used in two ways. The **nomothetic approach** (Greek *nomos*, law) studies many people on the same dimensions to find general laws. The **idiographic approach** (*idios*, one's own) studies one person intensively to grasp his or her unique pattern. The terms are Wilhelm Windelband's (Section 4.1). Cattell went nomothetic; Allport preferred the idiographic path.

> **EXAM TRAP**
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> The trait approach focuses on *specific psychological attributes along which individuals differ in consistent and stable ways*. "Broad patterns" describes the type approach; "expected behaviour based on similarities" describes stereotyping; "situational characteristics determine behaviour" is situationism, the critique of trait theory.

Allport: traits as real structures

Gordon Willard Allport (1897 to 1967) did more than anyone to make personality a recognised field. In 1920 he called on Sigmund Freud in Vienna and, to fill an awkward silence, described a small boy he had just seen on a tram who was terrified of dirt. Freud looked at his neat, meticulous visitor and asked, "Was that little boy you?" Allport decided that psychoanalysis digs too deep and that psychology should take conscious, present motives seriously. Back at Harvard, from 1924, he taught what was probably the first American college course on personality, and his *Personality: A Psychological Interpretation* (1937) established the field. It

defined personality as "the dynamic organization within the individual of those psychophysical systems that determine his unique adjustments to his environment" (Section 4.1).

For Allport a trait was not a convenient label but a real **neuropsychic structure**. It renders many stimuli functionally equivalent and guides consistent behaviour: a friendly man greets the stranger at the bus stop, the new colleague and the neighbour's guest with the same warmth. Traits therefore cause behaviour rather than merely summarising it. Allport distinguished **common traits**, shared in varying degrees by the members of a culture so that people can be compared on them, from individual traits, which he later renamed **personal dispositions** (*Pattern and Growth in Personality*, 1961): characteristics peculiar to one person.

Personal dispositions differ in how much of a life they govern, and Allport set them on three levels (Table 4.18). A **cardinal trait** is the most pervasive and powerful kind: a single ruling passion that dominates a whole life. Few people have one, and those who do often lend it a name: Machiavellian, narcissistic, Christlike. **Central traits** are the handful, five to ten, of outstanding traits that describe a person's behaviour, the words you would put in a reference letter. **Secondary traits** are the least important, displayed inconspicuously and inconsistently, such as a taste for filter coffee, and noticed only by close friends.

Table 4.18 Allport's three levels of personal dispositions.

Level	What it is	Who has it	Examples
Cardinal	The most pervasive and powerful trait; one ruling passion	Few people	Machiavellian, narcissistic, Christlike
Central	The handful (5 to 10) of outstanding traits that describe a person	Everyone	Honest, sociable, cynical: a reference letter
Secondary	The least important; shown inconspicuously and inconsistently	Everyone, many of them	Food or music preferences, passing attitudes

> EXAM TRAP
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> Allport's levels are **cardinal, central and secondary**. "Primary, secondary, tertiary" is invented; "cognitive, emotional, physiological" are components of emotion; "id, ego, superego" is Freud's structure (Section 4.2). Match the definitions: most pervasive and powerful, cardinal; handful of outstanding traits, central; least important and inconsistent, secondary; shared by members of a culture, common traits.

The idiographic approach

Because every personality is unique, Allport argued, its science must study the individual case and not only group averages, which describe nobody in particular. He therefore preferred the idiographic approach to the nomothetic search for laws, with its factorial designs (several variables crossed over many subjects) and group statistics such as chi-square (Chapter 9). Later he called his approach **morphogenic**, the study of how traits are patterned within one person. His main method was the **personal-document technique**: analysing diaries, letters and autobiographies for the traits they reveal.

KEY STUDY: LETTERS FROM JENNY (1965)

Over twelve years a woman Allport called Jenny, the mother of his college roommate, wrote more than 300 letters to Allport and his wife. Thirty-six judges read them and listed nearly 200 trait names, which Allport reduced to eight clusters, among them quarrelsome-suspicious, self-centred and dramatic-intense. A later computer count of her words by one of his students found much the same eight. Criticism: the method rests on judges' interpretations, and one case is hard to generalise.

Allport was not purely idiographic. With **Philip Vernon** (1931), and later with **Gardner Lindzey**, he built the **Study of Values**, a nomothetic self-report test of the relative strength of six values taken from Eduard Spranger's types: theoretical, economic, aesthetic, social, political and religious. With Vernon he also showed (1933) that expressive behaviour, such as handwriting, gesture and voice, is consistent across tasks.

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Allport preferred the **idiographic** approach. The nomothetic approach, factorial designs and chi-square are the group methods he criticised. Nor did he use factor analysis: that is the hallmark of Cattell and Eysenck.

Functional autonomy, the proprium and maturity

Freud explained adult motives by tracing them to childhood. Allport's reply was **functional autonomy**: the motives of a mature adult become independent of the experiences in which they began, as a grown tree no longer needs its seed. A girl made to practise the veena to escape scolding may, as a woman, play for hours for its own sake. *Perseverative* functional autonomy covers low-level routines that persist without reward, such as addictions and daily rituals. *Propriate* functional autonomy covers motives bound up with one's values, self-image and lifestyle, and is the key to adult motivation.

"Propriate" comes from the **proprium**, Allport's word for the self. It develops through seven aspects: (1) the bodily self, (2) self-identity and (3) self-esteem, in about the first three years; (4) self-extension ("my house, my school") and (5) the self-image, actual and ideal, at about four to six; (6) the self as a rational coper, from six to twelve; and (7) propriate striving, the long-range goals of adolescence. The self returns in Section 4.10.

If childhood needs for affection and security are met, the result is the mature adult, marked by six criteria: (1) extension of the sense of self beyond oneself; (2) warm relating to others; (3) emotional security and self-acceptance; (4) realistic perception, skills and commitment to work; (5) self-objectification, meaning insight together with humour; and (6) a unifying philosophy of life. In 1960 Allport described personality as an **open system**, taking in energy and information and growing, not merely reducing tension. Critics note that functional autonomy says that motives become independent but not how, that the sharp break between child and adult personality is doubtful, and that the theory produced little research.

Cattell: surface and source traits ●●

Raymond Bernard Cattell (1905 to 1998) studied chemistry in London and then took a PhD in psychology in 1929 under Charles Spearman. Spearman had applied factor analysis to mental

abilities; Cattell resolved to apply it to personality. He moved to the United States in 1937 and worked mostly at the University of Illinois. His goal was prediction: "Personality is that which permits a prediction of what a person will do in a given situation" (1950).

Factor analysis examines the correlations among many measures and finds the few underlying factors that account for them: if people rated talkative are also rated sociable and cheerful, one factor may lie beneath all three. Cattell took Allport and Odbert's 4,500 stable-trait words, merged synonyms to leave 171, found that ratings on these formed about 35 clusters, and factor-analysed ratings, questionnaires and tests down to the 16 factors of his Sixteen Personality Factor Questionnaire, the 16PF (Figure 4.9).

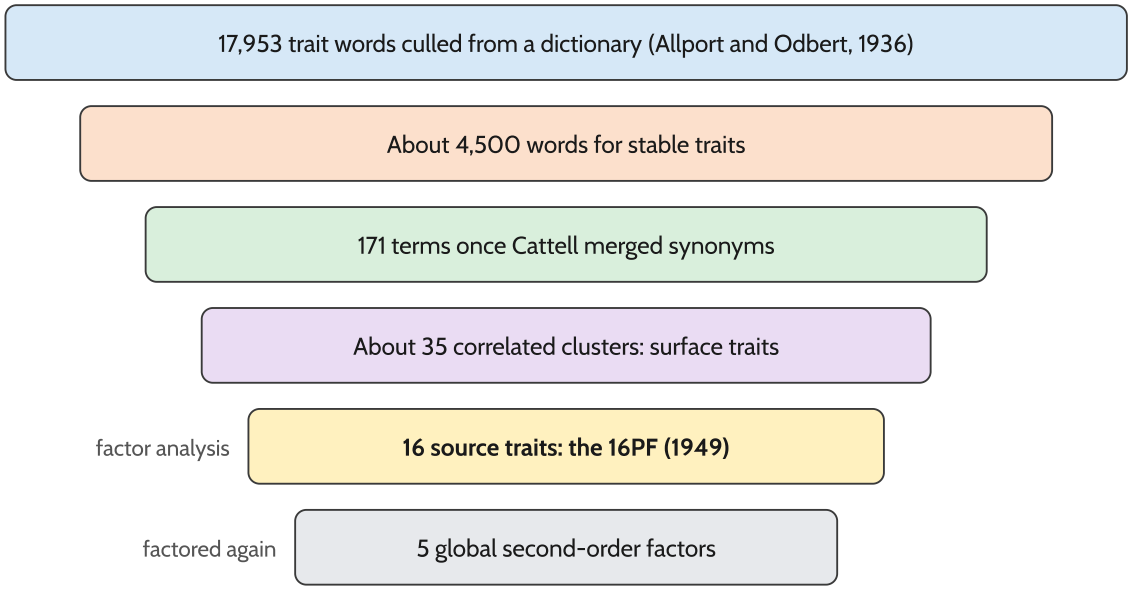

Figure 4.9 The lexical search, from dictionary to questionnaire. Each step keeps fewer, deeper units: the correlated surface clusters give way to the source traits beneath them.

The clusters and the factors give Cattell's most examined distinction. **Surface traits** are clusters of behaviours that correlate and appear together; anxiety, indecision and irrational fear, for example, form a surface trait one might call neuroticism. Because they arise from several sources, they are less stable and less important. **Source traits** are the stable, unitary factors beneath them, revealed by factor analysis: the building blocks of personality, as atoms are of matter. By origin, source traits are either **constitutional traits**, rooted in biology, or **environmental-mold traits**, learned from social and physical surroundings.

Cattell also separated common traits, which everyone has in some degree, from unique traits belonging to one or a few people (a passion for old coins), and divided traits by function into ability traits (how efficiently we pursue goals), temperament traits (emotional style) and **dynamic traits** (the motivating forces). Dynamic traits come in two kinds. **Ergs** are innate drives, such as hunger, sex, fear, curiosity, gregariousness and self-assertion. **Sentiments** are learned patterns of attitudes centred on an important object, such as career, religion, family or the self; the self-sentiment integrates the rest. A specific attitude ("I want to revise tonight") serves a sentiment, which in turn serves ergs, a chain Cattell called **subsidiation**. The web of such links is the **dynamic lattice** (Figure 4.10).

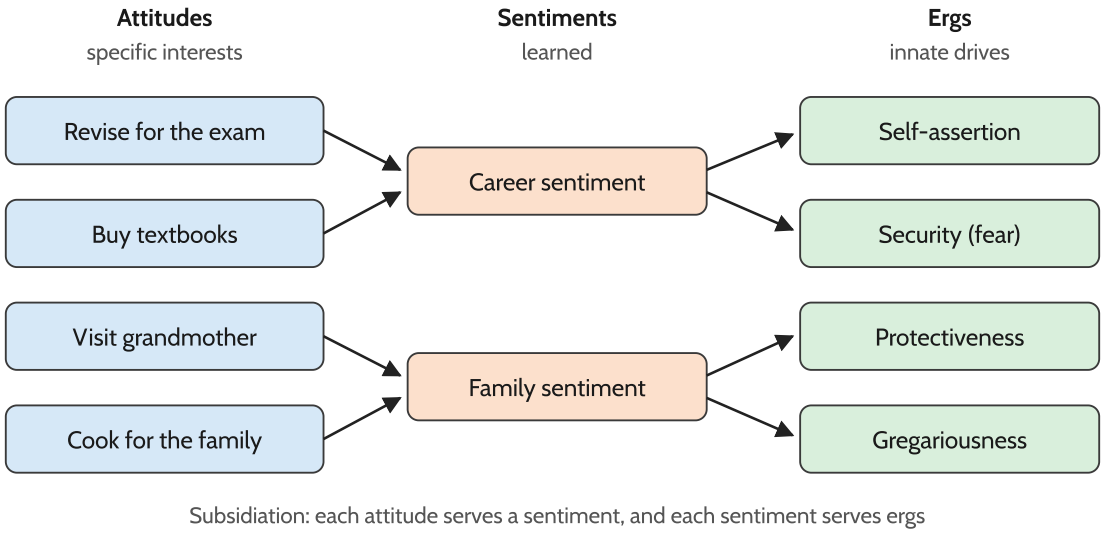

Figure 4.10 A simplified dynamic lattice. Read it from left to right: everyday attitudes are subsidiary to learned sentiments, which are subsidiary to innate ergs.

> **EXAM TRAP**
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> Cattell's pair is **surface and source traits**. Cardinal and central traits are Allport's; openness and conscientiousness are Big Five factors (Section 4.7); id and ego are Freud's. An Amity 2023 item says Cattell divided "attitudes" into source and surface: the key is right, but the stem garbles traits as attitudes. Its option "egos and sentiments" garbles Cattell's **ergs and sentiments**, the two kinds of dynamic trait. "Traits and types" are two approaches; defence and coping mechanisms belong to Sections 4.3 and 1.11.

L-data, Q-data and T-data ●●

A factor found in only one kind of data might be an artefact of method, so Cattell required source traits to reappear across three kinds (Table 4.19). **L-data** (life-record data) capture everyday behaviour through records or the ratings of observers who know the person: school performance, absences from work, a criminal record. **Q-data** (questionnaire data) are self-report ratings of one's own characteristics, attributes and interests, limited by poor self-knowledge and by faking. **T-data** (objective test data) come from controlled test situations whose purpose is hidden, so responses are hard to fake. The four 16PF factors labelled Q1 to Q4 appeared only in questionnaire data, hence the Q.

Table 4.19 Cattell's three sources of personality data.

Data	Source	Examples
L-data: life record	Records and observers' ratings of everyday behaviour	School performance, job absences, criminal record
Q-data: questionnaire	Self-report ratings of one's own characteristics and interests	Self-report personality inventories
T-data: objective test	Behaviour in a controlled situation whose purpose is hidden	Reaction-time and performance tasks

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There are exactly three: L, Q and T. "P-data" (projective data) and "Z-data" do not exist; Cattell classed hard-to-fake tasks such as inkblots as T-data, and projective responses are never L-data. Q-data is not the Q-sort, in which a person sorts self-descriptive cards from "most like me" to "least like me" (Rogers, Section 4.8). Nor is it the **P-technique**, a research design that factor-analyses one person measured on many occasions (the usual R-technique correlates many people measured once). Self-report is Q-data; school and criminal records are L-data; the 16 source traits are results, not data.

The Sixteen Personality Factor Questionnaire ●●●

Cattell packaged his source traits in the **Sixteen Personality Factor Questionnaire** (16PF), first published in 1949; the fifth edition (1993) has 185 items for people aged 16 and over. It measures normal personality on 16 primary factors (Table 4.20), lettered roughly by their size in Cattell's early rating studies; D, J and K are missing because those factors did not reappear reliably in adult questionnaire data. Factor B is a brief reasoning scale, but the instrument measures personality, not intelligence. Factoring the 16 again yields five second-order **global factors**: extraversion, anxiety, tough-mindedness, independence and self-control. These approximate the Big Five (extraversion, neuroticism, low openness, low agreeableness and conscientiousness; Section 4.7). Sten-score norms, reliability and validity are in Chapter 2.

Table 4.20 The 16 primary factors of the 16PF (fifth edition names).

Factor	Name	Low scorer	High scorer
A	Warmth	Reserved, aloof	Warm, outgoing
B	Reasoning	Concrete thinking	Abstract thinking
C	Emotional stability	Easily upset	Calm, stable
E	Dominance	Deferential, docile	Dominant, assertive
F	Liveliness	Serious, restrained	Lively, enthusiastic
G	Rule-consciousness	Expedient	Dutiful, rule-bound
H	Social boldness	Shy, timid	Bold, venturesome
I	Sensitivity	Tough-minded	Tender-minded
L	Vigilance	Trusting	Suspicious
M	Abstractedness	Practical	Imaginative
N	Privateness	Forthright	Discreet, shrewd
O	Apprehension	Self-assured	Worried, self-doubting
Q1	Openness to change	Traditional	Experimenting
Q2	Self-reliance	Group-oriented	Self-sufficient
Q3	Perfectionism	Lax, tolerates disorder	Controlled, exacting
Q4	Tension	Relaxed	Tense, driven

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Any stem with 16 factors, 16 source traits or the 16PF points to **Raymond B. Cattell**. Eysenck also used factor analysis but found three dimensions (the Eysenck Personality Questionnaire, EPQ); Costa and McCrae built the NEO-PI-R for five factors; Allport built the Study of Values; Sheldon rated body types; Murray devised the Thematic Apperception Test (TAT); Freud and Rogers proposed no trait dimensions at all. The 16PF measures normal personality traits, not fluid, crystallised or general intelligence (Cattell's fluid and crystallised intelligence and Spearman's *g* are in Section 1.9), and it is not a personality-disorder scale, a Q-sort or a projective test.

Heredity, development and the wider trait family

Comparing twins and ordinary siblings reared together and apart, Cattell concluded that about one third of personality is genetically based and two thirds environmental, though intelligence (factor B) and boldness (factor H) looked about 80 percent hereditary. He proposed six life-span stages: infancy (birth to 6, the major formative period), childhood (6 to 14), adolescence (14 to 23), maturity (23 to 50), late maturity (50 to 65) and old age. His **specification equation** predicts a response in a given situation as the sum of the person's trait scores, each weighted by its relevance to that situation. The main criticism is that factor analysis is less objective than it looks: the researcher chooses the measures and names the factors, which may be why others could not replicate all 16.

Two contemporaries complete the picture (Table 4.21). **Henry Alexander Murray** (1893 to 1988) built his personology around motives rather than traits: internal needs (for achievement, affiliation, dominance and more) meet environmental press, the situational forces that help or hinder them, and a recurring combination of need and press is a thema (Section 1.10). **Joy Paul Guilford** (1897 to 1987), known for his model of intellect (Section 1.9), factor-analysed temperament into ten traits, such as general activity, ascendance, sociability and emotional stability, measured by the Guilford-Zimmerman Temperament Survey (1949).

Table 4.21 Who built which trait or motive model.

Theorist	Core idea	Signature measure	Where
Gordon Allport	Cardinal, central, secondary dispositions; idiographic study	Personal documents; Study of Values	4.6
Raymond B. Cattell	16 source traits by factor analysis; L, Q and T data	16PF (1949)	4.6
Henry Murray	Personology: needs, press and thema	TAT, with Christiana Morgan (1935)	1.10; Chapter 2
J. P. Guilford	Factor-analytic temperament traits	Guilford-Zimmerman Temperament Survey	4.6
Hans Eysenck	Three superfactors: E, N and P	EPQ	4.7
Costa and McCrae	Five-Factor Model	NEO-PI-R	4.7

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Allport and Odbert (1936) culled _____ trait words from an English dictionary.
2. A single ruling passion that dominates a whole life is a _____ trait.
3. Allport's preference for the intensive study of one person is the _____ approach.
4. Adult motives that have become independent of the experiences in which they began show functional _____.
5. Allport's term for the self is the _____.
6. Cattell's stable underlying factors, found by factor analysis, are _____ traits.
7. School performance and criminal records are _____-data.
8. Cattell's innate drives are _____; his learned attitude patterns are _____.

Answers: (1) 17,953, (2) cardinal, (3) idiographic, (4) autonomy, (5) proprium, (6) source, (7) L, (8) ergs, sentiments.

4.7 Traits II: Eysenck, the Big Five and the Biology of Personality

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Section 4.6 ended with Raymond Cattell's sixteen factors. Hans Eysenck thought three were enough and looked for them in the brain; Paul Costa and Robert McCrae settled on five, the answer that now dominates. This section follows both, then asks how much of a trait is inherited and whether traits predict what you will actually do tomorrow.

Eysenck's three superfactors ●●

Hans Jurgen Eysenck (1916 to 1997) left Berlin for London in 1934 and spent his career at the Maudsley Hospital and Institute of Psychiatry. Like Cattell he used factor analysis, but he distrusted purely statistical factors: a real dimension of personality must also show up in laboratory experiments and have a biological cause. Factor analysis joined to experiment and biology is his signature.

His **hierarchical model** has four levels (Figure 4.11). At the bottom are **specific responses**: today you chat with a stranger at the bus stop. Responses that recur become **habitual responses**: you chat at every bus stop. Correlated habits form **traits** (sociability, liveliness, activity), and correlated traits form a **type** or superfactor, here extraversion. An Eysenck type is a broad continuous dimension, not a box people fall into.

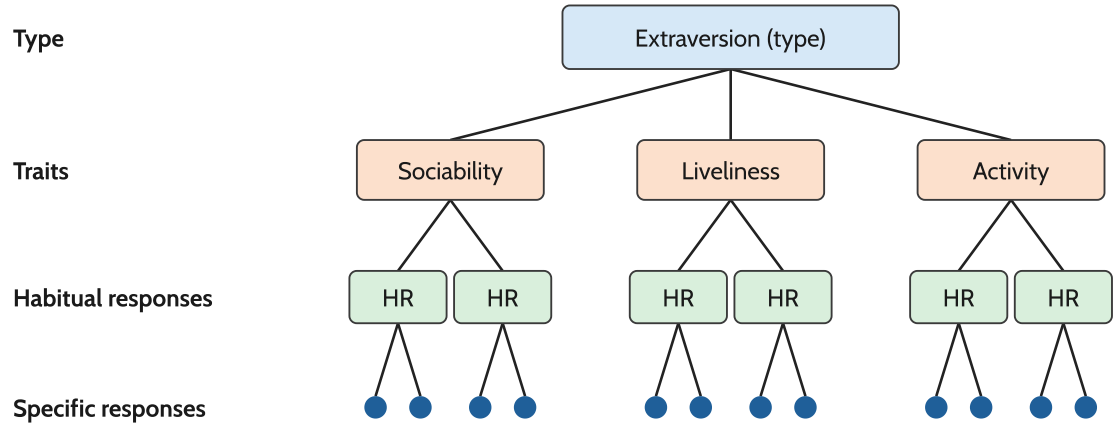

Figure 4.11 Eysenck's hierarchical model. Specific responses (dots) recur as habitual responses (HR), correlated habits form traits, and correlated traits form a type such as extraversion.

His early analyses (*Dimensions of Personality*, 1947) found two superfactors; a third was added in the 1970s, giving the **PEN model**:

- E, **extraversion** versus **introversion**. Extraverts are sociable, lively, active, talkative, assertive and sensation seeking; they enjoy parties and large gatherings and look for stimulation. Introverts are quiet, reserved and reflective; they prefer a few close friends, plan before acting and, in the popular phrase, gain energy from solitude while crowds drain them. Introversion is not unfriendliness, aggression or even shyness (Eysenck listed shyness under N).
- N, **neuroticism** versus emotional stability. High scorers are anxious, moody, tense, guilt-prone and emotionally reactive, with low self-esteem; low scorers are calm and even-tempered.
- P, **psychoticism** versus impulse control. High scorers are aggressive, cold, egocentric, impersonal, impulsive, antisocial and tough-minded, though sometimes creative. P is not psychosis; the scale is closer to psychopathy.

> EXAM TRAP
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> Eysenck's three are **E, N and P**. "Extraversion, sensing-intuition, thinking-feeling" smuggles in the Myers-Briggs dichotomies from Carl Jung's types (Section 4.4); "psychoticism-conscientiousness" borrows a Big Five factor. His **earliest** model had only E and N: not openness and conscientiousness (Big Five), not inferiority and superiority striving (Alfred Adler, Section 4.4), not internal and external locus of control (Julian Rotter, Section 4.9). "Aggressive and dominant" never describes the introvert: dominance goes with E, aggression with P.

Arousal, the visceral brain and the four temperaments

Why should one student crave the roar of a cricket stadium while her roommate hides in the library? Eysenck's answer (*The Biological Basis of Personality*, 1967) was arousal. The **ascending reticular activating system** (ARAS) in the brainstem sets the level of **cortical arousal**. Introverts, he proposed, have a *higher* baseline arousal, so extra stimulation overloads them and they avoid it; extraverts sit below their comfortable level and seek stimulation to climb

up to it. The laboratory bore him out: given a few drops of lemon juice on the tongue, introverts salivate more than extraverts, and they are more sensitive to faint stimuli, tolerate pain less and condition faster. Later work confirmed these differences more firmly than the arousal explanation.

Neuroticism rests on the **visceral brain**, an older name for the limbic system (hippocampus, amygdala, cingulum, septum, hypothalamus), which drives the sympathetic nervous system; in high-N people this alarm fires at mild stress and settles slowly. For P he suggested male hormones, since men score higher; this biology is the weakest.

Because E and N are independent, crossing them gives four quadrants, and Eysenck pointed out that these reproduce the four temperaments of Galen (Section 4.1): the stable extravert is **sanguine**, the unstable extravert **choleric**, the unstable introvert **melancholic** and the stable introvert **phlegmatic** (Figure 4.12).

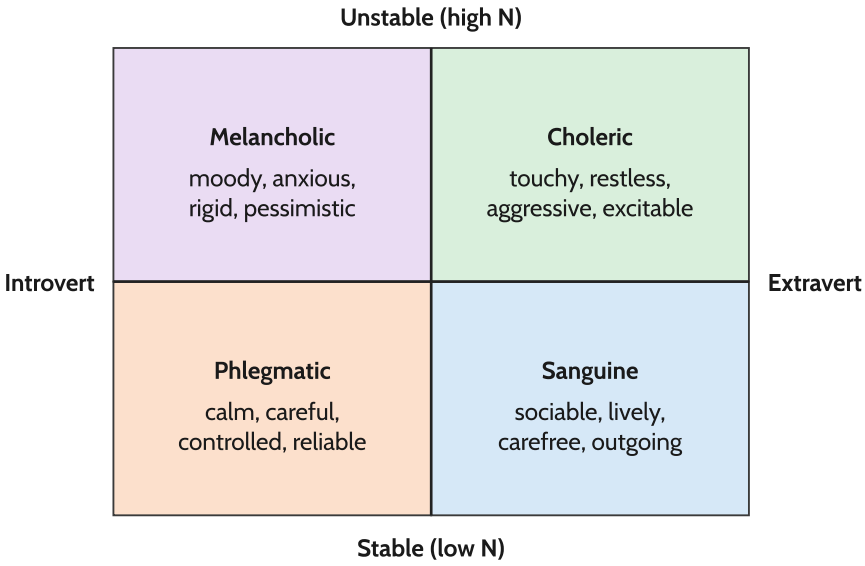

Figure 4.12 Eysenck's two independent dimensions crossed. Each quadrant matches one of Galen's temperaments; the trait words under each name are Eysenck's.

Eysenck's questionnaires, many built with his wife **Sybil B. G. Eysenck**, are the Maudsley Personality Inventory (MPI, 1959; E and N), the **Eysenck Personality Inventory** (EPI, 1964, or 1963 in some sources; E, N and a Lie scale), the **Eysenck Personality Questionnaire** (EPQ, 1975), which added P and kept the Lie scale (L) that catches faking good, and the EPQ-R (1985) with an improved P scale. Twin studies supported his view that the dimensions are largely inherited, E and N more than P, and all three appear in over 35 nations. In *Crime and Personality* (1964) he argued that offenders score high on E and N, adding P in later editions: extraverts condition poorly, so they are slow to acquire the conditioned anxiety that serves as conscience. The P link is the most consistent; the full theory has only modest support.

Other biological models of temperament

Eysenck opened the way to a biology of temperament, the inherited, early-appearing emotional style (Section 4.1) now seen as the biological core of personality. Table 4.22 sets out four successors. **Jeffrey Alan Gray** (1934 to 2004) rotated Eysenck's E and N axes in his **reinforce-**

ment sensitivity theory: anxiety runs from stable extravert to neurotic introvert, impulsivity from stable introvert to neurotic extravert. **C. Robert Cloninger** (1987, extended 1993) tied temperaments to neurotransmitters. **Jerome Kagan** (1929 to 2021) found that about one baby in five reacts to novelty with thrashing and crying, and these high-reactive infants tend to become shy, cautious children, showing **behavioural inhibition**. **Arnold Buss** and **Robert Plomin** (1984) proposed three heritable temperaments. Alexander Thomas and Stella Chess's infant categories are taught in Chapter 6.

Table 4.22 Biological models of temperament after Eysenck.

Model	Dimensions	Proposed basis
Gray: reinforcement sensitivity	behavioural inhibition system (BIS): anxiety, sensitivity to punishment; behavioural approach system (BAS, or activation system): impulsivity, sensitivity to reward	BIS: septo-hippocampal system; BAS: dopamine pathways
Cloninger: temperament and character	Temperament: novelty seeking, harm avoidance, reward dependence, persistence. Character: self-directedness, cooperativeness, self-transcendence	Dopamine, serotonin, noradrenaline for the first three; Temperament and Character Inventory
Kagan: behavioural inhibition	Inhibited (shy, withdrawn with the unfamiliar) versus uninhibited (bold, sociable) children	A highly reactive amygdala
Buss and Plomin: EAS	Emotionality, activity, sociability	Heritable; twin studies

The Big Five: discovery and the five factors ●●

If Cattell found sixteen factors and Eysenck three, who was right? Researchers who reanalysed trait ratings, following the lexical hypothesis of Section 4.6, kept arriving at five. **Donald Fiske** (1949) found five factors in ratings on Cattell's variables; **Ernest Tupes** and **Raymond Christal** (1961) found five recurrent factors in US Air Force ratings; **Warren Norman** (1963) replicated them; and **Lewis Goldberg** named them the **Big Five** (1981), "big" meaning broad, not important. Meanwhile **Paul Costa** (born 1942) and **Robert McCrae** (born 1949), at the US National Institutes of Health in Baltimore, reached the same five through questionnaires. The two roads met in the **Five-Factor Model** (FFM).

The five are remembered as **OCEAN** (or CANOE): **O**penness to experience, **C**onscientiousness, **E**xtraversion, **A**greeableness and **N**euroticism. Each is a continuous dimension with two poles, and most people fall near the middle. In the NEO-PI-R each factor, or domain, splits into six narrower **facets** (Table 4.23).

Table 4.23 The five factors, their poles and their six NEO-PI-R facets.

Factor	High scorer	Low scorer	Six facets
Openness (O)	Imaginative, curious, creative, unconventional	Conventional, practical, narrow interests	Fantasy, aesthetics, feelings, actions, ideas, values
Conscientiousness (C)	Organised, dependable, self-disciplined, dutiful, achievement-focused	Careless, disorganised, lax	Competence, order, dutifulness, achievement striving, self-discipline, deliberation
Extraversion (E)	Outgoing, sociable, talkative, assertive, energetic	Reserved, quiet, solitary	Warmth, gregariousness, assertiveness, activity, excitement-seeking, positive emotions

Factor	High scorer	Low scorer	Six facets
Agreeableness (A)	Trusting, kind, cooperative, softhearted	Suspicious, antagonistic, critical	Trust, straightforwardness, altruism, compliance, modesty, tender-mindedness
Neuroticism (N)	Anxious, moody, insecure, emotionally reactive	Calm, secure, even-tempered (emotional stability)	Anxiety, angry hostility, depression, self-consciousness, impulsiveness, vulnerability

To recognise a factor from a description, match it to the facets. Rohan is self-disciplined, achievement-focused and keen to do his duty: three facets of conscientiousness, so he scores high on C, not on E, A or N. Likewise "organised and dependable" is C, "outgoing and sociable" high E, "reserved and quiet" low E (introversion), and "imaginative and creative" O. Note that impulsiveness is a facet of N, and anxiety is a facet, not a factor.

Imposters and the NEO inventories ●●●

Table 4.24 lists the usual imposters and where each really belongs.

Table 4.24 Terms that are not Big Five factors.

Offered as a factor	What it really is
Psychoticism	Eysenck's third dimension (PEN); roughly low A plus low C
Intelligence	An ability, not a personality factor; it correlates with Openness
Intuition, sensing, thinking, feeling, judging, perceiving	Myers-Briggs Type Indicator (MBTI) dichotomies from Jung's types (Section 4.4)
Anxiety	One facet of Neuroticism
Sense of humour	An everyday trait word, not a factor
Honesty-humility	The sixth factor of HEXACO (below)
Sanguine, choleric, melancholic, phlegmatic	Galen's temperaments: the quadrants of Eysenck's E and N
Achievement, affiliation, power, intimacy, autonomy	Motives or needs, not traits (Section 1.10)

The standard measure is the NEO Personality Inventory. Costa and McCrae first measured only N, E and O, hence the name; the NEO-PI (1985) added A and C. The **NEO-PI-R** (1992) has 240 items, eight per facet, covering five domains with six facets each, 30 facets in all. The **NEO-FFI** (NEO Five-Factor Inventory) is a 60-item short form scoring domains only; the NEO-PI-3 (2005) simplified the wording for adolescents. The NEO belongs to the **trait approach**: it measures stable dispositions, not learned responses (behaviourist), thought processes (cognitive) or unconscious conflict (psychodynamic). Its psychometric properties are in Chapter 2.

> EXAM TRAP
>
> The NEO-PI-R rests on the **Five-Factor Model**: five domains, six facets each. It is not Cattell's 16 source traits (16PF, Section 4.6), Eysenck's three PEN dimensions (the EPQ), the MBTI's 16 types (Section 4.4), Henry Murray's needs and press (Section 4.6) or eleven personality-disorder scales (a clinical inventory, Chapter 2).

The Big Five across life and cultures

McCrae and Costa built the factors into **five-factor theory** (1996). The traits are **basic tendencies**: biologically based capacities that upbringing does not shape. What life shapes are **characteristic adaptations**: the habits, attitudes, skills, roles, relationships and self-concept through which a tendency is expressed in a particular culture. High E is a basic tendency; joining the college cultural committee in Mysuru is one adaptation of it.

The factors replicate in more than 50 nations, India among them, suggesting a common human structure. Critics note that most samples are urban and literate, that a remote Bolivian tribal group did not show the five clearly, and that lexical studies in some languages find six factors.

How stable are the factors? People keep their rank order, more so with age, but mean levels shift: from adolescence into middle age, conscientiousness and agreeableness rise while neuroticism falls, the **maturity principle**. Costa and McCrae borrowed **William James**'s 1890 phrase that by about 30 character is "set like plaster"; later work shows the plaster can still be chipped. Twin studies put the heritability of each factor at about 40 to 50 percent.

The factors also predict outcomes. **High conscientiousness and low neuroticism** go with better health and longer life. **Lewis Terman** began following about 1,500 bright Californian children in 1921; when **Howard Friedman** and colleagues (1993) traced their deaths, those rated conscientious and prudent as children had lived longer. Conscientious people exercise, wear seat belts, avoid smoking and follow medical advice. High N predicts distress and is the broadest personality risk factor for anxiety and depressive disorders (Chapter 8). Conscientiousness also predicted job performance in every occupational group in the meta-analysis of **Murray Barrick** and **Michael Mount** (1991).

> EXAM TRAP
>
> The health pair is **high C plus low N**. "Low agreeableness and low neuroticism" swaps in low A, which as hostility (the toxic core of the Type A pattern; Type A and Type D are in Section 1.11) predicts heart disease, not health. "Neuroticism exclusively" is backwards, since high N harms health, and "introversion exclusively" has no such link.

Beyond five: HEXACO and the Dark Triad

Not everyone accepts five. **Michael Ashton** and **Kibeom Lee** reanalysed trait words across many languages (from 2004) and found six factors, the **HEXACO** model: **H**onesty-Humility (sincere, fair, modest versus greedy, boastful, manipulative), **E**motionality (fearful, anxious, sentimental; close to N), e**X**traversion, **A**greeableness, **C**onscientiousness and **O**penness to experience. Honesty-Humility, the genuinely new factor, is low in offenders; the 100-item HEXACO-PI-R measures all six.

Delroy Paulhus and **Kevin Williams** (2002) described the **Dark Triad**, three overlapping, socially aversive but subclinical traits: **narcissism** (grandiosity, entitlement, hunger for admiration), **Machiavellianism** (cold, cynical, strategic manipulation) and **psychopathy** (callousness, impulsive thrill seeking, lack of remorse). All three share low agreeableness and low honesty-humility. The **Dark Tetrad** adds **everyday sadism**, enjoying others' suffering (Paulhus and colleagues, 2013).

Genes, environments and evolution

How can genes be separated from upbringing? **Behaviour genetics** uses two natural experiments. Identical (monozygotic) twins share all their genes and fraternal (dizygotic) twins about half, so greater similarity in identical pairs points to genes. In adoption studies, resemblance to the biological parents points to genes, resemblance to the adoptive parents to family environment. The **heritability** of a trait is the proportion of its variation *within a population* that is due to genetic differences; it says nothing about how "genetic" one person's trait is.

> **KEY STUDY: THE MINNESOTA STUDY OF TWINS REARED APART (FROM 1979)**
>
> Question: if identical twins grow up in different homes, how alike are they? Method: **Thomas Bouchard** and colleagues brought more than 100 sets of twins and triplets reared apart to the University of Minnesota for about a week of tests. Result (reported in 1990): identical twins reared apart were about as similar in personality as those reared together; heritability came to about 0.40 to 0.50 for personality and about 0.70 for IQ. Criticism: some pairs grew up in similar homes or had met before testing, and the sample was small and self-selected.

The same studies held a surprise: growing up in the same family adds almost nothing to personality similarity, and adoptive siblings reared together are scarcely more alike than strangers. Plomin and **Denise Daniels** (1987) asked why children in the same family are so different, and answered that what matters is the **non-shared environment**: different friends, teachers, illnesses, accidents and differential treatment by parents. The shared environment (the same home, parents, income and religion) has only a weak effect on adult personality.

Genes and environments are also entangled. **Sandra Scarr** and **Kathleen McCartney** (1983) described three kinds of **gene-environment correlation**. In the **passive** kind, parents give a child both genes and a matching home (musical parents, a house full of music); it is strongest in infancy. In the **evocative** kind, a child's inherited traits draw responses from others (a smiling baby is picked up more). In the **active** kind, or niche picking, people seek settings that suit them (the extravert joins the drama club); it grows with age. In **gene-environment interaction**, by contrast, a gene's effect depends on the environment: a vulnerable genotype may raise the risk of depression only after severe stress.

Why does variation persist at all? **David Buss** (born 1953) treats traits as **adaptive individual differences**: the Big Five summarise what our ancestors most needed to know about one another (who will lead, cooperate, work hard, panic or innovate), and different trait levels pay off in different environments. He is not Arnold Buss of the EAS model.

The person-situation debate

Traits assume consistency: an honest child should be honest in class, at play and at home. **Hugh Hartshorne** and **Mark May** tested this and found it largely missing (Key study below). Forty years later **Walter Mischel** (1930 to 2018) generalised the point in *Personality and Assessment* (1968). Trait measures, he found, correlated with behaviour in a given situation, and behaviour in one situation with behaviour in another, rarely above about 0.30, a ceiling he called the **personality coefficient**; it explains less than 10 percent of the variation in be-

haviour. His critique: **situational factors often override traits** in determining behaviour. So began the **person-situation debate**.

KEY STUDY: HARTSHORNE AND MAY'S HONESTY STUDIES (1928)

Question: is honesty a consistent trait? Method: in the Character Education Inquiry, thousands of American schoolchildren were given secret chances to cheat, lie or keep money in classroom, party, athletic and home settings. Result: honesty in one setting predicted honesty in another only weakly; a child who copied answers might still return a coin. Conclusion: honesty looked situation-specific. Criticism: single acts are unreliable samples; summed over many tests, the scores showed much more consistency.

Trait theorists replied in three ways. First, **aggregation**: **Seymour Epstein** (1979) showed that a single act is as unreliable as a single test item; averaged over many occasions, behaviour becomes far more stable, so traits predict most of the people much of the time. Second, a **strong situation** (a red traffic light, a funeral, an exam hall) has clear rules that make everyone behave alike, while a **weak situation** (a party, a free afternoon) lets traits show. Third, **interactionism** (**Norman Endler** and **David Magnusson**, 1976): behaviour depends on the person, the situation and their interaction. Mischel himself rebuilt consistency as stable if-then patterns in his cognitive-affective personality system (Section 4.9). Traits predict behaviour averaged across time and situations well, single acts poorly.

EXAM TRAP

Mischel's critique was that **situations often override traits**. He did not say that personality is entirely biological (closer to Eysenck), that traits outpredict situations (the opposite of his point), or that traits cannot be measured reliably: trait questionnaires were reliable; they simply predicted single behaviours poorly.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Eysenck's three dimensions are extraversion, neuroticism and _____.
2. Introverts have a higher baseline cortical arousal, regulated by the _____.
3. In Eysenck's scheme the unstable introvert corresponds to Galen's _____ temperament.
4. Cloninger linked harm avoidance to the neurotransmitter _____.
5. The label "Big Five" was coined by _____.
6. The NEO-PI-R has _____ items measuring 30 facets.
7. Plomin and Daniels showed that the _____ environment shapes personality more than the shared one.
8. Mischel's "personality coefficient" is a correlation of about _____.

Answers: (1) psychoticism, (2) ascending reticular activating system, (3) melancholic, (4) serotonin, (5) Lewis Goldberg, (6) 240, (7) non-shared, (8) 0.30.

4.8 Humanistic and Existential Theories: Rogers, Maslow and May

LOWER YIELD · 16 QS

Psychoanalysis pictured people driven by unconscious conflict; behaviourism pictured them shaped by reinforcement. By the 1950s a group of American psychologists objected that neither left room for what people experience most directly: choosing, hoping and trying to become someone. The humanists, Carl Rogers and Abraham Maslow, asked what a person could grow into. The existentialists, from Heidegger to Rollo May and Viktor Frankl, asked how a person faces freedom, anxiety and death.

The humanistic image of the person ●●

The **humanistic approach** to personality rests on four assumptions. People have **free will** and answer for their choices. Every person has an **innate tendency to grow** and develop, to reach his or her **maximum potential**. A person must be understood as a **whole**, not as a bundle of parts. What matters most is **conscious, present experience**, studied from the inside through the person's **phenomenological field**: the world as he or she perceives it, which may differ from the world as it is.

Table 4.25 sets this image beside its rivals. Humanism does not explain people by **unconscious conflict** (the psychodynamic view), **conditioned responses** (behaviourism) or information processing (the cognitive perspective), nor judge them against **statistical norms**, the "average is normal" yardstick of trait measurement and of one definition of abnormality (Chapter 8). Its founders were Abraham Maslow and Carl Rogers, with **Charlotte Buhler** (1893 to 1974), who studied the goals that organise the whole life course, and Rollo May, who brought in existential ideas. How it became psychology's "third force" is told in Section 1.2.

Table 4.25 Five images of the person.

Perspective	The person is	Behaviour explained by	Disorder seen as
Psychodynamic	Driven by hidden forces	Unconscious conflict, childhood	Repressed conflict
Behavioural	A product of learning	Conditioned responses, reinforcement	Faulty learning
Cognitive	An information processor	Thoughts, beliefs, schemas	Distorted thinking
Humanistic	Free, whole, growing	Conscious experience, the urge to grow	Blocked growth
Existential	Free, responsible, mortal	Choices made facing anxiety and death	Inauthentic, meaningless living

Disorder, on this view, is **blocked growth**: conditional love, unmet needs or a false self thwart the push to grow. Cure needs neither a reinforcement schedule nor the interpretation of buried conflict, but a relationship in which the therapist believes in the client's potential, offers positive regard and treats the client with respect (Chapter 7).

Critics reply that "actualisation" and "potential" are vague and nearly untestable, and that the focus on the individual's inner world misses the wider social and cultural contexts that shape experience and identity: the autonomous, self-made individual is a Western, individualist ideal.

EXAM TRAP

Maximum potential, an innate tendency to grow, free will and personal growth all point to **humanistic**. Maslow and Rogers propose the humanistic approach, not the existential (May, Frankl), cognitive or behavioural one. "Availability of reinforcement" is the behavioural view of disorder. The standard criticism is neglect of social and cultural context: "too much focus on cognition" fits the cognitive approach, "too much focus on past experience" fits psychoanalysis, and "neglecting personal experience" is the opposite of what humanists do.

Rogers: the self and its conditions of worth

Carl Ransom Rogers (1902 to 1987) built his theory from what clients told him about their own experience. Its single motive is the **actualising tendency**, the innate drive of every organism to maintain and enhance itself: it keeps the toddler walking despite falls, and later powers psychological growth. Once a self has formed, part of it becomes the **self-actualising tendency**, the push to actualise the self as perceived. From infancy, experience is judged by the **organismic valuing process**: what enhances the organism feels good and is sought, what hinders it feels bad and is avoided.

Each person lives in a private **phenomenal field**, also called the experiential world: all experience as the person perceives it, the only reality that counts for behaviour. Part of it is gradually set apart as "I" and "me": the **self-concept**, the organised picture of who one is. Beside it grows the **ideal self**, the self one would most like to be.

With the self comes a universal **need for positive regard** (warmth, love and acceptance from important others) and later a need for **positive self-regard**, approval from within. **Unconditional positive regard** values the child whatever he or she does: the act may be disapproved of, the child never is. **Conditional positive regard** is love with strings attached. The child then introjects (takes in as his or her own) the parents' standards as **conditions of worth**: "I am worthy only when I …". These work much like Freud's superego, and experience is now valued not by the organismic valuing process but by whether it earns approval.

The result is **incongruence**: a split between what the organism actually experiences and a self-concept burdened by conditions of worth, felt as tension, anxiety and confusion. Its opposite, **congruence**, is a close match between self-concept and experience. Threatening experience is kept out by **distortion** (admitting it in a changed form) or **denial** (keeping it out of awareness). Arjun, praised at home only when he tops the class in science, believes "I am the future doctor". When he finds he loves designing furniture and finds anatomy dull, he distorts it ("I am just tired") or denies it; tension builds and anxiety appears without a nameable cause. Figure 4.13 traces both routes.

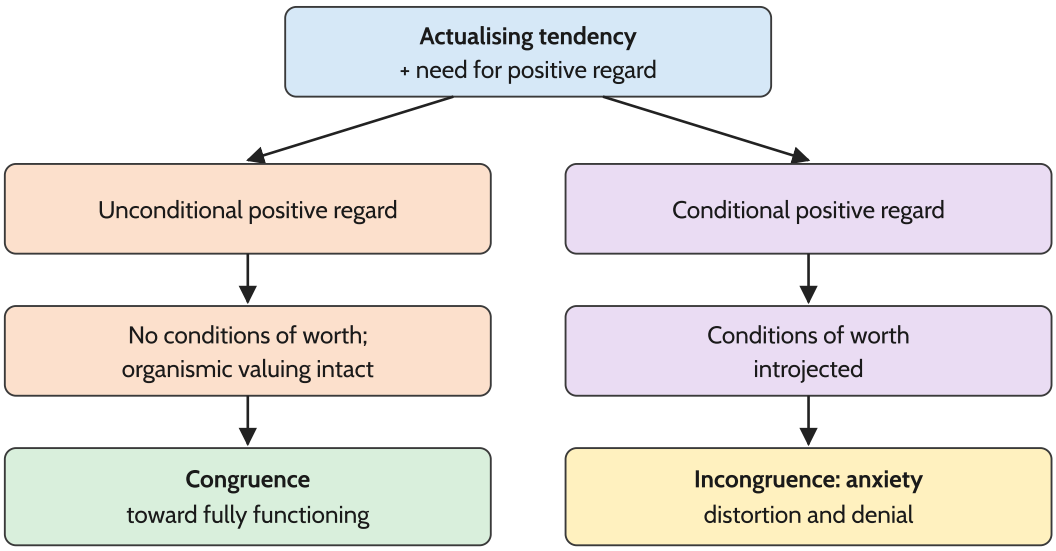

Figure 4.13 Two routes in Rogers's theory: the same need for positive regard leads to congruence when love is unconditional, and to incongruence and defence when it is conditional.

EXAM TRAP

"A split between the organismic valuing process and a self-concept burdened by conditions of worth, producing tension and confusion" is **incongruence**. Introjection is only how conditions of worth get in (and a defence mechanism, Section 4.3); conflict is a general term (id, ego and superego in Section 4.2, approach-avoidance in Section 1.10); the actualising tendency is the growth force that incongruence blocks.

The fully functioning person and the Q-sort

A child given unconditional positive regard stays congruent and moves toward what Rogers called the **fully functioning person**, "a direction, not a destination" (*On Becoming a Person*, 1961). Five features mark such a person: **openness to experience**, nothing distorted or denied; **existential living**, fully in each moment; **organismic trusting**, relying on one's own reactions; **experiential freedom**, a felt freedom to choose and responsibility for the choice; and **creativity**. Rogers stated the theory as **19 propositions** in *Client-Centered Therapy* (1951), beginning with the claim that each person lives at the centre of a continually changing world of experience and reacts to it as perceived. Therapy restores congruence; its core conditions are taught in Chapter 7.

Rogers also made the self measurable. The **Q-sort**, devised by the British psychologist **William Stephenson** (*The Study of Behavior*, 1953) and adopted by Rogers's group at Chicago, has the person sort self-descriptive statements ("I feel helpless", "I am emotionally mature") from "most like me" to "least like me" into a forced, roughly normal distribution: a few cards at each extreme, most in the middle. Because the statements are ranked against one another for the same person, the Q-sort captures within-person rather than between-person variation. Sorting once for the real self and once for the ideal self gives a self-ideal correlation, an index of congruence.

KEY STUDY: THE CASE OF MRS OAK (ROGERS, 1954)

Does successful therapy bring the perceived self closer to the ideal? Mrs Oak's self-ideal correlation rose from +.36 before client-centred therapy to +.79 a year after it, and in 25 clients studied by Rogers's associates **J. M. Butler** and **G. V. Haigh** (1954) the average rose from .01 before therapy to +.31 after it. Criticism: Q-sorts are self-reports, open to the wish to please the therapist, and a rising correlation can come from a lowered ideal as well as a changed self.

Research supports Rogers in part: conditionally loved children show unstable self-esteem, and large self-ideal gaps go with anxiety and depression. Critics note his reliance on conscious self-report and an actualising tendency assumed rather than shown; his lasting gifts were recorded therapy sessions as research data and the self-concept as a variable.

EXAM TRAP

The Q-sort was devised by Stephenson and used by Rogers to measure the self-concept and self-ideal congruence. It did not replace the MMPI (Chapter 2), and it is neither a test of infant reflexes nor a diagnostic test for schizophrenia.

Maslow: deficiency, being and the self-actualiser

Abraham Harold Maslow (1908 to 1970) asked what the healthiest people are like. His hierarchy of needs (physiological, safety, belongingness and love, esteem, self-actualisation, with lower needs stronger and prepotent) is taught in Section 1.10; here we follow what it implies for personality.

The four lower levels are **deficiency needs** (D-needs): a lack creates tension that drives us to fill it, and once filled they stop motivating. Behaviour driven by them is **D-motivation**. Once they are met, a person is moved differently. Maslow called this **metamotivation** (also B-motivation, from Being), the distinct motivation of self-actualisers: not making up a lack but growing and expressing oneself, even seeking tension. Its objects are being needs, the **metaneeds** or B-values, such as truth, goodness, beauty, justice, wholeness and meaningfulness. Frustrated metaneeds produce **metapathology**: cynicism, apathy and a sense of meaninglessness that the sufferer cannot trace to any lack (Table 4.26).

Table 4.26 Maslow's deficiency and being realms.

Feature	Deficiency (D)	Being (B)
Needs	Physiological, safety, belonging, esteem	Metaneeds (B-values): truth, beauty, justice
Motivation	D-motivation: fill a lack	Metamotivation: grow and express
Tension	Reduced	Sought and enjoyed
When thwarted	Illness, anxiety	Metapathology: cynicism, apathy
Typical of	Most people	Self-actualisers (1 percent or fewer)

Self-actualisation is the striving to fulfil one's potential, to become everything one is capable of becoming. The neurologist **Kurt Goldstein** (1878 to 1965) coined the term in *The Organism* (German 1934, English 1939); Maslow made it the summit of his hierarchy. Finding

almost none among students, he studied older people and historical figures, among them the later **Abraham Lincoln**, **Thomas Jefferson**, **Albert Einstein** and **Eleanor Roosevelt**, and estimated that 1 percent of people or fewer self-actualise. They showed efficient perception of reality; acceptance of self, others and nature; spontaneity; problem-centring (a mission outside themselves); a need for privacy; autonomy and resistance to enculturation; freshness of appreciation; peak experiences; social interest (Adler's term, Section 4.4); deep but few relationships; a democratic character; and creativity.

A **peak experience** is a brief, intense moment of ecstasy and awe in which the sense of self is transcended; late in life Maslow added the **plateau experience**, a calmer, longer-lasting and more voluntary serenity. Why do so few self-actualise? Besides unmet lower needs and poor upbringing, Maslow blamed the **Jonah complex**, the fear of one's own greatness, named after the prophet who ran from God's call. The **Personal Orientation Inventory** (POI) of **Everett Shostrom** (1964) later measured self-actualisation with 150 paired statements. Critics note that the sample was small and hand-picked by Maslow's own values, and that his terms are vague. The transpersonal "fourth force" he foresaw is in Section 1.2.

> EXAM TRAP
>
> The distinct motivation of self-actualisers is **metamotivation**; D-motivation drives everyone else, and "E-motivation" and "F-motivation" do not exist. "The striving for fulfilment of one's potential" is **self-actualisation**, not the self-concept (one's picture of oneself), the ideal self (the self one wishes to be) or self-efficacy (Bandura's belief that one can perform a task, Section 4.9).

Existential roots: Dasein and the three worlds ●●

Both schools reject determinism, but humanism assumes an inner nature, a seed of potential waiting to unfold, and existentialism denies even that. It grew from the Danish philosopher **Soren Kierkegaard** (1813 to 1855), who wrote of the anxiety of choosing oneself; the German **Martin Heidegger** (1889 to 1976), author of *Being and Time* (1927); and the French **Jean-Paul Sartre** (1905 to 1980), whose slogan was **existence precedes essence**: we exist first and then make ourselves by our choices. Nothing fixes the meaning of a life in advance, so each person must continually recreate himself or herself, and the meaning of existence keeps changing. This is existential psychology: not humanistic psychology, which assumes a potential to fulfil, nor experimental psychology, which seeks general laws by controlled experiment, nor philosophy alone, since it carries the philosophers' ideas into the study and treatment of persons.

Heidegger's key term was **Dasein** (German, "being there"): human existence as **being-in-the-world**. A person is not a mind looking out at a separate world but one unit with it, so Dasein expresses the basic unity of person and environment. Behaviourists, cognitivists and psychoanalysts all split person from world; existentialists refuse the split. The Swiss psychiatrists **Ludwig Binswanger** (1881 to 1966) and **Medard Boss** (1903 to 1990) applied Heidegger to patients as **Daseinsanalysis**, describing each patient's world rather than explaining symptoms.

Binswanger, and after him May, described three modes of being-in-the-world, lived at once (Figure 4.14). The **Umwelt** ("world around") is the biological and physical world: body, drives, sleep and death. The **Mitwelt** ("with-world") is the world of relationships with others. The

Eigenwelt ("own world") is one's relationship with oneself. Modern people, May argued, are alienated in all three.

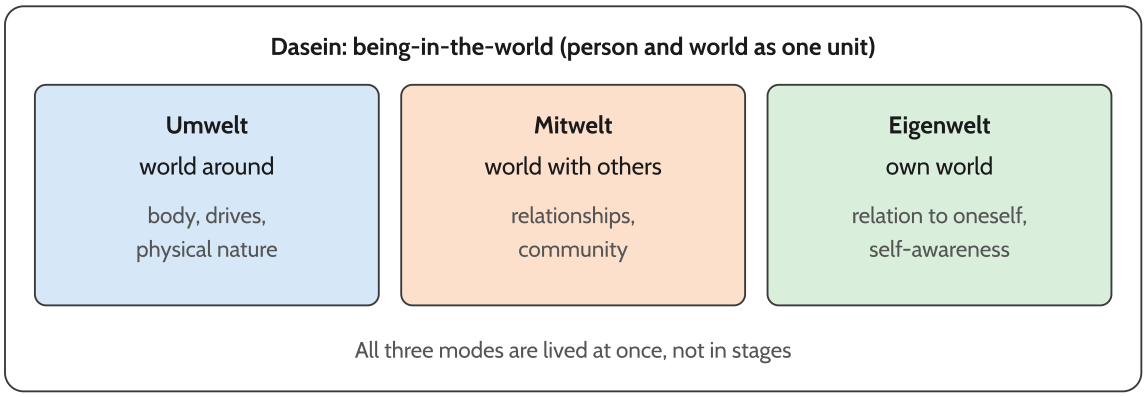

Figure 4.14 Dasein is the whole, the unity of person and world; Umwelt, Mitwelt and Eigenwelt are three sides of it that every person lives simultaneously.

> EXAM TRAP
>
> Dasein and being-in-the-world are **existential**, not behavioural, cognitive or psychoanalytic. Dasein is the unity of person and environment; Umwelt, Mitwelt and Eigenwelt are its three modes, so none of them is "the basic unity". The modes are not id, ego and superego, not conscious, preconscious and unconscious (both Section 4.2), and not Piaget's sensorimotor, preoperational and concrete operational stages (Chapter 6).

Rollo May: anxiety, the daimonic and love

Rollo May (1909 to 1994) brought existential psychology to America. After months in a tuberculosis sanatorium, unsure whether he would live, he wrote *The Meaning of Anxiety* (1950). For May anxiety is **ontological**, part of being itself: it arises whenever our existence, or a value we identify with, might be destroyed, and its root is awareness of **nonbeing**, death but also loss of self in conformity. **Normal anxiety** is proportionate to the threat, involves no repression and can be faced and used creatively; there is no freedom without it. **Neurotic anxiety** is disproportionate, involves repression and is handled by blocking off awareness.

In *Love and Will* (1969) May added three ideas. **Intentionality** is the structure that gives meaning to experience and lets a person decide about the future; it underlies will. The **daimonic** is any natural function that can take over the whole person, such as sex, anger or the craving for power: directed, it fuels creativity; unintegrated, it destroys. Love takes four forms: **sex** (biological release), **eros** (the longing to create and unite), **philia** (intimate, nonsexual friendship) and **agape** (selfless concern for another's welfare). In *Freedom and Destiny* (1981) he argued that freedom has meaning only against **destiny**, the givens of body, history and death; healthy people accept their destiny and use their freedom within it.

Frankl and the existential constructs

The Viennese psychiatrist **Viktor Emil Frankl** (1905 to 1997) survived Auschwitz and observed that prisoners who held on to a reason to live were the likelier to endure. In *Man's Search for Meaning* (German original 1946) he argued that the primary human motive is neither the will

to pleasure (Freud) nor the will to power (Adler) but the **will to meaning**. Frustrated, it leaves an **existential vacuum** of emptiness and boredom. If symptoms follow, the result is a **noogenic neurosis** (Greek *noos*, mind or spirit), arising from an existential problem such as loss of meaning or a clash of values rather than from conflict between drives; its anxiety is noogenic anxiety. Frankl's therapy, logotherapy, is taught in Chapter 7.

Across the existential writers five constructs recur: coming to terms with anxiety; confronting the universal fear of death; accepting freedom and the responsibility it brings; understanding our ultimate isolation, since no one can live or die for us; and facing meaninglessness by creating meaning. Irvin Yalom's four ultimate concerns restate the list for therapy (Chapter 7). Existentialists do not dig for unconscious fears and desires; that is the psychoanalytic aim. Critics find existential writing vague and hard to test.

> **EXAM TRAP**
>
> Noogenic neurosis and noogenic anxiety are Frankl's and belong to the **existential** school, not the humanistic, Gestalt (the school of perception in Section 1.2, or Perls's Gestalt therapy in Chapter 7) or psychodynamic school. "Uncovering our unconscious fears and desires" is psychoanalytic, the one option that is not an existential construct.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Rogers's term for the innate drive of every organism to maintain and enhance itself is the _____ tendency.
2. A split between experience and a self-concept burdened by conditions of worth is _____.
3. The Q-sort was devised by William _____.
4. Maslow called the distinct motivation of self-actualisers _____.
5. The fear of one's own greatness is Maslow's _____ complex.
6. Heidegger's term for being-in-the-world, the unity of person and environment, is _____.
7. The mode of being made up of relationships with others is the _____.
8. Frankl called a neurosis arising from loss of meaning a _____ neurosis.

Answers: (1) actualising, (2) incongruence, (3) Stephenson, (4) metamotivation, (5) Jonah, (6) Dasein, (7) Mitwelt, (8) noogenic.

4.9 Behavioural, Social-Cognitive and Cognitive Theories

LOWER YIELD · 16 QS

Why does one student freeze before every viva while his classmate, no cleverer, walks in calm? The theorists of this section ask what he has learned, what he expects and how he reads the situation. The story runs from Skinner, who denied any inner person, through the social learning theorists who put expectation and thought back in, to George Kelly, for whom personality is a way of construing the world.

Skinner: personality without a self

Burrhus Frederic Skinner (1904 to 1990) gave the most radical answer: personality is only a **collection of learned behaviour patterns**, each kept alive by its consequences. This is **radical behaviourism**: thoughts and feelings are behaviour obeying the same laws, and inner causes explain nothing ("he shouts because he is aggressive" is circular, since the only evidence of aggressiveness is the shouting).

Skinner distinguished two kinds of behaviour. **Respondent behaviour** is elicited by a specific stimulus: the knee-jerk to a tap, or salivation to Pavlov's bell after conditioning. **Operant behaviour** is emitted spontaneously or voluntarily and operates on the environment to change it: the rat presses a lever and food drops, the child cries and a parent comes. Most human behaviour is operant, so a personality is a history of reinforcement. Reinforcement and its schedules are taught in Section 1.6.

Shaping (successive approximation) builds a new response by reinforcing closer and closer approximations to it. **Superstitious behaviour** is behaviour strengthened by accidental reinforcement: pigeons fed every 15 seconds, whatever they did, kept repeating whatever they happened to be doing when food arrived, as a batsman clings to the bat that once scored a century. **Self-control**, for Skinner, is not willpower but control of the outside variables that control us, by four techniques: stimulus avoidance (studying away from the phone), self-administered satiation (eating sweets until sick of them), aversive stimulation (announcing a diet so that lapses bring ridicule) and self-reinforcement.

Walden Two (1948) is his novel of a utopian community run by positive reinforcement; *Beyond Freedom and Dignity* (1971) argued that freedom is an illusion and that cultures should design their contingencies deliberately. The approach is precise, testable and the source of behaviour modification and token economies; critics say it ignores cognition, heredity and inner experience and generalises too freely from rats and pigeons.

> EXAM TRAP
>
> "Behaviour emitted spontaneously or voluntarily that operates on the environment to change it" is **operant behaviour** (Skinner). The distractors are other theorists' terms: the **opposition principle** is Jung's (psychic energy arises from conflict between opposites; Section 4.4), **behaviour potential** is Rotter's (below) and **personal dispositions** are Allport's individual traits (Section 4.6).

Dollard and Miller: Freud in learning terms

Earlier, two Yale scholars had married psychoanalysis to learning theory. **John Dollard** (1900 to 1980) and **Neal Elgar Miller** (1909 to 2002) restated Freud's clinical insights in the drive-reduction terms of **Clark Leonard Hull** (1884 to 1952), in *Social Learning and Imitation* (1941) and *Personality and Psychotherapy* (1950): the **early social learning theory**. Learning needs four things ("want something, notice something, do something, get something"): a **drive**, a strong stimulus that impels action; a **cue**, which signals when, where and how to respond; a **response**; and **reinforcement**, the drive reduction that strengthens the cue-response link.

Their key idea was the **learned drive**. Fear is learned: Miller's rats, once shocked in a white box, learned to turn a wheel to escape it with no shock present, reinforced by fear reduction alone. Freud then translates easily. **Repression** is learned not-thinking: a forbidden thought

arouses fear, dropping it reduces fear, and the relief reinforces not-thinking. Neurotic conflict is competing approach and avoidance tendencies. Both grow stronger nearer the goal, but the avoidance gradient is steeper, so in an **approach-avoidance conflict** a person approaches a mixed goal from afar, retreats near it and hovers where the two gradients cross (the other conflict types are in Section 1.10). With Yale colleagues they also proposed the **frustration-aggression hypothesis** (1939): frustration always leads to some form of aggression, and aggression always presupposes frustration (Chapter 3). Table 4.27 shows the label each theorist carries.

Table 4.27 Who goes with which label.

Theorist	Label	Core ideas
B. F. Skinner	Radical behaviourism	Operant behaviour shaped by reinforcement
Dollard and Miller	Early social learning	Freud in Hull's terms: drive, cue, response, reinforcement
Bandura and Walters	Later social learning; social cognitive theory	Observational learning, reciprocal determinism, self-efficacy
Julian Rotter	Social learning (expectancy-value)	Behaviour potential, locus of control
Abraham Maslow	Self-actualisation (Section 4.8)	Humanistic hierarchy of needs

EXAM TRAP

Matching items pair Dollard and Miller with **early** social learning (Hull's drive theory) and Bandura and Walters with **later** social learning (observation and cognition); Skinner is **radical behaviourism**, Maslow **self-actualisation**.

Bandura: reciprocal determinism and self-regulation ●●

Albert Bandura (1925 to 2021) began where Dollard and Miller stopped. With **Richard Walters** he wrote *Social Learning and Personality Development* (1963): people learn much of what they do by watching models, without performing the act or being reinforced, which is **observational** or **vicarious learning** (its four processes and the Bobo doll studies are in Section 1.6). In *Social Foundations of Thought and Action* (1986) he renamed the approach **social cognitive theory**, stressing that thought, not reinforcement alone, guides action.

Its organising principle is **reciprocal determinism** (Figure 4.15): person (cognitive) factors such as beliefs and expectations, behaviour and the environment each influence the other two, in both directions. The master is not the environment (Skinner), the genes (biological theories) or the unconscious (psychodynamic theory). A student who believes she can master statistics (person) joins a study group (behaviour); its praise (environment) lifts her confidence, and she picks harder problems, changing the environment again.

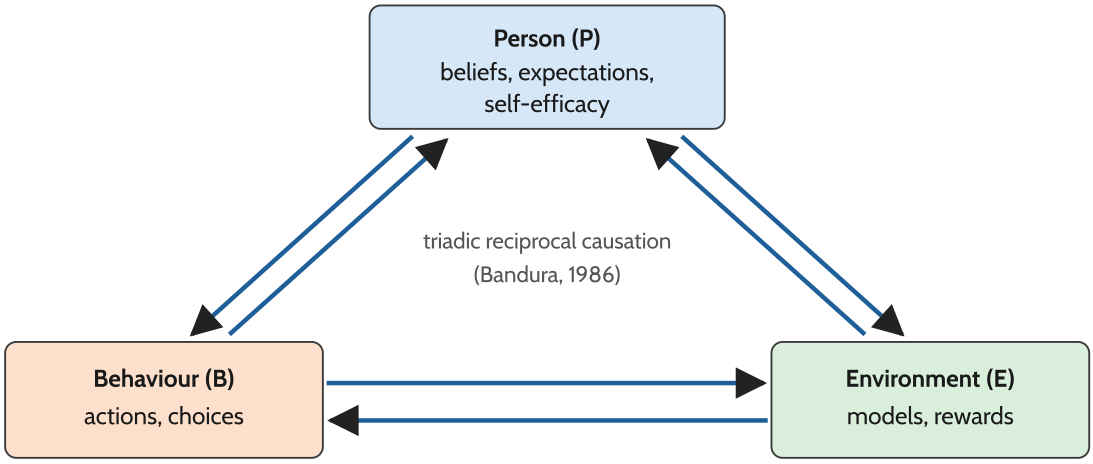

Figure 4.15 Reciprocal determinism. Each pair of factors is joined by arrows running both ways, so no single factor is the master cause.

EXAM TRAP

Reciprocal determinism is central to **social cognitive theory**, not psychodynamic, humanistic or biological theory, and it means two-way influence, not control by environment alone or genes alone, nor a denial of social reinforcement. Bandura's constructs are reciprocal determinism and observational (vicarious) learning, not **reciprocal inhibition** (Joseph Wolpe: a response incompatible with anxiety, such as relaxation, suppresses it; the basis of systematic desensitisation), **reciprocal altruism** (Robert Trivers: helping those who will help in return) or the invented "reciprocal interpretation" and "reciprocal dualism".

Bandura noticed that most behaviour occurs with no external reward or punishment in sight: nobody pays a novelist per page. Most behaviour must therefore stem from **self-regulation**. People adopt standards, largely from models such as parents and teachers, and run three processes: **self-observation** (watching their own performance), **judgement** (comparing it with the standard) and **self-response** (pride when they meet it, self-criticism when they fall short). Rewarding or punishing oneself this way is **self-reinforcement**. It is neither negative reinforcement (removal of an unpleasant stimulus, still an external consequence) nor Skinner's behavioural determinism. The machinery can also be switched off: **moral disengagement** silences self-condemnation during cruelty through moral justification, euphemistic labels, displaced responsibility and dehumanising the victim.

Self-efficacy: believing you can ●●

At the centre of Bandura's theory sits **self-efficacy**: a person's belief in his capability to organise and execute the actions needed to achieve a specific goal. Bandura introduced it in 1977 as a unifying theory of behavioural change, a cognitive-behavioural account of how people change in therapy (his concept, not Aaron Beck's, Albert Ellis's or Skinner's). It is the key motivational factor of social cognitive theory, ahead of external rewards, biological drives or goal difficulty (goal-setting theory, Section 1.10): people attempt what they believe they can do, aim higher and persist after failure.

Bandura separated two expectations (Figure 4.16). An **outcome expectancy** is the belief that a behaviour will produce an outcome ("steady revision gets a good rank"); an **efficacy ex-**

pectancy is the belief that one can perform it ("I can revise three hours a day"). Only the second is self-efficacy, and it is task-specific: confident at chess, helpless in a viva.

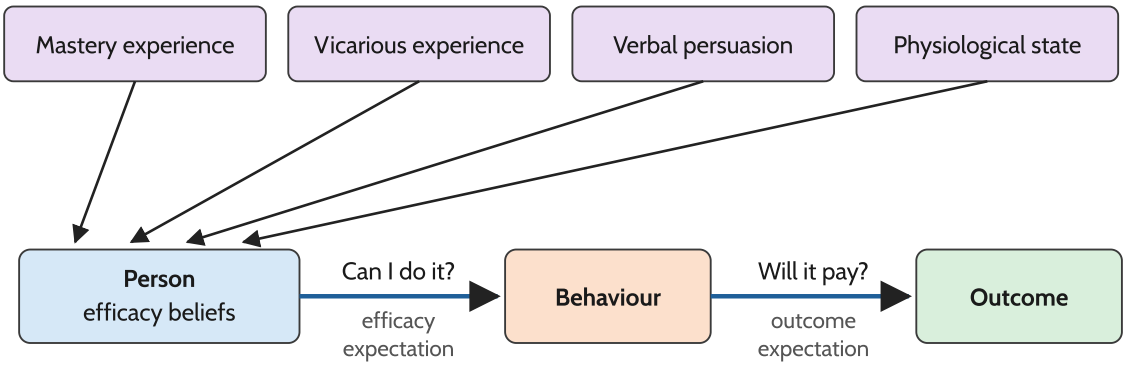

Figure 4.16 Efficacy versus outcome expectation. The four sources feed only the first, "Can I do it?".

Efficacy beliefs come from four sources, the first being the strongest:

1. **Performance (enactive) mastery**: one's own past successes; early failures weaken efficacy most.
2. **Vicarious experience**: seeing someone similar succeed ("if she can, so can I").
3. **Verbal persuasion**: being told "you can do it", which helps only if realistic.
4. **Physiological and emotional states**: calm is read as readiness, a pounding heart as inability.

KEY STUDY: MASTERING SNAKE PHOBIA (1969)

Bandura, Edward Blanchard and B. Ritter treated adults with intense snake phobia by a film of models approaching snakes, by systematic desensitisation, or by **participant modelling**: watching a live model handle a snake, then being guided, at first in gloves, to touch and hold it. Participant modelling, which gives direct mastery, worked best, removing the phobia in almost everyone. In later studies (1977) clients' efficacy beliefs after treatment predicted how close they would approach a snake, whichever treatment they had received. Critics reply that efficacy ratings may be predictions of behaviour rather than causes.

Collective efficacy is a group's shared belief that together it can succeed, as in a cricket team or a village self-help group. Bandura later framed the theory around **human agency**: people are agents who plan, anticipate, regulate and reflect on themselves, not puppets. Table 4.28 separates self-efficacy from its look-alikes.

Table 4.28 Self-efficacy and the concepts confused with it.

Concept	Theorist	What it is
Self-efficacy	Bandura	"I can do this task": a specific capability belief
Self-esteem, self-worth	Many	Global liking and acceptance of oneself (Section 4.10)
Locus of control	Rotter	Whether outcomes depend on me or on fate, chance, others
Inferiority complex	Adler	Exaggerated inadequacy behind striving for superiority (Section 4.4)

Concept	Theorist	What it is
Collective unconscious	Jung	Inherited layer of the psyche holding archetypes (Section 4.4)
Congruence	Rogers	Match between self-concept and experience (Section 4.8)

Rotter: expectancy, value and locus of control ●●

Julian Bernard Rotter (1916 to 2014) built his own **social learning theory** in *Social Learning and Clinical Psychology* (1954), asking which of several possible behaviours a person will perform in a situation. Four concepts answer:

- **Behaviour potential**: "the potentiality of any behaviour occurring in any given situation as calculated in relation to any single reinforcement or set of reinforcements".
- **Expectancy**: the subjective probability that the behaviour will bring that reinforcement.
- **Reinforcement value**: how much he prefers that reinforcement to others, expectancies being equal.
- **Psychological situation**: the situation as the person perceives it, which shapes both expectancy and value.

Behaviour potential is a function of expectancy and reinforcement value, BP = f(E & RV), often read as a product: a student who prizes a seat at NIMHANS but believes no study will win it will not study. There is no "punishment value" in the theory. Expectancies carried across many situations are **generalised expectancies**, which give personality its consistency; the best known are locus of control and interpersonal trust (belief that others' word can be relied on; Interpersonal Trust Scale, 1967).

Locus of control is a generalised expectancy about whether one's outcomes are controlled by internal efforts or external forces. A person with an **internal locus of control** believes outcomes depend on his own effort, ability and preparation, like the student who credits his exam result to his own work. One with an **external locus of control** credits luck, fate, chance, powerful others or an easy examiner. Rotter's **Internal-External (I-E) Scale** (1966) has 29 forced-choice pairs, 23 scored and 6 fillers; higher scores mean more externality. Internals, on average, achieve more, seek information, look after their health and cope actively; externals report more anxiety. Warm, consistent parents who praise achievement and encourage independence tend to raise internals. **Hanna Levenson** (1973) later split it into three separate dimensions: **internality**, **powerful others** and **chance** (her IPC scale).

> EXAM TRAP
>
> Locus of control has two poles, **internal** and **external**; there is no "neutral" or "situational" locus. It concerns **external forces versus internal efforts**, not cognitive schemas (Beck), genetic predisposition or social status. It is not self-efficacy: locus of control asks "do outcomes depend on what I do?", self-efficacy asks "can I do it?". Attribution theory, with its own locus dimension, is in Section 1.10.

Mischel's cognitive-affective personality system

Walter Mischel (1930 to 2018) is known for the person-situation critique of trait theory (Section 4.7): people behave less consistently across situations than traits predict. His positive answer, with **Yuichi Shoda**, was the **cognitive-affective personality system** (CAPS, 1995), a

stable network of five kinds of units: **encodings** (how one construes self, others and situations), **expectancies and beliefs**, **affects**, **goals and values**, and **competencies and self-regulatory plans**. Different situations activate different units, so what is stable is not an average level of behaviour but a **behavioural signature** of if-then relations: Ravi is aggressive *if* teased by classmates and meek *if* scolded by a teacher, every time. His delay of gratification research shows self-regulatory competencies at work (the marshmallow test is in Chapter 6).

Kelly: the person as scientist ●●

George Alexander Kelly (1905 to 1967) set out the most purely cognitive theory in *The Psychology of Personal Constructs* (1955). Ordinary people, he argued, behave like scientists: they form theories, predict events and revise the theories when predictions fail. The theories are **personal constructs**, bipolar ways of construing events, such as friendly versus hostile. A construct may be **permeable** (admitting new events, open to revision) or **impermeable** (rigid whatever the evidence, as in prejudice). Since no one is trapped in one construal, people can always re-construe their lives: **constructive alternativism**.

The theory has a **fundamental postulate**, "a person's processes are psychologically channelised by the ways in which he anticipates events", and eleven corollaries (Table 4.29). Since anticipation itself drives behaviour, Kelly needed no separate motivation theory, and he redefined emotions through constructs. **Anxiety** is recognising that events lie outside the range of one's constructs; **threat** is awareness of imminent comprehensive change in one's core constructs; **fear** is the same for an incidental change; **guilt** is perceiving oneself dislodged from one's core role; **hostility** is continued effort to extort evidence for a prediction that has already failed.

Table 4.29 Kelly's eleven corollaries.

Corollary	Meaning
Construction	We anticipate events by construing their replications
Individuality	People differ in how they construe events
Organisation	Constructs form a hierarchy, superordinate to subordinate
Dichotomy	Every construct is bipolar (honest versus dishonest)
Choice	We choose the pole that best extends our system
Range	Each construct fits a limited range of events
Experience	Constructs are revised as events unfold
Modulation	Revision is limited by the permeability of constructs
Fragmentation	A person may hold mutually inconsistent subsystems
Commonality	People who construe alike are psychologically alike
Sociality	To relate to someone we must construe his constructions

To find a person's constructs Kelly devised the **Role Construct Repertory Test** (Rep test). The client names people who fill role titles in his life (mother, father, a liked teacher, a disliked person), then takes them three at a time and says how two are alike and differ from the third; each answer yields a construct. Set out as a matrix of people by constructs, it becomes the **repertory grid**. To change constructs he used **fixed-role therapy**: from the client's self-

characterisation sketch, the therapist writes a fixed-role sketch of a new character with different constructs, and the client acts that character in daily life for about two weeks, discovering that new constructs work better. Kelly is praised for respecting the person's own view and criticised for neglecting emotion and development.

> EXAM TRAP
>
> Kelly is to the **Rep test** as Cattell is to the **16PF** (Section 4.6). The others: **California Psychological Inventory**, Harrison Gough; **Q-sort**, William Stephenson, used by Rogers; **MMPI** (Chapter 2). **Fixed-role therapy** is Kelly's; Aaron Beck developed cognitive therapy, Albert Ellis rational emotive behaviour therapy and Fritz Perls Gestalt therapy (Chapter 7).

Cognitive styles

A **cognitive style** is a habitual way of perceiving and processing information, distinct from ability. **Herman Witkin** (1916 to 1979) described **field dependence-independence**. In the **rod-and-frame test** a person in a dark room sets a luminous rod to vertical inside a tilted luminous frame: field-dependent people tilt the rod towards the frame, field-independent people trust their bodily sense of upright and also find shapes hidden in complex designs faster on the **Embedded Figures Test**. Field-independent people are more analytic, field-dependent people more global and socially attentive. **Jerome Kagan** (1929 to 2021) described **reflection-impulsivity**: on the Matching Familiar Figures Test, where a child picks which of several near-identical pictures matches a standard, impulsive children answer fast with many errors, reflective children slowly and accurately.

> SCAN YOUR BRAIN
>
> Fill in the blanks. The answers are below; cover them first.
>
> 1. Emitted behaviour that operates on the environment is _____ behaviour.
> 2. Dollard and Miller described learning by drive, cue, response and _____.
> 3. Bandura's principle that person, behaviour and environment influence one another is _____.
> 4. The strongest source of self-efficacy is _____.
> 5. For Rotter, behaviour potential is a function of expectancy and _____.
> 6. Mischel's stable if-then patterns are called _____.
> 7. Acting out a new character for about two weeks is Kelly's _____.
> 8. Witkin's rod-and-frame test measures _____.
>
> **Answers:** (1) operant, (2) reinforcement, (3) reciprocal determinism, (4) performance (enactive) mastery, (5) reinforcement value, (6) behavioural signatures, (7) fixed-role therapy, (8) field dependence-independence.

4.10 The Self, Positive Traits and Indian Approaches to Personality

LOWER YIELD · 16 QS

Who am I, how do I feel about that person, and what makes a life go well? This last section turns from personality seen from outside to the self as people experience it, then to narrow

traits and the strengths that positive psychology studies, and ends with India's oldest personality theory, the three gunas, and its bodily cousin, the Ayurvedic doshas.

The self: James, Cooley and Mead ●●

The modern study of the self begins with **William James** (1842 to 1910), whose *Principles of Psychology* (1890) split it in two. The **I** is the self as knower, the subject doing the thinking; the **me** is the self as known, everything a person can call his own. James divided the me into three parts. The **material self** is the body, clothes, family, home and possessions, which is why losing a phone can feel like losing a piece of yourself. The **social self** is the recognition one gets from others; a person has as many social selves as there are groups whose opinion he cares about. The **spiritual self** is the inner world of dispositions, conscience and faculties, the most intimate and enduring part.

James also gave a formula: **self-esteem = success / pretensions**. A student who aimed only to pass is content with 55 percent; one who pretended to the first rank is crushed by the second (James's example was a man ashamed to be only the second-best boxer in the world). So self-esteem rises in two ways: succeed more, or lower the pretensions.

For **Charles Horton Cooley** (1864 to 1929), writing in *Human Nature and the Social Order* (1902), the self is social from the start. His **looking-glass self** is the self-concept we build by seeing ourselves in other people as in a mirror, in three steps (Figure 4.17): we imagine how we appear to the other person, we imagine that person's judgement of that appearance, and we develop a self-feeling, such as pride or shame, from that imagined judgement. Every step is imagined, so the mirror can distort.

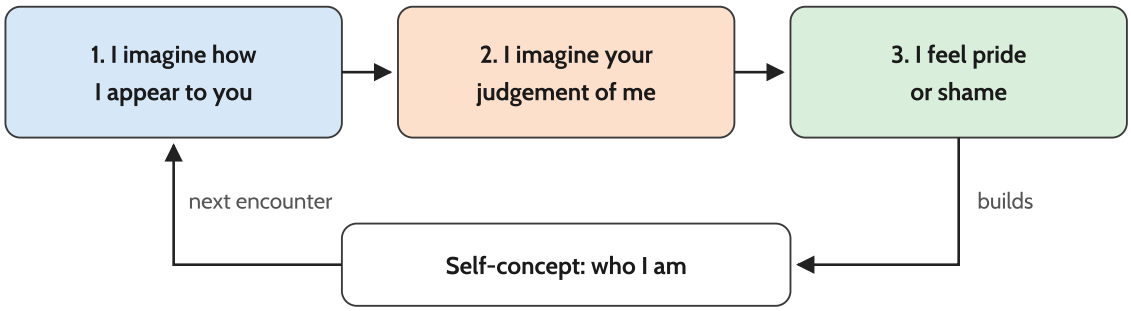

Figure 4.17 Cooley's looking-glass self. Notice that all three steps happen in the imagination: the self is built from what we think others think of us.

George Herbert Mead (1863 to 1931), whose lectures appeared after his death as *Mind, Self and Society* (1934), explained how the mirror gets inside us. With Cooley he founded the tradition later called **symbolic interactionism**: the self emerges through social interaction and shared symbols, above all language. Mead's "I" is the spontaneous, creative, unpredictable response of the person; his "me" is the organised set of others' attitudes that the person has taken in, the socialised self. The child builds the me by taking roles. In the **play stage** the child takes the role of one particular other at a time, playing at being mother or policeman. In the **game stage**, organised games demand holding everyone's role at once: a fielder in cricket must anticipate bowler, batter and captain within shared rules. The organised attitudes of

the whole group, and finally of the community, form the **generalised other**, the standpoint from which the person judges himself.

> **EXAM TRAP**
>
> Looking-glass self = imagined appearance, imagined judgement, self-feeling. The distractors are other theories: innate, unconscious drives (Freud, Section 4.2); comparing ourselves with similar others (Leon Festinger's social comparison theory, 1954, Chapter 3); resolving psychosocial crises (Erik Erikson, Section 6.10); operant reinforcement of self-statements (Section 4.9); solitary introspection or genetic inheritance (Cooley's self is neither private nor inborn). James's I is the knower, Mead's the spontaneous actor; the generalised other is Mead's.

Self-concept, self-esteem and the self-system

Two terms that are often blurred need separating. The **self-concept** is the cognitive description of who one is, the organised set of beliefs about one's own attributes: "I am a Kannadiga, a fast bowler, shy, good at maths." **Self-esteem** is the evaluative and affective component, how one feels about that self: "and I am worthwhile", or "and I am a failure." Two students may share the self-concept "I am shy", one at peace with it and one ashamed of it. The standard measure of global self-esteem is the ten-item Self-Esteem Scale (1965) of **Morris Rosenberg** (1922 to 1992), with statements such as "On the whole, I am satisfied with myself" rated from strongly agree to strongly disagree.

Nor is self-esteem the same as **self-efficacy** (Albert Bandura), the belief that one can carry out a specific task such as a statistics problem (Section 4.9), or **self-actualisation** (Kurt Goldstein's term, made famous by Abraham Maslow), the striving to fulfil one's potential (Section 4.8). Later researchers treated the self as a system that stores and uses information about itself (Table 4.30).

Table 4.30 The self as a cognitive and motivational system.

Concept	Proposed by	What it means
Self-schema	**Hazel Markus**, 1977	A generalisation about the self, drawn from past experience, that organises and guides how self-related information is processed
Possible selves	Markus and **Paula Nurius**, 1986	Selves one hopes to become (a clinical psychologist) or fears becoming (an unemployed graduate); they motivate present action
Self-discrepancy theory	**E. Tory Higgins**, 1987	Gaps between the actual self and the ideal self (hopes) or the ought self (duties) produce distinct emotions
Self-monitoring	**Mark Snyder**, 1974	High self-monitors tailor their self-presentation to each situation; low self-monitors act from inner attitudes and stay consistent

Higgins's theory has a sharp clinical edge. An actual-ideal gap ("I am not who I hoped to be") brings **dejection-related** emotions such as disappointment, sadness and depression; an actual-ought gap ("I am not who I should be") brings **agitation-related** emotions such as anxiety, guilt and fear. Elsewhere: Carl Rogers's self and ideal self (Section 4.8), Gordon Allport's proprium (4.6), Heinz Kohut's self psychology (4.5), the self-reference effect (1.7), social identity and self-presentation (Chapter 3), and the Indian jiva, ahamkara and atman (1.2).

> EXAM TRAP
>
> Self-concept is cognitive ("who am I?"); self-esteem is evaluative and affective ("how do I feel about me?"). The reversed pairing is wrong, the two are not identical, and the individual versus collective self (a self defined by personal traits versus one defined by group memberships) is a different distinction altogether.

Limited-domain traits: sensation seeking and explanatory style

A **limited-domain theory** studies one trait in depth rather than the whole person. **Marvin Zuckerman** (1928 to 2018) defined **sensation seeking** as the need for varied, novel, complex and intense sensations and experiences, and the willingness to take physical, social, legal and financial risks for them. His Sensation Seeking Scale (SSS; first form 1964, later the 40-item Form V) offers forced choices such as "I would like to try parachute jumping". Factor analysis yielded four components:

- **Thrill and adventure seeking:** a desire for speed, danger and defying gravity, as in bungee jumping or scuba diving.
- **Experience seeking:** novel experience through travel, music, art or a nonconformist lifestyle.
- **Disinhibition:** release through wild parties, drinking and risky sex.
- **Boredom susceptibility:** an aversion to routine, repetition and predictable people.

Sensation seeking is strongly heritable and belongs within Eysenck's extraversion, with its impulsive form also tied to psychoticism (Section 4.7). It rises in adolescence, falls with age and is higher in men. High sensation seekers have low levels of **monoamine oxidase** (MAO), the enzyme that breaks down dopamine and the other monoamines, which Zuckerman linked to a strong dopamine-driven approach system.

Martin E. P. Seligman (born 1942) reached optimism by way of learned helplessness (Section 1.6), in which dogs given inescapable shock later failed to escape avoidable shock. People, he found, differ in how far helplessness spreads, depending on how they explain bad events to themselves. This habitual way of explaining is the **explanatory style**, which *Learned Optimism* (1990) describes by three Ps. **Permanence**: is the cause lasting ("I will always fail") or temporary ("I was tired that day")? **Pervasiveness**: does it spoil everything ("I am useless") or one area ("I am weak in statistics")? **Personalisation**: is it me ("I am stupid") or outside me ("the paper was unusually hard")? The pessimist explains bad events as permanent, pervasive and personal, and good events as temporary, specific and lucky; the optimist does the reverse, and is less prone to depression, illness and giving up. The **Attributional Style Questionnaire** (ASQ; Christopher Peterson, Seligman and colleagues, 1982) measures it: the respondent names the main cause of imagined good and bad events and rates it as internal, stable and global. Being learned, the style can be retrained. Dispositional optimism and hardiness belong to stress (Section 1.11).

Well-being: hedonic and eudaimonic

Positive psychology (its founding, flow and PERMA are in Sections 1.2 and 1.10) asks what makes a life go well, and two traditions answer differently. **Hedonic well-being** equates the good life with pleasure: positive feelings, comfort, enjoyment and satisfaction with life. **Ed Diener** (1946 to 2021) measured it as **subjective well-being** (SWB): high life satisfaction, fre-

quent positive affect and infrequent negative affect, with life satisfaction assessed by his Satisfaction With Life Scale (1985).

Eudaimonic well-being goes back to **Aristotle** (384 to 322 BCE), for whom *eudaimonia*, flourishing, came from living by virtue and realising one's best potential. Here well-being means cultivating personal strengths, meaning and growth and contributing to the greater good, whether or not it feels pleasant at the moment. **Carol Ryff** (1989) gave it six dimensions of **psychological well-being**: self-acceptance, personal growth, purpose in life, environmental mastery (managing one's surroundings and demands), autonomy (self-determination and independence from social pressure) and positive relations with others. A holiday of films and good food yields hedonic well-being; one spent teaching village children to read may leave you tired but richer in eudaimonic well-being. The two correlate but are distinct.

> EXAM TRAP
>
> Eudaimonic = cultivating strengths, growth and contribution to the greater good. "Pleasure and relaxation" and "pleasure and happiness" describe hedonic well-being. "Income and material possessions" is a material yardstick that neither tradition accepts as well-being itself.

Character strengths, grit and sportsmanship

Christopher Peterson (1950 to 2012) and Seligman wrote *Character Strengths and Virtues* (2004) as a "manual of the sanities", a positive counterpart to the DSM. Surveying moral and religious traditions across the world, they found six virtues valued almost everywhere, each reached through specific **character strengths**, 24 in all (Table 4.31). These are measured by the VIA Inventory of Strengths (VIA stands for Values in Action); a person's top few are his **signature strengths**.

Table 4.31 The six virtues and 24 character strengths (Peterson and Seligman, 2004).

Virtue	Character strengths
Wisdom and knowledge	Creativity, curiosity, judgement (open-mindedness), love of learning, perspective
Courage	Bravery, perseverance (persistence), honesty (integrity), zest
Humanity	Love, kindness, social intelligence
Justice	Teamwork (citizenship), fairness, leadership
Temperance	Forgiveness, humility, prudence, self-regulation (self-control)
Transcendence	Appreciation of beauty and excellence, gratitude, hope (optimism), humour, spirituality

Angela Duckworth (born 1970) added **grit**: perseverance and passion for long-term goals, kept up for years despite failure and plateaus.

KEY STUDY: GRIT (DUCKWORTH AND COLLEAGUES, 2007)

Why do some equally able people achieve more? Duckworth's team gave a Grit Scale to adults, to cadets entering West Point's gruelling first summer and to Scripps National Spelling Bee finalists. Grit predicted educational attainment, which cadets completed the summer and how far spellers advanced, over and above measures of ability. Criticism: a later meta-analysis found grit so close to conscientiousness (Section 4.7) that it may be an old trait under a new name.

The same strengths define character in sport. **Sportsmanship** is playing hard within the rules and with respect for opponents: the true sportsman is persistent (perseverance), keeps self-control under provocation (self-regulation), stays optimistic after a defeat (hope) and plays fair (fairness). Focusing on winning at all costs, by cheating, injuring or gamesmanship, is its opposite: the batter who walks after edging the ball shows character; the bowler who tampers with the ball does not.

The triguna theory of personality ●●●

The **triguna** theory comes mainly from the Samkhya school, attributed to the sage **Kapila**, together with the Bhagavad Gita (chapters 14, 17 and 18), with earlier hints in the Upanishads. It does not come from Nyaya (the school of logic), Mimamsa (the school of Vedic ritual) or the materialist Charvaka. Samkhya divides reality into purusha, pure consciousness, the passive witness, and prakriti, primordial nature, from which body and mind evolve (Section 1.2 gives the full scheme, the inner instrument and the koshas). The three gunas (strands or qualities) belong to prakriti, not purusha; the Gita calls them "born of prakriti" (14.5). Because the mind evolves from prakriti, the gunas are the stuff of personality:

- **Sattva** is clarity, goodness, light, calm, balance and knowledge; the sattvic person is truthful, self-controlled, content and wise.
- **Rajas** is activity, passion, desire and restlessness; the rajasic person is ambitious, driven and attached to results, and tips into over-activity, greed, anger and anxiety.
- **Tamas** is inertia, dullness, darkness, lethargy and ignorance; the tamasic person is heavy, careless and confused, sleeps too much and cannot bring himself to act.

Every **jiva**, every embodied individual self, has all three in shifting proportions. None is ever absent; the dominant guna colours the personality (Figure 4.18), and the Gita traces that colouring through daily conduct (Table 4.32).

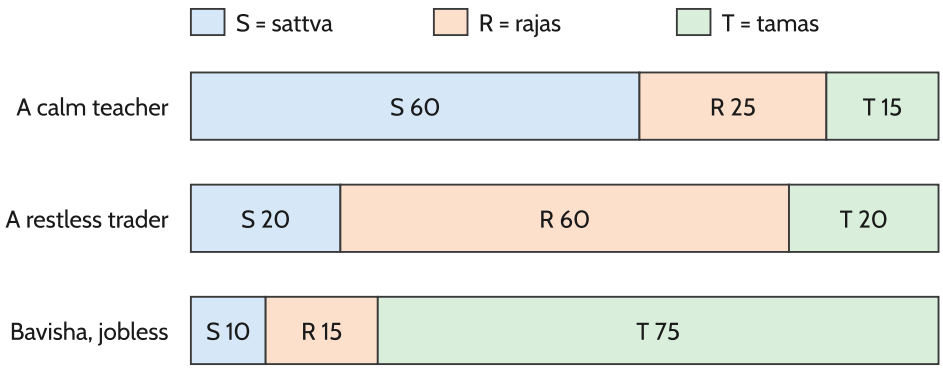

Figure 4.18 Everyone has all three gunas; personality follows the dominant one. The percentages are illustrative, not measured: what matters is that no bar lacks any guna.

Table 4.32 The sattvic, rajasic and tamasic person in the Bhagavad Gita (chapters 17 and 18).

Domain	Sattvic	Rajasic	Tamasic
Work	Done as duty, without craving for its fruit	Driven by desire and ego, with strain	Begun in delusion, careless of consequences
Food	Fresh, wholesome, nourishing	Very hot, sour, salty, pungent; brings pain and disease	Stale, tasteless, left over, impure
Knowledge	Sees one unity in all beings	Sees only separate, different things	Clings to one thing as if it were all
Giving	As duty, to the worthy, expecting nothing	For a return, or grudgingly	At the wrong time, to the unworthy, with contempt

A worked example. Bavisha has lost her job. Since then she is stressed, has stopped eating properly, has withdrawn and cannot get herself to do anything. Run her through the gunas. No calm clarity: not sattva. No restless over-activity, anger or frantic job-hunting: not rajas, which is how the same stress looks in someone who cannot sit still. Heaviness, withdrawal and inability to act: **tamas alone**. Yama cannot be her problem, alone or with tamas, because it is not a mental state but the first limb of Patanjali's yoga, the ethical restraints; pratyahara (withdrawal of the senses, the fifth limb) and jnana (the path of knowledge) are practices, not afflictions.

> **EXAM TRAP**
>
> The only gunas are sattva, rajas and tamas. Not gunas: ahimsa (non-violence, the first of the yamas); yama, pratyahara and samadhi (absorption, the eighth limb), all limbs of yoga; jnana (knowledge); samskara (the latent impression a past act leaves on the mind); ahamkara (the I-maker or ego, a product of prakriti); prakriti itself; vata (an Ayurvedic dosha). The gunas belong to prakriti, not to purusha, atman (the true self), Brahman (ultimate reality), ahamkara, samadhi or samskara. Distractor triads: prakriti, purusha and kshetra (the "field", the body that the self knows); atman, Brahman and maya (appearance), the Vedanta triad; karma, dharma and moksha (action, duty and liberation).

The gunas have been turned into questionnaires: **David Wolf**'s Vedic Personality Inventory (VPI, 1998) and Indian instruments such as the Mysore Triguna Scale score sattva, rajas and

tamas separately. Such studies tend to link sattva with mental health and life satisfaction, and tamas with depression and poor adjustment.

Ayurvedic constitution and Indian typologies

Ayurveda, India's classical medicine, adds a bodily typology. One of its foundational texts, the *Charaka Samhita* of **Charaka**, describes three doshas (the **tridosha**), bodily forces each built from two of the five elements (Table 4.33). Their proportions at conception fix a person's **prakriti** in the Ayurvedic sense: one's inborn bodily and mental constitution, a second meaning quite different from Samkhya's primordial nature. Ayurveda also describes a mental constitution in terms of the gunas, and calls rajas and tamas the two doshas of the mind.

Table 4.33 The three doshas and their Western parallels.

Dosha	Elements	Body	Temperament	Rough parallel
Vata	Air and space	Thin, light, dry	Quick, restless, anxious, creative	Ectomorph
Pitta	Fire and water	Medium, muscular, warm	Sharp, ambitious, irritable	Choleric; mesomorph
Kapha	Earth and water	Heavy, solid	Calm, steady, slow	Phlegmatic; endomorph

The parallels with the four humours of Greek medicine and with William Sheldon's somatotypes (Section 4.1) are real: all three systems tie a bodily fluid or build to a temperament. What are Indian typologies worth as personality science? They are holistic, joining body, diet, routine and mind; the guna model is dimensional, since everyone has all three in varying amounts, which is closer to modern trait thinking than to rigid boxes; and they point to change, since diet, routine, yoga and right action are meant to raise sattva. Their limits are those of all typologies: people get forced into types, the scales rest mostly on self-report with limited validation, what they add to the Big Five is unsettled, and body-temperament links, like Sheldon's, have little experimental support.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. For William James, self-esteem equals success divided by _____.
2. In Cooley's looking-glass self, the third step is a self-feeling such as pride or _____.
3. Mead's term for the organised attitudes of the whole community is the _____.
4. In Higgins's theory, an actual-ought discrepancy produces _____-related emotions.
5. Seligman's three Ps of explanatory style are permanence, pervasiveness and _____.
6. Ryff's six dimensions include self-acceptance, purpose in life and environmental _____.
7. The three gunas are strands of _____, not of purusha.
8. The dosha made of fire and water is _____.

Answers: (1) pretensions, (2) shame, (3) generalised other, (4) agitation, (5) personalisation, (6) mastery, (7) prakriti, (8) pitta.

The Boy on the Vienna Tram, Revisited

Return to Allport's tram story with the whole chapter behind you. Freud's question was psychoanalysis in one line: an innocent anecdote read as the surface of hidden conflict (Sections 4.2 and 4.3). Allport's reply became the trait approach: take conscious motives seriously and describe the stable dispositions a person actually shows (Sections 4.6 and 4.7). Adler would have seized on the "generalized inferiority feeling" Allport admitted to (Section 4.4); Rogers would have asked how the young man saw himself (Section 4.8); Bandura and Mischel would have pointed to the situation, an awkward silence in a famous man's study (Section 4.9). The modern answer is interactionism: a neat, orderly young man, in a situation that made him anxious, telling a story that revealed something of both.

Chapter summary

4.1 What Personality Is: Definitions, Approaches and Early Type Theories. Personality, from the Latin *persona* (the actor's mask), is the relatively enduring, distinctive pattern of thought, feeling and behaviour; **Allport** (1937) called it the dynamic organisation of psychophysical systems. **Temperament** is its biologically based, early-appearing emotional core; character is personality evaluated; traits are dimensions, types are categories. Theories describe, explain, predict and generate research, and differ on Schultz's six questions about human nature. Personality reflects heredity, environment and the situation in interaction, and is studied by clinical, experimental and correlational methods, nomothetically or idiographically (Windelband). The early type theories, the four humours, physiognomy, Kretschmer's body types and **Sheldon's somatotypes**, did not survive testing.

4.2 Freud's Psychoanalysis: Instincts, Structure and Psychosexual Stages. Sigmund Freud (1856 to 1939), a Viennese neurologist, built psychoanalysis from hysteria: Charcot's hypnosis, Bernheim's suggestion and Breuer's Anna O. (catharsis; *Studies on Hysteria*, 1895). Free association, dream analysis and parapraxes open the unconscious. Instincts (Eros and its libido; the death instinct, 1920) supply energy invested by cathexis. The topographic model divides conscious, preconscious and unconscious; the structural model (1923) sets the id (pleasure principle), ego (reality principle, the executive) and superego (conscience and ego-ideal) in conflict, and anxiety (reality, neurotic, moral) signals danger. Fixation at the oral, anal or phallic (Oedipus) stage shapes adult character; latency and the genital stage follow. Research supports unconscious processes, not the stages.

4.3 Defence Mechanisms: How the Ego Protects Itself. Defence mechanisms are the ego's unconscious, reality-distorting strategies against anxiety, described by Sigmund Freud and catalogued by Anna Freud in *The Ego and the Mechanisms of Defence* (1936); they are adaptive in moderation, pathological when rigid, and never rational. **Repression**, the involuntary blocking of threatening material from awareness, is the primary defence; suppression is its conscious, mature counterpart. Displacement changes the target, sublimation the activity, projection the owner, reaction formation the direction and rationalisation the stated reason. **George Vaillant** (1977) ranked defences as psychotic, immature, neurotic and mature. Obsessive-compulsive disorder favours isolation, undoing and reaction formation; phobias, displacement; hysteria, repression; borderline personality disorder, splitting.

4.4 Jung's Analytical Psychology and Adler's Individual Psychology. Jung and Adler were the first neo-Freudians to break with Freud (Adler in 1911, Jung in 1913). Jung's **analytical psychology** widened libido into general psychic energy and placed, beneath the personal unconscious and its complexes (detected with the word association test), an inherited **collective unconscious** of archetypes: persona, shadow, anima, animus and self. His two attitudes and four functions give eight types, the basis of the MBTI, and individuation belongs to midlife. Adler's **individual psychology** sees behaviour as goal-directed: universal inferiority feelings drive compensation and striving for superiority, shaped by fictional finalism, the style of life, social interest and birth order.

4.5 Horney, Fromm, Sullivan and the Object-Relations Theorists. **Karen Horney** traced neurosis to **basic anxiety**, the feeling of being lonely and helpless in a hostile world, managed by ten neurotic needs grouped into moving toward, against and away from people; she added the idealised self, the **tyranny of the shoulds** and **womb envy**. **Erich Fromm** described mechanisms of escape from freedom, five existential needs and character orientations; **Harry Stack Sullivan** saw personality as recurrent interpersonal situations. Ego psychology (Anna Freud, Hartmann, Erikson) strengthened the ego. **Object relations** are internalised images of self and others: Klein's splitting and two positions, Winnicott's good-enough mother. Kohut's **selfobjects** and Lacan's mirror stage complete the section.

4.6 Traits I: Allport, Cattell and the Lexical Search. The **trait approach** describes personality through stable, consistent dimensions of individual difference, unlike types, states or situationism. The **lexical hypothesis** led Allport and Odbert (1936) to cull 17,953 trait words from a dictionary. **Gordon Allport** treated traits as real neuropsychic structures, separated common traits from **personal dispositions** (cardinal, central, secondary), preferred the **idiographic** study of single cases (*Letters from Jenny*), and proposed **functional autonomy**, the proprium and six marks of maturity. **Raymond Cattell** factor-analysed the lexicon into **surface** and **source traits**, described ergs and sentiments, required L-, Q- and T-data, and built the **16PF** (1949).

4.7 Traits II: Eysenck, the Big Five and the Biology of Personality. **Hans Eysenck** joined factor analysis to experiment and biology: a hierarchy from specific responses to types, and three superfactors, extraversion (cortical arousal set by the ARAS), neuroticism (the visceral brain) and psychoticism; crossing E and N gives Galen's four temperaments. Gray, Cloninger, Kagan, and Buss and Plomin extended the biology of temperament. The **Big Five** (OCEAN) came from lexical and questionnaire research; Costa and McCrae's NEO-PI-R has 240 items and 30 facets. High conscientiousness and low neuroticism predict health. HEXACO adds honesty-humility. Heritability is about 40 to 50 percent, with the non-shared environment mattering most. **Mischel** (1968) argued that situations often override traits; aggregation and interactionism replied.

4.8 Humanistic and Existential Theories: Rogers, Maslow and May. The **humanistic approach** stresses free will, an innate tendency to grow toward one's maximum potential, the whole person and conscious experience; disorder is blocked growth, and critics call it vague and blind to culture. Rogers's **actualising tendency**, **organismic valuing process**, **conditions of worth** and **incongruence** explain how conditional love splits self from experience; the **fully functioning person** is the ideal, and the **Q-sort** measures self-ideal congruence. Maslow contrasted D-motivation with **metamotivation** and described self-actualisers, peak experiences and the **Jonah complex**. Existentialists

(Heidegger, Sartre, Binswanger, May, Frankl) describe Dasein in the Umwelt, Mitwelt and Eigenwelt, normal and neurotic anxiety, the will to meaning and noogenic neurosis.

4.9 Behavioural, Social-Cognitive and Cognitive Theories. Skinner's radical behaviourism treats personality as learned operant behaviour (emitted, acting on the environment) with no inner causes. Dollard and Miller (1950) recast Freud in Hull's drive, cue, response and reinforcement: early social learning. Bandura (later social learning, with Walters) proposed reciprocal determinism among person, behaviour and environment, self-regulation and self-efficacy (1977), fed by mastery, vicarious experience, persuasion and arousal. Rotter: behaviour potential = f(expectancy, reinforcement value); internal versus external locus of control (I-E Scale, 1966). Mischel's CAPS gives if-then signatures. Kelly (1955): the person as scientist, bipolar constructs, the Rep test and fixed-role therapy. Witkin: field dependence; Kagan: reflection-impulsivity.

4.10 The Self, Positive Traits and Indian Approaches to Personality. William James (1890) split the self into the knowing I and the known me (material, social and spiritual selves) and defined self-esteem as success divided by pretensions. Cooley's looking-glass self (1902) builds the self-concept from imagined appearance, imagined judgement and self-feeling; Mead (1934) added the play and game stages and the generalised other. Self-concept is cognitive, self-esteem evaluative (Rosenberg, 1965). Zuckerman's sensation seeking and Seligman's explanatory style are limited-domain traits. Hedonic well-being is pleasure; eudaimonic well-being is strengths, growth and contribution (Ryff's six dimensions). The triguna theory (Samkhya and the Gita) describes sattva, rajas and tamas, strands of prakriti; Ayurveda adds vata, pitta and kapha.

Who's who

Table 4.34 The people in this chapter.

Name	When	Examined for
Gordon Willard Allport	1897 to 1967	Definition of personality (1937, revised 1961); character is personality evaluated; idiographic study; met Freud in 1920
Alexander Thomas	New York Longitudinal Study, 1956 onwards	With Chess: easy, difficult and slow-to-warm-up temperaments (pointer)
Stella Chess	New York Longitudinal Study, 1956 onwards	With Thomas: easy, difficult and slow-to-warm-up temperaments (pointer)
Duane Schultz	Theories of Personality, 2017 edition	With Sydney Ellen Schultz: six questions about human nature
Sydney Ellen Schultz	Theories of Personality, 2017 edition	With Duane Schultz: six questions about human nature
Sigmund Freud	1856 to 1939	Psychoanalysis; the darkest image of human nature; built theory from case studies
Carl Gustav Jung	1875 to 1961	Analytical psychology; introvert and extravert types (Section 4.4)
Alfred Adler	1870 to 1937	Individual psychology; birth order (Section 4.4)
Karen Horney	1885 to 1952	Neo-psychoanalytic theory (Section 4.5)
Erich Fromm	1900 to 1980	Neo-psychoanalytic theory (Section 4.5)
Harry Stack Sullivan	1892 to 1949	Neo-psychoanalytic interpersonal theory (Section 4.5)
Anna Freud	1895 to 1982	Ego psychology; the ego's defences (Section 4.3)

Name	When	Examined for
Melanie Klein	1882 to 1960	Object relations theory (Section 4.5)
Erik Erikson	1902 to 1994	Life-span psychosocial stages (Chapter 6)
Raymond Bernard Cattell	1905 to 1998	Factor-analytic trait theory; 16PF (Section 4.6)
Hans Jurgen Eysenck	1916 to 1997	Factor analysis; extraversion and neuroticism; mapped the four temperaments (Section 4.7)
Paul T. Costa Jr.	NEO-PI, 1985	With McCrae: the Five-Factor Model (Section 4.7)
Robert R. McCrae	NEO-PI, 1985	With Costa: the Five-Factor Model (Section 4.7)
Abraham Maslow	1908 to 1970	Humanistic theory (Section 4.8)
Carl Ransom Rogers	1902 to 1987	Humanistic theory; free will, growth and optimism (Section 4.8)
Rollo May	1909 to 1994	Existential theory (Section 4.8)
Burrhus Frederic Skinner	1904 to 1990	Behavioural approach; experimental method (Section 4.9)
Albert Bandura	1925 to 2021	Social-cognitive theory; experimental method (Section 4.9)
Julian Rotter	1916 to 2014	Social-cognitive theory; locus of control (Section 4.9)
Walter Mischel	1930 to 2018	Person-situation critique (1968); social-cognitive theory (Section 4.9)
George Kelly	1905 to 1967	Personal construct theory, the cognitive approach (Section 4.9)
Marvin Zuckerman	1928 to 2018	Sensation seeking, a limited-domain theory
Martin Seligman	born 1942	Learned helplessness and optimism, a limited-domain theory
Wilhelm Windelband	1848 to 1915	Coined nomothetic and idiographic (1894)
Hippocrates	about 460 to 377 BC	The four bodily humours
Galen	AD 129 to about 216	Four temperaments from the dominant humour
Aristotle	384 to 322 BC	The Physiognomonica is attributed to him
Johann Kaspar Lavater	1741 to 1801	Popularised physiognomy with essays of the 1770s
Cesare Lombroso	1835 to 1909	Criminal Man (1876); criminal stigmata
Franz Joseph Gall	1758 to 1828	Founder of phrenology (skull bumps, localised faculties)
Johann Gaspar Spurzheim	1776 to 1832	Popularised phrenology under that name
Ernst Kretschmer	1888 to 1964	Physique and Character (1921): pyknic, asthenic, athletic, dysplastic
William Herbert Sheldon	1898 to 1977	Somatotypes (1940) and temperaments (1942): endomorph, mesomorph, ectomorph
Eduard Spranger	1882 to 1963	Six value types behind the Study of Values
Philip Ewart Vernon	1905 to 1987	Allport-Vernon-Lindzey Study of Values
Gardner Lindzey	1920 to 2008	Allport-Vernon-Lindzey Study of Values
Meyer Friedman	1910 to 2001	With Rosenman: Type A behaviour pattern (1959)
Ray Rosenman	1920 to 2013	With Friedman: Type A behaviour pattern (1959)
Johan Denollet	1990s	Type D (distressed) personality
Jean-Martin Charcot	1825 to 1893	Paris neurologist; Freud studied hysteria and hypnosis with him (1885 to 1886)
Hippolyte Bernheim	1840 to 1919	Nancy; showed that hypnosis works through suggestion
Josef Breuer	1842 to 1925	Treated Anna O.; catharsis; Preliminary Communication (1893) and Studies on Hysteria (1895) with Freud
Bertha Pappenheim	1859 to 1936	Anna O., Breuer's hysteria patient; named the talking cure
Granville Stanley Hall	1844 to 1924	Invited Freud to lecture at Clark University (1909)
Daniel Paul Schreber	1842 to 1911	Freud's case of paranoia (1911), analysed only from his published memoir
Sophocles	5th century BCE	Oedipus Rex, source of the name Oedipus complex
John Broadus Watson	1878 to 1958	Behaviourism, contrasted with psychoanalytic theory
Abraham Harold Maslow	1908 to 1970	Humanistic theory; needs, contrasted with Freud's instincts
Erik Homburger Erikson	1902 to 1994	Psychosocial stages; adolescent identity formation
Henry Alexander Murray	1893 to 1988	Psychogenic needs, contrasted with Freud's instincts
Clark Leonard Hull	1884 to 1952	Drives, contrasted with Freud's instincts
Aaron Temkin Beck	1921 to 2021	Cognitive distortions, contrasted with Freudian slips
Howard Shevrin	1977	Subliminal study: an unseen picture of a bee shaped free associations

Name	When	Examined for
Michael Motley	1987	Laboratory Freudian slips under anxiety (damn shock for sham dock)
Seymour Fisher	1977, 1996	With Greenberg, reviewed about 2,500 studies testing Freud's ideas
Roger Greenberg	1977, 1996	With Fisher, reviewed about 2,500 studies testing Freud's ideas
Girindrasekhar Bose	1887 to 1953	Founded the Indian Psychoanalytical Society (1922); corresponded with Freud; theory of opposite wishes
Elizabeth Loftus	born 1944	False memory research; critic of recovered repressed memories (2006 review)
Ernest Jones	1879 to 1958	Introduced the term rationalisation (1908)
Lawrence K. Frank	1939	Named projective methods; the projective hypothesis rests on the broad sense of projection
George Eman Vaillant	born 1934	Four-level hierarchy of defences (psychotic, immature, neurotic, mature); Adaptation to Life (1977); Grant Study
Otto Kernberg	born 1928	Borderline personality organisation; splitting and projective identification
Herbert Graf (Little Hans)	1909 case	Freud's case of horse phobia: fear of the father displaced onto horses
Joseph Wolpe	1915 to 1997	With Rachman (1960), a conditioning account of Little Hans's phobia
Stanley Rachman	1960	With Wolpe (1960), a conditioning account of Little Hans's phobia
Phebe Cramer	1987	Developmental sequence of defences: denial, then projection, then identification
Sam Glucksberg and L. J. King	1967	Laboratory analogue of repression: words linked with shock were forgotten
David S. Holmes	1990	Review of sixty years of studies: no convincing controlled evidence for repression
Hans Eysenck	1916 to 1997	Trait theorist (Section 4.7); not a neo-Freudian
Carl Rogers	1902 to 1987	Humanist (Section 4.8); empathic understanding; not a neo-Freudian
Eugen Bleuler	1857 to 1939	Directed the Burghölzli hospital, Zurich, where Jung worked from 1900
Franz Riklin	1878 to 1938	Jung's partner in the word association studies (1904)
Francis Galton	1822 to 1911	Studied word association before Jung
Wilhelm Wundt	1832 to 1920	Studied word association before Jung
Henry Murray	1893 to 1988	Thematic Apperception Test (Chapter 2); not the word association test
Donald Winnicott	1896 to 1971	Object relations: transitional object, good enough mother (Section 4.5); not the word association test
Katharine Cook Briggs	1875 to 1968	Co-creator of the Myers-Briggs Type Indicator from Jung's types (read Psychological Types in 1923)
Isabel Briggs Myers	1897 to 1980	Co-creator of the Myers-Briggs Type Indicator (first form 1943)
Raymond Cattell	1905 to 1998	16PF (Section 4.6); not based on Jung's types
Paul Costa	born 1942	NEO-PI-R and the Big Five, with McCrae (Section 4.7)
Robert McCrae	born 1949	NEO-PI-R and the Big Five, with Costa (Section 4.7)
Demosthenes	384 to 322 BC	Greek orator who overcame a stutter: Adler's example of compensation
Viktor Frankl	1905 to 1997	Logotherapy (Chapter 7); not the inferiority complex
Franz Alexander	1891 to 1964	Psychosomatic medicine; did not coin the inferiority complex
Gordon Allport	1897 to 1967	Trait theorist (Section 4.6); did not coin the inferiority complex
Hans Vaihinger	1852 to 1933	The Philosophy of As If (1911), source of Adler's fictional finalism
Albert Ellis	1913 to 2007	Rational emotive therapy (Chapter 7); influenced by Adler
Jean Piaget	1896 to 1980	Formal operations stage (Chapter 6)
Lawrence Kohlberg	1927 to 1987	Moral reasoning (Chapter 6); wrongly paired with formal operations
Sudhir Kakar	Born 1938	Psychoanalyst of Indian childhood and society (The Inner World, 1978); keyed for basic anxiety in the Indian context
Anand C. Paranjpe	2008 (Handbook)	Co-editor of the Handbook of Indian Psychology; distractor for basic anxiety in India
Durganand Sinha	1922 to 1998	Indigenisation of psychology; distractor for basic anxiety in India
Philip Wylie	1942	Coined momism in Generation of Vipers
Edward Strecker	1946	Psychiatrist who took up momism in Their Mothers' Sons; keyed when Wylie is absent

Name	When	Examined for
Edward Deci	1985	Self-determination theory (competence, autonomy, relatedness) with Ryan; distractor for Fromm's needs
Richard Ryan	1985	Self-determination theory with Deci
Heinz Hartmann	1894 to 1970	Father of ego psychology; conflict-free ego sphere and ego autonomy (1939)
Ronald Fairbairn	1889 to 1964	Libido is object-seeking, not pleasure-seeking
Margaret Mahler	1897 to 1985	Separation-individuation
John Bowlby	1907 to 1990	Attachment theory, an offshoot of object relations
Heinz Kohut	1913 to 1981	Self psychology; selfobjects; mirroring, idealising and twinship (alter ego); grandiose self; transmuting internalisation; fragmented self; failed mirroring behind narcissistic pathology
Jacques Lacan	1901 to 1981	The French Freud; return to Freud; mirror stage; unconscious structured like a language; Imaginary, Symbolic and Real
Jean-Étienne Esquirol	1772 to 1840	Modern meaning of hallucination; monomania; distractor for the French Freud
Jules Falret	1877	Folie à deux with Lasègue; distractor for the French Freud
Charles Lasègue	1877	Folie à deux with Falret
Paul Briquet	1859	Briquet's syndrome (now somatisation); distractor for the French Freud
Charles Spielberger	1927 to 2013	State versus trait anxiety; State-Trait Anxiety Inventory (1970)
William Sheldon	1940	Somatotypes, an example of a type approach (Section 4.1)
Henry Odbert	1936	With Allport, culled 17,953 trait words from a dictionary, about 4,500 of them stable traits
Jenny (pseudonym)	1965	Writer of the 300-plus letters in Allport's Letters from Jenny, the classic personal-document study
Philip Vernon	1931	Study of Values with Allport; consistency of expressive behaviour (1933)
Charles Spearman	1863 to 1945	Pioneer of factor analysis; supervised Cattell's PhD (1929)
Christiana Morgan	1935	Co-developer of the Thematic Apperception Test with Murray
Joy Paul Guilford	1897 to 1987	Factor-analytic temperament traits; Guilford-Zimmerman Temperament Survey (1949)
Sybil B. G. Eysenck	EPI 1964, EPQ 1975	Co-author of the Eysenck personality questionnaires
Carl Jung	1875 to 1961	Psychological types behind the MBTI dichotomies (Section 4.4), a distractor for Eysenck and the Big Five
Jeffrey Alan Gray	1934 to 2004	Reinforcement sensitivity theory: behavioural inhibition system (anxiety) and behavioural approach system (impulsivity)
C. Robert Cloninger	1987, extended 1993	Temperament and character model: novelty seeking (dopamine), harm avoidance (serotonin), reward dependence (noradrenaline), persistence
Jerome Kagan	1929 to 2021	Behavioural inhibition: inhibited and uninhibited children
Arnold Buss	1984	EAS temperaments (emotionality, activity, sociability), with Robert Plomin
Robert Plomin	1984; 1987	EAS temperaments with Arnold Buss; the non-shared environment with Denise Daniels
Donald Fiske	1949	First found five factors in ratings on Cattell's variables
Ernest Tupes	1961	Five recurrent factors in US Air Force ratings, with Raymond Christal
Raymond Christal	1961	Five recurrent factors in US Air Force ratings, with Ernest Tupes
Warren Norman	1963	Replicated the five factors
Lewis Goldberg	1981	Coined the label Big Five
William James	1842 to 1910	Character 'set like plaster' by about 30 (1890), a phrase borrowed for adult trait stability
Lewis Terman	1877 to 1956	Study of gifted children from 1921, the data in which conscientious children lived longer
Howard Friedman	1993	Showed that childhood conscientiousness predicted longevity in Terman's sample
Murray Barrick	1991	Meta-analysis with Michael Mount: conscientiousness predicts job performance in every occupational group

Name	When	Examined for
Michael Mount	1991	Meta-analysis with Murray Barrick: conscientiousness predicts job performance
Michael Ashton	2004	HEXACO six-factor model, with Kibeom Lee
Kibeom Lee	2004	HEXACO six-factor model, with Michael Ashton
Delroy Paulhus	2002; 2013	Dark Triad with Kevin Williams; everyday sadism, the fourth trait of the Dark Tetrad
Kevin Williams	2002	Dark Triad (narcissism, Machiavellianism, psychopathy), with Delroy Paulhus
Thomas Bouchard	1990	Minnesota Study of Twins Reared Apart: heritability of personality about 0.40 to 0.50
Denise Daniels	1987	Non-shared environment, with Robert Plomin
Sandra Scarr	1983	Passive, evocative and active gene-environment correlation, with Kathleen McCartney
Kathleen McCartney	1983	Passive, evocative and active gene-environment correlation, with Sandra Scarr
David Buss	born 1953	Evolutionary personality psychology: traits as adaptive individual differences
Hugh Hartshorne	1928	Honesty studies with Mark May: honesty looked situation-specific
Mark May	1928	Honesty studies with Hugh Hartshorne
Seymour Epstein	1979	Aggregation: behaviour averaged over many occasions is stable
Norman Endler	1976	Interactionism, with David Magnusson
David Magnusson	1976	Interactionism, with Norman Endler
Charlotte Buhler	1893 to 1974	Co-founder of humanistic psychology; goals across the life course
Kurt Goldstein	1878 to 1965	Coined the term self-actualisation (The Organism, German 1934, English 1939)
William Stephenson	1953	Devised the Q-sort (The Study of Behavior, 1953), later used by Rogers
J. M. Butler	1954	With Haigh, Q-sort study of 25 clients: average self-ideal correlation .01 before therapy, +.31 after
G. V. Haigh	1954	With Butler, Q-sort study of self-ideal congruence before and after client-centred therapy
Everett Shostrom	1964	Personal Orientation Inventory (POI), a measure of self-actualisation
Abraham Lincoln	1809 to 1865	One of Maslow's historical self-actualisers
Thomas Jefferson	1743 to 1826	One of Maslow's historical self-actualisers
Albert Einstein	1879 to 1955	One of Maslow's self-actualisers
Eleanor Roosevelt	1884 to 1962	One of Maslow's self-actualisers
Soren Kierkegaard	1813 to 1855	Danish philosopher at the root of existentialism; the anxiety of choice
Martin Heidegger	1889 to 1976	Being and Time (1927); Dasein, being-in-the-world
Jean-Paul Sartre	1905 to 1980	Existence precedes essence
Ludwig Binswanger	1881 to 1966	Daseinsanalysis; Umwelt, Mitwelt and Eigenwelt
Medard Boss	1903 to 1990	Daseinsanalysis
Viktor Emil Frankl	1905 to 1997	Man's Search for Meaning (German original 1946); will to meaning, existential vacuum, noogenic neurosis; logotherapy
Irvin Yalom	born 1931	Four ultimate concerns: death, freedom, isolation, meaninglessness
Fritz Perls	1893 to 1970	Gestalt therapy, a wrong option for noogenic anxiety
Ivan Petrovich Pavlov	1849 to 1936	Respondent (classical) conditioning, cited as the example of elicited behaviour
John Dollard	1900 to 1980	Early social learning theory with Miller (1941, 1950); frustration-aggression hypothesis (1939)
Neal Elgar Miller	1909 to 2002	Early social learning with Dollard; fear as a learned drive; approach-avoidance conflict
Edward B. Blanchard	1969	Co-author of the snake-phobia participant modelling study with Bandura and Ritter
B. Ritter	1969	Co-author of the snake-phobia participant modelling study with Bandura and Blanchard

Name	When	Examined for
Richard Walters	1963	Co-author with Bandura of Social Learning and Personality Development: later social learning
Robert Trivers	1971	Reciprocal altruism, a distractor for reciprocal determinism
Aaron Beck	1921 to 2021	Cognitive therapy and schemas; not the author of self-efficacy or fixed-role therapy
Julian Bernard Rotter	1916 to 2014	Social learning theory (1954): behaviour potential, expectancy, reinforcement value; locus of control and the I-E Scale (1966)
Hanna Levenson	1973	Three-part locus of control (IPC) scale: internality, powerful others, chance
Yuichi Shoda	1995	Co-author of the cognitive-affective personality system (CAPS) with Mischel
George Alexander Kelly	1905 to 1967	Personal construct theory (1955); Role Construct Repertory Test; fixed-role therapy
Harrison Gough	1921 to 2016	California Psychological Inventory (CPI)
Herman Witkin	1916 to 1979	Field dependence-independence; rod-and-frame test; Embedded Figures Test
Charles Horton Cooley	1864 to 1929	Looking-glass self (Human Nature and the Social Order, 1902)
George Herbert Mead	1863 to 1931	The I and the me, play and game stages, generalised other (Mind, Self and Society, 1934)
Leon Festinger	1919 to 1989	Social comparison theory (1954), a looking-glass distractor (Chapter 3)
Morris Rosenberg	1922 to 1992	Rosenberg Self-Esteem Scale (1965)
Hazel Markus	Key year 1977	Self-schemas; possible selves with Paula Nurius (1986)
Paula Nurius	Key year 1986	Possible selves, with Hazel Markus
E. Tory Higgins	Key year 1987	Self-discrepancy theory: actual, ideal and ought selves; dejection versus agitation
Mark Snyder	Key year 1974	Self-monitoring
Martin E. P. Seligman	Born 1942	Explanatory style and the three Ps (Learned Optimism, 1990); character strengths and virtues (2004)
Christopher Peterson	1950 to 2012	Attributional Style Questionnaire (1982); Character Strengths and Virtues with Seligman (2004)
Ed Diener	1946 to 2021	Subjective well-being; Satisfaction With Life Scale (1985)
Carol Ryff	Key year 1989	Six dimensions of psychological well-being
Angela Duckworth	Born 1970	Grit, perseverance and passion for long-term goals (2007)
Kapila	Ancient, date uncertain	Traditional founder of Samkhya, the main source of the triguna theory
Patanjali	Ancient, date uncertain	Yoga Sutras; the eight limbs, from yama to samadhi
David Wolf	Key year 1998	Vedic Personality Inventory (1998), measuring sattva, rajas and tamas
Charaka	Ancient, date uncertain	Charaka Samhita; the tridosha theory of vata, pitta and kapha

Key terms

personality, Temperament, Character, trait, type, Disposition, Habit, Attitude, personality theory, historical determinism, interactionism, clinical method, experimental method, correlational method, factor analysis, Nomothetic, idiographic, humours, Sanguine, Choleric, Melancholic, Phlegmatic, Physiognomy, criminal stigmata, Phrenology, Graphology, pyknic, asthenic, endomorphy, mesomorphy, ectomorphy, somatotype, Viscerotonia, Somatotonia, Cerebrotonia, psychoanalysis, hysteria, seduction theory, psychodynamic perspective, free association, manifest content, latent content, parapraxis, instincts, life instincts, libido, death instinct, Cathexis, anticathexis, psychic determinism, topographic model, preconscious, unconscious, id, pleasure principle, primary process thought, ego, reality principle, secondary process thought, superego, conscience, ego-ideal, Reality anxiety, Neurotic anxiety, Moral anxiety, psychosexual stages, Oedipus

complex, castration anxiety, penis envy, Electra complex, defence mechanisms, coping, repression, primal repression, repression proper, displacement, projection, sublimation, rationalisation, reaction formation, isolation of affect, intellectualisation, undoing, identification, identification with the aggressor, introjection, regression, dissociation, conversion, splitting, projective identification, projective hypothesis, empathy, denial, delusional projection, psychotic denial, distortion, schizoid fantasy, hypochondriasis, altruism, humour, anticipation, suppression, neo-Freudians, analytical psychology, principle of opposites, principle of equivalence, principle of entropy, personal unconscious, complex, word association test, collective unconscious, archetypes, persona, shadow, anima, animus, self, mandala, extraversion, introversion, rational functions, irrational functions, inferior function, individuation, Synchronicity, amplification, individual psychology, organ inferiority, compensation, inferiority complex, superiority complex, striving for superiority, fictional finalism, style of life, creative power of the self, social interest, early recollections, basic anxiety, neurotic needs, neurotic trends, idealised self-image, tyranny of the shoulds, womb envy, Momism, mechanisms of escape, existential needs, character orientations, interpersonal theory, self-system, security operations, personifications, prototaxic, parataxic, syntaxic, Ego psychology, conflict-free ego sphere, epigenetic principle, Object relations, part objects, paranoid-schizoid position, depressive position, good-enough mother, transitional object, true self, borderline personality organisation, self psychology, selfobjects, grandiose self, transmuting internalisation, mirror stage, trait approach, Situationism, lexical hypothesis, nomothetic approach, idiographic approach, common traits, personal dispositions, cardinal trait, Central traits, Secondary traits, morphogenic, personal-document technique, Study of Values, functional autonomy, proprium, open system, Surface traits, Source traits, constitutional traits, environmental-mold traits, dynamic traits, Ergs, Sentiments, subsidiation, dynamic lattice, L-data, Q-data, T-data, P-technique, Sixteen Personality Factor Questionnaire, global factors, specification equation, hierarchical model, PEN model, neuroticism, psychoticism, ascending reticular activating system, cortical arousal, visceral brain, Eysenck Personality Inventory, Eysenck Personality Questionnaire, reinforcement sensitivity theory, behavioural inhibition, behavioural inhibition system, behavioural approach system, Big Five, Five-Factor Model, NEO-PI-R, five-factor theory, basic tendencies, characteristic adaptations, maturity principle, HEXACO, Dark Triad, Behaviour genetics, heritability, nonshared environment, gene-environment correlation, gene-environment interaction, personality coefficient, person-situation debate, aggregation, strong situation, humanistic approach, actualising tendency, organismic valuing process, phenomenal field, self-concept, ideal self, need for positive regard, Unconditional positive regard, conditions of worth, incongruence, fully functioning person, Q-sort, deficiency needs, metamotivation, metaneeds, metapathology, Self-actualisation, peak experience, plateau experience, Jonah complex, Personal Orientation Inventory, existence precedes essence, Dasein, Daseinsanalysis, Umwelt, Mitwelt, Eigenwelt, Normal anxiety, Intentionality, daimonic, will to meaning, existential vacuum, noogenic neurosis, radical behaviourism, Respondent behaviour, Operant behaviour, Shaping, Superstitious behaviour, Self-control, drive, cue, learned drive, approach-avoidance conflict, frustration-aggression hypothesis, social cognitive theory, reciprocal determinism, self-regulation, self-reinforcement, moral disengagement, self-efficacy, outcome expectancy, efficacy expectancy, Collective efficacy, Behaviour potential, Expectancy, Reinforcement value, Psychological situation, generalised expectancies, Locus of control, cognitive-affective personality system, behavioural signature, personal constructs, constructive alternativism, fundamental postulate, Role Construct Repertory Test, fixed-role therapy, cognitive style, field dependence-independence, reflection-impulsivity, material self, social self, spiritual self, looking-glass self, symbolic interactionism, play stage, game stage, generalised other, Self-esteem, Self-schema, Possible selves, Self-discrepancy theory, Self-monitoring, limited-domain theory, sensation seeking, explanatory style, Hedonic well-being, subjective well-

being, Eudaimonic well-being, psychological well-being, character strengths, signature strengths, grit, Sportsmanship, triguna, Sattva, Rajas, Tamas, jiva, tridosha, prakriti, Vata, Pitta, Kapha

From the past papers

1. Biologically based characteristic way of reacting is known as (JEMAS 2025)

   (a) Disposition (b) Character (c) Trait (d) Temperament

   Answer: (d) Temperament. Temperament is the biologically based, early-appearing emotional style. Character is personality evaluated morally and learned, a trait is a descriptive dimension not defined by its biological origin, and disposition is too loose a term.

2. Sheldon's theory that people with different body types have different personalities has been: (RML 2020)

   (a) supported by research (b) discredited (c) shown to be accurate for thin people but not overweight people (d) shown to be accurate for women but not for men

   Answer: (b) discredited. Sheldon rated physique and temperament himself, so his expectations inflated the correlations, and independent studies found only weak links. It holds for no subgroup, and his subjects were men.

3. Physiognomy, an early approach to personality study, was based on : (IHBAS 2025)

   (a) Brain localization of functions (b) Dreams and unconscious motives (c) Handwriting style analysis (d) Facial features as indicators of character

   Answer: (d) Facial features as indicators of character. Brain localisation read from skull bumps is phrenology (Gall), handwriting analysis is graphology, and dreams as clues to unconscious motives belong to psychoanalysis.

4. According to Freud, fixation at which stage of development is associated with sarcasm, criticalness, overeating, and nail-biting in adults? (RML 2020)

   (a) Oral (b) Anal (c) Phallic (d) Latency

   Answer: (a) Oral. Overeating and nail-biting are oral-incorporative habits, and sarcasm and criticalness are oral-aggressive (biting) traits; both come from problems of feeding and weaning. Anal fixation gives orderliness, stinginess and obstinacy, or messiness and cruelty; latency is not a fixation point.

5. According to Freud, which term refers to the part of the personality that mediates between desires and social rules? (IHBAS 2024)

   (a) Id (b) Superego (c) Ego (d) Self

   Answer: (c) Ego. Working on the reality principle, the ego is the executive that mediates between the id's desires, the superego and reality or social rules. The id is pure desire (pleasure principle), the superego is internalised morality, and the self is not one of Freud's structures.

6. In psychoanalytic theory, what term describes the mental representation of psychic energy being invested or attached to a particular object, person, or idea? (Real PYQ 2025)

   (a) Libido (b) Cathexis (c) Fixation (d) Sublimation

   Answer: (b) Cathexis. Cathexis is the investment of psychic energy in an object, person or idea. Libido is the energy itself, fixation is libido left stuck at a psychosexual stage, and sublimation channels it into socially valued activity.

7. A woman harassed by her boss at work, initiates an argument with her husband. She is using which of the following ego's defense mechanisms: (RML 2024)

   (a) Projection (b) Identification (c) Sublimation (d) Displacement

   Answer: (d) Displacement. Anger at the boss, who could punish her, is redirected onto a safer target, her husband. Projection would attribute her anger to someone else, identification would take on another's qualities, and sublimation would channel the anger into a socially valued activity.

8. Which type of defence mechanism is dissociation? (NIMHANS 2026 recall)

   (a) Mature (b) Immature (c) Neurotic (d) Conscious

   Answer: (c) Neurotic. Vaillant (1977) placed dissociation at Level III with intellectualisation, isolation, repression, displacement and reaction formation. It is not immature (projection, acting out and the like), and defences are not conscious as a class; only suppression is deliberate.

9. Which of the following defense mechanisms is not involved in obsessive-compulsive disorder? (IHBAS 2020)

   (a) Reaction formation (b) Repression (c) Isolation (d) Undoing

   Answer: (b) Repression. In obsessional neurosis the disturbing idea stays conscious: its feeling is isolated, rituals undo it and reaction formation turns it into its opposite. Repression, which pushes the idea itself out of awareness, is the hallmark of hysteria.

10. Who believed that the universe of the collective unconscious could be accounted for by evolutionary theory, through the similarity of brain structure evident in all human races? (Amity 2023)

    (a) Freud (b) Jung (c) Horney (d) None of these

    Answer: (b) Jung. The collective unconscious is Jung's inherited, universal layer of the psyche, and he explained it by the same brain structure in all human races. Freud's unconscious was personal, and Horney stressed social and cultural forces.

11. The basic assumption of Adler's individual psychology is (JEMAS 2025)

    (a) human behaviour is purposeful and goal directed. (b) an optimistic view of human life. (c) human beings are social beings who could be understood in terms of their relationship with others. (d) development is a lifelong process and ego identity is granted a central place in the process.

    Answer: (a) Human behaviour is purposeful and goal directed. Adler explained behaviour by future goals that pull us forward. Optimism and the social emphasis are features of his view, not its basic assumption, and lifelong development with ego identity at the centre is Erikson's.

12. According to Adler, middle children tend to be: (RML 2022)

    (a) Overachieving (b) Competitive (c) Resentful of the freedom of the older child (d) Filled with feelings of inferiority

    Answer: (b) Competitive. With a pacesetter always ahead, the middle (second) child behaves as if in a race: ambitious and competitive. Such a child may also give up and underachieve, and inferiority feelings are universal, not the mark of the middle child.

13. Karen Horney emphasized personality development as influenced primarily by (JEMAS 2025)

    (a) Social and cultural factors. (b) Biological instincts exclusively. (c) Unconscious sexual conflicts exclusively. (d) Classical conditioning only.

    Answer: (a) Social and cultural factors. Horney broke with Freud's instinct theory: basic anxiety grows from parental indifference and the culture around the child. Instincts and sexual conflicts are Freud's emphasis, and conditioning belongs to behaviourism.

14. In paranoia, the ego's defense mechanism as given by Melaine Klein is seen as: (RML 2024)

    (a) Reaction formation (b) Splitting (c) Suppression (d) Projection

    Answer: (b) Splitting. Klein's paranoid-schizoid position rests on splitting good from bad objects; projection then puts the bad part outside. Projection is Freud's classic mechanism of paranoia, while reaction formation and suppression are unrelated defences.

15. A client tells you that they are "no one" and feels "empty inside." They report chronic fears of abandonment and have a history of intense, unstable relationships. According to psychoanalytic object relations theory, this pathology of the self is most likely a result of: (Real PYQ 2024)

    (a) Fixation at the oral stage of psychosexual development. (b) A failure in parental mirroring and idealizing, leading to a fragmented self. (c) Unresolved Oedipal complex and castration anxiety. (d) Harsh superego demands leading to constant guilt.

    Answer: (b) A failure in parental mirroring and idealizing, leading to a fragmented self. This is Kohut's self psychology: unmet selfobject needs leave an empty, fragmented self. Oral fixation, the Oedipus complex and superego guilt are classical Freudian conflict explanations, not deficits of the self.

16. Cattlell (sic) distinguished between which two types of traits? (JEMAS 2025)

    (a) Openness and Conscientiousness (b) Cardinal and central traits (c) Surface traits and Source traits (d) Id and Ego

    Answer: (c) Surface traits and source traits. Surface traits are clusters of correlated behaviours; source traits are the stable underlying factors that factor analysis reveals. Cardinal and central traits are Allport's, openness and conscientiousness are Big Five factors, and id and ego are Freud's.

17. Allport believed that traits could be organised into three levels: (RML 2020)

    (a) primary, secondary, tertiary (b) cognitive, emotional, physiological (c) id, ego, superego (d) cardinal, central, secondary

    Answer: (d) cardinal, central, secondary. A cardinal trait is a ruling passion found in few people, central traits are the five to ten that describe a person, and secondary traits are minor and inconsistent. Primary-secondary-tertiary is invented, and id-ego-superego is Freud's structure.

18. According to Raymond Cattell's classification of personality data sources, which of the following is NOT recognized as a legitimate data type? (Real PYQ 2025)

    (a) Q-data (Questionnaire data) (b) L-data (Life record data) (c) T-data (Objective test data) (d) P-data (Projective data)

    Answer: (d) P-data. Cattell used exactly three sources: L-data (life records), Q-data (questionnaires) and T-data (objective tests). He classed hard-to-fake tasks such as inkblots as T-data, and his P-technique is a research design, not a data type.

19. Hans Eysenck proposed which three dimension of personality? (JEMAS 2025)

    (a) Extraversion-Introversion, Neuroticism-Stability, Psychoticism (b) Extraversion-Introversion, Sensing-Intuition, Thinking-Feeling (c) Openness, Extraversion, Judging, Perceiving (d) Extraversion-Introversion, Neuroticism-Stability, Psychoticism-Conscientiousness

    Answer: (a) Extraversion-Introversion, Neuroticism-Stability, Psychoticism. These are Eysenck's PEN superfactors. Sensing-intuition, thinking-feeling, judging and perceiving are Myers-Briggs dichotomies from Jung's types, and conscientiousness is a Big Five factor, not part of Eysenck's model.

20. Which personality trait is linked to better health outcomes? (JEMAS 2025)

    (a) Low Agreeableness and Low Neuroticism (b) Neuroticism exclusively (c) Introversion exclusively (d) High conscientiousness and Low neuroticism

    Answer: (d) High conscientiousness and low neuroticism. Conscientious people look after their health and lived longer in Terman's sample, and low neuroticism means less distress. Low agreeableness (hostility) predicts heart disease, high neuroticism harms health, and introversion alone has no health advantage.

21. Mischel's critique of personality traits mainly emphasized that : (IHBAS 2025)

    (a) Situational factors often override traits in determining behavior (b) Personality is entirely biologically based (c) Traits are more important than situations in predicting behavior (d) Traits cannot be measured reliably

    Answer: (a) Situational factors often override traits in determining behaviour. In Personality and Assessment (1968) Mischel found trait-behaviour correlations rarely above about 0.30. He did not claim personality is biological, (c) is the opposite of his point, and trait questionnaires were reliable; they simply predicted single acts poorly.

22. In existential psychology, Dasein encompasses three fundamental psychological dimensions of human existence. Which of the following correctly identifies these three modes of being-in-the-world? (Real PYQ 2025)

    (a) Id, Ego, and Superego (b) Umwelt, Mitwelt, and Eigenwelt (c) Conscious, Preconscious, and Unconscious (d) Sensorimotor, Preoperational, and Concrete Operational

    Answer: (b) Umwelt, Mitwelt, and Eigenwelt. Binswanger and May describe the biological and physical world (Umwelt), the world of relationships (Mitwelt) and one's relation to oneself (Eigenwelt). Id, ego and superego is Freud's structural model, conscious to unconscious his topographic model, and the last option Piaget's stages.

23. Noogenic anxiety is described under: (IHBAS 2020)

    (a) Humanistic School (b) Gestalt school (c) Existential school (d) Psychodynamic school

    Answer: (c) Existential school. Viktor Frankl's noogenic neurosis and anxiety arise from a frustrated will to meaning, an existential problem, not from drive conflict (psychodynamic), blocked growth (humanistic) or Gestalt ideas.

24. A split between the organismic valuing process and a self-concept burdened by conditions of worth which results in feelings of tension and confusion: (Amity 2023)

    (a) Conflict (b) Incongruence (c) Introjection (d) Actualizing tendency

    Answer: (b) Incongruence. Rogers's term for the gap between experience and a self-concept shaped by conditions of worth. Introjection is how those conditions are taken in, conflict is a general term, and the actualising tendency is the growth force that incongruence blocks.

25. Which concept is central to Bandura's Personality Theory? (JEMAS 2025)

    (a) Self-efficacy (b) Inferiority complex (c) Collective unconscious (d) Congruence

    Answer: (a) Self-efficacy. Self-efficacy, the belief that one can organise and execute the actions needed for a specific goal, is the core of Bandura's social cognitive theory. The inferiority complex is Adler's, the collective unconscious Jung's and congruence Rogers's.

26. Which of the following best describes the concept of 'reciprocal determinism' as proposed by Albert Bandura? (RML 2025)

    (a) Individual behavior is solely controlled by environmental factors. (b) Cognitive processes influence both behavior and the environment in a bidirectional manner. (c) Personality develops strictly from genetic inheritance and biological predispositions. (d) Social reinforcement has no impact on self-concept formation.

    Answer: (b). In reciprocal determinism person (cognitive) factors, behaviour and environment influence one another in both directions. Environment alone is Skinner's position, genes alone a biological one, and Bandura never denied the role of social reinforcement.

27. Julian Rotter's concept of locus of control describes whether individuals believe that their outcomes are controlled by : (IHBAS 2024)

    (a) External forces or internal efforts (b) Their cognitive schemas (c) Genetic predisposition (d) Social status

    Answer: (a) External forces or internal efforts. Locus of control is a generalised expectancy that outcomes depend either on one's own effort (internal) or on luck, fate, chance and powerful others (external). Cognitive schemas belong to Beck; genes and social status are not part of the construct.

28. The "looking-glass self," a concept introduced by Charles Horton Cooley, posits that our self-concept is primarily formed by: (Real PYQ 2024)

    (a) Our innate, unconscious drives. (b) Comparing ourselves to others who are similar. (c) Imagining how we appear to others and how they judge us. (d) Successfully resolving psychosocial crises at each life stage.

    Answer: (c) Imagining how we appear to others and how they judge us. Cooley's three steps are imagined appearance, imagined judgement and a self-feeling such as pride or shame. Unconscious drives are Freud's, comparison with similar others is Festinger's social comparison, and psychosocial crises are Erikson's.

29. Which of the following best describes the concept of "self-esteem" in contrast to "self-concept"? (Real PYQ 2024)

    (a) Self-esteem is the cognitive appraisal of one's traits, while self-concept is the evaluative and emotional component. (b) Self-esteem is the evaluative and affective component of the self (how one feels about oneself), while self-concept is the cognitive description of who one is. (c) Self-esteem and self-concept are identical and can be used interchangeably. (d) Self-esteem is a collective term for the self in relation to others, while self-concept is the individual self.

    Answer: (b) Self-esteem is evaluative and affective; self-concept is the cognitive description of who one is. Option (a) reverses the two, they are not identical, and the individual versus collective self is a different distinction altogether.

30. Bavisha lost her job. Since then she is stressed, has stopped eating properly and cannot get herself to do anything. According to yogic philosophy, the main issue she is dealing with is: (NIMHANS 2026 recall)

    (a) Yama alone (b) Tamas alone (c) Both yama and tamas (d) Rajas alone

    Answer: (b) Tamas alone. Not eating, withdrawal and the inability to act are the inertia and lethargy of tamas. Rajas would show as restless over-activity and agitation, and yama is not a mental state at all but the first limb of Patanjali's yoga, the ethical restraints.

High-yield sheet

The facts this chapter's questions ask most, most-asked first. Last read before the exam.

1 The **id** works on the **pleasure principle** and the **ego** on the **reality principle**; the ego is the **executive** that mediates between the id's desires, the superego and reality or social rules. ASKED 7 TIMES

2 The **Big Five** are Openness, Conscientiousness, Extraversion, Agreeableness and Neuroticism (OCEAN); psychoticism, intelligence, intuition, sensing, anxiety (a facet of N) and sense of humour are not Big Five factors. ASKED 7 TIMES

3 The three gunas are **sattva** (clarity, calm, balance, knowledge), **rajas** (activity, passion, desire, restlessness) and **tamas** (inertia, darkness, lethargy, ignorance); a person sunk in withdrawn inertia who cannot act shows tamas alone. ASKED 7 TIMES

4 **Free association** (the fundamental rule: on the couch, say whatever comes to mind) is the central technique of psychoanalysis, and Freudian slips (**parapraxes**), mistakes that reveal the unconscious, are analysed through it. ASKED 6 TIMES

5 **Repression** is the primary defence: the involuntary, unconscious blocking of threatening memories, wishes and feelings from awareness. **Suppression** is its conscious, deliberate counterpart and is mature in Vaillant's scheme. ASKED 6 TIMES

6 **Vaillant (1977)** ranked defences in four levels: psychotic (delusional projection, psychotic denial, distortion), immature (projection, schizoid fantasy, hypochondriasis, passive aggression, acting out), neurotic (intellectualisation, isolation, repression, displacement, reaction formation, dissociation) and mature (altruism, anticipation, humour, sublimation, suppression). ASKED 6 TIMES

7 **Displacement** redirects feelings from a threatening target to a safer one (the woman harassed by her boss who picks a fight with her husband); it is the classic defence of phobias. ASKED 6 TIMES

8 **Alfred Adler** coined the **inferiority complex** (helplessness, hopelessness, lack of self-worth; from organ inferiority, pampering or neglect) and paired it with the striving for superiority; **organ inferiority** (1907) is also his. ASKED 6 TIMES

9 Incestuous fantasies, the **Oedipus complex** and the emergence of the superego belong to the **phallic stage**; the girls' counterpart was named the **Electra complex** by Jung in 1913. ASKED 5 TIMES

10 **Projection** attributes one's own unacceptable thoughts and impulses to others; in its broad sense it is any distorting effect of a person's inner world on his perception of the outer world. ASKED 5 TIMES

11 **Carl Jung**'s **collective unconscious** is the deepest, inherited, universal layer of the psyche, containing the archetypes; with the personal unconscious it forms his two levels of the unconscious. ASKED 5 TIMES

12 The **Sixteen Personality Factor Questionnaire (16PF)** was developed by **Raymond B. Cattell** (1949), who used factor analysis to derive 16 primary source traits; not Eysenck, Allport, Sheldon, Murray or Costa and McCrae. ASKED 5 TIMES

13 Smoking, drinking (alcohol dependence), overeating, nail-biting, sarcasm and criticalness indicate **oral** fixation, which arises from problems of feeding and weaning. ASKED 4 TIMES

14 The **id** is the wholly unconscious reservoir of unorganised instinctual drives; impulsive and irrational, it alone seeks immediate gratification of primitive needs ("Just do it"), whereas the superego says "Don't do it". ASKED 4 TIMES

15 **Sublimation** rechannels sexual or aggressive energy into a socially valued activity such as kickboxing, competitive sport or art; Freud's healthiest defence and mature in Vaillant's scheme. ASKED 4 TIMES

16 **Rationalisation** (Ernest Jones, 1908) is a logical, socially acceptable reason invented after the act for behaviour driven by unconscious or unacceptable motives. ASKED 4 TIMES

17 In Jung's theory the **anima** is the feminine side of a man and the **animus** the unconscious masculine side of a woman; the persona is the social mask and the shadow the dark, disowned side. ASKED 4 TIMES

18 Cattell's three sources of personality data are **L-data** (life records and observers' ratings, such as school performance and criminal records), **Q-data** (self-report questionnaires) and **T-data** (objective tests whose purpose is hidden); P-data and Z-data do not exist. ASKED 4 TIMES

19 The **NEO-PI-R** (Costa and McCrae, 1992) is a trait-approach inventory of the Five-Factor Model: 240 items, five domains with six facets each, 30 facets in all. ASKED 4 TIMES

20 The **humanistic** perspective (Maslow and Rogers) emphasises free will, personal growth and an innate tendency in everyone to grow toward their maximum potential, not unconscious conflict, conditioned responses or statistical norms. ASKED 4 TIMES

21 **Self-efficacy** (Bandura, 1977), a person's belief in his capability to organise and execute the actions needed to achieve a specific goal, is the central concept of **Albert Bandura**'s social cognitive theory and its key factor in motivation. ASKED 4 TIMES

22 The basic assumption of Adler's **individual psychology** is that human behaviour is purposeful and goal-directed; people are motivated primarily by social urges. ASKED 3 TIMES

23 Jung's two attitudes are **extraversion and introversion** (thinking, feeling, sensing and intuiting are his functions), and the **Myers-Briggs Type Indicator** is built directly on his psychological types. ASKED 3 TIMES

24 **Karen Horney** held that personality is shaped primarily by social and cultural factors; her **basic anxiety** is the all-pervading feeling of being lonely and helpless in a hostile world, the foundation of neurosis. ASKED 3 TIMES

25 **Dasein** (Heidegger's being-in-the-world, the basic unity of person and environment) is existential; its three modes are the **Umwelt** (biological and physical world), **Mitwelt** (relationships with others) and **Eigenwelt** (relation to oneself). ASKED 3 TIMES

26 Existential psychology holds that existence precedes essence, so meaning is never fixed; Frankl's **noogenic neurosis and anxiety** belong to the existential school, and uncovering unconscious fears and desires is psychoanalytic, not existential. ASKED 3 TIMES

27 **Reciprocal determinism** (Bandura): person (cognitive) factors, behaviour and environment influence one another in both directions; it is central to social cognitive theory and is not reciprocal inhibition (Wolpe) or reciprocal altruism (Trivers). ASKED 3 TIMES

28 **Julian Rotter**'s **locus of control** (I-E Scale, 1966) is either internal (outcomes depend on one's own effort and preparation) or external (luck, fate, chance, powerful others); there is no neutral or situational locus. ASKED 3 TIMES

29 Cooley's **looking-glass self** (1902): we imagine how we appear to others, imagine their judgement, and develop a self-feeling such as pride or shame. ASKED 3 TIMES

30 The gunas belong to **prakriti**, not purusha, and come mainly from the **Samkhya** school (Kapila) with the Bhagavad Gita (chapters 14, 17 and 18). ASKED 3 TIMES

CHAPTER 5

Physiological Psychology

IN THIS CHAPTER

5.1 ● Biopsychology and the Organisation of the Nervous System: Explain behaviour from the biological perspective, name the six divisions of biopsychology, and map the nervous system: central and peripheral, somatic and autonomic (sympathetic and parasympathetic), the reflex arc, spinal and cranial nerves, and the meninges, cerebrospinal fluid and blood-brain barrier that protect the brain.

5.2 ●● The Brain: Brainstem, Diencephalon and Cerebral Cortex: Name the brain's five developmental divisions and state where the medulla, pons, cerebellum, midbrain, reticular formation, thalamus, hypothalamus, limbic system, basal ganglia, cortical lobes, Brodmann areas and fibre systems lie and what each does.

5.3 ●● Neurons, Neural Conduction, Synapses and Neurotransmitters: Describe the neuron and the glial cells, explain the resting potential, the all-or-none action potential, refractory periods and saltatory conduction, follow synaptic transmission from Ca2+-triggered release to receptor binding and deactivation, and name the main neurotransmitters with their receptors, source nuclei and functions.

5.4 ●● Drugs, Reward and Psychopharmacology: Explain how psychoactive drugs reach and act on the brain, how tolerance, dependence and addiction develop, how the mesolimbic reward pathway and the four dopamine pathways work, and how antipsychotics, antidepressants, mood stabilisers, anxiolytics and drugs for alcohol dependence act and what side effects they cause.

5.5 ●● Research Methods, Behaviour Genetics and the Developing Brain: Choose the right method for studying brain structure, activity or timing, reason through Mendelian and behaviour-genetic designs, identify the main single-gene and chromosomal disorders, and trace how the nervous system develops, prunes itself and stays plastic.

5.6 ● Sensory Pathways and the Motor System: Trace each sense from receptor to cortex, locate a lesion from the visual field it destroys, and follow a movement down from association cortex through the motor cortex, descending tracts, basal ganglia, cerebellum and spinal circuits to the muscles.

5.7 ●●● The Endocrine System, Stress and the Emotional Brain: Explain how the hypothalamus, pituitary and other endocrine glands work, trace the two stress systems with their feedback and allostasis, and identify the limbic, amygdala and Klüver-Bucy findings that underlie emotion.

5.8 ●● Sleep, Biological Rhythms, Hunger, Thirst and Sex: Explain how the suprachiasmatic clock, the sleep and arousal switches of the brainstem and hypothalamus, and the hypothalamic circuits of hunger, thirst and sex work, and answer the classic lesion, EEG and hormone questions on them.

5.9 ●●● The Biology of Learning, Memory and Amnesia: Distinguish anterograde from retrograde amnesia and explain which brain structures (hippocampus, amygdala, diencephalon) and synaptic mechanisms (LTP, LTD, NMDA receptors, dopamine prediction errors) form and store memories, from H.M. to Korsakoff's syndrome.

5.10 ●● Lateralisation, the Split Brain and Language: Explain how the two hemispheres were found to differ (Dax, Broca, the Wada and dichotic listening tests, split-brain studies), list the functions each hemisphere favours, locate Broca's area, Wernicke's area, the arcuate fasciculus and the angular gyrus, trace the Wernicke-Geschwind model and its dual-stream successor, and tell the aphasias and related reading, writing and speech disorders apart by fluency, comprehension, repetition and lesion site.

5.11 ●● Brain Damage, Neurodegeneration and Neuropsychological Syndromes: Explain how the brain is damaged (tumours, stroke, head injury, infection, toxins, genes), identify the lesion, transmitter and signs of epilepsy, Parkinson's, Huntington's and Alzheimer's diseases, the other dementias and multiple sclerosis, and recognise agnosia, apraxia, hemispatial neglect and the prefrontal syndromes by their lesion sites.

Weightage. 364 of the 3180 questions mapped in this book (11.4%) test this chapter, 151 of them from real past papers. 2 of its 11 sections are High yield.

Graz, Austria, 1921. In the night before Easter Sunday the pharmacologist **Otto Loewi** woke with an idea, switched on the light and jotted a few notes on a slip of thin paper. At six in the morning he knew he had written down something important, but he could not read his own scrawl. The question behind it was an old one. Physiologists knew that stimulating the **vagus nerve** slows the heart, but how does a nerve tell a muscle what to do? Most assumed the message simply crossed as electricity. Loewi had long wondered whether the nerve might instead release a chemical, but he had never found a way to test it. The next night, at three o'clock, the idea came back. This time he got up, went straight to his laboratory and ran the experiment.

The design was simple enough to fit on that slip of paper. Loewi kept two frog hearts beating in small chambers of salt solution, the first with its vagus nerve attached. He stimulated the vagus, and the first heart slowed. Then he drew off the fluid that had bathed it and gave it to the second heart, which had been isolated without its nerves. The second heart slowed too. Something had passed from the nerve endings into the fluid, and that something, not an electric current, had carried the message. Loewi called it *Vagusstoff*, 'vagus substance'. It was later identified as **acetylcholine**, and in 1936 Loewi shared the Nobel Prize in Physiology or Medicine with the British scientist **Henry Dale** for discoveries about the chemical transmission of nerve impulses.

Loewi's two frog hearts hold this chapter in miniature. A change in what an organ does had a physical cause that could be isolated, carried from one heart to another and named. Physiological psychology studies behaviour in that spirit: through the nerves that carry messages, the cells and chemicals that pass them on, the drugs that interfere with them, and the brain systems behind sensation, movement, emotion, sleep, memory and language.

5.1 Biopsychology and the Organisation of the Nervous System

LOWER YIELD · 21 QS

Loewi's second frog heart slowed because a nerve had released a chemical. To see how such simple events add up to fear, memory and speech, you need a point of view that explains behaviour through the body, and a map of the machinery. This section supplies both; the structures inside the brain follow in section 5.2.

The biological perspective

The **biological perspective**, also called the neurobiological approach, holds that behaviour and mental processes are produced by the body: the brain and nervous system, hormones and genes. Every psychological event, from a memory to a mood, is assumed to have a physical basis in neural activity, so changing the body with a drug, a lesion or a hormone changes the mind.

Three neighbouring approaches (Chapter 1) sharpen the contrast. The *psychophysical* approach relates the physical intensity of a stimulus to the sensation it produces; the *cognitive* approach explains behaviour by mental processes such as perceiving, remembering and thinking; the *neo-behaviouristic* approach explains it by learned stimulus-response connec-

tions, with intervening variables such as habit strength inside the organism. The science built on this view, biopsychology, owes its emergence largely to **Donald Hebb**'s *The Organization of Behavior* (1949), the first comprehensive theory of how thought and memory could arise from brain activity (section 5.9).

> **EXAM TRAP**
>
> Asked which system the biological perspective links behaviour to, answer the **body**: the mind is the cognitive answer, the soul a dualist one and the unconscious the psychodynamic one. Behaviour "governed by the brain and nervous system" names the **neurobiological** approach.

Biopsychology as a neuroscience

Biopsychology is the scientific study of the biology of behaviour. For **John P. J. Pinel** and **Steven J. Barnes** (*Biopsychology*, 11th edition, 2021) it is a neuroscience: biopsychologists bring expertise in behaviour to what neuroanatomy, neurochemistry and neurophysiology discover. Their research falls into six divisions (Table 5.1).

Table 5.1 The six divisions of biopsychology (Pinel).

Division	What it studies	Typical subjects and methods
Physiological psychology	Neural mechanisms of behaviour, by manipulating and recording the brain	Laboratory animals; lesions, stimulation; mostly pure research
Psychopharmacology	Drug effects on brain and behaviour	Animals and humans; often applied
Neuropsychology	Psychological effects of brain damage	Patients; case studies; the most applied
Psychophysiology	Physiology of psychological processes	Humans; non-invasive EEG, heart rate, skin conductance
Cognitive neuroscience	Neural bases of cognition	Humans; brain imaging; the youngest division
Comparative psychology	Behaviour across species: evolution, genetics, adaptiveness	Many species, laboratory and wild

Biopsychologists study both humans, who can report what they experience, and nonhuman animals, whose simpler brains allow comparison across species and experiments that would be unethical in people. In an *experiment* the researcher assigns subjects to conditions and varies one independent variable, so only an experiment shows cause. A **quasi-experimental study** compares groups already exposed to a condition in real life, such as heavy drinkers and abstainers, so confounding variables remain. A *case study* examines one person in depth (Chapter 9).

KEY STUDY: CONVERGING OPERATIONS AND KORSAKOFF'S SYNDROME

Because each method has weaknesses, progress comes from **converging operations**: aiming several approaches at one problem so that each covers the others' weaknesses. Korsakoff's amnesia (section 5.9) was first blamed on alcohol from quasi-experiments with heavy drinkers; case studies of malnourished people who drank little, and experiments in which thiamine-deficient rats developed similar memory loss, then pointed to thiamine (vitamin B1) deficiency.

Behind all this lies the mind-brain problem. **René Descartes** (1596 to 1650) proposed **dualism**: the body, brain included, is a machine obeying natural laws, while a nonphysical mind controls it, the two meeting, he suggested, at the pineal gland. **Monism** holds that only one kind of substance exists; its materialist form, the working view of biopsychology, treats mental events as brain activity.

The divisions of the nervous system ●●

The nervous system is a system of twos (Figure 5.1). The **central nervous system** (CNS), the brain and spinal cord, lies inside the skull and spine; the **peripheral nervous system** (PNS) is all the nerves outside them. The PNS has two divisions. The **somatic nervous system** deals with the outside world: its **afferent** nerves carry sensory signals from skin, eyes and ears to the CNS, and its **efferent** nerves carry commands to the skeletal muscles, which are under voluntary control (afferent arrives, efferent exits). The **autonomic nervous system** (ANS) regulates the internal environment, controlling smooth muscle, cardiac muscle and glands largely involuntarily.

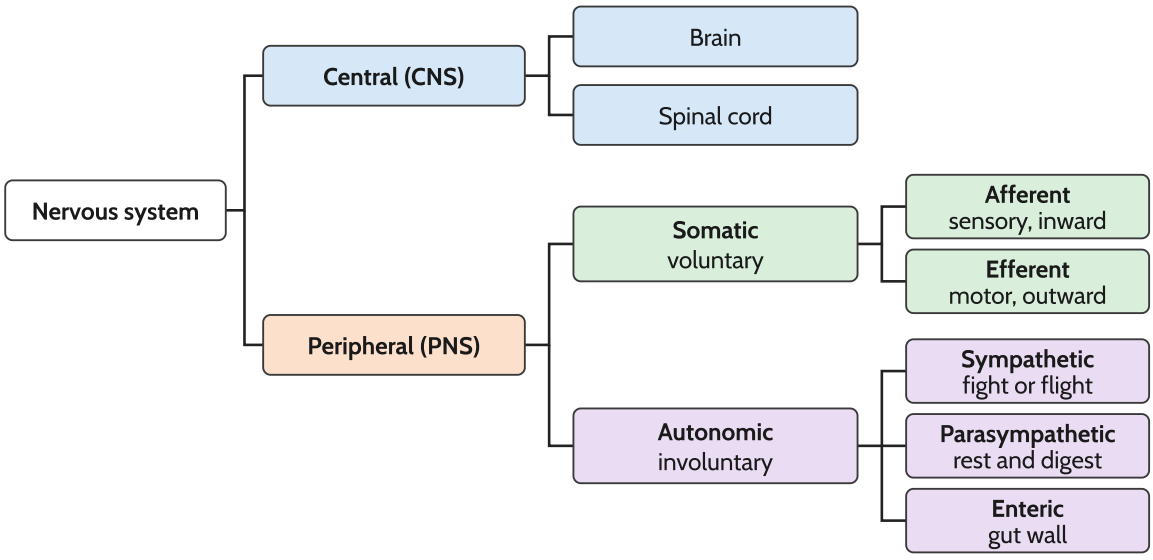

Figure 5.1 Somatic and autonomic are both peripheral; many books count the enteric network of the gut wall as a third autonomic division.

> **EXAM TRAP**
>
> The parts of the ANS are the **sympathetic and parasympathetic** divisions. "Central and peripheral" divide the whole nervous system, "voluntary and involuntary" describe somatic versus autonomic, and "cerebrospinal and cortical" is no division at all. The brain and spinal cord form the **CNS**.

The autonomic nervous system ●●

Each division of the ANS reaches its targets through two neurons: a preganglionic neuron from the CNS synapses in an autonomic ganglion on a postganglionic neuron that runs to the organ.

The **sympathetic division** mobilises and spends energy. **Walter Bradford Cannon** (1871 to 1945) described its emergency role in 1915 as the **fight-or-flight response**: the heart races, pupils and airways widen, blood shifts to the muscles and the liver releases glucose. Its preganglionic neurons leave spinal segments T1 to L2, the **thoracolumbar outflow**; their short fibres end mostly in the sympathetic chain, a string of ganglia beside the spine, from which long postganglionic fibres run to the organs. Acetylcholine is the transmitter at the ganglia, noradrenaline (norepinephrine) at most targets. Some preganglionic fibres go straight to the adrenal medulla, the core of the adrenal gland, which acts like a sympathetic ganglion and releases adrenaline (epinephrine) and some noradrenaline into the blood. This **sympathetic adrenal-medullary system** is the fast arm of the stress response.

The **parasympathetic division** serves rest and digest: it slows the heart, narrows the pupils and promotes digestion, conserving and storing energy. Its preganglionic neurons leave the brainstem in cranial nerves III, VII, IX and X and the spinal cord at S2 to S4, the **craniosacral outflow**. Their long fibres end in ganglia near or inside the organ, and both synapses use acetylcholine. The **vagus nerve** (X) carries most of these fibres, supplying the heart, lungs and most of the gut.

Most organs receive both divisions (*dual antagonistic innervation*), so your heart rate reflects the balance of sympathetic acceleration and vagal braking; sweat glands and most blood vessels are sympathetic only, and in sexual response the two cooperate (Table 5.2). The **enteric nervous system**, a network of neurons in the gut wall, runs local reflexes of gut movement and secretion on its own.

Table 5.2 The sympathetic and parasympathetic divisions compared.

Feature	Sympathetic	Parasympathetic
Outflow	Thoracolumbar (T1 to L2)	Craniosacral (III, VII, IX, X; S2 to S4)
Fibres	Short preganglionic, long postganglionic	Long preganglionic, short postganglionic
Transmitter at target	Noradrenaline (sweat glands: acetylcholine)	Acetylcholine
Heart	Faster, stronger	Slower
Pupils	Dilated	Constricted
Saliva	Reduced (dry mouth)	Increased
Bronchi	Dilated	Constricted
Digestion	Inhibited	Stimulated

Feature	Sympathetic	Parasympathetic
Bladder	Wall relaxed; urine held	Wall contracted; emptied
Sweating, skin conductance	Increased	No effect
Liver glucose	Released into blood	Stored as glycogen
Sexual response	Ejaculation, orgasm	Erection, lubrication

EXAM TRAP

Fight or flight is **sympathetic**; the fullest answer is "sympathetic nervous system plus adrenal medulla". Rest and digest and conserving energy are **parasympathetic**; allocating energy for expenditure is sympathetic. A stem naming the vagus and acetylcholine means parasympathetic, not the sympathetic adrenal-medullary system. The enteric system cannot slow the heart or narrow the pupils, and the pituitary-thyroid axis is endocrine, setting metabolic rate rather than the emergency response (section 5.7).

Grey matter, white matter and the spinal cord ●

Slice fresh brain or spinal cord and you see two tissues. **Grey matter** is mainly neuron cell bodies, dendrites, unmyelinated axons and synapses, where information is processed. **White matter** is mainly myelinated axons, the wiring between areas; myelin is a fatty insulating sheath (section 5.3), and its fat makes the tissue white. A cluster of cell bodies is a nucleus in the CNS but a **ganglion** in the PNS; a bundle of axons is a tract in the CNS but a nerve in the PNS. (The basal ganglia, despite the name, are CNS nuclei.)

In the spinal cord (Figure 5.2) a butterfly-shaped core of grey matter, with dorsal and ventral horns, is surrounded by white columns of ascending and descending tracts: the reverse of the cerebral cortex, which is grey outside. Thirty-one pairs of spinal nerves attach to the cord: 8 cervical, 12 thoracic, 5 lumbar, 5 sacral and 1 coccygeal. Each joins it by two roots. The dorsal root is purely sensory, its cell bodies sitting just outside the cord in the **dorsal root ganglion**; the ventral root is purely motor, its cell bodies lying in the ventral horn. This is the **Bell-Magendie law**, named after **Charles Bell** (1774 to 1842), who reported the motor role of the ventral roots in 1811, and **François Magendie** (1783 to 1855), who demonstrated both halves of the law in 1822.

EXAM TRAP

White matter is **myelinated axons**; cell bodies and unmyelinated axons are grey, and dendrites are never myelinated. Cell bodies in the PNS form **ganglia** (nuclei in the CNS); tracts and nerves are bundles of axons.

Neurons by function and the reflex arc ●●

Neurons fall into three functional classes. Sensory (afferent) neurons carry signals from receptors to the CNS. A **motor neuron** (efferent) carries commands from the brain and spinal cord to muscles and glands. An **interneuron** (connector or association neuron), by far the commonest kind, lies wholly within the CNS and links the two. Muscles and glands are effectors: targets, not neurons.

Together they form the **reflex arc** of an automatic response. Touch a hot pan and the signal runs: receptor in the skin, afferent neuron through the dorsal root, interneuron in the spinal grey matter, efferent neuron through the ventral root, effector muscle that pulls your hand away (Figure 5.2). The arc closes in the spinal cord, so you withdraw before you feel the pain. This withdrawal reflex is *polysynaptic*. The knee-jerk, a stretch reflex, is *monosynaptic*: the sensory neuron synapses directly on the motor neuron, with no interneuron.

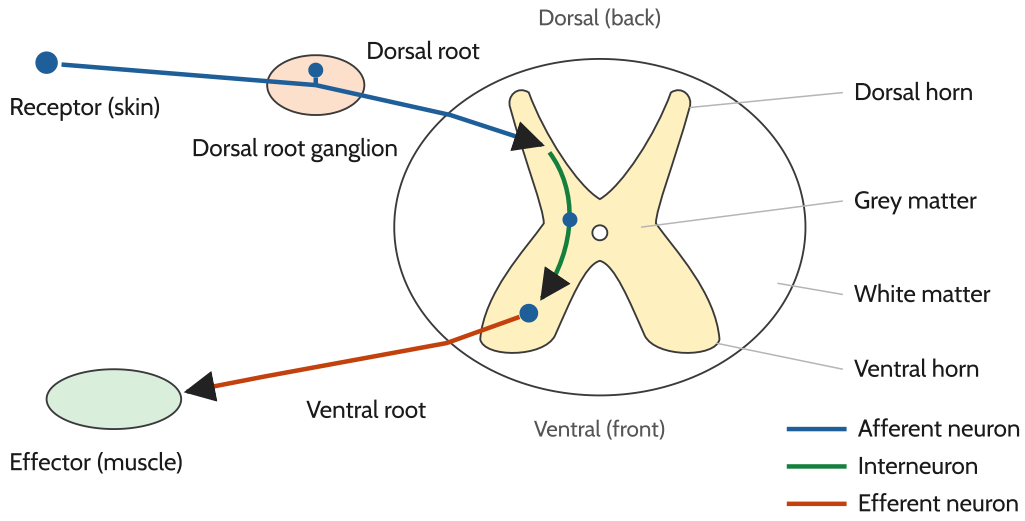

Figure 5.2 The spinal cord in cross section with a three-neuron withdrawal reflex: in by the dorsal root, across by an interneuron in the grey matter, out by the ventral root. The knee-jerk skips the interneuron.

> **EXAM TRAP**
>
> The order is **afferent neuron, interneuron, efferent neuron**; no arc runs afferent to efferent or starts with an efferent. A cell carrying information from the brain to muscles and glands is a **motor neuron**: "effector cell" names the target and "connector neuron" the interneuron.

The twelve cranial nerves

Twelve pairs of **cranial nerves** leave the brain directly, numbered from front to back. They serve mainly the head and neck, except the "wandering" vagus, which reaches the chest and abdomen (Table 5.3). Three are sensory (I, II, VIII), five are motor (III, IV, VI, XI, XII) and four are mixed (V, VII, IX, X). Four carry parasympathetic fibres: III (pupil), VII (tears, saliva), IX (saliva) and X (heart, lungs, gut).

Table 5.3 The twelve cranial nerves.

No.	Name	Type	Main functions
I	Olfactory	Sensory	Smell
II	Optic	Sensory	Vision
III	Oculomotor	Motor	Four of the six eye muscles; lifts eyelid; constricts pupil
IV	Trochlear	Motor	Superior oblique eye muscle

No.	Name	Type	Main functions
V	Trigeminal	Mixed	Facial sensation; chewing
VI	Abducens	Motor	Lateral rectus (turns eye outward)
VII	Facial	Mixed	Facial expression; taste, front of tongue; tears, saliva
VIII	Vestibulocochlear	Sensory	Hearing and balance
IX	Glossopharyngeal	Mixed	Taste, back of tongue; swallowing; saliva
X	Vagus	Mixed	Parasympathetic supply to heart, lungs, gut; throat
XI	Accessory	Motor	Neck and shoulder muscles
XII	Hypoglossal	Motor	Tongue movement

Nerve I is special: smell is the only sense whose main pathway reaches the cortex (the piriform cortex of the medial temporal lobe) without first relaying in the thalamus, as vision, hearing, taste and touch all do (section 5.6).

> REMEMBER
>
> Names: "On Old Olympus' Towering Tops, A Finn And German Viewed Some Hops" (A for acoustic, an old name for VIII). Types: "Some Say Marry Money, But My Brother Says Big Brains Matter More" (S sensory, M motor, B both).

Protecting the CNS ●●

The CNS is encased in bone and wrapped in three **meninges**: from outside in, the tough dura mater, the web-like arachnoid mater and the thin pia mater, which clings to the brain's surface. Between arachnoid and pia lies the subarachnoid space.

Cerebrospinal fluid (CSF), made by the **choroid plexus** (networks of capillaries in the ventricles), fills the brain's four ventricles, the central canal of the spinal cord and the subarachnoid space. It flows from the two lateral ventricles, the largest, to the third ventricle, through the cerebral aqueduct to the fourth, and out into the subarachnoid space, and is absorbed through the arachnoid granulations into the venous sinuses of the dura. CSF cushions the brain, which floats in it. If its flow is blocked, for example at the aqueduct, the ventricles swell and compress the brain: *hydrocephalus*, treated with a draining shunt.

The **blood-brain barrier** exists because the endothelial cells of brain capillaries are sealed together by tight junctions, supported by the end-feet of astrocytes (star-shaped glial cells). Small fat-soluble molecules such as oxygen, alcohol and nicotine pass; glucose is carried across by active transport; large or water-soluble molecules, many toxins and many drugs are kept out (section 5.4). Weak spots include the **area postrema** in the floor of the fourth ventricle, which detects poisons in the blood and triggers vomiting.

> EXAM TRAP
>
> The barrier is **tight junctions plus astrocyte end-feet**: not the pia mater (a meninx), microglia (immune scavengers) or oligodendrocyte myelin (around axons). CSF is made by the **choroid plexus** and absorbed by the arachnoid granulations, and the ventricles hold CSF, not blood, myelin or "synaptic fluid".

Directions and planes

Anterior (rostral) means toward the nose, posterior (caudal) toward the tail. Dorsal means toward the back or the top of the head, ventral toward the belly or the bottom of the head; because humans stand upright, this axis bends at the brainstem, so dorsal is up in the brain but back in the spinal cord. Medial means toward the midline, lateral away from it. Ipsilateral means on the same side, **contralateral** on the opposite side: each hemisphere controls the contralateral hand. A horizontal section is parallel to the ground, a frontal (coronal) section parallel to the face and a sagittal section parallel to the midline; the midsagittal section runs exactly between the hemispheres.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Explaining behaviour through the brain and nervous system is the _____ perspective.
2. Aiming several research approaches at one problem is called _____.
3. Sympathetic preganglionic neurons leave spinal segments _____.
4. Parasympathetic fibres reach most thoracic and abdominal organs through the _____ nerve.
5. A cluster of cell bodies in the PNS is a _____.
6. By the Bell-Magendie law, _____ roots are motor.
7. The knee-jerk reflex, with no interneuron, is _____.
8. Cerebrospinal fluid is made by the _____.

Answers: (1) biological (neurobiological), (2) converging operations, (3) T1 to L2, (4) vagus (X), (5) ganglion, (6) ventral, (7) monosynaptic, (8) choroid plexus.

5.2 The Brain: Brainstem, Diencephalon and Cerebral Cortex

MEDIUM YIELD · 27 QS

A lesion the size of a pea can stop your breathing in one place and merely blur a corner of your vision in another. Location is everything. This section tours the brain from the bottom up: the plan that names its five divisions, the hindbrain and midbrain that keep you alive and awake, the diencephalon that relays and regulates, and the cortex that perceives, plans and thinks.

Five divisions from three swellings

The brain's tongue-twisting names come from its development. The embryonic nervous system begins as a fluid-filled neural tube, and three swellings appear at its front end: the **forebrain** (prosencephalon), **midbrain** (mesencephalon) and **hindbrain** (rhombencephalon). Before birth the forebrain divides into the **telencephalon** and **diencephalon**, and the hindbrain into the **metencephalon** and **myelencephalon**, giving the five divisions of Table 5.4. A memory aid: the telencephalon is on top, and the other four lie below it in alphabetical order.

Table 5.4 The divisions of the brain.

Embryonic swelling	Division	Main adult structures
Forebrain (prosencephalon)	Telencephalon	Cerebral cortex, limbic system, basal ganglia, corpus callosum
Forebrain (prosencephalon)	Diencephalon	Thalamus, hypothalamus (and pineal gland)
Midbrain (mesencephalon)	Mesencephalon	Tectum, tegmentum, cerebral peduncles
Hindbrain (rhombencephalon)	Metencephalon	Pons, cerebellum
Hindbrain (rhombencephalon)	Myelencephalon	Medulla oblongata

The **brainstem** is the stalk on which the cerebral hemispheres sit: usually the midbrain, pons and medulla, though some texts use it for all four divisions below the telencephalon. Either way the cerebellum is treated separately (Figure 5.3).

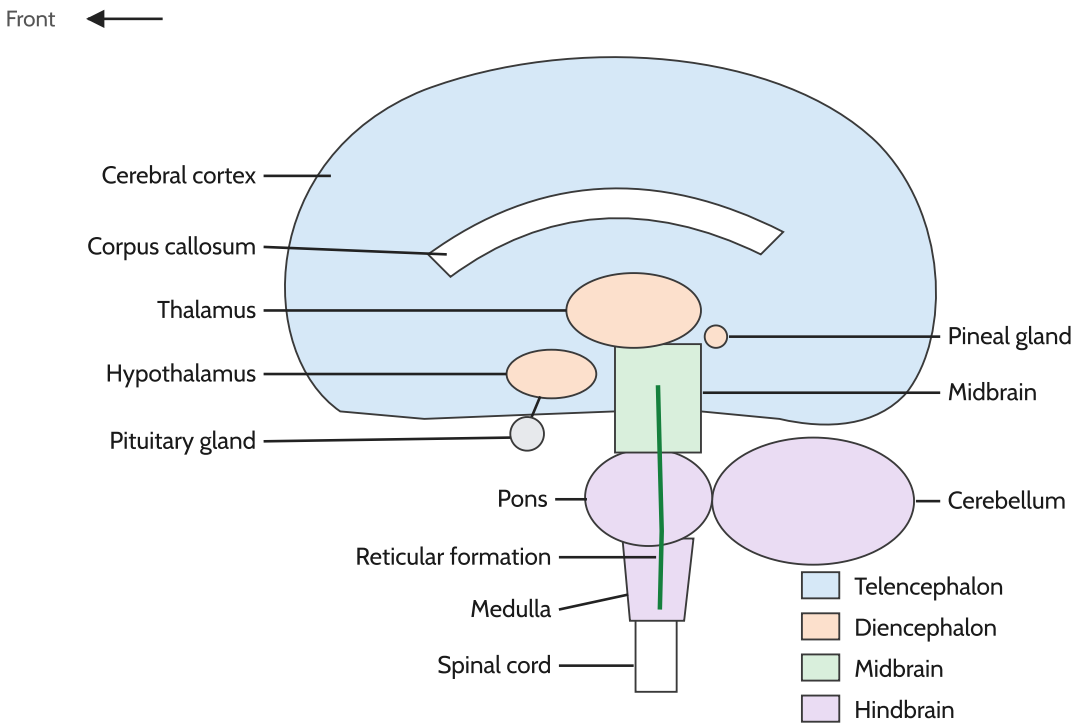

Figure 5.3 The divisions in a schematic midline section. The diencephalon lies beneath the corpus callosum; the reticular formation (dotted) runs through the brainstem core.

EXAM TRAP

The corpus callosum lies just above the thalamus, so it is often filed under the diencephalon. It is the white matter of the cerebral hemispheres, so it belongs to the **telencephalon**. The diencephalon is the thalamus and hypothalamus, the mesencephalon the midbrain, the metencephalon the pons and cerebellum.

The hindbrain: medulla, pons and cerebellum ●●●

The **medulla oblongata** is where the spinal cord widens into the brain. Much of it is tracts: on its front surface two ridges, the **pyramids**, carry the corticospinal motor tract, which crosses sides at the lower medulla (Section 5.6). It also holds the **vital centres**: the cardiovascular centre for heart rate and blood pressure, the respiratory centre that generates the

rhythm of breathing, and reflex centres for swallowing, coughing, sneezing and vomiting. So medullary damage is the brain injury most likely to kill, and an overdose of alcohol or opioids kills by silencing its respiratory centre.

Above it, the **pons** (Latin "bridge") bulges from the front of the brainstem; its nuclei relay signals from the cortex across to the cerebellum. It also helps set the rhythm of breathing, and its reticular nuclei contribute to arousal, sleep and REM sleep (Section 5.8).

Behind the pons sits the **cerebellum** ("little brain"): about 10 percent of the brain's mass but more than half its neurons. It does not start movements; the primary motor cortex does. It makes them accurate, comparing the intended movement with the actual one and correcting the error. So it coordinates and smooths movement, keeps balance and posture, times rapid sequences and supports motor learning, such as the practised timing of a cricket stroke. Damage produces **ataxia**: clumsy, uncoordinated movement and a wide, staggering gait, like drunkenness. It does not produce language (Broca's area), store declarative memories (hippocampus and cortex) or regulate emotion (amygdala); its circuitry is in Section 5.6.

> EXAM TRAP
>
> Heartbeat and respiration are modulated by the **hindbrain** (medulla and pons). The hypothalamus adjusts them for homeostasis but does not pace them; the amygdala is for fear; the cerebellum, though in the hindbrain, does not keep you breathing. Damage most likely to cause **death**: the medulla, not the pons, cerebellum or reticular formation (whose damage causes coma).

The midbrain and the reticular activating system ●●

Two words organise the midbrain. The **tectum** ("roof") is its back surface, with two pairs of bumps: the **superior colliculi** turn your eyes and head towards a sudden flash (visual orienting), and the **inferior colliculi** relay hearing (superior for sight, inferior for sound). The **tegmentum** beneath holds the **substantia nigra**, whose dopamine neurons feed the basal ganglia and die in Parkinson's disease; the **red nucleus**, a motor nucleus; the **periaqueductal grey**, which suppresses pain and mediates opioid analgesia; and the **ventral tegmental area**, source of the dopamine reward pathway (Section 5.4). On the front, the **cerebral peduncles** are thick stalks of axons descending from the cortex.

Through the core of the whole brainstem, from medulla to midbrain, runs the **reticular formation**, a net-like web of about 100 small nuclei (*reticulum*, "little net"). Its ascending part, the **reticular activating system** (RAS), projects through the thalamus and directly to the cortex. It regulates **arousal, wakefulness, attention and consciousness**, filtering stimulation so that a parent sleeps through traffic yet wakes at a baby's cry. Destroy it and the result is **coma**. Its descending parts help control muscle tone and cardiac and respiratory reflexes.

KEY STUDY: MORUZZI AND MAGOUN (1949)

Giuseppe Moruzzi and **Horace Magoun** stimulated the reticular formation of sleeping cats electrically. The cats woke, and their EEG changed from the slow, synchronised waves of sleep to the fast, desynchronised pattern of alertness. With brainstem transection and lesion findings, this led them to propose that low reticular activity produces sleep and high activity wakefulness. Later work showed that sleep is also actively generated elsewhere (Section 5.8).

EXAM TRAP

RAS questions offer its neighbours. Fine motor coordination, balance and posture belong to the cerebellum, emotional memory to the amygdala, episodic memory to the hippocampus, motor planning to premotor and prefrontal cortex, reading comprehension to Wernicke's area and the angular gyrus. The RAS means arousal, wakefulness, attention and consciousness.

The diencephalon: thalamus and hypothalamus ●●

The **thalamus** is the large two-lobed structure at the top of the brainstem, one lobe on each side of the third ventricle. It is the brain's **relay station**: every sense except smell synapses in a thalamic nucleus before reaching cortex. The **lateral geniculate nucleus** relays vision, the **medial geniculate nucleus** hearing (lateral for light, medial for music), and the **ventral posterior nucleus** touch, pain, temperature, body position and taste. Smell goes from the olfactory bulb directly to olfactory cortex. Nor is the thalamus a passive switchboard: each relay nucleus receives feedback from its cortex, and other nuclei relay motor loops (basal ganglia and cerebellum to motor cortex) and association loops (to prefrontal cortex).

Just below the front of the thalamus (*hypo*, "below") lies the almond-sized **hypothalamus**, master regulator of **homeostasis**, the maintenance of a stable internal environment. It controls hunger, thirst, body temperature, the sleep-wake cycle, sexual behaviour, and emotion and aggression, acting through the autonomic nervous system, through motivated behaviour and through the endocrine system. The **pituitary gland** hangs from its underside, and the hypothalamus directs it, with releasing hormones carried in local blood vessels to the anterior pituitary and axons to the posterior pituitary (Section 5.7). The pituitary is the "master gland", but the hypothalamus is its master: the control centre integrating the endocrine, autonomic and visceral systems.

Its nuclei are named here and taught later: the **suprachiasmatic nucleus**, the circadian clock above the optic chiasm; the lateral and ventromedial areas and arcuate nucleus in feeding (both Section 5.8); the preoptic area in temperature and sexual behaviour; and the supraoptic and paraventricular nuclei, which make oxytocin and vasopressin (Section 5.7). On its underside sit the two **mammillary bodies**, part of the limbic memory circuit and damaged in Korsakoff's syndrome (Section 5.9). Behind the thalamus, the epithalamus holds the **pineal gland**, which secretes melatonin at night.

EXAM TRAP

"Relays every sense except smell" is the **thalamus**. "Homeostasis", "master regulator", "controls the pituitary" and "integrates the endocrine, autonomic and visceral systems" are the **hypothalamus**. The brainstem runs autonomic reflexes without integrating them with hormones; the spinal cord carries messages and simple reflexes; the pineal makes melatonin; the corpus callosum, basal ganglia and hippocampus relay no senses.

The limbic system and basal ganglia

Beneath the cortex lie two loose groupings. The **limbic system** is a ring of midline structures circling the thalamus (*limbus*, "border"): the **hippocampus** of the medial temporal lobe (forming new declarative memories and spatial maps), the almond-shaped **amygdala** in front of it (emotion, especially fear, and emotional memory), the **cingulate cortex** above the corpus callosum, the **fornix** (the main limbic tract, arching from hippocampus to mammillary bodies and septum), the **septum** and the mammillary bodies. Together they serve emotion, motivation and memory (Sections 5.7 and 5.9).

The **basal ganglia** are large nuclei deep in each hemisphere. The tail-like **caudate nucleus** and the **putamen** receive cortical input; both look striped, so together they are the **striatum**. Its output goes to the **globus pallidus** ("pale globe"). The **subthalamic nucleus** and the substantia nigra, whose dopamine input to the striatum fails in Parkinson's disease, are functional partners; older texts also counted the amygdala. The basal ganglia select and initiate voluntary movements, build habits and support procedural learning (pathways in Section 5.6).

EXAM TRAP

"Which is NOT a basal ganglia structure?" The **red nucleus**, a motor nucleus of the midbrain tegmentum (source of the rubrospinal tract). The caudate and globus pallidus are core members; the subthalamic nucleus belongs to the circuit.

The cerebral cortex: layers and landmarks

The **cerebral cortex** ("bark") is a sheet of grey matter a few millimetres thick over the hemispheres, with white matter (myelinated axons) beneath. It is crumpled into ridges, **gyri**, and grooves, **sulci**; the deepest grooves are **fissures**. The **longitudinal fissure** separates the hemispheres. The **central sulcus** separates frontal from parietal lobe, with the **precentral gyrus** (primary motor cortex) in front and the **postcentral gyrus** (primary somatosensory cortex) behind. The **lateral fissure** (Sylvian fissure) separates the temporal lobe from the frontal and parietal lobes. The **calcarine sulcus**, on the medial surface of the occipital lobe, is lined by primary visual cortex (Figure 5.4).

About 90 percent of human cortex is **neocortex** ("new cortex", or isocortex), the most recently evolved part, with six layers numbered I to VI from the surface. Large **pyramidal cells** send long axons out of the area; small **stellate cells** are interneurons. **Layer IV** receives thalamic input, so it is thick in sensory cortex; **layer V** holds the large pyramidal output cells projecting to brainstem and spinal cord, so it is thick in motor cortex, where it contains the giant **Betz cells**. The neocortex is the seat of higher cognitive functions: perception, language, rea-

soning, planning and voluntary movement. The older **allocortex** has fewer layers: the hippocampus (three) and olfactory cortex.

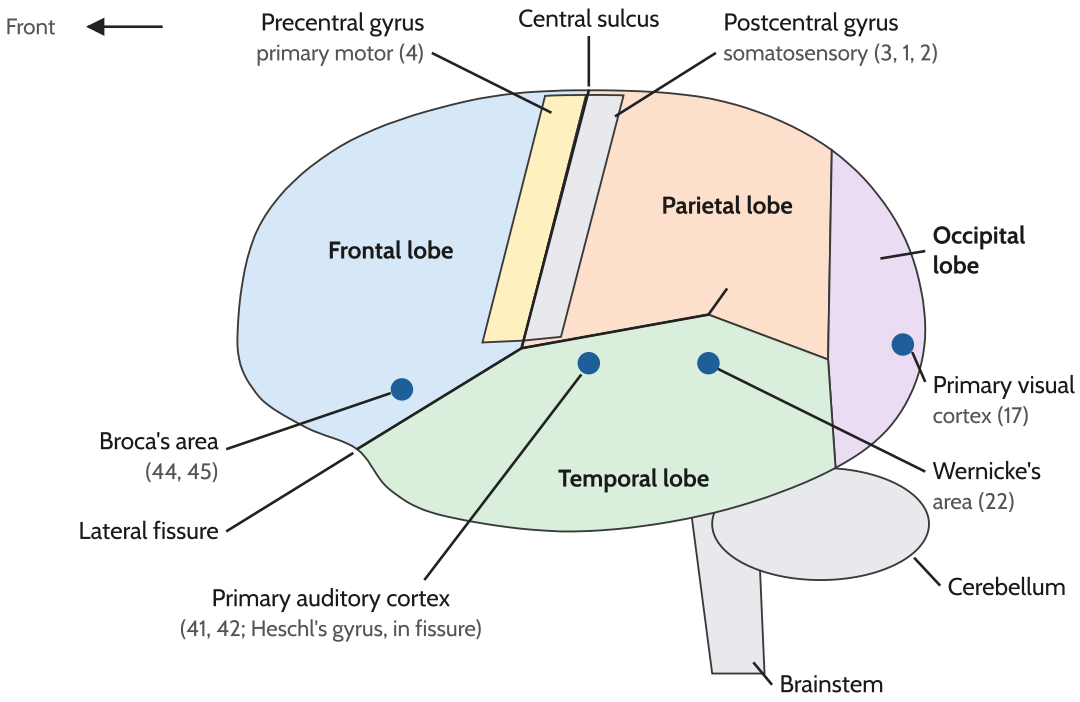

Figure 5.4 The left hemisphere from the side, with Brodmann numbers in brackets. Primary visual cortex lies mostly on the medial surface, along the calcarine sulcus.

EXAM TRAP

The neocortex means **higher cognitive functions**. Basic reflexes belong to the spinal cord and brainstem, olfactory processing to the old olfactory cortex, autonomic regulation to the hypothalamus and medulla.

The lobes, association cortex and Brodmann's map ●●●

The central sulcus and lateral fissure divide each hemisphere into four lobes, with the **insula** hidden inside the lateral fissure (Table 5.5). Lobes are folds of one continuous sheet, not functional units: each holds several different areas.

Table 5.5 The lobes of the cerebral cortex.

Lobe	Landmark and primary area	Main functions
Frontal	In front of the central sulcus; primary motor cortex in the precentral gyrus	Voluntary movement; prefrontal: planning, problem solving, reasoning, intellectual functioning, personality, social behaviour; Broca's area: speech production
Parietal	Behind the central sulcus; primary somatosensory cortex in the postcentral gyrus	Touch and body sense; interpreting and integrating sensory information; spatial attention
Temporal	Below the lateral fissure; primary auditory cortex in Heschl's gyrus	Hearing; Wernicke's area: language comprehension; object and face recognition; medial part: memory

Lobe	Landmark and primary area	Main functions
Occipital	Back of the brain; primary visual cortex (V1, striate cortex) along the calcarine sulcus	Vision
Insula	Inside the lateral fissure	Taste; interoception (sensing the body's internal state)

The frontal lobe, the largest, does two jobs. Its back strip, the **primary motor cortex**, commands voluntary movement of the opposite side of the body, and premotor cortex and the supplementary motor area plan it. Everything in front is **prefrontal cortex**: planning, problem solving, reasoning, working memory, intellectual functioning, personality and social behaviour. In the parietal lobe, posterior parietal cortex integrates touch, vision and body position into a sense of where objects and limbs are. Primary visual cortex is called **striate cortex** for the white stripe in its layer IV.

Cortex is also graded by distance from the senses. **Primary sensory cortex** receives most of its input directly from a thalamic relay nucleus; **secondary sensory cortex** mostly from primary cortex of the same sense. **Association cortex** receives input from more than one sensory system and integrates it; it makes up most of human cortex (prefrontal, posterior parietal, temporal).

In 1909 the German neuroanatomist **Korbinian Brodmann** (1868 to 1918) divided the cortex into numbered areas by **cytoarchitecture**, the size, density and layering of its cells. His numbers remain the common language of cortical maps (Table 5.6).

Table 5.6 Brodmann areas worth knowing.

Area	Region	Lobe
1, 2, 3	Primary somatosensory cortex	Parietal
4	Primary motor cortex	Frontal
6	Premotor cortex and supplementary motor area	Frontal
8	Frontal eye fields	Frontal
9 to 12	Prefrontal cortex	Frontal
17	Primary visual cortex (V1, striate)	Occipital
18, 19	Visual association cortex	Occipital
22	Wernicke's area (posterior part)	Temporal
39	Angular gyrus (reading)	Parietal
40	Supramarginal gyrus	Parietal
41, 42	Primary auditory cortex	Temporal
44, 45	Broca's area	Frontal

Four prefrontal regions return in later sections: **dorsolateral prefrontal cortex** (working memory, planning, mental flexibility), **orbitofrontal cortex** above the eye sockets (impulse control, social behaviour, reward value), **ventromedial prefrontal cortex** (emotion in decision making) and **anterior cingulate cortex** on the medial surface (detecting errors and conflict; the unpleasantness of pain). Their syndromes are in Section 5.11.

EXAM TRAP

"Interpreting sensory information" is **parietal**: every lobe processes some sense, but the phrase means integrating bodily sensation. "Intellectual functioning", "problem solving" and "cognitive functions and voluntary movement" are **frontal**. Vision and V1 are **occipital**; auditory processing is **temporal**; motor commands and language production are frontal.

Fibre systems: wiring the parts together

The white matter comes in three kinds. **Commissural fibres** cross the midline to connect the hemispheres. The largest commissure is the **corpus callosum**, an estimated 200 million axons linking the left and right hemispheres; the smaller **anterior commissure** links the temporal lobes and olfactory regions. **Association fibres** stay within one hemisphere: short U-fibres join neighbouring gyri, and long bundles such as the **arcuate fasciculus** link the frontal and temporal lobes (Broca's and Wernicke's areas). **Projection fibres** connect cortex with lower centres in both directions, from thalamus to cortex and from cortex to brainstem and spinal cord. They funnel through the **internal capsule** between the thalamus and basal ganglia, so a small stroke there can paralyse the opposite side of the body. Three pairs of **cerebellar peduncles** link the cerebellum to the brainstem. Cutting the corpus callosum produces the split brain (Section 5.10).

EXAM TRAP

The corpus callosum connects the **left and right hemispheres**. The wrong options name other fibre systems: brainstem and cerebellum (cerebellar peduncles), frontal and temporal lobes (arcuate fasciculus), thalamus and cortex (projection fibres).

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The corpus callosum belongs to the _____.
2. The vital centres for breathing and heart rate lie in the _____.
3. Cerebellar damage produces uncoordinated movement called _____.
4. Moruzzi and Magoun woke sleeping cats by stimulating the _____.
5. The thalamic relay for hearing is the _____ nucleus.
6. The hypothalamic circadian clock is the _____ nucleus.
7. The _____ nucleus is a midbrain motor nucleus, not a basal ganglion.
8. Primary visual cortex is Brodmann area _____.

Answers: (1) telencephalon, (2) medulla oblongata, (3) ataxia, (4) reticular formation, (5) medial geniculate, (6) suprachiasmatic, (7) red, (8) 17

5.3 Neurons, Neural Conduction, Synapses and Neurotransmitters

MEDIUM YIELD · 45 QS

Every word you read here is carried by cells that work like tiny batteries and talk in chemical whispers. This section follows one message from start to finish: how a neuron is built, how it

holds a charge at rest, how that charge flips into an action potential that races down the axon, and how chemicals carry the signal across the synapse.

The neuron: the basic unit ●●

In the early 1870s the Italian physician **Camillo Golgi** (1843 to 1926) found that silver chromate stains a few neurons in a slice completely black, showing single cells in silhouette for the first time: the **Golgi stain**. Golgi held that the nervous system is one continuous net of fused fibres, the **reticular theory**. The Spanish histologist **Santiago Ramón y Cajal** (1852 to 1934), using Golgi's own stain, showed that it is built of separate cells that touch but do not fuse: the **neuron doctrine**. The rivals shared the Nobel Prize in 1906, and Cajal was right. The **neuron** is the basic structural and functional unit of the nervous system; a dendrite is only part of one, the brain is an organ made of billions of them, and glial cells support neurons rather than carry their messages.

Figure 5.5 shows the parts. The **cell body** (soma) holds the nucleus and keeps the cell alive. Branching **dendrites**, studded with tiny **dendritic spines**, receive most incoming synapses. The cone-shaped **axon hillock** joins the soma to the **axon**, the long fibre that carries signals away from the soma; action potentials begin here (strictly, Pinel notes, in the axon initial segment just beyond it). A fatty **myelin sheath** wraps many axons, broken at intervals by bare gaps, the **nodes of Ranvier**. The axon ends in fine branches, the **telodendria**, each tipped by an **axon terminal** (terminal button) whose **synaptic vesicles** store and release the transmitter.

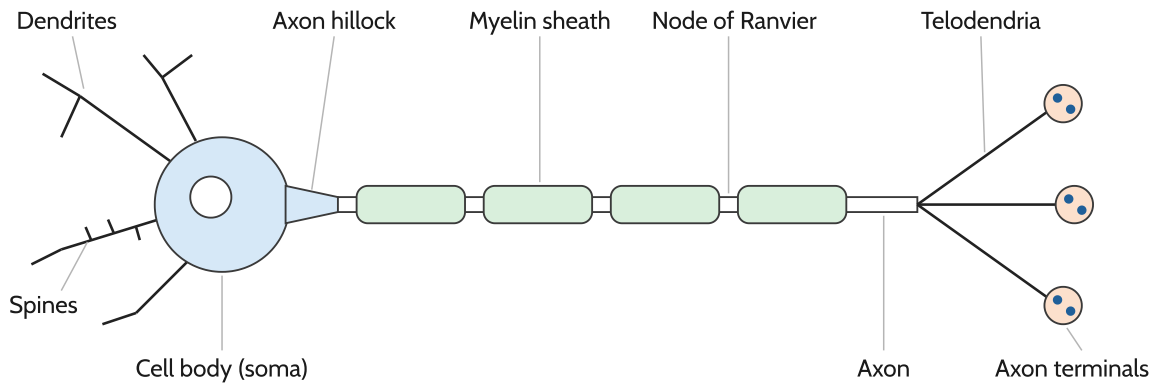

Figure 5.5 A multipolar neuron. Signals flow from the dendrites and soma, through the hillock and down the myelinated axon, to the terminals, whose vesicles (dots) hold the transmitter.

A **multipolar neuron** (one axon, many dendrites) is by far the commonest type and includes motor neurons. A **bipolar neuron** has two processes, one at each end, as in the retina. A **unipolar neuron** has a single process that splits in two, as in most somatosensory neurons.

> EXAM TRAP
>
> Transmitter is released from the **axon terminals**. The telodendria are the branches leading to them, the axon hillock is where the spike starts, and myelin only insulates. The basic unit is the neuron, not the dendrite, the brain or the glial cell.

Glial cells ●●

Neurons share the nervous system with **glial cells** (glia, "glue"). They are not ten times as numerous as neurons, as is often claimed: Pinel puts it at about two glia for every three neurons. Glia do not fire action potentials, but they do far more than hold neurons in place (Table 5.7).

Table 5.7 The glial cells.

Glial cell	Where	Main jobs
Astrocytes (star-shaped; largest)	CNS	End-feet on capillaries help form the blood-brain barrier; buffer extracellular K^+; recycle glutamate to neurons as glutamine (glutamate-glutamine cycle); supply fuel
Oligodendrocytes	CNS	Myelin; one cell wraps segments of many axons
Schwann cells	PNS: nerves outside the skull and spine	Myelin; one cell per segment; guide regrowth of cut nerves
Microglia (smallest)	CNS	Resident macrophages: immune surveillance, phagocytosis, synaptic pruning
Ependymal cells	Lining of the ventricles	Help make and circulate cerebrospinal fluid

EXAM TRAP

The **myelin sheath** is made by glia but is not a glial cell. Oligodendrocytes myelinate the CNS (brain, spinal cord, basal ganglia) and Schwann cells the PNS; options that swap them are wrong. Astrocytes never conduct action potentials.

The resting membrane potential ●●

Push a fine electrode into a resting neuron and its inside reads about 70 millivolts more negative than the outside: the **resting membrane potential**, about **-70 mV** (range about -60 to -80 mV). The neuron is **polarised** because ions are unevenly spread: sodium (Na^+) and chloride (Cl^-) ions are concentrated outside, potassium (K^+) ions and negatively charged proteins inside.

Two forces act on each ion: the **concentration gradient**, from crowded to scarce, and **electrostatic pressure**, as like charges repel and opposite charges attract. Both push Na^+ inwards, yet little enters, because at rest the membrane is far more permeable to K^+ than to Na^+: K^+ channels are open, Na^+ channels closed. K^+ leaks out until the negative charge inside holds the rest back. The **sodium-potassium pump**, powered by ATP, exchanges **three Na^+ out for every two K^+ in**, preserving the gradients against the leak. The Nernst equation gives one ion's equilibrium potential; the Goldman equation combines several ions.

KEY STUDY: HODGKIN AND HUXLEY'S SQUID GIANT AXON (1952)

Alan Hodgkin (1914 to 1998) and **Andrew Huxley** (1917 to 2012) recorded inside the giant axon of the squid, up to 1 mm across. In 1952 they described the action potential as a timed rise in Na^+ and then K^+ permeability, in equations that won a Nobel Prize in 1963. Criticism: the large, simple squid axon fits many mammalian neurons only roughly.

> **EXAM TRAP**
>
> The resting potential is about -70 mV, inside negative: not -40 mV and never positive. It rests on high K^+ permeability (not equal Na^+ and K^+ permeability) plus the pump. Open Na^+ channels, Ca^{2+} influx and Cl^- channels are distractors.

Postsynaptic potentials and the action potential ●●

Transmitter binding nudges a neuron's membrane potential. A depolarisation (say from -70 to -67 mV) makes firing more likely and is an **excitatory postsynaptic potential** (EPSP); a hyperpolarisation (to -72 mV) makes it less likely and is an **inhibitory postsynaptic potential** (IPSP). Both are **graded potentials**, sized in proportion to the input, and they spread fast but decrementally, fading like ripples on a pond.

Whether a neuron fires depends on the sum of thousands of PSPs reaching the axon initial segment, a process called **integration**. In **spatial summation**, PSPs arriving together from different synapses add up (an EPSP and an IPSP cancel); in **temporal summation**, PSPs arriving in quick succession at one synapse add up. If the total reaches the **threshold of excitation**, about -65 mV in Pinel (many texts say -55 mV), an action potential follows.

The **action potential** (AP, spike) is a massive reversal of the membrane potential lasting about 1 millisecond (Figure 5.6). At threshold, voltage-gated Na^+ channels open and Na^+ rushes in, driving the membrane from -70 to about **+50 mV**. The Na^+ channels then close and inactivate, while voltage-gated K^+ channels (the delayed rectifiers) let K^+ out, repolarising the membrane. They close slowly, so the membrane is briefly hyperpolarised before it returns to rest.

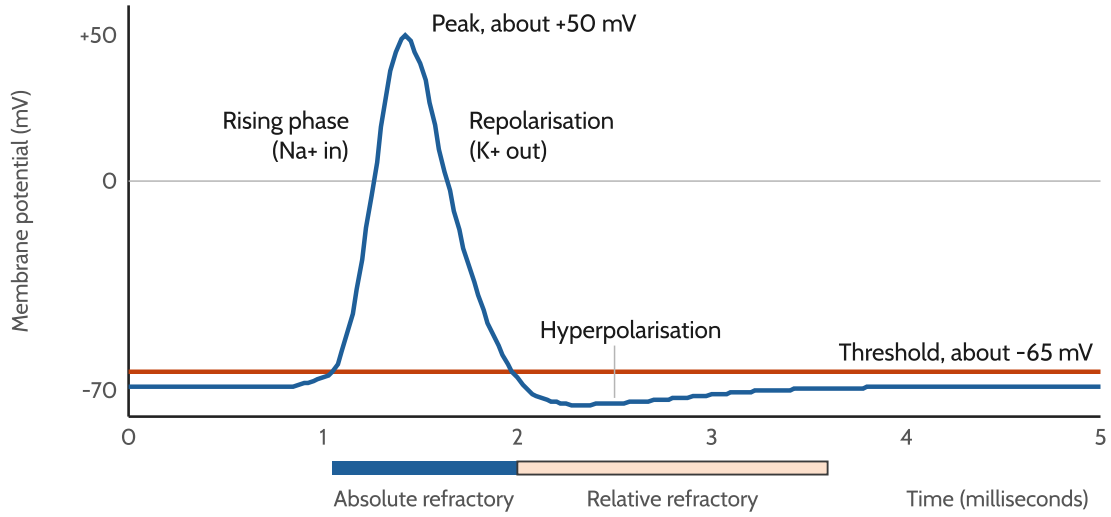

Figure 5.6 An action potential. A depolarisation that reaches the threshold line fires a full spike; the bars show when no stimulus can fire another (absolute) and when only a stronger one can (relative).

Unlike PSPs, action potentials obey the **all-or-none law**: once threshold is reached, the spike fires at full, constant size whatever the stimulus strength; below threshold, nothing fires. Squeeze a gun's trigger hard or gently and the bullet leaves at the same speed. A strong stimulus is signalled instead by firing more often: intensity is coded by frequency, the **rate law**.

EXAM TRAP

Only **action potentials** obey the all-or-none law. EPSPs and IPSPs are graded, and the resting potential is a steady baseline that does not vary with the stimulus. Stronger stimuli give more spikes, not bigger ones; sub-threshold stimuli give no "miniature" spikes; and the law says nothing about neighbouring neurons firing together or one spike per second.

Refractory periods and conduction ●●

For about 1 to 2 milliseconds after a spike begins, no stimulus, however strong, can fire another: the **absolute refractory period**, during which the voltage-gated Na^+ channels are inactivated and cannot reopen until the membrane repolarises. In the **relative refractory period** that follows, K^+ channels are still open and the membrane is hyperpolarised, so only a stronger than normal stimulus works. Two consequences follow. A spike cannot turn back onto the axon it has just left, so it travels one way, soma to terminals: **orthodromic conduction** (backward, antidromic travel is seen mainly under artificial stimulation). And firing is capped at about 1,000 spikes per second, intense stimulation refiring the cell as soon as the absolute period ends.

In an unmyelinated axon the spike is regenerated at each Na^+ channel in turn: **continuous conduction**, which never weakens but is slow, about 1 m per second in thin axons. In a myelinated axon ions can cross only at the nodes of Ranvier, where voltage-gated Na^+ channels are concentrated, so the spike is regenerated node by node and seems to jump: **saltatory conduction** (Latin *saltare*, to leap). The function of myelin is thus to increase conduction speed, up to about 100 m per second in the largest axons (Pinel cites up to 60 m per second for human motor neurons); wider axons are faster too. Multiple sclerosis destroys CNS myelin and slows or blocks conduction (Section 5.11).

EXAM TRAP

The absolute refractory period reflects **inactivated Na^+ channels**, not pump failure, inactivated K^+ channels or Cl^- influx. Myelin speeds conduction; it does not support the neuron, store transmitter or generate spikes. Saltatory conduction needs Na^+ channels clustered at the nodes, not spread evenly along the axon.

How a synapse works ●

A **synapse** is the junction where one neuron passes its message to another across a narrow gap, the **synaptic cleft**, in four steps.

1. **Synthesis and packaging.** Small-molecule transmitters are made and packed into vesicles in the terminal; neuropeptides are made in the soma and carried down the axon on microtubules.
2. **Release.** The arriving action potential opens **voltage-gated Ca^{2+} channels**; Ca^{2+} entry makes vesicles fuse with the membrane and spill their contents into the cleft: **exocytosis**.
3. **Binding.** Transmitter diffuses across the cleft and binds postsynaptic receptors.
4. **Deactivation.** Reuptake or an enzyme ends the message (below).

Most synapses are **axodendritic** (often onto a spine) or **axosomatic**. **Axoaxonic** synapses facilitate or inhibit a single terminal button, and **dendrodendritic** synapses can often transmit

both ways. Most are **directed synapses**, with release site and receptors side by side. At **nondirected synapses**, Pinel's string-of-beads synapses, transmitter is released from **varicosities**, swellings along the axon, and spreads to distant targets; most monoamine neurons work this way. Some neurons are linked by **gap junctions**, protein channels (connexins) joining their cytoplasm to form electrical synapses, faster than chemical ones. And one button often holds a small-molecule transmitter and a neuropeptide together: **coexistence**.

> EXAM TRAP
>
> Release is triggered by **Ca^{2+} influx**, not by K^+ efflux (which repolarises), closure of the pump, or Cl^- influx (which makes IPSPs). Varicosities are **non-directed** synapses that release **monoamines**.

Receptors and deactivation ●

A transmitter acts by binding a **receptor**, a membrane protein shaped for it, and most transmitters have several **receptor subtypes**, so one chemical can send different messages to different places. An **ionotropic receptor** is a ligand-gated ion channel that opens at once (Na^+ in gives an EPSP; Cl^- in or K^+ out an IPSP), so its effect is fast and brief. A **metabotropic receptor** is coupled to a G protein, which opens a nearby channel or triggers a **second messenger** such as cyclic AMP that can alter the cell's metabolism and even its genes; these receptors are more common, and their effects slower, longer lasting and more varied. Small-molecule transmitters bind both kinds, but virtually all neuropeptides bind only metabotropic receptors. Metabotropic receptors on the releasing neuron itself, **autoreceptors**, sense its own transmitter and cut release when levels are high.

Transmitter left in the cleft would jam the line. **Reuptake** by transporters, the commoner route, returns monoamines and amino acids to the button. In **enzymatic degradation** an enzyme breaks the transmitter apart: acetylcholine is deactivated solely by **acetylcholinesterase** in the cleft, and its choline is recycled. Monoamines are degraded by **monoamine oxidase** (MAO-A: serotonin, noradrenaline and dopamine; MAO-B: dopamine and phenylethylamine) and by catechol-O-methyltransferase (COMT).

> EXAM TRAP
>
> Ionotropic receptors bind **small transmitters only**. The transmitter cleared exclusively by an enzyme is **acetylcholine**; dopamine, GABA and glutamate are taken back up. MAO never acts on GABA, glutamate, glycine or acetylcholine.

Amino acids and acetylcholine ●●●

More than 100 transmitters are known, in five families (Table 5.8); only the neuropeptides are large molecules.

Table 5.8 The neurotransmitter families (after Pinel).

Family	Members	Notes
Amino acids	Glutamate, aspartate, glycine, GABA	Fast, directed synapses; GABA is made from glutamate

Family	Members	Notes
Monoamines: catecholamines	Dopamine, noradrenaline (norepinephrine), adrenaline (epinephrine)	Made from tyrosine, via L-dopa
Monoamines: indolamine	Serotonin (5-hydroxytryptamine, 5-HT)	Made from tryptophan
Monoamines: other	Histamine	Made from histidine
Acetylcholine	Acetylcholine (ACh)	A class by itself: choline plus an acetyl group
Unconventional	Nitric oxide, carbon monoxide; endocannabinoids such as anandamide (Sanskrit *ananda*, bliss)	Made on demand; often signal backwards to the presynaptic cell
Neuropeptides	Endorphins, enkephalins, substance P, neuropeptide Y, orexin (hypocretin), oxytocin, vasopressin	Made in the soma; metabotropic receptors only

The main excitatory transmitter of the central nervous system is **glutamate**, acting on three ionotropic receptors, AMPA, kainate and NMDA, and on metabotropic receptors. The **NMDA receptor** is plugged by a magnesium ion (Mg^{2+}): it opens only when glutamate is bound (glycine is a co-agonist) *and* the membrane is depolarised enough, usually by AMPA receptors, to expel the Mg^{2+}. It then admits Ca^{2+}. This coincidence detection underlies long-term potentiation in learning (Section 5.9). After a stroke, glutamate floods out, Ca^{2+} pours in through NMDA receptors and neurons die: **excitotoxicity**.

GABA (gamma-aminobutyric acid) is the main inhibitory transmitter of the brain: an amino acid, not a hormone, neuropeptide or catecholamine. The **GABA-A receptor** is an ionotropic Cl^- channel giving fast IPSPs, with binding sites for benzodiazepines, barbiturates, alcohol and neurosteroids, all of which enhance it (Section 5.4). The **GABA-B receptor** is metabotropic: through a G protein it opens K^+ channels and closes Ca^{2+} channels. **Glycine** is the main inhibitory transmitter of the spinal cord.

At the **neuromuscular junction** the transmitter that makes skeletal muscle contract is **acetylcholine** (ACh), which also serves the autonomic ganglia and parasympathetic endings and, in the brain, learning, memory, attention and REM sleep. Its **nicotinic receptors** (muscle, ganglia) are ionotropic, its **muscarinic receptors** metabotropic. The **nucleus basalis of Meynert** in the basal forebrain supplies the cortex with ACh and is depleted in Alzheimer's disease, hence treatment with cholinesterase inhibitors. Curare blocks nicotinic receptors and botulinum toxin blocks ACh release, both paralysing; in myasthenia gravis, antibodies attack muscle nicotinic receptors, causing weakness that worsens with use.

> EXAM TRAP
>
> Inhibitory: **GABA** (brain) and glycine (spinal cord). Excitatory: **glutamate**. GABA-A is the **ionotropic, ligand-gated Cl^- channel** (fast IPSPs), not a voltage-gated or excitatory channel; GABA-B is metabotropic. NMDA is ionotropic, needs depolarisation and admits Ca^{2+}, not Cl^-.

Monoamine systems and endogenous opioids ●●●

Monoamine neurons are few, clustered mostly in the brainstem, but their branching axons and varicosities reach almost the whole brain, tuning broad states such as arousal and mood.

Table 5.9 lists them with the other widely projecting systems.

Table 5.9 Where the modulating transmitters come from.

Transmitter	Main cell bodies	Main roles
Dopamine	Substantia nigra (nigrostriatal pathway); ventral tegmental area	Movement, reward, attention (Section 5.4)
Noradrenaline	Locus coeruleus (pons)	Arousal, vigilance; also the sympathetic postganglionic transmitter (heart, blood vessels)
Adrenaline	Adrenal medulla (as a hormone); a few medullary neurons	Fight or flight
Serotonin	Raphe nuclei, along the brainstem midline	Mood, appetite, sleep, impulse control
Histamine	Tuberomammillary nucleus (hypothalamus)	Wakefulness; older antihistamines sedate
Acetylcholine	Nucleus basalis of Meynert (basal forebrain); pons	Attention, memory; REM sleep
Orexin (hypocretin)	Lateral hypothalamus	Wakefulness (Section 5.8)
Oxytocin, vasopressin	Hypothalamus (supraoptic, paraventricular nuclei)	Released from the posterior pituitary (Section 5.7)

Low serotonin activity, shown by low cerebrospinal fluid levels of its breakdown product 5-hydroxyindoleacetic acid (**5-HIAA**), is linked to impulsive aggression and violent suicide.

> **EXAM TRAP: IS ADRENALINE A NEUROTRANSMITTER?**
>
> The IHBAS 2020 key marks **adrenaline** as "not a neurotransmitter" beside dopamine, serotonin and noradrenaline. Strictly it is one: Pinel lists it among the catecholamine transmitters, released by adrenergic neurons. It is the odd one out only because it acts mainly as a hormone of the adrenal medulla. Mark adrenaline, knowing why.

Why should the brain have receptors for morphine, a drug from the opium poppy? In 1973 **Candace Pert** (1946 to 2013) and **Solomon Snyder** (born 1938) demonstrated opiate receptors in the brain, plentiful in the periaqueductal grey, a pain-control area. A lock implies a natural key: in 1975 **John Hughes** and **Hans Kosterlitz** (1903 to 1996) isolated the **enkephalins** ("in the head"), and the **endorphins** ("endogenous morphine") followed. These **endogenous opioids**, the body's own morphine, are neuropeptides acting on metabotropic receptors; they control pain and give the "feel-good" glow often credited for the runner's high. Among other neuropeptides, substance P carries pain signals and neuropeptide Y stimulates eating.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Cajal's _____ doctrine replaced Golgi's reticular theory.
2. In the central nervous system myelin is made by _____; in the peripheral nervous system, by Schwann cells.
3. The sodium-potassium pump moves _____ Na^+ ions out for every two K^+ ions in.
4. At its peak the action potential reaches about _____ mV.
5. No stimulus can fire a neuron during the _____ refractory period, when its Na^+ channels are inactivated.
6. Transmitter release is triggered by the influx of _____ ions.
7. Acetylcholine is broken down in the cleft by _____.
8. Serotonin neurons lie in the _____ nuclei; noradrenaline neurons in the locus coeruleus.

Answers: (1) neuron, (2) oligodendrocytes, (3) three, (4) +50, (5) absolute, (6) calcium (Ca^{2+}), (7) acetylcholinesterase, (8) raphe.

5.4 Drugs, Reward and Psychopharmacology

MEDIUM YIELD · 40 QS

Every drug in this section works through the synaptic machinery of Section 5.3: filter coffee, a peg of whisky and a lithium tablet differ only in which synapses they touch. The section runs from the principles of drug action and addiction, through recreational drugs and the reward circuit they hijack, to the drugs prescribed for schizophrenia, depression, mania and anxiety.

Psychoactive drugs and their route to the brain

A **psychoactive drug** is a substance that alters mood, cognition, perception or behaviour by acting on the nervous system; the term does not mean a drug-induced hallucination, a particularly active psychopath or the psychology of physical activity. Drugs are taken in four ways. **Oral ingestion** is easy and safe but unpredictable, because food in the stomach alters absorption. **Injection** (subcutaneous, intramuscular or intravenous) is fast, strong and predictable, but an intravenous overdose cannot be recalled. **Inhalation** through the lung capillaries (tobacco, cannabis, anaesthetics) makes the dose hard to regulate. **Absorption through mucous membranes** of the nose, mouth or rectum is how snorted cocaine enters.

From the blood a drug must cross the blood-brain barrier (Section 5.3), which small, fat-soluble molecules pass most easily. Most drugs are then deactivated by liver enzymes, a process called **drug metabolism**, whose products usually can no longer penetrate the barrier.

Agonists, antagonists and synaptic mechanisms

Drugs that facilitate the effects of a particular neurotransmitter are its **agonists**; drugs that reduce them are its **antagonists**. A drug can act at any step in a transmitter's life. Agonists may increase synthesis (L-dopa, the precursor of dopamine), increase release (amphetamine), block reuptake (cocaine, SSRIs), block degradation (MAO inhibitors) or acti-

vate receptors (nicotine, morphine). Antagonists may block synthesis, empty the vesicles so the transmitter leaks out and is destroyed (reserpine), block release, or occupy receptors without activating them, as receptor blockers do (chlorpromazine, naloxone). Presynaptic autoreceptors normally slow release, so a drug that blocks them increases release and one that activates them reduces it.

An **allosteric modulator** binds a site on the receptor other than the transmitter's own and changes the receptor's response to the transmitter; benzodiazepines are positive allosteric modulators of the GABA-A receptor. A **partial agonist** activates a receptor only partly, raising activity when the transmitter is scarce but lowering it when the transmitter is plentiful; buspirone at the serotonin 5-HT1A receptor is an example.

Tolerance, withdrawal and addiction

Drug tolerance is decreased sensitivity to a drug that develops through exposure to it: a given dose has less effect, or more drug is needed for the same effect. The dose-response curve shifts to the right (Figure 5.7). **Metabolic tolerance** comes from changes that reduce the amount of drug reaching its sites of action (a heavy drinker's liver clears alcohol faster); **functional tolerance** comes from changes that reduce the reactivity of those sites, such as fewer receptors, and accounts for most tolerance to psychoactive drugs. Exposure to one drug also brings tolerance to others that act by the same mechanism, called **cross tolerance**: a heavy drinker needs more of a benzodiazepine or barbiturate, because all three enhance the GABA-A receptor. Tolerance may develop to some effects but not others, and some effects may instead grow with repeated use: **drug sensitisation** (amphetamine's motor effects in rats).

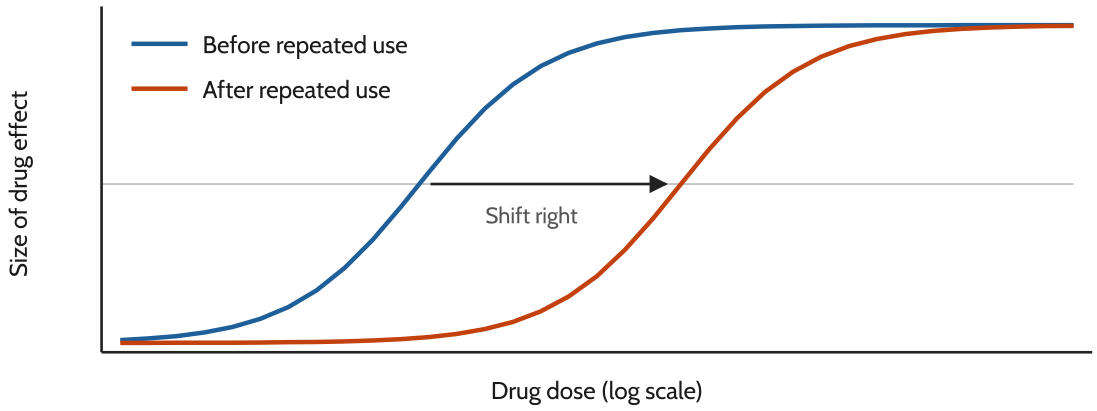

Figure 5.7 Tolerance as a rightward shift of the dose-response curve. After repeated use the same dose has less effect, and a larger dose is needed for the same effect.

When a drug that has been in the body for days is suddenly removed, a **withdrawal syndrome** follows whose effects are almost always opposite to the drug's: sleeping pills leave insomnia, alcohol leaves tremor, seizures and delirium tremens. The compensatory neural changes that produced tolerance are now unopposed. Suffering withdrawal on stopping defines **physical dependence**; **psychological dependence** is a compulsive craving for the drug's effects without bodily withdrawal. Neither defines **drug addiction**: habitual use that continues despite harm to health and social life and despite repeated efforts to stop.

KEY STUDY: SIEGEL'S HEROIN OVERDOSE RATS (1982)

Shepard Siegel treated each dose as a Pavlovian conditioning trial: the usual room and paraphernalia become conditioned stimuli that elicit compensatory responses opposite to the drug's effect, so tolerance is strongest where the drug is usually taken (**conditioned drug tolerance**). Siegel and colleagues (1982) made rats tolerant to heroin in one environment, then gave a large dose there or in a novel one. Many more died in the novel environment, where no compensatory responses opposed the drug, which may explain human overdoses in unfamiliar places. Criticism: some drug cues elicit drug-like rather than opposite responses.

EXAM TRAP

Cross tolerance means exposure to one drug reduces the effect of another acting by the same mechanism; it is not "no addiction to any drug". Withdrawal is usually the *opposite* of the drug's effect, and physical dependence is not addiction: detoxified users still relapse.

Sedative-hypnotics: alcohol, benzodiazepines, barbiturates ●●

Sedatives calm and **hypnotics** induce sleep; both are broad classes covering alcohol, barbiturates, benzodiazepines and newer sleeping pills, not one chemical family. These drugs depress the central nervous system mainly by enhancing GABA, the brain's main inhibitory transmitter, at the **GABA-A receptor**, a chloride (Cl-) channel whose opening hyperpolarises the neuron.

Alcohol is a positive modulator, functionally an agonist, at the GABA-A receptor, and also blocks the NMDA glutamate receptor, so it raises inhibition and lowers excitation. Drinking in pregnancy causes fetal alcohol syndrome (Chapter 6), and the thiamine deficiency of chronic drinkers causes Wernicke-Korsakoff syndrome (Section 5.9).

The **benzodiazepines** form one chemical class: diazepam (Valium), alprazolam (Xanax), flunitrazepam (Rohypnol), triazolam (Halcion) and chlordiazepoxide. They are used as anxiolytics, hypnotics and anticonvulsants. The older **barbiturates** (phenobarbital) bind another site on the same receptor. As Figure 5.8 shows, a benzodiazepine increases the *frequency* of Cl- channel opening and works only when GABA is present, so its effect has a ceiling; a barbiturate increases the *duration* of each opening and at high doses opens the channel by itself. Hence barbiturates, and benzodiazepines taken with alcohol, are dangerous in overdose, whereas a benzodiazepine overdose alone is rarely fatal and is reversed by the antagonist **flumazenil**. Withdrawal from any of these drugs is rebound excitation (anxiety, insomnia, tremor, seizures) and can kill.

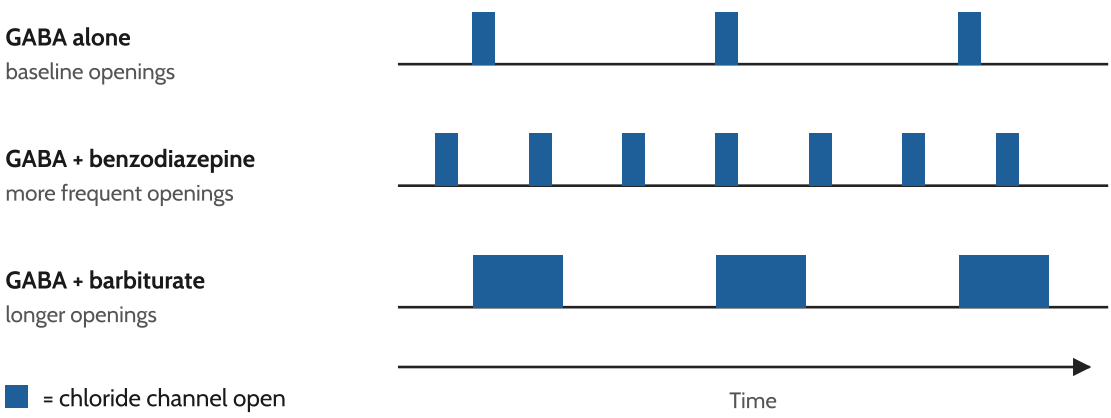

Figure 5.8 How the two drug families enhance the GABA-A receptor. Benzodiazepines make the channel open more often; barbiturates keep each opening longer.

EXAM TRAP

Frequency goes with benzodiazepines and **duration** with barbiturates, never the reverse. Both act on GABA-A, not GABA-B or NMDA receptors, and the receptor benzodiazepines modulate binds GABA, not glutamate, acetylcholine or glycine (the spinal cord's inhibitory transmitter). Valium, Xanax, Rohypnol and Halcion are all benzodiazepines; "sedatives" and "hypnotics" are too broad.

Opioids, stimulants, cannabis and hallucinogens ●●

Opium is the dried sap of the opium poppy's seed pods; its natural alkaloids are morphine, codeine, thebaine and papaverine. **Heroin** (diacetylmorphine) is not one of them: it is made by adding two acetyl groups to morphine, a *semisynthetic* opioid that enters the brain faster. Methadone and fentanyl are fully synthetic. Opioids act mainly at **mu opioid receptors**, the receptors for the body's endorphins (Section 5.3): they relieve pain, cough and diarrhoea, bring euphoria, and in overdose suppress breathing. **Naloxone** (short acting, the emergency reversal of overdose) and **naltrexone** (long acting, oral) are opioid antagonists. Opioid withdrawal mirrors the drug: dilated pupils, runny nose, sweating, gooseflesh and diarrhoea; miserable but rarely fatal. Table 5.10 covers the other classes.

Table 5.10 How stimulants, cannabis and hallucinogens act.

Drug	Main action	Note
Cocaine	Blocks reuptake of dopamine and noradrenaline	Binges; cocaine psychosis
Amphetamine, methamphetamine	Increase release of dopamine and noradrenaline	Psychosis like paranoid schizophrenia
Nicotine	Agonist at nicotinic acetylcholine receptors	Addictive agent of tobacco
Caffeine	Antagonist at adenosine receptors	Blocks adenosine's sleep signal
Cannabis (THC)	Agonist at CB1 receptors	Little risk of fatal overdose
LSD, psilocybin, mescaline	Agonists at serotonin 5-HT2A receptors	Classic hallucinogens
PCP, ketamine	NMDA glutamate receptor antagonists	Dissociative; mimic schizophrenia

The active ingredient of cannabis, delta-9-tetrahydrocannabinol (THC), acts at **CB1 receptors**, among the most common metabotropic receptors in the brain. They are densest in the hippocampus (memory impairment), basal ganglia and cerebellum (motor effects) and cortex, and present in reward circuits, but sparse in the brainstem centres that control breathing, so fatal overdose is unlikely. **CB2 receptors** lie mainly on immune cells. The brain makes its own ligands, the **endocannabinoids**; the first found, **anandamide**, is named from the Sanskrit *ananda*, bliss.

> EXAM TRAP
>
> Asked which is **not** a natural opium alkaloid, choose **heroin**: it is semisynthetic, made from morphine. Morphine, codeine, thebaine and papaverine all occur in the poppy. Cocaine blocks reuptake, whereas amphetamine increases release.

Reward circuits and theories of addiction ●●

> KEY STUDY: OLDS AND MILNER'S 'PLEASURE CENTRE' (1954)
>
> In 1954, at McGill University, **James Olds** and **Peter Milner** let rats press a lever that delivered brief electrical stimulation through an electrode in their own brains. With electrodes in the **septal area** and along the **medial forebrain bundle**, rats pressed thousands of times an hour, ignoring food, until exhausted. This **intracranial self-stimulation** seemed to reveal sites that mediate the pleasure of food, water and sex: 'pleasure centres'. Later work found no single centre; self-stimulation depends largely on dopamine axons running in the medial forebrain bundle.

Those axons form the **mesolimbic pathway**, from the ventral tegmental area (VTA) of the midbrain to the **nucleus accumbens** in the ventral striatum and other limbic sites. It is the core reward pathway of addiction: natural rewards and almost every addictive drug raise dopamine in the nucleus accumbens, directly (cocaine, amphetamine) or indirectly (opioids, alcohol, nicotine), and dopamine antagonists reduce self-stimulation. The reward transmitter is dopamine, not serotonin, noradrenaline or acetylcholine. With chronic use, control of drug taking shifts to the dorsal striatum (habit), and weakened **prefrontal control** makes stopping harder.

Physical-dependence theories said addicts take drugs to escape withdrawal, but detoxified users relapse months later, and cocaine causes little withdrawal distress. **Positive-incentive theories** replaced them: people take drugs for their anticipated pleasure. Yet addicts often crave a drug they no longer enjoy. The **incentive-sensitisation theory** of **Terry Robinson** and **Kent Berridge** (1993) explains this: in addiction-prone people repeated use sensitises the drug's incentive value ('wanting') while tolerance develops to its pleasure ('liking'). Mesolimbic dopamine signals wanting more than liking.

EXAM TRAP

The reward pathway is **mesolimbic** (VTA to nucleus accumbens), not nigrostriatal (movement), tuberoinfundibular (prolactin) or the raphe serotonergic pathway. Olds and Milner's sites were the **septal area and medial forebrain bundle**, not the cerebellar tonsils, the lateral geniculate nucleus (a visual relay) or the globus pallidus (a motor nucleus). The cerebellar vermis and vestibular nuclei serve balance, not addiction.

Dopamine pathways, schizophrenia and antipsychotics ●●●

Four dopamine pathways explain both what antipsychotic drugs fix and what they spoil (Table 5.11).

Table 5.11 The four dopamine pathways.

Pathway	Origin to target	Normal role	Disease or drug effect
Mesolimbic	VTA to nucleus accumbens and limbic structures	Reward, motivation	Overactive in positive symptoms; D2 blockade treats them
Mesocortical	VTA to prefrontal cortex	Cognition, motivation	Underactive in negative and cognitive symptoms
Nigrostriatal	Substantia nigra to caudate and putamen	Movement	Degenerates in Parkinson's disease; blockade gives extrapyramidal effects
Tuberoinfundibular	Hypothalamic arcuate nucleus to median eminence and pituitary	Inhibits prolactin	Blockade gives hyperprolactinaemia

The first antipsychotic, **chlorpromazine**, was developed as an antihistamine; **Jean Delay** and **Pierre Deniker** reported its calming effect in psychotic patients in 1952. **Arvid Carlsson** (Nobel Prize 2000) showed that dopamine is a transmitter, and in 1963 he and **Margit Lindqvist** found that chlorpromazine raises the levels of dopamine's metabolites and argued that it blocks dopamine receptors. Since amphetamine and L-dopa, which raise dopamine activity, can produce psychotic symptoms, the **dopamine hypothesis of schizophrenia** followed. Its key early support came in the mid-1970s, when **Philip Seeman**, **Solomon Snyder** and colleagues showed that the clinical potency of typical antipsychotics correlates with their affinity for the **D2 receptor**. In the revised form, **mesolimbic hyperactivity** produces positive symptoms (delusions, hallucinations) and **mesocortical hypoactivity** negative and cognitive symptoms, which L-dopa does not improve. The **glutamate hypothesis** adds NMDA receptor hypofunction (PCP and ketamine mimic both symptom types); a serotonin view rests on 5-HT2A hallucinogens and atypical drugs. Identical twin concordance is only about 50 percent, so genes are not everything.

EXAM TRAP

The classic hypothesis blames **excess mesolimbic dopamine (D2)**, not decreased serotonin, decreased cerebellar GABA, excess acetylcholine at the neuromuscular junction or glutamate excess (the glutamate view proposes NMDA *hypo*function). Its original support was the potency-D2 correlation, not antipsychotics raising serotonin, 100 percent twin concordance or L-dopa helping negative symptoms.

Antipsychotics (neuroleptics) are the front-line treatment for schizophrenia; CBT, stress management and social skills training are adjuncts (Chapter 7), and neuroleptics are not mainly for depression or anxiety. **Typical antipsychotics** (first generation: chlorpromazine, haloperidol) block D2 receptors; chlorpromazine also blocks D1, and blockade of muscarinic M1 and histamine receptors adds dry mouth and sedation. Because they block D2 in all four pathways, they also cause movement disorders, raise prolactin and may worsen negative symptoms. **Atypical antipsychotics** (second generation: clozapine, risperidone, olanzapine, quetiapine) block 5-HT2A as well as D2: fewer extrapyramidal effects, more weight gain and metabolic syndrome. **Clozapine**, kept for treatment-resistant cases, can cause agranulocytosis (loss of white blood cells), so blood counts are monitored.

The movement effects of nigrostriatal blockade are the **extrapyramidal side effects**, which appear in a fixed order (Table 5.12). **Akathisia** is subjective inner restlessness with an urge to pace; **tardive dyskinesia** comes last and may be irreversible; **neuroleptic malignant syndrome** is a rare emergency. Drug-induced conditions such as akathisia are **secondary movement disorders**, with an identifiable external cause; Huntington's chorea (inherited) and Parkinson's disease are primary.

Table 5.12 Antipsychotic movement side effects in order of onset.

Side effect	Onset	Features
Acute dystonia	Hours to days	Sustained spasms: torticollis, oculogyric crisis
Akathisia	Days to weeks	Inner restlessness, compelling urge to pace
Drug-induced parkinsonism	Weeks	Tremor, rigidity, bradykinesia
Tardive dyskinesia	Months to years	Involuntary lip-smacking, tongue and facial movements, grimacing
Neuroleptic malignant syndrome	Any time, often early	Fever, rigidity, autonomic instability, clouded consciousness

EXAM TRAP

Extrapyramidal effects come from **nigrostriatal** D2 blockade, not mesolimbic (the therapeutic action), mesocortical or tuberoinfundibular blockade. Restlessness soon after starting is **akathisia**; lip-smacking years later is **tardive dyskinesia**, caused by neuroleptics, not antidepressants, anxiolytics or barbiturates.

Antidepressants, mood stabilisers and anxiolytics ●●

The **monoamine hypothesis of depression** holds that depression reflects underactivity at serotonergic and noradrenergic synapses, because the antidepressants in Table 5.13 are all monoamine agonists.

Table 5.13 Antidepressant classes.

Class	Examples	Action	Note
MAO inhibitors	Iproniazid, phenelzine	Block monoamine oxidase	Tyramine 'cheese reaction': hypertensive crisis
Tricyclics	Imipramine, amitriptyline	Block noradrenaline and serotonin reuptake	Anticholinergic effects; dangerous in overdose

Class	Examples	Action	Note
SSRIs (selective serotonin reuptake inhibitors)	Fluoxetine, sertraline	Block serotonin reuptake	Fewer side effects
SNRIs (serotonin-noradrenaline reuptake inhibitors)	Venlafaxine	Block serotonin and noradrenaline reuptake	Some texts use 'SNRI' for selective noradrenaline reuptake inhibitors (reboxetine)
Atypical	Bupropion	Blocks noradrenaline and dopamine reuptake	Also aids quitting smoking

Reuptake blockade raises synaptic serotonin within hours, yet mood lifts only after 2 to 4 weeks. At first the extra serotonin stimulates **5-HT1A autoreceptors**, which damp the firing of serotonin neurons; over weeks these desensitise, firing recovers, and slower changes build up: more brain-derived neurotrophic factor (BDNF) and more hippocampal neurogenesis. The delay is not due to liver enzyme induction, slow entry into the brain or metabolite build-up. An antidepressant such as fluoxetine is no treatment for mania and can trigger it in bipolar patients.

For bipolar disorder the gold standard for treatment and prophylaxis is **lithium**, introduced by the Australian psychiatrist **John Cade** in 1949; it also lowers suicide risk. It has a **narrow therapeutic index**: the useful blood level is about 0.6 to 1.2 mEq/L, and toxicity (coarse tremor, ataxia, confusion) begins just above it and can overlap it. Lithium is cleared by the kidneys, not the liver, so dehydration and renal impairment raise levels, and blood levels, thyroid and kidney function are monitored. Long-term effects include hypothyroidism and nephrogenic diabetes insipidus (thirst, copious urine). Alternatives are the anticonvulsants valproate (divalproex), carbamazepine and lamotrigine. In acute mania, lithium, valproate and atypical antipsychotics such as olanzapine and risperidone are all first line, with haloperidol and the benzodiazepine lorazepam as adjuncts for agitation.

Among anxiolytics, benzodiazepines act within minutes but bring dependence and cross tolerance with alcohol. **Buspirone**, a 5-HT1A partial agonist, is neither sedating nor habit forming but takes weeks to act. **Beta blockers** such as propranolol quiet the bodily signs of anxiety (tremor, palpitations), as in stage fright.

Three drugs help in alcohol dependence. **Disulfiram** (Antabuse) inhibits **aldehyde dehydrogenase**, so acetaldehyde builds up and a drink brings flushing, nausea, headache and palpitations. **Naltrexone**, an opioid antagonist, dampens alcohol's reward and craving, which run partly through endorphin release. **Acamprosate** modulates glutamate (NMDA) transmission and reduces relapse.

> EXAM TRAP
>
> Disulfiram works by **aldehyde dehydrogenase inhibition**, not GABA-A partial agonism, NMDA blockade (closer to acamprosate) or reduced accumbens dopamine (closer to naltrexone). **Fluoxetine** is the odd one out among acute-mania treatments. Lithium toxicity (coarse tremor, ataxia, confusion) overlaps the therapeutic range; constipation and acute liver failure are not its problems, and renal impairment demands monitoring. Cade's lithium report dates from 1949; a 1954 date in some textbooks is an error.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. A heavy drinker who needs more diazepam for the same effect shows _____ tolerance.
2. Benzodiazepines increase the _____ of Cl- channel opening; barbiturates increase its _____.
3. Heroin is a _____ opioid made from morphine.
4. THC acts at _____ receptors, densest in the hippocampus.
5. The reward pathway runs from the _____ to the nucleus accumbens.
6. Extrapyramidal side effects follow D2 blockade in the _____ pathway.
7. The last extrapyramidal effect to appear, often irreversible, is _____.
8. Disulfiram inhibits the enzyme _____.

Answers: (1) cross, (2) frequency; duration, (3) semisynthetic, (4) CB1, (5) ventral tegmental area, (6) nigrostriatal, (7) tardive dyskinesia, (8) aldehyde dehydrogenase.

5.5 Research Methods, Behaviour Genetics and the Developing Brain

MEDIUM YIELD · 25 QS

Every fact in this chapter came from a method, and each method sees the brain differently: its shape, its activity, or the timing of that activity. This section surveys those methods, then asks how genes are inherited and studied, which genetic errors derail development, and how the brain builds and keeps rewiring itself.

Imaging the living brain ●●

Ordinary X-rays show little of the soft brain, so **contrast X-ray** techniques inject a substance that absorbs X-rays differently: in **cerebral angiography** a radio-opaque dye in a cerebral artery outlines the blood vessels, revealing vascular damage and, by displaced vessels, tumours. **Computed tomography** (CT), introduced in the early 1970s by **Godfrey Hounsfield**, passes X-rays through the head from many angles and computes horizontal slices. It is fast and shows fresh blood, so it is the first scan after acute head injury. **Magnetic resonance imaging** (MRI) uses a strong magnetic field and radio waves: hydrogen atoms then emit signals that differ between structures. With the highest **spatial resolution** (precision about where), MRI gives the most detailed view of brain structure. **Diffusion tensor imaging** (DTI), an MRI method, follows water diffusing along axons to map the white matter tracts.

CT and plain MRI are structural; **functional imaging** shows activity. **Positron emission tomography** (PET) injects a radioactive tracer such as 2-deoxyglucose (FDG in human PET), which active neurons absorb like glucose but cannot metabolise, so the scan maps metabolic activity. **Single-photon emission computed tomography** (SPECT) maps blood flow with a gamma-emitting tracer, more cheaply but less sharply. **Functional MRI** (fMRI) needs no injection: it detects the **BOLD signal** (blood-oxygen-level-dependent; **Seiji Ogawa**, 1990), which rises as active regions draw oxygenated blood. Its spatial resolution is good but its **temporal resolution** (precision about when) only moderate, because blood flow takes seconds to respond. **Functional near-infrared spectroscopy** (fNIRS) measures cortical oxygenation with

light shone through the scalp; it is portable and suits infants. Table 5.14 compares the methods.

Table 5.14 Where and when: the main methods compared (figures approximate).

Method	Spatial resolution	Temporal resolution
CT (structural)	Good	None (still image)
MRI (structural)	Best (under 1 mm)	None (still image)
PET (functional)	Moderate (4 to 5 mm)	Poor (about a minute or more)
SPECT (functional)	Poorer (about 1 cm)	Poor (minutes)
fMRI (functional)	Good (1 to 3 mm)	Moderate (seconds)
EEG	Poor (centimetres)	Best (milliseconds)
MEG	Better than EEG	Best (milliseconds)

A working brain does many things at once, so the **paired-image subtraction technique** images two tasks that differ in a single process (reading nouns aloud versus saying a related verb for each) and subtracts one image from the other. Subtraction revealed regions more active in the resting baseline than during tasks: the **default mode network** (DMN), after the "default mode" of brain function described by **Marcus Raichle** and colleagues in 2001. Its hubs are the medial prefrontal cortex, the posterior cingulate cortex and precuneus, and the lateral parietal cortex. It is most active at rest, during mind-wandering and self-referential thought, deactivates during externally focused, goal-directed tasks (moving, seeing, calculating), and is altered in depression, Alzheimer's disease and schizophrenia.

> EXAM TRAP
>
> The functional techniques are **PET and SPECT** (and fMRI), not CT or plain MRI, so "all of the above" is wrong. For brain structure after head injury choose **MRI**; CT is the quick emergency scan for bleeding. EEG, PET and fMRI record activity, not structure.

Recording electrical activity ●●

Imaging shows where; electrical recording shows when. The **electroencephalogram** (EEG), first reported in humans by **Hans Berger** in 1929, records through scalp electrodes the summed electrical activity of cortical neurons. The brain's counterpart of the electrocardiogram (ECG), it has the highest temporal resolution, in milliseconds, but poor spatial resolution, and it is the standard test for epilepsy and sleep (wave bands in Section 5.8). **Magnetoencephalography** (MEG) records the magnetic fields of the same currents: as fast, better at localising, far costlier. In animals, a microelectrode records one neuron's action potentials (**single-unit recording**) or a larger electrode the firing of many neighbours (**multiple-unit recording**).

The response to a single click is lost in the background EEG, so the EEG after hundreds of clicks is averaged (**signal averaging**): random activity cancels and the time-locked **event-related potential** (ERP) remains, named by polarity (P or N) and latency in milliseconds. In the **oddball paradigm**, a rare target among frequent standard stimuli evokes a large positive wave about 300 ms later, the **P300**, first described by **Samuel Sutton** and colleagues in 1965. It

reflects attention allocation and stimulus categorisation, and it is smaller and later in schizophrenia and Alzheimer's disease (Table 5.15).

Table 5.15 Event-related potentials to know.

Component	Timing	Reflects	Note
P50	50 ms, positive	Sensory gating: the response to a second click is normally suppressed	Gating deficient in schizophrenia
N100	100 ms, negative	Early sensory response	Larger when attended
P300	About 300 ms, positive	Attention allocation, categorisation (oddball)	Smaller, delayed in schizophrenia and Alzheimer's
Bereitschaftspotential (readiness potential)	Slow negative rise about 1 s before a voluntary movement	Preparation to move (**Hans Kornhuber** and **Lüder Deecke**, 1965)	Precedes the act; not the motor command

EXAM TRAP

Direct electrical recording "like the ECG" is **EEG**; MEG records magnetic fields, fMRI and PET blood and metabolism. No ERP marks REM sleep onset (Section 5.8) or issues motor commands (Section 5.6); the first visual cortical response is an early component, not the P300.

The **electromyogram** (EMG) records muscle tension and the **electrooculogram** (EOG) eye movements. **Skin conductance** (electrodermal activity, the **galvanic skin response**) rises as sympathetic arousal activates the sweat glands. Heart rate and blood pressure are other autonomic indices; the **polygraph** records several at once and, as a "lie detector", detects arousal, not lies.

Lesions, stimulation and genetic tools

Invasive methods, used mostly in animals, begin with **stereotaxic surgery**: a **stereotaxic atlas** gives each structure's coordinates, and a **stereotaxic instrument** holds the head and guides an electrode or cannula to them. A **lesion** destroys tissue to reveal its function by what is lost: **aspiration** sucks away cortex, **radio-frequency** current heats small subcortical areas, **knife cuts** sever tracts, and **cryogenic blockade** (cooling) makes a reversible lesion. Neurotoxins are selective: **kainic acid** kills cell bodies but spares passing axons, and **6-hydroxydopamine** destroys dopamine and noradrenaline neurons. **Bilateral** lesions usually have far larger effects than **unilateral** ones.

Conversely, **electrical stimulation** or **chemical stimulation** (a microinjected transmitter or agonist) can evoke behaviour. **Deep brain stimulation** (DBS) applies continuous current through implanted electrodes, chiefly in Parkinson's disease. **Transcranial magnetic stimulation** (TMS) induces current in the cortex from a coil on the scalp, a non-invasive "virtual lesion"; repeated TMS treats depression. **Optogenetics** inserts genes for light-sensitive ion channels into one type of neuron, so light switches those neurons on or off. **Microdialysis** samples extracellular fluid in behaving animals to measure transmitter release. In humans, neuropsychological tests (Chapter 2) also map damage.

Genetic methods alter genes. A **gene knockout** mouse lacks a working gene; a **knock-in** replaces or adds one; a **transgenic** animal carries another species' gene, such as a human disease gene. **CRISPR-Cas9**, developed by **Jennifer Doudna** and **Emmanuelle Charpentier** in 2012 (Nobel Prize in Chemistry, 2020), uses a guide RNA to lead the Cas9 protein to a chosen DNA sequence, where it edits the gene or switches it on or off. Other tools do not edit: the **polymerase chain reaction** (PCR) of **Kary Mullis** copies DNA millions of times, **gel electrophoresis** separates DNA or proteins by size, and **Western blotting** detects a particular protein.

Mendel, chromosomes and epigenetics ●

Gregor Mendel (1822 to 1884), a monk who crossed pea plants and published in 1866, found that each trait depends on a pair of inherited factors, now called **genes**, one from each parent. The alternative forms of a gene are **alleles**: two alike make an organism **homozygous** (DD or dd), two different **heterozygous** (Dd). The **genotype** is the alleles carried, the **phenotype** the trait shown. A **dominant** allele is expressed even in a heterozygote; a **recessive** trait appears only in the homozygote, as the **Punnett square** in Figure 5.9 shows. Blending occurs only in **incomplete dominance** (red and white snapdragons give pink) and **codominance** (blood group AB shows both), so with a stated dominant allele "blue-green" is wrong.

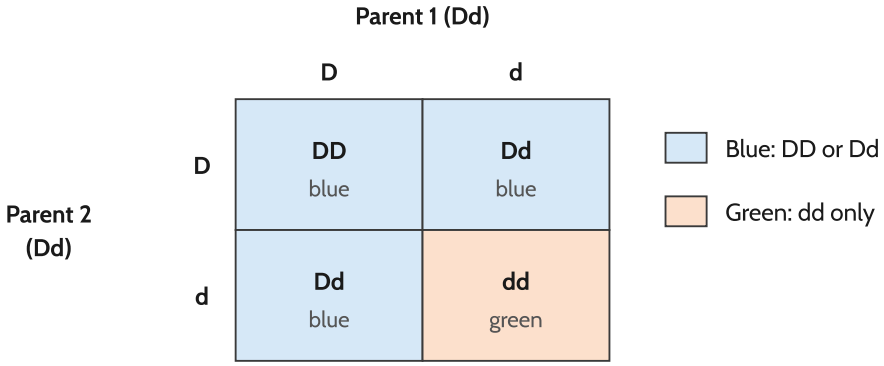

Figure 5.9 Two Dd parents, with blue (D) dominant over green (d): genotypes 1 DD : 2 Dd : 1 dd, phenotypes 3 blue : 1 green. Every Dd child has blue eyes. (The alleles are hypothetical.)

Genes lie on **chromosomes**, strands of **DNA** (deoxyribonucleic acid) whose base sequence codes for proteins. Human cells have 23 pairs: 22 pairs of **autosomes** and the **sex chromosomes**, XX in females and XY in males. **Meiosis** makes eggs and sperm with 23 single chromosomes, and **crossing over** (paired chromosomes swapping segments) makes each one unique. A **mutation** is an accidental change in a gene. X-linked recessive **sex-linked traits**, such as red-green colour blindness, are commoner in males, who have no second X to mask them. **Epigenetics** studies changes in gene expression without any change in the DNA sequence: **DNA methylation** usually silences genes, and **histone modification** loosens or tightens the packaging of DNA, making genes more or less readable. Experience, such as early stress or maternal care, alters these marks.

Separating heredity and environment ●●

Family studies ask whether risk rises with closeness of kinship, but relatives share homes as well as genes. **Twin studies** compare **monozygotic twins** (MZ), from one fertilised egg that

splits, who have the same heredity and share virtually all their genes, with **dizygotic twins** (DZ), who share about 50 percent like other siblings. The **concordance rate** is the percentage of pairs in which, when one twin has a disorder, the other has it too: for schizophrenia, about 48 percent in MZ and 17 percent in DZ twins in the pooled data of **Irving Gottesman** (1991). **Adoption studies** ask whether adoptees resemble their biological parents (genes) or adoptive parents (environment).

KEY STUDY: THE MINNESOTA STUDY OF TWINS REARED APART

From 1979, **Thomas Bouchard** and colleagues (main report in *Science*, 1990) gave adult MZ and DZ twins separated in infancy about 50 hours of intelligence and personality tests. MZ twins reared apart were about as alike as those reared together, and more alike than DZ twins, on almost every measure. Critics note that adoptive homes vary less than homes in general.

EXAM TRAP

Twins reared apart combine twin and adoption logic (Pinel even calls the Minnesota study an adoption study), but the keyed answer is **twin studies**, because comparing MZ with DZ pairs defines the design. Choose adoption studies only if twin studies is not offered.

Linkage studies trace a DNA marker inherited along with a disorder through affected families, locating the gene nearby. **Genome-wide association studies** (GWAS) compare hundreds of thousands of variants across the genome in thousands of patients and controls, usually finding many genes of tiny effect. **Heritability** is the proportion of variation in a trait, in a given population and environment, due to genetic differences; it describes populations, not individuals, and changes when environments change.

Diagnoses are too broad to map onto genes, so Gottesman and **Todd Gould** (2003) popularised the **endophenotype**: a measurable, heritable intermediate trait between genes and symptoms, through which genetic risk produces a disorder. P50 gating, P300 and smooth-pursuit eye-tracking deficits in schizophrenia are examples, also found in unaffected relatives. By contrast, the **phenotype** is the observable symptoms, the **genotype** the genes, and **molecular genetics** the study of DNA itself. The diathesis-stress model is taught in Chapter 8.

Single-gene and chromosomal disorders ●●●

Phenylketonuria (PKU), discovered in 1934 by the Norwegian **Ivar Asbjørn Følling**, is an autosomal recessive lack of **phenylalanine hydroxylase**, the enzyme that converts phenylalanine into **tyrosine**. Phenylalanine accumulates and damages the developing brain, but newborn screening and a low-phenylalanine diet prevent the intellectual disability. In a **trinucleotide repeat** disorder a three-base sequence is repeated too often; in **genomic imprinting** a gene is expressed only from one parent's copy, so one deletion causes different syndromes depending on its parent of origin (Table 5.16).

Table 5.16 Single-gene and chromosomal disorders.

Disorder	Cause	Inheritance	Key features
Phenylketonuria	No phenylalanine hydroxylase, so no tyrosine from phenylalanine	Autosomal recessive	Intellectual disability unless diet from birth

Disorder	Cause	Inheritance	Key features
Huntington's disease	CAG repeat in the HTT gene, chromosome 4 (marker 1983, gene 1993)	Autosomal dominant	Chorea, dementia, onset about 40 (Section 5.11)
Fragile X syndrome	CGG repeat on the X silences the FMR1 gene	X-linked	Commonest inherited intellectual disability
Down syndrome	Trisomy 21	Usually new; risk rises with maternal age	Intellectual disability, early Alzheimer's
Turner syndrome	45,X: one X missing	Sex chromosome error	Short, infertile female
Klinefelter syndrome	47,XXY: extra X	Sex chromosome error	Tall male, small testes, infertility, learning difficulties
Prader-Willi syndrome	Deletion at 15q11 to q13, paternal chromosome	Imprinting	Hyperphagia, obesity, short stature, hypotonia
Angelman syndrome	Same deletion, maternal chromosome	Imprinting	Severe intellectual disability, absent speech, seizures, laughter

EXAM TRAP

Asperger's syndrome (autism spectrum, Chapter 8) and transsexualism (a gender identity term) are not chromosomal syndromes. PKU blocks tyrosine: urea is a waste product of protein breakdown, tryptophan the precursor of serotonin, histidine of histamine. CGG and FMR1 mean fragile X; CAG on chromosome 4 means Huntington's.

Building the nervous system

After fertilisation the **zygote** divides into a solid ball (the **morula**) and then a hollow ball that implants (the **blastocyst**); these stages (Chapter 6) contain no nervous tissue. The nervous system begins at **gastrulation**, in the third week, when the embryo forms three germ layers: ectoderm, mesoderm and endoderm. Induced by the mesoderm beneath, a patch of ectoderm becomes the **neural plate**, which folds into the neural groove, whose lips fuse into the **neural tube**, forerunner of the brain and spinal cord. Failures of this **neurulation** cause **spina bifida** (tail end) and **anencephaly** (head end).

Development then proceeds in phases: **proliferation** in the **ventricular zone** lining the tube, **migration** of young neurons along **radial glial cells**, **aggregation** into structures, and axon growth led by **growth cones**. **Roger Sperry**, who rotated frogs' eyes and found that regenerating optic axons regrew to their original targets, proposed the **chemoaffinity hypothesis** (1963): each target releases a chemical label that attracts its axon. After synapses form, far more neurons exist than survive; in **apoptosis** (programmed cell death) those that fail to obtain **neurotrophins** from their targets die. **Rita Levi-Montalcini** discovered the first, **nerve growth factor** (Nobel Prize 1986). In **synapse rearrangement**, surviving axons take over the vacated sites.

EXAM TRAP

The nervous system begins forming at the **gastrula** stage. Zygote, morula and blastocyst come earlier.

Pruning, myelination and lifelong plasticity ●●

The young brain overproduces synapses; childhood synaptic density far exceeds the adult level. **Synaptic pruning** (neural pruning) then eliminates unused connections and keeps those in use: "use it or lose it". It is heaviest in childhood and continues in the prefrontal cortex through adolescence, partly carried out by **microglia**, which engulf unwanted synapses. **Myelination** also continues into early adulthood, the prefrontal cortex last, one reason judgement and impulse control mature late.

> EXAM TRAP
>
> Pruning is the **elimination of unused connections**, a process of neuronal development and plasticity. It is not memory consolidation, not long-term potentiation or any strengthening of pathways (both Section 5.9), not neurotransmitter release (Section 5.3), not the growth of new neurons and not an increase in brain size.

Neuroplasticity is the brain's ability to change its structure and function throughout life with experience, learning and injury; it is not a protective coating (skull, meninges or myelin), memory loss or cell death. In the 1960s **Mark Rosenzweig** and colleagues found that rats reared in **enriched environments** developed a thicker, heavier cortex than rats in barren cages. Plasticity peaks in **critical periods**: when **David Hubel** and **Torsten Wiesel** sewed one eye of a young kitten shut (**monocular deprivation**), the visual cortex permanently lost its responses to that eye, whereas adult cats were barely affected. Adults still make new neurons (**adult neurogenesis**) in the hippocampal dentate gyrus and, in rodents, the olfactory bulb; **Peter Eriksson** and colleagues confirmed it in the human hippocampus in 1998. **Brain-derived neurotrophic factor** (BDNF) supports synaptic plasticity, neuronal survival (it prevents apoptosis) and hippocampal neurogenesis, not action potential firing (Section 5.3) or myelin synthesis. Exercise and antidepressants raise it; chronic stress lowers it (Section 5.7).

> SCAN YOUR BRAIN
>
> Fill in the blanks. The answers are below; cover them first.
>
> 1. The brain's counterpart of the ECG, with the highest temporal resolution, is the _____.
> 2. CT and plain MRI are structural; PET and _____ are functional.
> 3. A positive wave about 300 ms after a rare oddball stimulus is the _____.
> 4. The tool that edits or switches genes at chosen DNA sequences is _____.
> 5. A heterozygous Dd child, with D dominant, shows the _____ trait.
> 6. The karyotype 47,XXY gives _____ syndrome.
> 7. In PKU, phenylalanine is not converted into _____.
> 8. The nervous system begins forming at the _____ stage.
>
> **Answers:** (1) EEG, (2) SPECT, (3) P300, (4) CRISPR-Cas9, (5) dominant, (6) Klinefelter, (7) tyrosine, (8) gastrula.

5.6 Sensory Pathways and the Motor System LOWER YIELD · 22 QS

Section 1.3 followed light, sound and touch to the sense organs. This section follows their signals into the brain, then traces motor commands back out to the muscles.

Principles of sensory systems

Every sensory system begins with **transduction**, the conversion of physical energy into neural signals by receptors. A neuron's **receptive field** is the area of the sensory surface (a patch of retina or skin) in which a stimulus changes its firing. The systems are hierarchical: receptors, thalamic relay nuclei, **primary sensory cortex** (input mainly from the thalamus), **secondary sensory cortex** (input mainly from primary cortex of the same sense) and **association cortex** (input from several senses). Each level contains areas with different jobs (**functional segregation**), and information flows along several routes at once (**parallel processing**).

Refractive errors •

The cornea and lens must focus the image on the retina; their failures are **refractive errors**, optical rather than neural faults. In **myopia** (near-sightedness) distant objects blur because the eyeball is too long, so their image focuses in front of the retina; a concave lens corrects it. In **hypermetropia** (far-sightedness) a short eyeball focuses near objects behind the retina; a convex lens corrects it. **Astigmatism** is an irregularly curved cornea that blurs some orientations, and **presbyopia** the stiffening of the ageing lens that brings reading glasses after about 40. Visual field defects, by contrast, are blind areas left by pathway damage (5.6.4).

Retinal circuits and lateral inhibition •

The retina is built inside out (the **inverted retina**): light passes the ganglion, amacrine, bipolar and horizontal cells, in that order, before reaching the rods and cones at the back. Signals return along the **through pathway**, receptors to bipolar cells to ganglion cells, whose axons form the optic nerve, while **horizontal** and **amacrine cells** carry signals sideways. Where the axons leave, at the optic disc, there are no receptors: the blind spot, filled in by **completion**.

Stephen Kuffler (1953) found that cat ganglion cells have circular receptive fields with an opposing centre and surround: an **on-centre cell** fires to light on its centre and is inhibited by light on its surround, and an off-centre cell does the reverse. Such cells signal contrast, not even illumination.

The mechanism is **lateral inhibition**, the suppression of a neuron by its active neighbours, analysed by **Haldan Keffer Hartline** in the eye of the horseshoe crab *Limulus* (Nobel Prize 1967). At a light-dark border, a receptor just on the bright side has dark neighbours, receives less inhibition and signals "brighter still"; one just on the dark side receives extra inhibition and signals "darker still". The result is the illusory bright and dark strips called **Mach bands** (**Ernst Mach**, 1865).

> **EXAM TRAP**
>
> Mach bands come from lateral inhibition, not the rod-cone dichotomy (dim-light versus colour vision), the opponent process (colour afterimages) or temporal summation (adding weak signals over time).

From retina to striate cortex

At the **optic chiasm**, fibres from the nasal half of each retina cross; temporal fibres do not. Because nasal hemiretinas see the outer visual fields, the **left visual field reaches the right hemisphere**. Each optic tract ends in the **lateral geniculate nucleus** (LGN) of the thalamus, the visual relay (hearing uses the medial geniculate nucleus, touch the ventral posterior nucleus, and smell no thalamic relay).

The LGN has six layers. Layers 1 and 2 are **magnocellular**: large, fast, colour-blind cells with large receptive fields, for motion, contrast and depth. Layers 3 to 6 are **parvocellular**: small, slower cells for colour and fine detail. Koniocellular cells lie between the layers. Optic radiations carry the output to the **primary visual cortex** (V1, striate cortex, area 17) around the calcarine fissure. V1 is **retinotopic**, and the tiny fovea claims about a quarter of it.

Field loss therefore locates damage (Figure 5.10). An optic nerve lesion blinds one eye. A pituitary tumour pressing on the chiasm cuts the crossing fibres: **bitemporal hemianopia**. Damage behind the chiasm (tract, radiations or V1) causes **contralateral homonymous hemianopia**, the same opposite half-field lost in both eyes. A posterior cerebral artery stroke of V1 often spares central vision (**macular sparing**), because the occipital pole also draws blood from the middle cerebral artery. A small V1 lesion leaves a **scotoma**, an island of blindness often hidden by completion; in **blindsight**, studied by **Lawrence Weiskrantz** in the patient D.B. (1974), people point accurately at stimuli in the scotoma that they deny seeing.

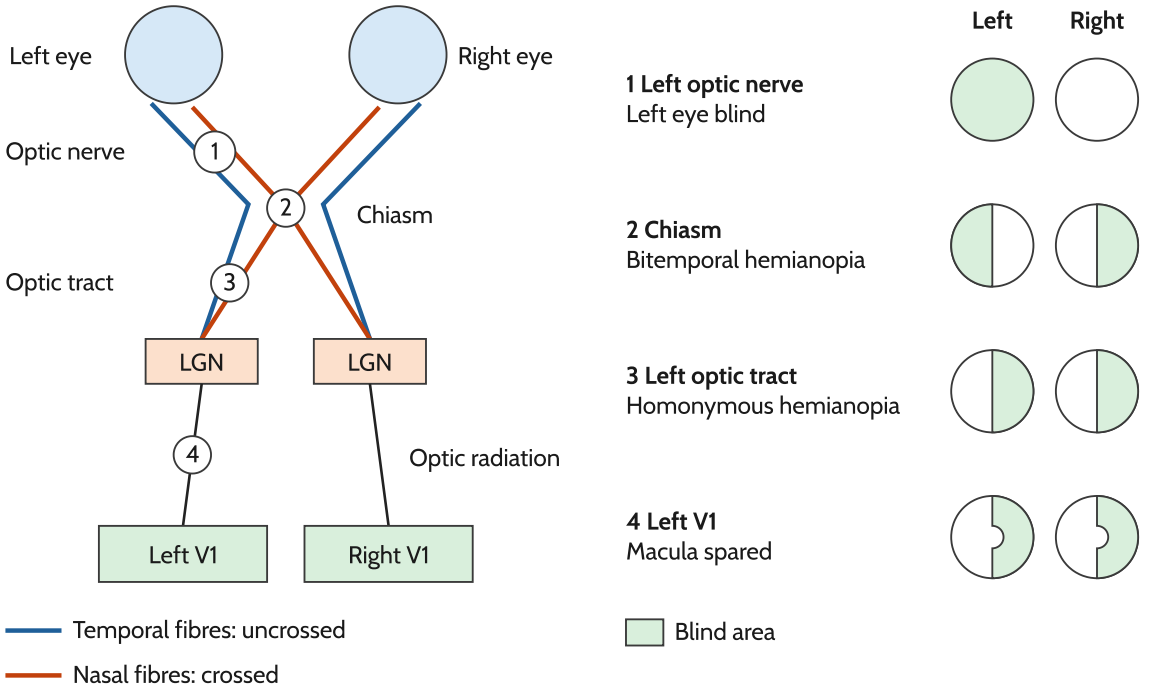

Figure 5.10 The visual pathway seen from above, and the field loss from damage at four sites. Circles show each eye's view as the patient sees it; shading marks blindness.

EXAM TRAP

The LGN relays vision only. Magnocellular is motion, parvocellular colour and detail; the two differ, and neither projects only to V4. Hemianopia with macular sparing means V1, not the chiasm, optic tract or optic nerve.

Visual cortex and the two streams

In V1, **David Hubel** and **Torsten Wiesel** (Nobel Prize 1981; Section 1.4) found **simple cells**, responding to a line of one orientation in one exact position, and **complex cells**, responding to that orientation anywhere in a larger field, often only when it moves. V2 (area 18) elaborates contours and feeds specialised areas: V4 for colour (damage causes **cerebral achromatopsia**, a grey world) and V5, or MT, for motion (damage causes **akinetopsia**, movement seen as still frames).

Leslie Ungerleider and **Mortimer Mishkin** (1982) grouped these areas into two streams leaving V1. The **dorsal stream** runs to the posterior parietal cortex: "where" things are. The **ventral stream** runs to the inferotemporal cortex: "what" they are, faces included (the fusiform face area, Section 5.11). **Melvyn Goodale** and **David Milner** (1992) recast the dorsal stream as "how", vision for guiding action such as grasping, and the ventral stream as vision for conscious perception.

EXAM TRAP

The streams are not colour against motion (motion is dorsal) or hearing against vision, and they diverge from V1 rather than converging on it.

Hearing, touch, pain, smell and taste ●●

The **hair cells** of the organ of Corti transduce mechanically: basilar membrane movement bends their stereocilia, opening cation channels through which potassium from the endolymph flows in and depolarises the cell. Inner hair cells supply almost all auditory nerve signals; outer hair cells amplify the vibration. The pathway runs from the cochlear nuclei to the **superior olives** on both sides, where the two ears' signals first meet and are compared to locate sounds, then to the inferior colliculi, the medial geniculate nucleus and the primary auditory cortex in Heschl's gyrus (areas 41 and 42), which is **tonotopic** (each frequency has its own place). **Conductive deafness** arises in the outer or middle ear, **sensorineural deafness** in the hair cells or auditory nerve.

Table 5.17 lists the skin's receptors; **proprioceptors**, the muscle spindles and Golgi tendon organs (5.6.8), signal body position.

Table 5.17 Skin receptors.

Receptor	Detects	Adapts
Free nerve endings (nociceptors)	Pain and temperature	Slowly
Pacinian corpuscles (deep)	Vibration, deep pressure	Fast
Meissner's corpuscles (fingertips)	Light touch	Fast
Merkel's discs	Sustained touch and pressure	Slowly
Ruffini endings	Skin stretch	Slowly

The **dorsal column-medial lemniscus system** carries touch, vibration and proprioception; it ascends uncrossed and crosses in the medulla. The **anterolateral system** (spinothalamic tract and others) carries nociceptive pain and temperature; it crosses in the spinal cord near the level of entry. Both relay in the ventral posterior nucleus of the thalamus to **primary somatosensory cortex** (SI; postcentral gyrus, areas 3, 1 and 2). **Wilder Penfield** and **Theodore Rasmussen** (1950) mapped its **somatosensory homunculus** by stimulating patients' cortex during surgery: the hands, lips and tongue, used for fine discrimination, have the largest areas, the trunk, hips and feet small ones. Secondary somatosensory cortex (SII), below SI, receives input from both sides of the body.

Pain has no single cortical centre; the anterior cingulate cortex tracks its unpleasantness. It can be blocked from above: **David Reynolds** (1969) operated on rats with no pain relief but electrical stimulation of the **periaqueductal grey** (PAG) (**stimulation-produced analgesia**). Opioid receptors were then found in the PAG, endorphins were isolated, and a **descending analgesia** circuit was traced from the PAG through the serotonergic raphe nuclei to the dorsal horn, where interneurons block incoming pain (compare the gate-control theory of Ronald Melzack and Patrick Wall, 1965, Section 1.3). **Phantom limbs** reflect cortical reorganisation: **Vilayanur Ramachandran** showed the face area of SI invading the lost hand's territory, and his mirror box eases phantom pain.

Smell alone reaches the cortex without a thalamic relay: the olfactory bulb projects to the **piriform cortex** and amygdala, and on to the orbitofrontal cortex. Taste travels in cranial nerves VII, IX and X to the solitary nucleus of the medulla, the ventral posterior thalamus and the gustatory cortex in the insula.

EXAM TRAP

Pain uses free nerve endings (Merkel, Meissner and Ruffini receptors serve touch) and the anterolateral system (not the route for proprioception, vibration and touch). Hair cells work by mechanical bending and potassium entry, not sodium-channel closure (the photoreceptor's response to light), GABA or chemical signalling. Phantoms are cortical remapping, not nerve regrowth, denial or cerebellar damage.

The sensorimotor hierarchy and mirror neurons ●●

The motor system is a hierarchy run in reverse, from goals down to muscles. At the top, posterior parietal cortex integrates where the body and nearby objects are (damage causes apraxia and neglect, Section 5.11), and dorsolateral prefrontal cortex decides to initiate voluntary responses. **Secondary motor cortex**, the premotor cortex and supplementary motor area (area 6), programmes patterns of movement and fires just before movement starts. **Primary motor cortex** (area 4, precentral gyrus), the main cortical exit, is laid out as a **motor homunculus** with outsized hands and mouth. Brainstem nuclei and spinal circuits execute. Primary, secondary and supplementary areas thus all control movement. The levels are functionally segregated and parallel, output is guided by sensory feedback, and practice hands control down to lower levels.

KEY STUDY: RIZZOLATTI'S MIRROR NEURONS (1990S)

Giacomo Rizzolatti and colleagues found neurons in area F5 of macaque ventral premotor cortex that fired both when the monkey grasped an object and when it watched an experimenter grasp it: **mirror neurons**, active for performing and for observing the same action. Humans show a mirror system in premotor cortex, Broca's area (inferior frontal gyrus) and the inferior parietal lobule, proposed to support imitation, action understanding and empathy. Criticism: human single-cell evidence is scarce, and the "broken mirror" theory of autism is debated.

EXAM TRAP

Mirror neurons fire for doing and watching, not one alone, and not in REM sleep. The system is not in the hippocampus, amygdala or cerebellum.

Descending pathways and spinal circuits ●●

Four tracts descend from the motor cortex (Table 5.18). The corticospinal, or pyramidal, tract passes through the pyramids of the medulla, and at the **pyramidal decussation** in the lower medulla about 85 to 90 percent of its fibres cross, so each hemisphere moves the opposite side of the body.

Table 5.18 The descending motor tracts.

Tract	Crossing	Controls
Dorsolateral corticospinal	Medullary pyramids	Limbs, fine finger movement
Dorsolateral corticorubrospinal (via red nucleus)	Brainstem	Distal limb muscles
Ventromedial corticospinal	Uncrossed	Trunk, proximal muscles

Tract	Crossing	Controls
Ventromedial cortico-brainstem-spinal	Bilateral	Posture, walking

Upper motor neurons run from the motor cortex down the corticospinal tract; lower motor neurons run from the anterior horn of the spinal cord to the muscle (Table 5.19). An upper lesion frees spinal reflexes from descending inhibition, so tone and reflexes rise and stroking the sole makes the big toe extend (an extensor plantar response), the **Babinski sign** (**Joseph Babinski**, 1896). Amyotrophic lateral sclerosis destroys both kinds of neuron and shows both patterns.

Table 5.19 Upper against lower motor neuron lesions.

Feature	Upper motor neuron	Lower motor neuron
Weakness	Spastic, tone raised	Flaccid, hypotonia
Reflexes	Brisk; Babinski sign	Reduced or absent
Muscle	Little wasting	Atrophy, fasciculations

Spinal circuits also work alone. A tendon tap stretches a muscle, and its muscle spindles, which detect length, excite its motor neurons through a single synapse: the **stretch reflex**, which holds limbs steady against loads. Golgi tendon organs detect tension and relax a dangerously contracting muscle. **Reciprocal innervation** relaxes a muscle's antagonist as it contracts. **Central pattern generators** produce rhythms: **Sten Grillner** (1985) showed that cats whose spinal cords were cut off from the brain could walk on a treadmill.

> EXAM TRAP
>
> The corticospinal tract crosses at the medullary pyramids, not at the optic chiasm (optic fibres), the spinal level of entry (pain fibres), the pons, the cerebral peduncles or the third ventricle, and "it does not cross" is false.

Basal ganglia and cerebellum

Neither the basal ganglia nor the cerebellum sends commands to the muscles; both modulate the cortex through the thalamus. In the basal ganglia, the striatum (caudate and putamen) receives cortical input, and the output nuclei, the internal globus pallidus (GPi) and substantia nigra pars reticulata (SNr), constantly inhibit the thalamus (Figure 5.11).

In the **direct pathway**, cortex excites striatal D1 neurons, which inhibit GPi and SNr; the thalamus, released, excites the cortex and movement is facilitated. In the **indirect pathway**, striatal D2 neurons inhibit the external globus pallidus (GPe), freeing the subthalamic nucleus (STN) to excite GPi; the thalamus is inhibited and movement suppressed. Dopamine from the substantia nigra pars compacta excites the direct and inhibits the indirect pathway. Its loss causes the slow, scarce movement of Parkinson's disease, a **hypokinetic** disorder; loss of indirect-pathway neurons causes the chorea of Huntington's disease, a **hyperkinetic** one (Section 5.11). The basal ganglia thus perform **action selection**, initiate and gate voluntary movement, and learn habits, motor sequences and rewarded actions. Depth perception, memory consolidation and cardiorespiratory control belong to the visual cortex, hippocampus and medulla.

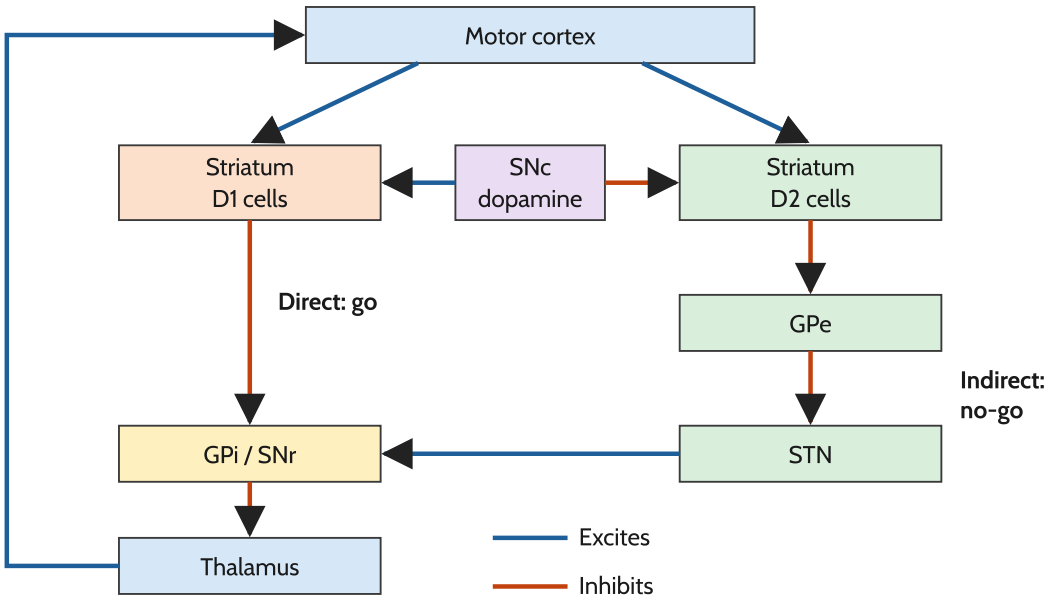

Figure 5.11 The basal ganglia loops. GPi and SNr brake the thalamus; the direct pathway lifts the brake, and the indirect pathway, through GPe and STN, tightens it. Dopamine favours movement through both.

The cerebellum compares intended with actual movement and corrects the error: **climbing fibres** from the inferior olive carry error signals to its Purkinje cells. It coordinates and times movement, maintains balance, and supports motor adaptation and eyeblink conditioning (Section 5.9). Damage causes **ataxia** (clumsy, unsteady movement), intention tremor (worse as the hand nears its target) and dysmetria (over- or undershooting).

Table 5.20 Cerebellum against basal ganglia.

Feature	Cerebellum	Basal ganglia
Core job	Error correction, coordination	Action selection, gating
Learning	Motor adaptation, eyeblink	Habits, procedural, reward-based
Damage	Ataxia, intention tremor	Parkinson's, Huntington's

> **EXAM TRAP**
>
> Climbing-fibre error correction is cerebellar; habit and reward-based selection are basal ganglia jobs. The route through GPe and STN is the indirect pathway, hippocampus-fornix-thalamus is a memory circuit, and cortex to cerebellum to spinal cord is no basal ganglia route.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Mach bands are explained by _____ in the retina.
2. A pituitary tumour pressing on the optic chiasm causes _____ hemianopia.
3. Motion and contrast are carried by the _____ layers of the LGN.
4. The ventral "what" stream runs from V1 to the _____ cortex.
5. Pain and temperature ascend in the _____ system.
6. Most corticospinal fibres cross at the _____ in the lower medulla.
7. Climbing fibres carry error signals to the cerebellum's _____ cells.

Answers: (1) lateral inhibition, (2) bitemporal, (3) magnocellular, (4) inferotemporal, (5) anterolateral, (6) pyramidal decussation, (7) Purkinje.

5.7 The Endocrine System, Stress and the Emotional Brain

HIGH YIELD · 46 QS

A bus swerves towards you on a Mysuru road. Before you have thought a word, your heart pounds and your palms are wet, and an hour later you are still shaky: nerves reacted in milliseconds, but hormones in your blood kept you on edge for hours. This section follows the hormones from the hypothalamus to the glands and the stress systems they build, then the brain circuits that turn a threat into fear.

Glands, ducts and hormones

Glands come in two kinds. **Exocrine glands** are **duct glands**: they deliver their secretions through ducts onto a surface or into a cavity, and those secretions are not hormones. Sweat, salivary, tear and digestive glands are exocrine. **Endocrine glands** are **ductless**: they release **hormones**, chemical messengers carried by the blood to distant cells that have receptors for them. The **pancreas** is both: most of it sends digestive enzymes down a duct, while islands of cells within it release insulin into the blood.

Hormones belong to three chemical classes. **Amino acid derivatives** are made in a few steps from one amino acid (adrenaline, thyroxine, melatonin). **Peptide and protein hormones** are short and long chains of amino acids (insulin, oxytocin, vasopressin, growth hormone, ACTH). **Steroid hormones** are made from cholesterol: the gonadal hormones (testosterone, oestrogens, progesterone) and the adrenal cortex hormones (cortisol, aldosterone). Most hormones act on receptors in the cell membrane, but steroids, small and fat-soluble, also enter the cell, bind receptors in the cytoplasm or nucleus and change **gene expression** directly, so their effects are the most diverse and long-lasting.

EXAM TRAP

Exocrine glands are **duct glands**; endocrine glands are **ductless**. "Primary" is no class of gland, and sweat, saliva and tears are never hormones.

The hypothalamus and the pituitary ●●●

The **pituitary gland**, a pea-sized gland hanging from the base of the brain, is called the **master gland** because most of its hormones are **tropic hormones**, which make other glands release theirs; it also releases growth hormone, which regulates growth. Yet the master gland has a master: the **hypothalamus** above it (Section 5.2), which integrates the endocrine, autonomic and visceral systems. In an emergency the hypothalamus supervises the fight-or-flight response, driving the sympathetic nervous system and the pituitary at once.

The pituitary is really two glands, controlled in two different ways (Table 5.21). The **anterior pituitary** (**adenohypophysis**) is glandular tissue that releases growth hormone, adrenocorticotropic hormone (ACTH), thyroid-stimulating hormone (TSH), the gonadotropins (follicle-stimulating hormone, FSH, and luteinising hormone, LH) and prolactin, yet no nerve fibres reach it. How, then, does the hypothalamus command it? **Geoffrey Harris** (1913 to 1971) proposed in 1955 that hypothalamic neurons release **releasing** and **release-inhibiting hormones** into a network of blood vessels, the **hypothalamo-hypophyseal portal system**, which carries them down the pituitary stalk; cutting the portal veins halted anterior pituitary release until they regrew. In the late 1960s **Roger Guillemin** and **Andrew Schally** separately isolated one, thyrotropin-releasing hormone (TRH), sharing the 1977 Nobel Prize.

The **posterior pituitary** (**neurohypophysis**) is neural tissue, an outgrowth of the hypothalamus, and makes nothing itself. Oxytocin and vasopressin are synthesised in neurons of the **supraoptic** and **paraventricular nuclei** of the hypothalamus, carried down their axons and released from the posterior pituitary into the blood. **Vasopressin**, or **antidiuretic hormone** (ADH), makes the kidneys reabsorb water; its failure causes **diabetes insipidus**, with copious dilute urine and constant thirst. **Oxytocin** contracts the uterus in labour, ejects milk during suckling and promotes bonding.

Table 5.21 How the hypothalamus runs the pituitary. The first five rows are anterior, the last two posterior.

Hypothalamic signal	Pituitary hormone	Target	Main effect
CRH	ACTH	Adrenal cortex	Cortisol release
TRH	TSH	Thyroid	Thyroxine release
GnRH	FSH and LH	Ovaries, testes	Sex hormones, eggs, sperm
GHRH; somatostatin inhibits	Growth hormone	Most tissues	Growth; excess gigantism, deficit dwarfism
Dopamine inhibits	Prolactin	Breasts	Milk production
Made in supraoptic and paraventricular nuclei; sent down axons	Oxytocin	Uterus, breasts, brain	Labour, milk ejection, bonding
Same nuclei, same route	Vasopressin (ADH)	Kidneys	Water reabsorption

EXAM TRAP

Master gland: the pituitary, controlled by the hypothalamus; not the hypothalamus, thalamus, pineal, thyroid or adrenal. **Supervisor** of fight or flight: the hypothalamus, not the pituitary, the adrenal gland or the parasympathetic system. The anterior pituitary is reached by **portal blood**, not by axons, cerebrospinal fluid or the vagus nerve. **Not anterior pituitary:** oxytocin (posterior) and melatonin (pineal). The posterior pituitary only stores and releases oxytocin and vasopressin; orexin (a hypothalamic wakefulness peptide, Section 5.8), noradrenaline and "vasodilators" are impostors. Oxytocin and vasopressin are made in the supraoptic and paraventricular nuclei, not the suprachiasmatic nucleus (body clock), the arcuate nucleus (feeding; the dopamine that inhibits prolactin) or the mammillary bodies. Prolactin makes milk; oxytocin ejects it.

The other endocrine glands ●●●

Three facts about the remaining glands (Table 5.22) recur: the **adrenal glands** run the stress response, the **thyroid** sets the metabolic rate and the **pancreas** controls blood glucose.

Each adrenal gland sits on a kidney and is two glands in one. The outer **adrenal cortex** makes steroids: **cortisol**, the chief glucocorticoid and the stress hormone; aldosterone, for salt and water balance; and adrenal androgens. The inner **adrenal medulla**, a modified sympathetic ganglion, releases **adrenaline** and noradrenaline when sympathetic nerves fire. Adrenaline and **epinephrine** are two names for one hormone (Latin and Greek for "on the kidney"), as noradrenaline is norepinephrine.

The **thyroid gland** releases thyroxine (T4) and triiodothyronine (T3), which set the body's metabolic rate and energy expenditure; the thyroid itself is controlled by TSH. In the islets of Langerhans of the pancreas, beta cells release **insulin**, which lowers blood glucose by letting cells take it up, and alpha cells release glucagon, which raises it. **Type 1 diabetes** is autoimmune destruction of the beta cells, usually in childhood. **Type 2 diabetes**, far commoner and widespread in India, is a disorder of **insulin resistance**: it runs in families, but it is most strongly associated with poor diet, obesity and a sedentary lifestyle.

Table 5.22 The other endocrine glands.

Gland	Hormones	Main functions	Disorders
Pineal	Melatonin, at night	Circadian rhythm (Section 5.8)	Used for jet lag
Thyroid	Thyroxine (T4), T3	Metabolic rate, energy use	Hypothyroidism: lethargy, depression; cretinism in infancy. Hyperthyroidism (Graves' disease): anxiety, agitation, weight loss
Parathyroids	Parathyroid hormone	Raises blood calcium	Low calcium: tetany
Adrenal cortex	Cortisol, aldosterone, adrenal androgens	Stress, glucose, immunity; salt and water	Cushing's (excess), Addison's (deficiency)
Adrenal medulla	Adrenaline, noradrenaline	Fast fight-or-flight arousal	Phaeochromocytoma: panic-like attacks
Pancreas (islets)	Insulin (beta cells), glucagon (alpha cells)	Lower and raise blood glucose	Type 1 and type 2 diabetes

Gland	Hormones	Main functions	Disorders
Gonads	Testosterone; oestrogens, progesterone	Sexual development and behaviour (Section 5.8)	Hypogonadism
Thymus	Thymosin	Matures T lymphocytes	Immune deficiency

Endocrine disorders often present as psychiatric ones. In **Cushing's syndrome**, excess cortisol (from a tumour or steroid medicines) brings a round "moon face" and central obesity and, most commonly, depression, anxiety, irritability and poor memory and concentration; a minority develop mania or psychosis (not catatonia or loss of smell). **Addison's disease**, failure of the adrenal cortex with too little cortisol, brings fatigue, apathy and depression. Thyroid disorders are in Table 5.22.

> EXAM TRAP
>
> Stress and stress-related metabolic change: **adrenal glands**. Overall metabolic rate: **thyroid**. Insulin: the **pancreas** only, not the kidney, thyroid or hypothalamus. Amity 2023's "Which controls the thyroid gland? Metabolism" means what the thyroid controls; the thyroid itself obeys TSH, glucose uptake is insulin's job and emotion the limbic system's. The adrenal secretes **both** cortisol and epinephrine; melatonin, thyroxine and insulin are distractors, and noradrenaline and dopamine are adrenaline's relatives, not its other name. For type 2 diabetes genetics matters, and stress and toxins contribute less, but the keyed answer is poor diet and sedentary lifestyle.

Two stress systems, feedback and allostasis ●●

The hypothalamus answers a threat through two systems (Figure 5.12). In the fast **sympathetic adrenal-medullary system** (SAM), sympathetic nerves fire and the adrenal medulla pours adrenaline and noradrenaline into the blood within seconds: the fight-or-flight response named by **Walter Bradford Cannon** (1871 to 1945; the autonomic wiring is in Section 5.1).

The slower **hypothalamic-pituitary-adrenal axis** (HPA axis) works by hormones. The paraventricular nucleus of the hypothalamus releases corticotropin-releasing hormone (CRH) into the portal blood, the anterior pituitary answers with ACTH, and ACTH in the general circulation makes the adrenal cortex release cortisol over minutes. Cortisol mobilises glucose and dampens inflammation and immunity. It also switches the axis off by **negative feedback** on the hypothalamus (less CRH) and the anterior pituitary (less ACTH), helped by the hippocampus, whose dense glucocorticoid receptors let it restrain the axis. The thyroid and gonadal axes work the same way. **Hans Selye**'s general adaptation syndrome, what follows when the axis stays switched on, is taught in Section 1.11.

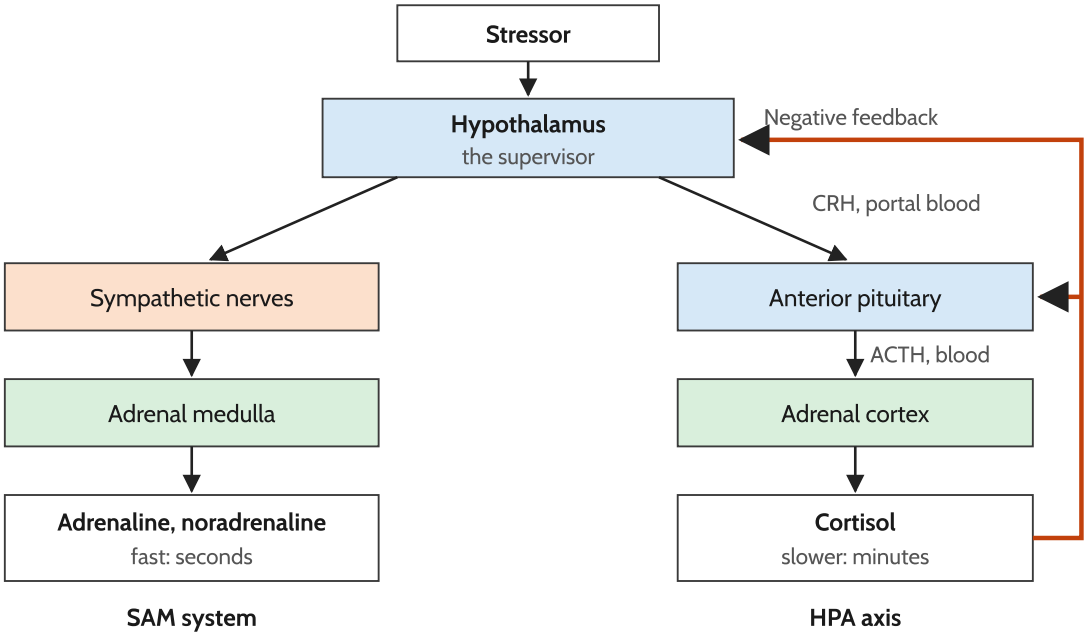

Figure 5.12 The two stress systems. Solid arrows drive; the dashed line is cortisol's negative feedback on the hypothalamus and anterior pituitary. The posterior pituitary and pineal gland play no part.

Brief cortisol surges help; chronic ones harm. Prolonged high cortisol shrinks hippocampal dendrites, and a damaged hippocampus brakes the axis less well, so cortisol climbs further. Chronic stress also suppresses immunity and slows wound healing. The study of such links between mind, nervous system and immune system is **psychoneuroimmunology**.

Cannon also named the goal of these systems: **homeostasis** (popularised in *The Wisdom of the Body*, 1932), keeping the internal state stable around fixed set points, as a thermostat holds a room's temperature. **Peter Sterling** and **Joseph Eyer** (1988) argued that the brain also predicts demands and shifts the set points in advance, raising blood pressure as you stand and cortisol before an examination. They called this **allostasis**, "stability through change": the brain regulating its own stress responses by controlling hormone release. **Bruce McEwen** and **Eliot Stellar** (1993) added **allostatic load**, the cumulative wear and tear when these responses are overused or fail to switch off: high blood pressure, abdominal fat, insulin resistance and hippocampal damage.

EXAM TRAP

The HPA chain is **hypothalamus (CRH), anterior pituitary (ACTH), adrenal cortex (cortisol)**. ACTH comes from the anterior pituitary, not from the adrenal gland it targets; the posterior pituitary and adrenal medulla are not in the axis, and "pineal, pituitary, thyroid" is no axis; the cerebellum plays no part. Cortisol feeds back on the **hypothalamus and anterior pituitary**, not the pineal, posterior pituitary, adrenal medulla or vagal motor nucleus (parasympathetic output). "The brain regulating its own stress responses by controlling hormone release" is **allostasis**, not homeostasis (fixed set points), neuroplasticity (rewiring with experience, Section 5.5) or circadian rhythm (Section 5.8).

CLINICAL LINK

In melancholic depression the HPA axis is often overactive: in the **dexamethasone suppression test**, a night dose of synthetic glucocorticoid fails to suppress next morning's cortisol (Chapter 8). Because endocrine disease mimics depression and anxiety, hormones are checked before a psychiatric diagnosis.

Emotion and the autonomic nervous system •

Fear and anger bring one broad sympathetic pattern: heart rate, blood pressure and breathing rise, the pupils dilate, blood shifts to the muscles, glucose enters the blood and sweating increases. Sweat conducts electricity, so skin conductance rises and skin resistance falls (the galvanic skin response). Pupil constriction is parasympathetic. The **polygraph** records heart rate, blood pressure, breathing and skin conductance together; it detects arousal, not lies. Whether each emotion has its own autonomic pattern divides the James-Lange, Cannon-Bard and Schachter-Singer theories (Section 1.11).

EXAM TRAP

Odd one out among signs of sympathetic activation: **pupil constriction**. "Lowered skin resistance" is sympathetic: it is the same event as raised skin conductance.

From sham rage to the limbic system •••

Which part of the brain makes emotion? In 1929 **Philip Bard** found that **decorticate** cats, their cortex removed, raged at the lightest touch, arching their backs, hissing and baring their teeth, but excessively and at no target. He called this **sham rage**. It survived removal of everything down to the hypothalamus but vanished once the hypothalamus went too. The hypothalamus organises rage; the cortex inhibits and directs it.

In 1937 **James Papez** (1883 to 1958) proposed a circuit for emotion (Figure 5.13). The **Papez circuit** runs from the hippocampus through the fornix to the mammillary bodies, by the mammillothalamic tract to the anterior thalamic nucleus, on to the cingulate cortex, through the cingulum to the entorhinal (parahippocampal) cortex and back to the hippocampus. Emotion, he argued, is expressed through the hypothalamus and experienced through the cortex. The loop is now thought to serve memory more than emotion (Section 5.9), but it began the search for an emotional brain.

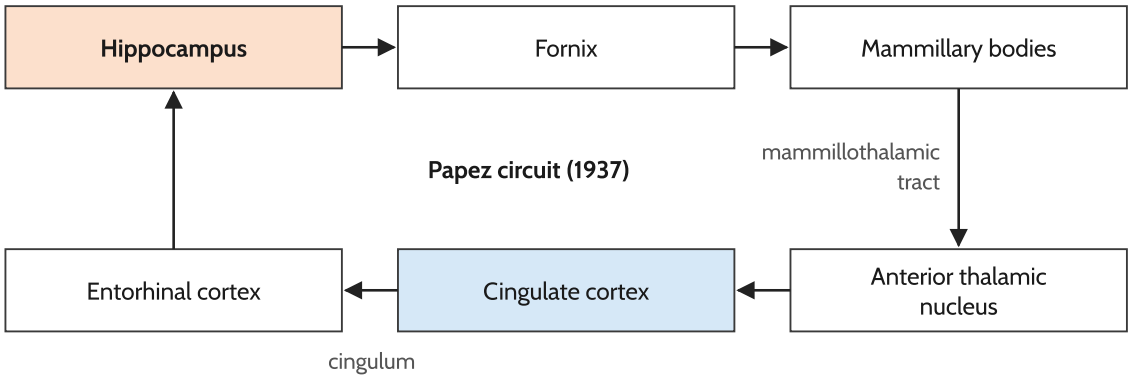

Figure 5.13 The Papez circuit, a closed loop. Loops through the caudate, putamen, globus pallidus or substantia nigra (basal ganglia), from retina to visual cortex, or through the cerebellum are other circuits.

Paul MacLean (1913 to 2007) enlarged the circuit in 1952, adding the amygdala, septum and other structures, and named it the **limbic system** (Latin *limbus*, border, because it rings the thalamus at the inner edge of the cortex); later it became the middle layer of his triune brain. Its members are the hippocampus, amygdala, cingulate gyrus, fornix, septum and mammillary bodies, with the hypothalamus and anterior thalamus closely tied in. It is the classic "seat of emotion", regulating emotion, motivation and memory.

EXAM TRAP

"Seat of emotion", "emotion and motivation", "emotional behaviour": **limbic system**; offered single parts (hippocampus, thalamus, hypothalamus) beside it, choose the system. Not the reticular formation (arousal), hindbrain (breathing, heart rate), whole forebrain (too broad), pituitary (hormones), cerebellum (motor coordination), corpus callosum (joins the hemispheres), frontal lobe (voluntary movement, decisions), thalamus and sensory cortex (sensory processing), visual cortex (acuity) or Broca's area (syntax). Hunger, thirst and hormone regulation are hypothalamic, so Amity 2023 keys "regulation of emotions", not "both".

The amygdala and fear ●●●

The **amygdala** (Greek for almond), a cluster of nuclei in each anterior medial temporal lobe, is the key emotional structure: it detects threat, gives events emotional salience, triggers fear and defensive aggression, and learns what predicts danger. Its role was mapped by fear conditioning: after a tone has preceded mild shocks, the tone alone makes a rat freeze, heart racing. Signals enter through the **lateral nucleus**, where tone and shock are associated, and leave through the **central nucleus** for the hypothalamus (autonomic and hormonal responses) and the **periaqueductal grey** of the midbrain (freezing). Amygdala lesions abolish conditioned fear.

Joseph LeDoux (born 1949) traced two routes in (Figure 5.14). The **low road**, from the sensory thalamus straight to the amygdala, is fast and crude: you jump back from a coiled shape before you know what it is. The **high road**, from the thalamus through the sensory cortex to the amygdala, is slower but accurate: it tells you the shape is a rope.

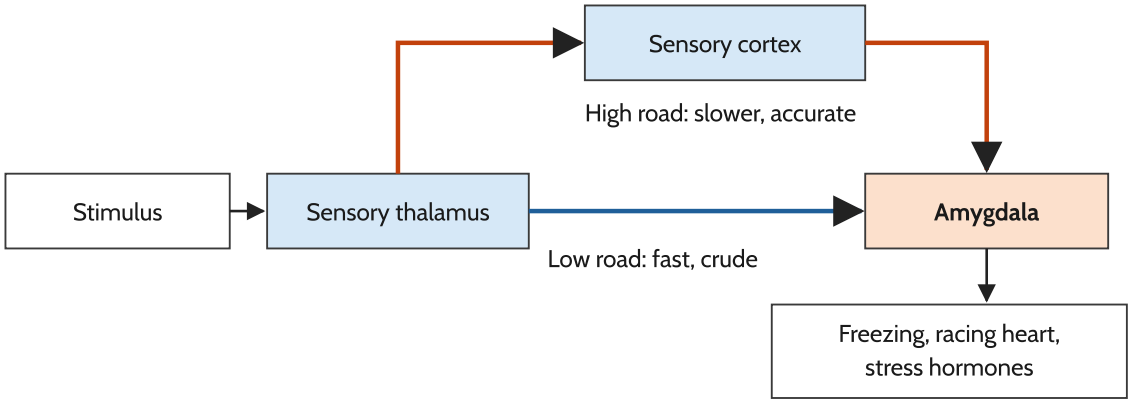

Figure 5.14 LeDoux's two roads to fear. The solid low road reacts before you consciously recognise the stimulus; the dashed high road identifies it.

KEY STUDY: PATIENT S.M. (ADOLPHS AND COLLEAGUES, 1994)

Does the human amygdala matter for emotion? **Ralph Adolphs** and colleagues tested S.M., whose amygdalae had been destroyed on both sides by Urbach-Wiethe disease, a rare genetic disorder that calcifies the amygdala and nearby medial temporal lobe. She recognised faces and most expressions but not fear, and could not produce a fearful expression herself. Later work found she barely feels fear, reaching for snakes in a pet shop (Justin Feinstein and colleagues, 2011). Criticism: one rare patient.

In **affective blindsight**, described by **Beatrice de Gelder** and colleagues in 1999, patients blinded in part of the visual field by damage to the primary visual cortex guess well above chance which emotion a face shown there expresses, while insisting they see nothing. The signal is thought to travel subcortically, from the superior colliculus through the pulvinar of the thalamus to the amygdala. "Affect neglect", "emotional neglect" and "motivational blindsight" are not established terms.

EXAM TRAP

Fear and aggression, emotion processing, threat detection, fear conditioning: **amygdala**. Not the hippocampus (conscious recall, declarative memory, Section 5.9), thalamus (relay), hypothalamus (executes the amygdala's orders), frontal lobe (verbal working memory), brainstem (vital functions), cerebellum and basal ganglia (starting movement, motor sequence learning) or suprachiasmatic nucleus (circadian rhythm). "Slower, more accurate identification" is the high road; conscious vision and voluntary motor output are cortical.

Klüver-Bucy syndrome and other influences ●●

What happens to an animal without its amygdalae? In 1937 and 1939 **Heinrich Klüver** and **Paul Bucy** removed both anterior temporal lobes, amygdalae included, in rhesus monkeys. Unmanageable monkeys became tame and placid, approaching snakes without fear. They showed **psychic blindness**, a visual agnosia (seeing objects without recognising their meaning); examined everything by mouth (**hyperorality**); ate almost anything, even meat, which rhesus monkeys normally refuse (**hyperphagia** and dietary change); directed sexual activity at inappropriate partners and objects (**hypersexuality**); and attended compulsively to every

sight (**hypermetamorphosis**). This **Klüver-Bucy syndrome** follows mainly from loss of the amygdala. Human cases, after herpes simplex encephalitis or head injury, are partial and often add aphasia and amnesia from wider damage (Table 5.23).

Table 5.23 Klüver-Bucy features and their look-alikes.

Option	Where it belongs
Tameness, placidity, loss of fear; psychic blindness, hyperorality, hyperphagia, hypersexuality	Klüver-Bucy (amygdala and temporal lobe)
Aphasia, amnesia	Human Klüver-Bucy with wider damage
Intense fear, attack posture, piloerection, hyperalertness	Amygdala stimulation; sham rage
Hyperphagia with obesity	Ventromedial hypothalamus (Section 5.8)
Pure anterograde amnesia	Hippocampus, as in H.M. (Section 5.9)
Avolition, alogia	Negative symptoms of schizophrenia (Chapter 8)
Spastic paralysis; loss of language comprehension	Motor pathways (Section 5.6); Wernicke's area (Section 5.10)

EXAM TRAP

JEMAS 2025 keys "agnosia, aphasia, hyperorality, hyperphagia, hypersexuality" (aphasia from human cases). Each rival cluster hides an impostor: hyperalertness, avolition and alogia, asexuality, hyposensitivity and delayed reaction. The structure is the **amygdala with adjacent temporal lobe**, not the cerebellum, hippocampus or caudate nucleus (Huntington's disease, Section 5.11).

Three other influences need a line each. The **ventromedial prefrontal cortex** regulates emotion, and its damage leaves people impulsive and socially inept (Section 5.11). **Testosterone** raises social aggression in the males of many species, less clearly in humans. Low **serotonin** activity is linked to impulsive aggression (Section 5.3).

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Exocrine glands release their secretions through _____; endocrine glands are ductless.
2. The hypothalamus reaches the anterior pituitary through the hypothalamo-hypophyseal _____ system.
3. Oxytocin and vasopressin are made in the supraoptic and _____ nuclei.
4. In the HPA axis, CRH makes the anterior pituitary release _____.
5. Sterling and Eyer's "stability through change" is _____.
6. Bard's decorticate cats showed _____ rage.
7. LeDoux's _____ road runs from the thalamus straight to the amygdala.
8. Tameness, hyperorality and hypersexuality after bilateral temporal lobectomy make up _____ syndrome.

Answers: (1) ducts, (2) portal, (3) paraventricular, (4) ACTH, (5) allostasis, (6) sham, (7) low, (8) Klüver-Bucy.

5.8 Sleep, Biological Rhythms, Hunger, Thirst and Sex

MEDIUM YIELD · 31 QS

Why does a student who revised all night still feel alert in the evening, hungry at lunchtime however large breakfast was, and thirsty after salty chaat? The answers lie in small clusters of neurons, mostly in the hypothalamus.

Biological rhythms and the master clock ●●●

A **circadian rhythm** (Latin *circa diem*, about a day) repeats about every 24 hours, like the sleep-wake cycle; **ultradian rhythms** are shorter (the 90-minute sleep cycle) and **infradian rhythms** longer (the menstrual cycle). Circadian rhythms are entrained, locked to the world, by **zeitgebers** ("time givers"), above all light. Without cues, in a cave or windowless laboratory, they persist as **free-running rhythms** slightly longer than 24 hours (about 24.2 in humans). Jet lag and shift work put the clock out of step with the day; eastbound flights, which demand a phase advance, are the harder.

Curt Richter found in 1967 that large medial hypothalamic lesions disrupted the daily cycles of eating, drinking and activity in rats; smaller lesions located the clock in the **suprachiasmatic nucleus** (SCN), just above the optic chiasm. SCN lesions abolish the circadian timing of sleep but not its total amount.

KEY STUDY: THE SCN TRANSPLANT (RALPH AND COLLEAGUES, 1990)

Martin Ralph and colleagues transplanted SCN tissue from fetal mutant hamsters with a 20-hour free-running cycle into normal adults whose 25-hour rhythms SCN lesions had abolished. Rhythms returned with the donor's 20-hour period (and the reverse transplant gave 25 hours): the SCN sets the period. It is the master clock, though most body cells carry clock machinery that it keeps in step.

Light reaches the SCN by the **retinohypothalamic tract**, from retinal ganglion cells containing the photopigment **melanopsin**, not from rods or cones. The clock itself is a molecular loop: clock genes such as *period*, *timeless* and *clock* are transcribed, and their proteins build up and switch off their own transcription over about 24 hours, as **Jeffrey Hall**, **Michael Rosbash** and **Michael Young** showed in fruit flies (Nobel Prize, 2017).

The SCN drives the **pineal gland**, a small endocrine gland of the epithalamus, to secrete **melatonin** at night, the "hormone of darkness"; light suppresses it. Melatonin promotes sleepiness, and taken at bedtime in the new time zone it eases jet lag.

> EXAM TRAP
>
> The master circadian pacemaker is the **SCN of the hypothalamus**, not the paraventricular or ventromedial nucleus or mammillary bodies, and no centre for hunger, temperature, anterior pituitary release, memory, language or pain (Table 5.25). Yet RML 2022 ("which gland regulates circadian rhythm") and RML 2023 ("which structure regulates the body's internal clock") keyed the **pineal gland**, which turns the clock's signal into melatonin. Choose the SCN whenever it is offered; for a gland, or "hypothalamus" against "pineal", these keys want the pineal. The pituitary (Section 5.7), cerebellum and thalamus keep no clock.

The EEG of sleep and REM sleep ●●●

When cortical neurons fire independently, the electroencephalogram (EEG) is desynchronised (small, fast waves); when they fire together it is synchronised (large, slow waves). Stages 3 and 4 together form **slow-wave sleep**, merged as N3 in the 2007 scoring rules of the American Academy of Sleep Medicine (Table 5.24). The NREM stages and a REM period make a cycle of about 90 minutes, four to six a night (Section 1.5).

Table 5.24 The EEG of waking and sleep.

State	EEG	Frequency	Marks
Alert waking	Beta: low amplitude, desynchronised	13 to 30 Hz	Eyes open, active thought
Relaxed waking	Alpha, occipital	8 to 12 (13) Hz	Eyes closed
N1 (stage 1)	Theta	4 to 7 Hz	Drowsy transition
N2 (stage 2)	Sleep spindles, K-complexes	Spindles 12 to 14 Hz	Light sleep
N3 (stages 3, 4)	Delta: high amplitude, synchronised	0.5 to 4 Hz	Slow-wave sleep; growth hormone surge
REM	Low voltage, fast, mixed; sawtooth waves	Waking-like	Eye movements, atonia, dreams

In 1953 **Eugene Aserinsky** and **Nathaniel Kleitman** found recurring bursts of rapid eye movements in sleepers, and with **William Dement** linked them to dreaming: about 80 per cent of REM awakenings yield a dream report. REM is **paradoxical sleep**: the EEG looks awake while the body is deeply asleep. Its signs are rapid eye movements; **atonia**, loss of tone in the core skeletal muscles; a low-voltage, fast, desynchronised EEG; penile and clitoral tumescence; irregular heart rate and breathing; and the most vivid, story-like dreams.

> **EXAM TRAP**
>
> REM is **not** marked by EEG synchrony, delta waves or the growth hormone surge: all three belong to N3. Spindles and K-complexes mark N2, theta N1, and occipital alpha relaxed waking with eyes closed. Slow-wave sleep is stages 3 and 4 only, not stages 1 to 4 or NREM plus REM.

The switches for sleep, waking and dreams

In 1936 **Frederic Bremer** cut cats' brainstems between the colliculi (the cerveau isolé, "isolated forebrain"), and the cortex showed almost constant slow-wave sleep; a cut at the bottom of the brainstem (the encéphale isolé, 1937) left the sleep-wake cycle intact. Something between the cuts keeps the forebrain awake. **Giuseppe Moruzzi** and **Horace Magoun** (1949) found that stimulating the brainstem reticular formation woke sleeping cats: it is the **reticular activating system**.

It is now seen as several ascending arousal systems named by transmitter: noradrenaline (locus coeruleus), serotonin (raphe nuclei), histamine (tuberomammillary nucleus), acetylcholine (basal forebrain and pons) and **orexin**, or hypocretin (lateral hypothalamus, in its posterior, wake-promoting part). Slow-wave sleep begins when GABA neurons of the **ventrolateral preoptic area** (VLPO) become active and inhibit these systems. As the two sides inhibit each other, **Clifford Saper** (2001) likened them to a flip-flop switch: one side wins, so we change state quickly (Figure 5.15).

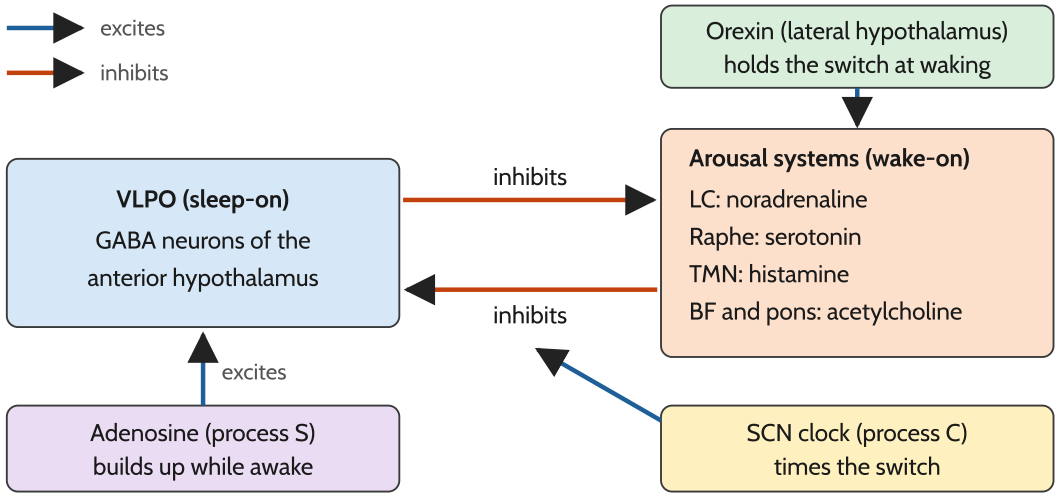

Figure 5.15 The sleep-wake flip-flop. The VLPO and the arousal systems inhibit each other, so one side wins; orexin holds it at waking, adenosine pushes it toward sleep, the SCN times it. LC, locus coeruleus; TMN, tuberomammillary nucleus; BF, basal forebrain.

The chemical of drowsiness is **adenosine**, a by-product of the brain's energy use that builds up during waking and falls during sleep: it is the homeostatic sleep pressure, and caffeine keeps you awake by blocking its receptors. Orexin, acetylcholine and serotonin promote arousal instead. **Alexander Borbély**'s **two-process model** (1982) joins this process S, rising with time awake, to the circadian process C from the SCN.

REM has its own switch: REM-on neurons of the sublaterodorsal nucleus in the pons and REM-off neurons of the ventrolateral periaqueductal grey inhibit each other. In REM the

sublaterodorsal nucleus drives medullary neurons that inhibit motor neurons, producing atonia.

In the activation-synthesis hypothesis of **J. Allan Hobson** and **Robert McCarley** (1977; Section 1.5), the cortex weaves a dream from random pontine activity. In REM the **dorsolateral prefrontal cortex** is relatively inactive, so dreams lack organisation, planning, logic and a sense of time; the amygdala is active, so they are emotional; and the **extrastriate** visual association cortex is active while the **striate** (primary visual) cortex is quiet, so we see vivid images without input from the eyes. The somatosensory cortex plays no special part.

Sleep disorders and why we sleep

In **narcolepsy** the orexin neurons of the lateral (posterior) hypothalamus are lost. Its signs are excessive daytime sleepiness with sleep attacks; cataplexy, a sudden loss of muscle tone triggered by emotion, in full consciousness; sleep paralysis; hypnagogic hallucinations while falling asleep; and sleep-onset REM. All but the sleepiness are REM breaking into waking.

In **REM sleep behaviour disorder** (RBD), REM atonia is lost and patients act out their dreams; it often precedes Parkinson's disease and Lewy body dementia by years (Section 5.11). Sleepwalking (somnambulism), night terrors and bedwetting are NREM parasomnias, arising from partial arousal out of N3 early in the night; nightmares are frightening REM dreams, mostly late in the night. In sleep apnoea, breathing stops repeatedly during sleep. Insomnia and its treatment are in Chapters 7 and 8.

> EXAM TRAP
>
> Sleepwalking and night terrors: **N3**, not REM, N1 or N2 (a body in REM is paralysed). RBD is loss of REM atonia, not an inability to enter REM, not bedwetting in N3 and not daytime sleepiness without night-time symptoms.

Why sleep at all? Recuperation theories say sleep restores what waking depletes; adaptation theories say it evolved to keep animals inactive and safe when they need not forage. Sleep also aids memory consolidation (Section 5.9).

Hunger: set points and positive incentives

Most people assume hunger works like a thermostat. On this **set-point assumption**, eating starts when energy falls below a set point and stops when it is restored: **Jean Mayer**'s glucostatic theory (1953) made blood glucose the regulated quantity, and the lipostatic theory made it body fat. **John Pinel** and colleagues argue instead for **positive-incentive theory**: we eat mainly for the anticipated pleasure of food, shaped by taste, learning and company, and body weight drifts to a settling point.

The circulating hormone that triggers hunger is **ghrelin**, from the empty stomach. Satiety signals come from the gut and from fat. In 1973 **James Gibbs**, **R. C. Young** and **Gerard Smith** injected **cholecystokinin** (CCK), released by the duodenum as food arrives, into hungry rats, which then ate smaller meals: the classic satiety peptide. Insulin, released by the pancreas as blood glucose rises, is both a satiety and a body-fat signal.

KEY STUDY: THE OB/OB MOUSE AND LEPTIN (1994)

Mice of the ob/ob strain overeat and grow up to three times normal weight. In 1994 **Jeffrey Friedman**'s group cloned the mutated gene: expressed only in fat cells, it codes a hormone they named **leptin**, which ob/ob mice lack, and leptin injections cut their eating and fat. Released in proportion to fat stores, leptin is the long-term regulator of body weight and fat. Yet obese humans usually have high leptin and do not respond to injections (leptin resistance). The long-term fat signal is leptin, not the hypothalamus (a brain area, not a hormone), oestrogen or cortisol.

The hypothalamus and the arcuate nucleus ●●●

Section 1.10 told how two lesion studies built the **dual-centre hypothesis**. In 1940 **Albert Hetherington** and **Stephen Ranson** destroyed the ventromedial hypothalamus (VMH) of rats, which overate (hyperphagia) and grew grossly obese, first in a **dynamic phase** of rapid gain, then in a **static phase** holding the new weight. In 1951 the Indian physiologist **Bal Krishan Anand** and **John Brobeck** found that lateral hypothalamus (LH) lesions caused aphagia, a fatal refusal to eat; **Philip Teitelbaum** and **Alan Epstein** (1962) showed that it comes with adipsia, a refusal to drink, and that tube-fed rats partly recover. **Eliot Stellar** (1954) made the LH the feeding centre and the VMH the satiety centre.

The model has since been revised: VMH lesions raise blood insulin, so food is turned into fat and the rats overeat because they are getting fat, not the reverse; cut fibres from the paraventricular nucleus explain much of the syndrome; and LH lesions cause broad motor and sensory deficits. The nuclei are now parts of a network (Table 5.25) whose hub is the arcuate nucleus.

Table 5.25 Hypothalamic nuclei met in this section.

Nucleus	Main role	Damage or note
Suprachiasmatic (SCN)	Circadian clock	Rhythms lost, total sleep kept
Ventrolateral preoptic	Switches on sleep	Insomnia
Medial preoptic area	Male sex; temperature	Sexually dimorphic nucleus
Ventromedial (VMH)	Satiety; female sex	Hyperphagia, obesity
Lateral (LH)	Feeding; orexin	Aphagia, adipsia
Arcuate	Appetite, energy balance	NPY and POMC neurons
Paraventricular	Stress (CRH); satiety	Hyperphagia
Supraoptic	ADH, oxytocin	Diabetes insipidus
Mammillary bodies	Memory	Amnesia (Section 5.9)

The **arcuate nucleus** is where hunger and fat signals meet, in two opposing groups of neurons (Figure 5.16). NPY/AgRP neurons release **neuropeptide Y** and agouti-related peptide; they are orexigenic, and NPY injected into the hypothalamus is among the strongest triggers of eating known. POMC/CART neurons release alpha-melanocyte-stimulating hormone (alpha-MSH) and cocaine-and-amphetamine-regulated transcript (CART); they are anorexigenic, and alpha-MSH acting on melanocortin receptors brings satiety. Leptin and insulin in-

hibit the NPY neurons and excite the POMC neurons; ghrelin excites the NPY neurons. Serotonin also promotes satiety.

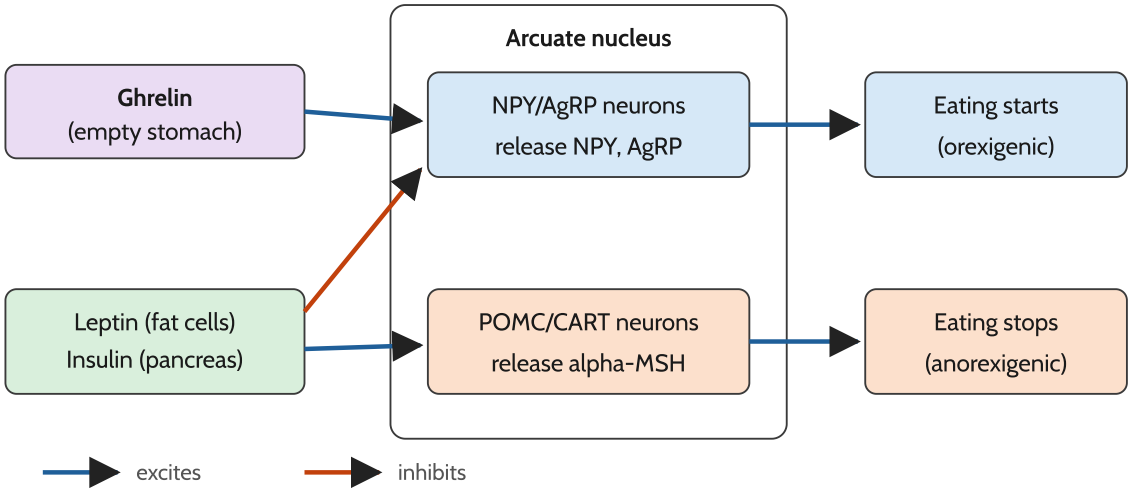

Figure 5.16 The arcuate nucleus. Ghrelin excites the NPY/AgRP (eating) neurons; leptin and insulin inhibit them and excite the POMC/CART (satiety) neurons.

EXAM TRAP

Satiety centre: **VMH**; its lesion gives hyperphagia and obesity, not excessive drinking, insomnia, diabetes insipidus or lost thermoregulation or circadian rhythm. Feeding centre: **LH**; its lesion gives aphagia, adipsia and weight loss (Lateral, Lose). "Ventromedial hippocampus" and "lateral thalamus" are invented; the cerebrum and basal ganglia are not hunger centres. The arcuate nucleus is not for motor coordination, memory or circadian rhythm. NPY triggers eating; leptin, CART and CCK are satiety signals.

Learned influences on eating, and thirst

What is on the plate, and who is at the table, matter too. In the **portion size effect**, people eat more when served larger portions: **Barbara Rolls** and colleagues (2002) served lunches of macaroni and cheese in different sizes, and the bigger the portion, the more people ate, without feeling fuller. In **sensory-specific satiety**, a food loses its appeal as you eat it while other foods stay attractive, so a new dish restarts eating. A small first course increases hunger (the appetiser effect), and a cafeteria diet of varied, palatable foods made rats eat 84 per cent more calories. Through social facilitation we eat more with others than alone, though we also eat less when companions do. Prader-Willi syndrome is in Section 5.5, eating disorders in Chapter 8.

Thirst has two triggers. In **osmotic thirst** the salt concentration of body fluid rises, as after a salty meal: water leaves the cells, and hypothalamic osmoreceptors sense this cellular dehydration, start thirst and trigger the release of antidiuretic hormone (ADH) so the kidneys keep water. In **hypovolaemic thirst** blood volume falls, as after bleeding: the kidneys release renin, which forms angiotensin II, and angiotensin II acts on the subfornical organ to start drinking. When ADH fails, in diabetes insipidus, the kidneys pass large volumes of dilute urine and the person is constantly thirsty.

Hormones, sexual development and sexual response

The **organisational effects** of hormones act early in development and permanently shape body and brain; activational effects act in adulthood, only while the hormone is present. **Charles Phoenix** and colleagues (1959) drew this distinction after female guinea pigs exposed to testosterone before birth mounted like males when given testosterone as adults.

The default course of a fetus is female. The **SRY gene** on the Y chromosome turns the primitive gonads into testes, which secrete testosterone, masculinising the body, and Müllerian-inhibiting substance, which makes the female ducts degenerate; **Alfred Jost** found that fetuses deprived of their gonads develop female ducts whatever their chromosomes. From puberty the pituitary gonadotropins, follicle-stimulating hormone (FSH) and luteinising hormone, make the gonads secrete androgens (chiefly testosterone), oestrogens (oestradiol) and progestins (progesterone). Both sexes make all three, in different amounts; hence Pinel's attack on the **mamawawa**, the men-are-men-and-women-are-women assumption that the sexes are separate, opposite categories.

In congenital adrenal hyperplasia, the adrenal glands of XX fetuses make excess androgens, partly masculinising girls' genitals; in androgen insensitivity syndrome, XY individuals with testes but unresponsive androgen receptors develop a female body without a uterus. The medial preoptic area, whose sexually dimorphic nucleus is several times larger in male rats (**Roger Gorski**, 1978), is critical for male sexual behaviour, the VMH for female. In 1991 **Simon LeVay** reported that INAH3, a nucleus of the anterior hypothalamus, was smaller in gay than in heterosexual men, nearer the size in women; the finding has not been consistently replicated, and reading it as a "female-like" brain is mamawawa thinking.

William Masters and **Virginia Johnson** (1966) observed volunteers in the laboratory and described a four-phase **sexual response cycle**: excitement, plateau (arousal levels off near its peak), orgasm and resolution, with a male refractory period after orgasm. **Helen Singer Kaplan** (1979) put desire first, in a three-phase model of desire, excitement and orgasm.

> **EXAM TRAP**
>
> Masters and Johnson: **excitement, plateau, orgasm, resolution**. A sequence that starts with plateau, or inserts desire anywhere, is not theirs: desire first is Kaplan, and the refractory period lies within the male resolution phase rather than being a phase of its own.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The master circadian pacemaker is the _____ nucleus of the hypothalamus.
2. Stages 3 and 4 (N3), with high-amplitude delta waves, form _____ sleep.
3. Slow-wave sleep begins when neurons of the _____ become active.
4. The chemical that builds up during waking and makes us drowsy is _____.
5. Narcolepsy follows the loss of _____ neurons in the lateral hypothalamus.
6. Lesions of the _____ hypothalamus produce hyperphagia and obesity.
7. In the arcuate nucleus, neurons releasing _____ directly trigger eating.

Answers: (1) suprachiasmatic, (2) slow-wave, (3) ventrolateral preoptic area, (4) adenosine, (5) orexin (hypocretin), (6) ventromedial, (7) neuropeptide Y.

5.9 The Biology of Learning, Memory and Amnesia HIGH YIELD · 47 QS

Every time you learn something, your brain physically changes. Where, and how? The answers come from patients whose memory broke in revealing ways, from animal lesion studies, and from the synapse itself. Section 1.7 treated memory as a psychological process; here the subject is its biology.

Amnesia: forward and backward in time ●●

Clinically, **amnesia** is a pathological loss of memory, far beyond ordinary forgetting. It usually follows brain damage or disease (head injury, stroke, surgery, encephalitis, thiamine deficiency); more rarely its cause is psychological, as in dissociative amnesia (Chapter 8). It is a disorder of memory, not a behavioural disturbance, a depressed mood or impaired perception.

Amnesias come in two directions in time from the onset (Figure 5.17). The first, **anterograde amnesia**, is the inability to form new long-term memories after the injury or onset: the patient chats normally, yet ten minutes later the conversation is gone. The second, **retrograde amnesia**, is the loss of memories formed before it. It usually has a **temporal gradient**: recent memories are lost most completely while remote ones survive, a pattern **Théodule Ribot** (1839 to 1916) described in *Les maladies de la mémoire* (1881), hence **Ribot's law**. The gradient shows that memories grow sturdier with age, through consolidation (5.9.4). In **source amnesia** you remember a fact but not where or when you learned it, a lapse common in old age and after frontal lobe damage.

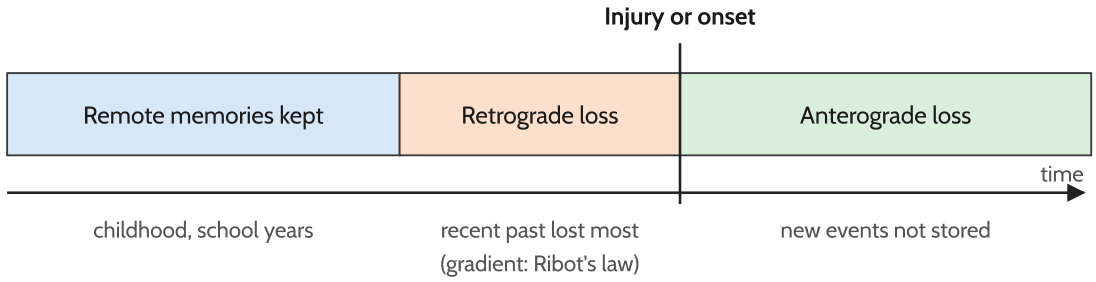

Figure 5.17 Amnesia is named by its direction from the onset. Retrograde loss is worst for the most recent past; anterograde loss blocks everything that follows.

> **EXAM TRAP**
>
> Anterograde means forward from the injury: **no new memories**. It is not "temporary memory loss" (transient global amnesia, Table 5.27), a visual deficit, or an inability to speak or recognise faces. Losing memories from **before** a concussion or stroke, or failing to retrieve old memories, is retrograde amnesia; dissociative amnesia is the psychological kind.

The search for the engram ●●

If learning changes the brain, the change must be somewhere. In 1904 the German zoologist **Richard Semon** (1859 to 1918) named it the **engram**: the physical trace an experience leaves in the brain. It is not a process of forgetting, a retrieval failure or a type of explicit memory.

Its most determined hunter, **Karl Spencer Lashley** (1890 to 1958), trained rats on mazes and then removed different parts of their cortex, expecting one lesion to erase the habit. None did. In *Brain Mechanisms and Intelligence* (1929) he stated two principles: **mass action** (the deficit depended on how much cortex was removed, not where) and **equipotentiality** (within a functional area, any part could do the work of any other). His review "In search of the engram" (1950) concluded that memories are not held in one place but distributed throughout the cortex.

His former student **Donald Olding Hebb** (1904 to 1985) explained why in *The Organization of Behavior* (1949). A memory is held by a **cell assembly**, a network of neurons linked because they fired together during an experience; spread across the cortex, it survives any single lesion in part. The links form at a **Hebbian synapse**: when neuron A repeatedly helps to fire neuron B, the connection from A to B strengthens ("cells that fire together wire together", a later summary). Long-term potentiation, found in 1973, bore him out (5.9.7). The Montreal neurosurgeon **Wilder Penfield** (1891 to 1976), stimulating the temporal lobes of awake epileptic patients, sometimes evoked vivid recollections, now regarded as reconstructions rather than stored replays.

> EXAM TRAP
>
> Lashley placed memories **throughout the cortex**: not in specific localised regions, the hippocampus alone or only the frontal and temporal lobes. The engram is the **physical trace** of a memory; the term is Semon's, the search Lashley's.

H.M. and the medial temporal lobes ●●●

Section 1.7 told H.M.'s story; here is the neural detail. **Henry Gustav Molaison** (1926 to 2008) had severe epilepsy that drugs could not control, apparently arising in both medial temporal lobes. In 1953, when he was 27, the neurosurgeon **William Beecher Scoville** (1906 to 1984) performed a **bilateral medial temporal lobectomy**, removing the inner parts of both temporal lobes: most of the hippocampus, the amygdala and adjacent cortex. The seizures eased. Scoville and the psychologist **Brenda Milner** (born 1918) reported the cost in 1957.

H.M. was left with severe anterograde amnesia for **declarative memory** (explicit memory: consciously recalled facts and events). He never learned his new address or recognised doctors he met after surgery, and he reread magazines with fresh interest. He had some retrograde amnesia for the few years before surgery. Spared were his intelligence (above average), language, personality, working memory (a normal digit span of about six) and remote memories, such as people and events from his childhood.

Most surprising, he could learn skills. Tracing a star while watching his hand only in a mirror, ten times a day on three days, he made fewer errors each day, yet each day denied ever doing the task. He also improved at rotary pursuit (keeping a stylus on a moving target), identified fragmented drawings sooner on the incomplete-pictures test (priming) and acquired a conditioned eyeblink: all implicit (nondeclarative) memory.

Three lessons followed. The medial temporal lobes consolidate new declarative memories but do not store remote ones, since his old memories survived. Short-term and long-term memory are separable. Explicit and implicit memory dissociate.

KEY STUDY: SCOVILLE AND MILNER (1957)

Their report on H.M. and other patients established medial temporal amnesia as a distinct syndrome. **Criticism:** the lesion also took the amygdala and nearby cortex, so the case alone could not show which structure mattered.

EXAM TRAP

H.M.'s most profound deficit was **anterograde amnesia** for declarative memory, not retrograde amnesia or a loss of procedural, semantic or working memory. Improving mirror drawing shows preserved **procedural** memory, not episodic, declarative or semantic memory. The surgery removed the **medial temporal lobes**, not prefrontal, parietal (angular gyrus) or occipital (lingual gyrus) cortex. It did not abolish motor coordination, selectively erase semantic memory, or cause "global retrograde amnesia with intact new learning", the reverse of his case.

The hippocampus: consolidation and space ●●●

Which structure mattered? Patient R.B., described by **Stuart Zola-Morgan**, **Larry Squire** and **David Amaral** (1986), suffered ischaemia (interrupted blood supply to the brain) at 52, when a pump failed during heart bypass surgery. His anterograde amnesia was milder than H.M.'s, and after his death the only obvious damage was to pyramidal cells in field CA1 of the hippocampus. Hippocampal damage alone, then, causes anterograde amnesia: a man who since a road accident remembers his past but not what he said ten minutes ago has probably injured his hippocampus.

The hippocampus is needed for **consolidation**, the conversion of fragile new memories into stable long-term ones, especially episodic and spatial declarative memories. In **standard consolidation theory**, it binds the scattered cortical pieces of an experience and replays them, partly during sleep (5.8), until cortical connections can hold the memory alone; remote memories then no longer need it. This explains both H.M.'s intact childhood and Ribot's gradient.

It is also the brain's map. In 1971 **John O'Keefe** (born 1939) and **Jonathan Dostrovsky** found **place cells** in the rat hippocampus, each firing only when the rat is in one particular spot; O'Keefe and **Lynn Nadel** argued in *The Hippocampus as a Cognitive Map* (1978) that the hippocampus builds a **cognitive map** of the environment. **May-Britt Moser** (born 1963) and **Edvard Moser** (born 1962) in 2005 found **grid cells** in the entorhinal cortex, the hippocampus's main input, each firing in a regular triangular grid of locations, a coordinate system for space. The three shared the 2014 Nobel Prize in Physiology or Medicine. **Eleanor Maguire** and colleagues (2000) found enlarged posterior hippocampi in London taxi drivers, larger the longer they had driven.

K.C., **Kent Cochrane** (1951 to 2014), studied by **Endel Tulving** (1927 to 2023), suffered diffuse brain damage including the medial temporal lobes in a motorcycle accident in 1981. He lost episodic, autobiographical memory for his whole life and could not imagine his future, yet his semantic knowledge, skills (chess, card games) and working memory survived: episodic and semantic memory are separable. **Clive Wearing** (born 1938), a British musician, lost both hippocampi and more to herpes simplex encephalitis in 1985; left with a memory span of seconds, he could still play the piano and conduct.

EXAM TRAP

The hippocampus **forms and consolidates new declarative memories** (especially episodic and spatial); remote memories end up in the wider cortex. It does not control temperature (hypothalamus), balance (cerebellum), speech (Broca's area, 5.10), procedural memory (basal ganglia, cerebellum), smell transduction (olfactory receptors, 5.6) or voluntary movement (motor cortex). Failure to form new memories after head injury points to the **hippocampus**, not the amygdala, prefrontal cortex, cerebellum, thalamus, hypothalamus, pituitary, brainstem, reticular formation (arousal) or spinal cord (reflexes).

Memory systems and the amygdala ●●

Memory, then, is several systems, each with its own structures (Table 5.26). Skills and habits depend on the cerebellum, basal ganglia and motor cortex, spared in H.M. but damaged in Parkinson's and Huntington's disease, which impair skill learning. **Mortimer Mishkin** (1926 to 2021) and **Jean Delacour** (1975) devised the **delayed nonmatching-to-sample test**: a monkey sees a sample object, then after a delay must choose the new object rather than the sample to find food. Monkeys with medial temporal lesions failed at longer delays, an animal model of H.M.; later lesions traced this object-recognition memory chiefly to the perirhinal cortex.

Table 5.26 Memory systems and the structures they depend on.

System	What it holds	Key structures	Classic evidence
Declarative (explicit)	Facts and events, consciously recalled	Hippocampus, medial temporal lobe; medial diencephalon	H.M., R.B., Korsakoff patients
Spatial	Places and routes	Hippocampus (place cells); entorhinal cortex (grid cells)	Water maze; taxi drivers
Object recognition	"Seen this before?"	Perirhinal cortex	Delayed nonmatching-to-sample
Semantic knowledge	Word meanings, general facts	Anterior and inferior temporal cortex	Semantic dementia (5.11)
Working memory	Items held and used now	Dorsolateral prefrontal cortex	H.M.'s normal digit span
Procedural	Motor and cognitive skills, habits	Cerebellum, basal ganglia (striatum), motor cortex	Mirror drawing; Parkinson's, Huntington's
Priming	Easier processing of a repeated stimulus	Neocortex	Incomplete-pictures test
Classical conditioning	Learned reflexes and fears	Cerebellum (eyeblink); amygdala (fear)	H.M.'s eyeblink conditioning
Emotional enhancement	Stronger memory for arousing events	Amygdala, acting on the hippocampus	Propranolol study (1994)

Why do you remember where you were when you heard shocking news, but not last Tuesday's lunch? **James McGaugh** (born 1931) showed that arousal hormones (adrenaline and cortisol) activate the amygdala, which then strengthens consolidation in the hippocampus and cortex: it boosts memories rather than storing them. **Larry Cahill**, McGaugh and colleagues (1994) told participants an emotional story; those given propranolol, a beta-adren-

ergic blocker, lost the usual memory advantage for its emotional part, while memory for a neutral version was unaffected. Flashbulb memories (1.7) are the everyday result.

> EXAM TRAP
>
> Procedural learning: **cerebellum, basal ganglia and motor cortex**, not the hippocampus with amygdala (declarative and emotional), the frontal association cortex (working memory, planning) or the hindbrain with spinal cord (vital functions, reflexes). Enhanced memory for emotional events: **amygdala**, not the hippocampus (which stores the event), prefrontal cortex, thalamus (sensory relay), hypothalamus (homeostasis) or suprachiasmatic nucleus (circadian clock, 5.8).

Korsakoff's syndrome and its look-alikes ●●●

The second classic site of amnesia is the medial diencephalon. **Korsakoff's syndrome** (Korsakoff's psychosis), described in the 1880s by the Russian neuropsychiatrist **Sergei Korsakoff** (1854 to 1900), is an amnesic disorder caused by deficiency of **thiamine (vitamin B1)**, classically in chronic alcoholism with a poor diet: alcohol supplies calories without vitamins and impairs thiamine absorption. It also occurs in malnourished non-drinkers, so the vitamin, not the alcohol, is the main cause. Thiamine is needed to metabolise glucose, so energy-hungry neurons die first.

It usually follows **Wernicke's encephalopathy**, described by **Carl Wernicke** (1848 to 1905) in 1881: an acute emergency with a triad of confusion (disorientation, lapses of attention), ataxia (loss of muscle coordination) and ophthalmoplegia (paralysis of eye movements), often with nystagmus (jerky eye movements). Attention and coordination are both impaired. Untreated survivors settle into Korsakoff's amnesia, and the continuum is called **Wernicke-Korsakoff syndrome**.

Damage centres on the mammillary bodies and the mediodorsal (dorsomedial) thalamus, a **diencephalic amnesia**, though it is diffuse. The result is dense anterograde amnesia, severe retrograde amnesia with a temporal gradient reaching back decades, confusion, apathy, poor insight and **confabulation**: detailed, plausible but false stories that fill memory gaps, told sincerely without intent to deceive. Malingering, by contrast, is deliberate faking for gain; delirium is an acute, fluctuating disturbance of consciousness; confusion is muddled thinking without invented stories. Thiamine supplements for heavy drinkers help prevent it. Treatment is high-dose thiamine given before any glucose (a glucose load can precipitate Wernicke's encephalopathy); it halts damage, but established amnesia is largely permanent.

> EXAM TRAP
>
> Korsakoff's: **vitamin B1 (thiamine)**, not B12, niacin, B6, folate or vitamin D, hypoxia or lead. Wernicke-Korsakoff syndrome impairs **both** attention and muscle coordination. Memory loss with confabulation and nystagmus in a heavy drinker is Wernicke-Korsakoff syndrome, not Alzheimer's, vascular or frontotemporal dementia. Sincere invented memories are **confabulation**, not malingering, delirium or confusion.

Table 5.27 Look-alike options: what each actually is or causes.

Option	What it is or causes

Option	What it is or causes
Vitamin B12 (cobalamin) deficiency	Anaemia (as in pernicious anaemia); subacute combined degeneration of the spinal cord
Niacin (B3) deficiency	Pellagra: dermatitis, diarrhoea, dementia
Pyridoxine (B6) deficiency	Peripheral neuropathy; seizures in infants
Folate (B9) deficiency	Neural tube defects in pregnancy; anaemia
Vitamin D deficiency	Rickets (soft bones)
Hypoxia	Loss of CA1 neurons: hippocampal amnesia, as in R.B.
Lead poisoning	Lowered IQ, inattention and anaemia in children
Alzheimer's disease	Progressive dementia; recent episodic memory fails first (5.11)
Parkinson's and Huntington's disease	Basal ganglia diseases (Huntington's is inherited, with chorea: jerky involuntary movements); procedural learning impaired
Pick's disease, frontotemporal dementia	Frontal and temporal atrophy; personality and language change before memory
Creutzfeldt-Jakob disease	Prion disease; rapidly progressive dementia with myoclonus (muscle jerks)
Vascular dementia	Repeated strokes; stepwise decline
Multiple sclerosis	Demyelination in brain and spinal cord; not a classic amnesic syndrome
Schizophrenia	Psychotic disorder (Chapter 8); not an amnesic disorder
Transient global amnesia	Sudden dense anterograde amnesia lasting hours, then full recovery
Concussion; electroconvulsive therapy	Short retrograde gap before the blow or treatment, some anterograde loss; mostly recovers

Memory at the synapse: Aplysia and LTP ●●

Learning was first traced to particular synapses in a sea slug. **Eric Kandel** (born 1929) chose *Aplysia* because its large neurons can be identified one by one. A touch to its siphon makes it withdraw its gill. Repeated harmless touches weaken the reflex (**habituation**) because sensory neurons release less transmitter onto motor neurons; a tail shock strengthens it (**sensitisation**) because interneurons release serotonin onto the sensory terminals, raising cyclic AMP and transmitter release. Long-term forms need the transcription factor CREB, protein synthesis and new synapses. Kandel shared the 2000 Nobel Prize.

In mammals the breakthrough came in 1973: **Timothy Bliss** (born 1940) and **Terje Lømo** (born 1935) stimulated the perforant path into the dentate gyrus of the hippocampus in anaesthetised rabbits. After a brief burst of high-frequency (tetanic) pulses, single test pulses produced larger responses for hours, and in later work for weeks or months. This **long-term potentiation** (LTP) is a lasting strengthening of synaptic transmission after high-frequency stimulation. Four properties fit a memory mechanism: **longevity** (it lasts); **input specificity** (only the stimulated synapses strengthen); **cooperativity** (several inputs must be active together); and **associativity** (a weak input active with a strong one is potentiated too, a cellular version of pairing a conditioned with an unconditioned stimulus).

The secret lies in the receptors for glutamate, the brain's main excitatory transmitter (Figure 5.18). **AMPA receptors** carry ordinary fast excitation. The **NMDA receptor** is special: at resting potential its channel is plugged by a magnesium ion (Mg^{2+}), so glutamate alone lets al-

most nothing through. Only when glutamate is bound *and* the postsynaptic membrane is already depolarised is the magnesium expelled, letting calcium (Ca^{2+}) flood in. The NMDA receptor is thus a **coincidence detector** of presynaptic and postsynaptic activity: Hebb's rule in molecular form. (Glycine must also bind as a co-agonist but cannot open it alone; the receptor is excitatory and postsynaptic, so it does not trigger vesicle release.)

In the dendritic spine, calcium activates CaMKII (calcium/calmodulin-dependent protein kinase II), which strengthens AMPA receptors and inserts extra ones, so the same glutamate release produces a larger response. Late LTP, lasting days or more, needs CREB-driven gene expression, new proteins and growth of more and larger dendritic spines; LTP does not involve fewer spines, glutamate depletion or neuron death (apoptosis). **Richard Morris** and colleagues (1986) tied it to behaviour: rats given AP5, an NMDA receptor blocker, showed no hippocampal LTP and were badly impaired at learning where a platform lay hidden in the Morris water maze, a pool of opaque water.

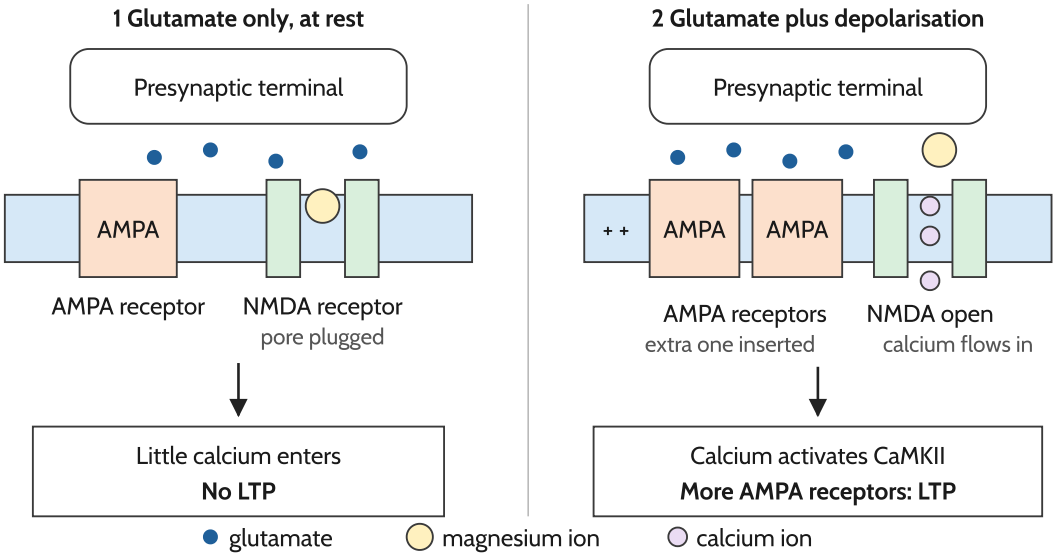

Figure 5.18 The NMDA receptor as a coincidence detector: magnesium plugs it when glutamate arrives alone (left) and leaves when the membrane is also depolarised (+ +), letting calcium in (right).

> **EXAM TRAP**
>
> LTP is a lasting **strengthening** after **high-frequency** stimulation, not a weakening (that is LTD), neuron death or myelination, and it is linked to **memory**, not thermoregulation, circadian rhythm or sleep. Its induction depends on **NMDA receptors**, not AMPA receptors alone, muscarinic M1 or GABA-A receptors. Magnesium blocks the NMDA channel, not the AMPA channel.

LTD and the dopamine teaching signal ●●

Synapses that could only strengthen would saturate. LTP's counterpart, **long-term depression** (LTD), is a lasting weakening of synaptic transmission. **Serena Dudek** and **Mark Bear** (1992) produced it in hippocampal field CA1 with prolonged low-frequency stimulation, about 1 Hz for 15 minutes. LTD there also needs NMDA receptors, but the modest, sustained calcium rise activates phosphatases (enzymes that remove phosphate groups) rather than kinases,

and AMPA receptors are pulled out of the membrane. A large, fast calcium surge gives LTP; a small, slow one gives LTD. LTD is not cell death, presynaptic GABA release, magnesium block of AMPA receptors or glutamate release without postsynaptic effect; high-frequency stimulation with NMDA receptors blocked produces neither LTP nor LTD.

Learning also needs a signal saying what is worth learning. **Wolfram Schultz** recorded from dopamine neurons of the ventral tegmental area and substantia nigra in monkeys learning that a light predicted juice. Before learning, the neurons burst when juice arrived unexpectedly. After learning, they burst at the light and ignored the expected juice; if the juice was withheld, firing paused when it should have come. Schultz, **Peter Dayan** and **Read Montague** (1997) identified this as a **reward prediction error**: reward received minus reward expected, positive for a pleasant surprise, zero for a predicted reward, negative for a disappointment. It is the surprise term of the Rescorla-Wagner model (Chapter 1), and the dopamine released strengthens the striatal synapses that were just active. Drug reward is covered in 5.4.

> **EXAM TRAP**
>
> The prediction-error signal comes from **dopamine neurons of the ventral tegmental area**, not serotonin release, GABA inhibition or acetylcholine release. LTD needs **prolonged low-frequency** stimulation; LTP needs brief high-frequency bursts.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Loss of memories from before an injury, worst for the most recent, is _____ amnesia.
2. Lashley concluded that memories are _____ throughout the cortex.
3. H.M.'s 1953 operation removed both _____ lobes.
4. R.B.'s amnesia followed damage to hippocampal field _____.
5. The beta blocker _____ abolished the memory advantage of emotional events.
6. Korsakoff's syndrome is caused by deficiency of _____ (vitamin B1).
7. At rest the NMDA channel is blocked by _____ ions.
8. Dopamine neurons that pause when an expected reward is omitted signal a reward _____ error.

Answers: (1) retrograde, (2) distributed, (3) medial temporal, (4) CA1, (5) propranolol, (6) thiamine, (7) magnesium, (8) prediction.

5.10 Lateralisation, the Split Brain and Language

MEDIUM YIELD · 28 QS

In 1861 a Paris surgeon found that a man who could say only one syllable had damage on the left side of his brain. That finding raised the two questions of this section: why do the hemispheres do different jobs, and where is language produced and understood?

Discovering the dominant hemisphere

Cerebral lateralisation is the tendency of a function to depend more on one cerebral hemisphere than on the other. In 1836 **Marc Dax**, a French country doctor, reported that none of his 40 or so patients with speech loss had damage confined to the right hemisphere, but nobody took notice. In 1861 **Paul Broca** (1824 to 1880) described Louis Victor Leborgne, nicknamed 'Tan' because it was almost the only syllable he could utter. Leborgne understood speech; at autopsy his lesion lay in the left inferior frontal cortex, and by 1864 Broca had seen the same in more patients. Because language, and later skilled movement, proved to depend on the left hemisphere, it was called the **dominant hemisphere** or major hemisphere, and the right the **minor hemisphere**. The second label is outdated.

Two tests reveal the language hemisphere of a living person. In the **Wada test** (sodium amytal test) of **Juhn Atsushi Wada** (1949), the barbiturate sodium amytal is injected into one carotid artery, anaesthetising that hemisphere for a few minutes; if it holds language, the patient falls mute. It is used before brain surgery. The harmless **dichotic listening** test plays different words or digits to the two ears at once. **Doreen Kimura** (1961) found that most people report more right-ear items, a right-ear advantage for words, because each ear's stronger pathway runs to the opposite hemisphere and the left hemisphere handles language; for melodies the advantage passes to the left ear.

About 90 percent of people are right-handed, and handedness is loosely tied to language. Language is in the left hemisphere in about 95 percent of right-handers and in most left-handers (roughly 70 percent); the rest have it on the right or in both hemispheres.

> EXAM TRAP
>
> The left hemisphere is the **major** (dominant) hemisphere. 'Spatial and pattern sense' and 'centre of musical ability' describe the right hemisphere, and 'minor hemisphere' is the old name for the right. Most left-handers are still left-dominant for language.

What each hemisphere does best

Table 5.28 summarises the asymmetries. Phonological processing means handling the sound structure of words; right-hemisphere control of spatial attention explains why neglect follows right parietal damage (5.11).

Table 5.28 Functional asymmetries in most right-handers. Each is a relative bias, not a monopoly.

Left hemisphere	Right hemisphere
Speech, phonology, grammar and syntax	Emotional prosody; understands simple words
Reading, writing, arithmetic	Visuospatial processing; spatial and pattern sense
Analytic, sequential processing	Holistic processing of whole patterns
Verbal memory	Non-verbal (visual, spatial) memory
Speech sounds	Music, melody, non-speech sounds
Skilled, complex movement	Faces; emotional expression; spatial attention

One right-hemisphere skill lies inside language itself. **Prosody** is the melody of speech: the intonation and emotional tone that let 'Oh, great' mean delight or sarcasm. Emotional

prosody depends on right-hemisphere regions homotopic to the language areas, that is, in their mirror-image positions. **Elliott Ross** (1981) described the **aprosodias**: damage opposite Broca's area leaves speech flat and monotonous, and damage opposite Wernicke's area leaves the patient deaf to the emotion in other voices, while words and grammar are spared.

The hemispheres differ in structure too. **Norman Geschwind** (1926 to 1984) and **Walter Levitsky** (1968) found the **planum temporale**, auditory cortex behind Heschl's gyrus that includes part of Wernicke's area, larger on the left in about two thirds of 100 brains. Popular talk of logical 'left-brained' and creative 'right-brained' people is a myth. Lateralisation is a matter of degree (the right-ear advantage is slight), every complex skill, creativity included, uses both hemispheres together, and imaging finds no 'left-brained' individuals.

> EXAM TRAP
>
> The function most reliably lateralised to the right is **visuospatial processing and prosody**; phonological processing, grammatical syntax and verbal memory are left-hemisphere. Speech intonation depends on **right-hemisphere regions homotopic to the language areas**, not Broca's area, the primary motor cortex or the occipital lobe.

The split brain ●●

The most striking evidence came from cutting the corpus callosum, the largest commissure linking the hemispheres (5.2). In 1953 **Ronald Myers** and **Roger Wolcott Sperry** (1913 to 1994) cut both the corpus callosum and the optic chiasm of cats, so each eye fed one hemisphere. A cat trained on a visual task with one eye patched had to relearn it from scratch with the other eye, as if it had two brains: the corpus callosum transfers learning between hemispheres.

From the 1960s surgeons cut the corpus callosum in people with intractable epilepsy, to stop seizures spreading between hemispheres. This **corpus callosotomy** (often called commissurotomy) reduced the seizures and left patients almost unchanged in daily life. Sperry and his student **Michael Gazzaniga** (born 1939) then tested them, work that won Sperry the Nobel Prize in 1981.

> KEY STUDY: SPERRY AND GAZZANIGA'S SPLIT-BRAIN PATIENTS (1960S)
>
> **Method:** the left visual field projects to the right hemisphere and the right field to the left (5.6). In **tachistoscopic presentation** the patient fixates a central dot while a picture is flashed to one side for about a tenth of a second, too fast for an eye movement. Each hand, hidden from view, reports to and is controlled by the opposite hemisphere. **Result:** objects flashed to the right visual field were named at once. Objects flashed to the left field reached only the non-speaking right hemisphere: the patient said he saw nothing, yet his left hand picked out the object by touch (Figure 5.19). Neither eye nor field was blind. **Criticism:** the patients were few, and long-standing epilepsy may have reorganised their brains.

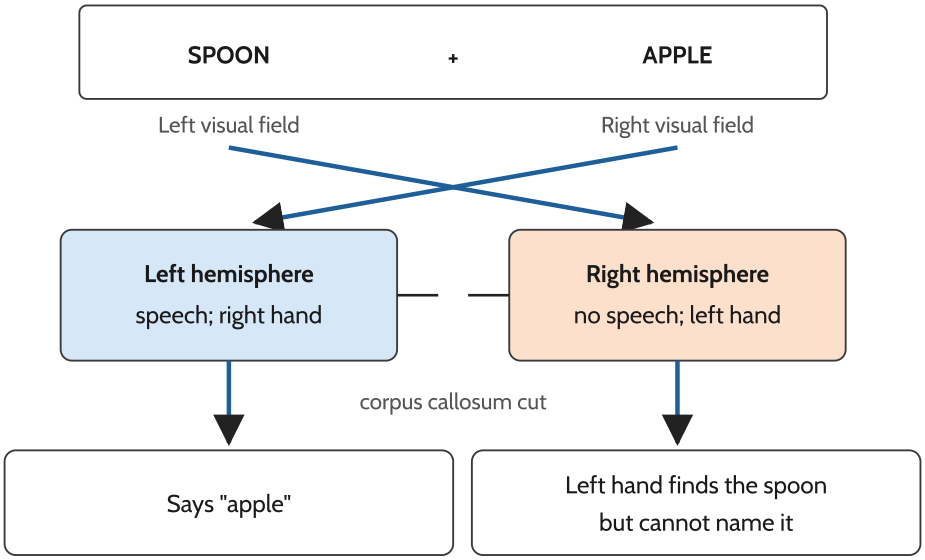

Figure 5.19 The split-brain test. Each visual field reaches only the opposite hemisphere, so the object seen by the right hemisphere can be found by the left hand but not named.

Four conclusions followed. Each hemisphere perceives, learns and remembers independently: shown two pictures at once, each hand fetches the object its own hemisphere saw. Language is not bilateral: in right-handers speech comes from the left hemisphere, and the right cannot speak, though it understands simple words. The right hemisphere is better at spatial tasks. And each hemisphere completes what it sees: shown **chimeric figures**, photographs joining halves of two different faces, patients of **Jerre Levy**, **Colwyn Trevarthen** and Sperry (1972) reported one whole face, naming the right-field half but pointing to the left-field half.

In Gazzaniga's best-known demonstration, one patient's left hemisphere saw a chicken claw and his right hemisphere a snow scene; his right hand chose a chicken, his left a shovel. Asked why, he said at once that the shovel was for cleaning out the chicken shed. Gazzaniga called this the **left-hemisphere interpreter**, which invents plausible reasons for behaviour whose real cause it does not know. A few patients show **intermanual conflict**: one man's left hand would switch off a television programme he was enjoying.

Outside the laboratory both hemispheres share the same eyes, ears and body, and some information crosses through the brainstem or by cross-cueing (one hemisphere signalling to the other with a frown). Cutting the corpus callosum therefore does not kill, blind or abolish language; its cost is difficulty coordinating activities that need both hemispheres.

> EXAM TRAP
>
> An object in the left visual field cannot be named but can be selected with the left hand; the patient is not blind in that eye or field and has not forgotten the object. The Nobel Prize went to Roger Sperry, not to **Karl Spencer Lashley** (1890 to 1958; the engram, 5.9), **Wilder Graves Penfield** (1891 to 1976; cortical stimulation mapping and the homunculus, 5.6) or **Donald Olding Hebb** (1904 to 1985; cell assemblies, 1949).

Broca's area and Wernicke's area ●●●

Lateralisation tells you which side; localisation tells you where within it. **Broca's area** lies in the left inferior frontal gyrus, in the pars opercularis and pars triangularis (areas 44 and 45, 5.2), just in front of the face area of the primary motor cortex. This frontal-lobe area serves speech production: programming articulation and assembling grammatical sentences. It is not the seat of comprehension, memory consolidation (the hippocampus, 5.9) or motor coordination (the cerebellum, 5.6).

In 1874 **Carl Wernicke** (1848 to 1905) described patients who spoke fluently but understood little, with damage further back. **Wernicke's area** is the posterior superior temporal gyrus of the dominant hemisphere (area 22), the classic centre for comprehending spoken and written language. In front of it, in Heschl's gyrus, the primary auditory cortex registers sounds without understanding them. The **arcuate fasciculus**, a bundle of association fibres, joins the two language areas. Behind Wernicke's area, where the temporal, parietal and occipital lobes meet, the **angular gyrus** (area 39) links a word's seen form to its sound and meaning, so it is vital for reading. The usual decoys have other jobs: the occipital pole holds primary visual cortex, the postcentral gyrus primary somatosensory cortex, the right parietal lobe spatial attention, and the anterior cingulate attention and error monitoring.

> EXAM TRAP
>
> Ria speaks slowly and laboriously in keywords that make sense. 'Frontal lobe' is true but too broad, and a temporal-lobe (Wernicke's area) lesion leaves speech fluent: the answer is **Broca's area**, also the site after a left frontal stroke with non-fluent speech. For comprehension the answer is **Wernicke's area** (posterior superior temporal gyrus), never the inferior frontal gyrus, Heschl's gyrus or the angular gyrus.

The Wernicke-Geschwind model and the dual streams

In 1965 Geschwind revived the ideas of Broca, Wernicke and Dejerine as the **Wernicke-Geschwind model**, a left-hemisphere circuit of seven parts (Figure 5.20). To repeat a heard word, the primary auditory cortex passes the sound to Wernicke's area, where it is understood; the arcuate fasciculus carries it to Broca's area, which selects an articulation programme; and the face area of the motor cortex drives the speech muscles. To read aloud, the primary visual cortex sends the written word to the angular gyrus, which converts it into its auditory code for Wernicke's area, and the same output path follows. Damage to each box or link predicts a particular aphasia.

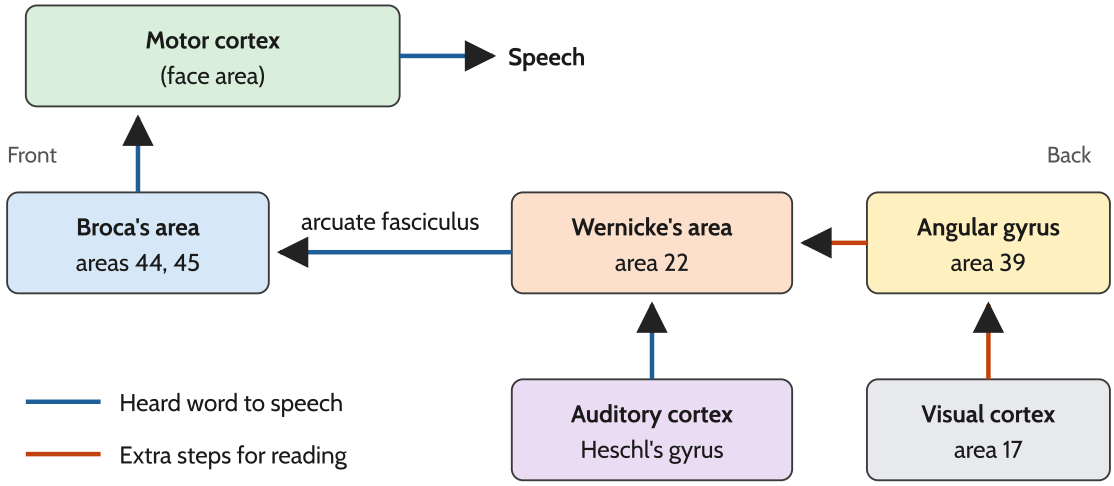

Figure 5.20 The Wernicke-Geschwind model, left hemisphere, front at the left. The solid route repeats a heard word; reading aloud enters by the dashed route and then follows the same output path.

The model's limits are serious. Removing Broca's area alone often leaves no lasting aphasia, real lesions are large and include white matter, and almost every aphasic patient has both expressive and receptive problems. Modern research favours the **dual-stream model** of **Gregory Hickok** and **David Poeppel** (2007). A ventral stream along the temporal lobe maps speech sounds onto meaning in both hemispheres; a dorsal stream from the temporoparietal junction to the frontal lobe, including the arcuate fasciculus, maps sounds onto articulation and is strongly left-lateralised. That is why damage to the dorsal route can spare understanding but ruin repetition.

Aphasia and its syndromes ●●●

Aphasia is an acquired disturbance of language function due to organic brain disease, most often a left-hemisphere stroke, and is the common term for all types of language loss. Dysphasia is often used for partial loss, dyslexia is a disorder of reading, and 'lexphasia' is not a term. Aphasia is a disorder of language, not of intellect, abstract thinking or memory, which fail in dementia and amnesia (5.9, 5.11). Clinicians classify aphasias by fluency, comprehension, repetition and naming (Table 5.29).

Broca's aphasia (expressive or non-fluent aphasia) is primarily a deficit of speech production. Speech is slow, effortful, poorly articulated and telegraphic, made of content words with the small grammatical words missing (**agrammatism**): 'Yes … Monday … Dad and Dick … Wednesday nine o'clock … doctors … and … teeth.' The keywords make sense, comprehension is relatively preserved, repetition is impaired, and the patient knows it and is frustrated. Right hemiparesis is common because the motor cortex lies next door. **Wernicke's aphasia** (receptive or fluent aphasia) is the mirror image: fluent, grammatical speech that is empty of meaning, full of **paraphasias** (wrong words or sounds) and neologisms (invented words), sometimes reaching word salad, with severely impaired comprehension, poor repetition and little awareness.

Table 5.29 The aphasia syndromes. Naming is impaired in all of them.

Aphasia	Speech	Comprehension	Repetition	Usual lesion

Aphasia	Speech	Comprehension	Repetition	Usual lesion
Broca's	Non-fluent, telegraphic	Relatively good	Poor	Broca's area
Wernicke's	Fluent, empty	Poor	Poor	Posterior superior temporal gyrus
Conduction	Fluent	Good	Very poor	Arcuate fasciculus, supramarginal gyrus
Global	Non-fluent	Poor	Poor	Whole perisylvian region
Transcortical motor	Non-fluent	Good	Good	Anterior or superior to Broca's area
Transcortical sensory	Fluent, echolalic	Poor	Good	Posterior watershed zone
Mixed transcortical	Non-fluent, echolalic	Poor	Good	Both watershed zones
Anomic	Fluent, word-finding pauses	Good	Good	Variable

A lesion of the arcuate fasciculus or the supramarginal region produces **conduction aphasia**: fluent speech and good comprehension but very poor repetition. A large lesion around the whole lateral fissure (the perisylvian region) causes **global aphasia**, with everything impaired. The transcortical aphasias follow lesions in the border (watershed) zones that spare the perisylvian loop, so repetition survives. In **transcortical motor aphasia** the lesion lies anterior or superior to Broca's area, sparing it: speech is non-fluent but repetition is preserved. **Transcortical sensory aphasia** resembles Wernicke's aphasia with intact repetition, often as echolalia (parroting what is heard); mixed transcortical aphasia combines both. In anomic aphasia the main problem is word finding, but **anomia** occurs in every aphasia.

EXAM TRAP

Damage to Broca's area gives non-fluent, effortful speech with relatively intact comprehension. Fluent aphasia with poor comprehension is Wernicke's, anomia occurs in every aphasia, failure to recognise faces is prosopagnosia (5.11), and hearing, reading and colour naming depend on other regions.

EXAM TRAP

Amity 2023 asked what '_____ is closely related to grammar and pronunciation' and keyed **aphasia**, the disorder in which grammar and pronunciation break down. The key is doubtful: the brain region concerned is **Broca's area**, which assembles grammar and programmes articulation, so expect the key 'aphasia' but know that the region is Broca's area. Wernicke's area serves comprehension, and agnosia is failure to recognise despite intact senses (5.11).

Reading, writing and speech beyond aphasia

Alexia is an acquired loss of reading and agraphia of writing. **Joseph Jules Dejerine** (1849 to 1917) linked alexia with agraphia to the left angular gyrus in 1891, and in 1892 described **alexia without agraphia** (pure alexia) after damage to the left occipital cortex and the splenium, the rear of the corpus callosum. His patient could write but not read what he had written: words seen by the intact right visual cortex could not cross to the left-hemisphere language areas. Such patients often cannot name colours either. Acalculia is an acquired loss of calcu-

lation. **Gerstmann syndrome** (**Josef Gerstmann**, 1920s) follows left angular gyrus damage and combines finger agnosia, left-right confusion, agraphia and acalculia.

Not every speech problem is aphasia. **Dysarthria** is faulty articulation from weak or uncoordinated speech muscles, and **apraxia of speech** is faulty programming of articulatory movements; in both, language itself is intact. **Developmental dyslexia** and other language-based learning difficulties are present from childhood despite adequate intelligence and teaching. Their core weakness is phonological, linked to the left perisylvian language network including Broca's area rather than to the occipital lobe (vision), cerebellum (coordination) or amygdala (emotion); the disorders are in Chapter 8. Finally, deaf signers with left-hemisphere damage develop sign-language aphasias of the Broca and Wernicke types, while right-hemisphere damage spares signing: the left hemisphere is specialised for language, not for sound.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The Wada test injects _____ into one carotid artery.
2. Dichotic listening shows a _____-ear advantage for words.
3. An object flashed to the left visual field reaches the _____ hemisphere and can be picked out with the _____ hand.
4. Emotional prosody depends on _____-hemisphere regions homotopic to the language areas.
5. Wernicke's area lies in the posterior _____ gyrus (area 22).
6. Fluent speech and good comprehension but very poor repetition is _____ aphasia.
7. Non-fluent speech with preserved repetition is _____ aphasia.

Answers: (1) sodium amytal, (2) right, (3) right; left, (4) right, (5) superior temporal, (6) conduction, (7) transcortical motor.

5.11 Brain Damage, Neurodegeneration and Neuropsychological Syndromes

MEDIUM YIELD · 32 QS

Much of what we know about the healthy brain comes from damaged ones: what a patient loses, and what survives, shows what the damaged part did. This section covers how brains are damaged, the major neurological diseases and the classic neuropsychological syndromes.

Causes of brain damage

Brain damage has six main causes. A **brain tumour** (neoplasm) is a mass of cells growing independently of the rest of the body. About 20 percent are **meningiomas**, encapsulated in their own membrane between the meninges, harmful only by pressure and almost always benign. Most others are **infiltrating**, growing diffusely into brain tissue and usually malignant; **gliomas**, from glial cells, are the commonest. About 10 percent are **metastatic**, seeded through the blood from cancers elsewhere, often the lungs.

A **stroke** is a sudden-onset cerebrovascular disorder. Around the dead tissue (the infarct) lies the **penumbra**, tissue that may recover or die over the next few days and that treatment tries

to save. **Cerebral haemorrhage** is bleeding from a ruptured vessel, often an aneurysm. **Cerebral ischaemia** is a blocked blood supply, from *thrombosis* (a plug formed on the spot), *embolism* (a plug carried from elsewhere) or *arteriosclerosis* (walls narrowed by fat deposits). Starved neurons release excess glutamate, which overactivates NMDA receptors and floods neighbouring neurons with Na^+ and Ca^{2+}, killing them over a day or two: **excitotoxicity** (Section 5.3).

A **closed-head traumatic brain injury** leaves the skull intact. A **contusion** (bruised brain) forms where the brain strikes the skull, at the site of the blow (coup) and often on the opposite side (**contrecoup injury**). **Concussion**, now called mild traumatic brain injury, disturbs consciousness without structural damage, but repeated ones produce **chronic traumatic encephalopathy**, the dementia of boxers and American football players. The confused period after a head injury in which new memories are not stored is **post-traumatic amnesia**; its length is the best single guide to severity.

Infection causes **encephalitis** (brain inflammation): herpes simplex encephalitis attacks the medial temporal lobes (Clive Wearing, Section 5.9), bacteria cause meningitis, neurosyphilis causes **general paresis** (psychosis and dementia) and HIV can cause dementia. **Neurotoxins** include lead, mercury, alcohol (Korsakoff's syndrome, Section 5.9) and antipsychotic drugs (tardive dyskinesia, Section 5.4). **Genetic** causes include recessive genes, the dominant gene of Huntington's disease and the extra chromosome 21 of Down syndrome. Whatever the cause, neurons die by **necrosis**, passive swelling and bursting within hours that inflames nearby tissue, or by **apoptosis**, programmed shrinkage over a day or two without inflammation.

Epilepsy

Epilepsy is a disorder of recurrent, spontaneous seizures arising from chronic brain dysfunction; a single seizure from a high fever does not count. Neurons discharge in synchronous bursts, often because GABA inhibition fails, and the EEG shows high-amplitude spikes. Many patients first sense an **aura** (a strange smell, déjà vu), which betrays where the seizure starts. **Generalised** seizures involve the whole brain, **focal** (partial) seizures only part of it (Table 5.30). Simple focal seizures are sometimes called Jacksonian after **John Hughlings Jackson** (1835 to 1911), whose **Jacksonian march** is jerking that spreads from fingers to arm to face as the discharge crosses the motor homunculus. Complex focal seizures usually begin in the temporal lobe (**temporal lobe epilepsy**) and bring **automatisms** such as fumbling with buttons. The International League Against Epilepsy (ILAE) classification of 2017 grades focal seizures by awareness instead: **focal aware** (simple) and **focal impaired awareness** (complex).

Table 5.30 The main seizure types.

Seizure	Consciousness	What you see	EEG
Tonic-clonic (grand mal)	Lost	Tonic stiffening, then clonic rhythmic jerking; tongue biting, cyanosis; postictal confusion	Spikes across the whole cortex
Absence (petit mal)	Lost for seconds	Blank stare, eyelid flutter; tone kept; mainly children	3-per-second spike-and-wave
Simple focal	Kept	Signs in one limb; Jacksonian march	Focal spikes
Complex focal	Impaired	Aura, automatisms, no memory of the episode	Temporal spikes

A seizure lasting more than five minutes, or repeated seizures without recovery of consciousness between them, is **status epilepticus** (older definitions said 30 minutes), a medical emergency. Anticonvulsants control seizures but cure none; resistant cases may have the focus removed (H.M., Section 5.9) or the corpus callosum cut (split brain, Section 5.10).

> EXAM TRAP
>
> Epilepsy is a neurological disorder but **not a neurodegenerative one**. Parkinson's, Alzheimer's and Huntington's diseases are progressive losses of neurons; epilepsy is episodic, often with normal function between seizures.

Parkinson's and Huntington's diseases ●●

Parkinson's disease, described in 1817 by **James Parkinson** (1755 to 1824) as the "shaking palsy", affects 1 to 2 percent of people over 65. Its signs are resting tremor, rigidity, **bradykinesia** (slowness) or akinesia (difficulty starting movement), postural instability, a masked face and a shuffling gait; loss of smell and depression may come years earlier, and dementia often follows. Dopamine neurons of the **substantia nigra pars compacta** die, so the **nigrostriatal pathway** no longer delivers dopamine to the striatum (Figure 5.21); in 1960 **Oleh Hornykiewicz** found the patients' striatum almost empty of dopamine. Surviving nigral neurons hold **Lewy bodies**, clumps of alpha-synuclein. **L-dopa**, the dopamine precursor, which unlike dopamine crosses the blood-brain barrier, and dopamine agonists relieve symptoms for some years; **deep brain stimulation** of the subthalamic nucleus helps when drugs fail. In 1982 young drug users in California who injected a "synthetic heroin" contaminated with **MPTP** developed permanent parkinsonism (reported by **J. William Langston** in 1983); MPTP destroys nigral dopamine neurons in monkeys too, giving the best animal model of the disease.

> EXAM TRAP
>
> Parkinson's disease is **dopamine** loss from the **substantia nigra pars compacta**. It is not degeneration of the locus coeruleus (noradrenaline), raphe nuclei (serotonin), spinal motor neurons (motor neuron disease, polio), hippocampal pyramidal cells (memory) or cholinergic basal forebrain (Alzheimer's), nor a serotonin, acetylcholine or GABA deficiency. The subthalamic nucleus is the stimulation target, not the lesion.

Huntington's disease, described by **George Huntington** in 1872, is rarer (about 1 in 7,500): **chorea** (jerky, dance-like involuntary movements), psychiatric symptoms and cognitive decline ending in dementia, beginning at about 40. An autosomal dominant expanded CAG repeat in the *huntingtin* gene on chromosome 4 (Section 5.5) kills the GABA neurons of the striatum, caudate first. Parkinson's is hypokinetic, Huntington's hyperkinetic.

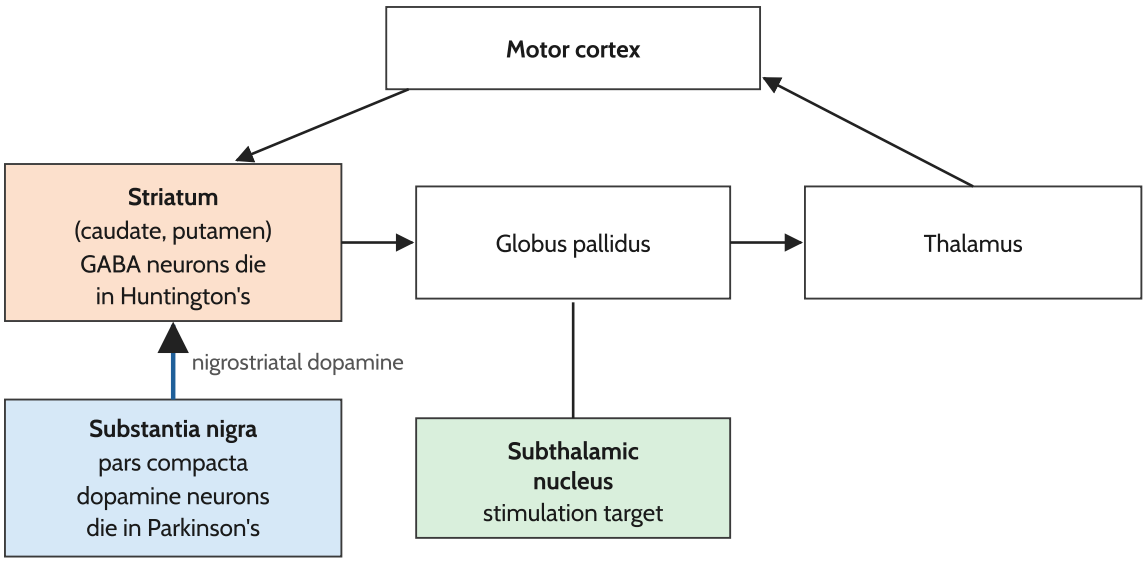

Figure 5.21 The basal ganglia motor loop. Parkinson's disease removes the nigrostriatal dopamine input (thick arrow); Huntington's disease destroys the striatum's own GABA neurons.

Alzheimer's disease and other dementias ●●

Dementia is an acquired, usually progressive decline of global intellect (memory, abstraction, judgement) that disrupts daily life; its criteria belong to Chapter 8, and "senile dementia" and "atrophy" are vague older labels. The commonest cause is **Alzheimer's disease**, described in 1906 by **Alois Alzheimer** (1864 to 1915) in his patient **Auguste Deter**. Onset is insidious and progression gradual: recent episodic memory fails first, then language (anomia), visuospatial and executive functions. The brain shows **amyloid plaques** (beta-amyloid, outside cells), **neurofibrillary tangles** (tau, inside cells) and neuron loss that begins in the entorhinal cortex and hippocampus. Death of cholinergic neurons in the **nucleus basalis of Meynert** leaves the cortex short of acetylcholine, so treatment uses **cholinesterase inhibitors** (donepezil, rivastigmine, galantamine) and **memantine**, an NMDA receptor antagonist. The **APOE e4** allele raises risk; rare early-onset families carry APP or presenilin mutations, and because APP lies on chromosome 21, almost everyone with Down syndrome has plaques and tangles by 40.

> EXAM TRAP
>
> Acetylcholine deficiency means **Alzheimer's disease**, not "senile dementia", "atrophy" or Korsakoff's syndrome (thiamine deficiency, Section 5.9). Alzheimer's is not a loss of GABA (Huntington's), noradrenaline or histamine (Section 5.3).

Table 5.31 compares the other dementias. Smell loss is an early sign of both Parkinson's and Alzheimer's disease.

Table 5.31 The dementias compared.

Dementia	Pathology	Course and distinctive signs
Alzheimer's disease	Plaques, tangles; cholinergic loss	Insidious, gradual; memory first, then language and visuospatial skills

Dementia	Pathology	Course and distinctive signs
Dementia with Lewy bodies	Cortical Lewy bodies	Fluctuating cognition; recurrent well-formed visual hallucinations; parkinsonism; REM sleep behaviour disorder; severe antipsychotic sensitivity
Vascular dementia	Multiple infarcts	Stepwise decline; patchy deficits
Frontotemporal (Pick's disease)	Frontal and temporal atrophy; Pick bodies	Early personality change, disinhibition, or language variants; memory spared early
Creutzfeldt-Jakob disease	Prions (infectious mis-folded protein)	Very rapid decline; myoclonic jerks

EXAM TRAP

Lewy body dementia means **visual** hallucinations, not auditory ones (typical of schizophrenia). Aphasia points to frontotemporal or late Alzheimer's disease, and smell loss is not specific.

Multiple sclerosis ●

Multiple sclerosis is an autoimmune attack on the myelin that oligodendrocytes make in the central nervous system. Hard scars, sclerotic **plaques**, form in the white matter, **disseminated in time and space**. It usually starts in early adulthood with a **relapsing-remitting** course of visual disturbance, weakness, numbness and fatigue, and is commoner in women and far from the equator. Contrast alpha-synuclein deposits (Parkinson's), beta-amyloid plaques (Alzheimer's) and degeneration of lower motor neurons (polio, and amyotrophic lateral sclerosis, which kills upper motor neurons too).

Agnosia, apraxia and other higher-function disorders ●●●

Damage to association cortex disturbs higher functions rather than simple sensation or movement. Most of their names begin with Greek "a-" (without) and are best learned together (Table 5.32).

Table 5.32 Disorders of higher function.

Disorder	What is lost	Note
Dementia	Global intellect and abstraction	Acquired, progressive
Amnesia	Memory	Section 5.9
Aphasia, alexia, acalculia	Language, reading, calculation	Section 5.10
Anomia	Naming	Word-finding failure
Agnosia	Recognition	Sensation intact
Apraxia	Learned skilled movement (motor skills)	Strength, coordination intact
Ataxia	Coordination	Cerebellum (Section 5.6)
Dyspraxia	Coordination, from childhood	Developmental coordination disorder
Anosmia	Smell	Olfactory damage

Agnosia, a term coined by **Sigmund Freud** in 1891, is failure to recognise despite intact sensation and intellect. **Heinrich Lissauer** (1890) split visual object agnosia into **apperceptive** (no whole percept forms, so the patient cannot copy a drawing) and **associative** (the drawing is copied but not recognised). In **prosopagnosia** (face blindness), faces are seen as faces but not recognised, even those of close family, who are known at once by voice. It follows bilateral or right-sided damage to the fusiform gyrus at the occipitotemporal junction, the **fusiform face area** identified by **Nancy Kanwisher** and colleagues in 1997. **Simultanagnosia**, seeing only one object at a time, is part of Bálint's syndrome after bilateral posterior parietal damage. **Anosognosia** (**Joseph Babinski**, 1914) is unawareness of one's own deficit. **Astereognosis** is failure to recognise objects by touch, and **auditory agnosia** failure to recognise sounds.

> EXAM TRAP
>
> Prosopagnosia is inability to **recognise faces**. The IHBAS 2024 key gives "distinguish tastes", which is wrong (loss of taste is ageusia); it is also not loss of facial expression or of object recognition. Face-recognition loss after a stroke is an **agnosia**, not aphasia, apraxia or ataxia, and prosopagnosia beats the broader "visual agnosia". The lesion is fusiform, not in the primary motor cortex (movement), dorsolateral prefrontal cortex (executive function) or cerebellar vermis (balance).

Apraxia (**Hugo Liepmann**, 1900) is inability to perform learned purposeful movements despite intact strength, sensation, coordination and comprehension. In **ideomotor apraxia** the patient cannot mime an act on command (show how you use a comb) though he does it naturally; in **ideational apraxia** he cannot sequence the steps of an act such as brushing his hair. **Constructional apraxia** spoils copying and block building, **dressing apraxia** putting on clothes. The lesion is usually left parietal or frontal.

> EXAM TRAP
>
> "Impairment in motor performance and coordination" is **dyspraxia** (developmental coordination disorder; Amity 2023 key). Apraxia is acquired loss of skilled movement with coordination intact; ataxia is cerebellar incoordination; anomia is failure to name, alexia to read.

Hemispatial neglect •

After right posterior parietal damage, patients with **hemispatial neglect** ignore the left side of space and of their own body. They eat from the right of the plate, shave only the right side of the face, bisect lines to the right and miss left targets on cancellation tests (Figure 5.22). In **extinction**, a left stimulus is noticed alone but missed when paired with a right one, and anosognosia is common. Left neglect is far commoner and more severe than right neglect, because the right hemisphere attends to both sides of space. Neglect is inattention, not blindness: in **homonymous hemianopia** the patient knows half the visual field is lost and compensates. Bilateral damage to primary visual cortex causes **cortical blindness** (bilateral homonymous hemianopia), sometimes denied in **Anton's syndrome**; **hemiplegia** (one-sided paralysis) follows motor cortex or internal capsule damage.

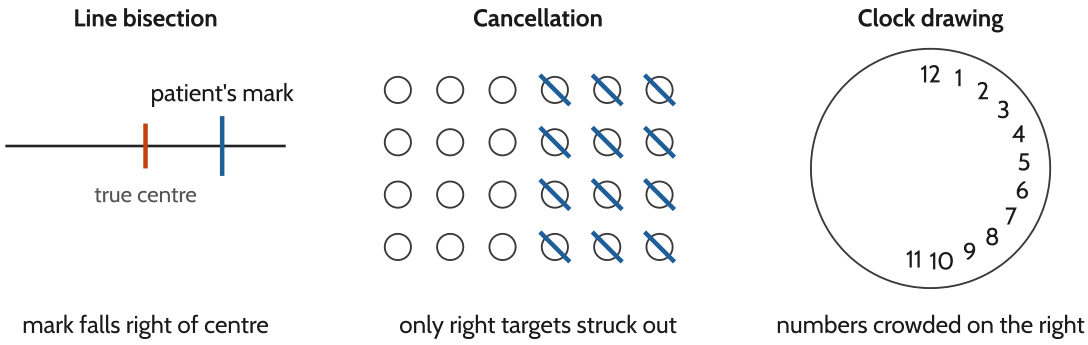

Figure 5.22 Left neglect after right parietal damage on three bedside tests (dashed line: true centre). Everything on the left is ignored.

EXAM TRAP

Right parietal lesion: **left** neglect. Anterograde amnesia (Section 5.9), expressive (Broca's) aphasia (Section 5.10), anosmia, prosopagnosia and bilateral homonymous hemianopia are distractors.

Phineas Gage and the prefrontal syndromes ●●●

KEY STUDY: PHINEAS GAGE (1848)

In 1848 an explosion drove a tamping iron through the left cheek and frontal lobe of **Phineas Gage** (1823 to 1860), a 25-year-old railway construction foreman in Vermont. His speech, memory and intelligence survived, but the reliable foreman became impulsive, profane and poor in judgement, and soon lost his job. His physician **John Martyn Harlow** reported the case in 1848 and, in 1868, that friends found him "no longer Gage". In 1994 **Hanna Damasio** and colleagues reconstructed the injury from his skull and placed it in the **ventromedial prefrontal cortex**. Criticism: the record is thin, and Gage later drove stagecoaches in Chile for several years before dying in San Francisco, so he may have partly recovered.

Gage is the classic evidence that the frontal lobes govern personality and social behaviour. Damage to different prefrontal regions produces different frontal lobe syndromes (Table 5.33). Dorsolateral damage causes the **dysexecutive syndrome**: **executive functions** (planning, sequencing, set-shifting and inhibition, using working memory) break down, and the patient shows **perseveration**, repeating a response after it has stopped working. In the **Wisconsin Card Sorting Test** of **David Grant** and **Esta Berg** (1948) cards are sorted by colour, shape or number, the rule being learned only from feedback and changed without warning; in 1963 **Brenda Milner** showed that frontal patients keep sorting by the old rule. Planning is tested by the Tower of London of **Tim Shallice** (1982), inhibition by the Stroop test, and verbal fluency by naming words that begin with a given letter.

Table 5.33 The prefrontal syndromes.

Region	Main deficits	Tests and cases
Dorsolateral	Working memory, planning, set-shifting, abstraction; perseveration	Wisconsin Card Sorting, Tower of London, Stroop, fluency

Region	Main deficits	Tests and cases
Orbitofrontal and ventromedial	Disinhibition, impulsivity, poor social judgement (pseudopsychopathy)	Iowa gambling task; Gage, E.V.R.
Medial frontal and anterior cingulate	Apathy, abulia (loss of will), akinetic mutism	Bilateral medial frontal damage

Orbitofrontal and ventromedial damage spares intelligence but wrecks conduct, as in patient E.V.R. (**Paul Eslinger** and **Antonio Damasio**, 1985), who kept a superior IQ after removal of an orbitofrontal tumour yet made ruinous decisions. Antonio Damasio's **somatic marker hypothesis** holds that this cortex tags choices with bodily emotional signals ("gut feelings"); without them, as the **Iowa gambling task** of **Antoine Bechara** and colleagues (1994) shows, patients keep choosing risky card decks. Prefrontal leucotomy or lobotomy (**António Egas Moniz**, Nobel Prize 1949; popularised by **Walter Freeman**) cut these connections and left patients apathetic.

> EXAM TRAP
>
> Gage means **ventromedial prefrontal cortex**, not the occipital lobe (vision), temporal lobe (hearing, memory), posterior parietal cortex (spatial attention), hippocampus and amygdala (memory, fear) or cerebellum and vermis (coordination). The Wisconsin test detects **dorsolateral** dysfunction, not hippocampal (long-term declarative memory), motor cortex or cerebellar damage, colour perception (visual cortex) or reflexes (spinal cord). Orbitofrontal damage gives disinhibition, not pure aphasia, cortical blindness or hemiplegia. Impaired executive function is the hallmark of the frontal lobe syndrome, not of obsessive-compulsive disorder, catatonic schizophrenia (Chapter 8) or temporal lobe epilepsy.

Recovery and plasticity after damage

When an axon is cut, its distal segment breaks up within days (**anterograde degeneration**, or Wallerian degeneration, after **Augustus Waller**, 1850), while **retrograde degeneration** creeps back to the cell body. Peripheral axons often **regenerate**, guided by Schwann cells; central axons rarely do, blocked by oligodendrocyte inhibitors and glial scars. Instead, healthy neighbouring axons grow branches into vacated synapses (**collateral sprouting**), and neighbouring cortex takes over lost territory (**cortical reorganisation**). Recovery is best with small lesions and young patients, and people with high **cognitive reserve** (roughly, education and intelligence) compensate better. In **constraint-induced movement therapy** (**Edward Taub**, 1993) a stroke patient's good arm is restrained while the weak arm is trained intensively, enlarging its cortical representation. Batteries such as the Halstead-Reitan, Luria-Nebraska and NIMHANS battery chart the losses (Chapter 2).

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Tissue around a stroke's infarct that treatment tries to save is the _____.
2. A 3-per-second spike-and-wave EEG marks the _____ seizure.
3. Parkinson's disease kills dopamine neurons of the _____ pars compacta.
4. Alzheimer's disease kills cholinergic neurons of the nucleus basalis of _____.
5. Recurrent well-formed visual hallucinations suggest dementia with _____.
6. Knowing one's wife by voice but not by face is _____.
7. Right posterior parietal damage causes neglect of the _____ side.
8. The Wisconsin Card Sorting Test is most sensitive to _____ prefrontal damage.

Answers: (1) penumbra, (2) absence, (3) substantia nigra, (4) Meynert, (5) Lewy bodies, (6) prosopagnosia, (7) left, (8) dorsolateral.

Loewi's Two Frog Hearts, Revisited

Every section of this chapter has passed through Loewi's laboratory. The nerve he stimulated was the vagus, the main parasympathetic nerve, whose rest-and-digest outflow slows the heart (5.1). The slowing was a chemical message crossing a synapse: acetylcholine binding to receptors and then broken down by acetylcholinesterase (5.3). Drugs act by imitating, boosting or blocking such messages (5.4), and Loewi's logic of changing one thing and measuring another runs through the methods of 5.5. The opposite state, the pounding heart of fear, belongs to the sympathetic division and the adrenal medulla (5.7). His idea surfaced at night, out of sleep (5.8). And when Alzheimer's disease strips the cortex of acetylcholine, the drugs prescribed are cholinesterase inhibitors, which keep Vagusstoff working longer (5.11).

Chapter summary

5.1 Biopsychology and the Organisation of the Nervous System. The biological perspective explains behaviour through the body, brain and nervous system. Biopsychology (Pinel) has six divisions and progresses by converging operations; Descartes's dualism opposes monism. The CNS is the brain and spinal cord; the PNS has somatic (voluntary) and autonomic divisions. The sympathetic division (fight or flight, Cannon 1915; T1 to L2; adrenal medulla) spends energy; the parasympathetic (rest and digest; craniosacral; vagus; acetylcholine) conserves it. A reflex arc runs afferent neuron, interneuron, efferent neuron. White matter is myelinated axons; PNS cell-body clusters are ganglia; dorsal roots are sensory, ventral roots motor. Twelve cranial nerves, the meninges, CSF and the blood-brain barrier complete the map.

5.2 The Brain: Brainstem, Diencephalon and Cerebral Cortex. The embryonic forebrain, midbrain and hindbrain become five divisions: telencephalon (cortex, limbic system, basal ganglia, corpus callosum), diencephalon (thalamus, hypothalamus), mesencephalon, metencephalon (pons, cerebellum) and myelencephalon (medulla). The medulla holds the vital centres for breathing and heart rate; the cerebellum coordinates movement and balance; the reticular activating system maintains arousal (Moruzzi and Magoun, 1949). The thalamus relays every sense except smell,

and the hypothalamus governs homeostasis through the pituitary. Six-layered neocortex, about 90 percent of cortex, serves higher cognition across frontal, parietal, temporal and occipital lobes, mapped by Brodmann (1909); commissural, association and projection fibres connect it.

5.3 Neurons, Neural Conduction, Synapses and Neurotransmitters. The neuron is the basic unit of the nervous system (Cajal's neuron doctrine; Nobel 1906, shared with Golgi); glia include astrocytes, microglia and the myelin makers, oligodendrocytes (CNS) and Schwann cells (PNS). The resting potential, about -70 mV, rests on K+ permeability and the pump (3 Na+ out, 2 K+ in). Summed EPSPs reaching threshold fire an all-or-none action potential (Na+ in, K+ out, peak about +50 mV), then refractory periods. Myelin gives fast saltatory conduction. Ca2+ influx triggers release; ionotropic and metabotropic receptors respond; reuptake or enzymes clear the transmitter. Glutamate excites, GABA inhibits, acetylcholine drives muscle, monoamines arise in brainstem nuclei, and endorphins control pain.

5.4 Drugs, Reward and Psychopharmacology. Psychoactive drugs act as agonists or antagonists at synapses. Tolerance (metabolic, functional, cross and conditioned, as in Siegel's 1982 heroin study) and withdrawal are distinct from addiction. Benzodiazepines raise the frequency and barbiturates the duration of GABA-A channel opening; heroin is a semisynthetic opioid; THC acts at CB1 receptors. Olds and Milner (1954) found self-stimulation sites in the septal area, and the mesolimbic pathway (VTA to nucleus accumbens) is the core reward circuit. Antipsychotics block D2, and nigrostriatal blockade causes extrapyramidal effects ending in tardive dyskinesia. SSRIs work after 2 to 4 weeks; lithium (Cade 1949) is the gold-standard mood stabiliser; disulfiram blocks aldehyde dehydrogenase.

5.5 Research Methods, Behaviour Genetics and the Developing Brain. Structural imaging (contrast X-ray, CT, MRI, DTI) shows anatomy; functional imaging (PET, SPECT, the BOLD signal of fMRI, fNIRS) shows activity, and paired-image subtraction revealed the default mode network, active at rest. EEG has the best temporal resolution, and averaging it yields ERPs such as the P300 to oddball targets. Lesions, stimulation (DBS, TMS, optogenetics), microdialysis, knockouts and CRISPR-Cas9 manipulate the brain. Mendelian inheritance and epigenetics underlie twin, adoption, linkage and GWAS designs, heritability and endophenotypes. PKU, Huntington's, fragile X, Down, Turner, Klinefelter and Prader-Willi syndromes show genetic errors. The neural tube forms at gastrulation; pruning, myelination and neuroplasticity continue for years.

5.6 Sensory Pathways and the Motor System. Sensory systems transduce energy, build receptive fields and climb from receptors through thalamic relays to primary, secondary and association cortex. In the inverted retina, lateral inhibition sharpens edges and creates Mach bands. Nasal fibres cross at the optic chiasm; the LGN's magnocellular layers carry motion, its parvocellular layers colour and detail, to V1, whose damage gives contralateral homonymous hemianopia, often with macular sparing. Dorsal 'where/how' and ventral 'what' streams leave V1. Hair cells, two somatosensory pathways and PAG analgesia complete the senses. Corticospinal fibres cross at the pyramidal decussation; mirror neurons fire for doing and watching; the basal ganglia select actions and the cerebellum corrects errors.

5.7 The Endocrine System, Stress and the Emotional Brain. Endocrine glands are ductless and release hormones into the blood; exocrine glands use ducts. Hormones are amino acid derivatives, peptides or steroids. The hypothalamus controls the pituitary, the master gland: releasing hormones reach the anterior lobe (GH, ACTH, TSH, FSH, LH, prolactin) through portal blood,

while oxytocin and vasopressin, made in the supraoptic and paraventricular nuclei, travel down axons to the posterior lobe. The adrenals run stress: the fast SAM system releases adrenaline, and the slower HPA axis (CRH, ACTH, cortisol) is braked by negative feedback. Allostasis is stability through change. Sham rage, the Papez circuit and the limbic system led to the amygdala, the hub of fear.

5.8 Sleep, Biological Rhythms, Hunger, Thirst and Sex. Circadian rhythms are timed by the suprachiasmatic nucleus (SCN), entrained by light through the retinohypothalamic tract, which drives night-time melatonin from the pineal gland. Sleep EEG runs from beta and alpha waking through theta (N1) and spindles (N2) to delta slow-wave sleep (N3); REM is paradoxical, with a waking-like EEG and atonia. Sleep begins when the VLPO inhibits the arousal systems; adenosine builds sleep pressure, and orexin, lost in narcolepsy, holds waking. Ghrelin starts hunger, CCK and leptin bring satiety, VMH lesions cause obesity and LH lesions aphagia, and arcuate NPY neurons trigger eating. Masters and Johnson described excitement, plateau, orgasm and resolution.

5.9 The Biology of Learning, Memory and Amnesia. Amnesia is a pathological loss of memory: anterograde (no new memories after the onset) or retrograde (loss of earlier memories, worst for recent ones, Ribot's law). Lashley's search for Semon's engram found memories distributed across the cortex, which Hebb explained with cell assemblies. H.M.'s bilateral medial temporal lobectomy (1953) showed that the hippocampus consolidates new declarative memories but not skills; R.B., place and grid cells, K.C. and Clive Wearing refined the picture, and the amygdala strengthens emotional memories. Korsakoff's syndrome follows thiamine deficiency and damages the mammillary bodies and mediodorsal thalamus. At the synapse, NMDA-dependent LTP strengthens and LTD weakens connections, while dopamine signals reward prediction errors.

5.10 Lateralisation, the Split Brain and Language. Dax (1836) and Broca (1861, patient Leborgne, 'Tan') tied language to the left, dominant hemisphere; the Wada test (1949) and Kimura's dichotic listening right-ear advantage (1961) confirm it in about 95 percent of right-handers and most left-handers. The right hemisphere leads in visuospatial processing, faces, music and emotional prosody. Split-brain studies (Myers and Sperry 1953; Sperry and Gazzaniga, Nobel 1981) showed each hemisphere working independently. Broca's area (inferior frontal gyrus, areas 44 and 45) serves speech production; Wernicke's area (posterior superior temporal gyrus, area 22) comprehension. Aphasia is acquired language loss from organic brain disease: Broca's, Wernicke's, conduction, global, transcortical and anomic types.

5.11 Brain Damage, Neurodegeneration and Neuropsychological Syndromes. Brain damage comes from tumours (encapsulated meningiomas, infiltrating gliomas, metastases), strokes (haemorrhage or ischaemia, with glutamate excitotoxicity in the penumbra), closed-head injury (contusions, contrecoup, chronic traumatic encephalopathy), infections, neurotoxins and genes; neurons die by necrosis or apoptosis. Epilepsy (tonic-clonic, absence, focal seizures) is neurological but not neurodegenerative. Parkinson's disease is dopamine loss from the substantia nigra pars compacta; Huntington's disease kills striatal GABA neurons; Alzheimer's disease brings plaques, tangles and acetylcholine loss. Multiple sclerosis is autoimmune demyelination. Agnosia (prosopagnosia), apraxia, right parietal left neglect and the prefrontal syndromes (Gage, Wisconsin Card Sorting Test) complete the section.

Who's who

Table 5.34 The people in this chapter.

Name	When	Examined for
Otto Loewi	1873 to 1961	Two frog hearts experiment (1921): vagus stimulation releases a chemical, Vagusstoff (acetylcholine); Nobel Prize 1936 with Henry Dale
Henry Hallett Dale	1875 to 1968	Acetylcholine and chemical transmission of nerve impulses; Nobel Prize 1936 shared with Loewi
Donald Olding Hebb	1904 to 1985	The Organization of Behavior (1949), which played a key role in the emergence of biopsychology
John P. J. Pinel	2021 (Biopsychology, 11th edition)	Biopsychology as a neuroscience; its six divisions; converging operations
Steven J. Barnes	2021	Co-author with Pinel of Biopsychology, 11th edition
René Descartes	1596 to 1650	Mind-body dualism; mind and body meeting at the pineal gland
Walter Bradford Cannon	1871 to 1945	Fight-or-flight response of the sympathetic division (1915)
Charles Bell	1774 to 1842	Bell-Magendie law: motor role of the ventral roots (1811)
François Magendie	1783 to 1855	Bell-Magendie law: dorsal roots sensory, ventral roots motor (1822)
Giuseppe Moruzzi	1949	With Magoun, showed that stimulating the brainstem reticular formation wakes sleeping cats and desynchronises the EEG: the reticular activating system
Horace Magoun	1949	With Moruzzi, proposed that reticular activity controls sleep and wakefulness (the reticular activating system)
Korbinian Brodmann	1868 to 1918	Numbered map of the cortex by cytoarchitecture (1909): areas 4 motor, 17 visual, 41 and 42 auditory, 44 and 45 Broca's area
Camillo Golgi	1843 to 1926	Golgi silver stain (early 1870s); reticular theory; shared the 1906 Nobel Prize with Cajal
Santiago Ramón y Cajal	1852 to 1934	Neuron doctrine: the nervous system is made of separate cells; shared the 1906 Nobel Prize
Alan Hodgkin	1914 to 1998	Squid giant axon; ionic model of the action potential (1952, with Huxley); Nobel Prize 1963
Andrew Huxley	1917 to 2012	Squid giant axon; ionic model of the action potential (1952, with Hodgkin); Nobel Prize 1963
Candace Pert	1946 to 2013	Demonstrated opiate receptors in the brain, with Snyder (1973)
Solomon Snyder	born 1938	Demonstrated opiate receptors in the brain, with Pert (1973)
John Hughes	1975 (key year)	Isolated the enkephalins, the first endogenous opioids, with Kosterlitz (1975)
Hans Kosterlitz	1903 to 1996	Isolated the enkephalins, with Hughes (1975)
Shepard Siegel	1982	Conditioned drug tolerance; heroin-tolerant rats died more often when dosed in a novel environment
James Olds	1954	Intracranial self-stimulation; the 'pleasure centre' in the septal area and medial forebrain bundle (with Peter Milner)
Peter Milner	1954	Co-discoverer of intracranial self-stimulation with James Olds
Terry Robinson	1993	Incentive-sensitisation theory of addiction with Kent Berridge: sensitised wanting, not liking
Kent Berridge	1993	Incentive-sensitisation theory; mesolimbic dopamine signals wanting more than liking
Jean Delay	1952	Reported chlorpromazine's antipsychotic effect with Pierre Deniker
Pierre Deniker	1952	Reported chlorpromazine's antipsychotic effect with Jean Delay
Arvid Carlsson	1963; Nobel Prize 2000	Dopamine as a transmitter; with Margit Lindqvist (1963) argued that chlorpromazine blocks dopamine receptors
Margit Lindqvist	1963	With Arvid Carlsson, found chlorpromazine raises dopamine metabolites and argued it is a dopamine receptor blocker
Philip Seeman	mid-1970s	Antipsychotic clinical potency correlates with D2 receptor affinity
John Cade	1949	Introduced lithium for mania

Name	When	Examined for
Godfrey Hounsfield	1919 to 2004	Introduced computed tomography (CT) in the early 1970s; Nobel Prize 1979
Seiji Ogawa	born 1934	Described the BOLD signal on which functional MRI depends (1990)
Marcus Raichle	born 1937	Described the brain's default mode (2001), which gave the default mode network its name
Hans Berger	1873 to 1941	First human electroencephalogram (EEG), reported in 1929
Samuel Sutton	1965	First described, with colleagues, the P300 event-related potential
Hans Kornhuber	1965	With Lüder Deecke, discovered the Bereitschaftspotential (readiness potential)
Lüder Deecke	1965	With Hans Kornhuber, discovered the Bereitschaftspotential (readiness potential)
Jennifer Doudna	born 1964	CRISPR-Cas9 gene editing with Emmanuelle Charpentier (2012); Nobel Prize in Chemistry 2020
Emmanuelle Charpentier	born 1968	CRISPR-Cas9 gene editing with Jennifer Doudna (2012); Nobel Prize in Chemistry 2020
Kary Mullis	1944 to 2019	Invented the polymerase chain reaction (PCR), which copies DNA
Gregor Mendel	1822 to 1884	Laws of inheritance from pea plants (published 1866): dominant and recessive traits
Irving Gottesman	1930 to 2016	Pooled schizophrenia twin concordance (about 48 percent MZ, 17 percent DZ; 1991); endophenotypes with Todd Gould (2003)
Todd Gould	2003	With Irving Gottesman, popularised the endophenotype concept in psychiatry
Thomas Bouchard	born 1937	Minnesota Study of Twins Reared Apart (begun 1979; main report in Science, 1990)
Ivar Asbjørn Følling	1888 to 1973	Discovered phenylketonuria (PKU) in 1934
Roger Sperry	1913 to 1994	Chemoaffinity hypothesis of axon growth (1963), from frog eye-rotation studies
Rita Levi-Montalcini	1909 to 2012	Discovered nerve growth factor, the first neurotrophin; Nobel Prize 1986
Mark Rosenzweig	1922 to 2009	Enriched environments produce a thicker, heavier rat cortex (1960s)
David Hubel	1926 to 2013	With Torsten Wiesel, monocular deprivation and the critical period in kitten visual cortex
Torsten Wiesel	born 1924	With David Hubel, monocular deprivation and the critical period in kitten visual cortex
Peter Eriksson	1998	Confirmed, with colleagues, adult neurogenesis in the human hippocampus
Stephen Kuffler	1913 to 1980	On-centre and off-centre receptive fields of retinal ganglion cells (1953)
Haldan Keffer Hartline	1903 to 1983	Lateral inhibition in the eye of Limulus, the horseshoe crab; Nobel Prize 1967
Ernst Mach	1838 to 1916	Mach bands, the illusory strips at light-dark edges (1865)
Lawrence Weiskrantz	1926 to 2018	Blindsight in the patient D.B. (1974)
Leslie Ungerleider	1946 to 2020	Dorsal 'where' and ventral 'what' visual streams (1982, with Mishkin)
Mortimer Mishkin	1926 to 2021	Dorsal 'where' and ventral 'what' visual streams (1982, with Ungerleider)
Melvyn Goodale	1992	Recast the dorsal stream as 'how', vision for action (with Milner)
David Milner	1992	Recast the dorsal stream as 'how', vision for action (with Goodale)
Wilder Penfield	1891 to 1976	Somatosensory and motor homunculi mapped by stimulating the cortex in surgery (1950, with Rasmussen)
Theodore Rasmussen	1910 to 2002	Somatosensory homunculus (1950, with Penfield)
Ronald Melzack	1929 to 2019	Gate-control theory of pain (1965, with Wall); taught in Section 1.3
Patrick Wall	1925 to 2001	Gate-control theory of pain (1965, with Melzack); taught in Section 1.3
David Reynolds	1969	Stimulation-produced analgesia: surgery on rats with only periaqueductal grey stimulation for pain relief
Vilayanur Ramachandran	born 1951	Phantom limbs explained by somatosensory cortical remapping; the mirror box
Giacomo Rizzolatti	born 1937	Mirror neurons in macaque ventral premotor area F5 (1990s)
Joseph Babinski	1857 to 1932	Babinski sign, the extensor plantar response of upper motor neuron lesions (1896)

Name	When	Examined for
Sten Grillner	1985	Spinal central pattern generators: cats with the spinal cord cut off from the brain walked on a treadmill
Geoffrey Harris	1913 to 1971	Proposed (1955) that the hypothalamus controls the anterior pituitary through hormones carried in portal blood vessels
Roger Guillemin	Nobel Prize 1977	Isolated thyrotropin-releasing hormone (TRH) from sheep hypothalami in the late 1960s
Andrew Schally	Nobel Prize 1977	Isolated TRH from pig hypothalami in the late 1960s, independently of Guillemin
Hans Selye	1907 to 1982	General adaptation syndrome (taught in Section 1.11)
Peter Sterling	1988	Allostasis, stability through change (with Joseph Eyer)
Joseph Eyer	1988	Allostasis, stability through change (with Peter Sterling)
Bruce McEwen	1993	Allostatic load (with Eliot Stellar)
Eliot Stellar	1993	Allostatic load (with Bruce McEwen)
Philip Bard	1929	Sham rage in decorticate cats: the hypothalamus organises rage
James Papez	1883 to 1958	Papez circuit of emotion (1937)
Paul MacLean	1913 to 2007	Named the limbic system (1952); triune brain
Joseph LeDoux	born 1949	Amygdala and fear conditioning; the low road and high road to the amygdala
Ralph Adolphs	1994	Patient S.M.: bilateral amygdala damage (Urbach-Wiethe disease) abolishes recognition of fear in faces
Justin Feinstein	2011	Showed that S.M. barely experiences fear
Heinrich Klüver	1937, 1939	Klüver-Bucy syndrome after bilateral anterior temporal lobectomy in rhesus monkeys (with Paul Bucy)
Paul Bucy	1937, 1939	Klüver-Bucy syndrome (with Heinrich Klüver)
Beatrice de Gelder	1999	Affective blindsight: recognising emotional faces in the blind field without awareness
Curt Richter	1967	Search for the circadian clock; medial hypothalamic lesions disrupt daily rhythms
Martin Ralph	1990	SCN transplants impose the donor's free-running period
Jeffrey Hall	Nobel 2017	Molecular clock genes (period) in fruit flies
Michael Rosbash	Nobel 2017	Molecular clock genes (period) in fruit flies
Michael Young	Nobel 2017	Molecular clock genes (timeless) in fruit flies
Eugene Aserinsky	1921 to 1998	Discovered REM sleep with Kleitman (1953)
Nathaniel Kleitman	1895 to 1999	Discovered REM sleep with Aserinsky (1953)
William Dement	1928 to 2020	Linked REM sleep to dreaming; the sleep cycle
Frederic Bremer	1936, 1937	Cerveau isolé and encéphale isolé preparations
Clifford Saper	2001	VLPO flip-flop switch for sleep and waking
Alexander Borbély	1982	Two-process model of sleep (process S and process C)
J. Allan Hobson	1977	Activation-synthesis hypothesis of dreams, with McCarley
Robert McCarley	1977	Activation-synthesis hypothesis of dreams, with Hobson
Jean Mayer	1953	Glucostatic theory of hunger
John Pinel	Biopsychology	Positive-incentive theory of eating; critique of the mamawawa
James Gibbs	1973	CCK reduces meal size (with Young and Smith)
R. C. Young	1973	CCK reduces meal size (with Gibbs and Smith)
Gerard Smith	1973	CCK reduces meal size (with Gibbs and Young)
Jeffrey Friedman	1994	Cloned the ob gene and discovered leptin
Albert Hetherington	1940	VMH lesions cause hyperphagia and obesity (with Ranson)
Stephen Ranson	1940	VMH lesions cause hyperphagia and obesity (with Hetherington)
Bal Krishan Anand	1951	LH lesions cause aphagia (with Brobeck); Indian physiologist
John Brobeck	1951	LH lesions cause aphagia (with Anand)
Philip Teitelbaum	1962	LH lesions cause adipsia as well as aphagia, with partial recovery (with Epstein)

Name	When	Examined for
Alan Epstein	1962	LH lesions cause adipsia as well as aphagia, with partial recovery (with Teitelbaum)
Barbara Rolls	2002	Portion size effect
Charles Phoenix	1959	Organisational and activational effects of hormones
Alfred Jost	1916 to 1991	Fetuses without gonads develop female reproductive ducts
Roger Gorski	1978	Sexually dimorphic nucleus of the medial preoptic area
Simon LeVay	1991	INAH3 and sexual orientation
William Masters	1915 to 2001	Four-phase sexual response cycle (1966), with Johnson
Virginia Johnson	1925 to 2013	Four-phase sexual response cycle (1966), with Masters
Helen Singer Kaplan	1929 to 1995	Desire, excitement and orgasm model (1979)
Théodule Ribot	1839 to 1916	Ribot's law (1881): in retrograde amnesia recent memories are lost before remote ones
Richard Semon	1859 to 1918	Coined the term engram, the physical trace of a memory (1904)
Karl Spencer Lashley	1890 to 1958	Search for the engram; mass action and equipotentiality (1929); 'In search of the engram' (1950): memories distributed throughout the cortex
Henry Gustav Molaison (H.M.)	1926 to 2008	Bilateral medial temporal lobectomy (1953); anterograde amnesia for declarative memory with preserved skill learning
William Beecher Scoville	1906 to 1984	Neurosurgeon who operated on H.M. in 1953; Scoville and Milner (1957)
Brenda Milner	born 1918	Studied H.M. (Scoville and Milner, 1957); mirror-drawing learning without recollection
Stuart Zola-Morgan	1986	Patient R.B.: damage to hippocampal field CA1 alone causes anterograde amnesia (with Squire and Amaral)
Larry Squire	1986	Patient R.B. (with Zola-Morgan and Amaral); memory systems of the medial temporal lobe
David Amaral	1986	Patient R.B. (with Zola-Morgan and Squire)
John O'Keefe	born 1939	Hippocampal place cells (1971, with Dostrovsky); The Hippocampus as a Cognitive Map (1978, with Nadel); Nobel Prize 2014
Jonathan Dostrovsky	1971	Discovered place cells with O'Keefe
Lynn Nadel	1978	The Hippocampus as a Cognitive Map, with O'Keefe
May-Britt Moser	born 1963	Grid cells of the entorhinal cortex (2005); Nobel Prize 2014
Edvard Moser	born 1962	Grid cells of the entorhinal cortex (2005); Nobel Prize 2014
Eleanor Maguire	2000	Enlarged posterior hippocampi in London taxi drivers
Kent Cochrane (K.C.)	1951 to 2014	Lost episodic, autobiographical memory after a 1981 accident; semantic, procedural and working memory spared
Endel Tulving	1927 to 2023	Episodic versus semantic memory; studied patient K.C.
Clive Wearing	born 1938	Musician left with profound amnesia by herpes simplex encephalitis (1985); musical skill preserved
Jean Delacour	1975	Delayed nonmatching-to-sample test (with Mishkin)
James McGaugh	born 1931	The amygdala modulates consolidation, so emotional events are remembered better
Larry Cahill	1994	Propranolol blocks the enhanced memory for an emotional story
Sergei Korsakoff	1854 to 1900	Described Korsakoff's syndrome (1880s): thiamine-deficiency amnesia with confabulation
Carl Wernicke	1848 to 1905	Described Wernicke's encephalopathy (1881): confusion, ataxia, ophthalmoplegia
Eric Kandel	born 1929	Habituation and sensitisation of the Aplysia gill-withdrawal reflex; Nobel Prize 2000
Timothy Bliss	born 1940	Discovered long-term potentiation with Lømo in the rabbit hippocampus (1973)
Terje Lømo	born 1935	Discovered long-term potentiation with Bliss (1973)
Richard Morris	1986	Morris water maze; the NMDA blocker AP5 blocks LTP and spatial learning
Serena Dudek	1992	Long-term depression by prolonged low-frequency stimulation (with Bear)

Name	When	Examined for
Mark Bear	1992	Long-term depression by prolonged low-frequency stimulation (with Dudek)
Wolfram Schultz	1997	Dopamine neurons of the ventral tegmental area signal reward prediction error
Peter Dayan	1997	Reward prediction error account of dopamine (with Schultz and Montague)
Read Montague	1997	Reward prediction error account of dopamine (with Schultz and Dayan)
Marc Dax	1836	First report linking speech loss to left-hemisphere damage; ignored in his lifetime
Paul Broca	1824 to 1880	Patient Leborgne ('Tan', 1861); left inferior frontal speech area (Broca's area)
Louis Victor Leborgne	1861	Broca's patient 'Tan': understood speech but could hardly speak; left frontal lesion
Juhn Atsushi Wada	1949	Wada (sodium amytal) test of language lateralisation
Doreen Kimura	1961	Dichotic listening; right-ear advantage for words, left-ear advantage for melodies
Elliott Ross	1981	Aprosodias after right-hemisphere damage homotopic to the language areas
Norman Geschwind	1926 to 1984	Larger left planum temporale (with Levitsky, 1968); Wernicke-Geschwind model (1965)
Walter Levitsky	1968	With Geschwind, left-right asymmetry of the planum temporale
Ronald Myers	1953	With Sperry, split-brain cats: the corpus callosum transfers learning
Roger Wolcott Sperry	1913 to 1994	Split-brain research in cats and humans; Nobel Prize 1981
Michael Gazzaniga	born 1939	Split-brain patients with Sperry; the left-hemisphere interpreter
Jerre Levy	1972	Chimeric figures test (with Trevarthen and Sperry)
Colwyn Trevarthen	1972	Chimeric figures test (with Levy and Sperry)
Wilder Graves Penfield	1891 to 1976	Cortical stimulation mapping and the homunculus; a distractor for split-brain research
Gregory Hickok	2007	Dual-stream (dorsal and ventral) model of speech processing, with Poeppel
David Poeppel	2007	Dual-stream (dorsal and ventral) model of speech processing, with Hickok
Joseph Jules Dejerine	1849 to 1917	Alexia with agraphia (angular gyrus, 1891); alexia without agraphia (1892)
Josef Gerstmann	1920s	Gerstmann syndrome: finger agnosia, left-right confusion, agraphia, acalculia
John Hughlings Jackson	1835 to 1911	Jacksonian (simple focal) seizures and the Jacksonian march
James Parkinson	1755 to 1824	Described the shaking palsy, Parkinson's disease (1817)
Oleh Hornykiewicz	1926 to 2020	Showed dopamine depletion in the striatum in Parkinson's disease (1960)
J. William Langston	1983	Reported MPTP parkinsonism in young drug users (1982 cases, 1983 report)
George Huntington	1850 to 1916	Described Huntington's disease (1872)
Alois Alzheimer	1864 to 1915	Described Alzheimer's disease in Auguste Deter (1906): plaques and tangles
Auguste Deter	1850 to 1906	Alois Alzheimer's first described patient with Alzheimer's disease
Sigmund Freud	1856 to 1939	Coined the term agnosia (1891)
Heinrich Lissauer	1861 to 1891	Apperceptive versus associative visual agnosia (1890)
Nancy Kanwisher	1997	Identified the fusiform face area with fMRI (1997)
Hugo Liepmann	1863 to 1925	Analysed apraxia and its link to left-hemisphere damage (1900)
Phineas Gage	1823 to 1860	Tamping-iron injury of the frontal lobe (1848): personality change, classic case for the ventromedial prefrontal cortex; later drove stagecoaches in Chile
John Martyn Harlow	1819 to 1907	Gage's physician; reported the case in 1848 and in 1868 (friends found him 'no longer Gage')
Hanna Damasio	1994	Reconstructed Gage's injury from his skull (1994), placing it in the ventromedial prefrontal cortex
David Grant	1948	Wisconsin Card Sorting Test (1948, with Esta Berg)

Name	When	Examined for
Esta Berg	1948	Wisconsin Card Sorting Test (1948, with David Grant)
Tim Shallice	1982	Tower of London test of planning (1982)
Antonio Damasio	1994	Somatic marker hypothesis of ventromedial prefrontal function
Paul Eslinger	1985	Described patient E.V.R. with Antonio Damasio (1985)
Antoine Bechara	1994	Iowa gambling task (1994): risky choices after ventromedial damage
E.V.R.	1985	Patient of Paul Eslinger and Antonio Damasio (1985): orbitofrontal damage, superior IQ but ruinous decisions
António Egas Moniz	1874 to 1955	Prefrontal leucotomy; Nobel Prize 1949
Walter Freeman	1895 to 1972	Popularised lobotomy in the United States
Augustus Waller	1816 to 1870	Wallerian (anterograde) degeneration of cut axons (1850)
Edward Taub	1993	Constraint-induced movement therapy for stroke
H.M. (Henry Molaison)	1926 to 2008	Amnesia after temporal lobe surgery for epilepsy (Section 5.9)

Key terms

biological perspective, Biopsychology, quasi-experimental study, converging operations, dualism, Monism, central nervous system, peripheral nervous system, somatic nervous system, afferent, efferent, autonomic nervous system, sympathetic division, fight-or-flight response, thoracolumbar outflow, sympathetic adrenal-medullary system, parasympathetic division, craniosacral outflow, vagus nerve, enteric nervous system, Grey matter, White matter, ganglion, dorsal root ganglion, Bell-Magendie law, motor neuron, interneuron, reflex arc, cranial nerves, meninges, Cerebrospinal fluid, choroid plexus, blood-brain barrier, area postrema, contralateral, telencephalon, diencephalon, brainstem, medulla oblongata, pons, cerebellum, ataxia, tectum, tegmentum, reticular formation, reticular activating system, thalamus, lateral geniculate nucleus, medial geniculate nucleus, hypothalamus, homeostasis, suprachiasmatic nucleus, mammillary bodies, limbic system, basal ganglia, striatum, cerebral cortex, central sulcus, lateral fissure, neocortex, allocortex, primary motor cortex, prefrontal cortex, Association cortex, cytoarchitecture, Commissural fibres, corpus callosum, Association fibres, Projection fibres, internal capsule, Golgi stain, neuron doctrine, neuron, dendrites, axon hillock, myelin sheath, nodes of Ranvier, telodendria, axon terminal, glial cells, resting membrane potential, sodium-potassium pump, excitatory postsynaptic potential, inhibitory postsynaptic potential, spatial summation, threshold of excitation, action potential, all-or-none law, absolute refractory period, relative refractory period, saltatory conduction, exocytosis, varicosities, gap junctions, coexistence, ionotropic receptor, metabotropic receptor, autoreceptors, acetylcholinesterase, monoamine oxidase, glutamate, NMDA receptor, GABA, acetylcholine, endorphins, psychoactive drug, drug metabolism, agonists, antagonists, allosteric modulator, partial agonist, Drug tolerance, Metabolic tolerance, functional tolerance, cross tolerance, drug sensitisation, withdrawal syndrome, physical dependence, psychological dependence, drug addiction, conditioned drug tolerance, benzodiazepines, barbiturates, CB1 receptors, endocannabinoids, intracranial self-stimulation, mesolimbic pathway, nucleus accumbens, incentive-sensitisation theory, dopamine hypothesis of schizophrenia, Antipsychotics, Typical antipsychotics, Atypical antipsychotics, extrapyramidal side effects, Akathisia, tardive dyskinesia, neuroleptic malignant syndrome, secondary movement disorders, monoamine hypothesis of depression, narrow therapeutic index, Computed tomography, Magnetic resonance imaging, spatial resolution, Diffusion tensor imaging, Positron emission tomography, Single-photon emission computed tomography, BOLD signal, temporal resolution, paired-image subtraction technique, default mode network, electroencephalogram, Magnetoencephalography, event-related potential, P300, galvanic skin response, stereotaxic surgery, Transcranial magnetic stimulation,

Optogenetics, gene knockout, CRISPR-Cas9, alleles, heterozygous, phenotype, Epigenetics, Twin studies, concordance rate, Heritability, endophenotype, Phenylketonuria, genomic imprinting, neural tube, chemoaffinity hypothesis, apoptosis, Synaptic pruning, Neuroplasticity, transduction, receptive field, functional segregation, myopia, on-centre cell, lateral inhibition, Mach bands, optic chiasm, magnocellular, parvocellular, primary visual cortex, bitemporal hemianopia, contralateral homonymous hemianopia, blindsight, simple cells, complex cells, dorsal stream, ventral stream, hair cells, superior olives, dorsal column-medial lemniscus system, anterolateral system, somatosensory homunculus, periaqueductal grey, mirror neurons, pyramidal decussation, Upper motor neurons, Babinski sign, stretch reflex, direct pathway, indirect pathway, action selection, climbing fibres, Exocrine glands, Endocrine glands, hormones, Steroid hormones, pituitary gland, tropic hormones, anterior pituitary, hypothalamo-hypophyseal portal system, posterior pituitary, Vasopressin, Oxytocin, adrenal cortex, cortisol, adrenal medulla, adrenaline, thyroid gland, insulin, Cushing's syndrome, Addison's disease, hypothalamic-pituitary-adrenal axis, negative feedback, psychoneuroimmunology, allostasis, allostatic load, sham rage, Papez circuit, amygdala, low road, high road, affective blindsight, Klüver-Bucy syndrome, circadian rhythm, zeitgebers, free-running rhythms, retinohypothalamic tract, melanopsin, pineal gland, melatonin, slow-wave sleep, paradoxical sleep, atonia, orexin, ventrolateral preoptic area, adenosine, two-process model, narcolepsy, REM sleep behaviour disorder, set-point assumption, positive-incentive theory, ghrelin, cholecystokinin, leptin, dual-centre hypothesis, arcuate nucleus, neuropeptide Y, portion size effect, sensory-specific satiety, osmotic thirst, hypovolaemic thirst, organisational effects, SRY gene, mamawawa, sexual response cycle, amnesia, anterograde amnesia, retrograde amnesia, temporal gradient, Ribot's law, source amnesia, engram, mass action, equipotentiality, cell assembly, Hebbian synapse, bilateral medial temporal lobectomy, declarative memory, consolidation, standard consolidation theory, place cells, cognitive map, grid cells, delayed non-matching-to-sample test, Korsakoff's syndrome, Wernicke's encephalopathy, Wernicke-Korsakoff syndrome, diencephalic amnesia, confabulation, habituation, sensitisation, long-term potentiation, AMPA receptors, coincidence detector, long-term depression, reward prediction error, Cerebral lateralisation, dominant hemisphere, minor hemisphere, Wada test, dichotic listening, Prosody, aprosodias, planum temporale, corpus callosotomy, tachistoscopic presentation, chimeric figures, left-hemisphere interpreter, intermanual conflict, Broca's area, Wernicke's area, arcuate fasciculus, angular gyrus, Wernicke-Geschwind model, dual-stream model, Aphasia, Broca's aphasia, agrammatism, Wernicke's aphasia, paraphasias, conduction aphasia, global aphasia, transcortical motor aphasia, Transcortical sensory aphasia, anomia, Alexia, alexia without agraphia, Gerstmann syndrome, Dysarthria, apraxia of speech, Developmental dyslexia, brain tumour, meningiomas, penumbra, excitotoxicity, contusion, contrecoup injury, chronic traumatic encephalopathy, post-traumatic amnesia, general paresis, necrosis, Epilepsy, Jacksonian march, status epilepticus, Parkinson's disease, Lewy bodies, Huntington's disease, Dementia, Alzheimer's disease, amyloid plaques, neurofibrillary tangles, Multiple sclerosis, Agnosia, prosopagnosia, Anosognosia, Apraxia, hemispatial neglect, extinction, dysexecutive syndrome, perseveration, somatic marker hypothesis, anterograde degeneration, collateral sprouting, cognitive reserve

From the past papers

1. Parts of the autonomic nervous system are: (Amity 2023)

   (a) Sympathetic and parasympathetic (b) Central and peripheral (c) Voluntary and involuntary (d) Cerebrospinal and cortical

   Answer: (a) Sympathetic and parasympathetic. These are the two main divisions of the ANS (many books add the enteric system of the gut wall). Central and peripheral divide the whole nervous system, voluntary and involuntary describe somatic versus autonomic control, and cerebrospinal and cortical is not a division at all.

2. Which of the following is the correct path of a reflex arc? (Amity 2023)

   (a) Afferent neuron to interneuron to afferent neuron (b) Afferent neuron to afferent neuron to interneuron (c) Afferent neuron to interneuron to efferent neuron (d) Efferent neuron to interneuron to the brain

   Answer: (c) Afferent neuron to interneuron to efferent neuron. The signal enters the spinal cord by a sensory (afferent) neuron, crosses an interneuron in the grey matter and leaves by a motor (efferent) neuron to the effector. An arc must end in an efferent neuron and closes in the spinal cord, not the brain; the knee-jerk even skips the interneuron.

3. White matter is primarily composed of (JEMAS 2025)

   (a) Myelinated axons. (b) Myelinated dendrites. (c) Cell bodies. (d) Unmyelinated axons.

   Answer: (a) Myelinated axons. The fat of the myelin sheath makes the tissue look white. Cell bodies and unmyelinated axons make up grey matter, and dendrites are not myelinated.

4. The reticular activating system (RAS), located in the brainstem, is crucial for the regulation of: (Real PYQ 2024)

   (a) Fine motor coordination (b) Arousal, wakefulness, and attention (c) Emotional memory (d) Balance and posture

   Answer: (b) Arousal, wakefulness, and attention. The RAS, the ascending part of the reticular formation in the brainstem core, keeps the cortex awake and alert; its destruction causes coma. Fine motor coordination and balance and posture belong to the cerebellum, emotional memory to the amygdala.

5. Which brain structure would most likely result in death if damaged? (Amity 2023)

   (a) Cerebellum (b) Medulla (c) Reticular formation (d) Pons

   Answer: (b) Medulla. The medulla oblongata holds the vital centres for breathing, heart rate and blood pressure, so its damage is the most likely to kill. Cerebellar damage causes ataxia and reticular damage coma; the pons helps set the breathing rhythm but is not the main vital centre.

6. Which brain structure serves as the primary control center for integrating the endocrine, visceral, and autonomic nervous systems? (Real PYQ 2025)

   (a) Cerebellum (b) Hypothalamus (c) Thalamus (d) Brainstem

   Answer: (b) Hypothalamus. It controls the autonomic nervous system and directs the pituitary, so it integrates neural and hormonal control of the body for homeostasis. The thalamus is the sensory relay, the cerebellum coordinates movement, and the brainstem carries out autonomic reflexes without integrating them with the endocrine system.

7. Which of the following is an inhibitory neurotransmitter? (JEMAS 2025)

   (a) Norepinephrine (b) GABA (c) Dopamine (d) Epinephrine

   Answer: (b) GABA. GABA is the main inhibitory transmitter of the brain: at GABA-A receptors it opens Cl- channels and produces IPSPs. Noradrenaline, dopamine and adrenaline are catecholamines that modulate arousal, movement and reward; none is the standard inhibitory transmitter.

8. Which of the following is NOT a type of glial cell? (Amity 2023)

   (a) Microglia (b) Astrocytes (c) Oligodendrocytes (d) Myelin sheath

   Answer: (d) Myelin sheath. Myelin is the fatty insulation that glial cells make (oligodendrocytes in the CNS, Schwann cells in the PNS); it is not a cell. Microglia (immune cells), astrocytes and oligodendrocytes are all glia.

9. Which of the following is not a neurotransmitter? (IHBAS 2020)

   (a) Dopamine (b) Serotonin (c) Nor-epinephrine (d) Adrenalin

   Answer: (d) Adrenalin (as keyed). Adrenaline is the odd one out because it acts mainly as a hormone of the adrenal medulla. Strictly it is also a catecholamine neurotransmitter, released by a few adrenergic brainstem neurons, so treat this as an odd-one-out question; dopamine, serotonin and noradrenaline are classic brain monoamine transmitters.

10. Valium, Xanax, Rohypnol, and Halcion are: (Amity 2023)

    (a) Benzodiazepines (b) Barbiturates (c) Sedatives (d) Hypnotics

    Answer: (a) Benzodiazepines. Diazepam, alprazolam, flunitrazepam and triazolam all belong to this one chemical class, which raises the frequency of GABA-A chloride channel opening. Barbiturates are a different class (they raise the duration of opening), and 'sedatives' and 'hypnotics' are broad functional labels that also cover alcohol and barbiturates.

11. Which of the following substances is NOT a natural alkaloid compound? (Amity 2023)

    (a) Morphine (b) Heroin (c) Codeine (d) Papaverine

    Answer: (b) Heroin. Heroin (diacetylmorphine) is semisynthetic, made by adding two acetyl groups to morphine. Morphine, codeine, papaverine and thebaine occur naturally in opium, the dried sap of the poppy.

12. A 45-year-old man with schizophrenia has been on antipsychotic medication for years. He presents with repetitive, involuntary, and purposeless movements of his mouth, tongue, and face, including lip-smacking and grimacing. This condition is most likely: (Real PYQ 2024)

    (a) Akathisia (b) Acute Dystonia (c) Tardive Dyskinesia (d) Parkinsonism

    Answer: (c) Tardive Dyskinesia. Involuntary orofacial movements after months or years of neuroleptic treatment are the late, often irreversible extrapyramidal effect. Acute dystonia (spasms) comes within hours to days, akathisia (inner restlessness, pacing) within days to weeks, and parkinsonism (tremor, rigidity, slowness) within weeks.

13. Which neuroscientific technique provides the highest temporal resolution for studying neural activity? (RML 2025)

    (a) fMRI (b) PET scan (c) EEG (d) CT scan

    Answer: (c) EEG. EEG records the brain's electrical activity directly, so it follows changes millisecond by millisecond. fMRI depends on a blood-flow response that takes seconds, PET on tracer uptake over a minute or more, and CT is a still structural image with no temporal resolution at all.

14. In Phenylketonuria (PKU), the body fails to convert phenylalanine into which substance due to a deficiency of phenylalanine hydroxylase? (Real PYQ 2025)

    (a) Urea (b) Tyrosine (c) Tryptophan (d) Histidine

    Answer: (b) Tyrosine. Phenylalanine hydroxylase converts phenylalanine into tyrosine; in autosomal recessive PKU the enzyme is missing, so phenylalanine accumulates and damages the brain unless a low-phenylalanine diet starts at birth. Urea is a waste product of protein breakdown, tryptophan the precursor of serotonin and histidine of histamine.

15. A male child presents with tall stature, small testes, and learning difficulties. Karyotype analysis shows an extra X chromosome. This condition is : (IHBAS 2025)

    (a) Klinefelter syndrome (b) Turner syndrome (c) Prader-Willi syndrome (d) Down syndrome

    Answer: (a) Klinefelter syndrome. An extra X in a male (47,XXY) gives tall stature, small testes, infertility and learning difficulties. Turner syndrome (45,X) gives a short, infertile female; Prader-Willi syndrome is a paternal deletion at 15q11 to q13; Down syndrome is trisomy 21.

16. The Mach bands phenomenon is best explained by : (IHBAS 2025)

    (a) Rod-cone dichotomy in the fovea (b) Opponent-process theory of color vision (c) Lateral inhibition in the retina (d) Temporal summation in photoreceptors

    Answer: (c) Lateral inhibition in the retina. At a light-dark border, cells on the bright side receive less inhibition from their dark neighbours and cells on the dark side receive more, so the edge is exaggerated into a brighter and a darker strip. The rod-cone dichotomy concerns dim-light versus colour vision, the opponent process colour coding and afterimages, and temporal summation the adding of weak signals over time.

17. Difficulty in seeing objects at a distance is called : (IHBAS 2024)

    (a) Myopia (b) Hypermetropia (c) Astigmatism (d) Visual Field Defects

    Answer: (a) Myopia. In myopia (near-sightedness) the eyeball is too long, so distant objects focus in front of the retina and blur; a concave lens corrects it. Hypermetropia blurs near objects, astigmatism comes from an irregular cornea, and visual field defects are blind areas caused by damage to the visual pathway, not focusing errors.

18. Which of the following areas of the brain is involved in controlling movements of the body? (Amity 2023)

    (a) Primary (b) Secondary (c) Supplementary (d) All of the above

    Answer: (d) All of the above. Movement is controlled by a hierarchy: primary motor cortex (area 4, precentral gyrus) is the main cortical output, and secondary motor cortex, the premotor cortex and supplementary motor area, programmes the patterns of movement it executes. No single area does the job alone.

19. In carrying out the 'fight or flight' response, the role of supervisor is assigned to the: (RML 2024)

    (a) Adrenal gland (b) Pituitary gland (c) Hypothalamus (d) Parasympathetic nervous system

    Answer: (c) Hypothalamus. It integrates the endocrine and autonomic systems and drives both the sympathetic nervous system and the pituitary. The pituitary and the adrenal gland carry out its orders, and the parasympathetic system calms the body instead.

20. Which cluster of symptoms is seen in Kluver-Bucy syndrome? (JEMAS 2025)

    (a) Agnosia, aphasia, hyper-orality, hyperphagia, hypersexuality (b) Agnosia, avolition, alogia, asexuality (c) Hypersexuality, altered emotional response, hyperorality, hyperalertness (d) Hypersexuality, hypo-sensitivity, delayed reaction

    Answer: (a). Bilateral anterior temporal lobe and amygdala damage gives psychic blindness (agnosia), hyperorality, hyperphagia and hypersexuality, with aphasia in human cases. Avolition and alogia are negative symptoms of schizophrenia, and the animals become placid, not hyperalert.

21. What is the term for the brain's process of regulating its own responses to stress by controlling hormone release? (IHBAS 2024)

    (a) Neuroplasticity (b) Homeostasis (c) Allostasis (d) Circadian rhythm

    Answer: (c) Allostasis. Sterling and Eyer (1988) defined it as stability through change: the brain anticipates demands and adjusts hormone release. Homeostasis is Cannon's stability around fixed set points, neuroplasticity is rewiring with experience, and circadian rhythm is the 24-hour cycle.

22. A lesion of the ventromedial hypothalamus (VMH) produces: (Recall set)

    (a) Loss of circadian rhythms (b) Aphagia and adipsia (c) Diabetes insipidus (d) Hyperphagia and obesity

    Answer: (d) Hyperphagia and obesity. The VMH was the classic satiety centre (Hetherington and Ranson, 1940). Aphagia and adipsia follow lateral hypothalamic lesions (Anand and Brobeck, 1951), circadian rhythms depend on the suprachiasmatic nucleus, and diabetes insipidus follows failure of ADH from the supraoptic and paraventricular nuclei.

23. Which gland regulates the circadian rhythm in the human beings? (RML 2022)

    (a) Hypothalamus (b) Cerebellum (c) Pituitary gland (d) Pineal gland

    Answer: (d) Pineal gland. The clock itself is the suprachiasmatic nucleus of the hypothalamus, but the question asks for a gland, and the pineal gland turns the clock's signal into night-time melatonin. The cerebellum coordinates movement and the pituitary is the master gland of other hormones; neither keeps a clock.

24. Masters and Johnson's sexual response cycle is distinct from Kaplan's model. Which of the following correctly reflects Masters and Johnson's sequence? (IHBAS 2025)

    (a) Excitement, Plateau, Orgasm, Resolution (b) Plateau, Excitement, Orgasm, Resolution (c) Excitement, Desire, Plateau, Orgasm, Refractory (d) Desire, Arousal, Plateau, Orgasm, Resolution

    Answer: (a) Excitement, Plateau, Orgasm, Resolution. Masters and Johnson (1966) described these four phases, with a male refractory period after orgasm. Putting desire first is Helen Singer Kaplan's (1979) change, and no model starts with plateau.

25. A person suffered a head injury in a road accident several years ago. Since the accident, he is unable to form new memories. He can remember many things which he did before the accident, but he cannot remember what he said ten minutes ago. The part of the brain which is most probably injured is: (IHBAS 2020)

    (a) Brain stem (b) Hippocampus (c) Spinal cord (d) Reticular formation

    Answer: (b) Hippocampus. Failing to form new memories while old ones survive is anterograde amnesia, the signature of hippocampal damage (H.M., R.B.). The brainstem and reticular formation serve vital functions and arousal, and the spinal cord carries reflexes and sensorimotor signals; none consolidates memory.

26. According to Karl Lashley's research on memory storage, where did he conclude that memories are primarily stored in the brain? (Real PYQ 2025)

    (a) In specific localized brain regions dedicated to particular memories (b) Distributed throughout the entire brain cortex (c) Exclusively in the hippocampus (d) Only in the frontal and temporal lobes

    Answer: (b) Distributed throughout the entire brain cortex. No cortical lesion erased a maze habit in Lashley's rats; the deficit depended on how much cortex was removed (mass action), not where (equipotentiality). He found no localised engram, and the hippocampus's role in forming memories emerged only later, with H.M.

27. Wernicke-Korsakoff syndrome results in: (Amity 2023)

    (a) Lapse of attention (b) Loss of muscle coordination (c) Both a & b (d) None

    Answer: (c) Both a and b. Thiamine deficiency first causes Wernicke's encephalopathy: confusion with lapses of attention, ataxia (loss of muscle coordination) and ophthalmoplegia with nystagmus. Survivors develop the amnesia of Korsakoff's syndrome, so the combined syndrome affects attention and coordination as well as memory.

28. The left hemisphere is: (RML 2024)

    (a) Concerned to a large extent with spatial and pattern sense (b) The minor hemisphere (c) The major hemisphere (d) Centre of musical ability

    Answer: (c) The major hemisphere. The left hemisphere is dominant for language in about 95 percent of right-handers and most left-handers, so it was called the major or dominant hemisphere. Spatial and pattern sense and musical ability are right-hemisphere strengths, and 'minor hemisphere' is the old label for the right.

29. Cutting through a person's corpus callosum causes the person to: (RML 2024)

    (a) Die (b) Become blind (c) Be unable to speak or understand language (d) Have difficulty coordinating certain kind of activities

    Answer: (d) Have difficulty coordinating certain kinds of activities. Corpus callosotomy for intractable epilepsy leaves patients alive, sighted and able to speak (the left hemisphere still speaks); the cost is that the hemispheres cannot share information, which shows up in tasks needing both.

30. Ria has difficulty enunciating words correctly and speaks in a slow and laboured way; her speech often makes sense, but it includes only keywords. Which part of her brain is damaged? (RML 2022)

    (a) Wernicke's area (b) Broca's area (c) Temporal lobe (d) Frontal lobe

    Answer: (b) Broca's area. Slow, effortful, telegraphic speech that still makes sense is Broca's (non-fluent) aphasia. 'Frontal lobe' is true but too broad; Wernicke's area and the temporal lobe produce fluent, empty speech with poor comprehension.

31. Prosopagnosia results in an inability to : (IHBAS 2024)

    (a) Recognize faces (b) Distinguish tastes (c) Express feelings through facial expressions (d) Recognize objects

    Answer: (a) Recognize faces. The source key gives (b), which is wrong: prosopagnosia is a visual agnosia for faces after fusiform gyrus damage, and loss of taste is ageusia. Loss of facial expression is a motor sign (the masked face of Parkinson's disease), and failure to recognise objects is object agnosia.

32. A patient presents with an inability to perform coordinated, purposeful movements, despite having intact motor function, sensation, and comprehension of the task. When asked to brush their hair, they cannot sequence the movements. This deficit is best described as: (Real PYQ 2024)

    (a) Agnosia (b) Ataxia (c) Apraxia (d) Alexia

    Answer: (c) Apraxia. Loss of learned purposeful movement with strength, sensation, coordination and comprehension intact is apraxia (here ideational: the steps cannot be sequenced). Agnosia is a failure of recognition, ataxia is cerebellar incoordination and alexia is loss of reading.

33. Which of the following neurological conditions is NOT classified as a neurodegenerative disorder? (Real PYQ 2025)

    (a) Parkinson's disease (b) Alzheimer's disease (c) Epilepsy (d) Huntington's disease

    Answer: (c) Epilepsy. Epilepsy is recurrent seizures from synchronous neural discharge, an episodic disorder with normal function between attacks. Parkinson's (substantia nigra dopamine neurons), Alzheimer's (cortical and hippocampal neurons) and Huntington's (striatal GABA neurons) are progressive losses of neurons.

High-yield sheet

The facts this chapter's questions ask most, most-asked first. Last read before the exam.

1 **Korsakoff's syndrome** is caused by thiamine (vitamin B1) deficiency, classically in chronic alcoholism; it damages the mammillary bodies and mediodorsal thalamus and causes dense anterograde amnesia, severe retrograde amnesia and confabulation. ASKED 13 TIMES

2 Extrapyramidal side effects come from D2 blockade in the **nigrostriatal** pathway and appear in order: acute dystonia (hours to days), akathisia (days to weeks, inner restlessness), parkinsonism (weeks) and **tardive dyskinesia** (months to years, orofacial movements, may be irreversible); akathisia is a secondary movement disorder. ASKED 10 TIMES

3 Damage to **Broca's area** (left inferior frontal gyrus, pars opercularis and triangularis, areas 44 and 45) causes Broca's aphasia: non-fluent, effortful, telegraphic speech with relatively preserved comprehension and impaired repetition. ASKED 9 TIMES

4 The **hippocampus** consolidates new declarative (especially episodic and spatial) memories; its damage, as after a head injury, causes anterograde amnesia while remote memories survive. ASKED 8 TIMES

5 The ANS has two divisions: the **sympathetic** mediates fight or flight (Walter Cannon, 1915) together with the adrenal medulla, which releases adrenaline, and spends energy; the **parasympathetic** (rest and digest) conserves energy, slowing the heart and constricting the pupils mainly through the vagus nerve and acetylcholine. ASKED 7 TIMES

6 The **limbic system**, named by Paul MacLean in 1952, is the classic seat of emotion, regulating emotion, motivation and memory; hunger and thirst belong to the hypothalamus. ASKED 7 TIMES

7 The **adrenal glands** run the stress response: the cortex releases cortisol at the end of the HPA axis (hypothalamus CRH, anterior pituitary ACTH, adrenal cortex cortisol, with negative feedback on the hypothalamus and anterior pituitary), and the medulla releases adrenaline, also called epinephrine. ASKED 7 TIMES

8 The **ventromedial hypothalamus** was the satiety centre: its lesions cause hyperphagia and obesity (Hetherington and Ranson, 1940), whereas **lateral hypothalamus** lesions cause aphagia and adipsia (Anand and Brobeck, 1951). ASKED 7 TIMES

9 **H.M.** (Henry Molaison) had both medial temporal lobes (hippocampus, amygdala, adjacent cortex) removed by William Scoville in 1953; he was left with severe anterograde amnesia for declarative memory but improved at mirror drawing (preserved procedural memory). ASKED 7 TIMES

10 The **dopamine hypothesis** of schizophrenia: mesolimbic dopamine (D2) hyperactivity produces positive symptoms and mesocortical hypoactivity negative symptoms; its original support was that typical antipsychotics' clinical potency correlates with their D2 affinity. ASKED 6 TIMES

11 The **amygdala** is the key structure for fear, aggression, threat detection and fear conditioning; in LeDoux's account the low road (thalamus straight to amygdala) is fast and crude, the high road through the cortex slower and accurate. ASKED 6 TIMES

12 **Parkinson's disease** is a **dopamine** deficiency caused by degeneration of the **substantia nigra pars compacta** (nigrostriatal pathway), with alpha-synuclein Lewy bodies; not the locus coeruleus, raphe nuclei or cholinergic basal forebrain. ASKED 6 TIMES

13 The **mesolimbic pathway**, from the ventral tegmental area to the nucleus accumbens (ventral striatum), is the core reward pathway of addiction, and dopamine is the reward transmitter. ASKED 5 TIMES

14 **Oxytocin and vasopressin** (ADH) are made in the supraoptic and paraventricular nuclei of the hypothalamus and released from the posterior pituitary; neither oxytocin nor melatonin is an anterior pituitary hormone. ASKED 5 TIMES

15 **Long-term potentiation** (Bliss and Lømo, 1973) is a lasting strengthening of synaptic transmission after high-frequency stimulation; its induction needs NMDA receptors, which pass calcium only when glutamate binds and the depolarised membrane expels their magnesium block. ASKED 5 TIMES

16 **Wernicke's area** is the posterior superior temporal gyrus of the dominant hemisphere (area 22), the classic region for comprehending spoken and written language; damage gives fluent, empty speech with poor comprehension. ASKED 5 TIMES

17 The **medulla oblongata** (hindbrain) holds the vital centres for breathing, heart rate and blood pressure; its damage is the brain injury most likely to cause death. ASKED 4 TIMES

18 **GABA** is the main inhibitory transmitter of the brain (glycine in the spinal cord); **glutamate** is the main excitatory transmitter of the CNS. ASKED 4 TIMES

19 **All-or-none law**: once threshold is reached, an action potential fires at full, constant amplitude whatever the stimulus strength, and intensity is coded by firing frequency; EPSPs, IPSPs and the resting potential do not obey it. ASKED 4 TIMES

20 **Saltatory conduction**: in myelinated axons the action potential jumps from one node of Ranvier to the next, where voltage-gated Na+ channels cluster; the function of myelin is to increase conduction speed. ASKED 4 TIMES

21 **Benzodiazepines** (Valium, Xanax, Rohypnol, Halcion) and alcohol enhance GABA at the GABA-A receptor; benzodiazepines increase the frequency of chloride channel opening, barbiturates its duration. ASKED 4 TIMES

22 The **pituitary** is the master gland, regulating most other glands and growth, and is itself controlled by the hypothalamus. ASKED 4 TIMES

23 **Klüver-Bucy syndrome** (Klüver and Bucy, 1937 and 1939), after bilateral removal of the anterior temporal lobes and amygdala, brings tameness and loss of fear, psychic blindness, hyperorality, hyperphagia and hypersexuality. ASKED 4 TIMES

24 The **suprachiasmatic nucleus** of the hypothalamus is the master circadian pacemaker, entrained by light through the retinohypothalamic tract; its lesions abolish circadian rhythms. ASKED 4 TIMES

25 **REM sleep** shows rapid eye movements, skeletal muscle atonia and a low-voltage, fast, desynchronised EEG; EEG synchrony, delta waves and the growth hormone surge belong to slow-wave sleep. ASKED 4 TIMES

26 Beta waves (13 to 30 Hz) mark alert waking and **alpha** (8 to 12 Hz) relaxed waking with eyes closed; **slow-wave sleep** is stages 3 and 4 (N3), with high-amplitude delta waves of 0.5 to 4 Hz. ASKED 4 TIMES

27 **Anterograde amnesia** is the inability to form new memories after the onset; **retrograde amnesia** is the loss of memories formed before it, usually worst for the most recent (Ribot's law). ASKED 4 TIMES

28 In split-brain patients (corpus callosotomy) an object flashed to the left visual field reaches only the right hemisphere, so it cannot be named but can be picked out with the left hand; each hemisphere works independently, and Roger Sperry won the 1981 Nobel Prize. ASKED 4 TIMES

29 **Prosopagnosia** is a visual **agnosia** for faces (recognised at once by voice), after bilateral or right-sided damage to the **fusiform gyrus** (fusiform face area); it is not loss of taste, despite the IHBAS 2024 key. ASKED 4 TIMES

30 **Cerebrospinal fluid** is made by the choroid plexus, fills the ventricles (the two lateral ventricles are the largest) and the subarachnoid space, and is absorbed through the arachnoid granulations; the **blood-brain barrier** is formed by tight junctions between capillary endothelial cells, supported by astrocyte end-feet. ASKED 3 TIMES

CHAPTER 6

Developmental Psychology

IN THIS CHAPTER

6.1 ● Foundations: The Lifespan View, Key Issues and Ecological Contexts: Define development, growth and maturation, explain the lifespan perspective, the principles and enduring issues of development and the difference between critical and sensitive periods, and place a child's settings in Bronfenbrenner's ecological systems.

6.2 ● Heredity, Prenatal Development, Birth and the Newborn: Explain how genes are inherited and expressed, how heredity and environment are separated and combined, how prenatal development unfolds and what threatens it, and how the newborn is assessed, including the Apgar scale and the newborn reflexes.

6.3 ● Infancy: Brain, Body, Motor Skills and Perception: Describe growth and brain development in infancy, apply the principles, milestones and theories of motor development, explain how infant perception is studied and what it reveals (senses, visual cliff, affordances), and order the early learning, imitation and milestone sequence.

6.4 ●●● Piaget's Theory of Cognitive Development: Explain Piaget's constructivist theory, from schemes, assimilation, accommodation and equilibration through the four stages and their landmark tasks (object permanence, conservation, the pendulum), and judge how later research and postformal theorists have revised it.

6.5 ●● Vygotsky, Bruner, Information Processing and Theory of Mind: Explain Vygotsky's sociocultural theory (the zone of proximal development, scaffolding, private speech) and how it differs from Piaget's, Bruner's modes of representation, the information-processing view of development, and how theory of mind and social perspective taking develop.

6.6 ● Emotional Development, Temperament and Self-Regulation: Trace emotional development from Bridges' first emotions through the self-conscious emotions, stranger and separation anxiety and the still face, and classify temperament (Thomas and Chess, Rothbart and Bates, Kagan) and self-regulation (Kopp, Mischel's marshmallow test).

6.7 ●● Attachment: Imprinting, Bowlby, Ainsworth and Beyond: Explain why infants form attachments, trace the phases of attachment over the first three years, classify Strange Situation patterns by reunion behaviour, and weigh the evidence on deprivation, culture and adult attachment.

6.8 ● Parenting, Family, Play and Gender in Childhood: Compare Baumrind's parenting styles on demandingness and responsiveness, recognise the forms of maltreatment and neglect, trace how play develops from Piaget to Parten, and explain how children acquire gender through socialisation, Kohlberg's stages and gender schemas.

6.9 • Moral Development: Piaget, Kohlberg, Gilligan and Beyond: Trace moral development from Piaget's heteronomous and autonomous morality through Kohlberg's three levels and six stages, place any moral reason at its level, and weigh the critiques from Gilligan, cross-cultural research and later theories.

6.10 •• Erikson's Psychosocial Stages, Identity and Adolescence: Name all eight of Erikson's psychosocial stages with their ages, crises, virtues and maldevelopments, and explain adolescent identity (Marcia's statuses), Elkind's adolescent egocentrism, puberty and the adolescent brain, and modern views of adolescence.

6.11 • Adulthood, Ageing, Death and Dying: Trace adult development from emerging adulthood through midlife and old age to death and grief, naming the theorist behind each stage model, type of marriage and theory of ageing, and telling the climacteric apart from general physical ageing.

Weightage. 230 of the 3180 questions mapped in this book (7.2%) test this chapter, 101 of them from real past papers. 1 of its 11 sections is High yield.

Southern France, January 1800. On a winter morning a boy walked out of the woods near the village of Saint-Sernin-sur-Rance, in the district of Aveyron, looking for food. He was naked but for the rags of a shirt, scarred, about twelve years old, and he could not speak. Local people had glimpsed a wild child in the forests for years, and more than once he had been caught and had escaped. This time he stayed. Within months he was in Paris, the most talked-about child in France, and learned men came to see what a human being is like when society has had no hand in shaping him.

The verdict of the greatest authority was harsh. **Philippe Pinel**, reformer of the Paris asylums, examined the boy and judged him an incurable idiot, perhaps abandoned because he was defective from birth. A 26-year-old physician at the Paris institution for deaf-mutes, **Jean-Marc-Gaspard Itard**, disagreed. The boy, he argued, was backward because he had grown up alone; give him the experiences he had missed and he would become fully human. Itard named him Victor and, with his carer Madame Guérin, worked with him daily for about five years, training his senses, his attention and his feelings, and trying above all to teach him language.

The results, reported in 1801 and 1806, were mixed. Victor came to recognise written words, follow instructions, show affection for Madame Guérin and protest when he was punished unjustly. But he never learned to speak more than a word or two, and Itard ended the lessons disappointed. Victor lived quietly with Madame Guérin in Paris until his death in 1828, aged about forty. Was his failure born or made, and had the teaching simply come too late? This chapter gives you the tools to answer.

6.1 Foundations: The Lifespan View, Key Issues and Ecological Contexts

LOWER YIELD · 13 QS

Before the story of how a fertilised cell becomes an ageing adult, you need its frame: what developmental change means, what drives it, the questions still argued over, and the contexts in which every child grows.

The science of change and stability

Developmental psychology is the scientific study of change and stability from conception to death. It takes the **developmental perspective**, the approach concerned with the changes that occur as people mature. The other perspectives of Section 1.2 ask other questions: the psychoanalytic about unconscious conflicts rooted in childhood, the biological about genes, brain and hormones, and the humanistic about free choice and self-actualisation.

Keep two kinds of change apart. **Growth** is quantitative change, an increase in size or amount: height, weight, vocabulary. **Development** is qualitative change in structure and organisation, an orderly, progressive series of changes towards maturity: the child does not just know more words but starts combining them into sentences. Change runs through three interacting domains: physical (body, brain, motor skills), cognitive (memory, language, thinking) and psychosocial, or socioemotional (emotions, personality, relationships).

The lifespan is divided into periods, but the divisions are a social construction, an idea a society shares rather than a fact of nature: adolescence became a recognised stage only with industrialisation. Indian courses follow **Elizabeth Hurlock** (1898 to 1988), whose *Developmental Psychology: A Life-Span Approach* marks ten periods (Table 6.1).

Table 6.1 Hurlock's ten periods of the lifespan.

Period	Age span	Period	Age span
1 Prenatal	Conception to birth	6 Puberty	10 or 12 to 13 or 14
2 Infancy	Birth to 2 weeks	7 Adolescence	13 or 14 to 18
3 Babyhood	2 weeks to 2 years	8 Early adulthood	18 to 40
4 Early childhood	2 to 6 years	9 Middle age	40 to 60
5 Late childhood	6 to 10 or 12	10 Old age	60 onward

Older texts treated development as ending with adolescence. **Paul Baltes** (1939 to 2006) countered in 1987 with the **lifespan perspective**: development is *lifelong* (no age holds supremacy), *multidimensional* (physical, cognitive and social strands), *multidirectional* (some abilities rise while others fall), *plastic* (open to change at every age), *contextual* (embedded in history and culture) and *multidisciplinary*, and every age brings both gains and losses (an older adult loses speed but may gain wisdom). Baltes named three kinds of influence. **Normative age-graded influences** are tied to age and shared by most people (puberty, retirement). **Normative history-graded influences** are shared by a cohort, a generation, because of when it lived (Partition, the COVID-19 pandemic). **Non-normative influences** are unusual events in one life (a parent's early death). Longitudinal, cross-sectional and cross-sequential designs, which separate age changes from cohort effects, are in Chapter 9; infant research methods are in Section 6.3.

From philosophers to baby biographies

Two philosophers framed the debate. **John Locke** (1632 to 1704) held that the newborn mind is a **tabula rasa**, a blank slate written on by experience: the root of the nurture view. **Jean-Jacques Rousseau** (1712 to 1778), in *Émile* (1762), saw the child as a noble savage, born good, whose natural plan of growth unfolds in stages: the root of the nature, maturation and stage views.

Scientific observation began with the **baby biography**, a detailed diary of one child. **Charles Darwin** (1809 to 1882) published one on his first son in 1877, and **Wilhelm Preyer** (1841 to 1897) followed with *The Mind of the Child* (1882), often taken as the start of scientific child psychology. **G. Stanley Hall** (1844 to 1924) led the child study movement with questionnaires given to large numbers of children; his *Adolescence* (1904) called the teens a time of storm and stress (Section 6.10) and proposed **recapitulation**, the idea that each child's development replays the evolutionary history of the species. Hall's student **Arnold Gesell** (1880 to 1961) took the **normative approach**, observing many children to find the typical age, or norm, of each behaviour. Against him stood the behaviourist **John B. Watson** (1878 to 1958), who boasted that he could train any healthy infant into any kind of specialist.

Maturation and learning ●●

Gesell's lasting idea is **maturation**, the biological unfolding of genetic potential: changes that appear on a genetic timetable with little dependence on practice. **Learning** is change through experience and practice. Maturation supplies readiness; learning builds on it. Three terms that depend on experience are not maturation: **socialisation**, acquiring the values, norms and behaviour of one's society; accommodation, Jean Piaget's term for changing a mental scheme to fit new information (Section 6.4); and conditioning, learning by association (Section 1.6).

KEY STUDY: GESELL AND THOMPSON'S CO-TWIN CONTROL (1929)

Can practice hurry an unready body? Gesell and **Helen Thompson** studied identical twin girls, T and C, holding heredity constant: the method of **co-twin control**. Twin T practised stair climbing daily for six weeks from 46 weeks of age. Twin C's training began only at 53 weeks, yet after two weeks she had caught up. Maturation, not early training, set the pace. Criticism: one pair of twins and a narrow set of skills.

Myrtle McGraw (1899 to 1988) refined the point in *Growth: A Study of Johnny and Jimmy* (1935). Intensive training from infancy did not hurry Johnny's *phylogenetic* skills, those common to the species (crawling, walking), but it advanced *ontogenetic* skills particular to the individual (swimming, roller skating) once he was ready; his twin Jimmy was left untrained. So a child's increasing skill at using his muscles is due mainly to maturation, not to the opportunity to exercise, pushing by parents or learning. Gesell turned such observations into age norms and developmental schedules for motor, adaptive, language and personal-social behaviour (the scales are in Chapter 2).

EXAM TRAP

"The biological unfolding of genetic potential" is **maturation** (Gesell), never socialisation, accommodation or conditioning. A child's growing muscular skill is credited to **maturation**, not exercise, parental pushing or learning. Growth is **quantitative** (size, amount); development is **qualitative** (structure, organisation).

Principles of development and developmental tasks

Development follows regularities that hold across cultures (Table 6.2). Two give the direction of physical growth: **cephalocaudal**, from head to tail (head control comes before walking), and **proximodistal**, from the centre outwards (arms before fingers). Motor milestones are in Section 6.3.

Table 6.2 Principles of development (Hurlock and standard texts).

Principle	Meaning and example
Orderly, predictable pattern	Same sequence in all children: sit, stand, walk
Cephalocaudal	Head to foot
Proximodistal	Centre to extremities
General to specific (mass to specific)	Whole-body wriggling comes before one hand reaching
Continuous but uneven in rate	Never stops; spurts in infancy and puberty

Principle	Meaning and example
Individual differences	Same sequence, different rates: walking at 10 or at 15 months
Maturation and learning interact	Reading needs both readiness and practice
Early foundations are critical	Early patterns tend to persist
Each stage has its own pattern	Characteristic behaviour, hazards and social expectations

Robert J. Havighurst (1900 to 1991), in *Developmental Tasks and Education* (1948), defined a **developmental task** as one that arises at about a certain period of life; success brings happiness and success with later tasks, while failure brings unhappiness, social disapproval and later difficulty. Tasks arise from physical maturation (walking), cultural pressure (reading) and personal values (choosing a vocation). Early childhood tasks include walking, talking, taking solid food and controlling elimination; middle childhood brings the skills of ordinary games, getting along with age-mates, and reading, writing and arithmetic. Adolescent and adult tasks are in Sections 6.10 and 6.11.

The enduring issues

Every theory takes a stand on the issues of Table 6.3. On **nature versus nurture** (heredity or environment?) today's answer is interaction: a genetic tendency shows itself only in some environments, and inherited traits help select the environments a child meets (Section 6.2). **Continuity versus discontinuity** asks whether development is gradual, quantitative accumulation, a ramp, or a series of qualitatively different stages, a staircase.

Table 6.3 The core issues of developmental psychology.

Issue	The question	Typical answers
Nature versus nurture	Heredity or environment?	Rousseau, Gesell (nature); Locke, Watson (nurture); today, interaction
Continuity versus discontinuity	Gradual change or stages?	Learning theory (continuous); Freud, Piaget, Erikson (stages)
Stability versus change	Do early traits persist?	Freud (personality set by 5 or 6); Baltes (plastic for life)
Early versus later experience	Do the first years decide?	Freud, Bowlby (early); Clarke and Clarke (later can repair)
Active versus passive child	Does the child shape development?	Piaget (active builder); Watson (moulded by environment)
Universal versus context-specific	One path everywhere?	Piaget (universal stages); Vygotsky, Bronfenbrenner (culture)

Try a worked example. In one Native American tribe, mothers answer an infant's distress at once and hold the baby almost constantly for three years; for generations the children have grown up quiet, even-tempered, obedient and respectful. Is the calm inborn? No: it follows a child-rearing practice handed down in the culture, so it is **learnt behaviour**, acquired through experience. **Innate behaviour** appears without experience, as reflexes do. The calm is also **overt behaviour** (observable action), not **covert behaviour** (private thoughts and feelings), and "complex behaviour" describes how many parts an act has, not where it comes from.

EXAM TRAP

Behaviour that follows a child-rearing practice handed down in a culture is **learnt**; stability across generations is not proof of heredity. Stage theorists (Freud, Piaget, Erikson) take the **discontinuity** side; learning and information-processing theorists take the **continuity** side.

Critical and sensitive periods ●●

A **critical period** is a fixed window in which a particular experience must occur for normal development; if it passes without that experience, the effects are largely irreversible. The model case comes from ethology, the study of behaviour in its evolutionary setting: **Konrad Lorenz** (1903 to 1989) reported in 1935 that greylag goslings imprint on the first moving object they see after hatching, even Lorenz himself (Section 6.7). In human development the clearest critical periods are in the embryo: each organ forms within a brief window, so a teratogen such as thalidomide or rubella seriously damages an organ only if it strikes while that organ is forming, mostly in weeks 3 to 8 (Section 6.2).

A **sensitive period** is an optimal but more flexible window: the experience matters most then, but some development remains possible later (Figure 6.1). First language is the standard case. **Eric Lenneberg** (1967) proposed a critical period for language from about age two to puberty (Section 1.8); the evidence now favours a sensitive period. Genie, found in Los Angeles in 1970 at 13 after years of isolation, learned many words but never mastered grammar. Such lasting limits after early language deprivation support sensitive periods. They say nothing about semantics (word meaning) or syntax (rules of word order), which are levels of language, or about nativism, the view of **Noam Chomsky** (born 1928) that language ability is innate.

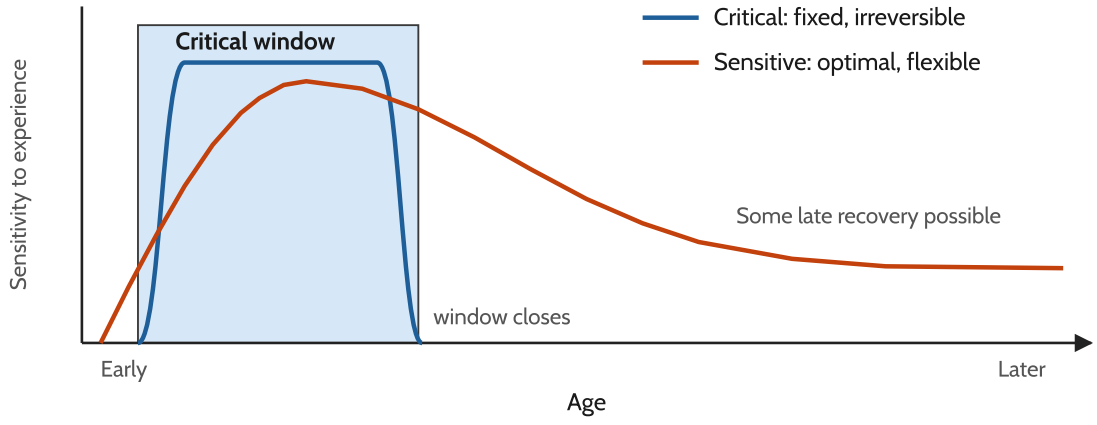

Figure 6.1 A critical period (solid line) opens and closes sharply; a sensitive period (dashed line) peaks early but never falls to zero.

Can later experience repair a poor start? **Ann Clarke** and **Alan Clarke** argued that it often can in *Early Experience: Myth and Evidence* (1976). The strongest test is the English and Romanian Adoptees study led by **Michael Rutter** (1933 to 2021): children adopted into British homes from Romanian orphanages after 1989 caught up almost fully if adopted before about six months, and greatly, though some kept lasting difficulties, if adopted later (Section 6.7).

Later experience compensating for earlier deficits contradicts the strict critical-period view. Sensitive periods allow late recovery by definition; continuity concerns gradual versus stepwise change, and stability whether traits persist.

> **EXAM TRAP**
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> RML 2024 asked for the strongest evidence for critical periods "in the case of human development" and keyed **imprinting**. Imprinting is the classic critical period, but it is animal evidence. Within human development the clearest critical periods are in **embryonic development** (teratogen timing); language shows a sensitive period, and vocational interests show none. Expect imprinting as the key; know why embryonic development is the better human answer.

The grand theories at a glance

Six families of theory answer these issues differently (Table 6.4), each taught in full elsewhere. Psychoanalytic and cognitive theories propose stages; learning theories see continuous, environment-driven change; ethology stresses evolved behaviour and critical periods; ecological and dynamic systems theories stress context and many interacting causes. In the dynamic systems theory of **Esther Thelen** (1941 to 2004), a new skill self-organises out of body, task and environment.

Table 6.4 The grand theories of development.

Family	Key theorists	Where taught
Psychoanalytic	Sigmund Freud (psychosexual stages); Erik Erikson (psychosocial stages)	Chapter 4; Section 6.10
Learning	John B. Watson, B. F. Skinner, Albert Bandura	Section 1.6
Cognitive	Jean Piaget; Lev Vygotsky; information processing	Sections 6.4, 6.5
Ethological	Konrad Lorenz, Nikolaas Tinbergen, John Bowlby	Section 6.7
Ecological	Urie Bronfenbrenner	Below
Dynamic systems	Esther Thelen	Section 6.3

Bronfenbrenner's ecological systems ●●

Most theories put the child under a microscope; **Urie Bronfenbrenner** (1917 to 2005) widened the lens. His **ecological systems theory**, set out in *The Ecology of Human Development* (1979), places the child inside environmental systems nested like Russian dolls (Figure 6.2), each affecting the child and affected in turn.

- The **microsystem** is the innermost layer: the settings in which the child takes part directly, face to face, such as family, school, peer group and neighbourhood. Influence is bidirectional: parents shape the child, and the child shapes the parents.
- The **mesosystem** is the set of links between microsystems: home and school (a parent-teacher meeting), family and peers. It is a relationship between settings, not a setting.
- The **exosystem** holds settings the child is not in that still affect the child: a parent's workplace (a night shift), the school board, the mass media, extended family.
- The **macrosystem** is the overarching culture: values, laws, ideology and the economic system.

- The **chronosystem**, added later, is time: transitions such as parental divorce and historical events such as a pandemic.

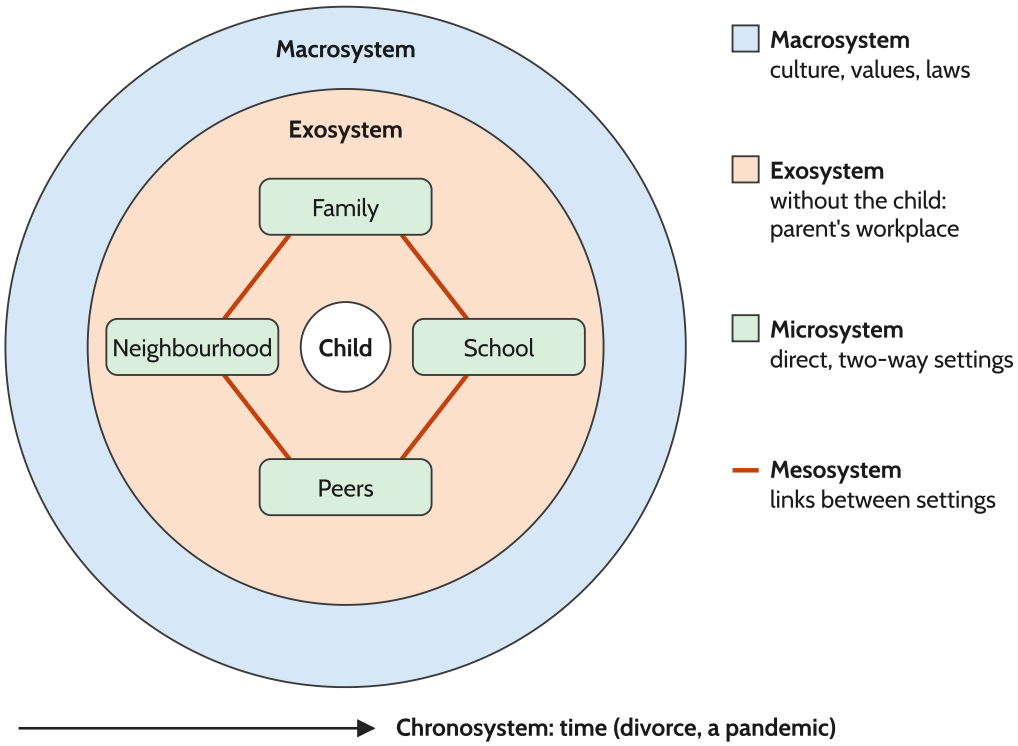

Figure 6.2 Bronfenbrenner's ecological systems. The dashed mesosystem lines are links between microsystem settings, not a setting; the exosystem holds settings without the child.

Later Bronfenbrenner gave biology more weight. In the **bioecological model**, developed with **Stephen Ceci** (1994) and later with **Pamela Morris** (1998), development is the product of Process-Person-Context-Time (PPCT). Its engine is the **proximal process**: regular, increasingly complex, two-way interaction between the child and nearby people, objects and symbols, such as a parent reading with a child every night. Person characteristics (temperament, abilities) shape these processes, context is the five systems, and time runs from moments to history. Bronfenbrenner also helped found Head Start (1965), the American preschool programme for poor children.

EXAM TRAP

Family, school and peer group are **all microsystem**; the immediate family is never meso-, exo- or macrosystem. The **mesosystem is a link** (school-family, parent-teacher), not a setting. The **exosystem does not contain the child** (the parent's workplace). Culture is macrosystem; divorce is chronosystem. Nested systems are **Bronfenbrenner's**, not Skinner's (operant learning), Gesell's (maturation) or Hall's (child study).

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Increase in size or amount is growth; qualitative change in structure and organisation is _____.
2. The biological unfolding of genetic potential is called _____.
3. Gesell and Thompson held heredity constant with the method of _____ control.
4. Head control before walking illustrates the _____ principle.
5. Events shared by a whole generation, such as a pandemic, are normative _____-graded influences.
6. A fixed window with irreversible effects is a _____ period.
7. Links between a child's home and school belong to the _____.
8. A parent's workplace belongs to the _____.

Answers: (1) development, (2) maturation, (3) co-twin, (4) cephalocaudal, (5) history, (6) critical, (7) mesosystem, (8) exosystem.

6.2 Heredity, Prenatal Development, Birth and the Newborn

LOWER YIELD · 7 QS

Every life begins as one cell; nine months later a newborn arrives able to suck, startle and grip. This section follows that journey, from genes to the newborn's first reflexes.

Genes, chromosomes and conception

The nucleus of each body cell holds 46 **chromosomes** in 23 pairs, one of each pair from each parent. A chromosome is a long strand of **DNA** (deoxyribonucleic acid); a **gene** is a segment of DNA that codes for a protein. Twenty-two pairs are **autosomes**; the twenty-third, the **sex chromosomes**, is XX in females and XY in males. Body cells multiply by **mitosis** into identical copies; sperm and ova form by **meiosis**, which halves the number to 23 and reshuffles genes, so siblings differ. At conception, usually in a fallopian tube, sperm and ovum unite into one cell, the **zygote**; since every ovum carries an X, the father's X or Y sperm decides the child's sex. A zygote that splits yields **monozygotic twins** (identical, sharing all their genes); two ova fertilised by two sperm yield **dizygotic twins** (fraternal, sharing about half, like any siblings).

Patterns of inheritance •

Each gene comes in alternative forms called **alleles**. The alleles a person carries are the **genotype**; the observable traits, shaped also by the environment, are the **phenotype**. Two identical alleles make a person **homozygous** for that gene; two different ones, **heterozygous**.

In **dominant-recessive inheritance**, the pattern **Gregor Mendel** (1822 to 1884) reported from pea plants in 1866, a heterozygote shows only the dominant allele's trait; the recessive allele is masked but can be passed on, so the person is a **carrier**. Let blue eyes (D) be dominant and green (d) recessive: DD and Dd children have blue eyes, dd children green. Figure 6.3 shows why two blue-eyed Dd parents can have a green-eyed child.

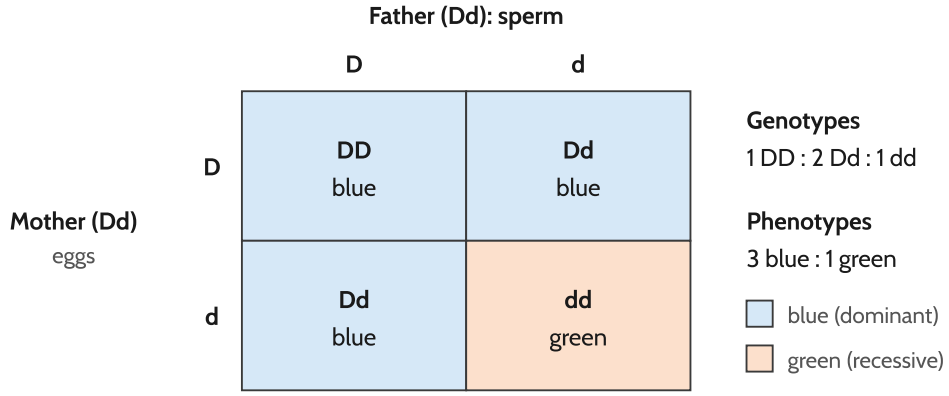

Figure 6.3 A Punnett square for two Dd parents, blue (D) dominant. Every Dd child is blue-eyed; only the dd child, one chance in four, has green eyes.

> **EXAM TRAP**
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> Heterozygous does not mean blended. A Dd child shows the dominant phenotype (blue). Blue-green would need incomplete dominance, and "cannot be determined" is wrong whenever dominance is stated.

Two carriers thus face a one-in-four risk, in each pregnancy, of a child with a recessive disorder: **phenylketonuria** (PKU), cystic fibrosis (thick mucus clogs lungs and pancreas), sickle-cell anaemia or Tay-Sachs disease (fatal nervous system degeneration in early childhood). In PKU the enzyme phenylalanine hydroxylase is missing, so phenylalanine builds up and damages the brain; newborn screening and a low-phenylalanine diet prevent the intellectual disability. A dominant disorder needs one copy: **Huntington's disease** (chorea and dementia) appears in midlife, often after it has been passed on. **Genetic counselling** estimates such risks from family history and tests.

In **incomplete dominance** the heterozygote is intermediate (sickle-cell carriers sickle mildly when oxygen is low); in **codominance** both alleles are fully expressed (A and B alleles give blood group AB). In **X-linked inheritance** a recessive allele on the X is usually masked by a female's second X, but a male has none, so haemophilia, red-green colour blindness and Duchenne muscular dystrophy strike mainly males, through carrier mothers. **Polygenic inheritance**, many genes of small effect, underlies continuous traits such as height and intelligence. In **genomic imprinting** one parent's allele is chemically silenced, so the effect depends on which parent it came from. A **mutation** is a sudden, permanent change in DNA, spontaneous or caused by radiation or chemicals.

Chromosomal abnormalities ●●

Whole chromosomes can be extra or missing, usually because a pair fails to separate in meiosis (nondisjunction), and parts can be lost (deletion). **Down syndrome**, described by **John Langdon Down** (1828 to 1896) in 1866, is usually **trisomy 21**, three copies of chromosome 21; its risk rises with maternal age. It is the commonest genetic cause of intellectual disability; fragile X is the commonest inherited one (Table 6.5).

Table 6.5 Chromosomal and related syndromes.

Syndrome	Karyotype or cause	Key features
Down	Trisomy 21	Intellectual disability, flat face, heart defects
Klinefelter	47,XXY male	Tall, small testes, infertility, language and learning difficulties
XYY	47,XYY male	Tall, usually typical; no proven link with crime
Turner	45,X female	Short, webbed neck, ovarian failure, spatial difficulties
Triple X	47,XXX female	Tall; mild verbal and learning difficulties
Fragile X	FMR1 gene on the X	Intellectual disability, long face, large ears; autism
Cri du chat	Deletion of 5p	Cat-like cry, small head, severe intellectual disability
Prader-Willi	Paternal 15q11-q13 deletion	Hypotonia, hyperphagia, obesity, short stature, hypogonadism
Angelman	Maternal 15q11-q13 deletion	Severe intellectual disability, no speech, seizures, frequent laughter
Williams	Deletion of 7q11.23	Very sociable, fluent speech, weak spatial skills

EXAM TRAP

Klinefelter (47,XXY): tall male, small testes. **Turner** (45,X): short female, webbed neck. **Prader-Willi** and **Angelman** share one region of chromosome 15: lost from the father's copy it gives Prader-Willi, the always-hungry child; from the mother's, Angelman. This is genomic imprinting.

Heredity and environment

How far is a trait inherited? (The nature-nurture debate itself is in Section 6.1.) **Family studies** are weak evidence, since relatives share homes as well as genes. **Adoption studies** ask whether adopted children resemble their biological or adoptive parents; **twin studies**, whether identical (MZ) twins are more alike than fraternal (DZ) twins. Most informative are **MZ twins adopted by different parents**: identical genes, different homes, so their similarities reflect heredity and their differences environment. MZ twins reared together share genes and home, confounding the two; DZ twins or siblings adopted apart share only about half their genes. **Albert Stunkard** and colleagues (1990) found identical twins reared apart nearly as alike in body-mass index as those reared together. **Heritability**, the proportion of variation in a trait within a population due to genetic differences, describes groups, not individuals (IQ figures: Section 1.9; behaviour genetics: Chapter 5).

Genes and environment also work together. **Sandra Scarr** and **Kathleen McCartney** (1983) described three kinds of **gene-environment correlation**: **passive** (parents supply both genes and a matching home: musical parents, a musical house), strongest in infancy; **evocative** (a child's traits evoke responses: a cheerful baby draws more play); and **active**, or **niche-picking** (children seek settings that suit them), which grows with age. In **gene-environment interaction** one environment affects genotypes differently: an ordinary diet harms only the child with PKU. **Irving Gottesman**'s (1963) **reaction range** is the band of outcomes a genotype permits, the environment deciding where in it a person falls. **Conrad Hal Waddington**'s (1905 to 1975) **canalisation** is heredity's channelling of some traits, such as early motor milestones, to the same outcome in almost any environment. **Epigenetics** studies how experience switches genes on or off (for example by DNA methylation) without changing the DNA sequence.

The three prenatal periods

Prenatal development lasts about 38 weeks from conception, in three periods (Figure 6.4). In the **germinal period**, about the first two weeks, the zygote divides (cleavage) as it travels down the fallopian tube, becoming a solid ball, the **morula** (about day 4), then a hollow **blastocyst** (about day 5), whose inner cell mass becomes the baby and outer layer the placenta. **Implantation** in the uterine wall ends the period.

The **embryonic period** (weeks 3 to 8) opens with **gastrulation**, when the inner cell mass forms three germ layers: ectoderm (nervous system, skin), mesoderm (muscle, bone, heart, blood) and endoderm (gut, lungs, glands). The nervous system begins here: the neural plate arises from ectoderm and folds into the **neural tube**, the future brain and spinal cord, by about week 4. Because every major organ forms now (**organogenesis**), this period is the most vulnerable to teratogens. The **fetal period** (week 9 to birth) brings growth and refinement: **quickening**, the first movements the mother feels, at about 16 to 20 weeks, and the **age of viability**, when a baby born early may survive with intensive care, at about 22 to 24 weeks.

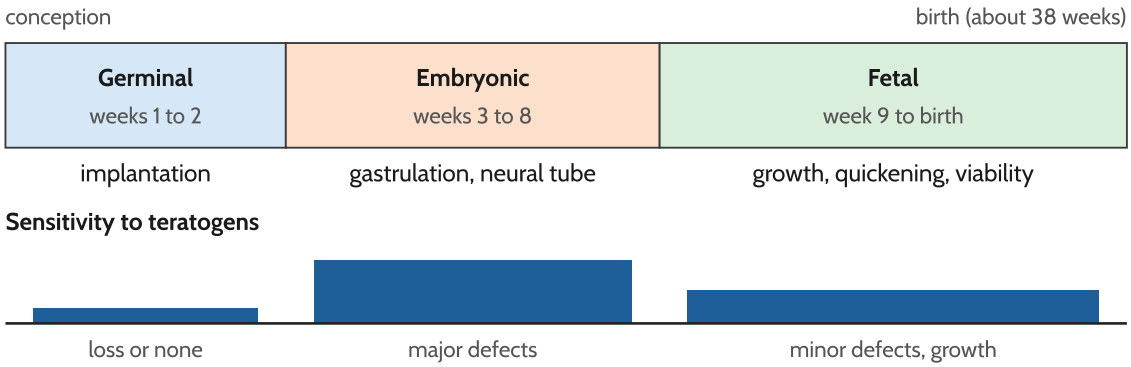

Figure 6.4 The three prenatal periods (not to scale). Bar height shows sensitivity to teratogens, greatest while the organs form.

> EXAM TRAP
>
> The nervous system begins at the **gastrula** stage, not at the zygote, morula or blastocyst stage, whose cells have not yet formed germ layers. The **embryonic**, not the fetal, period is the most vulnerable.

Teratogens, maternal factors and prenatal tests

A **teratogen** is any environmental agent that harms the developing child. Its effect depends on **timing** (each organ has its own critical period, as Figure 6.4 shows), **dose**, the **genetic susceptibility** of mother and child, and **delayed effects** that surface years later. Alcohol (first in Table 6.6) is the commonest preventable cause of intellectual disability: **fetal alcohol spectrum disorder** (FASD) ranges up to full fetal alcohol syndrome, with growth delay, a small head, learning problems and facial signs (short eye openings, a thin upper lip, a smooth philtrum).

Table 6.6 The main teratogens.

Agent	Main effects on the child

Agent	Main effects on the child
Alcohol	Fetal alcohol spectrum disorder
Nicotine	Low birth weight, prematurity, miscarriage
Cocaine, opioids	Prematurity, low birth weight, newborn withdrawal
Thalidomide (late 1950s)	Missing or shortened limbs
Valproate (antiepileptic)	Neural tube defects, lower IQ
Radiation	Small head, intellectual disability
Rubella	Deafness, cataracts, heart defects
Toxoplasmosis	Eye and brain damage
Syphilis	Stillbirth; eye, bone and brain damage
HIV	Infection before, during or after birth
Zika virus	Microcephaly (small head and brain)

Maternal factors matter too. Teenage mothers risk low birth weight; mothers over 35 face more chromosomal abnormalities and miscarriage. **Folic acid** before conception and in early pregnancy prevents most **neural tube defects**: **spina bifida** (an unclosed spinal column) and **anencephaly** (most of the brain missing). Poorly controlled diabetes raises birth defects and birth size; severe stress raises prematurity and low birth weight. In **Rh incompatibility** an Rh-negative mother's antibodies attack the red cells of an Rh-positive fetus, usually in a later pregnancy; an anti-D injection prevents it. Prenatal tests are compared in Table 6.7.

Table 6.7 Prenatal tests.

Test	How and when	Detects
Ultrasound	Sound-wave image; non-invasive	Age, growth, twins, structural defects
Amniocentesis	Amniotic fluid by needle; 15 to 20 weeks	Chromosomal and genetic disorders; small miscarriage risk
Chorionic villus sampling	Placental tissue; 10 to 13 weeks	The same, earlier; slightly higher risk
Maternal blood screening	Alpha-fetoprotein (AFP); 15 to 20 weeks	High: neural tube defect; low: Down syndrome risk

> **INDIAN CONTEXT**
>
> The **PCPNDT Act** (Pre-Conception and Pre-Natal Diagnostic Techniques Act; 1994, amended 2003) bans prenatal sex determination and sex selection; ultrasound centres must register and may not reveal the sex of the fetus.

Birth and assessing the newborn

Labour has three stages: **dilation** of the cervix (the longest), **delivery** of the baby, and delivery of the **placenta**. Its main danger is **anoxia**, too little oxygen (as when the cord is squeezed), which can damage the brain and cause cerebral palsy. A **preterm** baby is born before 37 weeks, perhaps at a weight right for its age; a baby **small for gestational age** weighs less than expected for its age, even at term, and usually fares worse. **Low birth weight** means

under 2,500 g. **Kangaroo care**, skin-to-skin holding on a parent's chest, steadies temperature, breathing and heart rate and improves survival.

The **Apgar scale**, devised in 1952 (published 1953) by the anaesthesiologist **Virginia Apgar** (1909 to 1974), rates five signs 0, 1 or 2 (total 10) at 1 and 5 minutes: **a**ppearance (colour), **p**ulse, **g**rimace (reflex irritability), **a**ctivity (muscle tone) and **r**espiration. A total of 7 or more is good; 3 or less is an emergency. The **Neonatal Behavioral Assessment Scale** (1973) of the paediatrician **T. Berry Brazelton** (1918 to 2018) assesses reflexes, muscle tone, state changes and responses to people.

Peter Wolff (1966) distinguished the newborn's **states of arousal**: regular (non-REM) sleep, irregular (REM) sleep, drowsiness, quiet alertness, and waking activity and crying. Newborns sleep about 16 to 18 hours a day, half of it REM sleep. **Sudden infant death syndrome** (unexplained death of a sleeping infant under one year) fell sharply after back-to-sleep campaigns taught parents to lay babies on their backs.

Newborn reflexes

A **reflex** is an inborn, automatic response to a specific stimulus (Table 6.8). **Survival reflexes** (breathing, blinking, sucking, swallowing, rooting) have clear value, and some last for life. **Primitive reflexes** (Moro, grasp, Babinski, stepping, tonic neck) are controlled by the brainstem and spinal cord and fade over the first year as the cortex matures. A reflex absent at birth, unequal on the two sides or persisting well past its age signals damage to the nervous system.

Table 6.8 Newborn reflexes.

Reflex	Stimulus	Response	Disappears
Rooting	Cheek stroked	Turns toward touch, opens mouth	3 to 4 months
Sucking	Object in mouth	Rhythmic sucking	About 4 months
Moro (startle)	Loss of support, loud noise	Arms fling out, then in as if embracing	4 to 6 months
Palmar grasp	Palm pressed	Fingers grip	3 to 6 months
Plantar grasp	Ball of foot pressed	Toes curl under	9 to 12 months
Babinski	Sole stroked	Big toe up, toes fan out	About 12 months
Stepping	Held upright, feet down	Walking movements	About 2 months
Swimming	Face down in water	Paddles, kicks, holds breath	4 to 6 months
Tonic neck	Head turned, lying on back	Fencing posture	4 to 6 months
Galant	Back stroked beside spine	Trunk curves to that side	4 to 6 months

EXAM TRAP

Texts differ, most of all for the Babinski (8 to 12 months, or up to 2 years). Answer **about 12 months**; an upgoing toe after about 2 years, or in an adult, signals an upper motor neuron lesion. Loss of support triggers the Moro; the sole, the Babinski; the cheek, rooting; the palm, the palmar grasp.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. If D is dominant, a Dd child shows the _____ phenotype.
2. A tall male with small testes and karyotype 47,XXY has _____ syndrome.
3. Prader-Willi syndrome is a 15q11-q13 deletion on the chromosome from the _____.
4. Heredity and environment are best separated in _____ twins adopted by different parents.
5. The nervous system begins forming at the _____ stage.
6. The period most vulnerable to teratogens is the _____ period.
7. The Babinski reflex is normal until about _____ months.

Answers: (1) dominant, (2) Klinefelter, (3) father, (4) monozygotic, (5) gastrula, (6) embryonic, (7) 12.

6.3 Infancy: Brain, Body, Motor Skills and Perception

LOWER YIELD · 12 QS

A newborn cannot hold up her own head. A year later she walks across the room and picks a grain of rice off the floor between thumb and finger. This section follows that change through brain, body, movement and the senses, and asks how we can know what a baby who cannot speak perceives and learns.

Physical growth in the first two years

Birth weight roughly doubles by about 5 months and triples by the first birthday. Growth has two directions. In the **cephalocaudal trend** (Latin, "head to tail") the head grows first and fastest; it is a quarter of body length at birth but an eighth in adulthood. In the **proximodistal trend** growth runs from the centre outward, trunk before arms and arms before hands. Breast milk is the ideal food. Severe malnutrition causes **marasmus**, wasting of the whole body from too little food of any kind, usually in the first year, or **kwashiorkor**, a protein deficiency after weaning in which belly and limbs swell with fluid.

The developing brain ●●

Neurogenesis, the birth of new neurons, is largely prenatal. Why, then, does the brain grow so much after birth? Because existing neurons form connections and gain insulation. In **synaptogenesis** the infant brain forms far more synapses than it will keep, and **synaptic pruning** then removes the rarely used ones and keeps those used often: "use it or lose it". Pruning is a process of neuronal development and plasticity that fits circuits to the child's own experience, continuing in the prefrontal cortex through adolescence (Section 5.5). **Myelination**, the wrapping of axons in fatty myelin that speeds conduction, reaches the prefrontal cortex last, continuing into early adulthood. Together these take the brain from about 25 percent of adult weight at birth to about half by 6 months, 75 percent by age 2 and 90 percent by age 5 to 6 (Figure 6.5).

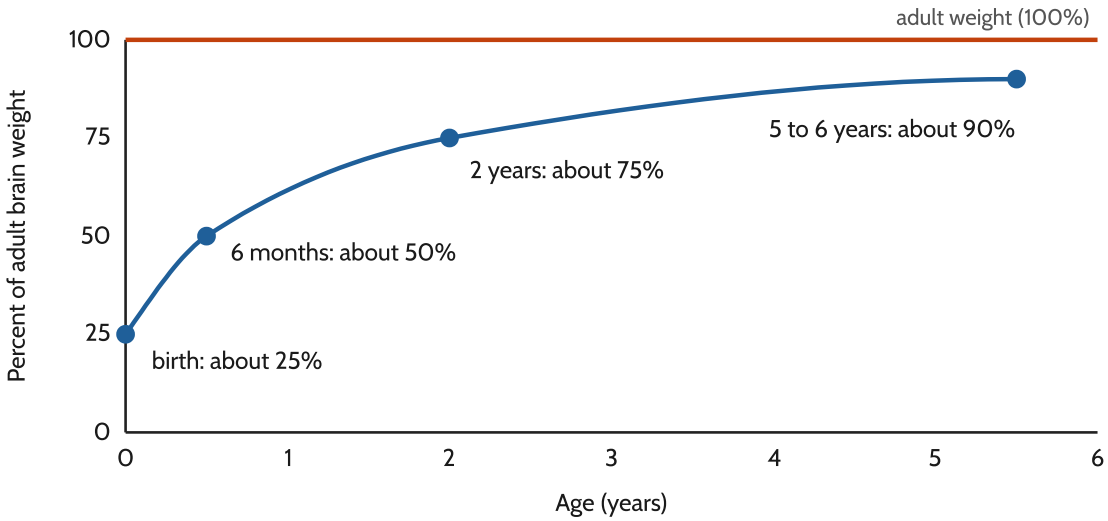

Figure 6.5 Brain weight as a percentage of adult weight (approximate values joined by a smooth line). The curve climbs most steeply in the first two years.

William Greenough and colleagues (1987) distinguished two kinds of plasticity. **Experience-expectant plasticity** depends on experiences every member of the species can count on, such as patterned light and speech; the brain overproduces synapses in readiness and keeps those the expected input uses. Hence sensitive periods: a cataract left untreated in an infant's eye leaves lasting poor vision. **Experience-dependent plasticity** builds connections from experiences unique to the individual and continues for life.

EXAM TRAP

Synaptic pruning belongs to **neuronal development and plasticity**, not memory consolidation (stabilising a new memory, Section 1.7), neurotransmitter release (the moment-to-moment signal at a synapse) or long-term potentiation (lasting strengthening of a synapse by repeated stimulation, Chapter 5). Half of adult brain size is keyed at **about 6 months**, not birth, 1 year or 3 years; some texts imply nearer the first year, so treat these percentages as approximate.

Principles and milestones of motor development

Motor control follows the same two directions as growth (Figure 6.6). By the **cephalocaudal** principle it spreads from head to foot: the infant holds up her head before she sits, and sits before she stands. By the **proximodistal** principle it spreads from the centre of the body to the extremities: she swings her whole arm at a toy months before her fingers can pick it up. Development also runs from **gross motor skills** (large movements of trunk and limbs, such as walking) to **fine motor skills** (precise movements of hands and fingers), and from **mass to specific**: shown a toy, a young infant waves, kicks and wriggles all at once, and only later reaches with one hand. Finally, motor development shapes social and behavioural expectations: a child who walks and feeds herself is treated differently and expected to do more.

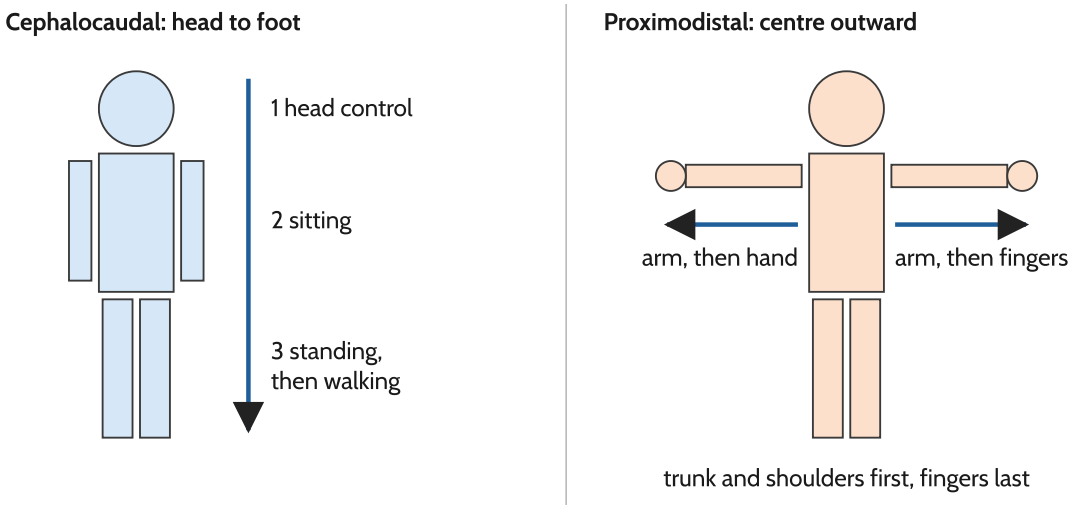

Figure 6.6 The two directions of motor development: head to foot (left) and centre to fingers (right). "Distal to proximal" runs the wrong way.

Table 6.9 gives average ages. Screening norms such as the Denver Developmental Screening Test (Chapter 2) show the ages by which 25, 50, 75 and 90 percent of babies pass each item: healthy babies vary by months.

Table 6.9 Gross motor milestones (approximate average ages).

Milestone	Approximate age
Holds head up steadily when held upright	6 weeks to 3 months
Rolls over	3 to 5 months
Sits without support	about 6 months
Crawls	7 to 9 months
Pulls to stand	8 to 9 months
Walks alone	about 12 months (9 to 17)

EXAM TRAP

Proximodistal means **centre to extremities**. "Distal to proximal" reverses it and is the false option; cephalocaudal, gross to fine, mass to specific, and shaping social and behavioural expectations are all true principles.

Fine motor skills: from reflex to pincer grasp

The hand tells the proximodistal story in miniature. The newborn's **palmar grasp reflex** closes her fingers round anything pressed into her palm. Her **prereaching** is poorly aimed swiping at objects in view, fading after about 2 months. At 3 to 4 months voluntary reaching brings the first voluntary grasp, the **ulnar grasp**: the fingers close clumsily against the palm, holding the object on the little-finger (ulnar) side. By 4 to 5 months she transfers objects from hand to hand. Control then shifts to the thumb side in **radial grasps** (the radius is the forearm bone on the thumb side), and by about 9 to 12 months the **pincer grasp**, thumb against forefinger, lifts that grain of rice.

EXAM TRAP

The keyed first grasp is the **ulnar grasp**, the first voluntary one; the palmar grasp at birth is a reflex. Texts order the middle steps differently (one even puts a clumsy palmar grip before the ulnar grasp), but the pincer grasp always comes last.

Explaining motor development

Why do babies sit, crawl and walk in this order? **Arnold Gesell** (1880 to 1961) gave the **maturational view**: motor development unfolds a genetically programmed sequence as nerves and muscles mature, so practice adds little (his co-twin study is in Section 6.1). Experience matters more than he thought. **Wayne Dennis** and Marsena Dennis (1940) found that Hopi infants bound to cradleboards walked at about the same age as unbound Hopi infants, which seemed to favour maturation; but in Iranian orphanages, where babies lay on their backs with almost no chance to move, Wayne Dennis (1960) found many 3- and 4-year-olds who could not walk alone. Cultures that train sitting and walking, such as the Kipsigis of Kenya, see these skills earlier. Maturation is necessary but not sufficient.

The modern account is **dynamic systems theory**, applied to motor skills by **Esther Thelen** (1941 to 2004): a skill is assembled from the body's current size and strength, the task and the support the environment offers. Take the newborn stepping reflex, which seems to vanish at about 2 months. It vanishes, Thelen showed, because the legs grow heavier faster than the muscles grow stronger: with her legs held in water, the infant steps again.

Methods for studying infant perception

How can you know what a baby perceives? Watch where she looks, how hard she sucks and how her brain responds. **Robert Fantz** (1961) used **preferential looking**: two stimuli are shown side by side and an observer times the looking at each. Infants from a few days to 6 months old looked longest at a face, then at patterns such as newsprint and a bull's-eye, and least at plain discs: they prefer patterns, and faces most of all.

The **habituation-dishabituation** method needs no favourite. Show a picture again and again and looking falls (**habituation**, the simplest form of learning); show a new one, and if looking recovers (**dishabituation**) the infant has noticed the difference. It is the standard test of discrimination and categorisation before language: habituate a baby to many cats, then show a new cat and a dog, and extra looking at the dog shows a category "cat". In **high-amplitude sucking**, strong sucks on a recording dummy switch on a sound, and sucking revives when a new sound is noticed. In the **violation-of-expectation** method of **Renée Baillargeon** (1985), infants look longer at an impossible event than a possible one, revealing what they expect of objects (Section 6.4). **Evoked potentials**, electrical brain responses recorded from the scalp, show whether the brain registers a difference when behaviour shows nothing.

EXAM TRAP

Habituation paradigms assess **discrimination and categorisation before language**, not long-term declarative memory (recognition here lasts minutes within one session), object permanence specifically (tested by violation of expectation) or moral reasoning (Section 6.9).

The infant's senses

Every sense works at birth, but vision is the least mature. Newborn **visual acuity**, the sharpness of sight, is poor (about 20/200 to 20/600, against the adult 20/20) but improves rapidly over the first year. Colour vision is present by about 2 to 3 months, and newborns already prefer faces.

Hearing works before birth. **Anthony DeCasper** and **William Fifer** (1980) found that newborns under 3 days old sucked in whichever pattern played their own mother's voice rather than another woman's. DeCasper and **Melanie Spence** (1986) had mothers read *The Cat in the Hat* aloud in late pregnancy; after birth the babies sucked to hear that story rather than a new one. Infants at first hear speech-sound contrasts their own language does not use, then lose them: **perceptual narrowing**. **Janet Werker** and **Richard Tees** (1984) found that most English-learning infants of 6 to 8 months heard the Hindi contrast between a dental and a retroflex "ta", but few did by 10 to 12 months.

Taste and smell are keen at birth: newborns relax at sweet tastes and grimace at bitter, and breastfed babies about a week old turn towards a pad their own mother has worn. Touch is well developed, and newborns feel pain. **Intermodal perception**, combining several senses into one event, appears early: **Elizabeth Spelke** (1976) showed 4-month-olds two films side by side with one soundtrack, and they looked longer at the film that matched the sound.

Depth, locomotion and affordances ●●

Eleanor Gibson and **Richard Walk** (1960) built the **visual cliff**, a glass-topped table with a checkerboard just under the glass on the shallow side and far below it on the deep side (full study in Section 1.4). Infants of crawling age, about 6 months onward, crossed the shallow side to their mothers but refused the deep side. Depth perception is therefore present by the time infants crawl: not fully formed at birth, not delayed until age 2 and not taught in words. Chicks and young goats tested within a day of birth also avoided the drop, which Gibson and Walk read as evidence that depth perception is largely innate.

Is the crawler's fear inborn? **Joseph Campos** and colleagues measured heart rate on the cliff. Two-month-olds on the deep side showed a slowing heart rate, a sign of interest; crawlers showed a quickening one, a sign of fear, and the more weeks an infant had crawled, the warier she was. Locomotor experience brings wariness of heights.

The **ecological view** of **James J. Gibson** (1904 to 1979) and Eleanor Gibson explains why: we perceive the world in terms of what we can do in it. An **affordance** is an action possibility the environment offers a particular perceiver: a slope affords crawling down to an experienced crawler but a fall to a novice. Because affordances depend on the perceiver's abilities, the change that most transforms them is **independent locomotion**: once the infant crawls and walks, distant toys become reachable and drops dangerous. Starting to speak, the attachment relationship and new caretaking reshape her social world, but none changes what her surroundings afford her actions nearly as much.

EXAM TRAP

The visual cliff tested **depth perception**, present **by crawling age**, not colour perception, object permanence (Section 6.4), language comprehension, imprinting (Section 6.7) or audio-visual integration (intermodal perception). Toddlers' affordances change most with **independent locomotion**.

Learning, imitation and the milestone sequence ●●

Infants learn from birth. Newborns can be classically conditioned: stroke a newborn's forehead just before giving sweet water and soon the stroke alone sets off sucking. In **Carolyn Rovee-Collier**'s operant studies a ribbon tied a 2- or 3-month-old's ankle to a mobile, so kicking made it move; days later infants kicked again at the sight of the same mobile, showing memory long before language.

KEY STUDY: MELTZOFF AND MOORE'S IMITATING NEWBORNS (1977)

Question. Can newborns imitate gestures they cannot see themselves make? **Method.** **Andrew Meltzoff** and **M. Keith Moore** showed infants 12 to 21 days old an adult protruding the tongue, opening the mouth and protruding the lips; coders blind to the gesture scored videotapes. **Result.** Infants made more of whichever gesture they had seen; imitation was later reported within hours of birth. **Conclusion.** Imitation is an important early feature of infant learning, appearing long before 1 year (and before the 8 to 12 months **Jean Piaget** expected), and not only for tongue protrusion. **Criticism.** Later studies often found only tongue protrusion matched, so the finding is debated. Deferred imitation is in Section 6.4.

Table 6.10 gathers the milestones of the first two years. Of the social smile, stranger anxiety, the pincer grasp and symbolic play, **symbolic play** (letting one object stand for another, a block for a phone) comes last, because it needs mental representation.

Table 6.10 The milestone sequence of the first two years (approximate ages).

Approximate age	Milestone
Birth	Reflexive smile, often in sleep
6 to 8 weeks	Social smile, at faces (Section 6.6)
About 2 months	Cooing (vowel-like sounds)
3 to 4 months	Laughter
About 6 months	Babbling ("ba-ba-ba")
About 8 months	Stranger anxiety (Section 6.6)
9 to 12 months	Pincer grasp
About 12 months	First word; first steps
18 to 24 months	Symbolic play; two-word speech

The language rows hide a gap: children understand about 50 words months before they can say 50. This **receptive-expressive gap** is the gap between what infants understand (receptive language) and what they can produce (expressive language); it concerns language, not cogni-

tive, motor or social skills. Language stages and theories (holophrases, telegraphic speech, overregularisation, the language acquisition device, motherese) are in Section 1.8.

> **EXAM TRAP**
>
> Imitation **may be seen as early as 12 to 21 days**, not only after 1 year and not only for tongue protrusion. Of the four milestones, **symbolic play** appears last.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The brain reaches about half its adult weight by about _____ months.
2. Removing rarely used synapses is synaptic _____.
3. Motor control spreading from the centre of the body outward is the _____ principle.
4. The first voluntary grasp is the _____ grasp.
5. Thelen's _____ theory explains the vanishing stepping reflex by heavier legs.
6. Looking less at a repeated picture and more at a new one underlies the _____ method.
7. English-learning infants losing a Hindi contrast by 10 to 12 months shows perceptual _____.
8. An action possibility the environment offers a perceiver is an _____.

Answers: (1) 6, (2) pruning, (3) proximodistal, (4) ulnar, (5) dynamic systems, (6) habituation-dishabituation, (7) narrowing, (8) affordance.

6.4 Piaget's Theory of Cognitive Development HIGH YIELD · 43 QS

Pour a five-year-old's juice from a short, wide glass into a tall, thin one and she will say she now has more, although she watched every drop. She is not careless; she is reasoning with the tools she has. A Swiss biologist built the most influential theory of children's thinking on mistakes like hers. This section follows it from method and core concepts through four stages and beyond, then weighs the evidence.

Piaget and his method ●●

Jean Piaget (1896 to 1980) was a biologist first, with a 1918 doctorate on molluscs. Around 1920, in the Paris laboratory of **Théodore Simon**, co-author with **Alfred Binet** (1857 to 1911) of the first intelligence scale (1905; Section 1.9), he scored children's answers to reasoning items. The wrong answers fascinated him: children of the same age made the same kinds of error, so younger children seemed not duller than older ones but to think differently.

In Geneva from 1921 he pursued one question for sixty years: how does knowledge grow? He called the field **genetic epistemology**, the study of the origins (genesis) of knowledge. His tool was the **clinical method** (*méthode clinique*), a flexible interview that follows each answer with further questions to uncover the reasoning behind it; for infancy he observed his own three children closely. His name-year books are *The Language and Thought of the Child* (1923),

The Moral Judgment of the Child (1932; Section 6.9), *The Origins of Intelligence in Children* (1936) and *The Construction of Reality in the Child* (1937); his stages of play are in Section 6.8.

Piaget treated **intelligence as biological adaptation**: as a species adapts through bodily structures, a child adapts through ever better mental structures. In his **constructivism** the child is an active "little scientist" who builds knowledge by acting on objects, first physically and later mentally.

> EXAM TRAP
>
> Piaget = intelligence as a biological adaptation, and the stages of cognitive development. Free association is **Sigmund Freud**'s (1856 to 1939), operant conditioning **Burrhus Frederic Skinner**'s (1904 to 1990) and aggression as learned social behaviour **Albert Bandura**'s (1925 to 2021). Psychosocial stages belong to **Erik Erikson** (1902 to 1994) and moral stages to **Lawrence Kohlberg** (1927 to 1987).

Schemes, adaptation and equilibration ●●

The unit of knowledge is the **scheme** (often called a schema), an organised pattern of action or thought for dealing with a class of experiences: a newborn's sucking and grasping, a ten-year-old's adding and classifying. Two **functional invariants** shape schemes at every age. **Organisation** combines simple schemes into larger systems, as looking and grasping merge into reaching for what one sees. **Adaptation** adjusts schemes to the environment through two complementary processes.

In **assimilation** the child fits new experience into an existing scheme: a two-year-old whose "doggie" scheme means "four-legged animal" calls a cow "doggie". In **accommodation** the child changes the scheme, that is, his way of thinking, to fit experience that will not fit: the cow moos, an adult says "cow", and the scheme splits in two. Assimilation keeps knowledge stable; accommodation makes it grow.

When schemes handle events, the child is in **equilibrium**. A misfit produces **disequilibrium**, an uncomfortable cognitive conflict, resolved by assimilating and accommodating until a new, more advanced balance is reached. This self-regulating process, **equilibration** (Figure 6.7), drives the child from stage to stage. Mature schemes become **operations**: internalised mental actions that are reversible (they can be undone in thought) and organised into logical systems; the stage names describe the child's relation to them.

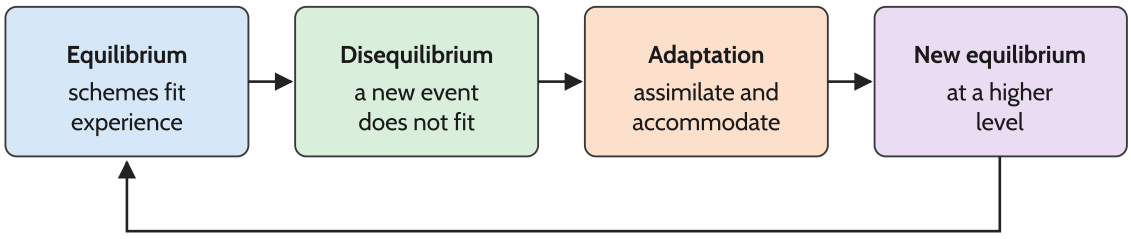

Figure 6.7 Equilibration. A misfit between schemes and experience creates disequilibrium; assimilation and accommodation restore balance at a more advanced level.

Piaget named four factors in development: **maturation** of the nervous system; **experience**, both physical (properties of objects, such as weight) and logico-mathematical (relations one's own actions create, such as ten pebbles staying ten in any order of counting); **social transmission** (being taught); and equilibration, which coordinates the other three. Development thus comes from maturation and learning together, never from either alone, and not from social and cultural practice alone, which is nearer Vygotsky's emphasis (Section 6.5).

> EXAM TRAP
>
> Piaget's two basic processes of learning and adaptation are **assimilation and accommodation**; encoding and retrieval are memory processes (Section 1.7), and attention and perception, or sensory input and motor output, are not his pair. Changing one's way of thinking to fit new experience is **accommodation**, not assimilation, conservation (below) or **habituation**, the decline in responding to a repeated stimulus (Section 6.3).

The four stages at a glance ●●

Piaget's stages are qualitatively different ways of thinking, not simply more knowledge. Every child passes through them in the same **invariant sequence**, none skipped or reversed, each built on the last. The ages are approximate; the order is what the theory fixes. Table 6.11 maps the rest of this section.

Table 6.11 Piaget's four stages of cognitive development.

Stage	Age (years)	Main achievements	Main limits
Sensorimotor	Birth to 2	Object permanence; intentional action; deferred imitation and first symbols by 18 to 24 months	Thinks through action; no mental symbols until the end
Preoperational	2 to 7	Symbolic function: language, pretend play, drawing	Egocentrism, animism, centration, irreversibility, no conservation
Concrete operational	7 to 11	Conservation, decentration, reversibility, seriation, transitivity, classification	Logic only about concrete, real things
Formal operational	11 or 12 onward	Abstract, hypothetical, propositional and hypothetico-deductive reasoning	Idealism, adolescent egocentrism; not reached by everyone

EXAM TRAP

The stages are sensorimotor, preoperational, concrete operational and formal operational. **Hypothetico-deductive reasoning is not a stage**; it is the hallmark of the formal operational stage. Preoperational = 2 to 7 years; concrete operational begins at about 7.

Sensorimotor stage and object permanence ●●●

For the first two years, Piaget argued, infants think with senses and actions rather than mental symbols: the **sensorimotor stage**. Its motor is the **circular reaction**, an action the infant hits on by chance and repeats because the result is interesting; its target moves outward from her own body (primary) to objects (secondary) to deliberate experiment (tertiary), as Table 6.12 shows. By 8 to 12 months acts are intentional, one scheme serving as the means to another (pushing a cushion aside to reach a toy). By 18 to 24 months she represents the world mentally: she solves problems by sudden insight, pretends, and shows **deferred imitation**, copying an action seen hours or days earlier.

Table 6.12 The six sensorimotor substages.

Substage	Months	What the infant does	Object permanence
1 Reflexes	0 to 1	Exercises inborn reflexes: sucking, grasping	None
2 Primary circular reactions	1 to 4	Repeats chance actions on own body (thumb sucking); coordinates schemes across senses (looks towards a sound)	Looks briefly where it vanished
3 Secondary circular reactions	4 to 8	Repeats interesting effects on objects (shakes a rattle)	Retrieves a partly hidden object
4 Coordination of secondary schemes	8 to 12	Intentional, goal-directed, means-end acts	Searches for a fully hidden object; A-not-B error
5 Tertiary circular reactions	12 to 18	Varies actions to see what happens	Follows visible moves; fails invisible ones
6 Mental representation (invention of new means)	18 to 24	Insight, deferred imitation, symbolic thought, pretend play	Complete: solves invisible displacements

The stage's best-known achievement is **object permanence**, the understanding that objects go on existing when they can no longer be seen, heard or touched. To a 4-month-old, Piaget claimed, a toy covered by a cloth has ceased to be. Search for a fully hidden object begins in substage 4, about 8 to 12 months, with a revealing mistake. Hide a toy under cloth A several times and let the infant find it; then, as she watches, hide it under cloth B. She searches under A. This **A-not-B error** shows that object permanence is still incomplete: the object belongs where her action last succeeded. At 12 to 18 months she follows visible moves but not invisible ones (the toy, hidden in a hand, is slipped under a cushion); only at 18 to 24 months does she infer where it must be, and object permanence is complete.

Adele Diamond (1985) showed that the A-not-B error grows with the delay between hiding and search, and later tied its disappearance to maturation of the dorsolateral prefrontal cortex, which supports working memory and inhibition of a *prepotent* (habitual, previously rewarded) reach.

KEY STUDY: BAILLARGEON'S DRAWBRIDGE (1985, 1987)

Renée Baillargeon used the **violation-of-expectation** method: infants look longer at events that surprise them. A screen rotated like a drawbridge towards a box placed behind it. In the possible event it stopped at the hidden box; in the impossible event it swung through the box's space. Five-month-olds (1985), and later some infants of 3.5 to 4.5 months (1987), looked longer at the impossible event, as if they knew the hidden box still existed. Criticism: longer looking may reflect novelty or perceptual preference rather than knowledge.

EXAM TRAP

Object permanence is a **sensorimotor** achievement, never preoperational, concrete or formal, and not to be confused with conservation. Keyed ages: it **begins about 8 to 12 months** (substage 4, A-not-B error) and is **fully established by 18 to 24 months**; 2 months, 4 to 6 months, 8 to 10 months (onset only) or 3 years are wrong for full mastery. The A-not-B error reflects incomplete object permanence with a prepotent reach, not centration or failed reversibility.

Preoperational thought ●●

Around the second birthday the **symbolic function** appears: the ability to make one thing (a word, image or gesture) stand for another, seen in language, pretend play (a banana becomes a telephone) and drawing. The **preoperational stage** (2 to 7) is named for what the child cannot yet do: perform operations. It has a **symbolic function substage** (2 to 4) and an **intuitive thought substage** (4 to 7), when children ask endless "why" questions and judge by intuition rather than logic. **Judy DeLoache** (1987) found that children under about 3 cannot use a scale model of a room to find a toy hidden in the real room: they lack **dual representation**, treating an object both as a thing and as a symbol for something else.

In the Piagetian view, then, preschoolers are unable to reason properly, and their thought shows a family of related limits:

- **Egocentrism**: the inability to distinguish one's own visual or conceptual perspective from another person's; a cognitive limit, not selfishness.
- **Animism**: attributing life and feelings to inanimate things ("the sun is following me").
- **Artificialism**: believing natural things were made by people ("someone painted the sky").
- **Centration**: focusing on one salient feature while ignoring the others. A 5-year-old who sorts a deck of cards by colour but cannot then re-sort the same deck by suit is centring on colour.
- **Irreversibility**: the inability to run an action backwards in the mind, with a focus on **states, not transformations**: the child compares before and after and ignores the pouring between.
- **Transductive reasoning**: reasoning from particular to particular with no general rule. Piaget's daughter, having missed her nap, announced, "It isn't afternoon."
- Failure of **class inclusion**: shown wooden beads, mostly brown, and asked "More brown beads or more wooden beads?", the child says "more brown", comparing subclasses instead of a subclass with the whole.
- Lack of conservation, the subject of the next subsection.

KEY STUDY: THE THREE MOUNTAINS TASK

Piaget and **Bärbel Inhelder** (1913 to 1997) seated children before a model of three mountains (one snow-capped, one with a house, one with a red cross) and asked them to pick the photograph showing the view of a doll sitting elsewhere. Four-year-olds usually picked their own view; correct choices became common only around 7 to 8. Piaget read this as egocentrism. Criticism: the task is unfamiliar and spatially hard; meaningful versions show perspective taking much earlier (Evaluating Piaget).

EXAM TRAP

Centration = focusing on one salient feature while ignoring others (the card sorter; judging liquid by height), not egocentrism, animism, transductive reasoning, irreversibility or the reversible operation behind conservation. Coordinating schemes across senses is a sensorimotor achievement, not centration. "Preschoolers cannot reason properly" is the **Piagetian** view, not that of **Arthur Robert Jensen** (1923 to 2012, Level I and Level II abilities), **Joy Paul Guilford** (1897 to 1987, structure of intellect) or a "holistic" view (Section 1.9).

Conservation ●●●

Conservation is the understanding that a quantity stays the same despite changes in shape or arrangement, so long as nothing is added or taken away. Every task follows one script (Figure 6.8). The child agrees that two identical short, wide glasses hold equal water. One is then poured, as she watches, into a tall, thin glass, where the water stands higher (it does not stay at the same height; that is the point). The preoperational child now says the tall glass has more: her judgement is swayed by one dimension, height, and she ignores the narrower width.

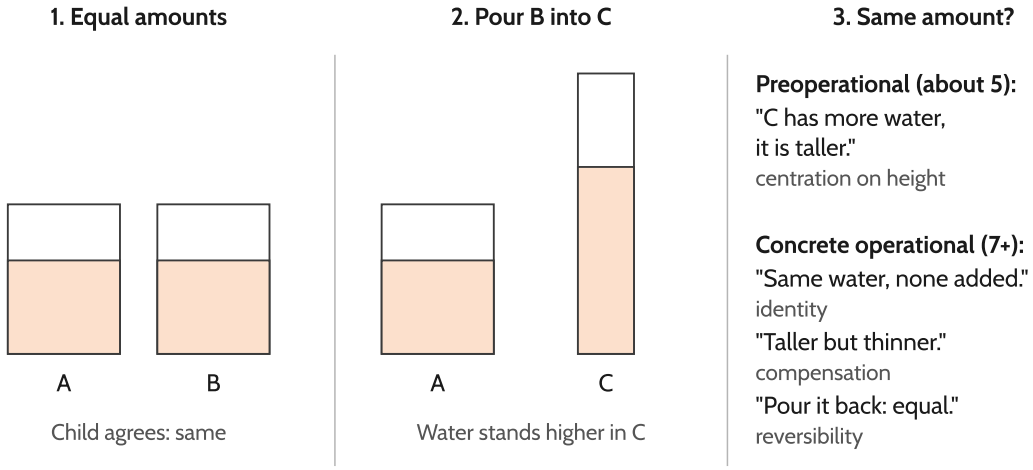

Figure 6.8 The liquid conservation task. Beaker C is half as wide as A, so the same water stands twice as high; only the conserving child sees that the amount is unchanged, and justifies it in one of three ways.

The conserving child gives one of three justifications, exactly the operations the preoperational child lacks:

- **Identity**: "It's the same water; you didn't add any or take any away." The principle of identity is the recognition that certain characteristics of an object remain fixed even when its appearance changes.
- **Compensation**: "It's taller, but it's thinner." A change in one dimension is offset by a change in another.
- **Reversibility** (negation): "If you pour it back, it will be the same as before."

Conservation is several achievements, mastered in a typical order (Table 6.13). A child who conserves substance at 7 or 8 may fail with weight until 9 or 10, though both need the same operation. Piaget called this **horizontal décalage** (French for "gap"): one operation, applied to different contents, mastered at different ages within a stage. **Vertical décalage** is the same achievement recurring at a higher level in a later stage: a toddler finds her way round the house by action but maps it only years later. Conservation of liquid, loosely called liquid volume, differs from volume by displacement, which comes last; all count as concrete operational achievements.

Table 6.13 Conservation tasks in their usual order of mastery (ages approximate; sources differ by a year).

Conservation of	The change the child sees	Usual age (years)
Number	One of two equal rows of counters spread out	6 to 7 (first)
Liquid quantity	Water poured into a taller, thinner glass	7 to 8
Mass (substance)	One of two equal clay balls rolled into a sausage	7 to 8
Length	One of two equal sticks moved ahead	7 to 8
Area	Equal blocks clustered on one field, scattered on another	8 to 9
Weight	Reshaped clay: will it still balance the scale?	9 to 10
Volume (displacement)	Reshaped clay: will it still raise the water as much?	11 to 12 (last)

EXAM TRAP

The **principle of identity** (certain characteristics of objects remain fixed when their appearance changes) is not recognising positive characteristics in oneself (self-esteem), identifying with one's gender (gender identity, Section 6.8) or bringing behaviour into harmony with the world. In the liquid task the liquid does **not** stay at the same height, and the test is not only for under-5s.

Concrete operational stage ●●

At about 7 children begin to perform true operations on real objects and events: the **concrete operational stage** (7 to 11). Shown the clay ball rolled into a sausage, the child now says it is still the same amount, and can say why. The gains behind this are:

- **Decentration**: attending to several features at once, such as height and width.
- Reversibility: mentally undoing an action, which first appears in this stage.
- **Seriation**: arranging items along a dimension (sticks from shortest to longest) systematically, not by trial and error.
- **Transitive inference** (transitivity): if A is longer than B and B is longer than C, then A is longer than C, without comparing A and C directly.

- Classification by more than one attribute, and success on class inclusion ("more wooden beads").
- Reduced egocentrism in perception and conversation.

The limit is in the name: the concrete thinker reasons logically about real, picturable things but balks at the purely hypothetical. Told "If all dogs were pink and Bruno is a dog, what colour is Bruno?", she objects that dogs are not pink.

> EXAM TRAP
>
> In the concrete stage reduced egocentrism, transitive inference and conservation of mass are present, object permanence is long established and deferred imitation dates from infancy. What the concrete operational child struggles with, and does **not** attain, is **hypothetico-deductive reasoning**.

Formal operations and postformal thought ●●

From about 11 or 12 the **formal operational stage** frees thought from the concrete. Adolescents reason about abstractions (justice, infinity), about the possible as well as the real, and about **propositions**, statements whose conclusions follow from their form ("If A then B; A; therefore B"). The signature is **hypothetico-deductive reasoning**: deducing hypotheses from a general theory and testing them systematically, one variable at a time. With it come **counterfactual reasoning** (from premises contrary to fact: "If the moon were made of cheese…"), **combinatorial reasoning** (generating every possible combination of factors) and metacognitive reflection, thinking about one's own thinking.

Inhelder and Piaget's tasks appear in *The Growth of Logical Thinking from Childhood to Adolescence* (1958). In the **pendulum problem** the child must find what controls how fast a pendulum swings: string length, weight, height of release or force of push. The concrete thinker changes several things at once; the formal thinker varies one at a time and finds that only length matters. In the chemical combination task the adolescent mixes colourless liquids in every combination to find which turns yellow. The new power brings idealism (measuring the real world against ideal ones) and adolescent egocentrism (Section 6.10). Nor is the stage universal: many adults do not reason formally on unfamiliar content.

Is formal logic the end point? Several theorists proposed **postformal thought**, adult reasoning that accepts contradiction, uncertainty and context (Table 6.14). Its core is **dialectical reasoning**: seeing that opposing views each hold part of the truth and integrating them in a synthesis.

Table 6.14 Postformal thought.

Theorist	Year	Idea
Klaus Riegel	1973	Dialectical operations, "the final period of cognitive development": integrating contradiction
William Perry	1970	College students move from dualism (right versus wrong) through relativism to commitment
Patricia Arlin	1975	Problem finding: a fifth stage beyond formal problem solving
Michael Basseches	1984	Dialectical thinking: seeing systems in constant change and interaction

Theorist	Year	Idea
Gisela Labouvie-Vief	1980	Pragmatic thought: logic adapted to real-life limits and emotion
Jan Sinnott	1998	Relativistic thought: choosing among logics, each valid in its context

EXAM TRAP

The stage of abstract thought is the **formal operational** stage. It includes hypothetical, hypothetico-deductive, propositional, counterfactual, combinatorial and metacognitive reasoning. **Dialectical reasoning is postformal** (Riegel, Basseches), not formal operational.

Evaluating Piaget

Piaget's broad picture survives: children actively construct knowledge, thinking changes in kind as well as amount, and his sequence recurs across cultures. His timing and stages have fared worse.

First, he **underestimated infants and preschoolers**. Baillargeon credited infants with object knowledge months early, and many preschool failures came from the task rather than the mind. **Margaret Donaldson** (1926 to 2020) drew the lesson in *Children's Minds* (1978): children reason well when a task makes "human sense" to them.

KEY STUDY: POLICEMEN AND A NAUGHTY TEDDY

Martin Hughes (1975) asked children of 3.5 to 5 to hide a boy doll from two policeman dolls among crossing walls; about 90 percent succeeded, though the task needs two perspectives at once, because hiding makes sense to them. **James McGarrigle** and Donaldson (1974) let a "naughty teddy" disarrange one row of the number-conservation task as if by accident; far more 4- to 6-year-olds conserved than when the adult changed the row on purpose. Criticism: some may have ignored the accidental change rather than truly conserved.

Second, development is less stage-like than claimed: décalage is the rule, and abilities grow gradually, domain by domain. Third, culture and schooling matter more than he allowed, the core of Lev Vygotsky's critique (Section 6.5). Fourth, he did not separate **competence**, the understanding a child has, from **performance**, what she shows on a task with its own language, memory and attention demands.

Neo-Piagetian theories keep the stages but explain them by growing processing capacity. **Juan Pascual-Leone** (1970) proposed **M-space**, the number of schemes a child can hold in mind at once, which grows with age. **Robbie Case** (1944 to 2000) traced stage progress to more efficient use of limited working memory as skills become automatic. **Kurt Fischer** (1943 to 2020) proposed skill theory (1980): skills develop through levels, separately in each domain, and run higher with support than alone.

EXAM TRAP

Spot the false statement: "Cognitive structures of children under 7 reflect abstract mental operations" is **false**; their thought is preoperational. True Piagetian statements: children construct knowledge by acting on objects physically or mentally; infant schemes are organised patterns of action; older children's operations let them see that quantities are conserved despite changes in appearance.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Changing one's schemes to fit a new experience is _____; fitting it into existing schemes is _____.
2. Piaget saw intelligence as a form of biological _____.
3. Object permanence begins at about _____ months and is complete by _____ months.
4. An infant who searches where a toy was last found, not where she saw it hidden, shows the _____ error.
5. A 5-year-old who sorts cards by colour but cannot re-sort them by suit shows _____.
6. Saying "it's the same clay; nothing was added" is the conservation justification called _____.
7. Mastering conservation of mass years before conservation of volume is _____ décalage.
8. Dialectical reasoning belongs to _____ thought, not formal operations.

Answers: (1) accommodation, assimilation; (2) adaptation; (3) 8 to 12, 18 to 24; (4) A-not-B; (5) centration; (6) identity; (7) horizontal; (8) postformal.

6.5 Vygotsky, Bruner, Information Processing and Theory of Mind

MEDIUM YIELD · 25 QS

Piaget's child (Section 6.4) works alone, a small scientist testing the world. Yet a real child learning to tie her shoelaces usually has an older sister kneeling beside her. Vygotsky and Bruner put that helper back into the picture; the information-processing approach treats the mind as a computer that gets faster and smarter; and research on theory of mind asks when children discover that other people have minds of their own.

Vygotsky's sociocultural theory

Lev Semyonovich Vygotsky (1896 to 1934), a Russian psychologist who died of tuberculosis at 37, saw cognitive growth as a socially mediated activity: children acquire new ways of thinking through cooperative dialogue with more skilled members of their culture. This is his **sociocultural theory**, set out in *Thinking and Speech* (1934) and known to English readers mainly through *Mind in Society* (1978), an edited collection of his essays.

Vygotsky separated elementary mental functions, natural capacities we share with animals (simple attention, perception, involuntary memory), from **higher mental functions** such as

voluntary attention, deliberate memory and verbal reasoning, which are human and cultural. What turns the first into the second is **mediation**: we act on the world, and on our own minds, through tools. Each culture supplies **tools of intellectual adaptation**, signs such as number systems, writing, maps and memory aids, that let children use their basic functions more adaptively; a knot tied in a handkerchief to remember an errand is a sign steering one's own memory. Language is the master tool.

How does a tool that exists between people get inside one head? The **general genetic law of cultural development** states that every higher function appears twice: first between people (interpsychological), then within the child (intrapsychological). This passage from outer to inner is **internalisation**. A mother who asks "Where did you last see your bag?" remembers together with her son; years later he asks himself silently. Joint work rests on **intersubjectivity**, the process by which two people who begin a task with different understandings arrive at a shared one. Inter-subjectivity therefore precedes intra-subjectivity.

Development, Vygotsky added, must be studied at four interrelated levels: microgenetic (change over minutes or days), ontogenetic (over a lifetime), phylogenetic (over evolution) and sociohistorical (change in the culture's values and technologies).

> EXAM TRAP
>
> For Vygotsky, **inter-subjectivity precedes intra-subjectivity**. "Intra precedes inter" reverses the law, and "children do not engage in inter-subjectivity" contradicts it. "The two are related" is true but empty: the law is about their order.

The zone of proximal development and scaffolding ●●●

Vygotsky described two children who, working alone, both solve test problems set for eight-year-olds. Given hints, one solves problems set for twelve-year-olds, the other only those for nine-year-olds. The distance from eight to twelve, or from eight to nine, is what he called the **zone of proximal development** (ZPD): the gap between the child's actual developmental level, shown by independent problem solving, and her level of potential development, shown by problem solving under adult guidance or with more capable peers. That helper is now called the **more knowledgeable other**. As a range of tasks, the ZPD holds those too difficult to master alone but achievable with help (Figure 6.9). Teaching should aim inside it, at functions still ripening.

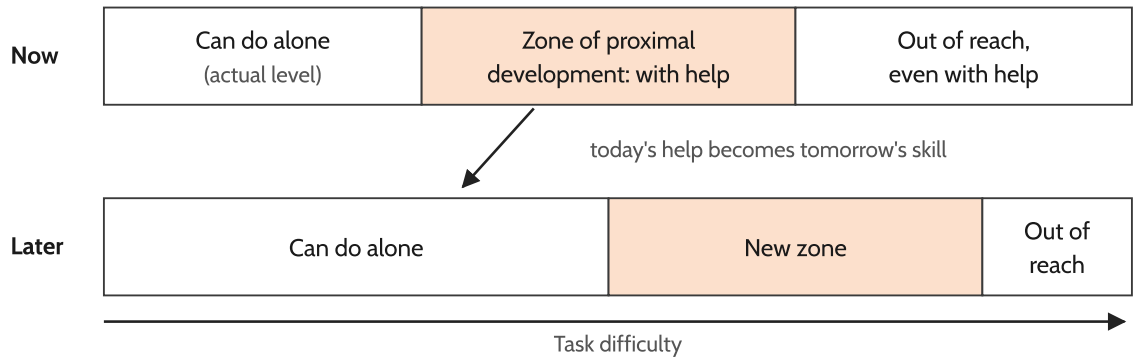

Figure 6.9 The zone of proximal development. As the child internalises what she first did with help, the zone moves towards harder tasks.

Teaching within the ZPD is now usually called **scaffolding**: support fitted to the learner's current level and withdrawn as competence grows. The term is not Vygotsky's: it comes from **David Wood**, Jerome Bruner and **Gail Ross** (1976).

KEY STUDY: WOOD, BRUNER AND ROSS (1976)

A tutor helped 3- to 5-year-olds build a pyramid of interlocking wooden blocks, a task too hard for them alone. Good tutoring recruited interest, simplified the task, kept the goal in view, marked critical features, controlled frustration and demonstrated, stepping in only as far as the child needed. **Criticism:** the metaphor implies a fixed structure built by the adult, whereas real tutoring is a two-way exchange.

Barbara Rogoff (*Apprenticeship in Thinking*, 1990) described development as an apprenticeship built on **guided participation**: children learn by taking part, beside adults and peers, in their community's valued activities. In **reciprocal teaching** (**Annemarie Palincsar** and **Ann Brown**, 1984), teacher and pupils take turns leading discussion of a text through questioning, summarising, clarifying and predicting, the teacher fading as pupils take over. **Dynamic assessment** (test, teach, retest) measures how much a child gains from help, as in **Reuven Feuerstein**'s Learning Potential Assessment Device (1979).

EXAM TRAP

The ZPD is the **gap between independent and assisted performance**, a term coined by **Vygotsky** (not Piaget, Bruner, Kohlberg, Bronfenbrenner, Erikson, Skinner or Freud). Wrong options name something else: tasks done alone (the actual level), too simple, too hard even with help, or learned without instruction; innate abilities, structures or the language acquisition device (Section 1.8); a critical, optimal or language-learning period (Section 6.1) or the usual age for a milestone; a stage of logical operations or object permanence (Section 6.4); a fixed capacity or genetic ceiling; mental age minus chronological age (Section 1.9); a brain or adolescent growth spurt; a neural region; a peer group.

Thought, speech and concepts ●●

On language the two theorists disagreed most clearly. For Piaget thought comes first, and language expresses concepts already built through action; the preschooler's talk to no one, his **egocentric speech** (*The Language and Thought of the Child*, 1923), mirrors egocentric thought and simply fades. For Vygotsky thought and language have separate roots: prelinguistic thought (a toddler using a stick to reach a toy) and preintellectual speech (babbling). At about age 2 they merge; thought becomes verbal, and from then on language shapes thought. So thought depends on language for Vygotsky, and language depends on thought for Piaget.

Speech then turns inward. Social speech, used to communicate, comes first. From about 3 children talk aloud to themselves as they work ("the red one goes… no, here"). Vygotsky saw this talk, now called **private speech**, as social speech turned to guiding one's own actions; at school age it goes underground as **inner speech**, silent verbal thought. **Laura E. Berk**'s studies in the 1980s and 1990s supported him: children use more private speech on hard tasks and after errors, those who use it more tend to improve, and it fades to whispers before going inward.

> EXAM TRAP
>
> **Vygotsky: thought depends on language; Piaget: language depends on thought.** The reversed option swaps the names, and "one studied early, the other later language" is false. Private speech guides and regulates behaviour and becomes inner speech. It is not a regression to "autistic" (undirected, wishful) thinking, a word Piaget used, nor mere imitation, a language disorder or egocentrism that fades with schooling.

Vygotsky traced concept formation with block-sorting tasks. Young children form heaps (syncretic groupings), objects lumped together by chance impression. Next come complexes, groups built on concrete, shifting links (a red block, then a red triangle, then any triangle). The most advanced complex, the **pseudoconcept**, looks like a true concept (every triangle gathered) but rests on visible likeness, not an abstract rule. True concepts mature in adolescence. Vygotsky linked them to the distinction between the acting "I" and the reflecting "I" (the self that does and the self that thinks about its doing), to the problems and developmental tasks society sets adolescents, and to participation in its institutions. Gender differences play no part.

Piaget and Vygotsky compared ●

Both men were born in 1896, both saw the child as active, and neither held that development is solely environmental or predetermined. Table 6.15 sets out where they differed.

Table 6.15 Piaget versus Vygotsky.

Issue	Piaget	Vygotsky
The child	Lone scientist	Social apprentice
Universality	Universal structures, invariant stages	Skills often specific to the culture, learned from more competent people
Learning and development	Development leads learning	Learning leads development

Issue	Piaget	Vygotsky
Engine	Equilibration through acting on objects	Social interaction and cultural tools
Language and thought	Thought first; language reflects it	Separate roots, merge at about 2; language shapes thought
Self-talk	Egocentric speech, fades	Private speech, becomes inner speech
Partners	Peers, whose differing views create conflict	Adults and more capable peers
Classroom	Discovery, readiness	Scaffolding in the ZPD

EXAM TRAP

Options often swap the theorists: it was **Piaget**, not Vygotsky, who held that all children pass through the same stages. "Solely environmental" and "predetermined" fit neither man.

Jerome Bruner: representation and discovery

Jerome Seymour Bruner (1915 to 2016), an American leader of the cognitive revolution, shared Vygotsky's interest in culture and teaching. In 1966 he proposed three **modes of representation**. The **enactive** mode represents knowledge through action (a baby knows a rattle as a shaking movement); the **iconic** mode, through images (picturing the route to school); the **symbolic** mode, through language, numbers and other arbitrary symbols. They appear roughly in that order from infancy to the school years, but unlike Piaget's stages none is left behind.

In *The Process of Education* (1960) Bruner argued that any subject can be taught in some intellectually honest form to any child at any stage. Hence the **spiral curriculum**, which returns to the same big ideas at rising levels of complexity, and **discovery learning**, in which learners find principles for themselves by working on problems. His language acquisition support system (LASS) is covered in Section 1.8.

Information processing: the mind as a computer

The **information-processing approach** likens the mind to a computer: information comes in, is held and transformed, and leaves as an answer. Development means better hardware (the brain and nervous system) and better software (attention, memory strategies and problem-solving routines), and it is continuous: many small quantitative gains, not qualitative stages.

The hardware speeds up. **Robert V. Kail** (1991) showed that processing speed rises steeply in childhood and more slowly in adolescence, at a similar rate across very different tasks, which suggests a general biological cause such as myelination. Attention grows more selective, working memory holds more, and the **executive functions** of the prefrontal cortex mature: inhibitory control (stopping a dominant response), working memory (holding and updating information) and cognitive flexibility (switching rules or viewpoints).

The software improves too. Memory strategies appear in order: rehearsal (repeating items) from about 7, organisation (grouping items by meaning) in middle childhood and elaboration (linking items through an image or sentence) mostly in adolescence. A child with a **produc-**

tion deficiency, **John H. Flavell**'s term, can use a strategy when told but does not produce it spontaneously; one with a **utilisation deficiency** (**Patricia H. Miller**) produces it but at first gains little from it.

Flavell (1979) also made **metacognition**, knowledge and control of one's own thinking, central; its memory part is **metamemory**, such as knowing that a long list needs more study. Knowledge matters as much as capacity: **Michelene Chi** (1978) found that child chess experts recalled board positions better than adult novices, who still recalled digit lists better. **Robert S. Siegler**'s **overlapping waves model** (1996) holds that children always have several strategies (for 3 + 5: counting on fingers, counting on from the larger number, simply retrieving 8) whose frequencies shift gradually, like overlapping waves, rather than in stages. In **fuzzy-trace theory** (**Charles Brainerd** and **Valerie Reyna**, 1990) we store separate verbatim and gist traces and rely more on gist with age.

Theory of mind ●●

A **theory of mind** is the understanding that people have mental states (beliefs, desires, intentions) that guide their actions and can differ from one's own and from reality. The term comes from **David Premack** and **Guy Woodruff** (1978), who asked whether a chimpanzee has one. Its precursors appear late in the first year: **joint attention** (following another's gaze or point to a shared object), social referencing (Section 6.6) and intention reading (treating a reach as aimed at a goal).

Gyorgy Gergely and **Gergely Csibra** (from 1995) call this early framework the **teleological stance**: around the end of the first year infants read an action as efficient and goal-directed, expecting an agent to take the most direct route its situation allows. In the philosopher **John Searle**'s (1983) terms, toddlers understand **intentions in action**, the goal visible in an ongoing act. They have moved beyond mere **contingencies** (this event follows that one) but not yet to **prior intentions** (plans formed before acting) or mental states such as belief and desire.

Henry Wellman (1990) traced the next steps: a desire psychology at about 2, explaining actions by wants, then a **belief-desire psychology** at 3 to 4, in which actions follow from what people want and what they believe, even falsely. The test is the **false-belief task**: the child must predict the action of someone whose belief the child knows is wrong. In **Heinz Wimmer** and **Josef Perner**'s original (1983), Maxi puts his chocolate in one cupboard and goes out; his mother moves it. Where will Maxi look? In the Smarties (unexpected-contents) task of Perner, **Susan Leekam** and Wimmer (1987), a child who expected sweets in a Smarties tube is shown pencils and asked what a friend will think is inside. Three-year-olds answer from reality; most children pass at about 4, in the preoperational years. Flavell's **appearance-reality distinction** comes at a similar age: shown a sponge painted like a rock, a 3-year-old says it looks like a sponge, while a 5-year-old says it looks like a rock but is really a sponge.

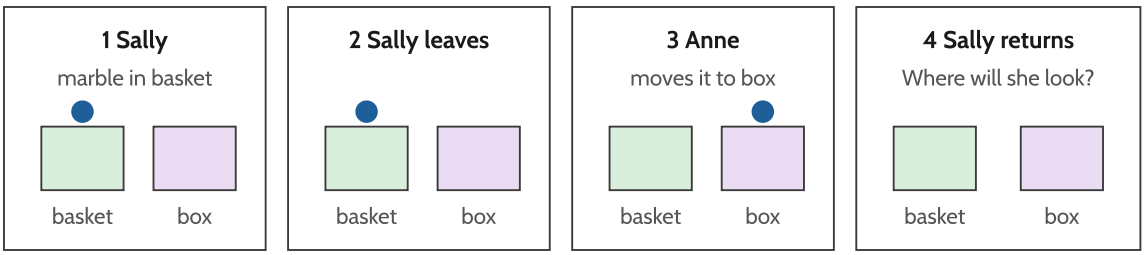

Figure 6.10 The Sally-Anne task. To pass, the child must answer from Sally's mistaken belief, not from what she herself knows.

KEY STUDY: BARON-COHEN, LESLIE AND FRITH (1985)

Simon Baron-Cohen, **Alan Leslie** and **Uta Frith** acted out the **Sally-Anne task** with dolls (Figure 6.10). About 85 percent of typically developing preschoolers and 86 percent of children with Down syndrome answered "the basket", but 80 percent of autistic children, though of higher verbal mental age than the Down syndrome group, pointed to the box. The authors proposed a specific theory-of-mind deficit in autism, later called mindblindness (Chapter 8). **Criticism:** the task also loads on language and memory, and some autistic children pass it.

EXAM TRAP

False-belief tasks (Maxi, Sally-Anne, Smarties) assess **theory of mind** and are passed at **about age 4**, in the preoperational period: not at 12 months (only the teleological stance), age 8 (concrete operations) or adolescence. They are not the A-not-B task (object permanence), a conservation task (both Section 6.4), the Strange Situation (attachment, Section 6.7) or a working memory test.

Recursive thought and social perspective taking

Passing a false-belief task is first-order reasoning: "Sally thinks the marble is in the basket." A **second-order theory of mind** nests one belief inside another ("John thinks that Mary thinks the ice-cream van is in the park"); Perner and Wimmer (1985) found it at about 6 to 7. The ability to nest perspectives is **recursive thought**, a form of perspective taking that views a situation from at least two perspectives, reasoning simultaneously about what two or more people are thinking ("I think that you think that I think..."). Patricia Miller, **Frank Kessel** and Flavell (1970) showed it growing through the school years. It is not **referential style**, **Katherine Nelson**'s term for toddlers whose first words mostly name objects; not **reversibility**, Piaget's mental undoing of an operation (Section 6.4); and not **reconstruction**, the rebuilding of memories (Section 1.7).

Robert L. Selman (1976, 1980) studied **social perspective taking**, understanding another person's thoughts and feelings, through dilemmas. In the best known, Holly has promised her father not to climb trees, then finds a friend's kitten stuck in one. From the answers

Selman described five levels (Table 6.16), a progression from egocentric to multiple, mutual perspective taking.

Table 6.16 Selman's levels of social perspective taking (ages approximate and overlapping).

Level	Age	What the child understands
0 Undifferentiated (egocentric)	3 to 6	Confuses others' views with her own: "He'll be happy; he likes kittens."
1 Social-informational (differentiated, subjective)	4 to 9	Views differ because information differs: "If he knows why, he won't be cross."
2 Self-reflective (reciprocal)	7 to 12	Sees her own thoughts from another's viewpoint, and knows others can do the same
3 Third-party (mutual)	10 to 15	Sees both views as a neutral third person would
4 Societal (in-depth)	Adolescence onward	Sees views as shaped by society's values and norms

EXAM TRAP

Selman's levels run from **egocentric to multiple, mutual perspective taking**, not from concrete to abstract number, inductive to deductive logic or procedural to declarative memory. Age bands differ across sources (level 1 is given as 4 to 9, 5 to 9 or 6 to 8), and one key places the move from level 1 to level 2 at **5 to 9 years**: choose it over 3 to 5 (level 0), 10 to 12 (level 3) or late adolescence (level 4).

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The gap between what a learner can do alone and with guidance is the _____.
2. Every higher function appears first between people, then within the child: the general _____ law of cultural development.
3. Self-guiding talk aloud that later becomes inner speech is _____ speech.
4. Bruner's modes of representation are enactive, iconic and _____.
5. Using a strategy when told but never spontaneously is a _____ deficiency.
6. Toddlers taking the teleological stance understand intentions in _____.
7. Reasoning simultaneously about what two or more people are thinking is _____ thought.

Answers: (1) zone of proximal development, (2) genetic, (3) private, (4) symbolic, (5) production, (6) action, (7) recursive

6.6 Emotional Development, Temperament and Self-Regulation

LOWER YIELD · 19 QS

Follow one baby through her first three years: at two months she smiles at your face, at eight months she cries when a stranger reaches for her, and at two years she hides her face in embarrassment. Alongside this shared timetable runs an individual style, temperament, and the growing capacity to manage it, self-regulation.

The first emotions

What can a newborn feel? In the **differentiation theory** of **Katharine Banham Bridges** (1932), built from observing infants of different ages, the newborn has only a diffuse *general excitement*. Within about three months this splits into two poles, **distress** and **delight**, which are therefore the first emotions infants express. Each pole then divides again, distress into anger, disgust and fear, delight into elation and affection, until by two years most adult emotions can be seen (Figure 6.11).

Two rival views differ. **John Broadus Watson** (1878 to 1958) held that only **fear, rage and love** are innate and all other emotions are conditioned onto them. **Carroll Izard** (1923 to 2017), in his **differential emotions theory**, held that each basic emotion is a separate innate system with its own facial expression, switched on by maturation (Section 1.11). Pairs such as "pain and anger" or "fear and euphoria" belong to no one's scheme.

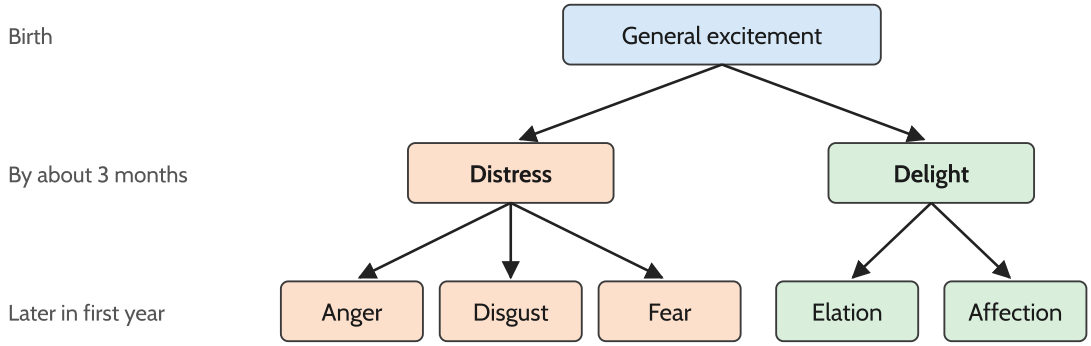

Figure 6.11 Bridges' differentiation of emotion (1932): general excitement divides first into distress and delight, then into specific emotions.

Modern research, much of it by **Michael Lewis**, places the **primary emotions** (joy, interest, surprise, sadness, disgust, anger and fear) within the first six months; none needs a sense of self. Is the first smile joy? Not yet. The **reflexive smile** (endogenous smile) of the first weeks comes from internal states, mostly in sleep. The **social smile**, a smile in response to a human face or voice, appears at about 6 to 10 weeks, and **laughter** at about 3 to 4 months (Figure 6.12).

Self-awareness and the self-conscious emotions

A second family of emotions must wait for a self. Embarrassment, pride, shame, guilt, envy and empathy are **self-conscious emotions** (secondary or self-evaluative emotions): each involves judging yourself through others' eyes or against a standard, so each needs *self-awareness* and appears only after 12 months, from the second half of the second year. Lewis places embarrassment, envy and empathy at about 18 to 24 months, once the child recognises himself, and pride, shame and guilt at about two and a half to three years, once he has rules to judge himself by. **Joseph Campos** (2009) and **Jerome Kagan** (1929 to 2021; 2010) agree on this timing.

KEY STUDY: THE ROUGE TEST (LEWIS AND BROOKS-GUNN, 1979)

Michael Lewis and **Jeanne Brooks-Gunn** secretly dabbed rouge on infants' noses and placed them before a mirror. A baby who touches her own nose, not the reflection's, knows the face is hers. Infants under a year treated the image as another baby; nose-touching was typical by 18 to 24 months. This **mirror self-recognition** marks the arrival of the self, just as embarrassment appears.

EXAM TRAP

Self-conscious emotions appear **after 12 months** (from the second half of the second year), never at 3 months, 6 to 9 months or "before 9 months". Those early ages belong to the primary emotions.

Strangers, separations and the still face

In the second half of the first year emotion turns sharply social. **Stranger anxiety**, wariness or crying when an unfamiliar adult approaches, emerges at about 6 to 8 months; **René Spitz** (1887 to 1974) called it eight-month anxiety (Section 6.7). **Separation protest** (separation anxiety), crying when the caregiver leaves, follows; Kagan found the same curve in several very different cultures, rising from about 8 months, peaking at about 13 to 15 months and then declining. Both fears therefore peak within roughly 8 to 15 months, because the baby now has a specific attachment figure and enough memory to notice an unfamiliar face or an absent caregiver. Excessive distress far beyond this age is separation anxiety disorder (Chapter 8).

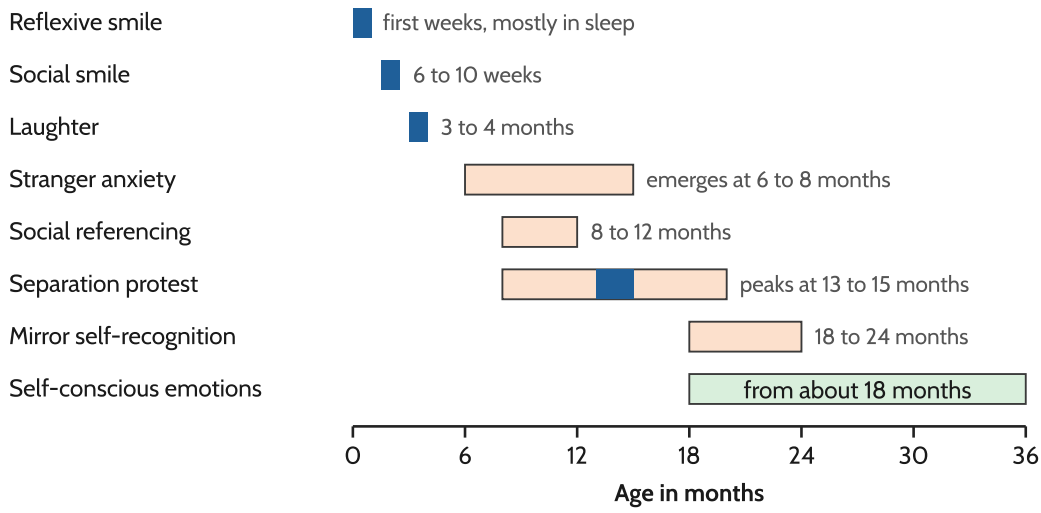

Figure 6.12 Emotional milestones of the first three years. The dark block marks the peak of separation protest; both fears overlap at 8 to 15 months.

EXAM TRAP

Stranger anxiety and separation anxiety peak at about **8 to 15 months**, not at 1 to 2 months (the age of the social smile), 3 years or 6 years.

Infants also learn to use others' emotions as information. In **social referencing**, from about 8 to 12 months, an infant facing an uncertain situation reads a trusted adult's face and lets it guide her. **James Sorce**, Campos and colleagues (1985) placed 12-month-olds on a visual cliff (Section 6.3) with a drop just deep enough to make them hesitate: when the mother looked happy, about three in four crossed; when she looked afraid, none did.

KEY STUDY: THE STILL-FACE PARADIGM (TRONICK AND COLLEAGUES, 1978)

Are young infants simply soaked in their caregivers' mood, or partners in the exchange? **Edward Tronick** and colleagues filmed mothers with infants a few months old in three episodes: normal face-to-face play; a still face, in which the mother keeps looking at the baby with a blank, unresponsive face; and reunion. Faced with the still face, babies first tried to restart the exchange with smiles, coos and reaching, then turned their gaze away, slumped and became distressed and disorganised. The **still-face paradigm** showed that even very young infants actively expect and seek affective reciprocity (emotional give-and-take); it is now used to study the effects of maternal depression.

EXAM TRAP

The still face shows infants as **active partners who expect reciprocity**, distressed and disorganised when it stops. They are not passive recipients of caregiver mood and do not habituate at once; the paradigm says nothing about fathers versus mothers, preferring strangers, or face recognition before 12 months.

From soothing to self-regulation

Caregivers first manage a baby's feelings by rocking and feeding. **Emotion regulation**, the processes by which we monitor and adjust the intensity and timing of our emotional reactions, develops as a handover to the child: by a few months babies avert their gaze from overstimulation or suck a thumb. **Claire Kopp** (1982) described three steps: *control* (about 12 to 18 months: complying with caregivers' demands), *self-control* (from about 2 years: complying without being watched) and *self-regulation* (from about 3 years: flexibly guiding one's own behaviour). Children also learn **display rules**, a culture's norms for when, where and how much emotion may be shown (Section 1.11): in many Indian homes a child learns early to hide anger before elders.

What temperament is

Some babies take a new face in their stride; others scream. **Temperament** names these differences. In **Mary Rothbart**'s definition, it is constitutionally based individual differences in *reactivity* (how quickly and strongly a child is aroused) and *self-regulation* (how she modulates that arousal). It appears early, within the first months. It is moderately stable: a fearful infant often becomes a cautious child, but many change. And it is generally considered biologically and genetically based: identical twins are more alike in temperament than fraternal twins. The environment does not create temperament but shapes its expression, the idea of goodness of fit (below). Temperament is the raw material of adult personality (Chapter 4).

> **EXAM TRAP**
>
> Asked what temperament is "generally considered to be", choose **genetically (biologically) based**. "Learned behaviour", "a result of parenting" and "environmentally influenced" are the distractors: the environment modulates temperament but is not its origin.

Thomas and Chess: the New York Longitudinal Study ●●

In 1956 the psychiatrists **Alexander Thomas** (1914 to 2003) and **Stella Chess** (1914 to 2007), husband and wife, launched the **New York Longitudinal Study** (NYLS), following children from infancy into adulthood. At a time when psychiatry blamed children's problems on parents, they asked whether babies arrive with behavioural styles of their own. They **interviewed parents**, asking not for impressions ("a good baby") but for concrete descriptions of feeding, bathing, a new food or a stranger's visit, and they **observed the babies** to check the reports. The nine dimensions in Table 6.17 were drawn from these records, not from clinical experience or earlier studies.

Table 6.17 Thomas and Chess's nine dimensions of temperament and the three types.

Dimension	What it describes	Easy	Difficult	Slow-to-warm-up
Activity level	Proportion of active to quiet periods	Not defining	Not defining	Low
Rhythmicity	Regularity of sleep, hunger, elimination	Regular	Irregular	Fairly regular
Approach-withdrawal	First response to a new food, toy or person	Approaches	Withdraws	Withdraws at first
Adaptability	Speed of adjusting to change	Quick	Slow	Slow but gradual
Threshold of responsiveness	Stimulus strength needed to evoke a response	Not defining	Not defining	Not defining
Intensity of reaction	Energy of a response, pleasant or not	Mild to moderate	Intense	Mild
Quality of mood	Pleasant versus unpleasant behaviour	Positive	Negative	Mildly negative
Distractibility	How easily outside stimuli interrupt an activity	Not defining	Not defining	Not defining
Attention span and persistence	Keeping at an activity despite obstacles	Not defining	Not defining	Not defining
Share of the sample	About 35 percent fitted no single type	About 40 percent	About 10 percent	About 15 percent

Easy, difficult and slow-to-warm-up children ●●●

The ratings clustered into three patterns, defined mainly by rhythmicity, approach-withdrawal, adaptability, intensity and mood.

- The **easy child** (about 40 percent) is regular in eating and sleeping, cheerful, approaches new things and adapts quickly.

- The **difficult child** (about 10 percent) has irregular biological rhythms, a generally negative mood and intense reactions, withdraws from new stimuli and is slow to adapt.
- The **slow-to-warm-up child** (about 15 percent) is low in activity, withdraws from new stimuli at first with mild rather than intense negativity, and adapts slowly but gradually with repeated exposure.

The remaining 35 percent or so showed mixed patterns: they are unclassified, not a fourth named type. Picture the first week at a Mysuru playschool. Ananya plays at once and naps on schedule: easy. Kabir screams, flings his snack and still protests a fortnight later: difficult. Meera clings quietly to her mother's dupatta, watches from the edge and joins in by the second week: slow-to-warm-up, because her withdrawal is mild and gives way.

The **difficult child syndrome** means that some infants are difficult even in the first few weeks after birth, before parenting could have produced it. It is a temperament, not a later behaviour problem (though such children run a higher risk of one), and has nothing to do with auditory discrimination (turning the head to one sound and not another) or a delay in voluntary control.

Why do some difficult children do well? Thomas and Chess's answer was **goodness of fit**: the match between a child's temperament and the demands and expectations of caregivers and the wider environment. A difficult child whose parents stay patient and consistent often thrives; the same child with impatient, punitive parents tends to develop problems.

> EXAM TRAP
>
> Thomas and Chess are known for **temperament**, not human sexuality, needs and drives or marital therapy. Their three types are **easy, difficult and slow-to-warm-up**. Distractors: sanguine, choleric, melancholic and phlegmatic (Hippocrates' humours, Galen's temperaments; Chapter 4); Type A, B and C (behaviour patterns linked to heart disease and cancer; Section 1.11); "inconsistent", "sensitive, resilient, withdrawn" and "inhibited only" (inhibition is Kagan's construct, below); and "insecurely attached", an attachment category (Section 6.7). Goodness of fit is never IQ differences between twins, the number of siblings or parent-child genetic concordance.

Rothbart and Bates, Kagan, Buss and Plomin

Later researchers condensed the overlapping nine dimensions. Mary Rothbart, who built the parent-report **Infant Behavior Questionnaire** (1981), and **John Bates** proposed a **structural model of temperament** (1998) with exactly three broad dimensions. **Surgency**, or extraversion, covers approach, activity, positive anticipation, smiling and sociability. **Negative affectivity** covers fear, frustration, sadness, discomfort and difficulty being soothed. **Effortful control** is the capacity to focus and shift attention voluntarily and to inhibit a dominant response; it is the self-regulatory side of temperament and matures in the preschool years.

Kagan studied reaction to the unfamiliar. Shown novel mobiles, voices and smells, about one 4-month-old in five thrashed and cried. These **high-reactive infants** tended to become *inhibited*, shy and wary children, while low-reactive infants tended to become *uninhibited*, bold and sociable. **Behavioural inhibition**, the tendency to withdraw from and be distressed by the unfamiliar, is Kagan's construct; he traced it to an easily excited amygdala. Its stability is moderate: many inhibited toddlers grow less shy. **Arnold Buss** and **Robert Plomin** (1984)

proposed the **EAS model** of three heritable temperaments: emotionality (how easily a child becomes distressed), activity (tempo and vigour) and sociability (preference for company). Table 6.18 matches each scheme to its authors.

Table 6.18 Who proposed which scheme of temperament.

Theorists	Units	Basis or method
Thomas and Chess (NYLS, 1956)	Nine dimensions; easy, difficult, slow-to-warm-up	Parent interviews, observation; goodness of fit
Rothbart and Bates	Surgency (extraversion), negative affectivity, effortful control	Infant Behavior Questionnaire; reactivity and self-regulation
Kagan	Inhibited versus uninhibited (behavioural inhibition)	Reactions to the unfamiliar; amygdala
Buss and Plomin (1984)	Emotionality, activity, sociability (EAS)	Heritable traits; twin studies
Hippocrates and Galen (ancient)	Sanguine, choleric, melancholic, phlegmatic	Bodily humours (Chapter 4)

EXAM TRAP

Asked which is **not** in Rothbart and Bates's structure, choose **inhibition to the unfamiliar**: it is Kagan's behavioural inhibition. Extraversion is on the list, as surgency.

CLINICAL LINK

Behavioural inhibition raises the later risk of social anxiety (Chapter 8), and difficult temperament with poor fit predicts behaviour problems: a vulnerability, not a sentence.

Effortful control and the marshmallow test

Effortful control is the core of **self-regulation**, the ability to manage attention, emotion and action in pursuit of a goal; its most famous test asks a child to wait.

KEY STUDY: THE MARSHMALLOW TEST (MISCHEL, FROM THE LATE 1960S)

Walter Mischel (1930 to 2018) tested preschoolers at the Bing Nursery School at Stanford University. A child could eat one treat now or have two if she waited, up to 15 minutes, for the experimenter to return; ringing a bell brought him back at once, but then she got only one. The **marshmallow test** measures **delay of gratification**: giving up a smaller immediate reward for a larger later one. Follow-ups found that longer waiters were rated more competent as adolescents and scored higher on the SAT, the American college admission test (**Yuichi Shoda**, Mischel and **Philip Peake**, 1990). A larger, more diverse replication by **Tyler Watts**, **Greg Duncan** and **Haonan Quan** (2018) found the link about half as strong, and weaker still once family background was controlled.

Waiting turned out to be a skill of attention. Children waited longer when the treats were covered or when they distracted themselves by singing. Dwelling on the treat's *hot* features (sweet, chewy) made waiting almost impossible; thinking of its *cool*, abstract features ("puffy, like a cloud") made it easier. Mischel later described a hot, emotional "go" system and a cool, cognitive "know" system that matures with age.

Keep delay of gratification apart from three neighbours. **Sustained attention** is keeping focus on one task over time. **Resistance to temptation** is obeying a prohibition when no one is watching, studied with the **forbidden-toy paradigm**: the child is told not to touch an attractive toy and left alone with it, with no bigger reward for complying. **Impulse control** in social settings is holding back an unacceptable act such as grabbing or interrupting. The marshmallow task draws on all three, but its defining feature is the choice between a smaller-sooner and a larger-later reward. Mischel's personality theory is in Chapter 4.

> EXAM TRAP
>
> The marshmallow test primarily measures **delay of gratification**, not sustained attention, resistance to temptation (the forbidden toy) or impulse control in social settings.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Bridges held that the newborn's general excitement first differentiates into _____ and _____.
2. Embarrassment, pride and guilt need _____, which the rouge test shows at about 18 to 24 months.
3. Separation protest peaks at about _____ months.
4. The still-face paradigm was devised by _____ and colleagues (1978).
5. A child who withdraws mildly from novelty but adapts gradually is _____.
6. The match between temperament and environmental demands is _____.
7. Rothbart and Bates's three dimensions are surgency, negative affectivity and _____.
8. Mischel's marshmallow test measures _____.

Answers: (1) distress; delight, (2) self-awareness, (3) 13 to 15, (4) Edward Tronick, (5) slow-to-warm-up, (6) goodness of fit, (7) effortful control, (8) delay of gratification.

6.7 Attachment: Imprinting, Bowlby, Ainsworth and Beyond

MEDIUM YIELD · 37 QS

Leave a one-year-old in a strange room and watch the door. She cries when her mother walks out, crawls to her the moment she returns and, within a minute, is back among the toys. That small drama is attachment, the first close relationship of life. This section asks why the bond forms, how it unfolds over the first three years, how its quality is measured, what its absence does and how it echoes into adulthood.

What attachment is: four theories

An **attachment** is a strong, enduring emotional bond to a specific person who cannot be swapped for anyone else. Four behaviours reveal it: the infant keeps close to the person (proximity seeking), explores from her as a **secure base**, runs back to her as a **safe haven** when frightened, and shows **separation distress** when she leaves. A baby who smiles at everyone is sociable; one who protests when one particular person leaves is attached.

Why does the bond form? Four answers compete (Table 6.19). For **Sigmund Freud** (1856 to 1939) it was feeding: the mother gratifies oral needs, so the infant loves her; **Erik Erikson** (1902 to 1994) added that responsive care builds basic trust (Section 6.10). The learning theorists **John Dollard** and **Neal Miller** (1950) and **Robert Sears** said the same in behavioural terms: the mother, paired again and again with hunger reduction, becomes a secondary reinforcer, so attachment is a learned **secondary drive**, nicknamed 'cupboard love'. The cognitive-developmental view adds a precondition: a baby must tell familiar people from strangers and know that people go on existing when out of sight, an achievement called **person permanence** (Section 6.4). The **ethological theory** of John Bowlby holds that attachment is an evolved behavioural system that kept helpless infants near a protector and so alive. Geese and monkeys, as the next two subsections show, undermined the feeding theories, and the ethological theory is the predominantly accepted account today.

Table 6.19 Four theories of why infants attach.

Theory	Key figures	Why the infant attaches	Status
Psychoanalytic	Freud; Erikson	Feeding gratifies oral needs; care builds trust	Feeding overrated
Learning (secondary drive)	Dollard and Miller; Sears	Mother becomes a secondary reinforcer ('cupboard love')	Refuted by Harlow
Cognitive-developmental	Builds on Piaget	Needs person permanence	Explains timing only
Ethological	Bowlby, after Lorenz	Evolved system for survival	Predominantly accepted

EXAM TRAP

The **ethological** theory is today's predominantly accepted account, not the psychoanalytic, learning or cognitive-developmental one. 'Cupboard love' is the learning view.

Lorenz and imprinting ●●

KEY STUDY: LORENZ'S GOSLINGS (1935)

Konrad Lorenz (1903 to 1989), an Austrian zoologist, split a clutch of greylag goose eggs. Half hatched under their mother; half hatched in an incubator, where the first moving thing the goslings saw was Lorenz. The incubator goslings followed him everywhere, and when the two broods were mixed and released, each gosling ran to its own 'mother', goose or man.

Lorenz called this **imprinting**: a rapid, largely irreversible following response to the first moving object, formed only within a narrow **critical period** after hatching (Section 6.1). It needs no practice and no food reward. It also lasts: birds imprinted on humans later courted humans, a long-term effect called sexual imprinting. **Eckhard Hess** (1916 to 1986) timed the window in mallard ducklings: imprinting peaked at about 13 to 16 hours after hatching and then fell away sharply. Later work found the window more flexible than Lorenz thought, so many now say sensitive period. Lorenz shared the 1973 Nobel Prize in Physiology or Medicine with **Nikolaas Tinbergen** and **Karl von Frisch**. Human babies do not imprint, but Lorenz gave Bowlby two ideas: early bonds are biologically prepared, and they have a critical period.

EXAM TRAP

Imprinting is **rapid, largely irreversible and limited to a critical period**: not slow, reversible lifelong learning, operant conditioning with food or a purely visual system unrelated to attachment. Nor is it habituation (Section 6.3), newborn reflexive smiling (Section 6.6) or same-sex playmate preference at 5 (Section 6.8). **Lorenz**, not Ainsworth, Bowlby or Erikson, argued that newborn creatures imprint during a critical period.

Harlow's surrogate mothers ●●

If feeding builds the bond, an infant should love whoever feeds it. **Harry Harlow** (1905 to 1981) tested this at the University of Wisconsin with his wife and colleague **Margaret Harlow**.

KEY STUDY: 'THE NATURE OF LOVE' (HARLOW, 1958)

Infant rhesus monkeys, taken from their mothers at birth, were raised with two surrogate mothers: a bare wire-mesh figure and one covered in terry cloth. For some infants only the wire mother held a feeding bottle. Whichever fed them, they spent most of the day clinging to the cloth mother and visited the wire one only to feed. Frightened by a mechanical toy, they fled to the cloth mother, calmed, then turned to explore; in a strange room they explored only when she was present. Conclusion: **contact comfort**, not feeding, is the primary basis of attachment, and the cloth mother served as safe haven and secure base. Criticism: the work is ethically troubling, and monkeys are not children.

Monkeys reared in total isolation for six months to a year grew up rocking, clasping themselves and unable to play or mate normally; isolate females who later had infants became 'motherless mothers', neglecting or abusing them. Even a cloth mother could not produce normal social development. Yet **Stephen Suomi** and Harlow (1972) found that housing isolates with younger 'therapist' monkeys, which clung and played without threat, restored much social behaviour. Early deprivation is serious but partly reversible, a theme the human studies below repeat.

EXAM TRAP

Harlow showed **contact comfort, not feeding**, as the basis of attachment: not conditioned reinforcement, operant shaping of clinging, classical conditioning of hunger or innate fear of strangers. He did not show imprinting in primates, critical periods for language or genetic determinism, and his isolates show that maternal absence harms even when nutrition is met.

Bowlby's attachment theory ●●

John Bowlby (1907 to 1990), a British psychiatrist and psychoanalyst, wove these threads into attachment theory. In *Forty-Four Juvenile Thieves* (1944) he found that 14 of the young thieves at his London clinic showed **affectionless psychopathy** (no guilt, no affection for others), and 12 of those 14 had suffered prolonged early separation from their mothers. His 1951 World Health Organization report, *Maternal Care and Mental Health*, set out the **maternal deprivation hypothesis**: an infant needs a warm, continuous relationship with the mother or a per-

manent substitute, and its prolonged disruption in the early years does lasting harm. The trilogy *Attachment* (1969), *Separation* (1973) and *Loss* (1980) completed the theory.

Its claims are these. Attachment is an evolved behavioural system: crying, smiling, clinging and following bring a caregiver close; the system switches on under threat or separation, and when the infant feels safe it explores from its secure base. **Monotropy** means that one bond, usually with the mother, is special and qualitatively different from the rest. The bond must form within a critical period, roughly the first two and a half years. And the infant builds an **internal working model**: a mental representation of the self (am I lovable?), of attachment figures (will they come?) and of relationships, built from early experience, that serves as a prototype, a schema, for later relationships yet can be revised by new experience.

What does a broken bond look like? Bowlby's colleague **James Robertson** filmed *A Two-Year-Old Goes to Hospital* (1952) and later, with **Joyce Robertson**, other brief separations. Young children passed through a **protest-despair-detachment** sequence: protest (crying, searching), despair (listless misery) and detachment (seeming recovery, but coolness to the mother on her return). Bowlby applied a similar sequence to grief (Section 6.11).

> EXAM TRAP
>
> Attachment theory is **Bowlby's**, not Freud's, Rogers's or Bandura's. The **internal working model** is a revisable representation of self, others and relationships. It is not the proximity-seeking mechanism or the attachment system itself (which the model guides), a 'bio-social relationship model' (not Bowlby's term), Tolman's cognitive map (Chapter 1), a hippocampus-amygdala circuit, a genetic fear template, a reflex repertoire, a language substrate or object permanence.

Phases of attachment: Bowlby, Schaffer and Emerson, Mahler ●●

Attachment assembles itself over the first two years. Bowlby described four phases (Figure 6.13):

- **Pre-attachment** (birth to 6 weeks): signals bring adults close; the baby knows her mother's smell and voice but does not mind strangers.
- **Attachment-in-the-making** (6 weeks to 6 to 8 months): she prefers familiar people but does not yet protest at separation.
- **Clear-cut attachment** (6 to 8 months to 18 months to 2 years): clear preference for the primary caregiver, separation protest, stranger anxiety (Section 6.6) and secure-base behaviour.
- **Formation of a reciprocal relationship**, the **goal-corrected partnership** (from 18 months to 2 years onwards): language lets the toddler understand and predict the parent's comings and goings, so protest gives way to negotiation.

Rudolph Schaffer and **Peggy Emerson** (1964) followed 60 Glasgow infants at home, monthly through the first year and again at 18 months, recording protest at everyday separations and fear of strangers. They found four stages: asocial (0 to 6 weeks), indiscriminate (6 weeks to about 7 months, accepting comfort from anyone), **specific attachment** (from about 7 months, with separation protest and stranger fear) and **multiple attachments** (soon after; most infants had several by 18 months). Infants attached most strongly to whoever responded most sensitively to them, not necessarily the person who fed them.

Margaret Mahler (1897 to 1985), a psychoanalyst, traced how the child separates psychologically within the bond (*The Psychological Birth of the Human Infant*, 1975, with **Fred Pine** and **Anni Bergman**). A normal autistic phase (the first weeks) and a symbiotic phase (to about 5 months, mother and infant felt as one) give way to **separation-individuation** in four subphases: differentiation or hatching (about 5 to 10 months: turning outward, first stranger reactions); practising (about 10 to 16 months: delighted exploration on new legs); **rapprochement** (about 15 to 24 months: aware now of being separate, the toddler alternates independent exploration with returns to the caregiver for **emotional refuelling**, and the clash of autonomy and closeness brings the clinging and tantrums of the rapprochement crisis); and consolidation (about 24 to 36 months), when **object constancy**, a stable inner image of the mother, lets the child bear her absence. Her developmental theory of schizophrenia and childhood psychosis belongs to Chapter 8.

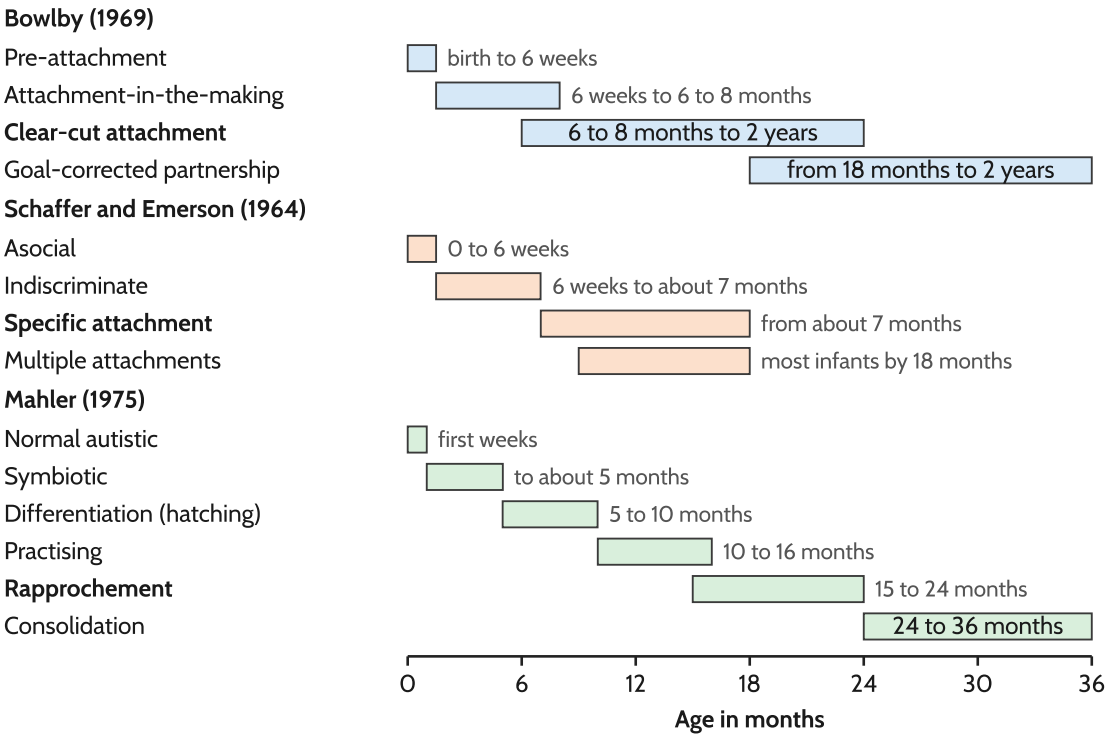

Figure 6.13 Three stage schemes of the first three years on one time line. Bars show usual age ranges; boundaries are approximate, and Schaffer and Emerson's bars stop where their study ended, at 18 months. Bold labels mark the key phases.

EXAM TRAP

Clear preference, separation protest and stranger anxiety mark Bowlby's **clear-cut** phase (6 to 8 months to 2 years; '6 months to 2 years' in keys), not pre-attachment, attachment-in-the-making or the goal-corrected partnership. Schaffer and Emerson's specific attachment starts at about **7 months**. **Rapprochement** (15 to 24 months) is exploration alternating with refuelling: not symbiotic merger, the first stranger reactions (differentiation) or full object constancy.

Ainsworth and the Strange Situation ●●

Bowlby's theory said every infant attaches; it did not say how well. That measure came from **Mary Ainsworth** (1913 to 1999), who worked with Bowlby in London and then watched mothers and babies at home in Uganda and Baltimore. To capture differences in security in one short session she devised the **Strange Situation** (Ainsworth and **Barbara Wittig**, 1969; reported in full in *Patterns of Attachment*, 1978, with **Mary Blehar**, **Everett Waters** and **Sally Wall**). Over eight episodes of about three minutes each, a 12- to 18-month-old explores a playroom with the caregiver, meets a stranger, is left with the stranger, is reunited, is left alone, is joined by the stranger and is reunited again. Observers score exploration, separation distress, reaction to the stranger and, above all, **reunion** behaviour, which best reveals what the infant expects of the caregiver.

Ainsworth's **maternal sensitivity hypothesis** explains the differences: security grows from noticing the baby's signals, reading them correctly and answering them promptly and appropriately. In Baltimore, mothers who answered crying promptly in the early months had securely attached one-year-olds; later studies confirm a real but more modest link.

> EXAM TRAP
>
> **Ainsworth** (not Bowlby, Harlow, Lorenz, Piaget or Erikson) devised the Strange Situation, and she is known for attachment, not cognitive, moral or language development. Bowlby is to attachment as Ainsworth is to the Strange Situation; the Bobo doll is Albert Bandura's, Little Albert is John Watson and Rosalie Rayner's, and the visual cliff is Eleanor Gibson and Richard Walk's (Section 6.3).

Four patterns of attachment, and culture ●●●

Ainsworth's infants fell into three groups, A, B and C; **Mary Main** and **Judith Solomon** (1986, 1990) added a fourth, D (Table 6.20). **Secure attachment** (B) is the commonest, about 60 to 70 percent of samples. **Insecure-avoidant attachment** (A, about 20 percent) follows rejecting care: an infant whose bids for comfort are rebuffed learns to suppress them. **Insecure-resistant attachment** (C), or ambivalent (about 10 to 15 percent), follows inconsistent care: never sure that comfort will come, the infant clings and protests. **Disorganised attachment** (D), or disoriented, abounds among maltreated infants and those with a frightened or frightening caregiver, who is both the source of fear and its only refuge.

Table 6.20 The four attachment patterns. Read the reunion column first.

Pattern	Separation	Reunion	Caregiving
Secure (B)	Explores freely; upset, often mildly	Actively seeks proximity, contact or interaction; readily comforted, returns to play	Sensitive
Insecure-avoidant (A)	Little distress; treats stranger like caregiver	Conspicuous avoidance: ignores or turns away	Rejecting
Insecure-resistant or ambivalent (C)	Explores little; very distressed	Seeks contact yet resists it, clinging then angry; cannot be soothed	Inconsistent
Disorganised or disoriented (D)	No consistent strategy	No one clear pattern; inconsistent, often bizarre: contradictory, freezing, dazed or apprehensive	Maltreating; frightened or frightening

EXAM TRAP

Read the reunion. Distressed, seeks interaction but rejects it when offered = **insecure-resistant (C)**. Little distress, ignores the caregiver = **avoidant (A)**. Contradictory, freezing or apprehensive, often after maltreatment = **disorganised (D)**, Main and Solomon's addition to Ainsworth's original A, B and C. 'Secure-resistant' and 'resistance without distress' are not types.

Is the pattern universal? **Marinus van IJzendoorn** and **Pieter Kroonenberg** (1988) pooled 32 studies from eight countries: secure was the commonest type everywhere, and variation within cultures was about one and a half times that between them. Northern German samples (**Klaus Grossmann** and **Karin Grossmann**, Bielefeld) held many avoidant infants because parents valued early independence, which lowers visible distress and reunion seeking. Japanese infants, rarely apart from their mothers, and Israeli kibbutz infants, who seldom met strangers, showed more resistant attachment.

EXAM TRAP

Northern Germany's excess of type A is **cultural** (an emphasis on early independence), not genetic, a modified protocol or a leap into adult attachment.

Deprivation and privation

René Spitz (1887 to 1974) compared babies in a foundling home, fed and clean but rarely held, with babies cared for by their own mothers in a prison nursery. The foundling babies fell behind, sickened and many died, a syndrome he called **hospitalism** (1945). He also described **anaclitic depression** (1946): infants separated from their mothers in the second half of the first year turned weepy, then withdrawn, lost weight and stopped developing, recovering if the mother returned within a few months.

Michael Rutter (1933 to 2021), in *Maternal Deprivation Reassessed* (1972), distinguished **deprivation** (losing a bond already formed) from **privation** (never forming one), the more harmful of the two. **Barbara Tizard** and **Jill Hodges** followed children raised in London residential nurseries by many changing carers: those later adopted formed close bonds with their new parents, closer than children returned to birth families, yet at 16 still had more difficulty with peers. Rutter's English and Romanian Adoptees study found that Romanian orphans adopted into British homes before about 6 months caught up almost fully, while later adoptees more often kept cognitive deficits and disinhibited attachment. Privation harms, but early rescue repairs much. **Donald Winnicott** (1896 to 1971) added the **good-enough mother**, reliably responsive rather than perfect, and the **transitional object**, a blanket or soft toy that comforts the child on the way to independence.

CLINICAL LINK

Severe early neglect can produce **reactive attachment disorder** (a withdrawn child who rarely seeks comfort) or **disinhibited social engagement disorder** (indiscriminate, overfamiliar approach to strangers); both are taught in Chapter 8.

EXAM TRAP

Hospitalism and anaclitic depression are **Spitz's**, not Piaget's, Skinner's or Eysenck's. Deprivation is losing a bond; privation is never having one.

Attachment beyond infancy

Mary Main's **Adult Attachment Interview** classifies adults by how coherently they discuss their childhood: **secure-autonomous** (balanced), **dismissing** (idealises parents, recalls little), **preoccupied** (still angrily entangled) and **unresolved** (incoherent about loss or abuse). These parallel infant types B, A, C and D, and a parent's classification predicts her infant's well above chance. **Cindy Hazan** and **Phillip Shaver** (1987) found secure, avoidant and anxious-ambivalent styles in adult romantic love, in proportions resembling the infant ones. **Kim Bartholomew** and **Leonard Horowitz** (1991) crossed positive or negative models of self and of others into four adult styles: secure, preoccupied, dismissing and fearful.

Monotropy, finally, has softened. Fathers are attachment figures in their own right, and most infants have several attachments by 18 months, as in any Indian joint family; responsiveness, not feeding, decides to whom. On day care, **Jay Belsky** (1986) warned that over 20 hours a week in the first year raises the risk of insecurity, but the large Study of Early Child Care run by the National Institute of Child Health and Human Development (NICHD) found that day care alone does not; risk rises only when poor or very long care meets insensitive mothering.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Lorenz's goslings followed the first moving object they saw during a critical period, a process called _____.
2. Harlow's monkeys clung to the cloth mother even when only the wire mother gave milk, showing the importance of _____.
3. Bowlby's mental representation of self, attachment figures and relationships, a prototype for later bonds, is the _____.
4. Clear preference, separation protest and stranger anxiety mark Bowlby's _____ phase.
5. The Strange Situation was devised by _____.
6. An infant very distressed at separation who on reunion seeks contact yet angrily resists it is classed as _____.
7. The disorganised pattern (D) was added by _____ and Judith Solomon.
8. In Mahler's _____ subphase the toddler alternates exploration with returns for emotional refuelling.

Answers: (1) imprinting, (2) contact comfort, (3) internal working model, (4) clear-cut attachment, (5) Mary Ainsworth, (6) insecure-resistant or ambivalent (C), (7) Mary Main, (8) rapprochement.

6.8 Parenting, Family, Play and Gender in Childhood LOWER YIELD · 12 QS

Watch a four-year-old for an afternoon. A parent sets her limits, siblings and grandparents crowd round, she plays beside and then with other children, and in almost everything she is

learning what it means to be a girl. This section follows those threads; attachment, which comes first, is Section 6.7.

Baumrind's parenting styles ●●

Why do some children grow up confident and self-controlled while others become anxious or unruly? In the 1960s **Diana Baumrind** (1927 to 2018) observed preschoolers at nursery school and at home and interviewed their parents. In papers from 1966 to 1971 she described three patterns of **parenting style**, the overall emotional climate in which a parent raises a child.

Authoritative parenting combines warmth and emotional support with clear boundaries: rules are explained, discipline is reasoned, children are held accountable for their behaviour and their growing autonomy is encouraged. **Authoritarian parenting** makes high demands with little warmth: strict obedience, rules without reasons ("because I said so") and often harsh punishment. **Permissive parenting**, also called indulgent, is warm and accepting but makes few demands and sets few rules.

In 1983 **Eleanor Maccoby** (1917 to 2018) and **John Martin** recast the styles as combinations of two independent dimensions: **demandingness** (control: expecting mature behaviour, supervising, disciplining) and **responsiveness** (warmth: acceptance and attention to the child's needs). The cell low on both became a fourth style, **neglectful parenting** (uninvolved): detached parents who do little beyond meeting basic needs. Figure 6.14 shows the four styles. Children of authoritative parents are the most self-reliant, competent and achievement-oriented; those of authoritarian parents tend to be anxious, withdrawn and unhappy, and those of permissive parents impulsive, with poor self-control; children of neglectful parents fare worst, with low self-esteem, poor schoolwork and later delinquency.

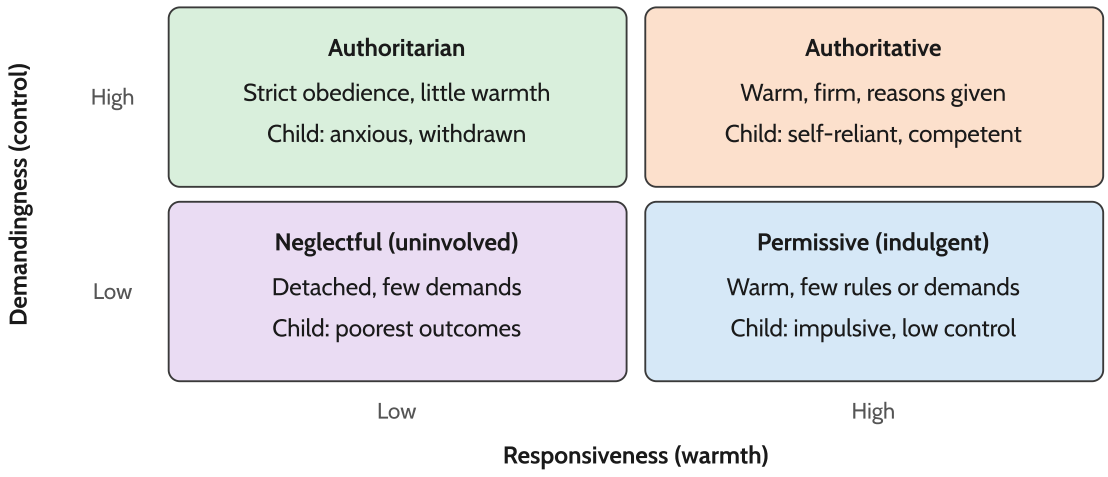

Figure 6.14 Maccoby and Martin's two dimensions give four parenting styles; only authoritative parenting is high on both.

Culture shifts the picture. **Ruth Chao** (1994) showed that Chinese-American parents score as authoritarian on Western scales, yet their children do well at school: their strictness expresses the Chinese idea of *training* (*chiao shun*), devoted involvement that children read as care. Many Indian families show a similar *warm control*, firm expectations of obedience inside close affection. Control harms mainly when it comes without warmth.

EXAM TRAP

Authoritative is the best style; **authoritarian** is strict and cold. "Emotional support, clear boundaries and accountability", "high demandingness and high responsiveness" and "reasoned discipline with autonomy support" all mean authoritative. Permissive is low demandingness with high responsiveness; neglectful is low on both and was added by Maccoby and Martin, not Baumrind.

New labels, discipline and the family

Helicopter parenting is over-involved parenting: parents hover, monitor and step in to solve problems the child could handle alone. The image comes from **Haim Ginott** (1922 to 1973), whose 1969 book *Between Parent and Teenager* quotes a teenager whose mother hovers over him like a helicopter. It has several roots at once: parental anxiety, parents' unfulfilled wishes lived out through the child, and parents' own personality traits such as perfectionism. **Tiger parenting** (after **Amy Chua**'s 2011 memoir) is strict and achievement-driven; **free-range parenting** grants age-appropriate independence with little supervision. **Martin Hoffman**'s (1967) three kinds of discipline, **induction** (explaining how the behaviour affects others), **power assertion** (force, threats) and **love withdrawal** (withholding affection), are compared in Section 6.9.

Parents are one part of a family system in which every member affects every other. Siblings are companions, teachers and rivals; the claim that birth order shapes personality belongs to Alfred Adler (Chapter 4). Only children, despite the "spoilt" stereotype, match others in achievement and adjustment. **E. Mavis Hetherington**, following families for decades (2002), found that most children adjust to divorce within two to three years, though about 25 percent show serious problems against about 10 percent from intact families; remarriage brings a second adjustment. Maternal employment does not in itself harm children, and day care matters through its quality.

INDIAN CONTEXT

In a **joint family** several generations share a home and its childcare, so the child has many attachment figures, more adults to obey and a strong sense of interdependence. Urbanisation has made the **nuclear family** (parents and children only) the commonest household, with day care or grandparents filling the gap.

Child maltreatment and neglect

In 1962 the paediatrician **C. Henry Kempe** (1922 to 1984) and colleagues described the **battered child syndrome**: young children with fractures at different stages of healing that their parents' accounts did not explain. **Child maltreatment** has four types: **physical abuse** (non-accidental injury), **sexual abuse** (any sexual activity with a child), **emotional abuse** (persistent rejecting, belittling or terrorising) and **neglect**, failure to meet a child's basic needs and the most often reported form. Neglect has five categories: **physical** (failure to provide proper nutrition, housing and clothing), **supervisory** (children left unattended), **health** or medical (failure to immunise, seek proper care or follow medical directions), **educational** (not enrolling the child, chronic absence) and **emotional** (withholding affection and attention).

EXAM TRAP

Using **licensed day care** is not neglect: it is approved, supervised care, the odd one out beside nutrition, supervision and immunisation.

Risk factors include parents who were themselves maltreated (about a third repeat it), young, depressed or misusing substances; premature, disabled or difficult children; and poverty, domestic violence and isolation. Effects include disorganised attachment (Section 6.7), aggression, school failure, depression and post-traumatic stress. Management and India's Protection of Children from Sexual Offences (POCSO) Act, 2012 are in Chapter 7.

The development of play ●●

Play is activity chosen freely and done for its own sake. **Jean Piaget** (1896 to 1980) saw it as pure assimilation, the child bending reality to fit existing schemes, so play follows the cognitive stages (Section 6.4). The first form, in infancy during the sensorimotor stage, is **functional play** (practice or sensorimotor play): simple actions repeated for the pleasure of mastering them. With the preoperational symbolic function comes **symbolic play** (pretend play), and in the concrete operational stage **games with rules**. In 1968 the Israeli psychologist **Sara Smilansky** added **constructive play**, making or building something. Table 6.21 gives the standard definitions.

Table 6.21 Types of play, in the order they appear.

Type	Definition	When	Example
Sensorimotor or practice (functional)	Deriving pleasure from exercising sensorimotor schemes	First; infancy	Shaking a rattle
Pretense or symbolic	Transforming the physical environment into a symbol	Preoperational, about 2 to 7	A banana becomes a phone
Constructive	Combining sensorimotor or practice play with symbolic representation of ideas	Preschool onwards	Building a palace from blocks
Social	Interaction with peers	Grows through preschool	Chasing, turn-taking
Games with rules	Play governed by rules agreed in advance	Concrete operational, about 7 to 11	Kho-kho, ludo, gully cricket

Sociodramatic play is shared pretend play in roles (house, shop, doctor), peaking around four to five. **Rough-and-tumble play** is friendly wrestling and chasing with a relaxed "play face"; it peaks in middle childhood and, unlike real aggression, ends with the children still together. For **Lev Vygotsky** (1896 to 1934), every imaginary situation has rules (a child playing "mother" must act as mothers do), so play creates a zone of proximal development in which the child acts "a head taller than himself" (Section 6.5).

EXAM TRAP

The first play of infancy is **functional (practice)** play, not symbolic play, which needs the preoperational symbolic function. "Transforming the physical environment into a symbol" is symbolic play; "combining sensorimotor or practice play with symbolic representation of ideas" is constructive play; "interaction with peers" is social play.

Social play and theories of play

Play also becomes more social. In 1932 **Mildred Parten** observed nursery children aged two to five and ranked their play by social involvement (Table 6.22), from solitary and **parallel play** (beside others, not with them) through **associative play** (together, without a shared goal) to **cooperative play** (organised, with roles and a common aim). Earlier forms persist: solitary and parallel play stay common in older preschoolers.

Table 6.22 Parten's stages of social play (1932).

Stage	What the child does	Rough age
Unoccupied	Not playing; watches whatever catches the eye	Infancy
Solitary	Plays alone, absorbed in own toys	About 2 to 3
Onlooker	Watches others play, may talk, does not join	About 2 to 3
Parallel	Plays beside others with similar toys	About 2 to 3
Associative	Plays with others, sharing, no common goal	About 3 to 4
Cooperative	Organised play, shared goal and roles	About 4 to 5

Why do children play at all? Table 6.23 compares the classical and modern answers; play therapy is in Chapter 7.

Table 6.23 Theories of play.

Theory	Proposed by	Children play to
Surplus energy	**Friedrich Schiller; Herbert Spencer**	Burn off energy left over once survival needs are met
Relaxation or recreation	**Moritz Lazarus**	Restore energy spent in work
Pre-exercise or practice	**Karl Groos**	Rehearse adult skills, as a kitten stalking string practises hunting
Recapitulation	**G. Stanley Hall**	Replay stages of human evolution, from climbing to team games
Psychoanalytic	Sigmund Freud	Fulfil wishes and master anxious experiences by replaying them
Cognitive	Jean Piaget; Lev Vygotsky	Consolidate schemes; create a zone of proximal development

Peers: status, friendship and bullying

Sociometric status is measured by asking classmates whom they like most and least. In 1982 **John Coie**, **Kenneth Dodge** and a colleague identified five groups: **popular** (many "like most"), **rejected** (many "like least"), **neglected** (few nominations of either kind), **controversial** (many of both) and **average**. Rejected children are most at risk of loneliness and later problems; neglected children usually do well once they find a friend. Friendship deepens from shared play to mutual trust and, in adolescence, intimacy. **Bullying**, as defined by the Norwegian psychologist **Dan Olweus** (1931 to 2020), is repeated, intentional aggression against someone less powerful; his whole-school programme roughly halved it in Norway.

Gender concepts and gender socialisation

Gender research needs exact terms. **Sex** is biological: chromosomes, hormones and anatomy. **Gender** is the psychological and social meaning a culture attaches to being male or female. **Gender identity** is one's own sense of being male, female or another gender; a **gender role** is the behaviour a culture expects of each sex. A **gender stereotype** is a rigid, oversimplified belief about what males and females are like ("boys don't cry"); gender stereotyping applies it to individuals. **Gender typing** is the process by which children adopt the preferences and behaviours their culture considers right for their sex.

Gender socialisation is the process by which children learn this "gender map" from the people and institutions around them. Parents come first: pink or blue, dolls or trucks, indoor or outdoor chores, much of it taught unconsciously because their own gender orientations are so deeply embedded. Peers, school and media reinforce the map. *Gender construction*, by contrast, is the sociological view that a society creates gender categories and their meanings, and *gender structuralisation* is not a standard developmental term. From about three, children prefer same-sex playmates; this **gender segregation** peaks in middle childhood, so that, as Maccoby put it, boys (large groups, rough play) and girls (small groups, talk) grow up in two cultures.

> EXAM TRAP
>
> Parents passing on the "gender map" (pink and blue, toys, chores), consciously or unconsciously, is **gender socialisation**. Gender stereotyping is a belief applied to people, not a learning process; gender construction and gender structuralisation are distractors.

Theories of gender development

The **biological** view points to prenatal hormones: girls with **congenital adrenal hyperplasia**, exposed to high androgen levels before birth, prefer boys' toys and rough play more than other girls, although raised as girls. In the **psychoanalytic** view the child of three to six identifies with the same-sex parent (Chapter 4). **Social learning theory**, applied to gender by **Walter Mischel** (1930 to 2018) in 1966, holds that children are reinforced for gender-appropriate behaviour and imitate same-sex models. **Kay Bussey** and **Albert Bandura** (1925 to 2021) extended this in 1999 into a social cognitive theory: reinforcement and modelling build internal gender standards by which children regulate themselves. Either way, shaping from outside comes first.

Lawrence Kohlberg (1927 to 1987) reversed the order in his cognitive-developmental theory of 1966: "I am a boy, therefore I want to do boy things." Understanding comes first, and children then socialise themselves. It grows in three steps (Figure 6.15): **basic gender identity** (about two to three: labelling oneself and others), **gender stability** (about three to five: a boy will become a man) and **gender constancy** or consistency (about five to seven: a boy in a frock is still a boy). These are his only stages; "gender-role achievement" is not one. **Ronald Slaby** and **Karin Frey** (1975) supported him: watching a split-screen film of a man and a woman doing the same things, children high in constancy looked longer at the same-sex adult. Yet children prefer gender-typed toys by about two, long before constancy.

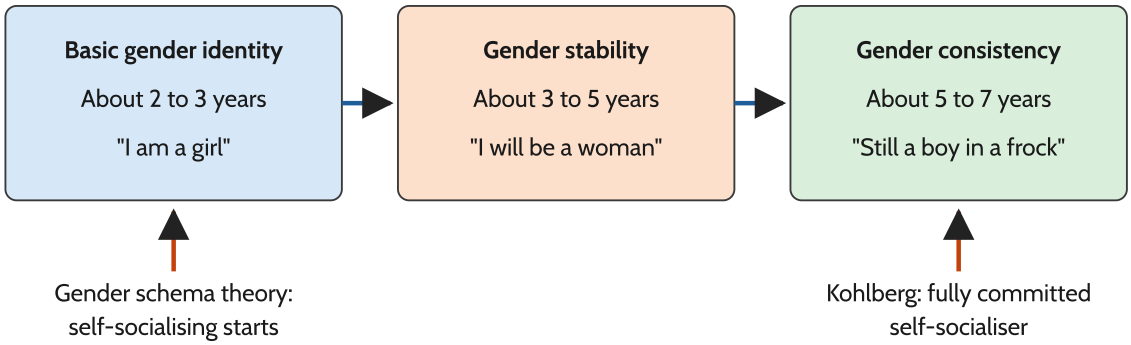

Figure 6.15 Kohlberg's three steps in understanding gender. Gender schema theory starts self-socialisation at the first step; on the stricter reading of Kohlberg, it is complete only at the third.

Gender schema theory, proposed in 1981 by **Sandra Bem** (1944 to 2014) and independently by **Carol Martin** and **Charles Halverson**, explains this. A gender schema is an organised set of beliefs about what goes with each sex. As soon as children have basic gender identity, they sort toys, activities and people into "for boys" and "for girls" and attend to and remember what fits their own group: shown a girl sawing wood, many later recall a boy doing it (Martin and Halverson, 1983). Earlier, Bem (1974) had treated masculinity and femininity as separate dimensions, so a person can be high on both: **androgyny**, measured by the Bem Sex Role Inventory (Chapter 2).

> EXAM TRAP
>
> IHBAS 2020 keyed **basic gender identity** as the starting point of self-socialisation in Kohlberg's theory: it is the first link in his chain. Many textbooks are stricter: for Kohlberg, children become fully committed self-socialisers only at gender consistency (constancy), and it is gender schema theory that starts self-socialisation at basic gender identity. Answer basic gender identity for that stem, but know both.

How large are the resulting differences? In 1974 Eleanor Maccoby and **Carol Jacklin** found only four well established: girls' greater verbal ability and boys' greater visual-spatial ability, mathematical ability and aggression. **Janet Shibley Hyde**'s **gender similarities hypothesis** (2005) went further: across 46 meta-analyses, 78 percent of differences were small or close to zero.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Warmth combined with clear, reasoned control is _____ parenting.
2. Maccoby and Martin's two dimensions are demandingness and _____.
3. In 1962 C. Henry Kempe described the _____ syndrome.
4. For Piaget, the first play of infancy is _____ play.
5. Play combining practice play with symbolic representation of ideas is _____ play.
6. Playing beside others with similar toys but not with them is Parten's _____ play.
7. Kohlberg's steps are basic gender identity, gender stability and gender _____.
8. Being high on both masculine and feminine traits is Bem's _____.

Answers: (1) authoritative, (2) responsiveness, (3) battered child, (4) functional (practice), (5) constructive, (6) parallel, (7) consistency (constancy), (8) androgyny

6.9 Moral Development: Piaget, Kohlberg, Gilligan and Beyond

LOWER YIELD · 20 QS

Read two stories to a six-year-old. John, called to dinner, opens a door and accidentally smashes fifteen cups standing on a tray behind it. Henry, reaching for jam while his mother is out, breaks one cup. Who is naughtier? The six-year-old usually says John, since fifteen cups are worse than one; a ten-year-old says Henry, who was doing wrong. What has changed is not knowledge about cups but reasoning about right and wrong. This section follows that change through Piaget, Kohlberg and Gilligan, then meets the researchers who widened the picture.

Three sides of morality

Moral development is the growth of the standards by which a person judges right and wrong, and of the feelings and actions that go with them. Morality has three sides, and each tradition studied one. The **emotional** side is guilt, shame and empathy (their emergence in toddlers is traced in Section 6.6). **Sigmund Freud** (1856 to 1939) placed it in the **superego**, the conscience formed at about five or six when the child resolves the Oedipus complex by identifying with the same-sex parent; it punishes misdeeds with guilt (Chapter 4). The **behavioural** side is what people do when tempted. Social learning theorists led by **Albert Bandura** (1925 to 2021) held that moral conduct is learned through reinforcement, punishment and above all **modelling**; in 1963 Bandura and **Frederick McDonald** even shifted children's judgments between consequences and intentions simply by letting them hear adults judge the other way. The **cognitive** side, **moral reasoning**, is how a person thinks a moral problem through; it belongs to Piaget and Kohlberg. The study below shows why reasoning and conduct need separate study.

KEY STUDY: THE CHARACTER EDUCATION INQUIRY (1928 TO 1930)

Hugh Hartshorne and **Mark May** asked whether children have a general trait of honesty. They gave more than 10,000 schoolchildren rigged chances to lie, cheat and steal at school, at home, at parties and in games. Most children cheated in some situations and not in others, and cheating in one setting only weakly predicted cheating in another. Honesty, they concluded, is largely **situation-specific**, not a fixed trait, and moral knowledge barely predicted honest conduct.

Piaget: from moral realism to autonomy

Jean Piaget (1896 to 1980) turned from logic (Section 6.4) to morality in *The Moral Judgment of the Child* (1932). He asked Swiss boys playing marbles where the rules came from and whether they could be changed, and he read children **paired stories** like John and Henry's, in which one child does more damage by accident and another less damage while misbehaving. He described a **premoral period**, before about age 5, when children pay little attention to rules, followed by two stages (Table 6.24).

In **heteronomous morality** (Greek for "ruled by another"), also called **moral realism**, from about 5 to 10 (some books say 9), rules come from adults and are sacred and unchangeable: a new marble rule is cheating even if every player agrees. Acts are judged by their consequences, not intentions, which Piaget called **objective responsibility**, so John is naughtier. The child believes in **immanent justice**, that the world itself punishes wrongdoing (a boy who stole apples falls through a rotten bridge "because he stole"), and favours **expiatory punishment**, severe and unrelated to the offence.

From about 10, cooperation among equal peers brings **autonomous morality**, or **moral relativism**. Rules become agreements that players may change by consent; intentions outweigh consequences, so Henry is naughtier; fairness means equality; and **reciprocal punishment** should fit the misdeed and repair it (the child who broke a window pays for it). Piaget underestimated preschoolers, who do use intentions when a story makes them clear, but his idea that moral reasoning grows with cognition and peer experience became Kohlberg's starting point.

Table 6.24 Piaget's two stages of moral reasoning.

Feature	Heteronomous (moral realism), about 5 to 10	Autonomous (moral relativism), from about 10
Rules	Fixed, sacred, made by authority	Social agreements, changeable
Acts judged by	Consequences (objective responsibility)	Intentions (subjective responsibility)
Justice	Immanent justice	Equality among peers
Punishment	Expiatory: severe, arbitrary	Reciprocal: fits the offence
Respect	One-sided, for adults (constraint)	Mutual, among peers (cooperation)

EXAM TRAP

Heteronomous children judge by **rules seen as unchangeable and the consequences of acts**. "Intentions and equality" describe Piaget's **autonomous** stage; "social contracts and individual rights" is Kohlberg's stage 5; "abstract principles of justice" is Kohlberg's stage 6. Piaget did not use the Heinz dilemma.

Kohlberg's method: the Heinz dilemma •

Lawrence Kohlberg (1927 to 1987) carried Piaget's approach into adolescence. For his 1958 doctoral study he interviewed 72 Chicago boys aged 10, 13 and 16, and re-interviewed many of them every three or four years for about 20 years. His method was to pose hypothetical moral dilemmas, stories in which two moral values collide, to children and adolescents of different ages. The most famous is the **Heinz dilemma**:

A woman is dying of cancer. A druggist has a drug that might save her, but he charges $2,000 for a dose that cost him $200 to make. Her husband, Heinz, can raise only about $1,000, and the druggist will not sell it cheaper or let him pay later. Heinz breaks in and steals the drug. Should he have done that? Why?

The decisive word is "why". Kohlberg scored the **reasoning**, not the choice: "steal" and "don't steal" can come from the same stage, and the same answer from different stages. His interview and scoring system became the **Moral Judgment Interview**. The theory describes moral reasoning, not moral behaviour.

> EXAM TRAP
>
> Kohlberg studied morality by **posing moral dilemmas to children of different ages**, not through daily journals, court records of delinquents, or making people perform legal and illegal acts (watching real cheating was Hartshorne and May's method). His stages concern **moral reasoning**; language acquisition (Chapter 1), motor skills (Section 6.3) and psychosexual conflict (Freud, Chapter 4) belong to other theories.

Three levels, six stages •••

Kohlberg grouped the reasons into three levels of two stages each (Table 6.25, Figure 6.16), named by the person's relation to **conventions**, society's rules and expectations.

At the **preconventional level** (Level I), typical of children under about 9, morality is under external control: the child obeys rules to avoid punishment or gain rewards, so rewarded acts are right and punished acts wrong. **Stage 1**, the **punishment-and-obedience orientation**, judges an act by the punishment it brings. **Stage 2**, **instrumental purpose and exchange** (naive hedonism), counts as right whatever serves one's own interest: what is in it for me?

At the **conventional level** (Level II), reached by most adolescents and where most adults stay, the person conforms to social rules to be a good group member and maintain the social order. **Stage 3**, the **good boy-nice girl orientation**, counts as right whatever wins the approval of close others; intentions matter, and the golden rule guides relationships. **Stage 4**, the **law-and-order orientation**, takes the whole society's view: do your duty, respect authority and obey laws, because society would collapse if everyone broke them.

At the **postconventional level** (Level III), reached by a minority of adults and rarely before 20, principles can stand above a particular law. **Stage 5**, the **social-contract orientation**, sees laws as instruments that express the will of the majority and further human welfare, to be changed by democratic agreement when they fail. **Stage 6**, **universal ethical principles**, follows self-chosen principles of justice and human dignity, even against an unjust law.

Table 6.25 Kohlberg's levels and stages, with the alternative labels textbooks use (the older ones come from Kohlberg's 1960s papers).

Level	Stage	Right means	Other labels
I Preconventional	1 Punishment and obedience	Avoiding punishment	Heteronomous morality (Kohlberg 1976)
I Preconventional	2 Instrumental purpose and exchange	Rewards; deals that pay me	Naive (instrumental) hedonism; individualism; self-interest
II Conventional	3 Good boy-nice girl	Approval of close others	Mutual interpersonal expectations; Papalia: maintaining mutual relations, approval of others, the golden rule
II Conventional	4 Law and order	Duty, authority, social order	Social-order-maintaining; social system and conscience; Papalia: social concern and conscience
III Postconventional	5 Social contract	Agreed laws serving rights and welfare	Papalia: morality of contract, of individual rights and of democratically accepted law
III Postconventional	6 Universal ethical principles	Self-chosen principles	Morality of individual principles of conscience; Papalia: morality of universal ethical principles

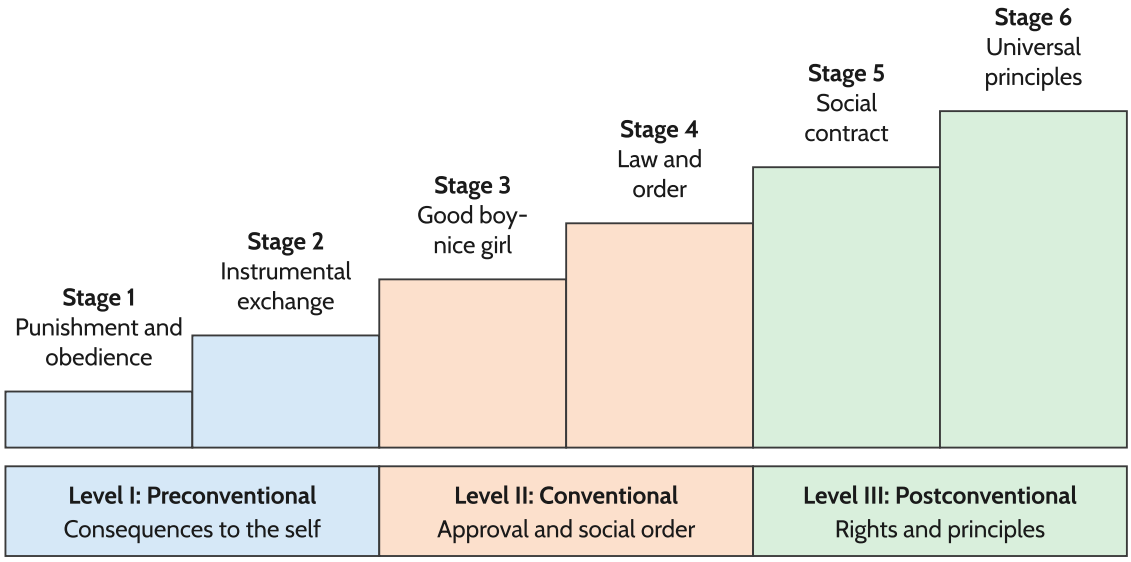

Figure 6.16 Kohlberg's stages as a staircase. Each level holds two stages, everyone climbs in the same order without skipping, and fewer people reach the higher steps.

EXAM TRAP

The levels are **preconventional, conventional and postconventional**; "unconventional", "formal conventional", "premoral" and "amoral" are not among them. "Premoral" is Piaget's period before rules matter (only Kohlberg's earliest papers used it for Level I), so treat it as not a level and never as the answer for reasoning about laws. Papalia calls Level III the **morality of autonomous moral principles**; when both names are offered for the third level, choose **postconventional**. Conformity to peers is stage 3, so conventional. In the four-statement item, "obedience, self-interest and conformity" (stages 1 to 3) and "law and order, human rights and universal ethics" (stages 4 to 6) are true; "law and order, self-interest and purpose" mixes levels, and "conformity, identity and competence" borrows Erikson's terms (Section 6.10). Answer: 1 and 3 only.

Classifying a reason: worked examples

To place a reason, ignore the choice and ask what makes the act right or wrong for the speaker. Consequences to the self mark Level I; approval, duty, the law and keeping society working mark Level II; rights and principles that could justify changing a law mark Level III (Table 6.26).

Table 6.26 Classifying moral reasons.

Reason given	Stage and level	The tell
"Don't steal; you will be punished."	1, preconventional	Punishment
"Help him; he will reward me."	2, preconventional	Reward, exchange
"Steal it; people will think him heartless otherwise."	3, conventional	Approval
"Don't steal; it is against the law, and laws keep society functioning."	4, conventional	Social order
"Keep your promise; society depends on people honouring agreements."	4, conventional	Social order
"Steal it; the right to life outweighs the right to property."	5, postconventional	Rights the law should serve
"Steal it and accept jail; every life has equal worth."	6, postconventional	Universal principle

EXAM TRAP

"Promises", "agreements" and "laws" do not by themselves mean stage 5. Keeping a promise **because society depends on it**, or obeying a law **because laws keep society functioning**, is stage 4: **conventional**. It becomes stage 5 only when the speaker asks whether the agreement serves people's rights and welfare, and would change it if not.

How sound is Kohlberg's theory?

Kohlberg held that the stages form an **invariant sequence**: everyone climbs them in the same order, without skipping, and regression is rare; the 20-year Chicago data largely agreed. Each stage needs a cognitive base that is necessary but not sufficient: postconventional reasoning needs formal operations (Section 6.4), yet many formal-operational adults never reach it. It also needs **role-taking**, seeing a problem from other people's viewpoints (Robert Selman's levels of social perspective taking, Section 6.5), which grows through discussion and education. Stage 6 proved so rare that Kohlberg later dropped it from his scoring manual. Because his interview needs trained scorers, **James Rest** built the **Defining Issues Test** in the 1970s, a questionnaire in which people rate and rank prepared statements about each dilemma.

Three criticisms dominate. The first is **cultural bias**. **John Snarey** (1985) reviewed 45 studies from 27 cultural areas: stages 1 to 4 appeared in order almost everywhere, but stage 5 mainly in urban, educated groups and hardly ever in village societies, whose communal reasoning the scoring missed. **Richard Shweder**, studying Hindu families in Bhubaneswar, Odisha, argued that the scale measures only an **ethic of autonomy** (harm, rights, justice) and misses an **ethic of community** (duty, hierarchy) and an **ethic of divinity** (purity, sanctity). The second is the **gap between reasoning and behaviour**: higher-stage reasoners act morally only somewhat more often. The third is **gender bias**: the founding sample was all male.

INDIAN CONTEXT

Shweder's Oriya Brahmin informants judged it a serious moral wrong for an eldest son to get a haircut and eat chicken the day after his father's death; Chicago informants saw a personal choice. **Joan Miller**, comparing Hindu Indians in Mysore with Americans, found that Indians treated helping family, friends and even strangers as **moral obligations**, where Americans saw many of the same acts as personal decisions. A scale built around individual rights scores such duty-based reasoning as merely conventional.

Gilligan: the ethic of care

Carol Gilligan (born 1936), who had worked with Kohlberg at Harvard, noticed that women often scored at stage 3, apparently below men. In *In a Different Voice* (1982) she argued that his scale rests on a male **ethic of justice**, reasoning from rights, rules and fairness, and misses the **ethic of care** more often heard in women, reasoning from relationships and responsibility, wanting no one hurt. Her 11-year-old Jake solved Heinz like "a math problem with humans": life outweighs property. Amy worried that if Heinz went to jail his wife would have no one, and wanted them to talk it out.

From interviews with 29 women deciding about abortion, Gilligan described three levels of care: an **orientation to individual survival** (caring only for oneself), **goodness as self-sacrifice** (caring for others at one's own expense) and the **morality of nonviolence** (caring for self and others, hurting no one).

Is there a sex difference? **Lawrence Walker**'s 1984 review found little difference in Kohlberg scores once education and occupation were allowed for, and later meta-analyses found only small differences in care and justice orientation: both sexes use both voices.

EXAM TRAP

The pairing is fixed: **females, ethic of care; males, ethic of justice**. Options that reverse it are wrong, and "males have an ethic of justice", though part of her argument, is what Kohlberg already measured; her departure is the female ethic of care. Gilligan added a voice; she did not replace Kohlberg.

Beyond the stage theories

Elliot Turiel's **social domain theory** (1983) holds that children build separate kinds of social knowledge from early on: the **moral domain** (harm, fairness, rights), the **social-conventional domain** (agreed customs such as dress codes or forms of address) and the **personal domain** (choices such as friends or hairstyle). Even preschoolers say hitting would be wrong in a school with no rule against it, but calling the teacher by her first name would then be fine: they already separate moral from conventional rules.

Nancy Eisenberg studied **prosocial moral reasoning**, about helping at a cost to oneself (should Mary, on her way to a party, stop to help a girl who has fallen?). Children move through five levels: hedonistic, needs-oriented, approval-oriented, empathic and finally internalised reasons.

Martin Hoffman saw **empathy** as the main motive of moral action and traced four stages: **global empathy** in the first year (an infant cries when another cries), **egocentric empathy** in

the second (a toddler fetches his own mother to comfort a crying friend), empathy for **another's feelings** from about 2 or 3, and in late childhood empathy for **another's life condition** (for the poor as a group). He also compared three kinds of discipline. Power assertion (force, threats, removing privileges) and love withdrawal (coldness, ignoring the child) produce compliance out of fear or anxiety. **Induction**, explaining how the child's act affects others ("Riya is crying because you snatched her book"), recruits empathy and guilt and so best fosters **internalisation**, behaving well when no one is watching (parenting styles are in Section 6.8).

Finally, **Jonathan Haidt**'s **social intuitionist model** (2001) reverses the cognitive tradition: moral judgments are fast gut intuitions, and reasoning mostly follows to justify them.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Hartshorne and May concluded that honesty is largely _____, not a general trait.
2. In Piaget's _____ morality, rules are unchangeable and acts are judged by their consequences.
3. The belief that the world itself punishes misdeeds is _____ justice.
4. Kohlberg posed moral _____, chiefly the Heinz story, and scored the reasoning, not the choice.
5. At the _____ level, rewarded acts are right and punished acts are wrong.
6. "Don't steal; laws keep society functioning" is stage 4, the _____ orientation.
7. Seeing laws as instruments that express the majority's will and further welfare is the _____ orientation.
8. Gilligan argued that women more often reason with an ethic of _____.

Answers: (1) situation-specific, (2) heteronomous, (3) immanent, (4) dilemmas, (5) preconventional, (6) law-and-order, (7) social-contract, (8) care.

6.10 Erikson's Psychosocial Stages, Identity and Adolescence

MEDIUM YIELD · 40 QS

Ask a nineteen-year-old in a Mysuru hostel who she is, and what she will make of her life, and you may get a long pause. Erik Erikson put that question at the centre of a theory that runs from the first feed to the last years, and he gave it a name now used everywhere: the identity crisis. His eight stages come first; then identity is followed into adolescence itself, with its identity statuses, its egocentrism, its changing body and brain, and the old argument over storm and stress.

Erikson and the psychosocial approach ●●

Erik Homburger Erikson (1902 to 1994) lived his own theory. Born in Frankfurt to a Danish mother who never told him who his father was, he carried his stepfather's surname, Homburger, and wandered Europe as an artist in his late teens. At 25 he began teaching at a small school in Vienna, was analysed by **Anna Freud** (1895 to 1982), and became a psychoanalyst without any university degree. He moved to the United States in 1933 and later treated

Second World War veterans whose trouble seemed to be a confused sense of who they were rather than repressed conflict. His key books are *Childhood and Society* (1950), which set out the eight stages, *Young Man Luther* (1958), *Identity: Youth and Crisis* (1968) and *Gandhi's Truth* (1969).

Erikson belonged to **ego psychology**, which treats the ego as a partly independent, adaptive part of the personality rather than a servant of the id. Development follows the **epigenetic principle**: like an embryo whose organs appear on a fixed timetable (*epi* means "upon"), the personality unfolds through a built-in sequence of stages, while society decides how each turns out. Each stage poses a **psychosocial crisis**, a turning point between two opposed tendencies set off by new social demands; it is a choice of direction, not a catastrophe. A favourable resolution leaves the ego with a basic strength, or **virtue**. Favourable means a balance tilted towards the positive pole, not the absence of the negative: a child who trusts everyone is gullible. An unresolved crisis makes later ones harder but not hopeless, because development continues across the whole lifespan.

Erikson coined the term **identity crisis** for the adolescent struggle to know who one is. His own labels for the eight stages of the life cycle run from the oral-sensory stage through the muscular-anal, locomotor-genital, latency, puberty and adolescence, young adulthood and adulthood stages to the stage of maturity. The first four echo Freud's oral, anal, phallic and latency stages; the adult stages are Erikson's own.

> **EXAM TRAP**
>
> The identity crisis, and the eight stages from the "oral-sensory stage" to the "stage of maturity", are Erikson's. Freud described five psychosexual stages, Harry Stack Sullivan developmental epochs from infancy to adulthood, and Melanie Klein the paranoid-schizoid and depressive positions of infancy (all Chapter 4). John Bowlby is attachment (Section 6.7); Eric Berne is transactional analysis (Chapter 7).

> **INDIAN CONTEXT**
>
> *Gandhi's Truth* (1969) is a psychohistory, a great life read through Erikson's stages and psychoanalytic ideas. Centred on the 1918 Ahmedabad textile mill strike, it asks how one man's identity struggle became a nation's method of nonviolent protest, as *Young Man Luther* had asked of Martin Luther. It won the Pulitzer Prize.

The eight stages ●●●

Table 6.27 is the map to memorise: for each stage an age, a crisis named by its two poles, the virtue a good resolution leaves, and the people who matter most, a "radius of significant relations" that widens from one mother to all of humankind.

Table 6.27 Erikson's eight psychosocial stages. Ages are approximate and vary between texts.

Stage (Erikson's label)	Age	Crisis	Virtue	Key relationships
1 Infancy (oral-sensory)	Birth to about 1 (some: 18 months)	Trust versus mistrust	Hope	Maternal person
2 Early childhood (muscular-anal)	About 1 or 1.5 to 3	Autonomy versus shame and doubt	Will	Parents

Stage (Erikson's label)	Age	Crisis	Virtue	Key relationships
3 Play age (locomotor-genital)	3 to 5 or 6	Initiative versus guilt	Purpose	Basic family
4 School age (latency)	6 to puberty	Industry versus inferiority	Competence	Neighbourhood and school
5 Adolescence (puberty and adolescence)	About 12 to 18	Identity versus role confusion	Fidelity	Peer groups; models of leadership
6 Young adulthood	About 18 to 40	Intimacy versus isolation	Love	Partners in friendship, sex, competition and cooperation
7 Adulthood (middle)	About 40 to 65	Generativity versus stagnation	Care	Shared household and work
8 Maturity (late adulthood)	65 onward	Ego integrity versus despair	Wisdom	Humankind, "my kind"

1. Trust versus mistrust. The helpless infant learns whether the world is dependable. Reliable feeding, comfort and contact build basic trust, an expectation of "consistency, continuity and sameness"; neglect or inconsistency breeds suspicion. The virtue is hope, a lasting confidence that wishes can be met.

2. Autonomy versus shame and doubt. The walking, talking toddler wants to do things alone. Erikson's key acts are holding on and letting go, and the classic battleground is toilet training. Patient parents leave the child with will, the determination to choose and to restrain oneself; harsh or overprotective ones leave shame and self-doubt.

3. Initiative versus guilt. The preschooler plans, explores, asks endless questions and acts out adult roles in play. Encouraged, the child gains purpose, the courage to pursue goals; punished or mocked for each venture, the child learns guilt.

4. Industry versus inferiority. School brings tools, tasks and classmates to compare oneself with. Finishing tasks and earning recognition build industry and competence; repeated failure, or being made to feel behind one's peers, brings inferiority.

5. Identity versus role confusion. With puberty and new choices, the young person must fuse childhood identifications, skills, values and future roles into one answer to "Who am I?" Success brings fidelity, loyalty to freely chosen values and people; failure brings role confusion (see Identity, moratorium and Marcia's statuses).

6. Intimacy versus isolation. With an identity in hand, the young adult can risk merging it with another's, in friendship and love, without losing it. The virtue is love, mutual devotion; failure brings isolation and self-absorption.

7. Generativity versus stagnation. Erikson's **generativity** is concern for establishing and guiding the next generation: raising children, mentoring juniors, teaching, building work that lasts (parenthood is neither necessary nor sufficient). Its failure is **stagnation**, a sense of having achieved little and having little to offer the next generation, with boredom and self-indulgence. The virtue is care.

8. Ego integrity versus despair. Looking back, the older person either accepts the one life lived, failures included, as meaningful (**ego integrity**), or feels regret, bitterness and fear of death because there is no time to start again (despair). The virtue is wisdom, a detached concern with life itself in the face of death.

Some personality texts put the adult boundaries at 35 and 55. The first four outcomes depend mainly on what others do to the child; in the last four the person increasingly chooses.

> **EXAM TRAP**
>
> An 18-month-old has been keyed to autonomy versus shame and doubt, yet another key gives trust versus mistrust as 0 to 18 months. Eighteen months is the boundary, so read the options; when both stages are offered for a child of exactly 18 months, the keyed answer has been autonomy.
>
> For sequence items, write the stage number under each option. In one mock item they run (a) 1, 2, 5, 7; (b) 2, 4, 5, 7; (c) 3, 4, 5, 8; (d) 1, 3, 6, 8. All rise, so the item is flawed; the key is (a), trust, autonomy, identity, generativity: of the two series that start at stage 1, it skips fewer stages. Virtues in order: hope, will, purpose, competence, fidelity, love, care, wisdom.

Applying the stages, and how they go wrong ●●

The stages are easiest to see in people. Mrs Rao, 52, finds her deepest satisfaction in training young engineers: generativity. Her colleague of the same age, who feels he has achieved little and has nothing to pass on, is stagnating. Her father, 78, would live his life the same way again, mistakes and all: integrity. His neighbour broods over wasted chances and dreads death: despair. Her niece in Class 4, measuring herself against classmates while mastering long division, is in industry versus inferiority.

> **EXAM TRAP**
>
> Erikson's term for the mid-life feeling of having achieved little and having little to offer the next generation is **stagnation**; generativity is the healthy pole. "Mid-life crisis" is a popular term, not Erikson's (Section 6.11), and presbyopia is the age-related loss of near focusing in the eye's lens.

A crisis can also be resolved too one-sidedly. In *The Life Cycle Completed* (1982) Erikson split this maldevelopment two ways. Too much of the positive pole, with none of the negative, gives a **maladaptive tendency**; too much of the negative pole gives a **malignant tendency**, the more serious (Table 6.28). He thought maladaptations could lead towards neurosis and malignancies towards psychosis, and that therapy could correct both, later experience the milder maladaptations.

Table 6.28 Erikson's maladaptive and malignant tendencies (1982).

Crisis	Maladaptive (too much positive)	Malignant (too much negative)
Trust versus mistrust	Sensory maladjustment: gullible	Withdrawal
Autonomy versus shame and doubt	Shameless wilfulness	Compulsion
Initiative versus guilt	Ruthlessness: goals at any cost	Inhibition: tries nothing
Industry versus inferiority	Narrow virtuosity: all skill, no childhood	Inertia
Identity versus role confusion	Fanaticism: one view only	Repudiation of adult roles
Intimacy versus isolation	Promiscuity: closeness without depth	Exclusivity: shuts others out
Generativity versus stagnation	Overextension: no time for self	Rejectivity: stops caring

Crisis	Maladaptive (too much positive)	Malignant (too much negative)
Integrity versus despair	Presumption	Disdain for life

Joan Erikson (1903 to 1997), his wife and collaborator, added a ninth stage in the book's extended edition (1997): in the eighties and nineties the negative poles of all eight crises return as the body fails, and a calm move beyond the self, gerotranscendence, becomes possible.

Erikson versus Freud

Erikson called his theory an extension of Freud's, but Table 6.29 shows how far it moved: from id to ego, from body zones to society, and from a personality set in early childhood to one still developing in old age. Freud's psychosexual stages are taught in Chapter 4.

Table 6.29 Freud and Erikson compared.

Point	Freud	Erikson
Kind of stage	Psychosexual: body zones	Psychosocial: crises with society
Driving structure	Id and instinct	Ego (ego psychology)
Span	Five stages; personality set by about 5 or 6	Eight stages, the whole lifespan
Culture and history	Little role; the family triangle	Shape every crisis
Poor outcome	Fixation	Weak virtue; maladaptive or malignant tendency
Adolescence	Genital stage	Identity crisis, the heart of the theory

Identity, moratorium and Marcia's statuses ●●●

For Erikson adolescence is the hinge of the life cycle. Childhood identifications (my father's son, a good student, a cricketer) must be sorted, kept or dropped, and fused with new abilities, sexual maturity and choices of work, beliefs and love into an **ego identity**: a sense of being the same person across time and situations, recognised by the people who count. Society helps by granting a **psychosocial moratorium**, a socially sanctioned delay of adult commitments in which the young can try out courses, friendships and causes before choosing; college years and Erikson's own wandering years are examples.

When the work fails the result is **identity confusion** (role confusion in Erikson's earlier wording; also identity diffusion): inner fragmentation, little or no sense of where one's life is headed, and an inability to gain the support of a social role or vocation. Some young people instead seize a **negative identity**, built from the roles their family or community holds up as most undesirable, such as the delinquent, on the logic that it is better to be somebody bad than nobody. Adolescence, then, is a true developmental stage and a moratorium, the fifth of eight and not the last.

> EXAM TRAP
>
> "Inner fragmentation, no sense of direction, no support from a social role or vocation" is **identity confusion**. Incongruence is Carl Rogers's gap between self-concept and experience (Chapter 4); dissociation is a break in the integration of memory, identity and consciousness (Chapter 8); "conflict" is too general.

KEY STUDY: MARCIA'S IDENTITY STATUS INTERVIEW (1966)

James E. Marcia asked whether Erikson's identity could be measured. He interviewed male college students about occupation, religion and politics and scored two dimensions: **crisis or exploration**, a period of actively weighing alternatives, and **commitment**, a firm investment in a choice. Crossing them gave four **identity statuses** (Figure 6.17) that predicted behaviour: achieved students coped best with a stressful task, foreclosed students endorsed authoritarian values most. Criticism: an all-male first sample, and a person can hold different statuses in different areas of life.

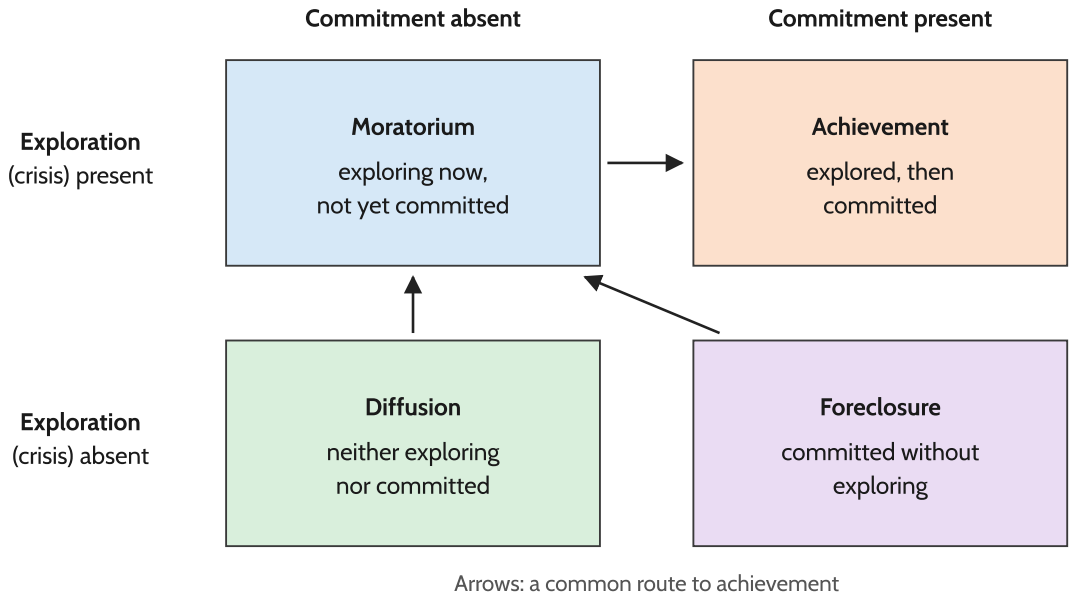

Figure 6.17 Marcia's identity statuses cross exploration (crisis) with commitment. Foreclosure and moratorium are mirror images: commitment without exploration, and exploration without commitment.

- Neither exploration nor commitment is **identity diffusion**. The young person drifts; adjustment is poorest.
- Commitment without exploration is **foreclosure**. Choices are taken over ready-made, often from parents, like the son who becomes a doctor because the family always assumed he would; such youth tend to be rigid and conforming.
- Exploration without commitment is **moratorium**. The person is in the middle of the crisis, anxious and actively trying options.
- Exploration followed by commitment is **identity achievement**, the most mature status, with the highest self-esteem.

Statuses are not fixed stages. Achievement grows through late adolescence and young adulthood, and adults reopen settled questions after a divorce, a job loss or migration and then close them again: the **MAMA cycles** (moratorium, achievement, moratorium, achievement) described by Marcia and colleagues in 1992. **Jean Phinney** applied the same logic to **ethnic identity**, from unexamined through exploration to achieved; a secure ethnic identity goes with higher self-esteem in minority youth.

EXAM TRAP

Foreclosure is commitment WITHOUT exploration; moratorium is exploration WITHOUT commitment; achievement has both; diffusion neither. Marcia's dimensions are crisis (exploration) and commitment, not IQ and emotional intelligence, locus of control (Chapter 4) or approach and avoidance motivation (Chapter 1).

Adolescent egocentrism

Formal operational thought (Section 6.4) lets adolescents think about thinking, including other people's. **David Elkind** argued in 1967 that this power brings a new **adolescent egocentrism**: the teenager fails to separate what others are thinking about from his or her own preoccupations. It shows in two mental constructions. The **imaginary audience** is the belief that everyone is watching and judging: the boy who will not enter class with a pimple on his chin, sure that all forty classmates will notice. The **personal fable** is the belief that one's experiences and feelings are unique, that no one, least of all a parent, could understand them ("You don't know what it's like to be me"), and with it a sense of invulnerability ("It won't happen to me") that feeds reckless riding. Some notes list the invulnerability strand separately as the "mythological fable". Both are strongest in early adolescence and fade by the late teens.

EXAM TRAP

Adolescent egocentrism = imaginary audience ("everyone is looking at me") plus personal fable ("my experiences are unique"). Conservation and reversibility are concrete operational (Section 6.4), inner speech and scaffolding are Vygotsky's (Section 6.5), and object permanence and stranger anxiety belong to infancy (Sections 6.4 and 6.6).

Puberty and the adolescent brain

Adolescence begins with **puberty**, the biological changes that bring sexual maturity; adolescence itself is the wider psychological and social transition. The hormones come in two waves. In **adrenarche**, from about 6 to 8, the adrenal glands raise their output of weak androgens. In **gonadarche**, from about 9 or 10, the hypothalamus and pituitary switch on the ovaries and testes, which pour out oestrogens and testosterone (Chapter 5). The visible result is the growth spurt, a rapid gain in height and weight that starts about two years earlier in girls (around 10) than in boys (around 12), so that for a while many girls are taller than boys of their age. The **primary sex characteristics** are the reproductive organs themselves (ovaries, uterus and vagina; testes, penis and scrotum); **secondary sex characteristics** are the other visible signs (breasts, body and facial hair, a deeper voice, wider hips or shoulders). In girls, **menarche**, the first menstrual period, comes late in the sequence, at about 12 or 13; **spermarche**, the first ejaculation, at about 13 or 14. Better nutrition and health have pulled menarche earlier, from about 16 or 17 in mid-nineteenth-century Europe to about 12 or 13 today, and made adults taller: the **secular trend**.

Timing matters too. Early-maturing boys tend to be confident and popular, though they try alcohol and delinquency earlier; late-maturing boys are often more anxious. Early-maturing girls carry the greatest risk: looking older than they feel, they attract older friends and show more depression, poor body image, eating problems and substance use.

Inside the skull, regions mature at different speeds. **Laurence Steinberg**'s **dual systems model** (2008) pairs a reward system in the limbic system and ventral striatum, which becomes sharply more responsive around puberty, with a cognitive control system centred on the prefrontal cortex, which matures slowly into the mid-twenties (Figure 6.18). In mid-adolescence the accelerator is pressed while the brakes are still being fitted, so risk-taking peaks, above all among peers.

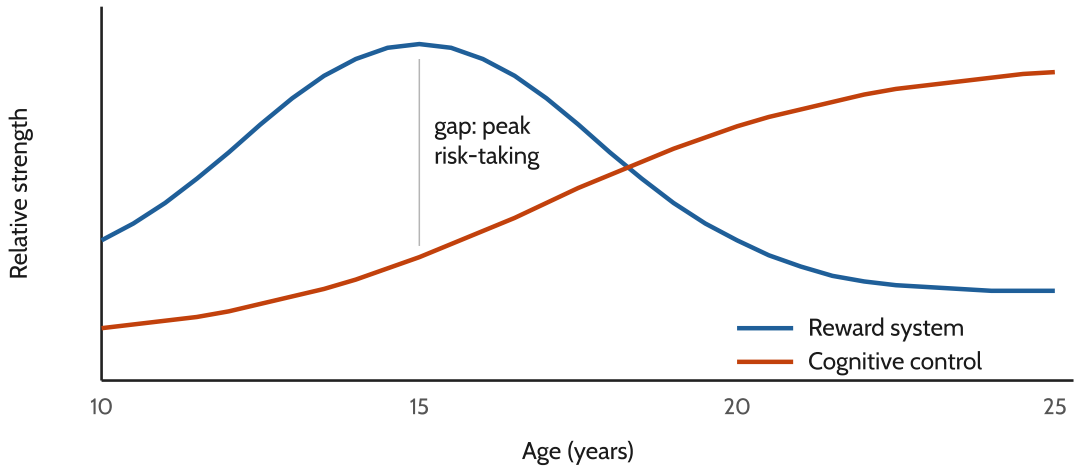

Figure 6.18 Steinberg's dual systems model (schematic). Reward sensitivity (solid line) peaks in mid-adolescence; cognitive control (dashed line) matures gradually into the mid-twenties. The gap marks the years of peak risk-taking.

KEY STUDY: RISK WITH FRIENDS WATCHING (GARDNER AND STEINBERG, 2005)

Margo Gardner and Steinberg had adolescents (13 to 16), college-age youths and adults play a video driving game in which they could brake at an amber light or gamble on getting through, alone or with two friends watching. Friends' presence roughly doubled the adolescents' risk-taking, raised the youths' by about half and left the adults' unchanged. Criticism: a laboratory game, though it matches teen drivers' higher crash rates with teen passengers.

Storm and stress, tasks, parents and peers ●●

Must adolescence be turbulent? **G. Stanley Hall** (1844 to 1924), in *Adolescence* (1904), said yes: the teens were a biologically driven period of **storm and stress**, of mood swings, conflict with parents and risk-taking. **Margaret Mead** (1901 to 1978) answered from the field. In *Coming of Age in Samoa* (1928) she described Samoan girls passing smoothly into adulthood in a relaxed culture with many caring adults, and concluded that turmoil is made by culture, not biology. **Derek Freeman** (1983) later attacked her fieldwork, but modern surveys support a middle view: most adolescents cope well and share their parents' basic values, while a minority run into serious trouble. Conflict and moodiness do rise in early adolescence; storm and stress is real but not universal.

Robert J. Havighurst (1900 to 1991) listed the **developmental tasks of adolescence** (the idea itself is in Section 6.1): new and more mature relations with age-mates of both sexes; a masculine or feminine social role; accepting one's body; emotional independence from parents and

other adults; preparing for marriage and family life; preparing for a career; a set of values and an ethical system to guide behaviour; and socially responsible behaviour. Emotional separation from parents, strong peer identification and heterosexual peer interest are therefore all tasks of adolescence.

Relations with parents are renegotiated, not cut. The older psychoanalytic view of Anna Freud and **Peter Blos** held that healthy adolescents must detach from their parents. Research now shows that autonomy grows best inside continued connection: teenagers who stay securely attached explore more confidently and adjust better, so it can be healthy for interdependence to continue. Detachment does not typically happen earlier, in middle childhood; peers do not simply replace parents, who keep their influence on values, education and career while peers sway dress, music and leisure; and parents' goals differ across families and cultures, many Indian families valuing lasting interdependence.

Dexter Dunphy (1963), observing teenagers in Sydney, distinguished **cliques**, small groups of close friends, from crowds, larger groups formed when several cliques join for parties and outings. He traced five stages: isolated same-sex cliques; same-sex cliques beginning to interact; their leaders forming the first mixed-sex clique; a full crowd of mixed-sex cliques; and the crowd loosening into groups of couples. Conformity to peers rises in early adolescence, peaks at about 14 and then declines as autonomy matures.

> **EXAM TRAP**
>
> Today's view of adolescents and parents: continued interdependence can be healthy. Not that detachment comes earlier in middle childhood, that the peer group matters more than parents, or that all parents share the same goals.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The virtue that emerges from initiative versus guilt is _____.
2. A toddler of 18 months who insists on feeding herself faces _____ versus shame and doubt.
3. A middle-aged man who feels he has achieved little and has nothing to offer the next generation shows Erikson's _____.
4. Erikson called the socially sanctioned delay of adult commitments a psychosocial _____.
5. Commitment without exploration is Marcia's status of _____.
6. The teenager convinced that no one has ever felt what she feels shows Elkind's _____.
7. Too much of the positive pole of the identity crisis gives the maladaptive tendency of _____.
8. In Steinberg's dual systems model the _____ cortex matures later than the reward system.

Answers: (1) purpose, (2) autonomy, (3) stagnation, (4) moratorium, (5) foreclosure, (6) personal fable, (7) fanaticism, (8) prefrontal.

6.11 Adulthood, Ageing, Death and Dying LOWER YIELD · 2 QS

Development does not stop at twenty. A Mysuru engineer of 24 unsure of his career, his mother of 50 nursing an ailing father, and his grandfather of 80 trimming his circle of friends are all still developing.

Emerging adulthood and adult development

In 2000 **Jeffrey Jensen Arnett** named the years from about 18 to 25 **emerging adulthood**, a period created in industrialised societies by later marriage, parenthood and settled work. Its five features are identity exploration (in love and work), instability, self-focus, feeling in-between (neither adolescent nor adult) and possibilities (life can still go many ways). Its Eriksonian task, intimacy versus isolation, is in Section 6.10.

Daniel Levinson (1920 to 1994) drew the best-known map of what follows. In *The Seasons of a Man's Life* (1978), built on interviews with 40 men aged 35 to 45, each man has a **life structure**, the underlying pattern of his life at a given time, built mainly around work and family. Overlapping eras alternate stable periods, when a structure is built, with transitions of about five years, when it is questioned and remade (Figure 6.19): the early adult transition (17 to 22) leaves adolescence behind, and the age-30 transition (28 to 33) revises the first adult structure. The young man pursues **the Dream**, a vision of the adult he hopes to become, helped by a mentor, an older guide at work. In the **midlife transition** (about 40 to 45) he reappraises the Dream against what he has achieved. Table 6.30 adds four other theorists.

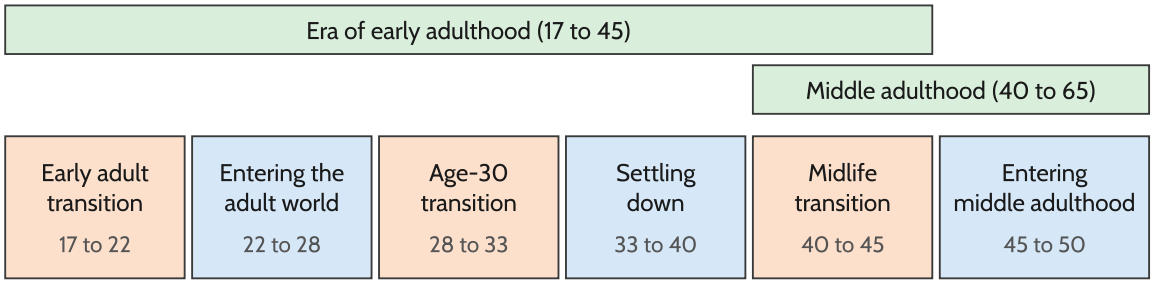

Figure 6.19 Levinson's seasons. Transitions alternate with stable periods; the midlife transition belongs to both eras.

Table 6.30 Other theories of adult development.

Theorist	Key idea
George Vaillant (born 1934), Grant Study	Followed 268 Harvard men from 1938; added career consolidation (after intimacy) and keeper of the meaning (after generativity) to Erikson's stages
Roger Gould, *Transformations* (1978)	Growth means shedding false childhood assumptions, such as "life is simple and controllable"
Bernice Neugarten (1916 to 2001)	Social clock: a cultural timetable for life events; off-time events (widowhood at 35) are harder than on-time ones
Robert Peck	Tasks of old age: ego differentiation, body transcendence and ego transcendence, versus work-role, body and ego preoccupation

Robert J. Havighurst (1900 to 1991) set adult developmental tasks (Section 6.1). Early adulthood: selecting a mate, learning to live with a partner, starting a family, rearing children, managing a home, starting an occupation, civic responsibility, a congenial social group. Middle age: civic and social responsibility, an economic standard of living, helping teenagers become adults, adult leisure, relating to the spouse as a person, accepting physical change, adjusting to ageing parents. Later maturity: adjusting to declining strength and health, to retirement and reduced income and to a spouse's death, affiliating with one's age group, civic obligations, satisfactory living arrangements.

Love, marriage and work

Why we love, including **Robert Sternberg**'s triangular theory, is social psychology (Chapter 3). Developmental psychology asks how a marriage divides roles. **Elizabeth Hurlock** (1898 to 1988), in *Developmental Psychology*, described three types (Table 6.31).

Table 6.31 Hurlock's types of marriage.

Type	Roles	What is emphasised
Traditional marriage	Sharply separate: husband provider, wife homemaker	Division of labour; husband heads the family
Companionship marriage	Shared and interchangeable	Romantic love, emotional intimacy, equality
Collegial marriage	Different but complementary; each spouse responsible for his or her own	Comradeship and sharing, as between colleagues

> **EXAM TRAP**
>
> Comradeship and sharing with different roles is a *collegial* marriage (JEMAS 2025). Companionship marriage shares the same roles; traditional marriage separates roles without stressing comradeship. "Satisfactory marriage" is not a type.

> **INDIAN CONTEXT**
>
> Most Indian marriages are still arranged by families, now usually with the couple's consent; love is expected to grow after the wedding, and the marriage joins two families.

Elsewhere more adults cohabit, divorce is commoner, and a first child usually lowers marital satisfaction for a while. Work is the other pillar. **Eli Ginzberg** (1911 to 2002; *Occupational Choice*, 1951) described three periods of career choice: fantasy (to about 11), wishing without regard to ability; tentative (about 11 to 17), weighing interests, capacities and values; and realistic (17 to the early twenties), exploring, crystallising and specifying one choice. **Donald Super**'s career stages are in Chapter 7.

Middle adulthood: the climacteric and midlife

From about 40 the body changes slowly. Near vision blurs as the lens stiffens (**presbyopia**), hearing for high pitches fades (presbycusis), bone mass falls (especially in women after menopause), hair recedes and greys, muscle strength and agility decline, and reaction time and manual dexterity slow. All of this is general physical ageing.

The **climacteric** is narrower: the end of ovulation and termination of reproductive capacity. In women it culminates in **menopause**, the last menstrual period (confirmed after 12 months without one), as ovarian oestrogen falls. The average age is about 51 in Western samples and about 46 to 47 in Indian surveys. The male climacteric, or **andropause**, is a slow decline in testosterone; most men stay fertile into old age.

> EXAM TRAP
>
> Climacteric means the end of ovulation and termination of reproductive capacity (JEMAS 2025). Lost strength, receding or greying hair, and slower reactions and dexterity are general ageing. Menopause is the event within the climacteric, not a synonym.

Elliott Jaques (1917 to 2003) coined **midlife crisis** in 1965, from creative artists confronting mortality in their mid-thirties. Large surveys find that only a minority report one, usually after illness or job loss. Midlife is more often a peak of competence and generativity (Section 6.10). The **empty nest**, when the last child leaves, more often brings relief than depression. The **sandwich generation** are middle-aged adults, usually women, caring for children and ageing parents at once. **K. Warner Schaie**'s Seattle Longitudinal Study (begun 1956) found that most mental abilities peak in middle adulthood, while perceptual speed declines from early adulthood.

Late adulthood and the biology of ageing

Gerontology studies the biological, psychological and social sides of ageing; geriatrics is the medicine of old age. **Robert Butler** (1927 to 2010) coined **ageism**, prejudice against people because they are old, in 1969. Older adults are divided into the young-old (65 to 74), old-old (75 to 84) and oldest-old (85 and over). Life span is the maximum possible length of life, about 120 years (**Jeanne Calment** lived 122); **life expectancy** is the average number of years a person born in a given year can expect to live, about 70 in India.

Primary ageing is gradual, inevitable and universal bodily decline. Secondary ageing comes from disease, disuse and lifestyle, so it is partly preventable. Tertiary ageing is the rapid loss shortly before death; its cognitive form, a sharp fall in test scores, is the **terminal drop** (**Robert Kleemeier**, 1962).

Programmed theories say the genes set ageing's timetable. **Leonard Hayflick** and **Paul Moorhead** (1961) found that human cells in culture divide only about 40 to 60 times (some textbooks say 75 to 80), the **Hayflick limit**, apparently because telomeres, the caps on chromosome ends, shorten with each division. Damage, or variable-rate, theories blame accumulated random injury instead (Table 6.32).

Table 6.32 Biological theories of ageing.

Family	Theory	Claim
Programmed	Cellular clock (Hayflick)	Cells divide a fixed number of times as telomeres shorten
Programmed	Programmed senescence	Genes switch on and off in sequence
Programmed	Immunological	Immune function declines on schedule, so disease rises
Programmed	Hormonal stress	An ageing hormonal system lowers resistance to stress
Damage	Wear and tear	Body parts, like machine parts, wear out with use

Family	Theory	Claim
Damage	Free radicals (**Denham Harman**, 1956)	Unstable oxygen molecules from metabolism damage DNA and cells
Damage	Rate of living	Faster metabolism, shorter life
Damage	Autoimmune	The immune system attacks the body's own cells

Psychosocial theories of ageing

Is a good old age a matter of stepping back or of staying busy? **Elaine Cumming** and **William Henry**'s **disengagement theory** (*Growing Old*, 1961) said old people and society withdraw from each other by natural, mutual consent. Research disagreed: withdrawal is usually forced by illness or loss, and it lowers well-being. Havighurst's **activity theory** replied that the more active older people stay, the more satisfied they are. Table 6.33 sets out the main theories.

Table 6.33 Psychosocial theories of ageing.

Theory	Proposed by	Core idea
Disengagement	Cumming and Henry, 1961	Mutual, natural withdrawal of the person and society
Activity	Havighurst	Stay active and replace lost roles
Continuity theory	**Robert Atchley**, 1989	Keep inner (values) and outer (roles, routines) continuity, using lifelong coping strategies
Socioemotional selectivity	Laura Carstensen	As time feels short, emotional goals beat knowledge goals; smaller, closer networks
Selective optimisation with compensation	Paul and Margret Baltes, 1990	Select fewer goals, optimise them, compensate for losses
Gerotranscendence	**Lars Tornstam**, 1989	From a materialistic, rational outlook to a cosmic, transcendent one
Successful ageing	**John Rowe** and **Robert Kahn**, 1997	Low disease and disability, high mental and physical functioning, active engagement

Laura Carstensen's **socioemotional selectivity theory** explains why the grandfather of 80 drops acquaintances but sees more of his family.**Paul Baltes** (1939 to 2006) and **Margret Baltes** (1939 to 1999) illustrated **selective optimisation with compensation** (SOC) with the pianist **Arthur Rubinstein**, who in old age played fewer pieces (selection), practised them more (optimisation) and slowed before fast passages so that they sounded faster (compensation). Butler's life review, used in therapy, is in Chapter 7.

> **EXAM TRAP**
>
> Disengagement says withdraw; activity, stay busy; continuity, stay yourself; socioemotional selectivity, prune the network for emotional reasons. SOC belongs to the Baltes couple, not Carstensen; the three criteria of successful ageing, to Rowe and Kahn.

Cognition, retirement and elder care

Fluid intelligence declines from early adulthood while crystallised intelligence holds or rises into old age (Section 1.9). **Timothy Salthouse**'s **processing-speed theory** (1996) traces much of the loss to a general slowing of processing. Episodic and working memory decline; semantic and procedural memory hold. Paul Baltes's **Berlin wisdom paradigm** defines wisdom as expert knowledge about the fundamental pragmatics of life, judged by factual and procedural knowledge, lifespan contextualism, value relativism and managing uncertainty; it reflects experience more than age. Dementia is disease, not normal ageing (Chapters 5 and 8).

Retirement is a process: Atchley described a honeymoon, then possible disenchantment, reorientation and stability; it goes best when chosen, healthy and funded. Widowhood falls mostly on women, who live longer and usually marry older men.

> **INDIAN CONTEXT**
>
> As joint families shrink, the law has stepped in. The Maintenance and Welfare of Parents and Senior Citizens Act 2007 obliges children (or, for the childless, heirs) to maintain parents and senior citizens (60 and over). Tribunals can order up to Rs 10,000 a month, and abandoning a senior citizen is punishable by up to three months in prison, a Rs 5,000 fine, or both.

Death, dying and grief

Death was once the stopping of heart and breathing. Life support broke that link, so India (Transplantation of Human Organs Act 1994) and most countries also recognise **brain death**: irreversible loss of all brain function or, in Indian law, of all brainstem function. In a persistent vegetative state the brainstem still works.

Maria Nagy (1948) questioned 378 Budapest children aged 3 to 10. Under about 5, death was a departure or sleep, and reversible; from 5 to 9 it was final but personified, a "death-man" who could be escaped; after 9 it was universal and inevitable. **Mark Speece** and **Sandor Brent** (1984) named three parts of a mature concept, irreversibility, universality and nonfunctionality (all functions stop), grasped by most children at about 5 to 7.

Elisabeth Kübler-Ross's five stages of dying (*On Death and Dying*, 1969) are denial, anger, bargaining, depression and acceptance (Section 7.9). **John Bowlby** (1907 to 1990) saw mourning as a broken attachment (Section 6.7) with four phases: numbness, yearning and searching, disorganisation and despair, and reorganisation. **Margaret Stroebe** and **Henk Schut**'s **dual process model** (1999) replaces fixed stages with oscillation between loss-oriented coping (grief work, yearning) and restoration-oriented coping (new tasks and roles, time off from grief).

Cicely Saunders (1918 to 2005) opened St Christopher's Hospice, London, in 1967; hospices give **palliative care**, aiming at comfort and dignity rather than cure (Section 7.9). Euthanasia is active when a lethal act ends life and **passive euthanasia** when life support is withheld or withdrawn; India permits only the passive kind. In *Aruna Shanbaug v Union of India* (2011), about a Mumbai nurse left vegetative by a 1973 assault, the Supreme Court refused the plea but allowed passive euthanasia with High Court approval. *Common Cause v Union of India* (2018) held that dying with dignity is part of the right to life (Article 21) and upheld the **living will**, an advance directive refusing life support.

EXAM TRAP

Kübler-Ross's stages describe the dying; Bowlby's phases, the bereaved; Stroebe and Schut's model is oscillation, not stages. Shanbaug (2011) allowed passive euthanasia; Common Cause (2018) added living wills.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Arnett called ages 18 to 25 _____ adulthood.
2. Levinson's vision of the adult a young man hopes to become is the _____.
3. Comradeship with different, complementary roles marks a _____ marriage.
4. The end of ovulation and of reproductive capacity is the _____.
5. Cumming and Henry proposed _____ theory.
6. Selection, optimisation and compensation is the model of Paul and Margret _____.
7. A sharp cognitive decline shortly before death is the _____.
8. The dual process model contrasts loss-oriented and _____-oriented coping.

Answers: (1) emerging, (2) Dream, (3) collegial, (4) climacteric, (5) disengagement, (6) Baltes, (7) terminal drop, (8) restoration

The Wild Boy of Aveyron, Revisited

Look at Victor again with this chapter behind you. Pinel and Itard were arguing nature versus nurture (Section 6.1), and the modern answer is that each may have been partly right: Victor may have had a developmental disability, perhaps autism, before he was abandoned (Sections 6.2 and 6.3). His failure to speak after years of teaching fits a **sensitive period** for language that had largely closed (Section 6.1), while his growing problem solving and his bond with Madame Guérin show cognition and attachment still developing (Sections 6.4 to 6.7). His protest at unjust punishment touches moral development (Section 6.9), the turmoil puberty brought him touches adolescence (Section 6.10), and his quiet life to about forty is a reminder that development runs across the whole lifespan (Section 6.11).

Chapter summary

6.1 Foundations: The Lifespan View, Key Issues and Ecological Contexts. Developmental psychology studies change and stability from conception to death. Growth is quantitative, development qualitative, and maturation is the biological unfolding of genetic potential (Gesell; co-twin control, 1929). Hurlock divides life into ten periods; Baltes's lifespan perspective (1987) sees development as lifelong, multidirectional and plastic, shaped by age-graded, history-graded and non-normative influences. Development runs cephalocaudally and proximodistally, with Havighurst's developmental tasks at each period. Core issues include nature versus nurture and continuity versus discontinuity. A critical period is fixed and irreversible (imprinting, embryonic organ formation); a sensitive period is flexible (language). Bronfenbrenner (1979) nests the child in micro-, meso-, exo- and macrosystems, with the chronosystem added later.

6.2 Heredity, Prenatal Development, Birth and the Newborn. Each body cell carries 46 chromosomes in 23 pairs; meiosis halves them in sperm and ovum, which unite as a **zygote**. In **dominant-recessive inheritance** a heterozygote shows the dominant trait, while incomplete dominance, codominance, X-linked, polygenic inheritance and genomic imprinting vary the pattern. Chromosomal errors include Down (trisomy 21), Klinefelter (47,XXY), Turner (45,X) and Prader-Willi (paternal 15q11-q13 deletion). **MZ twins reared apart** best separate heredity from environment, which also correlate and interact. Prenatal life runs through germinal, embryonic (gastrulation, neural tube, greatest vulnerability to **teratogens**) and fetal periods. Newborns are scored on the **Apgar scale** and show primitive reflexes such as the Moro and Babinski.

6.3 Infancy: Brain, Body, Motor Skills and Perception. Birth weight doubles by about 5 months and triples by 1 year, and growth and motor control run **cephalocaudal** and **proximodistal** (never distal to proximal). Neurons are mostly made before birth; synaptogenesis, **synaptic pruning** and myelination take the brain from about 25 percent of adult weight to half by about 6 months. The first voluntary grasp is ulnar; the pincer grasp follows by 9 to 12 months. Gesell stressed maturation, Dennis experience and Thelen dynamic systems. Looking methods reveal face preference and perceptual narrowing; the visual cliff shows depth perception by crawling age, when **independent locomotion** transforms affordances. Newborns imitate (Meltzoff and Moore, 1977), and comprehension leads production.

6.4 Piaget's Theory of Cognitive Development. Jean Piaget (1896 to 1980) saw intelligence as biological adaptation: the child, an active little scientist, builds **schemes** and adapts them by **assimilation** (fitting experience into existing schemes) and **accommodation** (changing schemes), driven by **equilibration**. Four stages follow in invariant order: **sensorimotor** (birth to 2; object permanence begins at 8 to 12 months with the A-not-B error and is complete by 18 to 24), **preoperational** (2 to 7; egocentrism, centration, no conservation), **concrete operational** (7 to 11; conservation, reversibility, transitivity) and **formal operational** (11 or 12 onward; hypothetico-deductive reasoning). Dialectical reasoning is postformal. Baillargeon, Donaldson and the neo-Piagetians showed he underestimated young children.

6.5 Vygotsky, Bruner, Information Processing and Theory of Mind. Lev Vygotsky's sociocultural theory holds that higher mental functions arise in social interaction, mediated by cultural tools (language the master tool): every function appears first between people, then within the child. The **zone of proximal development** is the gap between what a learner can do alone and with a more knowledgeable other; support there is **scaffolding** (Wood, Bruner and Ross, 1976). For Vygotsky language shapes thought and private speech becomes inner speech. Bruner described enactive, iconic and symbolic representation. Information-processing research tracks speed, strategies and metacognition. False-belief tasks are passed at about 4, and Selman traced perspective taking from egocentric to societal.

6.6 Emotional Development, Temperament and Self-Regulation. Bridges (1932) held that the newborn's general excitement differentiates first into **distress and delight**; primary emotions are present within six months, the social smile at 6 to 10 weeks, and the **self-conscious emotions** only after 12 months. Stranger and separation anxiety peak at about 8 to 15 months, and Tronick's still face shows that infants expect affective reciprocity. Temperament is early, moderately stable and biologically based: Thomas and Chess gave nine dimensions, the **easy, difficult and slow-to-warm-up** types and goodness of fit; Rothbart and Bates reduced it to surgency, negative affec-

tivity and effortful control; Kagan studied behavioural inhibition. Mischel's marshmallow test measures **delay of gratification**.

6.7 Attachment: Imprinting, Bowlby, Ainsworth and Beyond. Attachment is an enduring bond shown by proximity seeking, secure-base and safe-haven behaviour and separation distress. Lorenz's **imprinting** and Harlow's **contact comfort** undermined the feeding-based psychoanalytic and secondary-drive theories, leaving Bowlby's **ethological theory** (an evolved system, monotropy, a critical period, the **internal working model**) predominant. Bowlby's four phases, Schaffer and Emerson's stages and Mahler's separation-individuation chart the first three years. Ainsworth's **Strange Situation** sorts infants by reunion behaviour into secure (B), avoidant (A) and resistant (C), plus Main and Solomon's disorganised (D). Spitz, Rutter and the adoption studies show privation harms but is partly reversible.

6.8 Parenting, Family, Play and Gender in Childhood. Diana Baumrind described authoritative, authoritarian and permissive parenting; **Maccoby and Martin** (1983) crossed demandingness with responsiveness and added the neglectful style. Authoritative parenting, high on both, gives the best outcomes, though warm control also works in Chinese and Indian families. Helicopter parenting has several roots, and the family acts as a system. Neglect is physical, supervisory, medical, educational or emotional; licensed day care is not neglect. Piaget's play runs from functional to symbolic play to games with rules, Smilansky added constructive play, and Parten traced solitary to cooperative play. Children learn the gender map through socialisation; **Kohlberg** and gender schema theory explain how they then socialise themselves.

6.9 Moral Development: Piaget, Kohlberg, Gilligan and Beyond. Morality has emotional (guilt, empathy, the superego), behavioural (social learning; Hartshorne and May found honesty situation-specific) and cognitive sides. **Piaget** (1932) traced a shift from **heteronomous morality** (fixed rules, judging by consequences, immanent justice, expiatory punishment) to **autonomous morality** (rules as agreements, intentions, reciprocal punishment). **Kohlberg** posed dilemmas such as Heinz's and scored the reasoning: **preconventional** (punishment, exchange), **conventional** (approval, law and order) and **postconventional** (social contract, universal principles). Critics cite cultural bias (Snarey, Shweder, Miller), the reasoning and behaviour gap and gender bias; **Gilligan** added the female **ethic of care**. Turiel, Eisenberg, Hoffman and Haidt widened the field.

6.10 Erikson's Psychosocial Stages, Identity and Adolescence. Erik Erikson (1902 to 1994), an ego psychologist, proposed eight psychosocial stages governed by the epigenetic principle, each a crisis whose favourable resolution yields a virtue: trust (hope), autonomy (will), initiative (purpose), industry (competence), identity (fidelity), intimacy (love), generativity (care) and integrity (wisdom). One-sided resolution gives maladaptive or malignant tendencies. Adolescence is a psychosocial moratorium whose failure is identity confusion. **James Marcia** (1966) crossed exploration with commitment to give diffusion, foreclosure, moratorium and achievement. **David Elkind** described the imaginary audience and personal fable. Puberty, Steinberg's dual systems model and modern research temper Hall's storm and stress: connection with parents stays healthy.

6.11 Adulthood, Ageing, Death and Dying. Arnett's **emerging adulthood** (about 18 to 25) leads into Levinson's life structures, eras, transitions and Dream, joined by Havighurst's developmental tasks, Vaillant, Gould, Neugarten's social clock and Peck. Hurlock's marriages are traditional, companionship or **collegial** (comradeship with different roles). The **climacteric** is the end of ovulation and reproductive capacity, culminating in menopause; the midlife crisis (Jaques, 1965) is

not universal. Ageing is explained by programmed theories (Hayflick limit, telomeres) and damage theories (free radicals), and adapted to through disengagement, activity, continuity, socioemotional selectivity or SOC. Speed and fluid abilities fall; crystallised knowledge holds. Grief follows Bowlby's phases or Stroebe and Schut's dual process; India allows passive euthanasia and living wills.

Who's who

Table 6.34 The people in this chapter.

Name	When	Examined for
Victor of Aveyron	found 1800; died 1828	The wild boy of Aveyron, educated by Itard; nature versus nurture and language deprivation (chapter opening case)
Jean-Marc-Gaspard Itard	1774 to 1838	Physician who trained Victor of Aveyron (reports of 1801 and 1806); pioneer of special education
Philippe Pinel	1745 to 1826	Paris asylum reformer who judged Victor of Aveyron an incurable idiot (1800)
Elizabeth Hurlock	1898 to 1988	Developmental Psychology: A Life-Span Approach; ten periods of the lifespan; principles of development
Paul Baltes	1939 to 2006	Lifespan perspective (1987); normative age-graded, history-graded and non-normative influences
John Locke	1632 to 1704	Tabula rasa, the blank slate (nurture)
Jean-Jacques Rousseau	1712 to 1778	Emile (1762); the child as noble savage, natural unfolding in stages (nature)
Charles Darwin	1809 to 1882	Baby biography of his son (1877)
Wilhelm Preyer	1841 to 1897	The Mind of the Child (1882), start of scientific child psychology
G. Stanley Hall	1844 to 1924	Child study movement; Adolescence (1904); storm and stress; recapitulation
Arnold Gesell	1880 to 1961	Maturation; normative approach; co-twin control with Thompson (1929); developmental schedules
John B. Watson	1878 to 1958	Behaviourism; development moulded by environment (nurture)
Jean Piaget	1896 to 1980	Accommodation; active child; universal cognitive stages
Helen Thompson	1929	Co-twin control study of stair climbing with Gesell
Myrtle McGraw	1899 to 1988	Growth: A Study of Johnny and Jimmy (1935); training speeds ontogenetic, not phylogenetic, skills
Robert J. Havighurst	1900 to 1991	Developmental tasks; Developmental Tasks and Education (1948)
Sigmund Freud	1856 to 1939	Psychosexual stages; early experience decisive
Erik Erikson	1902 to 1994	Psychosocial stages
John Bowlby	1907 to 1990	Ethological attachment theory; importance of early experience
Ann Clarke	1976	Early Experience: Myth and Evidence, with Alan Clarke; later experience can repair early deficits
Alan Clarke	1976	Early Experience: Myth and Evidence, with Ann Clarke
Lev Vygotsky	1896 to 1934	Sociocultural theory; context-specific development
Urie Bronfenbrenner	1917 to 2005	Ecological systems theory, The Ecology of Human Development (1979); bioecological model and PPCT; Head Start (1965)
Konrad Lorenz	1903 to 1989	Imprinting in greylag goslings (1935); critical periods
Eric Lenneberg	1921 to 1975	Critical period hypothesis for language (1967)
Genie	found 1970	Isolated child who learned words but never mastered grammar; sensitive period for language
Noam Chomsky	born 1928	Nativism: language ability is innate
Michael Rutter	1933 to 2021	English and Romanian Adoptees study; recovery after early deprivation
Esther Thelen	1941 to 2004	Dynamic systems theory of motor development
B. F. Skinner	1904 to 1990	Operant conditioning; learning theory of development
Albert Bandura	1925 to 2021	Observational learning; learning theory of development

Name	When	Examined for
Nikolaas Tinbergen	1907 to 1988	Ethology
Stephen Ceci	1994	Bioecological model with Bronfenbrenner
Pamela Morris	1998	Bioecological model and PPCT with Bronfenbrenner
Gregor Mendel	1822 to 1884	Dominant-recessive inheritance, from pea plants (1866)
John Langdon Down	1828 to 1896	Described Down syndrome (1866)
Albert Stunkard	1990	Twins reared apart: body-mass index of identical twins nearly as alike apart as together
Sandra Scarr	1983	Passive, evocative and active (niche-picking) gene-environment correlation, with Kathleen McCartney
Kathleen McCartney	1983	Gene-environment correlation, with Sandra Scarr
Irving Gottesman	1963	Reaction range
Conrad Hal Waddington	1905 to 1975	Canalisation
Virginia Apgar	1909 to 1974	Apgar scale for the newborn (1952, published 1953)
T. Berry Brazelton	1918 to 2018	Neonatal Behavioral Assessment Scale (1973)
Peter Wolff	1966	Newborn states of arousal
William Greenough	1987	Experience-expectant versus experience-dependent plasticity
Wayne Dennis	1940; 1960	Hopi cradleboard study (1940, with Marsena Dennis); severe motor delays in Iranian orphanages (1960)
Marsena Dennis	1940	Co-author of the Hopi cradleboard study
Robert Fantz	1961	Preferential looking; infants prefer patterns, and faces most of all
Renée Baillargeon	1985	Violation-of-expectation method
Anthony DeCasper	1980; 1986	Newborns prefer the mother's voice (with Fifer, 1980) and a story heard in the womb (with Spence, 1986)
William Fifer	1980	With DeCasper, newborn preference for the mother's voice
Melanie Spence	1986	With DeCasper, The Cat in the Hat study of prenatal learning
Janet Werker	1984	With Tees, perceptual narrowing: English-learning infants lose a Hindi contrast by 10 to 12 months
Richard Tees	1984	Co-author with Werker of the perceptual narrowing study
Elizabeth Spelke	1976	Intermodal perception: infants look at the film that matches the soundtrack
Eleanor Gibson	1910 to 2002	Visual cliff with Walk (1960); ecological view of perception and affordances
Richard Walk	1960	Visual cliff with Eleanor Gibson
Joseph Campos	1970 onwards	Heart-rate studies on the visual cliff; locomotor experience brings wariness of heights
James J. Gibson	1904 to 1979	Ecological view of perception; affordances
Carolyn Rovee-Collier	1942 to 2014	Mobile-kicking (operant conditioning) studies of infant memory
Andrew Meltzoff	1977	With Moore, neonatal imitation at 12 to 21 days
M. Keith Moore	1977	Co-author with Meltzoff of the neonatal imitation study
Théodore Simon	1873 to 1961	Co-author of the Binet-Simon scale; Piaget scored test items in his Paris laboratory around 1920
Alfred Binet	1857 to 1911	First intelligence scale with Simon (1905)
Burrhus Frederic Skinner	1904 to 1990	Operant conditioning (a distractor in Piaget items)
Lawrence Kohlberg	1927 to 1987	Moral stages (a distractor for who proposed the cognitive stages)
Adele Diamond	1985	A-not-B error grows with delay; linked to prefrontal maturation and inhibition of a prepotent reach
Judy DeLoache	1987	Scale-model task; dual representation
Bärbel Inhelder	1913 to 1997	Three mountains task with Piaget; The Growth of Logical Thinking (1958)
Arthur Robert Jensen	1923 to 2012	Level I and Level II abilities (a distractor in the 'preschoolers cannot reason' item)
Joy Paul Guilford	1897 to 1987	Structure of intellect (a distractor in the 'preschoolers cannot reason' item)
Klaus Riegel	1973	Dialectical operations as the final period of cognitive development (postformal)

Name	When	Examined for
William Perry	1970	College students move from dualism through relativism to commitment
Patricia Arlin	1975	Problem finding as a fifth, postformal stage
Michael Basseches	1984	Dialectical thinking in adulthood
Gisela Labouvie-Vief	1980	Pragmatic postformal thought
Jan Sinnott	1998	Relativistic postformal thought
Margaret Donaldson	1926 to 2020	Children's Minds (1978): children reason well when tasks make human sense
Martin Hughes	1975	Policeman task: preschoolers take two perspectives when hiding makes sense
James McGarrigle	1974	Naughty teddy conservation study with Donaldson
Juan Pascual-Leone	1970	Neo-Piagetian M-space (mental capacity that grows with age)
Robbie Case	1944 to 2000	Neo-Piagetian: stage progress through more efficient use of working memory
Kurt Fischer	1943 to 2020	Neo-Piagetian skill theory (1980)
Lev Semyonovich Vygotsky	1896 to 1934	Sociocultural theory; zone of proximal development; general genetic law; private and inner speech; Thinking and Speech (1934), Mind in Society (1978)
David Wood	1976	Coined scaffolding with Bruner and Ross (the block-pyramid tutoring study)
Gail Ross	1976	Co-author of the scaffolding study with Wood and Bruner
Jerome Seymour Bruner	1915 to 2016	Enactive, iconic and symbolic modes of representation (1966); spiral curriculum and discovery learning (The Process of Education, 1960); scaffolding (1976); LASS
Barbara Rogoff	1990	Guided participation; development as apprenticeship (Apprenticeship in Thinking)
Annemarie Palincsar	1984	Reciprocal teaching, with Ann Brown
Ann Brown	1984	Reciprocal teaching, with Annemarie Palincsar
Reuven Feuerstein	1921 to 2014	Dynamic assessment; Learning Potential Assessment Device (1979)
Laura E. Berk	1980s to 1990s	Research supporting Vygotsky on private speech
Robert V. Kail	1991	Processing speed rises with age at a similar rate across tasks
John H. Flavell	1970, 1979	Production deficiency; metacognition and metamemory (1979); appearance-reality distinction; recursive thought study (1970)
Patricia H. Miller	1970, 1990	Utilisation deficiency; recursive thought study with Kessel and Flavell (1970)
Frank Kessel	1970	Recursive thought study with Miller and Flavell
Michelene Chi	1978	Child chess experts out-recall adult novices: the knowledge base
Robert S. Siegler	1996	Overlapping waves model of strategy development
Charles Brainerd	1990	Fuzzy-trace theory (verbatim and gist traces), with Valerie Reyna
Valerie Reyna	1990	Fuzzy-trace theory, with Charles Brainerd
David Premack	1925 to 2015	Coined theory of mind with Guy Woodruff (1978), asking whether the chimpanzee has one
Guy Woodruff	1978	Coined theory of mind with David Premack
Gyorgy Gergely	1995	Teleological stance (with Gergely Csibra): toddlers understand intentions in action
Gergely Csibra	1995	Teleological stance, with Gyorgy Gergely
John Searle	1983	Philosopher who distinguished prior intentions from intentions in action
Henry Wellman	1990	Desire psychology at about 2, belief-desire psychology at 3 to 4
Heinz Wimmer	1983	First false-belief task (Maxi and the chocolate) with Josef Perner; second-order belief (1985)
Josef Perner	1983	False-belief Maxi task with Wimmer (1983); second-order theory of mind (1985); Smarties task (1987)
Susan Leekam	1987	Smarties unexpected-contents task, with Perner and Wimmer
Simon Baron-Cohen	1985	Sally-Anne task with Leslie and Frith; theory-of-mind deficit (mindblindness) in autism
Alan Leslie	1985	Sally-Anne task, with Baron-Cohen and Frith

Name	When	Examined for
Uta Frith	1985	Sally-Anne task, with Baron-Cohen and Leslie
Katherine Nelson	1973	Referential style of early vocabulary (a wrong option for recursive thought)
Robert L. Selman	1976, 1980	Levels of social perspective taking, from egocentric to societal; the Holly dilemma
Katharine Banham Bridges	1932	Differentiation theory: general excitement splits first into distress and delight
John Broadus Watson	1878 to 1958	Three innate emotions: fear, rage and love
Carroll Izard	1923 to 2017	Differential emotions theory: basic emotions as separate innate systems
Michael Lewis	1979	Timetable of primary and self-conscious emotions; rouge test with Brooks-Gunn
Jeanne Brooks-Gunn	1979	Rouge (mirror self-recognition) test with Lewis, typical by 18 to 24 months
Jerome Kagan	1929 to 2021	Behavioural inhibition and high-reactive infants; cross-cultural separation protest peaking at 13 to 15 months
René Spitz	1887 to 1974	Eight-month anxiety (stranger anxiety)
James Sorce	1985	Social referencing on the visual cliff: no infant crossed when the mother looked afraid
Edward Tronick	1978	Still-face paradigm: infants expect affective reciprocity
Claire Kopp	1982	Control, then self-control, then self-regulation
Mary Rothbart	1981	Temperament as reactivity and self-regulation; Infant Behavior Questionnaire; structural model with Bates
John Bates	1998	Structural model of temperament with Rothbart: surgency, negative affectivity, effortful control
Alexander Thomas	1914 to 2003	New York Longitudinal Study (1956); nine dimensions; easy, difficult, slow-to-warm-up; goodness of fit
Stella Chess	1914 to 2007	New York Longitudinal Study with Thomas; difficult child syndrome; goodness of fit
Arnold Buss	1984	EAS model of temperament with Plomin: emotionality, activity, sociability
Robert Plomin	1984	EAS model of temperament with Buss
Walter Mischel	1930 to 2018	Marshmallow test of delay of gratification at Stanford's Bing Nursery School; hot and cool systems
Yuichi Shoda	1990	Follow-up linking preschool waiting to later competence and SAT scores
Philip Peake	1990	Co-author of the 1990 marshmallow follow-up
Tyler Watts	2018	Marshmallow replication: about half the original effect
Greg Duncan	2018	Co-author of the 2018 marshmallow replication
Haonan Quan	2018	Co-author of the 2018 marshmallow replication
Hippocrates	about 460 to 370 BC	Four humours, the ancient distractor to Thomas and Chess's types
Galen	AD 129 to about 216	Turned the humours into four temperaments: sanguine, choleric, melancholic, phlegmatic
John Dollard	1900 to 1980	With Neal Miller, secondary drive ('cupboard love') theory of attachment (1950)
Neal Miller	1909 to 2002	With John Dollard, secondary drive theory of attachment (1950)
Robert Sears	1908 to 1989	Learning (secondary drive) account of dependency and attachment
Eckhard Hess	1916 to 1986	Imprinting in mallard ducklings peaks about 13 to 16 hours after hatching
Karl von Frisch	1886 to 1982	Ethologist; shared the 1973 Nobel Prize with Lorenz and Tinbergen
Harry Harlow	1905 to 1981	Wire and cloth surrogate mothers: contact comfort, 'The Nature of Love' (1958); isolation and motherless mothers
Margaret Harlow	1918 to 1971	Harry Harlow's wife and colleague in the contact comfort research
Stephen Suomi	1972	With Harlow, rehabilitation of isolate monkeys by younger 'therapist' monkeys
James Robertson	1952	Film A Two-Year-Old Goes to Hospital; protest, despair and detachment
Joyce Robertson	1967 to 1973	Films of young children in brief separations, with James Robertson

Name	When	Examined for
Rudolph Schaffer	1964	With Peggy Emerson, 60 Glasgow infants; specific attachment from about 7 months; attachment to the most responsive person
Peggy Emerson	1964	With Rudolph Schaffer, stages of attachment in Glasgow infants
Margaret Mahler	1897 to 1985	Separation-individuation; rapprochement at about 15 to 24 months; object constancy
Fred Pine	1975	Co-author with Mahler of The Psychological Birth of the Human Infant
Anni Bergman	1975	Co-author with Mahler of The Psychological Birth of the Human Infant
Mary Ainsworth	1913 to 1999	Strange Situation; secure, avoidant and resistant types; maternal sensitivity hypothesis; Uganda and Baltimore studies
Barbara Wittig	1969	With Ainsworth, first report of the Strange Situation
Mary Blehar	1978	Co-author of Patterns of Attachment
Everett Waters	1978	Co-author of Patterns of Attachment
Sally Wall	1978	Co-author of Patterns of Attachment
Mary Main	1986, 1990	With Judith Solomon, the disorganised or disoriented (D) type; Adult Attachment Interview
Judith Solomon	1986, 1990	With Mary Main, the disorganised or disoriented (D) type
Marinus van IJzendoorn	1988	With Kroonenberg, cross-cultural meta-analysis: secure most common everywhere; more variation within than between cultures
Pieter Kroonenberg	1988	With van IJzendoorn, cross-cultural meta-analysis of the Strange Situation
Klaus Grossmann	1985	Many avoidant (A) infants in northern Germany, from a cultural emphasis on early independence
Karin Grossmann	1985	With Klaus Grossmann, northern German attachment samples
Barbara Tizard	1926 to 2015	With Jill Hodges, follow-up of children from London residential nurseries
Jill Hodges	1989	With Barbara Tizard, institution-reared children followed to age 16
Donald Winnicott	1896 to 1971	Good-enough mother; transitional object
Cindy Hazan	1987	With Phillip Shaver, attachment styles in adult romantic love
Phillip Shaver	1987	With Cindy Hazan, adult romantic attachment
Kim Bartholomew	1991	With Leonard Horowitz, four adult styles: secure, preoccupied, dismissing, fearful
Leonard Horowitz	1991	With Kim Bartholomew, four-category model of adult attachment
Jay Belsky	1986	Day care debate: more than 20 hours a week in the first year as a risk to security
John Watson	1878 to 1958	With Rosalie Rayner, Little Albert (conditioned fear); a distractor here
Rosalie Rayner	1920	With John Watson, Little Albert
Edward Tolman	1886 to 1959	Cognitive map (not the internal working model)
Hans Eysenck	1916 to 1997	Personality; a distractor for anaclitic depression
Carl Rogers	1902 to 1987	Person-centred theory; a distractor for attachment theory
Diana Baumrind	1927 to 2018	Parenting styles: authoritative, authoritarian, permissive (papers from 1966 to 1971)
Eleanor Maccoby	1917 to 2018	With John Martin (1983): demandingness and responsiveness, neglectful style; gender segregation; with Jacklin, sex differences (1974)
John Martin	1983	With Maccoby: two dimensions of parenting; added the neglectful (uninvolved) style
Ruth Chao	1994	Chinese parenting as training (chiao shun), not authoritarian control
Haim Ginott	1922 to 1973	Helicopter image in Between Parent and Teenager (1969)
Amy Chua	2011	Tiger parenting (her memoir of strict, achievement-driven parenting)
Martin Hoffman	1967	Discipline types: induction, power assertion, love withdrawal
Alfred Adler	1870 to 1937	Birth order and personality (Chapter 4)
E. Mavis Hetherington	2002	Long-term studies of divorce and remarriage
C. Henry Kempe	1922 to 1984	Battered child syndrome (1962)
Sara Smilansky	1968	Added constructive play; sociodramatic play
Mildred Parten	1932	Six stages of social play: unoccupied to cooperative
Friedrich Schiller	1759 to 1805	Surplus energy theory of play

Name	When	Examined for
Herbert Spencer	1820 to 1903	Surplus energy theory of play
Moritz Lazarus	1824 to 1903	Relaxation or recreation theory of play
Karl Groos	1861 to 1946	Pre-exercise or practice theory of play
John Coie	1982	Sociometric status groups (with Dodge)
Kenneth Dodge	1982	Sociometric status groups (with Coie)
Dan Olweus	1931 to 2020	Definition of bullying; whole-school prevention programme
Kay Bussey	1999	Social cognitive theory of gender (with Bandura)
Ronald Slaby	1975	With Frey: gender constancy and attention to same-sex models
Karin Frey	1975	With Slaby: gender constancy and attention to same-sex models
Sandra Bem	1944 to 2014	Gender schema theory (1981); androgyny and the Bem Sex Role Inventory (1974)
Carol Martin	1981	Gender schema theory (with Halverson); memory for gender-inconsistent pictures (1983)
Charles Halverson	1981	Gender schema theory (with Martin)
Carol Jacklin	1974	With Maccoby: review of sex differences
Janet Shibley Hyde	2005	Gender similarities hypothesis
Carol Gilligan	Born 1936	In a Different Voice (1982): ethic of care (women) versus ethic of justice (men); levels of care
Hugh Hartshorne	Character Education Inquiry, 1928 to 1930	With Mark May: honesty is largely situation-specific
Mark May	Character Education Inquiry, 1928 to 1930	With Hugh Hartshorne: honesty is largely situation-specific
Frederick McDonald	1963	With Bandura: modelling shifted children's judgments between consequences and intentions
James Rest	1970s	Defining Issues Test, a questionnaire measure of moral reasoning
John Snarey	1985	Cross-cultural review of Kohlberg research: 45 studies, 27 cultural areas; stage 5 mainly in urban, educated groups
Richard Shweder	1980s Bhubaneswar studies	Indian research: ethics of autonomy, community and divinity
Joan Miller	1980s to 1990s Mysore studies	Indian and American comparisons: Indians treat interpersonal duties as moral obligations
Lawrence Walker	1984	Review finding little sex difference in Kohlberg scores
Elliot Turiel	1983	Social domain theory: moral, social-conventional and personal domains
Nancy Eisenberg	1980s	Levels of prosocial moral reasoning: hedonistic, needs-oriented, approval-oriented, empathic, internalised
Jonathan Haidt	2001	Social intuitionist model: intuition first, reasoning follows
Diane Papalia	Human Development textbook	Alternative stage labels: social concern and conscience (stage 4); morality of autonomous moral principles (Level III)
Robert Selman	1976, 1980	Levels of social perspective taking (Section 6.5); role-taking underlies Kohlberg's stages
Erik Homburger Erikson	1902 to 1994	Eight psychosocial stages; coined identity crisis; Childhood and Society (1950), Young Man Luther (1958), Identity: Youth and Crisis (1968), Gandhi's Truth (1969)
Joan Erikson	1903 to 1997	Erikson's collaborator; added a ninth stage for the very old (1997)
Anna Freud	1895 to 1982	Analysed Erikson; older psychoanalytic view that healthy adolescents detach from parents
Harry Stack Sullivan	1892 to 1949	Developmental epochs from infancy to adulthood; not the oral-sensory to maturity scheme
Melanie Klein	1882 to 1960	Paranoid-schizoid and depressive positions; not the identity crisis
Eric Berne	1910 to 1970	Transactional analysis; a distractor for the identity crisis
Martin Luther	1483 to 1546	Subject of Erikson's psychohistory Young Man Luther (1958)

Name	When	Examined for
Mohandas Karamchand Gandhi	1869 to 1948	Subject of Erikson's psychohistory Gandhi's Truth (1969), centred on the 1918 Ahmedabad mill strike
James E. Marcia	1966	Four identity statuses from crisis (exploration) and commitment; MAMA cycles (1992)
Jean Phinney	1989	Ethnic identity: unexamined, exploration, achieved
David Elkind	1967	Adolescent egocentrism: imaginary audience and personal fable
Laurence Steinberg	2008	Dual systems model of adolescent risk-taking
Margo Gardner	2005	With Steinberg, peers watching doubled adolescents' risk-taking in a driving game
Margaret Mead	1901 to 1978	Coming of Age in Samoa (1928): adolescent turmoil is cultural
Derek Freeman	1983	Criticised Mead's Samoan fieldwork
Peter Blos	1967	Psychoanalytic view of adolescence as detachment (second individuation) from parents
Dexter Dunphy	1963	Cliques and crowds; five stages of adolescent peer group structure
Jeffrey Jensen Arnett	2000	Emerging adulthood (about 18 to 25): identity exploration, instability, self-focus, feeling in-between, possibilities
Daniel Levinson	1920 to 1994	The Seasons of a Man's Life (1978): life structure, eras and transitions, the Dream and the mentor, midlife transition at about 40 to 45
George Vaillant	born 1934	Grant Study of Harvard men; added career consolidation and keeper of the meaning to Erikson's stages
Roger Gould	1978	Transformations: adult growth as shedding false childhood assumptions
Bernice Neugarten	1916 to 2001	Social clock; on-time and off-time life events
Robert Peck	1950s	Tasks of old age: ego differentiation, body transcendence, ego transcendence
Robert Sternberg	born 1949	Triangular theory of love (Chapter 3)
Eli Ginzberg	1911 to 2002	Occupational Choice (1951): fantasy, tentative and realistic periods of career choice
Donald Super	1910 to 1994	Life-span career stages (Chapter 7)
Elliott Jaques	1917 to 2003	Coined the term midlife crisis (1965)
K. Warner Schaie	1956	Seattle Longitudinal Study: most mental abilities peak in middle adulthood
Robert Butler	1927 to 2010	Coined ageism (1969); life review
Jeanne Calment	1875 to 1997	Longest confirmed human life, 122 years
Robert Kleemeier	1962	Terminal drop in cognitive scores before death
Leonard Hayflick	1961	Hayflick limit on cell divisions; cellular clock theory of ageing
Paul Moorhead	1961	With Hayflick, showed that cultured human cells divide a limited number of times
Denham Harman	1956	Free-radical theory of ageing
Elaine Cumming	1961	Disengagement theory (Growing Old, with William Henry)
William Henry	1961	Disengagement theory (Growing Old, with Elaine Cumming)
Robert Atchley	1989	Continuity theory of ageing; phases of retirement
Lars Tornstam	1989	Gerotranscendence
John Rowe	1997	Successful ageing (with Robert Kahn)
Robert Kahn	1997	Successful ageing (with John Rowe)
Laura Carstensen	born 1953	Socioemotional selectivity theory
Margret Baltes	1939 to 1999	Selective optimisation with compensation (1990, with Paul Baltes)
Arthur Rubinstein	1887 to 1982	Pianist used by the Baltes couple to illustrate selective optimisation with compensation
Timothy Salthouse	1996	Processing-speed theory of cognitive ageing
Aruna Shanbaug	1948 to 2015	Nurse in a vegetative state; her 2011 Supreme Court case allowed passive euthanasia in India
Maria Nagy	1948	Children's understanding of death: three stages in 378 Budapest children
Mark Speece	1984	With Sandor Brent: irreversibility, universality and nonfunctionality in children's death concept

Name	When	Examined for
Sandor Brent	1984	With Mark Speece: irreversibility, universality and nonfunctionality in children's death concept
Elisabeth Kubler-Ross	1926 to 2004	On Death and Dying (1969): denial, anger, bargaining, depression, acceptance
Margaret Stroebe	1999	Dual process model of grief (with Henk Schut)
Henk Schut	1999	Dual process model of grief (with Margaret Stroebe)
Cicely Saunders	1918 to 2005	Founded St Christopher's Hospice, London (1967); modern hospice and palliative care

Key terms

Developmental psychology, developmental perspective, Growth, Development, lifespan perspective, Normative age-graded influences, Normative history-graded influences, Non-normative influences, tabula rasa, baby biography, recapitulation, normative approach, maturation, Learning, socialisation, co-twin control, cephalocaudal, proximodistal, developmental task, nature versus nurture, Continuity versus discontinuity, learnt behaviour, Innate behaviour, overt behaviour, covert behaviour, critical period, sensitive period, ecological systems theory, microsystem, mesosystem, exosystem, macrosystem, chronosystem, bioecological model, proximal process, chromosomes, gene, mitosis, meiosis, zygote, monozygotic twins, dizygotic twins, genotype, phenotype, dominant-recessive inheritance, phenylketonuria, incomplete dominance, codominance, X-linked inheritance, Polygenic inheritance, genomic imprinting, Down syndrome, Heritability, gene-environment correlation, gene-environment interaction, reaction range, canalisation, Epigenetics, germinal period, embryonic period, gastrulation, fetal period, teratogen, fetal alcohol spectrum disorder, preterm, small for gestational age, Kangaroo care, Apgar scale, Neonatal Behavioral Assessment Scale, Primitive reflexes, cephalocaudal trend, proximodistal trend, marasmus, kwashiorkor, Neurogenesis, synaptogenesis, synaptic pruning, Myelination, Experience-expectant plasticity, Experience-dependent plasticity, gross motor skills, fine motor skills, mass to specific, palmar grasp reflex, prereaching, ulnar grasp, radial grasps, pincer grasp, maturational view, dynamic systems theory, preferential looking, habituation-dishabituation, habituation, dishabituation, high-amplitude sucking, violation-of-expectation, Evoked potentials, visual acuity, perceptual narrowing, Intermodal perception, visual cliff, affordance, independent locomotion, receptive-expressive gap, genetic epistemology, clinical method, constructivism, scheme, functional invariants, Adaptation, assimilation, accommodation, equilibration, operations, sensorimotor stage, deferred imitation, object permanence, A-not-B error, symbolic function, preoperational stage, Egocentrism, Animism, Centration, Irreversibility, Transductive reasoning, class inclusion, Conservation, Identity, Reversibility, horizontal décalage, concrete operational stage, Decentration, Transitive inference, formal operational stage, hypothetico-deductive reasoning, counterfactual reasoning, postformal thought, dialectical reasoning, Neo-Piagetian theories, sociocultural theory, higher mental functions, mediation, tools of intellectual adaptation, general genetic law of cultural development, internalisation, intersubjectivity, zone of proximal development, more knowledgeable other, scaffolding, guided participation, reciprocal teaching, private speech, pseudoconcept, modes of representation, spiral curriculum, discovery learning, information-processing approach, executive functions, production deficiency, utilisation deficiency, metacognition, metamemory, overlapping waves model, fuzzy-trace theory, theory of mind, joint attention, teleological stance, belief-desire psychology, false-belief task, appearance-reality distinction, Sally-Anne task, second-order theory of mind, recursive thought, social perspective taking, differentiation theory, differential emotions theory, primary emotions, reflexive smile, social

smile, self-conscious emotions, mirror self-recognition, Stranger anxiety, Separation protest, social referencing, still-face paradigm, Emotion regulation, display rules, Temperament, New York Longitudinal Study, easy child, difficult child, slow-to-warm-up child, difficult child syndrome, goodness of fit, Infant Behavior Questionnaire, structural model of temperament, Surgency, Negative affectivity, Effortful control, high-reactive infants, Behavioural inhibition, EAS model, self-regulation, marshmallow test, delay of gratification, Sustained attention, Resistance to temptation, forbidden-toy paradigm, attachment, secure base, safe haven, separation distress, secondary drive, person permanence, ethological theory, imprinting, contact comfort, affectionless psychopathy, maternal deprivation hypothesis, Monotropy, internal working model, protest-despair-detachment, Clear-cut attachment, goal-corrected partnership, multiple attachments, separation-individuation, rapprochement, emotional refuelling, object constancy, Strange Situation, maternal sensitivity hypothesis, Secure attachment, Insecure-avoidant attachment, Insecure-resistant attachment, Disorganised attachment, hospitalism, anaclitic depression, deprivation, privation, good-enough mother, transitional object, Adult Attachment Interview, parenting style, Authoritative parenting, Authoritarian parenting, Permissive parenting, demandingness, responsiveness, neglectful parenting, Helicopter parenting, battered child syndrome, Child maltreatment, neglect, functional play, symbolic play, games with rules, constructive play, Sociodramatic play, Rough-and-tumble play, parallel play, associative play, cooperative play, Sociometric status, Bullying, Gender, Gender identity, gender role, gender stereotype, Gender typing, Gender socialisation, gender segregation, congenital adrenal hyperplasia, gender constancy, Gender schema theory, androgyny, Moral development, superego, moral reasoning, situation-specific, premoral period, heteronomous morality, moral realism, objective responsibility, immanent justice, expiatory punishment, autonomous morality, moral relativism, reciprocal punishment, Heinz dilemma, Moral Judgment Interview, preconventional level, punishment-and-obedience orientation, instrumental purpose and exchange, conventional level, good boy-nice girl orientation, law-and-order orientation, postconventional level, social-contract orientation, invariant sequence, role-taking, Defining Issues Test, ethic of justice, ethic of care, social domain theory, prosocial moral reasoning, empathy, Induction, social intuitionist model, ego psychology, epigenetic principle, psychosocial crisis, virtue, identity crisis, generativity, stagnation, ego integrity, maladaptive tendency, malignant tendency, ego identity, psychosocial moratorium, identity confusion, negative identity, identity statuses, identity diffusion, foreclosure, identity achievement, MAMA cycles, ethnic identity, adolescent egocentrism, imaginary audience, personal fable, puberty, adrenarche, gonadarche, primary sex characteristics, secondary sex characteristics, menarche, spermarche, secular trend, dual systems model, storm and stress, developmental tasks of adolescence, cliques, emerging adulthood, life structure, the Dream, midlife transition, keeper of the meaning, Social clock, Traditional marriage, Companionship marriage, Collegial marriage, presbyopia, climacteric, menopause, andropause, midlife crisis, empty nest, sandwich generation, Gerontology, ageism, life expectancy, Primary ageing, terminal drop, Hayflick limit, disengagement theory, activity theory, Continuity theory, Gerotranscendence, socioemotional selectivity theory, selective optimisation with compensation, processing-speed theory, Berlin wisdom paradigm, brain death, dual process model, palliative care, passive euthanasia, living will

From the past papers

1. Which term refers to the biological unfolding of genetic potential? (IHBAS 2025)

   (a) Maturation (b) Socialization (c) Accommodation (d) Conditioning

   Answer: (a) Maturation. Maturation (Arnold Gesell) is change on a genetic timetable, largely independent of practice. Socialisation, accommodation (Piaget's adjustment of a scheme) and conditioning all depend on experience.

2. The fact that people can use later experiences to compensate for earlier deficits is a contradiction of which element of earlier developmental psychology? (IHBAS 2020)

   (a) Critical periods (b) Continuity (c) Sensitive periods (d) Stability

   Answer: (a) Critical periods. A strict critical period makes a missed experience irreversible, so recovery through later experience (Clarke and Clarke, 1976; the English and Romanian Adoptees study) contradicts it. Sensitive periods allow late recovery by definition; continuity and stability concern other questions.

3. In the case of human development, the strongest evidence for the critical period hypothesis is found in: (RML 2024)

   (a) Imprinting (b) Language acquisition (c) Embryonic development (d) Development of vocational interests

   Answer keyed: (a) Imprinting. Imprinting (Lorenz) is the textbook critical period, but it is animal evidence; within human development the clearest critical periods are in embryonic development, where teratogen timing decides which organ is damaged. Language shows a sensitive period; vocational interests show none.

4. If a blue eye allele (D) is dominant and a green eye allele (d) is recessive, what is the phenotype of a heterozygous (Dd) child? (Real PYQ 2025)

   (a) Green eyes (b) Blue eyes (c) Blue-green eyes (d) Cannot be determined

   Answer: (b) Blue eyes. In dominant-recessive inheritance a heterozygote shows the dominant allele's trait. Green eyes need the dd genotype, a blue-green blend would need incomplete dominance, and the phenotype can be determined because dominance is stated.

5. During embryonic development, the nervous system begins forming at the : (IHBAS 2025)

   (a) Zygote stage (b) Morula stage (c) Gastrula stage (d) Blastocyst stage

   Answer: (c) Gastrula stage. At gastrulation, in week 3, the three germ layers form and the neural plate arises from ectoderm, folding into the neural tube by about week 4. The zygote, morula and blastocyst are earlier stages with no germ layers yet.

6. A child with hyperphagia, obesity, short stature, and hypotonia is found to have a deletion on chromosome 15 (15q11-q13). This syndrome is : (IHBAS 2025)

   (a) Fragile X syndrome (b) Turner syndrome (c) Prader-Willi syndrome (d) Klinefelter syndrome

   Answer: (c) Prader-Willi syndrome. Loss of the paternal 15q11-q13 region gives hypotonia, hyperphagia, obesity and short stature; loss of the maternal copy gives Angelman syndrome. Fragile X is an FMR1 disorder on the X chromosome, Turner is 45,X and Klinefelter 47,XXY.

7. Human motor development is governed by certain basic principles. Which of the following is Not a basic principle of motor development? (RML 2021)

   (a) Maturation of the motor system proceeds from gross motor to fine motor behaviours. (b) Motor development is from cephalo-to-caudal or from head to feet. (c) Motor development proceeds from the distal to the proximal or from the extremities to the central part of the body. (d) Motor development plays a major role in social and behavioural expectations.

   Answer: (c) Distal to proximal. The proximodistal principle runs from the centre of the body to the extremities, so (c) reverses it. Gross to fine, cephalocaudal and the social role of motor development are all genuine principles.

8. What is the principal factor that changes the affordances available to the toddler, compared with the infant? (RML 2022)

   (a) Independent locomotion (b) Starting to speak (c) The attachment relationship (d) New kinds of caretaking

   Answer: (a) Independent locomotion. Affordances are the action possibilities the environment offers a perceiver, so they change most when the child's own action abilities change: crawling and walking make toys reachable and drops dangerous. Speech, attachment and caretaking change the social world, not what the surroundings afford action.

9. The process of synaptic pruning is most closely associated with : (RML 2023)

   (a) Memory consolidation (b) Neuronal development and plasticity (c) Neurotransmitter release (d) Long-term potentiation

   Answer: (b) Neuronal development and plasticity. The young brain overproduces synapses and prunes the little-used ones, fitting circuits to experience. Memory consolidation, neurotransmitter release and long-term potentiation describe how memories and signals work at existing synapses, not the developmental sculpting of circuits.

10. Which is not attained in Piaget's concrete operational stage of cognitive development? (JEMAS 2025)

    (a) Reduced egocentrism (b) Deferred imitation (c) Hypothetico deductive reasoning (d) Transitive inference

    Answer: (c) Hypothetico deductive reasoning. It is the hallmark of the formal operational stage (11 or 12 onward). Reduced egocentrism and transitive inference are concrete operational gains, and deferred imitation was attained earlier, at 18 to 24 months in the sensorimotor stage, so it is present rather than missing.

11. A 5-year-old child is asked to sort a deck of cards. The cards can be sorted by color (red/black) or by suit (hearts, spades, etc.). The child successfully sorts by color but cannot then re-sort the same deck by suit. This inflexibility is an example of: (Real PYQ 2024)

    (a) Egocentrism (b) Animism (c) Centration (d) Transductive reasoning

    Answer: (c) Centration. The preoperational child focuses on one salient feature (colour) and ignores others (suit). Egocentrism is failure to take another's perspective, animism is giving life to objects, and transductive reasoning runs from particular to particular.

12. Formal operational reasoning involves all the following EXCEPT: (RML 2022)

    (a) hypothetical reasoning (b) counterfactual reasoning (c) dialectical reasoning (d) metacognitive reasoning

    Answer: (c) Dialectical reasoning. Integrating opposing views is postformal thought (Riegel 1973, Basseches 1984). Hypothetical, counterfactual and metacognitive reasoning all belong to Piaget's formal operational stage.

13. Vygotsky's concept of the 'zone of proximal development' (ZPD) refers to: (RML 2025)

    (a) The difference between what a learner can do independently and what they can achieve with guidance. (b) A measure of a person's fixed learning capacity across cognitive domains. (c) A developmental stage characterised by logical reasoning capability. (d) The optimal period during which specific cognitive skills must develop.

    Answer: (a) The difference between what a learner can do independently and what they can achieve with guidance. The ZPD runs from the actual developmental level (independent problem solving) to the level reached with a more knowledgeable other, and it moves as skills are internalised. It is not a fixed capacity, not a stage of logical operations and not a critical or optimal period.

14. What is the primary difference between Piaget and Vygotsky's theories of language development? (RML 2021)

    (a) Piaget argued thought depends on language while Vygotsky argued language depends on thought (b) Vygotsky argued thought depends on language while Piaget argued language depends on thought (c) Vygotsky was concerned with early language development while Piaget was concerned with later language development. (d) None of the above

    Answer: (b). For Piaget language expresses concepts already built through action, so it depends on thought; for Vygotsky thought and speech have separate roots that merge at about 2, after which language shapes thought. Option (a) swaps the names, and (c) is false: both studied the speech of young children (egocentric speech for Piaget, private speech for Vygotsky).

15. In the 'teleological stance', toddlers can understand: (RML 2022)

    (a) Contingencies (b) Intentions in action (c) Prior Intentions (d) Mental states such as belief and desire

    Answer: (b) Intentions in action. In Gergely and Csibra's teleological stance, infants near the end of the first year read an action as efficient and goal-directed, so they grasp the goal visible in an ongoing act. They have moved beyond mere contingencies but do not yet represent prior intentions (plans formed before acting) or beliefs and desires, which come with desire psychology at about 2 and belief-desire psychology at 3 to 4.

16. Which of the following is a key characteristic of the "slow-to-warm-up" temperament category in the Thomas and Chess model? (Real PYQ 2024)

    (a) Positive approach to new stimuli and high regularity of biological functions. (b) Negative withdrawal from new stimuli, slow adaptability, and mild intensity of reactions. (c) Intense and frequent negative emotional reactions with irregular routines. (d) A mix of easy and difficult characteristics with no clear pattern.

    Answer: (b) Negative withdrawal from new stimuli, slow adaptability, and mild intensity of reactions. The slow-to-warm-up child withdraws at first but mildly, and adapts gradually. Option (a) describes the easy child, (c) the difficult child, and (d) the roughly 35 percent who fitted no single type.

17. Which among the following is not included within the classification of structure of temperament given by Rothbart and Bates (RML 2022)

    (a) Extraversion (b) Inhibition to the unfamiliar (c) Negative affectivity (d) Effortful control

    Answer: (b) Inhibition to the unfamiliar. Rothbart and Bates have exactly three dimensions: surgency (extraversion), negative affectivity and effortful control. Inhibition to the unfamiliar is Jerome Kagan's behavioural inhibition.

18. According to experts on socioemotional development like Campos (2009) and Kagan (2010), self-conscious emotions appear at which age? (RML 2022)

    (a) 6-9 months (b) Before 9 months (c) After 12 months (d) 3 months

    Answer: (c) After 12 months. Embarrassment, pride, shame, guilt, envy and empathy need self-awareness, which arrives in the second half of the second year (mirror self-recognition at 18 to 24 months). The earlier ages belong to the primary emotions.

19. Children, who are distressed by their parent's absence, seek interaction but also reject it when offered. How can the reaction be categorised? (RML 2021)

    (a) Insecure-avoidant (b) Insecure-resistant (c) Insecure-disorganized (d) Secure-resistant

    Answer: (b) Insecure-resistant. Ainsworth's type C (ambivalent) infants are very distressed at separation and on reunion seek contact yet resist it. Avoidant (A) infants show little distress and ignore the caregiver, disorganised (D) infants show contradictory or freezing behaviour, and 'secure-resistant' is not a type.

20. According to the schema theory, which hypothetical construct provides a prototype for subsequent attachments and can be adapted through more experience? (RML 2021)

    (a) Proximity-seeking mechanism (b) Attachment system (c) Bio-social relationship model (d) Internal working model

    Answer: (d) Internal working model. Bowlby's internal working model is a schema of self, attachment figures and relationships, built from early experience, that acts as a prototype for later relationships and can be revised. Proximity seeking is a behaviour the attachment system produces, the attachment system is the evolved control system itself, and 'bio-social relationship model' is not Bowlby's term.

21. Which theory of attachment is currently considered the predominantly accepted theory by developmental psychologists? (IHBAS 2020)

    (a) Psychoanalytic (b) Learning (c) Cognitive-developmental (d) Ethological

    Answer: (d) Ethological. Bowlby's ethological theory treats attachment as an evolved behavioural system that keeps the infant near a protector. The psychoanalytic and learning (secondary drive, 'cupboard love') theories rest on feeding, which Harlow's contact comfort findings undermined, and the cognitive-developmental view explains only the timing of attachment.

22. Which parenting style is characterized by the combination of giving emotional support, setting clear boundaries, and maintaining accountability for children's behavior? (Real PYQ 2025)

    (a) Authoritarian parenting (b) Authoritative parenting (c) Permissive parenting (d) Neglectful parenting

    Answer: (b) Authoritative parenting. Warmth (emotional support) plus firm, reasoned control (boundaries, accountability) is Baumrind's authoritative style, high on both demandingness and responsiveness. Authoritarian is high control without warmth, permissive is warmth without control, and neglectful lacks both.

23. According to Jean Piaget's theory of play development, what is the first stage of child play that emerges during infancy? (Real PYQ 2025)

    (a) Symbolic play (b) Functional play (c) Games with rules (d) Constructive play

    Answer: (b) Functional play. Functional (practice or sensorimotor) play, repeating actions for the pleasure of exercising sensorimotor schemes, comes first in infancy. Symbolic play needs the preoperational symbolic function, games with rules belong to the concrete operational years, and constructive play was added later by Smilansky.

24. The starting point for self-socialization, according to Kohlberg's cognitive-developmental theory of gender-role development, is: (IHBAS 2020)

    (a) Gender consistency (b) Basic gender identity (c) Gender stability (d) Gender-role achievement

    Answer: (b) Basic gender identity (as keyed). Labelling oneself a boy or girl is the first link in Kohlberg's chain. Many textbooks say Kohlberg's children become full self-socialisers only at gender consistency, and that it is gender schema theory that starts self-socialisation at basic gender identity. Gender-role achievement is not one of his stages.

25. What is the focus of Kohlberg's preconventional level of moral development? (IHBAS 2025)

    (a) Social contract (b) Avoiding punishment and gaining rewards (c) Obeying laws (d) Ethical principles

    Answer: (b) Avoiding punishment and gaining rewards. At Level I morality is under external control: stage 1 obeys to avoid punishment and stage 2 seeks rewards and fair deals. Obeying laws is stage 4 (conventional); social contract and ethical principles are stages 5 and 6 (postconventional).

26. Jean Piaget's theory of moral development suggests that children in the stage of 'heteronomous morality' (typically ages 5-10) base their moral judgments on: (Real PYQ 2024)

    (a) Intentions and equality (b) Rules as being unchangeable and the consequences of actions (c) Social contracts and individual rights (d) Abstract principles of justice

    Answer: (b) Rules as being unchangeable and the consequences of actions. Heteronomous children (moral realism) treat adult rules as sacred and judge acts by the damage done (objective responsibility). Intentions and equality belong to Piaget's autonomous stage; social contracts and abstract justice are Kohlberg's stages 5 and 6.

27. Carol Gilligan departs from Kohlberg's theory on moral development by highlighting which of these differences? (RML 2022)

    (a) Males have ethics of care (b) Males have ethics of justice (c) Females have ethics care (d) Females have ethics of justice

    Answer: (c) Females have an ethic of care. In *In a Different Voice* (1982) Gilligan argued that Kohlberg's scale captures a male ethic of justice and misses the ethic of care, reasoning from relationships and responsibility, more often heard in women. Options (a) and (d) reverse the pairing; (b) is what Kohlberg already measured.

28. According to Erik Erikson's psychosocial development theory, a child who is 1.5 years old (18 months) would be in which developmental stage? (Real PYQ 2025)

    (a) Trust vs. Mistrust (b) Autonomy vs. Shame and Doubt (c) Initiative vs. Guilt (d) Industry vs. Inferiority

    Answer: (b) Autonomy vs. Shame and Doubt. Eighteen months is the boundary between the first two stages, and the keyed answer places the toddler, now walking and facing toilet training, in autonomy versus shame and doubt. Initiative versus guilt begins at about 3 and industry versus inferiority at school age.

29. According to Erikson, the state of _____ involves feelings of inner fragmentation, little or no sense of where one's life is headed, and an inability to gain the support of a social role or vocation. (Amity 2023)

    (a) Conflict (b) Identity confusion (c) Incongruence (d) Dissociation

    Answer: (b) Identity confusion. It is the negative pole of Erikson's fifth stage, identity versus role confusion. Incongruence is Carl Rogers's gap between self-concept and experience, dissociation is a break in the integration of memory and identity, and conflict is too general.

30. Which term was used by Erikson to refer to the feeling experienced by some individuals in mid-life that they have achieved very little or nothing and have little to offer to the next generation? (RML 2020)

    (a) Stagnation (b) Generativity (c) Presbyopia (d) Mid-life crisis

    Answer: (a) Stagnation. It is the negative pole of generativity versus stagnation in middle adulthood. Generativity is the healthy pole (guiding the next generation), presbyopia is age-related loss of near focusing in the lens, and mid-life crisis is a popular term, not Erikson's.

31. In adulthood climacteric refers to (JEMAS 2025)

    (a) loss of muscular strength and agility. (b) receding hairlines and greying of hair in adults. (c) the end of ovulation and termination of reproductive capacity. (d) gradual decline in reaction time and manual dexterity.

    Answer: (c) the end of ovulation and termination of reproductive capacity. The climacteric is the midlife winding down of reproductive capacity, culminating in menopause in women. Lost strength and agility, receding and greying hair, and slower reaction time and dexterity are real midlife changes, but they are general physical ageing, not the climacteric.

32. The type of marriage in which comradeship and sharing are emphasized, but husband and wife assume responsibility for different roles is known as (JEMAS 2025)

    (a) Traditional marriage. (b) Collegial marriage. (c) Companionship marriage. (d) Satisfactory marriage.

    Answer: (b) Collegial marriage. In Hurlock's typology the collegial marriage stresses comradeship and sharing while each spouse takes responsibility for different, complementary roles. A companionship marriage shares the same roles, a traditional marriage separates roles without stressing comradeship, and satisfactory marriage is not a type.

High-yield sheet

The facts this chapter's questions ask most, most-asked first. Last read before the exam.

1 **Object permanence** is a **sensorimotor** achievement: search for a fully hidden object begins at about 8 to 12 months (substage 4, with the A-not-B error) and object permanence is complete by 18 to 24 months. ASKED 10 TIMES

2 Erikson's crisis of adolescence (about 12 to 18) is **identity versus role confusion**, and its virtue is fidelity. ASKED 10 TIMES

3 The **zone of proximal development** is the gap between what a learner can do independently (the actual developmental level) and what the learner can achieve with guidance from a more knowledgeable other: tasks too hard to master alone but achievable with help. ASKED 8 TIMES

4 **Conservation** (quantity stays the same despite changes in shape or arrangement, as with the clay sausage or the tall glass) is mastered in the **concrete operational** stage (7 to 11), number first and volume last. ASKED 7 TIMES

5 **Hypothetico-deductive reasoning** and abstract thought define the **formal operational** stage (11 or 12 onward); it is not a stage itself, and the concrete operational child does not attain it. ASKED 7 TIMES

6 **Thomas and Chess** (New York Longitudinal Study, 1956) found three types: **easy** (about 40 percent: regular, positive, adaptable, approaches), **difficult** (about 10 percent: irregular rhythms, negative mood, intense reactions, withdrawal, slow adaptation) and **slow-to-warm-up** (about 15 percent: low activity, mild intensity, initial withdrawal, slow gradual adaptation); about 35 percent fitted no single type. ASKED 6 TIMES

7 **Harry Harlow** ('The Nature of Love', 1958): infant rhesus monkeys clung to a terry-cloth mother that gave no milk rather than a wire mother that fed them, so **contact comfort, not feeding**, is the primary basis of attachment. ASKED 6 TIMES

8 In the Strange Situation, reunion behaviour decides the type: **secure (B)** infants seek contact, are readily comforted and return to play; **insecure-avoidant (A)** infants show little distress and ignore the caregiver; **insecure-resistant or ambivalent (C)** infants are very distressed, seek contact yet resist it and cannot be soothed. ASKED 6 TIMES

9 **Authoritative** parenting (Baumrind) is high in demandingness and high in responsiveness: emotional support, clear boundaries, reasoned discipline and accountability, with the best child outcomes; authoritarian is high demand and low warmth, permissive low demand and high warmth, and neglectful (added by Maccoby and Martin, 1983) low on both. ASKED 6 TIMES

10 Kohlberg's **preconventional level** (stage 1 punishment-and-obedience, stage 2 instrumental purpose and exchange, or self-interest) is morality under external control: rules are obeyed to avoid punishment or gain rewards, so rewarded acts are right and punished acts wrong. ASKED 6 TIMES

11 Kohlberg's **conventional level** is conformity to social rules and maintenance of the social order: stage 3 good boy-nice girl (approval, interpersonal expectations, the golden rule) and stage 4 law and order (duty, respect for authority; Papalia's social concern and conscience). Obeying a law or keeping a promise because society depends on it is conventional. ASKED 6 TIMES

12 In **Bronfenbrenner**'s ecological systems theory (1979), the **microsystem** holds the child's direct settings (family, school, peers), the **mesosystem** is the links between them (home-school), the **exosystem** holds settings without the child (parent's workplace), the **macrosystem** is culture and the **chronosystem** (added later) is time. ASKED 5 TIMES

13 **Mary Ainsworth** devised the **Strange Situation** (with Barbara Wittig, 1969; Patterns of Attachment, 1978): Bowlby is to attachment theory as Ainsworth is to the Strange Situation. ASKED 5 TIMES

14 **Centration** is the preoperational focus on one salient feature while ignoring others, as in the 5-year-old who sorts cards by colour but cannot re-sort them by suit. ASKED 4 TIMES

15 The term **zone of proximal development** was coined by **Lev Vygotsky** (not Piaget, Bruner, Kohlberg, Bronfenbrenner, Erikson, Skinner or Freud); **scaffolding**, the support given within it, is a term from Wood, Bruner and Ross (1976). ASKED 4 TIMES

16 The fourth pattern, **disorganised or disoriented (D)**, was added by **Mary Main and Judith Solomon** (1986, 1990): no one clear pattern, with contradictory, freezing, dazed or apprehensive behaviour on reunion, often after maltreatment. ASKED 4 TIMES

17 Kohlberg's three levels are **preconventional, conventional and postconventional**; Level III (Papalia: morality of autonomous moral principles) holds stage 5, the social-contract orientation (laws as instruments expressing the majority's will and furthering welfare), and stage 6, universal ethical principles. Unconventional, formal conventional and amoral are not Kohlberg levels, and premoral is Piaget's period before rules matter. ASKED 4 TIMES

18 Erikson's virtues in order: trust gives **hope**, autonomy **will**, initiative **purpose**, industry **competence**, identity **fidelity**, intimacy **love**, generativity **care**, integrity **wisdom**. ASKED 4 TIMES

19 Marcia's identity statuses cross **crisis (exploration)** with **commitment**: diffusion (neither), foreclosure (commitment without exploration), moratorium (exploration without commitment), achievement (both). ASKED 4 TIMES

20 Middle adulthood is **generativity versus stagnation**: mentoring and guiding the next generation is generativity; feeling one has achieved little and has little to offer is stagnation. ASKED 4 TIMES

21 Erikson's first stage is **trust versus mistrust** in infancy (birth to about 1, some texts 18 months); a child of 18 months is keyed to autonomy versus shame and doubt. ASKED 4 TIMES

22 A **critical period** is a fixed window in which an experience must occur for normal development, with largely irreversible effects (Lorenz's imprinting; embryonic organ formation); later experience compensating for early deficits contradicts it. ASKED 3 TIMES

23 **Eleanor Gibson and Richard Walk** (1960) used the **visual cliff** to test depth perception: infants of crawling age (about 6 months onward) refused the deep side, so depth perception is present by the time infants crawl. ASKED 3 TIMES

24 Piaget's stage ages: **sensorimotor** birth to 2, **preoperational** 2 to 7, **concrete operational** 7 to 11, **formal operational** 11 or 12 onward. ASKED 3 TIMES

25 **False-belief tasks** (Wimmer and Perner's Maxi task, 1983; Baron-Cohen, Leslie and Frith's Sally-Anne task, 1985) assess **theory of mind** and are typically passed at **about age 4**, in the preoperational years. ASKED 3 TIMES

26 Bowlby's **internal working model** is a mental representation of self, attachment figures and relationships, built from early experience, that serves as a prototype for later relationships and can be revised by new experience. ASKED 3 TIMES

27 **Konrad Lorenz** (1935): greylag goslings **imprint** on the first moving object after hatching, a rapid, largely irreversible following response formed within a narrow critical period. ASKED 3 TIMES

28 Late adulthood (65 onward) is **ego integrity versus despair**: accepting the life one has lived versus regret and fear of death; the virtue is wisdom. ASKED 3 TIMES

29 **Maturation** (Arnold Gesell) is the biological unfolding of genetic potential; a child's increasing skill at using his muscles is due mainly to maturation, not exercise, parental pushing or learning.
ASKED 2 TIMES

30 **Klinefelter syndrome** (47,XXY) is a tall male with small testes, infertility and learning difficulties; **Turner syndrome** (45,X) is a short female with a webbed neck and ovarian failure. ASKED 2 TIMES

CHAPTER 7

Clinical Psychology

IN THIS CHAPTER

7.1 ●● The Profession: History, Models, Ethics and Indian Law: Define clinical psychology and tell it apart from neighbouring professions, trace its history in the West and in India, compare the models of abnormality, apply the APA ethical principles and the Tarasoff duty, and state the main provisions of India's mental health and disability laws.

7.2 ● The Clinical Interview, Mental Status Examination and Formulation: Conduct and record a clinical assessment: choose an interview format, use the core interviewing microskills, take a case history, carry out and write a mental status examination, assess suicide and violence risk, run a functional analysis, build a 5 Ps formulation, and use DSM-5-TR and ICD-11 with an eye on reliability and labelling.

7.3 ●●● Psychodynamic Therapies and Their Descendants: Explain how Freud's psychoanalysis works as a treatment and how its descendants (ego, object relations and self psychology, brief dynamic therapy, interpersonal psychotherapy, Jungian and Adlerian therapy, transactional analysis and supportive therapy) differ in theory, technique and evidence.

7.4 ●●● Behaviour Therapy: Trace the origins and assumptions of behaviour therapy and describe, attribute and choose among its classical, operant, modelling, relaxation and self-management techniques, from systematic desensitisation, flooding and ERP to token economies, habit reversal and parent training.

7.5 ●●● Cognitive and Cognitive-Behavioural Therapies, and the Third Wave: Explain and compare Ellis's REBT and Beck's cognitive therapy, describe the other cognitive-behavioural and third-wave therapies (DBT, ACT, MBCT and others) with their founders and techniques, and weigh the evidence for CBT.

7.6 ●●● Humanistic, Existential, Experiential and Brief Therapies: Explain person-centred, Gestalt, existential, reality, experiential and brief therapies, motivational interviewing with the stages of change, and the helping-skills models, naming each founder and signature technique.

7.7 ●● Family, Couples and Group Therapies: Explain why families are treated as systems, what the double bind and expressed emotion research found, how the Bowen, structural, strategic, Milan, experiential and later family, couples and sex therapies work, and how group therapy, psychodrama, self-help groups and therapeutic communities help, naming who devised each.

7.8 ● Does Psychotherapy Work? Evidence, Common Factors and the Alliance: Explain how psychotherapy outcome is evaluated, from Eysenck's challenge and meta-analysis to RCTs, empirically supported treatments and clinical significance, and weigh the evidence for common factors, the alliance and therapist and client variables.

7.9 • Health, Community, Crisis and Rehabilitation Psychology: Explain health psychology and the models of health behaviour, the psychology of physical and chronic illness, community mental health and prevention, crisis intervention and suicide prevention, and psychosocial, neuropsychological and addiction rehabilitation.

7.10 • Working with Children, Older Adults and Special Settings: Adapt assessment and therapy to children, adolescents, older adults and counselling, forensic, creative and digital settings, naming each method's originator and the Indian laws that apply.

Weightage. 366 of the 3180 questions mapped in this book (11.5%) test this chapter, 191 of them from real past papers. 4 of its 10 sections are High yield.

Paris in the 1790s. On the southern edge of the city stood La Bicêtre, a huge hospice for poor, old, sick and criminal men. In its wards for the insane, men were kept in chains, some of them for years, and treated more like dangerous animals than patients. The story most books tell is that in 1793, at the height of the French Revolution, the hospital's physician, **Philippe Pinel**, ordered the irons struck off, and that men who had not seen the sky for years walked calmly into the courtyard. Nineteenth-century painters made the scene famous.

The real story, pieced together by historians, is less heroic and more useful. The chains at Bicêtre were first removed not by the doctor but by **Jean-Baptiste Pussin**, the lay superintendent of the ward, who had seen what irons did to the men in his care and had begun to treat them as people: he spoke with them, kept them occupied and replaced irons with gentler restraint. Pinel's contribution was to recognise that Pussin's methods worked, to record his patients' histories carefully and count who recovered, and to take the approach, with Pussin himself, to the women's hospital, the Salpêtrière. Their *moral treatment* rested on three things: a relationship, a daily routine and careful observation of results.

Keep both men in mind: the attendant who knew how to be with a frightened, disturbed person, and the physician who wrote down what happened and checked whether it helped. Between them they hold the whole of clinical psychology: understanding and assessing a person, building a relationship that heals, choosing treatments by their evidence, caring for people in their communities and protecting their dignity in law. This chapter follows each of these threads, and at its end we return to Bicêtre.

7.1 The Profession: History, Models, Ethics and Indian Law

MEDIUM YIELD · 39 QS

Before learning how a clinical psychologist interviews, tests or treats, you need to know who this professional is, where the profession came from, which theories of abnormality it uses, and which ethical rules and Indian laws bind it.

What clinical psychology is

The American Psychological Association's Division 12, the Society of Clinical Psychology, describes a field that "integrates science, theory, and practice to understand, predict, and alleviate maladjustment, disability, and discomfort as well as to promote human adaptation, adjustment, and personal development". A **clinical psychologist** therefore researches, assesses, treats and promotes well-being. In India the profession is defined by qualification and registration: under the Rehabilitation Council of India (RCI) Act, a clinical psychologist holds an RCI-recognised qualification and is on the RCI's register, and the Mental Healthcare Act 2017 likewise defines one by an RCI-recognised qualification.

The work spans psychological assessment, diagnosis, psychotherapy, research, teaching and supervision, consultation and administration, in hospitals, rehabilitation centres, schools, universities, courts, prisons, the armed forces, private practice and online. Table 7.1 separates the clinical psychologist from four neighbours.

Table 7.1 The clinical psychologist and neighbouring professions in India.

Profession	Training	Main work	Prescribes?
Clinical psychologist	Psychology degree, then two-year M.Phil (now the RCI MA); RCI registration	Assessment, psychodiagnosis, psychotherapy, research	No
Psychiatrist	MBBS, then MD, DNB or DPM in psychiatry	Medical diagnosis, medication, ECT	Yes
Counselling psychologist	Master's in psychology or counselling	Adjustment, career and relationship problems of relatively healthy people	No
Psychiatric social worker	MSW, then M.Phil in psychiatric social work	Family and social assessment, casework, rehabilitation	No
Psychiatric nurse	Nursing degree, psychiatric specialisation	Ward care, medicines, monitoring, psychoeducation	No

From demons to moral treatment

How people explained strange behaviour decided how they treated it. The oldest explanation, the **supernatural model**, blamed possession, sin or sorcery. Some Stone Age skulls show **trephining**, a hole chipped through the skull, perhaps to let an evil spirit out. The Greek physician **Hippocrates** (460 to 377 BC) made the first break, finding natural causes in the brain and body. His **humoral theory**, developed by the Roman physician **Galen** (about AD 129 to 198), held that health is a balance of blood, yellow bile, black bile and phlegm, and that excess black bile causes melancholia. Medieval Europe swung back to demonology, and outbreaks of mass madness such as St Vitus's dance and tarantism were taken as signs of possession. From the sixteenth century the mentally ill were confined in **asylums** that were more prisons than hospitals; London's Bethlehem Hospital, shortened to **Bedlam**, was so notorious that its name now means uproar.

Reform came at the end of the eighteenth century. In Paris the physician **Philippe Pinel** (1745 to 1826) and the superintendent **Jean-Baptiste Pussin** (1746 to 1811) removed the chains from patients at La Bicêtre (in 1793, by the usual account) and then at the Salpêtrière. In England the Quaker **William Tuke** (1732 to 1822) opened the **York Retreat** in 1796, where patients lived and worked in a calm, family-like house. Both reforms practised **moral treatment** ("moral" meaning psychological): kindness, routine, work and social contact. From the 1840s the American schoolteacher **Dorothea Dix** (1802 to 1887) persuaded legislatures to build state hospitals, but they filled faster than they could be staffed and moral treatment decayed into custodial care. Then a former patient, **Clifford Beers** (1876 to 1943), described his abuse in hospital in *A Mind That Found Itself* (1908) and, through the mental hygiene societies he founded in 1908 and 1909, launched the **mental hygiene movement** for prevention, early treatment and child guidance.

Medicine meanwhile built the **biological** (medical) **model**. **Emil Kraepelin** (1856 to 1926), in a textbook first published in 1883, treated mental disorders as distinct syndromes with their own symptoms, course and outcome, separating dementia praecox (renamed schizophrenia by **Eugen Bleuler** in 1908) from manic-depressive insanity; modern classification descends from him.

EXAM TRAP

Pinel gets the credit, but Pussin had begun the humane regime before Pinel arrived at Bicêtre (1793; some textbooks print 1791), persuaded him to follow, and himself banned the chains there, a step historians date to 1797. "Mental hygiene movement" usually means Beers (1908 to 1909), though some textbooks also give the name to Dix's crusade; check the date.

Witmer to Vail: a profession is born ●●

The profession began in 1896, when **Lightner Witmer** (1867 to 1956), a former student of Wundt, opened the first **psychological clinic** at the University of Pennsylvania for children with learning problems. In 1907 he founded the journal *The Psychological Clinic* and in it first used the term **clinical psychology** in print. For decades the work was mainly testing: **Alfred Binet** and **Theodore Simon** published the first practical intelligence scale in 1905, and in 1917 **Robert Yerkes** (1876 to 1956) led the testing of about 1.75 million American recruits with the **Army Alpha**, a written test for literate men, and the non-verbal **Army Beta** for those who could not read English. After the Second World War, psychological casualties far outnumbered psychiatrists, and from 1946 the **Veterans Administration** (VA) paid to train clinical psychologists as therapists.

How should they be trained? The **Boulder conference** (Colorado, 1949) chose the **scientist-practitioner model**: a PhD training clinicians equally in research and practice, so that they consume research, evaluate their own work and create new knowledge. The **Vail conference** (Colorado, 1973) endorsed the **practitioner-scholar model** and the **PsyD** (Doctor of Psychology), a professional degree centred on practice. **Richard McFall**'s 1991 "Manifesto for a Science of Clinical Psychology" set out the stricter **clinical science model**: training grounded in research, and only scientifically supported methods in practice.

The APA formed a clinical section in 1919, which became **Division 12** when the APA reorganised into divisions in 1945. The newest debate is over **prescription privileges** for psychologists trained in psychopharmacology: supporters cite shortages of psychiatrists, opponents doubt the training and fear a loss of psychological identity. New Mexico (2002) and Louisiana (2004) were the first American states to grant them; in India clinical psychologists do not prescribe.

EXAM TRAP

The first clinic and the first printed "clinical psychology" are Witmer's, not Wundt's (first laboratory, 1879), G. Stanley Hall's (first APA president, 1892), James McKeen Cattell's (coined "mental test", 1890), Freud's, Erikson's, Rogers's or Skinner's. Boulder 1949 = scientist-practitioner, PhD; Vail 1973 = practitioner-scholar, PsyD. Salt Lake City hosted a later conference on graduate education (1987), and Adelphi is a university; neither names a training model.

Clinical psychology in India ●●

Girindrasekhar Bose (1887 to 1953) founded the **Indian Psychoanalytical Society** at Calcutta in 1922 (Chapter 1). Formal training began at the All India Institute of Mental Health, Bangalore (founded 1954), which began training clinical psychologists in 1955 and be-

came **NIMHANS** (National Institute of Mental Health and Neuro Sciences) in 1974; the Hospital for Mental Diseases, Ranchi, now the **Central Institute of Psychiatry** (CIP), had a clinical psychology department from the late 1940s and started its diploma course in 1962. Their post-MA training in medical and social psychology became the two-year **M.Phil in Clinical Psychology**, for decades the only qualifying degree. The **Indian Association of Clinical Psychologists** (IACP) was founded in 1968.

The **Rehabilitation Council of India** (RCI) began in 1986 as a registered society; the **RCI Act 1992** made it a statutory body (from 22 June 1993), and the Act was amended in 2000. The RCI regulates rehabilitation and disability professionals, clinical psychologists among them, so it sits under the Ministry of Social Justice and Empowerment rather than the Health Ministry. It recognises courses, sets syllabi and keeps the **Central Rehabilitation Register** (CRR); practising without registration is an offence under the Act. After the National Education Policy 2020 ended the M.Phil, the RCI replaced it with a two-year professional **MA in Clinical Psychology** from 2025-26.

> EXAM TRAP
>
> 1968 is the IACP. NIMHANS dates from 1974 and its community psychiatry unit from 1975; it was made an Institute of National Importance by an Act of Parliament in 2012; the PsyD is the American degree from Vail. RCI: society 1986, statutory body under the 1992 Act; Ministry of Social Justice and Empowerment, not Health. Registration is mandated by the RCI Act, not the Persons with Disabilities Act 1995 or the mental health Acts.

What is abnormal, and why? The models ●●●

No single criterion marks behaviour as abnormal. The **statistical criterion** calls rare behaviour abnormal, but genius is rare too. The **social norm criterion** flags rule-breaking, but norms vary: a devotee in trance at a temple festival is following custom. The **maladaptiveness criterion** asks whether behaviour stops a person meeting daily demands, the **personal distress criterion** whether the person suffers (a manic person may feel wonderful). **Marie Jahoda** (1907 to 2001) in 1958 defined **ideal mental health** instead: positive self-attitude, growth and self-actualisation, integration, autonomy, accurate perception of reality and environmental mastery. Textbooks now condense the criteria into the **four Ds**: **deviance**, **distress**, **dysfunction** and **danger** to self or others.

A **model** then says why a disorder arose and so what will help. Beyond the supernatural and biological models, the **psychodynamic model** of **Sigmund Freud** (1856 to 1939) blames unconscious childhood conflict; the **behavioural model** maladaptive learning; the **cognitive model** faulty beliefs; the **humanistic-existential model** blocked growth and lost meaning, answered with respect, empathy, unconditional positive regard and faith in human potential; the **sociocultural model** poverty, culture and social labels; and the **family systems model** disturbed family interaction, with the patient as symptom bearer. Table 7.2 lines them up.

Table 7.2 Models of psychopathology.

Model	Key names	Cause of disorder	Treatment
Supernatural	Ancient belief	Spirits, sin, sorcery	Exorcism

Model	Key names	Cause of disorder	Treatment
Biological	Hippocrates, Kraepelin	Brain, genes, neurotransmitters	Drugs, ECT
Psychodynamic	Freud	Unconscious conflict	Insight (7.3)
Behavioural	Pavlov, Skinner	Faulty learning, reinforcement	Relearning (7.4)
Cognitive	Ellis, Beck	Irrational, distorted thinking	Change thinking (7.5)
Humanistic-existential	Rogers, Maslow, Frankl	Blocked growth, lost meaning	Growth, choice (7.6)
Sociocultural	Szasz	Poverty, culture, labels	Social change (7.9)
Family systems	Bateson, Minuchin	Family interaction	Family therapy (7.7)

Since each model explains only some cases, two integrative models dominate. In a 1977 paper in *Science*, the psychiatrist **George L. Engel** (1913 to 1999) argued that the biomedical model cannot explain why two people with the same lesion fare so differently, and proposed the **biopsychosocial model**: every illness arises from interacting biological, psychological and social factors, and treatment must address all three.

The **diathesis-stress model** explains who falls ill and when. A **diathesis** is a predisposition, usually genetic but sometimes psychological (a pessimistic style, an early loss); **stress** is the triggering event. Neither alone suffices. **Paul Meehl** (1920 to 2003) applied the idea to schizophrenia in 1962, and **Joseph Zubin** and **Bonnie Spring** formalised it in 1977 as the **vulnerability-stress model**: everyone has a threshold, and the more vulnerable you are, the less stress it takes to cross it (Figure 7.1). Adding **protective factors** (social support, coping skills, continued medication), which raise the threshold, gives the **vulnerability-stress-protective factor model** used in relapse prevention.

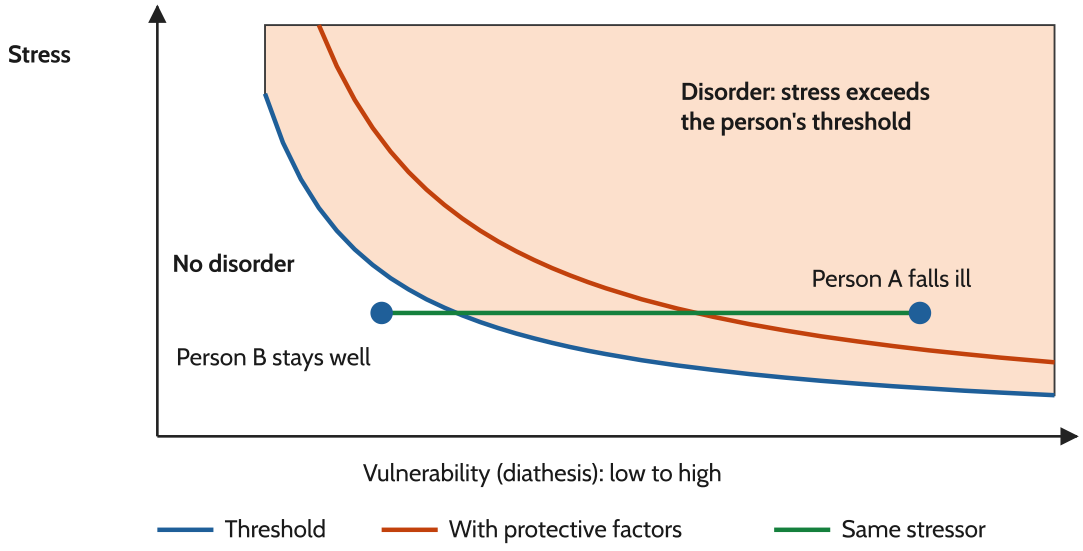

Figure 7.1 The diathesis-stress (vulnerability-stress) model. The same life event pushes highly vulnerable Person A over the threshold but not Person B; protective factors raise the threshold.

EXAM TRAP

"Predisposition plus trigger" is diathesis-stress, not the neurodevelopmental model (faulty early brain development in schizophrenia; Chapter 8); "clinical-reductionist" and "syndrome-sign" are not models. Single-cause options (genes, stress, learning, unconscious conflict or spirits alone) are wrong. The biopsychosocial model is Engel's, not Kraepelin's, Bleuler's, Beck's, Freud's, Walter Cannon's (fight or flight) or Richard Lazarus's (stress appraisal). Reinforcement is behavioural, never humanistic.

Ethical principles and practice ●●●

The APA's **Ethical Principles of Psychologists and Code of Conduct** (2002, amended 2010 and 2016) has two layers: five aspirational **General Principles** (A to E) that guide judgement but cannot be enforced, and ten sections of enforceable **Ethical Standards** (on competence, confidentiality, research, assessment, therapy and more). Table 7.3 pairs the principles with the four principles of biomedical ethics of **Tom Beauchamp** and **James Childress** (1979).

Table 7.3 The APA's five General Principles and Beauchamp and Childress's four.

APA General Principle	What it asks	Biomedical principle
A. Beneficence and Non-maleficence	Benefit clients; do no harm	Beneficence; non-maleficence
B. Fidelity and Responsibility	Trust; professional and social responsibility	None
C. Integrity	Accuracy, honesty, truthfulness	None
D. Justice	Fair access and equal quality of service	Justice
E. Respect for People's Rights and Dignity	Privacy, confidentiality, self-determination	Respect for autonomy

Behind any code lie two ways of reasoning. A **deontological** ethic (**Immanuel Kant**) judges an act by its intrinsic rightness: a duty such as keeping a confidence binds whatever the consequences. A **utilitarian** ethic (**Jeremy Bentham**, **John Stuart Mill**) weighs costs against benefits and chooses the greatest good for the greatest number. **Competence** means working only within one's training and supervised experience and referring beyond it; **ethical competence** means knowing the codes, recognising, analysing and resolving dilemmas, and taking responsibility for one's actions.

Informed consent is free agreement by a capable person who has been told, understandably, the nature and course of treatment, fees, the limits of confidentiality and the right to withdraw. A child cannot consent legally, so a guardian gives permission and the clinician seeks the child's **assent**. Research participants must be told the purpose, procedures, risks and benefits and their right to decline or withdraw (the consent form merely records this). Deception is allowed only when justified and must end in **debriefing**, a full explanation of the study's true purpose after participation. Research ethics go beyond legal compliance: protection from harm, confidential data, ethics committee approval and honest reporting; fabricating or manipulating data is misconduct.

Confidentiality is the clinician's ethical duty not to reveal a client's disclosures; **privileged communication** is the client's legal right to bar the clinician's testimony in court, which the client may waive. Neither is absolute. Disclosure is allowed or required with the client's consent; when there is serious risk of suicide or of harm to an identifiable person; when child abuse is suspected (India's POCSO Act 2012 makes reporting sexual offences against children

mandatory); under a court order, including evidence of guilt or innocence in a criminal case; when clients sue the clinician or put their mental state at issue; and in emergencies. The Mental Healthcare Act 2017 grants patients confidentiality with similar exceptions.

KEY STUDY: TARASOFF V. REGENTS OF THE UNIVERSITY OF CALIFORNIA (1974, 1976)

In 1969 **Prosenjit Poddar**, a graduate student from India at Berkeley, told his university therapist he intended to kill a young woman who had rejected him, readily identifiable as **Tatiana Tarasoff**. Campus police questioned and released him; no one warned her, and two months later he killed her. The California Supreme Court ruled in 1974 that therapists have a **duty to warn** an identifiable victim, and in 1976 widened it to a **duty to protect** (warning, informing police, hospitalisation): "The protective privilege ends where the public peril begins." Critics fear it deters dangerous clients from speaking openly.

EXAM TRAP

Duty to warn comes from Tarasoff. The M'Naghten (1843) and Durham (1954) rules are insanity tests, and the Daubert standard (1993) governs expert testimony (7.10). "There are never exceptions to confidentiality" is always false.

A **multiple (dual) relationship** exists when a psychologist in a professional role with someone also holds another role with that person or someone close to them, or promises one: treating a friend, employee or relative, or doing business with a client. Under APA Standard 3.05 it is unethical when it risks impaired objectivity, exploitation or harm. Co-therapy by two psychologists, a client seeing two therapists or a psychologist working in two clinics is not a dual relationship. **Sexual intimacy** with a current client is forbidden, and with a former client for at least two years after therapy and even then only in the most unusual circumstances. **Thomas Gutheil** and **Glen Gabbard** (1993) distinguished a **boundary crossing**, a harmless departure from the usual frame (a brief self-disclosure, seeing a client at the workplace for exposure work), from a **boundary violation**, which exploits or harms the client to serve the therapist, such as a romantic relationship or business deal.

Finally, **records** must be accurate, secure and open to clients. In **assessment**, psychologists use valid tests, explain results, release **test data** (raw scores and responses) to the client or whoever the client names, and guard **test security** by never releasing manuals or items to untrained people. **Teletherapy** requires technical competence, secure platforms, consent covering privacy risks, confirming the client's location and an emergency plan. Trainees work under **supervision**, clients are told so, and the supervisor carries responsibility. The RCI Act lets the Council prescribe a code of ethics, and misconduct can remove a professional from its register.

EXAM TRAP

A romance with a current client is a boundary violation; rapport and convenient appointments are just good practice. The APA has five General Principles; "equity" is not one. Integrity is APA's, not one of Beauchamp and Childress's four (autonomy, non-maleficence, beneficence, justice). Non-maleficence is refraining from harm; doing good is beneficence. Consent comes before a study, debriefing after.

Indian mental health law and policy ●●●

The **Indian Lunacy Act 1912** was custodial, built around detaining "lunatics" in asylums. The **Mental Health Act 1987** (in force 1993) modernised terms and licensing but gave few rights, exempted government hospitals from licensing and left attempted suicide a crime under section 309 of the Indian Penal Code. The **Mental Healthcare Act 2017** (MHCA, in force 2018) replaced it with a rights-based law aligned with the UN Convention on the Rights of Persons with Disabilities.

- **Rights.** A **right to access mental healthcare** (section 18) from government services, free below the poverty line, with rights to community living, dignity, equality (including insurance parity with physical illness, section 21(4)), confidentiality, information, legal aid and medical records.
- **Choices.** Every person, including one with mental illness, is presumed capable (section 4), and treatment needs informed consent. An adult may write an **advance directive** (section 5) on how to be treated, or not treated, in a future illness and appoint a **nominated representative** (NR, section 14). A minor can do neither; a minor's NR is the legal guardian unless the Board orders otherwise (section 15). Under section 87 a minor is admitted only on the NR's application after two independent examinations, with the minor's wishes taken into account, is kept apart from adults with the NR or an attendant always present, and is treated with the NR's consent; the Board is informed within 72 hours and must review any stay beyond 30 days.
- **Prohibitions (section 95).** No **ECT** without anaesthesia and muscle relaxant (unmodified ECT is never lawful, and section 94 bars any ECT as emergency treatment); no ECT for minors unless a psychiatrist judges it necessary, the guardian consents and the Board gives prior permission; no sterilisation as treatment; no chaining. Section 97 also bans seclusion and solitary confinement and allows physical restraint only as a last resort authorised by the psychiatrist.
- **Bodies.** The **Central Mental Health Authority** and **State Mental Health Authorities** register all mental health establishments, government or private, and professionals; **Mental Health Review Boards**, each for a district or group of districts, chaired by a serving or retired District Judge and including a psychiatrist, a medical practitioner and two service-user or carer members (section 74), register advance directives, appoint NRs, review admissions and hear complaints.
- **Section 115.** Anyone who attempts suicide is presumed, unless proved otherwise, to have **severe stress** and is not punished; in effect, decriminalisation, with a government duty to provide care.
- **Professionals.** Section 2 defines a clinical psychologist by an RCI-recognised qualification in clinical psychology, or a postgraduate degree plus a two-year M.Phil in Clinical or Medical and Social Psychology with supervised clinical training; registered with the State Authority, the clinical psychologist is a **mental health professional** under the Act.

The **Rights of Persons with Disabilities Act 2016** (RPwD, in force 2017) replaced the **Persons with Disabilities Act 1995** (7 disabilities) with 21 specified disabilities, including mental illness, intellectual disability, autism spectrum disorder and specific learning disabilities. At least 40 percent of a specified disability is a **benchmark disability**, qualifying for reservation (4 percent of government posts, 5 percent of higher education seats). Table 7.4 adds the other laws, programmes and surveys.

Table 7.4 Indian mental health law, policy and programmes.

Year	Law, programme or survey	Remember
1982	**National Mental Health Programme** (NMHP)	Minimum mental healthcare for all through primary care
1985	**NDPS Act** (Narcotic Drugs and Psychotropic Substances)	Controls drugs of abuse; addicts who seek treatment are spared prosecution
1996	**District Mental Health Programme** (DMHP)	District services on the **Bellary model**, a NIMHANS pilot in Karnataka
1999	**National Trust Act**	Welfare and guardianship for autism, cerebral palsy, mental retardation, multiple disabilities
2014	**National Mental Health Policy**	India's first
2015-16	**National Mental Health Survey** (NIMHANS)	Adults: current prevalence 10.6 percent; **treatment gap** 70 to 92 percent
2022	**Tele-MANAS**	Free 24-hour tele-mental-health helpline (14416)

EXAM TRAP

Advance directives, NRs, Review Boards, section 115 and the ECT ban belong to the MHCA 2017, never the 1987 Act, the RPwD Act or the RCI Act. RPwD = 2016 and 21; PwD = 1995 and 7. A 2024 paper keyed "a minor has the right to appoint a nominated representative, and their opinion must be considered". Pick it, since the other options are false, but only its second half is exact: a minor's NR is the legal guardian, and admitting a minor needs no prior Board consent. The NR cannot decide "irrespective" of the minor, whose views must be weighed, and a minor can be admitted on the NR's application even when unwilling. ECT is never allowed as emergency treatment.

Practising in India

Trained in an RCI-recognised programme with supervised hospital work, registered with the RCI and working as a scientist-practitioner, the Indian clinical psychologist also needs **cultural competence**, the awareness, knowledge and skills to work across cultures. Families usually share treatment decisions, so the clinician works with them; stigma delays help-seeking, and many people first visit faith healers, temples and dargahs, beliefs to respect while explaining what treatment offers. With a treatment gap above 70 percent, **task-shifting**, training lay counsellors and health workers to deliver simple psychological treatments under supervision, has become central (7.9).

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The first psychological clinic was opened in 1896 by _____.
2. The 1949 _____ conference adopted the scientist-practitioner model.
3. The biopsychosocial model was proposed in 1977 by _____.
4. The client's legal right to bar a therapist's testimony in court is _____ communication.
5. The duty to protect an identifiable victim comes from the _____ case.
6. Section _____ of the Mental Healthcare Act 2017 presumes severe stress in a person who attempts suicide.

Answers: (1) Lightner Witmer, (2) Boulder, (3) George L. Engel, (4) privileged, (5) Tarasoff, (6) 115.

7.2 The Clinical Interview, Mental Status Examination and Formulation

LOWER YIELD · 11 QS

Ravi, 24 (a composite case), has failed the bank examination for the third time, in the week his cousin was selected, and has told his mother that everyone would be better off without him. The clinician will interview him, take his history, examine his mental state, judge his safety and build a formulation: the steps this section follows.

Purposes and types of clinical interview

The **clinical interview** is a conversation with a professional purpose: it gathers information, tests hypotheses and begins a working relationship at once. Interviews are named by purpose: the **intake interview** (first contact: why now, how urgent?), the case-history interview, the **diagnostic interview** (symptoms checked against criteria), the mental status examination, the **crisis interview** (brief, active, focused on safety; Section 7.9) and the **termination interview** (reviewing progress and planning relapse prevention).

They also differ in structure. In an **unstructured interview** the clinician chooses the questions: flexible and good for rapport, but two clinicians may reach different diagnoses. A **semi-structured interview** fixes topics and core questions but lets a trained clinician rephrase, probe and judge. A **fully structured interview** fixes every word and scoring rule, so a lay interviewer or computer can give it. Structure buys **reliability** (interviewers agree) at the cost of flexibility and rapport, but not **validity**: interviewers can agree and still be wrong, and a schedule misses what it does not ask. Clinicians often open freely, then use a schedule (Table 7.5).

Table 7.5 The main diagnostic interview schedules.

Instrument	Structure	Use
SCID-5, Structured Clinical Interview for DSM-5 (First, Spitzer and colleagues)	Semi-structured, despite its name	DSM diagnosis
Present State Examination (PSE; Wing, 9th edition 1974); its WHO successor SCAN	Semi-structured	ICD diagnosis; cross-national research

Instrument	Structure	Use
Diagnostic Interview Schedule (DIS; Robins, 1981)	Fully structured, lay interviewers	Community surveys
MINI, Mini International Neuropsychiatric Interview (1998)	Structured, about 15 minutes	Rapid screening
K-SADS, Kiddie Schedule for Affective Disorders and Schizophrenia	Semi-structured, parent and child	Ages about 6 to 18
ADIS-5, Anxiety Disorders Interview Schedule (Brown and Barlow)	Semi-structured	Anxiety disorders, with severity

Related interview and observation instruments of developmental assessment are the ADI-R (Autism Diagnostic Interview-Revised, with caregivers), the ADOS-2 (Autism Diagnostic Observation Schedule) and the Vineland-3 (adaptive behaviour, by caregiver interview); see Chapter 2.

> EXAM TRAP
>
> "A structured interview" means the **SCID**. The **MMSE** (Mini-Mental State Examination, 1975) is a 30-point cognitive screen, not an interview and not the MSE; the **WAIS** is an intelligence test; the **TAT** a projective test.

Interviewing skills ●●

Allen Ivey's **microcounselling** (from 1968) teaches the craft one microskill at a time. **Rapport**, the trust that makes disclosure possible, grows from explaining the purpose and the limits of confidentiality and starting with neutral topics. **Attending behaviour** shows that you are listening; **Gerard Egan** (*The Skilled Helper*, 1975) summed it up as **SOLER**: face the client **S**quarely, **O**pen posture, **L**ean forward, **E**ye contact, **R**elaxed, though steady eye contact feels intrusive to many Indian clients.

An **open question** ("Tell me about...") invites a long answer in the client's own words; a **closed question** ("When did the sleep go?") pins down a fact. Move from open to closed. Table 7.6 shows the other skills. **Reflection of feeling** is the signature technique of **Carl Ransom Rogers** (1902 to 1987) in client-centred therapy (Section 7.6); interpretation belongs chiefly to psychodynamic therapy (Section 7.3). **Empathy**, understanding the client's world from inside and showing it, has two levels (Egan): primary empathy reflects what is said; advanced empathy voices what is only hinted, so it waits for trust.

Table 7.6 Microskills. Ravi says: "My father keeps comparing me to my cousin. I don't even go home for meals now."

Skill	What it does	Example
Minimal encourager (furthering)	Keeps him talking, without steering	A nod, leaning forward, "Mm-hmm"
Paraphrase	Restates content in different words	"Being compared makes home hard."
Reflection of feeling	Names the emotion	"You sound hurt."
Clarification	Makes the vague precise	"What does he actually say?"
Checking	Confirms your understanding	"You skip meals to avoid him; is that right?"

Skill	What it does	Example
Summarising	Pulls threads together	"The result, your cousin, your father: all at once."
Confrontation	Gently notes a discrepancy	"You say you don't care, yet you avoid home."
Interpretation	Offers a new meaning	"Avoiding him spares you feeling you failed him."
Silence	Gives room to feel	A pause after pain

EXAM TRAP

Nodding and "umm" are **furthering**, not **prompting** (steering to a topic: "What about work?") or **checking**. Restating **in different words** is **paraphrasing**, not questioning, empathy (the broader capacity) or nonverbal communication. "Tell me about" is an **open-ended question**, even inside an ethnographic interview (learning a culture in the insider's terms) or theory-guided inquiry (questions chosen from a theory).

Common errors are **leading questions** ("You've been depressed, haven't you?"), **double-barrelled questions** (two in one), accusing "why" questions, premature reassurance and premature closure on an early hypothesis. With children, use play, drawing and concrete words and add parents' and teachers' reports; with older adults, check hearing and vision before inferring confusion (Section 7.10); with resistant clients, acknowledge that coming was not their idea; with suicidal clients, ask directly, since asking does not plant the idea. Across cultures, elicit the client's **explanatory model** (his name for the problem, its cause, the cure he expects), as **Arthur Kleinman** (1980) urged, and respect **idioms of distress** such as "tension". Use a trained interpreter, never a relative or child.

The case history

The case history, the story of person and illness, follows a fixed order:

1. **Identifying data** (age, education, occupation, marital status, religion, socioeconomic status), then the **informant** and the account's reliability (relationship, closeness of contact, consistency).
2. **Chief complaints**, in the patient's own words, each with its duration ("Low mood, 3 months").
3. **History of present illness**: onset (abrupt, acute or insidious), course, **precipitating factors** and treatment so far.
4. **Past psychiatric and medical history**, and **family history** drawn as a **genogram** (three generations; squares for males, circles for females).
5. **Personal history**: birth and development, education, occupation, marital and sexual life, substance use and forensic record.
6. **Premorbid personality**: the person before the illness, the baseline for judging change.

The MSE: behaviour, mood, thought and perception

The **mental status examination** (MSE) is a snapshot of mental functioning during the interview. It records **appearance and behaviour**, **attitude towards the examiner** (cooperative, guarded, hostile), **psychomotor activity** (retardation, agitation) and **speech** (rate, volume, tone, reaction time, relevance, coherence). **Mood** is the sustained emotion the patient re-

ports: subjective, the climate. **Affect** is the emotion the examiner observes: objective, the weather. Affect is rated for range, reactivity, appropriateness and stability: it may be **blunted** (markedly reduced), **flat** (virtually absent), **labile** (shifting rapidly) or **inappropriate** (laughing about a death).

Thought has four headings: **stream** (speed and flow), **form of thought** (thought process: how ideas connect; Table 7.7), **content of thought** (what is thought: delusions, obsessions, overvalued ideas, phobias, suicidal or homicidal ideas) and **possession** (whose thoughts they feel to be). A **delusion** is a false, unshakeable belief out of keeping with the person's culture: persecutory, of reference (chance events carry a personal message), grandiose, nihilistic (the self, body or world does not exist), of jealousy, erotomanic (a person of higher status is secretly in love with one), of guilt or somatic (the body is diseased or changed). An **overvalued idea** is an understandable belief held with excessive emotion but without delusional fixity. An **obsession** is an intrusive thought recognised as one's own and resisted (Indian case formats file obsessions under possession, as lost control of thinking); a **phobia** is an excessive fear of a specific object, situation or idea. In **thought alienation** thoughts feel inserted, withdrawn or broadcast; these are among the first-rank symptoms of schizophrenia listed by **Kurt Schneider** (1887 to 1967; Chapter 8).

Table 7.7 Disorders of the stream and form of thought.

Term	What you hear	Typical of
Flight of ideas	Rapid jumps by rhyme or chance; links traceable	Mania
Loosening of associations	Slips to unrelated ideas; links lost	Schizophrenia
Tangentiality	Veers off; never reaches the point	Schizophrenia, mania
Circumstantiality	Needless detours, but the point is reached	Epilepsy, obsessional traits
Thought block	Sudden stop; mind goes blank	Schizophrenia
Perseveration	Same answer repeated to new questions	Brain damage
Neologism; word salad	New words; words strung without connection	Schizophrenia
Poverty of speech	Very little speech, brief replies	Schizophrenia, depression

EXAM TRAP

Circumstantiality is a disorder of the **form of thought** (thought process), not content, possession or perception (some notes file it under stream). The circumstantial patient arrives late; the tangential one never arrives.

A **hallucination** is a perception without an external stimulus, felt as real: auditory ones are commonest in schizophrenia, visual ones suggest delirium, olfactory and gustatory ones temporal lobe epilepsy, tactile ones cocaine use or alcohol withdrawal; those at sleep onset (hypnagogic) or waking (hypnopompic) are normal. An **illusion** misperceives a real stimulus (a rope seen as a snake). A **pseudohallucination** is perceived in inner space ("inside my head"). In **depersonalisation** the self feels unreal; in **derealisation**, the surroundings.

The MSE: cognition, insight and judgement

Cognitive functions are tested in order: **consciousness** (alert to coma); **orientation** to time, place and person (time is lost first); **attention and concentration** (digit span, **serial sevens** from 100, months backwards); **memory**, immediate (repeat three words), recent (recall them after five minutes) and remote (childhood events); **intelligence and general information**; **abstract thinking** (proverbs, similarities: literal answers are concrete); and **judgement**, personal (plans), social (conduct) and test ("What would you do if you saw smoke in a cinema?").

Insight is the patient's awareness and understanding of his own illness, graded on six levels: (1) complete denial; (2) slight awareness of illness while still denying it; (3) awareness, but blamed on others or physical causes; (4) awareness of something unknown within oneself; (5) **intellectual insight**, accepting the illness without changing behaviour; (6) **true emotional insight**, awareness that leads to change.

> EXAM TRAP
>
> Insight is **awareness of one's own illness**, even where a case format files it among the cognitive functions. It is not orientation, consciousness, immediate recall, serial subtraction (attention) or hallucinations (perception).

Write the MSE in the present tense, describing rather than labelling. For Ravi: *Slowed; rapport difficult. Mood "useless"; affect depressed, restricted. Hopelessness, passive death wishes, no plan. Oriented. Insight grade 4.*

Risk assessment: suicide and violence

Suicide risk factors include a previous attempt (the strongest predictor), depression, substance use, schizophrenia, hopelessness, male sex (women attempt more often), living alone, chronic illness, recent loss and access to means such as pesticides. **Protective factors** include family support, responsibility for children, religious objections and reasons for living. The **SAD PERSONS** mnemonic (Patterson and colleagues, 1983) scores one point each for Sex (male), Age (under 19 or over 45), Depression, Previous attempt, Ethanol abuse, Rational thinking loss, Social support lacking, Organised plan, No spouse and Sickness. It aids memory but predicts poorly: suicide is rare, so scales raise many false alarms yet miss many who act. Ask instead about ideation, intent, plan and means, and build a safety plan.

For violence, past violence is the best predictor, with youth, male sex, substance misuse and threats. Structured professional judgement such as the **HCR-20** (Historical, Clinical and Risk management items) beats unaided judgement; dynamic factors, unlike static history, can be treated.

Behavioural assessment

Behavioural assessment asks what a behaviour does for the person here and now. Its core, **functional analysis**, rests on the operant conditioning of **Burrhus Frederic Skinner** (1904 to 1990): consequences maintain behaviour. The target behaviour is defined operationally and its **ABC** recorded: **A**ntecedent, **B**ehaviour, **C**onsequence. A boy who throws his glass when told to clear it, and so escapes the chore, is being reinforced. The **SORC** model adds **O**rganism variables (biology, thoughts, learning history) between **S**timulus and **R**esponse.

Data come from **behavioural interviews** (what, when, how often), **self-monitoring** (the client's own diary, essential for private acts such as purging), **role play** (the client enacts a problem situation, such as refusing a request, while the clinician watches), and observation that is **naturalistic** (home, school), **analogue** (a simulated clinic situation) or **participant** (by a parent, teacher or nurse already present). **Event recording** counts occurrences; **interval recording** notes whether the behaviour occurred in each short interval; **time sampling** observes at chosen moments. **Reactivity** is behaviour changing because it is observed; **observer drift** is observers' gradual shift in applying definitions, corrected by retraining.

> EXAM TRAP
>
> Functional analysis is based on **operant conditioning**, not classical conditioning (Section 7.4), humanistic or psychodynamic principles. Its ABC is **Antecedent, Behaviour, Consequence**; Albert Ellis's ABC is **Activating event, Belief, Consequence** (Section 7.5).

Case formulation

A **diagnosis** places problems in a category: what the person has. A **case formulation** is an individualised, testable hypothesis about why this person has these problems now (causes, triggers, maintaining factors), revised as therapy proceeds. The commonest framework is the **5 Ps**: **presenting** problem, **predisposing** factors (early vulnerabilities), **precipitating** factors (triggers), **perpetuating** factors (what maintains it) and **protective** factors (strengths), as Figure 7.2 shows for Ravi. Filling each P with biological, psychological and social entries gives a **biopsychosocial formulation**, after **George Libman Engel** (1913 to 1999), who proposed the model in *Science* in 1977.

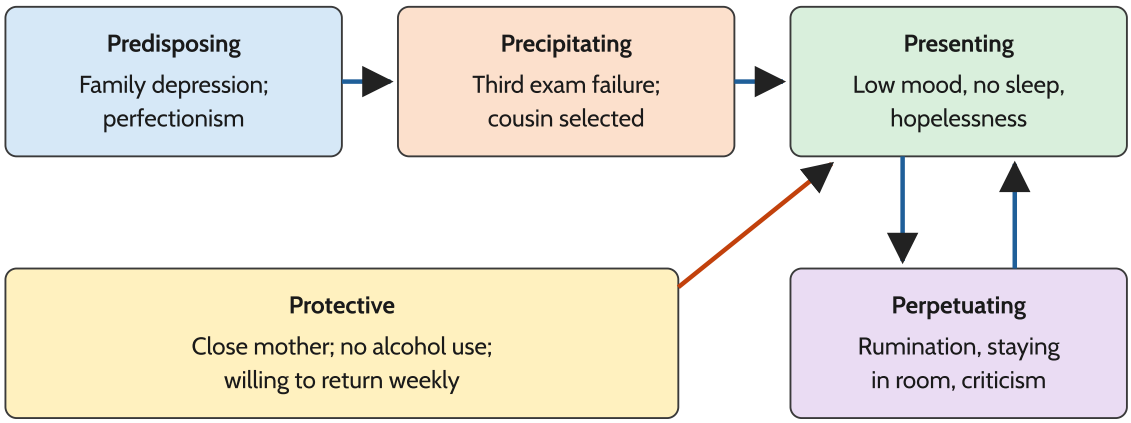

Figure 7.2 The 5 Ps applied to Ravi. Solid arrows run from vulnerability through trigger to the problem, which loops with its perpetuating factors; the dashed arrow marks protection.

In a cognitive-behavioural formulation, the **cognitive conceptualisation diagram** of **Judith S. Beck** (1995) links childhood data to **core beliefs** ("I am a failure"), rules ("If I fail the exam, I am worthless") and compensatory strategies, and traces each situation through automatic thought, meaning, emotion and behaviour. A psychodynamic formulation describes core unconscious conflicts, defences and repeating relationship patterns rooted in early life.

A **differential diagnosis** lists the conditions that could explain the picture and rules them in or out, physical and substance causes first (hypothyroidism can mimic depression). The **clinical report** answers the referral question in plain language (sources, history, MSE, results, formulation, diagnosis, recommendations); **feedback** begins with strengths and ends with an agreed plan.

> EXAM TRAP
>
> A formulation **explains**: it hypothesises causes, precipitants and maintaining factors. Classifying symptoms by DSM-5 is **diagnosis**, predicting course is **prognosis**, and ruling out organic causes is **differential diagnosis**.

Classification as a tool

Classification gives clinicians a shared language for communication, treatment, research and statistics (Table 7.8; disorders in Chapter 8).

Table 7.8 DSM-5-TR and ICD-11 compared.

Feature	DSM-5-TR	ICD-11
Publisher	American Psychiatric Association	World Health Organization
Dates	DSM-5 2013; text revision 2022	Adopted 2019; in force 2022
Scope	Mental disorders only	All diseases; mental disorders in Chapter 6
Structure	Three sections: basics; criteria and codes; emerging measures	Codes such as 6A20 (schizophrenia)
Style	Operational criteria; no multiaxial system	Flexible diagnostic requirements

Diagnostic agreement is measured by **Cohen's kappa** (**Jacob Cohen**, 1923 to 1998; 1960), which corrects raw agreement for chance: $\kappa = (P_o - P_e) / (1 - P_e)$. Kappa is 1 for perfect agreement and 0 for chance level; above about 0.6 is good. Even with explicit criteria, the DSM-5 field trials found kappa near 0.28 for major depressive disorder.

Labels can stigmatise, and they reflect values: **pathologisation** means classifying a condition as a mental disorder and **depathologisation** removing it, as when the American Psychiatric Association removed homosexuality from DSM-II in 1973 (the WHO followed in ICD-10, 1990).

> KEY STUDY: ROSENHAN, "ON BEING SANE IN INSANE PLACES" (1973)
>
> **David L. Rosenhan** (1929 to 2012) had eight sane volunteers, himself among them, seek admission to 12 US hospitals reporting voices saying "empty", "hollow" and "thud", then behaving normally. All were admitted, all but one diagnosed with schizophrenia, and stayed 7 to 52 days; note-taking was charted as "writing behavior". It became the classic case against labelling. **Robert Leopold Spitzer** (1932 to 2015) replied (1975) that faked symptoms say nothing about diagnosis; later inquiry questioned the study's conduct.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Nodding, leaning forward and saying "umm" are _____ responses.
2. The SCID, though named "structured", is strictly _____.
3. Reaching the point after needless detours is _____, a disorder of the _____ of thought.
4. Mood is _____; affect is _____.
5. Grade 6 insight is _____ insight.
6. Functional analysis records Antecedent, Behaviour and _____, and rests on _____ conditioning.
7. The five Ps are presenting, predisposing, precipitating, _____ and protective.

Answers: (1) furthering, (2) semi-structured, (3) circumstantiality; form, (4) subjective; objective, (5) true emotional, (6) Consequence; operant, (7) perpetuating.

7.3 Psychodynamic Therapies and Their Descendants HIGH YIELD · 39 QS

Every therapy in this section assumes that hidden forces pushing against one another (hence *dynamic*) shape what we feel and do, and that bringing them into awareness relieves suffering. The idea began with one patient in Vienna, became Freud's psychoanalysis, and then branched into a large family.

Freud's psychoanalysis as a treatment ●●●

KEY STUDY: BREUER'S ANNA O. (1880 TO 1882)

The physician **Josef Breuer** (1842 to 1925) treated Bertha Pappenheim, a young woman with hysterical paralyses and weeks in which she could not drink water. When, in a self-induced trance, she talked through the moment a symptom began and released the feeling she had swallowed then, the symptom eased: her "talking cure". Breuer and Freud published her as "Anna O." in *Studies on Hysteria* (1895). Criticism: she was not in fact cured, and spent months in sanatoria afterwards.

Breuer's procedure was the **cathartic method**: hypnotise the patient, take her back to the moment a symptom began and let her relive it. **Abreaction** is the discharge of the emotion attached to a particular repressed memory by reliving it; **catharsis** (Greek, "purging") is the release and relief that follow. **Sigmund Freud** (1856 to 1939), the father of psychoanalysis, then dropped hypnosis: many patients could not be hypnotised, its cures faded, and it went round resistance instead of exposing it. At Nancy in 1889 he had watched **Hippolyte Bernheim** (1840 to 1919) recover "forgotten" memories from waking patients.

The aim of analysis is to make the unconscious conscious, or, in Freud's formula of 1933, "Where id was, there ego shall be". Change comes through **insight**, an emotionally felt understanding of how present feelings repeat past conflicts, and the target is character, not just the symptom. The patient lies on a couch with the analyst out of sight behind, four or five times a

week for years; once- or twice-weekly, face-to-face work on the same principles is psychodynamic psychotherapy. Every analyst first undergoes a training analysis.

The patient's one duty is the **fundamental rule**: say whatever comes to mind, however trivial or embarrassing, without censoring it. Following the rule is **free association**, the defining technique of psychoanalysis. When conscious steering relaxes, Freud argued, thoughts follow paths laid down by unconscious wishes, and the places where association stalls point to what is repressed. The analyst brings evenly suspended attention; **neutrality**, taking no side in the patient's conflicts and giving no advice (the "blank screen"); and **abstinence**, declining to gratify the patient's wishes for love, praise or comfort so that they stay alive to be understood. Abstinence here has nothing to do with alcohol.

Dreams, Freud wrote in *The Interpretation of Dreams* (1900), are "the royal road" to the unconscious. The **manifest content**, the dream as remembered, disguises the **latent content**, the hidden wish, through the dream-work: condensation, displacement, symbolisation and secondary revision. The patient free-associates to each element to recover it.

Sooner or later the patient arrives late, forgets a session or changes the subject just as something painful approaches. This is **resistance**: everything in the patient that opposes the treatment and access to unconscious material. It is repression showing itself in the room, so in **analysis of resistance** the analyst points out the obstruction and explores with the patient what it protects against. **Interpretation** links what the patient says or does to its unconscious meaning; it is prepared by confrontation and clarification and aimed at the defence before the content it hides. Because one insight rarely changes a lifelong pattern, **working through** (Freud, 1914) re-examines the same conflicts and interpretations in context after context until lasting change in behaviour and character follows.

> REMEMBER
>
> Origin of disorder by school: psychodynamic, unconscious conflict from childhood emotional experience; behavioural, faulty learning (Section 7.4); cognitive, disturbed thought patterns (Section 7.5); humanistic, failure to fulfil one's potential (Section 7.6).

> EXAM TRAP
>
> Saying whatever comes to mind, uncensored, is free association; systematic desensitisation, flooding and modelling are behaviour therapy (Section 7.4), and paradoxical intention is Frankl's and Adler's. Obstructing the analysis while the therapist explores why is analysis of resistance. Working through is never "giving the interpretation once", advice or ending after one insight.

Transference and countertransference ●●●

In the Dora case (1905) Freud saw that patients come to feel about the analyst as they once felt about their parents. **Transference** is the patient's unconscious redirection onto the therapist of feelings and expectations that belong to significant figures from the past. It may be positive (affection, idealisation, erotic love) or negative (hostility, suspicion). When the patient's neurosis is re-created around the analyst, it becomes a **transference neurosis**, whose resolution is, in classical theory, the cure.

Countertransference, named by Freud in 1910, is the therapist's emotional reaction to the patient arising from the therapist's own unresolved conflicts, a blind spot to be recognised (hence the training analysis). Since **Paula Heimann**'s paper of 1950 the term also covers all the therapist's feelings towards the patient, used as information about the patient. **Negative countertransference** is the therapist's hostility, irritation or boredom towards a patient; positive countertransference is over-involvement or a wish to rescue. Both differ from the working alliance (**Ralph Greenson**, 1965), the patient's rational collaboration with the therapist (Section 7.8).

> EXAM TRAP
>
> Direction decides the term: patient to therapist is transference; therapist to patient is countertransference. "The therapist's emotional reaction to the client" is never transference, "the client's projection onto the therapist" is never countertransference, and a therapist's negative feelings are negative countertransference, not negative transference. Fear of the unknown or low motivation are sources of resistance, not countertransference. Managing both is the hallmark of psychodynamic models, not of behaviour modification, CBT or solution-focused therapy.

Defences and psychic energy ●●

In Freud's language of psychic energy, **libido** is the energy of the sexual (later, life) drive; **cathexis** is its investment in the mental representation of a particular object, person or idea; **fixation** is libido left attached to an earlier psychosexual stage; and **sublimation** redirects a drive into a socially valued aim. **Defence mechanisms** are the ego's unconscious strategies against anxiety. Their theory is in Chapter 4; Table 7.9 shows how a clinician meets them.

Table 7.9 Defences as the clinician meets them.

Defence	What happens	In the room
Repression	Threatening wish or memory kept out of awareness; the primary defence	A painful year cannot be recalled
Projection	Own unacceptable feeling attributed to another	Hostile patient sure the therapist dislikes him
Displacement	Feeling moved from a dangerous target to a safer one	Scolded by the boss, he shouts at his son
Reaction formation	Impulse turned into its exaggerated opposite	Gushing sweetness to a resented sister
Rationalisation	Respectable reasons for behaviour with other motives	"I failed because the paper was unfair"
Isolation of affect	Idea kept, its feeling split off	Flat account of a father's death
Splitting	Self or others seen as all good or all bad	Therapist idealised, then despised
Suppression	Conscious postponement of a worry	"I will think about it after the exam"

Repression is also the usual psychodynamic explanation of dissociative symptoms, and conversion turns a psychological conflict into a bodily symptom, as in Anna O.'s paralysis.

> **EXAM TRAP**
>
> Energy invested in a particular object, person or idea is cathexis, not libido, fixation or sublimation. Projection gives your feeling to someone else; displacement keeps it yours but moves its target. The primary defence, and the psychodynamic source of dissociation, is repression, not conversion, isolation or "ego state" (Paul Federn's term, later Berne's).

Ego psychology, object relations and self psychology ●●

Freud's successors kept the unconscious but changed what they listened for. **Ego psychology** turned from the id to the ego. **Anna Freud** (1895 to 1982), in *The Ego and the Mechanisms of Defence* (1936), made the analysis of defences central and pioneered child analysis; **Heinz Hartmann** (1894 to 1970) described a conflict-free ego sphere that develops for adaptation (1939); and **Erik Erikson** (1902 to 1994) added psychosocial stages and the adolescent identity crisis (Chapters 4 and 6).

Object relations theory holds that the mind is built from internalised representations of self and others ("objects") and of the relationships between them. **Melanie Klein** (1882 to 1960) analysed children through play. In her paranoid-schizoid position the infant manages terror by **splitting** the world into wholly good and wholly bad objects and by projective identification (placing unwanted parts of the self in another and then controlling that person); in the later depressive position good and bad are seen as one person, bringing guilt and reparation. Paranoia revives the first position, so its defence is splitting. **Donald Winnicott** (1896 to 1971) described the **good-enough mother**, closely attuned at first and then failing tolerably, who lets a true self grow; the **transitional object**, a blanket or soft toy that bridges inner and outer reality; and therapy as a **holding environment**, as reliable as a mother's arms. **Ronald Fairbairn** (1889 to 1964) held that libido is object-seeking, not pleasure-seeking.

Margaret Mahler (1897 to 1985) traced **separation-individuation**: a normal autistic phase, a symbiotic phase of fusion with the mother, then differentiation, practising, rapprochement and object constancy by about three years. Her theory of psychosis (1952) traces schizophrenia to these first phases: the person remains in the autistic phase and fails to develop an adequate, stable symbiosis (Silvano Arieti, Ping-Nie Pao and Edith Jacobson wrote other analytic accounts of psychosis). **Otto Kernberg** (born 1928) described borderline personality organisation (identity diffusion, splitting, intact reality testing) and developed **transference-focused psychotherapy** (TFP), which interprets the split images of self and others as they appear in the transference.

Heinz Kohut (1913 to 1981) founded **self psychology** with *The Analysis of the Self* (1971). The child needs selfobjects who provide **mirroring** (confirming the child's worth), an **idealised** figure to look up to and merge with and, in his later work, **twinship** (feeling like others). Chronic failure of mirroring and idealising leaves a fragile self that fragments into emptiness and rage. In therapy these needs return as mirror, idealising and twinship transferences, met by the analyst's **empathic attunement**.

Relational psychoanalysis (**Stephen Mitchell**, 1946 to 2000, with Jay Greenberg, 1983) treats therapy as a two-person process in which the analyst cannot be neutral. **John Bowlby**'s (1907 to 1990) attachment theory inspired attachment-based therapies in which the therapist is a secure base for revising internal working models. **Mentalisation-based treatment** (MBT), by **Anthony Bateman** and **Peter Fonagy** for borderline personality disorder (first trial 1999),

strengthens mentalisation, understanding one's own and others' behaviour in terms of feelings, wishes and intentions, a capacity that collapses under attachment stress.

> **EXAM TRAP**
>
> "Object relations" means internalised representations of self and others and their relationships, not the id-ego-superego relation, defences or the genital stage. A client who feels empty, "no one", afraid of abandonment and hungry for admiration is explained by failed parental mirroring and idealising leading to a fragmented self: that is Kohut's self psychology even when the stem says "object relations". Oral fixation, an unresolved Oedipus complex and a harsh superego are classical Freudian explanations of neurotic conflict and guilt (Chapter 4). For paranoia "as given by Klein", answer splitting; projection is Freud's classical explanation of paranoia.

Brief psychodynamic psychotherapies

Brief dynamic therapies keep the psychodynamic ideas but use one agreed focus, a limited number of sessions (usually 12 to 40), an active therapist who questions and confronts instead of waiting for free association, and early work on the transference. **David Malan** of the Tavistock Clinic (*The Frontier of Brief Psychotherapy*, 1976) supplied their map (Figure 7.3): the **triangle of conflict** (hidden feeling, anxiety, defence) and the **triangle of person** (others now, the therapist, parents then). Table 7.10 lists the models of Malan, **Peter Sifneos** (1920 to 2008), **Habib Davanloo** (born 1927), **James Mann** (1973), **Lester Luborsky** (1984), and **Hans Strupp** and Jeffrey Binder (1984).

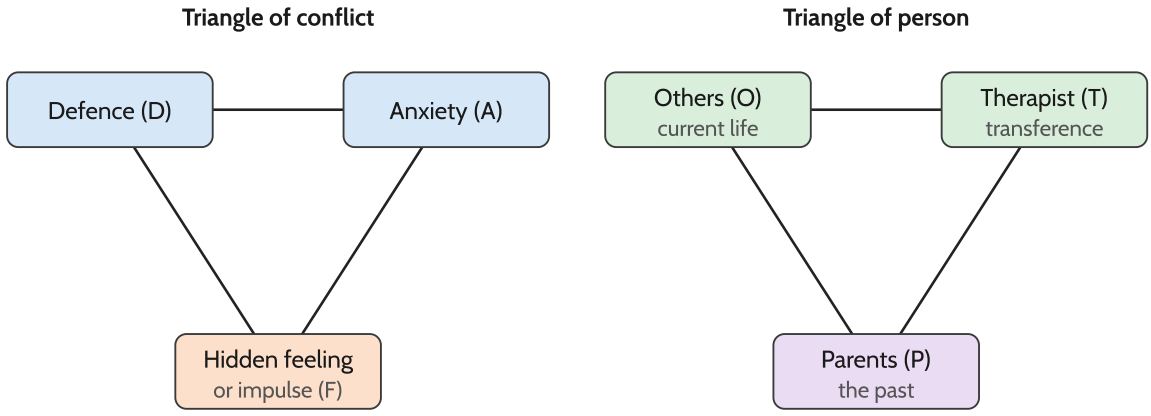

Figure 7.3 Malan's two triangles. The conflict on the left recurs at every corner of the triangle on the right; an interpretation linking therapist, others and parents (T-O-P) is the most powerful.

Table 7.10 The main brief psychodynamic psychotherapies.

Founder	Therapy	Hallmark
Malan	Brief focal psychotherapy	Two triangles; T-O-P links
Sifneos	Short-term anxiety-provoking psychotherapy (STAPP)	Strictly selected, motivated, psychologically minded patients; Oedipal focus
Davanloo	Intensive short-term dynamic psychotherapy (ISTDP)	Relentless pressure on defences

Founder	Therapy	Hallmark
Mann	Time-limited psychotherapy	Exactly 12 sessions; a central issue; time and loss
Luborsky	Supportive-expressive therapy	Core conflictual relationship theme: wish, response of other, response of self
Strupp and Binder	Time-limited dynamic psychotherapy	Cyclical maladaptive pattern

Interpersonal psychotherapy ●●

Interpersonal psychotherapy (IPT) was developed by **Gerald Klerman** (1928 to 1992) and his wife **Myrna Weissman** in the manual *Interpersonal Psychotherapy of Depression* (1984). Rooted in the ideas of Adolf Meyer, Sullivan and Bowlby, it works on current relationships rather than childhood conflict, does not interpret the transference, and treats depression as a medical illness, giving the patient the sick role.

IPT runs 12 to 16 weekly sessions in three phases. The initial phase diagnoses and explains depression, reviews key relationships (the interpersonal inventory) and picks one or two of four problem areas: **grief** (complicated bereavement), **interpersonal role disputes** (with a spouse, parent or employer), **role transitions** (marriage, a new job, retirement, illness) and **interpersonal deficits** (social isolation). The middle phase works on that area through encouragement of affect, clarification, communication analysis (who said what, exactly), role play and decision analysis. Termination treats ending as a small grief and credits the patient's own competence. IPT matched cognitive therapy in the NIMH depression trial (Section 7.8), meta-analyses rate it about equal to CBT, and adaptations treat depressed adolescents (IPT-A), perinatal depression and bulimia.

> EXAM TRAP
>
> Asked what IPT does NOT include beside grief, role conflict and role transition, choose "role playing and role reversal": the other three are its problem areas. The fine print: IPT therapists do use role play after communication analysis; role reversal is psychodrama's (Jacob Moreno, Section 7.7). IPT is Klerman and Weissman's; MBT, Bateman and Fonagy's; TFP, Kernberg's; **Lorna Smith Benjamin** developed interpersonal reconstructive therapy.

Jung, Adler, Horney and Sullivan as therapists ●●●

Freud's early dissenters built their own therapies (their personality theories are in Chapter 4). **Carl Gustav Jung** (1875 to 1961) founded analytical psychology. Beneath the personal unconscious he placed a **collective unconscious** of inherited archetypes: the persona (the social mask), the shadow, the **anima** (the unconscious feminine side of a man), the **animus** (the unconscious masculine side of a woman) and the self. Jung read dreams as direct, creative messages, not disguises. In **amplification** the analyst enlarges a dream image with parallels from myth and religion; in **active imagination** the patient, awake, holds a dialogue with a dream or fantasy figure. The goal is **individuation**, becoming a whole, integrated self.

Alfred Adler (1870 to 1937) left Freud in 1911 and called his school **individual psychology**. From **organ inferiority** (1907: compensation for a weak organ) he moved to universal feelings of inferiority that drive a striving for superiority; when they become overwhelming they form an **inferiority complex**. Therapy is an equal, educational partnership. Lifestyle assess-

ment explores the family constellation, birth order (the dethroned first-born, the competitive second, the pampered youngest) and early recollections, which reveal the present style of life. Its techniques are **encouragement**, the core one; **acting as if** (behaving as though one already had the feared ability); **spitting in the client's soup** (exposing a behaviour's hidden payoff so that it loses its appeal); the **push-button technique** (Harold Mosak: picturing a pleasant and then an unpleasant memory to show that we create our own feelings); and paradoxical intention. Adlerians do not analyse transference.

Karen Horney (1885 to 1952) traced neurosis to basic anxiety, met by moving towards, against or away from people; her therapy dismantled the idealised self-image and its "tyranny of the shoulds". **Harry Stack Sullivan** (1892 to 1949) made the therapist a participant observer of interpersonal patterns, including parataxic distortion (seeing others through the template of earlier relationships), and so influenced IPT.

> EXAM TRAP
>
> Analysis of transference is the technique an Adlerian counsellor would not use; acting as if, the push button and encouragement are all Adlerian. Collective unconscious, anima and animus: Jung. Individual psychology, inferiority complex, organ inferiority: Adler, not Jung, Horney, Frankl (logotherapy, Section 7.6) or Erikson.

Transactional analysis ●●

Transactional analysis (TA) was founded by **Eric Berne** (1910 to 1970), who took the term ego state from his analyst Paul Federn, in *Transactional Analysis in Psychotherapy* (1961) and *Games People Play* (1964). It proceeds through structural, transactional, game and script analysis.

Personality has three **ego states**. The **Parent** holds attitudes and behaviour copied from parents and parental figures, critical or nurturing; the **Adult** processes present reality objectively; the **Child** holds feelings and responses from childhood, free or adapted. A transaction is a stimulus from one ego state and a response from another (Figure 7.4). Complementary transactions run parallel and talk continues. Crossed ones stop it: "Where are my keys?" (Adult to Adult) answered by "Why do you always blame me?" (Child to Parent). Ulterior transactions carry a hidden message beneath the social one.

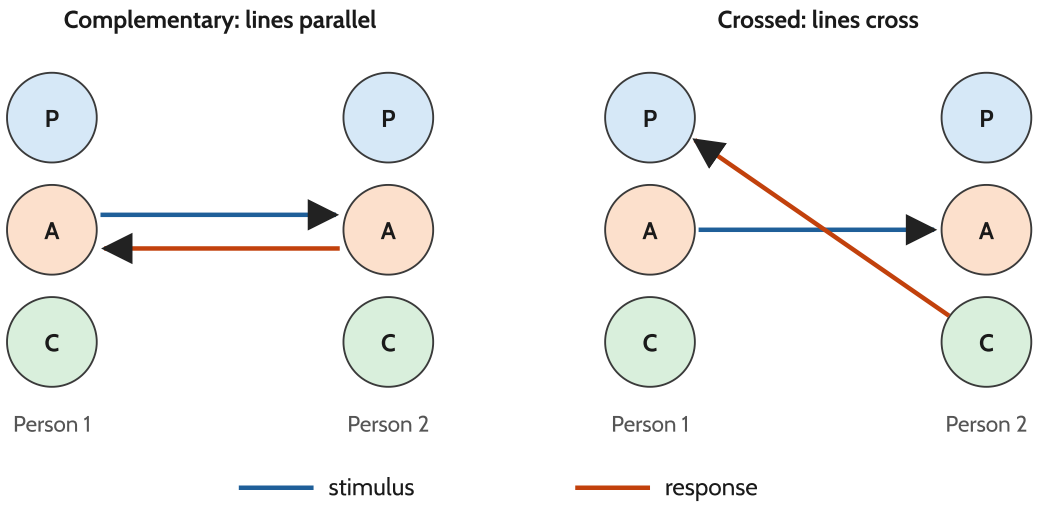

Figure 7.4 Ego states (P, Parent; A, Adult; C, Child). Left: Adult to Adult and back, so talk goes on. Right: the reply comes from Child to Parent, the lines cross and communication stops.

People need **strokes**, units of recognition, and prefer negative strokes to none. A **game** is a series of ulterior transactions with a predictable payoff of bad feelings, such as "Why Don't You, Yes But", in which every suggestion is rejected. A **life script** is an unconscious life plan decided in childhood under parental **injunctions** ("Don't be close", "Don't grow up", sent nonverbally from the parent's Child) and **counterinjunctions** ("Be perfect", "Work hard", spoken from the parent's Parent); Bob and Mary Goulding's redecision therapy helps clients revise them. Scripts rest on one of four **life positions**: I'm OK, you're OK (the healthy one); I'm OK, you're not OK; I'm not OK, you're OK; and I'm not OK, you're not OK, the first made famous by **Thomas Harris**'s best-seller *I'm OK, You're OK* (1969).

> **EXAM TRAP**
>
> Injunctions, counterinjunctions, games, scripts and strokes all mean transactional analysis, not Gestalt, REBT or person-centred therapy (Sections 7.5 and 7.6). The ego state copied from parents is the Parent; "persona" is Jung's archetype, not an ego state.

Supportive therapy, hypnosis, narcoanalysis and the evidence

Psychodynamic work runs along a continuum from expressive (uncovering conflict by interpretation, for patients with ego strength) to supportive. **Supportive psychotherapy** strengthens defences instead of dismantling them, through a warm relationship, reassurance, advice, encouragement, ventilation of feelings, education and practical changes in the environment. It is the mainstay when the ego is overwhelmed: in crisis management, bereavement, acute psychosis and severe physical illness.

Hypnotherapy uses hypnosis, focused attention with heightened suggestibility. It descends from **Franz Anton Mesmer**'s (1734 to 1815) "animal magnetism" and was named by James Braid in 1843. Today it assists the treatment of pain, habits and anxiety, but memories "recovered" under it are unreliable. **Narcoanalysis** gives a slow intravenous barbiturate (sodium amobarbital or thiopentone) so that a mute, stuporous or amnesic patient talks and abreacts,

as in Second World War battle neuroses. It is no "truth serum": drugged people can still lie and fantasise.

> **INDIAN CONTEXT**
>
> In *Selvi v. State of Karnataka* (2010) the Supreme Court held compulsory narcoanalysis, polygraph and brain-mapping unconstitutional (Articles 20(3) and 21); they need consent, and the statements made are not admissible as evidence.

Critics from **Hans Eysenck** (1952; Section 7.8) to the philosopher **Karl Popper**, who called its theory unfalsifiable, attacked psychoanalysis as unproven, long and costly. Yet meta-analyses reviewed by **Jonathan Shedler** (2010) show psychodynamic therapy matching other evidence-based therapies, with gains that grow after treatment ends; short-term dynamic therapy and IPT work for depression, and MBT and TFP for borderline personality disorder. Evidence for the specific mechanism, insight, remains weaker.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The patient's duty to say whatever comes to mind, uncensored, is the _____ rule.
2. Psychic energy invested in a particular person, object or idea is _____.
3. The therapist's emotional reaction to the patient, from the therapist's own unresolved conflicts, is _____.
4. Mann's time-limited psychotherapy lasts exactly _____ sessions.
5. IPT's four problem areas are grief, role disputes, role transitions and interpersonal _____.
6. In TA, a reply from an unexpected ego state that halts communication is a _____ transaction.

Answers: (1) fundamental, (2) cathexis, (3) countertransference, (4) 12, (5) deficits, (6) crossed.

7.4 Behaviour Therapy HIGH YIELD · 88 QS

A child terrified of rabbits, a woman who washes until her hands bleed, a boy with autism who cannot ask for water: behaviour therapy sees in each of them learned behaviour that can be unlearned or replaced. Building on the learning principles of Section 1.6, we trace where the approach came from, then take its techniques family by family.

Origins, assumptions and functional analysis ●●

Behaviour therapy is the application of experimentally established principles of learning to the assessment and treatment of psychological problems. Its spur was a 1952 paper by **Hans Jürgen Eysenck** (1916 to 1997) of the Maudsley Hospital, London, which claimed that about two-thirds of neurotic patients improved within two years whether or not they had psychotherapy (the debate belongs to Section 7.8). If talking did no better than time, Eysenck argued, the clinic needed a treatment derived from learning theory, aimed at the symptom and testable. His creed was blunt: get rid of the symptom and you have eliminated the neurosis.

The new therapy grew from three roots in the 1950s. In Johannesburg, Joseph Wolpe, **Arnold Allan Lazarus** (1932 to 2013) and **Stanley Rachman** worked from classical conditioning. At the Maudsley, Eysenck supplied the theory and **Monte Bernard Shapiro** the method of treating each patient as a single-case experiment. In the United States, **Burrhus Frederic Skinner** (1904 to 1990) and his students carried operant methods onto hospital wards and into schools, a strand later called **applied behaviour analysis** (ABA): operant principles, carefully measured, used to change socially important behaviour. The name *behaviour therapy* first appeared in a 1953 hospital report by Ogden Lindsley, Skinner and Harry Solomon, reached a journal in Lazarus's 1958 paper and was popularised by Eysenck's 1959 paper.

The shared assumptions all contradicted psychoanalysis. Problem behaviour is learned by the same principles as normal behaviour and can be unlearned. The behaviour is the problem, not the symptom of a hidden conflict, so therapy aims to eliminate it and build adaptive behaviour in its place. What maintains a problem now matters more than how it began, and everything is defined and measured. Applied to anxiety, this gives the **learning model of disorder**, best known as the **two-factor theory** of O. Hobart Mowrer (1947): a fear is acquired by classical conditioning and maintained because avoiding the feared object reduces fear and is therefore negatively reinforced. Avoidance stops the person ever learning that the object is harmless, which is why exposure works.

Every case starts with a **functional analysis**: identifying the observable events that trigger and maintain a specific problem behaviour. Its simplest form is the **ABC model** of Antecedents, Behaviour (defined so that two observers would agree) and Consequences; **Frederick Kanfer** and **George Saslow** (1965) expanded it into SORKC (stimulus, organism, response, contingency, consequence). After recording a **baseline**, the frequency, duration or intensity of the behaviour before treatment, the therapist asks what the behaviour achieves. If Arjun's tantrums at homework time get him sent to his room, they are rewarded by escape, and the "punishment" feeds them. Treatment follows the function, not the form.

> EXAM TRAP
>
> Behaviour therapies apply learning principles to **eliminate the unwanted behaviour**; "treat the underlying causes" is the psychodynamic aim, "improve learning and insight" that of insight therapies and "communication and social sensitivity" that of encounter groups (Section 7.7). Behaviour analysis rests on **operant conditioning**, not classical conditioning, modelling, cognitive learning or dream analysis.

Systematic desensitisation ●●●

Can a learned fear be unlearned? In 1924 **Mary Cover Jones** (1897 to 1987), a student of John Watson, treated Peter, a boy of almost three who was terrified of rabbits. He watched other children play happily with a rabbit (modelling), and while he ate a favourite food a caged rabbit was brought a little closer each day, never so close that he stopped eating, until he let it nibble his fingers. This is **counterconditioning**: attaching to a stimulus a new response that is incompatible with the old one, so that the person learns the stimulus is no longer threatening.

KEY STUDY: WOLPE'S CATS (LATE 1940S)

Joseph Wolpe (1915 to 1997), a South African psychiatrist, shocked cats in a small cage until they were too anxious to eat there even when hungry. He then fed them in a room very unlike the laboratory, then in rooms progressively more like it, and finally in the cage, until the anxiety vanished. A response incompatible with anxiety, evoked while the anxiety was weak, had eliminated it. Cats are not neurotic patients, which is why later trials with people mattered.

Wolpe borrowed a term from the physiologist Charles Sherrington. **Reciprocal inhibition** is the principle that two incompatible responses cannot occur together, so a response antagonistic to anxiety, evoked in the presence of anxiety-provoking stimuli, suppresses the anxiety and weakens its bond to them. His main inhibitor was deep relaxation; others were assertion, sexual arousal and, in children, eating. On it he built **systematic desensitisation** (*Psychotherapy by Reciprocal Inhibition*, 1958), which has three steps.

1. **Relaxation training.** The client learns Edmund Jacobson's progressive muscle relaxation over about six sessions, practising 10 to 15 minutes twice a day.
2. **Hierarchy construction.** An **anxiety hierarchy** ranks 10 to 20 feared situations from least to most frightening, each rated on the **subjective units of distress scale** (SUDS), Wolpe's self-rating from 0 (calm) to 100 (the worst imaginable), also called units of disturbance or discomfort.
3. **Desensitisation proper**, the step of gradual exposure. With the client relaxed (SUDS below about 25) the therapist describes the lowest item; at any anxiety the client raises a finger and relaxation is restored. Only when an item evokes no anxiety does the client move up (Figure 7.5).

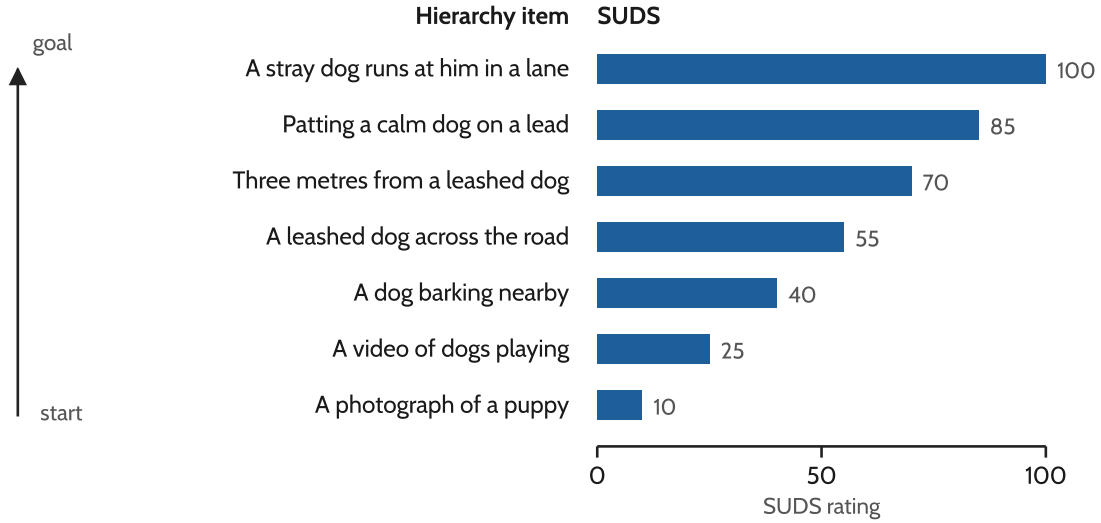

Figure 7.5 An anxiety hierarchy for Ravi, a Mysuru student bitten by a stray dog as a child. Treatment starts at the bottom and climbs only when the current item evokes no anxiety.

Wolpe worked in imagination, called **in vitro desensitisation** (Latin, "in glass", that is, in the consulting room); **in vivo desensitisation** ("in life") climbs the hierarchy in real situations, which generalises better, while imagination suits fears that cannot be staged, such as flying.

Gordon Paul (1966) found desensitisation better than insight therapy for fear of public speaking, but dismantling studies showed that it worked without relaxation or a strict hierarchy, never without exposure. Reciprocal inhibition is Wolpe's theory of why it works, not a proven mechanism.

> EXAM TRAP
>
> Systematic desensitisation is **Wolpe's** (not Skinner's, Pavlov's, Watson's, Rogers's, Perls's, Beck's, Freud's or Bandura's) and rests on **reciprocal inhibition**, not operant extinction, shaping, the law of effect, habituation alone, generalisation, vicarious reinforcement, punishment, adaptation to stress or catharsis. Its relaxation is **Jacobson's**; cognitive restructuring is not one of its steps; it treats **phobia**, not hallucination, delusion or depression. SUDS is keyed to desensitisation (IHBAS 2020), though all exposure therapies now use it; multimodal therapy, acceptance and commitment therapy and mindfulness-based stress reduction belong to Section 7.5. The Fear Survey Schedule (Wolpe and Peter Lang, 1964), State-Trait Anxiety Inventory and Hamilton Anxiety Rating Scale are questionnaires (Chapter 2), not the hierarchy scale.

Exposure, flooding and response prevention ●●●

If exposure is the active ingredient, why not go straight to it? **Exposure therapy** is repeated or prolonged confrontation with a feared stimulus, without escape, until the fear subsides. It works through **habituation**, the reduction in anxiety through repeated or prolonged encounter with a stimulus, and through extinction, as the stimulus keeps appearing without the feared outcome. Exposure may be graded or intense, in vivo or imaginal. **Interoceptive exposure** deliberately provokes feared bodily sensations (spinning in a chair, overbreathing) in panic disorder, and **virtual reality exposure** uses computer-generated scenes for fears of flying, heights or combat.

Flooding exposes the client at once, in real life or in imagination, to the most feared stimulus at full intensity, with no hierarchy and no relaxation, preventing escape until the anxiety peaks and subsides: the person afraid of heights stands on the balcony for as long as it takes. **Implosive therapy** (**Thomas Stampfl**, 1966; set out with **Donald Levis** in 1967) is imaginal flooding with exaggerated, horrific scenes enriched with presumed psychodynamic themes; it is now rare. Both work by extinction, and both are so unpleasant that clients must choose them knowingly.

Exposure and response prevention (ERP) is the first-line behavioural treatment for obsessive-compulsive disorder (OCD). **Victor Meyer** described it in London in 1966, and **Edna Foa** and colleagues refined and tested it. The client touches the "contaminated" object and is prevented from washing. The ritual brings quick relief, so it is negatively reinforced, and it stops the person learning that the anxiety would pass anyway (Figure 7.6). Blocking it allows habituation and extinction, and ERP helps more than 70 percent of patients. Foa also built **prolonged exposure** for post-traumatic stress disorder (PTSD): imaginal reliving of the trauma memory plus in vivo exposure to safe but avoided reminders.

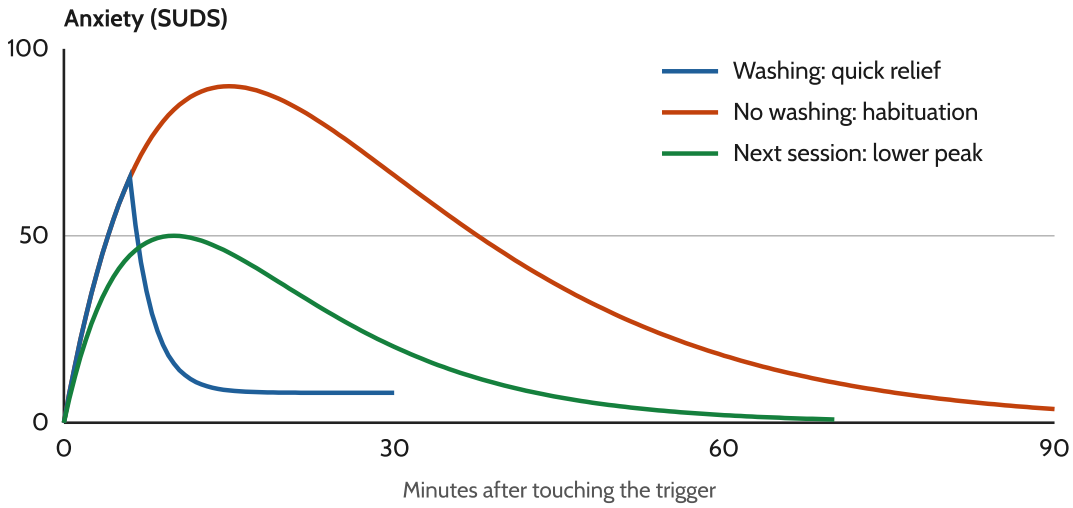

Figure 7.6 Why response prevention works. Washing ends anxiety within minutes and so negatively reinforces the ritual; without it, anxiety peaks and falls by itself, and the next session peaks lower (habituation within and between sessions).

A debated relative of exposure is **eye movement desensitisation and reprocessing** (EMDR), which **Francine Shapiro** (1948 to 2019) developed from a 1987 observation and published in 1989, initially for PTSD. The client holds a traumatic image, its negative belief and its body sensation in mind while following the therapist's fingers back and forth; this **bilateral stimulation** (eye movements, taps or tones) comes in sets of about 15 to 30. Shapiro's standard protocol has **eight phases**, always in this order: (1) history taking, (2) preparation, (3) assessment of the target image, negative belief and distress rating, (4) desensitisation with the eye movements, (5) installation of a positive belief, (6) body scan for leftover tension, (7) closure and (8) re-evaluation at the next session. EMDR works about as well as trauma-focused cognitive behaviour therapy, but the eye movements seem to add little, since holding the memory in mind is itself imaginal exposure. As critics put it, what is effective in EMDR is not new, and what is new is not effective.

EXAM TRAP

Flooding means full intensity with **no hierarchy and no relaxation**, in vivo or imaginal; pairing the feared stimulus with relaxation step by step is desensitisation, pairing it with shock is aversion therapy, and replacing irrational beliefs (Ellis) or cognitive restructuring is cognitive therapy (Section 7.5). ERP rests on **habituation and extinction**, not counterconditioning or reciprocal inhibition, and is first-line for **OCD**, not phobia, panic, depression, schizophrenia or personality disorder. EMDR is **Shapiro's**, for **PTSD**; prolonged exposure is Foa's, cognitive processing therapy Patricia Resick's and dialectical behaviour therapy Marsha Linehan's (Section 7.5). Bessel van der Kolk researches trauma, Pat Ogden founded sensorimotor psychotherapy and Joseph LeDoux maps the brain's fear circuits.

Aversion therapy, covert sensitisation and the bell and pad

Counterconditioning can also run in reverse. **Aversion therapy** pairs an unwanted behaviour, or the stimulus that tempts it, with an unpleasant stimulus, so that by classical conditioning it loses its appeal: a nausea-inducing drug is given just before the patient tastes al-

cohol (chemical aversion), or a brief shock accompanies a cigarette or a deviant sexual image (electrical aversion). It has been used for alcohol use, smoking and paraphilias. It inflicts pain, loses many clients, often fades outside the clinic and was misused against homosexuality, so today it needs informed consent and is tried only after milder methods fail.

Joseph Cautela (1967) moved the procedure into the imagination. In **covert sensitisation** the client imagines lifting a glass of whisky, then vividly imagines vomiting over himself in front of his family, then the relief of turning away; no real aversive stimulus is used.

> EXAM TRAP
>
> An undesirable behaviour imagined together with an imagined unpleasant consequence is **covert sensitisation** (Cautela), not flooding, implosion or desensitisation. Aversion therapy works by pairing (**classical conditioning**), not by intermittent reward, ignoring the behaviour (extinction) or restructuring beliefs.

The oldest classical technique is the **bell-and-pad method** for bed-wetting (nocturnal enuresis), built in 1938 by **Orval Hobart Mowrer** and **Willie Mae Mowrer**. Urine reaching a pad under the sheet completes a circuit and rings a bell; after repeated pairings, bladder tension alone wakes the child. Treatment takes 6 to 12 weeks, and the alarm remains first-line.

Operant techniques that build behaviour ●●●

Operant techniques work on consequences (the grid of reinforcement and punishment, and the schedules, are in Section 1.6). In therapy, **positive reinforcement** means giving something the person values immediately after a target behaviour, every time while the behaviour is new and then intermittently so that it lasts.

A behaviour that never occurs must be built. **Shaping** is reinforcing successive approximations to a target behaviour: for a silent child with autism, first eye contact, then any sound, then "wa", then "water". **Chaining** links the steps of a task, such as dressing, into a sequence; in backward chaining the last step is taught first, so every practice ends in success. A **prompt** is an extra cue (a hint, a gesture, a guiding hand) that makes the right response likely, and **fading** withdraws it gradually until the natural cue alone controls the behaviour.

The **Premack principle** of **David Premack** (1959) states that a more probable activity can reinforce a less probable one: video games after homework, or grandma's rule, vegetables before dessert. A **token economy** rewards target behaviours with tokens (points, stars, chips) that are later exchanged for backup reinforcers such as snacks or privileges. Tokens are **secondary reinforcers**, valued only for what they buy, and they bridge the delay between behaviour and reward.

> KEY STUDY: AYLLON AND AZRIN'S TOKEN ECONOMY (1968)
>
> **Teodoro Ayllon** and **Nathan Azrin** ran a ward of chronic psychotic women at Anna State Hospital, Illinois. Tokens earned for self-care and ward jobs bought privileges such as privacy or a walk in the grounds. When tokens were handed out regardless of work, the work almost stopped; when the contingency returned, so did the work. Their book *The Token Economy* (1968) launched the method, though gains often fade outside the ward, and basic needs must never become backup reinforcers.

Contingency management systematically arranges reinforcing consequences for target behaviours, as when clients with substance use disorders earn vouchers for drug-free urine samples. A **contingency contract** puts the arrangement in a signed agreement: "Ravi studies two hours each weekday; his parents allow cricket on Sunday."

> EXAM TRAP
>
> The token economy is **operant** (secondary reinforcers), so it is the treatment NOT derived from classical conditioning; flooding, desensitisation and aversion therapy are classical. Its developers were **Ayllon and Azrin**; Amity 2023 keyed Skinner, who supplied only the principles. Premack runs one way: the **high**-probability behaviour reinforces the **low**-probability one, never the reverse; it is a behaviour modification technique for children, not a theory of relationships, language or parenting. "Pleasant consequences make a behaviour likely to recur" is Thorndike's law of effect, and continuous reinforcement is the *least* resistant to extinction (Section 1.6).

Operant techniques that reduce behaviour, and ABA ●●●

To reduce a behaviour, therapists start with the least restrictive method. **Extinction** withholds the reinforcer that maintains a behaviour: parents who rewarded tantrums with attention now ignore them. Expect an **extinction burst**, a temporary rise when reinforcement stops; giving in at its peak teaches that louder tantrums pay. Extinction works best with **differential reinforcement**, which reinforces something else instead (Table 7.11).

Table 7.11 Operant techniques that reduce behaviour.

Technique	What is done	Category	Example
Extinction	Maintaining reinforcer withheld	Non-reinforcement	Ignoring attention-seeking tantrums
DRO (other behaviour)	Reinforce intervals free of the behaviour	Positive reinforcement	A star for every 10 minutes without hitting
DRI (incompatible behaviour)	Reinforce a behaviour that cannot co-occur	Positive reinforcement	Praise for staying seated, not wandering
DRA (alternative behaviour)	Reinforce a desirable alternative	Positive reinforcement	Praise for asking for help, not screaming
DRL (low rates)	Reinforce only a low rate	Positive reinforcement	Reward if questions stay under five
Time-out	Brief removal from all reinforcement	Negative punishment	Two minutes on a chair away from play
Response cost	Loss of a set amount of a reinforcer	Negative punishment	A fine of tokens; lost television time
Overcorrection	Restore the scene, or practise the right behaviour repeatedly	Positive punishment	Cleaning the whole room after throwing food
Satiation	Reinforcer given in excess	Reinforcer loses value	Ayllon's towel hoarder

Time-out removes the person briefly from all reinforcement and works only if "time-in" is rewarding; **response cost** takes away a set amount of a reinforcer already held, such as tokens. Both are **negative punishment**. **Overcorrection**, from **Richard Foxx** and Azrin (1972), re-

quires restitution (restoring the disturbed scene to better than before) or positive practice (repeating the correct behaviour many times); it is used for aggressive, disruptive and self-stimulatory behaviour in autism and intellectual disability. In **satiation** a reinforcer is given so freely that it loses its value: Ayllon (1963) gave a towel-hoarding patient towels until she threw them out. Positive punishment suppresses quickly but breeds fear, aggression and avoidance and teaches nothing new, so it is a last resort, used with consent and monitoring.

Combined into an intensive, data-driven programme, these methods are ABA, best known in autism. Its core is **discrete trial training**: a skill is broken into small units, each trial being an instruction, a prompt if needed, the child's response and an immediate consequence. In **functional communication training** a child whose tantrums win breaks from work learns to ask for a break with a word or a picture card instead.

KEY STUDY: LOVAAS'S EARLY INTERVENTION FOR AUTISM (1987)

Ole Ivar Lovaas (1927 to 2010) at the University of California, Los Angeles, gave 19 young children with autism 40 or more hours a week of one-to-one treatment for at least two years. Nine (47 percent) reached normal intellectual and educational functioning, against 2 percent of controls. Groups were not randomly assigned and aversives such as slaps were used, but the "Lovaas programme" became the best-known behavioural treatment for autism.

EXAM TRAP

Deliberately ignoring tantrums is **extinction** (not generalisation or discrimination, Section 1.6). Amity 2023 keyed "operant escape", which is wrong: escape ends an unpleasant stimulus already present and avoidance prevents one (Section 1.6). Time-out and response cost are **negative punishment**; negative reinforcement removes something unpleasant and *increases* behaviour. Overcorrection targets repetitive **aggressive** behaviour in autism; bed-wetting needs the bell and pad. Lovaas treated **autism**, not aphasia, depression or alcoholism, and a picture card to request a break is **ABA**, not CBT, play therapy or psychoanalysis.

Modelling, assertiveness and social skills

Albert Bandura (1925 to 2021) turned observational learning into therapy. In **live modelling** the client watches a real person perform the feared or missing behaviour; in **symbolic modelling** the model appears on film or in a story; in **participant modelling** the therapist models the behaviour and then physically guides the client through it. For snake phobia (Bandura, Edward Blanchard and Brunhilde Ritter, 1969), participant modelling beat both symbolic modelling and systematic desensitisation. In **behavioural rehearsal** the client practises the new behaviour in role-play, with feedback, before trying it in life.

Assertiveness training, pioneered by **Andrew Salter** (*Conditioned Reflex Therapy*, 1949) and developed by Wolpe and Lazarus, teaches people to express wishes and feelings directly while respecting others. **Passive** behaviour surrenders one's rights (silence when someone jumps the bus queue), **aggressive** behaviour violates other people's (shouting abuse), and **assertive** behaviour defends one's own without attack ("Please join the queue; we were here first"). **Social skills training** teaches eye contact, conversation and refusal through instruction, modelling, rehearsal, feedback and homework, in autism, social anxiety and schizophrenia, where antipsychotic medication remains front-line.

Relaxation, biofeedback and self-management ●●●

Several methods teach control of one's own body. **Edmund Jacobson**'s **progressive muscle relaxation** (1929, enlarged in 1938) tenses and releases muscle groups in turn; **Johannes Heinrich Schultz**'s **autogenic training** (1932) uses self-suggestions of heaviness and warmth; **breathing retraining** slows the overbreathing of panic; and the **applied relaxation** of **Lars-Göran Öst** (1987) trains rapid relaxation on cue, used at the first sign of anxiety in panic, generalised anxiety and phobias.

Biofeedback makes a bodily process the person cannot normally sense visible or audible, so that he can learn to control it. Its basis was **Neal Miller**'s 1960s report that rats could change heart rate and other autonomic responses through operant conditioning: the autonomic nervous system could come under voluntary control. Electromyographic (EMG) feedback of muscle tension treats tension-type headache and back pain, thermal feedback of hand temperature treats migraine and Raynaud's disease, and EEG neurofeedback of brain waves is used for ADHD and epilepsy. An early device was the perineometer of **Arnold Kegel** (1948), a pressure gauge that let a patient watch her pelvic floor squeeze.

Habit reversal training (HRT), from Azrin and **Gregory Nunn** (1973), is the treatment of choice for tics and habits such as hair pulling and nail biting. **Awareness training** teaches the client to detect each occurrence and the urge before it; **competing response training** teaches a response physically incompatible with the habit, inconspicuous and held for about a minute or until the urge passes (clenching the fists against hair pulling); **motivation** comes from reviewing the habit's costs and from family praise. A variant, regulated breathing, was applied to stuttering. In **negative practice** (massed practice), proposed by **Knight Dunlap** (1932), the habit is performed deliberately and repeatedly until fatigue makes it aversive. **Thought stopping**, popularised by Wolpe, interrupts an unwanted thought with "Stop!", first shouted, later silent; it has faded because suppressed thoughts rebound.

Self-monitoring, the systematic recording of one's own target behaviour, often changes that behaviour before treatment begins; this **reactivity** spoils the baseline but helps therapy. In **self-reinforcement** the client rewards himself on meeting his own goal, completing Kanfer's sequence of self-monitoring, self-evaluation and self-reinforcement.

Behavioural activation (BA) treats depression by re-engaging clients in meaningful, rewarding activities. **Peter Lewinsohn** proposed in the 1970s that depression follows a loss of response-contingent positive reinforcement. **Neil Jacobson** and colleagues (1996) found the activation component of cognitive therapy as effective as the whole package, and Jacobson and **Christopher Martell** developed BA in its own right: activity monitoring, scheduling and graded tasks that replace avoidance.

For insomnia, **stimulus control** (**Richard Bootzin**, 1972) re-links the bed with sleep: bed only when sleepy and only for sleep and sex, leave it if not asleep within about 10 to 20 minutes, a fixed rising time and no naps. **Sleep restriction** (**Arthur Spielman**, 1987) limits time in bed to the time actually slept, extending it as sleep consolidates. **Paradoxical intention**, from the logotherapist **Viktor Frankl** (Section 7.6), asks the person to do or wish for the very thing he fears: the insomniac tries hard to stay awake, which removes the performance anxiety that kept him awake. Frankl's **dereflection** instead turns attention away from the symptom.

EXAM TRAP

Operant control of autonomic functions is **biofeedback**, not hypnosis (Section 1.5), sublimation (a defence mechanism) or desensitisation; EMG feedback suits **tension-type headache**, and Kegel's perineometer was early biofeedback. Repeating a habit to fatigue is **negative practice**, not the competing response of habit reversal. HRT is for **tics**; OCD needs ERP. Doing or wishing the opposite of the goal is **paradoxical intention**, not stimulus control, sleep restriction, dereflection, negative practice or reciprocal inhibition. Behavioural activation means **engaging in meaningful activity**, not analysing childhood, relaxation or hypnosis.

Parents, classrooms and an evaluation

Gerald Patterson of the Oregon Social Learning Center described a **coercive cycle** at home: a parent makes a demand, the child whines or hits, the parent gives in (negatively reinforcing the aggression) and the child stops (negatively reinforcing the surrender). **Parent management training** (PMT) teaches parents to give clear commands, attend to and praise good behaviour, reward it with points or stickers, ignore minor misbehaviour, and use brief time-out or loss of privileges for serious misbehaviour; negative reinforcement is what it removes, not what it teaches. **Classroom management** applies the same principles at school: clear rules, praise, planned ignoring, token systems, daily report cards and group contingencies such as the Good Behaviour Game.

Behaviour therapy is a first choice for specific phobias, OCD, panic, enuresis, tics and habits, insomnia, conduct problems and skills in autism, and behavioural activation rivals cognitive therapy for depression; in psychosis and personality disorder it is an adjunct. Psychoanalysts predicted **symptom substitution**: remove a symptom without resolving its underlying conflict and another will take its place. Follow-ups from Paul's 1966 trial onwards found little sign of it. Aversive methods raise the sharpest ethical questions: they demand informed consent, the least restrictive alternative first, supervision and special protection for children and institutionalised patients.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Wolpe's systematic desensitisation rests on the principle of _____ inhibition.
2. The 0 to 100 scale used to rank hierarchy items is the subjective units of _____ scale.
3. Exposure to the most feared stimulus at full intensity, with no hierarchy and no relaxation, is called _____.
4. Exposure and response prevention for OCD works through habituation and _____.
5. The token economy was developed for psychiatric inpatients by Ayllon and _____.
6. Time-out and response cost are forms of _____ punishment.
7. In habit reversal training, a response physically incompatible with the habit is the _____ response.
8. Telling an insomniac to try to stay awake is Frankl's technique of _____ intention.

Answers: (1) reciprocal, (2) distress (or disturbance), (3) flooding, (4) extinction, (5) Azrin, (6) negative, (7) competing, (8) paradoxical.

7.5 Cognitive and Cognitive-Behavioural Therapies, and the Third Wave

HIGH YIELD · 69 QS

Two students score 48 out of 90 on the same mock test. One is disappointed and studies her mistakes; the other decides he was never good enough and stops opening his books. Same event, opposite reactions: what differed was what each made of the score. That observation underlies every therapy in this section, from the founders, Albert Ellis and Aaron Beck, to the "third wave", which asks clients not to argue with their thoughts but to change their relationship with them.

The cognitive turn and the three waves ●●

The term **cognitive-behavioural therapy** (CBT) covers a family of treatments that share three propositions: thinking affects emotion and behaviour; thinking can be monitored and changed; and changing maladaptive thought patterns relieves distress, even when the events that triggered it cannot be changed. The idea is old. The Stoic philosopher **Epictetus** (about 50 to 135 CE) wrote that people are disturbed not by things but by the views they take of them.

In the 1950s and 1960s two therapists, Albert Ellis and Aaron Beck, independently concluded that faulty thinking drives emotional disturbance, and both are counted as founders of the cognitive tradition. Behaviour therapists (Section 7.4) joined them in the 1970s; Donald Meichenbaum's *Cognitive-Behavior Modification* (1977) gave the hybrid its name. **Albert Bandura** (1925 to 2021) supplied a bridge in 1977 with **self-efficacy**, a person's belief that he or she can carry out the actions a situation demands. He argued that all effective treatments work largely by raising it, which made self-efficacy the classic cognitive-behavioural explanation of how people change.

In 2004 **Steven C. Hayes** (born 1948) described three waves. The first, behaviour therapy (1950s onwards), applied learning principles to behaviour. The second, the cognitive therapies of Ellis and Beck, changed the *content* of thoughts from irrational or distorted to realistic. The **third wave** (from the 1990s: dialectical behaviour therapy, acceptance and commitment therapy, mindfulness-based cognitive therapy, and compassion-focused, metacognitive and functional analytic therapies) changes a person's *relationship* to thoughts and feelings: noticing and accepting them while acting on one's values. All CBTs are time-limited, focused on present problems, structured, collaborative and educational; they rely on homework and measure symptoms throughout.

> EXAM TRAP
>
> "Third-wave CBT includes all EXCEPT": **psychoanalysis** (Section 7.3), which belongs to no wave. REBT and Beck's therapy are second wave; token economies are first wave. Self-efficacy is **Bandura**'s concept, not Beck's, Ellis's or Skinner's (Skinner rejected cognitive explanations).

Ellis's rational emotive behaviour therapy ●●●

Albert Ellis (1913 to 2007) treated his own shyness first: as a young man he made himself talk to a hundred women in New York's Bronx Botanical Garden in one month, and learned that rejection was unpleasant, not terrible. After a PhD at Columbia University (1947) he practised psychoanalysis until about 1953, frustrated that insight rarely made patients better. Drawing on Stoic philosophy, he began rational therapy in 1955; it became rational-emotive therapy (RET) and, in 1993, **rational emotive behaviour therapy** (REBT), a name that acknowledged its behavioural homework. His *Reason and Emotion in Psychotherapy* (1962) is the classic text, and REBT is usually called the first cognitive-behavioural therapy.

Its core is the **ABC model**: **A**, the activating event; **B**, the person's beliefs about A; **C**, the emotional and behavioural consequences. Common sense says A causes C; Ellis says A contributes, but B is the main cause. The two students shared an A, but "one mock does not decide my future" produced disappointment and study, while "I must score well, or I'm a failure" produced depression and avoidance (Figure 7.7). A rational belief is a flexible preference that fits reality, follows logically and serves the person's goals; an irrational belief is a rigid, absolute demand. A rational belief about a real loss still produces a healthy negative emotion (sadness, concern), whereas an irrational one produces an unhealthy emotion (depression, anxiety).

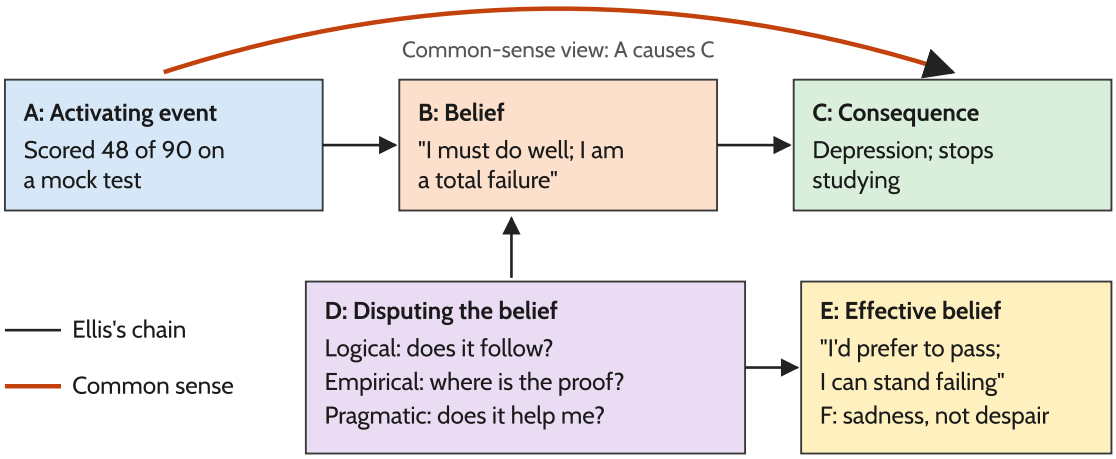

Figure 7.7 Ellis's ABC model extended to D (disputing), E (an effective new philosophy) and F (new feelings and behaviour). The dashed arrow is the common-sense view that events cause feelings directly; Ellis's chain runs through the belief.

Irrational thinking takes four forms. The root is demandingness, turning preferences into musts, shoulds and oughts, which Ellis called **musturbation** (also spelt musterbation). From it follow **awfulising** (rating a setback as more than 100 percent bad, close to Beck's catastrophising), **low frustration tolerance** ("I can't stand it") and global rating of self or others ("I failed, so I am a failure"). Ellis's three major musts concern the self ("I must do well and win approval, or I am no good"), others ("Others must treat me fairly, or they are rotten") and the world ("Life must give me what I want, quickly and easily"). They breed, respectively, anxiety and depression, anger and rage, and self-pity and procrastination.

The active ingredient is D, **disputing** the irrational belief. A **logical** dispute asks whether it follows ("You want to pass; why must you?"), an **empirical** dispute asks for evidence ("Where is the proof that you are a total failure?") and a **pragmatic** dispute asks whether it helps ("Where does this belief get you?"). Success yields E, an effective new philosophy, and F, new feelings and behaviour. The antidote to global rating is **unconditional self-acceptance**: one's acts can be rated, one's whole self cannot. Ellis extended it to others and to life; unlike Carl Rogers (Section 7.6), he thought therapist warmth unnecessary, and his style was famously blunt.

REBT mixes cognitive techniques (disputing, self-help forms, reading), emotive techniques and behavioural homework set at almost every session (in vivo exposure, rewards and penalties). In **rational emotive imagery**, adopted from the psychiatrist **Maxie C. Maultsby Jr.**, the client imagines the activating event, feels the unhealthy emotion, changes it to its healthy counterpart and notes the self-statement that did it. In a **shame-attacking exercise** the client does something harmless but embarrassing in public (Ellis's example was calling out station names in a subway train) to learn that disapproval is bearable.

REBT is clear, brief and teachable, but its evidence is thinner than Beck's: a 1991 meta-analysis of 70 studies found it better than control groups and baseline, and a 2008 trial found it comparable with cognitive therapy and medication for depression. Its chief criticism is that it can be direct and confrontational, so clients may find it intimidating; the therapist's idea of

"rational" may also impose his own values, and its philosophical claims are hard to test. A focus on sexuality is a criticism of Freud, not of Ellis.

> **EXAM TRAP**
>
> ABC stands for activating event, beliefs, consequences: B is the **belief about the activating event**, not behaviour, baseline, bias or "beliefs about oneself", and C is the consequence of that belief, not "critical analysis". D = **disputing**, not desensitisation, drama or "discuss"; E = effective new philosophy. The behavioural ABC of antecedent, behaviour, consequence is functional analysis (Section 7.4), and affect, behaviour, cognition is the ABC of attitudes (Chapter 3). Musts, musturbation and shame attacking signal Ellis, not Freud, William James, Rogers, Adler, Beck, DBT or Glasser's reality therapy (Section 7.6).

Beck's cognitive model and the distortions ●●●

Aaron Temkin Beck (1921 to 2021), a psychoanalytically trained psychiatrist at the University of Pennsylvania, noticed in the late 1950s that his patients had a second, barely noticed stream of quick, believable, self-critical thoughts ("He won't like me") that came before their feelings. He called them **automatic thoughts**. Depressed patients' dreams, expected to show anger turned inward, showed loss and defeat instead. Beck set out the model in *Depression* (1967) and the treatment in *Cognitive Therapy of Depression* (1979, with colleagues). He built cognitive therapy for depression and later extended it to anxiety, personality disorders, substance use and psychosis.

At its centre is the **cognitive triad**: negative views of the **self** ("I am worthless"), the **world** or current experience ("everything is against me") and the **future** ("it will never get better"). Remember it as self, world, future: not past, present, future; not self, others, environment; not the thoughts, feelings and behaviours that CBT links in its cognitive model; and not id, ego and superego. Cognition runs on three levels. The deepest are **core beliefs**, or schemas: rigid, global beliefs formed in childhood ("I am unlovable"). From them grow **intermediate beliefs**, also called dysfunctional assumptions: the rules, attitudes and assumptions that follow ("If I am not perfect, I will be rejected"). When a situation activates these, automatic thoughts appear and drive emotion and behaviour. A dormant schema can be woken by a critical incident such as a loss, so the model is a diathesis-stress one. Beck's daughter **Judith S. Beck** laid out these levels in *Cognitive Therapy: Basics and Beyond* (1995).

Negative thinking is kept going by **cognitive distortions**, systematic errors in processing information. Beck named arbitrary inference, selective abstraction, overgeneralisation, magnification and minimisation, personalisation and dichotomous thinking; the psychiatrist **David D. Burns** popularised the rest of Table 7.12 in *Feeling Good* (1980).

Table 7.12 The cognitive distortions, with an example of each.

Distortion	What happens	Example
Arbitrary inference	A conclusion without evidence	A friend does not greet you: "He dislikes me now"
Selective abstraction (mental filter)	One detail taken out of context, the rest ignored	A teacher sees two bored students: "My class dislikes me"
Overgeneralisation	A broad rule from one or a few events	"I failed statistics, so I fail at everything"

Distortion	What happens	Example
Magnification and minimisation	Faults enlarged, strengths shrunk	One slip "ruins" the talk; praise "means nothing"
Personalisation	Blaming oneself for events one did not cause	"The guests left early because of me"
Dichotomous (all-or-nothing) thinking	Only two extremes	"If I don't top the exam, I have failed"
Labelling	A global label for one act	"I'm a loser", not "I made a mistake"
Mind reading	Assuming one knows what others think	"She thinks I'm stupid"
Fortune telling	Predicting a bad outcome	"I'll freeze in the interview"
Emotional reasoning	A feeling taken as proof	"I feel useless, so I am useless"
Should statements	Rigid rules for self and others	"I should never make mistakes"
Catastrophising	Expecting the worst to happen	"If I fail this exam, my life is over"
Disqualifying the positive	Good events "don't count"	"They praised me only to be polite"

EXAM TRAP

A man whose friend fails to acknowledge him and who concludes "he doesn't like me any more" shows **arbitrary inference**; mind reading and fortune telling are later, narrower names for such jumps, so choose arbitrary inference when they are not offered. **Selective abstraction**, also called selective attention or the mental filter, is fixing on one detail and ignoring the whole, like the teacher with two bored students; overgeneralisation is a broad rule from a few events. **Catastrophising** is worrying that the worst will always happen; all-or-nothing thinking is extreme categories; personalisation is taking things personally.

Cognitive therapy in practice ●●●

Cognitive therapy treats thoughts as hypotheses to be tested, not facts to be argued away. In **collaborative empiricism** therapist and client investigate together, making objective assessments of beliefs against evidence. The main method is **Socratic questioning**, or guided discovery: open questions that help the client discover and attend to previously overlooked information, so that the discovery is the client's own rather than an interpretation from the therapist ("What is the evidence? How else could you see it? If it were true, what would it mean?"). The therapist neither interprets nor tells the client what to think. Collaborative empiricism, Socratic dialogue and guided discovery are the approach's fundamental concepts; free association and dream analysis belong to psychoanalysis (Section 7.3).

- The **dysfunctional thought record** (1979) has columns for the situation, the emotion (rated 0 to 100), the automatic thought, a balanced response and the outcome; with it clients learn to identify, examine and restructure their own automatic thoughts.
- Behavioural experiments test predictions in real life ("If I speak in class, everyone will laugh").
- Activity scheduling plans the week and rates each activity for mastery and pleasure; graded task assignment breaks a daunting task into small steps so that success comes early.

- The **downward arrow technique** asks repeatedly "If that were true, what would it mean to you?" to descend from an automatic thought to the core belief.
- Decatastrophising (the "what if" technique) weighs how likely and how bad the feared outcome is and how the client would cope; reattribution spreads responsibility fairly when the client blames himself for everything.

A session runs in a set order: mood check, bridge from the last session, agenda, homework review, work on the agenda, new homework, then summary and feedback. Progress is measured with Beck's scales: the Beck Depression Inventory (BDI, 1961; BDI-II, 1996), the Beck Anxiety Inventory (BAI, 1988) and the Beck Hopelessness Scale (BHS, 1974), described in Chapter 2.

> EXAM TRAP
>
> The BDI and BAI are **self-report**; the Hamilton depression and anxiety scales of **Max Hamilton** are **clinician-rated**. So "Which is clinician-rated?" is the Hamilton Anxiety Rating Scale, not "both". The downward arrow is not a 0 to 10 distress rating or an exposure hierarchy (Section 7.4).

Ellis and Beck share the central assumption that emotional disturbance arises mainly from how people think about events, not from the events themselves, and both therapies are active, brief and built on homework. Table 7.13 shows how they differ.

Table 7.13 Ellis's REBT and Beck's cognitive therapy compared.

Feature	Ellis (REBT)	Beck (cognitive therapy)
Origin	1955; a psychologist	1960s; a psychiatrist
Target	Irrational beliefs, above all musts	Automatic thoughts, distortions, core beliefs
Theory of disturbance	One core (demandingness) for all problems	Disorder-specific content
Style	Directive, persuasive, confrontational	Collaborative, Socratic
Beliefs are	Disputed as irrational	Tested as hypotheses
Signature methods	Disputing, shame attacking	Thought record, behavioural experiments
Research base	Smaller	Very large

Other cognitive-behavioural therapies ●●

Donald Meichenbaum (born 1940) worked on self-talk. His **self-instructional training**, first used with impulsive children in 1971, moves guidance inward in five steps: the therapist performs the task while talking aloud; the child performs it while the therapist instructs aloud; then the child instructs himself aloud, then in a whisper, then silently. His **stress inoculation training**, named after vaccination, builds resistance by coping with manageable doses of stress, in three phases: conceptualisation (how appraisals and self-talk create stress), skills acquisition and rehearsal (relaxation, cognitive restructuring, problem solving, coping self-statements) and application and follow-through (practice in imagery, role-play and real life, with relapse prevention).

Problem-solving therapy, introduced by **Thomas J. D'Zurilla** and **Marvin R. Goldfried** (1971) and extended to depression by **Arthur M. Nezu**, teaches a positive problem orientation

and a sequence: define the problem, brainstorm alternatives, choose one, act and check the result. **Arnold A. Lazarus** (1932 to 2013) built **multimodal therapy**, which assesses the **BASIC I.D.** (Behaviour, Affect, Sensation, Imagery, Cognition, Interpersonal relationships, Drugs and biology) and borrows techniques from any school (technical eclecticism, Section 7.8). In **fixed-role therapy**, from the personal construct theory (1955) of **George Alexander Kelly** (1905 to 1967), the client lives for about two weeks as a new character sketched by the therapist, to try out new constructs and new relationships.

For clients who failed standard CBT, especially those with personality disorders, **Jeffrey E. Young** developed **schema therapy** (1990). It targets 18 early maladaptive schemas, enduring themes such as abandonment, mistrust, defectiveness and unrelenting standards, formed when childhood needs went unmet, and schema modes, the states a person switches between (vulnerable child, angry child, punitive parent, detached protector, healthy adult). Its signature is **limited reparenting**, meeting some unmet childhood needs within professional limits, with imagery rescripting and chair work. **Anthony Ryle**'s cognitive analytic therapy (CAT, Britain, 1980s) is a brief therapy, often 16 sessions, joining cognitive therapy to psychoanalytic object relations through a reformulation letter and a diagram of the client's reciprocal roles.

> EXAM TRAP
>
> Fixed-role therapy is **George Kelly**'s, not Beck's, Ellis's or Fritz Perls's; BASIC I.D. is **Lazarus**'s; schema therapy, with its schema modes and limited reparenting, is **Jeffrey Young**'s, not DBT's.

Disorder-specific CBT: panic, social anxiety, trauma, psychosis ●●●

Beck's cognitive specificity hypothesis (each disorder has its own theme: loss in depression, danger in anxiety) led to disorder-specific models. **Panic. David M. Clark** (1986) traced panic attacks to the catastrophic misinterpretation of bodily sensations: a racing heart read as a heart attack raises anxiety, which intensifies the sensations, in a vicious circle. **Paul M. Salkovskis** (1991) added **safety behaviours**, such as gripping a shopping trolley or sitting down, which seem to avert disaster but stop the client learning that it would not have happened. Treatment provokes the feared sensations and gradually drops safety behaviours, as experiments that test the catastrophic beliefs. In the model of social anxiety by Clark and **Adrian Wells** (1995), self-focused attention and safety behaviours maintain the fear; effective CBT for social phobia combines exposure, cognitive restructuring and social skills training.

Trauma. Prolonged exposure (PE), developed by **Edna B. Foa**, combines imaginal exposure (repeatedly retelling the trauma) with in vivo exposure to avoided but safe reminders. Cognitive processing therapy (CPT), developed by **Patricia A. Resick** for rape survivors, challenges "stuck points" about blame, safety, trust, power, esteem and intimacy. Trauma-focused CBT (TF-CBT) for children, by **Judith A. Cohen**, **Anthony P. Mannarino** and **Esther Deblinger**, teaches coping skills first; its central aim is to build and process the trauma narrative and so desensitise the traumatic memories. With eye movement desensitisation and reprocessing (EMDR) of **Francine Shapiro** (Section 7.4), these are the first-line trauma treatments.

EXAM TRAP

PE comes from **Edna Foa and colleagues**, CPT from **Resick**, EMDR from **Shapiro**, DBT from Linehan. For survivors of sexual assault, pick the option naming trauma-focused CBT and EMDR (psychodynamic therapy is also commonly used, with less evidence); DBT, ACT, mindfulness-based therapy and interpersonal psychotherapy (Section 7.3) are not trauma-focused treatments. In panic, safety behaviours are dropped, never prescribed.

Psychosis. CBT for psychosis (CBTp), developed in Britain in the 1990s by **David Kingdon**, **Douglas Turkington** and others, normalises unusual experiences, tests delusional beliefs collaboratively and teaches ways of coping with voices. It is an adjunct, not an alternative: antipsychotic medication remains the front-line treatment for schizophrenia (Chapter 8), with CBTp, social skills training and stress management as psychosocial add-ons.

Dialectical behaviour therapy ●●●

Marsha M. Linehan (born 1943) developed **dialectical behaviour therapy** (DBT; manuals 1993) for chronically suicidal women, most with borderline personality disorder (BPD; Chapter 8). Standard CBT's constant push for change felt invalidating to them and they dropped out, yet acceptance alone left the self-harm untouched. DBT's dialectic synthesises the two: full acceptance of the client together with a firm push for change. The acceptance side, including core mindfulness, came from Zen Buddhism, which Linehan practised; the change side came from behaviour therapy.

Her **biosocial theory** sees BPD as pervasive emotion dysregulation arising when a biological emotional vulnerability (high sensitivity and reactivity, slow return to baseline) meets an invalidating environment that dismisses or punishes the child's emotions. Standard DBT has four modes: individual therapy, a weekly skills group (usually for a year), telephone coaching in crises and a therapist consultation team. Individual work targets life-threatening behaviour first, then therapy-interfering and then quality-of-life-interfering behaviour. The four skills modules are **core mindfulness** (observing without judgement and finding "wise mind" between reasonable mind and emotion mind), **distress tolerance** (surviving crises through distraction, self-soothing and radical acceptance), **emotion regulation** (naming emotions, acting opposite to unhelpful urges) and **interpersonal effectiveness** (asking and refusing while keeping relationships and self-respect).

KEY STUDY: LINEHAN'S DBT TRIALS (1991 AND 2006)

Linehan and colleagues randomly assigned chronically parasuicidal women with BPD to a year of DBT or treatment as usual; the DBT group made fewer and less medically serious parasuicidal acts, spent fewer days in hospital and dropped out less (1991). Critics cited the small sample and weak control, so in 2006 Linehan compared DBT with treatment by community experts in psychotherapy, and DBT still halved suicide attempts.

EXAM TRAP

DBT is Marsha Linehan's. Its four skills modules are mindfulness, distress tolerance, emotion regulation and interpersonal effectiveness; cognitive restructuring and relaxation are not among them. Family genogram construction is a Bowen technique (Section 7.7), not a DBT mode. DBT's mindfulness is Zen, not Stoic. DBT was built for borderline personality disorder with chronic suicidality, not for schizophrenia, phobias or OCD.

ACT, mindfulness and the rest of the third wave ●●

Hayes, with **Kirk D. Strosahl** and **Kelly G. Wilson**, developed **acceptance and commitment therapy** (ACT, said as the word "act"; their book appeared in 1999). It rests on **relational frame theory**, Hayes's account of language: because humans relate events to one another through words, a thought carries the sting of what it stands for, so remembering a failure hurts like failing. The usual response, experiential avoidance (trying to escape unwanted thoughts and feelings), backfires and narrows life. ACT therefore aims not at symptom elimination but at **psychological flexibility**, acting on one's values with whatever thoughts and feelings are present. It builds flexibility through the six processes of the hexaflex (Figure 7.8): acceptance, **cognitive defusion** (seeing thoughts as words, as in "I'm having the thought that I'm a failure"), contact with the present moment, self-as-context (the observing self), values and committed action.

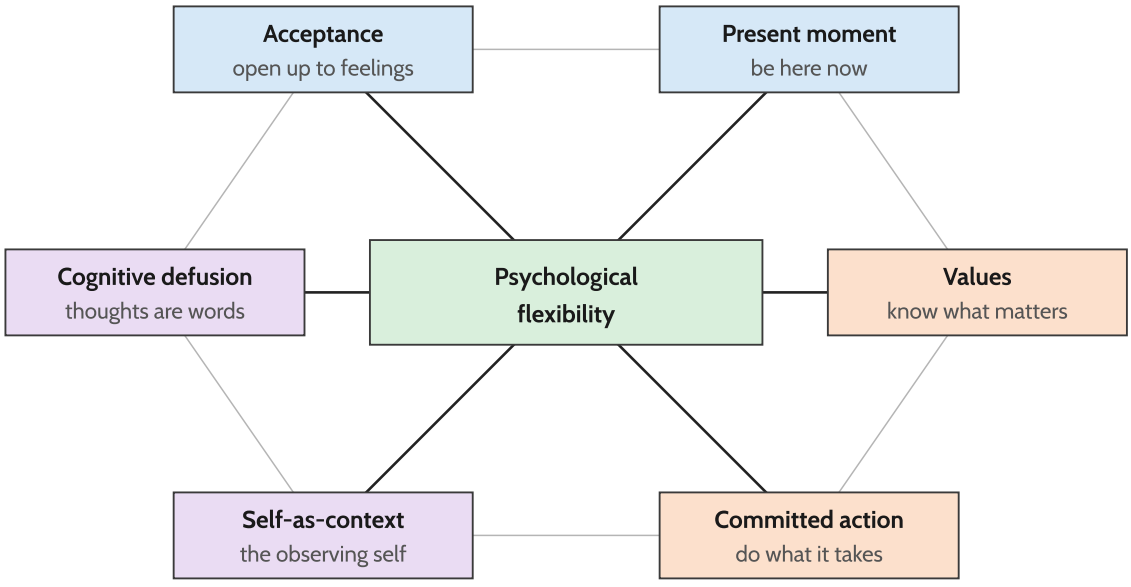

Figure 7.8 The hexaflex of acceptance and commitment therapy. Each process has an opposite that produces inflexibility: experiential avoidance, cognitive fusion, living in the past or future, attachment to a conceptualised self, unclear values, and inaction or impulsivity.

Jon Kabat-Zinn (born 1944) brought mindfulness (attention paid on purpose, in the present moment and non-judgementally) into medicine with **mindfulness-based stress reduction** (MBSR), founded in 1979 at the University of Massachusetts Medical School: an eight-week group course of body scan, meditation and gentle yoga for chronic pain and stress. **Zindel V. Segal**, **J. Mark G. Williams** and **John D. Teasdale** merged it with cognitive therapy as **mind-**

fulness-based cognitive therapy (MBCT; 2002), an eight-week group for recovered patients designed to prevent relapse in recurrent major depression by teaching decentering: seeing thoughts as passing mental events, not facts.

KEY STUDY: MBCT AND DEPRESSIVE RELAPSE (TEASDALE AND COLLEAGUES, 2000)

Recovered patients with recurrent depression were randomly assigned to usual treatment or usual treatment plus MBCT and followed for 60 weeks. For those with three or more past episodes, relapse fell from 66 to 37 percent; for those with two, MBCT made no difference. MBCT is a relapse-prevention treatment, not a treatment for acute illness.

Compassion-focused therapy (CFT) of **Paul Gilbert**, built for shame and self-criticism, describes three emotion-regulation systems: threat (protection, anxiety, anger), drive (seeking resources and achievement) and soothing (contentment, safeness, affiliation); compassionate mind training strengthens the soothing system, which self-critical people lack. **Metacognitive therapy** (MCT) of Adrian Wells targets not what people think but how they respond to thinking. Its central concept, the **cognitive attentional syndrome**, is persistent worry and rumination, threat monitoring and unhelpful coping (thought suppression, avoidance), driven by metacognitive beliefs such as "Worrying helps me cope" and "My worry is uncontrollable"; attention training and detached mindfulness loosen it. **Functional analytic psychotherapy** (FAP) of **Robert J. Kohlenberg** and **Mavis Tsai** (1991) uses the therapy relationship itself: the client's problem behaviours and improvements appear in session, and the therapist naturally reinforces the improvements. **Behavioural activation** grew from **Peter M. Lewinsohn**'s view that depression follows a loss of response-contingent positive reinforcement; when **Neil S. Jacobson** and colleagues (1996) found that the activation component of Beck's therapy worked as well as the whole package, scheduling valued activity and tackling avoidance became a treatment in its own right (Section 7.4).

EXAM TRAP

ACT comes from Steven Hayes and colleagues, not Linehan, Beck or Ellis. Its hexaflex (acceptance, defusion, present moment, self-as-context, values, committed action) is not DBT's list of skills, and ACT aims at psychological flexibility through six core processes, not symptom elimination, insight into childhood conflicts or disputing beliefs. MBSR is Kabat-Zinn's; MBCT prevents depressive relapse and is not a treatment for acute psychosis or phobias.

How well does CBT work?

CBT is the most researched psychotherapy. A 2006 review of 16 meta-analyses, co-authored by Beck, found large effects for depression, generalised anxiety disorder, panic disorder, social phobia, post-traumatic stress disorder and childhood depression and anxiety. CBT is first-line for most anxiety disorders and for depression, where it matches antidepressants and better protects against relapse after treatment ends; it is also first-line for bulimia nervosa and insomnia, and an adjunct to medication in bipolar disorder and schizophrenia. Its methods travel: in Goa, lay counsellors delivering behavioural activation helped depressed primary-care patients in a 2017 trial led by **Vikram Patel**.

It has limits. Many clients do not fully recover or later relapse; trial samples are more selected than clinic patients; it needs literacy and homework; critics say it neglects social causes such as poverty and assumes a Western, individualist outlook; and against other well-conducted therapies the differences are often small (Section 7.8). Computerised and internet-delivered CBT (iCBT), such as Beating the Blues in Britain and MoodGYM in Australia, delivers modules, thought records and homework on screen. Guided versions approach the results of face-to-face therapy, whereas unguided ones lose many users. It suits stepped care, in which low-intensity help comes first (telepsychology in India, Section 7.10).

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Ellis renamed his therapy rational emotive behaviour therapy in _____.
2. In Ellis's ABCDE model, D stands for _____ the irrational belief.
3. Beck's negative views of the self, the world and the future form the cognitive _____.
4. Concluding without evidence that a friend who did not greet you dislikes you is the distortion called _____.
5. Asking "If that were true, what would it mean to you?" to reach a core belief is the _____ technique.
6. The phases of stress inoculation training are conceptualisation, skills acquisition and rehearsal, and _____.
7. Linehan explains borderline personality disorder with her _____ theory.
8. The six processes of the ACT hexaflex build psychological _____.

Answers: (1) 1993, (2) disputing, (3) triad, (4) arbitrary inference, (5) downward arrow, (6) application and follow-through, (7) biosocial, (8) flexibility.

7.6 Humanistic, Existential, Experiential and Brief Therapies

HIGH YIELD · 49 QS

By the 1950s psychotherapy had two systems: psychoanalysis (Section 7.3) and behaviourism (Section 7.4). **Abraham Harold Maslow** (1908 to 1970) called the movement against both the **third force**. Its core, humanistic psychology, studies the whole person from his or her own viewpoint and assumes an inborn drive toward growth; Maslow (Chapter 4) and Carl Rogers, who built the therapy, were its leading proponents. Existential therapists share its respect for freedom but make the person face death, isolation and meaninglessness; experiential therapists work with feeling as it happens in the session. Together these humanistic-experiential psychotherapies aim to promote a more fluid and integrated organisation of the self, not to resolve deep-seated conflicts (the psychodynamic aim), change maladaptive behaviours (the behavioural aim) or train interpersonal skills.

Rogers: the self, conditions of worth and incongruence

Carl Ransom Rogers (1902 to 1987) gave his therapy three names: **non-directive therapy** in *Counseling and Psychotherapy* (1942), because the counsellor does not advise, interpret or steer; client-centred therapy in *Client-Centered Therapy* (1951), which added a theory of per-

sonality; and the person-centred approach from the 1970s. A "client", unlike a patient, keeps responsibility for his or her own life.

The theory starts from the **actualising tendency**, the inborn tendency of every organism to develop its capacities in ways that maintain and enhance it (not Maslow's self-actualisation). Infants judge experience by an *organismic valuing process*: food is good when hungry, pain is bad. Part of experience becomes the self-concept, the organised picture of "I" and "me", beside an ideal self, the self one would like to be. With the self comes a need for positive regard, and parents rarely give it without strings ("we love you when you top the class"). The child introjects these rules as **conditions of worth** and starts valuing experience by whether it earns approval.

The result is **incongruence**, a mismatch between self-concept and experience: a split between the organismic valuing process and a self burdened by conditions of worth, felt as tension, anxiety and confusion. Ananya, raised as "the doctor of the family", comes alive only when she draws, so she distorts the experience ("I am just tired") or denies it awareness (Figure 7.9). Rogers's 19 propositions (1951) state the theory: behaviour follows the world as perceived (the phenomenal field), and experience inconsistent with the self is denied or distorted until, in complete safety, it can be admitted. Growth heads toward the **fully functioning person**, a direction rather than a state: open to experience, living fully in each moment, trusting his or her own organism, free and creative.

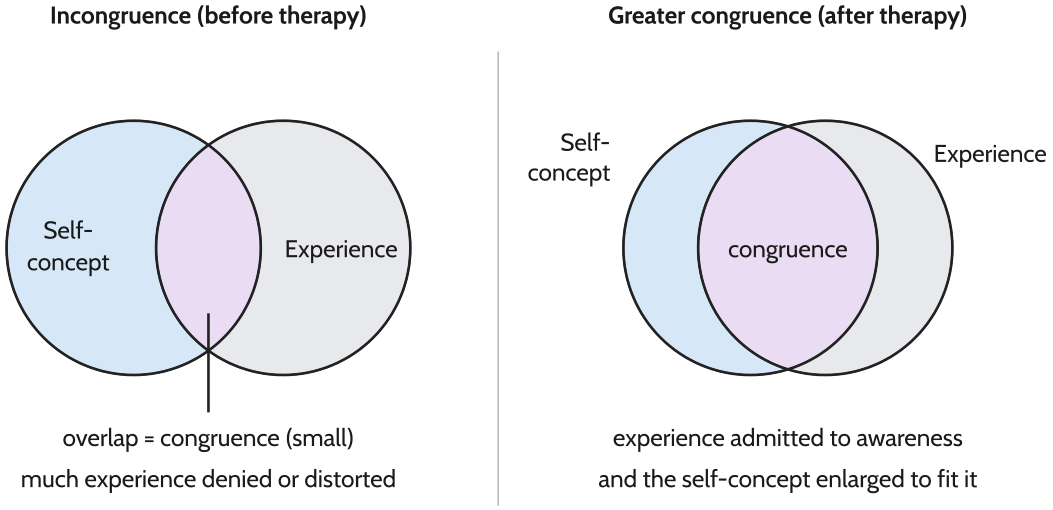

Figure 7.9 Rogers's model of incongruence. The overlap of self-concept and actual experience is the area of congruence; therapy enlarges it by letting denied and distorted experience into awareness.

EXAM TRAP

"A split between the organismic valuing process and a self-concept burdened by conditions of worth" is **incongruence**. *Introjection* is how conditions of worth get in, *conflict* has no special place in the theory, and the *actualising tendency* is the growth force that incongruence blocks.

The core conditions and person-centred practice ●●●

The cure, then, is a relationship without conditions of worth. In a 1957 paper Rogers listed six **necessary and sufficient conditions** of personality change: (1) two persons are in psychological contact; (2) the client is incongruent, vulnerable or anxious; (3) the therapist is congruent; (4) the therapist feels unconditional positive regard; (5) the therapist empathically understands the client's internal frame of reference and communicates this; (6) the client perceives this empathy and regard, at least minimally. "Necessary and sufficient" means no change without them and nothing more needed. Conditions 3 to 5 are the three core conditions (Table 7.14): **unconditional positive regard**, a term coined in 1954 by Rogers's student **Stanley Standal**; **congruence**, which Rogers later called the most basic; and empathy.

Table 7.14 Rogers's three core conditions.

Core condition	Also called	What it means	What it is not
Congruence	Genuineness, realness	Feelings, awareness and words match; no façade	Disclosing every feeling; playing the expert
Unconditional positive regard	Acceptance, prizing, warmth	Valuing the person without conditions of worth	Approving every act; praise; reassurance
Accurate empathic understanding	Empathy	Sensing the client's world "as if" one's own, never losing the "as if", and saying so	Sympathy; identification; interpretation

The therapist responds from inside the client's frame of reference, above all by **reflection of feeling**: putting into words the feeling the client is expressing and offering it back to be checked. To "I am not sure this course is mine", advice ("Talk to your parents") directs and analysis ("Perhaps you resent your father") imposes theory; a reflection ("You doubt whether this course is really yours") stays with the client's meaning. Interpretation, advice, diagnosis, testing and aversive conditioning (Section 7.4) have no place.

> EXAM TRAP
>
> The core conditions are empathy, unconditional positive regard and congruence ("all of the above" when all three are listed). Interpretation, transference and free association are psychoanalytic; reinforcement and shaping are behavioural; disputation and homework belong to Albert Ellis and Aaron Beck (Section 7.5); confrontation and paradoxical intention belong to Gestalt therapy and logotherapy. Edward Bordin's **working alliance** (1979: goals, tasks and bond; Section 7.8) is not Rogerian; person-centred therapy is the approach least likely to use the term, preferring the "real relationship" and the core conditions.

KEY STUDY: THE Q-SORT AND THE SELF-IDEAL GAP (CHICAGO, 1954)

Question. Does therapy bring the perceived self closer to the ideal self? **Method.** The **Q-sort** was devised by the British psychologist **William Stephenson** (*The Study of Behavior*, 1953) and adapted by Rogers's group. A person sorts about 100 self-statements ("I am a submissive person") from "most like me" to "least like me" into a forced, roughly normal distribution, once for the real and once for the ideal self; correlating the sorts captures within-person variation. **Results.** For John Butler and Gerard Haigh's clients (1954) the mean self-ideal correlation rose from about zero (minus .01) before therapy to .34 after it, and stood at .31 at follow-up. **Criticism.** Falling ideals or learning the "healthy" answer could shrink the gap; samples were small.

Using rating scales of empathy, warmth and genuineness, **Charles Truax** and **Robert Carkhuff** (1967) found that clients of high-scoring therapists improved more. The conditions help in every therapy (Section 7.8) but are rarely sufficient: a severe phobia still needs exposure.

Rogers also championed the **encounter group** (*Carl Rogers on Encounter Groups*, 1970): a small, loosely structured group whose principal role is personal growth and self-change through self-disclosure and honest feedback, not socialising, team building or diagnosis. Kurt Lewin's T-groups and other group formats are in Section 7.7.

Gestalt therapy ●●●

Fritz Perls (1893 to 1970), a German-born psychoanalyst, founded Gestalt therapy with his wife **Laura Perls** and the writer **Paul Goodman**; the founding book, *Gestalt Therapy* (1951), was by Perls, Ralph Hefferline and Goodman, and Perls's workshops at the Esalen Institute (1964 to 1969) made it famous. Therapy stays in the here and now. Its goal is awareness of what one is doing and feeling, and how, not why; with awareness comes responsibility, so clients say "I won't" rather than "I can't". Needs organise experience as *figure* and *ground*: the most pressing need stands out, is met and recedes. A feeling never expressed and completed (an unsaid goodbye to a dead parent) remains as **unfinished business**, an incomplete gestalt that intrudes on the present. Health needs good contact at the boundary between self and environment. Perls, Hefferline and Goodman's five disturbances of contact ended with egotism; the five of Erving and Miriam Polster (1973), the list most textbooks use, drop egotism and add deflection (Table 7.15).

Table 7.15 The contact boundary disturbances of Gestalt therapy.

Disturbance	What happens	Example
Introjection	Swallowing others' views whole, undigested	"Good girls never get angry", held as one's own rule
Projection	Disowning a part of oneself by placing it in others	Blaming the professor for an exam one did not study for
Retroflection	Doing to oneself what one wants to do to others	Biting one's nails instead of expressing anger
Deflection	Turning aside from direct contact	Joking, vagueness, talking about someone instead of to them

Disturbance	What happens	Example
Confluence	Self-other boundary blurred; a high need for acceptance and liking leads to enmeshment	"We feel exactly the same about everything"
Egotism	Excessive self-monitoring; watching oneself instead of engaging	Judging every sentence while talking

The techniques are experiments in awareness. In the **empty chair technique** the client speaks to an absent person, or a part of the self, imagined in an empty chair, often switching seats to reply: the classic way to finish unfinished business. In the two-chair technique each side of an inner split gets a chair, typically the *top dog* (the righteous "you should" voice) and the underdog (the apologetic voice that wins by stalling). On the *hot seat* one workshop member worked with Perls while the group watched. In dream work every person and object in a dream is a projection of a disowned part of the dreamer, so the client speaks as each part, in the present tense, instead of having the dream interpreted. The Gestalt prayer (1969) begins: "I do my thing and you do your thing. I am not in this world to live up to your expectations."

> EXAM TRAP
>
> Gestalt therapy is not Gestalt psychology, the perception school of Max Wertheimer, Wolfgang Köhler and Kurt Koffka (Chapter 1), which rejected Perls's borrowing of its terms. The empty chair always means Gestalt therapy, never person-centred therapy, CBT, REBT, dialectical behaviour therapy, psychoanalysis or Bowen's family systems therapy, though emotion-focused therapy later borrowed it.

Existential therapy and logotherapy ●●●

Existential therapy is more a view of life than a set of techniques. Its chief philosophical source is **Martin Heidegger**, whose *Being and Time* (1927) described human existence as **Dasein** ("being there"), a being-in-the-world: a person is never a thing apart from a world. It begins from the existential givens, which **Irvin Yalom** (born 1931) named in *Existential Psychotherapy* (1980) as four **ultimate concerns**: death, freedom (with its responsibility), isolation and meaninglessness. Anxiety about them is normal; trouble starts when people flee it into inauthentic living, and the therapist helps the client face the givens and choose.

The Swiss psychiatrists **Ludwig Binswanger** (1881 to 1966) and **Medard Boss** (1903 to 1990; *Psychoanalysis and Daseinsanalysis*, 1963) applied Heidegger to psychiatry as Daseinsanalysis. Binswanger described three simultaneous modes of being-in-the-world: the **Umwelt** (world around: body, drives and physical environment), the **Mitwelt** (with-world: relationships with others) and the **Eigenwelt** (own-world: one's relation to oneself); **Emmy van Deurzen** later added the Überwelt of spiritual values. **Rollo May** (1909 to 1994; *The Meaning of Anxiety*, 1950; *Existence*, co-edited 1958) brought these ideas to America. For May, existential anxiety is the anxiety of a being who knows it is free and will die: normal anxiety is proportionate, not repressed and usable; neurotic anxiety is disproportionate, repressed and paralysing. Existential-humanistic therapy (May, James Bugental) is the American blend.

Viktor Emil Frankl (1905 to 1997), a Viennese psychiatrist who survived Auschwitz and other Nazi camps, wrote *Man's Search for Meaning* (German original, 1946). His **logotherapy** (Greek *logos*, meaning), the "third Viennese school", holds that the primary human motive is not Freud's will to pleasure or Adler's will to power but the **will to meaning**. Its frustration

brings an existential vacuum of emptiness and boredom and sometimes a **noogenic neurosis** (Greek *noos*, mind or spirit): one rooted in an existential problem rather than in conflict between drives. Meaning comes from creating or doing (creative values), from experiencing or loving (experiential values), and from the stand one takes toward unavoidable suffering (attitudinal values).

In **paradoxical intention**, used for anticipatory anxiety in phobias, obsessions and insomnia, the client deliberately wishes for the feared symptom: a doctor afraid of sweating is told to show people how much he can really sweat, and the fear collapses. In **dereflection**, which counters hyper-reflection (excessive watching of oneself), attention is turned outward, as when a man anxious about his sexual performance attends to his partner instead. Socratic dialogue, the main technique, uses questions to help clients discover their own meanings.

> EXAM TRAP
>
> The will to meaning, noogenic neurosis or anxiety, paradoxical intention and dereflection all belong to Frankl's logotherapy, an existential therapy: not REBT, acceptance and commitment therapy (Section 7.5) or the broader existential-humanistic therapy. Dereflection is not sensate focus, the graded touching exercise of Masters and Johnson (Section 7.7). Yalom's four ultimate concerns are not his group therapeutic factors (Section 7.7).

Reality therapy, focusing, emotion-focused and positive therapies ●●

William Glasser (1925 to 2013), an American psychiatrist, refined reality therapy with delinquent girls at the Ventura School and published *Reality Therapy* in 1965. Its base, control theory, later renamed **choice theory** (*Choice Theory*, 1998), holds that we choose nearly all behaviour to meet five basic, genetically built-in needs: survival, love and belonging, power (achievement), freedom and fun. Behaviour is chosen "total behaviour" (acting, thinking, feeling, physiology), so Glasser spoke of "depressing" rather than being depressed; acting is the easiest part to change. The therapist is warm but firm: present behaviour, responsibility, no excuses, no punishment. **Robert Wubbolding** summed up the method as WDEP: Wants, Doing (and direction), Evaluation (is it working?) and Planning.

Eugene Gendlin (1926 to 2017), working with Rogers, found that improving clients attended to their inner bodily experiencing, and taught this skill as **focusing** (1978): attending to the vague bodily *felt sense* of a problem until a fitting word or image brings a "felt shift". **Leslie Greenberg** joined person-centred empathy to Gestalt tasks in process-experiential therapy, now **emotion-focused therapy**. Adaptive emotions are used, maladaptive ones changed by evoking a healthier emotion, and secondary ones (anger covering hurt) explored; two-chair work treats self-critical splits, the empty chair unfinished business. Greenberg and Susan Johnson also built emotionally focused couples therapy (Section 7.7).

"Positive psychotherapy" has two meanings. **Martin Seligman** (born 1942) launched positive psychology as president of the American Psychological Association in 1998; his positive psychotherapy builds strengths and well-being (PERMA: positive emotion, engagement, relationships, meaning, accomplishment) through exercises such as listing three good things daily. The idea of **flow**, total absorption when challenge matches skill, comes from **Mihaly Csikszentmihalyi** (1934 to 2021), not from Rogers, Maslow or Frankl. The Iranian-born German psychiatrist **Nossrat Peseschkian** (1933 to 2010) founded a separate, transcultural Positive Psychotherapy in the 1970s (Latin *positum*, the given): it starts from capacities, uses

stories from many cultures, balances four areas of life (body, achievement, contact, future) and runs in five stages (observation, inventory, situational encouragement, verbalisation, broadening of goals).

Narrative and solution-focused brief therapy •

These brief therapies grew from family therapy and social constructionism: no account of a life is the single true one, and the client is the expert. Narrative therapy was created by **Michael White** (1948 to 2008) in Adelaide, Australia, and **David Epston** in New Zealand (*Narrative Means to Therapeutic Ends*, 1990). Clients arrive with a thin, problem-saturated story; the therapist practises **externalising the problem** ("the person is not the problem; the problem is the problem"), searches for *unique outcomes* when the problem did not win, and helps the client *re-author* the self story around them. The **solution-focused brief therapy** (SFBT) of **Steve de Shazer** (1940 to 2005) and **Insoo Kim Berg** (died 2007) was developed at the Brief Family Therapy Center in Milwaukee (founded 1978). It ignores causes and builds, through questions (Table 7.16), on what the client wants and what already works; the *miracle question* (de Shazer, 1988) is its signature. Brief strategic therapy (the Mental Research Institute in Palo Alto; Jay Haley), with reframing and paradoxical directives such as prescribing the symptom, is in Section 7.7.

Table 7.16 Techniques of narrative and solution-focused therapy.

Technique	Approach	Example or purpose
Externalising	Narrative	"When did the Worry first move into your life?"
Unique outcome	Narrative	A time the problem did not dominate
Re-authoring	Narrative	Thickening that exception into a new self story
Miracle question	SFBT	"If a miracle solved the problem tonight, what would be the first sign tomorrow?"
Exception question	SFBT	"When is the problem absent or smaller?"
Scaling question	SFBT	"From 0 to 10, where are you now? What would one point higher look like?"
Coping question	SFBT	"With all this happening, how have you kept going?"
Compliments	SFBT	Genuine praise for strengths and what works

Motivational interviewing and the stages of change ••

Motivational interviewing (MI) was first described by **William R. Miller** in 1983 for problem drinkers and developed with **Stephen Rollnick** in *Motivational Interviewing* (1991; later editions 2002 and 2013). It is a collaborative, person-centred but goal-directed way of strengthening a person's own motivation by resolving ambivalence, wanting and not wanting change at once; arguing for change only makes the client defend the status quo. The spirit (2013 edition) is PACE: Partnership, Acceptance, Compassion and Evocation (drawing out the client's own reasons). The four processes are engaging, focusing, evoking and planning, and the core skills are OARS: Open questions, Affirmations, Reflections and Summaries. The 2002 edition gave four principles: express empathy, develop discrepancy between behaviour and the client's own goals, roll with resistance (now split into *sustain talk* for the status quo and *discord* in the relationship) and support self-efficacy. The counsellor strengthens **change talk**: DARN (Desire, Ability, Reasons, Need), then CAT (Commitment, Activation, Taking steps).

MI's home ground is substance use; it is also used for smoking, treatment adherence and other health behaviours. The meta-analysis of Burke, Arkowitz and Menchola (2003) found adapted MI effective for alcohol, drug, and diet and exercise problems (central in diabetes care), but not for smoking or HIV-risk behaviour; obsessive-compulsive disorder was not reviewed.

> **EXAM TRAP**
>
> MI was developed by Miller and Rollnick, not by Prochaska and DiClemente (stages of change), Albert Bandura (self-efficacy) or Alan Marlatt (relapse prevention, Section 7.9). Asked what MI is used to manage, choose substance abuse, its original field, over treatment adherence. Asked which problem Burke and colleagues (2003) did not list, choose obsessive-compulsive disorder: smoking was reviewed (with weak results), OCD not at all.

MI is often paired with the **transtheoretical model** of **James Prochaska** and **Carlo DiClemente**, built from studies of how smokers quit on their own (1983). People pass through stages of change (Figure 7.10): precontemplation (no intention to change within six months), contemplation (intending to, but ambivalent), preparation (acting within a month; small steps taken), action (changed under six months ago), maintenance (change kept over six months) and termination (no temptation, full confidence). Relapse is recycling to an earlier stage, not failure, so progress is a spiral. Ten processes of change drive movement: experiential ones (consciousness raising, dramatic relief, environmental and self-re-evaluation, social liberation) help most early, behavioural ones (self-liberation or commitment, counterconditioning, stimulus control, contingency management, helping relationships) in action and maintenance. Decisional balance (pros and cons) and self-efficacy track progress; the health belief model is in Section 7.9.

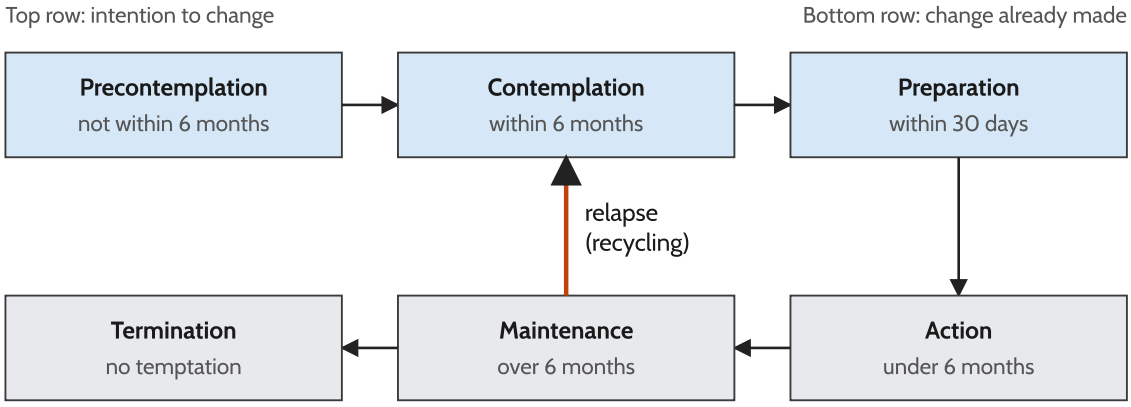

Figure 7.10 The stages of change (Prochaska and DiClemente). Time markers show when change is intended (top row) or how long it has lasted (bottom row); the dashed arrow shows relapse sending a person back to an earlier stage.

Counselling and the helping skills

Counselling and psychotherapy overlap so much that Rogers used the words interchangeably. Counselling usually helps fairly well-functioning people with developmental, educational, vocational or situational problems, briefly and in the present; psychotherapy treats

more severe or long-standing disorders, seeking deeper, slower personality change. **Allen Ivey**'s microcounselling (late 1960s) teaches interviewing one skill at a time. His **microskills hierarchy** rises from ethics and multicultural competence, through attending behaviour (culturally appropriate eye contact, vocal quality, verbal tracking, body language) and the basic listening sequence (questions, encouraging, paraphrasing, reflecting feeling, summarising), to focusing, empathic confrontation, reflection of meaning and influencing skills, and finally integration into a personal style. Carkhuff's model (*Helping and Human Relations*, 1969) has the helper attend, respond, personalise and initiate, moving the client through exploration, understanding and action; on his five-level empathy scale, level 3 is interchangeable with what the client said.

Gerard Egan's *The Skilled Helper* (1975) has three stages. Stage I asks "What is going on?" (the current picture: telling the story, seeing blind spots, choosing the key issue); Stage II asks "What do I want instead?" (the preferred picture: possibilities, goals, commitment); Stage III asks "How do I get there?" (strategies, a plan and action). The helper attends with SOLER: face the client Squarely, Open posture, Lean forward, Eye contact, Relaxed. Egan separates primary empathy, understanding what the client has said and felt, from **advanced empathy**, voicing what the client only hints at or half sees, which is a form of challenge.

> EXAM TRAP
>
> Advanced empathy is inappropriate in an initial session: without trust and enough information it feels like an intrusive interpretation. Primary empathy is right from the first minute, and culturally sensitive empathy is always appropriate.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Rogers's term for the mismatch between self-concept and actual experience is _____.
2. Rogers's three core conditions are empathy, congruence and _____.
3. In Gestalt therapy, a high need for acceptance that blurs the self-other boundary is _____.
4. Yalom's four ultimate concerns are death, freedom, isolation and _____.
5. Frankl's technique of turning attention away from oneself, used for hyper-reflection, is _____.
6. Glasser's five basic needs are survival, love and belonging, power, freedom and _____.
7. "If a miracle solved the problem tonight…" is the signature question of _____ therapy.

Answers: (1) incongruence, (2) unconditional positive regard, (3) confluence, (4) meaninglessness, (5) dereflection, (6) fun, (7) solution-focused brief.

7.7 Family, Couples and Group Therapies MEDIUM YIELD · 28 QS

For most of its history psychotherapy meant one patient and one therapist. In the 1950s clinicians noticed that patients who recovered in hospital often relapsed soon after going home, and that when one family member improved another sometimes fell ill. Perhaps the problem lay partly in the relationships. This section follows that idea from the family to the couple and the group.

The family as a system ●

The idea came from biology. **Ludwig von Bertalanffy** (1901 to 1972) proposed a **general systems theory**: a system is a set of interacting parts whose whole is more than their sum, so a change in one part changes the whole. **Norbert Wiener**'s cybernetics (1948) added self-regulation by feedback. A family tends toward **homeostasis**, a balance it defends as a thermostat defends a room temperature. Negative feedback restores the old balance and positive feedback amplifies change; neither word means good or bad.

Systems thinking replaces linear causality (A causes B, so someone is to blame) with **circular causality**, in which each person's behaviour is both cause and effect of the others' (Figure 7.11). The member brought for treatment is the **identified patient**, the visible sign of a troubled system rather than its cause; the symptom may even serve the system, as when a son's misbehaviour unites quarrelling parents, and a child made to carry the family's problems is its scapegoat. Families also pass through a **family life cycle**: **Betty Carter** and **Monica McGoldrick** described six stages (leaving home, the new couple, families with young children, with adolescents, launching children, and later life), and symptoms cluster at the transitions.

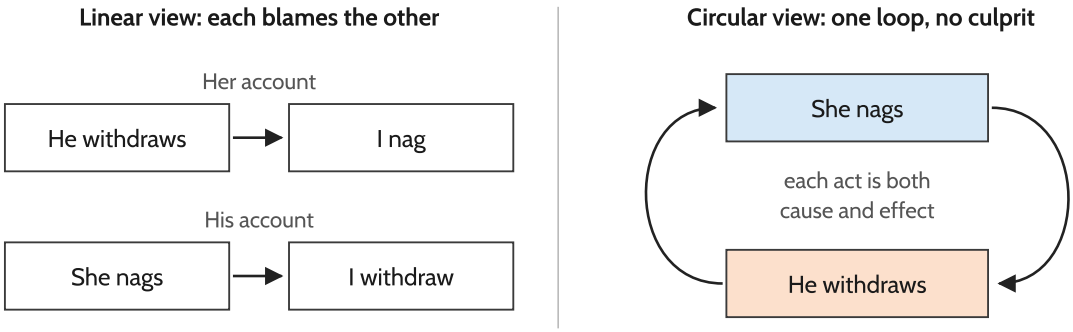

Figure 7.11 Linear and circular causality. Each partner says the other started it; the systems therapist treats the loop and blames no one.

Families and schizophrenia: from blame to expressed emotion ●●●

The first family research asked whether families cause schizophrenia. In 1948 **Frieda Fromm-Reichmann** (1889 to 1957) blamed a dominating, rejecting **schizophrenogenic mother**. In 1956 the anthropologist **Gregory Bateson** (1904 to 1980), with **Don Jackson** (1920 to 1968), Jay Haley and **John Weakland** at Palo Alto, proposed the **double bind**: a person in a vital relationship repeatedly receives two contradictory messages at different levels (typically words contradicted by manner) and can neither comment on the contradiction nor leave, so every response is wrong.

KEY STUDY: BATESON'S DOUBLE BIND (1956)

A young man recovering from an acute schizophrenic episode put his arm round his visiting mother, and she stiffened. He withdrew it; she asked, "Don't you love me any more?" When he blushed she said, "Dear, you must not be so easily embarrassed and afraid of your feelings." After she left he assaulted an aide. Whatever he did was wrong, and he could not say so. **Criticism:** anecdotal and never predictive, it is not accepted as a cause.

Theodore Lidz (1910 to 2001) at Yale described **marital schism**, open conflict in which each parent undermines the other and courts the child, and **marital skew**, in which one disturbed parent dominates while the other pretends all is normal. **Lyman Wynne** (1923 to 2007) described **pseudomutuality** (1958), a facade of harmony that hides distance and leaves no room for a separate identity. None of these survived as a cause of schizophrenia (Chapter 8), and the schizophrenogenic mother is discredited.

What survived concerned *relapse*. In London **George Brown** found that discharged patients who spent long hours with their relatives relapsed more often. The emotional climate of the home, the attitudes relatives express about the patient, is called **expressed emotion** (EE) and is rated from the Camberwell Family Interview. High EE has three components: **critical comments**, **hostility** and **emotional over-involvement** (intrusive, self-sacrificing concern). Brown, **James Birley** and **John Wing** (1972) showed that high EE predicts relapse.

KEY STUDY: VAUGHN AND LEFF (1976)

Christine Vaughn and **Julian Leff** followed patients with schizophrenia for nine months after discharge. About half of those returning to high-EE homes relapsed, against about one in eight in low-EE homes. In high-EE homes, regular medication and less than 35 hours a week of face-to-face contact each lowered the risk.

Hence **family psychoeducation**: teaching relatives about the illness, medication and warning signs, training calm communication and problem solving, and cutting contact hours where the climate will not change (Leff's programmes; **Ian Falloon**'s behavioural family management). It lowers relapse (Section 7.9).

EXAM TRAP

High EE predicts **relapse**, not onset, negative symptoms or violence. Warmth is rated in the interview but is **not** a high-EE dimension. Reducing contact hours manages high EE. One parent sending two conflicting messages is **double bind** communication; first-rank symptoms are Kurt Schneider's (Chapter 8).

INDIAN CONTEXT

In the 1950s Vidya Sagar at Amritsar Mental Hospital had families camp on the grounds and share in care. Indian families show the lowest rates of high EE in cross-cultural comparisons (Jenkins and Karno, 1992), and closeness that looks like over-involvement by Western norms may be ordinary support. A therapist unaware of a family's culture and customs will misread it and make wrong assessments.

Bowen's family systems therapy ●

Murray Bowen (1913 to 1990), who admitted whole families to hospital wards in the 1950s, founded the transgenerational school. Its master concept is **differentiation of self**: separating thinking from feeling, and staying oneself while emotionally connected; the poorly differentiated are fused and reactive. Under stress a two-person relationship draws in a third person, usually a child: **triangulation**. Table 7.17 lists all eight concepts. Bowen worked as a calm coach who stayed out of the family's triangles, and mapped three or more generations in a **genogram** (squares for males, circles for females, lines for marriages, conflicts and cutoffs).

Table 7.17 Bowen's eight concepts.

Concept	Meaning
Differentiation of self	Thinking separate from feeling
Triangles	A stressed pair draws in a third
Nuclear family emotional system	Fusion shows as marital conflict, an impaired spouse or a symptomatic child
Family projection process	Parents project anxiety onto one child
Multigenerational transmission	Low differentiation and anxiety pass down the generations
Emotional cutoff	Escaping fusion by distance from the family
Sibling position	Birth-order roles shape adult roles
Societal regression	Stressed societies lose differentiation too

> EXAM TRAP
>
> Transgenerational, multigenerational transmission, differentiation and the genogram mean **Bowen**. Diagrams differ: a genogram maps a family across generations; an **ecomap** (**Ann Hartman**, a social worker, in the 1970s) maps a family's links with school, work and services; a **sociogram** (Moreno, below) maps relations and communication within a group; a cognitive map is Tolman's mental map of a maze (Chapter 1).

Minuchin's structural family therapy ●●

Salvador Minuchin (1921 to 2017) worked with poor families of delinquent boys (*Families of the Slums*, 1967) and then at the Philadelphia Child Guidance Clinic (*Families and Family Therapy*, 1974). He studied the family's present *structure*: its subsystems (spousal, parental, sibling), a hierarchy with parents in charge, and boundaries that should be clear, firm yet permeable. Diffuse boundaries produce **enmeshment** (over-involvement, intrusion, little autonomy); rigid boundaries produce **disengagement** (isolation, little support), as Figure 7.12 shows. *Psychosomatic Families* (1978) described families of children with anorexia, diabetes and asthma as enmeshed, overprotective, rigid and unable to resolve conflict.

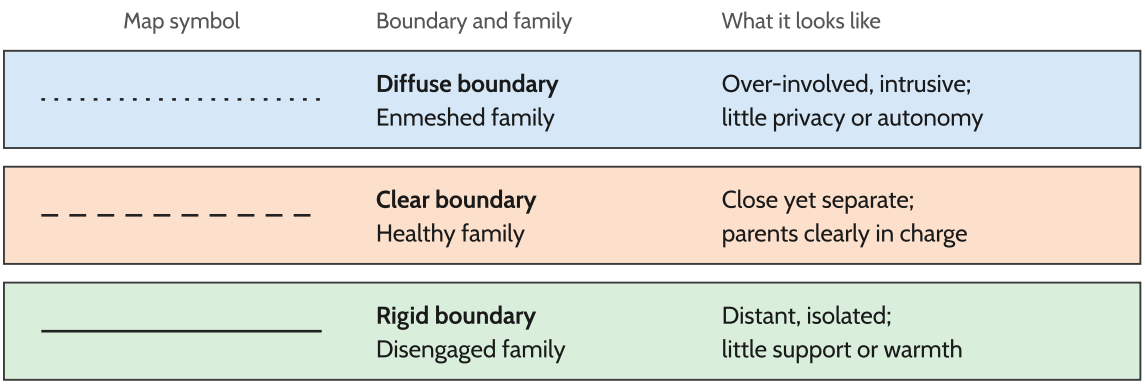

Figure 7.12 Minuchin's boundaries and their structural-map symbols (dotted, dashed, solid). Health is the middle row.

The therapist begins by **joining**, accommodating to the family's style, then maps its structure and asks for an **enactment**: the conflict acted out in the room rather than described. **Unbalancing** (siding with one member, often an overruled parent, to shift power), boundary making and reframing then restructure the family.

> EXAM TRAP
>
> Enmeshment means **diffuse** boundaries; rigid boundaries mean disengagement. Enmeshment is not a complete lack of communication, nor scapegoating. Structural is Minuchin, strategic is Haley, multigenerational is Bowen, experiential is Satir. A therapist's boundary violation with a client is professional ethics (Section 7.1).

Strategic, systemic and other family therapies

Jay Haley (1923 to 2007), influenced by the hypnotherapist **Milton Erickson**, made therapy strategic: the therapist plans how to remove the presenting problem through *directives* (tasks) and does not seek insight. With resistant families he used **paradoxical intervention**, such as prescribing the symptom (a quarrelling couple told to quarrel at a set hour); in **ordeal therapy** (1984) the client performs a harmless but tiresome ordeal every time the symptom occurs. **Cloe Madanes** (1981) added pretend techniques: the child pretends to have the symptom and the parent pretends to help.

At the Mental Research Institute in Palo Alto (founded by Jackson, 1959), **Paul Watzlawick**, Weakland and **Richard Fisch** offered brief therapy of about ten sessions. In *Change* (1974) they argued that *attempted solutions* keep problems going: a wife nags her husband to exercise, he resists, she nags more. Trying harder is **first-order change**, within unchanged rules; the goal is **second-order change**, a change in the rules themselves.

The Milan group (**Mara Selvini Palazzoli**, **Luigi Boscolo**, **Gianfranco Cecchin** and **Giuliana Prata**; *Paradox and Counterparadox*, 1978) worked as a team behind a one-way mirror. Their **positive connotation** recasts everyone's behaviour, symptom included, as serving the family's unity, and prescribed rituals interrupt the family's game. Their three guidelines were hypothesising (a provisional systemic explanation, revised each session), circularity (exploring relationships and differences) and neutrality (siding with no one), and **circular questioning**

asks one member about the relationship between others: "Who is most upset when Anu won't eat?"

Among the experiential therapists, **Virginia Satir** (1916 to 1988; *Peoplemaking*, 1972) named four stances that protect low self-esteem under stress: placating (always agreeing), blaming, computing or super-reasonable (cold logic) and distracting or irrelevant, against congruent, levelling communication in which words, feelings and body match. In **family sculpting** she posed members physically to show their relationships. **Carl Whitaker** (1912 to 1995) practised spontaneous, often absurd symbolic-experiential therapy with a co-therapist. Table 7.18 completes the map.

Table 7.18 Other family therapies.

Approach	Founders	Core idea and methods
Contextual	**Ivan Boszormenyi-Nagy** (*Invisible Loyalties*, 1973)	Relational ethics: loyalties and a ledger of fairness across generations
Behavioural family	**Gerald Patterson**; Falloon	Parent training (Section 7.10), contracts, communication and problem-solving skills
Narrative	**Michael White** and **David Epston** (1990)	Social constructionist: lives as stories; externalising the problem, unique outcomes, re-authoring
Solution-focused	**Steve de Shazer** and **Insoo Kim Berg**	Build on what works: miracle, exception and scaling questions (Section 7.6)
Functional family therapy	**James Alexander** and **Bruce Parsons**	A delinquent adolescent's behaviour serves a family function
Multisystemic therapy	**Scott Henggeler**	Intensive home-based work on youth offending across family, peers and school

> **EXAM TRAP**
>
> Prescribing the symptom is strategic paradox; exception questions are solution-focused; externalisation is narrative. "Individuals, families and cultures hold different power as creators of their realities" describes **narrative therapy**'s social constructionism, not transgenerational, strategic or structural therapy.

Couples and sex therapy

When a couple's distress keeps an individual problem going, the pair is treated together. For depression bound up with marital discord, marital and couple therapy is the better choice: behavioural marital therapy relieves the depression about as well as individual cognitive therapy and also mends the marriage.

Behavioural couples therapy (**Neil Jacobson** and **Gayla Margolin**, 1979) increases positive behaviour exchange and teaches communication and problem solving. Jacobson and **Andrew Christensen**'s integrative behavioural couple therapy adds *acceptance* of differences that will not change. **John Gottman**, filming couples in conflict, found that his **four horsemen** predict divorce: criticism (attacking character), contempt (mockery; the strongest predictor), defensiveness and stonewalling (withdrawal). Stable couples keep at least five positive interactions for every negative one, the 5:1 ratio, and his Sound Relationship House builds from love maps and fondness up to shared meaning, on walls of trust and commit-

ment. **Emotionally focused couple therapy** (**Sue Johnson**, 1980s) uses attachment theory: fights are protests at a threatened bond, locked in cycles such as pursue and withdraw.

William Masters and **Virginia Johnson** (*Human Sexual Inadequacy*, 1970) treated the couple as the patient, with a male and female co-therapist team. Their **sensate focus** is graded, non-demand touching with intercourse at first banned, removing performance anxiety; for premature ejaculation they taught the squeeze technique (firm pressure where the head of the penis joins the shaft, just before ejaculation). The stop-start technique, pausing stimulation before the point of no return, is **James Semans**'s (1956). **Helen Singer Kaplan** (*The New Sex Therapy*, 1974) blended these tasks with psychodynamic work in brief sex therapy. **Jack Annon**'s **PLISSIT model** (1976) grades help: Permission, Limited Information, Specific Suggestions, Intensive Therapy.

> EXAM TRAP
>
> Contempt, not criticism, is Gottman's best predictor of divorce. Squeeze is Masters and Johnson's; stop-start is Semans's (Masters and Johnson adapted Semans's method, so some textbooks credit Semans with both). Emotionally focused couple therapy (Johnson) is not Greenberg's individual emotion-focused therapy (Section 7.6).

Group therapy: Yalom's factors and group stages ●●

Why do groups help? **Irvin Yalom**'s *The Theory and Practice of Group Psychotherapy* (first edition 1970) sets out eleven **therapeutic factors** (Table 7.19). Of these, **universality**, discovering that one is not alone in one's suffering, is often a newcomer's first relief.

Table 7.19 Yalom's eleven therapeutic factors.

Factor	What happens
Instillation of hope	Seeing others improve
Universality	Others share my problem
Imparting information	Advice and teaching
Altruism	Helping others raises self-worth
Corrective recapitulation of the primary family group	Family conflicts relived and resolved differently
Development of socialising techniques	Social skills learned through feedback
Imitative behaviour	Modelling the leader and members
Interpersonal learning	Learning how one affects others
Group cohesiveness	Belonging; the group's therapeutic relationship
Catharsis	Expressing strong feeling
Existential factors	Accepting death, aloneness and responsibility

The engine of Yalom's group is the **here-and-now**. The group is a social microcosm, where members soon behave as they do in life, so the leader works with interactions in the session: first experiencing them, then illuminating the process by reflecting on what just happened. The leader creates and maintains the group, builds its norms, and activates and illuminates the here-and-now. **Bruce Tuckman** (1965) traced group development through forming (po-

lite testing), storming (conflict), norming (cohesion) and performing, adding adjourning with **Mary Ann Jensen** in 1977. A closed group keeps the same members for a fixed term and bonds quickly; an open group admits new members as others leave.

> EXAM TRAP
>
> "Not alone" is **universality**; seeing others recover is hope; helping is altruism; venting is catharsis. The here-and-now does not ban talk of outside life; it makes the group's own interactions, experienced and then reflected upon, the engine of change, and it is not didactic teaching or disputing irrational beliefs (Section 7.5). Free association, token reinforcement and schema work are not Yalom factors. Yalom's four ultimate concerns belong to existential therapy (Section 7.6).

Psychodrama, encounter and self-help groups, therapeutic communities ●●●

Jacob Levy Moreno (1889 to 1974), who coined the term group psychotherapy, opened a Theatre of Spontaneity in Vienna in 1921. In **psychodrama** the *protagonist* acts out scenes on a *stage*, guided by the therapist as *director*, with members as *auxiliary egos* playing significant others and an *audience* that shares afterwards. In role reversal the protagonist plays the other person; in doubling an auxiliary voices the protagonist's unspoken feelings; in mirroring someone replays the protagonist's behaviour for them to watch. Moreno's sociometry measures attraction and rejection in groups, and its **sociogram** draws members as points and choices as arrows, revealing the group's explicit and implicit communication patterns.

Kurt Lewin (1890 to 1947) inspired the **T-group** (training or sensitivity group, National Training Laboratories, 1947), in which members learn group dynamics and their effect on others. The **encounter group**, championed by **Carl Rogers**, encourages therapy and self-growth through open disclosure and interaction. **Self-help groups** are run by members, not professionals: Alcoholics Anonymous, founded in 1935 by **Bill Wilson** and **Bob Smith**, works through twelve steps, sponsors and anonymity, and Narcotics Anonymous adapted the steps to drugs.

In **milieu therapy** the whole ward environment, staff, routine and fellow patients, is the treatment. Its democratic form, the **therapeutic community**, was pioneered by **Maxwell Jones** (1907 to 1990) at Belmont Hospital in Surrey, whose unit later became the Henderson Hospital: staff and patients share decisions in daily community meetings, hierarchy is flattened, and every interaction becomes feedback. Addiction communities are in Section 7.9.

> EXAM TRAP
>
> Maxwell Jones means the **therapeutic community**. Milieu therapy is the broader idea; group analysis is **S. H. Foulkes**'s psychoanalytic group therapy; encounter groups are Rogers's and psychodrama Moreno's; the token economy, aversion therapy and EMDR belong to behaviour therapy (Section 7.4), and logotherapy is Frankl's (Section 7.6). Encounter groups aim at growth, not diagnosis or team building.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Treating the family as a whole with homeostasis and feedback applies the general systems theory of _____.
2. Contradictory messages that can be neither questioned nor escaped form a _____.
3. High expressed emotion comprises critical comments, hostility and _____.
4. Minuchin's diffuse boundaries produce an _____ family.
5. Gottman's strongest predictor of divorce is _____.
6. Yalom's "I am not alone" factor is _____.
7. The therapeutic community was pioneered by _____.

Answers: (1) Ludwig von Bertalanffy, (2) double bind, (3) emotional over-involvement, (4) enmeshed, (5) contempt, (6) universality, (7) Maxwell Jones.

7.8 Does Psychotherapy Work? Evidence, Common Factors and the Alliance

LOWER YIELD · 12 QS

Sections 7.3 to 7.7 described a crowded marketplace of therapies. Does any of it help, and if so, why? The story runs from Eysenck's attack, through the statistics that answered him, to a debate that continues: does the technique heal, or the relationship, the client and the therapist?

Eysenck's challenge and the meta-analytic answer

In 1952 **Hans Jürgen Eysenck** (1916 to 1997) reviewed the published outcome reports. Neurotic patients given psychoanalysis improved at about 44 percent and those given eclectic therapy at about 64 percent, but comparable patients without psychotherapy (insurance-disability claimants seen by general practitioners, and state-hospital discharges) improved at about 72 percent. Roughly two-thirds of neurotic patients, he concluded, improve within two years with or without therapy, through **spontaneous remission**: recovery driven by time, life events and ordinary support.

Critics attacked his method: the groups differed in severity, "improved" meant different things in different studies, and the "untreated" patients had received reassurance and sedatives. **Allen E. Bergin** (1971) showed that counting dropouts differently lifted psychoanalysis to about 85 percent improvement among those who completed it; **Michael J. Lambert** (1976) showed that Eysenck had inflated untreated improvement, putting spontaneous remission nearer 40 percent. The decisive answer came from a new statistic.

Gene V. Glass coined the term **meta-analysis** in 1976 for pooling many studies statistically instead of counting how many "worked". Each comparison of a treated with a control group yields an **effect size** (d): the difference between the group means divided by a standard deviation. Effect sizes, all in the same unit, can then be averaged across studies and compared across therapies. A d of 0.85 places the average treated client above about 80 percent of untreated people (Figure 7.13).

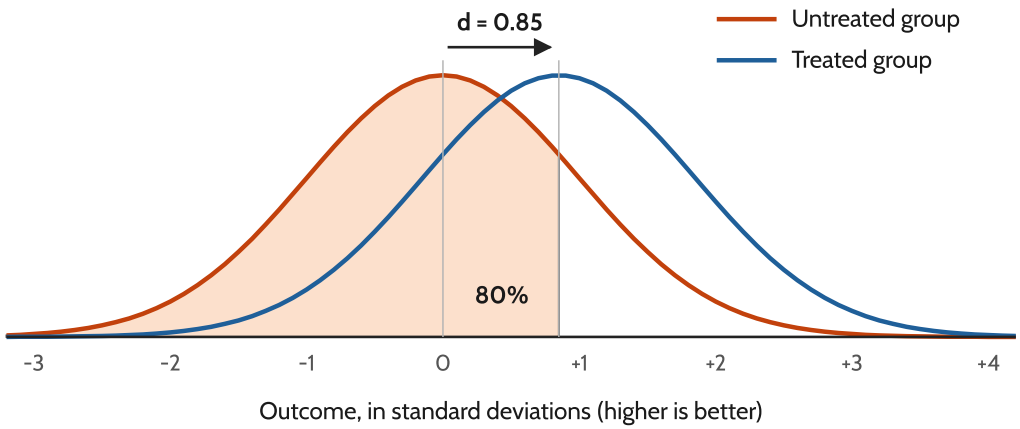

Figure 7.13 What an effect size of 0.85 means. The average treated client (solid curve) scores better than the shaded 80 percent of the untreated group (dashed curve).

KEY STUDY: SMITH AND GLASS (1977)

Mary Lee Smith and Gene Glass pooled 375 controlled studies. The average effect size was 0.68: the typical client was better off than 75 percent of untreated people, and types of therapy differed little. With Thomas Miller (*The Benefits of Psychotherapy*, 1980) they reached 475 studies and 0.85 (80 percent). Eysenck called it "mega-silliness": good and bad studies pooled ("garbage in, garbage out"), unlike outcomes averaged ("apples and oranges").

EXAM TRAP

Eysenck said therapy adds nothing to spontaneous remission, not that it harms. 0.68 and 75 percent is Smith and Glass (1977); 0.85 and 80 percent is Smith, Glass and Miller (1980).

The dodo bird verdict and common factors ●●

Smith and Glass also found the rival schools roughly equal. **Saul Rosenzweig** (1907 to 2004) had predicted this in 1936: diverse therapies succeed through implicit common factors, he argued, quoting the Dodo in *Alice in Wonderland*: "Everybody has won, and all must have prizes." In 1975 **Lester Luborsky**, Barton Singer and Lise Luborsky reviewed the comparative trials and named the **dodo bird verdict**: bona fide therapies, competently delivered, produce roughly equivalent outcomes, because change comes mainly from the **common factors** they share (the alliance, empathy, hope) rather than from specific techniques.

Specific-effects researchers, holding a medical model in which each therapy has active ingredients like a drug, point to exceptions: exposure beats non-exposure therapies for obsessive-compulsive disorder, phobias and panic. Common-factors researchers reply that most comparisons of bona fide therapies show differences near zero (at most about 0.2 in a 1997 meta-analysis) and that apparent winners often reflect allegiance (7.8.4).

Bruce E. Wampold formalised this view as the **contextual model** (*The Great Psychotherapy Debate*, 2001; in its fullest form in the 2015 edition with Zac Imel): therapy works through a real relationship; through expectations created by a convincing explanation and treatment plan; and through specific ingredients that help because they get the client doing healthy things, not because they correct a specific deficit.

Its root is **Jerome David Frank** (1909 to 2005), whose *Persuasion and Healing* (1961) compared psychotherapy with religious healing and placebo. Clients arrive *demoralised*, and every healing method restores morale through four shared features:

- The client has an emotionally charged, confiding relationship with a helper.
- Help takes place in a healing setting (a clinic, a temple, a consulting room).
- A rationale or "myth" plausibly explains the symptoms and how to overcome them.
- A ritual or procedure, believed in by both, requires the client's active participation.

> EXAM TRAP
>
> "Healing and Persuasion" is Frank's *Persuasion and Healing* with the title reversed. Distractors: **Gerald Klerman** co-developed interpersonal psychotherapy and **Donald Winnicott** was an object-relations analyst (both 7.3); **Alfred Pritz** is the Austrian psychotherapist who was founding president of the World Council for Psychotherapy (1995). The dodo verdict never says therapy is ineffective, that only psychoanalysis lasts or that medication always wins: bona fide therapies tie.

Lambert (1992) estimated, in his famous pie (Figure 7.14), that about 40 percent of improvement comes from client and extratherapeutic factors, 30 percent from relationship factors, 15 percent from expectancy (placebo and hope) and only 15 percent from model-specific techniques: rough judgements from the literature, not statistical results.

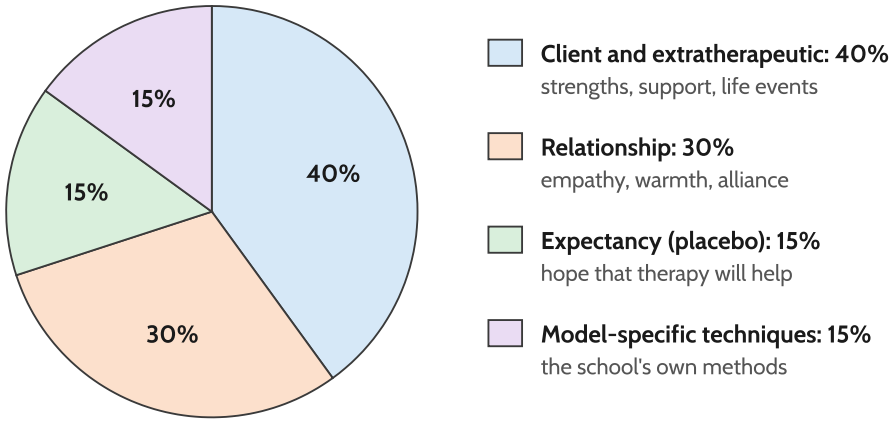

Figure 7.14 Lambert's (1992) rough estimates of what produces improvement. Technique, the slice the schools argue about, is the smallest.

The therapeutic alliance and the core conditions ●●

The alliance idea began in psychoanalysis. **Elizabeth Zetzel** (1956) spoke of the therapeutic alliance and **Ralph Greenson** (1965) of the working alliance, the patient's rational, cooperative tie to the analyst as distinct from transference; Greenson also described the *real relationship*, the genuine person-to-person bond. **Edward S. Bordin** (1979) made the **therapeutic alliance** pantheoretical: a collaborative relationship with three parts, agreement on **goals**, agreement on the **tasks** that reach them, and a **bond** of trust and respect.

A working relationship is thus a contract to work together realistically and collaboratively, visible in how well the two actually work together and resting on a committed belief in the helping relationship; it needs no particular theory. Adam Horvath and Leslie Greenberg's (1989) **Working Alliance Inventory** (36 items, with a 12-item short form) measures goals, tasks and bond. Across meta-analyses the alliance correlates about 0.28 with outcome: modest but robust.

Jeremy D. Safran and J. Christopher Muran described **alliance ruptures**: withdrawal (the client falls silent, goes vague, over-agrees) and confrontation (the client complains or attacks). Repair means noticing the strain, talking openly about what is happening between the two and owning one's part; repaired ruptures predict good outcomes.

Carl Ransom Rogers (1957) held that his core conditions (empathy, unconditional positive regard, congruence; 7.6) were necessary and sufficient. Research supports a weaker claim: empathy correlates about 0.3 with outcome, positive regard and congruence a little less (roughly 0.2 to 0.3): facilitative in every therapy, but not sufficient.

> EXAM TRAP
>
> Bordin's alliance is not a Rogerian concept; Rogers' terms are the actualising tendency, conditions of worth and unconditional positive regard (7.6). Person-centred therapy is the school least likely to say "therapeutic alliance": it treats the relationship and unconditional positive regard as the therapy itself, whereas psychoanalysis coined the term and cognitive-behavioural and interpersonal therapists use Bordin's version. The alliance is not an outcome score, not two clinicians agreeing on a diagnosis (inter-rater reliability, Chapter 2) and not the client's ability to do homework alone. What a working relationship does NOT need: "a psychodynamic theory base".

Therapist, client and allegiance effects

Consistent differences between individual therapists' outcomes, whatever their approach, are called **therapist effects**; they account for about 5 percent of outcome variance (roughly 3 in controlled trials, 7 in routine practice), far more than the differences between treatments. An **allegiance effect** appears when researchers find their own favoured therapy superior; Luborsky and colleagues (1999) showed that researcher allegiance predicted most of the outcome differences in comparative trials.

Client factors carry the biggest slice of the pie. **William Schofield** (*Psychotherapy: The Purchase of Friendship*, 1964) noted that therapists prefer, and do best with, the **YAVIS** client: young, attractive, verbal, intelligent and successful. YAVIS describes patient characteristics, not a technique, an illness or a therapist quality; its opposite, HOUND (homely, old, unattractive, nonverbal, dumb), names neglected clients. Readiness to change (stages of change, 7.6) and realistic expectations of outcome (hope) and of role (what a client does) also predict benefit.

In the **hello-goodbye effect**, described by **Starke R. Hathaway**, clients exaggerate distress at intake (to justify seeking help) and well-being at the end (to thank the therapist), so before-and-after self-reports overstate improvement.

Testing outcome: efficacy, effectiveness and ESTs ●●

Research on **efficacy** asks whether a treatment can work under ideal, controlled conditions (internal validity); research on **effectiveness** asks whether it does work in routine practice (external validity), as in **Martin E. P. Seligman**'s (1995) *Consumer Reports* survey, in which most readers reported benefit (Table 7.20).

Table 7.20 Efficacy and effectiveness studies.

Feature	Efficacy study	Effectiveness study
Clients	Homogeneous, one disorder	Mixed, often comorbid
Assignment	Random assignment to treatment or control	Often not random
Therapist	Controlled by a manual	Flexible
Sessions	Fixed number	Set by need

In **analogue research**, clinical questions are studied under conditions that only resemble therapy, usually with healthy, non-clinical participants (students with a mild fear of snakes standing in for phobic patients): control is high, generalisation uncertain. It is neither outcome research on a clinical population nor comparative research pitting two treatments against each other.

In the **randomised controlled trial** (RCT), chance decides who receives which condition, so the groups differ only in treatment. Its safeguard, **allocation concealment**, hides the upcoming allocation sequence from the recruiter until each participant is enrolled, preventing selection bias. Blinding hides the assigned condition after allocation; in psychotherapy only outcome assessors can be blind. The comparison group decides what a trial can show (Table 7.21).

Table 7.21 Control conditions in psychotherapy trials.

Control	Group receives	Controls for
Waiting list	Treatment after a delay	Time, spontaneous remission
Placebo-attention	Credible contact, no active ingredient	Attention and expectancy
Treatment as usual	Routine care	Usual care; shows added benefit

Trials follow treatment manuals (Aaron Beck's 1979 manual for cognitive therapy of depression was an early model) and check **treatment fidelity**: adherence to the manual and competence in delivering it.

In 1995 an APA Division 12 (Clinical Psychology) task force chaired by **Dianne L. Chambless** began listing **empirically supported treatments** (ESTs). A *well-established* treatment needed at least two good between-group experiments, by independent teams, showing it superior to a pill or psychological placebo or another treatment, or equal to an established one (or a large single-case series), with a manual and clearly described clients. A *probably efficacious* treatment needed only two experiments beating a waiting list, or strong trials from one team. Chambless and Steven Hollon (1998) refined the criteria; most listed treatments were cognitive-behavioural.

The APA's 2006 task force defined **evidence-based practice in psychology** more broadly: "the integration of the best available research with clinical expertise in the context of patient characteristics, culture, and preferences."

> **EXAM TRAP**
>
> Efficacy studies use manuals, homogeneous samples and random assignment; "therapist flexibility is encouraged" marks effectiveness research. Allocation concealment (before enrolment) is not blinding (after allocation).

Landmark trials, dose and clinical significance

> **KEY STUDY: NIMH TREATMENT OF DEPRESSION COLLABORATIVE RESEARCH PROGRAM (1989)**
>
> **Irene Elkin**'s trial randomly assigned about 250 depressed outpatients at three sites to 16 weeks of cognitive-behavioural therapy, interpersonal psychotherapy, imipramine or a pill placebo (the last two with clinical management). All groups improved and the two psychotherapies did not differ; only in severe depression did imipramine clearly beat placebo, with interpersonal therapy showing some advantage. Re-analyses found that the alliance predicted outcome in every arm, and that therapists differed more than treatments.

Project MATCH (1997) assigned about 1,700 people with alcohol problems to twelve-step facilitation, cognitive-behavioural coping-skills therapy or four-session motivational enhancement therapy, to test whether matching clients to treatments helps. All three worked about equally; matching effects were scarcely found.

Kenneth I. Howard and colleagues (1986), pooling about 2,400 outpatients from 15 studies, described the **dose-effect relationship** (Figure 7.15): about 50 percent had improved by 8 sessions, 75 percent by 26 and 85 percent by 52. The curve is negatively accelerated; Howard's later phase model reads it as quick remoralisation, then remediation of symptoms, then slow rehabilitation.

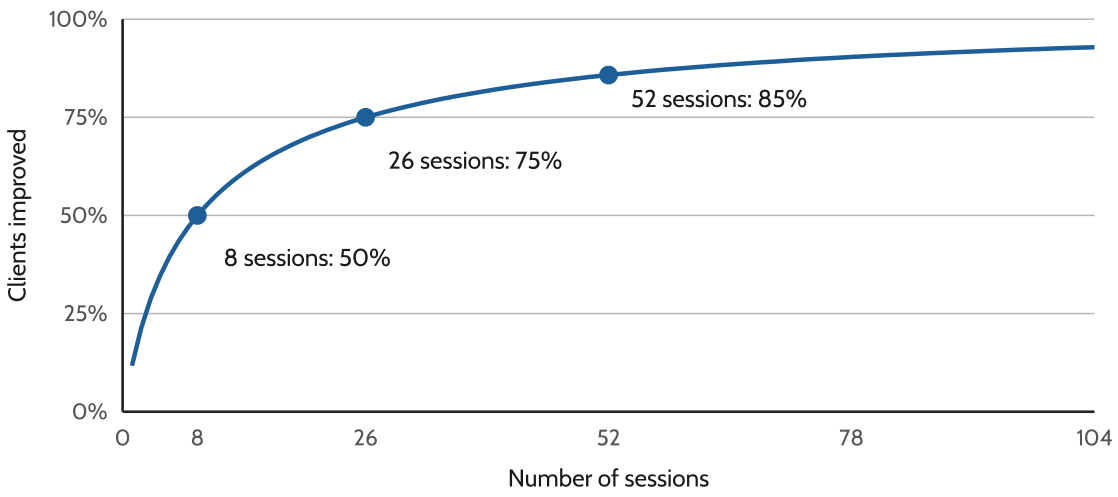

Figure 7.15 The dose-effect relationship (Howard and colleagues, 1986), a smooth curve through the published milestones: most of the benefit comes early.

A large sample can make a small average gain statistically significant while many clients stay unwell. **Neil S. Jacobson** and Paula Truax (1991) therefore judged **clinical significance** client by client, with two tests. The **reliable change index** (RCI), the change divided by the standard error of the difference (derived from the test's reliability), must exceed 1.96, so the change is unlikely to be measurement error; and the client must cross a cut-off from the dysfunctional into the functional range. Both tests passed means recovered; reliable change alone, improved; reliable worsening, deteriorated.

Take Meena, who falls from 32 to 12 on a depression scale with a standard deviation of 8 and reliability of 0.84. The standard error of measurement is $8 \times \sqrt{0.16} = 3.2$, the standard error of the difference $\sqrt{2} \times 3.2 \approx 4.5$, so her RCI is $20 \div 4.5 \approx 4.4$: reliable. With a functional cut-off of 14, she has recovered. A fall from 32 to 27 (RCI about 1.1) is not reliable change.

Therapy can also harm. Bergin (1966) named the **deterioration effect**: roughly 5 to 10 percent of adult clients get worse, which is why Lambert's Outcome Questionnaire-45 feeds each client's progress back to the therapist.

Process research and psychotherapy integration

Where outcome research asks whether therapy works, **process research** studies what happens within sessions and how it produces change. Therapy has a beginning (assessment, alliance, goals, contract, teaching the client's role), a middle or working phase (the main interventions and working through) and a planned termination (consolidating gains, preparing for setbacks, handling the ending).

Many clients end in **premature termination**, leaving before goals are met and against the therapist's advice: near half of outpatients in a 1993 meta-analysis, about one in five in a 2012 one, more often with young and poor clients, weak alliances and mismatched expectations. Classical analysts shunned **therapist self-disclosure** and humanistic therapists value it; it helps when brief, rare, client-focused and about resolved issues. Homework, central to cognitive and behavioural therapy, carries therapy into the client's week; completing it predicts better outcome.

Psychotherapy integration takes four routes (Table 7.22). In **technical eclecticism** techniques are borrowed without their theories; **theoretical integration** fuses theories into a new framework; **assimilative integration** keeps one home theory and absorbs techniques into it.

Table 7.22 Routes to psychotherapy integration.

Route	Example
Technical eclecticism	**Arnold A. Lazarus**: multimodal therapy, BASIC I.D. (7.5)
Theoretical integration	**Paul L. Wachtel**: cyclical psychodynamics; **James O. Prochaska**: transtheoretical approach
Common factors integration	Frank's four features; Wampold's contextual model
Assimilative integration	**Stanley B. Messer**: psychodynamic therapy using borrowed techniques

Wachtel's cycles are vicious circles (inner conflict breeds self-defeating behaviour that deepens the conflict), so he joins psychodynamic exploration to behavioural methods. Prochaska built his approach around the client's stage of readiness (7.6).

Culture-sensitive therapy and therapy in India

Western therapy prizes autonomy, insight and an equal, non-directive relationship. A therapist unaware of a family's culture and customs risks wrong assessments and misinterpretations, such as calling a joint family's shared decisions enmeshment. A **culturally adapted therapy** modifies an evidence-based treatment's language, metaphors, content, goals and methods to fit the client's culture (**Guillermo Bernal**'s ecological validity framework), and meta-analyses find adapted versions modestly more effective.

N. C. Surya, writing with S. S. Jayaram (1964), argued that Indian patients are more dependent than Western ones and less inclined to integrate inner processes, so the therapist should use the patient's own language and give direct, situational support rather than intrapsychic interpretation. **Jaswant Singh Neki** (1973) proposed the **guru-chela relationship** as an Indian therapeutic paradigm: a wise, warm, guiding therapist and a client who accepts a phase of dependence, normal in Indian relationships, which the guru gradually turns into self-reliance. Families usually accompany patients and join sessions, so therapists work with them as allies (7.1). Yoga-based programmes have been tested at NIMHANS as add-on treatment for depression and schizophrenia, and mindfulness is taught in familiar forms alongside pranayama and meditation.

Training, supervision and the therapist's self-care

Clinical supervision builds a trainee's skill and protects clients. Developmental models (Cal Stoltenberg's) trace trainees from anxious dependence to autonomy; psychotherapy-based models supervise in the language of the therapy taught; **Janine M. Bernard**'s (1979) discrimination model has the supervisor switch between teacher, counsellor and consultant roles across intervention, conceptualisation and personalisation. In parallel process the trainee re-enacts with the supervisor what is happening with the client.

Carl Gustav Jung's image of the **wounded healer**, after the wounded centaur Chiron, holds that a therapist's understood wounds deepen empathy while unexamined ones become countertransference. The risk of **burnout**, which **Christina Maslach** described as emotional exhaustion, depersonalisation and reduced personal accomplishment, makes self-care (supervision, personal therapy, boundaries, manageable caseloads, rest) an ethical duty.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Eysenck (1952) said two-thirds of neurotic patients improve without therapy, through _____.
2. Smith and Glass (1977) found an average effect size of _____.
3. The _____ verdict holds that bona fide therapies produce equivalent outcomes.
4. Frank's *Persuasion and Healing* (1961) says clients arrive _____.
5. Bordin's alliance has three parts: goals, tasks and _____.
6. Schofield's (1964) preferred client is _____.
7. Ideal-conditions research tests _____; routine-practice research tests _____.
8. Jacobson and Truax's _____ index must exceed 1.96.

Answers: (1) spontaneous remission, (2) 0.68, (3) dodo bird, (4) demoralised, (5) bond, (6) YAVIS, (7) efficacy; effectiveness, (8) reliable change.

7.9 Health, Community, Crisis and Rehabilitation Psychology

LOWER YIELD · 15 QS

Clinical psychology does not end at the consulting-room door. Most deaths in India now come from illnesses driven partly by behaviour, and most people with a mental illness never see a specialist. This section follows the psychologist onto the ward, into the community, into crises and into rehabilitation.

Health psychology and the sick role

In 1980 **Joseph D. Matarazzo**, first president of the American Psychological Association's Division of Health Psychology (Division 38, founded 1978), defined **health psychology** as the aggregate of psychology's educational, scientific and professional contributions to promoting and maintaining health, preventing and treating illness, and identifying the causes and correlates of health and illness. It is one discipline's field, leaning towards promotion and prevention. **Behavioural medicine** (**Gary Schwartz** and **Stephen Weiss**, 1978) is an interdisciplinary field that integrates behavioural and biomedical knowledge and techniques for prevention, diagnosis, treatment and rehabilitation. Both apply **George Engel**'s biopsychosocial model (1977; Section 7.1).

What a person does with a symptom, ignoring chest pain for a week or rushing to casualty, is **illness behaviour** (**David Mechanic**, 1962). Once accepted as ill, a person enters the **sick role** of **Talcott Parsons** (*The Social System*, 1951), with two rights (exemption from normal duties; no blame for the illness) and two obligations (to want to get well; to seek competent help and cooperate with it). It fits acute illness better than chronic or stigmatised illness. Seeking the sick role itself is the motive in factitious disorder; malingering seeks external gain (Chapter 8).

> **EXAM TRAP**
>
> Behavioural medicine, not health psychology, is the interdisciplinary field integrating behavioural and biomedical interventions (JEMAS 2025). Options such as "only uses medication" are nonsense. Health psychology is Division 38; clinical psychology, Division 12.

Models of health behaviour and adherence ●●

Models of health behaviour explain why knowing the risks is not enough (Table 7.23). The **health belief model** grew from a 1950s puzzle, the poor uptake of free tuberculosis X-rays, studied by **Godfrey Hochbaum**, **Irwin Rosenstock** and **Stephen Kegeles** in the United States: people acted when they felt susceptible, judged the disease severe, saw benefits outweighing barriers and met a cue to action. Rosenstock gave it its classic statement (1966), and **Marshall Becker** (1974) extended it to sick-role behaviour and adherence. **Martin Fishbein** and **Icek Ajzen**'s theory of reasoned action (1975) made intention the immediate cause of behaviour; Ajzen's **theory of planned behaviour** added perceived behavioural control (Figure 7.16). **Ronald Rogers**'s **protection motivation theory** (1975) shows that fear appeals work only when coping appraisal is also high, and **Ralf Schwarzer**'s health action process approach (1992) adds planning to turn intention into action.

Table 7.23 Models of health behaviour and their constructs.

Model	Proposed by	Key constructs
Health belief model	Hochbaum, Rosenstock, Kegeles (1950s); Rosenstock 1966	Susceptibility, severity, benefits, barriers, cues to action; self-efficacy added 1988
Theory of planned behaviour	Ajzen 1985, 1991	Attitude, subjective norm, perceived behavioural control
Protection motivation theory	Rogers 1975, revised 1983	Threat appraisal; coping appraisal (response efficacy, self-efficacy, costs)
Social cognitive theory	**Albert Bandura** (self-efficacy 1977)	Self-efficacy (confidence that one can act)
Transtheoretical model	Prochaska and DiClemente 1983	Stages of change (Section 7.6)
Health action process approach	Schwarzer 1992	Motivational, then volitional phase; action and coping plans

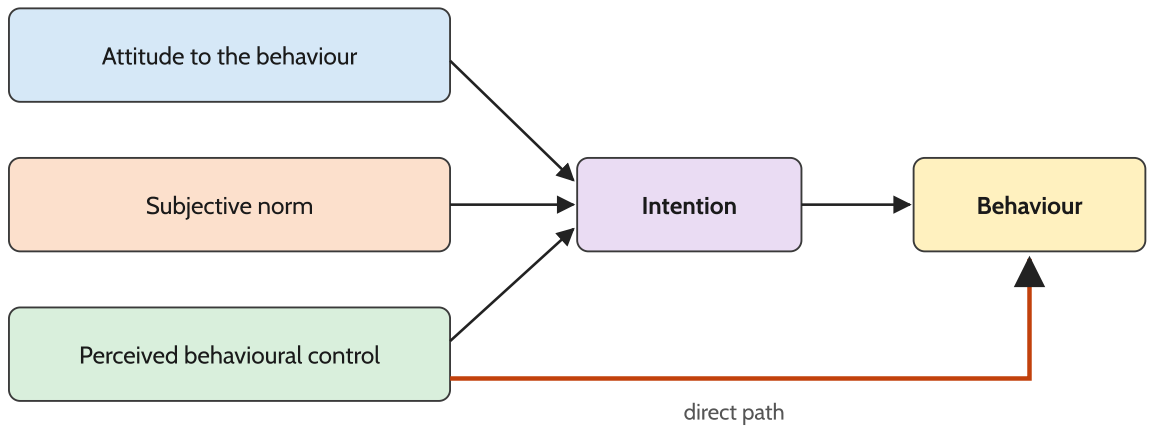

Figure 7.16 Ajzen's theory of planned behaviour; without the lowest box it is the theory of reasoned action.

EXAM TRAP

"Susceptibility, severity, benefits, barriers" is always the health belief model (JEMAS 2025). RML 2023 keyed the theory of planned behaviour as "attitude, subjective norms and self-efficacy": pick it, though strictly the third factor is perceived behavioural control, which Ajzen (1991) likened to Bandura's self-efficacy; the other options are not its constructs.

Medicine most needs **adherence**, following an agreed treatment; the older word compliance implied obedience (in social psychology it means yielding to a direct request, Chapter 3). Only about half of chronically ill patients in developed countries take treatment as prescribed (World Health Organization, 2003). Clear written instructions, simple regimens, reminders, family involvement and a good relationship all help. **Thomas Szasz** and **Marc Hollender** (1956) ranged patient-practitioner relationships from activity-passivity (the unconscious patient) through guidance-cooperation to mutual participation, best for chronic illness.

Stress and physical illness ●●

Stress itself is taught in Section 1.11: **Hans Selye**'s general adaptation syndrome (alarm, resistance, exhaustion), the life-events scale of **Thomas Holmes** and **Richard Rahe**, and psychoneuroimmunology, begun when **Robert Ader** and **Nicholas Cohen** (1975) conditioned immune suppression in rats. Health psychology adds that an acute stressor is brief with a clear end (a pop quiz), a chronic stressor persists (marital discord, HIV disease, colon cancer), and chronic stress is what suppresses immunity.

Franz Alexander (*Psychosomatic Medicine*, 1950) proposed that a specific unconscious conflict acting on a vulnerable organ caused each of seven **psychosomatic disorders**, the "holy seven": peptic ulcer, essential hypertension, bronchial asthma, rheumatoid arthritis, ulcerative colitis, neurodermatitis and thyrotoxicosis. The specificity idea failed (most ulcers proved bacterial, from *Helicobacter pylori*), but psychological factors are now known to shape the course of every illness (Table 7.24).

Table 7.24 Psychology in five physical illnesses.

Illness	Psychological link	Psychological care
Coronary heart disease	Type A (time urgency, competitiveness, hostility; **Meyer Friedman** and **Ray Rosenman** 1959), not relaxed Type B; hostility its toxic part	Anger and stress management
Cancer	Type C (**Lydia Temoshok**): unassertive, suppresses emotion; weak evidence	Distress, pain, conditioned nausea
HIV and AIDS	Spread by behaviour (unsafe sex, shared needles)	Test counselling, drug adherence, stigma
Type 2 diabetes	Poor diet and sedentary lifestyle	Self-management; lifestyle change cut new cases 58 percent (Diabetes Prevention Program, 2002)
Chronic pain (over three months)	Pain behaviours reinforced by sympathy and rest	Operant approach, CBT, pain clinics

In chronic pain, the gate control theory of **Ronald Melzack** and **Patrick Wall** (1965; Chapter 5) explains why attention and mood change pain. **Wilbert Fordyce** (1976) treated pain behaviours (grimacing, resting, asking for pills) as operants: his operant approach rewards well be-

haviour, raises activity by quotas and gives analgesics on a fixed schedule, not on complaint. Cognitive-behavioural therapy adds coping skills, pacing and challenges to catastrophising, usually in multidisciplinary pain clinics.

> EXAM TRAP
>
> Offered both, RML 2024 keyed **Type A** (the whole pattern) for coronary risk; later research found **hostility** its toxic component, so answer hostility only when the stem asks for the component. For Type 2 diabetes the key is diet and inactivity, the strongest modifiable cause, not genes, stress or toxins.

Behavioural medicine and lifestyle change

Biofeedback displays a bodily signal (muscle tension, finger temperature, skin conductance, brain waves) so that a person can learn to control it, as in tension headache or migraine, though it often does no better than relaxation: **Edmund Jacobson**'s progressive muscle relaxation, **Johannes Schultz**'s autogenic training or **Herbert Benson**'s relaxation response (1975). Stress management adds cognitive restructuring and problem solving (Section 7.5).

Chronic insomnia is treated first with **cognitive behavioural therapy for insomnia**: stimulus control (**Richard Bootzin**, 1972: the bed only for sleep and sex; leave it if awake for about 20 minutes), sleep restriction (**Arthur Spielman**, 1987: time in bed cut to time asleep) and cognitive work on sleep worries. Sleep hygiene (no evening caffeine, a dark quiet room, regular hours) is not enough alone.

Smokers are helped by the five As (ask, advise, assess, assist, arrange follow-up) and nicotine replacement. Drinkers are screened with the World Health Organization's **Alcohol Use Disorders Identification Test** (AUDIT, 1989): 10 items scored 0 to 40, with 8 or more suggesting hazardous or harmful use and 20 or more possible dependence. A positive screen earns a brief intervention, a few short sessions in the style of motivational interviewing (Section 7.6). Weight management relies on self-monitoring, stimulus control and a modest 5 to 10 percent loss; exercise lasts when it is enjoyable, shared and planned.

Chronic illness, dying and grief ●

Chronic illness is self-managed for life; depression is common and worsens self-care, so psychologists teach self-management and involve the family. When cure is impossible, **palliative care** seeks comfort and quality of life for patient and family. It began with **Cicely Saunders**'s St Christopher's Hospice, London (1967), which treated "total pain": physical, emotional, social and spiritual.

Elisabeth Kübler-Ross (*On Death and Dying*, 1969) described five stages in dying patients: denial, anger, bargaining, depression and acceptance. They are neither fixed nor universal, came from the dying rather than the bereaved, and have weak research support. **J. William Worden** (1982) proposed four **tasks of mourning**, in their current wording: accept the reality of the loss, process the pain of grief, adjust to a world without the deceased, and find an enduring connection with the deceased while embarking on a new life. Most grief is normal and runs its course with ordinary support; counselling adds little. **Complicated grief**, intense yearning that persists and disables (prolonged grief disorder, Chapter 8), needs intervention, usually one to one.

EXAM TRAP

"Crisis-related bereavement should always be considered traumatic grief" is the false statement (RML 2020): grief is usually normal, and "always" gives it away. Worden's are tasks, not stages.

Community mental health and prevention

President **John F. Kennedy**'s Community Mental Health Centers Act (1963) funded local centres to replace state hospitals. With antipsychotic drugs it sped **deinstitutionalisation**, but too few centres were built, and many discharged patients became homeless, imprisoned or caught in a "revolving door" of readmissions. Community psychology (Swampscott conference, 1965) turned to prevention and social change.

Gerald Caplan (*Principles of Preventive Psychiatry*, 1964) brought public health's three levels to mental health. **Primary prevention** lowers new cases (incidence) among the healthy: vaccination, health education. **Secondary prevention** lowers prevalence by early detection and prompt treatment. **Tertiary prevention** limits the disability of an established disorder: rehabilitation. **Robert Gordon** (1983) classified prevention by its target instead: universal (everyone), selective (high-risk groups, such as children of depressed parents) and indicated (individuals with early signs). Caplan (1970) also named four types of **mental health consultation**, helping another professional rather than the client: client-centred case, consultee-centred case, programme-centred administrative and consultee-centred administrative. **Julian Rappaport** (1981) preferred **empowerment**, people gaining control over their own lives.

Where specialists are scarce, **task-shifting** trains lay counsellors to deliver structured care under supervision. In Zimbabwe, **Dixon Chibanda**'s Friendship Bench has lay health workers, many of them grandmothers, offer problem-solving therapy on clinic benches. In Goa, **Vikram Patel**'s MANAS trial (2010) showed that lay counsellors in public primary-care clinics improved recovery from common mental disorders, and his group's PRIDE programme took lay-counsellor problem solving to adolescents in Indian schools. Stigma (**Erving Goffman**, 1963) deters help-seeking; contact with people who have recovered reduces it best. These attack the treatment gap, as do India's National Mental Health Programme (1982) and District Mental Health Programme (1996), modelled on a NIMHANS pilot in Bellary, Karnataka (Section 7.1).

EXAM TRAP

Promoting vaccination is primary prevention (JEMAS 2025). Treating a diagnosed illness, surgery for chronic disease and rehabilitation after injury come later (secondary or tertiary). Caplan's levels say *when* you intervene; Gordon's types say *whom* you target.

Crisis intervention and suicide prevention ●●

KEY STUDY: LINDEMANN AND THE COCOANUT GROVE FIRE (1944)

On 28 November 1942 a fire at Boston's Cocoanut Grove nightclub killed 492 people. **Erich Lindemann** interviewed 101 bereaved people and in 1944 described acute grief as a syndrome (bodily distress, preoccupation with the dead, guilt, hostility) that most work through within weeks by "grief work", while some develop delayed or distorted grief. Timely brief help, he concluded, could prevent lasting disorder: the seed of crisis intervention. Criticism: impressionistic data and too short a time course for grief.

Caplan defined a **crisis** as an obstacle to important life goals that is, for a time, insurmountable by customary problem solving, followed by disorganisation; it usually resolves, for better or worse, within four to six weeks. Crises are developmental (retirement) or situational. Situational events are large-scale (floods, earthquakes, tornadoes) or small-scale, striking one person or family: violence (child abuse, domestic abuse, crime), psychopathology that triggers a crisis (a suicide attempt, an overdose) and personal catastrophe (a car accident, a life-threatening illness).

Crisis intervention is brief, active, directive and focused on the here and now, aiming to restore previous functioning. Its stance is supportive therapy (listening, reassurance, problem solving, mobilising support), not exploratory work, and it is often named the treatment of choice for adjustment disorder (Chapter 8), a short reaction to an identifiable stressor. **Albert Roberts**'s seven stages are: assess lethality, establish rapport, identify the major problems, deal with feelings, explore alternatives, make an action plan, and follow up.

Suicide risk is assessed by asking directly about thoughts, plans and means (Chapter 2). The **safety plan** of **Barbara Stanley** and **Gregory Brown** (2012) is a written list of warning signs, coping strategies, distracting people and places, people to ask for help, professionals and crisis lines, and ways to make the environment safe. Means restriction (pesticide bans cut suicides sharply in Sri Lanka), gatekeeper training for teachers and police, and helplines such as Tele-MANAS (14416) work at population level.

After disasters, **psychological first aid** (World Health Organization, 2011) offers practical support: look (for safety and needs), listen (without forcing talk), link (to services and loved ones). It replaced **critical incident stress debriefing** (**Jeffrey Mitchell**, 1983), a single group session retelling the event, which does not prevent post-traumatic stress disorder and may harm (Cochrane review), so it is not recommended. **Trauma-informed care** (US Substance Abuse and Mental Health Services Administration, 2014) asks services to realise, recognise and respond to trauma and to resist re-traumatising people.

EXAM TRAP

Crisis management needs **supportive therapy** (IHBAS 2020), not acceptance and commitment, expressive or cognitive behaviour therapy. A tornado or flood is a **large-scale** event; personal events are small-scale.

Rehabilitation ●●

Rehabilitation aims at function and a valued life, not cure. Psychosocial rehabilitation of severe mental illness offers a ladder of services (Table 7.25); its **vocational rehabilitation** rungs, sheltered workshops and supported employment, aim at paid work.

Table 7.25 Services of psychosocial rehabilitation.

Service	What it offers
Day care	Daytime treatment and activity; the person sleeps at home
Halfway house	Supervised transitional home between hospital and independent living
Sheltered workshop	Paid work in a protected setting
Supported employment	A real job first, then support on the job (place, then train)
Assertive community treatment	Mobile team, about 10 clients per worker, round the clock
Social skills training	Modelling, role play, feedback and homework
Family psychoeducation	Teaches families about the illness, lowers expressed emotion and relapse
Clubhouse model	Member-run community with a work-ordered day (Fountain House, New York, 1948)

Leonard Stein and **Mary Ann Test** (1980, Madison, Wisconsin) randomly assigned patients to community treatment instead of hospital; they spent less time in hospital and worked more. This became **assertive community treatment**. **Case management** links a person with long-term, complex needs to services and coordinates them; it seeks skill training, environmental accommodations and empowerment, never cure. Its models are often contrasted as provider-driven, in which the agency sets the client's tasks and the worker controls access to services, and client-driven, in which the worker allies with the client in advocacy and empowerment.

After brain injury, neuropsychological rehabilitation takes two routes. The **restorative approach** retrains the damaged function by graded practice (cognitive retraining), as in the attention process training of **McKay Moore Sohlberg** and **Catherine Mateer** (1987), which drills sustained, selective, alternating and divided attention. The **compensatory approach** works around the deficit with memory aids (diaries, alarms, phone reminders). **Errorless learning** (**Alan Baddeley** and **Barbara Wilson**, 1994) prevents mistakes during learning, because amnesic patients cannot remember, and so cannot correct, their errors.

For substance users, detoxification (medically managed withdrawal) is only the start. **G. Alan Marlatt** and **Judith Gordon**'s **relapse prevention** (1985) sees relapse as failed coping in a high-risk situation (negative emotions, interpersonal conflict, social pressure). Coping raises self-efficacy; without it, self-efficacy falls, a lapse (first use) follows, and the **abstinence violation effect**, the guilt and "I have blown it" thinking after one slip, turns the lapse into relapse. Clients rehearse coping and treat lapses as lessons. **Therapeutic communities**, from Synanon (California, 1958), use the community itself (peer confrontation, graded responsibility, mutual help) as the method of change.

The Rehabilitation Council of India, a statutory body under the Ministry of Social Justice and Empowerment, regulates the training and registration of rehabilitation professionals, including clinical psychologists; disability certification under the Rights of Persons with Disabilities Act 2016 is covered in Section 7.1.

EXAM TRAP

Case management aims at function, skills, environmental accommodation and empowerment, **not cure** (RML 2020). The same paper's "task or role performance" model (structure-function theory at the organisational level; fiduciary and humanitarian obligations; authority to provide, deny or withdraw services as the client complies with tasks) is, whatever its label, provider-driven, so the exception is the **power alliance** with the client, a client-driven idea. The Rehabilitation Council of India is under Social Justice and Empowerment, not Health, Education or Labour.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The interdisciplinary field defined by Schwartz and Weiss (1978) is behavioural _____.
2. Ajzen turned the theory of reasoned action into the theory of planned behaviour by adding perceived behavioural _____.
3. Alexander's holy seven include peptic ulcer, essential hypertension and bronchial _____.
4. Caplan's _____ prevention lowers the number of new cases.
5. Psychological first aid rests on look, listen and _____.
6. Guilt and "I have blown it" thinking after one slip is the abstinence _____ effect.

Answers: (1) medicine, (2) control, (3) asthma, (4) primary, (5) link, (6) violation.

7.10 Working with Children, Older Adults and Special Settings

LOWER YIELD · 16 QS

Most therapies in this chapter assume an adult who talks and who chose to come. Many clients do not fit: a six-year-old who will not discuss his bed-wetting but will bury a toy tiger in a sand tray; a grandmother who asks each evening when her dead husband is coming home; a youth whose lawyer asks whether he was "of unsound mind". This section adapts therapy to them.

Children as clients, and play therapy •

Children differ from adult clients. Their developmental level limits what therapy can ask, so clinicians work through play and parents; adults make the referral; and school is a second home. Assessment is therefore **multi-informant assessment** (parents, teachers, the child, observation in clinic, home and class), not because parents over-report or teachers lack information but because *symptoms vary across contexts*: a child with attention-deficit hyperactivity disorder (ADHD) may be calm in a quiet clinic and unmanageable in a class of fifty. Clinic observation alone falls short for the same reason.

Play therapy uses the child's natural language. **Melanie Klein** (1882 to 1960) treated play as the equivalent of free association and interpreted it early and deeply; **Anna Freud** (1895 to 1982) began with a preparatory phase to win trust, involved parents and gave the analyst an educative role. **Virginia Axline** (1911 to 1988) carried Carl Rogers' ideas into the playroom

(*Play Therapy*, 1947; *Dibs: In Search of Self*, 1964). Her **non-directive play therapy** rests on eight principles: build warm rapport; accept the child exactly as he is; create permissiveness so that feelings flow freely; recognise feelings and reflect them back; respect the child's ability to solve his own problems; let the child lead; do not hurry; and set only the limits that anchor therapy to reality. Table 7.26 lists later forms, including **filial therapy** and **sandplay**.

Table 7.26 Later forms of play therapy.

Approach	Founder	Core idea
Relationship play therapy	**Clark Moustakas** (1950s)	The living relationship itself heals
Child-centred play therapy	**Garry Landreth**	The modern form of Axline's model
Filial therapy	**Bernard Guerney** and **Louise Guerney** (1964)	Parents are trained to hold the play sessions at home
Sandplay	**Dora Kalff** (1950s), from **Margaret Lowenfeld**'s World Technique (late 1920s)	The child builds a scene in a sand tray with miniatures; Jungian

EXAM TRAP

Axline is Rogerian, not Freudian. Klein, not Anna Freud, equated play with free association. Sandplay is Kalff's; the World Technique is Lowenfeld's. In filial therapy parents run the sessions.

Parent training and behaviour change at home and school ●●

Why does defiance worsen in loving homes? **Gerald Patterson** (1926 to 2016) answered with **coercion theory** (*Coercive Family Process*, 1982). A mother orders the television off, her son screams, and she gives in. His screaming is negatively reinforced (the demand goes away), and so is her giving in (the screaming stops; Figure 7.17). Each trains the other, and the exchanges escalate.

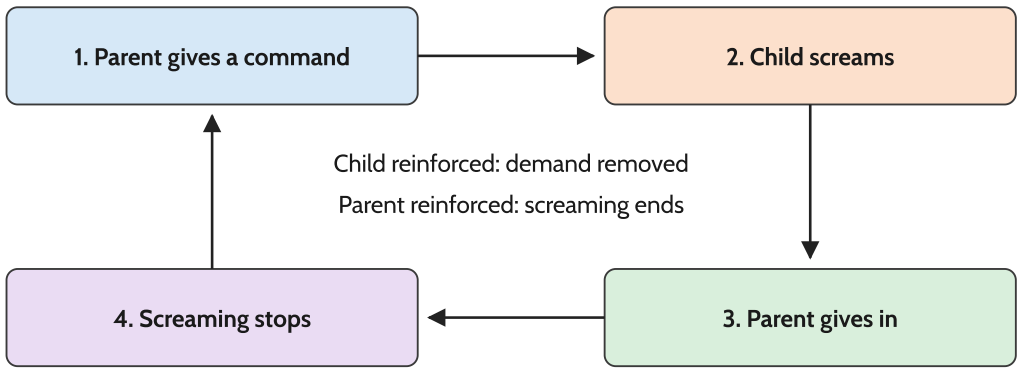

Figure 7.17 Patterson's coercion cycle. Both sides are negatively reinforced, so each round makes the next tantrum likelier.

The treatment, **parent management training** (PMT), reverses the trap. Parents give clear commands, attend to and praise good behaviour and reward it on a token chart, ignore mi-

nor misbehaviour (**extinction**; expect a brief extinction burst), and use time-out and response cost (Section 7.4). Named programmes include **Russell Barkley**'s *Defiant Children* (1987); **Carolyn Webster-Stratton**'s Incredible Years, which teaches parent groups with videotaped scenes; and **Sheila Eyberg**'s **parent-child interaction therapy** (PCIT, ages two to seven), which coaches the parent live through an earpiece, first in child-directed play, then in discipline.

At school, teachers add token economies (Section 7.4) and the daily report card: the teacher rates a few target behaviours daily and the child earns rewards at home. For repetitive aggression, often in autism or intellectual disability, **Richard Foxx** and **Nathan Azrin** devised **overcorrection**: restoring the damaged scene to better than before (restitution) and rehearsing the correct act (positive practice). For anxious children of 7 to 13, **Philip Kendall**'s 16-session Coping Cat (1990) teaches the FEAR plan (Feeling frightened? Expecting bad things? Attitudes and actions that help; Results and rewards), then graded exposure. Family work (Section 7.7) runs through all of these.

> EXAM TRAP
>
> Deliberately ignoring tantrums is **extinction**; some keys wrongly mark "operant escape". Escape (ending an aversive stimulus already present) and avoidance (preventing it) are negative reinforcement, which *strengthens* behaviour. Nor is negative reinforcement a PMT technique; PMT breaks that trap.

Elimination problems, school refusal and ADHD

Enuresis (wetting from age five) is treated best with the **bell-and-pad** of **O. Hobart Mowrer** and **Willie Mae Mowrer** (1938): urine completes a circuit and rings a bell until bladder tension alone wakes the child, and overlearning cuts relapse. Encopresis (soiling from age four) is usually retentive, with constipation and overflow; laxatives are combined with scheduled toilet sitting and rewards.

In **school refusal** the child refuses out of anxiety, with the parents' knowledge; in truancy the child hides the absence, shows no anxiety and often has conduct problems. **Christopher Kearney** and **Wendy Silverman** classified refusal by its function: avoiding distress caused by school, escaping social or evaluative situations, seeking the parents' attention, or gaining tangible rewards outside school. Treatment centres on an early, graded return to school.

> KEY STUDY: THE MTA STUDY (1999)
>
> The US National Institute of Mental Health randomly assigned 579 children with ADHD, aged 7 to 9, to 14 months of managed medication, intensive behavioural treatment, both, or community care. Medication and the combination beat the others on core symptoms; the combination added small social and family gains at lower doses. *Criticism*: the advantage faded after the trial. Hence the rule: behavioural treatment plus medication.

Autism, intellectual disability and learning disability ●●●

Autism was first described in 1943 by **Leo Kanner** (1894 to 1981) of Johns Hopkins, in eleven children with extreme aloneness and an anxious insistence on sameness (the disorder is in Chapter 8).

Treatment is early and behavioural, based on **applied behaviour analysis** (ABA), which applies operant principles to socially important behaviour. **Ole Ivar Lovaas** (1927 to 2010), at the University of California, Los Angeles, pioneered early intensive behavioural intervention using **discrete trial training**: a skill is broken into small steps, each taught in repeated trials (instruction, response, prompt, reinforcement), with prompts faded. In functional communication training, a child whose tantrums serve to escape demands learns an acceptable substitute, such as handing over a picture card that says "break". Table 7.27 lists other programmes.

KEY STUDY: LOVAAS (1987)

Nineteen autistic children under four received about 40 hours a week of one-to-one behavioural treatment for two years or more. Nine (47 percent) reached normal intellectual and educational functioning, against 2 percent of controls; a 1993 follow-up found the gains kept. *Criticism:* no random assignment, a narrow outcome measure, and early use of aversives.

Table 7.27 Other named programmes for autism.

Programme	Developer	Core idea
TEACCH	**Eric Schopler** (North Carolina, 1972)	Structured teaching with visual schedules
PECS (picture exchange)	**Andy Bondy** and **Lori Frost** (1985)	The child hands over a picture to get a wanted item
Social stories	**Carol Gray** (early 1990s)	Short stories explaining social situations
Early Start Denver Model	**Sally Rogers** and **Geraldine Dawson**	ABA within play and relationships, from 12 months

Intellectual disability is treated with task analysis, chaining, prompting and reinforcement to build adaptive skills, alongside special or inclusive education. India's **National Trust Act** 1999 covers exactly four conditions (autism, cerebral palsy, mental retardation, multiple disabilities) and provides legal guardians. The Rights of Persons with Disabilities Act 2016 (Section 7.1) lists 21 disabilities, including intellectual disability, specific learning disabilities and autism, but not ADHD. Autism is certified with the 40-item **Indian Scale for Assessment of Autism** (ISAA), developed in 2009 by the National Institute for the Mentally Handicapped (now NIEPID): each item is rated 1 to 5 across six domains, a total of 70 or more indicates autism, and the total grades severity (above 153 is severe) and the disability percentage. Specific learning disability is treated by remedial teaching with multisensory methods, such as Orton-Gillingham phonics and **Grace Fernald**'s tracing method (see, say, hear and trace each word), plus accommodations: extra time, a scribe, exemption from a third language.

EXAM TRAP

Kanner (1943) first described infantile autism; **Hans Asperger** (1944) described more able children, and **Eugen Bleuler** coined "autism" in 1911 for withdrawal in schizophrenia. The Indian autism tool is the **ISAA**; some keys mark the Childhood Autism Rating Scale (CARS), which is American (Schopler). The CBCL is **Thomas Achenbach**'s Child Behavior Checklist; the CGAS rates global functioning.

Child abuse, POCSO and the adolescent client

Child abuse (physical, sexual, emotional, neglect) entered medicine with **C. Henry Kempe**'s "battered child syndrome" (1962). Safety and reporting come first. The best-supported therapy is **trauma-focused cognitive behavioural therapy** (TF-CBT) of **Judith Cohen**, **Anthony Mannarino** and **Esther Deblinger**, for ages 3 to 18, with a non-offending parent. Its PRACTICE components (psychoeducation, relaxation, affect modulation, cognitive coping, trauma narrative, in vivo mastery, conjoint sessions, safety) build coping skills first, but the central goal is the trauma narrative: retelling the trauma desensitises the memories and corrects beliefs such as "it was my fault". (Emotion regulation and mindfulness skills define dialectical behaviour therapy, Section 7.5.)

> **INDIAN CONTEXT**
>
> The **POCSO Act** 2012 (Protection of Children from Sexual Offences) covers everyone under 18 and is gender-neutral. Reporting is mandatory (section 19): anyone, psychologists included, who knows of an offence or fears one is likely must inform the Special Juvenile Police Unit or the local police, and failing to do so is punishable under section 21 (up to six months' imprisonment, a fine or both). It also sets up Special Courts and child-friendly procedures.

Adolescence, Erik Erikson's stage of identity versus role confusion (Chapter 6), is the peak age for risk behaviour such as substance use, unsafe sex and self-harm. Anna Freud and **Peter Blos** held that healthy teenagers must detach from parents; research now shows that continuing interdependence is healthy, so parents remain allies. Engagement comes first: be collaborative and see the teenager alone for part of each session. The limits of **confidentiality with minors** are explained at the start: privacy holds except for risk to life, abuse or legal duties such as POCSO, and the teenager is told before anything is shared; guardians give consent. Schools offer counsellors and WHO life skills education. Adolescent depression is treated with CBT or interpersonal psychotherapy, adding fluoxetine with monitoring for suicidal thinking; self-harm calls for a safety plan, restricted means, family involvement and adolescent dialectical behaviour therapy.

Older adults: dementia care, depression and capacity

The field of **geropsychology** applies clinical psychology to older adults. **Robert Butler** (1927 to 2010) coined "ageism" (1969) and described reminiscence as a healthy life review (1963). In India families, mostly women, give most dementia care; their **caregiver burden**, objective (time, money) and subjective (distress, guilt), is measured with the Zarit Burden Interview and eased by psychoeducation, behaviour-management training, support groups and respite. Methods for the patient, from **reality orientation** to **validation therapy** and **cognitive stimulation therapy**, are in Table 7.28.

Table 7.28 Psychological methods in dementia care.

Method	Originator	What happens
Reality orientation	**James Folsom** (late 1950s)	Repeated orientation to time, place and person; can distress

Method	Originator	What happens
Reminiscence therapy	Butler's tradition	Sharing memories prompted by photographs and music
Life review	Robert Butler (1963)	A structured, evaluative review of the whole life
Validation therapy	**Naomi Feil** (developed 1963 to 1980)	Accept the person's reality and respond to feelings; never correct
Cognitive stimulation therapy	**Aimee Spector** and **Martin Orrell** (2003)	14 themed group sessions over 7 weeks; mild to moderate dementia

> EXAM TRAP
>
> To the grandmother awaiting her dead husband, reality orientation *corrects* ("He died four years ago"); validation *accepts* and explores the feeling ("You miss him"). Life review, unlike plain reminiscence, is evaluative and covers the whole life.

Late-life depression often shows as bodily, sleep and memory complaints rather than sadness. Depressive pseudodementia mimics dementia, but it begins abruptly, the person complains of poor memory and answers "I don't know", and it lifts with treatment. The 30-item Geriatric Depression Scale (**Jerome Yesavage**) omits bodily symptoms. CBT, problem-solving therapy, antidepressants and electroconvulsive therapy all help.

A person's **decision-making capacity** is presumed, and judged for one decision at one time. **Paul Appelbaum** and **Thomas Grisso** (1988) named four abilities: to understand the information, appreciate how it applies to oneself, reason about the options and communicate a choice. India's Mental Healthcare Act 2017 uses a similar test.

Counselling and career guidance

Guidance helps people make educational and vocational choices through information, tests and advice. Counselling is a face-to-face relationship that helps a basically healthy person resolve a problem of living and reach his own decision. Psychotherapy treats more disturbed clients, longer and deeper (Section 7.6). Counselling is called the heart of the guidance programme, and its instrument, the interview, was called the heart of the counselling process by **Ruth Strang**. Other tools are the cumulative record (a school-long file of each pupil's background, health, tests and achievement), the case study (an in-depth picture of one pupil drawn from all sources) and tests.

> EXAM TRAP
>
> One key names the aptitude test "the heart of the guidance programme". The sources are not uniform: Indian guidance texts give that title to counselling, Strang gave "heart of the counselling process" to the interview, and a few lesser sources use similar words for the cumulative record, but no standard text found gives it to the aptitude test, which only predicts trainability. If counselling is not offered, choose the interview, and expect some keys to differ.

Frank Parsons (1854 to 1908), father of vocational guidance, set out the **trait-factor approach** in *Choosing a Vocation* (1909): know yourself, know the world of work, and match

them by "true reasoning". The counselling process runs from relationship building through assessment, goals and intervention to termination (microskills are in Section 7.2). Areas include school, career, marital, rehabilitation, de-addiction, grief and HIV counselling (before and after testing, under the five Cs: consent, confidentiality, counselling, correct results, connection to care). Affirmative therapy for LGBTQ+ clients affirms their identities, addresses minority stress and rejects conversion therapy, which India's National Medical Commission called professional misconduct in 2022.

John Holland (1919 to 2008) sorted people and work environments into six **RIASEC** types (Figure 7.18); satisfaction follows from congruence between person and environment. **Donald Super** (1910 to 1994), in his **life-span, life-space theory**, saw a career as lifelong self-concept implementation through five stages (growth to 14, exploration to 24, establishment to 44, maintenance to 64, then decline) and many life roles, drawn as the Life-Career Rainbow; readiness for choice is career maturity.

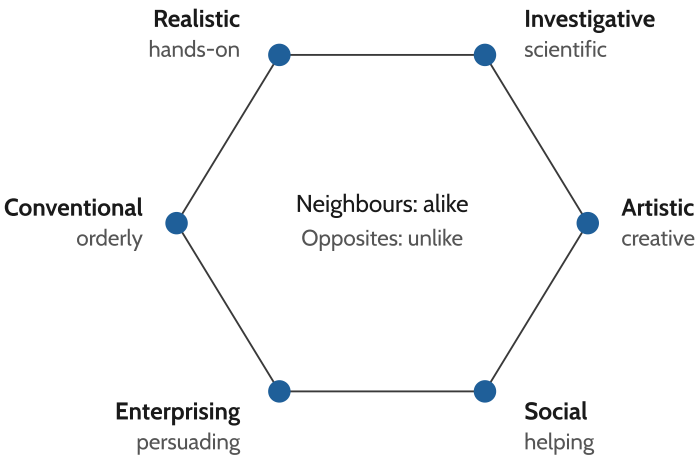

Figure 7.18 Holland's RIASEC hexagon. Neighbouring types are most alike; opposite types (Realistic and Social) are least alike.

Forensic clinical psychology

Forensic psychology answers legal questions for a client that is often the court, so confidentiality is limited. An accused must have **competence to stand trial**: the present ability to understand the proceedings and help his lawyer (*Dusky v. United States*, 1960). Criminal responsibility concerns his mind at the offence: the insanity defence (Table 7.29). It began in 1843, when **Daniel M'Naghten**, deluded that he was persecuted, shot Prime Minister Robert Peel's secretary; his acquittal produced the **M'Naghten rules**, which section 84 of the Indian Penal Code (now section 22 of the Bharatiya Nyaya Sanhita 2023) follows. This legal insanity is narrower than medical insanity, and the accused must prove it.

Table 7.29 Tests of insanity.

Test	Origin	Not responsible if he	Kind
M'Naghten rules	England, 1843	did not know the nature of the act, or that it was wrong	Cognitive
Irresistible impulse	US, 1880s	knew it was wrong but could not control himself	Volitional

Test	Origin	Not responsible if he	Kind
Durham rule	US, 1954 (Judge **David Bazelon**)	acted as a product of mental disease or defect (dropped in 1972)	Product
ALI test	American Law Institute, 1962	lacked substantial capacity to appreciate wrongfulness or to conform to the law	Both
IPC 84 / BNS 22	India, 1860 / 2023	could not know the nature of the act, or that it was wrong or contrary to law	Cognitive

EXAM TRAP

M'Naghten: did he *know*? Irresistible impulse: could he *control*? ALI: both, softened to "substantial capacity". Durham: was the act a *product* of illness? India's section 84 is M'Naghten-based. Competence concerns the mind at trial; insanity, the mind at the offence.

Risk assessment predicts violence or reoffending; unaided clinical prediction was right in at most one case in three (**John Monahan**, 1981), so structured tools such as the HCR-20 and **Robert Hare**'s Psychopathy Checklist-Revised are used. Custody evaluation follows the best interests of the child (in India, the child's welfare, under the Guardians and Wards Act 1890). An expert witness, unlike others, may give opinions (Section 45, Indian Evidence Act 1872; now Section 39, Bharatiya Sakshya Adhiniyam 2023).

The Juvenile Justice (Care and Protection of Children) Act 2015 sends a *child in conflict with law* to a Juvenile Justice Board and a *child in need of care and protection* to a Child Welfare Committee. A 16- to 18-year-old accused of a heinous offence (minimum seven years' imprisonment) may, after the Board's preliminary assessment of his mental and physical capacity to commit the offence and his grasp of its consequences, with psychologists' help, be tried as an adult. In correctional psychology, effective offender programmes follow the **risk-need-responsivity model** of **Donald Andrews** and **James Bonta** (intensity matched to risk, criminogenic needs targeted, cognitive-behavioural methods) and are community-based. The cognitive interview of **Ronald Fisher** and **Edward Geiselman**, which aids eyewitness recall, and offender profiling, which infers an unknown offender's traits from the crime, are investigative tools, not treatments.

Neuropsychology, creative therapies and telepsychology

Clinical neuropsychology, which assesses the effects of brain damage on behaviour, is covered in Chapters 2 and 5; rehabilitation psychology is in Section 7.9.

Art therapy has two founders: **Margaret Naumburg** (1890 to 1983) treated art as symbolic speech from the unconscious (art psychotherapy), and **Edith Kramer** (1916 to 2014) held that the creative process heals through sublimation (art as therapy). Music therapy uses listening and making music; dance and movement therapy (founded by **Marian Chace**, 1896 to 1970) uses movement and mirroring; drama therapy uses role play (psychodrama is in Section 7.7); bibliotherapy is guided reading. Modern **animal-assisted therapy** began in the 1960s with **Boris Levinson** (1907 to 1984), whose dog Jingles drew a withdrawn child into talk. In India yoga and meditation are common adjuncts, and NIMHANS trials support yoga in schizophrenia and depression.

Telepsychology delivers care by phone, video, messaging or apps. Guided internet CBT rivals face-to-face CBT for depression and anxiety, while apps vary widely in quality. India's **Tele-MANAS**, launched on 10 October 2022 with NIMHANS as nodal centre, gives free round-the-clock counselling on 14416. Its ethical issues are consent covering the technology's limits, data security, verifying identity and location, emergencies at a distance, jurisdiction and the digital divide.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. In Axline's _____ play therapy, the child leads.
2. Patterson's _____ theory explains how tantrums and giving in reinforce each other.
3. Deliberately ignoring tantrums is _____.
4. Infantile autism was first described in 1943 by _____.
5. Responding to a disoriented person's feelings without correcting her is _____ therapy.
6. Section 84 of the Indian Penal Code is now section _____ of the Bharatiya Nyaya Sanhita.

Answers: (1) non-directive, (2) coercion, (3) extinction, (4) Leo Kanner, (5) validation, (6) 22.

The Unchaining at Bicêtre, Revisited

Look back at Bicêtre with what this chapter has taught. Pinel, recording his cases and counting recoveries, was a scientist-practitioner a century and a half before Boulder, and his case notes were ancestors of the interview, mental status examination and formulation of section 7.2. Pussin's conversations and routines anticipated what outcome research found in section 7.8: much of what heals in psychodynamic, behavioural, cognitive and humanistic therapy alike is the relationship and the other common factors. Moral treatment collapsed when asylums grew too crowded to give each person attention, the lesson behind community care and rehabilitation in section 7.9. And the chains? India's Mental Healthcare Act 2017 still had to ban chaining in section 95, a reminder that ethics and law protect patients only when clinicians practise them.

Chapter summary

7.1 The Profession: History, Models, Ethics and Indian Law. Clinical psychology joins science and practice to understand, predict and relieve distress; in India it rests on an RCI-recognised qualification and RCI registration. Care moved from supernatural ideas and the humours through the asylums to Pinel and Pussin's moral treatment, Dix and Beers. Witmer opened the first clinic (1896); Boulder (1949) set the scientist-practitioner model and Vail (1973) the PsyD. In India the IACP dates from 1968 and the RCI Act from 1992. Engel's biopsychosocial and the diathesis-stress models integrate the older models. Ethics rests on five APA principles, consent, confidentiality

and its limits (Tarasoff) and boundaries; the Mental Healthcare Act 2017 adds rights, advance directives and section 115.

7.2 The Clinical Interview, Mental Status Examination and Formulation. The clinical interview may be intake, diagnostic, crisis or termination, and unstructured, semi-structured (SCID, PSE and SCAN, K-SADS, ADIS) or fully structured (DIS): structure buys reliability, not validity. Microskills include SOLER attending, open questions, furthering, paraphrasing and reflection of feeling. The case history runs from identifying data to premorbid personality. The MSE records appearance, speech, mood (subjective) and affect (objective), the stream, form, content and possession of thought, perception, cognition and insight (six grades). SAD PERSONS aids memory but cannot predict suicide. Functional analysis (ABC, SORC) is operant. A formulation (5 Ps, biopsychosocial) explains, a diagnosis classifies; kappa measures diagnostic reliability, and Rosenhan (1973) exposed labelling.

7.3 Psychodynamic Therapies and Their Descendants. Psychodynamic therapies hold that unconscious conflict drives symptoms. Freud's psychoanalysis grew from Breuer's cathartic method with Anna O.; it relies on the fundamental rule and free association, dream analysis, interpretation, analysis of resistance and of transference and countertransference, and working through, under neutrality and abstinence. Ego psychology, object relations (Klein, Winnicott, Mahler, Kernberg), Kohut's self psychology and mentalisation-based treatment changed what analysts listen for. Brief dynamic therapies (Malan, Sifneos, Davanloo, Mann, Luborsky, Strupp and Binder) and Klerman and Weissman's interpersonal psychotherapy shortened treatment. Jung, Adler, Horney, Sullivan and Berne's transactional analysis offered rival techniques, alongside supportive therapy, hypnosis and narcoanalysis.

7.4 Behaviour Therapy. Behaviour therapy applies learning principles to problem behaviour. Spurred by Eysenck's 1952 critique, it grew from Wolpe's Johannesburg, the Maudsley and Skinner's operant work, and every case starts with a functional analysis. Classical techniques include Jones's counterconditioning, Wolpe's systematic desensitisation (reciprocal inhibition, a SUDS-rated hierarchy), flooding, implosion, exposure and response prevention for OCD, aversion therapy, covert sensitisation, the bell and pad and the debated EMDR. Operant techniques build behaviour (shaping, chaining, the Premack principle, Ayllon and Azrin's token economy) or reduce it (extinction, time-out, response cost, overcorrection), as in ABA for autism. Modelling, assertiveness, relaxation, biofeedback, habit reversal, behavioural activation, insomnia methods and parent training complete the toolkit.

7.5 Cognitive and Cognitive-Behavioural Therapies, and the Third Wave. Cognitive-behavioural therapies hold that emotional disturbance follows mainly from how people interpret events. Albert Ellis's REBT (1955; renamed 1993) uses the ABC model, disputes irrational musts and aims at unconditional self-acceptance. Aaron Beck's cognitive therapy targets the cognitive triad, core and intermediate beliefs, automatic thoughts and distortions through collaborative empiricism, Socratic questioning and the thought record. Meichenbaum's stress inoculation, problem-solving and multimodal therapy, Kelly's fixed-role therapy, Young's schema therapy and disorder-specific models for panic, trauma and psychosis followed. The third wave (DBT, ACT, MBSR, MBCT, CFT, MCT, FAP, behavioural activation) changes the relationship to thoughts. CBT is the most researched psychotherapy, now also delivered online.

7.6 Humanistic, Existential, Experiential and Brief Therapies. Carl Rogers's person-centred therapy holds that conditions of worth create incongruence between self-concept and experience, and

that six necessary and sufficient conditions, above all **empathy, unconditional positive regard and congruence**, are enough for change. **Fritz Perls**'s Gestalt therapy works on here-and-now awareness, unfinished business and the **empty chair**. Existential therapists (Binswanger, Boss, May, Yalom) confront death, freedom, isolation and meaninglessness, and **Frankl**'s logotherapy rests on the will to meaning. Reality, focusing, emotion-focused, positive, narrative and solution-focused therapies follow. **Motivational interviewing** (Miller and Rollnick) resolves ambivalence alongside Prochaska and DiClemente's stages of change, and Ivey, Carkhuff and Egan model the helping skills.

7.7 Family, Couples and Group Therapies. Family therapy grew from **general systems theory**: families keep homeostasis through feedback, causality is circular, and the identified patient signals a troubled system. Theories that families cause schizophrenia (the schizophrenogenic mother, Bateson's **double bind**, Lidz's schism and skew, Wynne's pseudomutuality) failed, but high **expressed emotion** predicts relapse, which psychoeducation reduces. Bowen stressed differentiation and triangles, Minuchin boundaries, Haley directives and paradox, the MRI second-order change, the Milan team circular questioning. Couples work runs from behavioural therapy to Gottman's four horsemen; Masters and Johnson founded sex therapy. Groups heal through Yalom's eleven factors; Moreno created psychodrama and Maxwell Jones the therapeutic community.

7.8 Does Psychotherapy Work? Evidence, Common Factors and the Alliance. Eysenck (1952) claimed that about two-thirds of neurotic patients improve without therapy (**spontaneous remission**); Smith and Glass's meta-analysis (1977, effect size 0.68; 0.85 in 1980) answered him. Rival therapies tie (the **dodo bird verdict**: Rosenzweig 1936; Luborsky and colleagues 1975), which points to common factors: Frank's *Persuasion and Healing* (1961), Wampold's contextual model, Lambert's pie and Bordin's alliance of goals, tasks and bond. Outcome is tested in efficacy and effectiveness studies, RCTs with waiting-list, placebo-attention and usual-care controls, EST criteria and evidence-based practice (APA 2006), and judged by the dose-effect curve and the reliable change index. Integration, culturally adapted therapy (Neki's guru-chela relationship), supervision and self-care complete the picture.

7.9 Health, Community, Crisis and Rehabilitation Psychology. Health psychology (Matarazzo, 1980) is psychology's contribution to health; **behavioural medicine** (Schwartz and Weiss, 1978) is interdisciplinary. Parsons's **sick role** carries two rights and two duties. The health belief model, the theory of planned behaviour and protection motivation theory explain health behaviour and adherence. Alexander's holy seven, Type A hostility, Fordyce's operant pain treatment, CBT for insomnia and the AUDIT link mind and body; Kübler-Ross and Worden describe dying and grief. Caplan supplied prevention levels, consultation and crisis theory: crisis work is brief and supportive, and debriefing is not recommended. Rehabilitation uses assertive community treatment, case management, errorless learning and relapse prevention.

7.10 Working with Children, Older Adults and Special Settings. Children are assessed through several informants because **symptoms vary across contexts**, and treated with play therapy (Klein, Anna Freud, Axline's non-directive principles, filial therapy, sandplay), parent management training built on Patterson's **coercion theory**, the bell-and-pad for enuresis, behavioural treatment plus medication for ADHD (the MTA study) and applied behaviour analysis for autism (Lovaas 1987). TF-CBT treats abuse, and the POCSO Act 2012 makes reporting mandatory. Dementia care uses reality orientation, reminiscence, life review, validation and cognitive stimulation therapy.

Holland's RIASEC and Super's life-span, life-space theory guide careers. Forensic work covers competence, the M'Naghten-based section 84 (now BNS 22), risk assessment and the Juvenile Justice Act 2015.

Who's who

Table 7.30 The people in this chapter.

Name	When	Examined for
Hippocrates	460 to 377 BC	Natural causes of mental disorder; humoral theory
Galen	About AD 129 to 198	Developed the humoral theory
Philippe Pinel	1745 to 1826	Physician at La Bicetre from 1793; credited with unchaining patients there and at the Salpetriere; moral treatment
Jean-Baptiste Pussin	1746 to 1811	Superintendent at Bicetre who began humane care, persuaded Pinel and banned the chains there (1797)
William Tuke	1732 to 1822	Quaker founder of the York Retreat (1796); moral treatment in England
Dorothea Dix	1802 to 1887	Campaigned from the 1840s for American state hospitals
Clifford Beers	1876 to 1943	A Mind That Found Itself (1908); founded the mental hygiene movement (1908 to 1909)
Emil Kraepelin	1856 to 1926	Classification of mental disorders; separated dementia praecox from manic-depressive insanity
Eugen Bleuler	1857 to 1939	Named schizophrenia (1908)
Lightner Witmer	1867 to 1956	First psychological clinic (1896); first printed 'clinical psychology' (1907)
Wilhelm Wundt	1832 to 1920	First psychology laboratory, Leipzig (1879); Witmer's teacher
Alfred Binet	1857 to 1911	First practical intelligence scale (1905, with Simon)
Theodore Simon	1905	Binet-Simon intelligence scale
Robert Yerkes	1876 to 1956	Army Alpha and Army Beta testing (1917)
Richard McFall	1991	Manifesto for a Science of Clinical Psychology; clinical science model
Granville Stanley Hall	1844 to 1924	First APA president (1892)
James McKeen Cattell	1860 to 1944	Coined 'mental test' (1890)
Sigmund Freud	1856 to 1939	Psychodynamic model
Erik Erikson	1902 to 1994	Psychosocial development (distractor for 'clinical psychology')
Carl Rogers	1902 to 1987	Humanistic model; unconditional positive regard
Burrhus Frederic Skinner	1904 to 1990	Operant conditioning; behavioural model
Ivan Petrovich Pavlov	1849 to 1936	Classical conditioning; behavioural model
Albert Ellis	1913 to 2007	Irrational beliefs; cognitive model
Aaron T. Beck	1921 to 2021	Cognitive model of disorder
Abraham Maslow	1908 to 1970	Humanistic model; self-actualisation
Viktor Frankl	1905 to 1997	Existential model; search for meaning
Thomas Szasz	1920 to 2012	The Myth of Mental Illness (1961); sociocultural critique
Gregory Bateson	1904 to 1980	Family systems model; double bind
Salvador Minuchin	1921 to 2017	Family systems model; structural family therapy
Girindrasekhar Bose	1887 to 1953	Indian Psychoanalytical Society, Calcutta (1922)
Marie Jahoda	1907 to 2001	Ideal mental health (1958): six criteria
George L. Engel	1913 to 1999	Biopsychosocial model (Science, 1977)
Paul Meehl	1920 to 2003	Diathesis-stress account of schizophrenia (1962)
Joseph Zubin	1900 to 1990	Vulnerability-stress model (1977, with Spring)
Bonnie Spring	1977	Vulnerability-stress model (with Zubin)
Walter Cannon	1871 to 1945	Fight or flight (distractor for the biopsychosocial model)
Richard Lazarus	1922 to 2002	Appraisal theory of stress (distractor for the biopsychosocial model)
Tom Beauchamp	1979	Four principles of biomedical ethics (with Childress)

Name	When	Examined for
James Childress	1979	Four principles of biomedical ethics (with Beauchamp)
Immanuel Kant	1724 to 1804	Deontological ethics
Jeremy Bentham	1748 to 1832	Utilitarian ethics
John Stuart Mill	1806 to 1873	Utilitarian ethics
Prosenjit Poddar	1969	Client in the Tarasoff case
Tatiana Tarasoff	1969	Victim in Tarasoff v. Regents of the University of California (1974, 1976)
Thomas Gutheil	1993	Boundary crossings versus boundary violations (with Gabbard)
Glen Gabbard	1993	Boundary crossings versus boundary violations (with Gutheil)
Allen Ivey	1968 (microcoun-selling)	Microcounselling: training interview microskills one at a time
Gerard Egan	1975 (The Skilled Helper)	SOLER attending behaviour; primary versus advanced empathy
Carl Ransom Rogers	1902 to 1987	Client-centred therapy, whose signature technique is reflection of feeling
Arthur Kleinman	1980	The patient's explanatory model of illness in cross-cultural interviewing
Michael B. First	2015 to 2016 (SCID-5)	Lead author of the Structured Clinical Interview for DSM-5
Robert Leopold Spitzer	1932 to 2015	Led DSM-III (1980) and developed the SCID; critic of Rosenhan's study (1975)
John Kenneth Wing	1923 to 2010	Present State Examination (9th edition 1974), core of the WHO's SCAN
Lee Nelken Robins	1922 to 2009	Diagnostic Interview Schedule (1981), a fully structured interview for lay interviewers
Timothy A. Brown	ADIS-5 (DSM-5 edition)	Co-author of the Anxiety Disorders Interview Schedule for DSM-5
David H. Barlow	born 1942	Co-author of the Anxiety Disorders Interview Schedule (ADIS-5)
Kurt Schneider	1887 to 1967	First-rank symptoms of schizophrenia, including thought insertion, with-drawal and broadcasting
Patterson and colleagues	1983	SAD PERSONS mnemonic for suicide risk
George Libman Engel	1913 to 1999	Biopsychosocial model (Science, 1977)
Judith S. Beck	1995	Cognitive conceptualisation diagram for cognitive-behavioural formulation
Jacob Cohen	1923 to 1998	Cohen's kappa (1960), chance-corrected agreement between raters
David L. Rosenhan	1929 to 2012	'On being sane in insane places' (Science, 1973): sane pseudopatients di-agnosed with schizophrenia
Josef Breuer	1842 to 1925	Cathartic method; treated Anna O. (1880 to 1882); Studies on Hysteria (1895) with Freud
Bertha Pappenheim	1859 to 1936	Anna O., the first talking-cure patient; not in fact cured
Hippolyte Bernheim	1840 to 1919	Nancy school of suggestion; Freud visited him in 1889
Paula Heimann	1950	Broad view of countertransference as useful information
Ralph Greenson	1965	Working alliance
Anna Freud	1895 to 1982	Ego psychology; The Ego and the Mechanisms of Defence (1936); child analysis; analyst's neutrality
Heinz Hartmann	1894 to 1970	Ego psychology; conflict-free ego sphere (1939)
Melanie Klein	1882 to 1960	Child analysis through play; paranoid-schizoid and depressive positions; splitting; projective identification
Donald Winnicott	1896 to 1971	Good-enough mother, transitional object, holding environment, true and false self
Ronald Fairbairn	1889 to 1964	Object relations: libido is object-seeking, not pleasure-seeking
Margaret Mahler	1897 to 1985	Separation-individuation; autistic and symbiotic phases; theory of psy-chosis (On Childhood Psychosis and Schizophrenia, 1952)
Silvano Arieti	1955	Psychoanalytic account of schizophrenia (Interpretation of Schizophrenia)
Ping-Nie Pao	1979	Psychoanalytic account of schizophrenic disorders
Edith Jacobson	1964	The Self and the Object World; analytic work on psychosis and depression

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Otto Kernberg	born 1928	Borderline personality organisation; transference-focused psychotherapy (TFP)
Heinz Kohut	1913 to 1981	Self psychology (1971): mirroring, idealising and twinship selfobjects; empathic attunement
Stephen Mitchell	1946 to 2000	Relational psychoanalysis (with Jay Greenberg, 1983)
Jay Greenberg	1983	Object Relations in Psychoanalytic Theory, with Mitchell; relational psychoanalysis
John Bowlby	1907 to 1990	Attachment theory; therapist as a secure base
Anthony Bateman	1999	Mentalisation-based treatment (MBT) for borderline personality disorder, with Fonagy
Peter Fonagy	1999	Mentalisation-based treatment (MBT), with Bateman
David Malan	1976	Brief focal psychotherapy (Tavistock); triangle of conflict and triangle of person
Peter Sifneos	1920 to 2008	Short-term anxiety-provoking psychotherapy (STAPP)
Habib Davanloo	born 1927	Intensive short-term dynamic psychotherapy (ISTDP)
James Mann	1973	Time-limited psychotherapy: exactly 12 sessions
Lester Luborsky	1984	Core conflictual relationship theme (CCRT); supportive-expressive therapy
Hans Strupp	1984	Time-limited dynamic psychotherapy (TLDP), with Binder; cyclical maladaptive pattern
Jeffrey Binder	1984	Time-limited dynamic psychotherapy, with Strupp
Gerald Klerman	1928 to 1992	Interpersonal psychotherapy (IPT) for depression (manual 1984)
Myrna Weissman	1984	Interpersonal psychotherapy, with Klerman
Adolf Meyer	1866 to 1950	Psychobiology; an influence on IPT
Jacob Moreno	1889 to 1974	Psychodrama; role reversal
Lorna Smith Benjamin	2003	Interpersonal reconstructive therapy
Carl Gustav Jung	1875 to 1961	Analytical psychology: collective unconscious, archetypes (persona, anima, animus), amplification, active imagination, individuation
Alfred Adler	1870 to 1937	Individual psychology: organ inferiority (1907), inferiority complex, birth order, early recollections, encouragement, acting as if, spitting in the soup
Harold Mosak	1985	Adlerian push-button technique
Karen Horney	1885 to 1952	Basic anxiety; moving towards, against, away; idealised self; tyranny of the shoulds
Harry Stack Sullivan	1892 to 1949	Interpersonal psychiatry; participant observer; parataxic distortion; influence on IPT
Eric Berne	1910 to 1970	Transactional analysis: Transactional Analysis in Psychotherapy (1961), Games People Play (1964); Parent, Adult, Child ego states
Paul Federn	1871 to 1950	Coined ego states; Berne's analyst
Bob and Mary Goulding	1970s	Redecision therapy; script injunctions
Thomas Harris	1969	I'm OK, You're OK (Harper and Row, 1969): life positions popularised
Franz Anton Mesmer	1734 to 1815	Animal magnetism, forerunner of hypnosis
James Braid	1843	Coined the word hypnosis
Hans Eysenck	1916 to 1997	1952 challenge to the effectiveness of psychotherapy
Karl Popper	1902 to 1994	Called psychoanalytic theory unfalsifiable
Jonathan Shedler	2010	Review showing psychodynamic therapy as effective as other evidence-based therapies
Hans Jürgen Eysenck	1916 to 1997	1952 critique: about two-thirds of neurotics improve without psychotherapy; popularised behaviour therapy (1959)
Joseph Wolpe	1915 to 1997	Systematic desensitisation and reciprocal inhibition (Psychotherapy by Reciprocal Inhibition, 1958); SUDS; experiments on cats; thought stopping
Arnold Allan Lazarus	1932 to 2013	Wolpe's Johannesburg group; printed the term behaviour therapy (1958); assertiveness training; later multimodal therapy
Stanley Rachman	1950s	Member of Wolpe's Johannesburg group of behaviour therapists
Monte Bernard Shapiro	1950s	Maudsley Hospital; treated each patient as a single-case experiment

Name	When	Examined for
Ogden Lindsley	1953	First used the term behaviour therapy, in a hospital report with Skinner and Solomon (1953)
Harry Solomon	1953	First used the term behaviour therapy, in a hospital report with Lindsley and Skinner (1953)
Orval Hobart Mowrer	1907 to 1982	Bell-and-pad method for enuresis (1938, with Willie Mae Mowrer); two-factor theory of fear and avoidance (1947)
Willie Mae Mowrer	1938	Bell-and-pad method for enuresis, with O. H. Mowrer
Frederick Kanfer	1965	SORKC functional analysis (with Saslow, 1965); self-regulation as self-monitoring, self-evaluation and self-reinforcement
George Saslow	1965	SORKC functional analysis, with Kanfer
Mary Cover Jones	1897 to 1987	Removed Peter's fear of rabbits by counterconditioning and modelling (1924)
John Broadus Watson	1878 to 1958	Founder of behaviourism; teacher of Mary Cover Jones
Charles Scott Sherrington	1857 to 1952	Physiologist whose term reciprocal inhibition Wolpe borrowed
Edmund Jacobson	1888 to 1983	Progressive muscle relaxation (Progressive Relaxation, 1929; enlarged 1938), step 1 of desensitisation
Peter Lang	1964	Fear Survey Schedule, with Wolpe (1964)
Gordon Paul	1966	Desensitisation beat insight therapy for public-speaking anxiety, with no symptom substitution at follow-up (1966)
Thomas Stampfl	1966	Implosive therapy
Donald Levis	1967	Implosive therapy, with Stampfl (1967)
Victor Meyer	1966	First described exposure and response prevention for obsessional rituals (1966)
Edna Foa	1980s onwards	Refined and tested ERP for OCD; prolonged exposure for PTSD
Francine Shapiro	1948 to 2019	Eye movement desensitisation and reprocessing (observed 1987, published 1989), for PTSD; eight phases from history taking to re-evaluation
Patricia Resick	1990s	Cognitive processing therapy for PTSD
Marsha Linehan	1990s	Dialectical behaviour therapy
Bessel van der Kolk	1990s onwards	Trauma researcher
Pat Ogden	2000s	Sensorimotor psychotherapy
Joseph LeDoux	1990s	Neuroscience of fear circuits in the brain
Joseph Cautela	1967	Covert sensitisation
David Premack	1959	Premack principle: a more probable activity reinforces a less probable one
Teodoro Ayllon	1968	Token economy at Anna State Hospital (with Azrin, 1968); satiation of a towel hoarder (1963)
Nathan Azrin	1968 to 1973	Token economy (with Ayllon, 1968); overcorrection (with Foxx, 1972); habit reversal training (with Nunn, 1973)
Richard Foxx	1972	Overcorrection: restitution and positive practice, with Azrin
Ole Ivar Lovaas	1927 to 2010	Intensive early behavioural treatment for autism (1987): 47 percent reached normal functioning
Albert Bandura	1925 to 2021	Live, symbolic and participant modelling; participant modelling for snake phobia (1969); self-efficacy
Edward Blanchard	1969	Participant modelling study of snake phobia, with Bandura and Ritter
Brunhilde Ritter	1969	Participant modelling study of snake phobia, with Bandura and Blanchard
Andrew Salter	1949	Pioneer of assertiveness training (Conditioned Reflex Therapy, 1949)
Johannes Heinrich Schultz	1884 to 1970	Autogenic training (1932)
Lars-Göran Öst	1987	Applied relaxation
Neal Miller	1909 to 2002	Operant control of autonomic responses (1960s), the basis of biofeedback
Arnold Kegel	1948	Perineometer for pelvic floor exercises, an early biofeedback device
Gregory Nunn	1973	Habit reversal training, with Azrin
Knight Dunlap	1875 to 1949	Negative (massed) practice for habits (1932)
Peter Lewinsohn	1970s	Depression as a low rate of response-contingent positive reinforcement; roots of behavioural activation

Name	When	Examined for
Neil Jacobson	1949 to 1999	Showed the activation component of cognitive therapy works alone (1996); behavioural activation
Christopher Martell	2001	Developed behavioural activation for depression with Jacobson
Richard Bootzin	1972	Stimulus control treatment for insomnia
Arthur Spielman	1987	Sleep restriction treatment for insomnia
Gerald Patterson	1926 to 2016	Coercive family cycle; parent management training (Oregon Social Learning Center)
Edward Lee Thorndike	1874 to 1949	Law of effect (a wrong option for shaping and desensitisation)
Fritz Perls	1893 to 1970	Gestalt therapy (a wrong option for desensitisation; Section 7.6)
Aaron Beck	1921 to 2021	Cognitive therapy (a wrong option for desensitisation and EMDR; Section 7.5)
Epictetus	about 50 to 135 CE	Stoic philosopher: people are disturbed not by things but by their views of them; a root of REBT
Donald Meichenbaum	born 1940	Cognitive-Behavior Modification (1977); self-instructional training (1971); stress inoculation training
Steven C. Hayes	born 1948	Named the three waves (2004); acceptance and commitment therapy; relational frame theory
Maxie C. Maultsby Jr.	1970s	Psychiatrist; rational emotive imagery, adopted by Ellis
William James	1842 to 1910	Functionalism; a distractor for the founder of RET
William Glasser	1925 to 2013	Reality therapy (a distractor for REBT)
Aaron Temkin Beck	1921 to 2021	Cognitive therapy; automatic thoughts; cognitive triad; Cognitive Therapy of Depression (1979); BDI, BAI, BHS
David D. Burns	1980	Feeling Good; popularised the longer list of cognitive distortions
Max Hamilton	1959 and 1960	Clinician-rated Hamilton anxiety and depression scales
Thomas J. D'Zurilla	1971	Problem-solving therapy, with Goldfried
Marvin R. Goldfried	1971	Problem-solving therapy, with D'Zurilla
Arthur M. Nezu	1980s	Problem-solving therapy for depression
Arnold A. Lazarus	1932 to 2013	Multimodal therapy; BASIC I.D.; technical eclecticism
George Alexander Kelly	1905 to 1967	Personal construct theory (1955); fixed-role therapy
Jeffrey E. Young	1990	Schema therapy; early maladaptive schemas; schema modes; limited reparenting
Anthony Ryle	1980s	Cognitive analytic therapy (CAT)
David M. Clark	1986	Cognitive model of panic: catastrophic misinterpretation of bodily sensations; social anxiety model with Wells (1995)
Paul M. Salkovskis	1991	Safety behaviours in anxiety and panic
Adrian Wells	1995	Social anxiety model with Clark; metacognitive therapy; cognitive attentional syndrome
Edna B. Foa	1990s	Prolonged exposure for post-traumatic stress disorder
Patricia A. Resick	1990s	Cognitive processing therapy for post-traumatic stress disorder, first for rape survivors
Judith A. Cohen	1990s	Trauma-focused CBT for children, with Mannarino and Deblinger
Anthony P. Mannarino	1990s	Trauma-focused CBT for children, with Cohen and Deblinger
Esther Deblinger	1990s	Trauma-focused CBT for children, with Cohen and Mannarino
David Kingdon	1990s	CBT for psychosis, with Turkington
Douglas Turkington	1990s	CBT for psychosis, with Kingdon
Marsha M. Linehan	born 1943	Dialectical behaviour therapy (1993); biosocial theory of borderline personality disorder; trials of 1991 and 2006
Murray Bowen	1913 to 1990	Family systems therapy; the genogram (not a DBT mode)
Kirk D. Strosahl	1999	Acceptance and commitment therapy, with Hayes and Wilson
Kelly G. Wilson	1999	Acceptance and commitment therapy, with Hayes and Strosahl
Jon Kabat-Zinn	born 1944	Mindfulness-based stress reduction (1979, eight weeks)
Zindel V. Segal	2002	Mindfulness-based cognitive therapy, with Williams and Teasdale
J. Mark G. Williams	2002	Mindfulness-based cognitive therapy, with Segal and Teasdale
John D. Teasdale	2000 and 2002	Mindfulness-based cognitive therapy; the 2000 relapse-prevention trial

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Paul Gilbert	2000s	Compassion-focused therapy; threat, drive and soothing systems
Robert J. Kohlenberg	1991	Functional analytic psychotherapy, with Tsai
Mavis Tsai	1991	Functional analytic psychotherapy, with Kohlenberg
Peter M. Lewinsohn	1970s	Depression as loss of response-contingent positive reinforcement; behavioural activation
Neil S. Jacobson	1996	Component study showing behavioural activation alone works as well as full cognitive therapy
Vikram Patel	2017	Goa trial of lay-counsellor behavioural activation for depression
Abraham Harold Maslow	1908 to 1970	Named humanistic psychology the third force; with Rogers its leading proponent
Stanley Standal	1954	Coined the term unconditional positive regard
Edward Bordin	1979	Working alliance: goals, tasks and bond (not a Rogerian concept)
William Stephenson	1953	Devised the Q-sort (The Study of Behavior, 1953)
John Butler and Gerard Haigh	1954	Q-sort study: clients' mean self-ideal correlation rose from about zero (minus .01) before therapy to .34 after it (.31 at follow-up)
Charles Truax	1967	Rating scales of empathy, warmth and genuineness, with Carkhuff
Robert Carkhuff	1967; 1969	Core-condition rating scales with Truax; helping model in Helping and Human Relations (1969)
Kurt Lewin	1890 to 1947	T-groups, forerunners of encounter groups (Section 7.7)
Laura Perls	1905 to 1990	Co-founder of Gestalt therapy
Paul Goodman	1911 to 1972	Co-founder of Gestalt therapy; co-author of Gestalt Therapy (1951)
Ralph Hefferline	1951	Co-author of Gestalt Therapy (1951)
Erving Polster and Miriam Polster	1973	Gestalt Therapy Integrated (1973): five contact boundary disturbances, with deflection in place of egotism
Max Wertheimer	1880 to 1943	Gestalt psychology of perception, not Gestalt therapy (Chapter 1)
Wolfgang Köhler	1887 to 1967	Gestalt psychology of perception (Chapter 1)
Kurt Koffka	1886 to 1941	Gestalt psychology of perception (Chapter 1)
Martin Heidegger	1889 to 1976	Being and Time (1927): Dasein, being-in-the-world
Irvin Yalom	born 1931	Existential Psychotherapy (1980): four ultimate concerns (death, freedom, isolation, meaninglessness)
Ludwig Binswanger	1881 to 1966	Daseinsanalysis; Umwelt, Mitwelt and Eigenwelt
Medard Boss	1903 to 1990	Daseinsanalysis (Psychoanalysis and Daseinsanalysis, 1963)
Emmy van Deurzen	contemporary	Added the Überwelt (spiritual world) to the modes of being
Rollo May	1909 to 1994	The Meaning of Anxiety (1950); Existence (1958); existential anxiety, normal and neurotic anxiety
James Bugental	1915 to 2008	Existential-humanistic therapy
Viktor Emil Frankl	1905 to 1997	Logotherapy; Man's Search for Meaning (1946); will to meaning, noogenic neurosis, paradoxical intention, dereflection
William Masters and Virginia Johnson	1970	Sensate focus in sex therapy, not dereflection (Section 7.7)
Robert Wubbolding	1980s	WDEP summary of reality therapy
Eugene Gendlin	1926 to 2017	Focusing (1978) and the felt sense
Leslie Greenberg	1990s	Process-experiential and emotion-focused therapy
Susan Johnson	1980s	Emotionally focused couples therapy, with Greenberg (Section 7.7)
Martin Seligman	born 1942	Positive psychology (APA president 1998); positive psychotherapy; PERMA
Mihaly Csikszentmihalyi	1934 to 2021	Flow
Nossrat Peseschkian	1933 to 2010	Transcultural Positive Psychotherapy (1970s): balance model, five stages
Michael White	1948 to 2008	Narrative therapy: externalising, unique outcomes, re-authoring (with Epston, 1990)
David Epston	1990	Narrative therapy with Michael White (Narrative Means to Therapeutic Ends, 1990)
Steve de Shazer	1940 to 2005	Solution-focused brief therapy; the miracle question (1988)

Name	When	Examined for
Insoo Kim Berg	died 2007	Solution-focused brief therapy, Brief Family Therapy Center, Milwaukee (1978)
Jay Haley	1923 to 2007	Brief strategic therapy (Section 7.7)
William R. Miller	1983	Motivational interviewing (1983; book with Rollnick 1991, later editions 2002 and 2013)
Stephen Rollnick	1991	Motivational interviewing, with Miller
Burke, Arkowitz and Menchola	2003	Meta-analysis of motivational interviewing: effective for alcohol, drug, and diet and exercise problems; not supported for smoking or HIV-risk behaviour; OCD not reviewed
James Prochaska	1983	Transtheoretical model and stages of change, with DiClemente
Carlo DiClemente	1983	Transtheoretical model and stages of change, with Prochaska
Alan Marlatt	1941 to 2011	Relapse prevention (Section 7.9)
Ludwig von Bertalanffy	1901 to 1972	General systems theory, applied to families
Norbert Wiener	1894 to 1964	Cybernetics: self-regulation by feedback
Betty Carter	1980s	Family life cycle (with Monica McGoldrick)
Monica McGoldrick	1980s	Family life cycle (with Betty Carter); genograms
Frieda Fromm-Reichmann	1889 to 1957	The schizophrenogenic mother (1948), now discredited
Don Jackson	1920 to 1968	Double bind (1956); founded the Mental Research Institute, Palo Alto (1959)
John Weakland	1956	Double bind; MRI brief therapy
Theodore Lidz	1910 to 2001	Marital schism and marital skew
Lyman Wynne	1923 to 2007	Pseudomutuality (1958)
George Brown	late 1950s	Founded expressed emotion research on relapse in schizophrenia
James Birley	1972	Brown, Birley and Wing: high EE predicts relapse
John Wing	1972	Brown, Birley and Wing: high EE predicts relapse
Christine Vaughn	1976	Vaughn and Leff: EE, medication and 35-hour contact and relapse
Julian Leff	1976	Vaughn and Leff; family interventions to lower expressed emotion
Ian Falloon	1980s	Behavioural family management in schizophrenia
Vidya Sagar	1950s	Involved families in inpatient care at Amritsar Mental Hospital
Janis Jenkins	1992	Cross-cultural comparison of expressed emotion (with Marvin Karno)
Marvin Karno	1992	Cross-cultural comparison of expressed emotion (with Janis Jenkins)
Ann Hartman	1970s	The ecomap
Edward Tolman	1886 to 1959	Cognitive map (Chapter 1)
Milton Erickson	1901 to 1980	Hypnotherapist whose methods shaped strategic therapy
Cloe Madanes	1981	Strategic family therapy; pretend techniques
Paul Watzlawick	1974	MRI brief therapy; first-order and second-order change (Change, 1974)
Richard Fisch	1974	MRI brief therapy; attempted solutions
Mara Selvini Palazzoli	1916 to 1999	Milan systemic therapy; Paradox and Counterparadox (1978)
Luigi Boscolo	1978	Milan systemic therapy; circular questioning
Gianfranco Cecchin	1978	Milan systemic therapy; circular questioning
Giuliana Prata	1978	Milan systemic therapy
Virginia Satir	1916 to 1988	Experiential family therapy; communication stances; family sculpting
Carl Whitaker	1912 to 1995	Symbolic-experiential family therapy
Ivan Boszormenyi-Nagy	1920 to 2007	Contextual therapy; Invisible Loyalties (1973)
James Alexander	1970s	Functional family therapy (with Bruce Parsons)
Bruce Parsons	1970s	Functional family therapy (with James Alexander)
Scott Henggeler	1990s	Multisystemic therapy for youth offending
Gayla Margolin	1979	Behavioural couples therapy (with Jacobson)
Andrew Christensen	1996	Integrative behavioural couple therapy: acceptance
John Gottman	1990s	Four horsemen; the 5:1 ratio; Sound Relationship House
Sue Johnson	1980s	Emotionally focused couple therapy, based on attachment

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William Masters	1970	Sex therapy: sensate focus, squeeze technique (Human Sexual Inadequacy, 1970)
Virginia Johnson	1970	Sex therapy with William Masters
James Semans	1956	Stop-start technique for premature ejaculation
Helen Singer Kaplan	1974	Brief sex therapy (The New Sex Therapy, 1974)
Jack Annon	1976	PLISSIT model
Bruce Tuckman	1965	Stages of group development: forming, storming, norming, performing (1965); adjourning added with Mary Ann Jensen (1977)
Mary Ann Jensen	1977	Added the adjourning stage to Tuckman's model
Jacob Levy Moreno	1889 to 1974	Psychodrama; sociometry and the sociogram; coined group psychotherapy
Bill Wilson	1935	Co-founder of Alcoholics Anonymous
Bob Smith	1935	Co-founder of Alcoholics Anonymous
Maxwell Jones	1907 to 1990	The therapeutic community (Belmont Hospital, Surrey, later the Henderson Hospital)
S. H. Foulkes	1898 to 1976	Group analysis
Allen E. Bergin	1966; 1971	Showed (1971) that recounting dropouts lifted psychoanalysis to about 85 percent improvement among completers; named the deterioration effect (1966)
Michael J. Lambert	1976; 1992	Spontaneous remission nearer 40 percent, not two-thirds (1976); the pie of therapeutic factors (1992); Outcome Questionnaire-45
Gene V. Glass	1976; 1977	Coined meta-analysis (1976); Smith and Glass psychotherapy meta-analysis (1977)
Mary Lee Smith	1977; 1980	With Glass: 375 studies, effect size 0.68 (1977); 0.85 in The Benefits of Psychotherapy (1980)
Thomas I. Miller	1980	Co-author of The Benefits of Psychotherapy (effect size 0.85)
Saul Rosenzweig	1907 to 2004	Implicit common factors and the dodo bird quotation (1936)
Barton Singer	1975	Co-author of the dodo bird verdict review
Lise Luborsky	1975	Co-author of the dodo bird verdict review
Bruce E. Wampold	2001; 2015	Contextual model; The Great Psychotherapy Debate (2001; fullest form in the 2015 edition with Imel)
Zac E. Imel	2015	Co-author with Wampold of the 2015 edition of The Great Psychotherapy Debate (contextual model)
Jerome David Frank	1909 to 2005	Persuasion and Healing (1961): demoralisation and four common features of all healing
Alfred Pritz	born 1952	Austrian psychotherapist, founding president of the World Council for Psychotherapy (1995) (distractor)
Elizabeth Zetzel	1956	Coined the therapeutic alliance in psychoanalysis
Edward S. Bordin	1979	Pantheoretical working alliance: goals, tasks and bond
Adam Horvath	1989	Working Alliance Inventory (with Greenberg)
Jeremy D. Safran	2000	Alliance ruptures (withdrawal and confrontation) and their repair
J. Christopher Muran	2000	Alliance ruptures and repair (with Safran)
William Schofield	1964	YAVIS client; Psychotherapy: The Purchase of Friendship
Starke R. Hathaway	1903 to 1984	Described the hello-goodbye effect; co-author of the MMPI
Martin E. P. Seligman	1995	Consumer Reports effectiveness study
Dianne L. Chambless	1995; 1998	Chaired the APA Division 12 task force on empirically supported treatments; criteria with Hollon (1998)
Steven D. Hollon	1998	Refined the EST criteria with Chambless
Irene Elkin	1989	Led the NIMH Treatment of Depression Collaborative Research Program
Kenneth I. Howard	1986	Dose-effect relationship (50 percent improved by 8 sessions, 75 by 26); phase model
Paula Truax	1991	Clinical significance and the reliable change index (with Jacobson)
Paul L. Wachtel	1977	Cyclical psychodynamics; theoretical integration
James O. Prochaska	1979	Transtheoretical approach; stages of change

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Stanley B. Messer	1992	Assimilative integration
Guillermo Bernal	1995	Ecological validity framework for culturally adapted therapy
N. C. Surya	1964	Psychotherapy in the Indian setting (with S. S. Jayaram): Indian patients more dependent, need direct, situational support in their own language
S. S. Jayaram	1964	Co-author with Surya on psychotherapy in the Indian setting
Jaswant Singh Neki	1973	Guru-chela relationship as an Indian therapeutic paradigm
Cal Stoltenberg	1987	Integrated developmental model of supervision
Janine M. Bernard	1979	Discrimination model of supervision (teacher, counsellor, consultant)
Christina Maslach	1981	Burnout: emotional exhaustion, depersonalisation, reduced personal accomplishment
Joseph D. Matarazzo	1980	Defined health psychology; first president of APA Division 38
Gary Schwartz	1978	Defined behavioural medicine with Stephen Weiss
Stephen Weiss	1978	Defined behavioural medicine with Gary Schwartz
George Engel	1913 to 1999	Biopsychosocial model (1977)
David Mechanic	1962	Illness behaviour
Talcott Parsons	1902 to 1979	The sick role, in The Social System (1951)
Godfrey Hochbaum	1950s	Health belief model, from tuberculosis screening studies
Irwin Rosenstock	1966	Health belief model: 1950s origins with Hochbaum and Kegeles; classic statement 1966; self-efficacy added 1988
Marshall Becker	1974	Extended the health belief model to sick-role behaviour and adherence
Stephen Kegeles	1950s	Health belief model
Martin Fishbein	1936 to 2009	Theory of reasoned action with Ajzen (1975)
Icek Ajzen	born 1942	Theory of reasoned action (1975); theory of planned behaviour (1985, 1991)
Ronald Rogers	1975	Protection motivation theory (revised 1983)
Ralf Schwarzer	1992	Health action process approach
Marc Hollender	1956	Activity-passivity, guidance-cooperation and mutual participation, with Szasz
Hans Selye	1907 to 1982	General adaptation syndrome: alarm, resistance, exhaustion
Thomas Holmes	1967	Life-events scale (Social Readjustment Rating Scale) with Rahe
Richard Rahe	1967	Life-events scale (Social Readjustment Rating Scale) with Holmes
Robert Ader	1932 to 2011	Psychoneuroimmunology: conditioned immune suppression in rats (1975)
Nicholas Cohen	1975	Conditioned immune suppression with Ader
Franz Alexander	1891 to 1964	Psychosomatic Medicine (1950); the holy seven; specificity hypothesis
Meyer Friedman	1910 to 2001	Type A behaviour pattern with Rosenman (1959)
Ray Rosenman	1920 to 2013	Type A behaviour pattern with Friedman (1959)
Lydia Temoshok	1980s	Type C (cancer-prone) personality
Ronald Melzack	1929 to 2019	Gate control theory of pain with Wall (1965)
Patrick Wall	1925 to 2001	Gate control theory of pain with Melzack (1965)
Wilbert Fordyce	1976	Operant approach to chronic pain; pain behaviours
Johannes Schultz	1884 to 1970	Autogenic training
Herbert Benson	1975	The relaxation response
Cicely Saunders	1918 to 2005	St Christopher's Hospice (1967); total pain; modern palliative care
Elisabeth Kübler-Ross	1926 to 2004	Five stages of dying: denial, anger, bargaining, depression, acceptance (1969)
J. William Worden	1982	Four tasks of mourning
John F. Kennedy	1917 to 1963	Community Mental Health Centers Act (1963)
Gerald Caplan	1964	Primary, secondary and tertiary prevention; four types of consultation (1970); crisis theory
Robert Gordon	1983	Universal, selective and indicated prevention
Julian Rappaport	1981	Empowerment in community psychology

Name	When	Examined for
Dixon Chibanda	2010s	Friendship Bench: lay health workers deliver problem-solving therapy in Zimbabwe
Erving Goffman	1922 to 1982	Stigma (1963)
Erich Lindemann	1900 to 1974	Acute grief after the Cocoanut Grove fire (1944); founder of crisis intervention
Albert Roberts	1990s	Seven-stage model of crisis intervention
Barbara Stanley	2012	Safety planning intervention with Brown
Gregory Brown	2012	Safety planning intervention with Stanley
Jeffrey Mitchell	1983	Critical incident stress debriefing
Leonard Stein	1980	Assertive community treatment (Training in Community Living) with Test
Mary Ann Test	1980	Assertive community treatment with Stein
McKay Moore Sohlberg	1987	Attention process training with Mateer
Catherine Mateer	1987	Attention process training with Sohlberg
Alan Baddeley	born 1934	Errorless learning with Wilson (1994)
Barbara Wilson	1994	Errorless learning; memory rehabilitation
G. Alan Marlatt	1941 to 2011	Relapse prevention with Gordon (1985); abstinence violation effect
Judith Gordon	1985	Relapse prevention with Marlatt
Virginia Axline	1911 to 1988	Non-directive (client-centred) play therapy and its eight principles; Play Therapy (1947), Dibs: In Search of Self (1964)
Clark Moustakas	1950s	Relationship play therapy
Garry Landreth	1991	Child-centred play therapy, the modern form of Axline's model
Bernard Guerney	1964	Filial therapy, with Louise Guerney: parents trained to hold play sessions
Louise Guerney	1964	Filial therapy, with Bernard Guerney
Margaret Lowenfeld	late 1920s	World Technique (sand tray and miniatures), forerunner of sandplay
Dora Kalff	1950s	Jungian sandplay therapy
Russell Barkley	1987	Defiant Children parent-training programme
Carolyn Webster-Stratton	1980s	Incredible Years video-modelling parent programme
Sheila Eyberg	1970s	Parent-child interaction therapy (PCIT)
Philip Kendall	1990	Coping Cat CBT programme and FEAR plan for anxious children
O. Hobart Mowrer	1938	Bell-and-pad (urine alarm) for enuresis, with Willie Mae Mowrer
Christopher Kearney	1990s	Functional model of school refusal, with Wendy Silverman
Wendy Silverman	1990s	Functional model of school refusal, with Christopher Kearney
Leo Kanner	1894 to 1981	First described early infantile autism (1943)
Eric Schopler	1972	TEACCH programme for autism; Childhood Autism Rating Scale
Andy Bondy	1985	Picture Exchange Communication System (PECS), with Lori Frost
Lori Frost	1985	PECS, with Andy Bondy
Carol Gray	early 1990s	Social stories for autism
Sally Rogers	2010	Early Start Denver Model, with Geraldine Dawson
Geraldine Dawson	2010	Early Start Denver Model, with Sally Rogers
Grace Fernald	1879 to 1950	Multisensory tracing method for reading disability
Hans Asperger	1944	Described more able autistic children (autistic psychopathy)
Thomas Achenbach	1983	Child Behavior Checklist (CBCL)
C. Henry Kempe	1962	Battered child syndrome
Judith Cohen	2006	Trauma-focused CBT (TF-CBT), with Mannarino and Deblinger
Anthony Mannarino	2006	Trauma-focused CBT, with Cohen and Deblinger
Peter Blos	1904 to 1997	Held that healthy adolescents must detach from parents
Robert Butler	1927 to 2010	Life review (1963); coined ageism (1969)
James Folsom	late 1950s	Reality orientation for dementia
Naomi Feil	developed her method 1963 to 1980	Validation therapy for disoriented old people
Aimee Spector	2003	Cognitive stimulation therapy for dementia, with Martin Orrell

Name	When	Examined for
Martin Orrell	2003	Cognitive stimulation therapy, with Aimee Spector
Jerome Yesavage	1982	Geriatric Depression Scale (30 items)
Paul Appelbaum	1988	Four abilities of decision-making capacity, with Thomas Grisso
Thomas Grisso	1988	Four abilities of decision-making capacity, with Paul Appelbaum
Ruth Strang	1895 to 1971	Called the interview the heart of the counselling process
Frank Parsons	1854 to 1908	Father of vocational guidance; trait-factor approach, Choosing a Vocation (1909)
John Holland	1919 to 2008	RIASEC types and the hexagon; congruence
Donald Super	1910 to 1994	Life-span, life-space theory; five career stages; Life-Career Rainbow
Daniel M'Naghten	1843	His acquittal produced the M'Naghten rules of insanity
Robert Peel	1843	Prime Minister whose secretary M'Naghten shot
David Bazelon	1954	Durham rule (product test) of insanity
John Monahan	1981	Showed unaided clinical prediction of violence is right at most one time in three
Robert Hare	1991	Psychopathy Checklist-Revised
Donald Andrews	1990	Risk-need-responsivity model of offender rehabilitation, with James Bonta
James Bonta	1990	Risk-need-responsivity model, with Donald Andrews
Ronald Fisher	1980s	Cognitive interview for eyewitnesses, with Edward Geiselman
Edward Geiselman	1980s	Cognitive interview, with Ronald Fisher
Margaret Naumburg	1890 to 1983	Art psychotherapy: art as symbolic speech from the unconscious
Edith Kramer	1916 to 2014	Art as therapy: healing through sublimation in the creative process
Marian Chace	1896 to 1970	Founder of dance and movement therapy in the United States
Boris Levinson	1907 to 1984	Pioneer of animal-assisted therapy (his dog Jingles, 1960s)

Key terms

clinical psychologist, supernatural model, trephining, humoral theory, moral treatment, mental hygiene movement, psychological clinic, scientist-practitioner model, practitioner-scholar model, PsyD, clinical science model, Rehabilitation Council of India, ideal mental health, four Ds, biopsychosocial model, diathesis-stress model, diathesis, vulnerability-stress-protective factor model, General Principles, deontological, utilitarian, Informed consent, debriefing, Confidentiality, privileged communication, duty to warn, duty to protect, multiple (dual) relationship, boundary violation, advance directive, nominated representative, Mental Health Review Boards, benchmark disability, treatment gap, task-shifting, clinical interview, diagnostic interview, unstructured interview, semi-structured interview, fully structured interview, Rapport, SOLER, open question, Reflection of feeling, Empathy, explanatory model, genogram, Premorbid personality, mental status examination, Mood, Affect, form of thought, delusion, overvalued idea, thought alienation, hallucination, pseudohallucination, Insight, SAD PERSONS, functional analysis, SORC, self-monitoring, Reactivity, observer drift, case formulation, 5 Ps, biopsychosocial formulation, cognitive conceptualisation diagram, differential diagnosis, Cohen's kappa, cathartic method, Abreaction, catharsis, fundamental rule, free association, neutrality, abstinence, manifest content, latent content, resistance, Interpretation, working through, Transference, Countertransference, cathexis, Defence mechanisms, Ego psychology, Object relations, splitting, good-enough mother, transitional object, separation-individuation, self psychology, Mentalisation-based treatment, triangle of conflict, Interpersonal psychotherapy, collective unconscious, active imagination, individual psychology, inferiority complex, Transactional analysis, ego states, life script, Supportive psychotherapy, Narcoanalysis, Behaviour therapy, applied behaviour analysis, two-factor theory, counterconditioning, Reciprocal inhibition, systematic desensitisation, subjective units of distress scale,

Exposure therapy, habituation, Flooding, Implosive therapy, Exposure and response prevention, eye movement desensitisation and reprocessing, Aversion therapy, covert sensitisation, bell-and-pad method, Shaping, Chaining, Premack principle, token economy, Extinction, Time-out, response cost, Overcorrection, participant modelling, Assertiveness training, progressive muscle relaxation, Biofeedback, Habit reversal training, Behavioural activation, stimulus control, Paradoxical intention, Parent management training, symptom substitution, cognitive-behavioural therapy, self-efficacy, third wave, rational emotive behaviour therapy, ABC model, musturbation, awfulising, disputing, unconditional self-acceptance, rational emotive imagery, shame-attacking exercise, automatic thoughts, cognitive triad, core beliefs, intermediate beliefs, cognitive distortions, collaborative empiricism, Socratic questioning, dysfunctional thought record, downward arrow technique, stress inoculation training, multimodal therapy, fixed-role therapy, schema therapy, limited reparenting, safety behaviours, dialectical behaviour therapy, biosocial theory, acceptance and commitment therapy, relational frame theory, psychological flexibility, cognitive defusion, mindfulness-based stress reduction, mindfulness-based cognitive therapy, cognitive attentional syndrome, third force, non-directive therapy, actualising tendency, conditions of worth, incongruence, fully functioning person, necessary and sufficient conditions, unconditional positive regard, congruence, Q-sort, encounter group, unfinished business, empty chair technique, Dasein, ultimate concerns, Umwelt, Mitwelt, Eigenwelt, logotherapy, will to meaning, noogenic neurosis, dereflection, choice theory, focusing, emotion-focused therapy, flow, externalising the problem, solution-focused brief therapy, Motivational interviewing, change talk, transtheoretical model, microskills hierarchy, advanced empathy, general systems theory, homeostasis, circular causality, identified patient, family life cycle, schizophrenogenic mother, double bind, marital schism, marital skew, pseudomutuality, expressed emotion, family psychoeducation, differentiation of self, triangulation, enmeshment, disengagement, joining, enactment, paradoxical intervention, first-order change, second-order change, positive connotation, circular questioning, four horsemen, sensate focus, PLISSIT model, therapeutic factors, universality, here-and-now, psychodrama, sociogram, milieu therapy, therapeutic community, spontaneous remission, meta-analysis, effect size, dodo bird verdict, common factors, contextual model, therapeutic alliance, Working Alliance Inventory, alliance ruptures, therapist effects, allegiance effect, YAVIS, hello-goodbye effect, efficacy, effectiveness, analogue research, randomised controlled trial, allocation concealment, treatment fidelity, empirically supported treatments, evidence-based practice in psychology, dose-effect relationship, clinical significance, reliable change index, deterioration effect, process research, premature termination, therapist self-disclosure, technical eclecticism, theoretical integration, assimilative integration, culturally adapted therapy, guru-chela relationship, wounded healer, burnout, health psychology, Behavioural medicine, illness behaviour, sick role, health belief model, theory of planned behaviour, protection motivation theory, adherence, psychosomatic disorders, cognitive behavioural therapy for insomnia, Alcohol Use Disorders Identification Test, palliative care, tasks of mourning, Complicated grief, deinstitutionalisation, Primary prevention, Secondary prevention, Tertiary prevention, mental health consultation, empowerment, crisis, Crisis intervention, safety plan, psychological first aid, critical incident stress debriefing, Trauma-informed care, vocational rehabilitation, assertive community treatment, Case management, restorative approach, compensatory approach, Errorless learning, relapse prevention, abstinence violation effect, Therapeutic communities, multi-informant assessment, non-directive play therapy, filial therapy, sandplay, coercion theory, parent-child interaction therapy, bell-and-pad, school refusal, discrete trial training, TEACCH, PECS, National Trust Act, Indian Scale for Assessment of Autism, trauma-focused cognitive behavioural therapy, POCSO Act, confidentiality with minors, geropsychology, caregiver burden, reality orientation, validation therapy, cognitive stimulation therapy, decision-making capacity, trait-factor approach, RIASEC, life-span,

life-space theory, competence to stand trial, M'Naghten rules, ALI test, risk-need-responsivity model, animal-assisted therapy, Tele-MANAS

From the past papers

1. To describe a predisposition towards and risk triggers for a mental disorder, modern theorists use a (JEMAS 2025)

   (a) Diathesis-Stress model (b) Neurodevelopmental model (c) Clinical-Reductionist model (d) Syndrome-Sign model

   Answer: (a) Diathesis-stress model. The diathesis is the predisposition and stress is the trigger; disorder appears when stress exceeds the person's threshold. The neurodevelopmental model traces schizophrenia to faulty early brain development, and the other two options are not real models.

2. All of the following statements are TRUE about exceptions to the rule of confidentiality EXCEPT: (RML 2020)

   (a) a court order requires the release of information (b) the information is essential to safeguard the health of others (duty to protect; warn) (c) the information is probative of guilt or innocence in a criminal matter (d) there are never exceptions to the rule of confidentiality

   Answer: (d) There are never exceptions to the rule of confidentiality. Confidentiality is not absolute: court orders, the Tarasoff duty to warn and protect, and evidence a court needs in a criminal case are all recognised exceptions, so (a) to (c) are true.

3. Which of the following is a key legal provision under the Mental Healthcare Act, 2017? (IHBAS 2025)

   (a) Suicide attempts are decriminalized, recognizing mental illness as a contributing factor (b) Electroconvulsive therapy (ECT) can be administered without consent in all cases (c) Only psychiatrists can provide therapy; psychologists are excluded (d) Mental health establishments are exempt from registration if government-run

   Answer: (a). Section 115 presumes severe stress in anyone who attempts suicide, so the person is not punished. Treatment under the Act needs informed consent, and ECT is banned outright without anaesthesia and muscle relaxant; clinical psychologists are mental health professionals under the Act; and every establishment, government or private, must register (it was the 1987 Act that exempted government hospitals from licensing).

4. A structured interview commonly used in clinical psychology is : (IHBAS 2025)

   (a) MMSE (b) SCID (c) WAIS (d) TAT

   Answer: (b) SCID. The Structured Clinical Interview for DSM takes the clinician through the diagnostic criteria with standard questions (strictly it is semi-structured). The MMSE is a 30-point cognitive screening test, the WAIS an intelligence test and the TAT a projective test.

5. Leaning forward, saying umm, nodding, are examples of which one of the following interview techniques? (RML 2020)

   (a) Paraphrasing (b) Prompting (c) Furthering (d) Checking

   Answer: (c) Furthering. Furthering responses (minimal encouragers) keep the client talking without changing the topic. Paraphrasing restates content in new words, prompting steers the client to a particular topic, and checking asks the client to confirm the interviewer's understanding.

6. In the context of psychopathology, a "formulation" differs from a "diagnosis" in that a formulation primarily aims to: (Real PYQ 2024)

   (a) Classify the client's symptoms according to a standardized nosological system like DSM-5 (b) Provide a hypothesis about the cause, precipitants, and maintaining factors of an individual's difficulties (c) Determine the prognosis and likely course of the disorder over time (d) Rule out organic causes for the presenting psychological symptoms

   Answer: (b) A hypothesis about cause, precipitants and maintaining factors. A formulation (for example the 5 Ps) is an individualised explanation. Classifying symptoms is diagnosis, predicting course is prognosis, and ruling out organic causes belongs to differential diagnosis.

7. 'Counter transference' in psychotherapy refers to (JEMAS 2025)

   (a) therapists' emotional reactions to the client based on the therapists' own unresolved conflict (b) client's projection onto the therapist (c) client's resistance to therapy (d) behavioral reinforcement techniques

   Answer: (a) The therapist's emotional reactions to the client, based on the therapist's own unresolved conflict. Countertransference runs from therapist to client. The client's projection onto the therapist is transference, obstruction of therapy is resistance, and reinforcement belongs to behaviour therapy.

8. Which of the following technique or techniques would not typically be included within the repertoire of an Adlerian counsellor? (RML 2022)

   (a) Acting "as if" (b) Push button (c) Encouragement (d) Analysis of transference

   Answer: (d) Analysis of transference. Adlerian therapy is an equal, educational partnership that does not cultivate or analyse a transference. Acting as if, Mosak's push-button technique and encouragement (the core Adlerian technique) are all Adlerian.

9. Interpersonal Therapy (IPT) does NOT typically include which of the following techniques or focus areas? (Real PYQ 2025)

   (a) Role playing and role reversal techniques (b) Grief and bereavement work (c) Role conflict resolution (d) Role transition support

   Answer: (a) Role playing and role reversal techniques. Grief, role disputes and role transitions are three of IPT's four problem areas (the fourth is interpersonal deficits). Strictly, IPT does use role play after communication analysis, but role reversal is a psychodrama technique, so (a) is the best choice.

10. A technique in which a response that inhibits anxiety is paired with an anxiety-provoking stimulus, step by step, to break the conditioned link between that stimulus and anxiety is: (NIMHANS 2026 recall)

    (a) Systematic desensitization (b) Flooding (c) Aversion therapy (d) Exposure and response prevention

    Answer: (a) Systematic desensitization. Wolpe paired relaxation, a response incompatible with anxiety, with each step of a graded hierarchy (reciprocal inhibition). Flooding uses full-intensity exposure with no hierarchy or relaxation, aversion therapy pairs a behaviour with an unpleasant stimulus, and ERP blocks rituals so that anxiety habituates.

11. Exposure & response prevention procedure is based on behaviour therapy principal of : (IHBAS 2024)

    (a) Counter conditioning (b) Habituation and extinction (c) Reciprocal inhibition (d) Graded exposure

    Answer: (b) Habituation and extinction. Kept in contact with the obsessional trigger and prevented from performing the ritual, the patient's anxiety falls by itself (habituation) and the feared outcome fails to occur (extinction). Counterconditioning and reciprocal inhibition underlie systematic desensitisation; graded exposure describes how steps are arranged, not why fear falls.

12. The behavioral technique that involves removing a desirable stimulus following an undesirable behavior in order to reduce the frequency of that behavior is called: (Real PYQ 2025)

    (a) Negative reinforcement (b) Response cost (c) Positive punishment (d) Extinction

    Answer: (b) Response cost. Taking away a reinforcer the person already has (tokens, privileges) after a behaviour is response cost, a form of negative punishment. Negative reinforcement removes something unpleasant and increases behaviour, positive punishment adds something unpleasant, and extinction merely withholds the reinforcer that maintained the behaviour.

13. A patient sees a friend in the street who fails to acknowledge him; he thinks his friend does not like him anymore. He has the following type of cognitive error: (IHBAS 2020)

    (a) Selective abstraction (b) Arbitrary inference (c) Personalization (d) Magnification

    Answer: (b) Arbitrary inference. He draws a specific negative conclusion with no evidence for it, which is Beck's arbitrary inference (mind reading is a narrower name for the same jump). Selective abstraction would need him to fix on one detail while ignoring the rest, personalisation would blame himself for an unrelated event, and magnification would blow up a fault.

14. In the A-B-C-D-E model of cognitive behaviour therapy (RET) 'D' refers to _____ . (IHBAS 2025)

    (a) Drama (b) Desensitization (c) Discuss (d) Disputing

    Answer: (d) Disputing. In Ellis's model D is disputing the irrational belief with logical, empirical and pragmatic arguments, which leads to E, an effective new philosophy. Desensitisation is Wolpe's behavioural technique, and drama and discuss are fillers.

15. Distress tolerance is a major treatment modality in which of the following therapy? (RML 2024)

    (a) Behaviour therapy (b) Dialectical Behaviour therapy (c) Cognitive Behaviour Therapy (d) Psychodynamic therapy

    Answer: (b) Dialectical behaviour therapy. Distress tolerance is one of Marsha Linehan's four DBT skills modules, with mindfulness, emotion regulation and interpersonal effectiveness. Standard behaviour therapy, Beck's CBT and psychodynamic therapy have no such module.

16. A split between the organismic valuing process and a self-concept burdened by conditions of worth which results in feelings of tension and confusion: (Amity 2023)

    (a) Conflict (b) Incongruence (c) Introjection (d) Actualizing tendency

    Answer: (b) Incongruence. Rogers's term for the mismatch between self-concept and experience, felt as tension and anxiety. Introjection is only how conditions of worth get in, conflict has no special place in his theory, and the actualising tendency is the growth force that incongruence blocks.

17. In existential psychology, Dasein encompasses three fundamental psychological dimensions of human existence. Which of the following correctly identifies these three modes of being-in-the-world? (Real PYQ 2025)

    (a) Id, Ego, and Superego (b) Umwelt, Mitwelt, and Eigenwelt (c) Conscious, Preconscious, and Unconscious (d) Sensorimotor, Preoperational, and Concrete Operational

    Answer: (b) Umwelt, Mitwelt and Eigenwelt. Binswanger's three modes: the biological and physical world around, the with-world of relationships and the own-world of one's relation to oneself. The other options are Freud's structural and topographical models and Piaget's stages.

18. Frankl's technique for 'hyper-reflection', in which a person stops watching himself and turns his attention outward, for example from his own sexual performance to his partner, is: (NIMHANS 2026 recall)

    (a) Paradoxical intention (b) Sensate focus (c) Dereflection (d) Socratic dialogue

    Answer: (c) Dereflection. It counters hyper-reflection by turning attention away from the self. Paradoxical intention, also Frankl's, has the client wish for the feared symptom to defuse anticipatory anxiety; Socratic dialogue uncovers meaning; sensate focus is Masters and Johnson's sex-therapy exercise.

19. Which one of the following captures implicit and explicit group communication and interaction patterns? (RML 2020)

    (a) A genogram (b) A cognitive map (c) An ecomap (d) A sociogram

    Answer: (d) A sociogram. Moreno's sociometric diagram draws group members as points and their choices as arrows, exposing stars, isolates, cliques and lines of communication. A genogram maps a family across generations (Bowen), an ecomap maps a family's links with outside systems (Hartman), and a cognitive map is Tolman's mental map of space.

20. Expressed Emotion (EE) is a significant predictor of relapse in schizophrenia. Which of the following is NOT one of the three core dimensions of high Expressed Emotion? (Real PYQ 2024)

    (a) Hostility (b) Critical Comments (c) Emotional Over-involvement (d) Warmth and Positive Regard

    Answer: (d) Warmth and Positive Regard. High EE is defined by critical comments, hostility and emotional over-involvement. Warmth and positive remarks are also rated in the Camberwell Family Interview, but they do not count toward high EE.

21. Which of the following therapeutic modalities was pioneered by Maxwell Jones and is characterized by the democratic involvement of all community members (staff and patients) in decision-making and daily activities to promote rehabilitation? (Real PYQ 2024)

    (a) Encounter Groups (b) Therapeutic Community (c) Group Analysis (d) Milieu Therapy

    Answer: (b) Therapeutic Community. Maxwell Jones built the democratic therapeutic community at Belmont Hospital (later the Henderson Hospital), with shared decisions, daily community meetings and a flattened hierarchy. Milieu therapy is the broader idea of the ward as treatment, group analysis is Foulkes's psychoanalytic group therapy, and encounter groups belong to Rogers.

22. According to the "Dodo Bird Verdict," what is the primary factor responsible for equivalent outcomes across different therapeutic modalities? (Real PYQ 2025)

    (a) The specific techniques used by the therapist (b) Common factors such as the therapeutic alliance and empathy (c) The theoretical orientation of the clinician (d) The specific diagnosis of the patient

    Answer: (b) Common factors such as the therapeutic alliance and empathy. The verdict (Rosenzweig 1936; named by Luborsky, Singer and Luborsky 1975) holds that bona fide therapies produce roughly equal outcomes because they share common factors. Specific techniques and theoretical orientation are exactly what it says do not make the difference, and diagnosis is not the explanation offered.

23. 'Healing and Persuasion' is the work of: (IHBAS 2020)

    (a) Alfred Pritz (b) Jerome Frank (c) Charles Winnicot (d) Jeff Klerman

    Answer: (b) Jerome Frank. The book is *Persuasion and Healing* (1961), its title reversed in the question: all healing restores morale through a confiding relationship, a healing setting, a rationale and a ritual. Gerald Klerman co-developed interpersonal psychotherapy, Donald Winnicott was an object-relations analyst and Alfred Pritz was founding president of the World Council for Psychotherapy.

24. All of the following are the characteristics of efficacy studies EXCEPT (RML 2020)

    (a) Therapist actions are controlled by a protocol or treatment manual (b) Therapist flexibility is encouraged (c) The treatment is applied to a homogenous client population (d) The research design relies on random selection and random assignment to a treatment or control condition

    Answer: (b) Therapist flexibility is encouraged. Efficacy studies test a treatment under ideal, controlled conditions: manualised therapists, homogeneous samples and random assignment. Flexible, client-tailored therapy belongs to effectiveness research in routine practice.

25. Which model explains health behaviour by focussing on perceived susceptibility, severity, benefits and barriers? (JEMAS 2025)

    (a) Health Belief Model (b) Theory of Planned Behaviour (c) Social Cognitive Theory (d) Transtheoretical Model

    Answer: (a) Health Belief Model. Susceptibility, severity, benefits and barriers (with cues to action and, later, self-efficacy) are its signature constructs. The theory of planned behaviour uses attitude, subjective norm and perceived behavioural control; social cognitive theory centres on self-efficacy; the transtheoretical model describes stages of change.

26. Which of the following factor is most directly linked to a higher risk for coronary heart disease? (RML 2024)

    (a) Type A personality (b) Type B personality (c) Hostility (d) Self-efficacy

    Answer: (a) Type A personality (as keyed). Friedman and Rosenman's Type A pattern (time urgency, competitiveness, hostility) predicted coronary heart disease, but later research found hostility to be the toxic component, so choose hostility when a question asks for the component. Type B is relaxed and lower-risk; self-efficacy is not a risk factor.

27. Which therapy is mostly needed in 'Crisis Management'? (IHBAS 2020)

    (a) Cognitive behavior therapy (b) Acceptance & commitment therapy (c) Expressive therapy (d) Supportive therapy

    Answer: (d) Supportive therapy. Crisis intervention is brief, active and here-and-now: it listens, reassures, solves immediate problems and mobilises support to restore the previous level of functioning. CBT, acceptance and commitment therapy and expressive therapy are structured or exploratory treatments, not the first response to a crisis.

28. A parent deliberately ignores a child's temper tantrums in an attempt to discourage them. The parent's strategy is in accordance with: (Amity 2023)

    (a) Negative reinforcement (b) Extinction (c) Operant escape (d) Operant avoidance

    Answer: (b) Extinction. The source key gives (c), operant escape, which is wrong. Ignoring removes the attention that maintained the tantrums, so they weaken: that is extinction. Escape and avoidance are both forms of negative reinforcement, which strengthen behaviour.

29. Which of the following disabilities is explicitly covered under the scope of The National Trust Act (1999) in India for the welfare of persons with the disability? (Real PYQ 2024)

    (a) Autism Spectrum Disorder (b) Specific Learning Disability (c) Intellectual Disability (Mental Retardation) (d) Both (a) and (c)

    Answer: (d) Both (a) and (c). The National Trust Act 1999 covers exactly four conditions: autism, cerebral palsy, mental retardation (intellectual disability) and multiple disabilities. Specific learning disability is a specified disability under the RPwD Act 2016, not under the National Trust Act.

30. Infantile autism was first described by : (IHBAS 2025)

    (a) Hans Asperger (b) Leo Kanner (c) Eugen Bleuler (d) Kurt Schneider

    Answer: (b) Leo Kanner. Kanner described early infantile autism in eleven children in 1943. Asperger described more able children a year later, Bleuler coined the word autism in 1911 for withdrawal in schizophrenia, and Kurt Schneider is known for the first-rank symptoms of schizophrenia.

High-yield sheet

The facts this chapter's questions ask most, most-asked first. Last read before the exam.

1 **Systematic desensitisation** (Joseph Wolpe, *Psychotherapy by Reciprocal Inhibition*, 1958) rests on **reciprocal inhibition**: Jacobson's progressive relaxation is paired, step by step, with a graded anxiety hierarchy rated in SUDS (0 to 100); its main use is phobias. ASKED 21 TIMES

2 **Marsha Linehan**'s dialectical behaviour therapy (1993) was built for borderline personality disorder with chronic suicidality; its four skills modules are mindfulness, distress tolerance, emotion regulation and interpersonal effectiveness. ASKED 10 TIMES

3 **Carl Rogers**'s three core conditions are **empathy, unconditional positive regard and congruence** (genuineness), conditions 3 to 5 of his six necessary and sufficient conditions (1957); interpretation is not one of them. ASKED 10 TIMES

4 In **Albert Ellis**'s ABC model, A is the activating event, B the belief about it and C the emotional and behavioural consequence; B, not A, mainly causes C, and D is disputing the irrational belief. ASKED 9 TIMES

5 The **token economy** (Teodoro Ayllon and Nathan Azrin, 1968) is operant conditioning with tokens as secondary reinforcers, so it is the listed therapy NOT derived from classical conditioning. ASKED 8 TIMES

6 **Aaron Beck**'s cognitive triad in depression is a negative view of the self, the world and the future. ASKED 8 TIMES

7 The **empty chair** technique belongs to **Gestalt therapy** (Fritz Perls), where the client speaks to an absent person or a part of the self to finish unfinished business. ASKED 8 TIMES

8 **Exposure and response prevention** (Victor Meyer, 1966; refined by Edna Foa) is the first-line behavioural treatment for OCD and works through habituation and extinction. ASKED 7 TIMES

9 **EMDR** (Francine Shapiro, published 1989) was developed for PTSD: the client holds the traumatic memory in mind while following bilateral eye movements. ASKED 7 TIMES

10 **Transference** is the patient's unconscious redirection onto the therapist of feelings belonging to significant figures from the past; **countertransference** (named by Freud in 1910) is the therapist's emotional reaction to the patient arising from the therapist's own unresolved conflicts. ASKED 6 TIMES

11 **Viktor Frankl**'s **logotherapy** is an existential therapy built on the **will to meaning**; its frustration causes noogenic neurosis, and dereflection counters hyper-reflection. ASKED 6 TIMES

12 **Motivational interviewing** was developed by **William Miller and Stephen Rollnick** for substance use; its core skills are **OARS** (open questions, affirmations, reflections, summaries) and its spirit **PACE** (partnership, acceptance, compassion, evocation). ASKED 6 TIMES

13 The **diathesis-stress model** explains a disorder as a predisposing vulnerability (the diathesis) triggered by environmental stress; Zubin and Spring formalised it as the vulnerability-stress model in 1977. ASKED 5 TIMES

14 The **Mental Healthcare Act 2017** lets adults make an advance directive and appoint a nominated representative, presumes severe stress in anyone who attempts suicide (section 115) and bans ECT without anaesthesia and muscle relaxant (section 95). ASKED 5 TIMES

15 **Free association**, obeying the **fundamental rule** to say whatever comes to mind without censorship, is the central technique of Freud's psychoanalysis. ASKED 5 TIMES

16 **Interpersonal psychotherapy** (Klerman and Weissman, 1984) treats depression in 12 to 16 sessions through four problem areas: grief, role disputes, role transitions and interpersonal deficits. ASKED 5 TIMES

17 The **Premack principle** (David Premack, 1959): a high-probability behaviour reinforces a low-probability one, as when video games follow homework. ASKED 5 TIMES

18 Socratic questioning (guided discovery) helps the client discover previously overlooked information for himself rather than receive the therapist's interpretation; with collaborative empiricism it is a fundamental concept of Beck's cognitive therapy, whereas free association is not. ASKED 5 TIMES

19 **Salvador Minuchin**'s **structural family therapy** works on subsystems, hierarchy and boundaries: **diffuse** boundaries make an **enmeshed** family (over-involvement, little autonomy) and **rigid** boundaries a **disengaged** one; its techniques are joining, enactment and unbalancing. ASKED 5 TIMES

20 The **Rehabilitation Council of India**, a registered society from 1986, became a statutory body under the RCI Act 1992 (in force 1993); it works under the Ministry of Social Justice and Empowerment, and clinical psychologists must register with it to practise. ASKED 4 TIMES

21 **Informed consent** is agreement after being told a study's or treatment's purpose, procedures, risks and the right to withdraw; **debriefing** is the full explanation of a study's true purpose given after participation. ASKED 4 TIMES

22 **Alfred Adler**'s individual psychology rests on organ inferiority (1907), the inferiority complex and striving for superiority; its techniques (encouragement, acting as if, spitting in the client's soup, the push-button technique) do not include analysis of transference. ASKED 4 TIMES

23 **Object relations** are internalised representations of self and others and the relationships between them; Klein's paranoid-schizoid position rests on **splitting**, and Kohut traced an empty, fragmented self to failed parental mirroring and idealising. ASKED 4 TIMES

24 **Repression** is the primary defence and the psychodynamic explanation of dissociative symptoms; **projection** attributes one's own unacceptable feelings to someone else; **cathexis** is psychic energy invested in a particular object, person or idea. ASKED 4 TIMES

25 **Flooding** is prolonged exposure to the most feared stimulus at full intensity, with no hierarchy, no relaxation and no escape, until the anxiety subsides. ASKED 4 TIMES

26 Rational emotive therapy (1955), renamed rational emotive behaviour therapy in 1993, is **Albert Ellis**'s; challenging absolute musts (musturbation) is its signature. ASKED 4 TIMES

27 **Steven Hayes**'s acceptance and commitment therapy aims at psychological flexibility through the six hexaflex processes: acceptance, cognitive defusion, present moment, self-as-context, values and committed action. ASKED 4 TIMES

28 A person-centred therapist relies on **reflection of feeling**, not advice or interpretation, and speaks of the real relationship rather than Bordin's working alliance. ASKED 4 TIMES

29 **Maxwell Jones** pioneered the **therapeutic community** at Belmont Hospital (later the Henderson Hospital): staff and patients share decisions democratically in daily community meetings. ASKED 4 TIMES

30 The **double bind** (Bateson, Jackson, Haley and Weakland, 1956) is repeated contradictory messages at two levels from a vital figure, with no way to comment or escape; it was proposed as a cause of schizophrenia and is now rejected. ASKED 4 TIMES

CHAPTER 8

Psychiatric Disorders

IN THIS CHAPTER

8.1 ●● Abnormality, Causes and the Classification of Mental Disorders: Define abnormality and the basic terms of psychopathology, explain how biological, psychological and diathesis-stress models account for disorder, and use the DSM-5-TR, ICD-10 and ICD-11 classifications and the cultural concepts of distress.

8.2 ●●● Descriptive Psychopathology: Signs and Symptoms: Define and recognise the signs and symptoms of mental disorder function by function (consciousness, perception, thought, belief, memory, emotion, movement, insight and judgement), including the named delusional syndromes and the disorders each phenomenon points to.

8.3 ●● Neurodevelopmental Disorders: Recognise and tell apart the DSM-5-TR neurodevelopmental disorders (intellectual disability, communication disorders, autism, ADHD, specific learning disorder, and motor and tic disorders) by their criteria, history and causes.

8.4 ●●● Schizophrenia Spectrum and Other Psychotic Disorders: Trace the concept of schizophrenia from Kraepelin, Bleuler and Schneider, distinguish positive, negative and cognitive symptoms, apply the DSM-5-TR, ICD-10 and ICD-11 criteria and the 1 day, 1 month, 6 months duration ladder across the psychotic spectrum, and explain its genetic, dopaminergic, neurodevelopmental and family theories and the side effects of antipsychotic drugs.

8.5 ●●● Depressive and Bipolar Disorders, and Suicide: Define the major depressive, manic and hypomanic episodes, diagnose the depressive and bipolar disorders and their specifiers in DSM-5-TR, ICD-10 and ICD-11, explain the biological, psychological and social causes and first-line drugs, and judge suicide risk.

8.6 ●● Anxiety and Obsessive-Compulsive and Related Disorders: Distinguish fear from anxiety, state the DSM-5-TR features and duration rules of the anxiety disorders, panic attacks, OCD and its related disorders, and explain their biological, learning, cognitive and psychodynamic causes, including the defences of phobia and OCD.

8.7 ●●● Trauma, Dissociative and Somatic Symptom Disorders: Trace hysteria into its modern heirs and diagnose and tell apart PTSD, acute stress, adjustment, grief and attachment disorders, the dissociative disorders, and the somatic symptom disorders, factitious disorder and malingering.

8.8 ● Feeding, Eating, Elimination and Sleep-Wake Disorders: Diagnose and tell apart the DSM-5-TR feeding and eating, elimination and sleep-wake disorders by their criteria, durations, ages, medical signs and timing in the night.

8.9 • Sexual Dysfunctions, Paraphilic Disorders and Gender Dysphoria: Describe the normal sexual response cycle, classify the DSM-5-TR sexual dysfunctions, paraphilic disorders and gender dysphoria, and explain how ICD-11 and the history of homosexuality changed what counts as a sexual or gender disorder.

8.10 •• Substance-Related, Addictive and Neurocognitive Disorders: Define the core terms of substance use, classify substance-related and addictive disorders in ICD-10, DSM-5-TR and ICD-11, recognise the ten drug classes, and tell apart delirium, the dementias, pseudodementia, Wernicke-Korsakoff syndrome and psychiatric syndromes of medical illness.

8.11 •• Personality Disorders, Conduct and Impulse-Control Disorders: Define a personality disorder, place the ten DSM-5-TR personality disorders in their clusters with their criteria counts, recognise borderline, antisocial, histrionic, narcissistic and obsessive-compulsive personality and their look-alikes, describe the ICD-11 and AMPD dimensional models, and distinguish oppositional defiant disorder, conduct disorder, intermittent explosive disorder, pyromania and kleptomania.

Weightage. 411 of the 3180 questions mapped in this book (12.9%) test this chapter, 205 of them from real past papers. 4 of its 11 sections are High yield.

Heidelberg, 1891. **Emil Kraepelin**, aged thirty-five, takes up the chair of psychiatry at the university clinic. Years earlier he had worked in Wilhelm Wundt's laboratory in Leipzig, timing how alcohol and other drugs changed simple mental acts, and he brings a laboratory habit with him: record everything, then count. The psychiatry he inherits badly needs it. Diagnostic labels vary from one asylum to the next, and almost all of them describe how a patient looks on the day of admission. None says what will happen to the patient next.

Kraepelin's tool was humble: a card. For every patient admitted, a card recorded the history, the symptoms and a provisional diagnosis. The cards were not filed and forgotten. As months and years passed, the clinic added what became of each patient: recovery, relapse, or a slow slide into apathy and disordered thinking. Sorting the cards again, by outcome rather than by the picture at admission, Kraepelin could see which first impressions had been wrong, and the psychoses fell into two great families. In one, illness began in youth and ran a deteriorating course; he called it *dementia praecox*, premature dementia. In the other, attacks of mania or melancholia came and went, with recovery between them: *manic-depressive insanity*. The sixth edition of his textbook (1899) drew the line.

The idea on those cards is the idea of this chapter. A mental disorder, for Kraepelin, was a recognisable cluster of signs and symptoms with its own onset, course and outcome, and a classification was good if it predicted what would happen to the patient. When the American Psychiatric Association rebuilt its manual around explicit criteria in 1980, the psychiatrists behind the change were nicknamed the neo-Kraepelinians. Every disorder in the pages that follow is, in a sense, one of his piles of cards, redrawn by a century of research.

8.1 Abnormality, Causes and the Classification of Mental Disorders

MEDIUM YIELD · 30 QS

Every disorder in this chapter rests on three prior questions. When does unusual behaviour become a disorder? Why does one person fall ill while another does not? And how do clinicians everywhere agree on what to call what they see? This section answers each in turn, then shows how culture shapes distress itself. (The history of the asylums, and of the models as guides to treatment, is told in Section 7.1.)

What counts as abnormal ●●

No single test separates normal from abnormal, so clinicians weigh the **four Ds** together. **Deviance** takes two forms. The **statistical criterion** calls abnormal whatever deviates from typical patterns; its flaw is that rare talent, a genius IQ, is just as deviant. The **social-norm criterion** calls abnormal whatever violates a society's explicit and implicit norms; its flaw is **cultural relativity**, for a woman in possession trance at a temple festival is following custom, not breaking it. **Distress** is personal unhappiness and suffering, though a person in mania may feel wonderful. **Dysfunction** is maladaptiveness: behaviour that stops you working, studying or keeping relationships. **Danger** to self or others is the least common.

Marie Jahoda (1958) described ideal mental health positively, as self-acceptance, growth, integration, autonomy, accurate perception of reality and environmental mastery. **Jerome Wakefield** (1992) defined disorder as **harmful dysfunction**: a mental mechanism fails at the job evolution designed it for (a fact), and the failure harms the person by the standards of their culture (a value). In DSM-5-TR, a **mental disorder** is a syndrome of clinically significant disturbance in cognition, emotion regulation or behaviour that reflects a dysfunction in the underlying psychological, biological or developmental processes, usually with significant distress or disability. An expectable or culturally approved response to a common stressor or loss, such as the death of a loved one, is excluded, and so is socially deviant behaviour (political, religious or sexual) unless it results from a dysfunction in the person.

Daniel Offer and **Melvin Sabshin** (1966) described four views of normality. **Normality as health**, the medical view, is the absence of manifest psychopathology; judged from the person's own side it becomes the **subjective model**: no distress, no disability, no help-seeking. **Normality as utopia** is ideal, optimal functioning; **normality as average** is the statistical middle of the bell curve; **normality as process** is the product of systems that change across the lifespan.

> EXAM TRAP
>
> Statistical abnormality is deviation from typical patterns, not intense anxiety, loss of contact with reality or withdrawal. Optimal functioning is utopia; no distress, disability or help-seeking is the subjective model; absence of any disorder is the medical view; socially acceptable behaviour is the norm criterion.

The language of psychopathology ●●

Psychopathology is the scientific study of mental disorders; **abnormal psychology** is the branch of psychology that studies abnormal behaviour and experience, and the terms largely overlap. A **symptom** is what the patient reports (low mood), a **sign** what the clinician observes (tearfulness), and a **syndrome** a cluster of signs and symptoms that occur together (8.2). **Aetiology** is the study of causes. **Onset** is **acute** (sudden) or **insidious** (gradual). **Course** may be **acute** (short), **chronic** (long-lasting), **episodic** (recurring with recovery between, as in mood disorders) or **time-limited** (improving untreated within a short period). **Prognosis** is the likely course and outcome: a forecast, not the history, the presenting symptoms or the diagnostic criteria.

Prevalence is the proportion of a population that has a disorder, on one day (**point**), over a stated period such as a year (**period**) or ever (**lifetime**); **incidence** counts only new cases arising in a period. **Comorbidity** is the co-occurrence of two or more differently defined disorders in one person, such as depression with an anxiety disorder. In psychiatry it is the rule, not the exception.

> EXAM TRAP
>
> Comorbidity is not the frequency of mental illness (that is prevalence), not two or more causes of death, not the degree of suffering, and not dying together or fearing death. "Depression always in winter" garbles the seasonal pattern specifier (8.5).

Causes: from single models to diathesis-stress ●●●

Early theories were **one-dimensional**, tracing a disorder to a single cause. The modern view is **multidimensional** (integrative): biological, psychological, social and cultural influences interact. Table 8.1 gives each classical model's answer.

Table 8.1 Models of psychopathology and the causes they blame.

Model	Cause of disorder
Biological	Genes, brain structure, neurotransmitters
Psychodynamic	Unconscious conflict from emotional trauma in childhood; fixation; defences (Chapter 4)
Behavioural	Maladaptive (inappropriate) learning
Cognitive	Disturbed thought patterns and schemas (Beck and Ellis, 7.5)
Humanistic-existential	Failure to live up to one's potential; incongruence
Sociocultural	Poverty, gender, culture, stigma, labelling
Family systems	Disturbed family interaction

Biological research rests on **family**, **twin** and **adoption studies**. Twin studies compare **concordance**, the rate at which both twins are affected, in identical and fraternal pairs; adoption studies separate genes from upbringing. Together they estimate **heritability**, the share of variation in a population due to genetic differences. Disorders are **polygenic**: many small-effect genes add up to a polygenic risk score. In **gene-environment interaction** a gene matters only in some environments: **Avshalom Caspi** (2003) found that a short serotonin transporter variant raised adult depression mainly after childhood maltreatment. In **gene-environment correlation**, the reciprocal gene-environment model, genes make certain environments likelier, as when an impulsive temperament invites the very stressors that trigger the vulnerability. Neurotransmitter (dopamine) and brain-structure (enlarged ventricles) models complete the strand.

Because there is no single "schizophrenia gene", researchers seek **endophenotypes**: the genetic mechanisms that ultimately produce the problems behind the symptoms, measurable intermediate traits such as the eye-tracking and working-memory deficits of schizophrenia. **Irving Gottesman** and **James Shields** brought the term into psychiatry in the 1970s; Gottesman and **Todd Gould** revived it in 2003. Contrast the **genotype**, a person's genetic make-up, the **phenotype**, the observable disorder, and **molecular genetics**, a method that studies DNA directly.

The **diathesis-stress model** joins the strands: a predisposition, the **diathesis** (genetic, biological or psychological), lies dormant until a stressor sets it off, and neither alone suffices. **Paul Meehl** (1962) proposed an inherited **schizotaxia** that under stress could become schizophrenia; **Joseph Zubin** and **Bonnie Spring** (1977) recast it as the **vulnerability-stress model**: the more vulnerable you are, the less stress tips you into illness (Section 7.1 draws the threshold). Broadest is the **biopsychosocial model** of **George L. Engel** (1977): illness arises from interacting biological, psychological and social factors.

EXAM TRAP

Any "predisposition plus trigger" or "vulnerability plus stress" wording is diathesis-stress; genes, stress, learning, unconscious conflict or the supernatural alone are wrong. The **neurodevelopmental model** traces schizophrenia to faulty early brain development (8.4); "clinical-reductionist" and "syndrome-sign" are not models. Biopsychosocial is Engel's, not Kraepelin's, Bleuler's (8.4) or Beck's.

EXAM TRAP

Asked which disorders show strong biological and "little to no" psychosocial evidence, choose schizophrenia and bipolar disorder, the most heritable (bipolar heritability reaches about 90 percent in some twin studies), not bulimia nervosa, in which cultural pressure for thinness weighs heavily (8.8). But the premise is false: identical-twin concordance for bipolar disorder is only 40 to 70 percent and life events trigger relapses, while childhood adversity, migration and urban upbringing raise the risk of schizophrenia. Genes set the diathesis; stress still times the episodes.

Approaches to classification

Hippocrates (about 460 to 377 BC) sorted mental illness into **mania**, **melancholia** and **phrenitis** (feverish delirium), and coined the word **hysteria**. Modern classification begins with **Emil Kraepelin** (1856 to 1926), who classified descriptively by symptoms and above all by course, separating deteriorating dementia praecox from episodic manic-depressive insanity (8.4).

The **syndromal approach** classifies disorders by clinical signs and symptoms that cluster together; the **historical approach** (aetiological) classifies them by known cause, as medicine does with tuberculosis. Since most causes are unknown, the DSM and ICD are syndromal. A **categorical approach** says you have a disorder or you do not; a **dimensional approach** rates everyone on continuous scales; a **prototypical approach**, the DSM's, lists the features of a typical case and tolerates variation (five of nine symptoms make a major depressive episode, so two patients may share only one).

Reliability is agreement between clinicians who assess the same patient, measured by Cohen's kappa (agreement corrected for chance); **validity** is whether a category is real and useful, predicting course and treatment response. A diagnosis can be reliable but invalid, never the reverse. A **false positive** diagnoses a normal person as disordered; a **false negative** misses a real disorder. Loose criteria breed false positives, strict ones false negatives. Every diagnosis is also a **label** that can bring **stigma** (Section 7.2 tells the pseudopatient study).

EXAM TRAP

Syndromal means signs and symptoms; historical means known aetiology; option sets swap them. A high false-positive rate means normal people are diagnosed as disordered (criteria too loose, not too strict); missed cases are false negatives; low reliability is rater disagreement.

The DSM ●●

The **Diagnostic and Statistical Manual of Mental Disorders** (DSM) is published by the **American Psychiatric Association** (APA). Figure 8.1 sets its editions beside the ICD's. **DSM-I** (1952) and **DSM-II** (1968) were short glossaries of descriptions. **DSM-III** (1980), directed by **Robert Spitzer** (1932 to 2015), brought explicit diagnostic criteria, a descriptive stance neutral about causes, and the multiaxial system; **DSM-III-R** (1987), **DSM-IV** (1994) and **DSM-IV-TR** (2000) followed. **DSM-5** (2013) took an Arabic numeral to allow numbered updates. The edition in use, **DSM-5-TR** (2022), added **prolonged grief disorder** and codes for suicidal behaviour and non-suicidal self-injury.

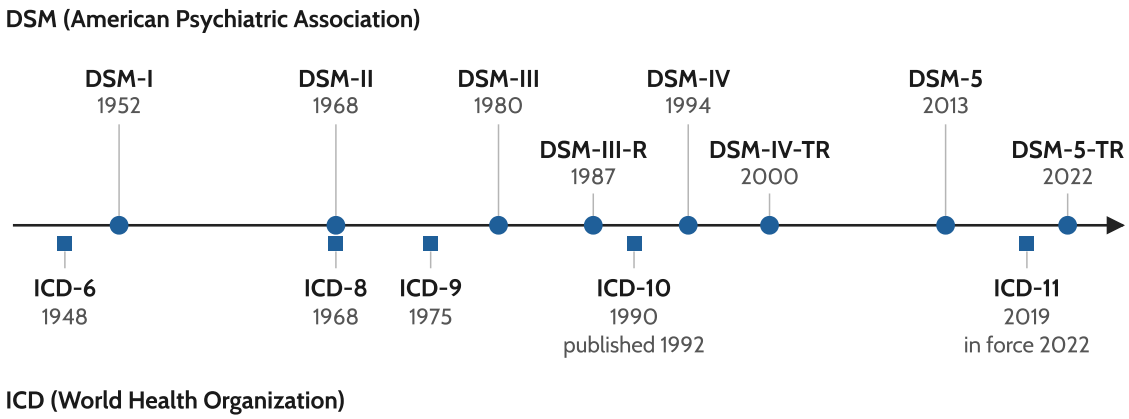

Figure 8.1 DSM editions (circles, above the line) and ICD editions (squares, below). ICD-11, adopted in 2019, took effect in 2022, the year of DSM-5-TR.

From DSM-III to DSM-IV-TR, diagnosis ran on five axes: **Axis I**, clinical disorders; **Axis II**, personality disorders and mental retardation; **Axis III**, general medical conditions; **Axis IV**, psychosocial and environmental problems; **Axis V**, the Global Assessment of Functioning (GAF). **DSM-5 abolished the multiaxial system**: Axes I to III are listed together, psychosocial problems take Z codes, and the GAF gave way to the WHO Disability Assessment Schedule (WHODAS 2.0).

DSM-5-TR has three sections. **Section I** explains how to use it. **Section II** holds the criteria and codes: 20 disorder chapters in rough lifespan order, from neurodevelopmental through paraphilic to other mental disorders, then medication-induced movement disorders and **other conditions that may be a focus of clinical attention**, problems that are not mental disorders, such as relational problems, malingering and borderline intellectual functioning. **Subtypes** are mutually exclusive ("specify whether"), **specifiers** can be combined ("specify if"), and **severity** is rated mild, moderate or severe. A presentation short of full criteria is **other specified** (reason recorded) or **unspecified** (reason not given). **Section III**, "Emerging Measures and Models", holds cross-cutting symptom measures, the WHODAS 2.0, the Cultural Formulation Interview, the Alternative Model for Personality Disorders (8.11) and the **conditions for further study**, proposed criteria sets not approved for clinical use.

REMEMBER: THE SEVEN CONDITIONS FOR FURTHER STUDY

Attenuated psychosis syndrome (milder psychotic symptoms at least weekly for a month); depressive episodes with short-duration hypomania; caffeine use disorder; internet gaming disorder; neurobehavioural disorder associated with prenatal alcohol exposure; suicidal behaviour disorder; and **non-suicidal self-injury disorder** (intentional cutting, burning, stabbing or hitting of one's own body on 5 or more days in a year, without suicidal intent). DSM-5's persistent complex bereavement disorder became DSM-5-TR's prolonged grief disorder (8.7).

EXAM TRAP

An option "DSM-V" means DSM-5: choose it for the version in use, knowing the current text is DSM-5-TR (2022). DSM-5 abolished the axes; it added no sixth axis and still gives diagnoses, with ICD-10-CM codes beside them. Axis III is general medical conditions, not personality disorders (II) or psychosocial problems (IV).

The ICD

The **International Classification of Diseases** (ICD) is the **World Health Organization**'s (WHO) classification of all diseases. **ICD-6** (1948) was the first to include mental disorders. **ICD-10** (endorsed in 1990, published in 1992, used from 1994) put them in Chapter V (F), with the *Clinical Descriptions and Diagnostic Guidelines* (the "blue book", also 1992) and the *Diagnostic Criteria for Research* (1993). Its blocks (Table 8.2) still fill Indian case files.

Table 8.2 ICD-10 Chapter V (F): the blocks.

Block	Disorders
F00–F09	Organic, including symptomatic, mental disorders (delirium, dementia)
F10–F19	Mental and behavioural disorders due to psychoactive substance use
F20–F29	Schizophrenia, schizotypal and delusional disorders
F30–F39	Mood (affective) disorders
F40–F48	Neurotic, stress-related and somatoform disorders
F50–F59	Behavioural syndromes associated with physiological disturbances and physical factors
F60–F69	Disorders of adult personality and behaviour
F70–F79	Mental retardation
F80–F89	Disorders of psychological development
F90–F98	Behavioural and emotional disorders with onset in childhood and adolescence
F99	Unspecified mental disorder

ICD-11, adopted in May 2019, took effect on 1 January 2022. Mental disorders form **Chapter 6**, "Mental, behavioural or neurodevelopmental disorders" (codes 6A00 to 6E8Z). Sleep-wake disorders moved to Chapter 7, and sexual dysfunctions and gender incongruence to Chapter 17, "Conditions related to sexual health" (8.9). New are complex PTSD, prolonged grief disorder, gaming disorder, compulsive sexual behaviour disorder and bodily distress disorder, and a single **personality disorder** graded by severity (8.11). Table 8.3 compares the systems.

Table 8.3 DSM and ICD compared.

Feature	DSM	ICD
Publisher	American Psychiatric Association	World Health Organization
Scope	Mental disorders only	All diseases
Purpose	Explicit criteria for clinicians and researchers	Global health statistics and clinical guidelines
Cost	Sold by the APA	Free
Codes	ICD-10-CM codes (United States)	F00 to F99; ICD-11 6A00 to 6E8Z
Use	United States; research worldwide	Official worldwide; India's health system and Mental Healthcare Act 2017

EXAM TRAP

Delirium and dementia are F00-F09 (organic), not F10-F19 (substances); mood is F30-F39. DSM and ICD are classification and diagnostic systems, not tests, therapies or intelligence scales.

Culture and diagnosis ●●

DSM-IV's glossary of **culture-bound syndromes** became, in DSM-5, **cultural concepts of distress**, because all distress is locally shaped. They are of three kinds: **cultural syndromes** (symptom clusters in particular groups), **cultural idioms of distress** (shared ways of voicing suffering, such as "thinking too much") and **cultural explanations** (perceived causes, such as spirits or semen loss). Section III's **Cultural Formulation Interview** (CFI) elicits them in 16 questions. **N. N. Wig** coined the term dhat syndrome in 1960; Table 8.4 lists it with the others.

Table 8.4 Cultural concepts of distress.

Syndrome	Region	Core feature
Dhat	Indian subcontinent, Sri Lanka	Anxiety and weakness blamed on semen loss
Koro	South and East Asia, north-east India	Genitals retracting into the body, feared fatal; epidemics
Suchi-bai	Bengal	Obsessional cleanliness
Possession syndrome	Rural India	Possession by a deity or dead relative's spirit
Bhanmati sorcery	South India	Illness blamed on sorcery
Gilhari syndrome	Rajasthan	Squirrel-like swelling creeping to the neck, feared fatal
Jhinjhinia	West Bengal	Tingling and numbness rising from the legs
Latah	Malaysia, Indonesia	Exaggerated startle to minimal stimuli; echolalia, echopraxia, automatic obedience, obscenities
Amok	Malaysia	Brooding, violent outburst, amnesia
Pibloktoq	Arctic and subarctic Inuit	Dissociative fit: screaming, tearing off clothes, running into snow; amnesia

Syndrome	Region	Core feature
Susto	Latin America	Fright drives the soul from the body
Ataque de nervios	Latin America, Caribbean	Shouting, crying, trembling; loss of control
Brain fag	West Africa (Nigeria)	Students: poor concentration and thinking, head pain
Taijin kyofusho	Japan	Fear that one's body or appearance offends others
Hwa-byung	Korea	Suppressed anger
Shenjing shuairuo	China	Neurasthenia: fatigue, poor concentration
Windigo	Algonquian peoples	Fear of becoming a cannibal
Zar	North Africa, Middle East	Spirit possession
Kufungisisa	Zimbabwe	"Thinking too much"
Ghost sickness	American Indian groups	Preoccupation with death and the dead
Falling out	Southern United States, Caribbean	Sudden collapse, unable to see
Mal de ojo	Mediterranean, Latin America	"Evil eye" illness in children

EXAM TRAP

Dhat, koro, suchi-bai, possession syndrome, bhanmati, gilhari and jhinjhinia are reported in India; **latah is not**. Startle with obscenities is latah, brooding then violence amok, Inuit running into the snow pibloktoq, soul loss susto. Together they are cultural concepts of distress, not anxiety subtypes, substance-induced or dissociative disorders.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The criterion of abnormality that centres on personal unhappiness is _____.
2. Wakefield defined a mental disorder as a harmful _____.
3. The co-occurrence of two or more differently defined disorders in one person is _____.
4. Measurable intermediate traits between genes and symptoms are _____.
5. Zubin and Spring's 1977 version of the diathesis-stress model is the _____ model.
6. The DSM edition that introduced explicit criteria and five axes, in 1980, was _____.
7. In ICD-10, delirium and dementia fall in block _____.
8. The Malaysian syndrome of exaggerated startle, echolalia and blurted obscenities is _____.

Answers: (1) distress, (2) dysfunction, (3) comorbidity, (4) endophenotypes, (5) vulnerability-stress, (6) DSM-III, (7) F00-F09, (8) latah.

8.2 Descriptive Psychopathology: Signs and Symptoms HIGH YIELD · 45 QS

A psychiatric diagnosis is built from small units: a voice nobody else hears, a belief no argument can shift, a smile at a funeral. This section names and defines those units, one mental function at a time. Section 7.2 shows how the mental status examination draws them out in an interview; the disorders they build begin in Section 8.3.

Jaspers and the descriptive method

Descriptive psychopathology is the precise description and classification of abnormal experiences and behaviour, without claims about their cause. Its founding text is *General Psychopathology* (1913) by the German psychiatrist and philosopher **Karl Jaspers** (1883 to 1969), who brought **phenomenology** into psychiatry: the clinician describes an experience as the patient lives it, setting theory aside. The method is **empathic understanding**: careful questioning until you grasp what the experience is like. Where empathy fails, where an experience cannot be understood from the person's life and mood, Jaspers suspected a disease process.

Three distinctions run through everything that follows. The **form** of a symptom is its type (a hallucination); its **content** is what it is about (a voice calling him a thief). Culture colours the content, but the form carries the diagnosis. A **primary symptom** arises directly from the illness; a **secondary symptom** is an understandable reaction to one, as when a man who hears accusing voices concludes he is being watched. A **symptom** is a subjective complaint the patient reports (low mood); a **sign** is an objective abnormality the clinician observes (flat affect, slowed movement).

Consciousness, attention and orientation

Clouding of consciousness is the mildest lowering of awareness: drowsiness, poor attention and an incomplete grasp of the surroundings. **Confusion** adds muddled thinking and disorientation. **Delirium**, in DSM-5-TR terms, is a disturbance of attention and awareness that develops over hours to days, **fluctuates** during the day (lucid in the morning, worse at night) and brings other cognitive change, often with illusions and visual hallucinations; it signals a medical cause (Section 8.10). In **stupor** the person is mute and motionless and responds only to strong stimuli (catatonia, severe depression, organic states); in **coma** even pain brings no response. A **twilight state** is a narrowed consciousness lasting hours or days, in which complex, seemingly purposeful acts are carried out and later forgotten (after epileptic seizures; alcohol and dissociative states).

Distractibility is attention pulled away by trivial outside stimuli (mania, delirium). **Disorientation** is loss of awareness of time, place or person: **time is lost first**, then place, and one's own identity last.

Disorders of perception ●●

In a **sensory distortion** a real object is perceived but altered in intensity (sounds painfully loud), quality (everything tinged yellow) or size: **micropsia** (objects seen smaller) and **macropsia** (larger). In a **sensory deception** a new perception appears (Figure 8.2).

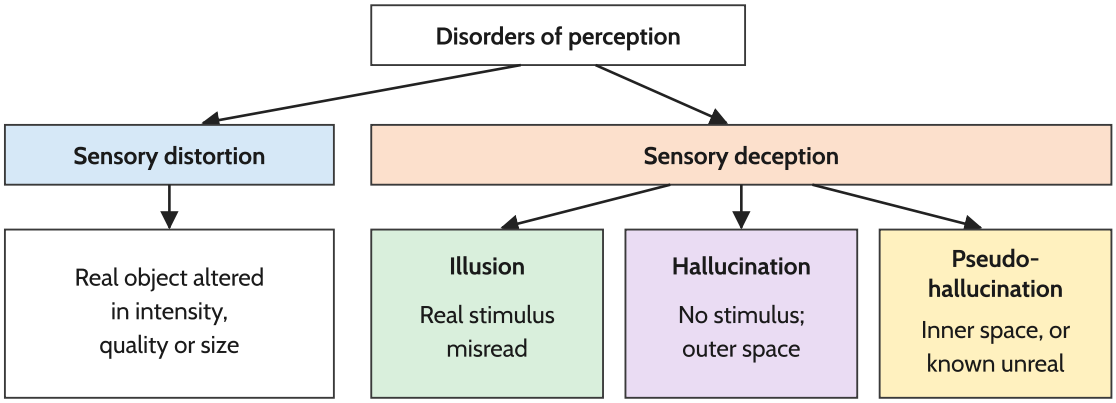

Figure 8.2 Disorders of perception. Only the hallucination needs no stimulus at all.

An **illusion** is a misperception of a real, existing stimulus. **Completion illusions** fill in missing parts (a misprint read as correct), **affect illusions** arise from emotion (a frightened person sees a coat on a hook as an intruder) and **pareidolic illusions** make vivid images from vague material (faces in clouds). A **hallucination** is a perception without an external stimulus, located in outer space and as vivid as a real perception. A **pseudohallucination** is located in inner space ("a voice inside my head") or is recognised as unreal. Hallucinations are most frequent in schizophrenia, and there mostly auditory; voices discussing or arguing about the patient in the third person, a running commentary on his actions and thought echo (hearing one's thoughts aloud) are first-rank symptoms (Section 8.4). Visual hallucinations suggest an organic cause (Table 8.5).

Table 8.5 Hallucinations by modality and special types.

Type	Experience	Typical of
Auditory, second person	A voice addresses the patient ("you are useless")	Psychotic depression, schizophrenia
Auditory, third person	Voices discuss, argue or comment; thought echo	Schizophrenia
Visual	Figures, animals, scenes	Delirium, Lewy body dementia, Charles Bonnet
Olfactory, gustatory	Smells, tastes	Temporal lobe epilepsy
Tactile; formication	Insects crawling under the skin	Cocaine, amphetamine, alcohol withdrawal
Somatic; kinaesthetic	Organs burning; limbs moving	Schizophrenia
Hypnagogic; hypnopompic	Falling asleep; waking	Normal people; narcolepsy
Functional	A real stimulus triggers one in the same sense (voices in a running tap)	Schizophrenia
Reflex	A stimulus in one sense, a hallucination in another	Schizophrenia, drugs
Extracampine	Outside the sensory field (someone behind one's head)	Schizophrenia, epilepsy
Autoscopic	Seeing oneself	Epilepsy, organic states
Lilliputian	Tiny people or animals	Delirium tremens

EXAM TRAP

A hallucination is a disorder of **perception**, not of thinking, judgement or intelligence. An illusion distorts a stimulus that exists; a hallucination has none. **Hypnagogic** hallucinations occur most commonly in **normal** people (DSM-5-TR counts them as normal experience), not in schizophrenia, delirium or drug dependence; narcolepsy also features them.

KEY STUDY: CHARLES BONNET'S GRANDFATHER (1760)

In 1760 the Swiss naturalist **Charles Bonnet** (1720 to 1793) described his elderly grandfather, almost blind from cataracts, who saw people, birds and buildings that were not there, while knowing they were unreal. **Charles Bonnet syndrome** now names complex visual hallucinations with significant visual loss, intact cognition and preserved insight. It is not a psychosis.

In **depersonalisation** the self feels unreal or changed ("as if I am watching myself in a film"); in **derealisation** the surroundings feel unreal or dreamlike. The person knows it only feels so. The disorder is in Section 8.7.

Disorders of thought ●●

Thought is described by its stream (speed and flow), form (how ideas connect), content (what is thought) and possession (whose the thoughts feel to be). In the stream, **pressure of thought** is a rapid crowding of ideas (mania); **poverty of thought** is a scarcity of slow, empty ones (depression, schizophrenia); **thought block** is a sudden stop in mid-sentence, the mind blank (schizophrenia). Diminished speech output (poverty of speech) is **alogia**.

Formal thought disorder is a disturbance in the links between ideas, inferred from speech (Table 8.6). In **flight of ideas** (mania) thought shifts rapidly from one idea to another before any is completed, but the links can be traced: chance, rhymes, puns and **clang associations** (words chosen for sound). In **loosening of associations**, or derailment (schizophrenia), the links are lost; the oblique jumps are called **knight's move thinking**.

Table 8.6 Disorders of the form of thought.

Term	What you hear	Typical of
Flight of ideas	Rapid jumps; links by chance, rhyme, pun, clang	Mania
Loosening of associations, derailment	Slips to unrelated topics; links lost	Schizophrenia
Tangentiality	Drifts off; never reaches the goal	Schizophrenia, mania
Circumstantiality	Needless detail; finally reaches the goal	Epilepsy, obsessional and organic states
Perseveration	Same idea repeated to new questions	Organic brain disease
Verbigeration	Senseless repetition of words or phrases	Schizophrenia
Neologism	New words, or old words with private meanings	Schizophrenia
Metonym	Approximate word for the exact one ("I ate three menus")	Schizophrenia
Word salad, incoherence	Words without connection; the most severe	Schizophrenia

Term	What you hear	Typical of
Concrete thinking	Literal; cannot abstract (proverbs)	Schizophrenia, organic states
Overinclusive thinking	Concept boundaries lost; irrelevant ideas included	Schizophrenia

Overinclusive thinking was described by **Norman Cameron** (1896 to 1975). Thought content includes preoccupations, **obsessions** (intrusive, unwanted thoughts recognised as one's own and resisted; Section 8.6), **phobias** (excessive fear of a specific object or situation), delusions and **overvalued ideas**, a term of **Carl Wernicke** (1848 to 1905): a single, understandable belief that dominates the person's life and is held with strong feeling but without delusional fixity, such as the anorexic patient's conviction that she is too fat.

In thought alienation, thoughts no longer feel one's own: in **thought insertion** they are put in from outside, in **thought withdrawal** taken out, and in **thought broadcasting** they pass out of the mind so that others know them. All three are first-rank symptoms of **Kurt Schneider** (1887 to 1967); the rest of his list is thought echo, voices arguing, running commentary, delusional perception, somatic passivity and made feelings, made impulses and made actions (Section 8.4). Many Indian texts also file **obsessions and compulsions** under possession: the thought is known to be one's own yet cannot be controlled.

> EXAM TRAP
>
> Flight of ideas, tangentiality, derailment, loosening and circumstantiality are disorders of **form** (process), not content (delusions, obsessions) or possession (insertion, broadcasting). Many Indian notes file flight of ideas, circumstantiality and tangentiality under stream; if form is offered, choose it. Word salad is the most severe formal thought disorder; repeating one idea is perseveration.

Delusions ●●●

A **delusion** is a false, firmly held belief, not shaken by reason or contrary evidence, and out of keeping with the person's cultural and educational background (a community's shared belief in the evil eye is not one). Jaspers named its marks: **certainty**, **incorrigibility** and **impossibility** of content. Delusions, hallucinations and grossly disorganised speech or behaviour mark **psychosis**, a loss of contact with reality. Table 8.7 separates the delusion from its neighbours.

A **primary delusion** (autochthonous) cannot be understood from any other experience. It appears as **delusional mood** (something sinister is going on), **delusional perception** (a normal perception suddenly carries a private meaning: the traffic light turns red, so he is chosen to save the world), **delusional memory** or a **sudden delusional idea**. A **secondary delusion** is understandable from another experience: guilt from depressed mood, persecution to explain hostile voices. **Systematised** delusions form a logically connected system around one theme. **Mood-congruent** delusions fit the mood (grandeur in mania); **mood-incongruent** ones do not. DSM-5-TR calls a delusion **bizarre** when it is clearly implausible, not understandable to same-culture peers and not drawn from ordinary life (his organs removed without a scar); a **non-bizarre** delusion could happen (being followed by the police).

A **delusion of persecution**, that one is being deliberately interfered with, plotted against, threatened or mistreated, is the **most common** type, mostly in schizophrenia and delusional disorder; **paranoid** means persecutory or self-referential, and **paranoia** was the old name for chronic, systematised delusional states. In a **delusion of reference** neutral events carry a personal message; an **idea of reference** is held without full conviction. In **delusions of control** (passivity) one's acts or feelings are made by an outside force. **Grandiose** delusions belong to mania; **nihilistic delusions** (oneself, one's organs or the world do not exist or are dead; DSM-5-TR words it as the conviction that a major catastrophe will occur) and delusions of **guilt** and **poverty** to severe depression. **Somatic** (hypochondriacal) delusions concern disease; jealousy, erotomania, infestation and misidentification have eponyms (Table 8.8).

Table 8.7 Delusion, overvalued idea and obsession compared.

Feature	Delusion	Overvalued idea	Obsession
Conviction	Absolute	Strong, emotional, not fixed	Recognised as senseless
Resisted	No	No	Yes
Understandable	No	Yes	Content often absurd
Example	Schizophrenia	Anorexia nervosa	Obsessive-compulsive disorder

EXAM TRAP

"Others are out to get me" is **persecution** (paranoid), the commonest delusion, most often in **schizophrenia**. Nihilistic delusions point to **depression**, not obsessive-compulsive or somatoform disorder.

Named delusional syndromes ●●●

Four delusional misidentification syndromes lead Table 8.8. **Joseph Capgras** (1873 to 1950) described the impostor delusion, **Capgras syndrome**, in 1923 with Jean Reboul-Lachaux as *l'illusion des sosies* ("the illusion of doubles"). **Jules Cotard** (1840 to 1889) described the nihilistic syndrome in 1880 and named it *délire des négations* in 1882; **Arnold Pick** (1851 to 1924) named reduplicative paramnesia in 1903. Erotomania carries the name of **Gaëtan Gatian de Clérambault** (1872 to 1934), whose classic account of it appeared in 1921. Paul Courbon and Gabriel Fail described Fregoli syndrome in 1927, naming it after the Italian quick-change actor Leopoldo Fregoli. John Todd and Kenneth Dewhurst named the Othello syndrome in 1955, after Shakespeare's jealous husband, and the Swedish neurologist **Karl-Axel Ekbom** (1907 to 1977) described delusional parasitosis in 1938.

Table 8.8 Named delusional and eponymous syndromes.

Syndrome	Core feature	Usual setting
Capgras	A familiar person, usually spouse or relative, replaced by an identical impostor	Schizophrenia; dementia, brain injury
Fregoli	Strangers are one familiar person (often a persecutor) in disguise	Schizophrenia, brain injury
Intermetamorphosis	People exchange identities, in appearance and personality	Psychosis, organic states
Subjective doubles	A double of oneself exists	Psychosis

Syndrome	Core feature	Usual setting
Reduplicative paramnesia	A familiar place or person exists in duplicate	Brain injury, dementia
Cotard	One is dead, does not exist, or has lost organs or blood	Severe psychotic depression
Othello	Morbid jealousy: the partner is unfaithful	Alcohol dependence
de Clérambault	Erotomania: a person of higher status secretly loves the patient	Delusional disorder
Ekbom	Delusional parasitosis: infested by insects or parasites	Delusional disorder, elderly
Couvade	A man develops pregnancy symptoms during his partner's pregnancy	Expectant fathers
Diogenes	Gross self-neglect, squalor and hoarding	Elderly people living alone
Ganser; folie à deux; Munchausen; Kleine-Levin	Approximate answers (Sigbert Ganser, 1898); shared delusion (Charles Lasègue and Jules Falret, 1877); factitious illness; recurrent hypersomnia	Sections 8.7, 8.4, 8.7, 8.8

EXAM TRAP

Capgras: a familiar person becomes an impostor. **Fregoli**: strangers become one familiar person. **Reduplicative paramnesia** duplicates a place. **Othello** is jealousy, **de Clérambault** love, **Cotard** being dead. Ekbom's name also labels **restless legs syndrome** (Section 8.8); in a delusion question, Ekbom means parasitosis.

Disorders of memory ●●●

Amnesia is a pathological loss of memory: **anterograde** (new memories cannot be stored after the onset) or **retrograde** (memories from before it are lost). Organic amnesia is in Section 8.10, **psychogenic** (dissociative) amnesia in Section 8.7, and normal memory and false memories in Section 1.7.

Among the paramnesias, **confabulation** fills memory gaps with false, often detailed and believable memories, without intent to deceive; it is the hallmark of Korsakoff syndrome (alcohol-related thiamine deficiency). **Pseudologia fantastica**, described by **Anton Delbrück** in 1891, is fantastic lying that the teller half believes (factitious disorder). **Retrospective falsification** unconsciously rewrites the past to fit present mood; in **cryptomnesia** a forgotten memory returns unrecognised, so it feels original; **hypermnesia** is exaggerated recall (mania, hypnosis).

Déjà vu ("already seen") is the illusion of recognition: a new situation feels already experienced. **Déjà entendu** is the illusion of having heard something before; **déjà vécu**, of having lived the whole moment before. **Jamais vu** is the reverse: the familiar feels unfamiliar. All occur in tired, healthy people and in temporal lobe epilepsy.

EXAM TRAP

Confabulation is **not lying**: there is no intent to deceive. **Malingering** is deliberate faking for external gain (Section 8.7); **confusion** and **delirium** are disturbances of consciousness, not invented memories.

Disorders of emotion ●●●

Mood is a sustained emotional state that the person reports; **affect** is the immediate, observed expression of emotion. Mood is the climate; affect is the weather. (Emotion is the broader, brief reaction, Chapter 1; feeling is its experienced side.) Affect is described by quality, range, reactivity, intensity, appropriateness and stability; persistence describes mood. Affect may be **restricted** (reduced range), **blunted** (severely reduced intensity), **flat** (almost no expression), **labile** (shifting rapidly without adequate cause) or **inappropriate** (incongruous with thought content: giggling about a death).

Anhedonia is the inability to experience pleasure or interest: a core symptom of depression and a negative symptom of schizophrenia. **Alexithymia**, a term of **Peter Sifneos** (1920 to 2008) in 1973, is difficulty identifying and describing one's own feelings. **Apathy** is lack of feeling and interest. **Ambivalence**, a term of **Eugen Bleuler** (1857 to 1939), is the coexistence of opposite feelings towards the same object. **Dysphoria** is an unpleasant mood; **euthymia** is normal mood. Elevated mood is graded by severity (Figure 8.3).

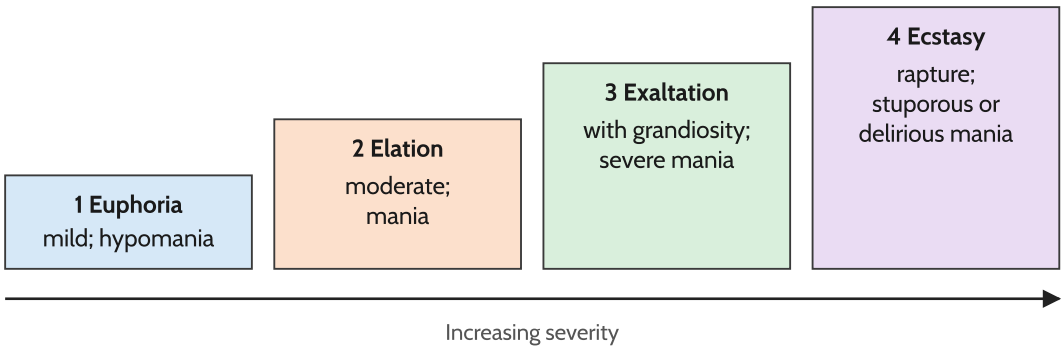

Figure 8.3 The four grades of elevated mood, mildest first.

Emotional incontinence is extreme lability that escapes voluntary control (vascular dementia). **Pseudobulbar affect** is sudden, brief, involuntary laughing or crying, out of proportion to or at odds with the underlying mood, in stroke, multiple sclerosis, amyotrophic lateral sclerosis, brain injury and dementia. *La belle indifférence* is in Section 8.7. A symptom is **egosyntonic** when experienced as consistent with one's self-image, and **ego-dystonic** when intrusive, unwanted and inconsistent with it (obsessions).

> EXAM TRAP
>
> "Our current mood and feelings", the emotion seen at this moment, is **affect**, not mood. Quality, range and reactivity describe affect; **persistence** describes mood. Severity: **euphoria, elation, exaltation, ecstasy**. Pseudobulbar affect is neurological, not the restricted affect of schizophrenia or histrionic emotional shifts.

Motor behaviour, insight and judgement ●

Psychomotor retardation slows movement, speech and thought (depression); **psychomotor agitation** is purposeless overactivity with inner tension (agitated depression, mania). **Catatonia** has twelve DSM-5-TR signs: stupor; **catalepsy** (a passively induced posture held

against gravity); **waxy flexibility** (slight, even resistance to positioning); mutism; **negativism** (opposing or ignoring instructions); posturing; **mannerisms** (odd caricatures of normal actions); **stereotypies** (repetitive, non-goal-directed movements); agitation; grimacing; **echolalia** (repeating others' words); and **echopraxia** (imitating others' movements). Older signs are **automatic obedience**, **Mitgehen** (moving with the lightest touch despite being told to resist) and **ambitendency** (offering, then withdrawing, a hand).

Compulsions are repetitive acts performed in response to an obsession, to reduce distress (Section 8.6); tics are in Section 8.3. **Avolition** is reduced self-initiated, purposeful activity (a negative symptom: the loss of a normal function; Section 8.4); **abulia** is lack of will, with difficulty initiating action or deciding, typically after frontal lobe damage.

Insight is awareness and understanding of one's own illness and need for treatment, graded on six levels: (1) complete denial; (2) slight awareness while still denying; (3) awareness, but blamed on others, external or physical causes; (4) awareness of something unknown in oneself; (5) **intellectual insight**, accepting the illness without changing; (6) **true emotional insight**, awareness that changes behaviour. **Judgement**, the ability to assess a situation and act appropriately, is personal (one's own plans), social (conduct with others) or test (a hypothetical problem, such as finding a stamped letter in the street). Section 7.2 shows how both are assessed.

> EXAM TRAP
>
> Insight means **awareness of one's own illness**, not orientation, serial subtraction (attention) or hallucinations (perception). Echolalia and echopraxia point to **catatonia**, not depression or panic.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Hallucinations while falling asleep, called _____, occur most often in normal people.
2. Needless detail that finally reaches the goal is _____, a disorder of the form of thought.
3. The belief that a spouse has been replaced by an identical impostor is _____ syndrome.
4. Filling memory gaps with false memories, without intent to deceive, is _____.
5. The observed, moment-to-moment expression of emotion is _____.
6. Elevated mood by severity: euphoria, elation, _____, ecstasy.

Answers: (1) hypnagogic, (2) circumstantiality, (3) Capgras, (4) confabulation, (5) affect, (6) exaltation.

8.3 Neurodevelopmental Disorders MEDIUM YIELD · 29 QS

The disorders in this section are differences in how the mind is built, not illnesses that strike a mind already formed. A child who cannot read a word he can say, or cannot sit still in class, has not lost a skill: his brain developed differently from the start.

The neurodevelopmental group ●●

DSM-5-TR (2022) defines **neurodevelopmental disorders** as conditions with onset in the developmental period, often before school entry, in which differences in brain development impair personal, social, academic or occupational functioning. They frequently co-occur: many autistic children have intellectual disability, and many children with ADHD have a specific learning disorder. ADHD belongs here; obsessive-compulsive and generalised anxiety disorder (Section 8.6) and borderline personality disorder (Section 8.11) do not.

DSM-IV-TR (2000) grouped them as "disorders usually first diagnosed in infancy, childhood or adolescence", which included the **pervasive developmental disorders**: autistic, Rett's, childhood disintegrative and Asperger's disorders and PDD not otherwise specified (PDD-NOS). ICD-10 (1992) spreads them over F70 to F98: mental retardation (F70 to F79), disorders of psychological development (F80 to F89) and childhood behavioural and emotional disorders such as hyperkinetic and tic disorders (F90 to F98).

Children behave differently at home, at school and in the clinic, so diagnosis uses **multi-informant assessment**: parents, teachers, clinician and child, across settings. ADHD even requires symptoms in two or more settings. The Vineland, CARS, ISAA and ADOS are described in Chapter 2.

> EXAM TRAP
>
> Several informants are needed because *symptoms vary across contexts*, not because parents over-report or teachers add nothing. "Observation alone is not enough" is true but is not the reason.

Intellectual disability: criteria and severity ●●●

Intellectual disability (DSM-5-TR: **intellectual developmental disorder**; DSM-IV-TR: mental retardation) has three criteria. (A) Deficits in intellectual functions such as reasoning, problem solving, abstract thinking and learning, confirmed by clinical assessment *and* individual standardised intelligence testing. (B) Deficits in **adaptive functioning**, meeting age and cultural standards of independence and social responsibility, in the conceptual (language, reading, money), social (communication, friendship, judgement) or practical (self-care, work) domain, severe enough to need ongoing support. (C) Onset in the developmental period.

Criterion A means an IQ about two standard deviations (SD) below the mean: 70 on tests with mean 100 and SD 15, a score only about 2.3 percent of people fall below (Figure 8.4). Allowing about 5 points of measurement error, the zone is 65 to 75. Prevalence is about 1 percent, higher in males.

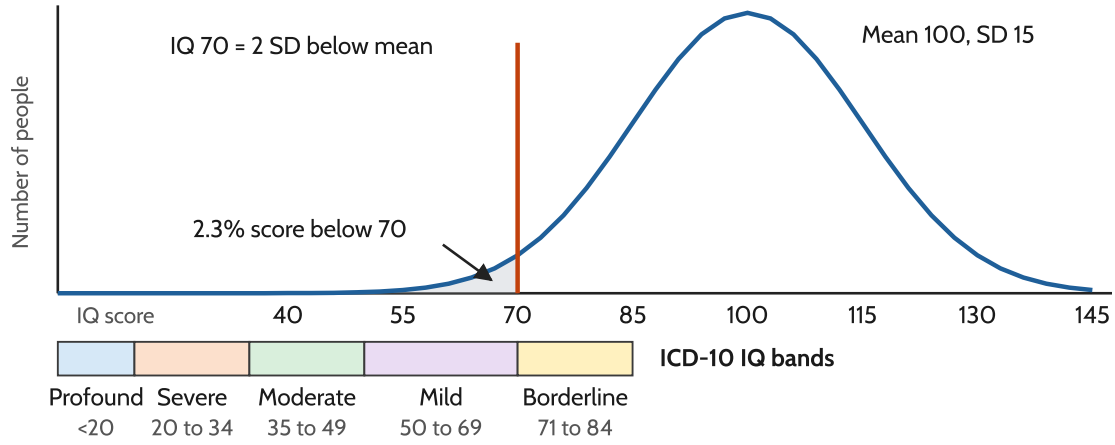

Figure 8.4 IQ is normally distributed; the intellectual criterion sits two SD below the mean. The bands are ICD-10's, with borderline intellectual functioning just above.

ICD-10 and older DSMs graded severity by IQ (Table 8.9; DSM-IV-TR used overlapping limits such as 50-55 to about 70 for mild), and about 85 percent of cases are mild. DSM-5 grades severity by *adaptive functioning*, which decides the support needed, because IQ is least valid at the low end. Severe levels show as delayed milestones in the first two years, mild ones often only at school; most people at every level now live in the community, not institutions.

Table 8.9 Severity levels with ICD-10 IQ bands and adult mental age (years).

Level	IQ	Mental age	Old label and outlook
Mild	50 to 69	9 to under 12	"Educable"; self-supporting adult with training
Moderate	35 to 49	6 to under 9	"Trainable"; partial independence in daily living, supervised work
Severe	20 to 34	3 to under 6	Basic self-care; close supervision
Profound	below 20	below 3	Constant care

An IQ near 70 is not enough: without adaptive deficits there is no diagnosis. **Borderline intellectual functioning** (traditionally IQ 71 to 84) is a condition that may be a focus of clinical attention, not a disorder. **Global developmental delay** is a provisional label for a child under 5 who misses milestones in several areas but whose severity cannot yet be reliably assessed or tested; it requires reassessment.

> EXAM TRAP
>
> Global developmental delay differs from intellectual disability by *age and testability*, not by social, language or motor deficits: a 7-year-old with full-scale IQ 65 and severe adaptive impairment has intellectual disability. On the WAIS-IV, 70 to 79 is labelled "borderline"; without adaptive deficits, FSIQ 70 is not intellectual disability.

Causes of intellectual disability ●●

Causes are grouped by timing, beginning with genes (Table 8.10). **Down syndrome**, described by **John Langdon Down** (1828 to 1896) in 1866, is the commonest chromosomal cause:

trisomy 21, usually from nondisjunction and more likely with older mothers. **Fragile X syndrome**, an expanded CGG repeat silencing the FMR1 gene on the X chromosome, is the commonest inherited cause. In **phenylketonuria** (PKU), found in 1934 by **Asbjørn Følling** (1888 to 1973), recessive deficiency of phenylalanine hydroxylase stops phenylalanine being converted to tyrosine, and the excess damages the brain; newborn screening and a low-phenylalanine diet prevent the disability.

Table 8.10 Genetic and chromosomal causes of intellectual disability.

Condition	Cause	Hallmarks
Down syndrome	Trisomy 21	Flat face, heart defects, early Alzheimer's
Fragile X	FMR1 CGG repeat	Long face, large ears, autistic features
Prader-Willi	Paternal 15q deletion	Overeating, obesity
Angelman	Maternal 15q deletion	No speech, jerky gait, laughter
Williams	7q11.23 deletion	Elfin face, very sociable, poor visuospatial skill
Cri-du-chat	5p deletion	Cat-like cry
Lesch-Nyhan	HPRT deficiency (X-linked)	Self-biting, excess uric acid
Tay-Sachs	Hexosaminidase A deficiency	Cherry-red spot, regression
Klinefelter	47,XXY	Tall male; language and learning problems
Turner	45,X	Short female; IQ usually normal

Prenatally, **fetal alcohol syndrome** from drinking in pregnancy gives a small head, smooth (flat) philtrum, thin upper lip, short palpebral fissures (eye openings), poor growth and intellectual disability; it is among the leading preventable causes, and caffeine, cannabis and mercury do not produce this face. Others are maternal **TORCH infections** (toxoplasmosis, other agents such as syphilis, rubella, cytomegalovirus, herpes) and iodine deficiency causing **cretinism** (congenital hypothyroidism), worldwide a leading preventable cause of brain damage; teratogens are in Chapter 6. Perinatal causes include asphyxia and extreme prematurity; postnatal ones, meningitis, encephalitis, lead, head injury and severe deprivation.

Communication disorders

Language disorder is persistent difficulty acquiring and using language: small vocabulary, simple sentences, poor discourse. **Speech sound disorder** is difficulty producing speech sounds, making speech hard to understand, without a physical cause such as cleft palate or deafness. **Childhood-onset fluency disorder** (stuttering) breaks the flow of speech with repetitions, prolongations, broken words and blocks; 80 to 90 percent of cases begin by age 6. **Social (pragmatic) communication disorder** is difficulty using language socially (adjusting to the listener, taking turns, grasping hints and idioms) without the repetitive behaviour of autism.

Ask why a child does not communicate. In **mutism**, as in an inherently mute person, there is no capacity for speech: language itself is the barrier. In **selective mutism**, an anxiety disorder (Section 8.6), the child speaks at home but not in feared settings, so anxiety is the barrier. In autism the barrier is the social use of language; in specific learning disorder, reading and writing.

Autism spectrum disorder ●●●

Eugen Bleuler (1857 to 1939) coined autism (Greek *autos*, self) in 1911 for the withdrawal seen in schizophrenia. In 1943 **Leo Kanner** (1894 to 1981) published "Autistic disturbances of affective contact", on eleven children with extreme aloneness and insistence on sameness, and soon called the condition *early infantile autism*. In 1944 **Hans Asperger** (1906 to 1980) described boys with fluent speech but odd social behaviour and narrow interests. **Bruno Bettelheim**'s (1903 to 1990) "refrigerator mother" theory (1967) blamed cold parenting; it was discredited when autism proved largely genetic. **Lorna Wing** (1928 to 2014) and **Judith Gould** (1979) described a **triad of impairments**: social interaction, communication and imagination.

DSM-5 (2013) merged autistic disorder, Asperger's disorder, PDD-NOS and **childhood disintegrative disorder** (two or more years of normal development, then loss of skills; **Theodor Heller**, 1908) into **autism spectrum disorder** (ASD). **Rett syndrome** (**Andreas Rett**, 1966) was left out as a genetic condition (MECP2): girls regress after early normal development, losing speech and hand use, with hand-wringing.

Criterion A: persistent deficits in social communication and social interaction across multiple contexts, in all three of social-emotional reciprocity, nonverbal communication (eye contact, gesture) and relationships. **Criterion B**: **restricted, repetitive patterns of behaviour**, interests or activities, at least two of stereotyped movements or speech (hand-flapping, lining up toys, echolalia or echoing others' words), insistence on sameness, intense restricted interests, and sensory hyper- or hyporeactivity. Symptoms start early (though they may show only as social demands rise) and impair functioning. Severity is rated per domain: level 1 "requiring support", level 2 "requiring substantial support", level 3 "requiring very substantial support". Specifiers note accompanying intellectual impairment or language impairment, a known genetic or medical condition, and catatonia; many former Asperger's cases are now ASD without either impairment.

> EXAM TRAP
>
> ASD needs social-communication deficits plus repetitive behaviour; delayed speech alone, IQ below 70 and hyperactivity are not required. No eye contact, delayed language and repetitive movements spell ASD, not ADHD, conduct disorder or SLD. Kanner first described infantile autism; Bleuler only coined the word, and Kurt Schneider (Section 8.4) and Cameron, Goldstein and Jaspers (Section 8.2) are distractors. The research deficit is in *encoding social cues*; trouble deceiving follows from mindblindness, and anger control and memory retrieval are not core deficits.

ASD affects about 1 to 2 percent of people, three to four males per female (DSM-5-TR says 3:1; girls are under-recognised). Outside the communication disorders themselves, language development is compromised more often in ASD than in any other neurodevelopmental disorder. The commonest co-occurring conditions are intellectual disability and ADHD (diagnosable alongside ASD only since DSM-5), then anxiety, depression and epilepsy.

Heritability is about 80 percent (estimates 37 to over 90); some cases carry *de novo* (new, uninherited) mutations, and older parents, extreme prematurity and prenatal valproate add risk. Autistic people show deficits in *encoding social cues* such as faces, gaze and tone of voice, and find deliberate deception hard, while rote memory is often a strength. **Simon Baron-Cohen**

(born 1958), **Alan Leslie** and **Uta Frith** (born 1941) proposed a deficit in **theory of mind** (Chapter 6), or "mindblindness". Frith's **weak central coherence** (1989) is a bias to detail over the whole; executive dysfunction explains rigidity; Baron-Cohen's *extreme male brain* theory casts autism as strong systemising with weak empathising.

KEY STUDY: THE SALLY-ANNE TEST (1985)

Baron-Cohen, Leslie and Frith tested 20 autistic, 14 Down syndrome and 27 typical children. Sally puts a marble in her basket and leaves; Anne moves it to her box. Where will Sally look? About 85 percent of typical and Down syndrome children said the basket; 80 percent of autistic children, despite higher verbal mental age than the Down group, said the box, failing to represent Sally's false belief. Criticism: a fifth passed, the deficit is not unique to autism, and it cannot explain repetitive behaviour.

In 1998 **Andrew Wakefield** linked the MMR vaccine to autism in a 12-child *Lancet* paper; it was retracted in 2010, he was struck off the medical register, and large studies find no link. Treatment is behavioural and educational: **Ivar Lovaas**'s applied behaviour analysis (1987), TEACCH and PECS (Sections 7.4 and 7.10).

Attention-deficit/hyperactivity disorder ●●

In 1902 **George Frederic Still** (1868 to 1941) described restless, inattentive children with a "defect of moral control". Later labels were minimal brain dysfunction, hyperkinetic reaction of childhood (DSM-II, 1968), attention deficit disorder (DSM-III, 1980) and ADHD (DSM-III-R, 1987). ICD-10's **hyperkinetic disorder** (F90) is overactivity, distractibility, short attention span and impulsivity; social withdrawal is not a feature.

DSM-5-TR **attention-deficit/hyperactivity disorder** (ADHD) needs six of nine inattention and/or six of nine hyperactivity-impulsivity symptoms (Table 8.11; five from age 17) for at least 6 months; several present before age 12 (DSM-IV: 7) and in two or more settings; and impairment. The *presentation* is combined, predominantly inattentive or predominantly hyperactive-impulsive.

Table 8.11 The 18 DSM-5-TR symptoms of ADHD (each "often").

Inattention (A1)	Hyperactivity-impulsivity (A2)
Careless mistakes	Fidgets or squirms
Cannot sustain attention	Leaves seat
Does not seem to listen when spoken to directly	Runs about or climbs
Does not follow through	Cannot play quietly
Disorganised	"On the go", "driven by a motor"
Avoids sustained mental effort	Talks excessively
Loses things	Blurts out answers
Easily distracted	Cannot wait turn
Forgetful	Interrupts or intrudes

> EXAM TRAP
>
> "Does not seem to listen when spoken to" is an *inattention* item; squirming, not playing quietly and talking excessively are hyperactivity items. The onset age is now 12, not 7. ADHD is not a learning disability (dyslexia, dysgraphia and auditory processing disorder are), though the two often co-occur.

About 7 percent of children and 2.5 percent of adults have ADHD, boys about twice as often as girls; hyperactivity fades, but inattention and impulsivity often persist into adulthood. Heritability is about 74 percent. Dopamine and noradrenaline in frontostriatal circuits are implicated; risks include prenatal smoking and alcohol and low birth weight. **Russell Barkley**'s (born 1949) **behavioural inhibition model** (1997) makes poor response inhibition the core deficit. Stimulants such as methylphenidate are first-line drugs, with the non-stimulant atomoxetine as the main alternative (Section 7.10).

Specific learning disorder ●●

DSM-5 merged reading, mathematics and written expression disorders into one **specific learning disorder** (SLD): difficulty in reading, spelling, writing or mathematics persisting at least 6 months *despite targeted intervention*, achievement well below age expectation on standardised tests, onset in the school years, and no better explanation in intellectual disability, sensory loss or poor teaching. Intelligence is usually average (IQ above about 70 plus or minus 5), hence "unexpected underachievement". It affects 5 to 15 percent of schoolchildren.

The specifiers are impairment in reading (word-reading accuracy, fluency and comprehension), whose core pattern of poor word recognition, decoding and spelling is called **dyslexia**; in written expression, often called **dysgraphia**; and in mathematics, **dyscalculia**.

The Glasgow eye surgeon **James Hinshelwood** (1859 to 1919) described "congenital word-blindness" (a book of that title in 1917), and **Samuel Torrey Orton** (1879 to 1948) in the 1920s blamed incomplete cerebral dominance. The modern **phonological deficit theory** locates dyslexia in linking speech sounds (phonemes) to letters. The old **IQ-achievement discrepancy** model waited for a child to fail; **response to intervention** (RTI) identifies children who do not respond to good teaching. **Auditory processing disorder**, difficulty interpreting heard sounds despite normal hearing, is grouped with the learning difficulties though DSM-5-TR omits it. India's RPwD Act 2016 covers specific learning disabilities (Section 7.1).

> EXAM TRAP
>
> SLD means roughly average intelligence with persistent academic difficulty; IQ below 50 points to intellectual disability, "only attention deficit" to ADHD and "motor coordination deficit only" to dyspraxia. Dyslexia is difficulty reading and decoding words, not understanding speech (language disorder) or concentrating in lectures (ADHD).

Motor disorders and tics ●●

Developmental coordination disorder (**dyspraxia**) is impaired acquisition and execution of coordinated motor skills: clumsiness and slowness in catching, using scissors, handwriting or cycling, without a neurological cause. It differs from syndromes acquired through brain damage (Chapter 5): **apraxia**, loss of learned purposeful movement despite intact

strength; **agnosia**, failure to recognise despite intact sensation; **anomia**, inability to name. **Stereotypic movement disorder** is repetitive, driven, purposeless movement (rocking, head-banging, self-biting), with or without self-injury.

A **tic** is a sudden, rapid, recurrent, non-rhythmic motor movement or vocalisation (blinking, shrugging, sniffing, throat-clearing). Seizures are abnormal electrical discharges that cannot be held back, dissociation disrupts consciousness or identity (Section 8.7), and Parkinson's tremor is rhythmic. Tics are often preceded by a **premonitory urge** and can be briefly suppressed; they begin at 4 to 6 years and peak at 10 to 12. All tic disorders need onset before 18 (Table 8.12). **Tourette's disorder** was described in 1885 by **Georges Gilles de la Tourette** (1857 to 1904); **coprolalia** (involuntary obscenities) occurs in a minority and is not required. Tics often accompany OCD and ADHD; habit reversal training treats them (Section 7.4).

Table 8.12 The DSM-5-TR tic disorders.

Disorder	Tics	Duration
Tourette's disorder	Multiple motor and one or more vocal, not necessarily together	More than 1 year
Persistent (chronic) motor or vocal tic disorder	Motor or vocal, not both	More than 1 year
Provisional tic disorder	Motor and/or vocal	Less than 1 year

EXAM TRAP

Tourette's needs motor plus vocal tics for more than 1 year, onset before 18; coprolalia is not required. Developmental clumsiness is dyspraxia; apraxia, agnosia and anomia are acquired.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The IQ criterion for intellectual disability is about two _____ below the mean.
2. DSM-5 grades intellectual disability by _____ functioning.
3. Global developmental delay is for children under age _____.
4. In PKU, phenylalanine is not converted to _____.
5. Infantile autism was first described in 1943 by _____.
6. The Sally-Anne task tests understanding of a _____ belief.
7. Several ADHD symptoms must be present before age _____.
8. Tourette's disorder requires tics for more than _____.

Answers: (1) standard deviations, (2) adaptive, (3) 5, (4) tyrosine, (5) Leo Kanner, (6) false, (7) 12, (8) 1 year.

8.4 Schizophrenia Spectrum and Other Psychotic Disorders

HIGH YIELD · 79 QS

Psychosis means losing contact with shared reality: hearing voices no one else hears, or holding a belief no evidence can shake. Schizophrenia is the central psychotic disorder; its neighbours differ from it mainly in duration. We follow the concept's history, symptoms, criteria, causes and drug treatment.

From dementia praecox to schizophrenia ●●●

In the 1850s the French physician **Benedict Morel** (1809 to 1873) used *démence précoce*, early loss of mind, for young patients who deteriorated. **Ewald Hecker** described **hebephrenia** (silly, disorganised behaviour beginning in youth) in 1871, and **Karl Ludwig Kahlbaum** described **catatonia** (stupor, posturing and excitement) in 1874. **Emil Kraepelin** (1856 to 1926) united these pictures, with paranoid forms, under the Latin **dementia praecox**, 'premature dementia', in 1893. In the sixth edition of his textbook (1899) he separated it from **manic-depressive insanity** by course: dementia praecox began early and deteriorated, while manic-depressive insanity came in episodes with recovery between them. These two clinical types still divide Sections 8.4 and 8.5. Despite the name, dementia praecox is the ancestor of schizophrenia, not of Alzheimer's disease or frontotemporal dementia (Section 8.10).

The Swiss psychiatrist **Eugen Bleuler** (1857 to 1939) saw that not every patient deteriorated or fell ill young. In a 1908 lecture and his 1911 monograph, *Dementia Praecox or the Group of Schizophrenias*, he coined **schizophrenia** (Greek *schizein*, to split, and *phren*, mind) for a splitting between thought, emotion and behaviour, not a split personality. His fundamental symptoms, the **four As**, were present in every case:

- Loosening of **Associations**: thoughts lose their logical links.
- Blunted or incongruous **Affect**, such as smiling at sad news.
- **Ambivalence**: opposite feelings or wishes held at once towards the same object, a term Bleuler coined.
- **Autism**: withdrawal into an inner world (not the developmental disorder of Section 8.3).

Hallucinations and delusions were only **accessory symptoms**, because they occur in other illnesses too.

> EXAM TRAP
>
> Auditory hallucinations are **not** one of the four As. Ambivalence was coined by Bleuler, not Kraepelin, Freud or Fromm-Reichmann. Alogia, avolition and anhedonia are later negative-symptom As (8.4.2); apathy, anxiety, aphasia, amnesia, attention and anosognosia are not Bleuler's.

Bleuler's symptoms were hard to rate reliably. The German psychiatrist **Kurt Schneider** (1887 to 1967) listed instead, in *Clinical Psychopathology* (English edition 1959), **first-rank symptoms** (FRS) that, in the absence of brain disease, strongly suggest schizophrenia. Their common theme is a failure of the boundary between self and world:

- **Three kinds of voices**: audible thoughts (*Gedankenlautwerden*, thought echo), voices arguing or discussing the patient in the third person, and voices giving a running commen-

tary on the patient's actions.

- **Thought alienation**: thought insertion, thought withdrawal and thought broadcasting.
- **Passivity phenomena** (delusions of control): made feelings, made impulses, made volitional acts, and somatic passivity (bodily sensations imposed from outside).
- **Delusional perception**: a normal perception suddenly given a private delusional meaning (a traffic light turns red, and he 'knows' he has been chosen).

His **second-rank symptoms** were other hallucinations, perplexity, sudden delusional ideas, mood changes and emotional blunting. FRS are typical of schizophrenia but not specific: they also occur in mania, so DSM-5 gives them no special weight.

> EXAM TRAP
>
> Not first-rank: **depersonalisation**, **loose associations** (Bleulerian), **delusions of self-reference**, catatonic signs such as echolalia, negative symptoms, olfactory hallucinations and memory loss. Delusions of control **are** first-rank. FRS are Schneider's: **Karl Jaspers** (*General Psychopathology*, 1913) described primary delusions (Section 8.2), and Bateson and Wynne are family theorists (8.4.7).

Positive, negative and cognitive symptoms ●●●

Positive symptoms are the presence of something abnormal, an excess or distortion of normal function: delusions, hallucinations (commonest in schizophrenia, mostly voices, though mania and psychotic depression can bring them too), disorganised speech and grossly disorganised behaviour. **Negative symptoms** are the absence or reduction of something normal: **affective flattening** (less emotional expression in face, voice and gesture), **alogia** (poverty of speech), **avolition** (little self-started purposeful activity), **anhedonia** (reduced pleasure) and **asociality**. ICD-10 lists its negative symptoms as marked apathy, paucity of speech, and blunting or incongruity of emotional responses. A third, **cognitive** dimension (poor attention, working memory and executive function) best predicts work and social functioning.

Timothy Crow (1980) proposed **Type I schizophrenia** (positive symptoms, acute onset, good response to neuroleptics, normal brain structure, dopamine D2 overactivity) and **Type II** (negative symptoms, chronic course, poor drug response, enlarged ventricles, intellectual decline). **Peter Liddle** (1987) found three syndromes: **reality distortion** (delusions, hallucinations), **disorganisation** (thought disorder, inappropriate affect) and **psychomotor poverty** (poverty of speech, flat affect, slowed movement). Symptoms are rated with **Nancy Andreasen**'s SANS and SAPS (Scales for the Assessment of Negative and of Positive Symptoms) and the 30-item PANSS (Positive and Negative Syndrome Scale, 1987); see Chapter 2.

> EXAM TRAP
>
> Delusions, hallucinations and disorganised speech or behaviour are **positive**; flat affect is **negative**, never positive. Ambivalence is Bleulerian, not a negative symptom. Positive means the presence of something unusual, negative the absence of something normal, never the reverse.

Diagnostic criteria: DSM-5-TR, ICD-10 and ICD-11 ●●●

DSM-5-TR (2022) diagnoses **schizophrenia** when these are all met:

- **A.** Two or more of five symptoms, each present for a significant part of a 1-month period (less if successfully treated): delusions, hallucinations, disorganised speech, grossly disorganised or catatonic behaviour, and negative symptoms. At least one must be one of the first three.
- **B.** Functioning in work, relationships or self-care falls markedly below the earlier level.
- **C.** Continuous signs last at least 6 months, including at least 1 month of active-phase symptoms; the rest may be **prodromal** or **residual** phases, showing only negative or attenuated symptoms (odd beliefs, unusual perceptions).
- **D to F.** Schizoaffective and mood disorders, substances and medical conditions are excluded; after childhood autism, prominent delusions or hallucinations must last 1 month.

DSM-5 (2013) dropped the DSM-IV subtypes (paranoid, disorganised, catatonic, undifferentiated, residual) and the rule that one bizarre delusion or one Schneiderian voice was enough. ICD-10 (F20) needs one very clear symptom from groups (a) to (d), or symptoms from two of groups (e) to (h), present most of the time for **1 month** or more: (a) thought echo, insertion, withdrawal or broadcasting; (b) delusions of control or passivity, and delusional perception; (c) voices commenting on or discussing the patient; (d) persistent, culturally impossible delusions; (e) persistent hallucinations with delusions; (f) breaks in the train of thought, incoherence or neologisms; (g) catatonic behaviour; (h) negative symptoms. Its subtypes were paranoid (the commonest), hebephrenic, catatonic, undifferentiated, post-schizophrenic depression, residual and **simple schizophrenia** (insidious negative symptoms without overt psychosis). Table 8.13 compares the three systems.

Table 8.13 Schizophrenia in the three classification systems.

Feature	DSM-5-TR	ICD-10 (F20)	ICD-11 (6A20)
Minimum duration	6 months, including 1 month active phase	1 month	1 month
Symptoms needed	2 of 5, one of the first three	1 clear from (a) to (d), or 2 from (e) to (h)	2, one a core positive symptom
Functional decline	Required	Not required	Not required
Subtypes	Dropped	Seven, from paranoid to simple	Replaced by symptom and course specifiers

> **EXAM TRAP**
>
> ICD-10 or ICD-11: **1 month**. DSM-5 or DSM-5-TR: **6 months**, including 1 month of active-phase symptoms. Catastrophic thinking is a cognitive distortion (Section 7.5), never a criterion of schizophrenia.

The spectrum: duration, delusions, mood and catatonia ●●●

Symptoms identical to those of schizophrenia can be short-lived, so DSM-5-TR separates the disorders largely by duration (Figure 8.5). **Brief psychotic disorder** lasts at least 1 day but less than 1 month, with full return to premorbid functioning; one symptom suffices if it is

delusions, hallucinations or disorganised speech. Specifiers: with marked stressors (brief reactive psychosis), without marked stressors, or with peripartum onset (in pregnancy or within 4 weeks after delivery). **Schizophreniform disorder** has the full Criterion A picture for at least 1 month but less than 6 months and needs no decline in functioning. Six months or more is schizophrenia.

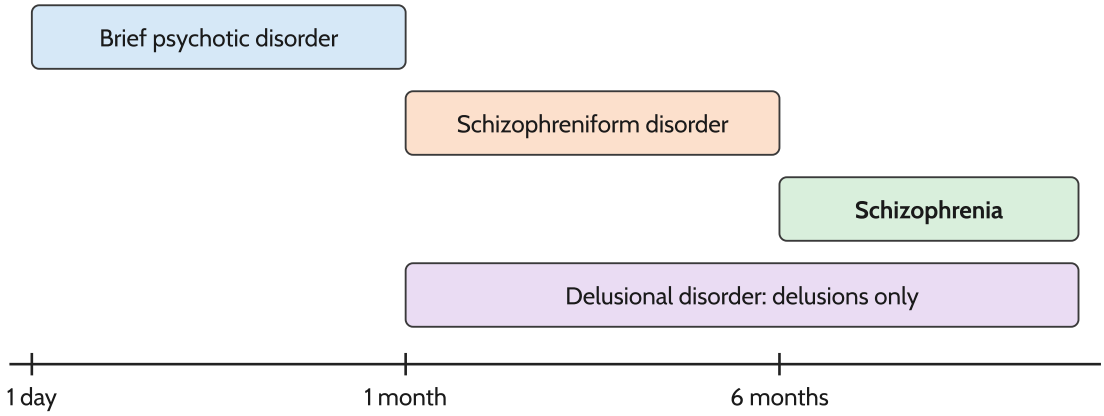

Figure 8.5 The duration ladder: 1 day, 1 month, 6 months. The same psychotic symptoms are named by how long they last; delusional disorder also starts at 1 month.

EXAM TRAP

One keyed answer (IHBAS 2025) calls psychosis lasting 1 to 6 months 'brief psychotic disorder'. It is **schizophreniform disorder**: brief psychotic disorder ends within a month.

Delusional disorder, Kraepelin's *paranoia*, is one or more delusions lasting 1 month or more (ICD-10 and ICD-11: about 3 months) in someone who has never met Criterion A for schizophrenia; hallucinations are not prominent, and apart from the delusion functioning is not markedly impaired. Types follow the theme: **erotomanic** (another person, usually of higher status such as a film star, is secretly in love with the patient), **grandiose** (an unrecognised great talent or discovery), **jealous** (the partner is unfaithful), **persecutory** (being plotted against, spied on or poisoned; the commonest), **somatic** (bodily functions, such as emitting a foul odour), mixed and unspecified.

Schizoaffective disorder needs an uninterrupted period in which a major mood episode coincides with Criterion A of schizophrenia, plus delusions or hallucinations for **2 weeks or more** without a major mood episode, with mood episodes present for most of the illness; it has bipolar and depressive types. Psychotic depression and mania belong to Section 8.5. **Schizotypal disorder**, eccentricity below the psychotic threshold, is listed in this DSM-5-TR chapter but described in Section 8.11; ICD-10 placed it beside schizophrenia as F21, and ICD-11 keeps it there as 6A22. Psychosis caused directly by a drug, medication or toxin is **substance/medication-induced psychotic disorder**; psychosis caused by a medical illness is **psychotic disorder due to another medical condition**.

Catatonia requires three or more of twelve signs: stupor, catalepsy, waxy flexibility, mutism, negativism, posturing, mannerism, stereotypy, agitation, grimacing, echolalia and echopraxia. DSM-5 makes it not an independent disorder but a **specifier** across several disor-

ders (catatonia associated with schizophrenia, depression and others), plus catatonic disorder due to another medical condition and unspecified catatonia. **Attenuated psychosis syndrome** (Section III, for further study) means weaker psychotic symptoms with insight retained, at least weekly for a month. ICD-10 **acute and transient psychotic disorders** (F23) begin within 2 weeks and usually resolve within 2 to 3 months; ICD-11 6A23 keeps the 2-week onset, with rapidly changing symptoms lasting at most 3 months. They are common in India and recover well.

Folie à deux (**Charles Lasègue** and **Jules Falret**, 1877) is a delusion transmitted from a dominant **primary** case with an established psychotic disorder to a more passive, often socially isolated **secondary** partner in a close relationship, usually a relative. Separation often lets the secondary partner's belief fade. In *folie imposée* the delusion is imposed on a healthy partner; in *folie simultanée* two predisposed people fall ill together; *folie à trois* involves three. ICD-10 coded it **F24 induced delusional disorder**, DSM-IV **shared psychotic disorder**. DSM-5 dropped the category: the partner receives delusional disorder if its criteria are met, otherwise other specified schizophrenia spectrum disorder ('delusional symptoms in the context of relationship with an individual with prominent delusions'). ICD-11 has no separate code either: it describes 'shared or induced delusional disorder' (folie à deux) as a rare form of **6A24 delusional disorder**, in its block 'schizophrenia or other primary psychotic disorders'.

> EXAM TRAP
>
> A belief shared by a whole cultural or religious group is not folie à deux; nor are two independent psychotic episodes, or a stress psychosis in two relatives. ICD-10 F24 is induced delusional disorder; F21 is schizotypal disorder. ICD-11 puts it under neither schizoaffective nor schizotypal disorder: it is a form of delusional disorder among the other primary psychotic disorders.

Epidemiology, course and prognosis

The lifetime risk is usually quoted as about **1 percent**; DSM-5-TR gives 0.3 to 0.7 percent. Schizophrenia is above all a disorder of **young adults**: psychosis emerges between the late teens and the mid-30s, peaking in the early to mid-20s in men and the late 20s in women, with a second peak in women in midlife, around the menopause. Prevalence is about equal in the two sexes, but men have more negative symptoms and worse outcomes. A prodrome of withdrawal and odd ideas precedes the active phase. In the WHO **International Pilot Study of Schizophrenia** (1973, nine countries including India) and the later Determinants of Outcome study, patients in developing countries had a better course, often linked to family support. Schizophrenia is commoner in the lowest social classes: **social causation** says poverty and stress produce it; **social drift** says the illness pushes people down the social ladder. Urban birth, migration and heavy adolescent cannabis use raise the risk. Table 8.14 lists the classic prognostic factors.

Table 8.14 Prognostic factors in schizophrenia.

Good prognosis	Poor prognosis
Acute onset	Insidious onset
Late onset	Early onset
Clear precipitating stressor	No precipitant

Good prognosis	Poor prognosis
Mood symptoms; mainly positive symptoms	Prominent negative symptoms
Good premorbid functioning	Poor premorbid adjustment
Female, married, good support	Male, single, withdrawn
Family history of mood disorder	Family history of schizophrenia

Biological causes: genes, dopamine and development ●●●

Irving Gottesman (1991) pooled about 40 family and twin studies (Figure 8.6): risk climbs with shared genes, to about 48 percent, roughly half, for an identical twin. Because it is not 100 percent, environment matters too. The **Genain quadruplets**, identical sisters who all developed schizophrenia but with different onset, severity and outcome, were studied by **David Rosenthal** (1963). Adoption studies separate genes from upbringing: **Leonard Heston** (1966) found schizophrenia in children of mothers with schizophrenia adopted away at birth but in none of the control adoptees, and the Danish adoption studies of **Seymour Kety**, Rosenthal and **Paul Wender** (from 1968) found it concentrated in biological, not adoptive, relatives. Risk is **polygenic**: many common variants of small effect, partly shared with bipolar disorder and autism.

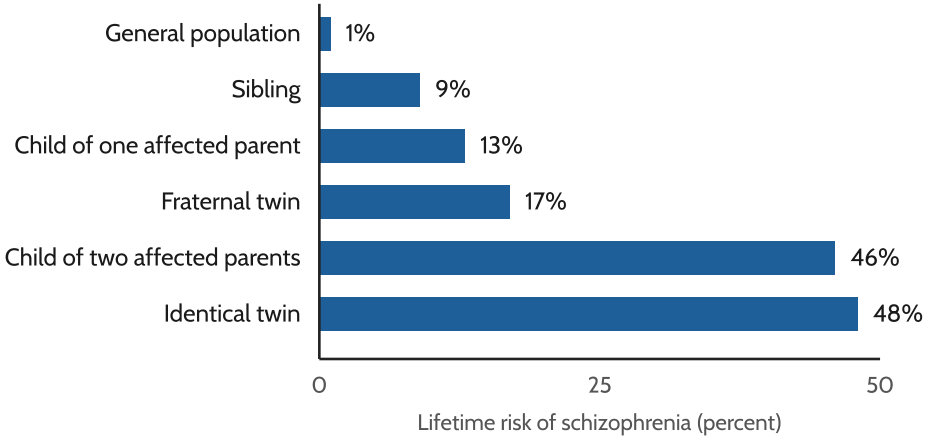

Figure 8.6 Gottesman's (1991) risk figures. Risk rises with the genes shared with an affected relative, but even an identical co-twin falls ill only about half the time.

The **dopamine hypothesis** holds that positive symptoms reflect excess dopamine activity at D2 receptors in the **mesolimbic** pathway. **Arvid Carlsson** showed that dopamine is a neurotransmitter and that antipsychotics act on its system (Nobel Prize, 2000); amphetamine, which releases dopamine, can cause a paranoid psychosis, and L-dopa for Parkinson's disease can cause hallucinations (it does not relieve negative symptoms). The original support came in 1976, when **Philip Seeman** and colleagues, and Ian Creese, David Burt and **Solomon Snyder**, showed that the clinical potency of typical antipsychotics correlates with their affinity for D2 receptors. Dopamine runs in four pathways, so a drug that blocks D2 everywhere both helps and harms (Table 8.15).

Table 8.15 The four dopamine pathways.

Pathway	Route	Role in schizophrenia and treatment

Pathway	Route	Role in schizophrenia and treatment
Mesolimbic	Ventral tegmental area (VTA) to nucleus accumbens (ventral striatum) and limbic areas; reward	Overactive: positive symptoms; D2 blockade treats them
Mesocortical	VTA to prefrontal cortex	Underactive: negative and cognitive symptoms
Nigrostriatal	Substantia nigra to caudate and putamen; voluntary movement	D2 blockade: extrapyramidal symptoms, parkinsonian rigidity
Tuberoinfundibular	Hypothalamus to median eminence and pituitary	D2 blockade: raised prolactin

Kenneth Davis and colleagues (1991) revised the hypothesis: too much subcortical dopamine for positive symptoms, too little prefrontal dopamine for negative ones. **Oliver Howes** and **Shitij Kapur** (2009) located the core fault in increased presynaptic dopamine synthesis in the striatum, a final common pathway to psychosis. Phencyclidine and ketamine block **NMDA glutamate receptors** and produce positive and negative symptoms, suggesting NMDA receptor hypofunction; atypical antipsychotics add blockade of **5-HT2A serotonin** receptors.

The **neurodevelopmental hypothesis** (**Daniel Weinberger**, 1987; **Robin Murray** and Shon Lewis, 1987) proposes that an early brain lesion stays silent until adolescent maturation unmasks it. The evidence: enlarged **lateral ventricles**, first shown on CT by **Eve Johnstone** and colleagues (1976); reduced grey matter; **hypofrontality**, low prefrontal activity during tasks; obstetric complications, prenatal influenza and winter-spring birth; and, later, adolescent cannabis use.

Family and psychological theories ●●●

Early family theories blamed parents. **Frieda Fromm-Reichmann** (1948) described the cold, dominating **schizophrenogenic mother**; the idea is discredited and caused families needless guilt. **Gregory Bateson**, with Don Jackson, Jay Haley and John Weakland (1956), proposed the **double bind**: a person repeatedly receives contradictory messages at two levels from a significant other, and can neither comment on the contradiction nor escape.

KEY STUDY: BATESON'S DOUBLE BIND (1956)

A young man, fairly well recovered from an acute schizophrenic episode, put his arm round his visiting mother; she stiffened. He withdrew, and she asked, 'Don't you love me any more?' He blushed, and she said, 'Dear, you must not be so easily embarrassed and afraid of your feelings.' After she left he assaulted an aide. Bateson called this the **symptomatic double bind**: every response was wrong. No controlled study has shown that double binds cause schizophrenia. The group's cybernetic and homeostatic views of the family belong to family therapy (Section 7.7).

Theodore Lidz (1957) described **marital schism** (parents in open conflict, undermining each other) and **marital skew** (one dominant, disturbed parent whose distortions the other accepts). **Lyman Wynne** (1958) described **pseudomutuality**, a façade of harmony that forbids individuality, and later **communication deviance**, vague and fragmented parental speech. **Ronald David Laing** led **anti-psychiatry**, which saw schizophrenia as an understandable response to an unliveable family situation.

Where these theories sought causes and failed, **expressed emotion** (EE) predicts relapse. EE is the emotional climate of a key relative towards the patient, rated from the **Camberwell Family Interview** on three dimensions: **critical comments**, **hostility** and **emotional over-involvement**. Warmth and positive remarks are also rated, but they are not high-EE dimensions.

KEY STUDY: EXPRESSED EMOTION AND RELAPSE (1972, 1976)

George Brown, James Birley and **John Wing** (1972) followed patients discharged to their families: relapse was about four times commoner in high-EE homes. **Christine Vaughn** and **Julian Leff** (1976) replicated this and found relapse highest when a patient spent more than about 35 hours a week face to face with a high-EE relative and took no medication. Criticism: the data are correlational; a more disturbed patient may provoke more criticism. In WHO studies in Chandigarh, Indian relatives were far less often high in EE than London relatives, one suggested reason for better outcome in India.

High EE is managed by family psychoeducation and by reducing face-to-face contact hours. Table 8.16 summarises the psychodynamic theories; the vulnerability-stress model that integrates these causes is in Section 8.1.

EXAM TRAP

High EE predicts **relapse** of psychotic illness, not its onset, negative symptoms or violence; relapse is more precise than a vague 'poor treatment outcome'. It means critical, hostile or over-involved relatives, not overwhelmingly positive comments, a family without emotion, or a patient who cannot talk about feelings.

Table 8.16 Psychodynamic theories of schizophrenia.

Theorist	Core idea
Sigmund Freud	Libido withdrawn from the world into the self, with regression; symptoms try to restore contact
Paul Federn	Weak ego boundaries: inner and outer reality blur
Harry Stack Sullivan	Interpersonal: overwhelming anxiety in early relationships
Silvano Arieti	Palaeologic thinking under the von Domarus principle: identity from shared predicates ('the Virgin Mary was a virgin; I am a virgin; so I am the Virgin Mary')
Margaret Mahler	Fixation at the autistic phase, with failure to form an adequate and stable symbiosis with the mother (symbiotic psychosis)
Ping-Nie Pao	Symptoms as the best possible adaptation to overwhelming 'organismic panic'
Edith Jacobson	Psychotic regression fuses self and object representations

Antipsychotic drugs and their side effects ●●●

Antipsychotic medication (the **neuroleptics**) is the front-line treatment for schizophrenia; CBT, social skills training, family work and stress management are added to it, not used instead (Sections 7.4, 7.5, 7.7 and 7.9). Neuroleptics are used chiefly for schizophrenia (and in mania), not for depression or anxiety. The era began in 1952, when **Jean Delay** and **Pierre Deniker** in Paris gave **chlorpromazine** to psychotic patients. **Typical antipsychotics** (first-

generation), such as chlorpromazine and haloperidol, act mainly by blocking dopamine **D2** receptors (not D1); chlorpromazine also blocks muscarinic (M1) receptors, causing dry mouth and constipation. **Atypical antipsychotics** (second-generation), such as risperidone, olanzapine and **clozapine**, block 5-HT2A as well as D2 and cause fewer movement effects; clozapine is reserved for treatment-resistant schizophrenia.

Because a typical drug blocks D2 in the nigrostriatal pathway too, it produces **extrapyramidal symptoms** (EPS), in a fixed order:

- **Acute dystonia**, within hours to days: sustained muscle spasms, a twisted neck, eyes forced upwards.
- **Akathisia**, within days to weeks: a subjective inner restlessness with a compelling need to pace or move.
- **Drug-induced parkinsonism**, within weeks: tremor, rigidity and slowed movement.
- **Tardive dyskinesia**, after months to years: involuntary orofacial movements such as lip-smacking, tongue movements and grimacing; the last to appear and the most likely to be irreversible.

Other dangers are **neuroleptic malignant syndrome**, a rare emergency of high fever, 'lead-pipe' rigidity, unstable pulse and blood pressure and clouded consciousness; **hyperprolactinaemia** from tuberoinfundibular blockade (breast enlargement, milk secretion, missed periods); **metabolic syndrome** (weight gain, diabetes, raised lipids) with atypicals, especially clozapine and olanzapine; and **agranulocytosis** with clozapine, which demands regular blood counts. DSM-5-TR lists these reactions as medication-induced movement disorders and other adverse effects of medication.

> EXAM TRAP
>
> Akathisia and tardive dyskinesia are **secondary** (drug-induced) movement disorders; Huntington's chorea (inherited) and Parkinson's disease (nigral degeneration) are **primary**. Tardive dyskinesia follows neuroleptics, not antidepressants, anxiolytics or barbiturates. Typicals cause EPS by blocking nigrostriatal D2 receptors, not by blocking serotonin reuptake (SSRIs), stimulating GABA-A receptors (benzodiazepines) or blocking mesolimbic receptors selectively.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Kraepelin separated dementia praecox from _____ insanity by its deteriorating course.
2. Bleuler's four As are associations, affect, ambivalence and _____.
3. Thought echo, voices giving a running commentary and delusional perception are Kurt Schneider's _____ symptoms.
4. Psychosis lasting 3 months with no decline in functioning is _____ disorder.
5. ICD-10 and ICD-11 need symptoms for _____; DSM-5-TR needs 6 months.
6. High expressed emotion consists of critical comments, hostility and emotional _____.
7. Involuntary lip-smacking after years on haloperidol is _____.

Answers: (1) manic-depressive, (2) autism, (3) first-rank, (4) schizophreniform, (5) 1 month, (6) over-involvement, (7) tardive dyskinesia.

8.5 Depressive and Bipolar Disorders, and Suicide HIGH YIELD · 49 QS

Sadness after a loss and elation after a win are normal. A **mood disorder** (affective disorder) is a low or high mood, with changed sleep, energy, thinking and behaviour, that takes over a life for weeks or years. Such disorders are built from three kinds of episode, so the episodes come first.

Mood episodes: the building blocks ●●●

A disorder with depressive episodes only is **unipolar**; one in which mood swings between the extremes, mania or hypomania as well as depression, is **bipolar**. DSM-5 (2013) split DSM-IV's "mood disorders" into two chapters, **bipolar and related disorders** and **depressive disorders**, placing bipolar disorders between the psychotic and depressive chapters as a bridge. Table 8.17 defines the three episodes.

Table 8.17 The three mood episodes (DSM-5-TR).

Feature	Major depressive episode	Manic episode	Hypomanic episode
Core mood	Depressed mood or loss of interest or pleasure (one required)	Elevated (often euphoric), expansive or irritable mood, plus raised energy	As in mania
Duration	2 weeks, most of the day, nearly every day	1 week, or any duration if hospitalised	4 consecutive days
Symptoms	5 or more of 9	3 or more of 7; 4 if mood is only irritable	As in mania
Severity	Distress or impairment	Marked impairment, hospitalisation or psychosis	Observable change; no marked impairment, hospitalisation or psychosis

The nine depressive symptoms are depressed mood plus SIG E CAPS: **S**leep change, loss of **I**nterest or pleasure (**anhedonia**), **G**uilt or worthlessness, low **E**nergy, poor **C**oncentration,

Appetite or weight change, **P**sychomotor agitation or retardation, and **S**uicidal thoughts. The seven manic symptoms are DIG FAST: **D**istractibility, **I**ndiscretion (spending sprees and other risky pleasures), **G**randiosity, **F**light of ideas, increased goal-directed **A**ctivity, decreased need for **S**leep and **T**alkativeness or pressured speech. Extremely elevated mood and grandiose ideas are the two hallmarks of mania.

Mania and hypomania differ in degree, not kind: hypomania is shorter, and though others notice it, it never causes marked impairment or hospitalisation; psychotic features make an episode manic by definition. DSM-5 also dropped the **bereavement exclusion**, which had barred a diagnosis of depression in the 2 months after a death, because grief-triggered depression proved to be like any other.

> EXAM TRAP
>
> Specific phobias and euphoric mood are not symptoms of a major depressive episode; anhedonia and appetite change are. Social phobia and fear of weight gain (anorexia, Section 8.8) are not manic symptoms, and hallucinations, though possible in severe mania, are not among its seven. Mania lasts longer than hypomania and need not follow depression; hypomania must be observable by others.

Depressive disorders ●●●

Major depressive disorder (MDD), one or more major depressive episodes with no mania or hypomania, is marked by persistent sadness and loss of interest; it may be a single episode or recurrent. Called the **common cold of psychiatry**, it affects about 16 percent of people in a lifetime (estimates run 15 to 20 percent), and rates are rising, fastest among the young. Onset grows likelier at puberty, after which women are affected about twice as often as men (in children the ratio is about equal). Episodes are often triggered by a stressful event, and the more severe the depression, the higher the suicide risk.

> EXAM TRAP
>
> Asked which statement least accurately describes depression, pick "men are more vulnerable than women": the reverse is true. Widespread and rising incidence, stress triggers and the severity-suicide link are all accurate.

Persistent depressive disorder (PDD), formerly dysthymia, merges dysthymic disorder and chronic major depression. Mood is depressed most of the day, more days than not, for at least **2 years in adults or 1 year in children and adolescents**, with two or more of six symptoms: appetite change, sleep change, low energy, low self-esteem, poor concentration and hopelessness. The person is never symptom-free for more than 2 months and has never had mania or hypomania. Dysthymia is persistent mild depression, **less severe than MDD but longer**. A major depressive episode on top of it is **double depression**.

Disruptive mood dysregulation disorder (DMDD), new in DSM-5, means severe temper outbursts grossly out of proportion to the provocation, three or more times a week, with persistently irritable mood between them, for 12 months, in two of three settings (home, school, peers). Onset is before age 10 and diagnosis between 6 and 18. It was added to curb the overdiagnosis of bipolar disorder in children, who in fact grow up to have depression and anxiety.

EXAM TRAP

A 12-year-old whose temper outbursts have just begun cannot have DMDD, because onset must come before age 10. Bipolar disorder, conduct disorder and ADHD (Sections 8.3 and 8.11) all remain possible.

Premenstrual dysphoric disorder (PMDD) needs five or more symptoms, at least one of them mood swings, irritability, depressed mood or anxiety, in the week before menses in most cycles, easing within a few days after menses begin. Depression caused by a drug, or by an illness such as hypothyroidism, is substance-induced or due to another medical condition (Section 8.10).

The "hybrid" disorder is ICD-10's **mixed anxiety and depressive disorder** (F41.2): anxiety and depression both present, neither enough for its own diagnosis. ICD-11 keeps it among the depressive disorders; DSM-5 does not. (Multiple personality, borderline and schizoid disorders belong to Sections 8.7 and 8.11.) **René Spitz** (1887 to 1974) described **anaclitic depression** in 1946: infants separated from their mothers in the second half of the first year grew weepy, then withdrawn, lost weight and stopped developing, recovering if the mother soon returned. His term **hospitalism** named the graver, sometimes fatal, decline of infants reared in impersonal institutions. (Piaget, Skinner and Eysenck are known for cognitive stages, operant conditioning and personality dimensions.) **Masked depression** presents as bodily complaints while sadness is denied, common in Indian primary care; depressive pseudodementia is covered in Section 8.10.

Specifiers ●●

A specifier adds detail to a diagnosis without making a new disorder (Table 8.18).

Table 8.18 Specifiers for depressive and bipolar disorders.

Specifier	Defining features	Applies to
With anxious distress	2 or more of: keyed up, restless, worried, fearing disaster or loss of control	Depressive and bipolar
With mixed features	3 or more symptoms of the opposite pole	Manic, hypomanic or depressive episode
With rapid cycling	4 or more mood episodes in 12 months	Bipolar I and II only
With melancholic features	No pleasure or mood reactivity; worse in the morning, early waking, psychomotor change, weight loss, guilt	Depressive episode
With atypical features	Mood reactivity; weight gain, hypersomnia, leaden paralysis, rejection sensitivity	Depressive episode
With psychotic features	Delusions or hallucinations, mood-congruent or incongruent	Depressive or manic episode
With catatonia	Catatonic features (Section 8.4)	Depressive or manic episode
With peripartum onset	Onset in pregnancy or within 4 weeks of delivery	Any episode
With seasonal pattern	Onset and remission at a set time of year, for 2 years	Recurrent MDD; bipolar I and II

A manic episode **with mixed features** has concurrent depressive symptoms, and a depressive one concurrent manic symptoms; the specifier replaced DSM-IV's mixed episode. In mood disorders, psychosis occurs only during episodes and is usually **mood-congruent**: guilt, disease or deserved punishment in depression, grandiosity in mania. Mood congruence, with this timing, best separates a mood disorder with psychotic features from a primary psychotic disorder (Section 8.4); delusions as such, episode length and anxiety comorbidity do not.

After childbirth, postpartum blues (tearful, labile mood in the first days, gone within a week) is common and not a disorder; postpartum depression is a major depressive episode with peripartum onset; postpartum psychosis (1 to 2 per 1,000 births, often bipolar) risks infanticide and is an emergency. **Norman Rosenthal** and colleagues described **seasonal affective disorder** in 1984: depression each autumn or winter, lifting in spring, with oversleeping and carbohydrate craving, commoner at high latitudes. It responds to bright light therapy.

> EXAM TRAP
>
> Seasonal affective disorder is the "with seasonal pattern" specifier of recurrent major depressive or bipolar disorder: not a separate, anxiety or personality disorder, and not confined to women. "With limited prosocial emotions" belongs to conduct disorder (Section 8.11), not mania.

Bipolar and related disorders ●●●

The two-pole idea came from 1850s France: **Jean-Pierre Falret** (1794 to 1870) described *folie circulaire* and **Jules Baillarger** (1809 to 1890) *folie à double forme*, mania and melancholia alternating in one patient (Falret's son Jules gave us folie à deux). **Emil Kraepelin** (1856 to 1926) united recurrent mood illness as **manic-depressive insanity** in 1899, apart from dementia praecox, and **Karl Leonhard** split unipolar from bipolar forms in 1957.

Bipolar I disorder needs at least one manic episode; major depressive and hypomanic episodes are common but not required, and subthreshold depressive periods are allowed. **Bipolar II disorder** needs at least one hypomanic and one major depressive episode and never a full mania, which would make it bipolar I. Bipolar II is not milder, since its depressions are frequent and disabling, and it needs no psychotic features. **Cyclothymic disorder** is the chronic low-grade form: for at least 2 years (1 in the young), numerous hypomanic and depressive symptoms that never meet episode criteria, never absent for more than 2 months. Swinging between the poles is what makes a disorder bipolar (Figure 8.7). Bipolar disorder affects about 1 percent of people, men and women equally, usually beginning around age 20.

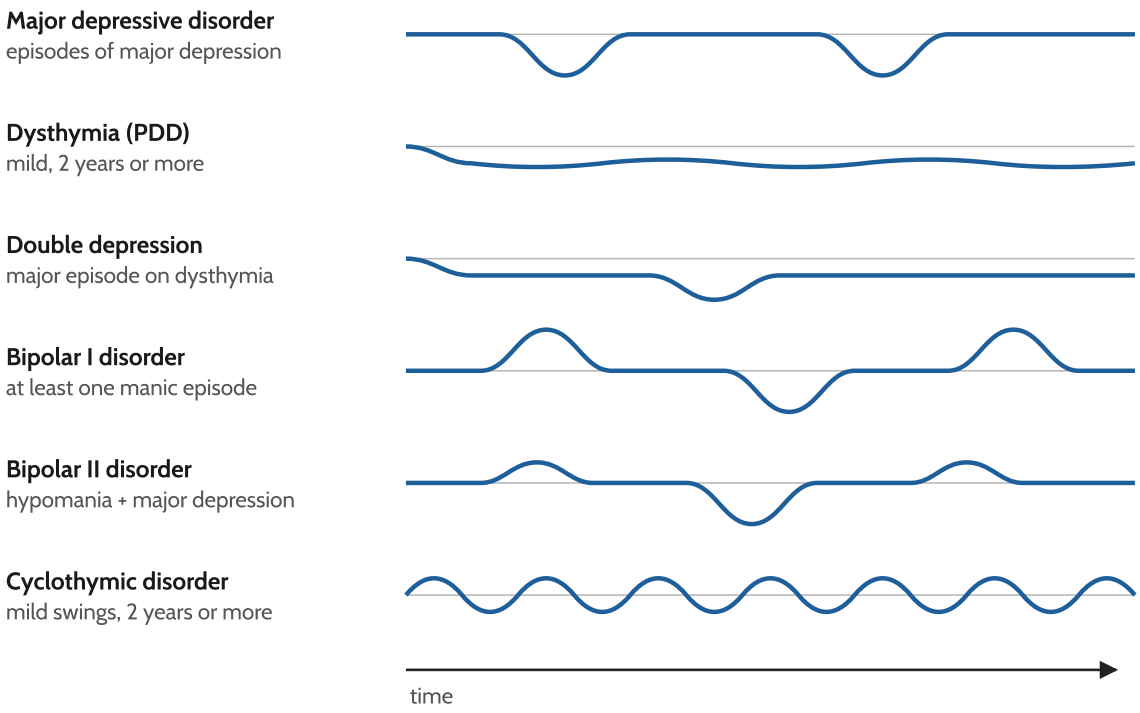

Figure 8.7 Mood over time. The grey line is normal mood: tall peaks are mania, short peaks hypomania, deep troughs major depression and shallow troughs milder depression.

> EXAM TRAP
>
> Bipolar I is defined by mania, so "bipolar I includes hypomania" is the incorrect statement: hypomania defines bipolar II and is optional in bipolar I, which may also include depressive periods, even subthreshold ones. Rapid cycling and psychotic depressions occur in both, and bipolar II has no manic episodes at all, only shorter hypomanias; a full manic episode is what separates I from II.

ICD-10 and ICD-11 compared

ICD-10 codes mood (affective) disorders at F30 to F39 (Table 8.19). Its depressive episode lasts 2 weeks and is graded mild, moderate or severe by how many symptoms are present, three of them typical (depressed mood, loss of interest, reduced energy). Mild and moderate episodes are coded with or without the **somatic syndrome**, ICD-10's counterpart of melancholic features. ICD-11 (in use from 2022) keeps "mood disorders" as an umbrella over bipolar and depressive groupings, retains the mixed episode that DSM-5 dropped, and lets one manic or mixed episode establish bipolar type I.

Table 8.19 Mood disorders in ICD-10, ICD-11 and DSM-5-TR.

ICD-10	ICD-11	DSM-5-TR
F30 Manic episode (including hypomania)	Manic, hypomanic, mixed and depressive episodes	Manic and hypomanic episodes
F31 Bipolar affective disorder: at least two episodes, one elevated	Bipolar type I, type II	Bipolar I, bipolar II
F32 Depressive episode	Single episode depressive disorder	MDD, single episode

ICD-10	ICD-11	DSM-5-TR
F33 Recurrent depressive disorder	Recurrent depressive disorder	MDD, recurrent
F34 Persistent: F34.0 cyclothymia, F34.1 dysthymia	Cyclothymic disorder; dysthymic disorder	Cyclothymic disorder; PDD
F41.2 Mixed anxiety and depressive disorder	Mixed depressive and anxiety disorder	None

Causes ●●●

Mood disorders run in families: heritability is about 40 percent for MDD and up to about 90 percent for bipolar disorder in twin studies, among the highest in psychiatry, though identical-twin concordance of 40 to 70 percent leaves room for the environment. **Joseph Schildkraut**'s **catecholamine hypothesis** (1965) linked depression to a deficit of noradrenaline at brain synapses and mania to an excess; the serotonin **permissive hypothesis** adds that low serotonin lets other transmitters swing towards either pole. Both are now thought too simple. Depression also brings overactivity of the hypothalamic-pituitary-adrenal (HPA) axis with raised cortisol: about half of patients fail to suppress cortisol on the **dexamethasone suppression test**, though the finding is too unspecific for diagnosis. Sleep shows **shortened REM latency**, dream sleep arriving sooner. **Robert Post**'s **kindling hypothesis** (1992) holds that each episode sensitises the brain and lowers the threshold for the next, so early episodes follow major stress and later ones need little or none.

Table 8.20 summarises the psychological and social theories.

Table 8.20 Psychological and social theories of depression.

Approach	Theorist and year	Key idea
Psychodynamic	**Sigmund Freud**, *Mourning and Melancholia* (1917); **Karl Abraham** (1911)	Loss of a loved object; anger at it turned inward as self-reproach
Ego psychology	**Edward Bibring** (1953)	The ego's helplessness to meet its aspirations; loss of self-esteem
Behavioural	**Peter Lewinsohn**	Low rate of response-contingent positive reinforcement
Learned helplessness	**Martin Seligman** (1975)	Uncontrollable bad events teach passivity (dogs given inescapable shock later stopped trying to escape)
Attributional	**Lyn Abramson**, Seligman and **John Teasdale** (1978)	Bad events blamed on internal, stable, global causes ("I am stupid")
Hopelessness	Abramson, **Gerald Metalsky** and **Lauren Alloy** (1989)	Expecting bad outcomes that nothing can change
Cognitive	Aaron Beck (1967)	Negative cognitive triad, schemas, automatic thoughts
Response styles	**Susan Nolen-Hoeksema** (1991)	Rumination (dwelling on one's distress) prolongs depression; women ruminate more, part of the gender gap
Social	George Brown and Tirril Harris (1978)	Vulnerability factors plus provoking life events

The most examined theory is that of **Aaron Temple Beck** (1921 to 2021), who argued in 1967 that depression is maintained by the **negative cognitive triad**: negative views of the **self** ("I am worthless"), the **world** or present experience ("nothing goes right") and the **future** ("it will never improve"). Underlying negative schemas, early core beliefs such as "I am unlovable", are activated by stress and generate quick, unbidden **automatic thoughts**, kept

believable by cognitive distortions such as overgeneralisation and personalisation; the conditional rules linking them ("If I am not liked by all, I am worthless") are dysfunctional assumptions. Cognitive therapy targets all of these (Section 7.5).

EXAM TRAP

Beck's triad is self, world and future, not id-ego-superego, past-present-future, mother-father-self, self-others-environment, thoughts-feelings-behaviours, body-mind-spirit or mood-behaviour-somatic state. Automatic thoughts, distortions and dysfunctional assumptions are other parts of the model, not the triad. Repressed sexual conflict, too little reinforcement (Lewinsohn) and insecure attachment belong to other theories.

KEY STUDY: SOCIAL ORIGINS OF DEPRESSION (BROWN AND HARRIS, 1978)

George Brown and **Tirril Harris** interviewed women in Camberwell, south London. Severe losses and disappointments provoked depression mainly in women with **vulnerability factors**: no confiding relationship, three or more children under 14 at home, loss of the mother before age 11, and no job outside the home. Criticism: one city, women only, and life events recalled after the fact.

First-line drug treatment ●●

For depression, selective serotonin reuptake inhibitors (SSRIs) such as fluoxetine are first line, safer in overdose than **tricyclics** (imipramine) and **monoamine oxidase inhibitors** (MAOIs, which need a low-tyramine diet). For bipolar disorder the gold standard is **lithium** (**John Cade**, 1949), used for acute mania and for **prophylaxis** (preventing further episodes), and it lowers suicide risk. Its therapeutic range is narrow, so blood levels are monitored; toxicity brings coarse tremor, vomiting, unsteadiness and confusion. Valproate, carbamazepine and lamotrigine are alternatives, and atypical antipsychotics (olanzapine, risperidone) are first line for acute mania; haloperidol and lorazepam are adjuncts for agitation only. Electroconvulsive therapy and the psychotherapies are in Chapter 7.

EXAM TRAP

Fluoxetine is not a first-line treatment for acute mania: an antidepressant given alone in bipolar disorder can trigger a switch into mania or rapid cycling. Lithium, valproate and atypical antipsychotics are first line.

Suicide ●●

Suicidal ideation is thinking of killing oneself; a plan fixes method, place and time; intent is how much the person wants to die; an attempt is a self-injurious act with some intent to die that is survived; completed suicide ends in death. **Lethality** is the medical danger of the method, so the mode of an attempt (hanging or pesticide against a few tablets) shows its lethality, while intent is inferred from planning and precautions against rescue. **Norman Kreitman** coined **parasuicide** (1969) for non-fatal deliberate self-injury or overdose whatever its intent, later called deliberate self-harm. **Non-suicidal self-injury** (cutting or burning without intent to die) is a DSM-5 condition for further study.

The World Health Organization counts over 700,000 suicides a year. India's National Crime Records Bureau recorded about 1.71 lakh in 2022, about seven in ten of them men and most aged 18 to 45, with family problems (about a third) and illness the leading recorded causes. Women attempt suicide more often, but men die by it more often, using more lethal methods: the gender paradox.

Émile Durkheim (1858 to 1917), in *Suicide* (1897), traced suicide to social ties: **egoistic** (too little integration, the isolated person), **altruistic** (too much, dying for the group as in harakiri), **anomic** (too little regulation, sudden upheaval such as losing a fortune) and **fatalistic** (too much regulation, as for slaves or prisoners). **Edwin Shneidman** (1918 to 2009), founder of suicidology, saw suicide as escape from **psychache**, unbearable psychological pain. In **Thomas Joiner**'s interpersonal theory (2005) the wish to die comes from **perceived burdensomeness** and **thwarted belongingness**, but lethal attempts also need an **acquired capability**, fearlessness of pain and death built by exposure such as self-harm. **Marie Åsberg** (1976) found that depressed patients with low cerebrospinal fluid **5-HIAA**, the serotonin metabolite, made more frequent and more violent suicide attempts.

The strongest single predictor of suicide is a **previous attempt**. Other risk factors are depression and other mental disorders, alcohol, **hopelessness** (Beck's Hopelessness Scale predicts better than depression severity), male sex, older age, living alone, chronic illness, a family history of suicide and access to means such as pesticides. Social support, children at home, religious objection and restricted means protect. Risk is highest when high lethality, a specific plan, frequent ideation and family history converge. SAD PERSONS (Patterson and colleagues, 1983) is in Section 7.2, and the decriminalisation of attempted suicide by the Mental Healthcare Act 2017 in Section 7.1.

> **EXAM TRAP**
>
> The mode or method of an attempt indicates **lethality**; intentionality is the wish to die, and conviction and survival are not standard dimensions. A recent attempt of low lethality with no family history carries less risk than high lethality, a specific plan, frequent ideation and family history together.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. A major depressive episode needs five of nine symptoms in the same _____ period.
2. Persistent depressive disorder requires depressed mood for at least _____ years in adults.
3. Disruptive mood dysregulation disorder requires onset before age _____.
4. Four or more mood episodes in 12 months earn the specifier "with _____".
5. Beck's cognitive triad is a negative view of the self, the world and the _____.
6. The kindling hypothesis of mood disorders was proposed by _____.
7. The strongest single predictor of suicide is a _____.

Answers: (1) 2-week, (2) 2, (3) 10, (4) rapid cycling, (5) future, (6) Robert Post, (7) previous attempt.

8.6 Anxiety and Obsessive-Compulsive and Related Disorders

MEDIUM YIELD · 32 QS

A racing heart before an exam is anxiety doing its job. A man who has avoided buses for three years since he panicked on one, or a woman who washes for three hours a day to cancel a thought of germs, has crossed into disorder. This section sets out the anxiety disorders and why they develop, then obsessive-compulsive disorder and its relatives.

Fear, anxiety and the old neuroses

Fear is the alarm response to danger that is present now: a surge of autonomic arousal and the urge to escape. **Anxiety** is apprehension about a future threat, felt as muscle tension, vigilance and cautious avoidance. Both are normal, and moderate anxiety even helps performance (the Yerkes-Dodson law, Section 1.10). Anxiety becomes pathological when it is out of proportion to the danger, developmentally inappropriate, persistent (typically six months or more) and impairing.

The Scottish physician **William Cullen** coined the word neurosis in 1769 for disorders of sense and motion arising from the nervous system without fever or local disease. **Sigmund Freud** (1856 to 1939) narrowed **neurosis** to a non-psychotic disorder of unconscious conflict, with insight preserved, and in 1895 described an "anxiety neurosis". DSM-III (1980) dropped the term because it implied Freud's theory. DSM-5 (2013) split the old anxiety group into three chapters: anxiety disorders, obsessive-compulsive and related disorders, and trauma- and stressor-related disorders (Section 8.7). ICD-10 kept the umbrella block F40 to F48, "neurotic, stress-related and somatoform disorders", running from phobic anxiety (F40), other anxiety (F41) and OCD (F42) to somatoform (F45) and other neurotic disorders (F48). ICD-11 follows DSM-5, with separate "anxiety or fear-related" and "obsessive-compulsive or related" groupings.

The anxiety disorders and their time rules ●●●

DSM-5-TR orders the anxiety disorders by typical age of onset and separates them by what is feared (Table 8.21); six months is the usual duration. Anxiety caused by a drug or its withdrawal, or by an illness such as hyperthyroidism, is diagnosed instead as substance/medication-induced anxiety disorder or anxiety disorder due to another medical condition.

Table 8.21 The DSM-5-TR anxiety disorders and their duration rules.

Disorder	Feared or key sign	Duration
Separation anxiety disorder	Separation from attachment figures (3 of 8 signs)	4 weeks in children and adolescents; typically 6 months in adults
Selective mutism	Silent at school, speaks at home	1 month, not only the first month of school
Specific phobia	A specific object or situation	Typically 6 months
Social anxiety disorder	Scrutiny and embarrassment	Typically 6 months
Panic disorder	Recurrent unexpected panic attacks	1 month of worry or behaviour change after an attack
Agoraphobia	2 of 5 situations where escape is hard	Typically 6 months

Disorder	Feared or key sign	Duration
Generalised anxiety disorder	Many events; 3 of 6 symptoms (1 in children)	More days than not, at least 6 months

Separation anxiety disorder (excessive fear of separation from attachment figures) and **selective mutism** (consistent failure to speak in specific social situations while speaking in others) moved here from DSM-IV's childhood disorders, and separation anxiety can now begin in adulthood. The selectively mute child's silence at school comes from social anxiety, not a language problem.

A **specific phobia** is a persistent, irrational fear of a specific object or situation, out of proportion to the danger, provoked almost every time and avoided or endured with dread: an anxiety disorder, not a mild stress reaction, a personality disorder or a passing hallucination. Its five types are animal, natural environment (heights, storms, water), blood-injection-injury, situational (aeroplanes, lifts, enclosed places) and other (choking, vomiting). Most phobics show sympathetic arousal, but the blood-injection-injury type shows a diphasic **vasovagal response**: heart rate and blood pressure rise briefly, then drop, and the person may faint.

Social anxiety disorder is fear of situations of possible scrutiny (conversation, eating in public, a speech) because one may act in a way that is negatively evaluated or embarrassing; in children it must occur with peers. The **performance only** specifier covers fear confined to speaking or performing in public.

Generalised anxiety disorder (GAD) is excessive anxiety and worry about a number of events or activities, more days than not for at least six months, that the person finds hard to control, with three or more of six symptoms (only one in children): restlessness, easy fatigue, poor concentration, irritability, muscle tension and sleep disturbance. Hopping from health to money to a child's journey home, it is continued, **free-floating** apprehension about unspecified things. ICD-10 asks for anxiety on most days for several weeks at a time, usually several months; its research criteria fix six months. ICD-11 asks for at least several months, more days than not, with no fixed number.

> EXAM TRAP
>
> GAD is **continued apprehension about unspecified things**, defined by persistent, excessive worry. "Fear of fear" is panic disorder, flashbacks PTSD and safety checking OCD; mania and mood swings are mood symptoms (Section 8.5), hallucinations psychotic (Section 8.4). GAD needs at least **6 months** in adults and specific phobia typically 6 months, not 1 week or 1, 3 or 12 months.

Panic attacks, panic disorder and agoraphobia ●●

A **panic attack** is an abrupt surge of intense fear or discomfort that peaks **within minutes** (literally a few minutes; not seconds, half an hour or hours), with four or more of thirteen symptoms: palpitations, sweating, trembling, shortness of breath, choking, chest pain, nausea, dizziness, chills or heat sensations, paraesthesias (numbness or tingling), derealisation or depersonalisation, fear of losing control or "going crazy", and fear of dying. **Expected** attacks have an obvious cue (a spider); **unexpected** ones come out of the blue. A panic attack is not itself a disorder; in other disorders it is a specifier ("PTSD with panic attacks").

Panic disorder requires recurrent unexpected attacks, at least one followed by a month or more of persistent worry about further attacks or their consequences (a heart attack, going mad), a maladaptive change in behaviour (avoiding exercise), or both. One attack is not panic disorder.

Agoraphobia, literally "fear of the marketplace", is marked fear of two or more of five situations: public transport, open spaces, enclosed places (shops, cinemas), queues or crowds, and being outside the home alone, because escape might be difficult or help unavailable if panic-like symptoms strike. The situations are avoided, endured with dread or faced only with a companion, typically for six months. Since DSM-5, agoraphobia is diagnosed independently of panic disorder; a person meeting both receives both.

In 1978 **Alan Goldstein** and **Dianne Chambless** argued that after a few attacks people come to fear the bodily sensations of anxiety themselves. This **fear of fear** marks panic disorder and agoraphobia.

> EXAM TRAP
>
> Agoraphobia is the **irrational fear of public places**. Hearing voices is psychotic, nervousness in a lift is claustrophobia, fear of tall trees or buildings a fear of heights, and re-experiencing trauma PTSD. Panic disorder is **recurrent unexpected** attacks, not one attack, worry about many things (GAD) or fear of panicking in enclosed spaces.

Naming the phobias ●

Phobia names join a Greek word for the feared thing to *phobos*, fear: *algos* (pain) gives algophobia (Table 8.22). Clinically, glossophobia is social anxiety disorder (performance only) and trypanophobia the blood-injection-injury type.

Table 8.22 Phobia names and their objects.

Name	Fear of	Name	Fear of
Algophobia	Pain	Acrophobia	Heights
Nosophobia	Disease	Claustrophobia	Enclosed spaces
Cynophobia	Dogs	Agoraphobia	The marketplace (public places)
Haematophobia	Blood	Arachnophobia	Spiders
Ophidiophobia	Snakes	Aerophobia	Flying
Hydrophobia	Water	Nyctophobia	Darkness
Thanatophobia	Death	Xenophobia	Strangers
Mysophobia	Dirt and germs	Astraphobia	Thunder and lightning
Glossophobia	Public speaking	Trypanophobia	Needles

Why anxiety develops ●●

Biology. Anxiety is moderately heritable, though what is inherited is a general anxious tendency. Benzodiazepines calm anxiety by boosting GABA, the main inhibitory transmitter, so the **GABA-benzodiazepine system** is the classic anxiety system; noradrenaline from the locus coeruleus drives the alarm, serotonin modulates it, and the amygdala is the hub of the

fear circuit. **Jerome Kagan** (1929 to 2021) found that **behavioural inhibition**, shyness and withdrawal from novelty visible by four months of age, predicts later anxiety.

Learning. The **two-factor theory** of **O. Hobart Mowrer** (1907 to 1982), set out in 1947, explains why fears last. A neutral stimulus paired with a frightening one becomes a conditioned stimulus for fear (classical conditioning, as with Little Albert, Section 1.6); avoiding it then reduces fear, so avoidance is negatively reinforced and the fear is never unlearned. **Martin Seligman** (born 1942) explained why snakes and heights are feared more than electric sockets with **preparedness** (1971): evolution has prepared us to learn ancestral fears quickly and keep them. **Stanley Rachman** (1977) explained phobias with no frightening memory through three pathways: direct conditioning, vicarious learning and information.

Cognition. David M. Clark (1986) traced panic to the **catastrophic misinterpretation** of bodily sensations: a skipped heartbeat is read as a heart attack, and fear and sensations feed each other (Figure 8.8). **Catastrophic thinking** in general means worrying that the worst will always happen. It differs from other distortions (Section 7.5): dichotomous thinking (everything extremely good or extremely bad), selective abstraction (seeing only the negative) and personalisation (taking things personally). **Steven Reiss**'s **anxiety sensitivity**, the belief that anxiety symptoms are harmful, predicts panic. For GAD, **Thomas Borkovec** held that verbal worry avoids the vivid imagery and arousal of fear and is reinforced by that avoidance; **Adrian Wells**' metacognitive model adds positive and negative beliefs about worry itself; and **Michel Dugas** stressed **intolerance of uncertainty**.

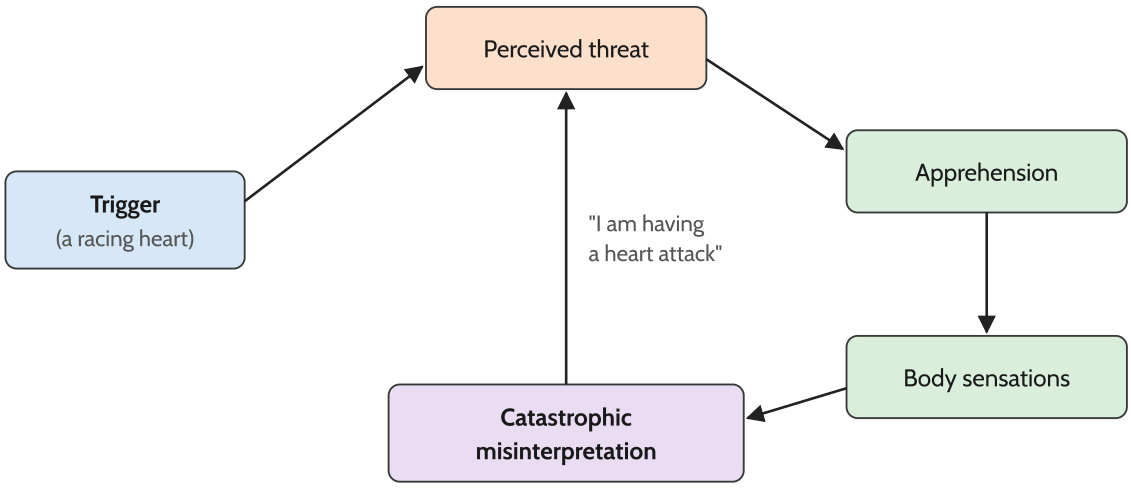

Figure 8.8 Clark's (1986) vicious circle of panic: misread bodily sensations raise the threat, which strengthens the sensations.

The suffocation alarm. Donald F. Klein's **suffocation false alarm theory** (1993) proposes an evolved suffocation monitor that in panic disorder is oversensitive, firing at harmless rises in carbon dioxide; carbon dioxide inhalation and lactate infusion do trigger attacks far more readily in panic patients. It explains panic disorder, not Down syndrome, autism (Section 8.3) or delirium (Section 8.10). **David H. Barlow**'s **triple vulnerability** theory then combines a generalised biological vulnerability (an anxious temperament), a generalised psychological vulnerability (a sense that events are uncontrollable) and a specific psychological vulnerability (learning that certain things are dangerous).

Psychodynamic view. Freud later treated anxiety as **signal anxiety** (1926), a warning to the ego that a forbidden impulse is near, which sets defences going. The defence of phobia is **displacement**, shifting feeling from a threatening object to a safer one. Each classic neurosis had a signature defence: repression in hysteria, displacement in phobia, isolation, undoing and reaction formation in OCD, introjection in depression.

KEY STUDY: LITTLE HANS (FREUD, 1909)

Hans, aged five, would not go out for fear that a horse would bite him. Freud, working mainly through the boy's father, read the fear as an Oedipal fear of the father displaced onto horses, which, unlike a father, could be avoided. Critics noted that the evidence came from a father devoted to Freud, and that Hans had seen a horse fall in the street, which suggests simple conditioning.

EXAM TRAP

Displacement is the defence of **phobias**, not of hysteria, OCD or depression. Catastrophic thinking is **expecting the worst**, not dichotomous thinking, selective abstraction or personalisation.

Obsessive-compulsive disorder ●●●

Obsessions are recurrent, persistent, intrusive and unwanted thoughts, urges or images that cause anxiety and that the person tries to ignore, suppress or neutralise. **Compulsions** are repetitive behaviours (washing, checking, ordering, tapping) or mental acts (praying, counting) performed in response to an obsession or by rigid rules, to reduce anxiety or prevent a dreaded event, though not realistically connected to it. The cardinal DSM-5 feature is obsessions and/or compulsions that are **time-consuming (more than 1 hour a day) or cause significant distress or impairment**. Merely having obsessions is not enough, and there is no six-hour or two-week rule.

Priya, 22, after touching a door handle, fears she will carry germs that kill her mother and washes three hours a day: this is OCD. GAD worry concerns realistic matters without rituals, illness anxiety disorder is fear of having a disease (Section 8.7), and a specific phobia simply avoids an external object.

Obsessions are **ego-dystonic**: intrusive, unwanted and inconsistent with the self-concept, neither part of one's identity nor pleasurable. Obsessive-compulsive personality disorder (Section 8.11), a separate disorder, is **ego-syntonic**: perfectionism and control felt as right, without true obsessions or compulsions. The **insight specifier** runs from good or fair through poor to absent insight/delusional beliefs (certain the house will burn unless the stove is checked 30 times; 4 percent or fewer). A **tic-related** specifier marks a current or past tic disorder.

Symptoms cluster in four dimensions: contamination and washing, doubt or harm and checking, symmetry with ordering and counting, and taboo (aggressive, sexual, religious) thoughts. Hoarding is no longer an OCD compulsion, since DSM-5 made hoarding disorder separate. Twelve-month prevalence is about 1.2 percent, and onset is in childhood or early adulthood (mean 19.5 years; boys earlier).

EXAM TRAP

Tapping, washing and checking are compulsions; **hoarding is not**. OCD is **ego-dystonic**, OCPD ego-syntonic; obsessions are not always reality-grounded, OCD is not confined to childhood, and total unawareness that obsessions are unreasonable is absent insight. Grandiosity and need for admiration mark narcissistic personality disorder (Section 8.11).

Biology. OCD responds to clomipramine and the SSRIs, which block serotonin reuptake, implicating the **central serotonin (5-HT) system**; GABA is the anxiety system, and glutamate and dopamine play lesser parts. Imaging implicates the **cortico-striato-thalamo-cortical circuit**: the orbitofrontal cortex and anterior cingulate signal "something is wrong", the caudate nucleus fails to filter the signal, and the thalamus loops it back. **Susan Swedo** (1998) described **PANDAS**, sudden childhood OCD or tics after streptococcal infection, thought to be an autoimmune attack on the basal ganglia. First-degree relatives have about twice the rate, ten times after childhood onset.

Learning. Mowrer's two-factor theory applies directly (Figure 8.9): the obsession acquires anxiety by classical conditioning, the compulsion relieves it, and the relief negatively reinforces the compulsion. Compulsive acts are thus **escape or avoidance responses** (safety behaviours): not conditioned stimuli (the contaminating thought is the stimulus), not unconditioned responses (unlearned reflexes), not positively reinforced (distress is removed, nothing pleasant added) and not involuntary. Conditioning is the general mechanism; escape is the pattern. Exposure and response prevention blocks the ritual (Section 7.4).

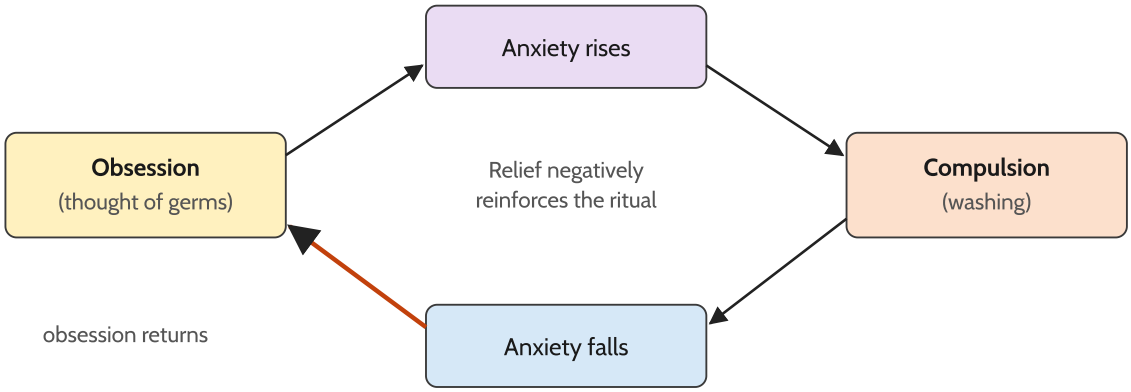

Figure 8.9 The obsessive-compulsive cycle: relief reinforces the ritual, but the obsession returns (dashed line).

Cognition and psychodynamics. Nearly everyone has odd intrusive thoughts; **Paul Salkovskis** (1985) argued that OCD begins when they are appraised as signs of **inflated responsibility** for harm, which must be neutralised. Rachman added **thought-action fusion**, the belief that thinking of an act is as bad as doing it or makes it likelier. Freud's Rat Man (1909), tormented by the thought of rats gnawing into his father, is the classic case of "obsessional neurosis", which Freud traced to regression to anal-stage conflicts over control, dirt and aggression. Its defences are **isolation of affect** (the thought stays, its feeling goes), **undoing** (an act that symbolically cancels a forbidden thought, the most characteristic), **reaction**

formation (scrupulous cleanliness against a wish to soil) and intellectualisation. Repression is the defence of hysteria.

> EXAM TRAP
>
> Isolation, undoing and reaction formation are **all** OCD defences. For a single answer choose **undoing**, the defence of obsessive-compulsive neurosis (not depression, mania or schizophrenia). The defence NOT used in OCD is **repression**.

The related disorders ●●

Body dysmorphic disorder (BDD) is preoccupation with perceived defects or flaws in appearance that are not observable or appear slight to others, with repetitive behaviours (mirror checking, grooming, skin picking, reassurance seeking) or mental acts (comparing). Specifiers record insight and **muscle dysmorphia**, the belief that one's body is too small or insufficiently muscular. **Enrico Morselli** called it dysmorphophobia in 1891. DSM-IV listed BDD among the somatoform disorders; DSM-5 placed it in **obsessive-compulsive and related disorders**, not with somatic symptom, anxiety or personality disorders. It concerns appearance, unlike somatic symptom disorder (distressing bodily symptoms), illness anxiety disorder (fear of serious illness) and conversion disorder (neurological symptoms without matching disease), all in Section 8.7.

Hoarding disorder is persistent difficulty discarding possessions regardless of value, from a perceived need to save them, so that clutter blocks the use of living spaces. **Trichotillomania** (hair-pulling disorder), named by **François Hallopeau** in 1889, is recurrent hair pulling causing hair loss, with repeated attempts to stop; **excoriation disorder** is the same with skin picking and lesions. These body-focused repetitive behaviours are not driven by obsessions. Substance/medication-induced OCRD and OCRD due to another medical condition complete the chapter. ICD-11's obsessive-compulsive or related group also includes olfactory reference disorder (a conviction of emitting a foul body odour) and hypochondriasis (health anxiety), and calls trichotillomania and excoriation body-focused repetitive behaviour disorders. Finally, an obsession is resisted as intrusive and usually recognised as excessive; a GAD worry concerns real-life matters and brings no ritual; a delusion is a false belief held as simply true.

> SCAN YOUR BRAIN
>
> Fill in the blanks. The answers are below; cover them first.
>
> 1. Panic disorder needs at least _____ of worry or behaviour change after an attack.
> 2. The phobia type with a diphasic vasovagal fainting response is the _____ type.
> 3. Donald Klein's _____ theory explains panic disorder.
> 4. In Little Hans, fear of the father was shifted onto horses by _____.
> 5. In learning theory, compulsions are _____ responses, negatively reinforced by relief.
> 6. DSM-5 classifies body dysmorphic disorder under _____ disorders.
>
> **Answers:** (1) 1 month, (2) blood-injection-injury, (3) suffocation false alarm, (4) displacement, (5) escape (avoidance), (6) obsessive-compulsive and related.

8.7 Trauma, Dissociative and Somatic Symptom Disorders HIGH YIELD · 50 QS

A soldier who relives a shelling every night, a woman who finds clothes she does not remember buying, a girl whose arm goes limp after a quarrel: all three were once called hysterical. This section follows hysteria as it split into the trauma-related, dissociative and somatic symptom disorders, and ends with illness that is feigned.

Hysteria, from Hippocrates to DSM-III ●●

The word **hysteria** comes from the Greek *hystera*, womb: **Hippocrates** (about 460 to 370 BCE) and his followers blamed its fits and paralyses on a "wandering womb". At the Salpêtrière in Paris, **Jean-Martin Charcot** (1825 to 1893) showed that hysterical phenomena such as loss of feeling in a limb or fainting could be produced and removed by hypnosis and suggestion, so their cause was psychological, not a lesion.

His pupil **Pierre Janet** (1859 to 1947) named the mechanism *désagrégation* (1889), now called **dissociation**: under strain, a cluster of memories and functions splits off from consciousness and runs on its own. The phrase **la belle indifférence**, the calm unconcern some patients show about a dramatic symptom, is usually credited to Janet. In Vienna **Josef Breuer** (1842 to 1925) relieved "Anna O." of her paralyses by letting her talk through their origins under hypnosis, her "talking cure", and with **Sigmund Freud** (1856 to 1939) published *Studies on Hysteria* (1895).

For Freud, hysteria was physical reactions to emotional trauma: a painful memory is pushed out of awareness by **repression** and its energy converted into a bodily symptom, hence **conversion**. Most psychodynamic theorists still see repression behind both conversion and dissociative symptoms. The symptom gives **primary gain**, relief from the anxiety of the conflict, and often **secondary gain**, attention or escape from a duty.

> KEY STUDY: HILGARD'S HIDDEN OBSERVER
>
> **Ernest Ropiequet Hilgard** (1904 to 2001) gave hypnotised volunteers a suggestion that they would feel no pain and put their hands in iced water. They reported little pain, yet a **hidden observer**, addressed by a prearranged cue, reported nearly normal pain. His **neodissociation theory** (*Divided Consciousness*, 1977) holds that awareness can split into parallel streams. Critics such as Nicholas Spanos argued that the instructions create the observer.

In 1980 DSM-III divided hysteria into somatoform and dissociative disorders (Figure 8.10); ICD-10 kept them together as dissociative (conversion) disorders, F44.

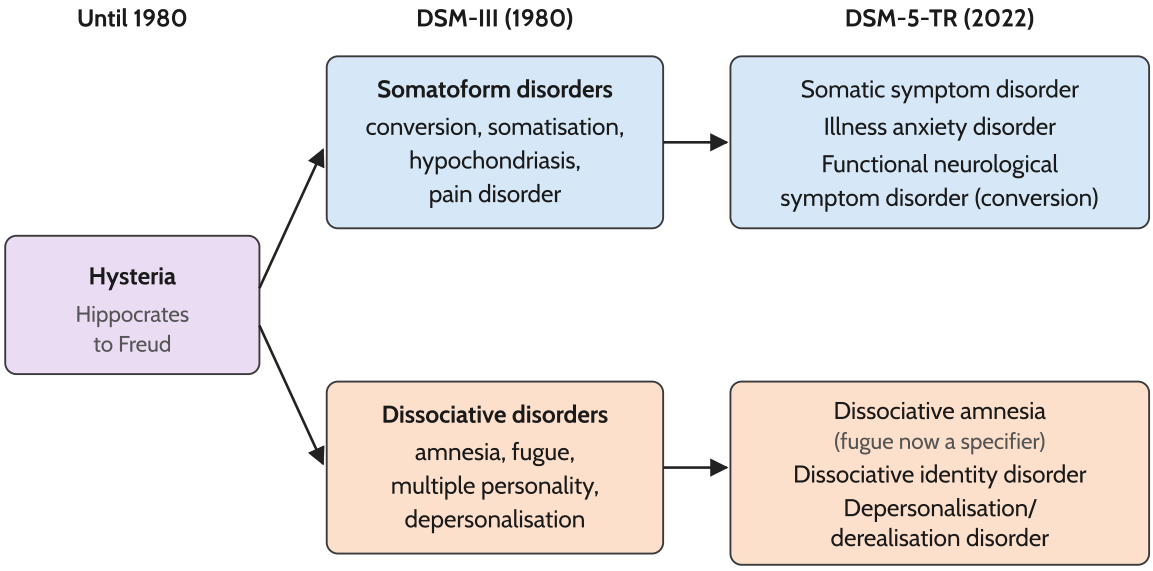

Figure 8.10 How hysteria was divided: DSM-III split it into two classes, which DSM-5 renamed.

EXAM TRAP

Freud's hysteria is **physical reactions to emotional trauma**; a phobia of horses is Little Hans, and the Oedipus complex and oral traits belong to Chapter 4. The psychodynamic mechanism of dissociative symptoms is **repression**: conflict is what gets repressed. Charcot used **both** hypnosis and suggestion. La belle indifférence is credited to **Janet**, not Freud, Anna Freud (who catalogued the defence mechanisms, Chapter 4) or Hippocrates.

Posttraumatic and acute stress disorders ●●●

First World War soldiers who shook, froze or went mute were said to have shell shock; **Abram Kardiner** (1891 to 1981) described *The Traumatic Neuroses of War* (1941), and DSM-III named **posttraumatic stress disorder** (PTSD) in 1980. DSM-5 moved it from the anxiety disorders into a new trauma- and stressor-related chapter.

The event must involve actual or threatened death, serious injury or sexual violence, experienced directly, witnessed in person, learned of as happening to a close family member or friend, or met through repeated exposure to aversive details, as by police officers hearing about child abuse. Symptoms fall into four clusters:

- **Intrusion** (1 of 5): distressing memories, nightmares, flashbacks (acting or feeling as if the event were recurring), distress at reminders.
- **Avoidance** (1 of 2) of trauma-related thoughts and feelings or of outside reminders.
- **Negative alterations in cognitions and mood** (2 of 7): amnesia for part of the event, self-blame, "no one can be trusted", guilt, detachment, no positive emotions.
- **Alterations in arousal and reactivity** (2 of 6): irritability, recklessness, hypervigilance, exaggerated startle, poor concentration, poor sleep.

Symptoms must last **more than 1 month**. Specifiers are the dissociative subtype (persistent depersonalisation or derealisation) and delayed expression (full criteria met only 6 months or

more after the event); children 6 and younger have separate criteria. **Acute stress disorder** covers **3 days to 1 month** after the trauma and needs nine of fourteen symptoms from five categories. DSM-IV required dissociative symptoms for it; DSM-5 dropped that rule, so timing now separates it from PTSD, and both sit in the trauma chapter.

ICD-11 places PTSD among disorders specifically associated with stress, which require an extremely threatening or horrific event, and keeps three core elements: re-experiencing in the here and now, avoidance, and a persistent sense of current threat. **Complex PTSD** adds disturbances in self-organisation (affect dysregulation, negative self-concept, relationship difficulties), usually after prolonged, inescapable trauma such as torture or childhood abuse. An acute stress reaction is a normal response, not a disorder.

Who develops PTSD depends first on the dose and type of trauma (most prisoners of war do; interpersonal violence is worse than accidents), then on vulnerability: earlier anxiety, family instability and **peritraumatic dissociation** (feeling dazed or unreal during the event) raise risk, while social support protects. Conditioning attaches fear to reminders and avoidance blocks extinction. The amygdala is overactive, the hippocampus smaller and the HPA (hypothalamic-pituitary-adrenal) stress axis dysregulated. Prolonged exposure, cognitive processing therapy and EMDR are covered in Sections 7.4 and 7.5. Trauma can also bring **post-traumatic growth**, which **Richard Tedeschi** and **Lawrence Calhoun** (1996) measured in five domains: appreciation of life, relating to others, personal strength, new possibilities and spiritual change.

> EXAM TRAP
>
> PTSD has **four** clusters (DSM-IV had three). Magical thinking and ideas of reference belong to schizotypal personality and psychosis (Sections 8.4 and 8.11), and "denial, anger, bargaining" are Elisabeth Kübler-Ross's stages of dying (Chapter 6). PTSD lasts more than 1 month; acute stress disorder 3 days to 1 month; delayed expression begins at 6 months.

Adjustment, grief and attachment disorders

An **adjustment disorder** is an emotional or behavioural reaction to an identifiable stressor, such as divorce or job loss, with disproportionate distress or impaired functioning; it begins within 3 months of the stressor and ends within 6 months after the stressor or its consequences end. Its subtypes are with depressed mood, with anxiety, with mixed anxiety and depressed mood, with disturbance of conduct, with mixed disturbance of emotions and conduct, and unspecified. Being brief and tied to a present stressor, it is best treated by brief psychotherapy and **crisis intervention** (Section 7.9).

Prolonged grief disorder, added in DSM-5-TR (2022), is intense yearning for or preoccupation with the deceased, with symptoms such as identity disruption, disbelief and meaninglessness, at least 12 months after the death (6 months in children and adolescents; ICD-11 needs 6), beyond cultural norms. Ordinary bereavement is not a disorder.

After severe neglect or institutional rearing, a child with **reactive attachment disorder** is inhibited and emotionally withdrawn, rarely seeking comfort, while one with **disinhibited social engagement disorder** shows indiscriminate sociability, readily going off with strangers. Both need a developmental age of at least 9 months (Table 8.23).

Table 8.23 Trauma- and stressor-related disorders (DSM-5-TR).

Disorder	Trigger	Time rule	Core picture
PTSD	Death, injury or sexual violence	More than 1 month	Four symptom clusters
Acute stress disorder	As for PTSD	3 days to 1 month	9 of 14 symptoms
Adjustment disorder	Any stressor	Starts within 3 months; ends within 6 months of stressor's end	Disproportionate distress
Prolonged grief disorder	Death of someone close	12 months (children 6)	Yearning, preoccupation
Reactive attachment disorder	Severe neglect	Developmental age 9 months; before age 5	Withdrawn, seeks no comfort
Disinhibited social engagement disorder	Severe neglect	Developmental age 9 months	Indiscriminate with strangers

Dissociative identity disorder ●●●

Dissociative disorders disrupt the normal integration of consciousness, memory, identity, emotion, perception and behaviour, and most follow trauma. **Dissociative identity disorder** (DID), called multiple personality disorder until DSM-IV (1994), requires two or more distinct personality states, or an experience of possession, **and** recurrent gaps in the recall of everyday events, personal information or trauma beyond ordinary forgetting, not explained by accepted religious practice.

The identity most often encountered, carrying the person's real name, is the **host**; the others are **alters**, which may differ in age, gender and voice, and changing between them is a **switch**. The amnesia shows in daily life: a woman finds clothes she does not remember buying. About 90 percent of patients report childhood abuse or neglect.

Corbett Thigpen and **Hervey Cleckley**'s *The Three Faces of Eve* (1957) and *Sybil* (1973), a case later criticised as shaped by the therapist, frame the debate on causes. The trauma model sees identities as states split off to escape unbearable abuse. The **sociocognitive model** of **Nicholas Spanos** (1942 to 1994) sees them as roles shaped in suggestible people by therapists' questions, hypnosis and media; in a 1985 experiment, over 80 percent of students playing an accused murderer produced an "alter" when interviewed.

> EXAM TRAP
>
> "Multiple personalities in one body" is **DID, not schizophrenia**, whose "split" is between thought and emotion (Section 8.4). Bipolar disorder alternates moods, and borderline personality disorder an unstable self-image, not separate identities with amnesia. The identity with the real name is the **host**; "ghost" and "mixed" identities do not exist.

Dissociative amnesia, fugue and depersonalisation ●●●

Dissociative amnesia is an inability to recall important autobiographical information, usually traumatic or stressful, too extensive to be ordinary forgetting and not due to a substance or a neurological or medical condition. It is a potentially reversible, mainly retrograde failure of retrieval, in five forms:

- **Localised**: a circumscribed period, such as the hours of an assault (the commonest form).

- **Selective**: some, but not all, events of that period.
- **Generalised amnesia**: one's entire life history, often with identity.
- **Systematised**: one category, such as everything about a violent sibling.
- **Continuous**: each new event, as it occurs.

Fugue, a separate diagnosis in DSM-IV, is now the specifier *with* **dissociative fugue**: apparently purposeful travel or bewildered wandering with amnesia for identity, as in the man found working in another city with no memory of who he is. Organic amnesias differ. **Transient global amnesia** is a sudden, hours-long loss of new learning in an older adult who keeps their identity but repeats the same questions; **Korsakoff syndrome** is thiamine-deficiency amnesia with confabulation in chronic alcohol use; **Alzheimer's disease** begins with gradually failing new learning (Section 8.10); amnesia from head injury or another medical condition is mainly anterograde. In all of these, personal identity is lost last; in dissociative amnesia it can go first.

Depersonalisation/derealisation disorder means persistent or recurrent experiences of either kind, or both. **Depersonalisation** is detachment from oneself: being an outside observer of one's own thoughts, feelings, body or actions, or feeling no longer one's normal self. **Derealisation** is unreality of the surroundings, which seem dreamlike, foggy or distorted. **Reality testing stays intact** ("it is *as if* I were dreaming"), which separates it from psychosis.

> EXAM TRAP
>
> Self unreal is **depersonalisation**; world unreal is **derealisation**. Multiple personality, fugue and amnesia are hysterical dissociative states, but **déjà vu** (a new situation feels already lived) is not: it is a disorder of recognition memory, seen in healthy people and temporal lobe epilepsy. A hallucination is perception without a stimulus (Section 8.2).

Other specified dissociative disorder covers possession without amnesia, identity change after brainwashing, acute reactions lasting under a month, and **dissociative trance**, an acute narrowing or loss of awareness of the surroundings with profound unresponsiveness, outside accepted cultural practice. ICD-10 adds **dissociative stupor** and **trance and possession disorders**, in which a spirit or deity seems to take over; possession states are among the commonest dissociative presentations in India, though within an accepted ritual they are not disorders. **Ganser syndrome**, described by **Sigbert Ganser** (1853 to 1931) in 1898 in prisoners awaiting trial, combines **approximate answers** (a cow has "five" legs), clouding of consciousness, pseudohallucinations and conversion symptoms.

Somatic symptom and illness anxiety disorders ●●

Somatoform disorders (DSM-III and DSM-IV) were conditions with symptoms suggesting a medical condition without evidence of physical pathology. They included **somatisation disorder**, earlier **Briquet's syndrome** after **Paul Briquet** (1796 to 1881), who in 1859 described patients returning with endless lists of complaints; **hypochondriasis**; **pain disorder**; and conversion. **Neurasthenia** ("lack of nerve strength"), named by **George Miller Beard** (1839 to 1883) in 1869 for unexplained exhaustion, survives in ICD-10.

DSM-5 renamed the class somatic symptom and related disorders and defined it by positive features. **Somatic symptom disorder** needs one or more distressing somatic symptoms with

excessive thoughts, feelings or behaviours: disproportionate thoughts about their seriousness, high health anxiety, or excessive time and energy spent on them, typically for more than 6 months. The symptoms may have a medical cause: a woman with real arthritis who spends hours a day researching it qualifies.

> EXAM TRAP
>
> "Physical symptoms without a medical cause" is the old **somatoform** idea, which DSM-5-TR calls "problematic". Against hallucinations, delusions and obsessions it is still the keyed answer for somatic symptom disorder, but the DSM-5-TR feature is **excessive response** to distressing symptoms.

Illness anxiety disorder (formerly hypochondriasis) is preoccupation with having or acquiring a serious illness when somatic symptoms are absent or mild, with high health anxiety and excessive checking or avoidance, for at least 6 months. It has **care-seeking** and **care-avoidant** types. ICD-11 keeps the name hypochondriasis but places it among obsessive-compulsive and related disorders, and replaces somatoform disorders with **bodily distress disorder**, distressing bodily symptoms with excessive attention to them. **Psychological factors affecting other medical conditions** covers stress or behaviour that worsens a real disease (Section 7.9).

> EXAM TRAP
>
> Chronic fatigue syndrome, or **systemic exertion intolerance disease**, has **no DSM-5-TR criteria**. The US Institute of Medicine (2015) requires fatigue for 6 months or more, not relieved by rest; post-exertional malaise; unrefreshing sleep; and cognitive impairment or orthostatic intolerance. **Weight change is not a criterion.**

Conversion: functional neurological symptom disorder ●●●

Hysteria's direct heir is **functional neurological symptom disorder** (conversion disorder): one or more symptoms of altered voluntary motor or sensory function, with clinical findings incompatible with recognised neurological or medical conditions. The symptoms are not intentionally produced. They include paralysis; abnormal movements such as **astasia-abasia** (cannot stand or walk, yet moves the legs normally in bed); **globus**, a lump in the throat; **aphonia**; **psychogenic non-epileptic seizures**; **glove anaesthesia**, ending sharply at the wrist where no nerve territory ends; and tunnel vision or **functional blindness**, explained by psychological factors with no damage to eye, optic nerve or cortex.

Diagnosis rests on positive signs. In **Hoover's sign** a "paralysed" leg cannot press down on request, yet presses down when the patient lifts the healthy leg against resistance. A woman whose arm becomes paralysed after an argument, with a normal examination and the remark "it's not really bothering me", is the classic picture: sudden onset after a stressor, motor or sensory loss, and la belle indifférence. DSM-5-TR no longer requires a stressor and calls la belle indifférence non-specific: in one review 21 percent of conversion patients showed it, against 29 percent with organic disease. ICD-11 calls the condition **dissociative neurological symptom disorder**. Conversion differs from hypochondriasis: neurological-like symptoms without organic basis, versus fear of having a serious illness.

EXAM TRAP

La belle indifférence points to **conversion disorder**, not malingering, factitious disorder, somatic symptom disorder or illness anxiety. Conversion symptoms "affect motor or sensory function, begin suddenly and follow a stressor" (the classic picture, though DSM-5-TR no longer requires the stressor); "suggest a medical disorder" fits any somatic disorder, and normal investigations alone never make the diagnosis.

Factitious disorder and malingering ●●●

In **factitious disorder** the person falsifies signs or symptoms, or induces injury or disease, with deception and presents as ill even without obvious external rewards; the motive is to assume the sick role. The severe form **imposed on self**, with dramatic stories and many admissions, was named **Munchausen syndrome** by **Richard Asher** (1912 to 1969) in 1951, after the tall tales of **Baron Munchausen**. In factitious disorder **imposed on another** (Munchausen by proxy, **Roy Meadow**, 1977) a carer induces illness in a dependent person, like the nursing assistant who secretly injects insulin into elderly patients; the perpetrator receives the diagnosis. DSM-5 lists it in the somatic symptom chapter, but it was never a somatoform disorder: DSM-IV gave it a chapter of its own.

Malingering is intentionally producing false or grossly exaggerated symptoms for external incentives such as compensation, drugs, or escape from work, military duty or prosecution. It is not a mental disorder but a condition that may be a focus of clinical attention (Table 8.24).

Table 8.24 Telling the conditions apart by intention and motive.

Condition	Intentional?	Motive	Mental disorder?
Conversion disorder	No	Unconscious conflict	Yes
Somatic symptom disorder	No	Excessive response to symptoms	Yes
Illness anxiety disorder	No	Fear of serious illness	Yes
Factitious disorder	Yes	Sick role, no external reward	Yes
Malingering	Yes	External reward	No

EXAM TRAP

Factitious disorder and malingering differ in the **presence or absence of external rewards**, not severity, age of onset or belief in the symptoms. Confabulation (filling memory gaps without intent to deceive), false identification (misrecognising a person or place) and altered identities are not faking, and the insulin-injecting carer has factitious disorder imposed on another, not antisocial personality disorder.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Charcot produced hysterical symptoms by hypnosis and _____.
2. PTSD symptoms must last more than _____.
3. In DID, the identity carrying the real name is the _____.
4. Loss of memory for one's entire life is _____ amnesia.
5. Surroundings feeling unreal or dreamlike is _____.
6. A "paralysed" leg pressing down when the healthy leg lifts shows _____ sign.
7. Faking illness for an external incentive is _____.

Answers: (1) suggestion, (2) 1 month, (3) host, (4) generalised, (5) derealisation, (6) Hoover's, (7) malingering.

8.8 Feeding, Eating, Elimination and Sleep-Wake Disorders

LOWER YIELD · 20 QS

Eating, bladder and bowel control, and sleep are among the first things a child learns to regulate, and DSM-5-TR gives each its own family of disorders. The eating disorders, above all anorexia nervosa, threaten life most and come first; the elimination and sleep-wake disorders follow.

Pica, rumination and ARFID

DSM-5-TR's **feeding and eating disorders** are persistent disturbances of eating that impair health or psychosocial functioning. Three of the six begin mostly in childhood and involve no body-image concern.

Pica is the persistent eating of non-nutritive, non-food substances (chalk, soil, paper, hair, paint) for **at least 1 month**, inappropriate to the developmental level and not culturally sanctioned, and is diagnosed from about age 2. It is linked with iron or zinc deficiency, pregnancy, intellectual disability and autism. In **rumination disorder** food is repeatedly regurgitated for at least 1 month, then re-chewed, re-swallowed or spat out, without a medical cause such as reflux; it occurs in infants and in intellectual disability. **Avoidant/restrictive food intake disorder** (ARFID) is food avoidance (for texture or smell, lack of interest, or fear of choking) causing weight loss, nutritional deficiency, dependence on supplements or tube feeding, or psychosocial interference; mere picky eating does not qualify. ARFID has **no disturbance in how body weight or shape is experienced**, which separates it from anorexia.

Anorexia nervosa: history and criteria ●●

In 1873 two physicians described young women who starved themselves while insisting they were well. In London **Sir William Withey Gull** (1816 to 1890) named the illness *anorexia nervosa* (in a paper read in 1873 and printed in 1874); in Paris **Ernest-Charles Lasègue** (1816 to 1883) called it *anorexie hystérique*. The name misleads: *anorexia* means loss of appetite, yet most patients are hungry and driven by fear of fatness. **Hilde Bruch** (1904 to 1984), in *The Golden Cage* (1978), described a compliant, high-achieving girl who feels ineffective and finds in control of her body the one thing truly hers.

Anorexia nervosa (AN) has three DSM-5-TR criteria:

1. Restriction of energy intake leading to a **significantly low body weight** for age, sex, development and physical health.
2. **Intense fear of gaining weight** or becoming fat, or persistent behaviour that interferes with weight gain, even at a low weight.
3. Disturbance in how weight or shape is experienced, undue influence of weight or shape on self-evaluation, or persistent failure to recognise the seriousness of the low weight.

Subtypes describe the last 3 months. The **restricting type** has had no recurrent binge eating or purging; weight is lost by dieting, fasting or excessive exercise. The **binge-eating/purging type** has had recurrent binges or purging (vomiting, or misuse of laxatives, diuretics or enemas). Adult severity uses body mass index (BMI, weight in kilograms divided by height in metres squared): **mild** 17 or more, **moderate** 16 to 16.99, **severe** 15 to 15.99, **extreme** below 15.

> EXAM TRAP
>
> DSM-5 **dropped amenorrhoea** as a criterion and sets **no BMI cut-off**: a BMI of 17 marks mild severity, not eligibility. ICD-10 used BMI 17.5 or less; ICD-11 uses below 18.5. Purging does not exclude anorexia; it defines a subtype. The core is fear of weight gain with restriction, not obsession with cleanliness (obsessive-compulsive disorder, Section 8.6).

The starving body and the anorexic personality ●●

Most physical signs are the body's defences against starvation, so think "slow and low". Metabolism slows, giving **bradycardia** (slow heart rate), **hypotension** and **hypothermia**. Fine downy hair called **lanugo** grows on the body. Falling hormones produce **amenorrhoea**, the absence of menstrual periods, and in time **osteoporosis**. Vomiting adds **hypokalaemia** (low potassium), which can trigger fatal arrhythmias. Hyperthyroidism, tachycardia, hypercalcaemia and polyphagia (overeating) all point the wrong way.

Anorexia nervosa is the **most lethal eating disorder** and has one of the highest mortality rates of any psychiatric disorder. DSM-5-TR puts its crude mortality at about 5 per cent per decade, from medical complications (cardiac arrhythmia among them) and from suicide. It usually begins in adolescence, far more often in women.

The personality behind it is strikingly consistent: **perfectionism**, obsessionality, rigidity and anxiety; the patient is over-controlled. Impulsivity belongs to bulimia, while avoidance and dependence define the avoidant and dependent personality disorders (Section 8.11).

> EXAM TRAP
>
> Amenorrhoea is no longer *required* but is still *common*, so it remains a correct "finding in anorexia". It means absence of menstrual periods, not fear of weight gain or excessive exercise.

Bulimia nervosa and binge-eating disorder ●●●

The London psychiatrist **Gerald Russell** named **bulimia nervosa** (BN) in 1979, calling it "an ominous variant of anorexia nervosa". DSM-5-TR requires:

1. Recurrent **binge eating**: eating, in a discrete period (about 2 hours), an amount definitely larger than most people would, with a **sense of loss of control**.
2. Recurrent **inappropriate compensatory behaviour** against weight gain: self-induced vomiting, misuse of laxatives, diuretics or other medicines, fasting or excessive exercise.
3. Both on average **at least once a week for 3 months** (DSM-IV asked for twice a week).
4. Self-evaluation unduly influenced by body shape and weight.
5. Not occurring only during episodes of anorexia nervosa.

Secret binges, often set off by low mood, are followed by shame and purging, in repeating cycles. Because purging undoes only part of a binge, people with bulimia are usually of **normal weight or overweight**. Vomiting leaves marks: eroded dental enamel, swollen parotid glands, hypokalaemia and **Russell's sign**, calluses on the back of the hand from the teeth. The typical trait is **impulsivity**.

Binge-eating disorder (BED), a full diagnosis only since DSM-5, is binge eating at least once a week for 3 months, with **marked distress** and three of five features: eating very rapidly, until uncomfortably full, when not hungry, alone from embarrassment, and with disgust or guilt afterwards. There is **no regular compensatory behaviour**. Figure 8.11 shows how the three disorders are told apart.

Anorexia and bulimia share a psychology: a **distorted body image**, an **intense fear of being fat**, and eating **regulated by psychological rather than physiological processes** (rules, fear and mood instead of hunger and fullness).

> EXAM TRAP
>
> **Significantly low body weight** separates anorexia from bulimia. Purging, fear of fatness and damage to health occur in both. Binges plus compensation is bulimia; binges with distress and no compensation is binge-eating disorder. Only one of the three is diagnosed within an episode.

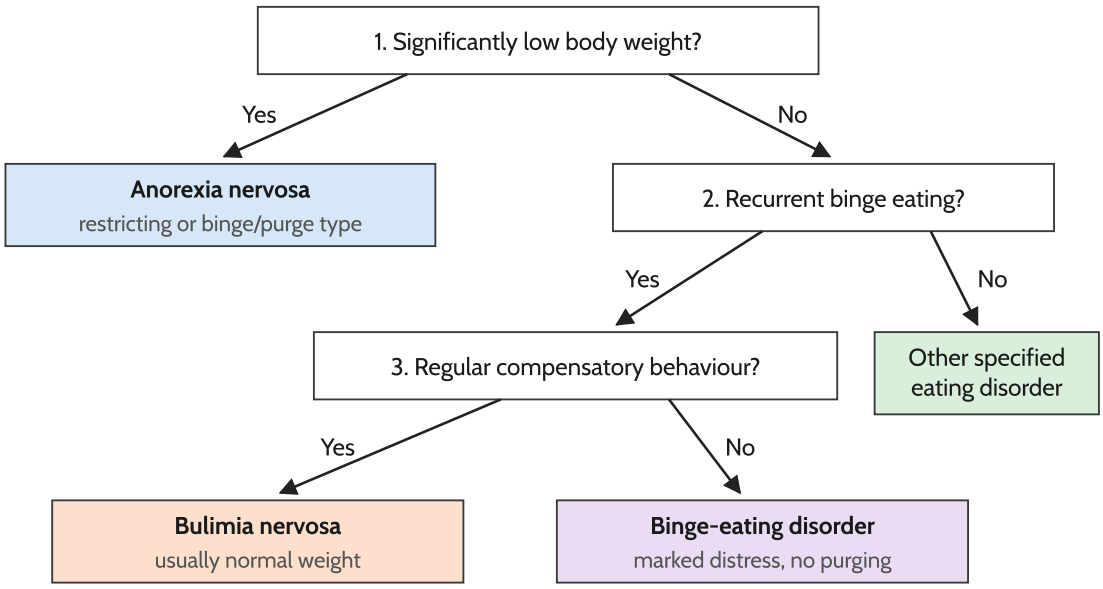

Figure 8.11 Telling the eating disorders apart. Weight is asked first, so a low-weight person who binges and purges has anorexia (binge-eating/purging type), not bulimia.

Causes and other specified eating disorders

Risks pile up. **Sociocultural**: they cluster where a thin ideal is prized, as in modelling and ballet. **Dieting** starts most histories, and restraint provokes bingeing as the body defends its weight (set-point theory, Section 1.10). **Family**: Salvador Minuchin's enmeshed, overprotective, rigid, conflict-avoiding **psychosomatic family** (Section 7.7) features in anorexia. **Personality**: perfectionism in anorexia, impulsivity in bulimia. **Biology**: twin studies put heritability near one half, and low serotonin activity is linked with impulsivity and binge eating.

Presentations fitting no full pattern are **other specified feeding or eating disorder** (OSFED), for example **atypical anorexia nervosa** (every criterion met except low weight: despite significant weight loss, weight is within or above the normal range), bulimia or binge-eating disorder of low frequency or short duration, **purging disorder** (purging without binge eating) and **night eating syndrome** (eating after waking at night, or excessively after the evening meal, with recall).

Enuresis and encopresis

The **elimination disorders** are the repeated passage of urine or faeces into inappropriate places by a child developmentally old enough to control them, voluntary or not. **Enuresis** is repeated voiding of urine into bed or clothes, at least **twice a week for 3 consecutive months** or with significant distress or impairment, at an age of at least **5 years**, and not caused by a substance (a diuretic) or a medical condition (diabetes, epilepsy). It is nocturnal only (the commonest), diurnal only, or both. In **primary** enuresis the child has never been dry; **secondary** enuresis returns after a dry period, usually between 5 and 8 years, often after stress. About 5 to 10 per cent of 5-year-olds wet the bed; most outgrow it. The bell-and-pad alarm is described in Section 7.4.

Encopresis is repeated passage of faeces into inappropriate places (clothing, the floor) **at least once a month for 3 months**, at age **4 or older**. It is specified **with constipation and**

overflow incontinence, the commoner form, in which painful defecation leads to withholding and leakage around impacted stool, or **without constipation**, often with oppositional or conduct problems. It too may be primary or secondary.

> EXAM TRAP
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> Lack of **bowel** control is **encopresis**; lack of bladder control is **enuresis**. Anorexia is an eating disorder, and "analism" is not a recognised elimination disorder. Ages: urine 5, faeces 4.

The sleep-wake disorders and insomnia

The sleep stages, non-rapid eye movement (NREM) stages N1 to N3 and rapid eye movement (REM) sleep, are taught in Section 1.5. DSM-5-TR's **sleep-wake disorders**, ten disorders or groups (Table 8.25), share a complaint about the quality, timing or amount of sleep with daytime distress. ICD-11 moved them out of the mental disorders into their own chapter, Chapter 7. Four follow a rule of threes, at least three times a week for 3 months: insomnia, hypersomnolence, narcolepsy and restless legs syndrome.

Insomnia disorder is difficulty initiating or maintaining sleep, or early waking, on at least 3 nights a week for 3 months **despite adequate opportunity for sleep**. Its treatment, cognitive behaviour therapy with stimulus control, sleep restriction and paradoxical intention, is in Section 7.4. In **hypersomnolence disorder** the person is excessively sleepy despite a main sleep of at least 7 hours, with daytime lapses into sleep, unrefreshing sleep of over 9 hours, or difficulty waking fully.

In the **breathing-related sleep disorders** breathing falters in sleep: the upper airway collapses (obstructive sleep apnoea hypopnoea), the brain stops driving breathing (central sleep apnoea), or shallow breathing lets carbon dioxide rise (sleep-related hypoventilation). In the **circadian rhythm sleep-wake disorders** the body clock is out of step with the required schedule: sleep comes too late or too early (delayed or advanced phase), is scattered across the 24 hours (irregular type), drifts later each day (non-24-hour type, common in blind people) or clashes with working hours (shift work type).

Table 8.25 The sleep-wake disorders of DSM-5-TR.

Disorder	Core feature	Key fact
Insomnia disorder	Poor onset or maintenance, early waking	3 nights a week, 3 months
Hypersomnolence disorder	Sleepy despite 7 hours' sleep	3 times a week, 3 months
Narcolepsy	Irrepressible sleep attacks	Cataplexy, low hypocretin or short REM latency
Obstructive sleep apnoea hypopnoea	Airway collapse: snoring, gasping, sleepiness	5 events an hour with symptoms, or 15
Central sleep apnoea	Breathing effort stops	Cheyne-Stokes breathing; opioids
Sleep-related hypoventilation	Shallow breathing, raised carbon dioxide	Polysomnography
Circadian rhythm sleep-wake disorders	Body clock out of step with schedule	Delayed (adolescents), advanced (elderly), irregular, non-24-hour, shift work
NREM sleep arousal disorders	Sleepwalking, sleep terrors	First third of night; amnesia

Disorder	Core feature	Key fact
Nightmare disorder	Frightening, remembered dreams	Second half of night
REM sleep behaviour disorder	Acting out dreams	Loss of REM atonia
Restless legs syndrome	Urge to move legs at rest, evenings	3 times a week, 3 months
Substance/medication-induced sleep disorder	Intoxication or withdrawal	Insomnia, sleepiness, parasomnia or mixed

Narcolepsy

The French physician **Jean-Baptiste-Édouard Gélineau** (1828 to 1906) coined the term narcolepsy in 1880. DSM-5-TR's **narcolepsy** is recurrent periods of irrepressible need to sleep or napping within a day, at least three times a week for 3 months, plus one of three markers: **cataplexy**, brief, sudden, bilateral loss of muscle tone with consciousness preserved, triggered by laughter or joking; **hypocretin (orexin) deficiency** in the cerebrospinal fluid; or a short REM latency (REM within 15 minutes of sleep onset, or a mean sleep latency of 8 minutes or less with two or more sleep-onset REM periods on the multiple sleep latency test). Type 1 has cataplexy or hypocretin deficiency; type 2 has neither.

Hypocretin neurons in the lateral hypothalamus stabilise waking; when they are lost, fragments of REM sleep break into it. Hence the classic **narcoleptic tetrad**: sleep attacks, cataplexy (REM atonia while awake), **sleep paralysis** (awake but unable to move on falling asleep or waking) and vivid **hypnagogic hallucinations** while falling asleep.

> EXAM TRAP
>
> **Cataplexy** is a sudden loss of muscle tone triggered by emotion, in full consciousness, and it marks narcolepsy. **Catalepsy** is the reverse, a posture held for long periods with the waxy flexibility of catatonia (Section 8.4).

Parasomnias and restless legs

Parasomnias are abnormal behaviours or experiences arising from sleep, and timing identifies them: deep N3 sleep fills the first third of the night, while REM periods lengthen towards morning. The **NREM sleep arousal disorders** are incomplete awakenings, usually from N3, in the **first third of the night**, with little dream recall and **amnesia**; they are commonest in children. In **sleepwalking** (**somnambulism**) the person sits up, walks about or performs complex acts while asleep, blank-faced and hard to wake. In **sleep terrors** (*pavor nocturnus*) the sleeper wakes abruptly with a panicky scream, intense fear and autonomic arousal (racing heart, rapid breathing, sweating, dilated pupils), and cannot be comforted.

REM parasomnias come later. **Nightmare disorder** is repeated, extended, extremely dysphoric dreams, usually in the second half of the night; the dreamer wakes alert and **remembers vividly**. In **REM sleep behaviour disorder** the normal muscle atonia of REM fails, so the person **acts out dream content**, shouting, punching or kicking. Commonest in men over 50, it often precedes Parkinson's disease, dementia with Lewy bodies or multiple system atrophy, often by more than a decade (about 75 per cent of idiopathic cases within 10 to 15 years).

Restless legs syndrome (Willis-Ekbom disease) is an urge to move the legs, with creeping sensations, that starts at rest, is relieved by movement and worsens in the evening. A **substance/medication-induced sleep disorder** follows intoxication or withdrawal. **Kleine-Levin syndrome**, recurrent hypersomnia with hyperphagia and hypersexuality, mostly in adolescent boys, is a separate diagnosis in the International Classification of Sleep Disorders (ICSD-3); DSM-5-TR names it only as an example of other specified hypersomnolence disorder.

EXAM TRAP

Sitting, walking or complex behaviour while asleep is **somnambulism**; screaming terror is sleep terrors; acting out dreams is **REM sleep behaviour disorder**, from loss of REM atonia, not absence of REM; uncomfortable leg sensations are restless legs. Bedwetting is enuresis; daytime sleepiness alone suggests hypersomnolence.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Pica requires eating non-food substances for at least _____ month.
2. DSM-5 dropped _____ as a criterion for anorexia nervosa.
3. An adult with anorexia and a BMI of 15.5 is rated _____.
4. In bulimia, binges and compensatory behaviour occur at least once a week for _____ months.
5. Calluses on the back of the hand from induced vomiting are _____ sign.
6. The minimum age is _____ years for enuresis and _____ years for encopresis.
7. Sudden loss of muscle tone triggered by laughter, in narcolepsy, is _____.
8. Sleepwalking usually arises from _____ sleep in the first third of the night.

Answers: (1) 1, (2) amenorrhoea, (3) severe, (4) 3, (5) Russell's, (6) 5; 4, (7) cataplexy, (8) N3 (slow-wave NREM).

8.9 Sexual Dysfunctions, Paraphilic Disorders and Gender Dysphoria

LOWER YIELD · 9 QS

Few corners of psychiatry have been redrawn as often. A man who ejaculates too soon, a person aroused by shoes, a child who insists she is a boy and a contented gay man in 1970 have all been diagnosed; only some would be today. We start with normal sexual response, take the three DSM-5-TR groups in turn, and end with homosexuality's exit from the manuals.

The sexual response cycle and the language of gender

Disorder is judged against the normal picture drawn by the gynaecologist **William Howell Masters** (1915 to 2001) and his research partner **Virginia Eshelman Johnson** (1925 to 2013). In *Human Sexual Response* (1966), from laboratory recordings of nearly 700 volunteers, they described a four-phase **sexual response cycle**. In **excitement**, blood engorges the genitals, producing erection or vaginal lubrication. In **plateau**, arousal levels off just before climax.

Orgasm is a peak of rhythmic muscular contractions, with ejaculation in men. In **resolution** the body returns to rest, and men pass through a **refractory period** in which another orgasm is impossible; many women have none and can have multiple orgasms.

Helen Singer Kaplan (1929 to 1995) saw patients whose bodies worked but who did not want sex, and in *Disorders of Sexual Desire* (1979) put a **desire** phase first (desire, excitement, orgasm), giving the linear sequence of Figure 8.12; dysfunctions are still sorted by the phase they disturb. **Rosemary Basson** argued in 2000 that this fits many women in long relationships poorly. In her **circular model** a woman often starts from sexual neutrality, is receptive for the sake of intimacy, is aroused by sexual stimuli and only then feels **responsive desire**; satisfaction feeds back into intimacy. DSM-5 merged women's desire and arousal disorders partly for this reason.

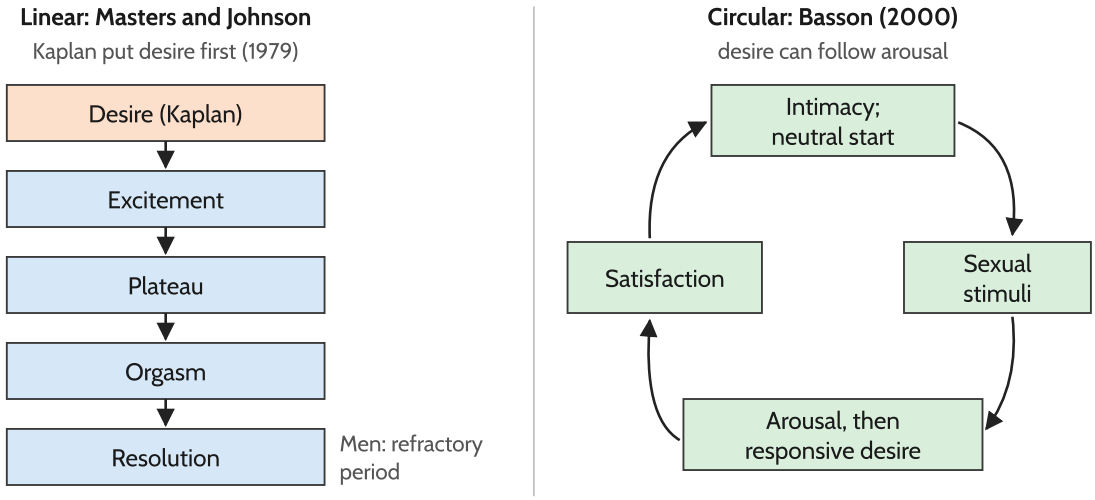

Figure 8.12 Left: Masters and Johnson's four phases with Kaplan's desire phase in front; only men have a refractory period. Right: Basson's circle, in which desire can follow arousal.

Four terms must be kept apart. **Sex** is biological (chromosomes, gonads, hormones, genitals); **gender identity** is the inner sense of being a man, a woman or another gender; **gender role** is the behaviour a culture expects of each gender; **sexual orientation** is the gender of those one is attracted to. **Androgyny**, in the 1974 work of **Sandra Lipsitz Bem** (1944 to 2014) with her Bem Sex Role Inventory, means scoring high on both masculine and feminine traits; it describes a flexible personality, not a disorder.

Sexual dysfunctions ●●

A **sexual dysfunction** is a clinically significant disturbance in the ability to respond sexually or to experience sexual pleasure. Most of DSM-5-TR's eight share one template: the problem occurs on almost all occasions (about 75 to 100 percent), has lasted at least about 6 months, causes clinically significant distress, and is not better explained by another mental disorder, severe relationship distress, a substance or a medical condition. Specifiers give the course: **lifelong** (present since sexual activity began) or **acquired** (after a period of normal function), and **generalised** or **situational** (only with certain partners, stimulation or situations). Table 8.26 gives the eight.

Table 8.26 The DSM-5-TR sexual dysfunctions.

Disorder	Core feature	Note
Delayed ejaculation	Marked delay, infrequency or absence of ejaculation	Older name: inhibited male orgasm
Erectile disorder	Difficulty getting or keeping an erection, or reduced rigidity (1 of 3)	DSM-IV: male erectile disorder
Female orgasmic disorder	Delay, infrequency or absence of orgasm, or weaker orgasm	Specifier: never had an orgasm
Female sexual interest/arousal disorder	At least 3 of 6 signs of low interest or arousal	Merges old desire and arousal disorders
Genito-pelvic pain/penetration disorder	1 or more of 4: penetration difficulty, pain, fear of pain, pelvic-floor tensing	Merges vaginismus and dyspareunia; lifelong or acquired only
Male hypoactive sexual desire disorder	Deficient or absent sexual thoughts, fantasies and desire	Judged against age and context
Premature (early) ejaculation	Within about 1 minute of vaginal penetration and before the person wishes	Severe: under about 15 seconds
Substance/medication-induced	Begins during or soon after intoxication, withdrawal or a medication	No 6-month rule

Vaginismus is involuntary tightening of the vaginal muscles that makes penetration painful or impossible; **dyspareunia** is pain during intercourse. Erectile disorder has the strongest medical ties: diabetes, vascular disease, smoking and age. The substance/medication-induced form follows drugs: selective serotonin reuptake inhibitors (SSRIs) most often delay orgasm or ejaculation, and antipsychotics, antihypertensives and chronic alcohol use are frequent culprits. Masters and Johnson (*Human Sexual Inadequacy*, 1970) described **performance anxiety**, fear of failing, and **spectatoring**, mentally watching and judging one's own performance, so that attention leaves erotic sensation and one failure breeds the next. Relationship conflict, past abuse, depression and religious prohibitions add to the picture. Premature ejaculation is the dysfunction most tied to anxiety, and is commoner in men with anxiety disorders, especially social anxiety disorder.

Treatment is in Section 7.7; here, match tool to problem. The **stop-start technique** of **James Semans** (1956), halting stimulation just before the point of no return, and Masters and Johnson's **squeeze technique**, firm pressure where the head of the penis meets the shaft, teach ejaculatory control. **Sensate focus**, graded non-demanding touch with intercourse at first forbidden, removes performance anxiety in many dysfunctions. Vaginismus is treated with relaxation and graded dilators, erectile disorder often with phosphodiesterase-5 inhibitors such as sildenafil.

> EXAM TRAP
>
> Stop-start and squeeze treat **premature ejaculation**, not erectile disorder, inhibited male orgasm (delayed ejaculation, the opposite problem) or vaginismus. Premature ejaculation is most commonly associated with an **anxiety state**: masturbation does not cause it (a myth that also fuels dhat syndrome, Section 8.1), diabetes is the classic link of erectile disorder, and personality disorder has no special tie.

Paraphilias and paraphilic disorders ●●

DSM-5-TR calls any intense and persistent sexual interest other than in genital stimulation or fondling with physically mature, consenting human partners a **paraphilia**. The key DSM-5 move was to separate interest from disorder. A **paraphilic disorder** is a paraphilia that currently causes distress or impairment, or whose satisfaction has involved harm, or risk of harm, to others. Criterion A gives the interest, recurrent and intense for at least 6 months in fantasies, urges or behaviours; Criterion B the consequences. A shoe fetish that troubles no one is a paraphilia but not a disorder; exposing oneself to strangers meets Criterion B even without distress.

The eight named disorders (Table 8.27) fall into two groups. **Anomalous activity preferences** are the **courtship disorders**, distorted pieces of courtship (looking, showing, touching), and the **algolagnic disorders**, which involve pain. **Anomalous target preferences** are pedophilic, fetishistic and transvestic disorder. Most sets allow the specifiers *in a controlled environment* (such as prison) and *in full remission* (no acting on urges and no distress for 5 years outside such settings).

Table 8.27 The eight DSM-5-TR paraphilic disorders.

Disorder	Recurrent, intense arousal from	Group and notes
Voyeuristic	Observing an unsuspecting person naked, undressing or having sex	Courtship; person at least 18
Exhibitionistic	Exposing one's genitals to an unsuspecting person	Courtship; to children, adults or both
Frotteuristic	Touching or rubbing against a non-consenting person	Courtship
Sexual masochism	Being humiliated, beaten, bound or made to suffer	Algolagnic; with asphyxiophilia
Sexual sadism	Physical or psychological suffering of another	Algolagnic
Pedophilic	Prepubescent children (generally 13 or younger)	Person at least 16 and 5 years older
Fetishistic	Non-living objects or specific non-genital body parts	Not cross-dressing clothes or vibrators
Transvestic	Cross-dressing	With fetishism; with autogynephilia

The fetish object is non-living (shoes, rubber, undergarments) or a specific non-genital body part such as feet, a focus called **partialism**. Other paraphilias go under other specified paraphilic disorder: **necrophilia** (corpses), **zoophilia** (animals), coprophilia (faeces), urophilia (urine), klismaphilia (enemas) and **telephone scatologia** (obscene calls). **Hypoxyphilia**, arousal from restricted breathing, is the 'with asphyxiophilia' specifier of sexual masochism disorder and can kill by accident.

Ray Blanchard coined **autogynephilia** in 1989: a male's recurrent, intense sexual arousal by the thought or image of himself as a woman. It anchors his controversial typology of male-to-female gender dysphoria, and in men with transvestic disorder it raises the likelihood of gender dysphoria.

EXAM TRAP

Autogynephilia is arousal by the image of **oneself as a woman**; arousal by prepubescent children is pedophilia, by being humiliated is masochism, and mere arousal by oneself is not autogynephilia. An **inanimate object** means fetishism: necrophilia needs a corpse, frotteurism a non-consenting person, voyeurism an unsuspecting one.

Why do paraphilias form? Learning theorists cite **classical conditioning**: **Stanley Rachman** (1966) conditioned mild arousal to a picture of women's boots in the laboratory. **John Money** (1921 to 2006) proposed the **lovemap**, a childhood template of the ideal lover and erotic scene that early experience can distort. **Kurt Freund** (1914 to 1996), in his **courtship disorder hypothesis**, saw voyeurism, exhibitionism, frotteurism and preferential rape as disturbances of successive courtship phases (searching, pre-touch interaction, touching, intercourse), which is why they cluster.

Gender dysphoria and gender incongruence ●●

Gender dysphoria in DSM-5-TR is a marked incongruence between one's experienced or expressed gender and one's assigned gender, lasting at least 6 months and causing clinically significant distress or impairment. A child needs six of eight indicators, one of which must be a strong desire to be, or insistence that one is, the other gender; others include cross-gender clothing, play and playmates and dislike of one's anatomy. An adolescent or adult needs two of six, such as a strong desire to be rid of one's sex characteristics or to be treated as the other gender. Specifiers: 'with a disorder/difference of sex development' and, for adults, 'posttransition'.

DSM-III (1980) introduced transsexualism for adults (a historic term for a person seeking social and often bodily transition to the other sex) and gender identity disorder of childhood. DSM-IV (1994) merged them into **gender identity disorder**, a condition in which one's experienced gender identity does not match one's assigned sex. That made the identity itself the disorder, so DSM-5 (2013) renamed it gender dysphoria, moving the focus from identity to distress. ICD-10 listed transsexualism (F64.0), dual-role transvestism (F64.1) and gender identity disorder of childhood (F64.2) under F64.

ICD-11, in force from 1 January 2022, renamed the condition **gender incongruence**, of adolescence or adulthood and of childhood (about 2 years' persistence in children), and moved it out of the mental disorders chapter into a new Chapter 17, **Conditions related to sexual health**: being transgender is not a mental illness, but a code lets people obtain gender-affirming care. The sexual dysfunctions and sexual pain disorders moved there too. The paraphilic disorders stayed in Chapter 6, now built around non-consent: exhibitionistic, voyeuristic, pedophilic, **coercive sexual sadism disorder** and frotteuristic, plus other paraphilic disorder involving non-consenting individuals and paraphilic disorder involving solitary or consenting individuals with marked distress or risk. Consensual fetishism, fetishistic transvestism and sadomasochism were removed (Table 8.28). DSM-5's gender dysphoria corresponds to ICD-11's gender incongruence.

Table 8.28 Where the sexuality and gender groups went in ICD-11.

Group	ICD-10	ICD-11
Gender identity disorders	F64 (Chapter V)	Gender incongruence, Chapter 17

Group	ICD-10	ICD-11
Sexual dysfunctions	F52 (Chapter V)	Chapter 17
Vaginismus, dyspareunia	F52.5, F52.6	Sexual pain-penetration disorder, Chapter 17
Disorders of sexual preference	F65	Paraphilic disorders, stay in Chapter 6
Fetishism, fetishistic transvestism, sadomasochism	F65.0, F65.1, F65.5	Removed
Egodystonic sexual orientation	F66.1	Removed

EXAM TRAP

ICD-11 gender incongruence is in **Chapter 17, Conditions related to sexual health**: not Chapter 6, not Chapter 5 (endocrine, nutritional or metabolic diseases), and not deleted. Only the paraphilic disorders stayed in Chapter 6. The same mismatch is gender identity disorder in DSM-IV and ICD-10 and gender dysphoria in DSM-5.

Gender identity is distinct from a paraphilia (what arouses a person), a sexual dysfunction (the response cycle) and androgyny (a blend of traits). Practice today is gender-affirmative, supporting identity rather than trying to change it (Chapter 7).

Homosexuality: pathologised, then depathologised

Pathologisation is classifying a behaviour or identity as a mental disorder; **depathologisation** is removing that classification (not changing orientation, the aim of discredited conversion therapies, nor coming out). Homosexuality is the clearest case. DSM-I (1952) placed it under **sociopathic personality disturbance** as a sexual deviation; DSM-II (1968) kept it among the sexual deviations. **Alfred Charles Kinsey** (1894 to 1956), in *Sexual Behavior in the Human Male* (1948), rated behaviour on the **Kinsey scale** from 0 (exclusively heterosexual) to 6 (exclusively homosexual), showing same-sex experience to be common and graded.

KEY STUDY: HOOKER'S BLIND JUDGES (1957)

Evelyn Hooker (1907 to 1996) asked whether homosexual men are more maladjusted. She matched 30 homosexual and 30 heterosexual men, none in therapy, for age, IQ and education and gave them projective tests, including the Rorschach. Expert judges blind to group could not tell them apart beyond chance, and rated their adjustment alike. Criticism: a small, non-random sample; but earlier claims had rested on patients in treatment.

With this evidence and activist pressure, the board of the American Psychiatric Association voted in December 1973 to remove homosexuality. The seventh printing of DSM-II (1974) substituted **sexual orientation disturbance** for people distressed by their orientation; DSM-III (1980) renamed it **ego-dystonic homosexuality**, and DSM-III-R (1987) dropped it. ICD-10 (endorsed in 1990) no longer listed homosexuality, stating that orientation by itself is not a disorder, but kept F66, including egodystonic sexual orientation; ICD-11 deleted F66.

INDIAN CONTEXT

In 2018 the Indian Psychiatric Society stated that homosexuality is not a psychiatric disorder. On 6 September 2018 the Supreme Court, in *Navtej Singh Johar v. Union of India*, read down Section 377 of the Indian Penal Code, decriminalising consensual sex between adults of the same sex.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. After orgasm, men pass through a _____ period.
2. In Basson's model, many women feel _____ desire, which follows arousal.
3. Most sexual dysfunctions need the problem on about _____ percent of occasions.
4. Stop-start and squeeze techniques treat _____.
5. Arousal from a specific non-genital body part is _____.
6. Blanchard's term for a male's arousal by the image of himself as a woman is _____.
7. ICD-11 places gender incongruence in Chapter _____.
8. Ego-dystonic homosexuality was dropped in _____ (1987).

Answers: (1) refractory, (2) responsive, (3) 75 to 100, (4) premature ejaculation, (5) partialism, (6) autogynephilia, (7) 17, (8) DSM-III-R.

8.10 Substance-Related, Addictive and Neurocognitive Disorders

MEDIUM YIELD · 36 QS

A man who has drunk heavily for twenty years is admitted for surgery. On his third dry day he is trembling, picking at the sheets and swatting at insects only he can see. Months later, sober, he forgets a nurse within minutes and fills the gap with invented stories. He spans both halves of this section: the disorders drugs produce, and the failures of cognition that drugs, disease and age produce.

The language of substance use ●●

A **psychoactive substance** is a drug that alters mood, cognition, perception or behaviour by acting on the brain; the word has nothing to do with a drug-induced hallucination or an "active psychopath". **Intoxication** is the reversible syndrome of recent use. **Withdrawal** is the syndrome that follows stopping or cutting down after heavy, prolonged use, and it usually mirrors intoxication: a sedative leaves an agitated brain behind, a stimulant a slowed one.

Tolerance is needing markedly more of a drug for the same effect: metabolic tolerance when the liver breaks the drug down faster, pharmacodynamic tolerance when receptors and neurons adapt. **Cross-tolerance** is tolerance that extends to other drugs, especially of the same class, so a heavy drinker needs large doses of benzodiazepines or barbiturates. In **cross-dependence** one drug suppresses another's withdrawal. Together these are why alcohol-dependent patients are detoxified with a long-acting benzodiazepine such as chlordiazepoxide, and

why methadone can replace heroin. **Sensitisation** (reverse tolerance) is the opposite: repeated cocaine or amphetamine use brings paranoia or seizures at ever lower doses.

Physical dependence shows itself in a withdrawal syndrome; psychological dependence is a compelling need to use for pleasure or relief. Addiction, compulsive drug seeking despite harm, is different again: a cancer patient on morphine is physically dependent but not addicted. **Craving** is an intense urge to use, often cued by places and objects linked to the drug, and relapse is a return to use after abstinence (Chapter 7).

ICD-10 codes these disorders F10 (alcohol) to F19 (multiple drugs), and the digit after the point gives the clinical pattern: .0 acute intoxication, .1 **harmful use** (use that is actually damaging health), .2 dependence syndrome, .3 withdrawal state, .4 withdrawal with delirium, .5 psychotic disorder, .6 amnesic syndrome and .7 residual and late-onset psychotic disorder. The **dependence syndrome** was described for alcohol in 1976 by **Griffith Edwards** and **Milton Gross**: a narrowed drinking repertoire, drink given priority over everything else, tolerance, repeated withdrawal, drinking to relieve it, a felt compulsion, and rapid reinstatement after abstinence.

DSM-IV split this ground into substance abuse and substance dependence. DSM-5 (2013) and DSM-5-TR (2022) merge them into one **substance use disorder**: at least 2 of 11 criteria within 12 months, in four groups. Impaired control covers using more or longer than intended, failed efforts to cut down, much time spent, and craving; social impairment covers failed roles, interpersonal problems and activities given up; risky use covers hazardous situations and use despite known harm; the pharmacological criteria are tolerance and withdrawal. Severity is mild with 2 or 3, moderate with 4 or 5, and severe with 6 or more. Craving was added and legal problems dropped. "Addiction" is not a DSM-5-TR diagnosis.

DSM-5-TR also includes **gambling disorder** (four or more of nine criteria in 12 months, such as chasing losses) as its only behavioural addiction; internet gaming disorder is a condition for further study. ICD-11's block "disorders due to substance use or addictive behaviours" codes each drug as an episode of harmful use, a harmful pattern of use, or dependence, and adds gambling disorder and **gaming disorder**: gaming with impaired control, given priority over other interests despite harm.

> EXAM TRAP
>
> "Drug abuse" is not an ICD-10 pattern: abuse was a DSM-IV category, and ICD-10's nearest term is harmful use. Gaming disorder sits in ICD-11's substance use and addictive behaviours block, not under stress-associated, mood or impulse control disorders. Cross-tolerance is tolerance carried over to a related drug, not freedom from addiction.

Why drugs capture the brain ●●

Every drug of abuse, by its own route, ends up raising dopamine in the **mesolimbic dopamine system**, from the ventral tegmental area to the nucleus accumbens in the ventral striatum (Chapter 5). It is the circuit most consistently altered in substance use disorders; the cerebellar vermis (gait), lateral geniculate nucleus (vision) and vestibular nuclei (balance) are not reward structures. Drugs are positive reinforcers because they give pleasure and negative reinforcers because they remove withdrawal, anxiety or pain (Chapter 1).

In the **opponent-process theory** of **Richard Solomon** and **John Corbit** (1974), every drug high (the a-process) triggers an opposing low (the b-process). With repetition the b-process strengthens: the high shrinks (tolerance), the crash deepens (withdrawal), and the user takes the drug to escape misery rather than to feel good.

Using Swedish adoption records in 1981, **C. Robert Cloninger** and colleagues separated type I alcoholism (onset after about 25, guilt, binges with loss of control, both sexes, shaped by environment) from type II (early onset, novelty seeking, antisocial behaviour, inability to abstain, mainly men, highly heritable). **Sigmund Freud** (1856 to 1939) saw alcohol-dependent people as fixated at the oral stage (Chapter 4). **Elvin Morton Jellinek** (1890 to 1963) argued in *The Disease Concept of Alcoholism* (1960) that alcoholism is a disease with five "species": alpha (psychological reliance, no loss of control), beta (physical damage without dependence), gamma (tolerance, withdrawal and loss of control: once started, the drinker cannot stop), delta (inability to abstain) and epsilon (periodic binges).

KEY STUDY: KANDEL'S STAGES OF DRUG USE (1975)

Denise Kandel surveyed New York State high-school students twice and found a regular order: beer or wine, then cigarettes or spirits, then marijuana, then other illicit drugs. Few reached heroin or cocaine without marijuana, so in this **gateway hypothesis** marijuana is the drug most often cited as the gateway, although alcohol and tobacco come first. Criticism: a sequence is not a cause, and shared risk factors could explain it.

Drugs differ in addictive liability, the chance that a user becomes dependent. Nicotine (tobacco) ranks highest: a larger share of people who try it become dependent than of those who try heroin, cocaine, alcohol or marijuana.

The ten substance classes ●●

DSM-5-TR recognises ten classes (Table 8.29).

Table 8.29 The ten DSM-5-TR substance classes.

Class	Examples	Intoxication	Withdrawal
Alcohol	Arrack, toddy, beer	Slurred speech, ataxia, nystagmus	Tremor, sweating, seizures, hallucinations
Caffeine	Coffee, tea	Restlessness, insomnia, tachycardia	Headache, fatigue
Cannabis	Bhang, ganja, charas	Red eyes, hunger, dry mouth	Irritability, poor sleep, low appetite
Hallucinogens; phencyclidine	LSD, psilocybin, mescaline; PCP	Perceptual changes, dilated pupils; PCP: nystagmus, violence	Not defined
Inhalants	Glue, petrol, whitener	Dizziness, nystagmus, euphoria	Not defined
Opioids	Morphine, heroin	Pinpoint pupils, slowed breathing	Yawning, tears, dilated pupils, diarrhoea
Sedatives, hypnotics, anxiolytics	Benzodiazepines, barbiturates	As for alcohol	As for alcohol; seizures
Stimulants	Amphetamines, cocaine	Dilated pupils, agitation, paranoia	Crash: dysphoria, fatigue, hunger

Class	Examples	Intoxication	Withdrawal
Tobacco	Cigarettes, bidis, gutka	No diagnosis	Irritability, restlessness, hunger
Other or unknown	Anabolic steroids, betel nut	Varies	Varies

Alcohol is a positive modulator (functionally an agonist) at GABA-A receptors and an antagonist at NMDA glutamate receptors; it is not a serotonin, acetylcholine or endorphin agonist. DSM-5-TR notes that withdrawal begins within about 4 to 12 hours of the last drink, peaks on the second day and eases markedly by the fourth or fifth. Its complications keep only a rough clock (Figure 8.13): tremor, sweating and insomnia within the first day (roughly 6 to 24 hours); seizures at about 12 to 48 hours; **alcoholic hallucinosis**, hallucinations (usually voices) in clear consciousness, at about 12 to 48 hours; and **delirium tremens** at about 48 to 72 hours, with clouded consciousness, fever, racing pulse, coarse tremor and vivid visual and tactile hallucinations of insects, small animals or tiny (Lilliputian) people. Fewer than 1 in 10 people in withdrawal reach this stage, but it can kill, so benzodiazepines are given to prevent it, with thiamine against Wernicke's encephalopathy.

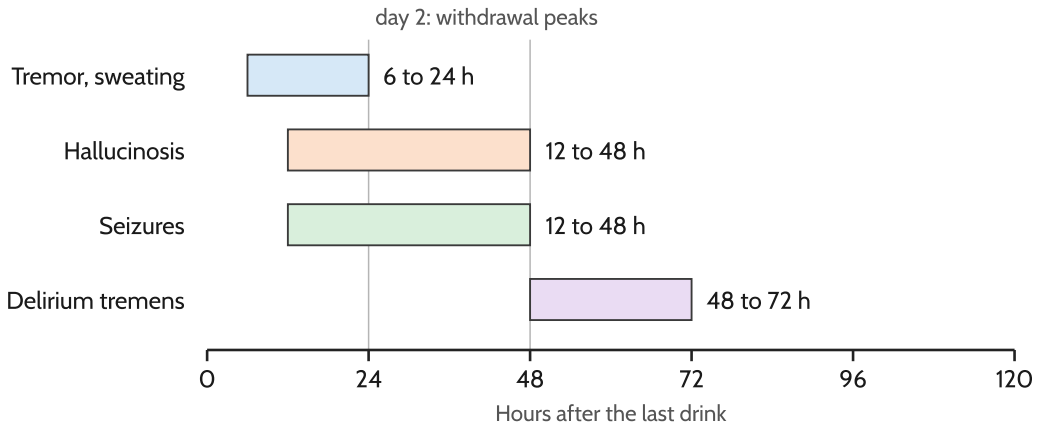

Figure 8.13 Approximate onset windows of alcohol withdrawal features. Tremor comes first and delirium tremens, the most dangerous, last; timing varies between people.

Chronic drinkers may develop alcoholic jealousy, a delusion that the partner is unfaithful; fetal alcohol syndrome is in Section 8.3, and the CAGE and AUDIT questionnaires in Chapter 2. **Disulfiram** inhibits aldehyde dehydrogenase, so acetaldehyde accumulates after a drink and causes flushing, nausea and palpitations. Naltrexone, an opioid antagonist, blunts alcohol's reward and craving, and acamprosate, acting on glutamate, eases protracted withdrawal.

Opium's natural alkaloids are morphine, codeine, thebaine, papaverine and noscapine. Heroin (diacetylmorphine) is made from morphine, so it is semi-synthetic; methadone, pethidine, fentanyl and tramadol are synthetic. Intoxication brings pinpoint pupils and respiratory depression, reversed by naloxone. Withdrawal is miserable but rarely dangerous: yawning, lacrimation, rhinorrhoea, dilated pupils, piloerection ("cold turkey"), diarrhoea and aches. Methadone or buprenorphine substitution stabilises dependent users.

Cocaine blocks dopamine reuptake and amphetamines release dopamine, giving euphoria, energy and, at high doses, paranoia; amphetamine psychosis can resemble paranoid schizophrenia, and a dysphoric crash follows a binge. MDMA (ecstasy) is an amphetamine derivative that releases serotonin. **Formication**, tactile hallucinations of insects crawling on or under the skin ("cocaine bugs"), is most characteristic of stimulant intoxication and of alcohol or sedative withdrawal, not of depression, phobia or generalised anxiety.

Cannabis acts through delta-9-tetrahydrocannabinol (THC); in India it is used as bhang (leaves), ganja (flowering tops) and charas (resin). Heavy use is linked to an **amotivational syndrome** of apathy and to a raised risk of psychosis. LSD (lysergic acid diethylamide), whose effects **Albert Hofmann** (1906 to 2008) discovered in 1943, psilocybin and mescaline alter perception in clear consciousness; some users later re-experience trails and halos (hallucinogen persisting perception disorder, or flashbacks). Inhalants intoxicate like alcohol and damage organs. Nicotine acts on nicotinic acetylcholine receptors; caffeine use disorder is only a condition for further study. Benzodiazepines and barbiturates are cross-tolerant with alcohol, and their withdrawal can also bring seizures and delirium.

> EXAM TRAP
>
> Alcohol is a GABA agonist. Heroin is the odd one out among opium alkaloids because it is semi-synthetic. Opioid intoxication gives pinpoint pupils, withdrawal dilated ones. Disulfiram works by aversion through aldehyde dehydrogenase; reducing reward is naltrexone's action and glutamate modulation acamprosate's.

Neurocognitive disorders: terms and domains ●●

ICD-10 groups disorders of failing cognition as organic mental disorders (F00 to F09). DSM-5 renamed them **neurocognitive disorders** (NCDs), acquired declines from a previous level. It kept delirium and replaced dementia and amnestic disorder with **major neurocognitive disorder**, a significant decline in one or more of six domains (complex attention, executive function, learning and memory, language, perceptual-motor function, social cognition) that costs independence in daily life, and mild NCD, a modest decline with independence kept. One impaired domain is enough, so pure amnesia now counts as major NCD.

Older terms name the function each disorder chiefly affects: dementia, a global decline that includes abstraction and judgement; amnesia, memory; aphasia, alexia and acalculia, language and symbolic function; apraxia, skilled purposeful movement (motor skills); and agnosia, recognition despite intact senses (Chapter 5).

Delirium ●●●

Delirium is a disturbance of attention and awareness (clouded consciousness) that develops over hours to days, fluctuates through the day, and brings disorientation, memory loss or perceptual disturbance, often visual hallucinations. The sleep-wake cycle may reverse, and confusion worsens in the evening (**sundowning**). DSM-5-TR specifies acute or persistent delirium and hyperactive, hypoactive (quiet and easily missed) or mixed types. Motor signs include **asterixis** (a flapping tremor of outstretched hands), multifocal myoclonus, tremor, abnormal tone and reflexes, and **carphologia** or floccillation, picking at the bedclothes: all can occur. The electroencephalogram (EEG) shows diffuse slowing.

Delirium always has a physical cause: infection, metabolic or electrolyte disturbance, hypoxia, drugs, intoxication or withdrawal, or surgery. It affects 11 to 51 percent of older people after operations, and it is the commonest psychiatric syndrome after cardiac surgery, ahead of depression, anxiety and schizophrenia. Table 8.30 compares it with dementia and with depressive pseudodementia.

Table 8.30 Delirium, dementia and depressive pseudodementia.

Feature	Delirium	Dementia	Pseudodementia
Onset	Acute, hours to days	Insidious	Fairly abrupt
Course	Fluctuating, worse at night	Progressive	Follows the mood
Attention, consciousness	Impaired, clouded	Clear until late	Clear
Memory and answers	Impaired; distractible	Impaired; near misses, hides deficits	Complains of poor memory; "don't know"
Hallucinations	Common, visual	Can occur	Uncommon
EEG	Diffuse slowing	Normal or mildly slow	Normal
Outcome	Usually reversible	Usually irreversible	Improves with antidepressants

EXAM TRAP

The most reliable distinction is acute onset, fluctuating course and disturbed attention or consciousness in delirium versus impaired memory with clear consciousness in dementia. Hallucinations, delusions and memory loss occur in both, and both are commonest in the elderly, so age does not separate them. Delirium, not dementia, is usually reversible. Confabulation points to Korsakoff's syndrome, early aphasia and apraxia to Alzheimer's.

Subtypes of major and mild NCD ●●●

KEY STUDY: AUGUSTE D. (1901 TO 1906)

In 1901 **Alois Alzheimer** (1864 to 1915) examined **Auguste Deter** (1850 to 1906), 51, admitted to the Frankfurt asylum with failing memory, jealousy and disorientation. Asked to write her name, she said "I have lost myself". After her death he described, in 1906, plaques between her neurons and tangles inside them. Her youth made the disease seem presenile; it is now the commonest cause of dementia at any age.

Alzheimer's disease begins insidiously with loss of episodic memory, then aphasia, apraxia and visuospatial deficits. Loss of cholinergic neurons in the nucleus basalis of Meynert causes acetylcholine deficiency, the target of cholinesterase inhibitors (Chapter 5). "Senile dementia" (onset after 65) and cerebral atrophy are older descriptive labels, not specific diseases. Table 8.31 gives the other subtypes.

Table 8.31 Subtypes of major and mild neurocognitive disorder.

Subtype	Hallmarks	Pathology or cause

Subtype	Hallmarks	Pathology or cause
Alzheimer's disease	Insidious; memory, then language and visuospatial loss	Amyloid plaques, tau tangles; apolipoprotein E4 gene; early in Down syndrome
Vascular	Sudden or stepwise; focal signs	Multiple infarcts; Binswanger's small-vessel disease
Frontotemporal (Pick's disease)	Onset often in 50s; personality change, disinhibition or language loss; memory relatively spared	Frontal and temporal degeneration
Lewy bodies	Fluctuating cognition, well-formed visual hallucinations, parkinsonism; REM sleep behaviour disorder; antipsychotic sensitivity	Alpha-synuclein Lewy bodies
Parkinson's disease	Parkinsonism a year or more before cognitive decline	Loss of nigral dopamine neurons
Huntington's disease	Chorea, depression, subcortical dementia	Autosomal dominant CAG repeat, chromosome 4; caudate atrophy
Prion disease	Rapid decline; myoclonus	Creutzfeldt-Jakob disease
Others	HIV, brain injury, substances or medication	Infection, trauma, toxicity

Pick's disease was described by **Arnold Pick** (1851 to 1924) in 1892, and Huntington's disease by **George Huntington** (1850 to 1916) in 1872. Huntington's is autosomal dominant, unlike Down syndrome (trisomy 21), phenylketonuria (recessive) or fragile X (an X-linked repeat), which are in Section 8.3. Cortical dementias (Alzheimer's, frontotemporal) produce aphasia, apraxia, agnosia and amnesia; **subcortical dementia** (Huntington's, Parkinson's, HIV) produces slowing, apathy, depression and motor signs without aphasia. Alzheimer's, Parkinson's, Huntington's, frontotemporal degeneration and amyotrophic lateral sclerosis are neurodegenerative, with progressive neuron death. Epilepsy, a tendency to recurrent seizures, is not; multiple sclerosis is a demyelinating disease.

Pseudodementia, a term popularised by **Leslie Kiloh** in 1961, is cognitive impairment that mimics dementia but has a psychiatric cause, most commonly depression (not schizophrenia, mania or hypochondriasis). Clues: fairly abrupt onset, depressed mood, a patient who complains of poor memory and answers "I don't know", and improvement with antidepressants.

> EXAM TRAP
>
> Lewy body NCD means visual, not auditory, hallucinations; aphasia belongs to Alzheimer's, and loss of smell is only a non-specific early sign. Insidious memory, language and visuospatial decline is Alzheimer's; stepwise decline is vascular.

Wernicke-Korsakoff syndrome ●●●

Return to the opening drinker. Heavy drinkers eat poorly and absorb thiamine (vitamin B1) badly, and neurons need thiamine to use glucose. Acute deficiency causes **Wernicke's encephalopathy**, described by **Carl Wernicke** (1848 to 1905) in 1881: confusion, ataxia, and ophthalmoplegia with nystagmus, an emergency treated with thiamine before glucose. Untreated, it leaves **Korsakoff's syndrome** (Korsakoff's psychosis), described by **Sergei Korsakoff** (1854 to 1900) in 1887: severe anterograde and retrograde amnesia, apathy, and **confabulation**, fluent false accounts filling memory gaps without intent to deceive.

Immediate memory and intelligence are relatively spared. Together these form Wernicke-Korsakoff syndrome, with lesions in the mammillary bodies and dorsomedial thalamus.

> **EXAM TRAP**
>
> Memory loss, confabulation and nystagmus in a heavy drinker is Wernicke-Korsakoff syndrome. The cause is thiamine deficiency, not vitamin B12, vitamin D, folate, hypoxia or lead. Alzheimer's, Pick's, Parkinson's, Huntington's and Creutzfeldt-Jakob diseases are not caused by drinking.

Disorders due to another medical condition

Physical disease can present psychiatrically, so DSM-5-TR has diagnoses such as depressive disorder due to another medical condition (ICD-10 F06 and F07, the "organic" syndromes). Cushing's syndrome (excess cortisol) mainly causes depression, anxiety and cognitive impairment, occasionally mania or psychosis; Addison's disease (cortisol deficiency) causes apathy and depression. Hypothyroidism causes depression and, when severe, a psychosis called "myxoedema madness"; hyperthyroidism causes anxiety. Vitamin B12 and folate deficiency, neurosyphilis (general paresis, with grandiose delusions), HIV and systemic lupus erythematosus can all cause mood change, psychosis or dementia. Epilepsy can bring an interictal psychosis, and head injury lasting personality change. Frontal damage produces the **frontal lobe syndrome**, an organic personality disorder of disinhibition, apathy and poor judgement (Chapter 5).

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Tolerance that extends to a related drug is called _____.
2. A substance use disorder with 6 or more criteria is rated _____.
3. The ventral tegmental area to nucleus accumbens pathway is the _____ dopamine system.
4. Heroin is not a natural opium alkaloid because it is _____.
5. Disulfiram inhibits _____.
6. Picking at the bedclothes in delirium is called _____.
7. Visual hallucinations, fluctuating cognition and parkinsonism define NCD with _____.
8. Korsakoff's syndrome results from deficiency of _____.

Answers: (1) cross-tolerance, (2) severe, (3) mesolimbic, (4) semi-synthetic, (5) aldehyde dehydrogenase, (6) carphologia, (7) Lewy bodies, (8) thiamine (vitamin B1).

8.11 Personality Disorders, Conduct and Impulse-Control Disorders

MEDIUM YIELD · 32 QS

Most disorders in this chapter come as episodes. The conditions here describe how a person habitually thinks, feels and relates, year after year. We define personality disorder, sort the

ten types into clusters, study borderline and antisocial personality closely, meet the dimensional models, and end with the disorders of conduct and impulse control.

When personality becomes a disorder ●

Everyone has personality traits, enduring ways of perceiving, relating to and thinking about the world and oneself; they become a **personality disorder** (PD) only when inflexible and maladaptive. DSM-5-TR defines a PD as an enduring pattern of inner experience and behaviour that deviates markedly from the expectations of the person's culture and shows in two or more of four areas: **cognition** (how self, others and events are perceived), **affectivity** (the range, intensity, lability and appropriateness of emotion), **interpersonal functioning** and **impulse control**. The pattern is pervasive and inflexible, stable, traceable to adolescence or early adulthood, causes distress or impairment, and is not better explained by another disorder, a substance or a medical condition. A PD can therefore disturb thinking, mood and relationships alike.

The traits are usually **ego-syntonic**: they feel like "just me", so informants become essential. The median prevalence of any PD is about 10 percent (10.5 percent in DSM-5-TR's review). Diagnosis before 18 needs features present for at least a year, because childhood traits often fade; antisocial PD is never diagnosed before 18. DSM-IV recorded PDs on **Axis II** (Section 8.1). DSM-5-TR keeps the **categorical model**, PDs as distinct syndromes, while noting the **dimensional model**, PDs as extreme variants of normal traits that shade into normality (see the newer models).

The ten disorders in three clusters ●●

DSM-5-TR groups its ten specific PDs into three **clusters** by descriptive similarity (Table 8.32): **Cluster A**, odd or eccentric; **Cluster B**, dramatic, emotional or erratic; **Cluster C**, anxious or fearful (weird, wild, worried). The clusters are poorly validated: one person often shows traits from several.

Table 8.32 The ten DSM-5-TR personality disorders.

Cluster	Disorder	Core pattern	Criteria needed
A: odd, eccentric	Paranoid	Pervasive distrust and suspiciousness; motives read as malevolent	4 of 7
A	Schizoid	Detachment from relationships, restricted emotional expression; prefers solitude	4 of 7
A	Schizotypal	Acute discomfort in close relationships, cognitive and perceptual distortions, eccentric behaviour	5 of 9
B: dramatic, emotional, erratic	Antisocial	Disregard for and violation of others' rights since age 15	3 of 7
B	Borderline	Instability of relationships, self-image and affects; marked impulsivity	5 of 9
B	Histrionic	Excessive emotionality and attention seeking	5 of 8
B	Narcissistic	Grandiosity, need for admiration, lack of empathy	5 of 9
C: anxious, fearful	Avoidant	Social inhibition, inadequacy, hypersensitivity to negative evaluation	4 of 7

Cluster	Disorder	Core pattern	Criteria needed
C	Dependent	Excessive need to be taken care of; submissive, clinging, fears separation	5 of 8
C	Obsessive-compulsive	Orderliness, perfectionism and control, at the expense of flexibility and efficiency	4 of 8

The schizotypal person shows **ideas of reference** (believing that unrelated events, such as strangers laughing, refer to oneself, short of delusion), **magical thinking** (telepathy, a "sixth sense"), odd speech and eccentric dress. It is genetically linked to schizophrenia, so DSM-5-TR also lists it in the schizophrenia spectrum (Section 8.4) and ICD-10 classes it there. DSM-5-TR adds **personality change due to another medical condition** (a lasting change caused by, say, a frontal lobe lesion), and other specified and unspecified PD.

> EXAM TRAP
>
> There is **no Cluster D. Avoidant PD is Cluster C**, not A: the schizoid person does not want company; the avoidant person longs for it but fears rejection. Among paranoid, schizoid, schizotypal and borderline, the odd one out is **borderline** (Cluster B).

Borderline personality disorder ●●●

Borderline personality disorder (BPD) is a pervasive pattern of instability of interpersonal relationships, self-image and affects, with marked impulsivity, beginning by early adulthood. Its signature is emotional instability and impulsivity, not detachment, perfectionism or paranoid ideation. Five or more of nine criteria are needed:

1. Frantic efforts to avoid real or imagined abandonment.
2. Unstable, intense relationships alternating between idealisation and devaluation.
3. Identity disturbance: a markedly and persistently unstable self-image.
4. Impulsivity in at least two self-damaging areas (spending, sex, substances, reckless driving, binge eating).
5. Recurrent suicidal behaviour, gestures or threats, or self-mutilation.
6. Affective instability from marked reactivity of mood, episodes lasting hours, rarely days.
7. Chronic feelings of emptiness.
8. Inappropriate, intense anger or difficulty controlling anger.
9. Transient, stress-related paranoid ideation or severe dissociation.

Adolph Stern coined "borderline" in 1938 for patients between neurosis and psychosis. **Otto F. Kernberg** (born 1928) recast it as **borderline personality organisation**: identity diffusion, primitive defences and generally intact reality testing. The primary defence is **splitting**: self and others are seen as all good or all bad, so today's idealised rescuer becomes tomorrow's betrayer. With it come **projective identification** (disowned feelings are placed in another person, who is pressed into feeling them), primitive idealisation and devaluation. Repression (pushing wishes out of awareness), sublimation (channelling them into valued activity), rationalisation (plausible excuses), regression (retreat to earlier behaviour) and undoing (an act that symbolically cancels another) are not the BPD defence (Chapter 4).

Heinz Kohut (1913 to 1981), founder of **self psychology**, held that a child needs **mirroring** (a parent's delighted reflection of its worth) and a calm figure to **idealise**; when both fail, the self stays fragmented, felt as emptiness and being "no one". He built the theory mainly to explain narcissism. **Marsha M. Linehan** (born 1943) proposed the **biosocial theory**: biological emotional vulnerability meets an invalidating environment that dismisses the child's feelings, so regulation is never learned. Treatments (Chapter 7): her dialectical behaviour therapy, mentalisation-based treatment, Kernberg's transference-focused psychotherapy and schema therapy. ICD-10 calls BPD **emotionally unstable personality disorder**, impulsive or borderline type.

Two differentials. BPD mood shifts last hours and follow interpersonal triggers; bipolar episodes last days to weeks, and both may coexist. Conduct disorder is diagnosed under 18 and centres on rule violation and aggression; BPD is an adult pattern of identity and affective instability and fear of abandonment, and childhood conduct disorder leads to ASPD, not BPD.

CLINICAL LINK

BPD is diagnosed more often in women in clinical samples, though not in community samples. Self-harm is common; about 6 percent of hospitalised patients died by suicide over 24 years.

EXAM TRAP

BPD mood is reactive and unpredictable, never "regular and predictable"; eccentric, odd thinking is schizotypal; cruelty to animals and rule-breaking before 15 are conduct disorder. A 2023 paper asking which is NOT a BPD feature offered both "regular and predictable changes in mood" and "eccentric behaviour with odd thinking"; both are strictly correct, but the key is the mood option, the direct inversion of BPD's affective instability. A 2024 paper keys "failure of parental mirroring and idealising, leading to a fragmented self" under *object relations theory*: strictly it is Kohut's **self psychology**, a cousin of the object relations school of **Melanie Klein** (1882 to 1960) and Kernberg. Choose it; oral fixation, Oedipal conflict and a harsh superego are classical Freudian answers.

Antisocial personality disorder and psychopathy ●●

Antisocial personality disorder (ASPD) is a pervasive pattern of disregard for and violation of the rights of others, occurring since age 15, shown by three or more of seven criteria: failure to conform to laws; deceitfulness and conning; impulsivity; irritability and aggressiveness; reckless disregard for safety; consistent irresponsibility; and lack of remorse. The person must be at least 18, with evidence of **conduct disorder beginning before 15** (Figure 8.14). ICD-10 calls it **dissocial personality disorder**. Antisocial behaviour often remits by about age 40.

Philippe Pinel (1745 to 1826) described **manie sans délire** (mania without delirium) in 1801: impulsive rages with intact reasoning. **James Cowles Prichard** (1786 to 1848) named **moral insanity** in 1835. **Hervey Cleckley** (1903 to 1984), in *The Mask of Sanity* (1941), listed 16 characteristics of the psychopath, among them superficial charm, absence of nervousness, untruthfulness and lack of remorse: a normal mask over an emotional void. **Robert D. Hare** (born 1934) developed the **Psychopathy Checklist-Revised** (PCL-R, 1991), built on Cleckley: 20

items scored 0 to 2 from interview and records (total 40; 30 or more is the usual cut-off for psychopathy), with an interpersonal-affective factor (glibness, grandiosity, callousness) and an antisocial-lifestyle factor (impulsivity, irresponsibility, early behaviour problems).

Psychopathy is defined by traits, ASPD mostly by behaviour: most men in prison meet ASPD criteria, but only a minority are psychopaths. Causes combine genes with adverse rearing (abuse, neglect, erratic discipline). In *Deviant Children Grown Up* (1966), **Lee Nelken Robins** (1922 to 2009) found that adult sociopathy almost never arose without childhood antisocial behaviour: the root of the conduct-disorder rule. The underarousal hypothesis adds that psychopaths have low autonomic arousal (low resting heart rate and skin conductance) and seek stimulation.

KEY STUDY: LYKKEN'S FEARLESS PSYCHOPATHS (1957)

David Thoreson Lykken (1928 to 2006) had primary psychopaths, other prisoners and controls learn a "mental maze" in which some wrong choices brought a shock. Psychopaths learned the maze as well but avoided the shocked errors less, reported less anxiety and showed smaller skin conductance responses to a signal paired with shock. Conclusion, the **fearlessness hypothesis**: punishment teaches psychopaths little. Criticism: small samples, one laboratory task.

EXAM TRAP

The PCL-R is **Hare's**, built on Cleckley's description; not Cleckley alone, **Aaron Temkin Beck** (1921 to 2021; cognitive therapy) or **Hans Jürgen Eysenck** (1916 to 1997; the PEN model). In ASPD the pattern dates from 15, the diagnosis waits until 18, and conduct disorder must begin before 15, not 18. An option reading "conduct disorder before 15 and disregard for others' rights from 18" is loosely worded but still the answer: 18 is the minimum age for the diagnosis.

Histrionic, narcissistic and the rest of Cluster B ●●

Histrionic personality disorder is excessive emotionality and attention seeking: discomfort when not the centre of attention, seductive or provocative behaviour, shallow and rapidly shifting emotion, impressionistic speech lacking detail, theatricality, suggestibility, and seeing relationships as more intimate than they are. **Narcissistic personality disorder** rests on a core triad of **grandiosity**, **need for admiration** and **lack of empathy**, with fantasies of unlimited success, the belief that one is special and understood only by high-status people, entitlement, exploitation, envy and arrogance. Kohut saw narcissism as arrested development after parents failed to mirror, to be met with empathy; Kernberg saw a pathological grandiose self defending against rage and envy toward cold parents, to be confronted. Table 8.33 separates the look-alikes.

Table 8.33 Telling the attention-drawing disorders apart.

Disorder	Core	Wants	Tell-tale sign
Histrionic	Excessive emotionality	Attention	Seductive, theatrical, shallow, suggestible
Narcissistic	Grandiosity, no empathy	Admiration	Entitlement, envy, feels "special"
Borderline	Instability, impulsivity	Not to be abandoned	Self-harm, emptiness, idealise then devalue

Disorder	Core	Wants	Tell-tale sign
Antisocial	Disregard for rights	Profit or pleasure	Deceit, aggression, no remorse
Dependent (C)	Need to be cared for	Reassurance	Submissive, clinging, cannot decide alone

EXAM TRAP

An admiration seeker who is emotional and seductive rather than grandiose is **histrionic**. Pervasive instability of relationships and self-image is **borderline**, not narcissistic.

Cluster C, and OCPD versus OCD

Avoidant personality disorder is social inhibition, inadequacy and hypersensitivity to negative evaluation in someone who wants relationships; it overlaps with social anxiety disorder (Section 8.6). **Dependent personality disorder** is an excessive need to be taken care of: needing reassurance for everyday decisions, never disagreeing for fear of losing support, urgently seeking a new relationship when one ends.

Obsessive-compulsive personality disorder (OCPD) is a preoccupation with orderliness, perfectionism and mental and interpersonal control at the expense of flexibility, openness and efficiency: lists swallow the task, perfectionism stops it being finished, work crowds out friends, morals are rigid, worthless objects are kept and money is hoarded. The motive is **control**, not achievement or autonomy; efficiency is what it sacrifices. **Sigmund Freud** (1856 to 1939) described the same trio in the **anal character** (1908): orderly, parsimonious and obstinate. OCPD is ego-syntonic, with limited insight and no true obsessions or compulsions; OCD is ego-dystonic, its intrusive obsessions and compulsions felt as alien and resisted (Section 8.6). Both can be diagnosed together.

Newer models: ICD-11 and the AMPD

ICD-11 abolished the ten types: one diagnosis of personality disorder is graded **mild, moderate or severe**, with **personality difficulty** below that threshold. Five **trait domain qualifiers** describe its style: negative affectivity, detachment, dissociality, disinhibition and **anankastia** (rigid perfectionism and control), plus a **borderline pattern specifier**. DSM-5's Section III offers the **Alternative Model for Personality Disorders** (AMPD). Criterion A rates the **level of personality functioning** from 0 to 4 on identity and self-direction (self) and empathy and intimacy (interpersonal); at least moderate impairment is needed. Criterion B requires one or more pathological traits, organised in five domains: negative affectivity, detachment, antagonism, disinhibition and psychoticism. Six types survive: antisocial, avoidant, borderline, narcissistic, obsessive-compulsive and schizotypal, plus personality disorder-trait specified.

EXAM TRAP

Dissociality and anankastia are ICD-11 domains; antagonism and psychoticism are AMPD domains.

Oppositional defiant and conduct disorder

The disruptive, impulse-control and conduct disorders, failures of emotional and behavioural self-control, mostly begin in childhood and are commoner in boys. **Oppositional defiant disorder** (ODD) is angry or irritable mood, argumentative or defiant behaviour, or vindictiveness for at least 6 months, with four or more of eight symptoms shown with someone other than a sibling. Most children with ODD never develop conduct disorder.

Conduct disorder (CD) is a repetitive, persistent pattern violating the basic rights of others or major age-appropriate norms: three or more of 15 criteria in the past 12 months, one in the past 6 months, from four groups: **aggression to people and animals**, **destruction of property**, **deceitfulness or theft** and **serious violations of rules** (staying out at night or truancy before 13, running away). The **childhood-onset type** shows a symptom before age 10, the **adolescent-onset type** none. The specifier **with limited prosocial emotions**, applied to CD alone, needs two or more of four traits for 12 months: lack of remorse or guilt, callous lack of empathy, unconcern about performance, and shallow or deficient affect. These callous-unemotional traits mark more severe, planned aggression.

Terrie E. Moffitt (1993) explained the two onsets. **Life-course-persistent** antisocial behaviour is rare, starts in early childhood from neuropsychological deficits meeting adverse homes, and continues into adult crime. **Adolescence-limited** antisocial behaviour is common, begins at puberty as teenagers copy older peers, and fades in early adulthood.

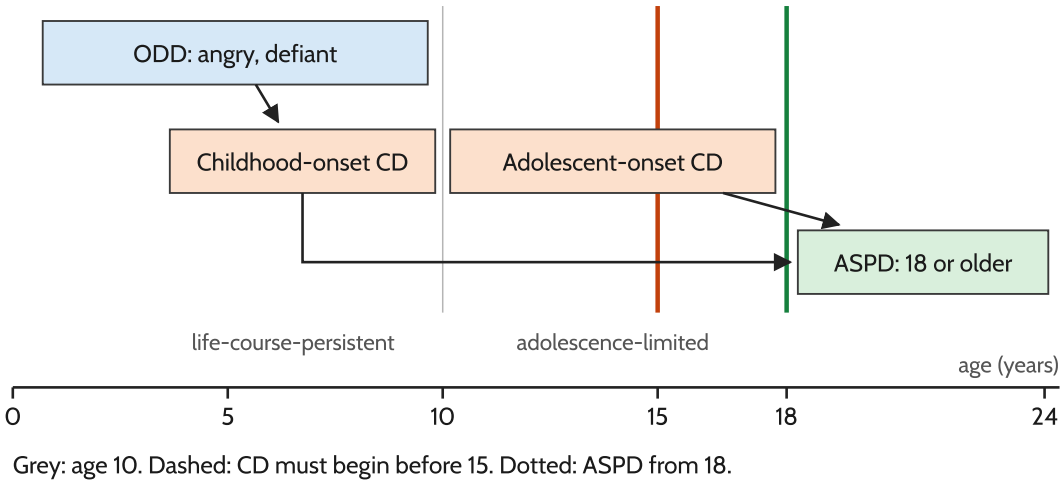

Figure 8.14 From defiance to antisocial personality. ASPD needs conduct disorder before 15 and age 18; the childhood-onset path is the one that usually persists.

Explosive, fire-setting and stealing impulses

Intermittent explosive disorder (IED) is recurrent impulsive, verbal or physical, aggressive outbursts grossly out of proportion to the provocation (twice weekly for 3 months, or three destructive or injurious outbursts in a year), unplanned and not aimed at money or power, from age 6. In **pyromania** the person deliberately and purposefully sets fires on more than one occasion, with tension or arousal before, fascination with fire, and pleasure, gratification or relief after; fires set for money, revenge, ideology or to hide a crime do not count. Pyromania is planned fire setting for gratification; IED is impulsive aggression beyond the

trigger. In **kleptomania** the person repeatedly fails to resist impulses to steal objects not needed for personal use or their value, with rising tension before and pleasure or relief during the theft; women outnumber men three to one.

Charles Chrétien Henri Marc and **Jean-Étienne Esquirol** (1772 to 1840) named kleptomania in 1838, renaming the *klopemanie* (stealing madness) of the Swiss physician **André Matthey** (1816). Not the distractors: **Eugen Bleuler** (1857 to 1939) named schizophrenia and described ambivalence and autism; **Enrico Morselli** coined dysmorphophobia (1891); **Alistair Munro** described monosymptomatic hypochondriacal psychosis. ICD-10's habit and impulse disorders (F63) were pathological gambling, fire-setting and stealing, and trichotillomania. ICD-11's impulse control disorders are pyromania, kleptomania, compulsive sexual behaviour disorder and intermittent explosive disorder; gambling moved to addictive behaviours (Section 8.10), trichotillomania to the obsessive-compulsive group (Section 8.6).

> EXAM TRAP
>
> "With limited prosocial emotions" is a specifier of **conduct disorder** only, not ASPD, ODD or IED. IED aggression may be verbal or physical. "Kleptomania" is **Marc's** term (with Esquirol), not Bleuler's, Munro's or Morselli's.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. A personality disorder shows in two or more of cognition, affectivity, interpersonal functioning and _____.
2. Avoidant, dependent and obsessive-compulsive PDs form Cluster _____.
3. The primary defence of borderline personality disorder is _____.
4. Borderline PD needs _____ of nine criteria.
5. ASPD requires age 18 and conduct disorder beginning before age _____.
6. The Psychopathy Checklist-Revised was developed by _____.
7. The specifier "with limited prosocial emotions" applies only to _____.
8. ICD-11 grades personality disorder as mild, moderate or _____.

Answers: (1) impulse control, (2) C, (3) splitting, (4) five, (5) 15, (6) Robert Hare, (7) conduct disorder, (8) severe.

Kraepelin's Cards, Revisited

Kraepelin's cards hold this chapter in miniature. Each began with signs and symptoms, the vocabulary of Section 8.2, and ended with onset, course and prognosis, the terms of Section 8.1. The line the cards drew still divides schizophrenia (8.4) from the bipolar and depressive disorders (8.5), and DSM-III turned his method into explicit criteria, which is why every disorder from 8.3 to 8.11 arrives as a criterion set, often with a minimum duration. Yet his deeper hope, that each pile would prove to be a separate brain disease, has not come true. Risk genes for schizophrenia are shared with bipolar disorder and depression, comorbidity is the rule, and the diathesis-stress model explains more than any single cause. So DSM-5 added

severity ratings and spectra to his categories, and ICD-11 grades personality disorder by severity.

Chapter summary

8.1 Abnormality, Causes and the Classification of Mental Disorders. No single criterion defines abnormality: clinicians weigh the **four Ds**, Wakefield's **harmful dysfunction** and the DSM-5-TR definition, which excludes expectable grief and mere social deviance. Psychopathology has its own vocabulary of aetiology, onset, course, prognosis, prevalence, incidence and **comorbidity**. Genes, endophenotypes and stress meet in the **diathesis-stress model** and Engel's **biopsychosocial model**. Classification is syndromal and prototypical, judged by reliability and validity. The **DSM** (APA; DSM-5 dropped the axes; DSM-5-TR, 2022, is current) and the **ICD** (WHO; ICD-10 F codes; ICD-11 Chapter 6) are the two systems, and cultural concepts of distress such as dhat and latah show culture shaping distress.

8.2 Descriptive Psychopathology: Signs and Symptoms. **Descriptive psychopathology**, founded by Karl Jaspers (*General Psychopathology*, 1913), describes experiences phenomenologically, separating form from content, primary from secondary symptoms and signs from symptoms. An **illusion** misreads a real stimulus; a **hallucination**, a disorder of perception, has no stimulus; hypnagogic hallucinations are normal. Flight of ideas, loosening, tangentiality and circumstantiality are disorders of the **form** of thought. Persecution is the commonest **delusion**; Capgras, Fregoli, Cotard, Othello and de Clérambault name delusional themes. Confabulation fills memory gaps without intent to deceive. Affect is observed and momentary, mood sustained; elevated mood runs euphoria, elation, exaltation, ecstasy. **Insight** has six grades.

8.3 Neurodevelopmental Disorders. **Neurodevelopmental disorders** begin in the developmental period and often co-occur, so assessment uses several informants. **Intellectual disability** needs intellectual deficits (IQ about two SD below the mean, 65 to 75), adaptive deficits and early onset; DSM-5 grades severity by adaptive functioning, and global developmental delay is used under age 5. Causes include trisomy 21, fragile X, PKU and fetal alcohol syndrome. **Autism spectrum disorder** (Kanner, 1943) needs social-communication deficits plus restricted, repetitive behaviour, not low IQ. **ADHD** needs six of nine symptoms in a domain, several before 12, in two settings. **Specific learning disorder** occurs with average intelligence; Tourette's needs motor and vocal tics for over a year.

8.4 Schizophrenia Spectrum and Other Psychotic Disorders. Kraepelin's **dementia praecox** (1893), separated from manic-depressive insanity by its deteriorating course, became Bleuler's **schizophrenia** (1908, 1911), defined by four As; Schneider's **first-rank symptoms** stress voices, thought alienation, passivity and delusional perception. Symptoms are positive, negative or cognitive. DSM-5-TR needs 6 months including 1 month of active symptoms; ICD-10 and ICD-11 need 1 month. Brief psychotic disorder lasts 1 day to under 1 month, schizophreniform disorder 1 to under 6 months. Risk is about 48 percent for an identical twin. Positive symptoms reflect mesolimbic dopamine excess; high **expressed emotion** predicts relapse. Antipsychotics are first-line; nigrostriatal D2 blockade causes dystonia, akathisia, parkinsonism and, last, tardive dyskinesia.

8.5 Depressive and Bipolar Disorders, and Suicide. Mood disorders are built from three episodes: a **major depressive episode** (5 of 9 symptoms for 2 weeks), a **manic episode** (1 week; marked impair-

ment or psychosis) and a hypomanic episode (4 days, no marked impairment or psychosis). MDD affects women twice as often as men; dysthymia is milder but lasts 2 years; DMDD begins before age 10. Specifiers include mixed features, rapid cycling, melancholic, atypical, peripartum and seasonal. Bipolar I needs mania; bipolar II needs hypomania plus major depression; cyclothymia is subthreshold. Causes range from monoamines and kindling to Beck's cognitive triad. Lithium is the gold-standard mood stabiliser. A previous attempt is the strongest predictor of suicide.

8.6 Anxiety and Obsessive-Compulsive and Related Disorders. Fear answers present danger, anxiety a future threat. DSM-5 split the old neuroses into anxiety, obsessive-compulsive and trauma chapters. Most anxiety disorders need about 6 months; panic disorder and selective mutism need 1 month. A panic attack peaks within minutes with 4 of 13 symptoms; agoraphobia needs 2 of 5 situations and GAD 3 of 6 symptoms. Causes include the GABA system, Mowrer's two-factor theory, Clark's catastrophic misinterpretation, Klein's suffocation alarm and displacement (Little Hans). OCD needs ego-dystonic obsessions or compulsions taking over an hour a day or causing distress; compulsions are negatively reinforced escape responses and undoing its typical defence. BDD, hoarding, trichotillomania and excoriation are its relatives.

8.7 Trauma, Dissociative and Somatic Symptom Disorders. Charcot, Janet, Breuer and Freud explained hysteria by hypnosis, dissociation and repression, and DSM-III (1980) split it into somatoform and dissociative disorders. PTSD needs a qualifying trauma and symptoms in four clusters for more than 1 month; acute stress disorder lasts 3 days to 1 month, and ICD-11 adds complex PTSD. DID combines two or more personality states with amnesic gaps; dissociative amnesia, fugue and depersonalisation/derealisation complete the class. Somatic symptom disorder is an excessive response to symptoms, conversion a neurological-like loss incompatible with disease, and factitious disorder and malingering are deliberate, differing in external reward.

8.8 Feeding, Eating, Elimination and Sleep-Wake Disorders. DSM-5-TR's feeding and eating disorders include pica (non-food eating for at least 1 month), rumination disorder and ARFID, which restricts food with no body-image concern. Anorexia nervosa (Gull and Lasègue, 1873) needs restriction to a significantly low weight, fear of weight gain and disturbed body experience; amenorrhoea was dropped, severity is graded by BMI, and it is the most lethal eating disorder. Bulimia nervosa (Russell, 1979) pairs binges with compensation weekly for 3 months at normal weight; binge-eating disorder lacks compensation. Enuresis starts at age 5, encopresis at 4. Among sleep-wake disorders, narcolepsy brings cataplexy, N3 brings sleepwalking and sleep terrors, and REM brings nightmares and REM sleep behaviour disorder.

8.9 Sexual Dysfunctions, Paraphilic Disorders and Gender Dysphoria. Masters and Johnson's sexual response cycle (1966: excitement, plateau, orgasm, resolution, with a male refractory period) gained Kaplan's desire phase (1979) and Basson's circular model (2000). Most DSM-5-TR sexual dysfunctions need symptoms on 75 to 100 percent of occasions for about 6 months with distress; premature ejaculation is tied to anxiety and treated with stop-start and squeeze. A paraphilic disorder is a paraphilia plus distress, impairment or harm to others; DSM-5-TR names eight. Gender dysphoria replaced gender identity disorder, and ICD-11 moved gender incongruence to Chapter 17 while paraphilic disorders stayed in Chapter 6. The APA removed homosexuality in 1973.

8.10 Substance-Related, Addictive and Neurocognitive Disorders. A **psychoactive substance** alters mood, cognition or behaviour; tolerance, **cross-tolerance**, withdrawal and dependence describe how use becomes disorder. ICD-10 codes F10 to F19 patterns (harmful use, Edwards and Gross's **dependence syndrome**; 'drug abuse' is not one), DSM-5-TR merges abuse and dependence into **substance use disorder** (2 of 11 criteria), and ICD-11 adds gambling and gaming disorder. Drugs converge on the mesolimbic dopamine system. Alcohol is a GABA-A agonist; heroin is semi-synthetic. **Delirium** is acute, fluctuating and clouded, unlike dementia; NCD subtypes differ in course and pathology; depression causes pseudodementia; thiamine deficiency causes **Wernicke-Korsakoff syndrome**.

8.11 Personality Disorders, Conduct and Impulse-Control Disorders. A **personality disorder** is an enduring, pervasive, inflexible pattern deviating from cultural expectations in two or more of cognition, affectivity, interpersonal functioning and impulse control, beginning by adolescence or early adulthood; usually ego-syntonic, it affects about 10 percent. DSM-5-TR's ten types form Clusters A (odd), B (dramatic) and C (anxious). **Borderline PD** (5 of 9 criteria) is instability of relationships, self-image and affect with impulsivity, its defence splitting. **Antisocial PD** needs age 18 and conduct disorder before 15; Hare's PCL-R measures psychopathy. ICD-11 and the AMPD grade severity and traits. Conduct disorder alone takes the specifier **with limited prosocial emotions**; pyromania and kleptomania follow tension with relief.

Who's who

Table 8.34 The people in this chapter.

Name	When	Examined for
Marie Jahoda	1907 to 2001	Ideal (positive) mental health, 1958
Jerome Wakefield	1992	Mental disorder as harmful dysfunction
Daniel Offer	1966	With Sabshin, four perspectives on normality: health, utopia, average, process
Melvin Sabshin	1966	With Offer, four perspectives on normality
Avshalom Caspi	2003	Serotonin transporter gene and childhood maltreatment in adult depression (gene-environment interaction)
Irving Gottesman	1970s; 2003	Brought the endophenotype concept into psychiatry with Shields and revived it with Gould
James Shields	1970s	Endophenotype concept and schizophrenia twin research with Gottesman
Todd Gould	2003	Endophenotype concept paper with Gottesman
Paul Meehl	1920 to 2003	Schizotaxia (1962): an early diathesis-stress account of schizophrenia
Joseph Zubin	1977	Vulnerability-stress model, with Bonnie Spring
Bonnie Spring	1977	Vulnerability-stress model, with Joseph Zubin
George L. Engel	1913 to 1999	Biopsychosocial model (1977)
Hippocrates	about 460 to 377 BC	Classified mania, melancholia and phrenitis; coined the word hysteria
Emil Kraepelin	1856 to 1926	Descriptive classification by course; dementia praecox versus manic-depressive insanity
Robert Spitzer	1932 to 2015	Directed DSM-III (1980): explicit criteria and the multiaxial system
N. N. Wig	1960	Coined the term dhat syndrome
Jacob Cohen	1960	Cohen's kappa, the chance-corrected agreement index used for diagnostic reliability
Aaron T. Beck	1921 to 2021	Cognitive model: disturbed thought patterns and schemas (Section 7.5)
Albert Ellis	1913 to 2007	Irrational beliefs in the cognitive model (Section 7.5)

Name	When	Examined for
Eugen Bleuler	1857 to 1939	Named schizophrenia (Section 8.4); a distractor for the biopsychosocial model
Karl Jaspers	1883 to 1969	General Psychopathology (1913); phenomenology and empathic understanding; primary delusions; the marks of a delusion (certainty, incorrigibility, impossibility)
Charles Bonnet	1720 to 1793	Described (1760) visual hallucinations in his nearly blind grandfather: Charles Bonnet syndrome
Norman Cameron	1896 to 1975	Overinclusive thinking in schizophrenia
Carl Wernicke	1848 to 1905	Introduced the term overvalued idea
Kurt Schneider	1887 to 1967	First-rank symptoms of schizophrenia, including thought insertion, withdrawal and broadcasting
Joseph Capgras	1873 to 1950	Capgras syndrome (1923, 'l'illusion des sosies'): a familiar person replaced by an identical impostor
Jules Cotard	1840 to 1889	Cotard syndrome (described 1880, named 'délire des négations' 1882): nihilistic delusion of being dead or not existing
Arnold Pick	1851 to 1924	Named reduplicative paramnesia (1903)
Gaëtan Gatian de Clérambault	1872 to 1934	de Clérambault syndrome (erotomania; classic account 1921): a person of higher status is secretly in love with the patient
Jean Reboul-Lachaux	1923 (key year)	Co-author with Joseph Capgras of the 1923 description of l'illusion des sosies (Capgras syndrome)
Paul Courbon	1927 (key year)	Described Fregoli syndrome with Gabriel Fail (1927)
Gabriel Fail	1927 (key year)	Described Fregoli syndrome with Paul Courbon (1927)
Leopoldo Fregoli	1867 to 1936	Italian quick-change actor after whom Fregoli syndrome is named
John Todd	1955 (key year)	Named the Othello syndrome (morbid jealousy) with Kenneth Dewhurst (1955)
Kenneth Dewhurst	1955 (key year)	Named the Othello syndrome (morbid jealousy) with John Todd (1955)
Karl-Axel Ekbom	1907 to 1977	Described delusional parasitosis (1938), Ekbom syndrome; his name also labels restless legs syndrome
Sigbert Ganser	1853 to 1931	Ganser syndrome (1898): approximate answers
Charles Lasègue	1816 to 1883	Described folie à deux (shared delusion) with Jules Falret (1877)
Jules Falret	1824 to 1902	Described folie à deux (shared delusion) with Charles Lasègue (1877)
Anton Delbrück	1891 (key year)	Pseudologia fantastica: fantastic lying the teller half believes
Peter Sifneos	1920 to 2008	Introduced the term alexithymia (1973)
Leo Kanner	1894 to 1981	First described early infantile autism, 'Autistic disturbances of affective contact' (1943)
Hans Asperger	1906 to 1980	Described 'autistic psychopathy' in fluent, intelligent boys (1944)
Bruno Bettelheim	1903 to 1990	Discredited 'refrigerator mother' theory of autism (The Empty Fortress, 1967)
Lorna Wing	1928 to 2014	Triad of impairments in autism, with Judith Gould (1979)
Judith Gould	1979	Triad of impairments in autism, with Lorna Wing
Theodor Heller	1908	Dementia infantilis, later called childhood disintegrative disorder
Andreas Rett	1966	Described Rett syndrome (MECP2; girls; regression and hand-wringing)
Simon Baron-Cohen	born 1958	Theory of mind deficit in autism and the Sally-Anne test (1985); mind-blindness; extreme male brain theory
Alan Leslie	1985	Sally-Anne false-belief study with Baron-Cohen and Frith
Uta Frith	born 1941	Sally-Anne study (1985); weak central coherence theory of autism (1989)
Andrew Wakefield	1998	Retracted Lancet paper falsely linking the MMR vaccine to autism
Ivar Lovaas	1927 to 2010	Early intensive applied behaviour analysis for autism (1987)
George Frederic Still	1868 to 1941	First description of ADHD-like children, a 'defect of moral control' (1902)
Russell Barkley	born 1949	Behavioural inhibition model of ADHD (1997)
John Langdon Down	1828 to 1896	Described Down syndrome (1866)
Asbjørn Følling	1888 to 1973	Discovered phenylketonuria (1934)
James Hinshelwood	1859 to 1919	Congenital word-blindness (dyslexia); book of that title (1917)
Samuel Torrey Orton	1879 to 1948	Explained dyslexia by incomplete cerebral dominance (1920s)

Name	When	Examined for
Georges Gilles de la Tourette	1857 to 1904	Described Tourette's disorder (1885)
Kurt Goldstein	1878 to 1965	Concrete versus abstract attitude after brain damage; a distractor in autism questions (Section 8.2)
Benedict Morel	1809 to 1873	Démence précoce (1850s), the first name for the illness
Ewald Hecker	1871	Described hebephrenia
Karl Ludwig Kahlbaum	1828 to 1899	Described catatonia (1874)
Timothy Crow	1980	Type I (positive) and Type II (negative) schizophrenia
Peter Liddle	1987	Three syndromes: reality distortion, disorganisation, psychomotor poverty
Nancy Andreasen	1980s	SANS and SAPS rating scales for negative and positive symptoms
David Rosenthal	1963	Genain quadruplets; Danish adoption studies with Kety and Wender
Leonard Heston	1966	Adoption study: schizophrenia in adopted-away children of mothers with schizophrenia
Seymour Kety	1968	Danish adoption studies
Paul Wender	1968	Danish adoption studies
Arvid Carlsson	1923 to 2018	Dopamine as a neurotransmitter; dopamine hypothesis; Nobel Prize 2000
Philip Seeman	1976	Antipsychotic potency correlates with D2 receptor affinity
Ian Creese	1976	With Burt and Snyder: antipsychotic potency correlates with D2 binding
David Burt	1976	With Creese and Snyder: antipsychotic potency correlates with D2 binding
Solomon Snyder	1976	With Creese and Burt: antipsychotic potency correlates with D2 binding
Kenneth Davis	1991	Revised dopamine hypothesis: subcortical excess, prefrontal deficit
Oliver Howes	2009	With Kapur: presynaptic striatal dopamine as the final common pathway
Shitij Kapur	2009	With Howes: dopamine hypothesis version III
Daniel Weinberger	1987	Neurodevelopmental hypothesis; hypofrontality
Robin Murray	1987	With Shon Lewis: neurodevelopmental hypothesis
Shon Lewis	1987	With Robin Murray: neurodevelopmental hypothesis
Eve Johnstone	1976	First CT evidence of enlarged lateral ventricles in schizophrenia
Frieda Fromm-Reichmann	1889 to 1957	The schizophrenogenic mother (1948), now discredited
Gregory Bateson	1904 to 1980	Double bind theory of schizophrenia (1956); the symptomatic double bind
Don Jackson	1956	Co-author of the double bind theory; family homeostasis
Jay Haley	1956	Co-author of the double bind theory
John Weakland	1956	Co-author of the double bind theory
Theodore Lidz	1957	Marital schism and marital skew
Lyman Wynne	1958	Pseudomutuality; communication deviance
Ronald David Laing	1927 to 1989	Anti-psychiatry; schizophrenia as a response to an unliveable family situation
George Brown	1972	Expressed emotion predicts relapse (with Birley and Wing)
James Birley	1972	Expressed emotion and relapse (with Brown and Wing)
John Wing	1972	Expressed emotion and relapse (with Brown and Birley)
Christine Vaughn	1976	With Leff: expressed emotion, relapse and 35 hours of face-to-face contact
Julian Leff	1976	With Vaughn: expressed emotion and relapse; Chandigarh studies of low EE in India
Sigmund Freud	1856 to 1939	Schizophrenia as withdrawal of libido and regression
Paul Federn	1871 to 1950	Weak ego boundaries in schizophrenia
Harry Stack Sullivan	1892 to 1949	Interpersonal theory of schizophrenia
Silvano Arieti	1914 to 1981	Palaeologic thinking; the von Domarus principle
Eilhard von Domarus	1940s	The von Domarus principle: identity from shared predicates
Margaret Mahler	1897 to 1985	Schizophrenia as fixation at the autistic phase with failure of stable symbiosis; symbiotic psychosis

Name	When	Examined for
Ping-Nie Pao	1979	Symptoms as adaptation to organismic panic
Edith Jacobson	1897 to 1978	Psychotic fusion of self and object representations
Jean Delay	1907 to 1987	With Deniker, introduced chlorpromazine for psychosis (1952)
Pierre Deniker	1917 to 1998	With Delay, introduced chlorpromazine for psychosis (1952)
René Spitz	1887 to 1974	Anaclitic depression in separated infants (1946); hospitalism
Norman Rosenthal	1984	Described seasonal affective disorder and its light therapy
Jean-Pierre Falret	1794 to 1870	Folie circulaire (1850s): alternating mania and melancholia
Jules Baillarger	1809 to 1890	Folie a double forme (1850s), the same alternating illness
Karl Leonhard	1957	Separated unipolar from bipolar mood disorders
Joseph Schildkraut	1965	Catecholamine hypothesis: noradrenaline deficit in depression, excess in mania
Robert Post	1992	Kindling (sensitisation) hypothesis: each episode lowers the threshold for the next
Karl Abraham	1911	Early psychoanalytic account of depression as hostility turned against the self
Edward Bibring	1953	Depression as the ego's awareness of its helplessness; loss of self-esteem
Peter Lewinsohn	1970s	Behavioural theory: low rate of response-contingent positive reinforcement
Martin Seligman	1975	Learned helplessness theory of depression
Lyn Abramson	1978, 1989	Attributional reformulation (internal, stable, global); hopelessness theory
John Teasdale	1978	Co-author of the attributional reformulation of learned helplessness
Gerald Metalsky	1989	Co-author of the hopelessness theory of depression
Lauren Alloy	1989	Co-author of the hopelessness theory of depression
Aaron Temple Beck	1921 to 2021	Cognitive theory of depression (1967): negative cognitive triad of self, world and future
Susan Nolen-Hoeksema	1959 to 2013	Response styles theory (1991): rumination and the gender gap in depression
Tirril Harris	1978	Co-author of Social Origins of Depression
John Cade	1912 to 1980	Introduced lithium for mania (1949)
Norman Kreitman	1969	Coined the term parasuicide
Emile Durkheim	1858 to 1917	Suicide (1897): egoistic, altruistic, anomic and fatalistic suicide
Edwin Shneidman	1918 to 2009	Founder of suicidology; psychache
Thomas Joiner	2005	Interpersonal theory of suicide: burdensomeness, thwarted belongingness, acquired capability
Marie Asberg	1976	Low CSF 5-HIAA in depressed patients who make violent suicide attempts
W. M. Patterson and colleagues	1983	SAD PERSONS suicide risk scale (Section 7.2)
Jean Piaget	1896 to 1980	Cognitive developmental stages (Chapter 6); a distractor to Spitz for anaclitic depression
B. F. Skinner	1904 to 1990	Operant conditioning (Chapter 1); a distractor to Spitz for anaclitic depression
Hans Eysenck	1916 to 1997	Personality dimensions (Chapter 4); a distractor to Spitz for anaclitic depression
William Cullen	1769	Coined the term neurosis (1769)
Alan Goldstein	1978	With Chambless, 'fear of fear' in panic disorder and agoraphobia (1978)
Dianne Chambless	1978	With Goldstein, 'fear of fear' in panic disorder and agoraphobia (1978)
Jerome Kagan	1929 to 2021	Behavioural inhibition, an inherited shy temperament visible by four months that predicts anxiety
O. Hobart Mowrer	1907 to 1982	Two-factor theory (1947): fear acquired by classical conditioning, avoidance kept by negative reinforcement; explains compulsions as escape responses
Stanley Rachman	1977	Three pathways to fear (1977): conditioning, vicarious learning, information; thought-action fusion in OCD
David M. Clark	1986	Catastrophic misinterpretation of bodily sensations in panic (1986)

Name	When	Examined for
Steven Reiss	Contemporary	Anxiety sensitivity, the belief that anxiety symptoms are harmful
Thomas Borkovec	Contemporary	Avoidance theory of worry in GAD
Adrian Wells	Contemporary	Metacognitive model of GAD (beliefs about worry)
Michel Dugas	Contemporary	Intolerance of uncertainty in GAD
Donald F. Klein	1993	Suffocation false alarm theory of panic disorder (1993)
David H. Barlow	Contemporary	Triple vulnerability theory of anxiety
Susan Swedo	1998	PANDAS: childhood OCD or tics after streptococcal infection (1998)
Paul Salkovskis	1985	Inflated responsibility in OCD (1985)
Enrico Morselli	1891	Named dysmorphophobia, now body dysmorphic disorder (1891)
François Hallopeau	1889	Named trichotillomania (1889)
Jean-Martin Charcot	1825 to 1893	Salpêtrière: produced and removed hysterical symptoms by hypnosis and suggestion
Pierre Janet	1859 to 1947	Dissociation (désagrégation, 1889); la belle indifférence usually credited to him
Josef Breuer	1842 to 1925	Treated Anna O.; Studies on Hysteria with Freud (1895)
Anna O. (Bertha Pappenheim)	1859 to 1936	Breuer's hysterical patient; named the 'talking cure'
Anna Freud	1895 to 1982	Defence mechanisms; a distractor for who coined la belle indifférence
Ernest Ropiequet Hilgard	1904 to 2001	Neodissociation theory and the hidden observer (Divided Consciousness, 1977)
Abram Kardiner	1891 to 1981	The Traumatic Neuroses of War (1941)
Elisabeth Kübler-Ross	1969	Stages of dying (denial, anger, bargaining); not PTSD clusters
Richard Tedeschi	1996	Post-traumatic growth, with Calhoun: five domains
Lawrence Calhoun	1996	Post-traumatic growth, with Tedeschi
Corbett Thigpen	1957	The Three Faces of Eve, with Cleckley
Hervey Cleckley	1957	The Three Faces of Eve, with Thigpen
Nicholas Spanos	1942 to 1994	Sociocognitive model of DID; critic of the hidden observer
Paul Briquet	1796 to 1881	Briquet's syndrome (1859), later somatisation disorder
George Miller Beard	1839 to 1883	Named neurasthenia (1869)
Richard Asher	1912 to 1969	Named Munchausen syndrome (1951)
Baron Munchausen	1720 to 1797	Teller of tall tales after whom Munchausen syndrome is named
Roy Meadow	1977	Described Munchausen syndrome by proxy (factitious disorder imposed on another)
Sir William Withey Gull	1816 to 1890	Named anorexia nervosa (paper read 1873, printed 1874)
Ernest-Charles Lasègue	1816 to 1883	Described anorexie hystérique (1873)
Hilde Bruch	1904 to 1984	The Golden Cage (1978): anorexia as a struggle for control by an ineffective-feeling, compliant girl
Gerald Russell	1979	Named bulimia nervosa, 'an ominous variant of anorexia nervosa' (1979); Russell's sign
Salvador Minuchin	1921 to 2017	The psychosomatic family in anorexia (Section 7.7)
Jean-Baptiste-Édouard Gélineau	1828 to 1906	Coined the term narcolepsy (1880)
William Howell Masters	1915 to 2001	Human Sexual Response (1966): four-phase sexual response cycle; Human Sexual Inadequacy (1970): performance anxiety, spectatoring, squeeze technique
Virginia Eshelman Johnson	1925 to 2013	Masters's research partner; co-author of the sexual response cycle (1966) and of sex therapy methods (1970)
Helen Singer Kaplan	1929 to 1995	Added a desire phase to the sexual response cycle (Disorders of Sexual Desire, 1979)
Rosemary Basson	2000	Circular model of female sexual response; responsive desire
Sandra Lipsitz Bem	1944 to 2014	Androgyny; Bem Sex Role Inventory (1974)
James Semans	1956	Stop-start technique for premature ejaculation
Ray Blanchard	1989	Coined autogynephilia: a male's arousal by the image of himself as a woman
John Money	1921 to 2006	Lovemaps: childhood templates of the ideal lover, distorted in paraphilias

Name	When	Examined for
Kurt Freund	1914 to 1996	Courtship disorder hypothesis: voyeurism, exhibitionism and frotteurism as disturbed courtship phases
Alfred Charles Kinsey	1894 to 1956	Kinsey scale (0 to 6); Sexual Behavior in the Human Male (1948)
Evelyn Hooker	1907 to 1996	1957 study: blind judges could not tell homosexual from heterosexual men on projective tests; no difference in adjustment
Griffith Edwards	1976	Described the alcohol dependence syndrome with Milton Gross (1976)
Milton Gross	1976	Co-described the alcohol dependence syndrome with Griffith Edwards (1976)
Richard Solomon	1974	Opponent-process theory of drug use, with John Corbit (1974)
John Corbit	1974	Opponent-process theory, with Richard Solomon (1974)
C. Robert Cloninger	1981	Type I and type II alcoholism, from Swedish adoption studies
Elvin Morton Jellinek	1890 to 1963	The Disease Concept of Alcoholism (1960); alpha to epsilon alcoholism, gamma loss of control, delta inability to abstain
Denise Kandel	1975	Stages of adolescent drug use; the gateway hypothesis (marijuana)
Albert Hofmann	1906 to 2008	Discovered the effects of LSD (1943)
Alois Alzheimer	1864 to 1915	Described Alzheimer's disease (1906): plaques and tangles in Auguste Deter
Auguste Deter	1850 to 1906	Alzheimer's first patient, Auguste D., admitted in 1901
George Huntington	1850 to 1916	Described Huntington's chorea (1872)
Leslie Kiloh	1961	Popularised the term pseudodementia; depression is its common cause
Sergei Korsakoff	1854 to 1900	Korsakoff's syndrome (1887): amnesia with confabulation
Adolph Stern	1938	Coined the term borderline for patients between neurosis and psychosis
Otto F. Kernberg	born 1928	Borderline personality organisation (identity diffusion, primitive defences such as splitting, intact reality testing); transference-focused psychotherapy; narcissism as a pathological grandiose self
Heinz Kohut	1913 to 1981	Self psychology: failed parental mirroring and idealising leave a fragmented self; narcissism as arrested development
Marsha M. Linehan	born 1943	Biosocial theory of borderline personality disorder; dialectical behaviour therapy
Melanie Klein	1882 to 1960	Object relations theory, the school to which splitting belongs
Philippe Pinel	1745 to 1826	Manie sans délire (mania without delirium), 1801, an early description of antisocial personality
James Cowles Prichard	1786 to 1848	Moral insanity, 1835
Robert D. Hare	born 1934	Psychopathy Checklist-Revised (PCL-R, 1991), 20 items scored 0 to 2, cut-off 30 of 40, two factors
Lee Nelken Robins	1922 to 2009	Deviant Children Grown Up (1966): adult sociopathy almost always preceded by childhood antisocial behaviour
David Thoreson Lykken	1928 to 2006	Fearlessness hypothesis of psychopathy (1957): poor avoidance of shock, low anxiety, low skin conductance
Aaron Temkin Beck	1921 to 2021	Cognitive therapy; a distractor for the PCL-R
Hans Jürgen Eysenck	1916 to 1997	PEN model of personality; a distractor for the PCL-R
Terrie E. Moffitt	1993	Life-course-persistent versus adolescence-limited antisocial behaviour
Charles Chrétien Henri Marc	1838	With Esquirol, gave kleptomania its name
Jean-Étienne Esquirol	1772 to 1840	With Marc, named kleptomania (1838)
André Matthey	1816	Swiss physician whose klopemanie (stealing madness) Marc and Esquirol renamed kleptomania
Alistair Munro	Late twentieth century	Monosymptomatic hypochondriacal psychosis (a distractor for kleptomania)

Key terms

four Ds, statistical criterion, cultural relativity, harmful dysfunction, mental disorder, subjective model, Psychopathology, abnormal psychology, syndrome, Aetiology, Prognosis, Prevalence, inci-

dence, Comorbidity, concordance, heritability, gene-environment interaction, gene-environment correlation, endophenotypes, diathesis-stress model, vulnerability-stress model, biopsychosocial model, syndromal approach, historical approach, prototypical approach, Reliability, validity, false positive, false negative, conditions for further study, non-suicidal self-injury disorder, culture-bound syndromes, cultural concepts of distress, Cultural Formulation Interview, Descriptive psychopathology, phenomenology, sign, Delirium, twilight state, illusion, hallucination, pseudohallucination, Charles Bonnet syndrome, depersonalisation, derealisation, thought block, Formal thought disorder, flight of ideas, loosening of associations, overvalued ideas, thought broadcasting, delusion, primary delusion, delusional perception, delusion of persecution, nihilistic delusions, Capgras syndrome, Amnesia, confabulation, Déjà vu, Jamais vu, Mood, affect, Anhedonia, Alexithymia, Pseudobulbar affect, ego-dystonic, Catatonia, Insight, neurodevelopmental disorders, pervasive developmental disorders, multi-informant assessment, Intellectual disability, adaptive functioning, Borderline intellectual functioning, Global developmental delay, Down syndrome, Fragile X syndrome, phenylketonuria, fetal alcohol syndrome, Childhood-onset fluency disorder, Social (pragmatic) communication disorder, selective mutism, triad of impairments, childhood disintegrative disorder, autism spectrum disorder, Rett syndrome, restricted, repetitive patterns of behaviour, theory of mind, weak central coherence, hyperkinetic disorder, attention-deficit/hyperactivity disorder, behavioural inhibition model, specific learning disorder, dyslexia, phonological deficit theory, response to intervention, Auditory processing disorder, Developmental coordination disorder, dyspraxia, apraxia, tic, Tourette's disorder, coprolalia, Psychosis, dementia praecox, schizophrenia, four As, Ambivalence, first-rank symptoms, Positive symptoms, Negative symptoms, affective flattening, alogia, avolition, Brief psychotic disorder, Schizophreniform disorder, Delusional disorder, Schizoaffective disorder, acute and transient psychotic disorders, Folie à deux, social drift, dopamine hypothesis, neurodevelopmental hypothesis, schizophrenogenic mother, double bind, pseudomutuality, expressed emotion, Typical antipsychotics, Atypical antipsychotics, extrapyramidal symptoms, Akathisia, Tardive dyskinesia, neuroleptic malignant syndrome, mood disorder, unipolar, bipolar, bereavement exclusion, Major depressive disorder, Persistent depressive disorder, double depression, Disruptive mood dysregulation disorder, Premenstrual dysphoric disorder, mixed anxiety and depressive disorder, anaclitic depression, hospitalism, Masked depression, mood-congruent, seasonal affective disorder, Bipolar I disorder, Bipolar II disorder, Cyclothymic disorder, somatic syndrome, catecholamine hypothesis, permissive hypothesis, dexamethasone suppression test, shortened REM latency, kindling hypothesis, negative cognitive triad, automatic thoughts, lithium, Suicidal ideation, Lethality, parasuicide, Non-suicidal self-injury, psychache, Fear, Anxiety, neurosis, Separation anxiety disorder, specific phobia, Social anxiety disorder, Generalised anxiety disorder, panic attack, Panic disorder, Agoraphobia, fear of fear, behavioural inhibition, two-factor theory, preparedness, catastrophic misinterpretation, Catastrophic thinking, anxiety sensitivity, suffocation false alarm theory, triple vulnerability, signal anxiety, displacement, Obsessions, Compulsions, insight specifier, PANDAS, thought-action fusion, isolation of affect, undoing, reaction formation, Body dysmorphic disorder, Hoarding disorder, Trichotillomania, excoriation disorder, hysteria, dissociation, la belle indifférence, repression, conversion, primary gain, secondary gain, hidden observer, posttraumatic stress disorder, Acute stress disorder, Complex PTSD, post-traumatic growth, adjustment disorder, Prolonged grief disorder, reactive attachment disorder, disinhibited social engagement disorder, Dissociative identity disorder, host, sociocognitive model, Dissociative amnesia, Generalised amnesia, dissociative fugue, Ganser syndrome, Somatoform disorders, Somatic symptom disorder, Illness anxiety disorder, bodily distress disorder, functional neurological symptom disorder, functional blindness, Hoover's sign, factitious disorder, Munchausen syndrome, Malingering, feeding and eating disorders, Pica, rumination disorder, Avoidant/restrictive food intake disorder,

Anorexia nervosa, restricting type, binge-eating/purging type, lanugo, amenorrhoea, bulimia nervosa, binge eating, inappropriate compensatory behaviour, Russell's sign, Binge-eating disorder, other specified feeding or eating disorder, purging disorder, night eating syndrome, elimination disorders, Enuresis, Encopresis, sleep-wake disorders, Insomnia disorder, hypersomnolence disorder, narcolepsy, cataplexy, sleep paralysis, hypnagogic hallucinations, Parasomnias, NREM sleep arousal disorders, sleepwalking, sleep terrors, Nightmare disorder, REM sleep behaviour disorder, Restless legs syndrome, Kleine-Levin syndrome, sexual response cycle, refractory period, circular model, Sex, gender identity, sexual orientation, Androgyny, sexual dysfunction, Vaginismus, dyspareunia, performance anxiety, spectatoring, stop-start technique, squeeze technique, Sensate focus, paraphilia, paraphilic disorder, courtship disorders, algolagnic disorders, partialism, necrophilia, zoophilia, telephone scatologia, autogynephilia, lovemap, courtship disorder hypothesis, Gender dysphoria, gender identity disorder, gender incongruence, coercive sexual sadism disorder, Pathologisation, depathologisation, Kinsey scale, sexual orientation disturbance, ego-dystonic homosexuality, psychoactive substance, Intoxication, Withdrawal, Tolerance, Cross-tolerance, cross-dependence, Sensitisation, Physical dependence, Craving, harmful use, dependence syndrome, substance use disorder, gambling disorder, gaming disorder, mesolimbic dopamine system, opponent-process theory, gateway hypothesis, alcoholic hallucinosis, delirium tremens, Disulfiram, Formication, amotivational syndrome, neurocognitive disorders, major neurocognitive disorder, sundowning, asterixis, carphologia, subcortical dementia, Pseudodementia, Wernicke's encephalopathy, Korsakoff's syndrome, frontal lobe syndrome, personality disorder, ego-syntonic, categorical model, dimensional model, clusters, ideas of reference, magical thinking, Borderline personality disorder, borderline personality organisation, splitting, projective identification, self psychology, biosocial theory, Antisocial personality disorder, manie sans délire, moral insanity, Psychopathy Checklist-Revised, Psychopathy, fearlessness hypothesis, Histrionic personality disorder, Narcissistic personality disorder, Avoidant personality disorder, Dependent personality disorder, Obsessive-compulsive personality disorder, anal character, personality difficulty, anankastia, Alternative Model for Personality Disorders, Oppositional defiant disorder, Conduct disorder, with limited prosocial emotions, Intermittent explosive disorder, pyromania, kleptomania

From the past papers

1. Which of the following culture-bound syndromes is NOT reported in India? (Amity 2023)
   (a) Dhat Syndrome (b) Koro (c) Suchi-bai (d) Latah

   Answer: (d) Latah. Latah, an exaggerated startle with echolalia, echopraxia, automatic obedience and blurted obscenities, belongs to Malaysia and Indonesia. Dhat (anxiety over semen loss), koro (fear that the genitals are retracting, with epidemics in north-east India) and suchi-bai (obsessional cleanliness in Bengal) are all reported in India.

2. The diathesis-stress model of psychopathology suggests that mental disorders result from: (RML 2023)
   (a) Genetic factors only (b) Environmental factors only (c) A combination of genetic predisposition and environmental stressors (d) Neurobiological abnormalities only

   Answer: (c) A combination of genetic predisposition and environmental stressors. The diathesis, a genetic, biological or psychological vulnerability, produces disorder only when a stressor sets it off; Zubin and Spring (1977) called this the vulnerability-stress model. Every option ending in 'only' names a single cause, the one-dimensional thinking the model replaced.

3. A high false positive rate in diagnosis means: (IHBAS 2025)

   (a) Diagnosis is too strict (b) Normal people are diagnosed as disordered (c) Diagnosis has low reliability (d) Disordered people go undetected

   Answer: (b) Normal people are diagnosed as disordered. A false positive wrongly labels a healthy person as ill, which happens when criteria are too loose. Disordered people going undetected are false negatives, the result of over-strict criteria, and low reliability means clinicians disagree with each other, a separate problem.

4. Hypnagogic hallucinations most commonly occur in: (RML 2024)

   (a) Schizophrenia (b) Delirium (c) Drug dependence (d) Normal

   Answer: (d) Normal. Hallucinations while falling asleep (hypnagogic) or waking (hypnopompic) lie within normal experience and are commonest in healthy people; narcolepsy also features them. The hallucinations of schizophrenia are mainly auditory and occur in clear consciousness, while delirium and drugs bring visual and tactile ones.

5. Our current mood and feelings are best described by the term: (RML 2024)

   (a) Emotion (b) Mood (c) Affect (d) Feeling

   Answer: (c) Affect. Affect is the immediate, observed, moment-to-moment expression of emotion (the weather); mood is the sustained state the person reports (the climate). Emotion is the broader term for a brief reaction, and feeling is its experienced side.

6. Choose the correct arrangement of these stages of elevated mood based upon severity: i) Elation, ii) Exaltation, iii) Euphoria, iv) Ecstasy. (Amity 2023)

   (a) i, ii, iii & iv (b) iii, i, ii & iv (c) iv, iii, ii & i (d) None of the above

   Answer: (b) iii, i, ii & iv. Elevated mood rises from euphoria (mild; hypomania) through elation (moderate; mania) and exaltation (intense, with grandiosity; severe mania) to ecstasy (rapture; stuporous or delirious mania). Option (a) puts elation first, and (c) reverses the scale.

7. Which is NOT a core symptom to diagnose as autism in children as per DSM- 5? (IHBAS 2024)

   (a) Communication skills (b) Unusual repetitive behavior (c) Social reciprocity (d) I.Q. is below 70

   Answer: (d) I.Q. is below 70. DSM-5 autism rests on two core domains, social communication and reciprocity, and restricted, repetitive behaviour, so (a) to (c) are core. Intellectual impairment is only a specifier (with or without).

8. In which of the following situations would a diagnosis of "global developmental delay" be inappropriate? (RML 2022)

   (a) The patient is a child who is too young to fully manifest specific symptoms or to complete requisite assessments. (b) The patient, a 7-year-old boy, has a full-scale IQ of 65 and severe impairment in adaptive functioning. (c) The patient's scores on psychometric tests suggest intellectual disability, but there is insufficient information about the patient's adaptive functional skills. (d) The patient's impaired adaptive functioning suggests intellectual developmental disorder, but there is insufficient information about the level of cognitive impairment measured by standardized instruments.

   Answer: (b) The 7-year-old with full-scale IQ 65 and severe adaptive impairment. Global developmental delay is a provisional label for children under 5 whose severity cannot yet be reliably assessed, as in (a), (c) and (d); a testable 7-year-old who meets both criteria has intellectual disability.

9. Within attention deficit/hyperactivity disorder, which of the following is NOT classified as a problem relating to hyperactivity/impulsivity? (RML 2021)

   (a) Experiences difficulties in playing quietly. (b) Rather than sitting still, squirms in seat and fidgets. (c) Talks excessively. (d) Doesn't appear to listen when being told something or spoken to

   Answer: (d) Doesn't appear to listen when spoken to. Not seeming to listen when spoken to directly is one of the nine inattention items; not playing quietly, squirming and fidgeting, and talking excessively are hyperactivity-impulsivity items.

10. "First rank symptoms" of schizophrenia given by Schneider include all except (JEMAS 2025)

    (a) Depersonalization (b) Hallucinatory voice in the form of running commentary (c) Delusional perception (d) Thought broadcasting

    Answer: (a) Depersonalization. Running-commentary voices, delusional perception and thought broadcasting are all on Kurt Schneider's list; depersonalisation, like loose associations and delusions of self-reference, is not a first-rank symptom.

11. The ICD-10 duration criterion for schizophrenia requires symptoms to persist for at least: (IHBAS 2025)

    (a) 6 months (b) 2 weeks (c) 3 months (d) 1 month

    Answer: (d) 1 month. ICD-10 and ICD-11 require 1 month. Six months is the DSM-5-TR total (including 1 month of active-phase symptoms); 2 weeks is the onset window of acute and transient psychotic disorders.

12. High EE (Expressed Emotion) is known to cause: (IHBAS 2020)

    (a) Poor treatment outcome (b) Relapse of psychotic illness (c) Negative symptoms of schizophrenia (d) Violent behavior in schizophrenic patients

    Answer: (b) Relapse of psychotic illness. Brown, Birley and Wing (1972) and Vaughn and Leff (1976) showed that critical, hostile or over-involved relatives predict relapse after discharge; EE does not cause the illness, its negative symptoms or violence.

13. Which of the following statements concerning Bipolar disorder is INCORRECT? (RML 2022)

    (a) Bipolar-I Disorder includes mania (b) Bipolar-II Disorder includes hypomania (c) Bipolar-I Disorder includes hypomania (d) Bipolar-I Disorder includes periods of depression even if these do not reach the threshold for major depressive disorder

    Answer: (c) Bipolar-I Disorder includes hypomania. Bipolar I is defined by at least one manic episode; hypomania is the defining high of bipolar II and is optional in bipolar I. Bipolar I may include depressive periods, even below the major depressive threshold, so (a), (b) and (d) are correct statements.

14. A 12-year-old boy begins to have new episodes of temper outbursts that are out of proportion to the situation. Which of the following is NOT a diagnostic possibility in this patient? (RML 2022)

    (a) Disruptive Mood Dysregulation Disorder (b) Bipolar Disorder (c) Conduct Disorder (d) Attention Deficit Hyperactivity Disorder

    Answer: (a) Disruptive Mood Dysregulation Disorder. DMDD requires onset before age 10 (diagnosis between 6 and 18), so outbursts that begin at 12 rule it out. Bipolar disorder, conduct disorder and ADHD can all begin or present at this age.

15. Suicidal risk is assessed to be highest when the following criteria are met in comparison to others, the 'time' factor being the same for all (JEMAS 2025)

    (a) High lethality, specific plan, frequent suicidal ideas, family history (b) Disrupted biological functioning, no recent attempt, no suicidal ideation (c) Adequate biological functioning, no previous attempt, moderate intent (d) Frequent suicidal ideation, recent suicide attempt, low lethality, absence of family history

    Answer: (a). Risk is highest when several markers converge: a highly lethal method, a specific plan, frequent ideation and a family history of suicide. Options (b) and (c) lack ideation or attempts, and (d) has low lethality and no family history.

16. The following defense mechanisms are used in obsessive-compulsive disorder: (RML 2024)

    (a) Isolation (b) Undoing (c) Reaction formation (d) Displacement

    Answer: (b) Undoing. Isolation, undoing and reaction formation are all classic OCD defences, but undoing, an act that symbolically cancels a forbidden thought, is the most characteristic and is the key. Displacement is the defence of phobias (Little Hans).

17. Generalized Anxiety Disorder (GAD) is a pervasive condition in which the sufferer experiences: (IHBAS 2024)

    (a) Continued apprehension about unspecified things (b) Fear of fear (c) Continual flashbacks to past events (d) A desire to check that the environment is safe

    Answer: (a) Continued apprehension about unspecified things. GAD is excessive, hard-to-control, free-floating worry about many matters for at least 6 months. Fear of fear marks panic disorder and agoraphobia, flashbacks PTSD and checking OCD.

18. A client with OCD washes their hands excessively to neutralize the anxiety caused by intrusive thoughts of contamination. This hand-washing behavior is BEST conceptualized as a(n): (Real PYQ 2024)

    (a) Conditioned stimulus (b) Unconditioned response (c) Avoidance response / Safety behavior (d) Operant behavior maintained by positive reinforcement

    Answer: (c) Avoidance response / Safety behavior. In Mowrer's two-factor account the washing removes anxiety, so it is negatively reinforced. The contamination thought is the conditioned stimulus, an unconditioned response is an unlearned reflex, and nothing pleasant is added, so it is not positive reinforcement.

19. The term 'la Bella indifference' seen in hysteria was first used by: (RML 2024)

    (a) Sigmund Freud (b) Anna Freud (c) Pierre Janet (d) Hippocrates

    Answer: (c) Pierre Janet. The phrase for the calm unconcern of some conversion patients is usually credited to Janet, who explained hysteria as dissociation. Freud explained hysteria by repression and conversion, Anna Freud catalogued the defences, and Hippocrates blamed a wandering womb.

20. When an individual has dissociative amnesia it is usually associated with several types of memory disturbances. In contrast, in generalized amnesia: (RML 2022)

    (a) There is a failure of recall of the entire past week (b) There is a failure of recall that encompasses the person's entire life (c) There is a failure of recall that encompasses the person's childhood (d) There is a failure of recall for traumatic events

    Answer: (b) Failure of recall that encompasses the person's entire life. Generalised amnesia covers the whole life history, often with identity. Loss of a set period such as a week is localised amnesia, loss of one category of information is systematised, and loss for traumatic events describes dissociative amnesia in general.

21. The difference between malingering and factitious disorder lies in: (IHBAS 2025)

    (a) The presence or absence of external rewards (b) Whether the individual believes the symptoms are real (c) The severity of symptoms (d) Age of onset

    Answer: (a) The presence or absence of external rewards. Both deliberately falsify symptoms and know they are false; the malingerer seeks an external incentive such as money or escape from duty, whereas factitious disorder persists without obvious external reward, to occupy the sick role.

22. Unlike people with anorexia, individuals with bulimia (RML 2022)

    (a) Often purge themselves to stay thin (b) Are obsessed with being too fat (c) Can do damage to their health (d) May appear normal in weight

    Answer: (d) May appear normal in weight. Significantly low body weight is a criterion of anorexia nervosa; people with bulimia are usually of normal weight or overweight. Purging (the binge-eating/purging type of anorexia), fear of fatness and damage to health occur in both disorders, so they do not distinguish them.

23. In narcolepsy, which of the following is most characteristically present? (IHBAS 2025)

    (a) Both cataplexy and catalepsy equally (b) Cataplexy, sudden loss of muscle tone triggered by emotions (c) Neither cataplexy nor catalepsy (d) Catalepsy, maintenance of imposed posture with waxy flexibility

    Answer: (b) Cataplexy. Narcolepsy type 1 shows brief, sudden, bilateral loss of muscle tone with preserved consciousness, triggered by laughter or joking, from loss of hypocretin (orexin) neurons. Catalepsy, a held posture with waxy flexibility, belongs to catatonia.

24. Lack of bowel control in children is known as: (RML 2024)

    (a) Encopresis (b) Analism (c) Enuresis (d) Anorexia

    Answer: (a) Encopresis. Encopresis is repeated passage of faeces into inappropriate places at least once a month for 3 months from age 4. Enuresis is the urinary counterpart (from age 5), anorexia is an eating disorder, and analism is not a recognised elimination disorder.

25. Premature ejaculation is most commonly associated with: (RML 2024)

    (a) Diabetes mellitus (b) Masturbation (c) Personality disorder (d) Anxiety state

    Answer: (d) Anxiety state. Premature ejaculation is mostly psychogenic and most tied to performance anxiety; it is commoner in men with anxiety disorders. Masturbation does not cause it (a myth), diabetes is the classic link of erectile disorder, and personality disorder has no special association.

26. Autogynephilia is defined by: (RML 2022)

    (a) Recurrent, and intense sexual attraction to prepubescent children (b) Recurrent, and intense sexual attraction to thoughts, images, or thoughts about oneself as a woman (c) Recurrent, and intense sexual attraction to oneself (d) Recurrent, and intense sexual attraction to being humiliated

    Answer: (b) Attraction to thoughts or images of oneself as a woman. Ray Blanchard coined the term in 1989, and DSM-5-TR uses it as a specifier of transvestic disorder. Attraction to prepubescent children is pedophilia and to being humiliated is masochism; mere attraction to oneself is not autogynephilia.

27. When gender incongruence moved out of the mental disorders chapter in ICD-11, which of these stayed in Chapter 6 (Mental, behavioural or neurodevelopmental disorders)? (NIMHANS 2026 recall)

    (a) Sexual dysfunctions (b) Paraphilic disorders (c) Sexual pain disorders (d) Gender incongruence of childhood

    Answer: (b) Paraphilic disorders. ICD-11 moved gender incongruence (of childhood as well as of adolescence or adulthood), the sexual dysfunctions and the sexual pain disorders into Chapter 17, Conditions related to sexual health; only the paraphilic disorders, now centred on non-consent, stayed in Chapter 6.

28. A patient presents with severe memory deficits, confabulation, and nystagmus, with a history of chronic alcohol use. This clinical picture is classic for: (Real PYQ 2024)

    (a) Alzheimer's Disease (b) Wernicke-Korsakoff Syndrome (c) Vascular Dementia (d) Frontotemporal Dementia

    Answer: (b) Wernicke-Korsakoff Syndrome. Thiamine deficiency in chronic drinkers damages the mammillary bodies and dorsomedial thalamus: nystagmus belongs to Wernicke's encephalopathy, amnesia with confabulation to Korsakoff's syndrome. Alzheimer's, vascular and frontotemporal dementia are not caused by alcohol and do not produce this triad.

29. Which of the following substances is NOT a natural alkaloid compound? (Amity 2023)

    (a) Morphine (b) Heroin (c) Codeine (d) Papaverine

    Answer: (b) Heroin. Heroin (diacetylmorphine) is made by chemically modifying morphine, so it is semi-synthetic. Morphine, codeine and papaverine all occur naturally in opium.

30. A key of distinguishing feature of dementia vs. delirium is : (IHBAS 2025)

    (a) Delusions in dementia but not delirium (b) Onset gradual in delirium vs. acute in dementia (c) Hallucinations in dementia but not delirium (d) Impaired memory in dementia vs. fluctuating consciousness in delirium

    Answer: (d) Impaired memory in dementia, fluctuating consciousness in delirium. Delirium is an acute, fluctuating disturbance of attention and awareness; dementia is an insidious decline in memory and cognition in clear consciousness. Delusions and hallucinations occur in both, and option (b) reverses the two onsets.

31. Borderline Personality Disorder is most characteristically associated with which primary defense mechanism? (Real PYQ 2025)

    (a) Repression (b) Splitting (c) Sublimation (d) Rationalization

    Answer: (b) Splitting. Splitting, seeing self and others as all good or all bad, is the primitive defence Kernberg placed at the centre of borderline personality organisation. Repression is the classic neurotic defence, sublimation a mature one and rationalisation a plausible excuse; none defines BPD.

32. Identify the 'odd one out' from the groups: (Amity 2023)

    (a) Paranoid personality disorder (b) Schizoid personality disorder (c) Schizotypal personality disorder (d) Borderline personality disorder

    Answer: (d) Borderline personality disorder. Paranoid, schizoid and schizotypal form Cluster A (odd or eccentric); borderline belongs to Cluster B (dramatic, emotional or erratic).

33. A client reports a persistent pattern of seeking admiration, but their behavior is characterized by excessive emotionality, attention-seeking, and seductive behavior, rather than grandiosity. This presentation is more characteristic of: (Real PYQ 2024)

    (a) Narcissistic Personality Disorder (b) Histrionic Personality Disorder (c) Borderline Personality Disorder (d) Dependent Personality Disorder

    Answer: (b) Histrionic Personality Disorder. Excessive emotionality, attention seeking and seductive behaviour define histrionic PD. Narcissistic PD also seeks admiration but rests on grandiosity and lack of empathy; borderline centres on instability and fear of abandonment; dependent on a need to be taken care of.

High-yield sheet

The facts this chapter's questions ask most, most-asked first. Last read before the exam.

1 Positive symptoms reflect excess dopamine D2 activity in the **mesolimbic** pathway (ventral tegmental area to nucleus accumbens); typical antipsychotics block D2, and blockade in the **nigrostriatal** pathway causes extrapyramidal symptoms. ASKED 13 TIMES

2 **Kurt Schneider's first-rank symptoms**: audible thoughts, voices arguing or commenting, thought insertion, withdrawal and broadcasting, made feelings, impulses and acts, somatic passivity and delusional perception; depersonalisation, loose associations and delusions of self-reference are not first-rank. ASKED 10 TIMES

3 **DSM-5-TR** schizophrenia needs 6 months of continuous signs including 1 month of active-phase symptoms; **ICD-10 and ICD-11** need 1 month. ASKED 10 TIMES

4 **Borderline personality disorder** is a pervasive pattern of instability of interpersonal relationships, self-image and affects, with marked impulsivity; DSM-5-TR requires five or more of nine criteria, including frantic efforts to avoid abandonment, identity disturbance, recurrent self-harm and chronic emptiness. ASKED 9 TIMES

5 **Positive symptoms** are the presence of something abnormal (delusions, hallucinations, disorganisation); **negative symptoms** are the absence of something normal (affective flattening, alogia, avolition, anhedonia, asociality). ASKED 8 TIMES

6 **Aaron Beck**'s cognitive theory of depression (1967): depression is maintained by the **negative cognitive triad**, negative views of the **self, the world (experience) and the future**, arising from negative schemas, automatic thoughts and cognitive distortions. ASKED 8 TIMES

7 **Korsakoff's syndrome** (anterograde and retrograde amnesia with confabulation) follows Wernicke's encephalopathy (confusion, ataxia, ophthalmoplegia with nystagmus); both result from **thiamine (vitamin B1) deficiency**, classically in chronic alcoholism, with lesions in the mammillary bodies and dorsomedial thalamus. ASKED 8 TIMES

8 **Persistent depressive disorder (dysthymia)**: depressed mood most of the day, more days than not, for at least **2 years in adults or 1 year in children and adolescents**, with 2 or more of 6 symptoms and never symptom-free for more than 2 months; less severe than major depression but longer. ASKED 7 TIMES

9 **Bipolar I** needs at least one **manic episode** (1 week or any duration if hospitalised, with marked impairment, hospitalisation or psychotic features); **bipolar II** needs at least one **hypomanic episode** (4 days, no marked impairment or psychosis) plus at least one major depressive episode and never a full mania. ASKED 7 TIMES

10 DSM-5-TR **PTSD** needs symptoms in all **four clusters** (intrusion, avoidance, negative alterations in cognitions and mood, alterations in arousal and reactivity) lasting **more than 1 month**; **acute stress disorder** covers 3 days to 1 month after the trauma. ASKED 7 TIMES

11 **Dissociative identity disorder** (formerly multiple personality disorder) requires two or more distinct personality states, or possession, plus recurrent amnesic gaps; the identity carrying the real name is the **host**, not schizophrenia's 'split'. ASKED 7 TIMES

12 **Cultural concepts of distress** (DSM-5's replacement for culture-bound syndromes): dhat (Indian subcontinent; anxiety over semen loss; named by N. N. Wig, 1960), koro (fear that the genitals are retracting into the body, causing death), pibloktoq (Inuit dissociative fits) and latah (Malaysia and Indonesia; exaggerated startle, echolalia and obscenities; not reported in India). ASKED 6 TIMES

13 Flight of ideas (mania), loosening of associations or derailment (schizophrenia), tangentiality and circumstantiality are disorders of the **form** of thought, not its content; circumstantial speech reaches the goal, tangential speech never does, and **word salad** is the most severe formal thought disorder. ASKED 6 TIMES

14 **Capgras syndrome** (Joseph Capgras, 1923) is the delusion that a familiar person, usually a spouse or relative, has been replaced by an identical impostor; in **Fregoli syndrome** strangers are a familiar person in disguise. ASKED 6 TIMES

15 Antipsychotic movement effects appear in order: acute dystonia (hours to days), akathisia (days to weeks), parkinsonism (weeks), **tardive dyskinesia** (months to years; orofacial, often irreversible); akathisia and tardive dyskinesia are secondary, drug-induced movement disorders. ASKED 6 TIMES

16 **Folie à deux** (Lasègue and Falret, 1877) is a delusion transmitted from a dominant psychotic primary case to a passive partner in a close relationship; ICD-10 F24 induced delusional disorder, DSM-IV shared psychotic disorder; ICD-11 describes it under 6A24 delusional disorder. ASKED 6 TIMES

17 A **major depressive episode** needs **5 or more of 9 symptoms** in the same **2-week** period, at least one being depressed mood or loss of interest or pleasure; phobias and euphoric mood are not among them. ASKED 6 TIMES

18 **La belle indifférence**, calm unconcern about a dramatic symptom, is the classic but non-diagnostic sign of **conversion disorder** (functional neurological symptom disorder); the term is usually credited to **Pierre Janet**. ASKED 6 TIMES

19 **Factitious disorder** (Munchausen syndrome, Asher 1951; imposed on another, Meadow 1977) and **malingering** both feign illness on purpose; they differ in the **absence or presence of external rewards**, and malingering is not a mental disorder. ASKED 6 TIMES

20 The **diathesis-stress model** explains disorder as a predisposition (a genetic, biological or psychological diathesis) set off by stress; Meehl applied it to schizophrenia in 1962, and Zubin and Spring named it the vulnerability-stress model in 1977. ASKED 5 TIMES

21 **Dissociative amnesia** is inability to recall important, usually traumatic autobiographical information beyond ordinary forgetting; **generalised** amnesia covers the whole life history, and **fugue** (purposeful travel with loss of identity) is now a specifier. ASKED 5 TIMES

22 **Bulimia nervosa** needs recurrent binges with loss of control plus compensatory behaviour (vomiting, laxatives, diuretics, fasting, exercise), both at least **once a week for 3 months**; **binge-eating disorder** needs binges once a week for 3 months with marked distress, three of five features and no regular compensation. ASKED 5 TIMES

23 **Cotard syndrome** (Jules Cotard, 1880) is a nihilistic delusion that one is dead, does not exist, or has lost organs or blood; nihilistic delusions are characteristic of severe psychotic depression. ASKED 4 TIMES

24 Depression, the **common cold of psychiatry**, is characterised by persistent sadness and loss of interest; it is widespread and increasing, often triggered by stress, linked to suicide risk by severity, and about **twice as common in women** as in men. ASKED 4 TIMES

25 OCD obsessions are **ego-dystonic** (intrusive, unwanted, inconsistent with the self-concept); obsessive-compulsive personality disorder is **ego-syntonic** perfectionism without true obsessions or compulsions. ASKED 4 TIMES

26 **Generalised anxiety disorder** is excessive, hard-to-control worry about many events, more days than not for at least **6 months**, with 3 of 6 symptoms (restlessness, fatigue, poor concentration, irritability, muscle tension, sleep disturbance; 1 in children): continued apprehension about unspecified things. ASKED 4 TIMES

27 **Undoing** is the most characteristic OCD defence, alongside isolation of affect and reaction formation; **repression** is the defence of hysteria and **displacement** that of phobias. ASKED 4 TIMES

28 **Depersonalisation** is feeling detached from or unreal in oneself ('not my normal self'); **derealisation** is the surroundings feeling unreal or dreamlike; reality testing stays intact. ASKED 4 TIMES

29 DSM-5-TR **anorexia nervosa**: restriction leading to significantly low body weight, intense fear of weight gain (or behaviour blocking gain) and disturbed experience of weight or shape; **amenorrhoea was dropped** and no BMI cut-off is required; the restricting type has no recurrent bingeing or purging in the last 3 months. ASKED 4 TIMES

30 Anorexia nervosa is the **most lethal eating disorder** (crude mortality about 5 per cent per decade in DSM-5-TR), with lanugo, bradycardia, hypotension, hypothermia, amenorrhoea and hypokalaemia, and the trait of **perfectionism**. ASKED 4 TIMES

CHAPTER 9

Research Methodology

IN THIS CHAPTER

9.1 ●● Science and the Research Process: Problems and Hypotheses: Distinguish science from other ways of knowing, inductive from deductive reasoning and the main types of research, and turn a research problem and literature review into a well-formed hypothesis of the right type.

9.2 ●● Variables, Experimental Control and the Validity of Research: Identify and classify the variables in a study, define them operationally, choose the right technique of control, recognise participant and experimenter artefacts and what blinding controls, and judge a study's statistical conclusion, internal, construct and external validity and the threats to each.

9.3 ●● Sampling: Populations, Probability and Non-Probability Methods: Define population, sample, parameter and statistic, tell probability from non-probability sampling, recognise every major sampling method from a description (simple random, systematic, stratified, cluster and multistage, quota, purposive, convenience, snowball), and distinguish sampling error from bias.

9.4 ●● Research Designs: Experimental, Single-Case and Non-Experimental: Distinguish experimental, quasi-experimental, single-case, non-experimental and developmental designs, and judge which conclusions each one licenses.

9.5 ●● Clinical Epidemiology, Randomised Trials and Evidence Synthesis: Calculate and interpret prevalence, incidence, relative risk, odds ratio, NNT, sensitivity, specificity and predictive values, tell the epidemiological designs and trial analyses apart, and explain meta-analysis and the reporting guidelines.

9.6 ● Methods of Data Collection and Qualitative Research: Choose and critique the main tools of data collection (observation, interview, questionnaire and schedule, content analysis, sociometry), place a study in its research paradigm, and tell the qualitative traditions, their analysis and their quality criteria apart.

9.7 ● Research Ethics and Writing the Research Report: Trace research ethics from the abuses that produced it to its codes and core principles (informed consent, protection from harm, confidentiality, deception and debriefing, the 3Rs, honest publication), and describe the structure and style of an APA research report and of a research proposal.

9.8 ●●● Measurement Scales, Organising Data and Descriptive Statistics: Classify any variable by Stevens's four levels of measurement, process, tabulate and graph raw data, choose and compute averages and measures of variability, read skewness and kurtosis, and use the normal curve to convert scores through z into percentiles and standard scores.

9.9 ●● Sampling Distributions and the Logic of Hypothesis Testing: Explain how psychologists reason from samples to populations: the sampling distribution and standard error, the central limit theorem, confidence intervals, the steps of a significance test, the p-value, Type I and Type II errors, power and effect size.

9.10 ●●● Comparing Groups: t-Tests, ANOVA and Non-Parametric Tests: Choose the right test for comparing groups from the design and the level of measurement (t-tests, one-way and factorial ANOVA with post hoc tests, ANCOVA and MANOVA, chi-square and the rank tests) and work out its degrees of freedom.

9.11 ●● Correlation, Regression and Factor Analysis: Choose and interpret the right correlation coefficient, read and use a regression equation, and decide how many factors to retain and how to rotate them in a factor analysis.

Weightage. 320 of the 3180 questions mapped in this book (10.1%) test this chapter, 175 of them from real past papers. 2 of its 11 sections are High yield.

Berlin, 1904. A retired schoolteacher named Wilhelm von Osten puts a question to his horse: what is three plus five? Hans taps his hoof eight times and stops. Von Osten had spent years teaching him, and Hans now seemed able to add, subtract, multiply and divide, to work with fractions, tell the time and answer questions about German words, all by tapping. He answered strangers as well as his owner, so deliberate fraud seemed unlikely. In September 1904 a commission that included the psychologist Carl Stumpf watched him perform and found no trick. The newspapers called him Clever Hans.

Stumpf passed the puzzle to his student Oskar Pfungst, who treated it as a research problem rather than a marvel. Pfungst set out rival explanations (trickery, genuine reasoning, cues from the people present) and tested them by changing one condition at a time. When the questioner knew the answer, Hans was nearly always right; when the questioner did not, Hans was nearly always wrong. With blinkers that hid the questioner, or with the questioner standing far off, Hans failed again. The horse was not counting; he was watching. Pfungst found the signal: once the question was put, the questioner bent forward slightly to watch the hoof, and Hans began to tap; when the right count was reached, the questioner's head lifted a fraction and the tension left his body. Hans stopped at that release. None of the questioners, von Osten included, knew they were doing it.

Pfungst published his study in 1907. He even took the horse's part himself in the laboratory, tapping out answers for human questioners, and found that they gave him the same small, unwitting signals. The lesson has outlived the horse: what an observer expects can leak into the very data meant to test it. This chapter is about the methods science has built to stop that from happening.

9.1 Science and the Research Process: Problems and Hypotheses

MEDIUM YIELD · 21 QS

Why should you believe a psychologist's claim? Not because the psychologist is famous or certain, but to the degree that the claim was reached by a method that could have shown it to be wrong. This section describes that method, the reasoning it uses, the kinds of research it produces, and the first two steps of every study: finding a problem and stating a hypothesis.

Ways of knowing

In "The Fixation of Belief" (1877) the American philosopher **Charles Sanders Peirce** (1839 to 1914) described four ways in which people settle what they believe, and **Fred N. Kerlinger** (1910 to 1991) made them standard in *Foundations of Behavioral Research* (1964). By the **method of tenacity** you hold a belief because you always have; by the **method of authority**, because a respected source says so; by the **a priori method**, because it seems self-evidently reasonable (intuition). Only the method of science settles belief by public observation that anyone can repeat, so only science corrects its own mistakes.

Older traditions stand behind the list (Table 9.1). **Socrates** (about 470 to 399 BCE) used the **elenchus**, a question-and-answer cross-examination that draws out a person's beliefs until they contradict one another (the Socratic method). **Aristotle** (384 to 322 BCE) observed and

classified nature; **René Descartes** (1596 to 1650) trusted reason and intuition (rationalism); **Galileo Galilei** (1564 to 1642) experimented; and **Francis Bacon** (1561 to 1626) argued in *Novum Organum* (1620) for building knowledge inductively from observations.

Table 9.1 Ways of knowing. Only the last corrects itself.

Way of knowing	Linked with	Belief is settled by
Elenchus	Socrates	Questioning that exposes contradictions
Observation, classification	Aristotle	Watching and sorting nature
Intuition, reason	Descartes	Thinking alone
Experiment, induction	Galileo; Bacon	Varying conditions; generalising
Tenacity	Peirce	Habit and repetition
Authority	Peirce	An expert or text
A priori	Peirce	What "stands to reason"
Science	Peirce	Public, repeatable observation

What makes research scientific

Kerlinger defined scientific research as "systematic, controlled, empirical and critical investigation of hypothetical propositions about the presumed relations among natural phenomena". Unpacked, that gives its characteristics.

- **Empiricism**: claims are settled by observation, not by argument or rank.
- Objectivity through systematic, controlled observation: the result should not depend on who observes it.
- **Replicability** and reproducibility: the study can be repeated, and gives the same results when it is. When the Open Science Collaboration (2015) repeated 100 published psychology studies, about 97 percent of the originals but only about 36 percent of the repetitions were significant.
- **Parsimony**: prefer the simpler of two explanations that fit equally well. This is Occam's razor, after **William of Ockham** (about 1287 to 1347); in animal psychology it is Lloyd Morgan's canon (1894), after **Conwy Lloyd Morgan** (1852 to 1936): never explain an action by a higher mental faculty if a lower one will do.
- **Falsifiability**: **Karl Popper** (1902 to 1994), in *Logik der Forschung* (1934), noted that a thousand white swans cannot prove "all swans are white" but one black swan disproves it. A statement is scientific only if some possible observation could show it false; psychoanalysis, which explains any behaviour after the event, was his example of a theory that risks nothing.
- Determinism and tentativeness: behaviour has natural causes and so is lawful, yet every conclusion is provisional.

A. K. Singh adds three assumptions of science (nature is lawful, behaviour is determined, nature is understandable) and six attitudes of the scientist: open-minded, tolerant of uncertainty, aware of being fallible, cautious (a finding is evidence, never "proof"), sceptical and ethical. The goals of science are to describe, explain, predict and control behaviour.

> EXAM TRAP
>
> "A characteristic of good research" is **reproducibility of results**. The distractors are the opposites of the real criteria: good research has clear objectives, unbiased findings, adequate scope and ethical conduct.

Science builds with the terms in Table 9.2. A hypothesis is a guess about one relation, a law a confirmed regularity, a theory an explanatory system; "just a theory" misuses the word.

Table 9.2 The building blocks of scientific knowledge.

Term	Meaning	Example
Fact	An observation others can verify	Pavlov's dogs salivated before feeding
Concept	An abstraction naming a class of observations	Reward, aggression
Construct	A concept invented for science, inferred and not directly observed	Anxiety, intelligence
Theory	Interrelated constructs and propositions that explain and predict (Kerlinger)	Beck's cognitive theory of depression
Law	A highly confirmed, usually narrow regularity	Weber's law
Model	A simplified representation or analogy	Atkinson and Shiffrin's memory model

Popper said how science ought to work; **Thomas S. Kuhn** (1922 to 1996) described how it does, in *The Structure of Scientific Revolutions* (1962). Scientists usually do normal science inside a **paradigm**, a shared package of theory, methods and assumptions. Anomalies pile up, crisis follows, and a scientific revolution installs a new paradigm (a paradigm shift), as when the cognitive revolution displaced behaviourism.

Induction, deduction and abduction ●●●

Induction runs from specific observations to broad generalisations, bottom-up. You interview twenty hostel students in exam week, notice that the poorest sleepers worry most, and generalise. Induction builds theory, so it belongs to the theory-development stage and is typical of exploratory and qualitative research. Its weakness, noted by **David Hume** (1711 to 1776), is that no number of cases guarantees the rule.

Deduction runs top-down, from theory to hypothesis to the observation that tests it: if the cognitive theory of depression is right, depressed patients should recall more negative words, so you test them. It is typical of quantitative, experimental research. **Abduction**, Peirce's term, is inference to the best explanation: from a surprising fact to the hypothesis that would best account for it, as when Oskar Pfungst guessed that Clever Hans was reading his questioners (see the chapter opening).

The **hypothetico-deductive method** joins them: propose a hypothesis, deduce a prediction, test it, then keep, revise or drop the hypothesis. **Clark L. Hull** (1884 to 1952) built *Principles of Behavior* (1943) this way. Figure 9.1 shows the cycle.

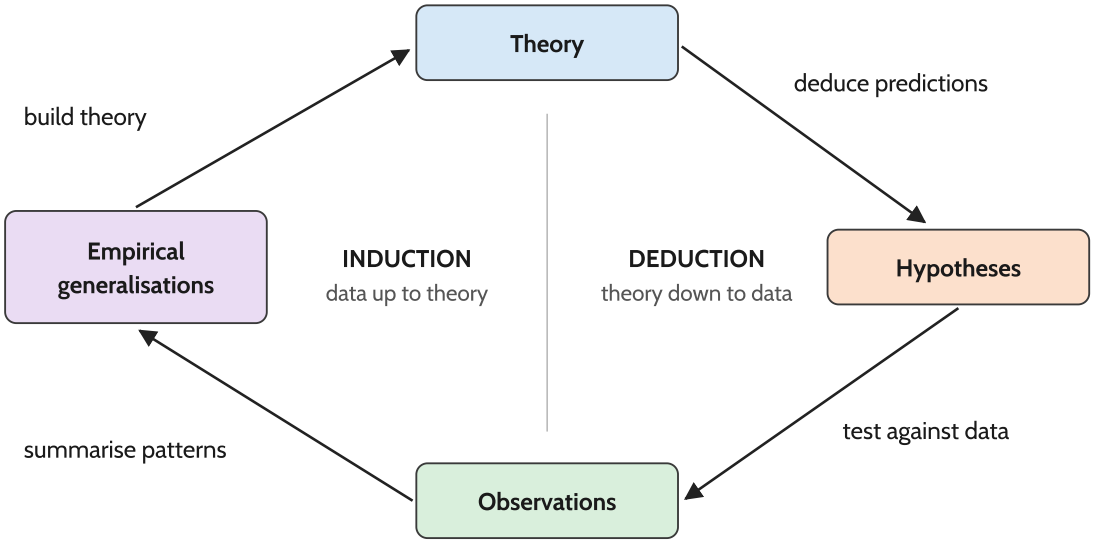

Figure 9.1 The research cycle, after Walter Wallace's "wheel of science" (1971). Deduction carries theory down to data (right); induction carries data back up to theory (left), at the theory-development stage.

EXAM TRAP

"Starts with a theory and tests it with observations" is **deductive reasoning**; the hypothetico-deductive method (the whole conjecture-and-test cycle) is the distractor. "Gathers data, finds patterns, then generates a theory", "specific observations to broader generalisations" and "associated with qualitative research" are all **inductive reasoning**, used at the **theory-development** stage, not in formulating hypotheses, collecting data or reviewing literature. Experimental design and quantitative analysis go with deduction; qualitative analysis is a way of handling data, not a direction of reasoning.

Methodology, methods and the steps of research ●●

Research methods are the techniques for collecting and analysing data: an interview, an experiment, a t-test. **Research methodology** is the wider study of the logic behind choosing them: why this method suits this problem, and what its results can show. As **C. R. Kothari** puts it in *Research Methodology: Methods and Techniques*, methods are part of methodology, so methodology has the wider scope.

Research advances theory, informs government policy, guides practice (which therapy works) and, in business, provides insights for decision-making, a role broader than profit and far from merely academic. Kothari's criteria of good research are a clearly defined purpose, a procedure described well enough to be repeated, a design planned for objective results, frankly reported flaws, adequate analysis, and conclusions confined to what the data justify.

Singh's eight steps are: (1) identifying the problem, with the literature review; (2) formulating hypotheses; (3) identifying, manipulating and controlling variables (9.2); (4) the research design (9.4); (5) devices for observation and measurement (9.6); (6) summarising results (9.8); (7) statistical analysis (9.9 to 9.11); and (8) drawing conclusions. Kothari's flow runs: problem, literature survey, hypotheses, research design, sample design (9.3), data collection, execution, analysis, hypothesis testing, generalisation and interpretation, and the report (9.7), which raises the next problem.

Types of research ●●●

One study has a place on each axis of Table 9.3: a randomised trial of yoga for anxiety in a Bengaluru hospital is applied, explanatory, quantitative, experimental, field-based and longitudinal.

Table 9.3 Ways of classifying research.

Basis	Types	What separates them
Purpose (also Kothari)	Basic (pure, fundamental) versus applied	Knowledge for its own sake versus a practical problem
Depth	Exploratory, descriptive, explanatory	Generates hypotheses; describes; tests causes
Kothari	Descriptive versus analytical	Describes things as they are versus critically evaluates available facts
Kothari	Quantitative versus qualitative	Amounts versus qualities and meanings (9.6)
Kothari	Conceptual versus empirical	Abstract ideas versus observation or experiment
Kerlinger	Experimental versus non-experimental	Independent variable manipulated versus measured
Setting	Laboratory versus field	Control versus realism
Time	Cross-sectional versus longitudinal	One occasion versus followed over time (9.4)
Focus	Nomothetic versus idiographic	General laws versus the unique individual

Basic research seeks knowledge for its own sake; **applied research** solves a practical problem. **Action research**, an applied form, was introduced by **Kurt Lewin** (1890 to 1947), often called the father of modern social psychology, in 1946. It is research done by or with the people in a setting to solve a specific local problem there, in cycles now drawn as plan, act, observe and reflect: a ward team studies why patients miss follow-up visits, changes the reminders and measures again (participatory action research is in 9.6). Evaluation research judges whether a programme works; **Michael Scriven** (1967) separated formative evaluation, which improves a running programme, from summative evaluation, which judges it at the end.

Exploratory research probes a poorly understood area without firm hypotheses, so it is hypothesis-generating. **Descriptive research** observes and describes what exists without manipulating variables. **Explanatory research** (causal research) tests hypotheses about why, most strongly by experiment, a variable-manipulating study. A case study is a design for one case (9.4).

Wilhelm Windelband (1848 to 1915) in 1894 separated sciences seeking general laws from those seeking to understand the unique case. The **nomothetic approach** (Greek *nomos*, law) finds laws across many people through experimental and quasi-experimental methods; the **idiographic approach** (*idios*, one's own) studies one person in depth through case studies, ethnography and other qualitative methods. **Gordon W. Allport** (1897 to 1967) brought the pair into personality psychology in 1937 and preferred the idiographic approach (Chapter 4).

> **EXAM TRAP**
>
> Both experimental and quasi-experimental methods are nomothetic; when both are offered, the expected answer is the **experimental method**. Qualitative methods and ethnography are idiographic. Allport preferred the **idiographic** approach; factorial design (9.4) and chi-square (9.10) are a design and a test, not approaches. Action research is "**research carried out to solve a specific problem**": it is applied but need not be longitudinal, so "all of the above" fails. Lewin founded action research, not psychoanalysis, psychophysics or functionalism (Chapter 1).

The research problem

A **research problem** is a felt difficulty sharpened until data can answer it: "students seem stressed" becomes "Is sleep duration related to test anxiety in first-year students?" Kerlinger's criteria are that it relates two or more variables, is stated clearly as a question, and implies empirical testing. A good research question is clear, answerable, connected to earlier work and worth answering.

Problems come from experience, theory and earlier findings, which reveal a gap, contradictory results, or an isolated unexplained fact, such as the apparent movement that **Max Wertheimer** (1880 to 1943) studied as the phi phenomenon, which led to Gestalt psychology. **Pauline V. Young** (1956) named documentary sources, key informants and library sources; Baker and Keller add programmed activities, research proposals, government policies and social values.

A problem should be novel, significant, feasible (time, money, participants), within the researcher's interest and competence, supported by available data and instruments, and ethical. It must also be solvable by empirical observation. An **unsolvable problem** ("Is there a divinity that shapes our ends?") is metaphysical: unstructured, with terms that cannot be operationally defined and data that cannot be collected. Singh's steps in formulating a problem are: choose a broad area, divide it, select a subarea, raise questions, turn them into objectives, assess their feasibility, and double-check.

Reviewing the literature ●●

The **literature** is the body of prior work related to a research problem; it is not an archive (a store of historical records) or a repository (a storage place for documents and data). The **review of literature** evaluates it. Its primary purpose is to provide a theoretical framework, from which hypotheses are deduced; it also avoids duplication, locates gaps, refines the problem and suggests variables, designs and tools.

Singh's five types are primary (traces a topic's development), content (links a study to related work), methodological (compares methods), integrative or systematic (combines empirical results; see 9.5) and theoretical (compares theories). Sources include peer-reviewed journals, books, review articles, abstracts (*Psychological Abstracts*, begun in 1927, now PsycINFO), indexes, PubMed, dissertations (Shodhganga in India), the internet and **grey literature**, material without an ISBN or ISSN such as theses and government reports. A review is written as critical synthesis organised by theme, with each source annotated, not as a string of summaries.

EXAM TRAP

The primary objective of a literature review is to **provide a theoretical framework.** Establishing the researcher's credibility is a secondary purpose Singh mentions; summarising your own findings belongs in the results, and ignoring earlier research is the opposite of a review.

The hypothesis: meaning, criteria and functions ●

The problem asks; the **hypothesis** answers. Kerlinger (1973) called it "a conjectural statement of the relation between two or more variables", always declarative: a tentative, testable statement. "Is sleep related to test anxiety?" is a problem, interrogative and not directly testable; "Students who sleep under six hours will report higher test anxiety" is a hypothesis, declarative and testable. Hypotheses can be recast as "if (antecedent), then (consequent)".

A good hypothesis is testable and falsifiable, conceptually clear, specific, consistent with known facts, parsimonious, and related to theory and to available techniques. **William J. Goode** and **Paul K. Hatt** (*Methods in Social Research*, 1952) named three difficulties in formulating one: no clear theoretical framework (or ignorance of it), inability to use that framework logically (Singh renders this as not knowing what theoretical evidence is available), and unfamiliarity with research techniques.

Hypotheses come from theory, observation, earlier research, analogy (learned helplessness in dogs suggested a model of human depression), intuition and culture. A hypothesis guides data collection and analysis (what to collect, what to ignore), delimits the study, suggests the design and links theory to observation. It is never meant to prove the researcher's view or steer the results, and it does not replace the literature review it grows from. A supported hypothesis is retained, never proved.

Types of hypotheses ●●

Table 9.4 lists the types. The **null hypothesis** (H_0) states no difference or relationship in the population; the **alternative hypothesis** (H_1) states that one exists, and is accepted when the null is rejected. An alternative is a **directional hypothesis** if it names the direction (one-tailed test) and a **non-directional hypothesis** if it predicts only a difference (two-tailed); statistical hypotheses and their testing are in 9.9. A **universal hypothesis** holds for all values of the variables at all times and places; an **existential hypothesis** holds for at least one case. Universal hypotheses are preferred because they predict more.

Table 9.4 Types of hypotheses.

Type	States	Example
Universal	All values, all times and places	Adequate light increases reading efficiency
Existential	At least one case	At least one patient improves untreated
Causal	One variable causes another	Boring content causes channel changing
Descriptive	A feature of behaviour, no cause	Viewers alone change channels more
Simple	One independent, one dependent variable	Reward improves learning
Complex	More than two variables	Class and gender jointly affect adjustment

Type	States	Example
Research	Prediction derived from the theory	CBT beats usual care for depression
Working	Provisional guess, revised as data arrive	Dropout may depend on distance
Null (H_0)	No difference or relationship	No difference between male and female incomes in the population
Alternative (H_1)	A difference or relationship	Male and female incomes differ
Directional	Names the direction	Reinforcement will have a positive effect on performance
Non-directional	A difference only	Boys and girls differ in aggression

EXAM TRAP

"Reinforcement will have a positive effect on performance" predicts an effect (**alternative**, not null) and its direction (**directional**). The statement that "sounds like a null" is the one of **no difference**: "there is a correlation", "the coin is not fair" and "the defendant is guilty" all claim an effect (the courtroom null is "innocent"). For universal hypotheses, options about "two" or "some" values are invented: universal means all values everywhere.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Peirce's four methods of fixing belief are tenacity, authority, a priori and _____.
2. Socrates' cross-examination that exposes contradictions is the _____.
3. For Popper, a scientific statement must be _____.
4. Specific observations to general theory is _____ reasoning.
5. Allport preferred the _____ approach.
6. Action research was founded by _____.
7. A literature review chiefly provides a _____.
8. "Reinforcement will improve performance" is an alternative, _____ hypothesis.

Answers: (1) science, (2) elenchus, (3) falsifiable, (4) inductive, (5) idiographic, (6) Kurt Lewin, (7) theoretical framework, (8) directional

9.2 Variables, Experimental Control and the Validity of Research

MEDIUM YIELD · 22 QS

Suppose a counsellor at a Mysuru college runs a six-week yoga course for forty anxious students, and their anxiety falls. Did the yoga do it, or did the examinations end, or did the students relax because someone was watching? Deciding needs the language of variables, measurement, control, guards against bias and the idea of validity. (Hypotheses are Section 9.1; designs, Section 9.4.)

Variables and their kinds ●●

A **variable** is any characteristic that can take two or more values; one with a single value in a study is a **constant**. A **concept** is an abstraction from observed events (honesty, courage), and

a **construct** is a concept deliberately invented for scientific use, such as intelligence (**Fred N. Kerlinger**, 1910 to 1991, *Foundations of Behavioral Research*, 1964). Measurability is what separates a variable from a concept.

The **independent variable** (IV) is the presumed cause, manipulated by the experimenter or, where that is impossible, selected. The **dependent variable** (DV) is the measured outcome, expected to change with the IV. Find the IV by asking what is presumed to cause what, not by word order: in "old age home residents who have heart disease die earlier than those who do not", the IV is having heart disease, the DV is age at death, and the old age home is a constant.

Heart disease cannot be handed out, which illustrates Kerlinger's split between **active variables**, which the researcher manipulates (dose, instructions), and **attribute variables**, which participants bring and which can only be measured or selected (age, sex, diagnosis). With an attribute IV random assignment is impossible, so causal claims are weaker. Attribute variables are also called organismic, the O in **Robert S. Woodworth**'s (1869 to 1962) stimulus-organism-response scheme. Table 9.5 lists the other contrasts.

Table 9.5 Ways of classifying variables.

Distinction	Meaning	Example
Stimulus, organismic, response	Feature of the task or setting; of the person; the behaviour measured	Noise; age; reaction time
Explanatory and response	IV and DV in observational studies	Smoking; lung disease
Predictor and criterion	IV and DV in correlation and regression	Entrance score; MA marks
Continuous and discrete	Any value in a range; separate, countable values	Reaction time; siblings
Quantitative and qualitative	Differ in amount; differ in kind (categorical)	Anxiety score; diagnosis
True and artificial dichotomy	Two natural categories; a continuum cut in two (biserial correlation, Section 9.11)	Alive or dead; pass or fail

EXAM TRAP

The IV is the presumed cause (having heart disease), not the first-mentioned variable (dying early) or the setting (the old age home, a constant). "Manipulated by the experimenter" defines the IV; "measured as the outcome" defines the DV; "randomly assigned" describes participants, not a variable.

Extraneous, confounding, moderator and mediator variables

An **extraneous variable** is any variable other than the IV that may affect the DV: **subject** or organismic variables (age, intelligence, motivation), **task** variables (difficulty, instructions), **environmental** variables (noise, light) and **situational** variables (time of day, the experimenter's manner). One that varies randomly only adds noise. One that varies systematically with the IV and also affects the DV is a **confounding variable**: it influences both, and its effect cannot be separated from the IV's. If the yoga volunteers were also more motivated, and motivation lowers anxiety, motivation is confounded with treatment.

Reuben M. Baron and **David A. Kenny** (1986) separated two other third variables (Figure 9.2). A **moderator** changes the direction or strength of the IV-DV relationship, saying when or for

whom: if yoga helps mild but not severe anxiety, severity moderates the effect (statistically an interaction, Section 9.10). A **mediator** is the mechanism through which the IV acts: yoga slows breathing, which lowers anxiety. An **intervening variable** is such a link that can only be inferred, never observed, such as learning or fatigue.

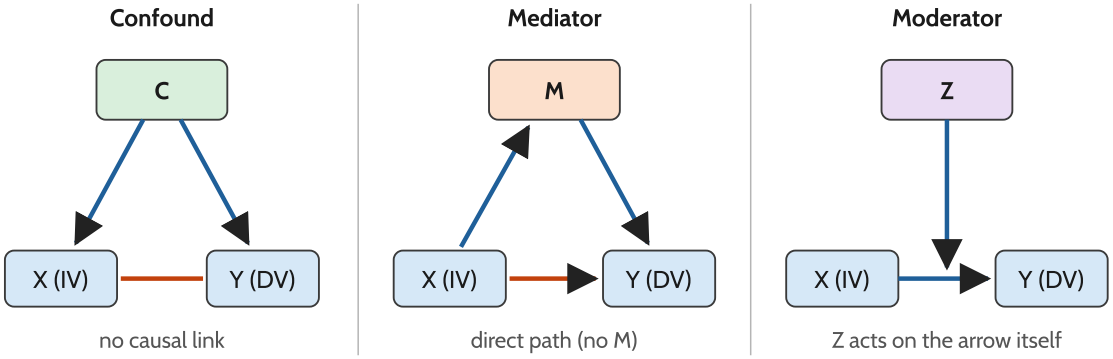

Figure 9.2 A confound (C) drives both X and Y; a mediator (M) carries the effect from X to Y; a moderator (Z) changes how strongly X affects Y.

> EXAM TRAP
>
> A variable that "only affects the DV" is merely extraneous; a confound must also vary with the IV, which is why it "influences both". Hook: mediator = mechanism (how); moderator = more for some (when, for whom).

Operational definitions and measuring the DV

An **operational definition** defines a variable by the operations used to manipulate or measure it: anxiety is the score on a named inventory; hunger is hours without food. This **operationism** comes from the physicist **Percy Williams Bridgman** (1882 to 1961) in *The Logic of Modern Physics* (1927); **Stanley Smith Stevens** (1906 to 1973) brought it into psychology in 1935. It makes research objective and replicable, though no single operation captures a whole construct. The DV can be measured by direct observation of behaviour, by indirect physiological measures or reaction time, by judgements about a stimulus, or by self-report. A good DV is reliable, valid and sensitive. Scores bunched at the bottom of a scale because the task is too hard show a **floor effect**; scores bunched at the top because it is too easy show a **ceiling effect**. Both hide real differences.

Control and the MAXMINCON principle ●●

Control means holding constant, or otherwise neutralising, the influence of extraneous variables on the DV, so that conditions differ only in the IV. Kerlinger summed up the aim of design in the **MAXMINCON principle**: **maximise** systematic (experimental or **primary**) variance, the variance due to the IV, by making its levels as different as possible; **minimise error variance**, the random variance from individual differences and measurement error, by reliable measures and steady conditions; and **control** extraneous (**secondary**) variance, the systematic variance from other variables, by the techniques that follow.

EXAM TRAP

MAXMINCON reads primary, error, secondary; "true variance" belongs to test theory (Chapter 2). One real paper defines control as holding constant "the possible influence of the *dependent* variable". The stem is careless (extraneous variables are what is held constant), but control is still the answer. Standard deviation is a statistic (Section 9.8); an experiment is the whole procedure (Section 9.4).

Techniques of control ●●

Elimination removes an extraneous variable (a sound-proof room removes noise). **Constancy of conditions** holds it at one value for all (same room, time and instructions), at some cost to generality. **Matching** pairs participants on a relevant variable and splits each pair between groups. **Balancing** spreads a known variable equally across groups, as with equal numbers of men and women in each.

When the same people serve in every condition, **order effects** arise: practice, fatigue and carry-over. **Counterbalancing** varies the order so that each condition appears equally often in each position: **complete** (every order: 2 for two conditions, 6 for three, 24 for four), **partial** (a subset, such as a Latin square, Section 9.4) or **ABBA** (one order, then its reverse). Psychophysics controls **primacy and recency effects** the same way, by counterbalancing the order of presentation.

The most powerful technique is **randomisation**, or random assignment: every participant has an equal chance of being placed in any group. It makes the groups alike at the start on known and unknown variables, so that the main difference between them is the treatment, and is the chief guard against selection bias and confounding. That is why an HIV vaccine trial randomly assigned over 16,000 volunteers to vaccine or placebo: not to reduce non-response or sampling variability (Section 9.3), and not to blind anyone. It does nothing about measurement error (reliable measures reduce that), recall bias (faulty memory of past exposures) or publication bias (selective publishing of positive results; both Section 9.5). Figure 9.3 separates it from random sampling.

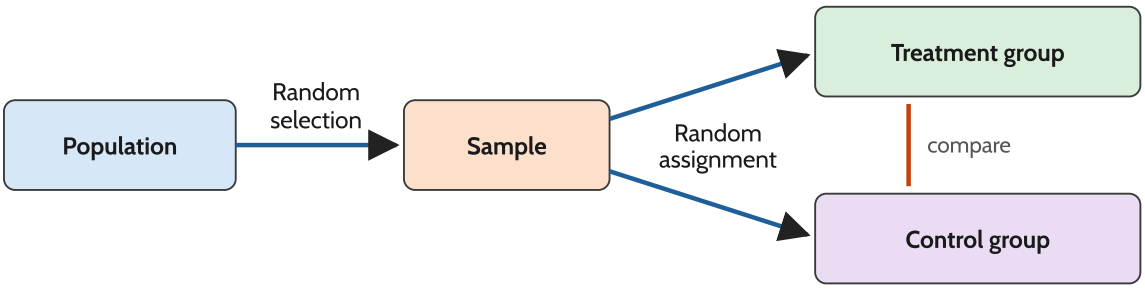

Figure 9.3 Random selection decides who enters the study; random assignment decides who gets which condition.

Finally, a within-subjects design uses **participants as their own control**: stable individual differences drop out, but order effects enter. **Statistical control** removes a measured extra-

neous variable in the analysis (ANCOVA, Section 9.10; partial correlation, Section 9.11). A **control group**, untreated or given a placebo, shows what would have happened anyway.

EXAM TRAP

Equal probability of being assigned to a treatment group is randomisation. **Random sampling** decides who enters the study, **stratification** splits a population into subgroups before sampling, and **randomness** is mere unpredictability. In the psychophysics item, "pairing" is matching, and randomisation only scatters order effects by chance. "Double-bind technique" is most likely a misprint for double-blind; the double bind proper is a theory of contradictory family messages in schizophrenia (Chapter 8). None of these balances order, so counterbalancing is the answer.

Bias and artefacts ●●

Bias is any influence that distorts a study's results away from the truth. (Control is its remedy; sample and population are the people studied, Section 9.3.) Distortions created by the research situation itself are called artefacts; the lesser ones are in Table 9.6. The best known is the **Hawthorne effect**: people change their behaviour because they know they are being observed, whatever the intervention.

KEY STUDY: THE HAWTHORNE STUDIES (1924 TO 1932)

The Western Electric Hawthorne Works near Chicago studied workers' output. In the illumination experiments output rose when lighting was increased, but also in the control group, and sometimes when the light was dimmed. The studies, associated with **Elton Mayo** (1880 to 1949), were reported by **Fritz J. Roethlisberger** (1898 to 1974) and **William J. Dickson** in *Management and the Worker* (1939); **Henry A. Landsberger** coined the term in 1958. Reanalyses find the data weaker than the legend.

Martin T. Orne (1927 to 2000) argued in 1962 that participants are active problem-solvers. **Demand characteristics** are the cues in a setting (rumours, instructions, procedure, the experimenter's manner) that tell participants what is expected and so shape their behaviour. **Stephen J. Weber** and **Thomas D. Cook** (1972) described good (confirming), faithful (literal), negativistic (opposing) and apprehensive subjects, the last showing **evaluation apprehension**, anxiety to be judged favourably (**Milton J. Rosenberg**, 1965). From the other side, **Robert Rosenthal** (born 1933), in *Experimenter Effects in Behavioral Research* (1966), showed that researchers' expectations change how participants are treated and responses scored: the **experimenter expectancy effect**, or experimenter bias.

KEY STUDY: PYGMALION IN THE CLASSROOM (1968)

Rosenthal and **Lenore Jacobson** gave the pupils of Oak School (a pseudonym), a Californian elementary school, an IQ test and told teachers that it had picked out certain pupils, in fact chosen at random (about 20 percent), as about to bloom intellectually. Retested on the same IQ test at the end of the year, these children, especially in grades one and two, had gained more IQ points than their classmates: the **Pygmalion effect**, a self-fulfilling prophecy. Replications find it real but smaller.

The **halo effect**, described by **Edward Lee Thorndike** (1874 to 1949) in 1920, is one salient attribute colouring ratings of others. When a patient's anxiety improves and the rater therefore scores depression as improved too, that is halo: not a ceiling effect (a scale limit), not Hawthorne (a change in the participant) and not a therapeutic effect (a genuine benefit of treatment).

Table 9.6 Further artefacts.

Effect	What happens	Main control
Placebo effect	Improvement from expecting treatment	Placebo control; blinding
Nocebo effect	Symptoms from negative expectations	Placebo control; blinding
John Henry effect	Control group competes harder (compensatory rivalry)	Keep groups apart
Novelty effect	Gain because a treatment is new, fading later	Longer follow-up
Social desirability	Answers chosen to look good (Chapter 2)	Anonymity; lie scales

EXAM TRAP

Being observed: Hawthorne. Guessing the hypothesis from cues: demand characteristics. The researcher's expectation: experimenter bias. Others' expectations lifting performance: Pygmalion. Expecting treatment: placebo. Demand characteristics are cues, not properties of the hypothesis, statistical demands or the effort an item requires.

Blinding ●●

In a **single-blind** study participants do not know their condition, which controls their expectancy and demand, but the experimenter can still bias the results. In a **double-blind** study neither the participants nor the experimenters who treat and assess them know who is in which group, so it minimises both experimenter bias and expectancy effects. A **triple-blind** study also blinds those who analyse the data. (Allocation concealment is a different safeguard; Section 9.5.)

EXAM TRAP

Double-blinding is what controls experimenter bias. Random assignment equalises groups, but the experimenter still knows who got what; a control group hides nothing; demand characteristics are a bias, not a control; single-blinding leaves the experimenter free; pretest instructions hide nothing. Double-blind does not mean two control groups or running the study twice.

The four kinds of validity of a study ●●

Here validity belongs to a study, not a test (Chapter 2). **Donald T. Campbell** (1916 to 1996) first set internal against external validity in 1957 and, with **Julian C. Stanley** (1918 to 2005), listed the threats to each in 1963; Cook and Campbell (1979) and **William R. Shadish**, Cook and Campbell (2002) gave four types:

- **Statistical conclusion validity**: is the IV-DV relationship real, or did low power, violated assumptions or many tests mislead?

- **Internal validity**: did the IV cause the change in the DV? Campbell and Stanley called it the basic minimum for interpreting an experiment.
- **Construct validity**: do the manipulation and measures represent the intended constructs? Giving plain versus barbecue-flavoured chips does not manipulate anxiety.
- **External validity**: the extent to which findings generalise to other people (population validity), settings (**ecological validity**) and times (temporal validity). **Glenn H. Bracht** and **Gene V Glass** (1968) split it into population and ecological validity.

The two main kinds trade off: tight laboratory control raises internal validity but makes the setting unlike life, while a busy clinic is realistic but loosely controlled. Researchers establish an effect under control, then replicate it in the field. **Alan Bryman**, in *Social Research Methods* (2001), judges studies by reliability (measures of concepts stable on different occasions), **replicability** (methods stated clearly enough to repeat) and validity (generalising to other social settings is external validity). The researcher's reputation is no criterion.

> EXAM TRAP
>
> In the chips item, internal validity tempts, but no rival variable explains the result; the manipulation simply is not anxiety, so construct validity fails. "Statistical validity" loosely means statistical conclusion validity; "empirical validity" is an old name for a test's criterion-related validity (Chapter 2). Causation is internal validity; internal consistency and reliability belong to measures.

Threats to internal and external validity

A **threat to internal validity** is any rival explanation for a change in the DV. Campbell and Stanley (1963) listed eight, easiest to see in the counsellor's original study, a pretest and posttest with no control group (Table 9.7); Cook and Campbell (1979) added, among others, four that arise when groups influence each other. A control group measured at the same times cancels history, maturation, testing and instrumentation; only random assignment cancels selection. Campbell and Stanley also listed four threats to external validity, to which experimenter effects are usually added (Table 9.8).

Table 9.7 Threats to internal validity, in the yoga study.

Threat	What it is	Example
History	Outside events between pretest and posttest	Examinations end mid-course
Maturation	Change within people over time: growth, fatigue, recovery	Students settle in
Testing	The pretest changes posttest scores	The inventory feels familiar
Instrumentation	The measure or the raters change	Counsellor rates leniently later
Statistical regression	Extreme scorers drift towards the mean on retest	Only the most anxious enrolled
Selection	Groups differ before treatment	Volunteers against non-volunteers
Experimental mortality	Differential drop-out (attrition), not death	The most anxious stop coming
Selection-maturation interaction	Groups would change at different rates anyway	First-years against final-years

Threat	What it is	Example
Diffusion or imitation (1979)	Controls pick up the treatment	Controls copy the breathing
Compensatory equalisation (1979)	Staff give controls something extra	Warden counsels the controls
Compensatory rivalry (1979)	Controls try harder (John Henry effect)	Controls vow to stay calm
Resentful demoralisation (1979)	Controls who miss out give up	Wait-listed students lose heart

Table 9.8 Threats to external validity.

Threat	What it is
Reactive or interaction effect of testing	The pretest sensitises people to the treatment
Interaction of selection and treatment	The effect holds only for the kind of people sampled
Reactive arrangements	The study setting itself changes behaviour (Hawthorne)
Multiple-treatment interference	Earlier treatments carry over into later ones
Experimenter effects	The result depends on who runs the study

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. MAXMINCON says: minimise _____ variance.
2. A variable that varies with the IV and also affects the DV is a _____ variable.
3. The mechanism through which an IV acts is a _____.
4. Bridgman (1927) defined concepts by operations: the _____ definition.
5. Primacy and recency in psychophysics are controlled by _____.
6. Changing behaviour because one is observed is the _____ effect.
7. Blinding both participants and experimenters is a _____ design.
8. Generalisation to other people, settings and times is _____ validity.

Answers: (1) error, (2) confounding, (3) mediator, (4) operational, (5) counterbalancing, (6) Hawthorne, (7) double-blind, (8) external.

9.3 Sampling: Populations, Probability and Non-Probability Methods

MEDIUM YIELD · 24 QS

Suppose you want to know how anxious Indian school students feel before their board examinations. You will test a few hundred of the crores who exist, then speak about all of them. Whether that leap is justified depends almost entirely on how the few hundred were chosen.

Populations, samples and parameters ●●

A **population** (or universe) is the whole, well-defined group about which you want to draw conclusions. The **target population** is the group you would like to generalise to (all school students in India); the **accessible population** is the part you can actually reach (students in the Mysuru schools that admit you). Each member about whom data are collected is an **element**; a **sampling unit** is whatever is selected at a step of sampling, either an element or a

whole group such as a school. The **sampling frame** is the actual list used for selection (a school register, a voters' list), and the **sample** is the smaller set chosen from it by some rule or plan. Studying every element is a **census**; studying a part in order to describe the whole is a sample survey.

A number describing a population is a **parameter**, written with Greek letters (mean μ, standard deviation σ); the same number computed from a sample is a **statistic**, written with Roman letters (M or $\bar{X}$, s). A parameter is fixed but usually unknown; a statistic is known but varies from sample to sample. Sampling exists to estimate the first from the second.

> EXAM TRAP
>
> "A statistical measure based upon a population" is a **parameter**; one based upon a sample is a **statistic**. Pair the initials: Population with Parameter, Sample with Statistic.

Sampling is needed because it saves time and money, often measures more accurately (a few trained investigators beat an army of hasty ones), is the only option for huge or infinite populations, and is unavoidable when testing destroys what is tested. Its principles: a statistic usually differs somewhat from its parameter; parameters are estimated through sampling distributions; and a random sample mirrors its population more closely as it grows. Its aims: a representative sample, unbiased estimates and a known margin of error. A good sample therefore shows **representativeness** (population characteristics in the same proportions), **adequacy** (enough cases), **independence** (choosing one element does not affect choosing another) and **homogeneity** (its units are of the same kind as the population's).

Probability versus non-probability sampling ●●

Following the sociologist **Hubert M. Blalock** (1926 to 1991), whose *Social Statistics* appeared in 1960, sampling methods fall into three families (Figure 9.4). **Probability sampling** methods clearly specify the likelihood of inclusion of each element: every element has a known, non-zero chance of selection, and a random mechanism does the choosing. In **non-probability sampling** that chance cannot be known, because a person (researcher, interviewer, volunteer or friend) does the choosing. **Mixed sampling** combines features of both: Indian texts following Blalock place systematic sampling here, since only its start is random, and the term also covers designs that join a purposive stage to a random one.

Why does the random mechanism matter? It removes the researcher's personal bias from selection, and because selection follows the laws of chance, the size of the chance error can be calculated (Section 9.9), which is what licenses inferential statistics. Probability sampling is therefore the method most likely to yield a representative sample: likely, not certain, since a random draw can still come out lopsided. Non-probability samples allow neither randomisation nor a calculated sampling error.

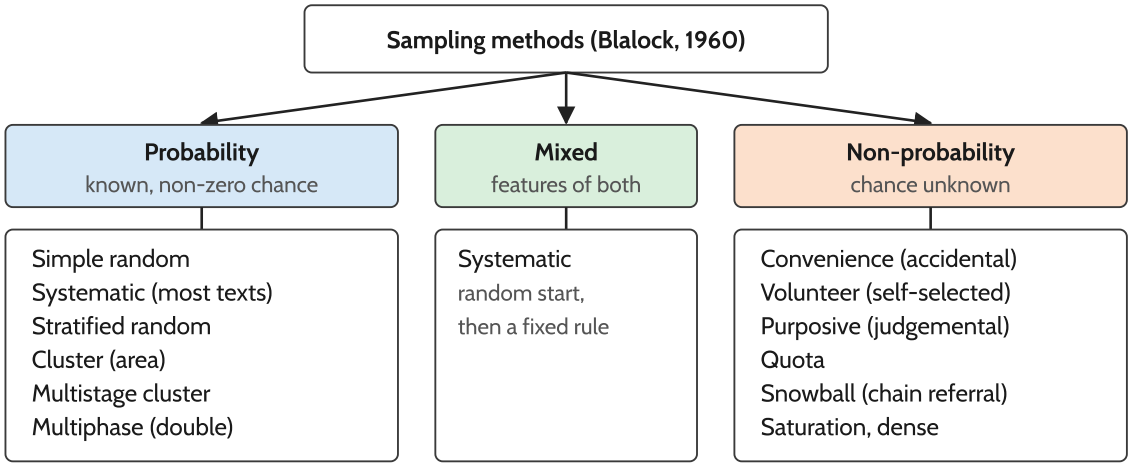

Figure 9.4 The three families of sampling methods. Systematic sampling appears twice: most texts call it probability sampling, Indian texts following Blalock call it mixed.

EXAM TRAP

A past paper keys "random sampling guarantees a representative sample". Choose it, since the other options (it ensures personal bias, reduces the sample needed, makes data collection harder) are plainly wrong, but know the exact fact: it makes representativeness *likely* and error *calculable*, never guaranteed. "Statistical" and "empirical" sampling are not Blalock's families; methods that specify each element's chance of inclusion are **probability** methods.

Simple random and systematic sampling

Simple random sampling (unrestricted random sampling) is the foundation of the others: every element has an equal and independent chance of selection. Equal means none is favoured; independent means one draw does not alter another's chance. Randomness comes from the **lottery or fishbowl method** (names on identical slips, drawn blind), a **table of random numbers** (the most popular: elements are numbered 1 to N, and numbers above N and repeats are skipped) or computer generation for large populations. Returning each drawn element before the next draw is **sampling with replacement**; psychologists usually sample **without replacement**. The method needs no knowledge of the population's make-up and its error is easy to compute, but it requires a complete frame and may, by bad luck, miss small subgroups.

Systematic sampling takes every kth (or nth) element after a random start, with the interval k = N/n. For 100 students from a register of 2,000, k = 20, so a random start of 7 gives the 7th, 27th, 47th student and so on. It is quick and spreads the sample evenly, but beware **periodicity**: if a hostel list runs room by room, four to a room with the prefect always first, k = 4 yields every prefect or none. Chance decides only the start, hence the label mixed in Indian texts following Blalock; it is never a non-probability method.

Stratified random sampling

Suppose a university has 1,000 PhD, 2,000 MA and 7,000 BA students, and research anxiety may differ by level. **Stratified random sampling** first divides the population into **strata**,

mutually exclusive and exhaustive subgroups formed on a variable related to the study, then draws a random sample from every stratum. In **proportionate stratified sampling** each stratum gets the same sampling fraction, so a sample of 100 holds 10 PhD, 20 MA and 70 BA students and mirrors the population. In **disproportionate stratified sampling** small strata are deliberately over-sampled (say 40, 30 and 30) so that each can be analysed on its own, and results are weighted back for the whole. **Optimum allocation** takes more cases from strata that are larger, more variable and cheaper to sample, giving the most precise estimate for a fixed cost; **Jerzy Neyman** (1894 to 1981) gave its classic form, weighing stratum size and variability, in 1934.

Stratification guarantees that every important subgroup appears. When strata are internally homogeneous and differ from one another, it is also more precise than a simple random sample of the same size, because chance can no longer hand you too many BA students. Its costs: every element's stratum must be known in advance, and in a homogeneous population it gains nothing.

> EXAM TRAP
>
> Stratified sampling is preferred over simple random sampling when **important subgroups must be represented**. It does not help a homogeneous population, is not chosen to minimise sample size, and needs more knowledge of the frame, not less (with no list of individuals, use cluster sampling). Every nth case is systematic; intact schools are cluster; volunteers are non-probability.

Cluster, multistage and multiphase sampling ●●

No list of every school student in India exists, and a simple random sample would scatter interviewers across thousands of villages. **Cluster sampling** (area sampling) solves both problems: the population is divided into naturally occurring groups (districts, villages, schools, wards), a random sample of these clusters is drawn, and units inside the chosen clusters are studied. Only a list of clusters is needed. It began in agricultural research and now drives large surveys.

In **one-stage** cluster sampling everyone in each chosen cluster is studied; in **two-stage** cluster sampling a random sample is taken within each chosen cluster; in **multistage sampling**, associated with the British statistician **Claus Moser** (1922 to 2015), selection descends through more levels. Randomly select 5 districts from each state, then 10 schools per district, then 30 students per school: districts and schools are intact groups picked at random, so this is multistage cluster sampling, the logic of India's National Family Health Surveys.

The price is precision. Students in one school are alike, so each extra student from a chosen school adds less new information than one drawn from the whole country. For the same n, a cluster sample usually has a larger sampling error than a simple random sample, and a stratified sample a smaller one; cluster sampling buys cheapness, not a larger sample. Ideal clusters are miniatures of the population, the reverse of ideal strata (Table 9.9, Figure 9.5).

Table 9.9 Stratified versus cluster sampling.

Feature	Stratified random	Cluster (area)
Groups are	Strata defined by a characteristic (sex, class)	Natural units (schools, districts)

Feature	Stratified random	Cluster (area)
Groups sampled	Every stratum	Some clusters, chosen at random
Within a group	A random sample	Everyone, or a random subset (two-stage)
Ideal groups	Alike inside, different from each other	Varied inside, alike to each other
List needed	Every element, with its stratum	Only the clusters
Precision versus simple random	Usually higher	Usually lower
Main reason to use	Represent subgroups	Cost; no list of individuals
Family	Probability	Probability

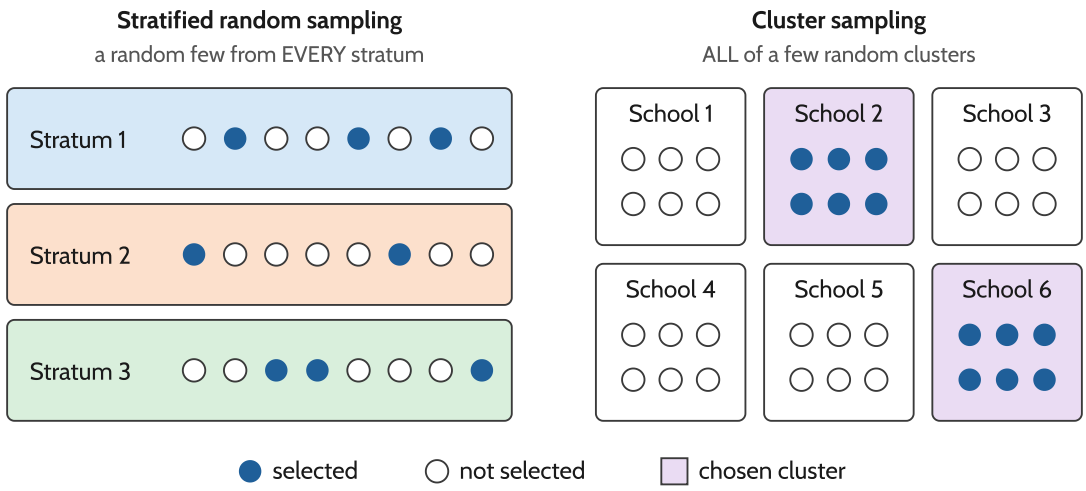

Figure 9.5 Stratified sampling draws a few people at random from every stratum; cluster sampling picks some whole clusters at random (shaded), studies everyone in them and skips the rest.

Two relatives remain. In **multiphase sampling** (with two phases, **double sampling**) the same kind of unit is sampled repeatedly: a large first-phase sample answers a cheap screening question, and a smaller subsample drawn from it gets the detailed study (screen 2,000 students with a brief anxiety scale, then interview 200). Units change across the stages of a multistage design but stay the same across phases. In **panel sampling** one sample is studied repeatedly over time (longitudinal designs are in Section 9.4).

EXAM TRAP

A stem saying researchers "divide a population into smaller groups, and then randomly select from these groups" is a stock definition of **cluster** sampling and is keyed so: the groups themselves are picked at random. Some groups taken and the rest skipped means cluster; every group contributing a random sample means stratified. In the districts, schools, students stem every state contributes, so states act as strata (strictly, stratified multistage cluster sampling), but the keyed answer is **cluster**. "Two-stage" points to cluster sampling, not to proportionate or disproportionate stratified sampling, and two-stage is not two-phase (double) sampling.

Non-probability methods ●●●

Non-probability methods give up the random mechanism for speed, low cost or access. In all of them the chance of selection is unknown, randomisation is impossible, sampling error cannot be calculated, and generalisation rests on judgement (Table 9.10).

Convenience sampling (accidental or incidental sampling) takes people who are available, volunteer or can easily be recruited: your own classmates, shoppers passing a mall. It is the quickest, cheapest and least controlled method. In a **volunteer** or **self-selected sample** people answer an advertisement or online poll; volunteers are usually more interested and motivated than others. In **purposive sampling** (judgemental sampling) the researcher's knowledge and judgement hand-pick cases thought typical, expert or information-rich, such as senior nurses she considers typical of their wards.

Quota sampling, "a common and poor version of stratified sampling", fixes how many people of each kind are needed to mirror population proportions (60 men and 40 women, say), then lets interviewers fill each quota with whoever they find. The interviewer, not chance, picks people within quotas, so bias enters there. Cheap and needing no frame, it is widely used in market research and opinion polls.

In **snowball sampling** (chain-referral sampling) a few people with the characteristic of interest recruit others like them, so existing participants recruit future participants. It is the route into **hidden populations** that have no list, such as drug users or sex workers. In **saturation sampling** the researcher includes as many available members of a limited group as possible (900 of a school's 1,000 students); in **dense sampling** all or nearly all of the accessible population is taken. Both are rare. Qualitative purposive strategies (theoretical sampling, maximum variation, typical and extreme case, criterion sampling, sampling to saturation) belong to Section 9.6.

Table 9.10 The non-probability methods.

Method	Who is chosen	Main weakness
Convenience (accidental, incidental)	Whoever is available	Least representative
Volunteer (self-selected)	People who offer themselves	Volunteers differ from others
Purposive (judgemental)	Cases the researcher judges typical	Researcher bias
Quota	Proportional quotas filled non-randomly	Interviewer bias within quotas
Snowball (chain referral)	Participants recruit others	Over-represents the well connected
Saturation, dense	Nearly all of a small accessible group	Costly near-census

> **EXAM TRAP**
>
> A 2024 paper (repeated in mocks) asks which method relies on "the researcher's judgment to select participants who are typical or representative" and allows no calculation of sampling error, and keys **quota**. That description is purposive sampling; quota wins only as the one non-probability option beside simple random, stratified and cluster. Rule: judgement of typical cases, purposive; fixed numbers per category, quota; whoever is available, convenience; participants recruit participants, snowball. Offered both purposive and quota for a judgement stem, choose purposive.

Sampling error, bias and the Literary Digest

Sampling error is the chance difference between a statistic and its parameter; **Everet Franklin Lindquist** (1901 to 1978) defined it in 1940 as "the difference between a parameter and an estimate of that parameter which is derived from a sample". It implies no mistake. It falls as n rises (the standard error of the mean is σ divided by the square root of n, so quadrupling n halves it) and as population variability falls, is zero in a census, and can be calculated only for probability samples.

Non-sampling error afflicts a census too: coverage error (the frame misses part of the population), non-response error (non-repliers differ from repliers), measurement error (poor questions, interviewer effects) and processing error (coding and entry mistakes). Error that pushes results consistently one way is **sampling bias**. Random error shrinks as n grows; bias does not, and a bigger biased sample is only a more confident wrong answer.

KEY STUDY: THE LITERARY DIGEST POLL (1936)

Question. Who would win the 1936 United States presidential election? **Method.** The magazine *Literary Digest* mailed about 10 million ballots to names from its subscriber lists, telephone directories and car registrations; over 2 million came back. **Result.** It predicted a comfortable win for the Republican challenger; the sitting president won by a landslide, carrying all but two states. A young pollster, **George Horace Gallup** (1901 to 1984), with a far smaller sample, called it correctly. **Why.** In the Depression, telephone and car owners were richer than average (coverage bias), and those who returned ballots differed from those who did not (non-response bias). **Lesson.** Size cannot cure bias.

EXAM TRAP

Random sampling decides who enters the study and supports generalisation (external validity). **Random assignment** decides who gets which condition and supports causal conclusions (internal validity; Section 9.2). A true experiment needs random assignment and can survive a convenience sample.

How large should a sample be?

The size needed rises with population variability, with the precision and confidence level wanted, with smaller effects to detect, with higher power (conventionally .80), with the number of subgroups to analyse separately and with expected non-response; cost, time and the planned analysis set the ceiling. Rules of thumb call 30 or more a large sample and about 100 adequate for most educational research, but the modern answer is a **power analysis** before data collection (Section 9.9). A parameter can be estimated as a single value (**point estimation**) or as a range likely to contain it (**interval estimation**, the confidence interval); both rest on the sampling distribution, taught in Section 9.9.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. A statistical measure based upon a population is a _____.
2. In probability sampling every element has a known, _____ chance of selection.
3. Every kth name after a random start is _____ sampling; its danger is _____.
4. Sampling randomly within every mutually exclusive subgroup is _____ sampling.
5. Random districts, then schools, then students is _____ sampling.
6. Interviewers filling proportional quotas non-randomly is _____ sampling.
7. Existing participants recruiting future participants is _____ sampling.
8. Sampling error shrinks as n rises; sampling _____ does not.

Answers: (1) parameter, (2) non-zero, (3) systematic; periodicity, (4) stratified random, (5) multistage cluster, (6) quota, (7) snowball, (8) bias.

9.4 Research Designs: Experimental, Single-Case and Non-Experimental

MEDIUM YIELD · 21 QS

A counsellor in Mysuru believes that slow breathing before class lowers exam anxiety. She could ask 500 students whether they practise it, follow one anxious student as the practice starts and stops, or let chance decide which half of a class practises. Each is a **research design**, a plan fixing who is studied, what is done to them and when they are measured, and each licenses different conclusions. We begin with the only design that can establish cause.

What makes an experiment ●●●

An **experiment** has three features: **manipulation** of the independent variable (IV) by the researcher (variables are defined in Section 9.2), **control** of everything else, and **random assignment**, in which chance (a lottery, random numbers), not the researcher or the participant, decides who enters which condition. The **experimental group** receives the treatment; the **control group** is treated identically but receives no treatment, and shows what would have happened anyway.

Random assignment makes the groups **equivalent** at the start, spreading every characteristic, even unmeasured ones, evenly by chance. If equivalent groups end unequal on the dependent variable (DV), only the IV can explain it. Creating equivalence among participants is therefore vital for experimental research; the issue does not arise in correlational or descriptive research, where nobody is assigned. **John Stuart Mill** (1806 to 1873) gave the logic in *A System of Logic* (1843). By his **method of agreement**, a circumstance shared by every case of an effect is probably its cause; by his **method of difference**, if cases with and without the effect differ in only one circumstance, that circumstance is the cause, which is exactly what an experiment arranges. Causal inference needs **covariation**, **temporal precedence** (cause before effect) and **elimination of alternative explanations**. Only the experiment secures all three, so only it licenses cause-and-effect conclusions.

Ronald Aylmer Fisher (1890 to 1962), in *The Design of Experiments* (1935), gave three principles of experimental design: **replication** (several units per treatment, so that chance variation can be estimated), **randomisation** (every participant has an equal chance of assignment to any group) and **local control** (blocking: grouping similar units so that their differences leave the error term). Homogeneity is not one of the three.

> EXAM TRAP
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> **Random selection** (sampling, Section 9.3) decides who enters a study; **random assignment** decides who gets which condition. A survey of 10,000 randomly selected adults finding regular meditators less stressed has no assignment: people chose to meditate, and calmer people may be the ones who do. Large or random samples never turn an observational study into an experiment, so neither this study nor one linking exercise and depression in 30,000 adults supports a causal conclusion.

Campbell and Stanley's designs

Donald Thomas Campbell (1916 to 1996) and **Julian C. Stanley** (1918 to 2005) catalogued designs in 1963 in a notation still used: **R** is random assignment, **X** the treatment and **O** an observation; each row is a group and time runs left to right. They grouped designs by how well they rule out threats to internal validity such as history and maturation (Section 9.2), as Table 9.11 shows. **Pre-experimental designs** have no random assignment and little or no comparison; **true experimental designs** have R and a control group.

A pretest has a hidden cost, **pretest sensitisation**: a questionnaire on mental illness may make students more responsive to an anti-stigma film that follows. **Richard L. Solomon** (1949) answered with the **Solomon four-group design**, which runs the pretest-posttest and posttest-only designs together with four random groups, so the effect of pretesting and its interaction with treatment are measured and controlled. Maturation and selection are already handled by any randomised control group, and researcher bias by blinding (Section 9.2).

A **quasi-experiment** manipulates the IV but uses intact groups (classes, wards) without random assignment. Its main forms are the **nonequivalent control group design**, the **interrupted time-series design** (one group measured many times before and after the intervention) and the **regression discontinuity design** (Donald Thistlethwaite and Campbell, 1960), which assigns treatment by a cut-off score and looks for a jump in the regression line at the cut-off.

Table 9.11 Campbell and Stanley's designs (/ separates the groups).

Family	Design	Notation	Key point
Pre	One-shot case study	X O	Nothing to compare with
Pre	One-group pretest-posttest	O X O	No control group
Pre	Static-group comparison	X O / O	Groups may differ from the start
True	Pretest-posttest control group	R O X O / R O O	Classic trial; pretest may sensitise
True	Posttest-only control group	R X O / R O	Randomisation alone equates
True	Solomon four-group	Both above, four groups	Controls pretest sensitisation
Quasi	Nonequivalent control group	O X O / O O	Pretest shows how unequal the groups are

Family	Design	Notation	Key point
Quasi	Interrupted time series	O O O X O O O	Change judged against the prior trend
Quasi	Regression discontinuity	Assignment by cut-off	Jump in the line at the cut-off

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Two fourth-grade classes take a test, one with a radio on. Noise was manipulated, so it is not a correlational study or naturalistic observation; no child was randomly assigned, so it is not a true experiment. It is a **quasi-experiment**.

Between, within and mixed designs ●●

In a **between-subjects design** each participant receives one condition: two randomised groups; a **multigroup design** (one IV at three or more levels, which shows the shape of its effect); or the **matched-groups** or **randomised block design**, in which participants are matched on a variable correlated with the DV and the members of each block are randomly assigned to conditions, Fisher's local control in practice.

In a **within-subjects** or **repeated-measures design** everyone receives every condition and serves as his or her own control, so fewer participants are needed and individual differences leave the error term. The cost is **order effects** (practice, fatigue) and **carry-over effects** (a drug still acting in the next condition), handled by **counterbalancing** the order (Section 9.2). A **Latin square** arranges orders so that each condition appears once in every position: four conditions need four orders instead of 24. Latin squares thus control order effects in repeated-measures designs; they are not correlational, and "equal group sizes" or "one IV with several levels" describe balanced and multigroup designs. A **mixed design** has a between factor and a within factor (group × time). In a **crossover design** each participant receives both treatments in sequence, half in each order.

EXAM TRAP

Patients take a placebo for four weeks and a viral medication for four more, or the reverse. The researchers assigned the treatments, so this crossover study is an **experiment**, and the **explanatory variable** (the one that explains changes in the response) is the type of treatment; symptom improvement is the response.

Factorial designs ●●

A **factorial design** studies two or more IVs at once, every level of each combined systematically with every level of the others. Each IV is a **factor**, each of its values a **level**, and each combination a **treatment combination** or **cell**. Count the numerals for factors and multiply them for cells: 2 × 3 is two factors and six cells; 2 × 2 × 2 is three factors and eight cells.

A **main effect** is one factor's effect averaged over the levels of the others; a **simple effect** is its effect at one level of another; an **interaction effect** exists when one factor's effect depends on the level of another, shown by non-parallel lines (Figure 9.6). There noise adds 2 errors on an easy task and 8 on a hard one: a main effect of 5 errors that hides an interaction only a factorial design can reveal. Two factors give two main effects and one interaction; three give three main effects, three two-way and one three-way interaction (factorial ANOVA, Section 9.10).

In a **nested design** a factor's levels occur within only one level of another: therapists who each deliver only one therapy are nested within it.

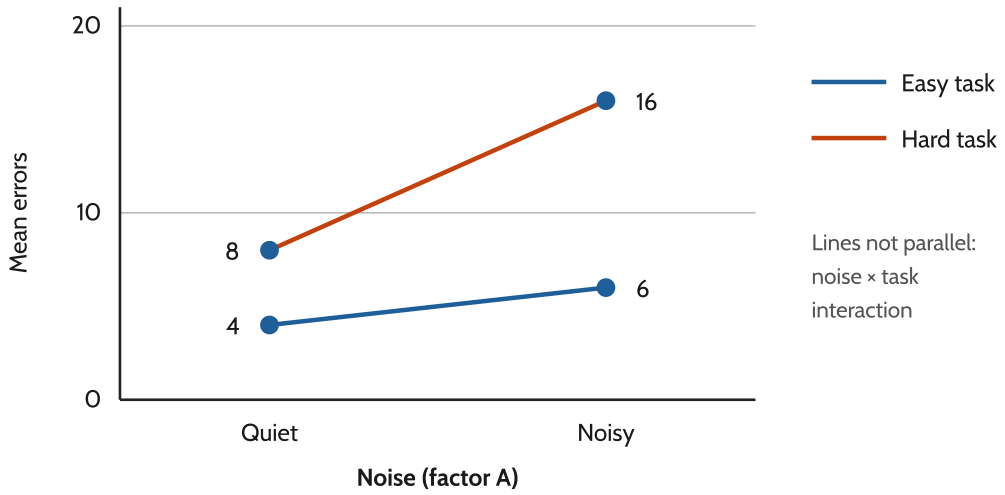

Figure 9.6 A 2 × 2 factorial design (illustrative means). Noise adds 2 errors on the easy task but 8 on the hard one; non-parallel lines signal an interaction.

> EXAM TRAP
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> IQ (100 or 120), age (30 or 60) and sex make a **three-factor** (2 × 2 × 2) design: count the IVs, not the levels. "Eight factor" counts cells and "six factor" adds levels.

Laboratory, field and natural experiments

A **laboratory experiment**, as defined by **Leon Festinger** and **Daniel Katz** (1953), is one in which the investigator creates exactly the conditions he wants, controlling some variables and manipulating others: high internal validity, but limited external validity, that is, generalisation to real life (Section 9.2). A **field experiment** manipulates an IV in a natural setting, as when Irving Piliavin and colleagues (1969) staged a collapse on New York subway trains to see who helped. Its advantage is higher external validity; the laboratory keeps higher internal validity and easier control, and both can use between-participants designs. A **field study** manipulates nothing but studies relations as they exist in a factory or school; Katz (1953) divided field studies into exploratory (finding the variables) and hypothesis-testing types. In a **natural experiment** nature or policy does the manipulating: people conceived at the height of the Dutch famine of 1944 to 1945 later showed about twice the usual risk of schizophrenia. With no random assignment, it remains quasi-experimental. Analogue studies are covered in Section 9.5.

Single-case and small-N designs

The **single-subject design** (single-case or N = 1 design) asks whether a treatment changes one person's behaviour, by measuring it repeatedly while treatment is switched in planned phases; with a few participants it becomes a **small-N design**. Such designs grew from the intensive study of a few animals by **Burrhus Frederic Skinner** (1904 to 1990). **Murray Sidman**'s *Tactics of Scientific Research* (1960) argued that a group average may fit no individual and that

generality comes from replication, and **Alan E. Kazdin** (1982) systematised the designs for clinical work.

Behaviour is recorded (by frequency, duration or time intervals) through a **baseline**, phase A, until stable, which predicts what would happen untreated; B is the treatment and C a second treatment. An A-B design shows one change and cannot rule out coincidence. The **withdrawal design** (A-B-A) removes the treatment to see behaviour return; the **reversal design** (A-B-A-B) reinstates it, the clearest demonstration (Figure 9.7). Withdrawal is unethical for self-injury and useless for skills, such as reading, that do not reverse; the **multiple-baseline design** avoids it by starting treatment at staggered times across behaviours, participants or settings (Table 9.12).

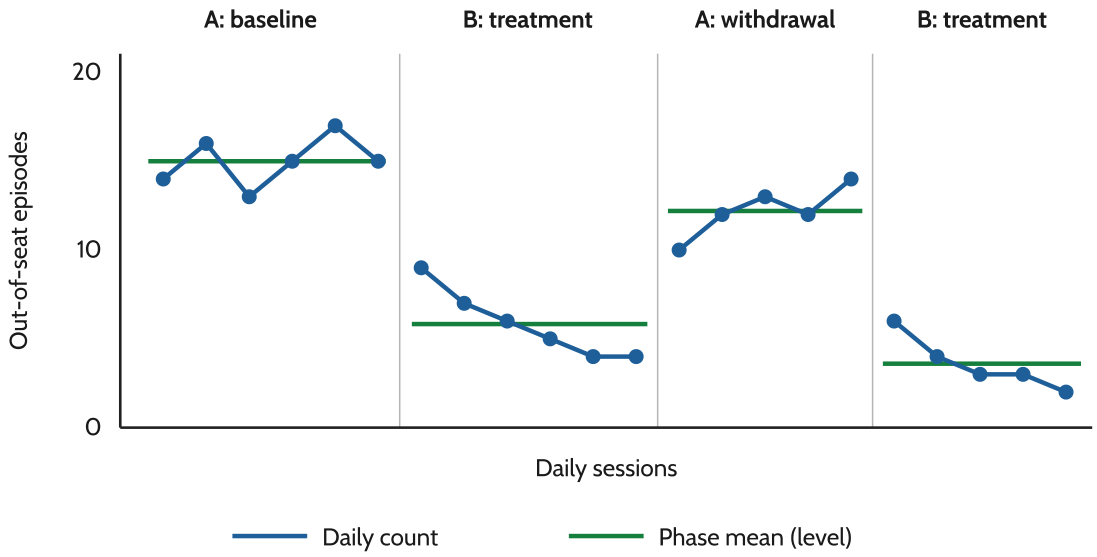

Figure 9.7 An A-B-A-B (reversal) design for one child (illustrative data). Behaviour drops each time treatment starts and climbs when it is withdrawn, so the treatment, not time or chance, controls it.

Table 9.12 The single-case designs (A = baseline; B, C = treatments).

Design	How it works	Strength or limit
A-B	Baseline, then treatment	Cannot rule out coincidence
A-B-A (withdrawal)	Treatment removed	Shows control; ends untreated
A-B-A-B (reversal)	Treatment reinstated	Strongest; ends on treatment
B-A-B	Starts with treatment	Dangerous behaviour; no initial baseline
A-B-A-C-A	Two treatments, each against baseline	Cannot rank B against C
Multiple baseline	Staggered starts across behaviours, participants or settings	No withdrawal; baselines must be independent
Changing criterion	Reward criterion tightened step by step (fewer cigarettes)	Behaviour should track each criterion
Alternating treatments	Two treatments rapidly alternated at random	Quick comparison; treatments may interfere

Data are judged by **visual analysis** of **level** (the phase mean), **trend** (the slope) and **variability**; Kazdin also weighs the **latency** of change after each phase shift. Compared with large-N research, single-case designs show causation within an individual and suit rare disorders and clinical practice; their weakness is external validity, built by replication rather than by sampling.

Non-experimental designs ●●

In **non-experimental** or **observational research** variables are measured as they exist and nothing is assigned. The commonest form, **correlational research**, measures two or more variables in the same people (Section 9.11) and allows prediction but not causation. The **directionality problem** (does poor sleep cause depression, or the reverse?) means it cannot determine the direction of influence; the **third-variable problem** means an unmeasured factor such as loneliness may cause both. It needs no random assignment and is among psychology's most used designs. No sample, however large or random, cures these two problems.

Fred N. Kerlinger (1910 to 1991) defined **survey research** as studying populations by selecting samples, to discover the relative incidence, distribution and interrelations of sociological and psychological variables, but not the causal factors of an event. A survey of an entire population is a **census** (the Census of India is the classic survey); of a part, a **sample survey**. Modes are interview, mail, telephone, online and the panel technique (the same respondents surveyed repeatedly). In Kerlinger's **ex post facto research** ("after the fact"), also called **causal-comparative research**, the IV has already occurred or cannot be manipulated, so pre-existing groups such as smokers and non-smokers are compared; self-selection and the post hoc fallacy are its risks.

Historical research uses **primary sources** (eyewitness records) and **secondary sources** (accounts by people not present). Its evidence faces **external criticism**, which tests authenticity (is the document genuine?), and **internal criticism**, which tests credibility (is it accurate, its author honest and informed?). Table 9.13 gives the other descriptive designs.

Table 9.13 Other descriptive designs.

Design	What it is	Limit
Naturalistic observation	Behaviour recorded in its natural setting without intervening	Reactivity, observer bias
Case study	Intensive study of one person or group (Freud's Little Hans, 1909)	No control; cannot generalise
Archival research	Records kept for other purposes (Durkheim's *Suicide*, 1897)	No control over what was recorded
Sociometry	Jacob Moreno's (1934) mapping of choices within a group (Section 9.6)	Describes, does not explain

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A **census** is a survey of everyone, not ex post facto research, a field study or sociometry. What survey research does NOT do is study the causal factors of an event.

Developmental designs ●●●

A **cohort** is a group born in the same period and sharing its history. The **longitudinal design** studies the same people repeatedly over time (one group across ten years) and is the best design for change over time; its problems are **attrition** (selective drop-out), practice effects, cost and time. The **cross-sectional design** tests different age groups at one time: quick, cheap and covering several ages at once, but limited by the **cohort effect**, generational differences (less schooling in older groups) mistaken for development. The **time-lag design** tests one age in different years. **K. Warner Schaie** (1965) combined these in **sequential designs**, which test several cohorts on several occasions so that age changes can be told apart from cohort differences; in the commonest, the **cross-sequential design**, several age groups are tested and then retested over years (Figure 9.8). A case-control design instead compares people with and without an outcome on past exposures (Section 9.5).

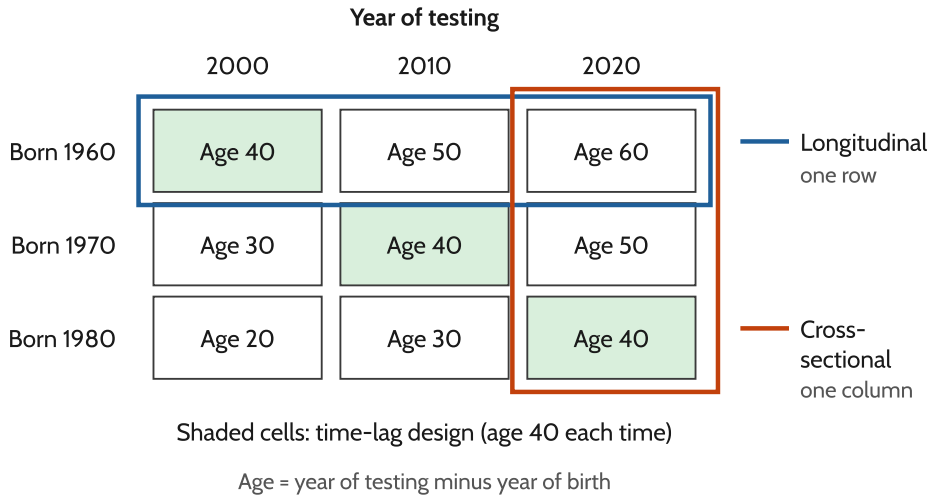

Figure 9.8 Age, cohort and year of testing. A longitudinal study reads one row, a cross-sectional study one column (so age is tangled with cohort) and a time-lag study the diagonal.

KEY STUDY: TERMAN'S GIFTED CHILDREN (FROM 1921)

Question: are gifted children frail and maladjusted? **Method: Lewis Madison Terman** (1877 to 1956) followed about 1,500 California schoolchildren with very high Stanford-Binet IQs from 1921 into old age, the classic longitudinal study. **Result:** they grew up healthier, better adjusted and more successful than average. **Criticism:** the sample was teacher-nominated and largely middle class, and Terman helped his participants' careers.

EXAM TRAP

Cohort effects limit cross-sectional designs; **attrition** and **practice effects** limit longitudinal ones. Socio-economic factors can confound any comparison but are not the characteristic flaw, age cannot be randomly assigned in either design, and a cross-sectional design studies many ages at once.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Fisher's principle giving every participant an equal chance of assignment to any group is _____.
2. The design that controls pretest sensitisation is the _____ design.
3. Two intact classes, one tested with the radio on, form a _____.
4. Latin squares control _____ effects in repeated-measures designs.
5. IQ, age and sex, each at two levels, make a _____-factor design.
6. The single-case design that staggers treatment across behaviours, participants or settings, with no withdrawal, is the _____ design.
7. Survey research cannot identify the _____ of an event.
8. The main weakness of the cross-sectional design is the _____ effect.

Answers: (1) randomisation, (2) Solomon four-group, (3) quasi-experiment, (4) order, (5) three, (6) multiple-baseline, (7) causal factors, (8) cohort.

9.5 Clinical Epidemiology, Randomised Trials and Evidence Synthesis

MEDIUM YIELD · 27 QS

How common is a disorder, does a test detect it, and does a treatment work? The designs and numbers come from **epidemiology**, the study of how disorders are distributed in populations and what determines them; evidence synthesis pools the answers. In epidemiology the independent variable is called the **exposure** and the dependent variable the **outcome**.

Counting cases: prevalence and incidence

A disorder's **prevalence** counts existing cases, old and new together, as a proportion of a population. Its narrowest form, **point prevalence**, is the number of existing cases on a specific date divided by the total population on that date (40 cases among 2,000 Mysuru adults on 1 March: 2 percent). **Period prevalence** counts everyone who was a case at any time during a stated period, such as the past 12 months, and **lifetime prevalence** everyone who has ever met the criteria. By contrast, **incidence** counts only new cases arising during a period among people free of the disorder at its start, either as **cumulative incidence** (new cases divided by the population at risk) or as an **incidence rate** (new cases per person-time at risk, such as per 1,000 person-years).

For a stable condition, **prevalence ≈ incidence × average duration**. Schizophrenia has a low incidence but lasts for years, so its prevalence is relatively high; a brief adjustment disorder can have many new cases yet a low point prevalence. Incidence is used to find causes, prevalence to plan services. A **morbidity rate** expresses illness (its incidence or prevalence) per unit of population over a stated time, and a **mortality rate** gives deaths per unit of population, usually per 1,000 or 100,000 a year.

EXAM TRAP

"Existing cases on a specific date out of the total population" is **point prevalence**. Plain "prevalence" is less specific, "periodic prevalence" covers a stretch of time, and "incidence" counts new cases only.

Epidemiological designs ●●

The designs differ in where the researcher starts and which way the inquiry runs in time (Table 9.14). An **ecological study** correlates group-level data, such as suicide and unemployment rates across Indian states; inferring individual behaviour from such correlations is the **ecological fallacy**, demonstrated by **William S. Robinson** in 1950. A **cross-sectional study** measures exposure and outcome at one moment, as prevalence surveys do, so it cannot show which came first.

A **case-control study** starts from the outcome: cases who have the disorder and matched controls who do not are compared on past exposure. For a rare dissociative disorder and childhood trauma, a cohort would need thousands of children followed for decades, whereas a case-control study starts from the cases a hospital already has: for rare outcomes it is the most feasible design. It cannot measure incidence, so it yields the odds ratio, and it is prone to **recall bias**, because ill people search their memories harder than healthy controls.

A **cohort study** classifies people who are free of the outcome as exposed or unexposed and follows them to see who develops it, which gives incidence and the relative risk. A **prospective cohort** runs forward from now. In the British Doctors Study, begun by **Richard Doll** (1912 to 2005) and **Austin Bradford Hill** (1897 to 1991), British doctors were questioned about smoking in 1951, again in 1957, 1966 and repeatedly up to 2001, and their deaths were traced: lung cancer mortality rose with the number of cigarettes smoked. Nobody was assigned to smoke, so the study is observational; it ran forward, so it is prospective. The **Framingham Heart Study** of a Massachusetts town (from 1948) is another. A **retrospective cohort** (historical cohort) uses existing records but still reasons from exposure to outcome. Cohorts are slow and costly, and they lose people to attrition.

Table 9.14 Epidemiological designs at a glance.

Design	Starts from	Direction	Yields
Cross-sectional	One sample, one moment	Snapshot	Prevalence
Case-control	Cases and controls	Outcome back to exposure	Odds ratio
Cohort	Exposed and unexposed	Exposure forward to outcome	Incidence, relative risk
Randomised trial	Randomly allocated groups	Forward, exposure assigned	Risk difference, NNT

EXAM TRAP

Retrospective is not case-control: only case-control reasons from outcome back to exposure. The British Doctors Study is a **prospective observational** study, not retrospective, not a sample survey (9.3) and not an experiment. "Rare outcome plus past exposure" means **case-control**: trauma cannot be randomised, and a cross-sectional study cannot show which came first.

Measures of association and effect ●●

Any exposure and yes-or-no outcome fit a 2 × 2 table. In a trial, 200 people recovering from depression receive maintenance cognitive behaviour therapy (CBT) and 200 receive treatment as usual (TAU); 50 and 80 of them relapse. **Risk** is the proportion who develop the outcome: 0.25 with CBT, 0.40 with TAU (the control event rate). The **relative risk** or risk ratio (RR) divides the risks: 0.25/0.40 = 0.625. The **odds ratio** (OR) divides the odds, odds being those with the outcome over those without: (50/150)/(80/120) = 0.50. A ratio of **1.0 means no association**. Above 1 the exposure increases risk (an RR of 2.0 means double the incidence); below 1 it is protective, as CBT is here. A case-control study gives only the OR, there phrased as the odds of exposure among cases over that among controls; it approximates the RR when the outcome is rare.

Absolute measures show size. The **attributable risk** (risk difference) is the risk in the exposed minus the risk in the unexposed; in a trial it becomes the **absolute risk reduction** (ARR): 0.40 − 0.25 = 0.15. The **relative risk reduction** (RRR = ARR/control risk = 1 − RR) is 37.5 percent, a grander-sounding figure. The **number needed to treat** (**NNT = 1/ARR**) is how many patients must receive the treatment instead of the control for one additional patient to benefit: 1/0.15 = 6.7, rounded up to 7. An NNT of 8 for an antidepressant means treating 8 patients for one extra to benefit; the ideal is 1. The **number needed to harm** (NNH) is 1 divided by the absolute risk increase of an adverse effect: a side effect in 30 percent on the drug and 10 percent on placebo gives NNH = 5.

> EXAM TRAP
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> **NNT = 1/ARR.** The distractors 1/RRR, 1/RR, RR × control event rate (that gives the treated group's event rate), OR − 1, OR × 100, RRR × 100 and "enrolled ÷ events" are all wrong. An NNT of 8 does not mean the drug works in 8 percent of patients or that 8 must drop out, and there is no NNT threshold for drug approval. **RR or OR = 1.0 means no association**: not protective (below 1), not "doubles the risk" (2.0), not "causes the outcome" and not "perfect correlation" (r = 1, 9.11).

Screening and diagnostic accuracy ●●

Against a gold standard such as a clinical interview, a screening test sorts people into four cells (Table 9.15): true positives, **false positives** (healthy people flagged), **false negatives** (cases missed) and true negatives. A test's **sensitivity**, TP/(TP + FN), is the **true positive rate**: the proportion of real cases it catches. Its **specificity**, TN/(TN + FP), is the proportion of non-cases it correctly clears. A depression test with sensitivity 90 percent and specificity 70 percent therefore identifies 90 percent of depressed people as depressed and 70 percent of healthy people as healthy. Both are computed among people whose true status is known, so they do not change with prevalence.

Table 9.15 A screening test against the true diagnosis.

Test result	Disorder present	Disorder absent	Row index
Positive	True positive (TP)	False positive (FP)	PPV = TP/(TP + FP)
Negative	False negative (FN)	True negative (TN)	NPV = TN/(TN + FN)
Column index	Sensitivity = TP/(TP + FN)	Specificity = TN/(TN + FP)	Rows depend on prevalence

The patient asks the reverse: given a positive result, am I ill? The **positive predictive value** (PPV), the proportion of positive scorers who truly have the disorder, and the **negative predictive value** (NPV), the proportion of negative scorers who truly lack it, answer it, and they depend on the prevalence or **base rate**, the pre-test probability. Apply the 90/70 test to 1,000 clinic patients, 20 percent depressed: 180 true and 240 false positives give PPV = 180/420 = 43 percent. Where only 5 percent are depressed, 45 true and 285 false positives give PPV = 14 percent. The test is unchanged; the base rate fell. **Paul Everett Meehl** (1920 to 2003) and **Albert Rosen** warned in 1955 that with rare conditions a test's cutting score can make more errors than predicting from the base rate alone.

Lowering a cut-off score raises sensitivity and lowers specificity. The **receiver operating characteristic (ROC) curve**, from signal detection theory, plots sensitivity (y-axis) against 1 – specificity (x-axis) across all possible cut-offs (Figure 9.9). The **area under the curve** (AUC) runs from 0.5 (chance) to 1.0 (perfect discrimination). Finally, **likelihood ratios** combine both indices: sensitivity/(1 – specificity) shows how much a positive result raises the odds of the disorder (3 for the 90/70 test), and (1 – sensitivity)/specificity how much a negative result lowers them (0.14).

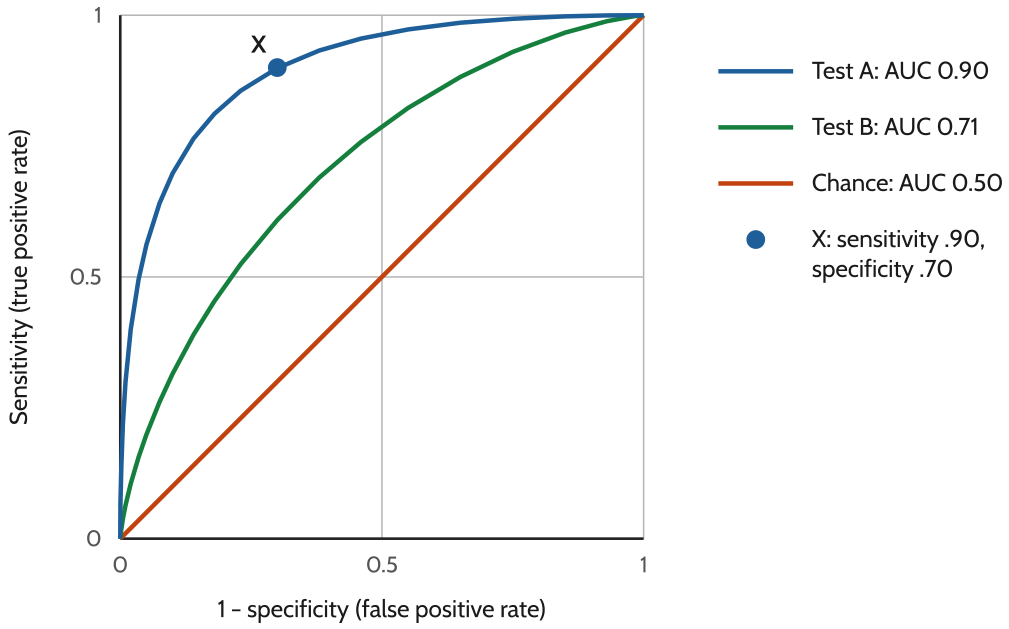

Figure 9.9 ROC curves of two tests (binormal model). Each point is one cut-off; the better test bows towards the top-left corner and has the larger area under the curve.

EXAM TRAP

Sensitivity is not PPV. "A positive test reflects true disease" is PPV, "probability of disease before testing" is prevalence, "true negatives correctly identified" is specificity, and TN/(TN + FN) is NPV. PPV is most affected by **prevalence**, not by sensitivity or specificity alone or by the number of items (Chapter 2). ROC curves do not plot Type I against Type II error (9.9), predictor against outcome (9.11), item difficulty against discrimination, validity against reliability, score against age, or means against standard deviations.

The randomised controlled trial

A **randomised controlled trial** (RCT) is a true experiment on patients: participants are randomly allocated to a treatment or a comparison condition and compared on an outcome named in advance. **Randomisation** gives everyone the same chance of each arm, balancing measured and unmeasured characteristics. Next, **allocation concealment** hides the upcoming allocation sequence from those enrolling participants until each is irreversibly enrolled, using central randomisation or sealed opaque envelopes, so that a clinician cannot steer a promising patient into the treatment arm; it prevents selection bias. **Blinding** works after allocation, hiding the arm from the participant (single-blind), also from those treating and assessing (double-blind), and also from the analysts (triple-blind); see 9.2. Psychotherapy trials cannot blind patients or therapists, only assessors and analysts. Controls may receive a **placebo** (an inert, identical pill) or an **active** established treatment; psychotherapy trials also use waiting-list, attention-placebo and treatment-as-usual controls (Chapter 7).

> EXAM TRAP
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> **Allocation concealment is not blinding.** Concealment hides the *upcoming* allocation from recruiters before enrolment; blinding hides the *actual* arm after it. "Blinding the patient to the assigned arm", "hiding outcome assessors' identity", "sealing the analysis dataset", "concealing the primary outcome from the statistician" and "keeping investigators unaware of the hypothesis" are all wrong.

Analysing, designing and registering trials

In **intention-to-treat analysis** (ITT) every participant is analysed in the group to which they were randomised, regardless of adherence, drop-out or crossover. A **per-protocol analysis** keeps only those who completed the treatment as planned (completers), and **as-treated analysis** groups people by the treatment they actually received. Drop-outs differ between arms, so discarding them reintroduces selection bias and usually flatters the treatment; ITT preserves randomisation and gives a conservative, real-world estimate. **Attrition**, the loss of participants, reduces power, and when it differs between arms it destroys the balance that randomisation created.

> EXAM TRAP
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> ITT analyses **everyone in their randomised group**. "Only completers" and "excluding protocol deviators" are per-protocol; "by actual treatment received" is as-treated; "only responders" or "only the control group" is no analysis at all.

Most trials are **parallel-group**: each person receives one condition. In a **crossover trial** each person receives both conditions in random order, with a washout between, which fails when effects carry over (9.4). A **cluster randomised trial** randomises whole groups such as schools or villages. A **superiority trial** asks whether the new treatment is better; a **non-inferiority trial** whether it is no worse than the standard by more than a preset margin; an **equivalence trial** whether the difference lies within that margin in both directions.

Drug trials run in phases: **phase I** tests safety and dose in about 20 to 100 mostly healthy volunteers; **phase II** seeks first evidence of efficacy in up to several hundred patients; **phase III** runs confirmatory RCTs in roughly 300 to 3,000 patients, the basis for approval; **phase IV** is

post-marketing surveillance for rare and long-term harms. **Trial registration** records planned outcomes publicly before enrolment; the Clinical Trials Registry-India (CTRI), run by the Indian Council of Medical Research since 2007, has been mandatory since 2009. Finally, **efficacy** is whether a treatment works under ideal conditions (an explanatory trial) and **effectiveness** whether it works in routine care (a pragmatic trial; Chapter 7).

Analogue research

In **analogue research**, a clinical phenomenon is modelled under conditions that only resemble the clinical situation, usually with healthy, non-clinical participants (often students) or simulated conditions: snake-fearful undergraduates standing in for phobic patients, or sadness induced in the laboratory. It offers control, easy recruitment and fewer ethical obstacles, but its findings may not generalise to patients (external validity, 9.2). It is not research on clinical populations, nor an efficacy or comparative effectiveness trial.

Systematic reviews, meta-analysis and the evidence hierarchy

A **systematic review** uses explicit, pre-specified, reproducible methods to find, appraise and synthesise all the studies on one question, unlike a narrative review whose author picks the studies. A **meta-analysis** is its statistical step: it summarises the results of many studies by pooling their effect sizes (9.9) into one weighted average. **Gene V. Glass** (born 1940) coined the term in 1976.

KEY STUDY: SMITH AND GLASS (1977)

Hans Eysenck (1916 to 1997) had claimed in 1952 that neurotic patients improve about as often without psychotherapy as with it. **Mary Lee Smith** and Glass pooled 375 controlled studies of psychotherapy: the average effect size of 0.68 meant that the average treated client was better off than about 75 percent of untreated controls. Critics, Eysenck foremost, replied that pooling poor studies gives a precise but wrong answer ("garbage in, garbage out").

EXAM TRAP

One real paper (IHBAS 2024) keys "to conduct a single large experiment" as the purpose of meta-analysis. That is wrong: a single experiment, however large, is primary research. Meta-analysis exists **to summarise findings from multiple studies**; it collects no primary data and does not analyse individual case studies.

A **forest plot** shows each study as a square (its effect, sized by its weight) on a line (its 95 percent confidence interval), with a vertical line of no effect and a diamond for the pooled estimate. **Heterogeneity**, variation between studies beyond chance, is tested with Cochran's Q and expressed as I^2. Pooled effects are inflated by **publication bias**, the preferential publication of significant results; **Robert Rosenthal** (born 1933) named its extreme form the **file-drawer problem** in 1979: null results stay in researchers' drawers. A **funnel plot** of effect size against precision is symmetrical without bias and lopsided when small null studies are missing.

The **hierarchy of evidence** for treatment questions runs from systematic reviews and meta-analyses of RCTs at the top, through single RCTs, cohort studies, case-control studies and case series or reports, down to expert opinion.

Reporting guidelines ●●

Reporting guidelines are design-specific checklists telling authors how to write a study up so that readers can judge it (Table 9.16). **CONSORT** (Consolidated Standards of Reporting Trials) adds a flow diagram accounting for every randomised participant; **PRISMA** (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) governs a systematic review, such as one on CBT for insomnia.

Table 9.16 Reporting guidelines matched to designs.

Guideline	Use it for
CONSORT	Randomised controlled trials
SPIRIT	Trial protocols, before the trial runs
PRISMA	Systematic reviews and meta-analyses
STROBE	Observational studies: cohort, case-control, cross-sectional
STARD	Diagnostic accuracy studies
CARE	Case reports
COREQ	Qualitative research (interviews, focus groups)

> **EXAM TRAP**
>
> CONSORT is for **randomised controlled trials**, not systematic reviews (PRISMA), observational studies (STROBE), diagnostic accuracy (STARD), qualitative research (COREQ) or case reports (CARE).

> **SCAN YOUR BRAIN**
>
> Fill in the blanks. The answers are below; cover them first.
>
> 1. Existing cases on one date over the total population: _____.
> 2. Rare outcome, past exposures: the _____ design.
> 3. NNT = 1 divided by the _____.
> 4. Sensitivity = TP/(TP + _____).
> 5. PPV depends heavily on the _____.
> 6. Hiding the upcoming allocation from recruiters: _____.
> 7. Systematic reviews are reported using _____.
>
> **Answers:** (1) point prevalence, (2) case-control, (3) absolute risk reduction, (4) false negatives, (5) prevalence (base rate), (6) allocation concealment, (7) PRISMA.

9.6 Methods of Data Collection and Qualitative Research LOWER YIELD · 15 QS

A design (Section 9.4) settles when and on whom you measure; a method of data collection settles how the information reaches you: by watching people, questioning them, handing them a form, or studying what they have produced. This section takes those tools and their errors, then the paradigms beneath them, and ends with qualitative research.

Data sources and observation

Primary data are collected first-hand for the present study, by observation, interview, questionnaire or test. **Secondary data** were collected by others for another purpose and are reused: census records, hospital files, archives, published surveys. They are cheap, large and non-reactive, but the researcher controls neither what was recorded nor how accurately.

Observation is the systematic watching and recording of behaviour without manipulating it; it alone does not depend on what people say they do. In **participant observation** the observer joins the group's life; in non-participant observation the observer stays outside. Structured observation codes predefined categories; unstructured observation keeps open field notes. Naturalistic observation records behaviour in its usual setting (play in a school yard); controlled observation arranges standard conditions (mother and child behind a one-way mirror). Overt observers are known to the group; covert observers are not, which raises consent problems (Section 9.7). **James P. Spradley** (1933 to 1982), in *Participant Observation* (1980), described three narrowing phases: **descriptive** observation, broad, to grasp the field; **focused** observation, on the processes found relevant; and **selective** observation, near the end, seeking further evidence for specific practices.

Its two errors are **reactivity**, behaviour changing because people know they are watched (reduced by habituation and one-way mirrors), and **observer bias**, seeing what one expects (reduced by blind observers and inter-observer agreement). The participant observer may also "go native", losing critical distance. Time and event sampling of behaviour are covered in Section 9.4.

Interviews and ethnographic inquiry ●●

An **interview** is a purposeful conversation in which the interviewer asks questions and records the answers. A **structured interview** fixes the wording and order of every question: reliable and comparable, but rigid. A semi-structured interview follows a topic guide but lets the interviewer rephrase and probe; an unstructured interview fixes only the aim, trading reliability for depth. An in-depth interview is a long one-to-one exploration of experience; a **focus group** is a moderated discussion among about 6 to 10 similar people whose interaction is itself the data, though dominant members can sway it.

An **open-ended question** lets respondents answer in their own words ("Tell me about..."); a closed-ended question restricts the answer to yes or no or a fixed choice ("Did you sleep well?"). Probes are neutral follow-ups ("Could you say more?", an expectant pause) that draw out a fuller answer without suggesting one.

Spradley's *The Ethnographic Interview* (1979) treats the respondent as an informant, an expert on their own cultural world, asked to explain its meanings to an outsider through descriptive, structural and contrast questions. "Can you tell me why this band is so popular among your age group?" is an **ethnographic interview** question: the teenager explains the culture of the peer group. Other styles of inquiry differ: **psychodynamic inquiry** (feelings, early relationships, hidden motives: "What does this remind you of?"), **task-focused inquiry** (completing a defined task), **strength-based inquiry** (resources and past successes: "What has helped you cope?") and **theory-guided inquiry** (questions set by one theory, such as cognitive therapy).

The interviewer is part of the instrument, so errors arise from the interviewer's attitudes and expectations, from their sex, age, caste or status, from confusing or overlong questions, coldness and unmotivated respondents, and from interviewers recording similar answers differently. Training and standard wording reduce them.

> EXAM TRAP
>
> "Tell me about..." is an **open-ended question**, not an ethnographic technique. The band question is **ethnographic inquiry** because it asks an insider to explain a culture.

Questionnaires and schedules

A **questionnaire** is answered by respondents themselves; a **schedule** is asked and filled in by the investigator or enumerator face to face, as in the census, so it suits respondents who cannot read. Items are open (own words; rich but slow to code) or closed: dichotomous (yes or no), multiple choice, ranking, rating, and the contingency question that routes respondents ("If no, go to question 12"). Attitude scales (Likert, Thurstone, Guttman) are taught in Chapter 2. Table 9.17 lists the wording faults; presuming and hypothetical questions are avoided too.

Table 9.17 Faults in question wording.

Fault	Example
Double-barrelled question: two questions, one answer	"How often and how much time do you exercise?"
Leading: suggests the answer	"Don't you agree therapy should be free?"
Loaded: emotive words	"Should lunatics roam the streets?"
Ambiguous or vague	"Do you drink regularly?"
Double negative	"Do you not agree patients should not be told?"

The **funnel sequence** moves from broad, easy questions to narrow ones, with sensitive items last. A good questionnaire is short and relevant, states its purpose, asks one idea per item in simple, neutral words, follows a logical order, looks neat and has been pilot-tested. Mailed questionnaires are cheap and anonymous but cannot clarify questions, and by one survey estimate about 70 percent are never returned; because non-responders differ from responders, a low response rate biases the sample.

Every tool thus has its own error: reactivity and observer bias, interviewer bias, response sets and non-response, records of unknown accuracy. Careful studies combine tools (triangulation).

Content analysis, sociometry and Verstehen

Content analysis studies communications (letters, diaries, case notes, films) rather than people. **Bernard Berelson** (1912 to 1979) defined it in 1952 as "the objective, systematic and quantitative description of the manifest content of communication". **Manifest content** is what is plainly there; latent content, the underlying meaning, is left to qualitative versions. His method, as set out in his 1954 handbook chapter (the date A.K. Singh gives): specify the universe (the content and its categories), choose the unit of analysis (word, theme, item, character, or space and time) and quantify by counting, ranking or rating.

Sociometry measures an individual's social behaviour and relations within a group. **Jacob Levy Moreno** (1889 to 1974), founder of psychodrama, introduced it in *Who Shall Survive?* (1934). Members name whom they would choose for an activity, and the **sociogram** draws the choices as arrows, showing stars (chosen by many), isolates (chosen by none) and mutual choices.

Verstehen ("understanding") is the interpretive grasp of the meaning an action has for the actor. **Wilhelm Dilthey** (1833 to 1911) argued in 1883 that the natural sciences explain from outside while the human sciences must understand from inside; **Max Weber** (1864 to 1920) made Verstehen the method of interpretive sociology. It is an interpretive stance, not a measure of social behaviour; nor is the "sample method", which is simply sampling (Section 9.3). Q-methodology (**William Stephenson**, 1935), which factor-analyses persons through Q-sorts and bridges the two traditions, is covered in Chapter 2.

Research paradigms

Every method rests on a paradigm, a set of basic beliefs. **Egon Guba** (1924 to 2008) and **Yvonna Lincoln** (1994) defined paradigms by three questions: **ontology** (what is the nature of reality?), **epistemology** (what is the relation between the knower and the known?) and methodology (how should the inquirer find out?). Table 9.18 gives their four paradigms.

Table 9.18 Guba and Lincoln's (1994) paradigms.

Paradigm	Ontology	Quality criteria
Positivism	Naive realism: a "real" reality that is apprehendable	Internal and external validity, reliability, objectivity
Postpositivism	Critical realism: real, but only imperfectly and probabilistically apprehendable	The same conventional benchmarks
Critical theory	Historical realism: a virtual reality shaped by social, political, cultural, economic, ethnic and gender values	Historical situatedness; erosion of ignorance; stimulus to action
Constructivism	Relativism: local and specific constructed realities	Trustworthiness and authenticity

Knowledge grows in positivism and postpositivism by accretion, building blocks added to an edifice of knowledge; in critical theory by historical revisionism and generalisation by similarity; and in constructivism through more informed reconstructions and vicarious experience. Positivists observe from a detached distance; for critical theorists and constructivists inquirer and participant are interactively linked, so findings are value-mediated (critical theory) or created (constructivism). Quantitative research rests on positivism and postpositivism, qualitative research on constructivism and critical theory, and mixed methods on pragmatism, which lets the question choose the methods.

> **EXAM TRAP**
>
> Positivism's reality is real and **apprehendable**; "only imperfectly and probabilistically" is postpositivism. Historical situatedness and generalisation by similarity belong to critical theory, vicarious experience to constructivism; none is a positivist answer.

Qualitative versus quantitative research ●●

Qualitative research studies meaning, experience, process and context through words, images and observed action, analysed by interpretation; quantitative research measures variables and tests hypotheses statistically. Qualitative work seeks the insider's view and treats the researcher as the instrument, which calls for reflexivity, an open account of the researcher's influence. It values language and discourse, process and human individuality, and claims neither objective truth nor the researcher's authority over participants' meanings (Table 9.19).

Table 9.19 Quantitative and qualitative research compared.

Feature	Quantitative	Qualitative
Logic	Deductive, top-down	Inductive, bottom-up
Setting and sample	Controlled; probability samples, standardised tools	Natural, context-specific; purposive samples
Aim	Objective truth, prediction	Meaning, process, individuality
Quality	Validity, reliability, objectivity	Trustworthiness
Methods	Experiment, survey	Ethnography, case study, grounded theory

Its roots include Verstehen, phenomenology and **symbolic interactionism**, the name **Herbert Blumer** (1900 to 1987) gave in 1937 to the ideas of **George Herbert Mead** (1863 to 1931), and set out in his 1969 book: people act toward things according to meanings that arise in social interaction and are modified through interpretation. **Norman Denzin** and Lincoln trace its history through eight "moments" (2005), from Malinowski and the Chicago school to the present; **Michael Quinn Patton** (2002) grouped its twelve themes under design (emergent, purposeful, naturalistic), fieldwork and analysis (inductive, holistic, context-sensitive). In India, qualitative research, from ethnography to action research, has been practised since the late nineteenth century.

> EXAM TRAP
>
> The **survey** is quantitative; case study, ethnography and grounded theory are qualitative. The "odd one out" is whatever implies objective truth, standardised tools, probability samples, top-down evidence or the researcher's authority.

The qualitative traditions ●●

Ethnography ("writing about a people") describes a culture or group from the inside through long immersion in its setting, mainly by participant observation and key informants; the researcher's role is to understand from the insider's perspective, not to manipulate variables, test large samples or reanalyse official statistics. The linguist **Kenneth Pike** (1912 to 2000) coined **emic** (insiders' own categories) and etic (outsiders' comparative categories) in 1954: "tension" as an idiom of distress is emic, "generalised anxiety disorder" etic. **Clifford Geertz** (1926 to 2006) made **thick description** the aim (1973): behaviour recorded with the context that gives it meaning, which alone tells a wink from a twitch of the eyelid.

KEY STUDY: MALINOWSKI IN THE TROBRIAND ISLANDS (1915 TO 1918)

Stranded in Australian territory by the First World War, **Bronislaw Malinowski** (1884 to 1942) pitched his tent in a Trobriand village off New Guinea, learned the language and joined daily life for about two years. *Argonauts of the Western Pacific* (1922) set ethnography's goal: "the native's point of view". Criticism: his diary, published in 1967, revealed contempt for the people he studied.

Grounded theory generates theory from data. **Barney Glaser** (1930 to 2022) and **Anselm Strauss** (1916 to 1996) launched it in *The Discovery of Grounded Theory* (1967). Constant comparison of each new incident with earlier ones drives it; theoretical sampling picks whom to study next from the emerging categories; collection stops at theoretical saturation, when new data add nothing new; and memos record the analyst's developing ideas. The coding stages are those of Strauss and **Juliet Corbin** (1990), shown in Figure 9.10: open coding breaks the data apart and labels concepts; **axial coding** weaves the data back together around categories; selective coding integrates everything around one core category.

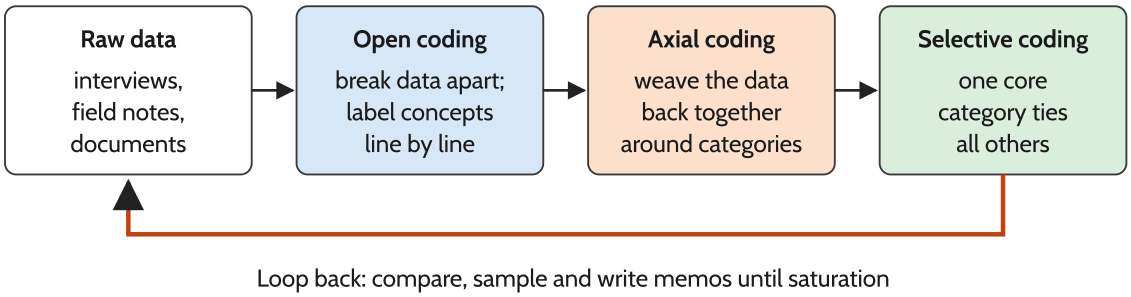


Figure 9.10 Strauss and Corbin's (1990) coding stages: open coding fractures the data, axial coding reassembles it, selective coding integrates it.

EXAM TRAP

"Weaving the data back together around groups of concepts" is **axial coding**. "Early coding" and "core coding" are not stages; the core category is found in selective coding.

Phenomenology studies lived experience as it appears to the person. **Edmund Husserl** (1859 to 1938) asked researchers to describe it after **bracketing** (epoché), suspending their own assumptions; his former assistant **Martin Heidegger** (1889 to 1976) replied (1927) that we are always immersed in the world, so all understanding is interpretation (hermeneutic phenomenology). **Jonathan Smith**'s interpretative phenomenological analysis (IPA, 1996) follows the hermeneutic line: the researcher makes sense of participants making sense of their experience. Most methods texts call the two lines **descriptive** (Husserl) and **interpretive** or hermeneutic (Heidegger) phenomenology. One source used for this book renames them, contrasting a **direct approach** (Husserlian: a detached researcher lights up the foreground, participants' mental representations) with an **indirect approach** (Heideggerian: an involved observer, immersed in the everyday world, lights up its background).

EXAM TRAP

"Direct" and "indirect" are not standard labels, and sources disagree on them. The safe fact is that the detached, bracketing approach is Husserl's (descriptive) and the involved one Heidegger's (interpretive). This book's source calls detached **direct** and involved **indirect** ("direct; indirect"); one practice key reverses them ("indirect; direct"). "Semi-direct" is not an approach.

Table 9.20 completes the family. Thematic analysis runs in six phases: familiarising with the data, generating initial codes, searching for themes, reviewing themes, defining and naming themes, and writing the report.

Table 9.20 Other qualitative approaches.

Approach	Key name	Focus
Case study	**Robert Yin** (1984)	One bounded case in depth, from several sources
Narrative and biographical	Several	Life stories kept whole; lives reconstructed retrospectively
Hermeneutics	Dilthey, Heidegger	Interpretation; the circle of parts and whole
Ethnomethodology	**Harold Garfinkel** (1967)	Everyday methods of making social order; breaching experiments
Conversation and discourse analysis	Several	Turn-taking in talk; what talk does and constructs
Thematic analysis	**Virginia Braun** and **Victoria Clarke** (2006)	Patterns of meaning (themes) across a data set
Framework analysis	**Jane Ritchie** and **Liz Spencer** (1994)	Policy research; a matrix of cases by themes
Co-operative inquiry	**John Heron**	Every participant is co-researcher and co-subject
Participatory action research	**Kurt Lewin**'s action research (1946)	Those affected share power; aims at change

Analysis, trustworthiness and triangulation

Matthew Miles and **A. Michael Huberman** (1994) saw analysis as three interwoven activities: data reduction (coding and summarising), data display (matrices and networks) and conclusion drawing and verification.

Lincoln and Guba's (1985) **trustworthiness** replaces the experimental criteria with four parallels: credibility (internal validity), transferability (external validity, judged by the reader), dependability (reliability) and confirmability (objectivity). Credibility comes from prolonged engagement, peer debriefing, triangulation and member checking (returning findings to participants); transferability from thick description; dependability and confirmability from an audit trail of decisions and from reflexivity.

Triangulation, a surveyor's term, studies one question from several positions so that each method's errors are offset by the others. Denzin (1970) named four types: data (several sources, times and places), investigator (several researchers), theory (several theoretical lenses) and methodological (several methods). Uniting qualitative with quantitative methods is therefore triangulation; "coalition" is no research term, a "multidimensional approach"

merely studies several aspects of a problem, and co-operative inquiry is Heron's participatory method.

Mixed-methods research integrates both kinds of data on a pragmatist footing. **John Creswell**'s three core designs are convergent (both strands at once, then merged), explanatory sequential (quantitative first, qualitative to explain it) and exploratory sequential (qualitative first, building a tool that a quantitative phase tests).

> EXAM TRAP
>
> Uniting qualitative and quantitative methods is **triangulation**. Explanatory designs start quantitative; exploratory designs start qualitative.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. A form asked and filled in by an enumerator is a _____.
2. "How often and how much time do you exercise?" is a _____ question.
3. Moreno's method of measuring social relations in a group is _____.
4. Positivism's ontology is naive realism; constructivism's is _____.
5. Weaving data back together around categories is _____ coding.
6. Uniting qualitative and quantitative methods is called _____.

Answers: (1) schedule, (2) double-barrelled, (3) sociometry, (4) relativism, (5) axial, (6) triangulation.

9.7 Research Ethics and Writing the Research Report LOWER YIELD · 10 QS

A study is finished twice: once when the last participant leaves unharmed, and again when its findings are written up for others to check. This section follows the scandals that produced ethical rules, the rules themselves, and honest reporting.

Why research needs ethics

Modern research ethics was written in response to abuse. After Nazi physicians' lethal experiments on concentration camp prisoners, the verdict of the Doctors' Trial at Nuremberg (1947) set out ten principles, the **Nuremberg Code**. Its first point reads: "The voluntary consent of the human subject is absolutely essential"; another lets the subject stop at any time. In 1964 the World Medical Association adopted the profession's own code, the **Declaration of Helsinki**, which puts the participant's welfare above the interests of science and later required independent committee review.

From 1932 to 1972 the United States Public Health Service ran the Tuskegee syphilis study: about 600 poor Black men, 399 with syphilis, were told they had "bad blood", and penicillin was withheld even after it became the standard cure. Its exposure in 1972 led to the **Belmont Report** (1979) and its three principles, each with one application. *Respect for persons* (treat people as autonomous and protect those with reduced autonomy) is applied through informed consent; *beneficence* (do no harm, maximise benefits and minimise harms) through

risk-benefit assessment; and *justice* (share research's burdens and benefits fairly) through fair selection of participants, the principle Tuskegee violated most plainly.

Psychology's own debated cases matter here for their ethics only (Section 1.6, Chapter 3). **John B. Watson** (1878 to 1958) and **Rosalie Rayner** (1898 to 1935) conditioned fear in an infant, Little Albert (1920), and never removed it. **Philip Zimbardo** (1933 to 2024) cast 24 male students as guards or prisoners in his Stanford prison study (1971); planned for two weeks, it was stopped after six days as guards turned abusive, with Zimbardo himself acting as superintendent.

KEY STUDY: MILGRAM'S OBEDIENCE STUDY (1963) AND BAUMRIND'S CRITIQUE

Stanley Milgram (1933 to 1984) ordered 40 men, told the study concerned punishment and learning, to give rising "shocks" up to 450 volts to a learner who was really an unharmed confederate; 26 (65 per cent) went to the maximum. **Diana Baumrind** (1927 to 2018) argued in 1964 that the deception and stress were unjustified, and critics add that prods such as "you have no other choice, you must go on" undercut the right to withdraw. Milgram replied that all were debriefed and 84 per cent were glad to have taken part.

Codes and ethics committees

Psychologists follow the **APA Ethics Code**, the American Psychological Association's *Ethical Principles of Psychologists and Code of Conduct* (2002, amended 2010 and 2016). Its five aspirational general principles are (A) beneficence and nonmaleficence, (B) fidelity and responsibility, (C) integrity, (D) justice, and (E) respect for people's rights and dignity; beneath them, enforceable standards set specific rules, Standard 8 covering research and publication. In India, the Indian Council of Medical Research issued the **ICMR National Ethical Guidelines** for Biomedical and Health Research Involving Human Participants (2017), which cover behavioural research in health. Under either system no study may begin until an independent **institutional ethics committee** (in the United States, an **institutional review board** or IRB) has approved its protocol. The Rehabilitation Council of India's code for clinical psychologists is in Section 7.1.

EXAM TRAP

The APA has five general principles. Competence and confidentiality are enforceable standards, the usual intruders in "which is NOT a general principle".

Informed consent and voluntary participation ●●

Informed consent is the principle that participants are told enough about a study (its purpose, procedures, duration, risks, benefits, the protection of their data, and their rights) to decide voluntarily whether to take part, and agree before it begins. It needs understandable *information*, *voluntariness* and *capacity*, the ability to understand, weigh and communicate a choice. The **consent form** is only the signed record of that agreement; a signature extracted with jargon or pressure is not informed consent.

Suppose a Mysuru student reads that your exam-anxiety study takes 20 minutes, may stir up worry (a counsellor's number is given), codes answers by number and can be left at any point

without effect on marks. If she then agrees, she has given informed consent; her signed form merely documents it.

Children cannot give legal consent, so a parent or guardian consents and the child gives **assent**, an age-appropriate agreement of his or her own; a child's refusal is normally respected. An adult lacking capacity needs a legally authorised representative's consent, though a diagnosis alone does not remove capacity. **Voluntary participation** rules out coercion (a threat to grades, treatment or parole) and undue inducement (a payment large enough to distort judgement about risk). It continues as the **right to withdraw** at any time, without a reason or penalty, including withdrawal of one's data.

> EXAM TRAP
>
> "Told enough about the research to decide whether to take part" and "permission obtained after informing them of the purpose and procedures" both define **informed consent**, given before participation. The *consent form* is the record, not the principle. *Expectation management* (a business phrase) and *de-concealment* are not research-ethics terms; revealing a deception is dehoaxing, part of debriefing. Random assignment is a design procedure (Section 9.2), not an ethical principle.

Harm, risk and confidentiality

Participants should leave no worse off than they came. **Protection from harm** covers physical harm but in psychology chiefly psychological harm (anxiety, guilt, lowered self-esteem) and social harm such as stigma. Above minimal risk (no more than in daily life), the committee makes a **risk-benefit analysis**: are risks as low as possible, outweighed by benefits to participants or society, and covered by safeguards?

The duty most often used to sum up research ethics is ensuring the **confidentiality** of data: the researcher knows who gave the data but does not disclose it, coding names, storing data securely and reporting only group results. **Anonymity** is stricter: no one, not even the researcher, can link data to a person, because no identity is recorded. Confidentiality has limits, stated at consent, such as a serious risk of harm to self or others. Ethics asks more than following the law, and manipulating data to fit a hypothesis (falsification) or ignoring consent is its opposite.

Vulnerable populations, people less able to protect their own interests, need extra safeguards: children, prisoners, people with severe mental illness or intellectual disability, pregnant women, the very poor, and anyone dependent on the researcher, such as students or patients. Justice forbids loading risky research onto them, but also excluding them from research that could benefit them.

> EXAM TRAP
>
> Confidential means the researcher could identify you but will not; anonymous means nobody can. A face-to-face interview or a longitudinal study can be confidential but never anonymous.

Deception and debriefing •

Obedience or conformity cannot be studied if participants know the hypothesis. **Deception** is misleading participants about a study's purpose or procedures, actively (a cover story, a confederate, false feedback) or passively (withholding the hypothesis). APA Standard 8.07 permits it only when the study has significant scientific, educational or applied value; no effective non-deceptive method is feasible; nobody is deceived about anything expected to cause physical pain or severe emotional distress; and the deception is explained as early as feasible, no later than the end of data collection, with the option to withdraw one's data.

That explanation is **debriefing**: after participation, participants receive full information about all aspects of the study, including its true purpose and any deception, and are checked for distress. **David S. Holmes** (1976) named its two parts. **Dehoaxing** convinces participants that the false information was false (the learner received no shocks); **desensitising** helps them deal with distress or unwelcome self-knowledge (that they obeyed). Debriefing is good practice even without deception, but cannot excuse an unjustified deception. It is not description or explanation (goals of science, Section 9.1), nor interpretation, which belongs to a report's Discussion. Figure 9.11 places each safeguard on the participant's path.

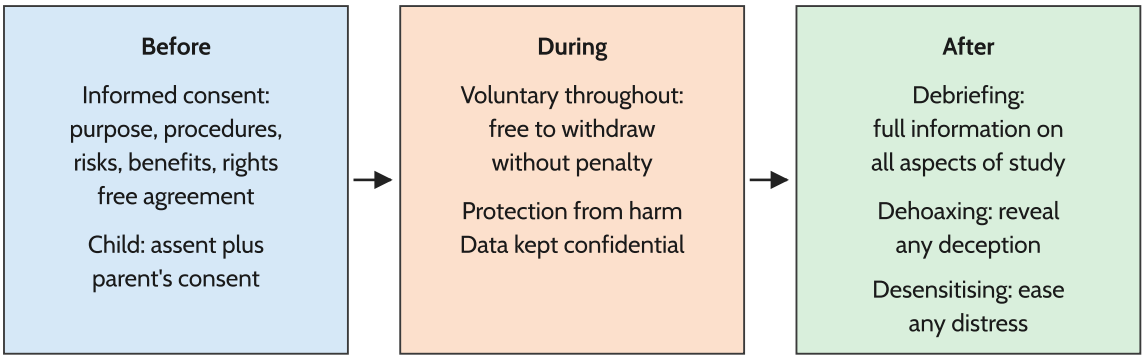


Figure 9.11 The participant's path. Informed consent comes before participation, the right to withdraw and protection last throughout, and debriefing comes after.

Animal research and the 3Rs

Animals cannot consent, so animal research ethics rests on justification and welfare. **William Russell** and **Rex Burch**, in *The Principles of Humane Experimental Technique* (1959), proposed the **3Rs**: *replacement* (non-animal methods, such as cell cultures or computer models, wherever possible), *reduction* (the fewest animals that give a valid answer) and *refinement* (less pain and distress, through anaesthesia and better housing).

Misconduct, plagiarism and giving credit ••

Research misconduct means **fabrication** (inventing data or results), **falsification** (manipulating materials, procedures or data, or omitting results, so that the record misrepresents the research) and **plagiarism** (presenting another person's ideas, words or results as one's own). Honest error is not misconduct. Table 9.21 gives the forms of plagiarism.

Table 9.21 Forms of plagiarism, and the one practice that is not plagiarism.

Form	What it is

Form	What it is
Copying	Using someone's words verbatim without quotation marks or citation
Patchwriting	Reworking a source's sentences by swapping synonyms and word order
Mosaic plagiarism	Stitching phrases from several sources together without credit
Collusion	Submitting unauthorised joint work as individual work
Ghost-writing	Submitting work someone else wrote
Self-plagiarism	Reusing one's own published text or data as if new
Citation (not plagiarism)	Openly acknowledging the source: the cure for plagiarism

A **citation** is the brief acknowledgement in the text where an idea is used, such as (Milgram, 1963). Its primary purpose is to give credit to the original authors, and it also lets readers trace the idea; avoiding legal trouble is a side effect. **Referencing** is listing the full details of every source used at the end of the report, so that readers can find each one; it is not adding footnotes, presenting raw data or sorting data into categories (Section 9.8). Authorship credit (APA Standard 8.12) goes only to substantial contributors, in order of contribution; adding a head of department who did nothing is gift authorship. Under data sharing (Standard 8.14), published data must not be withheld from competent professionals who wish to verify the findings by reanalysis.

> EXAM TRAP
>
> "Which is NOT a form of plagiarism?" Citation. Collusion, patchwriting and ghost-writing are all plagiarism. Citing gives credit to the original authors; referencing lists the sources used.

Writing the research report ●●

The purpose of a **research report** is to communicate the findings: what was done, why, what was found and what it means, so that others can evaluate and replicate the work. Collecting and analysing data come before it, and it summarises raw data rather than presenting them. In psychology the format is set by the APA Publication Manual, now in its 7th edition (released in 2019, dated 2020). Its sections form an hourglass (Figure 9.12); Table 9.22 lists their contents.

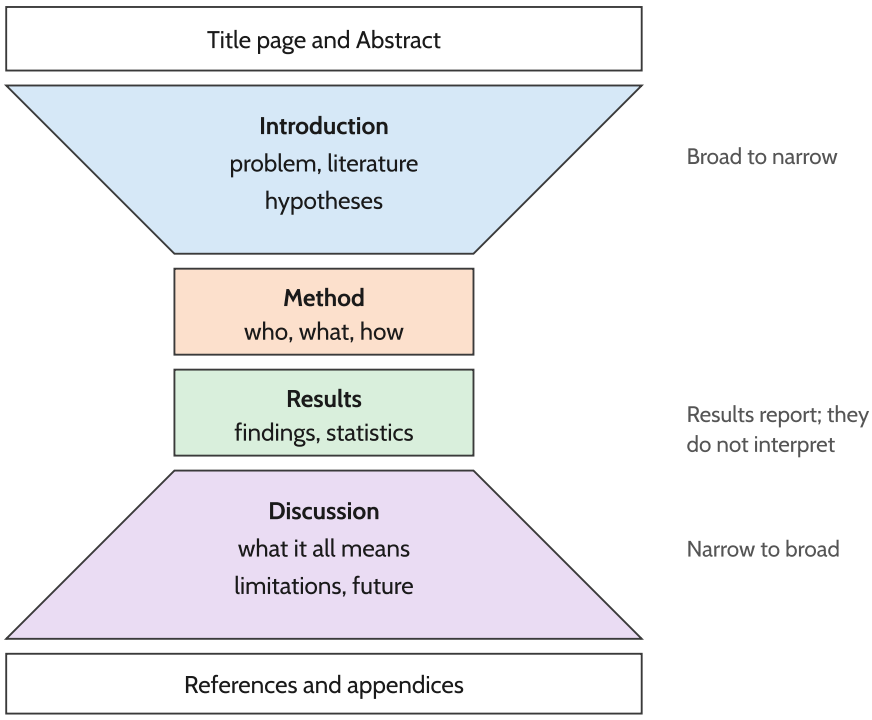


Figure 9.12 The hourglass shape of a research report: the Introduction narrows to the hypotheses, and the Discussion widens again to interpretation.

Table 9.22 The parts of an APA research report.

Part	Contents	Point to remember
Title page	Title, author(s), affiliation, author note; running head (short title) in professional papers	No "A study of"; running head at most 50 characters
Abstract	Problem, method, results, conclusions	About 150 to 250 words (A. K. Singh, following an older edition: 100 to 150); no citations
Introduction	Problem, review of literature, hypotheses	Not headed "Introduction"
Method	Participants (subjects), apparatus or materials, design, procedure	Enough detail to replicate
Results	Findings, statistics, tables and figures, including results against the hypotheses	Reported without interpretation
Discussion	Interpretation, implications, limitations, future research, conclusions	Where findings are interpreted
References	Every work cited in the text, alphabetical	Not a bibliography
Appendices	Tests, stimuli, instructions, long tables	Labelled A, B and so on

A. K. Singh lists the author note (affiliations, funding, acknowledgements) as a ninth part. **References** include only the works cited in the text; a **bibliography** lists everything consulted, cited or not, and is not normally part of an APA report. An APA journal reference runs: Milgram, S. (1963). Behavioral study of obedience. *Journal of Abnormal and Social Psychology*, 67(4), 371-378. https://doi.org/10.1037/h0040525. For three or more authors, the text citation uses the first author and "et al." from the start.

In APA style, numbers below 10 are words and 10 and above numerals, but statistics and measurements always take numerals and a number opening a sentence is spelled out; an abbreviation is spelled out at first use; headings have five levels (Level 1, used for Method, Results and Discussion, is centred and bold; Levels 2 and 3 are flush left, bold and then bold italic; Levels 4 and 5 are indented and run into the paragraph); bias-free language avoids reducing people to labels ("people with schizophrenia", not "schizophrenics"); and procedures take the past tense. Evaluating a report asks whether the hypothesis is testable, the methods sound and the conclusions supported by the data.

> EXAM TRAP
>
> "In which step of a research report are findings interpreted?" One real paper (RML 2023) keys *Results*. The correct answer is **Conclusion**, that is the Discussion: Results report what was found, statistics included, without interpretation, and the Discussion interprets.

The research proposal

A **research proposal** is the detailed plan of a study, an architect's blueprint, written to win approval and often funding. **A. K. Singh** gives nine steps: (1) the problem, in testable terms; (2) definitions (operational definitions, Section 9.2), assumptions, limitations and delimitations; (3) a selective review of related literature; (4) hypotheses; (5) methods: participants, procedure and data analysis; (6) a time schedule; (7) expected results, with alternative outcomes; (8) references; (9) an appendix of instruments and instructions.

Assumptions are statements the researcher believes but cannot prove. **Limitations** are conditions beyond the researcher's control that may weaken the conclusions, such as a low response rate. **Delimitations** are boundaries the researcher deliberately sets, such as studying only first-year students in Mysuru colleges, and they decide to whom the findings apply.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The Nuremberg Code of _____ declared voluntary consent "absolutely essential".
2. The Belmont Report's principles are respect for persons, beneficence and _____.
3. Telling participants enough to decide whether to take part is _____.
4. Full information about all aspects of the study, given after it, is _____.
5. Holmes's two parts of debriefing are dehoaxing and _____.
6. Data the researcher can link to a person but will not disclose are _____.
7. The 3Rs are replacement, reduction and _____.
8. Findings are interpreted in the _____ section, not the Results.

Answers: (1) 1947, (2) justice, (3) informed consent, (4) debriefing, (5) desensitising, (6) confidential, (7) refinement, (8) Discussion.

9.8 Measurement Scales, Organising Data and Descriptive Statistics

HIGH YIELD · 58 QS

Two hundred students in Mysuru have filled in an anxiety questionnaire, and you now own a notebook of numbers that, on their own, say nothing. Before any hypothesis can be tested (Sections 9.9 to 9.11), you must decide what kind of numbers you have, put them in order and describe them. This section does those jobs, ending with the normal curve and the z-score.

Levels of measurement ●●●

Measurement is the assignment of numbers to objects or events according to rules. But numbers mean different things: the 2 on a cricketer's shirt, 2nd rank in class, 2°C and 2 kg are all "2", yet only the last can be doubled meaningfully. In 1946 the Harvard psychophysicist **Stanley Smith Stevens** (1906 to 1973) sorted these meanings into four **levels of measurement** ("On the theory of scales of measurement", *Science*). Each level adds one property: **identity** (a number names a category), **magnitude** (order), **equal intervals** (a unit means the same amount anywhere on the scale) and an **absolute zero** (zero means none of the attribute).

A **nominal scale** has identity only: labels name unordered categories such as marital status, eye or car colour and gender, and coding "married" as 2 does not make it more than "single" at 1. An **ordinal scale** adds magnitude: cases are ranked, but the distances between ranks are unknown. Class ranks, socioeconomic status categories and any task in which respondents place items along a continuum by preference or degree of agreement give ordinal data. A single Likert item (strongly disagree 1 to strongly agree 5) is strictly ordinal, though a total over many items is conventionally treated as interval.

An **interval scale** adds equal intervals (20°C to 30°C is the same rise as 30°C to 40°C), but its zero is *arbitrary*, a reference point rather than the absence of the quantity: 0°C is just the freezing point of water. Differences are meaningful, ratios are not; 40°C is not twice as hot as 20°C. Celsius temperature, calendar years and most test scores, IQ included, are treated as interval. A **ratio scale** adds a *true* zero: height, weight, reaction time and the number of books a person owns (zero books is no books, and 6 is twice 3). The level decides which statistics are permissible (Table 9.23), and each level keeps every statistic of the levels below it (memory hook: NOIR).

Table 9.23 Stevens's four levels of measurement, from least to most informative.

Scale	Properties	Examples	Permissible statistics
Nominal	Identity only	Marital status, eye or car colour, gender	Frequency, percentage, mode, chi-square
Ordinal	Adds magnitude	Class rank, SES category, one Likert item	Adds median, percentiles, rank correlation
Interval	Adds equal intervals; arbitrary zero	Celsius temperature, IQ, most test scores	Adds mean, SD, Pearson r, t and F tests
Ratio	Adds a true zero	Height, weight, reaction time, books owned	All, including coefficient of variation

EXAM TRAP

One real paper keys "ratio scale" for "Arbitrary zero is the characteristic of which scale?". The key is wrong: an arbitrary zero defines the **interval** scale, and the ratio scale has a **true** zero. Some Indian texts cite "Stevens (1951)", his handbook chapter; the scales date from 1946.

KEY STUDY: STEVENS AND THE SCALES OF MEASUREMENT (1946)

Can sensations be measured at all? A committee of the British Association for the Advancement of Science had debated it for years without agreeing, its physicists doubting it (final report, 1940). Stevens replied by defining measurement broadly, as assigning numbers by rule, and by classing scales according to the transformations that leave their information intact: any one-to-one relabelling (nominal), any order-preserving change (ordinal), a linear change, aX + b (interval), and multiplication by a constant (ratio). The result was the four levels and their permissible statistics. Critics such as **Frederic Lord** (1953, "On the statistical treatment of football numbers") replied that the numbers do not remember where they came from, so scale type should not ban a statistic outright; the debate continues.

Two further classifications cut across the levels. A **categorical variable** (qualitative) sorts cases into classes; a quantitative one records an amount. A **discrete variable** takes separate values, usually counts (books owned, so a count can be discrete yet ratio); a **continuous variable** can take any value in a range (height, reaction time). Nominal variables are always categorical. Ordinal variables are strictly ordered categories, but because ranks are taken to lie on an underlying continuum (anxiety, agreement) they are usually analysed as continuous.

EXAM TRAP

"Ordered variables are considered ____, nominal variables ____" is keyed **continuous, categorical**. Take the key, knowing that some books call nominal and ordinal variables alike qualitative. "Discontinuous" simply means discrete.

A good measure is **reliable**, that is, consistent (it gives the same result when the same thing is measured again), **valid** (it measures what it claims to), **practical** (cheap, quick, easy to interpret) and **sensitive** to small differences. Objectivity belongs here too; high cost, low reliability and subjectivity mark a poor measure. Test reliability and validity are treated in Chapter 2.

Processing data and statistical software ●●●

Filled-in questionnaires become analysable data through four processing operations, in this order, as set out in **C. R. Kothari**'s *Research Methodology*. **Editing** examines raw data to detect and correct errors, omissions and inconsistencies (an age of 210, a blank item). Field editing is done by the investigator soon after the interview; central editing is done at the office, by one editor or team, once all forms are in. **Coding** assigns numerals or other symbols to answers so that responses fall into a limited number of exhaustive, mutually exclusive categories (male 1, female 2); the key to the codes is the codebook. **Classification** arranges the data into groups or classes sharing a characteristic, by attribute (urban or rural) or by class interval (ages 20 to 29). **Tabulation** arranges the classified data in rows and columns: a simple (one-way) table shows one characteristic, a complex table two or more. Graphs, charts and dia-

grams present the same data visually instead. On computer, data entry is followed by data cleaning: hunting for out-of-range values (7 on a 1-to-5 scale), duplicates and missing-value codes.

> **EXAM TRAP**
>
> Coding *assigns symbols* (the "assigning numerical values" option); classification *groups* data into classes. One real paper keys "organising data into categories or groups" for both: it is exact for classification and the nearest option for coding. Editing corrects errors; tabulation uses rows and columns.

IBM SPSS (Statistical Package for the Social Sciences), first released in 1968 and owned by IBM since 2009, is psychology's standard package; its primary purpose is data analysis, from descriptive statistics to factor analysis. R, SAS, Stata and the free JASP are alternatives. **Microsoft Excel** is a spreadsheet for data entry, tabulation, charts and calculation: =AVERAGE(A1:A10) gives the mean; SUM, MAX, MIN, MEDIAN and MODE do what they say; STDEV gives the sample SD, CORREL Pearson's r and COUNT the number of numeric cells. Word (report writing), Photoshop (photo editing), Illustrator (drawing) and InDesign (page layout) are not analysis tools.

Frequency distributions and graphs ●●

A list of raw scores, the scores as first recorded, says little. A **frequency distribution** shows how many cases (f) obtained each score (ungrouped) or fell in each class interval (grouped), the intervals being of equal width, usually 10 to 20 of them. An interval's apparent limits (25 to 29) are those written down; its **exact limits** (24.5 to 29.5) extend half a unit beyond, because a score of 29.4 is recorded as 29. The midpoint (27) stands for every score in it. **Cumulative frequency** (cf) is the running total up to an interval's top, and percentages restate f and cf out of 100 (Table 9.24).

Table 9.24 A grouped frequency distribution of 40 test scores.

Class interval	Exact limits	Midpoint	f (%)	cf (cumulative %)
40 to 44	39.5 to 44.5	42	2 (5)	40 (100)
35 to 39	34.5 to 39.5	37	4 (10)	38 (95)
30 to 34	29.5 to 34.5	32	8 (20)	34 (85)
25 to 29	24.5 to 29.5	27	12 (30)	26 (65)
20 to 24	19.5 to 24.5	22	8 (20)	14 (35)
15 to 19	14.5 to 19.5	17	4 (10)	6 (15)
10 to 14	9.5 to 14.5	12	2 (5)	2 (5)

A set of raw scores does not give a unique set of grouped scores: the interval width and starting point are the researcher's choice, and grouping treats every score as lying at its midpoint (grouping error). Pooled scores, by contrast, are simply scores from several groups combined. Raw counts also mislead when groups differ in size: 30 of 200 urban students against 15 of 50 rural ones. **Relative frequencies**, each f divided by N as a proportion (.15 against .30) or a percentage, make such distributions comparable.

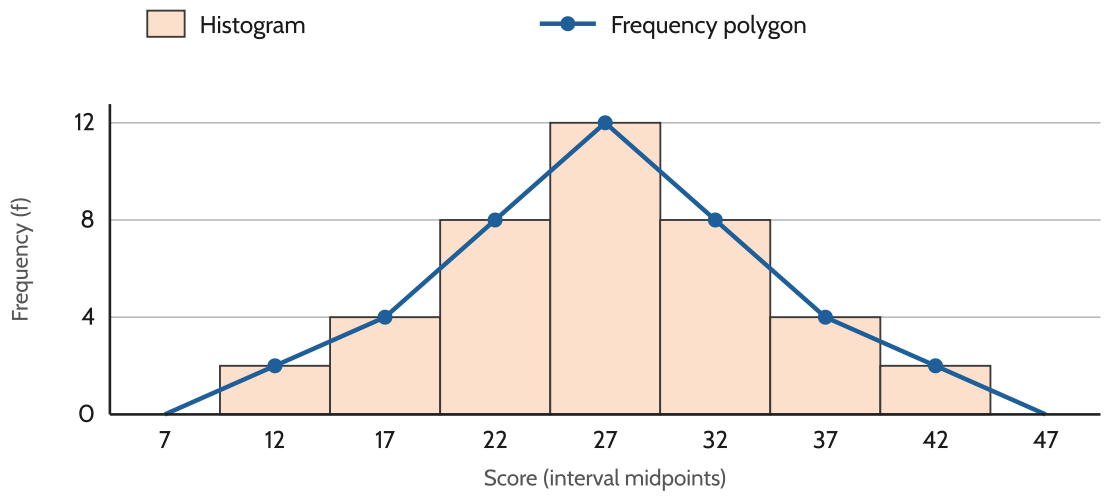


Figure 9.13 The 40 scores of Table 9.24 as a histogram and frequency polygon. The bars touch because the scale is continuous; the polygon joins the interval midpoints and closes on the baseline.

Categorical data take a **bar graph** (separate bars with gaps) or a pie chart (parts of one whole). Grouped continuous data take a **histogram**, adjoining rectangles each standing on a class interval with height equal to its frequency; joining the midpoints of the bar tops gives the **frequency polygon** (Figure 9.13), best for overlaying two distributions. Cumulative frequencies plotted against upper exact limits give the **ogive**, an S-shaped curve from which percentiles are read. **John Wilder Tukey** (1915 to 2000), in *Exploratory Data Analysis* (1977), added the stem-and-leaf display, which keeps every raw score, and the box plot (a box from Q_1 to Q_3, a median line and whiskers), good for comparing groups and spotting outliers. A line graph shows change over time; a **scatter plot** shows the relationship and spread of two continuous variables (Section 9.11).

Measures of central tendency ●●●

A **measure of central tendency** is one score that represents the whole distribution. The arithmetic **mean** ($M = \Sigma X / N$) is the balance point: deviations from it sum to zero, the sum of squared deviations about it is a minimum (least squares), and adding a constant to every score adds it to the mean, while multiplying every score by a constant multiplies the mean by it. Using every score makes it the most stable average across samples, but also lets one extreme score pull it: 4, 7, 7, 9 and 13 have a mean of 8, and adding a 40 lifts it to 13.3. It cannot be computed for an open-ended top class. A combined mean weights each group by its size: 30 boys averaging 60 and 20 girls averaging 70 give (1800 + 1400) / 50 = 64.

The **median** is the middle score of the ordered set, the 50th percentile (for 4, 7, 7, 9, 13, 40 it is 8). Depending only on position, it resists extreme scores, so it is the average for skewed data: community income, where a few rich households inflate the mean, and anxiety or symptom scores, where most people score low and a few very high. It also suits ordinal data.

The **mode** is the most frequent score and the only average for nominal data. One peak makes a distribution unimodal and two bimodal; when two or more scores share the highest frequency it is multimodal. The geometric mean (the Nth root of the product of N scores) suits growth rates; the harmonic mean (N divided by the sum of the reciprocals) suits rates such as speeds.

REMEMBER

Symmetrical interval or ratio data: **mean**. Skewed data, outliers or ordinal data: **median**. Nominal data: **mode**.

EXAM TRAP

Range, standard deviation, variance and **mean deviation** measure variability, not central tendency. Mean deviation is the favourite decoy because it contains the word "mean".

Measures of variability ●●●

Two classes may share a mean of 50 while one spans 45 to 55 and the other 10 to 90; variability (dispersion) describes this spread. The range (highest minus lowest score) rests on the two most extreme scores. The interquartile range (IQR = $Q_3 - Q_1$) covers the middle 50 per cent; half of it is the **quartile deviation** or semi-interquartile range, $Q = (Q_3 - Q_1) / 2$, the partner of the median. The mean deviation is the average absolute deviation from the mean, $\Sigma|X - M| / N$.

The **variance** squares the deviations instead: for a population $\sigma^2 = \Sigma(X - \mu)^2 / N$, and the **standard deviation** (SD, σ) is its square root, back in the original units: the typical distance of a score from the mean. For 2, 4, 4, 4, 5, 5, 7, 9 the mean is 5; the deviations −3, −1, −1, −1, 0, 0, 2, 4 square to 9, 1, 1, 1, 0, 0, 4, 16, a sum of squares (SS) of 32. So $\sigma^2 = 32 / 8 =$ **4** and $\sigma = 2$ (32 is the SS, 2 the SD, 5 the mean).

To estimate a population's variance from a sample, divide SS by N − 1: $s^2 = \Sigma(X - M)^2 / (N - 1)$. Deviations from a sample's own mean run slightly small, so dividing by N underestimates. The fix is Bessel's correction, after the astronomer **Friedrich Wilhelm Bessel** (1784 to 1846), and N − 1 is the **degrees of freedom**: once the mean is fixed, only N − 1 deviations can vary freely. For the sample 3, 5, 7: SS = 8 and $s^2 = 8 / 2 =$ **4**, so s = 2 (dividing by N gives 2.67). The **coefficient of variation**, CV = (SD / M) × 100, compares spread across different units but needs a ratio scale.

Adding a constant to every score shifts the mean but leaves the SD and variance unchanged; multiplying by c multiplies the SD by c and the variance by c^2. With variance 4 and scores multiplied by 3 and then increased by 10, the new SD is 6 and the new variance $4 \times 9 =$ **36** (not 12, which multiplies the variance by 3 only, nor 46, which also adds the 10).

REMEMBER

Affected by extreme scores: range, mean deviation, variance, SD and the mean. Resistant: IQR, quartile deviation and the median.

Skewness and kurtosis ●●

Skewness is asymmetry, and it is named after the long tail. In positive (right) skew most scores are low and a few high ones trail right (income, a very hard test); in negative (left) skew most are high and a few low ones trail left (a very easy test). The mean, which uses every score's value, is dragged furthest into the tail, the mode stays at the hump and the median lies between. So positive skew gives **mode < median < mean** and negative skew **mean < median <**

mode, the mean lying left of the median (Figure 9.14). **Karl Pearson** (1857 to 1936) measured it as Sk = 3(Mean – Median) / SD, zero for a symmetrical curve.

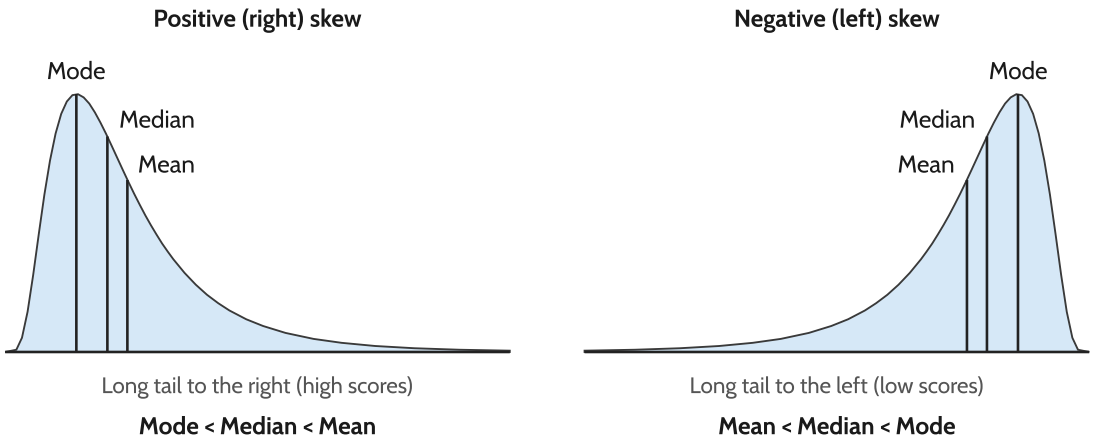


Figure 9.14 Skew is named after the tail. The mean is pulled furthest into the tail, the mode stays at the peak and the median lies between them.

Kurtosis is peakedness relative to the normal curve: leptokurtic (more peaked), mesokurtic (normal) or platykurtic (flatter). The percentile coefficient Ku = Q / (P_{90} – P_{10}) is 0.263 for the normal curve; above it is platykurtic, below it leptokurtic. On the moment measure the normal curve scores 3, and SPSS prints excess kurtosis (minus 3), so 0 means mesokurtic.

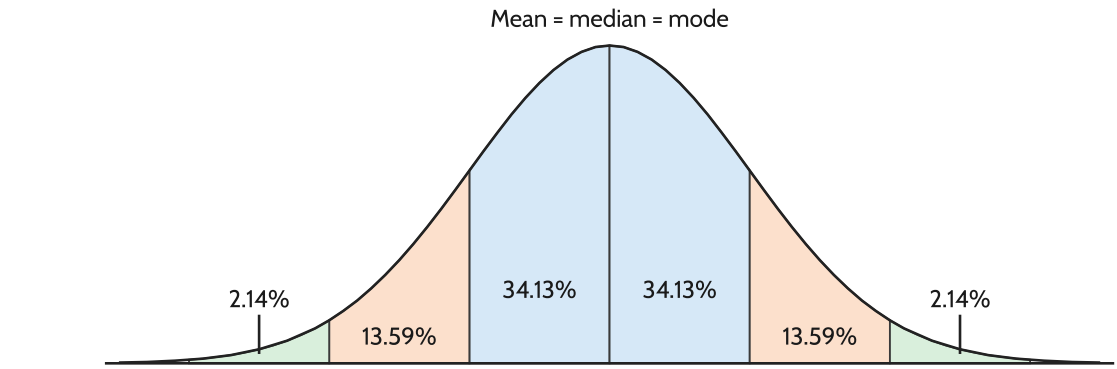
The normal probability curve ●●●

The **normal probability curve** was derived by **Abraham de Moivre** (1667 to 1754) in 1733 as the limit of the binomial distribution. **Pierre-Simon Laplace** (1749 to 1827) developed it, and **Carl Friedrich Gauss** (1777 to 1855) used it in 1809 for errors in astronomical measurement, hence the Gaussian curve. **Adolphe Quetelet** (1796 to 1874) applied it to human characteristics, and **Francis Galton** (1822 to 1911) to human ability.

> EXAM TRAP
> "Who propounded the normal curve?" is keyed **Carl Gauss**, after whom it is named; mark Gauss, but know that de Moivre came first (1733). **Siméon Denis Poisson** (1781 to 1840) gave the distribution of rare events, **Jacob (James) Bernoulli** the law of large numbers (1713), and **Wilhelm Reich** (1897 to 1957) was a psychoanalyst.

The curve is bell-shaped, symmetrical and unimodal, with mean, median and mode coinciding at the centre; a two-peaked curve cannot be normal. Its tails are asymptotic, nearing the baseline without touching it. The total area under it is 1 (100 per cent), so areas read as probabilities. Its points of inflection lie at ±1 SD, and it is defined by its mean and SD alone.

The areas are therefore fixed (Figure 9.15). On each side, 34.13 per cent of cases lie between the mean and 1 SD, 13.59 per cent between 1 and 2 SD and 2.14 per cent between 2 and 3 SD. About **68 per cent** fall within ±1 SD, **95 per cent** within ±2 SD (exactly 95 within ±1.96) and **99.7 per cent** within ±3 SD.



z score	−3	−2	−1	0	+1	+2	+3
T score	20	30	40	50	60	70	80
Deviation IQ	55	70	85	100	115	130	145
Percentile	0.1	2	16	50	84	98	99.9

Figure 9.15 The normal curve with the percentage of cases in each SD band. The rows beneath give each point as z, T, deviation IQ (SD 15) and percentile rank.

z-scores, percentiles and standard scores ●●●

You score 65 in statistics (class mean 50, SD 10) and 70 in English (mean 60, SD 20). Which is better? A **z-score** states how many SDs a score lies from its mean: z = (X − M) / SD. Statistics gives z = +1.5 and English z = +0.5, so the lower raw score is the better performance. That is the purpose of z-scores: they let you compare results from different distributions. A z of +1 lies one SD above the mean, −1 one SD below, and 0 at the mean itself. A set of z-scores has mean 0 and SD 1, and because the conversion is linear, a skewed set stays skewed. A z-score shows relative position. It is not a measure of variability (the SD, variance or quartile deviation) or of central tendency, nor the coefficient of variation, nor the standard error (the SD of a sampling distribution, Section 9.9). The z table, derived from the normal distribution, converts z into percentiles.

A **percentile rank** is the percentage of cases below a score. Quartiles split the distribution into four ($Q_1 = P_{25}$, Q_2 = the median, $Q_3 = P_{75}$), deciles into ten. On the normal curve z = 0 is the 50th percentile, +0.5 about the 69th, +1 the 84th, +1.28 the 90th, +1.5 the 93rd, +1.96 the 97.5th, +2 about the 98th and +3 the 99.9th; z = −1 is the 16th and −2 the 2nd. Percentile units are unequal (crowded in the middle), so they are never averaged.

Deviation IQs have mean 100 and SD 15 (**David Wechsler**'s system, Chapter 2): IQ 100 is the 50th percentile, 85 the 16th, 115 the 84th and 130 (z = +2) about the 98th. Older Binet-type tests used SD 16, so an IQ of 136 is z = 36 / 16 = +2.25; the table puts .9878 below it, so about 98.7 to 98.8 per cent score lower.

CLINICAL LINK

An IQ of 70 lies two SD below the mean, near the 2nd percentile. DSM-5 places the intellectual deficits of intellectual disability at about this point, roughly 65 to 75 once measurement error is allowed for (Chapter 8).

Other **standard scores** avoid the signs and decimals of z: new score = new mean + (new SD × z), so z = +2 is a T of 70, an IQ of 130 and a stanine of 9 (Table 9.25). Such linear standard scores keep the raw distribution's shape; normalised standard scores (normalised T scores, stanines) convert percentile ranks into normal-curve z values, forcing a normal shape. Norms are treated in Chapter 2.

Table 9.25 Common standard-score systems.

System	Mean	SD	Where you meet it
z-score	0	1	The base of all the others
T score	50	10	MMPI and most personality inventories
Deviation IQ	100	15	Wechsler scales; Stanford-Binet 5
Old Stanford-Binet IQ	100	16	Earlier Binet-type tests
Stanine	5	2	Nine bands, 1 to 9
Sten	5.5	2	Ten bands; the 16PF
Scaled score	10	3	Wechsler subtests

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Equal intervals with an arbitrary zero define the _____ scale.
2. The only average usable with nominal data is the _____.
3. The representative income of a community is best given by the _____.
4. In negative skew the mean lies to the _____ of the median.
5. The population variance of 2, 4, 4, 4, 5, 5, 7, 9 is _____.
6. Multiplying every score by 3 multiplies the variance by _____.
7. About _____ per cent of normal-curve cases lie within ±2 SD of the mean.
8. A deviation IQ of 85 lies at about the _____ percentile.

Answers: (1) interval, (2) mode, (3) median, (4) left, (5) 4, (6) 9, (7) 95, (8) 16th.

9.9 Sampling Distributions and the Logic of Hypothesis Testing

MEDIUM YIELD · 37 QS

At the Rothamsted research station in the 1920s, the biologist **Muriel Bristol** claimed she could taste whether the milk or the tea had gone into her cup first. **Ronald Aylmer Fisher** (1890 to 1962) turned the claim into the opening example of *The Design of Experiments* (1935): eight cups, four made each way, in random order. A pure guesser would sort all eight correctly only 1 time in 70 (p = .014). That is the whole logic of this section: assume nothing is going on, work out how probable the result would be by chance, and reject chance when the result would be rare.

From describing to inferring ●●

Descriptive statistics summarise the scores you have (section 9.8). **Inferential statistics** reason from a sample to the population it came from, drawing on two things: the sample data, and knowledge of how random processes behave. Their question is how plausible it is that a random process alone, drawing samples from a population in which nothing is going on, could produce results as extreme as those obtained. Inference always runs from the observed sample to the unobserved population, never the reverse.

If a therapy group improves by 6 points and a control group by 4, part of the gap is chance, because two random samples almost never have identical means. What statistics (applied to health data, **biostatistics**) tells the psychologist is whether an observed difference is larger than chance would usually produce. Only a statistical test can show a reliable difference between groups; norms (a reference group's typical scores), reliability (consistency) and validity (measuring what is claimed) are properties of tests (Chapter 2).

> EXAM TRAP
> "Inferential statistics make statements about samples based on observing populations" is the FALSE option: it reverses the logic. Statistics does not choose patients for therapy, design interviews or plan therapeutic exercises; it judges whether differences are due to chance.

Probability: the language of chance

A probability runs from 0 (impossible) to 1 (certain). **Classical probability** is worked out in advance from equally likely outcomes, favourable outcomes divided by all outcomes: a die shows a six with probability 1/6. **Empirical probability** is the observed relative frequency over many trials: if 30 of 200 discharged patients relapse, p(relapse) is about .15. The **addition rule** handles "or": for mutually exclusive events p(A or B) = p(A) + p(B), so a one or a two on a die is 2/6. The **multiplication rule** handles "and": for independent events p(A and B) = p(A) × p(B), so ten heads in a row has probability 1/1024. For dependent events you multiply by a **conditional probability**, p(B given A); remember it when you meet the p-value.

The **binomial distribution** gives the probability of each number of successes in N independent two-outcome trials with fixed probability p (mean Np, SD √Npq): ten coin tosses average 5 heads, yet exactly 5 occurs only about one time in four. As N grows the binomial becomes bell-shaped, as **Abraham de Moivre** (1667 to 1754) showed in 1733. The normal distribution is therefore the probability model of inference: area under it is probability, and since 95 percent lies within ±1.96 SD (section 9.8), a chance value beyond those points has probability .05.

The sampling distribution and the standard error ●●

Draw a random sample of 36 students, compute their mean, replace them, and repeat thousands of times. The means vary, and the theoretical distribution they form is the **sampling distribution of the mean**. You never build it, but three facts about it let you judge your one real sample. Its mean equals the population mean, μ. Its standard deviation is the **standard error of the mean**, SE = σ/√N (estimated as s/√N): for IQ (σ = 15), samples of 25 give SE = 3 and samples of 100 give 1.5, so halving the SE takes four times the sample. And its shape is normal once N is large.

That last fact is the **central limit theorem**: with a sufficiently large sample, conventionally 30 or more (A.K. Singh says above 50), the sampling distribution of the mean is approximately normal whatever the shape of the population, centred on μ, with a spread that shrinks as N grows (Figure 9.16). De Moivre's coin result was its first case and **Pierre-Simon Laplace** (1749 to 1827) generalised it. It is why tests on means can use the normal curve even though psychological scores are often skewed.

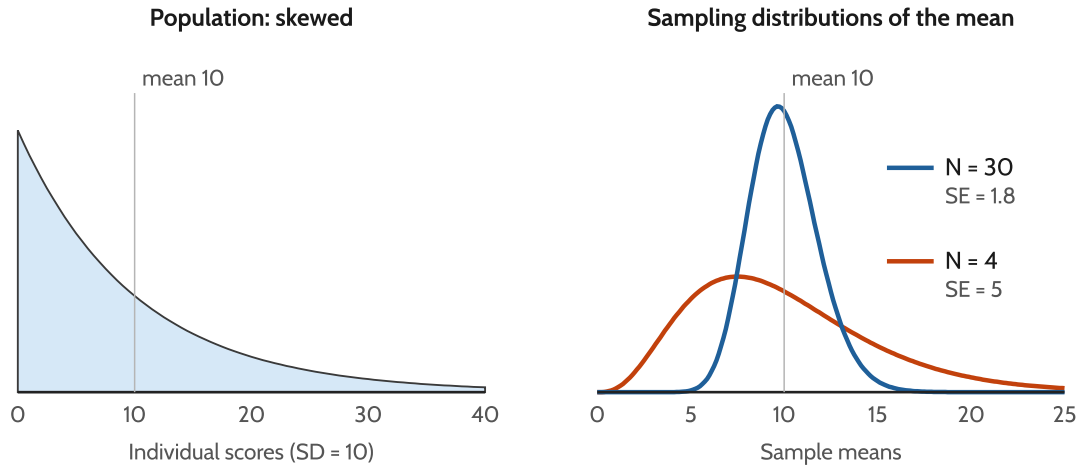


Figure 9.16 The central limit theorem. Means of samples from a skewed population cluster around its mean; with N = 30 they are nearly normal and far narrower (SE about 1.8) than with N = 4.

Sampling error is the gap between one sample's statistic and the parameter (M − μ); it is a random error, unsystematic and averaging to zero. The standard error is the typical size of that gap across all samples. The **law of large numbers** (**Jacob Bernoulli**, 1655 to 1705, published 1713) says only that a sample mean gets closer to the population mean as N grows; it says nothing about the shape of the sampling distribution.

> EXAM TRAP
>
> The standard error is **the SD of the sampling distribution of the mean**: not the SD of the sample (which does not shrink with N) and not M − μ (sampling error). As N rises the sampling distribution approaches normal; it does not grow skewed, does not become identical to the population distribution, and its SE gets smaller. "Normal distribution theorem" and "sampling distribution theorem" are invented names.

Estimation and confidence intervals

A **point estimate** is a single value, such as a sample mean of 52 for μ. An interval estimate, the **confidence interval** (CI), adds a margin: M ± 1.96 SE for 95 percent and M ± 2.58 SE for 99 percent. With M = 52, SD = 12 and N = 36, SE = 2, so the 95 percent CI runs from 48.08 to 55.92 and the 99 percent CI, wider, from 46.84 to 57.16. Higher confidence, a larger SD and a smaller N all widen the interval.

The population mean is fixed, so the confidence belongs to the procedure: if the sampling were repeated many times, 95 percent of the intervals so built would contain the true mean. If a 95 percent CI for a difference includes zero, the difference is not significant at α = .05 (two-

sided): a mean difference of 0.4 with a CI of −0.1 to 0.9 is non-significant. That is not proof of no effect, since the interval also allows 0.9; a wide CI signals an imprecise estimate. For ratios the null value is 1 (section 9.5).

> **EXAM TRAP**
>
> A 95 percent CI is NOT "a 95 percent probability that μ lies in this interval", NOT "95 percent of individual scores fall in it" and NOT "a 95 percent chance of significance". A CI crossing zero means non-significant, not a difference of exactly zero.

The logic and steps of a significance test ●

A significance test assumes the null hypothesis (H_0: no difference or effect), asks how probable the data would be if H_0 were true, and rejects H_0 in favour of the alternative hypothesis (H_1) if they would be rare (wording hypotheses: section 9.1). The steps:

1. State H_0 and H_1; choose a one-tailed or two-tailed test.
2. Set the **significance level** (α), usually .05 or .01, before seeing data.
3. Compute the test statistic (z, t, F, chi-square): the observed effect over its standard error.
4. Compare it with the **critical value** bounding the critical region, or compare p with α.
5. Reject H_0 or fail to reject (retain) it; report the effect size and CI.

A stress scale has a population mean of 50 and SD of 12; after a mindfulness course 36 students average 55. Then $z = (55 - 50)/(12/\sqrt{36}) = 2.5$: beyond 1.96, so significant at .05 two-tailed (p about .012), but short of 2.58, so not at .01. Table 9.26 gives the cut-offs. They explain the rough rule that a score is significant when it lies **more than two standard deviations from the mean**: about 95 percent of a normal distribution lies within two SDs, so a value beyond is in the outer 5 percent, while values within one SD, or between one and two, are ordinary chance results.

Table 9.26 Critical values of z.

α	Two-tailed	One-tailed	Matching CI
.05	±1.96	1.645	95 percent
.01	±2.58	2.33	99 percent

A **two-tailed test** (non-directional H_1) splits α between the tails, .025 each. A **one-tailed test** (directional H_1) puts all of α in one tail, so z = 1.80 is significant one-tailed but not two-tailed. It is more powerful in the predicted direction, blind in the other, and legitimate only if the direction was fixed in advance. Tests other than z also need **degrees of freedom** (df), the number of values free to vary once a constraint is fixed: five scores with a set mean of 10 leave four free, df = N − 1 = 4. Critical values of t, F and chi-square depend on df (section 9.10).

The p-value and what it is not ●●●

The **p-value** is the probability of obtaining the observed result, or one more extreme, if H_0 is true: the probability, revealed by a statistical test, that a random process (random samples of a given size from a given population) could produce that outcome. If $p \le \alpha$, reject H_0: with α = .05, p = .03 is statistically significant and H_0 is rejected.

Because p assumes H_0 and asks about the data, it is not the probability that H_0 (or H_1) is true, which would need Bayes' theorem. Nor is it the effect size (a huge sample makes a trivial effect give tiny p), the probability of replication, or power. It is continuous: p = .049 and p = .051 are essentially the same evidence, because .05 is a convention, not a natural boundary. Two distractors share a letter: the **q-value**, the false-discovery-rate analogue of p used when thousands of tests are run (brain imaging, genetics), and Tukey's studentised range q (section 9.10).

> EXAM TRAP
>
> The key is always "the probability of the data (or more extreme) **if H_0 is true**"; the favourite wrong option reverses it, "the probability that H_0 is true". Nor is p the magnitude of a correlation or the sample size, and p = .001 does not mean a large, clinically important or certain-to-replicate effect.

Type I and Type II errors ●●●

A decision from a sample can go wrong in two ways (Table 9.27). A **Type I error** (alpha error) is rejecting a true null hypothesis: a false positive. Its probability is α, so the significance level is the probability of a Type I error, in analysis of variance as in every test; it does not certify that F is correct, and the Type II risk is β, not α. A **Type II error** (beta error) is retaining, or loosely "accepting", a false null hypothesis: a false negative, a real effect missed. Its probability, β, is not chosen; it follows from N, the true effect, α and variability.

Table 9.27 The four outcomes of a significance test.

Decision	H_0 true	H_0 false
Reject H_0	Type I error (α), false positive	Correct: power (1 − β)
Retain H_0	Correct (1 − α)	Type II error (β), false negative

In court terms, a Type I error convicts the innocent and a Type II error frees the guilty. With N fixed, a stricter α (.05 to .01) cuts Type I errors but raises β; only a larger sample or a more precise measure reduces both (Figure 9.17). A non-significant result is a failure to reject H_0, never proof of it.

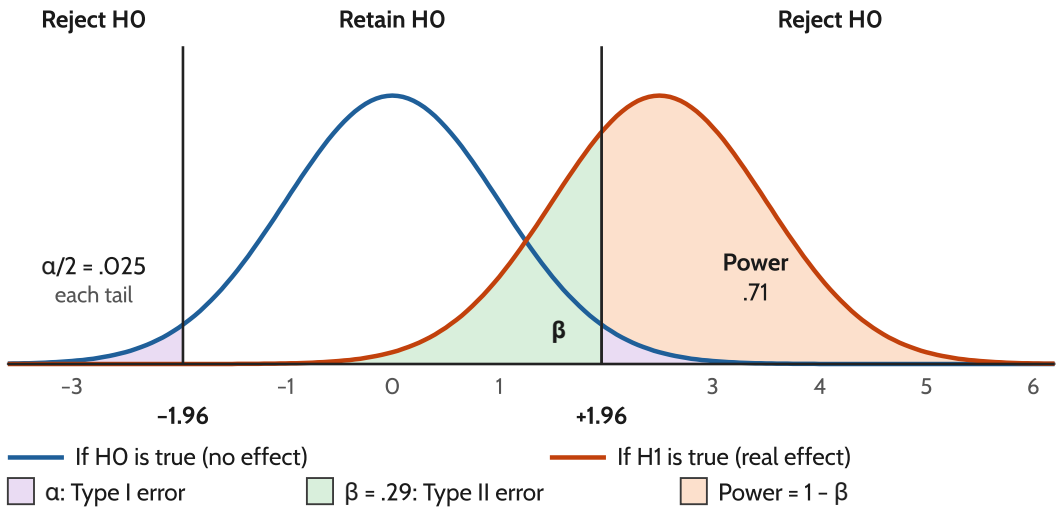


Figure 9.17 Errors and power in a two-tailed z test at α = .05. Under H_0 (solid curve) a result beyond ±1.96 is a Type I error; if the true effect is 2.5 standard errors (dashed curve), the test misses it 29 percent of the time (β) and detects it 71 percent of the time (power).

The procedure fuses two rival traditions. Fisher (*Statistical Methods for Research Workers*, 1925) treated p as graded evidence against a single null hypothesis and offered .05 as a convenient cut-off. **Jerzy Neyman** (1894 to 1981) and **Egon Pearson** (1895 to 1980), in papers of 1928 and 1933, recast testing as a decision between H_0 and H_1 with fixed error rates, giving us Type I and II errors and power.

> EXAM TRAP
>
> Swapped wordings ("accepting a false null" offered as Type I) are the commonest wrong options. **Type III error** is informal: the right answer to the wrong question, or rejecting H_0 but in the wrong direction; "Type IV" is rarer still. Low power, sample size (too small, too large or "large sample bias"), a wide CI, a misused one-tailed test, sampling error, standard error and confounding (section 9.2) are not definitions of these errors, though several change their risk.

Statistical power

Power (1 − β) is the probability of correctly rejecting a false null hypothesis, of finding an effect that is really there; the convention is at least 0.80. Power rises with a larger true effect, a larger N (it shrinks the SE, pulling the curves of Figure 9.17 apart), a larger, less strict α (at the price of more Type I errors), lower variability (homogeneous samples, standard procedures, repeated measures, reliable measures) and a correctly predicted one-tailed test. It falls when N shrinks, α becomes stricter or variability grows. **Power analysis**, done before a study, fixes α, the desired power and the smallest effect worth detecting, and solves for N. **Jacob Cohen** (1923 to 1998) made it routine in *Statistical Power Analysis for the Behavioral Sciences* (2nd edition, 1988): detecting a medium effect (d = 0.5) with an independent-samples t-test at α = .05, two-tailed, and power .80 needs about 64 people per group.

Effect size and practical significance ●●

Significance says an effect is probably not zero; **effect size** says how big it is, independent of N. **Cohen's d** is the standardised difference between two means, $(M_1 - M_2)$/pooled SD, with benchmarks 0.2 small, 0.5 medium and 0.8 large. The mindfulness example gives d = 5/12 = 0.42, between small and medium, despite p = .012. The correlation r is also an effect size (.10, .30, .50); its sign gives only direction, so r = −.30 is a medium negative relationship, not "none", not perfect (−1.00) and not causal (section 9.11). Table 9.28 adds the variance-explained indices; the odds ratio belongs to section 9.5.

Table 9.28 Common effect sizes and Cohen's benchmarks.

Index	Expresses	Small	Medium	Large
Cohen's d	Mean difference in SD units	0.2	0.5	0.8
r	Strength of a linear relationship	.10	.30	.50
Eta squared (η^2)	Share of total variance explained (ANOVA)	.01	.06	.14

Eta squared is SS_{effect}/SS_{total}; **partial eta squared**, $SS_{effect}/(SS_{effect} + SS_{error})$, drops the other factors from the denominator, so values across effects can sum past 1; omega squared (ω^2) is a less biased, smaller estimate. Statistical significance (data surprising under H_0, driven by N) differs from practical significance (an effect big enough to matter) and **clinical significance** (patients really function better, often back in the normal range): with 10,000 participants r = .03 is significant and meaningless. **p-hacking**, analysing data many ways (extra measures, dropped cases, stopping once p dips below .05, a one-tailed test chosen afterwards) until something is significant, inflates Type I errors; preregistration, power analysis and reporting effect sizes with CIs are the remedies.

KEY STUDY: THE OPEN SCIENCE COLLABORATION (2015)

About 270 researchers coordinated by **Brian Nosek** repeated 100 studies published in 2008 in three leading psychology journals. Of the originals, 97 percent were significant; only 36 percent of the replications were, with effect sizes about half as large. The project made the **replication crisis** famous: small, underpowered studies plus flexible analysis yield fragile findings, although critics noted that some replications departed from the originals.

EXAM TRAP

d of 0.2, 0.5 and 0.8 = small, medium, large (not "trivial, weak, moderate"). d measures standardised effect size for a difference between two means, not significance, sample-size adequacy or correlation strength.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The SD of the sampling distribution of the mean is the _____.
2. The _____ theorem says that with a large sample the sampling distribution of the mean is approximately normal, whatever the population's shape.
3. The 99 percent confidence interval is the mean plus or minus _____ standard errors.
4. The p-value is the probability of the observed result, or one more extreme, if the _____ is true.
5. Rejecting a true null hypothesis is a _____ error.
6. Power equals 1 minus _____.
7. A Cohen's d of 0.5 is a _____ effect.

Answers: (1) standard error of the mean, (2) central limit, (3) 2.58, (4) null hypothesis, (5) Type I, (6) β, (7) medium.

9.10 Comparing Groups: t-Tests, ANOVA and Non-Parametric Tests

HIGH YIELD · 57 QS

A counsellor at a Mysuru college tries three ways of revising for examinations: practice testing, flashcards and plain rereading. A month later the groups' recall scores differ a little. Is the difference real, or the wobble any three groups would show by chance? Section 9.9 gave the logic of significance testing; here are the tools. Choosing one always comes down to three questions: what kind of data, how many groups, and are the groups independent or made of the same (or matched) people?

Parametric or non-parametric? ●●

A **parametric test** estimates population parameters (means and variances) and rests on four assumptions: the scores come from a **normally distributed** population; the groups have roughly equal variances (**homogeneity of variance**); observations are **independent** of one another; and the dependent variable (DV) is measured on an **interval or ratio** scale, so that means make sense. The z-test, t-tests, ANOVA's F-test and Pearson's r are parametric. When data are normal with equal variances, a parametric test is the right choice, because it uses the actual distances between scores and so has the most power.

A **non-parametric test**, also called a **distribution-free test**, makes few assumptions about the population's shape and works on ranks, signs or frequency counts. **Sidney Siegel** (1916 to 1961) brought these tests into psychology with *Nonparametric Statistics for the Behavioral Sciences* (1956). They suit ordinal data (ranks), nominal data (counted categories), very small samples and clearly violated assumptions (heavy skew, outliers, very unequal variances). Rank order is thus the classic non-parametric data, whereas height, weight and Celsius temperature are interval or ratio measures (9.8). Chi-square is not a third family: it is the non-parametric test for nominal frequencies.

The price is power, because ranking discards the distances between scores. Siegel measured the cost as **power-efficiency**: the sample a parametric test needs, as a percentage of the sam-

ple a non-parametric test needs for equal power. If a t-test needs 95 cases to match what the Mann-Whitney test does with 100, Mann-Whitney is 95 percent power-efficient. Under normality the Mann-Whitney and Wilcoxon tests reach about 95 percent; the sign test only about 63 percent in large samples.

> EXAM TRAP
>
> "Which is non-parametric data?" Rank order, not height, weight or Celsius temperature (Celsius has an arbitrary zero but equal intervals: interval data). "Distribution-free" is just another name for non-parametric. For normal data with equal variances the answer is parametric tests, never "chi-square tests" or "distribution-free tests".

Checking the assumptions •

Normality is tested with the **Shapiro-Wilk test** (the usual choice) or the **Kolmogorov-Smirnov test**, which compares the sample's cumulative distribution with a normal one; it is inspected with a **Q-Q plot** (quantile-quantile plot), where normal data hug a straight diagonal, and with skew and kurtosis values near zero (9.8). **Homogeneity of variance**, or **homoscedasticity**, is tested with **Levene's test** (**Howard Levene**, 1960), which stays accurate when data are not normal; **Bartlett's test**, more powerful but upset by non-normality; or **Hartley's F-max**, the largest group variance divided by the smallest (near 1 is fine). These checks precede a t-test, an ANOVA or a regression. **Independence** has no test: it is secured by random sampling and assignment, and its violation is the most damaging.

Repeated-measures ANOVA adds **sphericity**: equal variances of the difference scores between every pair of conditions. It matters only with three or more conditions, is tested by **Mauchly's test**, and when violated the df are reduced by the **Greenhouse-Geisser** correction (more conservative) or the **Huynh-Feldt** correction (less so). ANCOVA adds homogeneity of regression slopes (9.10.6). Read these tests backwards: their null hypothesis is that the assumption holds, so a non-significant result ($p > .05$) is good news.

> EXAM TRAP
>
> Shapiro-Wilk tests normality, NOT homogeneity of variance, so it is the odd one out beside Levene's and Bartlett's tests. Levene's test checks equal variances, not normality, independence or equal group sizes. Equal group sizes are not an assumption, though they make ANOVA more forgiving of unequal variances.

The t-tests •••

A **z-test** compares means using the population standard deviation, which you almost never know (older texts allow it above about 30 cases). Estimating the standard deviation from the sample adds a second source of error, and the statistic then follows the t distribution.

KEY STUDY: "STUDENT" AT THE GUINNESS BREWERY (1908)

William Sealy Gosset (1876 to 1937), a chemist at the Guinness brewery in Dublin, had to judge barley and stout from tiny samples. He worked out the exact distribution of a mean divided by its estimated standard error. Guinness barred staff from publishing under their own names, so his 1908 *Biometrika* paper appeared as "Student", hence Student's t. (Some Indian texts print 1915; the paper is from 1908.)

The **t distribution** is a family of curves, one for each number of degrees of freedom (df). Like the normal curve it is symmetrical and bell-shaped around zero; unlike it, it has heavier tails and a lower peak, so its variability and kurtosis are greater. As df rise it closes in on the normal curve (Figure 9.18), which is why critical t values shrink towards 1.96 in large samples.

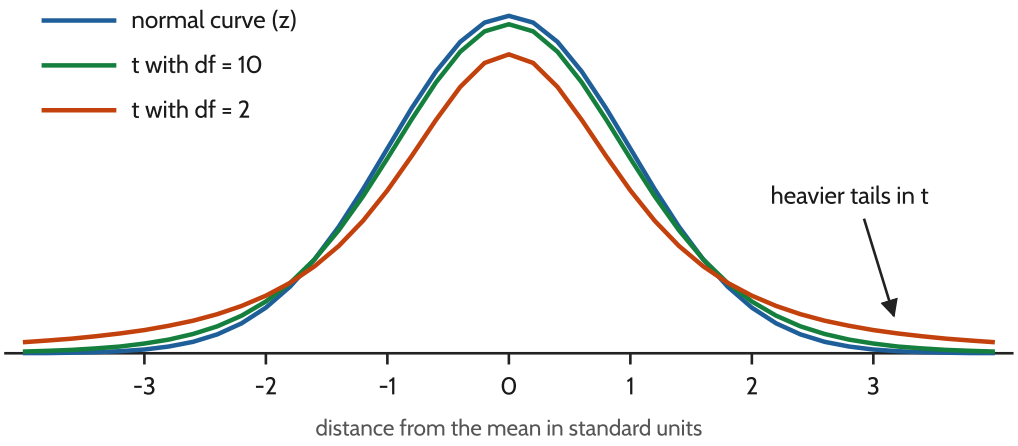


Figure 9.18 The t distribution against the normal curve. All are symmetrical, but with few df t has a lower peak and heavier tails, so a larger t is needed for significance; by df = 10 it is already close to normal.

There are three t-tests, each with its own df.

- The **one-sample t-test** compares one sample mean with a known population value; df = N - 1, since once the mean is fixed the last score cannot vary. N = 20 gives 19; N = 30 gives 29.
- The **independent-samples t-test** (unpaired t) compares the means of two separate groups, using their pooled variance. Its df = N1 + N2 - 2: 25 per group gives 48. If Levene's test shows unequal variances, **Welch's t-test** is used; it does not pool the variances and lowers the df.
- The **paired-samples t-test** (dependent or matched-pairs t) is for the same people measured twice, pre and post, or for matched pairs. It tests whether the mean difference score is zero; df = pairs - 1. Each person acts as their own control, so individual differences leave the error term and power rises.

Why not run three t-tests on three groups? Each carries a 5 percent Type I risk (9.9), and the risks accumulate. With c tests the chance of at least one false positive, the **family-wise error rate**, is $1 - (1 - .05)^c$: about .14 for three tests and .26 for the six that four groups need. ANOVA tests all the means at once at one alpha.

EXAM TRAP

One-sample df: N - 1. Independent: N1 + N2 - 2 (25 per group gives 48, not 24, 49 or 50). Paired: pairs - 1. The t curve shares only its symmetry with the normal curve; its kurtosis and variability are larger.

One-way ANOVA: partitioning variance ●●●

Analysis of variance (ANOVA) was developed by **Ronald Aylmer Fisher** (1890 to 1962) for agricultural trials at Rothamsted Experimental Station and published in *Statistical Methods for Research Workers* (1925); **George Waddel Snedecor** (1881 to 1974) named its statistic F in Fisher's honour. A **one-way ANOVA** compares the means of independent groups formed by one independent variable (IV, or factor) on one continuous DV. It is used when more than two means are compared, though it works for two as well, where $F = t^2$.

ANOVA compares means by analysing variances. The deviations of all scores from the **grand mean** sum to zero, so they are squared and added to give the **total sum of squares**. Fisher split it into exactly two parts. The **between-groups sum of squares** reflects how far each group mean lies from the grand mean: treatment effect plus chance. The **within-groups sum of squares** reflects how far each person lies from their own group mean: since a group was treated alike, this is pure chance or error. SS total = SS between + SS within, and each part has its own df: between = k - 1 (k groups), within = N - k, total = N - 1. Three groups of 10 give 2, 27 and 29.

Each SS divided by its df is a **mean square** (MS), a variance. MS within is the **pooled error variance**, and it is the denominator of the **F ratio**: F = MS between / MS within. If the null hypothesis is true, both estimate the same chance variance and F is about 1; a real effect swells only the numerator. So an F below 1, such as 0.99, can never be significant whatever the df, and as variability due to chance falls, F rises. Figure 9.19 and Table 9.29 work through the revision study.

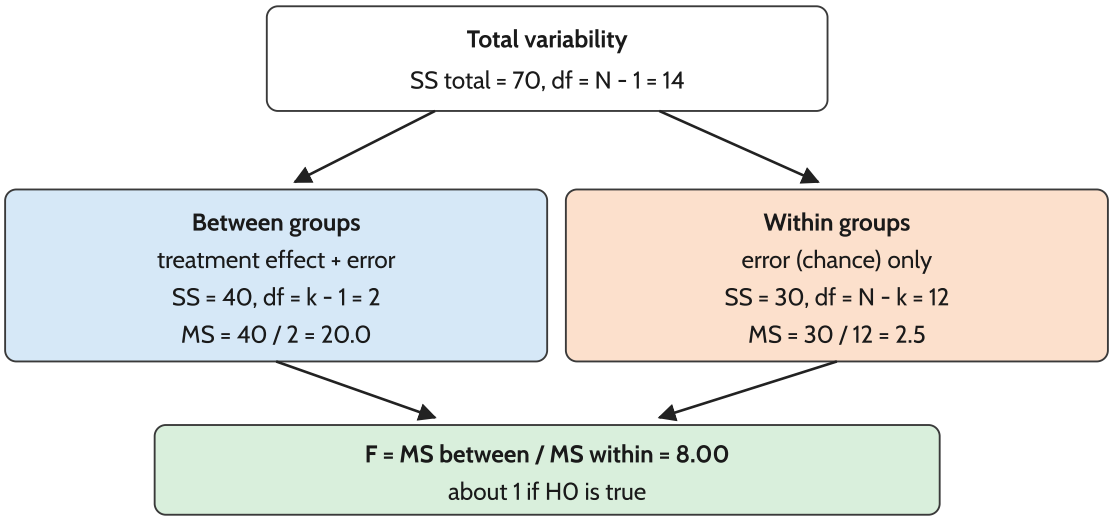


Figure 9.19 Fisher's partition of variance. The two sums of squares add up to the total, each has its own df, and the within-groups (error) mean square is always the denominator of F.

Table 9.29 A worked one-way ANOVA. Practice testing 9, 8, 10, 7, 6 (mean 8); flashcards 6, 5, 7, 4, 8 (mean 6); rereading 4, 3, 5, 2, 6 (mean 4); grand mean 6.

Source	SS	df	MS	F
Between groups	$5(2)^2 + 5(0)^2 + 5(-2)^2 = 40$	3 - 1 = 2	20.0	8.00
Within groups (error)	10 + 10 + 10 = 30	15 - 3 = 12	2.5	
Total	70	15 - 1 = 14		

The critical F for df 2 and 12 is 3.89 at .05 and 6.93 at .01, so the methods differ ($p < .01$), and condition accounts for 40/70, or 57 percent, of the variance (eta squared, 9.9). But F is an **omnibus test**: it says at least one mean differs, not which. Being a ratio of two variances, the **F distribution** is never negative and is positively skewed, its shape set by two df (numerator and denominator); the test is one-tailed, because only a large F counts against the null.

> EXAM TRAP
>
> F's denominator is MS within (the pooled error variance), not MS between. One-way ANOVA has two sources of variance, each with its own df; "error variance" is the within-groups part, not a third source, and "information variance" does not exist. F = 0.99 is never significant; F is never negative. ANOVA tests means, not "variance only", and unequal variances violate its assumption.

After a significant F: post hoc tests ●●

Post hoc tests (a posteriori or multiple comparison tests) compare groups pair by pair while holding the family-wise error rate down. They are used only after a significant F; after a non-significant F you stop. With k groups there are k(k - 1)/2 pairs: 3 for three groups, 6 for four, 10 for five. The tests differ in how conservative they are, that is, how hard they make it to call a difference significant (Table 9.30).

Table 9.30 Common post hoc tests. The first five run from most liberal to most conservative.

Test	What it does	Conservatism
Fisher's LSD (least significant difference)	Plain t-tests, allowed only after a significant F	Least; most power
Duncan's multiple range test	Stepwise comparison of ordered means	Liberal
Newman-Keuls	Stepwise; critical value depends on how far apart the means rank	Between LSD and Tukey
John Wilder Tukey's HSD (honestly significant difference)	All pairs, using the studentised range q; the standard choice with equal n	Moderate
Henry Scheffé's test	Any comparison, including complex ones (group 1 against groups 2 and 3)	Most conservative
Dunnett's test	Each treatment against one control group	Powerful for that job
Games-Howell	Pairs when variances or group sizes are unequal	Moderate

Planned contrasts (a priori comparisons) are instead fixed from theory before data collection; they are few (at most k - 1 independent ones), need no significant F and keep more power. The simplest guard for any set of comparisons is the **Bonferroni correction**, after **Carlo Emilio Bonferroni** (1892 to 1960): divide alpha by the number of comparisons. Six compar-

isons are each tested at .05/6 = .0083, keeping the family-wise Type I error rate at .05 or below. The cost is power: Bonferroni is conservative, raising Type II risk as comparisons multiply. A single planned comparison needs no correction.

> EXAM TRAP
>
> Duncan, Newman-Keuls, Tukey and Scheffé are used only when F is significant. Bonferroni divides alpha, never multiplies it (software may multiply each p-value by the number of tests, the same rule). It controls family-wise Type I error; it does not raise power, fix unequal sample sizes or transform skewed data (skew calls for a log or square-root transformation, or a rank test).

Factorial, repeated-measures and covariance designs ●●

A **factorial ANOVA** (two-way or higher) crosses two or more IVs so that every level of one meets every level of the other; a 2 × 3 design has two factors and six cells. It tests a **main effect** of each IV (averaged over the other) and an **interaction effect**, present when the effect of one IV depends on the level of another: plotted, the lines are not parallel. Of the familiar techniques only factorial ANOVA detects interactions; a t-test has one two-level IV, chi-square tests association between two categorical variables, and correlation measures how two variables move together. A significant interaction is interpreted before the main effects.

Repeated-measures ANOVA tests the same people under three or more conditions, extending the paired t-test. It accounts for the correlation among each person's scores by removing individual differences from the error term, so it needs fewer people; but it is still parametric, assuming normality, plus sphericity. A **randomised block design** uses different people grouped into homogeneous blocks (by pre-test, age or IQ) and randomly assigned to treatments within each block, so block differences leave the error term. With one person per treatment per block, error df = (treatments - 1)(blocks - 1): 4 treatments and 5 blocks give 3 × 4 = 12. A **mixed (split-plot) ANOVA** combines a between-subjects and a within-subjects factor.

> CLINICAL LINK
>
> Most therapy trials are mixed designs: group (therapy or waitlist) by time (pre, post, follow-up). Evidence that therapy worked is the group × time interaction, not either main effect.

Analysis of covariance (ANCOVA), also Fisher's, builds a regression into ANOVA. A **covariate** (a continuous variable such as pre-test score or IQ) predicts the DV, its effect is removed, and groups are compared on adjusted means. This raises power and statistically equates groups that started unequal, though it cannot replace random assignment. ANCOVA keeps ANOVA's assumptions and adds a linear covariate-DV relation, a covariate measured reliably before treatment, and above all **homogeneity of regression slopes**: the covariate must predict the DV equally in every group, so the regression lines are parallel. **Multivariate analysis of variance** (MANOVA) tests groups on two or more DVs at once, avoiding the Type I error of many separate ANOVAs; its statistics are Wilks' lambda (most reported; smaller means a bigger effect), Pillai's trace (most robust), Hotelling's trace and Roy's largest root. MANCOVA adds covariates. In all these names, M means more than one DV and C means a covariate.

EXAM TRAP

The assumption ANCOVA adds to ANOVA is homogeneity of regression slopes. Homogeneity of variance and homoscedasticity (one idea, two names) belong to both; equal sample sizes to neither. Repeated-measures ANOVA is not non-parametric, still assumes normality and is not limited to two conditions.

Chi-square: comparing frequencies ●●●

Nominal data offer no means, only counts of people in categories. For these, **Karl Pearson** (1857 to 1936) introduced the **chi-square test** (χ^2) in 1900. It compares the observed frequency (O) in each category with the frequency expected (E) under the null hypothesis: $\chi^2 = \Sigma (O - E)^2/E$. Being squared, it is never negative and its distribution is positively skewed. It is a non-parametric test for categorical frequencies, not for interval, ratio or parametric data.

The **goodness-of-fit test** asks whether one variable's frequencies match a hypothesised pattern (do students prefer four brands of chai equally?); df = k - 1 for k categories. The **test of independence** asks whether two categorical variables in a **contingency table** are associated (is gender related to treatment preference?). Each expected frequency is row total × column total / N, and df = (rows - 1)(columns - 1), because once the margins are fixed only that many cells can vary: 2 × 2 gives 1, 3 × 4 gives 6, 4 × 5 gives 12.

Chi-square needs raw counts (not percentages), each person in one cell only, and an **expected frequency** of at least 5 per cell. For 2 × 2 tables with small expected values, **Frank Yates** (1902 to 1994) proposed the **Yates' correction for continuity** in 1934, subtracting 0.5 from each |O - E| before squaring, while **Fisher's exact test** computes the exact probability of a small table. Paired yes-or-no data (the same people before and after) need the **McNemar test** (**Quinn McNemar**, 1947), which counts only those who changed. Chi-square shows whether an association exists, not its strength; for that report phi (2 × 2), the contingency coefficient or Cramér's V (9.11).

EXAM TRAP

Contingency df is a product, (rows - 1)(columns - 1): 3 × 4 gives 6, not 12, 7 or 11; 4 × 5 gives 12, not 20 or 9. It is not N - 1 (one-sample t) or N - 2 (independent t), and it subtracts 1, not 2.

Rank tests and choosing the right test ●●●

Each parametric test has a rank-based partner. The **Mann-Whitney U test** (**Henry Berthold Mann** and **Donald Ransom Whitney**, 1947) pools and ranks the scores of two independent samples and asks whether one group's ranks run higher. It is the non-parametric equivalent of the independent-samples t-test, and identical to the **Wilcoxon rank-sum test**. For two related samples, the **Wilcoxon matched-pairs signed-rank test** (**Frank Wilcoxon**, 1945) ranks the size of each pair's difference and reattaches the signs, using both direction and magnitude; the **sign test** counts only pluses and minuses (direction) and is weaker. For three or more independent groups, the **Kruskal-Wallis H test** (**William Kruskal** and **W. Allen Wallis**, 1952) is the one-way ANOVA by ranks, H being read against chi-square with k - 1 df. For three or more related samples or repeated measures, the **Friedman test** of the economist **Milton Friedman** (1912 to 2006), devised in 1937, ranks conditions within each person: the two-way ANOVA by ranks.

The **median test** splits independent groups at the combined median and applies chi-square. The **Kolmogorov-Smirnov two-sample test** compares two samples' whole cumulative distributions, so it also catches differences in shape and spread. The **Wald-Wolfowitz runs test** counts runs of the same group in the pooled, ordered scores; few runs mean the groups differ. **Cochran's Q** extends McNemar to three or more related yes-or-no samples. Rank correlations (Spearman, Kendall), Pearson's r and regression belong to 9.11. Table 9.31 gathers the choices.

Table 9.31 Which test when, by design and level of measurement.

Design	Interval or ratio	Ordinal	Nominal
One sample against a known value	One-sample t (z if SD known)	Sign test; one-sample Wilcoxon	Chi-square goodness of fit
Two independent groups	Independent t (Welch if variances differ)	Mann-Whitney U; median, Kolmogorov-Smirnov, runs tests	Chi-square; Fisher's exact
Two related samples	Paired t	Wilcoxon signed-rank; sign test	McNemar
Three or more independent groups	One-way ANOVA	Kruskal-Wallis H	Chi-square (r × c)
Three or more related samples	Repeated-measures ANOVA	Friedman	Cochran's Q
Two or more IVs	Factorial ANOVA	None in common use	None in common use
Association of two variables	Pearson r (9.11)	Spearman, Kendall	Chi-square; phi, Cramér's V

EXAM TRAP

Two names, two independent groups: Mann and Whitney. Wilcoxon signed-rank is for related pairs (but "Wilcoxon rank-sum" is Mann-Whitney in disguise). Kruskal-Wallis is for independent groups, Friedman for related ones. Ordinal differences with magnitude call for Wilcoxon, not the sign test; McNemar is for paired nominal data.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Non-parametric tests are also called _____ tests.
2. Levene's test checks homogeneity of _____; Shapiro-Wilk checks _____.
3. An independent-samples t-test with 25 per group has _____ df.
4. In one-way ANOVA, F = MS between / _____.
5. An IV with 4 levels needs _____ pairwise post hoc comparisons.
6. ANCOVA adds the assumption of homogeneity of _____.
7. Chi-square on a 3 × 4 table has _____ df.
8. The non-parametric equivalent of repeated-measures ANOVA is the _____ test.

Answers: (1) distribution-free, (2) variance; normality, (3) 48, (4) MS within, (5) 6, (6) regression slopes, (7) 6, (8) Friedman.

9.11 Correlation, Regression and Factor Analysis

MEDIUM YIELD · 28 QS

Does stress steal sleep? Is a satisfied worker a productive one? Such questions concern quantities that rise and fall together, not the group means of Section 9.10. Correlation measures how strongly two variables go together, regression turns that association into a prediction, and factor analysis asks how many underlying dimensions produced a whole table of correlations.

Correlation: direction, strength and form ●●

A **correlation** is an association between two variables measured on the same people. It has a *direction*, positive when both rise together (study hours and marks) and negative when one rises as the other falls (stress and sleep quality); a *strength*, how closely the scores keep to the pattern; and a *form*, straight or curved. **Sir Francis Galton** (1822 to 1911), studying heredity, described such "co-relations" in 1888, and his protégé **Karl Pearson** (1857 to 1936) gave them an exact formula in 1896: the **Pearson product-moment correlation coefficient**, r. So r measures association; it is not a mean, a variance or a ratio.

Correlational data are graphed as a **scatterplot** (scatter diagram), with one variable on each axis and each person a single dot (Figure 9.20). A rising cloud is a positive correlation, a falling cloud a negative one, and a shapeless cloud no relationship. The last panel is a warning: a clear inverted U gives $r = 0$, because r detects only straight-line relations. Do not confuse the scatterplot with the bell curve or a skewed distribution, which describe a single variable (Section 9.8), or with chi-square, which is a test, not a graph (Section 9.10).

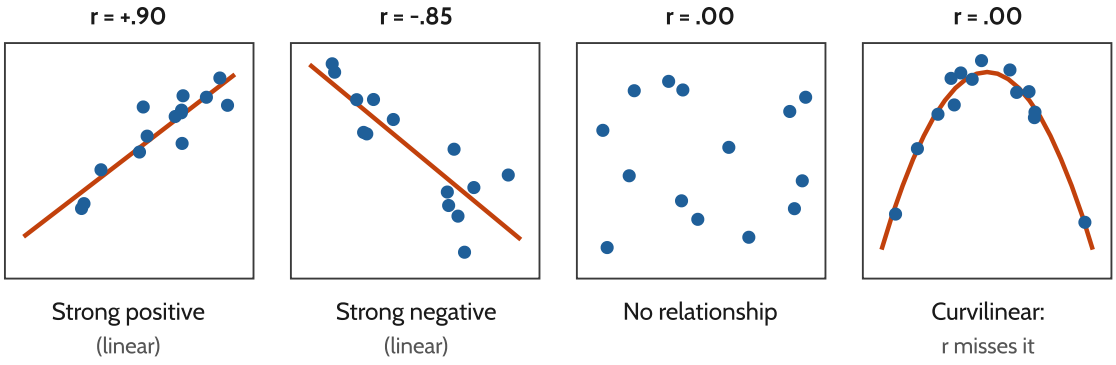


Figure 9.20 Four scatterplots; each dot is one person. The curved relation in the last panel is strong, yet r is zero.

Reading Pearson's r ●●●

Pearson's r runs from -1.00 through 0 to +1.00. Its *sign* gives the direction; its *size*, ignoring the sign, gives the strength. So $r = -0.85$ between stress and sleep quality is a strong negative (inverse) relationship, exactly as strong as +0.85: the more stress, the worse the sleep, and the scores keep closely to that pattern. The range of r is never 0 to 1 (that is the range of r^2 and of multiple R), never -0.04 to +0.04, and never 0 to 100 or -100 to +100. Table 9.32 gives the usual verbal labels; effect-size conventions are in Section 9.9.

Table 9.32 Interpreting the size of r (the sign gives only direction).

Size of r	Usual label	Shared variance (r^2)
.00 to .20	Negligible	0 to 4 per cent
.20 to .40	Low, weak	4 to 16 per cent
.40 to .70	Moderate	16 to 49 per cent
.70 to .90	High, strong	49 to 81 per cent
.90 to 1.00	Very high	81 to 100 per cent

Squaring *r* gives the **coefficient of determination**, the proportion of variance the two variables share. If study hours and marks correlate +0.90, r^2 = 0.81: 81 per cent shared variance, not 90. This is why correlation is related to prediction: the nearer *r* is to plus or minus 1, the better one variable predicts the other, and at 0 it predicts nothing. Correlation does not, however, imply causation. Y may cause X, or a third variable may drive both (ice-cream sales and drownings both follow summer heat). Only a randomised experiment (Section 9.4) licenses a causal claim, so *r* = +0.90 shows a strong positive linear association: it neither proves that X causes Y nor shows that they are causally unrelated.

KEY STUDY: THE HAPPY, PRODUCTIVE WORKER (1985, 2001)

Managers long assumed that satisfied workers perform better. In 1985 **Michelle Iaffaldano** and **Paul Muchinsky** pooled the studies in a meta-analysis (Section 9.5) and found a mean correlation of only about .17. In 2001 **Timothy Judge** and colleagues, pooling more than 300 samples and correcting for measurement error, put it at about .30. Job satisfaction and performance thus show a weak (modest) positive correlation: not zero, not negative, not strong, and silent on which causes which.

EXAM TRAP

A minus sign never means weak: -0.85 is strong. And *r* = +0.90 is neither "90 per cent of the variance" (it is 81 per cent) nor proof that X causes Y.

Assumptions, limits and testing r ●●

Pearson's *r* is the technique for two continuous (interval or ratio) variables, such as stress level and sleep duration. Chi-square (association between categories), the t-test, analysis of variance (ANOVA) and the Mann-Whitney U test (differences between groups) answer other questions (Section 9.10). Pearson's *r* assumes *linearity*; *homoscedasticity*, an equal scatter of Y at every value of X, whose violation, **heteroscedasticity**, shows as a fan-shaped cloud; *bivariate normality*, for significance tests; and *independence* of cases. Observations that depend on earlier ones, such as daily mood ratings, show **autocorrelation**: a variable correlated with itself over time. Homogeneity of variance is the group version of homoscedasticity, equal variances across groups. *r* is also distorted by **restriction of range** (entrance scores predict grades poorly among admitted students alone, whose scores span a narrow band), by outliers, by curvilinear relations and by pooling unlike groups.

A sample *r* is tested with $t = r\sqrt{(N - 2)} / \sqrt{(1 - r^2)}$ on N - 2 degrees of freedom: *r* = .30 fails with 20 people but passes with 100. To compare two correlations, **Sir Ronald Aylmer Fisher** (1890 to 1962) devised *Fisher's r-to-z transformation*, $z' = 0.5 \ln[(1 + r)/(1 - r)]$, which removes the skew in

the sampling distribution of r. **Fisher's z-test** then divides the difference between the two z' values by its standard error, $\sqrt{[1/(N_1 - 3) + 1/(N_2 - 3)]}$. A t-test compares two means, ANOVA several, and Pearson's r measures a single association; none compares two correlations.

Matching the coefficient to the data ●●

When the data are ranks or categories, another coefficient is used. The key distinction is between a *true dichotomy*, two natural categories (alive or dead), and an *artificial dichotomy*, a continuous variable cut in two (pass or fail from marks). **Charles Spearman** (1863 to 1945) introduced the rank-difference method in 1904. **Spearman's rho**, $\rho = 1 - 6\Sigma D^2 / N(N^2 - 1)$, where D is each person's difference in ranks, correlates ranks; it therefore suits ordinal data and monotonic (steadily rising or falling) but non-linear relations, and it resists outliers. *Kendall's tau*, from **Maurice Kendall** (1938), counts agreeing and disagreeing pairs of ranks and suits small samples with many ties. Table 9.33 gives the rest; agreement between raters uses the intraclass correlation (Chapter 2).

Table 9.33 Choosing a correlation coefficient.

Coefficient	Variable 1	Variable 2	Example or note
Pearson r	Continuous	Continuous	Stress and sleep duration
Spearman rho, Kendall tau	Ranks	Ranks	Two judges' rankings
Point-biserial	True dichotomy	Continuous	Item right or wrong and total score
Biserial	Artificial dichotomy	Continuous	Pass or fail (cut from marks) and IQ
Phi (φ)	True dichotomy	True dichotomy	2 by 2 table
Tetrachoric	Artificial dichotomy	Artificial dichotomy	Both split from continuous scores
Contingency C, Cramér's V	Nominal	Nominal	Built on chi-square
Eta (correlation ratio)	Categories	Continuous	Curvilinear relations

> **EXAM TRAP**
> "One continuous and one dichotomous variable" means point-biserial unless the dichotomy was made by splitting a continuous variable (biserial). Two continuous variables call for Pearson's r, not Spearman, which is for ranks or monotonic non-linear relations.

Partial, semi-partial and multiple correlation ●

Ordinary r is a zero-order correlation: nothing is held constant. Marital stress may rise with length of marriage partly because long marriages have more children. The **partial correlation**, $r_{12.3}$, is the correlation between two variables after a third variable's influence has been removed statistically from *both*; partialling out the number of children gives the stress and length relation among couples with equally many children. Holding one variable constant gives a first-order partial, two a second-order partial. A **semi-partial correlation** (part correlation) removes the third variable from only *one* of the two, usually the predictor; its square is a predictor's unique contribution in multiple regression, and it is never larger than the partial. The **multiple correlation** (R, from 0 to 1) links one variable with the best weighted combination of several others. Linear (ordinary) correlation, biserial correlation and multiple correlation remove nothing.

EXAM TRAP

"Controlling for", "holding constant" or "subtracting out the influence of" a third variable always means partial correlation.

Simple linear regression ●●

Regression also began with Galton. In 1886 he found that children of very tall parents were tall, but less extremely so, and called it "regression towards mediocrity". This **regression to the mean** follows whenever two measures correlate imperfectly: extreme scores on one pair with less extreme scores on the other (a threat to validity, Section 9.2).

Simple linear regression predicts a criterion Y from a predictor X with the line Y = a + bX (Figure 9.21). The *intercept* a is the predicted Y when X is zero; the **regression coefficient** b is the slope, the change in Y for each one-unit rise in X. $b = r(SD_Y/SD_X)$ and $a = M_Y - bM_X$, where SD is the standard deviation and M the mean. If hours and marks correlate .60, the SD of marks is 10 and the SD of hours 2, then b = 3 marks per hour; with means of 5 hours and 50 marks, a = 35, so 8 hours predicts 35 + 24 = 59 marks.

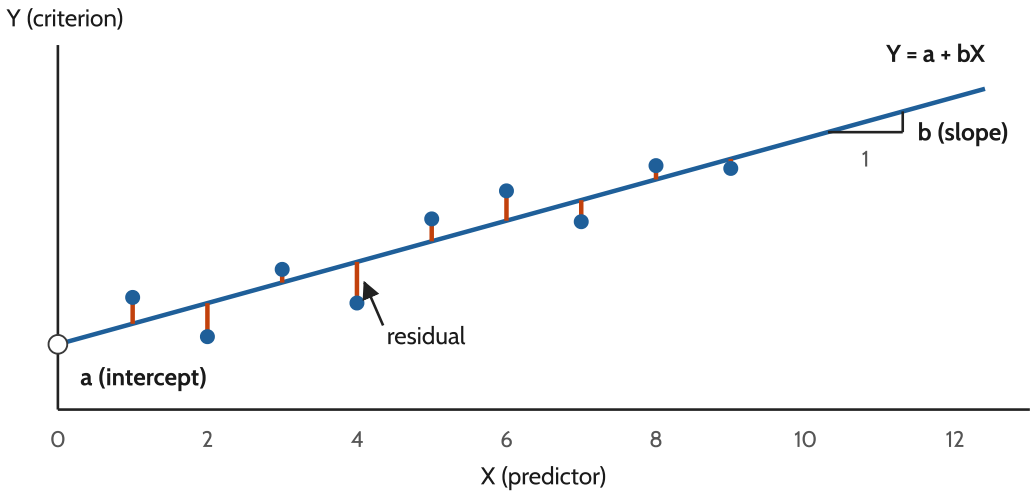


Figure 9.21 A least-squares regression line. The dashed verticals are the residuals whose squared sum the line minimises.

The **method of least squares**, published by **Adrien-Marie Legendre** (1752 to 1833) in 1805 and developed by **Carl Friedrich Gauss** (1777 to 1855), chooses the line that minimises the sum of squared *residuals* (actual minus predicted Y). Their typical size is the **standard error of estimate**, $SD_Y\sqrt{(1 - r^2)}$. Regression assumes linearity, independence of residuals, normality of residuals and equal residual variance (LINE). Correlation is symmetrical and unit-free; regression is directional and gives an equation in real units. A researcher therefore uses regression to predict the value of one variable from another, not to find a mean, count frequencies or estimate sampling error.

EXAM TRAP

In Y = a + bX the slope is **b**; a is the intercept, X the predictor and Y the predicted score.

Multiple regression and multicollinearity ●●

Multiple regression uses several predictors, $Y = a + b_1X_1 + b_2X_2 + \ldots$, each b holding the others constant. Standardised **beta weights** (β) let predictors be compared. R^2 is the variance all predictors explain together, and adjusted R^2 corrects its rise with every added predictor. Entry may be simultaneous (all at once), hierarchical (theory-ordered blocks, testing the change in R^2) or stepwise (chosen by the computer, which capitalises on chance).

When predictors correlate highly with each other, the result is **multicollinearity**. Anxiety and neuroticism scores correlating .90 carry the same information, so R^2 stays high but the weights become unstable, their standard errors inflate and signs may flip. A **variance inflation factor** (VIF) above 10 (above 5 on a stricter rule) or a tolerance (1/VIF) below .10 flags it; the cure is to drop or combine predictors.

> EXAM TRAP
>
> Multicollinearity is correlation among predictors, not between a predictor and the outcome. Heteroscedasticity is unequal residual spread, autocorrelation is residuals correlated over time, homogeneity of variance is equal group variances, a categorical outcome calls for logistic regression, and "assumptions are violated" defines nothing.

A dichotomous outcome (relapsed or not) calls for **logistic regression**, which models the log of the odds, estimates weights by maximum likelihood, expresses each as an odds ratio (above 1 raises the odds, below 1 lowers them) and tests it with the Wald test. Regression, the t-test, ANOVA and analysis of covariance (ANCOVA) are all cases of the general linear model: factorial designs are analysed by ANOVA, and ANCOVA blends ANOVA with regression, using the covariate's correlation with the outcome (Section 9.10). Mediation and moderation are tested by regression, following **Reuben Baron** and **David Kenny** (1986), with the Sobel test of **Michael Sobel** (1982) for indirect effects (Section 9.2). *Discriminant analysis* predicts membership of known groups from continuous variables. *Path analysis*, devised by **Sewall Wright** (1889 to 1988) in 1921, chains regressions to test a diagram of directional paths, and *structural equation modelling* (SEM) adds latent variables, joining a confirmatory factor analysis to a path model.

Factor analysis: the dimensions behind correlations

Twenty tests yield 190 correlations, and **factor analysis** asks how few underlying factors could produce them: if verbal tests correlate among themselves and numerical tests among themselves, two factors may explain the whole table. It serves data reduction and construct validation (Chapter 2). Spearman invented it in 1904 to argue for a general factor, g (Chapter 1). **Louis Leon Thurstone** (1887 to 1955) extended it in the 1930s into multiple factor analysis, with the goal of **simple structure**: each variable loading highly on one factor and near zero on the others.

A **factor loading** is a variable's correlation with a factor; about .30 to .40 or more counts. A variable's squared loadings summed across factors give its **communality** (h^2), and $1 - h^2$ is its uniqueness. A factor's squared loadings summed down the column give its **eigenvalue**, the variance it explains in units of one variable. Before factoring, the Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy should be at least .60 and **Bartlett's test of sphericity** significant (the correlation matrix is not an identity matrix), with at least 10 cases per variable and

ideally 300 or more. Of the extraction methods, **principal components analysis** (**Harold Hotelling**, 1933) analyses total variance for data reduction, whereas principal axis factoring and maximum likelihood analyse only common variance, to model latent factors.

Retaining, rotating and confirming factors ●●

How many factors? The **Kaiser-Guttman rule** (**Louis Guttman**, 1916 to 1987, and **Henry Kaiser**) keeps eigenvalues above 1.0, since a factor below 1 explains less than one variable; it tends to keep too many. The **scree test** of **Raymond Cattell** (1905 to 1998), published in 1966, keeps the factors above the elbow of the eigenvalue plot, where the slope flattens into "scree", the rubble below a cliff. **John Horn**'s **parallel analysis** (1965), the most accurate, keeps factors whose eigenvalues beat those of random data. With eigenvalues of 3.5, 1.8, 0.9, 0.7 and 0.5 (Figure 9.22), two exceed 1.0 and two lie above the elbow at factor 3: retain two.

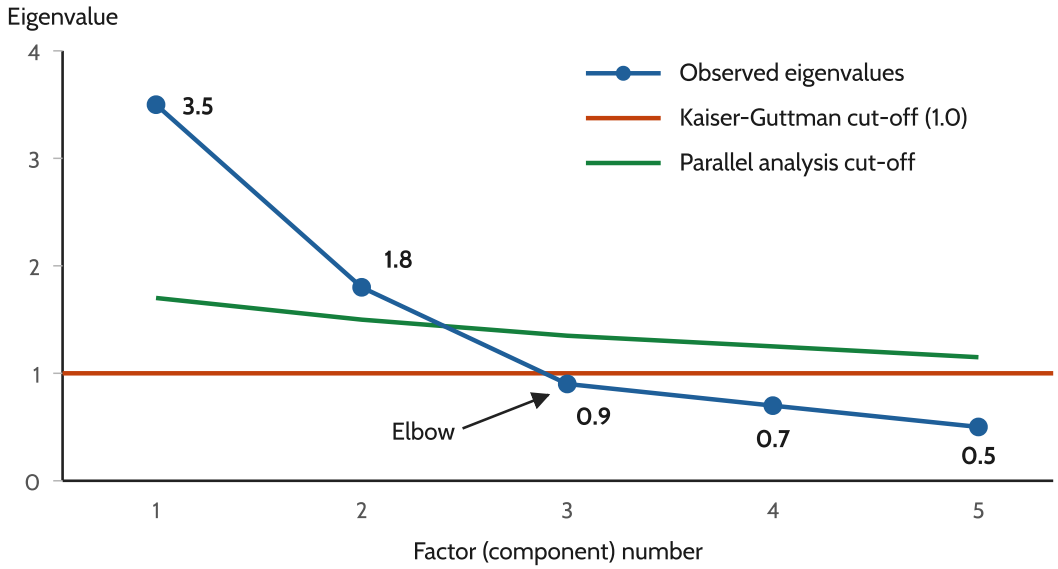


Figure 9.22 Scree plot of the worked example: two eigenvalues lie above both cut-off lines and before the elbow, so every rule retains two factors.

> **EXAM TRAP**
>
> The elbow factor itself is dropped: an elbow at factor 3 means retain two, not three. Five counts every eigenvalue, and one keeps only the largest.

Factor rotation turns the factor axes towards simple structure without changing the variance explained. An **orthogonal rotation** keeps factors uncorrelated: varimax (Kaiser, 1958) simplifies each factor, quartimax each variable, and equimax compromises. An **oblique rotation** (promax, direct oblimin) lets factors correlate, as psychological constructs usually do. A **second-order factor** is extracted from the correlations among first-order factors (general ability above verbal, numerical and spatial factors, a hierarchical model), so it needs an oblique rotation such as promax; orthogonal factors leave no correlations to factor.

EXAM TRAP

Second-order factor analysis needs promax (oblique). Varimax, quartimax and equimax are all orthogonal.

All of this is **exploratory factor analysis** (EFA): data-driven, with no prior model. By contrast, **confirmatory factor analysis** (CFA), developed by **Karl Jöreskog** in 1969, tests a prespecified structure, fixing which items load on which factor, and judges fit by the comparative fit index (CFI) and Tucker-Lewis index (TLI), about .95 or above, the root mean square error of approximation (RMSEA), about .06 or below, and the standardised root mean square residual (SRMR), .08 or below, with modification indices suggesting changes. It needs no rotation, takes continuous or categorical items, and is the measurement model of SEM. *Cluster analysis* groups people rather than variables, and *multidimensional scaling* maps judged similarities between objects in as few dimensions as possible.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Correlational data are graphed as a _____.
2. If r = +0.90, the shared variance is _____ per cent.
3. The significance of r is tested on _____ degrees of freedom.
4. A continuous and an artificially dichotomised variable call for the _____ correlation.
5. Removing a third variable's influence from both variables gives the _____ correlation.
6. In Y = a + bX, the slope is _____.
7. Highly intercorrelated predictors show _____.
8. Second-order factors need an _____ rotation such as promax.

Answers: (1) scatterplot, (2) 81, (3) N - 2, (4) biserial, (5) partial, (6) b, (7) multicollinearity, (8) oblique.

The Case of Clever Hans, Revisited

Pfungst's investigation is a compact course in this chapter. He turned a marvel into a research problem with rival, falsifiable hypotheses and preferred the simplest explanation that fitted, as Lloyd Morgan's canon advises (9.1). He varied one condition at a time while holding the others constant, and exposed the questioner's expectation as a hidden variable that undermined every earlier demonstration (9.2). His study of one horse was a single-case experiment (9.4); he observed systematically (9.6) and honestly reported a result that deflated a celebrity (9.7). His verdict rested on counts of right and wrong answers compared across conditions (9.8 to 9.10). And the blinding of modern clinical trials (9.5), in which those who assess outcomes do not know who was treated, exists so that no one plays von Osten again.

Chapter summary

9.1 Science and the Research Process: Problems and Hypotheses. Science differs from tenacity, authority and the a priori method (Peirce, 1877) because it fixes belief by public, repeatable observation. Scientific research is empirical, objective, replicable, parsimonious (Occam's razor, Lloyd Morgan's canon), falsifiable (Popper, 1934) and tentative; Kuhn (1962) described paradigm shifts. **Induction** builds theory from observations (theory development, qualitative research); **deduction** tests theory against data; abduction infers the best explanation. Methodology is wider than methods. Research is basic or applied (including Lewin's action research), exploratory, descriptive or explanatory, nomothetic or idiographic (Allport). A clear problem and a literature review (which provides a theoretical framework) yield hypotheses: null or alternative, directional or non-directional, universal or existential.

9.2 Variables, Experimental Control and the Validity of Research. A **variable** takes two or more values; the **independent variable** is the presumed cause (manipulated or selected) and the **dependent variable** the measured outcome. A **confound** varies with the IV and affects the DV; a **moderator** changes the strength of an effect and a **mediator** carries it (Baron and Kenny, 1986). **Control** follows Kerlinger's MAXMINCON principle (maximise primary, minimise error, control secondary variance), above all through **random assignment**. Hawthorne, demand (Orne), expectancy and Pygmalion (Rosenthal) and halo (Thorndike) effects distort results; **double-blinding** controls expectancy on both sides. Studies are judged by statistical conclusion, internal, construct and external validity (Campbell and Stanley, 1963; Cook and Campbell, 1979).

9.3 Sampling: Populations, Probability and Non-Probability Methods. A **parameter** describes a population and a **statistic** a sample. Following Blalock (1960), sampling methods are probability, non-probability or mixed. In **probability sampling** every element has a known, non-zero chance of selection, so sampling error can be calculated and the sample is likely, though not guaranteed, to be representative: simple random, systematic (every kth after a random start), stratified (random sampling within every stratum) and cluster sampling (random choice of whole natural groups, multistage when repeated down levels). In **non-probability sampling** (convenience, purposive, quota, snowball, saturation, dense) the chance is unknown. Sampling error shrinks as n rises; bias, as in the 1936 *Literary Digest* poll, does not.

9.4 Research Designs: Experimental, Single-Case and Non-Experimental. Only an **experiment**, with manipulation, control and random assignment that makes the groups equivalent, licenses cause-and-effect conclusions; Fisher's (1935) principles are replication, randomisation and local control. Campbell and Stanley (1963) classed designs as pre-experimental, true experimental (including Solomon's four-group design, which controls pretest sensitisation) and quasi-experimental (intact groups, no random assignment). Experiments may be between-subjects, within-subjects (order effects controlled by counterbalancing and Latin squares), mixed or crossover; factorial designs reveal interactions. Single-case designs (A-B-A-B, multiple baseline) test a treatment in one person. Correlational, survey (a census is a survey) and ex post facto designs cannot establish cause; longitudinal designs suffer attrition, cross-sectional designs cohort effects.

9.5 Clinical Epidemiology, Randomised Trials and Evidence Synthesis. Epidemiology counts disorder as **prevalence** (existing cases: point, period, lifetime) or **incidence** (new cases); prevalence is roughly incidence times duration. Cross-sectional studies give prevalence, **case-control** studies

(best for rare outcomes) give odds ratios, and cohort studies, such as Doll and Hill's prospective British Doctors Study, give relative risk; a ratio of 1 means no association, and NNT = 1/ARR. Sensitivity and specificity belong to the test, but predictive values depend on the base rate (Meehl and Rosen, 1955). Trials rely on randomisation, allocation concealment, blinding and intention-to-treat analysis. Meta-analysis (Glass, 1976) pools many studies; CONSORT reports trials and PRISMA systematic reviews.

9.6 Methods of Data Collection and Qualitative Research. Data are primary or secondary. Observation is participant or non-participant, structured or unstructured, naturalistic or controlled, overt or covert. Interviews run from structured to unstructured, with open-ended questions, probes and Spradley's ethnographic inquiry. A questionnaire is self-answered, a schedule filled in by the enumerator; items must not be double-barrelled, leading or loaded. Berelson's content analysis, Moreno's sociometry and Dilthey's Verstehen follow. Guba and Lincoln's paradigms run from positivism (naive realism) to constructivism (relativism). Qualitative research is naturalistic, inductive and emic; ethnography, grounded theory (open, axial, selective coding) and phenomenology lead its traditions, judged by trustworthiness and strengthened by triangulation.

9.7 Research Ethics and Writing the Research Report. Research ethics grew out of abuse: the Nazi experiments produced the Nuremberg Code (1947), and the Tuskegee study (1932 to 1972) led to the Belmont Report (1979: respect for persons, beneficence, justice). The APA Ethics Code (2002) has five general principles, and in India the ICMR guidelines (2017) require ethics committee approval. Informed consent tells participants enough to decide voluntarily before taking part; debriefing gives full information afterwards (dehoaxing and desensitising). Confidentiality differs from anonymity, deception needs strict justification, and animal research follows the 3Rs. Misconduct means fabrication, falsification and plagiarism; citation is not plagiarism. A research report communicates findings, and its Discussion, not its Results, interprets them.

9.8 Measurement Scales, Organising Data and Descriptive Statistics. S. S. Stevens (1946) gave four levels of measurement: nominal (unordered categories), ordinal (ranks), interval (equal intervals, arbitrary zero, as in Celsius) and ratio (true zero, as in height). Raw data are edited, coded, classified and tabulated, analysed in SPSS or Excel, and shown as frequency distributions and graphs. The mean suits symmetrical data, the median skewed data and the mode nominal data; the SD, the square root of the variance, measures spread. The normal curve holds about 68, 95 and 99.7 per cent of cases within 1, 2 and 3 SD, and z = (X − M)/SD converts any score into a percentile or another standard score.

9.9 Sampling Distributions and the Logic of Hypothesis Testing. Inferential statistics use sample data and the laws of chance to judge whether a result could be due to chance. The sampling distribution of the mean centres on the population mean, its SD is the standard error ($\sigma/\sqrt{N}$), and by the central limit theorem it is near normal once N is about 30. A 95 percent CI is M ± 1.96 SE. A test assumes H0 and rejects it when $p \le \alpha$. A Type I error rejects a true H0 (α); a Type II error retains a false one (β). Power is 1 − β; Cohen's d of 0.2, 0.5 and 0.8 is small, medium and large.

9.10 Comparing Groups: t-Tests, ANOVA and Non-Parametric Tests. Parametric tests (z, t, F) assume normality, equal variances, independence and interval or ratio data; non-parametric or distribution-free tests (Siegel, 1956) handle ranks, categories and small or skewed samples. Gosset's t tests one or two means (df N - 1, N1 + N2 - 2 or pairs - 1). Fisher's ANOVA splits variance into between and within parts, F = MS between / MS within, about 1 under H0; a significant F is followed

by post hoc tests or a Bonferroni correction. Factorial ANOVA detects interactions; ANCOVA adds homogeneity of regression slopes. Chi-square handles frequencies, with df (r - 1)(c - 1), and rank tests partner each parametric test.

9.11 Correlation, Regression and Factor Analysis. Correlation (Pearson's *r*, 1896) measures the direction (sign) and strength (size) of a linear association from -1 to +1. It is graphed as a scatterplot; *r* squared is shared variance (+0.90 means 81 per cent), and correlation aids prediction but never proves causation. Spearman's rho, point-biserial, biserial, phi and tetrachoric suit other data; partial correlation removes a third variable from both. **Regression** (Y = a + bX) predicts one variable from another; multiple regression is undermined by multicollinearity among predictors. **Factor analysis** (Spearman, 1904) reduces correlated variables to a few factors, retained by eigenvalues above 1 or the scree elbow and rotated orthogonally (varimax) or obliquely (promax).

Who's who

Table 9.34 The people in this chapter.

Name	When	Examined for
Charles Sanders Peirce	1839 to 1914	Four methods of fixing belief (1877): tenacity, authority, a priori, science; named abduction
Fred N. Kerlinger	1910 to 1991	Foundations of Behavioral Research (1964); definitions of scientific research, problem, hypothesis and theory
Socrates	about 470 to 399 BCE	The elenchus, the Socratic method of question-and-answer cross-examination
Aristotle	384 to 322 BCE	Knowledge through observation and classification
René Descartes	1596 to 1650	Rationalism: knowledge through reason and intuition
Galileo Galilei	1564 to 1642	Knowledge through experimentation
Francis Bacon	1561 to 1626	Inductive method; Novum Organum (1620)
William of Ockham	about 1287 to 1347	Occam's razor (parsimony)
Conwy Lloyd Morgan	1852 to 1936	Lloyd Morgan's canon (1894): never explain behaviour by a higher faculty if a lower one will do
Karl Popper	1902 to 1994	Falsifiability as the mark of science; Logik der Forschung (1934)
A. K. Singh	Indian textbook (several editions)	Tests, Measurements and Research Methods in Behavioural Sciences: eight steps of research, assumptions of science, attitudes of scientists, types of hypotheses and reviews
Thomas S. Kuhn	1922 to 1996	Paradigms, normal science and scientific revolutions (1962)
David Hume	1711 to 1776	The problem of induction
Clark L. Hull	1884 to 1952	Hypothetico-deductive theory of learning; Principles of Behavior (1943)
C. R. Kothari	2004 (2nd revised edition)	Research Methodology: Methods and Techniques; methodology wider than methods; types of research; criteria of good research; research process
Kurt Lewin	1890 to 1947	Father of modern social psychology; founded action research (1946)
Michael Scriven	1967	Formative versus summative evaluation
Wilhelm Windelband	1848 to 1915	Nomothetic versus idiographic sciences (1894)
Gordon W. Allport	1897 to 1967	Brought nomothetic and idiographic into personality psychology (1937); preferred the idiographic approach
Max Wertheimer	1880 to 1943	Phi phenomenon: an isolated unexplained fact that became a research problem and led to Gestalt psychology
Pauline V. Young	1956	Sources of research problems: documentary sources, key informants, library sources
Baker and Keller	cited by A. K. Singh	Sources of research problems: programmed activities, research proposals, government policies, social values

Name	When	Examined for
William J. Goode	1952 (with Hatt)	Methods in Social Research: criteria of good hypotheses and three difficulties in formulating them
Paul K. Hatt	1952 (with Goode)	Methods in Social Research: criteria of good hypotheses and difficulties in formulating them
Walter L. Wallace	1971	The wheel of science: the research cycle of deduction and induction
Ivan Petrovich Pavlov	1849 to 1936	Example of a scientific fact: dogs salivating before feeding (conditioning, Chapter 1)
Aaron T. Beck	1921 to 2021	Cognitive theory of depression, cited as an example of a theory
Ernst Heinrich Weber	1795 to 1878	Weber's law, cited as an example of a scientific law
Richard Atkinson and Richard Shiffrin	1968	Multi-store memory model, cited as an example of a model
Wilhelm von Osten	1904	Owner and trainer of Clever Hans (chapter opening)
Carl Stumpf	1848 to 1936	Sat on the 1904 commission that examined Clever Hans; Pfungst's teacher
Oskar Pfungst	1907	Showed that Clever Hans responded to questioners' unwitting cues; a classic demonstration of experimenter expectancy
Robert S. Woodworth	1869 to 1962	S-O-R scheme: stimulus, organismic and response variables
Reuben M. Baron	1986	Moderator versus mediator distinction (with David Kenny)
David A. Kenny	1986	Moderator versus mediator distinction (with Reuben Baron)
Percy Williams Bridgman	1882 to 1961	Operationism; The Logic of Modern Physics (1927)
Stanley Smith Stevens	1906 to 1973	Brought operational definition into psychology (1935)
Elton Mayo	1880 to 1949	Associated with the Hawthorne studies (1924 to 1932)
Fritz J. Roethlisberger	1898 to 1974	Reported the Hawthorne studies in Management and the Worker (1939, with Dickson)
William J. Dickson	1939	Co-author of Management and the Worker (Hawthorne studies)
Henry A. Landsberger	1958	Coined the term Hawthorne effect
Martin T. Orne	1927 to 2000	Demand characteristics (1962)
Stephen J. Weber	1972	Participant roles: good, faithful, negativistic and apprehensive subjects (with Cook)
Thomas D. Cook	1972; 1979	Participant roles (with Weber, 1972); four types of validity and added threats to internal validity (with Campbell, 1979)
Milton J. Rosenberg	1965	Evaluation apprehension
Robert Rosenthal	born 1933	Experimenter expectancy effect (1966); Pygmalion effect (with Jacobson, 1968)
Lenore Jacobson	1968	Pygmalion in the Classroom (with Rosenthal)
Edward Lee Thorndike	1874 to 1949	Halo effect (1920)
Donald T. Campbell	1916 to 1996	Internal versus external validity (1957); their threats (with Stanley, 1963; with Cook, 1979)
Julian C. Stanley	1918 to 2005	Campbell and Stanley (1963): eight threats to internal validity, four to external validity
Glenn H. Bracht	1968	Split external validity into population and ecological validity (with Gene Glass)
Gene V Glass	1968	Split external validity into population and ecological validity (with Glenn Bracht)
William R. Shadish	2002	Shadish, Cook and Campbell: the four types of validity in current form
Alan Bryman	2001	Social Research Methods: reliability, replicability and validity of a study
Hubert M. Blalock	1926 to 1991	Social Statistics (1960); classification of sampling into probability, non-probability and mixed methods
Jerzy Neyman	1894 to 1981	Neyman (optimum) allocation in stratified random sampling by stratum size and variability (1934)
Claus Moser	1922 to 2015	Associated with multistage sampling in area (cluster) surveys
Everet Franklin Lindquist	1901 to 1978	Definition of sampling error (1940): the difference between a parameter and its estimate from a sample
George Horace Gallup	1901 to 1984	Correctly forecast the 1936 US presidential election with a far smaller sample than the Literary Digest

Name	When	Examined for
John Stuart Mill	1806 to 1873	A System of Logic (1843): methods of agreement and difference, the logic of causal inference
Ronald Aylmer Fisher	1890 to 1962	The Design of Experiments (1935): replication, randomisation and local control
Donald Thomas Campbell	1916 to 1996	With Stanley (1963): R, X, O notation; pre-experimental, true and quasi-experimental designs; regression discontinuity with Thistlethwaite (1960)
Richard L. Solomon	1918 to 1995	Solomon four-group design (1949), which controls pretest sensitisation
Donald Thistlethwaite	1960	Regression discontinuity design, with Campbell
Leon Festinger	1953	Definition of the laboratory experiment (with Daniel Katz)
Daniel Katz	1953	Field studies: exploratory versus hypothesis-testing
Irving Piliavin	1969	Subway field experiment on helping a collapsed passenger (with colleagues)
Burrhus Frederic Skinner	1904 to 1990	Experimental analysis of behaviour, the origin of single-subject designs
Murray Sidman	1960	Tactics of Scientific Research: group averages can mislead; generality comes from replication
Alan E. Kazdin	1982	Single-Case Research Designs: systematised single-case designs for clinical work
Jacob Moreno	1934	Sociometry: mapping choices and relations within a group
Sigmund Freud	1909	Little Hans, a classic case study
Émile Durkheim	1897	Suicide: archival research built from official statistics
K. Warner Schaie	1965	Sequential developmental designs (several cohorts tested on several occasions, the commonest being cross-sequential) to tell age changes from cohort differences
Lewis Madison Terman	1877 to 1956	Genetic Studies of Genius (from 1921), the classic longitudinal study of gifted children
William S. Robinson	1950	Demonstrated the ecological fallacy with US state-level census data
Richard Doll	1912 to 2005	British Doctors Study of smoking (prospective cohort, 1951 to 2001), with Bradford Hill
Austin Bradford Hill	1897 to 1991	British Doctors Study of smoking (from 1951), with Richard Doll
Paul Everett Meehl	1920 to 2003	With Albert Rosen (1955): base rates limit the predictive value of test signs and cutting scores
Albert Rosen	1955	With Meehl: base rates (antecedent probability) and the efficiency of psychometric signs and cutting scores
Hans Eysenck	1916 to 1997	1952 claim that neurotic patients improve as often without psychotherapy; critic of meta-analysis
Mary Lee Smith	1977	With Glass, pooled 375 controlled psychotherapy studies: average effect size 0.68
James P. Spradley	1933 to 1982	The Ethnographic Interview (1979): the informant explains their own cultural world; phases of participant observation (1980): descriptive, focused, selective
Bernard Berelson	1912 to 1979	Content analysis (1952): objective, systematic, quantitative description of manifest content; method of universe, unit of analysis, quantification (1954 handbook chapter)
Jacob Levy Moreno	1889 to 1974	Sociometry and the sociogram (Who Shall Survive?, 1934): stars, isolates, mutual choices; founder of psychodrama
Wilhelm Dilthey	1833 to 1911	Verstehen (understanding) versus explanation in the human sciences, 1883
Max Weber	1864 to 1920	Made Verstehen the method of interpretive sociology
William Stephenson	1935	Q-methodology: factor analysis of persons through Q-sorts
Rensis Likert	1903 to 1981	Summated rating attitude scale (taught in Chapter 2)
Louis Leon Thurstone	1887 to 1955	Equal-appearing interval attitude scale (taught in Chapter 2)
Louis Guttman	1916 to 1987	Cumulative attitude scale (taught in Chapter 2)
Egon Guba	1924 to 2008	Research paradigms with Lincoln (1994); trustworthiness criteria (Naturalistic Inquiry, 1985)
Yvonna Lincoln	1985 and 1994	Paradigms and trustworthiness with Guba; historical moments of qualitative research with Denzin (2005)

Name	When	Examined for
Herbert Blumer	1900 to 1987	Coined symbolic interactionism (1937); Symbolic Interactionism (1969)
George Herbert Mead	1863 to 1931	Ideas on meaning and social interaction behind symbolic interactionism
Norman Denzin	1970	Four types of triangulation (The Research Act): data, investigator, theory, methodological; historical moments with Lincoln (2005)
Michael Quinn Patton	2002	Twelve themes of qualitative inquiry under design, fieldwork and analysis
Kenneth Pike	1912 to 2000	Coined emic and etic (1954)
Clifford Geertz	1926 to 2006	Thick description (The Interpretation of Cultures, 1973)
Bronislaw Malinowski	1884 to 1942	Trobriand Islands fieldwork (1915 to 1918) and Argonauts of the Western Pacific (1922): participant observation, the native's point of view
Barney Glaser	1930 to 2022	Grounded theory with Strauss (The Discovery of Grounded Theory, 1967)
Anselm Strauss	1916 to 1996	Grounded theory (1967); open, axial and selective coding with Corbin (1990)
Juliet Corbin	1990	Basics of Qualitative Research with Strauss: open, axial, selective coding
Edmund Husserl	1859 to 1938	Founder of phenomenology; bracketing (epoche); descriptive phenomenology, the detached approach (called direct in one source)
Martin Heidegger	1889 to 1976	Hermeneutic (interpretive) phenomenology (Being and Time, 1927); the involved approach (called indirect in one source)
Jonathan Smith	1996	Interpretative phenomenological analysis (IPA)
Robert Yin	1984	Case study research
Harold Garfinkel	1917 to 2011	Ethnomethodology (Studies in Ethnomethodology, 1967); breaching experiments
Virginia Braun	2006	Thematic analysis in six phases, with Clarke
Victoria Clarke	2006	Thematic analysis in six phases, with Braun
Jane Ritchie	1994	Framework analysis, with Spencer
Liz Spencer	1994	Framework analysis, with Ritchie
John Heron	Late 20th century	Co-operative inquiry: participants as co-researchers and co-subjects
Matthew Miles	1994	Qualitative analysis with Huberman: data reduction, data display, conclusion drawing and verification
A. Michael Huberman	1994	Qualitative analysis with Miles: data reduction, data display, conclusion drawing and verification
John Creswell	2018	Three core mixed-methods designs (with Vicki Plano Clark): convergent, explanatory sequential, exploratory sequential
John B. Watson	1878 to 1958	Little Albert study (1920) with Rosalie Rayner: conditioned fear in an infant, an ethically debated case
Rosalie Rayner	1898 to 1935	Co-author of the Little Albert study (1920)
Philip Zimbardo	1933 to 2024	Stanford prison study (1971), stopped after six days
Stanley Milgram	1933 to 1984	Obedience study (1963): deception and stress; 26 of 40 (65 per cent) went to 450 volts
Diana Baumrind	1927 to 2018	Ethical critique of Milgram's obedience study (1964)
David S. Holmes	1976	Named the two parts of debriefing: dehoaxing and desensitising
William Russell	1959	The 3Rs of animal research (replacement, reduction, refinement), with Rex Burch
Rex Burch	1959	The 3Rs of animal research, with William Russell
Frederic Lord	1912 to 2000	On the statistical treatment of football numbers (1953), a critique of Stevens's permissible statistics
John Wilder Tukey	1915 to 2000	Exploratory Data Analysis (1977): stem-and-leaf display and box plot
Friedrich Wilhelm Bessel	1784 to 1846	Bessel's correction: dividing by N − 1 for the unbiased sample variance
Karl Pearson	1857 to 1936	Coefficient of skewness, 3(Mean − Median)/SD
Abraham de Moivre	1667 to 1754	First derived the normal curve (1733)
Pierre-Simon Laplace	1749 to 1827	Developed the normal curve
Carl Friedrich Gauss	1777 to 1855	Normal curve of error (1809), hence Gaussian; the keyed propounder of the normal curve
Adolphe Quetelet	1796 to 1874	Applied the normal curve to human characteristics (the average man)

Name	When	Examined for
Francis Galton	1822 to 1911	Applied the normal curve to human ability
Siméon Denis Poisson	1781 to 1840	Poisson distribution of rare events; distractor for the normal curve
Jacob (James) Bernoulli	Ars Conjectandi published 1713	Law of large numbers and Bernoulli trials; distractor for the normal curve
Wilhelm Reich	1897 to 1957	Psychoanalyst; distractor in a normal-curve question
David Wechsler	1896 to 1981	Deviation IQ with mean 100 and SD 15
Muriel Bristol	1920s	The lady tasting tea at Rothamsted; Fisher's opening example of a significance test
Jacob Bernoulli	1655 to 1705	Law of large numbers (published 1713)
Egon Pearson	1895 to 1980	With Jerzy Neyman: Type I and Type II errors, beta and power
Jacob Cohen	1923 to 1998	Power analysis and effect-size benchmarks (d = 0.2, 0.5, 0.8; r = .10, .30, .50); Statistical Power Analysis for the Behavioral Sciences (2nd edition, 1988)
Brian Nosek	2015	Coordinated the Open Science Collaboration's replication of 100 psychology studies (36 percent replicated)
William Sealy Gosset	1876 to 1937	The t-test, published as 'Student' in 1908 while at the Guinness brewery
George Waddel Snedecor	1881 to 1974	Named the F ratio in Fisher's honour
Sidney Siegel	1916 to 1961	Nonparametric Statistics for the Behavioral Sciences (1956); power-efficiency
Howard Levene	1914 to 2003	Levene's test of homogeneity of variance (1960)
Henry Scheffé	1907 to 1977	Scheffé's test, the most conservative post hoc test
Carlo Emilio Bonferroni	1892 to 1960	Bonferroni correction: alpha divided by the number of comparisons
Frank Yates	1902 to 1994	Correction for continuity in 2 x 2 chi-square tables (1934)
Quinn McNemar	1900 to 1986	McNemar test for paired nominal data (1947)
Henry Berthold Mann	1905 to 2000	Mann-Whitney U test for two independent samples (1947)
Donald Ransom Whitney	1915 to 2007	Mann-Whitney U test for two independent samples (1947)
Frank Wilcoxon	1892 to 1965	Wilcoxon matched-pairs signed-rank test (1945); rank-sum test
William Kruskal	1919 to 2005	Kruskal-Wallis H test, one-way ANOVA by ranks (1952)
W. Allen Wallis	1912 to 1998	Kruskal-Wallis H test, one-way ANOVA by ranks (1952)
Milton Friedman	1912 to 2006	Friedman test for three or more related samples, two-way ANOVA by ranks (1937)
Sir Francis Galton	1822 to 1911	Described co-relations (1888); regression towards mediocrity, now regression to the mean (1886)
Michelle Iaffaldano	1985	With Muchinsky, meta-analysis: job satisfaction and performance correlate only about .17
Paul Muchinsky	1985	With Iaffaldano, meta-analysis of job satisfaction and performance
Timothy Judge	2001	With colleagues, put the satisfaction and performance correlation at about .30
Sir Ronald Aylmer Fisher	1890 to 1962	r-to-z transformation; Fisher's z-test for the difference between two correlations
Charles Spearman	1863 to 1945	Rank-difference correlation (rho) and factor analysis, both 1904
Maurice Kendall	1938	Kendall's tau, a rank correlation based on agreeing and disagreeing pairs
Adrien-Marie Legendre	1752 to 1833	Published the method of least squares (1805)
Reuben Baron	1986	With Kenny, regression tests of mediation and moderation
David Kenny	1986	With Baron, regression tests of mediation and moderation
Michael Sobel	1982	Sobel test of an indirect (mediated) effect
Sewall Wright	1889 to 1988	Path analysis (1921)
Harold Hotelling	1933	Principal components analysis
Henry Kaiser	1958	Varimax rotation; the eigenvalue-above-1 rule; KMO measure
Raymond Cattell	1905 to 1998	Scree test for the number of factors (1966)
John Horn	1965	Parallel analysis for the number of factors
Karl Jöreskog	1969	Confirmatory factor analysis

Key terms

method of tenacity, method of authority, a priori method, elenchus, Empiricism, Replicability, Parsimony, Falsifiability, paradigm, Induction, Deduction, Abduction, hypothetico-deductive method, Research methods, Research methodology, Basic research, applied research, Action research, Exploratory research, Descriptive research, Explanatory research, nomothetic approach, idiographic approach, research problem, unsolvable problem, literature, review of literature, grey literature, hypothesis, null hypothesis, alternative hypothesis, directional hypothesis, non-directional hypothesis, universal hypothesis, existential hypothesis, variable, construct, independent variable, dependent variable, active variables, attribute variables, extraneous variable, confounding variable, moderator, mediator, intervening variable, operational definition, floor effect, ceiling effect, Control, MAXMINCON principle, order effects, Counterbalancing, randomisation, Statistical control, Bias, Hawthorne effect, Demand characteristics, evaluation apprehension, experimenter expectancy effect, Pygmalion effect, halo effect, Placebo effect, double-blind, Statistical conclusion validity, Internal validity, Construct validity, External validity, ecological validity, threat to internal validity, population, target population, accessible population, element, sampling unit, sampling frame, sample, census, parameter, statistic, Probability sampling, non-probability sampling, Mixed sampling, Simple random sampling, sampling with replacement, Systematic sampling, periodicity, Stratified random sampling, proportionate stratified sampling, disproportionate stratified sampling, Optimum allocation, Cluster sampling, multistage sampling, multiphase sampling, panel sampling, Convenience sampling, purposive sampling, Quota sampling, snowball sampling, saturation sampling, dense sampling, Sampling error, Non-sampling error, sampling bias, experiment, random assignment, control group, method of difference, replication, local control, pretest sensitisation, Solomon four-group design, quasi-experiment, between-subjects design, randomised block design, repeated-measures design, Latin square, crossover design, factorial design, main effect, interaction effect, field experiment, field study, single-subject design, reversal design, multiple-baseline design, correlational research, third-variable problem, survey research, ex post facto research, external criticism, internal criticism, longitudinal design, cross-sectional design, cohort effect, cross-sequential design, epidemiology, prevalence, point prevalence, incidence, morbidity rate, mortality rate, ecological fallacy, case-control study, cohort study, prospective cohort, retrospective cohort, relative risk, odds ratio, attributable risk, absolute risk reduction, number needed to treat, sensitivity, specificity, positive predictive value, base rate, receiver operating characteristic (ROC) curve, likelihood ratios, randomised controlled trial, allocation concealment, intention-to-treat analysis, per-protocol analysis, analogue research, systematic review, meta-analysis, publication bias, file-drawer problem, hierarchy of evidence, CONSORT, PRISMA, Primary data, Secondary data, participant observation, reactivity, observer bias, structured interview, focus group, open-ended question, ethnographic interview, questionnaire, schedule, Double-barrelled question, funnel sequence, Content analysis, Manifest content, Sociometry, sociogram, Verstehen, ontology, epistemology, Positivism, Postpositivism, Constructivism, Qualitative research, symbolic interactionism, Ethnography, emic, thick description, Grounded theory, axial coding, Phenomenology, bracketing, trustworthiness, Triangulation, Mixed-methods research, Nuremberg Code, Declaration of Helsinki, Belmont Report, APA Ethics Code, ICMR National Ethical Guidelines, institutional ethics committee, institutional review board, Informed consent, consent form, assent, Voluntary participation, right to withdraw, Protection from harm, risk-benefit analysis, confidentiality, Anonymity, Vulnerable populations, Deception, debriefing, Dehoaxing, desensitising, 3Rs, Research misconduct, fabrication, falsification, plagiarism, citation, Referencing, research report, References, bibliography, research proposal, Assumptions, Limitations, Delimitations, levels of measurement, nominal scale, ordinal scale,

interval scale, ratio scale, categorical variable, discrete variable, continuous variable, Editing, Coding, Classification, Tabulation, frequency distribution, exact limits, Cumulative frequency, Relative frequencies, bar graph, histogram, frequency polygon, ogive, scatter plot, measure of central tendency, mean, median, mode, quartile deviation, variance, standard deviation, coefficient of variation, Skewness, Kurtosis, normal probability curve, z-score, percentile rank, standard scores, Descriptive statistics, Inferential statistics, biostatistics, Classical probability, Empirical probability, addition rule, multiplication rule, conditional probability, binomial distribution, sampling distribution of the mean, standard error of the mean, central limit theorem, law of large numbers, point estimate, confidence interval, significance level, critical value, two-tailed test, one-tailed test, degrees of freedom, p-value, q-value, Type I error, Type II error, Power, Power analysis, effect size, Cohen's d, Eta squared, partial eta squared, clinical significance, p-hacking, replication crisis, parametric test, non-parametric test, power-efficiency, Shapiro-Wilk test, homoscedasticity, Levene's test, sphericity, t distribution, one-sample t-test, independent-samples t-test, Welch's t-test, paired-samples t-test, family-wise error rate, Analysis of variance, one-way ANOVA, mean square, F ratio, F distribution, Post hoc tests, Planned contrasts, Bonferroni correction, factorial ANOVA, Repeated-measures ANOVA, Analysis of covariance, homogeneity of regression slopes, Multivariate analysis of variance, chi-square test, test of independence, McNemar test, Mann-Whitney U test, Wilcoxon matched-pairs signed-rank test, Kruskal-Wallis H test, Friedman test, correlation, Pearson product-moment correlation coefficient, scatterplot, coefficient of determination, heteroscedasticity, autocorrelation, restriction of range, Fisher's z-test, Spearman's rho, partial correlation, semi-partial correlation, multiple correlation, regression to the mean, regression coefficient, method of least squares, standard error of estimate, beta weights, multicollinearity, variance inflation factor, logistic regression, factor analysis, simple structure, factor loading, communality, eigenvalue, Bartlett's test of sphericity, principal components analysis, Kaiser-Guttman rule, scree test, parallel analysis, orthogonal rotation, oblique rotation, second-order factor, exploratory factor analysis, confirmatory factor analysis

From the past papers

1. 'Reinforcement will have positive effect on performance,' which type of hypothesis is this? (RML 2024)

   (a) Null and directional (b) Alternate and directional (c) Null and non-directional (d) Alternate and non-directional

   Answer: (b) Alternate and directional. The statement predicts an effect, so it is the alternative hypothesis (the null would predict no effect), and it says which way the effect goes (positive), so it is directional and would be tested one-tailed.

2. Nomothetic approach of study is involved in (JEMAS 2025)

   (a) Qualitative method (b) Experimental method (c) Quasi-experimental method (d) Ethnography

   Answer: (b) Experimental method. The nomothetic approach seeks general laws across many people, and the experiment is its purest method. Quasi-experiments are nomothetic too but are the weaker choice; qualitative methods and ethnography study particular cases in depth and are idiographic.

3. In which method does the researcher start with a theory and then test it with observations? (RML 2023)

   (a) Inductive reasoning (b) Deductive reasoning (c) Abductive reasoning (d) Hypothetico-deductive method

   Answer: (b) Deductive reasoning. Deduction runs top-down from theory to hypothesis to observation. Induction runs the other way (observations to theory), abduction infers the best explanation for a surprising fact, and the hypothetico-deductive method is the whole cycle of conjecture and test, not the direction of reasoning asked for.

4. Maximization of ____ variance, minimization of ____ variance, control of ____ variance. Select the appropriate option. (JEMAS 2025)

   (a) secondary, true, error (b) true, primary, secondary (c) primary, error, secondary (d) primary, true, error

   Answer: (c) primary, error, secondary. This is Kerlinger's MAXMINCON principle: maximise primary (experimental) variance due to the IV, minimise error variance, control secondary (extraneous) variance. True variance belongs to test theory, not to this principle.

5. Which of the following can help control experimenter's bias? (RML 2022)

   (a) Double-blinding (b) Randomly assigning subjects to groups (c) Using control groups (d) Demand characteristics

   Answer: (a) Double-blinding. When the experimenters who treat and assess participants do not know who is in which condition, their expectations cannot shape the results. Random assignment equalises groups and a control group gives a comparison, but the experimenter still knows who got what; demand characteristics are a source of bias, not a control.

6. A researcher is interested in the impact of anxiety on performance. To manipulate anxiety, some subjects were given plain chips to eat while others were given barbecue flavoured chips to eat. The most obvious problem with this experiment is a lack of: (RML 2022)

   (a) Statistical Validity (b) Empirical Validity (c) Internal Validity (d) Construct Validity

   Answer: (d) Construct Validity. Chip flavour does not manipulate anxiety, so the operation does not represent the construct named. No rival variable explains the result (internal validity), and statistical or empirical validity (an old name for a test's criterion-related validity) do not name this flaw.

7. A researcher wants to study the prevalence of test anxiety among all school students in India. Due to feasibility, they first randomly select 5 districts from each state, then randomly select 10 schools from each district, and finally randomly select 30 students from each school. This sampling technique is known as: (Real PYQ 2024)

   (a) Stratified Random Sampling (b) Quota Sampling (c) Cluster Sampling (d) Purposive Sampling

   Answer: (c) Cluster Sampling. Intact groups (districts, then schools) are chosen at random at successive levels, which is multistage cluster sampling. Stratified sampling would draw individuals from every stratum, and quota and purposive sampling involve no random selection at all.

8. Which of the following sampling methods relies on the researcher's judgment to select participants who are "typical" or "representative" of the population, and does NOT allow for the calculation of sampling error? (Real PYQ 2024)

   (a) Simple Random Sampling (b) Stratified Random Sampling (c) Quota Sampling (d) Cluster Sampling

   Answer: (c) Quota Sampling. It is the only non-probability method offered, so its sampling error cannot be calculated; the other three are probability methods. Strictly, a judgement-based choice of typical cases describes purposive sampling, which would be the better answer if it were offered.

9. A statistical measure based upon a population is known as: (IHBAS 2020)

   (a) Statistic (b) Parameter (c) Sample (d) None of the above

   Answer: (b) Parameter. A parameter (Greek letters such as mu and sigma) describes a population; a statistic (Roman letters) is computed from a sample, and a sample is a group of people, not a measure.

10. In experimental design, the principle that ensures every participant has an equal chance of being assigned to any group is: (Real PYQ 2025)

    (a) Replication (b) Local control (c) Randomization (d) Homogeneity

    Answer: (c) Randomization. Randomisation, one of Ronald Fisher's three principles (*The Design of Experiments*, 1935), gives every participant an equal chance of assignment to any group. Replication and local control (blocking) are the other two principles; homogeneity is not one of them.

11. Researchers studying meditation contacted over 10000 randomly selected adults in 10 nations. The results showed that those who meditated on a regular basis were less likely to suffer from stress than adults who meditate less regularly or not at all. Can we conclude that meditation lowers stress? Why? (RML 2021)

    (a) Yes, because the sample size was large (b) Yes, because the adults were randomly selected (c) Yes, because the adults were randomly assigned (d) Not necessarily, because this wasn't an experiment

    Answer: (d) Not necessarily, because this wasn't an experiment. The researchers only measured who already meditated; nobody was randomly assigned, so meditators may differ in other ways that lower stress. A large or randomly selected sample helps generalisation, not causal inference, and option (c) describes something the study did not do.

12. Longitudinal and cross-sectional designs can be used to study how behaviour changes as a function of age. A disadvantage of cross-sectional studies is that the result can be influenced by : (IHBAS 2024)

    (a) Random assignment (b) Socio-economic factors (c) Cohort difference (d) Participant attrition

    Answer: (c) Cohort difference. A cross-sectional study compares different age groups at one time, so generational (cohort) differences can masquerade as age changes. Attrition is the problem of longitudinal studies, random assignment plays no part in age comparisons, and socio-economic differences are a general confound rather than the design's characteristic flaw.

13. A famous study examined the smoking habits of British doctors and whether or not the doctors development various types of cancers. Participants responded to questionnaires about their smoking habits and health in 1951, 1957, 1966 and many more times all the way up to 2001. What type of observation study is this? (RML 2021)

    (a) A retrospective observational study (b) A prospective observational study (c) A sample survey (d) None of these. It's an experimental study.

    Answer: (b) A prospective observational study. This is Doll and Hill's British Doctors Study. Nobody was assigned to smoke, so it is observational rather than experimental, and the doctors were followed forward from exposure (smoking in 1951) to outcome, so it is a prospective cohort, not a retrospective one. A sample survey describes a population at one point in time.

14. What is the main purpose of using meta-analysis in clinical psychology? (IHBAS 2024)

    (a) To summarize findings from multiple studies (b) To conduct a single large experiment (c) To analyze individual case studies (d) To collect primary data

    Answer: (a) To summarize findings from multiple studies. The paper's key gives (b), which is wrong: a single experiment, however large, is primary research. Meta-analysis statistically pools the effect sizes of many existing studies; it neither collects primary data nor analyses individual case studies.

15. In a cohort study investigating the association between a new psychological intervention and relapse rates in schizophrenia, the risk ratio is calculated as 1.0. This finding indicates that: (Real PYQ 2024)

    (a) The incidence of relapse in the exposed group is double that of the unexposed group. (b) The intervention has a significant protective effect against relapse. (c) There is no difference in the risk of relapse between the intervention and control groups. (d) The odds of exposure are equal in both the relapse and non-relapse groups.

    Answer: (c) There is no difference in the risk of relapse between the groups. A risk ratio divides the risk in one group by the risk in the other, so 1.0 means equal risks. Double the incidence would be 2.0 and a protective effect below 1.0; (d) is how a case-control study's odds ratio is phrased, not what a cohort's risk ratio measures.

16. 'Can you tell me why this band is so popular among your age group?' is an example of: (RML 2020)

    (a) psychodynamic inquiry (b) a task-focused inquiry (c) an ethnographic inquiry (d) a strength-based inquiry

    Answer: (c) An ethnographic inquiry. The question treats the teenager as an informant and asks them to explain the culture of their own peer group, as in Spradley's ethnographic interview. Psychodynamic inquiry probes feelings and hidden motives, task-focused inquiry a defined task, and strength-based inquiry the person's resources.

17. Uniting various qualitative methods with quantitative methods can be called as: (IHBAS 2020)

    (a) Coalition (b) Triangulation (c) Multidimensional approach (d) Co-operative inquiry

    Answer: (b) Triangulation. Combining methods so that each one's errors are offset by the others is Denzin's (1970) methodological triangulation. Coalition is not a research term, a multidimensional approach merely studies several aspects of a problem, and co-operative inquiry is Heron's participatory method.

18. Which one of the following does not apply to qualitative research? (IHBAS 2020)

    (a) Recognition of the role of language and discourse (b) Emphasis on objective truth (c) Understanding of processes (d) Concern for human individuality

    Answer: (b) Emphasis on objective truth. Qualitative research rests on constructivism, which sees realities as constructed and multiple; language and discourse, processes and individuality are its hallmarks, while objective truth belongs to positivist, quantitative work.

19. 'The ethical principle that research participants should be told enough about a piece of research to be able to make a decision about whether to participate in it.' Which research principle is this a glossary definition of? (RML 2021)

    (a) Informed consent (b) Debriefing (c) Expectation management (d) De-concealment

    Answer: (a) Informed consent. Being told enough to decide, before taking part, is informed consent. Debriefing comes after participation; expectation management and de-concealment are not research-ethics principles (revealing a deception is called dehoaxing).

20. The principle of ethical research under which the subjects should be given full information about all aspects of the research study after it is done is known as: (IHBAS 2020)

    (a) Debriefing (b) Description (c) Explanation (d) Interpretation

    Answer: (a) Debriefing. Debriefing gives full information after participation, including any deception (dehoaxing) and help with distress (desensitising). Description and explanation are goals of science, and interpretation belongs to a report's Discussion.

21. Which of the following is NOT a form of plagiarism? (RML 2023)

    (a) Collusion (b) Citation (c) Patchwriting (d) Ghost-writing

    Answer: (b) Citation. Citing openly acknowledges a source and gives credit to its authors, so it is the cure for plagiarism. Collusion (unauthorised joint work), patchwriting (lightly reworded source text) and ghost-writing (work written by someone else) are all plagiarism.

22. Arbitrary zero is the characteristic of which scale? (RML 2024)

    (a) Ratio scale (b) Ordinal scale (c) Interval scale (d) Nominal scale

    Answer: (c) Interval scale. The source keys (a), which is wrong: an interval scale such as Celsius has equal intervals but an arbitrary zero, so ratios are meaningless, while a ratio scale is defined by a true zero. Ordinal and nominal scales have no meaningful zero or equal intervals at all.

23. A person wants to determine the representative income of the people living in a community. He would be best advised to use the (JEMAS 2025)

    (a) Mean (b) Median (c) Mode (d) Standard deviation

    Answer: (b) Median. Income is positively skewed: a few very rich households pull the mean upward, while the median, depending only on position, stays with the typical family. The mode is unstable, and the standard deviation measures spread, not central tendency.

24. In a negatively skewed distribution, the position of the mean with reference to the median is: (RML 2022)

    (a) Left of the median (b) Right of the median (c) Same as median (d) Same as a mode

    Answer: (a) Left of the median. In negative skew the long tail points to low scores and drags the mean furthest into it, so from low to high the order is mean, median, mode. The mean lies right of the median in positive skew, and the three coincide only in a symmetrical distribution.

25. Type II error occurs : (RML 2022)

    (a) whenever a true null hypothesis is accepted (b) whenever a false null hypothesis is rejected (c) whenever a false null hypothesis is accepted (d) whenever a true null hypothesis is rejected

    Answer: (c) whenever a false null hypothesis is accepted. A Type II error (β) misses a real effect: a false negative. Rejecting a true null (d) is a Type I error; (a) and (b) are the two correct decisions.

26. The statistical theorem states that given a sufficiently large sample size, the sampling distribution of the mean for a variable will be approximately normally distributed, regardless of that variable's distribution in the Population. What is this definition of? (RML 2021)

    (a) The normal distribution theorem (b) The central limit theorem (c) The sampling distribution theorem (d) The law of large numbers

    Answer: (b) The central limit theorem. With N of about 30 or more, the distribution of sample means is near normal whatever the population's shape. The law of large numbers says only that a sample mean approaches the population mean as N grows; (a) and (c) are not standard theorems.

27. What is the p-value in statistical testing? (IHBAS 2024)

    (a) The probability of the null hypothesis being true (b) The probability of obtaining the observed data if the null hypothesis is true (c) The effect size of the experiment (d) The power of the study

    Answer: (b). p is the probability of data at least this extreme *assuming* H0 is true. Option (a) reverses the conditional; effect size and power (1 − β) are separate quantities.

28. Which non-parametric test is equivalent to the independent samples t-test? (RML 2023)

    (a) Mann-Whitney U test (b) Kruskal-Wallis test (c) Wilcoxon signed-rank test (d) Friedman test

    Answer: (a) Mann-Whitney U test. It ranks the pooled scores of two independent groups and is the rank partner of the independent t-test. Wilcoxon signed-rank is for two related samples, Kruskal-Wallis for three or more independent groups, and Friedman for three or more related samples.

29. Which of the following is a pooled variance estimate that constitutes the denominator of an F-ratio? (RML 2021)

    (a) The between-cells mean square (MSB). (b) The mean of the sampling distribution. (c) The sum of the deviations from the grand mean. (d) The within-cells mean square (MSw).

    Answer: (d) The within-cells mean square (MSw). F = MS between / MS within, and MS within pools the error variance inside the groups. MS between is the numerator; deviations from the grand mean always sum to zero, which is why they are squared; the mean of the sampling distribution plays no part in F.

30. A researcher wants to calculate the degrees of freedom (df) for a 3 x 4 contingency table in a Chi-square test. The df is: (Real PYQ 2025)

    (a) 12 (b) 7 (c) 6 (d) 11

    Answer: (c) 6. For a test of independence df = (rows - 1)(columns - 1) = 2 x 3 = 6. Twelve multiplies the raw dimensions, 7 adds them, and 11 is the number of cells minus 1.

31. A researcher uses a Scree plot to determine the number of factors to retain in an EFA. The plot shows a sharp drop from eigenvalue 3.5 to 1.8, followed by a gradual plateau where eigenvalues are 0.9, 0.7, and 0.5. According to the Kaiser-Guttman rule and the Scree plot, how many factors should ideally be retained? (Real PYQ 2024)

    (a) 5 (b) 3 (c) 2 (d) 1

    Answer: (c) 2. Only 3.5 and 1.8 exceed the Kaiser-Guttman cut-off of 1.0, and both lie above the elbow at the third factor, so the two rules agree. Three wrongly keeps the elbow factor, five counts every eigenvalue and one keeps only the largest.

32. When two variables that are being used as predictors in a statistical analysis have a high correlation with each other, this statistical problem is known as: (Real PYQ 2025)

    (a) Heteroscedasticity (b) Multicollinearity (c) Autocorrelation (d) Homogeneity of variance

    Answer: (b) Multicollinearity. Highly intercorrelated predictors make regression weights unstable and inflate their standard errors (flagged by a VIF above 10). Heteroscedasticity is unequal residual spread, autocorrelation is residuals correlated over time, and homogeneity of variance is equal variance across groups.

33. Which of the following correlation methods is used when one variable is continuous and the other is dichotomous? (RML 2023)

    (a) Pearson correlation coefficient (b) Spearman's rank correlation coefficient (c) Point-biserial correlation coefficient (d) Biserial correlation coefficient

    Answer: (c) Point-biserial correlation coefficient. A continuous variable paired with a true (natural) dichotomy calls for point-biserial. Biserial is for an artificial dichotomy cut from a continuous variable, Pearson for two continuous variables and Spearman for ranks.

High-yield sheet

The facts this chapter's questions ask most, most-asked first. Last read before the exam.

1 A **z-score**, (X − M)/SD, states a score's distance from the mean in SD units (mean 0, SD 1) so scores from different distributions can be compared; on the normal curve z = 0, +1, +1.5 and +2 lie at about the 50th, 84th, 93rd and 98th percentiles, so deviation IQs of 85, 100 and 130 sit at the 16th, 50th and 98th. ASKED 12 TIMES

2 A **Type I error** (α) is rejecting a true null hypothesis (false positive); a **Type II error** (β) is retaining, or accepting, a false null hypothesis (false negative). The significance level α is the probability of a Type I error, in ANOVA as in every test. ASKED 11 TIMES

3 **Stevens (1946)**: nominal = unordered categories (marital status, car colour); ordinal = rank order (preference or agreement); interval = equal intervals with an **arbitrary zero** (Celsius); ratio = equal intervals with a **true zero** (height, number of books). ASKED 9 TIMES

4 **One-way ANOVA** compares the means of three or more independent groups on one IV with a continuous DV (it also works for two, where F = t squared); repeated t-tests would inflate the family-wise Type I error. ASKED 9 TIMES

5 The **Mann-Whitney U test** (Mann and Whitney, 1947; the same as the Wilcoxon rank-sum test) is the non-parametric equivalent of the independent-samples t-test; Wilcoxon signed-rank partners the paired t, Kruskal-Wallis the one-way ANOVA and Friedman the repeated-measures ANOVA. ASKED 9 TIMES

6 The **p-value** is the probability of obtaining the observed result, or one more extreme, if H0 is true; it is not the probability that H0 is true. If $p \leq \alpha$ ($p = .03$ with $\alpha = .05$), reject H0. ASKED 8 TIMES

7 **Chi-square** (Karl Pearson, 1900) is a non-parametric test for nominal frequency data; the test of independence asks whether two categorical variables are associated, with df = (rows - 1)(columns - 1): 3 x 4 gives 6, 4 x 5 gives 12. ASKED 8 TIMES

8 The **standard deviation** is the square root of the variance and measures variability; it is affected by extreme scores (the IQR and quartile deviation are not), and multiplying every score by c multiplies the variance by c^2 while adding a constant changes nothing (variance 4, scores × 3 + 10: variance 36). ASKED 7 TIMES

9 **Cluster sampling** randomly selects whole natural groups (districts, schools) and studies all or some units inside them; districts, then schools, then students is multistage cluster sampling, whereas **stratified random sampling** samples randomly within every stratum. ASKED 6 TIMES

10 In **non-probability sampling** (convenience, purposive, quota, snowball) the chance of selection is unknown, randomisation is impossible and sampling error cannot be calculated; quota sampling, the commonest option-list answer, fills proportional quotas non-randomly. ASKED 6 TIMES

11 The **normal curve** is symmetrical and unimodal with mean = median = mode, and about 68, 95 and 99.7 per cent of cases lie within ±1, ±2 and ±3 SD of the mean; a bimodal curve is not normal. ASKED 6 TIMES

12 The **median** is the average for skewed data (community income, anxiety scores), the **mode** the only one for nominal data; range, SD and mean deviation are measures of variability, not central tendency. ASKED 6 TIMES

13 **Cohen's d**, the standardised difference between two means, is small at 0.2, medium at 0.5 and large at 0.8; for r the benchmarks are .10, .30 and .50, so r = −.30 is a medium negative relationship. ASKED 6 TIMES

14 **F = MS between / MS within**; MS within, the pooled error variance, is the denominator. F is about 1 when H0 is true, so an F below 1 (0.99) is never significant; F rises as chance variability falls, can never be negative, and the test is one-tailed. ASKED 6 TIMES

15 **Post hoc tests** (Tukey's HSD, Scheffé the most conservative, Newman-Keuls, Duncan, Fisher's LSD the least conservative) are used only after a significant F; k levels give k(k - 1)/2 pairs (4 levels give 6); the **Bonferroni correction** divides alpha by the number of comparisons to control family-wise Type I error. ASKED 6 TIMES

16 **Induction** moves from specific observations to broad generalisations and builds theory (the theory-development stage; typical of qualitative research); **deduction** starts with a theory and tests it against observations (typical of quantitative, experimental research). ASKED 5 TIMES

17 **Degrees of freedom**: one-sample t = N - 1 (N 20 gives 19); independent t = N1 + N2 - 2 (25 per group gives 48); one-way ANOVA within = N - k (3 groups of 10 give 27); randomised block error = (treatments - 1)(blocks - 1) (4 and 5 give 12). ASKED 5 TIMES

18 **r = -0.85** is a strong negative relationship: the sign of *r* gives the direction and its size gives the strength. ASKED 5 TIMES

19 A **double-blind** study keeps both the participants and the experimenters who treat and assess them unaware of the conditions, so it controls both experimenter bias and participant expectancy effects. ASKED 4 TIMES

20 In **probability sampling** every element has a known, non-zero chance of selection, so sampling error can be calculated and the sample is likely, not guaranteed, to be representative. ASKED 4 TIMES

21 Only a true **experiment** (manipulation, control, random assignment) licenses cause-and-effect conclusions; correlational and observational studies cannot, however large or randomly selected the sample, because of the directionality and third-variable problems. ASKED 4 TIMES

22 **NNT = 1/ARR**, where the absolute risk reduction is the control event rate minus the treated event rate; an NNT of 8 means treating 8 patients for one additional patient to benefit. ASKED 4 TIMES

23 **Sensitivity** = TP/(TP + FN), the true positive rate; **specificity** = TN/(TN + FP). Sensitivity 90 percent and specificity 70 percent means the test correctly identifies 90 percent of ill and 70 percent of healthy people. ASKED 4 TIMES

24 **Qualitative research** is naturalistic, context-specific and inductive (bottom-up evidence), seeks meaning, process and the insider's view, and claims neither objective truth nor the researcher's authority; the **survey** is quantitative, whereas case study, ethnography and grounded theory are qualitative. ASKED 4 TIMES

25 **Random assignment** (randomisation) gives every participant an equal chance of being placed in any group, making groups equivalent on known and unknown variables so the main difference is the treatment; it controls selection bias and confounding and is not random sampling. ASKED 3 TIMES

26 The **independent variable** is the presumed cause, manipulated or selected (having heart disease in 'residents with heart disease die earlier'); the **dependent variable** is the measured outcome expected to change. ASKED 3 TIMES

27 **Stratified random sampling** divides the population into mutually exclusive strata and samples randomly within each; it is preferred when important subgroups must be represented and is more precise than simple random sampling when strata are internally homogeneous. ASKED 3 TIMES

28 A **longitudinal design** studies the same people repeatedly over time and is the best design for examining change over time; its problems are attrition, practice effects, cost and time. ASKED 3 TIMES

29 A **relative risk or odds ratio of 1.0** means no association between exposure and outcome; above 1 marks a risk factor, below 1 a protective one. ASKED 3 TIMES

30 **Intention-to-treat analysis** analyses every participant in the group to which they were randomised, regardless of adherence, drop-out or crossover, and so preserves randomisation.
ASKED 3 TIMES

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